The size Ramsey number of a pair of star forests: a proof of the Burr–Erdős–Faudree–Rousseau–Schelp formula (Erdős Problem #561)

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Abstract

We prove that for all positive integers s, t and nonincreasing lists $n_1 \geq \cdots \geq n_s \geq 1$ and $m_1 \geq \cdots \geq m_t \geq 1$, the size Ramsey number of the pair of star forests $F_1 = \bigsqcup_{i=1}^{s} K_{1,n_i}$ and $F_2 = \bigsqcup_{j=1}^{t} K_{1,m_j}$ equals $\sum_{k=2}^{s+t} \ell_k$, where $\ell_k = \max\{n_i + m_j - 1 \mid 1 \leq i \leq s,\ 1 \leq j \leq t,\ i + j = k\}$; this settles Erdős Problem #561 [EP561] and the conjecture of Burr, Erdős, Faudree, Rousseau, and Schelp [BEF+78]. The upper bound comes from a disjoint union of stars indexed by antidiagonals of the two component lists. For the lower bound, certificate costs reduce a host below the proposed edge budget to a minimal obstruction of maximum degree 4, 6, 8, 10, or 12, and we exclude all five possibilities. The complete proof is published alongside this paper as a verified record of 315 statements in 26 chapters [Ans26]; this paper proves the reduction and the assembly in full and states every result it cites from the record. The main result of this paper was obtained with the assistance of the Ansatz system, an automated proof-search and verification system.

AI-use disclaimer

This paper uses AI models to assist with the generation, development, and presentation of mathematical arguments. AI-generated content may contain errors or omissions, and automated review does not by itself establish mathematical correctness. The authors are responsible for the final manuscript, and its mathematical claims remain subject to independent verification.

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Reading this paper. Sections 1–7 contain the main exposition. The complete 315-item proof record cited there as [Ans26] is reproduced in Appendix A, with its original item numbering and a separate contents list. References to the record's original page count and chapter descriptions are retained from the supplied texts.

1 Introduction

The problem. The size Ramsey problem for two star forests asks how many edges force one of two prescribed monochromatic forests. The proposed answer was formulated in [BEF+78, p. 194] and is recorded as Erdős Problem #561 in [EP561]. It depends on the full ordered lists of component sizes. The uniform case, in which each of the two lists is constant, was proved in [BEF+78, Theorem 1]. Further cases appear in [GySc02, DJK+21]. These results motivate the formula stated below.

Definition 1.1 (Size Ramsey number)**.** For finite simple graphs H, G_1, G_2, write $H \to (G_1, G_2)$ if every function $E(H) \to \{\text{red}, \text{blue}\}$ gives a red subgraph isomorphic to G_1 or a blue subgraph isomorphic to G_2. Copies are ordinary subgraphs and need not be induced. The size Ramsey number $\widehat{R}(G_1, G_2)$ is the least nonnegative integer $e(H)$ among such hosts, whenever the set of hosts is nonempty. Write $\widehat{R}(G) = \widehat{R}(G, G)$. For a positive integer h, the star $K_{1,h}$ has one center, h distinct leaves, and exactly the h edges joining the center to the leaves.

The formula. The following is the asserted general formula, with both component lists retained in full.

Theorem 1.2 (Size Ramsey number of unions of stars)**.** *Let s, t be positive integers, and let*

$$n_1 \geq \cdots \geq n_s \geq 1, \qquad m_1 \geq \cdots \geq m_t \geq 1$$

be lists of positive integers. Let

$$F_1 = \bigsqcup_{i=1}^{s} K_{1,n_i}, \qquad F_2 = \bigsqcup_{j=1}^{t} K_{1,m_j},$$

where each union is vertex-disjoint. For every integer $2 \leq k \leq s + t$, set

$$\ell_k = \max\{n_i + m_j - 1 : 1 \leq i \leq s,\ 1 \leq j \leq t,\ i + j = k\}.$$

Then

$$\widehat{R}(F_1, F_2) = \sum_{k=2}^{s+t} \ell_k. \tag{1.1}$$

The statement includes repeated component sizes, stars with one edge, and targets consisting of a single component. The upper-bound construction in Theorem 2.5 also establishes the existence of the minimum in Definition 1.1. The final assembly of the equality is given in Section 6; its accompanying certificate statement is [Ans26, Chapter 26].

Remark 1.3 (Proofs in the accompanying record). Theorem 1.2 is proved in the accompanying record [Ans26, Chapter 26]. The arguments below distinguish proofs reproduced here from auxiliary results cited to that record. The improving certificate construction used for degrees eight, ten, and twelve is proved in [Ans26, Chapter 21].

The record [Ans26] presents the 315 statements in dependency order with complete proofs; each chapter citation in this paper refers to the chapter of the record that proves the cited statement.

Relation to known cases. Burr, Erdős, Faudree, Rousseau and Schelp [BEF+78, Theorem 1] proved the formula when both forests are uniform, that is, when all components of each forest have the same size, and conjectured it in general. Further special cases were established by Győri and Schelp [GySc02] (as restated in [DJK+21, Theorem 1.3], under a largeness condition on the components) and by Davoodi, Javadi, Kamranian and Raeisi [DJK+21, Theorems 2.3–2.6], whose results carry additional hypotheses on the component sizes (for instance all components of size at least two, or all component sizes odd). Fu, Luo and Ni [FLN26, Theorem 1.5] classified the size-Ramsey-minimal graphs for uniform star forests. The general formula remained open; see [EP561].

Method. The upper bound uses one host star for each antidiagonal $i + j = k$. In a coloring of that star, either the next required red component or the next required blue component can be selected. Selections lie in distinct host components. After at most $s + t - 1$ selections, one target is complete. This gives a host with precisely the edge count in (1.1).

The lower bound uses integer certificate costs, defined in Section 3. A certificate partitions all edges into two parts of bounded maximum degree and two parts of bounded matching number. For fixed degree capacities, its cost is the least total matching budget for which every division of that total is feasible. The maximum cost over capacity pairs with a fixed sum gives a cost row. The sum of the rows is compared with the edge count through the excess, positive-level slack, and signed gap. Proposition 3.4 preserves the integer allocation condition in a finite representation. Theorem 3.7 proves that the positive-level slack is nonnegative.

The reduction must retain the original host until the component estimates have been applied. For a host below the appropriate tail budget, the small-deficit estimate forces a matching deficit of at least four. Component localization then gives an actual component of positive gap and excess at least four. Hereditary goodness implies that this component contains a failure of the property T_- from Definition 3.8. Choosing a smallest failing subgraph gives minimality under both edge and vertex deletions. Theorem 4.16 carries out this choice and its consequences.

The structural results identify such a minimal failure as a connected, regular graph of odd order with a perfect matching after every vertex deletion. Its degree is even. Large-degree goodness and the gap-two restriction leave only degrees $4, 6, 8, 10, 12$. The excess estimate on the original component is not transferred to the contained minimal graph. This distinction is retained in both the statement and proof of Theorem 4.16.

The five exclusions are assembled in Section 5. At degrees eight, ten, and twelve, the quoted construction supplies certificates at a total one below the row cost it is supposed to attain. The contradiction concerns all capacity splits and all budget splits on the same graph. At degree six, the excess and the associated integer parameters separate the argument into three exhaustive branches. At degree four, a uniform family of independent-set frames has remainders containing an odd cycle, while the auxiliary restrictions exclude every allowed odd cycle. The construction used in the first mechanism is proved in [Ans26, Chapter 21].

Organization. Section 2 fixes the graph conventions, proves normalization, and gives the upper bound. Section 3 develops the finite costs and the slack inequality. Section 4 gives the structural reduction and contained-obstruction theorem. Section 5 states and assembles all five exclusions. Section 6 proves the lower bound from those results and completes the main assembly. Section 7 describes how the accompanying certificate and its published inputs enter the argument.

Acknowledgements

This work was developed with the assistance of the Ansatz system.

Remark 1.4 (Automated assistance and review status). The main result of this paper was obtained with the assistance of the Ansatz system, an automated proof-search and verification system. Each statement in the accompanying record was checked mechanically by the system's verifier and reviewed independently by two automated auditors; neither the record nor this paper has been checked by a human referee. A further automated check of this paper as written confirmed its own deductions but reported that the results cited to the record are not re-proved here; the authors have chosen to publish the paper in this form, with the record serving as the certificate of those results. Because automated systems have limitations, it is possible that we have missed related references in the literature, and we welcome comments from experts.

2 Star profiles and the upper bound

In this section, we prove the normalization statements and the antidiagonal upper bound used in Theorem 1.2.

2.1 Preliminaries

We use the matching terminology of [LP86]. The graph conventions below also fix the treatment of empty graphs and of edge sets that occur in certificates.

Notation 2.1 (Graph conventions)**.** All graphs are finite and simple. Write $V(G)$, $E(G)$, and $e(G) = |E(G)|$ for the vertex set, edge set, and edge count. The maximum degree is $\Delta(G)$, with value zero for an edgeless graph, including the empty graph. A matching is a set of pairwise vertex-disjoint edges. Its maximum cardinality is $\nu(G)$. The independence number is $\alpha(G)$. A vertex cover meets every edge, and its minimum cardinality is $\tau(G)$.

For $U \subseteq V(G)$, the graph $G[U]$ is induced, and $G - U = G[V(G) \setminus U]$. For an edge e, the graph $G - e$ retains all vertices and deletes that edge. A proper subgraph may delete vertices, edges, or both. In particular, spanning edge deletions are allowed.

The maximum degree and matching number of an edge set $P \subseteq E(G)$ mean those of the spanning graph $(V(G), P)$. Both are zero for the empty edge set. All capacities, matching budgets, cost indices, and universal totals used below belong to $\mathbb{N}_0 = \{0, 1, 2, \ldots\}$. Differences of parameters, slack, and gap are allowed to be signed unless a bound is asserted.

Definition 2.2 (Positive star forests and profiles)**.** A positive star forest is a nonempty vertex-disjoint union of stars $K_{1,h}$ with $h \geq 1$. It has no isolated vertices. A profile is a nonincreasing list of positive integers. Its entries record component edge counts with multiplicity.

Setup 2.3 (Profiles, suffixes, and budgets)**.** Fix the profiles and targets of Theorem 1.2. Put

$$M = s + t - 1, \qquad L = \sum_{k=2}^{s+t} \ell_k. \tag{2.1}$$

For integers $0 \leq a < s$ and $0 \leq b < t$, let F_1^a and F_2^b be the actual suffix forests consisting of components $a+1, \ldots, s$ and $b+1, \ldots, t$, respectively. For an integer $0 \leq q < M$, put

$$\mathcal{A}_q = \{(F_1^a, F_2^b) : 0 \leq a < s,\ 0 \leq b < t,\ a + b = q\}, \qquad L_q = \sum_{k=q+2}^{s+t} \ell_k. \tag{2.2}$$

Thus the cuts delete leading whole components. The canonical q-star union is

$$\mathcal{P}_q = \bigsqcup_{k=2}^{q+1} K_{1,\ell_k},$$

with the empty union when $q = 0$. Whenever it is joined disjointly to another graph, new vertices are used.

Proposition 2.4 (Star profiles and normalization)**.** *With Setup 2.3, the following statements hold.*

1. *Every positive star forest has a unique profile. Sorting an arbitrary component enumeration preserves the isomorphism type, the arrowing problem, and the set of possible host edge counts. The budget computed from the sorted profiles is independent of the original enumeration. Repeated sizes and entries equal to one are retained.*
2. *Every maximum defining ℓ_k is over a nonempty finite set, and*
$$\ell_2 \geq \ell_3 \geq \cdots \geq \ell_{s+t} \geq 1, \qquad \ell_2 = n_1 + m_1 - 1, \qquad \ell_{s+t} = n_s + m_t - 1.$$
3. *Every $\mathcal{A}_q$ in (2.2) is nonempty and consists of pairs of nonempty forests. Each member has $s + t - q$ components in total. The budget L_q is a positive integer. In particular, $\mathcal{A}_0 = \{(F_1, F_2)\}$ and $L_0 = L$.*
4. *If G° is obtained by deleting all isolated vertices of G, then*
$$e(G^\circ) = e(G), \qquad G^\circ \to (F_1, F_2) \iff G \to (F_1, F_2).$$
Exchanging the two profiles preserves every ℓ_k, their sum L, and the possible host edge counts. It exchanges the entries of every pair in $\mathcal{A}_q$.

This is the normalization statement of [Ans26, Chapter 1].

Proof. An isomorphism maps connected components to connected components and preserves their edge counts. Two stars with the same positive edge count are isomorphic by a bijection of their leaves and a choice of centers. For a one-edge star either endpoint may serve as the center. Thus a forest is determined, up to isomorphism, by its multiset of component edge counts. That finite multiset has a unique nonincreasing enumeration. Composing a monochromatic embedding with a target isomorphism preserves the copy in either direction. Sorting therefore preserves arrowing for every host and preserves the possible edge counts. The sorted lists, and hence the budget computed from them, do not depend on the initial enumerations.

For $2 \leq k \leq s + t$, choose $i = \max(1, k - t)$ and $j = k - i$. The bounds $k - t \leq s$ and $1 \leq s$ give $i \leq s$. Both entries in the maximum defining i are at most $k - 1$, so $j \geq 1$. Also $i \geq k - t$, so $j \leq t$. The defining set for ℓ_k is therefore nonempty. It is finite and consists of positive integers.

Choose a pair attaining ℓ_{k+1}. If its first index is at least two, decreasing that index by one gives a valid pair at sum k whose value is at least ℓ_{k+1}. Otherwise its second index is at least two, and decreasing that index has the same effect. This proves the monotonicity. The only pairs at the two endpoints are $(1,1)$ and (s,t). When $s = t = 1$, the endpoint identities coincide and there is no monotonicity comparison to make.

For $0 \le q < M$, set $a = \max(0, q - (t-1))$ and $b = q - a$. Since $q \le s + t - 2$, these satisfy $0 \le a \le s - 1$ and $0 \le b \le t - 1$. They give a member of $\mathcal{A}_q$. Every legal cut leaves positive lengths $s - a$ and $t - b$, with sum $s + t - q$. The sum for L_q contains $M - q \ge 1$ positive terms. At $q = 0$, the only cuts are $a = b = 0$.

Deleting isolates removes no edges, so restriction and extension give a bijection of edge colorings of G and G°. Every vertex in either target is incident with a target edge. Its image in a copy cannot be an isolated host vertex. Hence the copies in corresponding colorings are the same.

Finally, interchanging red and blue is an involution on the colorings of a fixed host. It exchanges the two arrowing assertions. The bijections $(i,j) \mapsto (j,i)$ and $(a,b) \mapsto (b,a)$ preserve the relevant sums of indices. The first also preserves the antidiagonal values, and the second exchanges the suffix entries. □

2.2 The antidiagonal construction

Theorem 2.5 (Antidiagonal star-union upper bound)**.** *With Setup 2.3, let*

$$G^* = \bigsqcup_{k=2}^{s+t} K_{1,\ell_k}.$$

Then $G^* \to (F_1, F_2)$ and $e(G^*) = L$. Consequently the set of edge counts of arrowing hosts is nonempty and $\widehat{R}(F_1, F_2) \le L$. The assertion includes $s = 1$, $t = 1$, repeated component sizes, and stars with one edge.

The construction is recorded in [Ans26, Chapter 2].

Proof. Proposition 2.4 makes every component a finite simple star of positive integer size. The disjoint union therefore has exactly L edges.

Fix an edge coloring. Start with $i = j = 1$. Before either target is complete, maintain selected red stars of sizes $n_1, \dots, n_{i-1}$ and selected blue stars of sizes $m_1, \dots, m_{j-1}$. Require all selected stars to lie in distinct host components, with exactly the components indexed $2, \dots, i + j - 1$ used. These requirements hold initially, when both lists are empty.

If both targets are incomplete, then $1 \le i \le s$ and $1 \le j \le t$. The next component has index $k = i + j$, which lies between 2 and $s + t$. Let r_k, u_k be its red and blue edge counts. They satisfy

$$r_k + u_k = \ell_k \ge n_i + m_j - 1.$$

If $r_k \ge n_i$, select n_i red edges at the center and increase i by one. If $r_k < n_i$, integrality gives $r_k \le n_i - 1$, so $u_k \ge m_j$. Select m_j blue edges at the center and increase j by one.

Each choice uses exactly one new host component and one new target entry. It increases $i + j$ by one. The disjointness of host components preserves the disjointness of all selected stars, and the interval of used indices extends by one. Thus the requirements continue to hold until $i = s + 1$ or $j = t + 1$. These two terminal cases give a red F_1 and a blue F_2, respectively.

If $s + t - 1$ selections were made without reaching either terminal case, then $i \le s$, $j \le t$, and $i + j = 2 + (s + t - 1) = s + t + 1$, a contradiction. Every required component before termination has already been checked to have an allowed index. Hence every coloring contains a target.

The possible host edge counts form a nonempty subset of $\mathbb{N}_0$ containing L. Its least element exists and is at most L. Only positivity and the specified index ranges were used, so no additional restriction is needed for the stated boundary cases. □

3 Certificate costs

In this section, we define the integer costs used in Theorem 4.16 and prove their finite representation and nonnegative positive-level slack.

3.1 Integer allocation costs

Definition 3.1 (Certificates and universal costs)**.** For a graph H and capacities A, B, let $\mathcal{C}_{A,B}(H; x, y)$ be the set of ordered partitions (Q_R, Q_B, P_R, P_B) of $E(H)$ satisfying

$$\Delta(Q_R) \le A, \qquad \Delta(Q_B) \le B, \qquad \nu(P_R) \le x, \qquad \nu(P_B) \le y.$$

Every part may be empty. Define

$$c_{A,B}(H) = \min\{c \in \mathbb{N}_0 : \mathcal{C}_{A,B}(H; x, c-x) \neq \varnothing \text{ for every integer } 0 \le x \le c\}, \tag{3.1}$$

$$g_{A,B,H}(x) = \min\{\nu(P_B) : (Q_R, Q_B, P_R, P_B) \text{ partitions } E(H),\ \Delta(Q_R) \le A,\ \Delta(Q_B) \le B,\ \nu(P_R) \le x\}. \tag{3.2}$$

The existence of these minima is part of Proposition 3.4.

A certificate table of row index θ and total T means a choice of an element of $\mathcal{C}_{A,B}(H; x, y)$ for every $A + B = \theta$ and every $x + y = T$. Its cost is $\mathrm{cost}(\mathcal{Q}) = T$. All its entries concern the entire edge set of the same graph H.

Definition 3.2 (Rows, excess, slack, and gap)**.** For $\theta \in \mathbb{N}_0$, put

$$c^*_\theta(H) = \max_{A+B=\theta} c_{A,B}(H), \qquad c_\theta(H) = c^*_\theta(H), \qquad c_0(H) = c_{0,0}(H).$$

Define

$$\Sigma(H) = \sum_{\theta \ge 0} c_\theta(H), \qquad \mathrm{ex}(H) = c_0(H) - \nu(H), \tag{3.3}$$

$$\mathrm{ps}(H) = e(H) - \nu(H) - \sum_{\theta \ge 1} c_\theta(H), \qquad \mathrm{gap}(H) = \Sigma(H) - e(H). \tag{3.4}$$

The finiteness of the sums is proved below. At this point $\mathrm{ps}(H)$ and $\mathrm{gap}(H)$ are signed integers. Call H *LP-good* if $\mathrm{gap}(H) \le 0$. This definition concerns the integer costs.

Definition 3.3 (Missing allocations and their runs)**.** For $t \in \mathbb{N}_0$, let

$$Z_H(t) = \{x \in \{0, \dots, t\} : \mathcal{C}_{0,0}(H; x, t-x) = \varnothing\}.$$

Let $\rho_t(H)$ be the greatest cardinality of an integer interval contained in $Z_H(t)$, with value zero if the set is empty. Set $\mathrm{tr}(H) = 0$ when $c_0(H) = 0$. Otherwise set

$$Z_H = Z_H(c_0(H) - 1), \qquad \mathrm{tr}(H) = \max\{v - u + 1 : 0 \le u \le v < c_0(H),\ [u, v] \cap \mathbb{Z} \subseteq Z_H\}.$$

The endpoints u, v in this maximum are integers. Thus $\mathrm{tr}(H) = \rho_{c_0(H)-1}(H)$ when $c_0(H) > 0$. At capacities zero the two degree-bounded parts are empty. Consequently this is also the run of infeasible two-part matching allocations.

Proposition 3.4 (Finite integer representation)**.** *The minima in Definition 3.1 exist. For fixed H, A, B, abbreviate $g_{A,B,H}$ to g. Then*

$$c_{A,B}(H) = \max\big(\{0\} \cup \{x + g(x) : 0 \le x < \nu(H),\ g(x) > 0\}\big). \tag{3.5}$$

Every budget pair of total c is feasible if and only if $c \ge c_{A,B}(H)$. The minimum in (3.1) may therefore be restricted to $0 \le c \le \tau(H)$.

Each feasibility test is the following binary integer program. For every $e \in E(H)$ and $L \in \{Q_R, Q_B, P_R, P_B\}$, introduce $z_{e,L} \in \{0, 1\}$. Its constraints are

$$\sum_L z_{e,L} = 1 \qquad (e \in E(H)), \tag{3.6}$$

$$\sum_{e \ni v} z_{e,Q_R} \le A, \qquad \sum_{e \ni v} z_{e,Q_B} \le B \qquad (v \in V(H)), \tag{3.7}$$

$$\sum_{e \in N} z_{e,P_R} \le x, \qquad \sum_{e \in N} z_{e,P_B} \le y \qquad (N \text{ a matching of } H). \tag{3.8}$$

Under the edge-indicator correspondence, its feasible family is precisely $\mathcal{C}_{A,B}(H; x, y)$. For $g(x)$, replace the last blue bound by $\sum_{e \in N} z_{e,P_B} \le b$ for every matching N, and minimize the integer b over $0 \le b \le \nu(H)$. Replacing the binary variables by real variables in $[0, 1]$ defines a separate relaxation; equality of feasible sets or optima with the integer problems is not asserted.

The quantities in Definitions 3.2 and 3.3 are finite integers. They satisfy

1. $c_{A,B}(H) = c_{B,A}(H)$, *and* $c_{A,B}(H) \le c_{A',B'}(H)$ *if* $A' \le A$ *and* $B' \le B$;
2. $0 \le c_{\theta+1}(H) \le c_\theta(H) \le c_0(H)$, *and*
$$c_\theta(H) = 0 \quad \text{for } \theta \ge \max\{1, 2\Delta(H) - 1\};$$
3. $$\nu(H) \le c_0(H) \le \tau(H) = |V(H)| - \alpha(H), \qquad 0 \le \mathrm{ex}(H) \le \tau(H) - \nu(H);$$
4. $$\mathrm{gap}(H) = \mathrm{ex}(H) - \mathrm{ps}(H), \tag{3.9}$$
so LP-goodness is equivalent to $\mathrm{ps}(H) \ge \mathrm{ex}(H)$;
5. *if* $c_0(H) > 0$, *then* $Z_H \ne \varnothing$ *and* $1 \le \mathrm{tr}(H) \le c_0(H)$.

The statement and the integer formulation are recorded in [Ans26, Chapter 4].

Proof. Putting every edge in P_B gives a candidate for $g(x)$. There are finitely many partitions, so the minimum exists and lies in $[0, \nu(H)]$. Putting all edges in P_R gives $g(x) = 0$ for $x \ge \nu(H)$. Attainment of the minimum gives $\mathcal{C}_{A,B}(H; x, y) \ne \varnothing$ exactly when $y \ge g(x)$.

Let T be the right side of (3.5). For $x + y = c \ge T$, an allocation with $g(x) = 0$ is feasible. If $g(x) > 0$, then $x < \nu(H)$ and $x + g(x) \le T \le c$, so $y \ge g(x)$. Hence every total at least T is feasible.

Universal feasibility is upward closed in the integer total. For a pair $x + y = c + 1$ with $x \ge 1$, retain a certificate at $(x - 1, y)$ from total c. For $x = 0$, retain one at $(0, c)$. Increasing a matching budget preserves the certificate. Iteration gives the assertion for all larger totals. If $T > 0$, choose x attaining T with $g(x) > 0$. The nonnegative pair $(x, g(x) - 1)$ has total $T - 1$ and is infeasible. Upward closure rules out every smaller universal total. If $T = 0$, zero is feasible and is already the smallest allowed total. This proves (3.5) and the threshold characterization.

Let W be a minimum vertex cover. For $x + y = |W|$, partition $W = W_R \sqcup W_B$ with cardinalities x, y. Put every edge meeting W_R in P_R, and all remaining edges in P_B. Every remaining edge meets W_B. A matching in either part uses distinct vertices of its corresponding cover set. Taking $Q_R = Q_B = \varnothing$ gives a certificate, so $c_{A,B}(H) \le \tau(H)$.

The edge equations of the integer program assign each edge to exactly one part. The vertex constraints are precisely the two degree bounds. For a matching N of H, the red sum in (3.8) is $|N \cap P_R|$. A red matching bound implies every such inequality. Conversely, every matching in P_R is a matching of H, so these inequalities imply the red bound. The blue argument has the same two implications. For a fixed partition, the least allowed integer b is $\nu(P_B)$. Minimizing over the finite partition family proves the objective assertion for g.

Exchanging both pairs of red and blue parts proves the capacity symmetry. Increasing capacities preserves every certificate. For a capacity split of total $\theta + 1$, decrease a positive coordinate by one. Its cost is then bounded by $c_\theta(H)$, which proves row monotonicity after maximizing over the splits.

If $\Delta(H) = 0$, all edge sets and all costs are zero. Otherwise put $d = \Delta(H) \ge 1$. For $\theta \ge 2d - 1$, every split $A + B = \theta$ has a capacity at least d, since two capacities at most $d - 1$ sum to at most $2d - 2$. Put every edge in that degree-bounded part. This gives total zero and proves finiteness of all row sums.

At capacities $A = B = 0$, an allocation with first budget zero forces Q_R, Q_B, P_R to be empty. A nonempty simple edge set has both positive maximum degree and a matching with one edge. Thus $(0, c)$ fails if $c < \nu(H)$, proving the lower bound on $c_0(H)$. The cover construction gives its upper bound. Complementation interchanges vertex covers and independent sets, so $\tau(H) = |V(H)| - \alpha(H)$. Subtracting $\nu(H)$ gives the excess bounds. Expanding (3.3)–(3.4) gives

$$\mathrm{ex}(H) - \mathrm{ps}(H) = c_0(H) - \nu(H) - e(H) + \nu(H) + \sum_{\theta \ge 1} c_\theta(H) = \Sigma(H) - e(H).$$

Finally, if $c_0(H) > 0$, its minimality implies failure of at least one allocation at total $c_0(H) - 1$. Thus Z_H is nonempty. A member of Z_H gives a singleton interval, and every permitted interval has cardinality at most $c_0(H)$. The maximum exists because the family is finite. For $c_0(H) = 0$, the explicit convention avoids any negative total. □

3.2 Restriction, deletion, and slack

Lemma 3.5 (Restriction and one-vertex extension)**.** *If J is a subgraph of H, then $c_{A,B}(J) \le c_{A,B}(H)$ and $c_i(J) \le c_i(H)$. For every vertex v and every $i \in \mathbb{N}_0$,*

$$0 \le c_i(H) - c_i(H - v) \le 1.$$

Proof. Restricting the four parts to a subgraph preserves degree and matching bounds. For extension, put $c = c_{A,B}(H - v)$ and take $x + y = c + 1$. If $x \ge 1$, choose a certificate of $H - v$ at $(x - 1, y)$ and put all edges at v in P_{R}. These added edges form a star, so a matching uses at most one of them. If $x = 0$, use a certificate at $(0, c)$ and put the added star in P_{B}. Both constructions assign each edge once and meet the required bounds. The threshold property in Proposition 3.4 therefore gives $c_{A,B}(H) \le c_{A,B}(H - v) + 1$. Taking row maxima and using restriction proves the assertion. □

The slack proof uses the following degree-splitting input.

Lemma 3.6 (Degree splitting with a covering matching)**.** *Let $D \ge 1$, assume $\Delta(H) \le D$, and assume that a matching covers every vertex of degree D. For every capacity split $A + B = D$, there is a partition $E(H) = Q_{\mathrm{R}} \sqcup Q_{\mathrm{B}}$ with $\Delta(Q_{\mathrm{R}}) \le A$ and $\Delta(Q_{\mathrm{B}}) \le B$.*

This is the splitting input invoked in [Ans26, Chapter 6, proof of nonnegative positive-level slack].

Theorem 3.7 (Nonnegative positive-level slack)**.** *For every graph H,*

$$c_i(H) = 0 \quad (i \ge \Delta(H) + 1), \qquad \mathrm{ps}(H) \ge 0, \qquad \Sigma(H) \le e(H) + \mathrm{ex}(H).$$

If $\tau(H) = \nu(H)$, then $\Sigma(H) \le e(H)$. If $D \ge 1$, $\Delta(H) \le D$, and a matching covers every degree-D vertex, then $c_D(H) = 0$.

More precisely, assume H has an edge and v has degree $D = \Delta(H)$. Put

$$h = \nu(H) - \nu(H - v), \qquad a_i = c_i(H) - c_i(H - v) \quad (1 \le i \le D).$$

Then $h, a_1, \dots, a_D \in \{0, 1\}$, and

$$\mathrm{ps}(H) - \mathrm{ps}(H - v) = D - h - \sum_{i=1}^{D} a_i. \tag{3.10}$$

The difference is nonnegative if $h = 0$, or if a matching covers every degree-D vertex. In the latter case $a_D = 0$, and the difference is at least $1 - h$.

Every H admits a sequence of maximum-degree vertex deletions, ending at an edgeless graph after at most $|V(H)|$ deletions, along which ps *never increases upon deletion. The assertions include degree one, isolates, and the empty graph.*

This is the slack theorem of [Ans26, Chapter 6].

Proof. Lemma 3.6, with the two matching parts empty, certifies total zero for every split of D. Hence $c_D(H) = 0$ under its covering hypothesis. For $i \ge \Delta(H) + 1$, there are no degree-i vertices. The empty matching covers that set, so the same assertion with $D = i$ gives $c_i(H) = 0$. An edgeless graph has all costs zero directly.

Lemma 3.5 gives $a_i \in \{0, 1\}$. Deleting a vertex removes at most one edge from a matching. Every matching in $H - v$ is also one in H, so $h \in \{0, 1\}$. Rows above D vanish in both graphs, and deletion removes D edges. Subtracting the definitions of ps gives (3.10). If $h = 0$, the D terms a_i sum to at most D. If a matching covers every degree-D vertex, then $c_D(H) = 0$, and restriction gives $c_D(H - v) = 0$. Thus $a_D = 0$ and the sum is at most $D - 1$.

An eligible deletion always exists while an edge remains. If some maximum matching misses a maximum-degree vertex v, that matching lies in $H - v$, so $h = 0$. If no such pair exists, every maximum matching covers every maximum-degree vertex. A maximum matching exists by finiteness, and any maximum-degree vertex then satisfies the second condition. Delete an eligible vertex and repeat. The number of vertices decreases at each step, so the sequence ends after at most $|V(H)|$ steps. The terminal edgeless graph has slack zero. Composing the deletion inequalities proves $\mathrm{ps}(H) \ge 0$. An initially edgeless graph uses an empty sequence.

By (3.9), $\Sigma(H) - e(H) = \mathrm{ex}(H) - \mathrm{ps}(H) \le \mathrm{ex}(H)$. If $\tau(H) = \nu(H)$, the excess bounds in Proposition 3.4 give $\mathrm{ex}(H) = 0$. This proves the remaining inequality. □

3.3 The obstruction property

Definition 3.8 (The properties and minimal classes)**.** Let $h\colon \mathbb{N}_0 \to \mathbb{Z}$ be nondecreasing with $h(D) \geq 3$. A graph G satisfies T_h if

$$\operatorname{gap}(G) > 0 \quad \Longrightarrow \quad \big[\operatorname{gap}(G) = 1 \text{ and } \operatorname{tr}(G) + \operatorname{ex}(G) \leq h(\Delta(G))\big].$$

Put

$$h_-(D) = \max\{4, \lfloor D/2 \rfloor + 1\}, \qquad \mathrm{T}_- = T_{h_-}.$$

For $d \in \{4, 6, 8, 10, 12\}$, let $\mathcal{M}_d$ be the entire class of graphs of maximum degree exactly d that violate T_- while every proper subgraph satisfies T_-. Let N_d denote the assertion $\mathcal{M}_d = \varnothing$. There is no additional restriction on order, matching number, excess, or origin of a graph in this definition.

A *closed d-lump* is a nonempty connected d-regular graph of odd order for which deleting any vertex leaves a perfect matching. It is *totally critical* if every vertex deletion strictly lowers every cost row with index $1, \ldots, d$. By Lemma 3.5, each such strict loss equals one. A graph is *well-covered* if every maximal independent set has cardinality $\alpha(G)$.

4 Minimal obstructions

In this section, we identify the possible minimal failures of T_- and derive the contained obstruction needed for the lower bound.

4.1 Regularity and perfect matchings

The regularity argument uses three matching and cost inputs. Their conclusions are stated separately so that the later deduction specifies exactly what is required.

Lemma 4.1 (Initial restrictions on a minimal failure)**.** *Assume F violates T_h and every proper induced subgraph satisfies T_h, including the empty graph. Then F is connected, has an edge and positive gap, and has no matching covering every maximum-degree vertex. If $d = \Delta(F)$, v has degree d, and $\nu(F - v) = \nu(F)$, then*

$$c_i(F - v) = c_i(F) - 1 \qquad (1 \leq i \leq d).$$

These are the initial minimality and criticality inputs used in [Ans26, Chapter 10, proof of regularity and factor-criticality].

Lemma 4.2 (Positive gap in maximum degree at most three)**.** *If $\Delta(G) \leq 3$ and $\operatorname{gap}(G) > 0$, then*

$$\operatorname{gap}(G) = 1, \qquad \operatorname{tr}(G) + \operatorname{ex}(G) \leq 3.$$

This is the low-degree input used in [Ans26, Chapter 10, proof of regularity and factor-criticality].

Notation 4.3 (Marked matching deficiency)**.** For a marked vertex set $S \subseteq V(G)$, let $\kappa_S(G)$ be the minimum number of vertices of S uncovered by a matching. For $X \subseteq V(G)$, let $q_S(G - X)$ count the odd components of $G - X$ whose vertex sets are wholly contained in S.

Lemma 4.4 (Marked matching maximization)**.** *Assume $\kappa = \kappa_S(G) \geq 1$. Choose X with maximum cardinality among the sets maximizing $q_S(G - X) - |X|$. Then*

$$q_S(G - X) = |X| + \kappa,$$

and every counted component is factor-critical. That is, deleting any vertex of such a component leaves a perfect matching.

This is the marked-set matching statement used in [Ans26, Chapter 10, proof of regularity and factor-criticality].

Lemma 4.5 (Cost transfer across a nonempty maximizing set)**.** *Let G be connected, let $d = \Delta(G) \geq 4$, and let S be its degree-d vertices. Assume $\kappa_S(G) \geq 1$. Choose X as in Lemma 4.4, and assume $X \neq \varnothing$. Assume every component Q of $G - X$ satisfies $\operatorname{gap}(Q) \leq 1$. If d is even, then $\operatorname{gap}(G) \leq 0$. If d is odd and $\operatorname{gap}(G) > 0$, there is a component Z of $G - X$ such that*

$$\operatorname{gap}(G) = \operatorname{gap}(Z) = 1, \qquad \operatorname{ex}(G) = \operatorname{ex}(Z), \qquad \operatorname{tr}(G) + |X| \leq \operatorname{tr}(Z).$$

This is the transfer input used in [Ans26, Chapter 10, proof of regularity and factor-criticality].

Proposition 4.6 (Regularity and factor-criticality of minimal failures). *Assume F violates T_h and every proper induced subgraph of F satisfies T_h. Then F is connected and d-regular for an even integer $d \geq 4$. It has order $2k+1$, where $k = \nu(F)$, and deleting any vertex leaves a perfect matching. For every vertex v,*

$$c_i(F - v) = c_i(F) - 1 \qquad (1 \leq i \leq d). \tag{4.1}$$

Moreover $\mathrm{gap}(F) \in \{1, 2\}$. *If* $\mathrm{gap}(F) = 2$, *then every vertex v satisfies*

$$c_0(F - v) = c_0(F) - 1, \qquad \mathrm{gap}(F - v) = 1, \qquad \mathrm{ex}(F - v) = \mathrm{ex}(F) - 1.$$

Every member of each class $\mathcal{M}_d$ in Definition 3.8 has these structural and row-loss properties. The class definition retains minimality under arbitrary proper subgraphs, including spanning edge deletions.

The proposition is recorded in [Ans26, Chapter 10].

Proof. Lemma 4.1 gives connectedness, an edge, positive gap, and absence of a matching covering all maximum-degree vertices. Put $d = \Delta(F)$. If $d \leq 3$, Lemma 4.2 gives gap one and $\mathrm{tr}(F) + \mathrm{ex}(F) \leq 3 \leq h(d)$, contrary to failure of T_h. Hence $d \geq 4$.

Let S be the degree-d vertices and $\kappa = \kappa_S(F) \geq 1$. Choose X with maximum cardinality among maximizers of $q_S(F - X) - |X|$. The finite nonempty choice domain makes both maximizations possible. The chosen set is also inclusionwise maximal among maximizers. Lemma 4.4 gives $q_S(F - X) = |X| + \kappa$ and factor-criticality of the counted components.

Suppose $X \neq \varnothing$. Each component of $F - X$ is a proper induced subgraph, so satisfies T_h and has gap at most one. For even d, Lemma 4.5 contradicts the positive gap of F. For odd d, that lemma gives a proper induced component Z with gap one, equal excess, and the stated run transfer. Monotonicity of h then yields

$$\mathrm{tr}(F) + \mathrm{ex}(F) \leq \mathrm{tr}(Z) + \mathrm{ex}(Z) - |X| \leq h(\Delta(Z)) - |X| \leq h(d) - 1.$$

Together with gap one this asserts $T_h(F)$, again a contradiction. Thus $X = \varnothing$.

Connectedness and $q_S(F) = \kappa \geq 1$ force the sole component F to have odd order and all its vertices in S. Hence F is d-regular. The degree sum $d|V(F)| = 2e(F)$ makes d even. Lemma 4.4 gives factor-criticality. Writing $|V(F)| = 2k + 1$, a perfect matching of $F - v$ has k edges, and no matching on $2k + 1$ vertices has more. Therefore $\nu(F - v) = \nu(F) = k$ for every v. Lemma 4.1 now gives (4.1) at every vertex.

Fix v and put $a_0 = c_0(F) - c_0(F - v)$. Lemma 3.5 gives $a_0 \in \{0, 1\}$. Theorem 3.7 makes rows above d zero in both graphs. Deletion removes d edges and, by (4.1), exactly d units of positive-row cost. Consequently

$$\mathrm{gap}(F) - \mathrm{gap}(F - v) = a_0. \tag{4.2}$$

Since $F - v$ satisfies T_h, its gap is at most one. Positive integer gap for F and $a_0 \leq 1$ give $1 \leq \mathrm{gap}(F) \leq 2$. If $\mathrm{gap}(F) = 2$, (4.2) forces $a_0 = 1$ and $\mathrm{gap}(F - v) = 1$. The equality of matching numbers then gives the excess loss.

A proper induced subgraph is a proper subgraph. Thus a member of any $\mathcal{M}_d$ has the induced-minimality hypothesis. The function h_- is nondecreasing, integer-valued, and at least four, so it is an allowed choice of h. □

4.2 Gap one and the degree restriction

Lemma 4.7 (Hereditary bounds on missing runs). *Assume every induced subgraph of J, including J and the empty graph, satisfies* T_-. *If* $\mathrm{gap}(J) > 0$, *then*

$\mathrm{ex}(J)$	1	2	3
upper bound on $\mathrm{tr}(J)$	2	4	5.

If $\mathrm{ps}(J) = 0$, *then* $\rho_{\nu(J)}(J) \leq 2$.

These are the hereditary run bounds used in [Ans26, Chapter 11, proofs of the minimal-failure degree restrictions].

Lemma 4.8 (The gap-two degree restriction)**.** *Assume F has the connected, even-regular, odd-order, factor-critical, and total-criticality conclusions of Proposition 4.6. Assume every proper induced subgraph J has the following local properties: positive gap and excess $1, 2, 3$ give the respective run bounds $2, 4, 5$, and zero slack gives $\rho_{\nu(J)}(J) \le 2$. If $\mathrm{gap}(F) = 2$, then $\Delta(F) \ge 20$.*

This is the gap-two input used in [Ans26, Chapter 11, proofs of the minimal-failure degree restrictions].

Proposition 4.9 (Goodness at large even degrees)**.** *Every closed d-lump of even degree $d \ge 14$ is LP-good. No order, excess, or hereditary hypothesis is imposed.*

Proof from the degree-specific certificate results. The degree-fourteen result and the degree-sixteen result invoked in [Ans26, Chapter 11, large-degree goodness] each give nonpositive gap under exactly the closed-lump hypotheses. For even degrees at least eighteen, the cited development covers degrees $18, 20, 22, 24$ separately and every even degree at least 26. These five ranges exhaust all even integers at least eighteen. Together with degrees fourteen and sixteen, they exhaust the stated domain, and each gives $\mathrm{gap}(H) \le 0$. □

Proposition 4.10 (Gap one and the preliminary degree bound)**.** *Assume F violates* T_- *and every proper induced subgraph satisfies* T_-*. Then F is connected, has order $2k+1$ with $k = \nu(F) \ge 2$, and is d-regular for*

$$d \in \{4, 6, 8, 10, 12, 14, 16\}.$$

Deleting any vertex leaves a perfect matching. Moreover,

$$\mathrm{gap}(F) = 1, \qquad \mathrm{tr}(F) + \mathrm{ex}(F) \ge h_-(d) + 1,$$

and (4.1) holds for every vertex.

Every graph violating T_- *contains such a graph as an induced subgraph. The conclusions also hold when every proper subgraph of F satisfies* T_-*. In particular they apply to every member of the entire classes $\mathcal{M}_4, \mathcal{M}_6, \mathcal{M}_8, \mathcal{M}_{10}, \mathcal{M}_{12}$.*

This is the preliminary gap-one statement of [Ans26, Chapter 11].

Proof. Proposition 4.6, with $h = h_-$, supplies the structural conclusions, even degree $d \ge 4$, exact row losses, and gap in $\{1, 2\}$. Simplicity gives $2k \ge d \ge 4$, so $k \ge 2$.

Every induced subgraph of a proper induced subgraph J of F is again a proper induced subgraph of F. Lemma 4.7 therefore gives all the local run bounds required by Lemma 4.8. If $\mathrm{gap}(F) = 2$, that lemma gives $d \ge 20$. The large-degree results in Proposition 4.9 then give nonpositive gap, a contradiction. Thus $\mathrm{gap}(F) = 1$. Using just the degree-18, degree-20, degree-22, degree-24, and even-degree-at-least-26 cases of that proposition also rules out $d \ge 18$. This leaves the seven degrees in the statement. Failure of T_- at gap one, with integer run and excess, gives the strict bound in its displayed form.

For a graph G violating T_-, choose a violating induced subgraph with the fewest vertices. The choice is over a finite nonempty family containing G. Every proper induced subgraph of the chosen graph has fewer vertices and is induced in G, so it satisfies T_-. The preceding argument applies. The stronger proper-subgraph hypothesis implies the required proper-induced hypothesis, which also proves the assertion about the classes $\mathcal{M}_d$. □

Theorem 4.11 (Minimal-obstruction criterion)**.** *Assume F violates* T_- *and every proper induced subgraph satisfies* T_-*. Then F is a totally critical closed d-lump with*

$$d \in \{4, 6, 8, 10, 12\}, \qquad |V(F)| = 2k+1, \qquad k = \nu(F) \ge 2,$$

and

$$\mathrm{gap}(F) = 1, \qquad \mathrm{tr}(F) + \mathrm{ex}(F) \ge h_-(d) + 1, \qquad c_i(F - v) = c_i(F) - 1 \quad (1 \le i \le d). \tag{4.3}$$

Every graph violating T_- *contains such an induced subgraph; equality with the original graph is allowed.*

Every graph violating T_- *also contains a subgraph in one of $\mathcal{M}_4, \mathcal{M}_6, \mathcal{M}_8, \mathcal{M}_{10}, \mathcal{M}_{12}$. This contained subgraph need not be induced. Every member of each class has all the preceding structural and exact cost properties. Consequently every finite simple graph satisfies* T_- *if and only if all five classes are empty.*

The large-degree assertion of Proposition 4.9 and this criterion form the corresponding statement in [Ans26, Chapter 11].

Proof. Proposition 4.10 gives gap one and the seven preliminary degrees. Proposition 4.9 rules out fourteen and sixteen. Equivalently, the argument excluding gap two first supplies the hereditary run bounds on every proper induced subgraph, then uses Lemma 4.8 to force degree at least twenty, where large-degree goodness contradicts positive gap. The remaining structural conclusions and row identities are those of Proposition 4.6. The bound $k \geq 2$ follows from $2k \geq d \geq 4$. Failure of T_- gives the integer inequality in (4.3).

The minimum-vertex choice in the proof of Proposition 4.10 gives the induced containment. For proper-subgraph containment, choose a violating subgraph F minimizing $|V(F)| + e(F)$. The finite family is nonempty. Every proper subgraph has a strictly smaller sum and is still a subgraph of the original graph. It therefore satisfies T_-. The induced-minimal classification just proved puts the maximum degree of F in the displayed set. The full proper-subgraph minimality then puts this same F in the corresponding $\mathcal{M}_d$.

Conversely, each member of a class has the proper-induced minimality needed for the structural conclusion. If every graph satisfies T_-, every class is empty by its failure clause. If the five classes are empty and a graph violates T_-, the minimum-subgraph choice gives a member of one of them, a contradiction. □

4.3 Localization on the actual host

The following three inputs connect the certificate quantities to the star-forest budget. Their suffix families and budgets are exactly those of Setup 2.3.

Proposition 4.12 (The small-deficit budget bound)**.** *Put $K = M - q$. If H arrows every prescribed-color member of $\mathcal{A}_q$ and $\nu(H) \geq K - 3$, then $e(H) \geq L_q$.*

This is the small-deficit input invoked in [Ans26, Chapter 24, proof of contained minimal obstructions].

Proposition 4.13 (Component localization below the budget)**.** *Put $K = M - q$. If H arrows every prescribed-color member of $\mathcal{A}_q$ and $e(H) < L_q$, then H has an edge. Every connected component C of largest matching number satisfies*

$$\mathrm{ex}(C) \geq \mathrm{ex}(H) \geq K - \nu(H), \qquad \mathrm{gap}(C) \geq 1.$$

These simultaneous component conclusions are invoked in [Ans26, Chapter 24, proof of contained minimal obstructions].

Proposition 4.14 (Hereditary goodness at excess at least four)**.** *If every induced subgraph of J, including J itself, satisfies T_-, then*

$$\mathrm{ex}(J) \geq 4 \quad \Longrightarrow \quad \Sigma(J) \leq e(J).$$

Consequently, if every proper subgraph of F satisfies T_-, then every proper subgraph J with $\mathrm{ex}(J) \geq 4$ is LP-good.

The hereditary implication is invoked in [Ans26, Chapter 24, proof of contained minimal obstructions]. For its stated consequence, every induced subgraph of a proper subgraph J is itself a proper subgraph of F, including when J is a spanning edge deletion.

Proposition 4.15 (The proper-subgraph bound in degree four)**.** *If $F \in \mathcal{M}_4$, then every proper subgraph J satisfies*

$$\mathrm{gap}(J) > 0 \quad \Longrightarrow \quad \big[\mathrm{gap}(J) = 1 \text{ and } \mathrm{tr}(J) + \mathrm{ex}(J) \leq 3\big].$$

Every such J with $\mathrm{ex}(J) \geq 3$ is LP-good. These assertions include spanning edge deletions.

The strengthened run bound is the degree-four input invoked in [Ans26, Chapter 24, proof of contained minimal obstructions]. The goodness consequence follows because positive gap forces $c_0(J) > 0$, and then Proposition 3.4 gives $\mathrm{tr}(J) \geq 1$.

Theorem 4.16 (Contained minimal obstructions and conditional avoidance)**.** *Use Setup 2.3 and the certificate conventions of Section 3. Put $K = M - q$.*

(I) *Assume H arrows every entire prescribed-color member of $\mathcal{A}_q$ and $e(H) < L_q$. There are an actual connected component C of H and a subgraph $F \subseteq C$ such that*

$$\mathrm{ex}(C) \geq K - \nu(H) \geq 4, \qquad \mathrm{gap}(C) \geq 1. \tag{4.4}$$

The graph F is a totally critical closed d-lump with

$$d \in \{4, 6, 8, 10, 12\}, \qquad d \leq \Delta(H), \qquad \mathrm{gap}(F) = 1, \qquad \mathrm{tr}(F) + \mathrm{ex}(F) \geq h_-(d) + 1.$$

Every proper subgraph of F satisfies T_-*, so* $F \in \mathcal{M}_d$*. Every proper subgraph J of F with* $\mathrm{ex}(J) \geq 4$ *is LP-good. The graph F need not be induced in H, and no lower bound* $\mathrm{ex}(F) \geq K - \nu(H)$ *is asserted.*
Write $|V(F)| = 2k+1$*, where* $k = \nu(F) \geq 2$*. If* $\mathrm{ex}(F) \geq 4$*, then for every edge e and every* $\theta \in \mathbb{N}_0$*,*

$$r_\theta(e) = c_\theta(F) - c_\theta(F-e) \in \{0,1\}.$$

These differences vanish for $\theta \geq d+1$*, and* $r_d(e) = 1$*. Moreover,*

$$\nu(F-e) = k, \qquad \mathrm{ex}(F-e) = \mathrm{ex}(F) - r_0(e), \tag{4.5}$$

$$R(e) := \sum_{\theta=0}^{d} r_\theta(e) \geq 2, \qquad \mathrm{gap}(F-e) = 2 - R(e) \leq 0. \tag{4.6}$$

Thus every edge strictly lowers some row below d, although that row may depend on the edge. If $d = 4$*, every proper* $J \subsetneq F$ *satisfies the strengthened positive-gap implication in Proposition 4.15, and every proper J with* $\mathrm{ex}(J) \geq 3$ *is LP-good.*

(II) *Assume every totally critical closed d-lump subgraph F of H with* $d \in \{4,6,8,10,12\}$*, gap one, and every proper subgraph satisfying* T_-*, obeys*

$$\mathrm{tr}(F) + \mathrm{ex}(F) \leq h_-(d).$$

If $e(H) < L_q$*, then H has a coloring avoiding both entire prescribed targets of some actual member of* $\mathcal{A}_q$*. Every such avoiding coloring extends over the unchanged canonical q-star union $\mathcal{P}_q$ to avoid both full original targets. It also extends across any specified actual q-vertex set X in any graph G with* $H = G - X$*, to avoid both full original targets in G. Each extension preserves every edge color of H. For* $q = 0$*, the star union and the outer vertex set are empty.*

No further degree or budget-boundary hypothesis is imposed on H. The containment assertion does not assert emptiness of the five classes or equality between the degree of F and the degree of H.

The statement is recorded in [Ans26, Chapter 24].

Proof. For part (I), Proposition 4.12 and $e(H) < L_q$ give, by integer contraposition, $K - \nu(H) \geq 4$. The graph and the suffix family have not changed. Proposition 4.13 gives an edge of H, so its component family is nonempty. Choose a component C of largest matching number. The same proposition gives

$$\mathrm{ex}(C) \geq \mathrm{ex}(H) \geq K - \nu(H) \geq 4, \qquad \mathrm{gap}(C) \geq 1.$$

If every induced subgraph of C satisfied T_-, then Proposition 4.14 would make C LP-good, contrary to its positive gap. Hence a subgraph of this actual C violates T_-.

Among all violating subgraphs of C, choose F first with the fewest edges and then with the fewest vertices. The family is finite and nonempty. A proper subgraph J either has fewer edges, or has the same edges and fewer vertices. If J violated T_-, it would contradict the respective part of the choice. Thus every proper subgraph of F satisfies T_-. This includes spanning edge deletions and deletions of isolated vertices; no degree or inducedness restriction was imposed on the choice.

Theorem 4.11 applies to this same F. It gives all the closed-lump, degree, gap, run, and exact row-loss conclusions. Containment in H gives $d \leq \Delta(H)$. The failure clause and full proper-subgraph minimality put F in $\mathcal{M}_d$. Proposition 4.14 gives the stated goodness of its proper subgraphs. The component excess estimate has been used only on C.

Assume $\mathrm{ex}(F) \geq 4$, and fix an edge $e = uv$. Restriction gives nonnegative row differences. Conversely, a certificate of $F - e$ at total c extends to every allocation at total $c+1$ by putting e in a matching part whose budget is increased by one. The two choices according to whether the first budget is positive are those in Lemma 3.5. Hence each row difference is at most one. Theorem 3.7 makes the differences zero above d.

A perfect matching of $F - v$ covers all its degree-d vertices, so $c_d(F-v) = 0$ by Theorem 3.7. The exact vertex loss gives $c_d(F) = 1$. A perfect matching of $F - u$ is also a matching in $F - e$. It covers every degree-d vertex of $F - e$, since the only vertices whose degrees changed are u, v. Therefore $c_d(F-e) = 0$, giving $r_d(e) = 1$. The same matching has k edges, and no matching in $F - e$ has more than k. This proves the matching identity in (4.5); the excess identity follows by subtraction.

The row differences and the deletion of one edge give $\mathrm{gap}(F-e) = \mathrm{gap}(F) + 1 - R(e) = 2 - R(e)$. If $r_0(e) = 1$, the distinct rows $0, d$ already give $R(e) \geq 2$. If $r_0(e) = 0$, then $\mathrm{ex}(F-e) = \mathrm{ex}(F) \geq 4$, and

proper-subgraph goodness gives $\mathrm{gap}(F-e) \le 0$, again forcing $R(e) \ge 2$. Together with $r_d(e) = 1$ and vanishing above d, this proves that some row below d drops. The degree-four refinement is Proposition 4.15. This completes part (I).

For part (II), suppose $e(H) < L_q$. If H arrowed every member of $\mathcal{A}_q$, part (I) would give a contained graph satisfying the hypotheses of the assumed bound but with run plus excess at least $h_-(d)+1$. Thus some actual pair $(F_1^a, F_2^b) \in \mathcal{A}_q$ is not arrowed. By Definition 1.1, H has a coloring avoiding both of its prescribed targets.

We give the extension argument explicitly; it is also the suffix extension used in [Ans26, Chapter 24]. A collection of $s-a$ components remaining after any a components of F_1 are discarded contains a copy of F_1^a. To see this, write their retained indices as $j_1 < \cdots < j_{s-a}$. Then $j_r \le a+r$, so $n_{j_r} \ge n_{a+r}$, and each retained star may be restricted to the required smaller star. The same assertion holds for F_2^b.

Color a of the $q = a+b$ stars in $\mathcal{P}_q$ entirely red and the other b entirely blue. Vertex-disjoint positive stars inside one host star can contribute at most one target component, because any two of its edges meet the center. Hence a red full target would leave at least $s-a$ red components in H, containing F_1^a. A blue full target would leave a blue F_2^b in H. Both contradict the given coloring.

For the outer-set assertion, partition the specified set as $X = X_{\mathrm{R}} \sqcup X_{\mathrm{B}}$, with cardinalities a, b. Color every edge meeting X_{R} red. Color every remaining edge meeting X_{B} blue. Retain the colors inside $H = G - X$. Every red target component not wholly in H uses a vertex of X_{R}, so at most a vertex-disjoint red components can lie partly outside H. Every blue component not wholly in H uses a vertex of X_{B}, giving the corresponding bound b. The retained-index argument again gives an avoided suffix if a full target existed. All assertions concern the specified vertices and the unchanged original targets. When $q = 0$, both extensions add nothing. □

5 The five whole-degree exclusions

In this section, we state all five exclusions and give their supplied assembly arguments using the same integer costs and the entire classes of Definition 3.8.

5.1 Deletion identities and the improving certificate

The common input for degrees eight, ten, and twelve concerns all maximum independent sets and all admissible choices in the subsequent vertex deletions.

Definition 5.1 (Complete maximum-degree deletion sequences)**.** A complete maximum-degree deletion sequence starting from J is a finite sequence

$$J = J_0, J_1, \dots, J_\ell, \qquad J_{r+1} = J_r - v_r,$$

where v_r has degree $\Delta(J_r) \ge 3$, and $\Delta(J_\ell) \le 2$. At each step any current maximum-degree vertex is allowed. A finite graph always reaches such a terminal graph after at most its number of vertices.

Setup 5.2 (The independent-set deletion hypotheses)**.** Let F be connected, d-regular, factor-critical, and well-covered, with

$$|V(F)| = 2k+1, \qquad k = \nu(F) \ge 1, \qquad a = \alpha(F), \qquad E = \mathrm{ex}(F) \ge 1.$$

The following are the deletion hypotheses.

1. For every maximum independent set S,
$$c_i(F-S) = c_i(F) - a \quad (1 \le i < d), \qquad c_0(F) = a + \nu(F-S) + \mathrm{ps}(F-S). \tag{5.1}$$
2. For every such S and every complete deletion sequence beginning at $F-S$, each deletion parent J, of actual maximum degree $D \ge 3$, has a matching covering all its degree-D vertices. Deleting its chosen maximum-degree vertex v gives
$$c_i(J-v) = c_i(J) - 1 \qquad (1 \le i < D). \tag{5.2}$$
Every terminal graph has slack zero.

Proposition 5.3 (Universal deletion conclusions in degrees eight, ten, and twelve)**.** *For $d \in \{8, 10, 12\}$, every $F \in \mathcal{M}_d$ has all the structural properties and deletion hypotheses of Setup 5.2. It also satisfies*

$$\operatorname{gap}(F) = 1, \qquad \operatorname{tr}(F) + E \geq d/2 + 2.$$

The assertions hold for every maximum independent set and every complete deletion sequence, without restricting the order, excess, matching number, or cost values of F.

These conclusions are proved in [Ans26, Chapter 17]. Together with Proposition 5.4, they are the two inputs to each of the degree-eight, degree-ten, and degree-twelve exclusions [Ans26, Chapter 24].

Proposition 5.4 (Improving certificates at the penultimate even row)**.** *Assume Setup 5.2, let $d \in \{8, 10, 12\}$, and assume*

$$\operatorname{gap}(F) = 1, \qquad \operatorname{tr}(F) + E \geq d/2 + 2.$$

Then $T = c_{d-2}(F) - 1$ is a nonnegative integer, and there is a certificate table $\mathcal{Q}$ of row index $d-2$ and total T on the entire graph F. Explicitly, for every $A + B = d - 2$ and every $x + y = T$, there is an ordered partition of all edges of F satisfying

$$\Delta(Q_{\mathrm{R}}) \leq A, \quad \Delta(Q_{\mathrm{B}}) \leq B, \quad \nu(P_{\mathrm{R}}) \leq x, \quad \nu(P_{\mathrm{B}}) \leq y.$$

This is proved in [Ans26, Chapter 21] (the improving certificate under the failed run bound); its two inputs, the deletion-process results of [Ans26, Chapter 17] and the exact costs and native rows of [Ans26, Chapter 16], are also proved there.

The distinction between a table entry and a universal total is essential here. One entry proves feasibility for one pair of budgets and capacities. Proposition 5.4 requires an entry for every pair in both finite index sets. Only that universal assertion contradicts the row maximum.

Theorem 5.5 (Exclusion in degree twelve)**.** *The entire class $\mathcal{M}_{12}$ is empty. Equivalently, every graph of maximum degree exactly twelve whose every proper subgraph satisfies T_- also satisfies T_-. There is no additional restriction on order, excess, matching number, or certificate costs.*

The exclusion is stated in [Ans26, Chapter 24].

Proof from the stated construction input. Suppose $F \in \mathcal{M}_{12}$. The certificate quantities in Propositions 5.3 and 5.4 agree with Definitions 3.1–3.3. They partition every edge and use integer capacities and matching budgets. At capacities zero the degree-bounded parts are empty, so the missing-allocation run is precisely $\operatorname{tr}(F)$, including its zero-cost convention. Also (3.9) identifies the signed gap with $\operatorname{ex}(F) - \operatorname{ps}(F)$.

Proposition 5.3 gives connectedness, twelve-regularity, factor-criticality, and well-coveredness. With its k, a, E, the graph has order $2k + 1$, $E \geq 1$, and gap one. Failure of T_-, together with $h_-(12) = 7$, gives $\operatorname{tr}(F) + E \geq 8$. Every maximum independent set satisfies (5.1) for $1 \leq i < 12$. Every deletion parent has the covering matching and the exact lower-row losses in (5.2), and every terminal has zero slack. Thus both universal hypotheses of the improving construction hold on this same graph.

Proposition 5.4 supplies a nonnegative total $T = c_{10}(F) - 1$. For every $A + B = 10$, every allocation of T has a certificate of all F-edges. Definition 3.1 gives $c_{A,B}(F) \leq T$ for each split. Taking their maximum yields

$$c_{10}(F) \leq T = c_{10}(F) - 1, \tag{5.3}$$

a contradiction.

A degree-twelve graph satisfying the proper-subgraph hypothesis but failing T_- would belong to $\mathcal{M}_{12}$. Conversely, the stated universal implication excludes every member of that class. This proves the equivalence. □

The final calculation (5.3) completes the degree-twelve exclusion using the universally feasible total supplied by Proposition 5.4.

Theorem 5.6 (Exclusion in degree ten)**.** *The entire class $\mathcal{M}_{10}$ is empty. The conclusion holds for every order, excess, matching number, and cost configuration allowed by its definition.*

This exclusion is stated in [Ans26, Chapter 24].

Proof from the stated construction input. Suppose $F \in \mathcal{M}_{10}$. The capacity, budget, row, gap, and run definitions agree with the common inputs for the reasons given in the degree-twelve proof. Proposition 5.3 gives the connected ten-regular, factor-critical, well-covered graph of Setup 5.2, with odd order $2k+1$, $E \geq 1$, and gap one. Here $h_-(10) = 6$, so failure of T_- gives $\mathrm{tr}(F) + E \geq 7$. For every maximum independent set, the exact row identities hold for $1 \leq i < 10$. The covering-matching, row-loss, and zero-terminal-slack assertions hold for every complete deletion sequence.

Apply Proposition 5.4 with $d = 10$. Its total is the nonnegative integer $T = c_8(F) - 1$. Every split $A + B = 8$ and every split $x + y = T$ has a certificate on the entire original F. Hence $c_{A,B}(F) \leq T$ for every capacity split, and $c_8(F) \leq T = c_8(F) - 1$, a contradiction. The input is unrestricted over the costs of members of $\mathcal{M}_{10}$, so no further case distinction is needed. □

Theorem 5.7 (Exclusion in degree eight)**.** *The entire class $\mathcal{M}_8$ is empty. No additional excess, matching-number, order, or certificate hypothesis is imposed.*

This exclusion is stated in [Ans26, Chapter 24].

Proof from the stated construction input. Suppose $F \in \mathcal{M}_8$. Its integer certificate definitions again coincide with the common inputs, and $\mathrm{gap}(F) = \mathrm{ex}(F) - \mathrm{ps}(F)$. Proposition 5.3 supplies the connected eight-regular, factor-critical, well-covered graph in Setup 5.2, with odd order $2k+1$, $E \geq 1$, and gap one. Since $h_-(8) = 5$, its failure of T_- gives $\mathrm{tr}(F) + E \geq 6$. The independent-set row identities hold for $1 \leq i < 8$. The deletion conclusions hold for every allowed independent set and every complete deletion sequence.

Proposition 5.4 now gives a table at the nonnegative total $T = c_6(F) - 1$. It supplies certificates of all F-edges for every $A + B = 6$ and every $x + y = T$. Thus each $c_{A,B}(F) \leq T$, and maximizing gives $c_6(F) \leq T = c_6(F) - 1$. This contradiction excludes F. □

Degree d	$h_-(d)$	Required $\mathrm{tr}(F) + E$	Improved row	Table total
8	5	≥ 6	6	$c_6(F) - 1$
10	6	≥ 7	8	$c_8(F) - 1$
12	7	≥ 8	10	$c_{10}(F) - 1$

Table 1: The parameter substitutions in the three common improving-certificate exclusions.

5.2 Degree six

The degree-six assembly uses the same universal deletion identities, together with three branches determined by the excess and the associated parameters.

Proposition 5.8 (Universal deletion conclusions in degree six)**.** *If $F \in \mathcal{M}_6$, then Setup 5.2 holds with $d = 6$. In addition,*

$$\mathrm{tr}(F) + E \geq 5, \qquad c_5(F) = a.$$

The identities (5.1) hold for every maximum independent set and every index $1 \leq i \leq 5$. Every complete sequence deleting arbitrary current maximum-degree vertices while their degree is at least three has the covering matchings and exact positive-row losses in (5.2). Every terminal has zero slack.

These are the degree-six structural and deletion conclusions used in [Ans26, Chapters 23–24].

Proposition 5.9 (The small-excess exclusion in degree six)**.** *Under Setup 5.2 with $d = 6$, assume*

$$1 \leq E \leq 3, \qquad \mathrm{tr}(F) + E \geq 5.$$

No graph satisfies these hypotheses.

This is the small-excess branch used in the degree-six assembly [Ans26, Chapter 23].

Proposition 5.10 (The parameter alternative in degree six)**.** *Let F be connected, six-regular, and factor-critical, with*

$$|V(F)| = 2k+1, \quad k = \nu(F) \geq 1, \quad a = \alpha(F), \quad E = c_0(F) - k \geq 4,$$

and assume $c_5(F) = a$. *Set*

$$g = k - a, \qquad p = g - E + 1, \qquad \lambda = 2E - 3 - g. \tag{5.4}$$

Then either $p \geq 2$, *or*

$$|V(F)| = 27, \qquad \alpha(F) = 9, \qquad c_0(F) = 17. \tag{5.5}$$

This is the parameter alternative used in [Ans26, Chapter 23].

Proposition 5.11 (The improving table in degree six)**.** *Under Setup 5.2 with* $d = 6$, *assume* $E \geq 4$ *and* $p \geq 2$, *with* (5.4). *Set*

$$\delta = c_0(F) - c_2(F), \qquad \epsilon = c_0(F) - c_1(F), \qquad \beta = p + \delta - \epsilon, \qquad r = k - \lambda - 1 + \delta. \tag{5.6}$$

Then

$$c_4(F) = r - \beta, \qquad T = r - \beta - 1 \geq 0.$$

There is a certificate table of row index four and total T *on the entire graph* F. *Thus every capacity split* $A + B = 4$ *and every budget split* $x + y = T$ *admits a four-part partition with the stated degree and matching bounds, including all edges incident with vertices removed in the intermediate deletions.*

This is the improving-table branch in [Ans26, Chapter 23]. The differences in (5.4) and (5.6) are integers; their nonnegativity is not assumed unless explicitly asserted.

Proposition 5.12 (The exceptional degree-six exclusion)**.** *Under Setup 5.2 with* $d = 6$, *no graph satisfies* (5.5).

This is the exceptional-order branch used in [Ans26, Chapter 23].

Theorem 5.13 (Exclusion in degree six)**.** *The entire class* $\mathcal{M}_6$ *is empty. The conclusion imposes no additional excess, deficit, matching-number, order, cost-row, or component hypothesis.*

The complete branch assembly is recorded in [Ans26, Chapter 24].

Proof. Suppose $F \in \mathcal{M}_6$. Every proper induced subgraph is proper, so the hypotheses of Proposition 5.8 hold. The certificate conventions agree: all parts partition the original edge set, all capacities and budgets are nonnegative integers, and each cost is universal over the budget allocation. The row sum is finite; already for $i \geq 2\Delta(G)$, every capacity split has one capacity at least $\Delta(G)$, so all edges may be placed in that degree-bounded part. Expanding the definitions gives

$$\mathrm{ex}(G) - \mathrm{ps}(G) = c_0(G) - \nu(G) - \left(e(G) - \nu(G) - \sum_{i \geq 1} c_i(G) \right) = \sum_{i \geq 0} c_i(G) - e(G).$$

At capacities zero, the run is the two-part matching run, including the zero-cost convention.

Put $k = \nu(F)$, $a = \alpha(F)$, and $E = c_0(F) - k$. Proposition 5.8 gives connectedness, six-regularity, factor-criticality, well-coveredness, order $2k + 1$, $E \geq 1$, $\mathrm{tr}(F) + E \geq 5$, and $c_5(F) = a$. It gives both universal deletion hypotheses on this same F.

If $1 \leq E \leq 3$, Proposition 5.9 rules out F. Assume $E \geq 4$ and use (5.4). The structural hypotheses and $c_5(F) = a = \alpha(F)$ allow Proposition 5.10. It gives $p \geq 2$ or (5.5).

In the first branch, $p = E - 2 - \lambda$ by (5.4). Use the parameters in (5.6). All hypotheses of Proposition 5.11 are the already established structural and universal deletion conclusions. It gives $c_4(F) = r - \beta$ and a table $\mathcal{Q}$ on all edges of F with nonnegative total $T = r - \beta - 1$. Thus $c_{A,B}(F) \leq T$ for every $A + B = 4$, and

$$c_4(F) \leq \mathrm{cost}(\mathcal{Q}) = T = r - \beta - 1 = c_4(F) - 1,$$

a contradiction.

In the second branch, F retains connectedness, six-regularity, factor-criticality, and well-coveredness. Substituting $a = 9$ in (5.1) gives the independent-set hypotheses of Proposition 5.12. The universal deletion hypotheses are unchanged. That proposition rules out (5.5).

The integer cases $1 \leq E \leq 3$ and $E \geq 4$ are exhaustive. The two alternatives in the latter case are also exhaustive. Hence no member of $\mathcal{M}_6$ remains. □

5.3 Degree four

The degree-four argument retains a family of frames for every maximum independent set. All restrictions on the remainder must hold uniformly over that family.

Definition 5.14 (Independent-set frames)**.** For a maximum independent set S in F, let $C(S)$ be the subgraph of $F - S$ induced by its degree-three vertices. For a positive integer m, a frame is a pair (S, U), where S is maximum independent and U is an independent m-element subset of $C(S)$. For a frame put

$$W = S \cup U, \qquad R = F - W, \qquad r = \nu(R).$$

When $F[W]$ consists of disjoint edges and isolates, let P be the set of endpoints of those edges and let $Z = W \setminus P$ be the isolates. For $p \in P$, put $K_p = F[V(R) \cup \{p\}]$. The graphs P_j and C_j denote a path and a cycle on j vertices, respectively. A whole component means a connected component of R itself.

Proposition 5.15 (Uniform frame structure in degree four)**.** *If $F \in \mathcal{M}_4$, then F is connected, four-regular, and well-covered, with $a = \alpha(F) \geq 5$. There are integers m, b, q, σ satisfying*

$$m \geq 1, \qquad 0 \leq b < m, \qquad q = m - b \geq 1, \qquad \sigma \geq b,$$

such that

$$c_2(F) = a + 2m - b, \qquad \max_S \alpha(C(S)) = m. \tag{5.7}$$

The maximum is over maximum independent sets S. At least one frame exists. Every frame has the following properties.

1. *The graph $F[W]$ consists of m disjoint edges and $a - m$ isolates, and $\alpha(R) = a$.*
2. *The whole components of R are exactly q odd cycles from C_3, C_5, C_7, exactly b even paths from P_2, P_4, P_6, and otherwise P_3, C_4, or isolates.*
3. *For every $p \in P$,*
$$\nu(K_p) = r + 1, \qquad c_1(K_p) = r + 1 - b. \tag{5.8}$$
4. *There is one fixed pair of nonnegative integer budgets (u, v), independent of the frame, for which the two-part matching allocation on F is infeasible and*
$$u + v = |W| + r - 1, \qquad |W| - u \geq q + b + \sigma. \tag{5.9}$$

This is the uniform frame input from the degree-four development [Ans26, Chapter 22], used in the exclusion assembly of [Ans26, Chapter 24].

Proposition 5.16 (The frame exclusion with zero even-path count)**.** *Let F be connected, four-regular, and well-covered, with $a = \alpha(F) \geq 4$. Assume $m \geq 1$ and $\alpha(C(S)) \leq m$ for every maximum independent set S. Assume the following properties hold for every frame: $F[W]$ consists of m disjoint edges and $a - m$ isolates; $\alpha(R) = a$; the remainder has exactly m whole odd cycles, each from C_3, C_5, C_7, and all other components are P_3, C_4, or isolates; and every frame-edge endpoint satisfies*

$$\nu(K_p) = c_1(K_p) = r + 1.$$

Then no frame exists.

This is the zero-count frame exclusion invoked in [Ans26, Chapter 22].

Proposition 5.17 (Restrictions when the even-path count is positive)**.** *Assume the uniform frame conclusions of Proposition 5.15, with $b \geq 1$. It suffices in (5.9) to assume the fixed missing pair satisfies $|W| - u \geq q + 2$ for every frame. Then every frame has these additional properties:*

1. *No whole component of R is P_6.*
2. *Every $p \in P$ has at most one degree-two neighbor in R. It cannot be adjacent both to the center and to a leaf of one whole P_3.*
3. *Each whole odd cycle of R has at most one edge to P.*
4. *For every $T \subseteq W$ with $|T| \leq q + 2$, there are edges tv_t, for $t \in T$, into pairwise distinct whole odd-order components C_t of R, such that $C_t - v_t$ has a perfect matching.*

These are the positive-count frame restrictions from [Ans26, Chapter 22].

Proposition 5.18 (Exclusion of triangular remainder components)**.** *Assume the connected, four-regular, well-covered frame setting of Proposition 5.15. Assume every frame has the component list there with P_6 removed, both endpoint-neighbor restrictions in Proposition 5.17, and its simultaneous assignments to distinct odd-order components. Assume the exact second-cost identity (5.7). Then no frame remainder has a whole C_3 component.*

This is the triangular-component exclusion invoked in [Ans26, Chapter 22].

Proposition 5.19 (Exclusion of the remaining odd cycles)**.** *Assume the connected, four-regular, well-covered frame setting with $a \geq 4$, $m \geq 1$, and $0 \leq b < m$. Assume every frame has the component list of Proposition 5.15 with P_6 removed, both endpoint-neighbor restrictions in Proposition 5.17, and at most one edge from each whole odd cycle to P. Assume (5.7). Then no frame remainder has a whole C_5 or C_7 component.*

These are the two remaining cycle exclusions invoked in [Ans26, Chapter 22].

Theorem 5.20 (Exclusion in degree four)**.** *The entire class $\mathcal{M}_4$ is empty. Thus no graph of maximum degree exactly four violates T_- while every proper subgraph satisfies T_-. There is no further order, matching-number, excess, deficit, frame, or remainder hypothesis in this conclusion.*

The assembly is recorded in [Ans26, Chapter 24].

Proof. Suppose $F \in \mathcal{M}_4$. The certificate definitions, signed gap, top missing-allocation run, and full proper-subgraph minimality are the same as in the frame inputs. Proposition 5.15 therefore applies to this F. It gives connectedness, four-regularity, well-coveredness, $a \geq 5$, the integer bounds on m, b, q, σ, and the exact value $c_2(F) = a + 2m - b$. It also gives $\max_S \alpha(C(S)) = m$, so some maximum independent set has an independent m-set in $C(S)$. Hence a frame exists.

For every frame, retain W, R, r, P, Z from Definition 5.14. The graph $F[W]$ consists of m edges and $a - m$ isolates. The remainder has independence number a, the stated whole component list, and the endpoint identities (5.8). The single missing pair (u, v) works in (5.9) for every frame.

If $b = 0$, then $q = m$. The remainder has exactly m of the listed odd cycles and otherwise only P_3, C_4, or isolates. Every endpoint extension has $\nu(K_p) = c_1(K_p) = r + 1$. Moreover $a \geq 4$ and every $C(S)$ has independence number at most m. All hypotheses of Proposition 5.16 hold uniformly, so no frame exists, a contradiction.

Therefore $b \geq 1$. Since $\sigma \geq b$, the fixed missing pair satisfies

$$|W| - u \geq q + b + \sigma \geq q + 2$$

in every frame. Proposition 5.17, with its missing budgets taken to be (u, v), rules out P_6. It gives both endpoint-neighbor restrictions and at most one edge from each whole odd cycle to P.

For each $T \subseteq W$ with $|T| \leq q + 2$, that proposition also gives edges tv_t into distinct whole odd-order components C_t, with a perfect matching after deleting v_t. Such a matching has $(|V(C_t)| - 1)/2$ edges. No matching in an odd-order C_t can have more, so

$$\nu(C_t - v_t) = \nu(C_t).$$

Thus the simultaneous assignment condition holds with matching number preserved in every chosen component.

The component list with P_6 removed, the two endpoint-neighbor restrictions, these assignments, and the exact second cost are the hypotheses of Proposition 5.18. It excludes C_3 from every remainder. The same component list and endpoint restrictions, the at-most-one edge condition for odd cycles, and the same second-cost identity give all hypotheses of Proposition 5.19. It excludes C_5 and C_7.

Choose an existing frame. Its remainder has exactly $q = m - b \geq 1$ whole odd cycles, each from C_3, C_5, C_7. All three possibilities have been excluded. This contradiction proves the theorem. □

6 The lower bound and the proof of the main theorem

In this section, we combine Theorem 4.16 with the five whole-degree exclusions and then with the explicit upper-bound host.

Theorem 6.1 (Lower bound for two star forests)**.** *With the full positive profiles and ordinary copies of Setup 2.3, put*

$$B = \sum_{k=2}^{s+t} \ell_k = L.$$

Every finite simple graph G satisfying $G \to (F_1, F_2)$ has $e(G) \geq B$.

The deduction is recorded in [Ans26, Chapter 25].

Proof. Suppose $G \to (F_1, F_2)$ and $e(G) < B$. Take $q = 0$. Since $s, t \geq 1$, the integer $M = s + t - 1$ is positive and $0 \leq q < M$. Here $K = M$ and $L_0 = B$. The only nonnegative cuts with sum zero are $a = b = 0$, and both are legal. Thus $\mathcal{A}_0 = \{(F_1, F_2)\}$, so G arrows every member of this same suffix family.

Theorem 4.16 gives a subgraph F of an actual component of G, with degree $d \in \{4, 6, 8, 10, 12\}$. It is nonempty and d-regular, so $\Delta(F) = d$. It has gap one and $\mathrm{tr}(F) + \mathrm{ex}(F) \geq h_-(d) + 1$. Hence it violates T_-. Every proper subgraph satisfies T_-, including arbitrary edge deletions, vertex deletions, and their combinations. Therefore this same F belongs to the entire class $\mathcal{M}_d$.

For $d = 4$, Theorem 5.20 contradicts this membership. For $d = 6$, use Theorem 5.13. For $d = 8$, use Theorem 5.7. For $d = 10$, use Theorem 5.6. For $d = 12$, use Theorem 5.5. These are all possible degrees supplied by containment. Thus $e(G) < B$ is impossible, proving the lower bound from the stated inputs. □

Proof of Theorem 1.2. Fix the arbitrary full lists in the theorem and put $B = \sum_{k=2}^{s+t} \ell_k$. Proposition 2.4 uses those same positive profiles and prescribed-color ordinary copies. Its budget L is B, since both sum the same antidiagonal maxima.

By Theorem 2.5, the vertex-disjoint union $\bigsqcup_{k=2}^{s+t} K_{1,\ell_k}$ arrows (F_1, F_2) and has exactly B edges. The set of edge counts of finite simple arrowing hosts is therefore a nonempty subset of $\mathbb{N}_0$. It has a least element, and $\widehat{R}(F_1, F_2) \leq B$.

Theorem 6.1 applies to every finite simple arrowing host for the same full targets and gives at least B edges. Hence $\widehat{R}(F_1, F_2) \geq B$. Combining the two inequalities gives (1.1), including all endpoints and repetitions allowed in the statement. □

7 Concluding remarks

The accompanying certificate is described as a complete record of 315 items in 26 chapters, arranged in dependency order and occupying 607 pages [Ans26]. Chapters 1 and 2 contain the profile normalization and upper bound. The certificate framework is developed in Chapters 3–10, and the degree reduction is assembled in Chapter 11. The degree-specific developments occupy Chapters 12–23. Chapter 24 assembles containment and the five exclusions. Chapters 25 and 26 give the lower bound and the final equality. The reported automated review status, including the absence of human checking, is disclosed in Remark 1.4.

A chapter citation in this article specifies the mathematical statement needed at that point. To check its use, a reader first compares the four-part certificate definition, the integer domains, and the universal allocation quantifiers. The next comparison concerns the graph on which the certificate is asserted. In the improving-table steps, every entry must partition all edges of the original graph, including edges incident with vertices deleted during the intermediate argument. A table for one chosen allocation would not imply the universal cost inequality.

For the containment argument, the component C and the minimal subgraph F have different roles. The lower bound on excess belongs to C. The structural and proper-subgraph conclusions belong to F. The proof of Theorem 4.16 records the finite choice that connects them, and it retains edge minimality as well as vertex minimality. These distinctions are needed when matching the cited hypotheses.

The construction in Proposition 5.4 is proved in [Ans26, Chapter 21]. It derives the terminal matching ranks and the odd-cycle remainder from the exact cost identities and constructs partitions for every capacity split and every budget split at the improved total. The contradictions in Section 5 establish the consequences of that construction.

7.1 Published inputs recorded for the certificate

The published inputs listed below have distinct roles. The coloring and domination assertions are quoted at the specializations supplied for the certificate. The weighted greedy reference supplies methodological provenance.

Proposition 7.1 (The four-coloring input)**.** *Let G be connected and four-regular, with $G \ncong K_5$. Assign the constant list $\{1, 2, 3, 4\}$ to every vertex. Then G has a proper vertex coloring from these lists.*

This is the constant-list specialization of [HKS09, Corollary 6]. Only its list-coloring assertion is used in the recorded degree-four development.

Definition 7.2 (Ordinary domination)**.** A set $D \subseteq V(G)$ dominates G if every vertex belongs to D or is adjacent to a vertex of D. The minimum cardinality of such a set is the ordinary closed-neighborhood domination number $\gamma(G)$.

Proposition 7.3 (The cubic domination input)**.** *If G is a connected cubic graph of integer order $n > 8$, then*

$$\gamma(G) \leq \frac{5n}{14}.$$

This is [KS09, Theorem 4]. Its stated use is restricted to simple connected components of these orders. The accompanying development treats orders four, six, and eight directly [Ans26]; the displayed published bound is not being applied at those orders.

Remark 7.4 (Weighted greedy provenance). The weighted greedy potential used in the accompanying development takes methodological provenance from [BK14, Theorem 1]. Its minimum-degree domination theorem is not assumed for the prescribed-maximum-degree argument. The latter variant and its sufficient gain inequalities are separate arguments in the accompanying certificate [Ans26].

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A Complete proof record

About this document

This document contains a complete, self-contained proof of the size Ramsey formula for pairs of star forests (Erdős Problem #561), rewritten in compressed form from the system's full verified record: every statement needed for the main theorem, with its full proof, arranged in 26 chapters so that each statement relies only on statements that precede it or on cited published theorems. Items are numbered consecutively from 1 to 315; the main theorem is the last item. Supporting results are called Lemmas; the main theorem and the principal results it rests on (the upper bound, the contained-obstruction theorem, the five whole-degree exclusions and the lower bound) are called Theorems. Each item ends with the list of earlier items its proof depends on.

The proofs were produced within the Ansatz system, an automated proof-search and verification system. Each statement was checked mechanically by that system's verifier and reviewed by automated auditors. **No statement in this document has been checked by a human.** The text has been converted mechanically from the system's working notation into standard mathematical typesetting; a small number of items whose notation could not be converted automatically are printed in plain notation and marked as such. Nothing has been reworded, shortened or added.

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Lemma 1 (Weighted greedy domination of prescribed maximum-degree vertices). *Let $d \geq 2$ be an integer. Let $a > 0$ and $\sigma > 0$ be real numbers, and let b_0,..., b_d be real numbers with $b_0 = 0$ and*

$0 \leq b_d - b_{d-1} \leq \cdots \leq b_2 - b_1 \leq b_1, b_d \leq a.$

Assume

$a + d\, b_1 \geq \sigma,$

$(d+1)a - d\, b_{d-1} \geq \sigma,$

and, for every integer $2 \leq k \leq d$,

$k\, a - (k-1)b_{k-2} + k(d+1-k)(b_k - b_{k-1}) \geq \sigma.$

Let H be a finite simple graph, let $n = |V(H)|$, and assume its maximum degree is at most d. Let S be any subset of $\{vinV(H) : d_H(v) = d\}$. Write $N_H(v)$ for the open neighborhood of v and $N_H[v] = N_H(v)$ union $\{v\}$ for its closed neighborhood. A set X dominates S in closed neighborhoods when S is contained in $\bigcup_{x \in X} N_H[x]$.

Fix any ordering of $V(H)$. Start with X empty and $W = S$. While W is nonempty, choose a vertex maximizing $|N_H[v]$ intersect $W|$, p refer a vertex in W whenever one attains the maximum, and use the fixed ordering to break remaining ties. Adjoin this vertex to X and replace W by W minus $N_H[v]$.

Every execution of this rule terminates after at most $|S|$ selections, uses distinct vertices, and produces a set X dominating S with

$|X| \leq floor((a|S| + \sum_{(} v \in V(H) minus S) b_{(} |N_H(v) intersect S|))/\sigma)$

$\leq floor((a|S| + b_d(n - |S|))/\sigma).$

These conclusions hold for every initial ordering. If S is the full set of degree-d vertices, the same X also satisfies $\Delta(H - X) \leq d - 1$, where H-X is the full induced graph after deleting the vertices of X.

The graph may be empty, S may be empty, and all orders, degrees, cardinalities, neighbor counts, step numbers and indices are nonnegative integers in their stated ranges. The weights a, σ, b_j are real numbers in the displayed domains. No minimum-degree, connectivity, parity or matching assumption is imposed.

Proof. We prove the prescribed-set construction directly. Its weighting method follows Csilla Bujtas and Sandi Klavzar, "Improved upper bounds on the domination number of graphs with minimum degree at least five", arXiv:1410.4334 (2014), Theorem 1. That theorem concerns a minimum-degree hypothesis and domination of the entire graph; it is not an input to the argument below.

1. The evolving set and potential.

At each stage W consists precisely of the vertices of S not in any selected closed neighborhood. If W is nonempty, every vertex of W belongs to its own closed neighborhood, so

$k = \max_{(} v \in V(H)) |N_H[v]$ intersect $W|$

is an integer between one and $d + 1$. The specified maximizer exists in the finite vertex set and removes exactly k vertices from W. A previously selected vertex has no member of W in its closed neighborhood, so it cannot be selected again. Each selection strictly decreases $|W|$; there are at most $|S|$ selections and the terminal W is empty. This proves termination, distinctness, and domination, including the empty-S case.

For a current W give every vertex in W weight a. Give a vertex outside W with j neighbors in W weight b_j. The index j lies in {0,...,d}. Let $P(W)$ be the sum of all these weights. The weight conditions imply

$0 = b_0 \leq b_1 \leq \cdots \leq b_d \leq a.$

When W decreases, a vertex remaining outside W has a nonincreasing neighbor count and hence a nonincreasing weight. A vertex leaving W changes from a to some $b_j \leq a$. Every individual change is therefore a nonnegative loss.

Suppose $1 \leq k \leq d$, an old outside vertex has $t \leq k$ neighbors in W, and ℓ of these neighbors are removed in the step. Then $0 \leq \ell \leq t$ and

$b_t - b_{t-\ell} \geq \ell(b_k - b_{k-1})$. (1)

For $\ell = 0$ both sides are zero. For $\ell > 0$ the left side is the sum of the ℓ differences with indices $t - \ell + 1$,..., t, each of which is at least the kth difference by the assumed ordering. This inequality counts losses by incidences, including when several removed vertices share an outside neighbor.

At a stage with maximum closed count k, a vertex in W has at most k-1 neighbors in W, and a vertex outside W has at most k neighbors in W. Every member of W has total H-degree exactly d, because W is a subset of S. We prove that every step loses at least σ in potential.

2. Selections with $2 \le k \le d$.

First suppose the selected vertex v lies in W. The removed set consists of v and its k-1 neighbors in W. The new weight of v is $b_0 = 0$. Each other removed vertex had at most k-1 old neighbors in W and loses the neighbor v, so it has at most k-2 new neighbors in W. The combined loss of these k vertices is at least

$k\,a - (k-1)b_{k-2}$.

Each removed vertex had at least $d - (k-1) = d + 1 - k$ neighbors outside the old W. The number of incidences between the removed set and the old outside set is thus at least $k(d+1-k)$. Apply (1) to each old outside vertex, with ℓ equal to its actual number of removed neighbors, and sum. Its t is at most k by maximality of the chosen closed count. The additional loss is at least $k(d+1-k)(b_k - b_{k-1})$. The old outside set and the removed set are disjoint, so these two lower bounds add. All uncounted losses are nonnegative. The kth assumed inequality makes their sum at least σ.

Now suppose the selected vertex v is outside W. The preference rule implies that no vertex in W has closed count k. Thus each member of W has at most k-2 neighbors in W. The selected vertex has exactly k neighbors in W, and these are precisely the k removed vertices. Its own weight falls from b_k to zero. Each removed vertex ends with weight at most b_{k-2}, giving a loss of at least $k(a - b_{k-2})$ among them.

Each removed vertex had at least $d + 2 - k$ neighbors outside W. Excluding its neighbor v leaves at least $d + 1 - k$ other old outside neighbors. Applying (1) to those old outside vertices other than v yields another loss of at least $k(d+1-k)(b_k - b_{k-1})$. Each such vertex's full loss accounts for all its removed-neighbor incidences, so shared neighbors introduce no duplicate charge. The three counted sets are disjoint. The total loss is at least

$b_k + k(a - b_{k-2}) + k(d+1-k)(b_k - b_{k-1})$.

This is the kth assumed gain plus $b_k - b_{k-2} \ge 0$, and is therefore at least σ.

3. Endpoint selections.

If $k = 1$, every vertex of W has no neighbor in W and attains the maximum. The rule selects some v in W. It loses weight a. Its d distinct neighbors are outside W and each has exactly one old neighbor in W, namely v, since no outside vertex has closed count greater than one. Each loses b_1. Thus the potential loss is at least $a + d\,b_1 \ge \sigma$.

If $k = d + 1$, an outside vertex cannot attain k because its degree is at most d. The selected vertex lies in W and has d neighbors in W. It loses weight a; each of these neighbors loses v and ends with at most d-1 remaining neighbors in W. Their total loss is at least $d(a - b_{d-1})$, so the step loses at least $(d+1)a - d\,b_{d-1} \ge \sigma$. These are all possible values of k.

4. Sum of losses and the induced deletion.

Initially the potential is

$P(S) = a|S| + \sum_{(} v \in V(H) minus S) b_{(}|N_H(v) intersect S|)$.

At termination every vertex has weight $b_0 = 0$. Adding the per-step losses gives $\sigma|X| \le P(S)$. Divide by $\sigma > 0$ and use integrality of $|X|$ to obtain the first floor bound. Every summand in the displayed sum is at most b_d, proving the second.

If S is the full degree-d set, a vertex of S remaining after deletion of X has a neighbor in X, because it is dominated and does not itself belong to X. It loses at least one incident edge. Every vertex outside S originally had degree at most d-1, and induced deletion cannot increase degree. Hence every remaining vertex has degree at most d-1. The empty graph is assigned maximum degree zero, which also obeys this bound. The argument uses only the stated preference among maximizers and works for every ordering that resolves the other ties. □

Remark. Concavity turns each removed uncovered incidence into a controlled loss of potential. A finite list of possible closed-neighborhood sizes suffices to verify every greedy step.

Depends on. Nothing earlier; this item is self-contained.

Lemma 2 (Supported credits for a decreasing integer profile). *Let D, d, r, E be integers with $D \geq 4$, $1 \leq d < D$, D-d even, and r, $E \geq 0$. Put $u = (D-d)/2$ and $T = floor((D-1)/2)$. Let c_0,..., c_{D-1} be a nonincreasing sequence of nonnegative integers satisfying*

$c_0 = r + E$,

$c_i \leq r$ *for every integer* $T < i < D$,

and $c_i + c_{D-i} \leq 2r + 1$ *for every integer* $1 \leq i < D$.

Define the integer

$\psi = (D-1)r + u - \sum_{i=1}^{D-1} c_i$

and suppose $\psi \geq 1$ *and* $E \leq \psi + 1$.

For integers $1 \leq i < D$ *put* $h_i = \max(c_i - r, 0)$ *and* $s_i = \max(r - c_i, 0)$. *Impose the following additional bound, according to D:*

if D is 4 or 5, then $\psi + \sum_{i=1}^{T} h_i \geq 2u$;

if $D \geq 6$, *then* $r - c_{D-1} \geq u$.

Put $\delta = \max(2u - \psi, 0)$. *Then*

$\delta \leq h_1 + \sum_{i=2}^{T} \min(h_i, 1)$.

Define integers a_1,..., a_{D-1} *by*

$a_1 = \min(\delta, h_1)$,

$a_i = 1$ *when* $2 \leq i \leq T$ *and* $i \leq 1 + \delta - a_1$,

and $a_i = 0$ *for every other integer* $2 \leq i < D$.

These numbers satisfy

$0 \leq a_i \leq h_i$ *for* $1 \leq i < D$,

$\sum_{i=1}^{D-1} a_i = \delta$,

$a_1 \leq u$,

and $a_i \leq 1$ *for every* $2 \leq i < D$.

At every index i with $a_i > 0$ one has $E \leq D - d - i + 1$. At every integer j with $2 \leq j \leq T$ and $a_j > 0$ one also has

$0 \leq r - c_{D-1} \leq u + T - j$.

In addition,

$\psi \geq \max(1, ceil((D-d-T)/2))$.

All sums and index intervals in this statement contain integers only. Empty sums equal zero. In particular $T = 1$ is allowed, and when $\delta = 0$ all credits are zero. The nonnegative integers $\mu_i = c_i - a_i$ satisfy

$\mu_i \geq \min(c_i, r)$,

$\sum_{i=1}^{D-1} (c_i - \mu_i) = \delta$.

Thus the displayed construction reduces only entries strictly above r, and pays precisely δ using the stated coordinate bounds.

Proof. The parity and strict inequalities on D, d give $u \geq 1$. Since $D \geq 4$, one has $T \geq 1$ and $D - 1 > T$. The sequence h_i is nonincreasing, because the sequence c_i is nonincreasing. The upper-half hypothesis gives $h_i = 0$ whenever $i > T$. Moreover $c_1 \leq c_0 = r + E$, so $h_1 \leq E$.

For each i, the definitions give $c_i = r + h_i - s_i$. Substitution in the definition of ψ gives the exact identity

$\psi = u + \sum_{i=1}^{D-1} s_i - \sum_{i=1}^{T} h_i$. (1)

We first prove the asserted bound on δ.

If $D = 4$, the parity and strict defect bounds force $d = 2$, $u = 1$ and $T = 1$. The assumed capped bound gives $2u - \psi \leq h_1$. Since h_1 is nonnegative, taking the maximum with zero proves $\delta \leq h_1$, as required.

If $D = 5$, then $T = 2$ and $u \leq 2$. Since $\psi \geq 1$, one has $\delta \leq 3$. The assumed capped bound gives $\delta \leq h_1 + h_2$. If $h_2 \leq 1$ this is the required inequality. If $h_2 \geq 2$, then $h_1 \geq h_2 \geq 2$, so

$\delta \leq 3 \leq h_1 + 1 = h_1 + \min(h_2, 1)$.

This proves the bound when $D = 5$.

Suppose $D \geq 6$. For every integer i with $2 \leq i \leq T$, the reflected index D-i exceeds T and is different from D-1. Thus $c_{D-i} \leq r$. If $h_i \geq 1$ then $c_i = r + h_i$ and the paired bound yields $c_{D-i} \leq r + 1 - h_i$; hence $s_{D-i} \geq h_i - 1$. If $h_i = 0$, nonnegativity gives $s_{D-i} \geq 0$. In both cases

$s_{D-i} \geq \max(h_i - 1, 0)$.

These reflected indices are distinct. Also $s_{D-1} = r - c_{D-1} \geq u$ by the assumed tail bound. Retaining these nonnegative terms in (1) gives

$\psi \geq 2u - h_1 - \sum_{i=2}^{T} \min(h_i, 1)$.

Rearrange this inequality and use the nonnegativity of its right-hand credit bound to obtain

$\delta = \max(2u - \psi, 0) \leq h_1 + \sum_{i=2}^{T} \min(h_i, 1)$.

This completes the proof of the thin support bound in every degree.

If $\delta \leq h_1$, the construction has $a_1 = \delta$. Its condition for a later positive credit would be $i \leq 1$, so all other credits are zero. Their sum is δ and each lies below its h_i.

If $\delta > h_1$, then $a_1 = h_1$. Put $q = \delta - h_1$, a positive integer. The support bound just proved gives

$q \leq \sum_{i=2}^{T} \min(h_i, 1)$.

Each h_i is a nonnegative integer, so this sum counts the indices in {2,...,T} with $h_i > 0$. Because the h_i are nonincreasing, these positive entries form an initial interval of that index set. It follows that $q \leq T - 1$ and $h_i \geq 1$ for every $2 \leq i \leq q + 1$. Exactly these q indices receive $a_i = 1$. The credits therefore lie below h_i and sum to $h_1 + q = \delta$. All later credits vanish. This proves the full support and total assertions, including the possibility $T = 1$, in which case this second case cannot occur.

To bound a_1, split into two exhaustive integer cases. If $\psi \leq u - 1$, then

$a_1 \leq h_1 \leq E \leq \psi + 1 \leq u$.

If $\psi \geq u$, then

$a_1 \leq \delta = \max(2u - \psi, 0) \leq u$.

Thus $a_1 \leq u$ always, and the definition bounds every later credit by one.

If $a_1 > 0$, then $\delta > 0$, so $\psi \leq 2u - 1$. Hence $E \leq \psi + 1 \leq 2u = D - d$, which is the claimed active-index bound for $i = 1$. Now let $2 \leq i \leq T$ and $a_i > 0$. The construction forces $\delta > h_1$, $a_1 = h_1$, and $\delta - h_1 \geq i - 1$. Its already proved support gives $h_i \geq 1$, and monotonicity gives $h_1 \geq h_i \geq 1$. Consequently $\delta \geq i$. Since $\delta > 0$, its definition gives $\psi = 2u - \delta$. We obtain

$E \leq \psi + 1 = 2u - \delta + 1 \leq 2u - i + 1 = D - d - i + 1$.

There are no positive credits at indices greater than T.

Let $2 \leq j \leq T$ with $a_j > 0$. As just shown, $\delta > h_1$ and $\delta - h_1 \geq j - 1$. Thus

$\psi + h_1 = 2u - \delta + h_1 \le 2u - j + 1$. (2)

The reflected-index argument used earlier applies for every D for which this j exists, without using the additional tail bound. Using (1), the distinct indices D-i for $2 \le i \le T$, and the separate term s_{D-1}, it gives

$\psi \ge u + s_{D-1} - h_1 - \sum_{i=2}^{T} \min(h_i, 1)$

$\ge u + s_{D-1} - h_1 - (T-1)$.

Combine this with (2) to obtain $s_{D-1} \le u + T - j$. Since $D - 1 > T$, the upper-half hypothesis gives $c_{D-1} \le r$ and hence $s_{D-1} = r - c_{D-1} \ge 0$. This proves the terminal-tail assertion.

The thin support bound and $h_1 \le E \le \psi + 1$ imply

$\delta \le \psi + 1 + (T-1) = \psi + T$.

Since $2u - \psi \le \delta$ even when 2u-psi is negative, it follows that $2\psi \ge 2u - T = D - d - T$. Integrality of ψ gives $\psi \ge ceil((D - d - T)/2)$, and its assumed lower bound gives the stated maximum with one.

Finally $0 \le a_i \le h_i = \max(c_i - r, 0)$ implies $c_i - a_i \ge \min(c_i, r) \ge 0$: if $c_i \le r$, then $h_i = 0$ and $a_i = 0$; if $c_i > r$, subtracting at most $c_i - r$ leaves at least r. The sum of the decreases $c_i - \mu_i$ is the already established sum of the credits, namely δ. When $\delta = 0$, the definition gives $a_1 = 0$ and no later index can satisfy $i \le 1$, so all credits are zero. Every boundary case in the statement is therefore included. □

Remark. Complementary rows turn every overflow after the first into at most one available credit. The first coordinate supplies the remaining amount, and its size is controlled by the excess bound.

Depends on. Nothing earlier; this item is self-contained.

Lemma 3 (Cubic domination from the connected bound). *For a finite simple graph G and $D \subseteq V(G)$, let $N_G[D]$ be D together with all vertices having a neighbor in D, and define $\gamma(G) = min\{|D| : N_G[D] = V(G)\}$. Every vertex may be selected, and the minimum for the empty graph is zero. If every vertex of G has degree exactly three, then*

$8\gamma(G) \le 3|V(G)| = 2e(G)$,

where $e(G) = |E(G)|$. Empty and disconnected graphs are permitted. All orders, degrees, edge counts and set cardinalities are nonnegative integers.

Proof. We use the following published theorem: for every connected cubic graph C of integer order $n > 8$, $\gamma(C) \le 5n/14$, where a dominating set contains or has a neighbor at every vertex. This is Theorem 4 of A. V. Kostochka and C. Stocker, A new bound on the domination number of connected cubic graphs, Siberian Electronic Mathematical Reports 6 (2009), 465–504. Here cubic means degree exactly three; the theorem has no girth, bridgelessness or higher-connectivity hypothesis. Its domination convention, defined in the paper's introduction, is the closed-neighborhood convention in the statement. The precise connected $n \ge 9$ version is also recorded as Lemma 2 of R. Gomez and J. Gutierrez, Domination and packing in graphs, arXiv:2402.05088. We apply the original theorem only to finite simple connected components with more than eight vertices.

First let C be a nonempty connected component of G and put $n = |V(C)|$. No edge joins distinct components, so C is itself cubic. Counting the two endpoints of each edge gives $3n = 2e(C)$; hence n is even. Simplicity and degree three give $n \ge 4$. We handle $n = 4$, 6, 8 directly.

If $n = 4$, every vertex is adjacent to the other three vertices, and any single vertex dominates C. Thus $8\gamma(C) \le 8 \le 3n$.

If $n = 6$, choose a vertex x. Its closed neighborhood $N_C[\{x\}]$ has four vertices, so the remaining set U has exactly two vertices u, v. If u and v are adjacent, {x, u} dominates C. If they are not adjacent, each of u and v has all three of its neighbors in the three-element set $N_C(x)$: neither is adjacent to x, and there is no edge uv. Thus every vertex w in $N_C(x)$ neighbors both u and v, and {x, w} dominates C. In either case $\gamma(C) \le 2$, giving $8\gamma(C) \le 16 \le 18 = 3n$.

If $n = 8$, choose a vertex x, put $A = N_C(x)$, and put $U = V(C)$ minus $N_C[\{x\}]$. Then $|A| = 3$, $|U| = 4$, and there is no edge from x to U. Every vertex in A uses one of its three incident edges on x, so the number q of edges joining A to U is at most six. Write $h = e(C[U])$ for the edge count induced by U. Counting degrees at

U yields $12 = 2h + q$, so $h \geq 3$. A graph on four vertices with maximum degree at most one has at most two edges, by the degree sum. Consequently C[U] has a vertex y with at least two neighbors in U. Selecting y covers at least three vertices of U. If any vertex of U remains uncovered, select it as well. Together with x these selections dominate C and use at most three vertices. Hence $8\gamma(C) \leq 24 = 3n$.

For every remaining connected component $n > 8$, the stated Kostochka–Stocker theorem applies: C is finite, simple, connected and cubic, and the selected vertices can be any vertices of C. It gives $\gamma(C) \leq 5n/14 \leq 3n/8$, since $40 \leq 42$.

In each component choose a dominating set satisfying $8|D_C| \leq 3|V(C)|$. Such sets exist by the cases just proved, and there are only finitely many components. Their union D dominates G, as each vertex is covered within its component. Disjoint component vertex sets give

$8\gamma(G) \leq 8|D| = \sum_C 8|D_C| \leq 3sum_C\ |V(C)| = 3|V(G)|$.

If G is empty, choose D empty and the same inequality is $zero \leq zero$. Finally the degree-sum identity for G gives $3|V(G)| = 2e(G)$, including the empty graph. This proves the assertion for all permitted graphs. □

Remark. The published connected bound is slightly stronger than the needed cubic bound; explicit small-order covers and union over components restore the full domain.

Depends on. Nothing earlier; this item is self-contained.

Lemma 4 (Binary remnant bounds at arbitrary plateau width). *All graphs are finite and simple. A positive star forest is a nonempty vertex-disjoint union of stars $K_{1,d}$ with positive integer edge sizes d. Copies are ordinary subgraphs and different target components have disjoint vertex sets. Write $e(F)$ for an edge count and $\nu(C)$ for maximum matching cardinality.*

Let F, J have full nonincreasing positive integer component sizes $n_1,\ldots, n_S$ and $m_1,\ldots, m_T$, where $S, T \geq 1$ are integers and their heads are $B = n_1$ and $A = m_1$. Put $M = S + T - 1$ and

$\ell_k = max\{n_i + m_j - 1 : 1 \leq i \leq S, 1 \leq j \leq T, i + j = k\}$

for every integer $2 \leq k \leq S + T$. Fix an integer $0 \leq q < M$. Put $D = \ell_{q+2} - 1$ and $P_q = \sum_{k=q+2}^{S+T}(\ell_k - 1)$. Assume $4 \leq A, B \leq D - 1$. Let R be the number of initial entries equal to $D + 1$ in the natural tail $\ell_{q+2},\ldots, \ell_{S+T}$.

Put $W = \sum_{j=1}^{T} \max(m_j - 2, 0)$. Let λ, l be nonnegative integers satisfying

$\lambda \geq e(F) + 1, 2\lambda \geq 3S, T \geq \lambda + 1, W \leq S,$

$\lambda - S \geq l - 1.$

For $j = 1, 2$ define $T_j = |\{i : m_i \geq j\}|$ and

`a_j=max{|U|:` $U \subseteq \{i : m_i \geq j\}, |U| \leq \lambda,$

sum_(i in U)(m_i+1)<=2lambda+l}.

Put $c_j = T_j - a_j$. Then $0 \leq c_2 \leq c_1$ and $c_1 \geq 1$. Let J_{bin} consist of c_2 two-edge stars and $c_1 - c_2$ unit stars. It is a nonempty componentwise subforest of J.

Write $m'_1 \geq \cdots \geq m'_{c_1}$ for its complete positive size list and $M_{bin} = S + c_1 - 1$. If $q < M_{bin}$, put

$P_q^b in = \sum_{k=q+2}^{S+c_1}$

$(max\{n_i + m'_j - 1 : 1 \leq i \leq S, 1 \leq j \leq c_1, i + j = k\} - 1).$

If $q \geq M_{bin}$, put $P_q^b in = 0$; in that case the new nonempty actual total-q suffix family is empty. Then

$0 \leq P_q - P_q^b in \leq (D - 1)\lambda + 2R - 1.$

For every integer $0 \leq \beta < c_1$, let J^β and $J_b in^\beta$ denote the forests obtained by deleting exactly β leading whole components from their respective full sorted lists. For every pair of disjoint finite simple graphs C, Q with $|V(C)| \leq 2\lambda + l$ and $\nu(C) \leq \lambda$, every copy of J^β in C disjoint union Q leaves a copy of $J_b in^\beta$ entirely in Q.

All list entries, indices, cuts, component counts, fitting maxima, capacities, matching sizes, D and R are integers in the stated domains. No restriction on parity, repeated sizes, original unit multiplicities, or the degree of C is imposed.

Proof. Each fitting maximum exists because its finite set of candidate subsets contains the empty subset. One has $0 \le a_j \le \min(\lambda, T_j)$. An eligible subset for a_2 is eligible for a_1, so $a_2 \le a_1$. Put $v = T - T_2$, the number of original blue units. Removing the units from a subset attaining a_1 leaves an eligible subset for a_2 with at least $a_1 - v$ members. Hence $a_1 - a_2 \le v$, which gives $0 \le c_2 \le c_1$. Also $c_1 = T - a_1 \ge T - \lambda \ge 1$. Choose c_2 distinct nonunit J-components and trim them to size two, then choose $c_1 - c_2$ further components and trim them to size one. There are enough because $c_2 \le T_2$ and $c_1 \le T$. This proves componentwise containment and the existence of the complete list of J_{bin}.

Fix a legal $\beta < c_1$ and a copy of J^β in C disjoint union Q. A copied component is connected and lies wholly in one of the two parts. Selecting one edge from each component lying in C shows that their number is at most λ. Their total vertex count is at most $2\lambda + l$. Thus at most a_1 of all the copied components, and at most a_2 of the copied nonunits, lie in C. The full suffix has $\max(T_j - \beta, 0)$ components of size at least j, for $j = 1, 2$. Therefore Q retains at least $c_1 - \beta$ total components and at least $\max(c_2 - \beta, 0)$ nonunits. Select and trim the required nonunits first, and then select enough further disjoint components and trim them to edges. This yields exactly $J_b in^\beta$ in Q. The argument applies to the same binary forest for every β and does not require compatible subsets attaining a_1 and a_2.

We now count profile widths directly. For all positive integers x, y put $N_x = |\{i : n_i \ge x\}|$ and $T_y = |\{j : m_j \ge y\}|$, with value zero beyond the respective heads. For a positive integer z the number V_z of full excess entries $\ell_k - 1$ at least z is

$$V_z = max\{N_x + T_y - 1 : x + y = z + 2, N_x T_y > 0\}, \quad (1)$$

with maximum zero when there is no candidate. For each positive threshold rectangle, all index sums from two through $N_x + T_y$ occur, giving an initial interval of profile positions of length $N_x + T_y - 1$. Conversely any pair with $n_i + m_j \ge z + 2$ dominates positive thresholds x, y with that sum: take $x = \max(1, z + 2 - m_j)$, $y = z + 2 - x$. The union of the intervals is therefore exactly the superlevel set. This proves (1) and in particular shows that the full excess profile is nonincreasing.

After q initial positions its height-z width is $\max(V_z - q, 0)$. Summing over positive heights gives P_q. Define $V_z^b in$ by the same formula for the full binary pair. Componentwise containment ensures $V_z^b in \le V_z$: its sorted component sizes are at most the corresponding initial original sizes, so every new diagonal is at most the old one. Define

$$\Delta_z = \max(V_z - q, 0) - \max(V_z^b in - q, 0).$$

These numbers are nonnegative and at most $V_z - V_z^b in$. Their sum is $P_q - P_q^b in$. This remains true when $q \ge M_{bin}$, since $V_z^b in \le M_{bin}$ and all its natural widths then vanish. By the definition of D and R, the old natural height-D width is R and all higher natural widths vanish. Thus $\Delta_D \le R$ and $\Delta_z = 0$ for $z > D$.

Put $U = e(F) - S - (B - 1) \ge 0$ and $h_0 = \lambda - S$. Then

$$U + W \le U + S = e(F) - B + 1 \le \lambda - B < \lambda < T. \quad (2)$$

For $x \ge 2$ one has $N_x - 1 \le U/(x - 1)$, because each counted component other than one fixed head contributes at least x-1 to U. For $y \ge 3$, one has $T_y \le W/(y - 2)$. In particular, any rectangle with $x \ge 2$, $y \ge 3$ has width at most $U + W < \lambda$.

For each integer $1 \le j \le B - 1$ put $f_j = |\{i : n_i = j\}|$, $F_j = \sum_{a=1}^{j} f_a$, and $d_j = |\{i : m_i \le j\}|$; thus $d_1 = v$. At height j the blue-unit rectangle has width $T + N_{j+1} - 1 \ge T$ and dominates all mixed rectangles with $x \ge 2, y \ge 3$ by (2). The blue-threshold-two rectangle and the red-unit rectangle are the only other possible improvements. Their differences from that baseline are $f_j - v$ and $F_j - d_j$, respectively. If the red-unit rectangle does not exist because $j + 1 > A$, then $d_j = T$ and $F_j - T < 0$, so that expression has no effect. Hence

$$V_j = T + N_{j+1} - 1 + \Psi_j,$$

$$\Psi_j = \max(0, f_j - v, F_j - d_j). \quad (3)$$

The new full unit rectangle has width $c_1 + N_{j+1} - 1$. With $a = a_1$ and $\delta_0 = \lambda - a$, the total loss at these heights is consequently at most

$(B-1)a + \sum_{j=1}^{B-1} \Psi_j$. (4)

The same bound applies after natural truncation, since truncation cannot increase a width difference.

We bound the last sum using the smallest blue components. Put $p = S - 1$. The nonnegative vector $(f_1, \dots, f_{B-1})$ has coordinate sum at most p. If $p = 0$, all its coordinates vanish and the sum in (4) is zero. Suppose $p > 0$. For fixed blue counts, each Ψ_j is the maximum of three affine functions of this vector. For any nonnegative weights totaling one, such a maximum at the weighted average is at most the weighted average of the maxima, by applying the affine equality to each of the three functions separately and then taking their maximum. Summing over j preserves that inequality.

Write the f-vector as a weighted average of zero and vectors having p in one coordinate and zero elsewhere: the weight of the coordinate-j vector is f_j/p and the weight of zero is one minus their sum. At the vector whose nonzero coordinate is a_0, the sum of Ψ_j equals

$\max(p - v, 0) + \sum_{j=a_0+1}^{B-1} \max(p - d_j, 0)$.

It is at most $\Xi = \sum_{j=1}^{B-1} \max(p - d_j, 0)$, since $d_1 = v$ and all terms are nonnegative. At zero the sum is zero. The weighted-average inequality therefore gives

$\sum_{j=1}^{B-1} \Psi_j \le \Xi$. (5)

List the blue sizes increasingly as $x_1 \le \cdots \le x_T$. We have $p < T$ because $\lambda \ge e(F) + 1 \ge S + B$ and $T \ge \lambda + 1$. Among the first p sizes, the number greater than j is $\max(p - d_j, 0)$. Counting by heights yields

$\Xi = \sum_{i=1}^{p} \min(x_i - 1, B - 1)$. (6)

The a smallest components have the least vertex cost of any a-element subset. Some subset of size a attains the fitting maximum, so these a smallest components also fit. Therefore

$\sum_{i=1}^{a} (x_i - 1) \le 2\lambda + l - 2a = 2\delta_0 + l$. (7)

First suppose $a \ge p$. If $\Xi = 0$, its bound by $3\delta_0$ follows from $\delta_0 \ge 0$. If $\Xi > 0$, then $p \ge 1$ and $x_p \ge 2$. Every one of the a-p components after position p has excess $x_i - 1$ at least one. Equations (6), (7) imply

$\Xi \le 2\delta_0 + l - (a - p) = 3\delta_0 - h_0 + l - 1 \le 3\delta_0$, (8)

using $h_0 \ge l - 1$.

Now suppose $a < p$. If $x_p \le 2$, then $\Xi \le p \le S$. If $x_p \ge 3$, all T-p later blue components contribute at least one to W, whence

$\sum_{i=1}^{p} \max(x_i - 2, 0) \le W - (T - p)$.

As $x_i - 1 \le 1 + \max(x_i - 2, 0)$, equation (6) yields

$\Xi \le 2p + W - T \le 3S - \lambda - 3$. (9)

Thus in either case $\Xi \le \max(S, 3S - \lambda)$. The assumption $2\lambda \ge 3S$ gives both $3(\lambda - S) \ge S$ and $3(\lambda - S) \ge 3S - \lambda$. Since $a < p = S - 1$, $\delta_0 = \lambda - a > \lambda - S$. It follows that $\Xi \le 3\delta_0$ in this case as well. Combining (5), (8), (9) and $B \ge 4$ gives

$\sum_{j=1}^{B-1} \Psi_j \le 3\delta_0 \le (B-1)\delta_0$.

Equation (4) shows that the total loss at heights 1,..., B-1 is at most (B-1)lambda.

Consider the middle heights $B \le z \le D - 1$. There is no blue-unit rectangle because its red threshold $z + 1$ would exceed B. Mixed rectangles with $x \ge 2$, $y \ge 3$ have full width below λ by (2). A red-unit rectangle has $y = z + 1 \ge B + 1$ and width at most

$S + W/(B-1) - 1 \le 4S/3 - 1 < \lambda$,

where $B \ge 4$, $W \le S$, and $2\lambda \ge 3S$ were used. The only remaining possible rectangle has thresholds (B,2), at height B.

If $c_2 > 0$, this rectangle also exists in the binary pair and its full width loses $a_2 \le \lambda$; its natural width loses no more. All other candidates have width below λ, so their presence does not increase the natural loss beyond λ. If $c_2 = 0$, then $T_2 = a_2 \le \lambda$. The old head sum is at least $D + 2$ because $\ell_{q+2} = D + 1$. Thus the original pair $(B, D + 2 - B)$ of thresholds exists. Its blue threshold is at least three since $B \le D - 1$. At height D its width is at most $V_D = q + R$, giving $N_B \le q + R$. The natural width of (B,2) is therefore at most $\lambda + R - 1$. All other middle heights lose at most λ. There are D-B middle heights, and hence their total loss is at most

$(D - B)\lambda + R - 1.$

The remaining height D loses at most R, and higher natural heights contribute zero. Adding the low, middle and top bounds gives

$P_q - P_q^b in \le (B - 1)\lambda + (D - B)\lambda + (R - 1) + R$

$= (D - 1)\lambda + 2R - 1.$

Nonnegativity was proved by containment. If the new family is exhausted at q, all its natural widths vanish and the same proof applies to the defined value $P_q^b in = 0$. Finally, that family is empty exactly when no nonnegative legal cuts $\alpha < S$, $\beta < c_1$ total q, equivalently $q \ge S + c_1 - 1 = M_{bin}$; this justifies its stated exhaustion convention. All arguments count full positive components and permit unit stars and repeated sizes. □

Remark. The excess of the smallest S-1 blue components controls every low-height profile correction. Their vertex cost pays this error even under the weaker three-halves component ratio, while the last possible missing nonunit rectangle costs only one additional plateau-width term.

Depends on. Nothing earlier; this item is self-contained.

Lemma 5 (A maximum independent row dominating its host). *All graphs are finite and simple. Write $V(G)$, $E(G)$, $\nu(G)$, and $\alpha(G)$ for the vertex set, edge set, maximum matching cardinality, and maximum independent-set cardinality of G. An independent set is maximal if no vertex can be added while preserving independence. A graph is well-covered if every maximal independent set has size $\alpha(G)$. Graph deletion and restriction below are induced unless edge sets are explicitly partitioned.*

Let F be well-covered, let R be an induced subgraph, and put $U = V(F)$ minus $V(R)$, $u = |U|$, and $m = \nu(R)$. Suppose every connected component C of R satisfies $\nu(C) = floor(|V(C)|/2)$. For an odd-order component C, call z in $V(C)$ an exposed vertex if C-z has a perfect matching. Suppose every exposed vertex of every nonisolated odd-order component C belongs to a maximum independent set of C. Let s be the number of nonisolated odd-order components of R.

Let x, y be nonnegative integers with $x + y = u + m - 1$ such that $E(F)$ has no partition into two edge sets with matching numbers at most x and y, respectively. Suppose $j = u - x \ge s + 1$. All orders, component counts, and matching and independence cardinalities are nonnegative integers.

Let I_0 be the set of isolated vertices of R and let U_0 consist of those vertices of U having no neighbor in I_0. Then $|U0| \le s$. There is one matching assigning every vertex of U_0 to an exposed vertex in a distinct nonisolated odd component of R, compatible with a maximum matching in every component. There is a maximum independent set I of R containing all the assigned exposed vertices and all of I_0, and this I is maximal independent in F. Consequently $\alpha(R) = \alpha(F)$. Every maximum independent set of R is maximum in F, and every induced graph G with $V(R)$ contained in $V(G)$ contained in $V(F)$ has $\alpha(G) = \alpha(F)$.

The component matching and exposed-vertex assumptions hold whenever every component of R is a path or a cycle. In particular this includes isolated vertices and the empty graph. No regularity or source-minimality assumption is imposed.

Proof. We first prove the matching assignment needed for one simultaneous independent-set choice. Let T be any subset of U with $h = |T| \le j$, and put $K = F[V(R)$ union T]. Since $u - h \ge x$, choose an x-vertex set Z in U minus T. Assigning all edges incident with Z to the first part gives that part matching number at most x: different edges of a matching require distinct vertices of Z. The missing allocation therefore forces $\nu(F - Z) \ge y + 1$. Every matching of F-Z has at most $\nu(K)$ edges lying wholly in K and at most u-h-x other edges, since every other edge uses a distinct vertex outside K and Z. Thus

$\nu(K) \geq y + 1 - (u - h - x) = m + h$.

Conversely, a matching of K has at most m edges within R and at most h edges meeting T. Hence $\nu(K) \leq m + h$. For a maximum matching equality holds in both counts: there are m internal edges and h edges meeting T. Those h edges saturate T and each has exactly one T endpoint, because otherwise they would require more than h distinct T vertices. The internal matching is maximum in every component of R, since the matching number of a disjoint union is the sum of its component matching numbers. By the component hypothesis it leaves no vertex unused in an even component and one unused vertex in each odd component. Thus the T endpoints are matched to exposed vertices of distinct odd components. Finiteness guarantees that all the maximum matchings just used exist.

If $|U0| \geq s + 1$, take an $(s + 1) - vertex$ set T in U_0. Its size is at most j, so the preceding argument matches it into $s + 1$ distinct odd R-components. None can be an isolate, because vertices of U_0 have no isolate neighbor. Only s nonisolated odd components exist, a contradiction. Therefore $|U0| \leq s < j$. Apply the same matching construction to the entire set U_0, including the case it is empty. No receiving component is an isolate. In each receiving component choose a maximum independent set containing its assigned exposed vertex, as permitted by the hypothesis. Only one vertex is prescribed per component, so the choices are compatible. In each unassigned component choose any maximum independent set. Their union I is independent and has size $\alpha(R)$, since independent sets restrict componentwise and component maxima can be united. It contains every isolate.

The set I is maximal within R: an addable R-vertex would enlarge a maximum independent set in its component. Every vertex of U_0 has its assigned neighbor in I. Every vertex of U minus U_0 has a neighbor in I_0 contained in I. Therefore every vertex of F outside I has an I-neighbor, so I is maximal in F. Well-coveredness implies $|I| = \alpha(F)$, yielding $\alpha(R) = \alpha(F)$. Any maximum independent set of R is independent in F with this cardinality and hence is maximum in F. If R is induced in G and G is induced in F, restriction and inclusion give $\alpha(R) \leq \alpha(G) \leq \alpha(F)$, proving the last equality.

Finally consider path and cycle components. A path on n vertices has matching number $\lfloor n/2 \rfloor$, attained by alternate edges and bounded above because matching edges have distinct endpoints. A cycle has the same matching number by alternate edges and the same upper bound. Only odd-order components matter for the exposed-vertex condition. Label an odd path on $2a + 1$ vertices consecutively, where a is a nonnegative integer. Deleting vertex i leaves paths of orders i-1 and $2a + 1 - i$. A path has a perfect matching exactly when its order is even, including zero, by parity and alternate edges. Hence the exposed vertices of the odd path are precisely those at odd positions. All these vertices belong to the independent set of odd positions, of size $a + 1$. This set is maximum, since the first 2a vertices can be partitioned into a adjacent pairs and an independent set takes at most one from each pair plus the remaining vertex.

For an odd cycle of order $2a + 1$, where $a \geq 1$, deletion of any vertex gives an even path and hence a perfect matching. Any prescribed cycle vertex belongs to an independent a-set obtained by alternating around the cycle and stopping before its other neighbor. An independent cycle set has size at most a: map each selected vertex to its next vertex in a fixed cyclic orientation; the images are distinct and unselected, so twice the independent-set size is at most $2a + 1$. Thus the prescribed set is maximum. Even components impose no exposed-vertex condition, and an isolated vertex imposes none because the assumption concerns nonisolated components. This establishes all component assertions, including the empty family of components. □

Depends on. Nothing earlier; this item is self-contained.

Lemma 6 (The residual of a matching that isolates a vertex). *All graphs are finite and simple. A matching consists of edges with pairwise disjoint endpoints; $V(M)$ is the set of its covered vertices. A matching is maximal if no edge can be added to it, maximum if its cardinality is largest, and perfect if it covers every vertex. The empty graph has its empty perfect matching. A nonempty graph is factor-critical if deleting any one vertex leaves a perfect matching. It is equimatchable if every maximal matching is maximum. A graph is 2-connected if it has at least three vertices, is connected, and remains connected after deletion of any one vertex. Write $N_G(v)$ for the open neighborhood of v. Vertex deletion is induced.*

Let G be a factor-critical equimatchable graph, let v be any vertex, and let M be any matching in G-v covering every vertex of $N_G(v)$. Such a matching exists. Put $R = G - (V(M) union \{v\})$. Every connected component of R is either an even complete graph K_{2a} or a balanced complete bipartite graph $K_{a,a}$, for some positive integer a which may depend on the component. In particular R has a perfect matching and has no isolated vertex.

If, in addition, G is 2-connected and M is inclusion-minimal among matchings in G-v covering $N_G(v)$, then R has at most one connected component. Equivalently R is empty, or R is K_{2a}, or R is $K_{a,a}$, with $a \geq 1$. Inclusion-minimality means that no proper edge subset of M still covers $N_G(v)$; it is equivalent to requiring that every edge of M have at least one endpoint adjacent to v. An inclusion-minimal such matching always exists.

No regularity, degree bound, maximum-cardinality condition on M, or uniqueness of M is imposed. All graph orders, matching cardinalities, component counts and parameters are nonnegative integers, with a positive when used to describe a nonempty component. The first assertion includes the one-vertex graph G and an empty R.

Proof. Since G is nonempty and factor-critical, its order is odd and its maximum matching cardinality is $(|V(G)| - 1)/2$. Every matching extends to a maximal matching by repeatedly adding a disjoint edge; finiteness guarantees termination. Equimatchability therefore says that every matching of G extends, without changing its existing edges, to one leaving exactly one vertex uncovered. We use this extension property throughout. A perfect matching of G-v covers $N_G(v)$, proving existence of the required M. Among its finitely many edge subsets still covering $N_G(v)$, choose an inclusion-minimal one when needed. Because matching edges have disjoint endpoints, deleting an edge uncovers exactly its two endpoints. Thus an edge can be deleted without uncovering a neighbor of v exactly when neither endpoint neighbors v. This proves the stated minimality equivalence.

We first prove the elementary classification needed for the residual components. Let H be a nonempty connected finite simple graph in which every maximal matching is perfect. A maximal matching exists, so fix a perfect matching with edges $x_i\, y_i$, $1 \leq i \leq m$, where $m \geq 1$. Every matching of H extends to a perfect one by adding edges. Consequently the two uncovered vertices of any matching covering all other vertices must be adjacent.

For two different fixed pairs, any edge between them forces its complementary cross edge. For example, $x_i\, x_j$ together with all fixed edges outside pairs i, j leaves only y_i, y_j uncovered, so $y_i\, y_j$ must be present. The same argument works for every choice of endpoint names.

Form a graph on the m fixed pairs, joining two pairs if any cross edge exists. It is connected because H is connected. Consider a path on three distinct pairs i, j, k. Choose an edge $a_i\, a_j$ between the first two pairs and denote their mates by b_i, b_j. There is an edge from pair j to pair k. Replacing it by its complementary cross edge if necessary makes its pair-j endpoint b_j; write it as $b_j\, a_k$, with mate b_k. The two disjoint edges $a_i\, a_j$ and $b_j\, a_k$, together with all fixed edges outside these three pairs, leave only b_i, b_k uncovered. Their forced edge short cuts the path. A shortest path of length at least two could therefore be shortened. The graph on fixed pairs is complete.

If $m \geq 2$, fix the names on pair 1 and name the endpoints of every other pair so that $x_1\, y_i$ and $y_1\, x_i$ are present. This is possible by pairwise adjacency and complementary cross edges. All edges $x_i\, y_j$ now exist: equal indices are matching edges; indices involving 1 were arranged; for distinct i, j different from 1, the edges $x_i\, y_1$ and $x_1\, y_j$ together with the remaining fixed edges force $y_i\, x_j$. Interchanging i, j gives the other cross edge. Thus H contains the complete bipartite graph with parts $X = \{x_1, \ldots, x_m\}$, $Y = \{y_1, \ldots, y_m\}$.

If neither part has an internal edge, H is $K_{m,m}$. Otherwise exchange X, Y if necessary and take an edge $x_i\, x_j$ within X. Its complementary edge $y_i\, y_j$ is also present. If $m = 2$ these are all the remaining edges, so H is K_4. If $m \geq 3$ and k differs from i, j, use $x_i\, x_j$ and $y_i\, x_k$ together with the fixed edges outside these three pairs. The uncovered y_j, y_k are adjacent, and complementary cross edges then give $x_j\, x_k$. Hence x_j neighbors every other vertex of X. For distinct k, l both different from j, the edges $x_j\, x_k$ and $y_j\, x_l$ with the remaining fixed edges force $y_k\, y_l$ and then $x_k\, x_l$. Every pair in X, and therefore every pair in Y, is adjacent. Thus H is K_{2m}. For $m = 1$, H is K_2. Conversely every nonperfect matching of K_{2m} leaves two adjacent uncovered vertices, while every nonperfect matching of $K_{m,m}$ leaves uncovered vertices in both parts. Such a matching is not maximal. This completes the classification.

Return to G, v, M, R. If Z is a maximal matching of R, then M union Z is maximal in G. Indeed its uncovered vertices are v and the vertices of R uncovered by Z. The latter have no edge between them by maximality of Z, and none is adjacent to v because M covers $N_G(v)$. The extension property implies that M union Z leaves only v uncovered. Therefore Z is perfect on R.

For each nonempty component of R, take any maximal matching there and choose maximal matchings independently in all other components. Their union is maximal in R and hence perfect. Thus every maximal

matching of each component is perfect. The classification just proved makes each component an even clique or a balanced complete bipartite graph, and proves the entire first assertion.

Now assume 2-connectivity and inclusion-minimality of M, and suppose for contradiction that R has at least two components. An attachment to a component means an edge from an endpoint of M into that component. There is no edge from v to R.

For an edge xy of M, its two endpoints cannot have attachments xc and yd into distinct components C and D of R. If they did, replace xy by these two disjoint edges. Every old M-endpoint remains covered, while v is uncovered. The available graph is the disjoint union of v, the odd-order graphs C-c and D-d, and the other components of R. These are separated because C, D were components and all M-endpoints remain covered. Any extension leaves v and at least one vertex in each of C-c and D-d uncovered. This contradicts the extension property.

Suppose xy belongs to M and xc is an attachment into C. Then y neighbors v or some vertex of C other than c. Otherwise replace xy by xc. The only old M-endpoint now uncovered is y. It has no neighbor in C-c by the supposition, no neighbor in any other R-component by the preceding restriction, and no edge to v. Also v has no remaining neighbor, since every neighbor except possibly y stays covered and y was assumed not to neighbor v. Thus v and y are both isolated in the available graph, so every extension leaves at least two vertices uncovered, a contradiction.

It follows that every M-edge attached to C has two disjoint attachments, one to v and one to C. For if y neighbors v, use yv and xc. Otherwise y has a neighbor in C-c, while x must neighbor v because each M-edge meets $N_G(v)$; use those two edges.

We next show that every M-edge attached to C has two disjoint attachments into C itself. If not, choose an attachment xc such that its mate y has no neighbor in C-c. Choose a different residual component D. The vertices outside D having a neighbor in D include at least two distinct vertices. If there were none, connectedness of G would fail. If there were only one, deleting it would disconnect the nonempty D from the other nonempty component C, contradicting 2-connectivity. All these outside neighbors belong to $V(M)$, because D has no neighbor in v or any other component of R.

The first attachment restriction excludes an edge from y to D. Hence at most one of the two outside neighbors just found belongs to {x, y}. Some different edge pq of M is attached to D. By the preceding paragraph its endpoints may be named so that pd is an edge to d in D and qv is an edge. Replace xy, pq by xc, pd, qv. These three edges are disjoint. The vertex v is covered, and the only old M-endpoint left uncovered is y. This y has no neighbor in C-c by assumption and none in other R-components by the first restriction, so it is isolated in the available graph. The available C-c and D-d both have odd order and have no edge to one another or to any other available component. Every extension leaves y and a vertex in each of these odd sets uncovered, again a contradiction. This proves the two disjoint attachments into C.

In particular, both endpoints of every M-edge attached to C have neighbors in C. The first restriction now shows that neither endpoint has any attachment to another residual component. Assign the M-edge to its unique attached component. An M-edge with no residual attachment is called unassigned.

Edges assigned to distinct components have no edge between their endpoint sets. Indeed, suppose x_1y_1 is assigned to C, x_2y_2 to D, and x_1x_2 is an edge, after renaming endpoints if needed. Choose attachments y_1c into C and y_2d into D. Replace the two M-edges by x_1x_2, y_1c, y_2d. All old M-endpoints remain covered, and the available graph contains isolated v and the separated odd sets C-c, D-d. Every extension leaves at least three vertices uncovered, a contradiction.

Nor can an unassigned M-edge have an edge to an assigned one. Suppose uw is unassigned, x_1y_1 is assigned to C, and ux_1 is an edge. Choose another residual component D. Connectedness supplies an M-edge x_2y_2 attached to D. It is distinct from uw and x_1y_1 by their assignments. By minimality one of its endpoints neighbors v; name it x_2. The two-attachment conclusion gives a neighbor d in D of y_2. Choose also c in C adjacent to y_1. Replace uw, x_1y_1, x_2y_2 by ux_1, y_1c, x_2v, y_2d. All four edges are disjoint: their named M-endpoints came from distinct matching edges, while c, d, v lie outside $V(M)$ and are distinct.

The sole uncovered old M-endpoint is w, which has no residual attachment because uw was unassigned. It is isolated in the available graph, since v is covered and all other M-endpoints are covered. The sets C-c and D-d are separated odd-order components of that available graph. Every extension leaves at least three vertices uncovered, the same contradiction.

For each residual component C, form the block consisting of C and all endpoints of M-edges assigned to C. There are no edges between distinct residual components, no attachments of an assigned pair to a different component, no edges between pairs assigned to different components, and no edges from assigned to unassigned pairs, by the preceding arguments. Every vertex outside these blocks other than v belongs to an unassigned pair. Therefore each block has no edge to any vertex outside it in G-v. There are at least two nonempty blocks under the current supposition, so G-v is disconnected. This contradicts 2-connectivity.

Thus R has at most one component. The first assertion already classified every nonempty component, giving exactly the three stated possibilities. The arguments allow an empty R and do not require M to have maximum cardinality. □

Remark. A neighbor-covering matching isolates the chosen vertex, so every maximal matching of the remaining graph must be perfect. If two residual components survived minimal isolation in a 2-connected graph, an edge connecting their attachment groups would create three vertices that no matching extension could cover.

Depends on. Nothing earlier; this item is self-contained.

Lemma 7 (Matching constraints from a missing allocation). *All graphs are finite and simple. Write $V(G)$, $E(G)$, and $\nu(G)$ for the vertex set, edge set, and maximum matching cardinality of a graph G. A matching is a set of edges with disjoint endpoints. For an edge subset P, $\nu(P)$ is the matching number of its spanning graph. Write F[L] for the full induced graph on L and F-Z for deletion of Z.*

Let F be a finite simple graph. Let x, y be nonnegative integers such that there is no edge partition $E(F) = P_R$ dot-union P_B satisfying $\nu(P_R) \le x$ and $\nu(P_B) \le y$. Put $n = |V(F)|$, $t = x + y + 1$, and $q = \min(x, y)$. For every vertex subset L whose complement has at least q vertices,

$\nu(F[L]) \ge |L| + t - n$.

Let A be any vertex subset, $U = V(F)$ minus A, $u = |U|$, and $m = \nu(F[A])$. Suppose $t = m + u$ and $u \ge q$. For every W contained in U with $h = |W| \le u - q$, put $H = F[A \text{ union } W]$. Then $\nu(H) = m + h$. Every maximum matching of H has exactly m edges inside A and exactly h edges from W to A, saturates W, and has no edge inside W.

If each connected component C of $F[A]$ has $\nu(C) = floor(|V(C)|/2)$, every maximum matching of H uses a maximum internal matching in each component and sends the vertices of W into distinct odd-order components. Its endpoint z in any receiving component C satisfies that C-z has a perfect matching. Conversely, any collection of actual edges matching all of W to vertices z in distinct odd-order components C for which C-z has a perfect matching, together with those perfect matchings and maximum matchings in all unassigned components, is a maximum matching of H. All cardinalities are nonnegative integers; $q = 0$, $h = 0$, and empty A are allowed. Deleting an isolated component's vertex leaves the empty perfect matching.

Proof. Color symmetry allows us to use the missing allocation (q, t-q-1). For any vertex set Z of size q, assign every edge incident with Z to the first part and every other edge to the second. Every matching in the first part has size at most q: choosing a Z endpoint for each edge gives distinct vertices. Since the allocation is missing, $\nu(F - Z) > t - q - 1$, hence $\nu(F - Z) \ge t - q$ by integrality.

For the stated L, choose Z of size q outside L. Every matching in F-Z has at most $\nu(F[L])$ edges wholly in L. Each remaining edge uses a distinct vertex outside L and Z, so there are at most $n - |L| - q$ such edges. Thus $t - q \le \nu(F - Z) \le \nu(F[L]) + n - |L| - q$. Rearranging proves the lower bound.

Apply this bound to A union W. Its complement has size $u - h \ge q$, so $\nu(H) \ge |A| + h + t - n = m + h$. For the reverse bound take any matching M of H and let a_M, b_M, c_M count its edges inside A, crossing between A and W, and inside W, respectively. Then $a_M \le m$ and $b_M + 2c_M \le h$. Consequently $|M| = a_M + b_M + c_M \le m + h - c_M \le m + h$. A maximum matching exists by finiteness and attains $m + h$. Equality forces $c_M = 0$, $a_M = m$, and $b_M = h$. This proves the claimed structure on W.

Under the component hypothesis, matching number adds across the connected components of F[A]: restriction bounds each part above, and the union of maximum component matchings attains the sum. Since M has exactly m internal edges, its restriction to every component C is maximum. The number of vertices of C not used internally is $|V(C)| - 2\nu(C)$, zero for even C and one for odd C. Therefore crossing endpoints occupy distinct odd components. In a receiving component the unique internally unmatched vertex is the crossing endpoint z, and the internal matching is perfect in C-z.

For the converse, the chosen crossing edges have distinct W endpoints and distinct receiving components. Their internal perfect matchings avoid the crossing endpoints; the maximum matchings in all other components are disjoint from them. The union is therefore a matching with m internal edges and h crossing edges. Its size is $m + h = \nu(H)$. The argument includes empty sets and uses the empty matching in a component after its isolated vertex is deleted. □

Depends on. Nothing earlier; this item is self-contained.

Lemma 8 (Regular equimatchable graphs with a balanced bipartite vertex deletion). *Let $D \geq 6$ be an even integer, and let C be a finite connected simple D-regular graph. A matching is a set of pairwise vertex-disjoint edges, its maximum cardinality is denoted by $\nu(C)$, and it is inclusionwise maximal if no further edge can be added. Assume C is equimatchable, meaning that every inclusionwise maximal matching has cardinality $\nu(C)$. Suppose x is a vertex such that $V(C)$ minus {x} is the disjoint union of independent sets A and B with $|A| = |B|$. Then $|A| = |B| = D$. There are subsets A_0 of A and B_0 of B, each of cardinality $D/2$, and a perfect matching W between A_0 and B_0 such that the induced graph C-x is exactly the complete bipartite graph on A, B with the edges of W deleted, and the open neighborhood of x is exactly A_0 union B_0.*

Conversely, for every even integer $D \geq 6$, start with the complete bipartite graph $K_{D,D}$, delete a matching W of $D/2$ edges, and add a new vertex x adjacent to all D endpoints of W. The resulting graph C is finite, simple, connected, D-regular and equimatchable, has $2D + 1$ vertices and $\nu(C) = D$, and is factor-critical: deleting any vertex leaves a perfect matching. Deleting x leaves the stated balanced bipartite graph. All orders, degrees, part sizes and matching cardinalities are integers in the indicated domains; all vertex deletions are induced.

Proof. We first establish two elementary matching assertions used below. In a finite bipartite graph, a matching covers a designated part L if every subset S of L has at least $|S|$ neighbors. Indeed, if a maximum matching misses L, start alternating paths at all its unmatched L-vertices, following nonmatching edges toward the other part and matching edges back. No reached vertex in the other part is unmatched, since reversing an alternating path to it would enlarge the matching. The reached L-set has all their distinct matched partners and at least one unmatched starting vertex. Its full neighborhood is exactly the reached set in the other part, which is smaller. This contradicts the subset condition. The converse follows by counting distinct matching partners.

The second assertion is that a finite bipartite graph in which every maximal matching is perfect is a disjoint union of balanced complete bipartite graphs. Choose a perfect matching, which exists by extending the empty matching to a maximal one. Index its edges as $a_i\ b_i$ in the respective parts. If $a_i\ b_j$ is an edge for $i \neq j$, replace $a_i\ b_i$ and $a_j\ b_j$ by that edge, retaining every other matching edge. Unless $a_j\ b_i$ exists, this is a nonperfect maximal matching. Thus off-diagonal adjacency is symmetric. For three distinct indices i, j, l with edges $a_i\ b_j$ and $a_j\ b_l$, replace their three matching edges by these two. The only uncovered vertices are a_l and b_i, which must be adjacent by the same argument. Symmetry then gives $a_i\ b_l$. Consequently adjacency of the indices is an equivalence relation, and each equivalence class induces a balanced complete bipartite graph with no edge to another class. This includes the empty graph.

Write $D = 2k$, where $k \geq 3$ is an integer, and put $|A| = |B| = \mu$. For vertex sets S let $N_C(S)$ be their open neighborhood union. Put $A0 = N_C(x)$ intersect A and $B0 = N_C(x)$ intersect B. Counting degrees on the equal independent parts gives

$Dmu = e_C(A, B) + |A0| = e_C(A, B) + |B0|$,

where $e_C(A, B)$ counts edges between the parts. Since $|A0| + |B0| = D$, both sizes are k. A vertex of A_0 has D-1 neighbors in B, so $\mu \geq D - 1 > k$. Some vertex of A is outside A_0 and has D neighbors in B. Hence $\mu \geq D$.

There is a perfect matching of C-x. For a nonempty subset S of A, the number of edges from S to B is $D|S| - |S$ intersect $A0| \geq D|S| - k$. If its B-neighborhood had size at most $|S| - 1$, that edge count would be at most $D(|S| - 1)$, contradicting $D > k$. The first matching assertion applies, and equality of the part sizes makes the matching perfect. Thus $\nu(C) = \mu$ by its order $2\mu + 1$. Every matching of C extends to one of size μ by adding edges until maximal and using equimatchability.

Fix distinct a, a′ in A_0, and put $Y = N_C(a)$ intersect B, of size D-1. There is no matching from Y into A minus {a, a′} covering Y. If J were such a matching, J union {xa′} would have D edges and would leave a uncovered with all its neighbors covered. The other available vertices would consist of mu-D-1 vertices in A

and $\mu - D + 1$ vertices in B. These counts are nonnegative if J exists. Every additional edge must run between these two sets, so every extension would have at most $D + (\mu - D - 1) = \mu - 1$ edges, a contradiction. The same assertion holds with A and B exchanged.

Define

$A1 = (N_C(B0) intersect A)$ minus A_0, $B1 = (N_C(A0) intersect B)$ minus B_0,

$A2 = A$ minus (A0 union A1), $B2 = B$ minus (B0 union B1).

We claim A_0 is complete to B_1. Otherwise some $y0$ in B_1 neighbors a in A_0 but not a' in A_0. For $Y = N_C(a)$ intersect B, consider its bipartite edges into A minus {a, a'}. Every vertex of Y retains at least D-3 neighbors, and $y0$ retains D-1, since $y0$ has neither an apex edge nor an edge to a'. At least $k - 1 \geq 2$ members of Y are outside B_0. Choose one $y1$ different from $y0$; it retains at least D-2 neighbors. Any nonempty subset of Y of size at most D-3 has enough neighbors by its minimum degree. A subset of size D-2 contains $y0$ or $y1$, so has at least D-2 neighbors. All of Y has at least D-1 neighbors from $y0$. The first matching assertion gives the prohibited matching covering Y. This proves the claim. Exchanging the roles of the two independent parts proves that B_0 is complete to A_1. Both A_1 and B_1 are nonempty because $D - 1 > k$.

For every a in A_0 we next show

$|N_C(N_C(a) intersect B)$ intersect $A| \leq D$. (1)

Choose a' in A_0 minus {a} and again set $Y = N_C(a)$ intersect B. The prohibited matching implies that a nonempty subset S of Y has fewer than $|S|$ neighbors in A minus {a, a'}. If the left side of (1) exceeded D, all of Y would retain at least D-1 neighbors, so S would be proper and $|S| \leq D - 2$. Each member of Y retains at least D-3 neighbors. Deficiency therefore forces $|S| = D - 2$ and a full remaining neighborhood Z of size D-3. Every y in S then has all D of its C-neighbors in the D-element set {x, a, a'} union Z. In particular every y neighbors x, giving S contained in B_0. This would imply $D - 2 \leq k$, whereas $2k - 2 > k$. Thus (1) holds. Its counterpart holds with the parts exchanged.

By the proved complete blocks, every a in A_0 has $D - 1 - |B1|$ neighbors in B_0, and every b in B_0 has $D - 1 - |A1|$ neighbors in A_0. Counting the edges between A_0 and B_0 and using their common size k gives $|A1| = |B1| =: t$. The A_0, B_0 block is regular of degree $h = D - 1 - t$, with $0 \leq h \leq k$ and $k - 1 \leq t \leq 2k - 1$.

Whenever a matching J of C-x covers A_0 union B_0, put $R = C - (V(J) union \{x\})$, where $V(J)$ is its endpoint set. Every maximal matching Z of R unites with J to a maximal matching of C: x has no remaining neighbor, and all uncovered vertices of R are mutually nonadjacent. Consequently $|Z| = \mu - |J|$, exactly half the order of R. The second elementary assertion shows that each nonempty component of R is balanced complete bipartite in its inherited A, B parts. In particular, R has no isolates.

Suppose $h = k$, so $t = k - 1$. Choose a perfect matching J between A_0 and B_0. In R, vertices of A_1 and B_1 have degree k, whereas vertices of A_2 and B_2 retain degree 2k. A component containing a vertex of the nonempty A_1 is a balanced complete bipartite graph of degree k. Its B-side can contain no B2-vertex of degree 2k, so all its k vertices belong to B_1. This contradicts $|B1| = k - 1$.

Suppose $h = 0$, so $t = 2k - 1$. Each a in A_0 has B-neighborhood B_1. Bound (1) says that B_1 has at most 2k neighbors in A, including all k vertices of A_0. Each B1-vertex has degree 2k and no apex edge. Thus their neighbors outside A_0 form one common k-element set Z_A, complete to B_1. The exchanged argument gives a k-element set Z_B outside B_0, complete to A_1, containing every neighbor of A_1 outside B_0. There is no A_1, B_1 edge: its A_1 endpoint would belong to Z_A and thus neighbor all 2k-1 vertices of B_1 and all k vertices of B_0, exceeding degree 2k. Therefore Z_A is contained in A_2 and Z_B in B_2.

Choose k-element sets U contained in B_1 and V contained in A_1. Let J pair A_0 with U and B_0 with V, using the complete blocks. This matching covers the apex neighborhood. In R, the set B_1 minus U has $k - 1 \geq 2$ vertices, each with precisely the k neighbors Z_A. All lie in a single balanced complete bipartite component, with A-side Z_A and a B-side of size k. Besides B_1 minus U its B-side has exactly one vertex b, necessarily in B_2. If b belonged to Z_B, its neighbors would include both Z_A and all 2k-1 vertices of A_1, exceeding degree 2k. Thus b lies outside Z_B and has no neighbor in A_1. It has none in A_0 or at x by membership in B_2. The deleted A-vertices are A_0 union V, so b loses no neighbor in forming R and retains degree 2k. This contradicts degree k throughout its component.

We have $1 \leq h \leq k - 1$ and hence $t \geq k$. Choose an edge ab between A_0 and B_0. The set $Y = N_C(a)$ intersect B contains B_1 and b. Its A-neighborhood contains A_0 because B_1 is nonempty and complete to A_0, and contains A_1 because b is complete to A_1. Bound (1) gives $k + t \leq 2k$. Thus $t = k$ and $h = k - 1$. That bound is

saturated by A_0 union A_1, so B_1 has no neighbor in A_2. Exchanging the parts gives no A_1, B_2 edge. Every A1-vertex has all its D neighbors in B_0 union B_1, a set of size D, and is complete to it; the counterpart holds for B_1. The definitions also exclude A_2, B_0 edges and A_0, B_2 edges, and neither A_2 nor B_2 neighbors x. Hence no edge joins A_2 union B_2 to its complement. Connectedness makes both sets empty. Therefore $\mu = 2k = D$. All A, B edges are present except in A_0, B_0, where each vertex misses exactly one opposite vertex. The missing edges are a perfect matching W between A_0 and B_0. This proves the asserted necessity.

For the converse construction, each endpoint of a deleted edge loses one edge and gains its edge to x. All other old degrees stay D and x has degree D. Both old parts have vertices not incident with W, complete to the opposite part, so the graph is connected; x has neighbors in both parts.

Consider a maximal matching. If it leaves x uncovered and has fewer than D edges, equal positive numbers of vertices in the old parts remain uncovered. Each old vertex misses at most one vertex of the opposite part. An uncovered set containing vertices of both parts can therefore be independent only when exactly one vertex remains in each part and their pair is in W. Both are adjacent to the uncovered x, again contradicting maximality. If x is matched to A and the remaining cross matching has $p \leq D - 2$ edges, at least one A-vertex and at least two B-vertices remain uncovered. Some two of these are adjacent, so this matching is not maximal. Thus $p = D - 1$. Interchanging the parts treats an apex edge to B. Every maximal matching has D edges. One exists by finiteness, and the order $2D + 1$ bounds every matching by D. This proves equimatchability and $\nu(C) = D$.

Finally, a balanced bipartite graph with part size $m \geq 2$ in which each vertex misses at most one opposite vertex has a perfect matching. A nonempty subset of one part of size at most m-1 has at least m-1 neighbors by looking at one vertex. The whole part has all m neighbors because every opposite vertex has at least $m - 1 \geq 1$ neighbors. Apply the first matching assertion. Deleting x in the constructed graph gives such a graph with $m = D$. If an old A-vertex is deleted, match x to any vertex of B_0; the remaining old graph has equal parts of size $D - 1 \geq 2$ and each vertex misses at most one opposite vertex. Its perfect matching, together with that apex edge, is perfect in the vertex-deleted graph. The exchanged construction handles deletion of an old B-vertex. Every vertex deletion thus has a perfect matching, proving factor-criticality and completing the converse. □

Remark. Equimatchability forbids isolating a vertex while leaving unequal bipartite parts. Degree counts force four complete neighbor classes, and matching deletions rule out their extreme sizes, leaving a complete balanced bipartite graph with one matching replaced by an apex.

Depends on. Nothing earlier; this item is self-contained.

Lemma 9 (Independent extension counts in small octic terminals). *Let H be a finite simple eight-regular graph on 27 vertices with independence number $\alpha(H) = 7$. Assume H is well-covered: every inclusionwise maximal independent vertex set has cardinality seven. Write C_3 for a triangle, P_3 for a path on three vertices, K_2 for a single edge, and K_1 for an isolate; sums denote disjoint unions with no edges between summands. Then H has neither an induced subgraph $4C3 + 2K1$ nor an induced subgraph $3C3 + P3 + 2K1$. In addition, there is no partition $V(H) = W$ disjoint union R such that $H[W]$ is $P3 + 5K2$ and the maximum degree of $H[R]$ is at most two. Here $V(H)$ is the vertex set and $H[X]$ denotes its induced graph on a vertex subset X. All orders, degrees, independent-set sizes and component multiplicities are integers with the displayed values.*

Proof. Every independent six-set in H has an extending vertex, meaning a vertex outside it adjacent to none of its members. Otherwise it would be inclusionwise maximal of size six, contrary to the hypothesis. We contradict this requirement in each of the three configurations.

First suppose R is an induced vertex set giving one of the two fourteen-vertex graphs in the statement, and put $W = V(H)$ minus R. Write $e(X)$ for the number of edges induced on a vertex set X. Since H is eight-regular, it has 108 edges. Summing its degrees on R gives 112-2e(R) edges between R and W, and therefore $e(W) = e(R) - 4$. Thus $e(W)$ is eight for $4C3 + 2K1$ and seven for $3C3 + P3 + 2K1$. One has $|W| = 13$. Every vertex cover of H[W] has size at least six: its complement is an independent set of H and has size at most seven. For any vertex of W-degree d, choosing that vertex and one endpoint of each edge not incident with it gives a cover of size at most $1 + e(W) - d$. Hence $d \leq e(W) - 5$. The two cases consequently give maximum W-degree at most three and at most two, respectively.

Let u, v be the two isolates of H[R]. Each has all eight neighbors in W. At least eight vertices of W neighbor u or v, so at most five vertices of W avoid both. Only these five or fewer vertices can extend an independent set containing u, v.

For $R = 4C3 + 2K1$, select u, v and one vertex from each of the four triangles. There are $3^4 = 81$ independent six-sets, each maximal within H[R]. If a vertex w in W avoids u, v, then its W-degree at most three forces at least five neighbors among the twelve triangle vertices. Let a_j be its number of nonneighbors in triangle j. Thus $\sum_{j=1}^{4} a_j \leq 7$, with $0 \leq a_j \leq 3$. The number of the 81 sets extended by w is the product of these four counts. If some count is zero this number is zero. Otherwise $a_j \leq 2^{a_j-1}$, valid for the three integer values 1, 2, 3, gives $product_j\, a_j \leq 2^{7-4} = 8$. The at most five eligible W-vertices together extend at most forty of the eighty-one sets. This uses only that the cardinality of a union of finite sets is at most the sum of their cardinalities. One six-set has no extension in W and none in R, a contradiction.

For $R = 3C3 + P3 + 2K1$, select u, v, the center of the path, and one vertex from each triangle. There are 27 independent six-sets, all maximal within H[R]. An extending vertex w in W must avoid u, v and the path center. Its W-degree at most two gives at least six R-neighbors, at most two of them at the path leaves. Therefore it has at least four neighbors among the nine triangle vertices. Its three triangle nonneighbor counts sum to at most five. The same product argument bounds the number of choices extended by w by $2^{5-3} = 4$, including zero when a factor vanishes. There are at most five eligible vertices, so at most twenty of the twenty-seven sets have an extension. Again one has none anywhere in H, a contradiction.

For the final partition, $H[W] = P3 + 5K2$ gives $|W| = 13$ and $|R| = 14$. Select the path center and one endpoint from each of the five edges of H[W]. These are 32 independent six-sets, each maximal within H[W]. Any vertex r in R has at least six neighbors in W, since its degree within R is at most two and its H-degree is eight. If r neighbors the path center, it extends none of the selected sets. If it neighbors both endpoints of one edge, it also extends none. In every other case at most two W-neighbors of r are path leaves, so at least four are in distinct edge components. For each such component, extension forces the choice of its other endpoint. At most one of the five edge choices is free, and r extends at most two of the thirty-two sets. The fourteen vertices of R together extend at most twenty-eight sets. One selected six-set is unextendable in both W and R, contradicting well-coveredness. This proves all three exclusions. □

Remark. There are more independent six-sets inside each small terminal configuration than the exterior can extend. One is therefore maximal in the whole graph, contradicting well-coveredness.

Depends on. Nothing earlier; this item is self-contained.

Lemma 10 (Prescribed saturation by a maximum matching). *Let H be a finite simple graph, including the empty graph and graphs with isolated vertices, and let U be any subset of $V(H)$. A matching is a set of edges with pairwise disjoint endpoints; it saturates a vertex when an edge of the matching is incident with that vertex. Write $V(M)$ for the set of vertices saturated by a matching M, and $\nu(H)$ for the largest cardinality of a matching of H. For $S \subseteq V(H)$, let H-S be the induced graph on $V(H)$ minus S and define $o_U(H - S)$ to be the number of connected components C of H-S such that $|V(C)|$ is odd and $V(C) \subseteq U$. All cardinalities and matching sizes are nonnegative integers. The following conditions are equivalent: (i) $o_U(H - S) \leq |S|$ for every $S \subseteq V(H)$; (ii) H has a matching saturating every vertex of U; (iii) H has a matching of cardinality $\nu(H)$ saturating every vertex of U. More precisely, for every matching M_0 of H there is a sequence $M_0, M_1, \ldots, M_t$ of matchings, where $t = \nu(H) - |M_0|$ is a nonnegative integer, such that $|M_i| = |M_0| + i$ for every integer $0 \leq i \leq t$ and $V(M_i) \subseteq V(M_{i+1})$ for every integer $0 \leq i < t$. Thus maximum cardinality can be achieved while preserving saturation of all vertices saturated by M_0.*

Proof. For any finite graph K, let $o(K)$ count its odd-order connected components. The sum of the component orders shows that $o(K)$ and $|V(K)|$ have the same parity.

We first establish an elementary bipartite matching criterion for use inside the proof. If a finite bipartite graph with parts A, B satisfies $|N(X)| \geq |X|$ for every $X \subseteq$ A, where $N(X)$ is the set of neighbors of X in B, then it has a matching saturating A. Choose a matching L of largest cardinality. If a vertex a in A is unmatched, follow alternating paths from a, using nonmatching edges from A to B and matching edges from B to A. Let A′ and B′ be the reached vertices in the two parts. A reached unmatched vertex of B would give an alternating path whose edge exchange increases $|L|$, a contradiction. Thus every vertex of B′ has a matching partner in A′. Conversely every vertex of A′ other than a was reached along its matching edge from B′. All neighbors

of A′ lie in B′: a nonmatching edge is a permitted next step, and the matching neighbor of a reached vertex of A′ other than a is already reached. Consequently $|N(A')| = |B'| = |A'| - 1$, contradicting the assumed inequality. This proves the criterion, including empty parts.

We next prove that a finite simple graph K has a perfect matching whenever $o(K - X) \le |X|$ for every $X \subseteq V(K)$. A perfect matching means a matching saturating all vertices. Necessity follows because each odd component of K-X needs a distinct matching endpoint in X. To prove sufficiency, use induction on the number n of vertices. The condition at X empty implies $o(K) = 0$ and hence n is even. For $n = 0$ the empty matching suffices. Suppose $n \ge 2$ and assume the assertion for all smaller graphs. For any vertex v, the graph K-v has odd order, so $1 \le o(K - v) \le 1$. Hence some nonempty set S satisfies $o(K - S) = |S|$. Choose S of maximum cardinality among all sets satisfying that equality; this is possible because there are finitely many subsets.

Every component of K-S has odd order. Indeed, if C were an even-order component and v belonged to C, deleting v would leave at least one odd component in C-v. The other $|S|$ odd components of K-S would be unchanged, so $o(K - (Sunion\{v\})) \ge |S| + 1$. The assumed inequality makes this an equality, contradicting the maximum cardinality of S. There are therefore exactly $|S|$ components of K-S, all odd.

Fix any such component C and any vertex v in it. For every $X \subseteq V(C)$ minus {v}, apply the assumed inequality to S union {v} union X. The other $|S| - 1$ components of K-S remain unchanged, giving $o(C - v - X) \le |X| + 2$. This odd-component count has the same parity as $|X|$, because $|V(C)| - 1$ is even. If it were larger than $|X|$, it would equal $|X| + 2$. Then S union {v} union X would again satisfy equality in the assumed inequality and would be larger than S, a contradiction. Thus $o(C - v - X) \le |X|$ for every such X. The graph C-v has fewer than n vertices, so induction gives a perfect matching of C-v.

Form the bipartite graph with parts S and the components of K-S, joining a vertex to a component when K has an edge between them. For any $A \subseteq S$, the components with no neighbor in A remain distinct odd components after deleting S minus A. If $N(A)$ denotes the components adjacent to A, this gives $|S| - |N(A)| \le o(K - (SminusA)) \le |S| - |A|$. Hence $|N(A)| \ge |A|$. The bipartite criterion gives a matching of S to distinct components, and it uses every component because there are exactly $|S|$ of them. Realize each selected incidence by an actual edge sv with v in its assigned component C, and choose a perfect matching of C-v, whose existence was proved for every v. The union of those edges and internal matchings is a perfect matching of K. This completes the induction and proves the perfect-matching criterion without external premises.

We now prove (ii) implies (i). Let M saturate U and fix $S \subseteq V(H)$. Every odd component C of H-S lying wholly in U must have an M-edge from C to S. Otherwise all vertices of C would have to be saturated by internal edges, which cover an even number of vertices. These edges for distinct components have distinct endpoints in S, because M is a matching. Therefore $o_U(H - S) \le |S|$.

Assume (i). If $n = |V(H)|$ is zero, the empty matching proves (ii). For $n > 0$ construct J by adjoining n new vertices Z, joining every two distinct vertices of Z, and joining each vertex of Z to every vertex of $V(H)$ minus U. Keep every H-edge and add no edge between Z and U. This is a finite simple graph on 2n vertices.

For any $T \subseteq V(J)$, put $S = T$ intersect $V(H)$ and $Y = T$ intersect Z. If $Y = Z$, the graph J-T is H-S and has $n - |S|$ vertices, so $o(J - T) \le n - |S| \le n + |S| = |T|$. Suppose instead that Z minus Y is nonempty. Its vertices lie in one component together with every component of H-S that contains a vertex outside U, since all such vertices are joined to every remaining vertex of Z. The other components are exactly those components of H-S lying wholly in U. Thus $o(J - T) \le o_U(H - S) + 1 \le |S| + 1$. If $|Y| \ge 1$ this is at most $|T|$. If Y is empty, $o(J - T)$ has the same parity as $2n - |S|$, and therefore as $|S|$; an integer of this parity bounded above by $|S| + 1$ is at most $|S| = |T|$. The perfect-matching criterion just proved gives a perfect matching of J. Restrict it to edges with both endpoints in $V(H)$. The result is a matching of H saturating U, since vertices of U have no neighbors in Z. This proves (i) implies (ii).

We prove the sequence assertion directly. Let M be any matching with $|M| < \nu(H)$, and choose a matching N with $|N| > |M|$. Such a choice exists because H has finitely many edge subsets. Form the graph with edge set (M minus N) union (N minus M). Each vertex is incident with at most one edge from each matching, so every nontrivial connected component is a path or a cycle with alternating edge membership. Every cycle has even length and equally many edges from M and N. On a path the difference between the numbers of N-edges and M-edges is -1, 0, or 1. The sum of these differences is $|N| - |M| > 0$. Some path therefore has one more N-edge than M-edge and begins and ends with N-edges.

Its endpoints are unmatched by M. An M-edge incident with an endpoint would either belong to the edge-difference graph, contradicting that this vertex is an endpoint, or belong to both matchings, contradicting the other incident N-edge on the path. Every internal vertex has its M-edge on the path. Delete the path's M-edges from M and insert its N-edges. No path vertex has an M-edge to the complement of the path. Each internal vertex remains matched and both endpoints become matched. The result M′ is consequently a matching with $|M'| = |M| + 1$ and $V(M) \subseteq V(M')$. Repeating this operation while cardinality is less than $\nu(H)$ gives exactly $t = \nu(H) - |M_0|$ steps and proves the asserted sequence.

Applying the sequence to a matching saturating U proves (ii) implies (iii). Condition (iii) implies (ii) by forgetting the size requirement. If U is empty, every o_U count is zero and augmentation can start from the empty matching. If an isolated vertex belongs to U, the condition with S empty fails and no matching saturates U. Unprescribed isolated vertices cause no change in the arguments. These observations include all the stated degenerate cases. □

Remark. The only obstruction to covering a prescribed vertex set is too many odd components that cannot leave that set; a dummy clique converts this to the perfect-matching condition, and augmentation preserves covered vertices.

Depends on. Nothing earlier; this item is self-contained.

Lemma 11 (Reduction of weighted marked domination to cubic domination). *For a finite simple graph X and $D \subseteq V(X)$, let $N_X[D]$ be the union of the closed neighborhoods of the vertices of D, and define $\gamma(X) = min\{|D| : N_X[D] = V(X)\}$. Every graph vertex may be selected; the minimum for the empty graph is zero. Let $N \geq 0$ be an integer and assume that*

$8\gamma(H) \leq 3|V(H)|$

for every finite simple cubic graph H with $|V(H)| \leq N$; cubic means that every vertex has degree three.

Let G be any finite simple graph of maximum degree at most three and order at most N. Let $R \subseteq V(G)$ contain every vertex of degree three. A vertex in R is required, and a vertex outside R is optional. Every vertex may be selected. Define

$\gamma_R(G) = min\{|D| : D \subseteq V(G), R \subseteq N_G[D]\}$.

Assign required vertices of degrees zero, one, two and three the respective weights 8, 6, 4, 3, and assign optional vertices of degrees zero, one and two the respective weights 0, 2, 2. Write $W(G, R)$ for the total weight. Then

$8\gamma_R(G) \leq W(G, R)$.

In particular, write $S_3(G) = \{vinV(G) : d_G(v) = 3\}$, $\tau_3(G) = \gamma_(S_3(G))(G)$, $n_1(G) = |\{vinV(G) : d_G(v) = 1\}|$, and $e(G) = |E(G)|$. Under the same finite-order cubic hypothesis,

$4\tau_3(G) \leq e(G) + n_1(G)/2$.

Thus $4\tau_3(G) \leq e(G)$ when G has no degree-one vertex.

Empty graphs, disconnected graphs, isolates, leaves and an empty required set are allowed. All degrees, weights, orders and set cardinalities are nonnegative integers.

Proof. Fix an integer $N \geq 0$ satisfying the stated cubic hypothesis. Suppose a violating marked graph of order at most N exists. Choose such a graph G, R minimizing vertex order and then edge count; these integer minima exist by well-ordering. Write $\gamma = \gamma_R(G)$, $W = W(G, R)$. Every marked graph of smaller order has order at most N and satisfies the inequality.

For a smaller marked graph H, R_H, if each cover lifts with at most s extra selections and $W(G, R) - W(H, R_H) \geq 8s$, then

$8\gamma_R(G) \leq 8\gamma_{R_H}(H) + 8s \leq W(H, R_H) + 8s \leq W(G, R)$,

contrary to the choice of G. Labels are retained unless explicitly changed. Every survivor newly made optional loses an edge, so has degree at most two; every smaller instance therefore has all cubic vertices required.

First checkpoint: normalization and the short-path reduction.

The cover number and weight are additive over whole components. A cover restricts to a cover in each component, and component covers can be united, because closed neighborhoods do not cross components. A disconnected violation would therefore have a smaller violating component. The empty graph is not a violation, so G is nonempty and connected.

An isolated required vertex has weight eight and requires exactly its own selection; deleting it is a paid one-selection lift. An isolated optional vertex can be deleted with zero weight change and zero extra selections.

If x is an optional leaf with neighbor v, deleting x preserves the cover number. Indeed a selection x can be replaced by v without increasing the size, since v is the only possible required vertex covered by x; covers of the smaller graph also work in G. The removed weight is two. If v is required, its degree changes from one to zero, two to one, or three to two, raising its weight by two, two, or one respectively. If v is optional, its weight does not increase. Thus the total weight does not increase under deletion, excluding this case.

If x is a required leaf with neighbor v, delete x and v and make all surviving neighbors of v optional. Adjoining v to a cover of the resulting graph covers every removed or newly optional requirement. The deleted x weighs six and the nonisolated v weighs at least two. At another neighbor of v a required old degree one, two, or three changes to an optional degree zero, one, or two; the weight changes from six, four, or three to zero, two, or two. An originally optional neighbor also has no weight increase. Only these surviving degrees change. The weight drop is at least eight, excluding this case. Consequently the minimum degree of G is at least two.

An edge with two optional endpoints can be deleted without changing which required vertices any selected set covers: neither endpoint is required, so no adjacency relevant to required coverage is removed. Optional weights cannot increase when a degree drops. This would preserve a violation at the same order with fewer edges. Hence the optional vertices form an independent set.

Consider a path a-x-y-z-b on five distinct vertices whose three internal vertices have degree two, and suppose ab is absent. Delete x, y, z and add the edge ab, retaining surviving labels, to obtain H. All surviving degrees are unchanged and H is simple. Every cover D of H lifts with one extra selection. If a is selected and b is not, add z: a covers x, z covers y, z, b, including any requirement at b which had used ab. If b is selected and a is not, add x, which covers a, x, y, while b covers z. If both endpoints are selected, or neither is selected, add y. Then the internal path is covered; each endpoint in the first case covers itself, while in the second case neither endpoint used ab for coverage. All other coverage is unchanged. At most two of x, y, z are optional by independence. Their total weight is at least $2 + 4 + 2 = 8$, with no surviving weight change. This is a paid lift. Therefore such a path cannot occur in G.

We also exclude the possibility that every vertex has degree two. A finite connected simple two-regular graph is a cycle: following unused forward neighbors closes a cycle, and no cycle vertex has an edge leaving it, so connectedness leaves no other vertices. Let $n \geq 3$ be its order and let o be its optional count. Its weight is 4n-2o. A path on m vertices can be dominated by $\lceil m/3 \rceil$ selections: partition consecutive vertices into triples, select each middle vertex, and select an endpoint of the final block of one or two vertices if present. A cycle can therefore be dominated by $\lceil n/3 \rceil$ selections using a spanning path. If $o = 0$, $\lceil n/3 \rceil \leq n/2$ for $n \geq 3$: the cases $n = 3$, 4 give one and two; for $n \geq 4$ use $\lceil n/3 \rceil \leq (n+2)/3 \leq n/2$. This gives eight times the cover number at most 4n.

If $o > 0$, remove one optional vertex and dominate the remaining path, using at most $ceil((n-1)/3)$ selections. These cover every required cycle vertex. Optional independence gives $o \leq \lfloor n/2 \rfloor$, by injecting each optional vertex into its cyclic successor, which is required. It suffices that

$8ceil((n-1)/3) \leq 4n - 2floor(n/2) = 3n + (nmod2)$.

For $n = 3a$ the left side is 8a and the right side is at least 9a. For $n = 3a + 1$ the left side is 8a and the right side is at least $9a + 3$. For $n = 3a + 2$ it is $8a + 8$, and the difference on the right is $a - 2 + (nmod2)$. Here $a \geq 1$. If $a = 1$ then $n = 5$ and the difference is zero; if $a \geq 2$ it is nonnegative. This proves the cycle bound.

The cubic case is excluded by the hypothesis for this N: the graph has order at most N and every vertex is required, so $W = 3|V(G)|$ and the assumed cubic inequality gives $8\gamma \leq W$. G therefore has both degree-two and degree-three vertices.

The number of degree-three vertices is even, since the degree sum is twice the edge count. With minimum degree two, W is consequently even. Since G violates the inequality, $W \leq 8\gamma - 2$. If x is optional, its two neighbors are distinct and required. Delete x with their labels retained. A cover of the smaller required set

still covers every original requirement in G, so its cover number is at least γ. The removed weight is two; each cubic neighbor increases its weight by one and each degree-two neighbor increases its weight by two. If at most one neighbor has degree two, the new total weight is at most $W + 1 \leq 8\gamma - 1$, leaving a smaller violation. Thus both neighbors of every optional vertex have degree two.

Let V_2 be the degree-two vertex set. Each component of $G[V_2]$ is a path, including a one-vertex path: a cycle component would have no exterior edge, contrary to connectedness and the presence of cubic vertices. Each path endpoint has a cubic exterior neighbor. For a component p_1,..., p_r with $r \geq 4$, let u be the exterior neighbor of p_1. Apply the forbidden-path conclusion to $u - p_1 - p_2 - p_3 - p_4$: it forces up_4. If $r \geq 5$, p_4 already has its two path neighbors, impossible. Hence $r = 4$, and the exterior neighbor at the other end is also u. Such a component forms a five-cycle with u. For $r = 3$ the two exterior neighbors either coincide or are adjacent, by applying that conclusion to the path through the three internal vertices. These are all possibilities needed next.

Second checkpoint: elimination of optional vertices.

Suppose a component P of $G[V_2]$ contains an optional vertex. No endpoint can be optional, because an optional vertex cannot neighbor a cubic vertex. Optional independence and the length bound therefore give the following cases.

If $P = p_1 - p_2 - p_3$, the middle vertex is optional and both ends required. Let a, b be its cubic exterior neighbors. If a and b are distinct, delete the path, retaining all surviving labels. Its removed weight is $4 + 2 + 4 = 10$. Each of a, b loses one edge and its required weight rises from three to four. The net drop is eight. Add p_2 to a cover of the smaller graph to cover p_1, p_3, all the removed requirements. This is a paid lift.

If $a = b = u$, then u and the three path vertices form a four-cycle. Let q be the third neighbor of u, which lies outside that cycle. Delete the cycle and make q optional. Add u to any smaller cover: u covers the removed requirements u, p_1, p_3 and the newly optional q, while p_2 was optional. The removed weight is $3 + 4 + 2 + 4 = 13$. The required q has degree two or three before deletion, since it neighbors a cubic vertex and G has minimum degree two. Its change to optional degree one or two drops its weight from four or three to two. Thus the drop is at least thirteen, enough for the one-selection lift.

If $P = p_1 - p_2 - p_3 - p_4$, its exterior neighbor is the same cubic vertex u at both ends. Exactly one of p_2, p_3 is optional, because at least one is optional and they are adjacent. The five-cycle has total weight $3 + 3 \cdot 4 + 2 = 17$. Let q be the third neighbor of u, outside the cycle. It is required and has degree two or three.

If q is cubic, delete the five-cycle retaining all remaining labels. Only q has a surviving degree change, increasing its weight by one. The drop is sixteen. Add p_1, p_3 to a smaller cover; together they dominate the entire deleted five-cycle. This is a paid two-selection lift.

If q has degree two, let r be its other neighbor. Every cycle vertex already has its full displayed degree, so r is outside the cycle. Delete the cycle and q, retaining remaining labels. The removed weight is $17 + 4 = 21$. The only surviving change is loss of the edge rq. A required r gains at most two weight, and an optional r gains none. The drop is at least nineteen. Add u, p_2: u covers u, p_1, p_4, q and p_2 covers p_1, p_2, p_3. All deleted requirements are covered at a cost of sixteen. Again the lift is paid.

Every possibility has contradicted minimality, so all vertices of G are required.

Third checkpoint: elimination of the remaining degree-two vertices.

If three consecutive vertices p_1, p_2, p_3 occur in $G[V_2]$, delete them with all surviving labels retained. Their removed weight is twelve, and at most two edges join them to surviving vertices. Loss of one edge raises a required weight by at most two; if one survivor loses both edges, its old degree two to zero changes four to eight, or its old degree three to one changes three to six, also at most four in total. Thus the surviving increase is at most four and the weight drop at least eight. Add p_2 to cover the three deleted vertices. This is a paid lift, so every component of $G[V_2]$ has at most two vertices.

For a two-vertex component $p_1 - p_2$ let a, b be its exterior cubic neighbors. Delete p_1, p_2 and make a optional. Adding p_1 covers the deleted vertices and a, precisely all removed or newly optional requirements. The removed weight is eight. If $a \neq b$, the weight at a drops from three to two, while the weight at b rises from three to four, giving net drop eight. If $a = b$, it loses two edges and changes from required cubic weight three to optional degree-one weight two, giving drop nine. Both lifts are paid. Therefore V_2 is independent.

Choose v in V_2; such a vertex was proved to exist. Its two neighbors u, w are cubic. Write $N_G(u) = \{v, a, b\}$ with distinct a, b. Delete u, v and make a, b optional. Adding u covers u, v, a, b, all requirements no longer covered in the smaller marked graph. The removed weight is $3 + 4 = 7$.

If w is different from a, b, each of a, b loses one edge and its weight drops by at least one: required degrees two or three become optional degrees one or two. The required w loses its edge to v and its weight rises by one. There are no other surviving degree changes, so the net drop is at least $7 + 1 + 1 - 1 = 8$.

If w equals one of a, b, label it a. This cubic vertex loses edges to u and v and becomes optional of degree one, dropping its weight by one. The weight of b drops by at least one. No further survivor loses an edge, so the drop is at least nine. Both cases give a paid one-selection lift, the final contradiction.

Under the stated cubic hypothesis, no marked instance of order at most N violates the inequality. All smaller graphs used in the proof, including disconnected graphs, isolates and leaves, remain in the asserted class and within the order bound.

Take $R = S_3(G)$ and let n_i count the degree-i vertices, for integers $0 \le i \le 3$. Then $W = 3n_3 + 2n_2 + 2n_1$ and $2e(G) = 3n_3 + 2n_2 + n_1$. Hence $8\tau_3(G) \le 2e(G) + n_1(G)$; division by two proves the final bounds. □

Remark. Every minimum weighted marked counterexample has a paid local reduction unless it is cubic. The stated finite-order cubic hypothesis excludes precisely that final case.

Depends on. Nothing earlier; this item is self-contained.

Lemma 12 (Two complementary level sums and their exact degree transfer). *Let $d \ge 4$ be an even integer and let k, E be nonnegative integers. Let c_0,..., c_d be nonnegative integers such that $c_0 = k + E$, $c_i \le k + E$ for $1 \le i \le d$, $c_d \le 1$, $c_i + c_{d+1-i} \le 2k + 1$ for $1 \le i \le d/2$, and $c_i + c_{d-i} \le 2k + 2$ for $1 \le i \le d - 1$. Define $g = \sum_{i=0}^{d} c_i - d(2k + 1)/2$. For every integer t with $1 \le t \le d/2 - 1$ and every real vector $b = (b_1, \dots, b_t)$ satisfying $c_{d-j} \le b_j$ for $1 \le j \le t$, put $U_{d,t}(k, E; b) = -tk + \sum_{j=1}^{t} b_j + min\{(t + 2)E - t, (t + 1)E + d/2 - 2t\}$. Then $g \le U_{d,t}(k, E; b)$. The second expression in this minimum is no larger than the first exactly when $E \ge d/2 - t$. For every even integer $d \ge 6$, integer $1 \le t \le d/2 - 2$, nonnegative integers k, E, and two real vectors b, b' of length t, the exact transfer identity is $U_{d-2,t}(k, E; b') - U_{d,t}(k, E; b) = \sum_{j=1}^{t}(b'_j - b_j) - 1_{E \ge d/2-t}$, where 1_P is one if P holds and zero otherwise. Consequently, if a second sequence indexed from 0 to d-2 satisfies the displayed sequence hypotheses with d replaced by d-2 and the same k, E, and has top bounds b', its corresponding integer g' is at most zero whenever $U_{d,t}(k, E; b) + \sum_j (b'_j - b_j) - 1_{E \ge d/2-t} < 1$. In particular, for every even $d \ge 10$, every sequence satisfying the original hypotheses with $k \ge 20$ and $E \le floor((k + 1)/3)$ has $g \le 0$ if there are bounds b_1,..., b_4 with $c_{d-j} \le b_j$ and $\sum_{j=1}^{4} b_j < 4 + 97k/50$.*

Proof. Fix t in the stated range. First partition the indices 1,..., d into the pairs $\{i, d + 1 - i\}$, $1 \le i \le d/2$. In the pair $i = 1$, use $c_1 \le k + E$ and $c_d \le 1$. In each pair $i = j + 1$ with $1 \le j \le t$, use $c_{j+1} \le k + E$ and retain $c_{d-j} \le b_j$. These are distinct pairs because $t + 1 \le d/2$. There are $d/2 - t - 1$ remaining pairs; their total is at most $(d/2 - t - 1)(2k + 1)$ by the first family of paired inequalities. Including $c_0 = k + E$ gives $\sum_i c_i \le (t + 2)(k + E) + 1 + \sum_j b_j + (d/2 - t - 1)(2k + 1)$. Subtract $d(2k + 1)/2$ and collect the k, E, and constant terms to obtain $g \le (t + 2)E - tk - t + \sum_j b_j$.

For the second bound, partition 0,..., d into the low indices 0,..., t, the pairs {i, d-i} for $t + 1 \le i \le d/2 - 1$, the singleton $d/2$, the top indices d-t,..., d-1, and the singleton d. The intervals are disjoint and cover every index; when $t = d/2 - 1$ the family of pairs is empty. The low indices contribute at most $(t + 1)(k + E)$, the top indices at most $\sum_j b_j$, and c_d at most one. Each of the $d/2 - t - 1$ pairs contributes at most $2k + 2$. The second paired inequality at $i = d/2$ reads $2c_{d/2} \le 2k + 2$, so the middle singleton contributes at most $k + 1$. Thus $\sum_i c_i \le (t + 1)(k + E) + \sum_j b_j + 1 + (d/2 - t - 1)(2k + 2) + (k + 1)$. Subtracting $d(2k + 1)/2$ gives $g \le (t + 1)E - tk + d/2 - 2t + \sum_j b_j$. Taking the smaller bound proves $g \le U$. The difference of the first expression inside the minimum and the second is $E - d/2 + t$, proving the asserted cross over condition, including equality.

For the transfer identity, write the minimum as $(t + 1)E + min\{E - t, d/2 - 2t\}$. Put $x = E - t$ and $y = d/2 - 2t$. Both are integers. If $x \ge y$ then $min\{x, y - 1\} - min\{x, y\} = -1$. If $x < y$, integrality gives

$x \le y-1$ and the difference is zero. On replacing d by d-2, y becomes y-1; all other terms of U are unchanged except the sum of the b_j. Therefore the change in U is exactly $\sum_j (b'_j - b_j) - 1_{x \ge y}$, and $x \ge y$ is equivalent to $E \ge d/2 - t$. The range on t ensures that the preceding sum bounds apply at degree d-2. Apply that bound to the second sequence and then the identity. The assumed strict upper bound by one forces $g' \le 0$, since its c_i are integers and d-2 is even, making g' an integer.

Finally take $t = 4$ in the first sum bound; $d \ge 10$ makes this a permitted value. Using the two assumed upper bounds gives $g \le 6E - 4k - 4 + \sum_{j=1}^{4} b_j < 6(k+1)/3 - 4k - 4 + 4 + 97k/50 = 2 - 3k/50$. For $k \ge 20$ this is at most $4/5$, which is less than one. The number g is an integer because d is even, so $g \le 0$. □

Remark. Pairing levels with indices summing to $d+1$ or to d gives two bounds. Their minimum has a one-unit change when the degree drops by two, and the integrality of the total converts any bound below one into nonpositive excess.

Depends on. Nothing earlier; this item is self-contained.

Lemma 13 (Incompatibility of two component tail bounds). *Let D, m, k, r, q be integers with $D \ge 4$, $q \ge 0$, $m \ge 9 + 8q$, $k \ge 3m$, and $r \ge 3m - 1$. Put $T = floor((D-1)/2)$, and suppose $m \le T + q + 1$. For every real p in [0,1] and nonnegative integer a define*

$f_D(p) = p + (1-p)^D$, $B_D(a,p) = a - (2a+1) f_D(p)$.

Then at least one of the following strict inequalities holds:

$B_D(k, 1/10) > 2m + q - 1$,

$B_D(r, 1/10) > T + q$,

$B_D(r, 1/6) > T + q$.

Consequently there are no real numbers s, t satisfying simultaneously

$s \le 2m + q - 1$, $t \le T + q$,

$s \ge B_D(k, 1/10)$, $t \ge B_D(r, 1/10)$, $t \ge B_D(r, 1/6)$.

No parity restriction on D or graph interpretation of the parameters is imposed.

Proof. The assumptions $m \ge 9 + 8q$ and $m \le T + q + 1$ give $T \ge 8 + 7q$. In particular $m \ge 9$, $k \ge 27$, and $r \ge 26$. We establish three strict rational bounds on f_D.

First,

$(9/10)^6 = 531441/1000000 < 8/15$,

because $7971615 < 8000000$. Therefore

$(9/10)^{36} < (8/15)^6 < 1/40$.

The last comparison follows from $40 \cdot 8^6 = 10485760 < 11390625 = 15^6$. Since a power of $9/10$ decreases as its integer exponent increases, $f_D(1/10) < 1/8$ for every $D \ge 36$.

Second, $(5/6)^5 = 3125/7776 < 5/12$ because $37500 < 38880$. Consequently

$(5/6)^{17} < (5/12)^3 \cdot (25/36) = 3125/62208 < 1/18$,

where the last comparison is $56250 < 62208$. It follows that $f_D(1/6) < 2/9$ for every $D \ge 17$.

Third, the first six-power bound gives

$(9/10)^{31} < (8/15)^5 \cdot (9/10) = 294912/7593750 < 3/70$,

because $20643840 < 22781250$. Hence $f_D(1/10) < 1/7$ for every $D \ge 31$.

Suppose $D \ge 36$. Since $2k+1$ is positive, the first bound yields

$B_D(k, 1/10) > 3k/4 - 1/8 \ge 9m/4 - 1/8$.

Subtracting $2m + q - 1$ from the final expression gives

$m/4 - q + 7/8 \geq 25/8 + q > 0$,

using $m \geq 9 + 8q$. This proves the first asserted strict inequality.

Suppose $D < 31$. Then $T \leq 14$. Combining this with $T \geq 8 + 7q$ and integer $q \geq 0$ forces $q = 0$; it also gives $T \geq 8$ and $D \geq 17$. The second rational bound and $r \geq 26$ yield

$B_D(r, 1/6) > 5r/9 - 2/9 \geq 128/9 > 14 \geq T + q$.

Thus the third asserted inequality holds.

The remaining range is $31 \leq D \leq 35$. Here $T \leq 17$ and $T \geq 8 + 7q$ imply $q \leq 1$. Thus $T + q \leq 18$. The third rational bound gives

$B_D(r, 1/10) > 5r/7 - 1/7 \geq 129/7 > 18 \geq T + q$.

This proves the second asserted inequality.

The three degree ranges cover every admissible D. Each range proves one of the stated strict inequalities. If s, t satisfied all five proposed bounds, the first strict inequality would contradict $s \leq 2m + q - 1$, and either of the other two would contradict $t \leq T + q$. Therefore no such pair s, t exists. □

Remark. At large degree the root tail is already too large. At the remaining degrees the count of unfinished indices is at most one, so the active component's separate tail gives a contradiction.

Depends on. Nothing earlier; this item is self-contained.

Lemma 14 (Indexed matching transition across a vertex cover). *All graphs are finite and simple. Write $V(G)$, $E(G)$, $\nu(G)$, and $\Delta(G)$ for the vertex set, edge set, matching number, and maximum degree of a graph G. Edge sets are regarded as spanning graphs. For nonnegative integers A_c, B_c, define $c_{A_c,B_c}(G)$ as the least nonnegative integer T such that every pair of nonnegative integers x, y with $x + y = T$ admits a partition $E(G) = Q_R$ dot-union Q_B dot-union P_R dot-union P_B satisfying $\Delta(Q_R) \leq A_c$, $\Delta(Q_B) \leq B_c$, $\nu(P_R) \leq x$, and $\nu(P_B) \leq y$. For each nonnegative integer i, put $c_i(G) = \max_{A_c+B_c=i} c_{A_c,B_c}(G)$, with nonnegative integer capacities.*

Let F be a graph, S a subset of $V(F)$, $H = F - S$ the induced graph obtained by deleting S, and $N = |S|$. Fix a positive integer k and pairwise edge-disjoint matchings M_0,..., M_{k-1} in H. Define $P_i = (V(H), E(H) \text{minus} \bigcup_{0 \leq l < i} M_l)$ for $0 \leq i \leq k$, and $L_i = (V(H), \bigcup_{0 \leq l < i} M_l)$. Thus these operations delete edges and retain every vertex of H.

Let I be a subset of $V(H)$ such that every edge of $H[I]$ belongs to L_k. Put $C = V(H)$ minus I and $b = |C|$. Assume $c_k(F) \geq N + b$ and $N \geq \max(0, b - 2)$. Let B be any subset of C that is independent in H. Suppose each v in B has exactly $k + 1$ neighbors in I in H and, for each integer l with $0 \leq l < k$, the matching M_l contains an edge from v to I. The set B may be empty.

Then $\nu(P_k) = b = |V(H)| - |I|$. Every maximum matching M_k of P_k matches each vertex of C to a vertex of I and has no other edges. Such a maximum matching exists. Each v in B has exactly one edge to I in P_k, and that edge belongs to every such M_k. After defining $P_{k+1} = P_k - M_k$ by deleting only its matching edges, the enlarged set I union B is independent in P_{k+1}. Moreover $P_{k+1}[C] = P_k[C]$: no edge with both endpoints in C is deleted.

For any integer i with $1 \leq i \leq k$, put $\rho_i = \nu(P_i)$. If $N \geq \max(0, \rho_i - 1)$, then for every nonnegative integer capacity pair $A_c + B_c = i$ and every nonnegative integer budget pair $x + y = N + \rho_i$ there is a certificate of F whose degree-bounded edge parts have union exactly $E(L_i)$. Every edge meeting S, including an edge with both endpoints in S, belongs to a matching-bounded part. Consequently $c_i(F) \leq N + \rho_i$. In particular, if $N \geq \max(0, b - 1)$, then $c_k(F) = N + b$.

All indices, capacities, budgets, orders, cardinalities, and matching sizes in this statement are integers in the stated nonnegative or positive domains. No maximality or maximum-cardinality assumption is imposed on the supplied prefix M_0,..., M_{k-1}.

Proof. First record the explicit certificate construction used to force the rank. For any finite graph G the costs in the statement are finite: if $\nu(G) = 0$ then G is edgeless, while if $\nu(G) > 0$ then any pair of nonnegative integer budgets totaling 2nu(G)-1 has one budget at least $\nu(G)$. Put every edge in that matching-bounded part and leave the degree-bounded parts empty.

Fix any integer i with $1 \le i \le k$, let $m = \nu(P_i)$, and fix nonnegative integer capacities $A_c + B_c = i$. Put the first A_c of the matchings M_0,..., M_{i-1} into Q_R, and the remaining B_c matchings into Q_B. A matching contributes at most one edge at any vertex, so their respective maximum degrees are at most A_c and B_c. Empty groups also satisfy these bounds. Their union is exactly $E(L_i)$.

Order the N vertices of S. Assign each original edge meeting S to its first endpoint in that order that lies in S. The assigned edges form N pairwise edge-disjoint, possibly empty star pieces. Each piece has matching number at most one. An edge with both endpoints in S is assigned exactly once. Because H is induced, every other original edge belongs to precisely one of L_i and P_i.

Suppose a nonnegative integer total T satisfies $T \ge N + m$ and, when $m > 0$, $T \ge 2m - 1$. For any nonnegative integer split $x + y = T$, one budget is at least m. If $m = 0$ this is also true, and P_i is empty. Relabel the two matching-bounded parts temporarily so that $x \ge m$. Set $a = \min(N, x - m)$. Put all P_i edges and any a of the star pieces into the first matching-bounded part, and the remaining N-a pieces into the other. The first matching number is at most $m + a \le x$. The second is at most $N - a \le y$: if $a = N$ there is nothing to check, while otherwise $a = x - m$ and $N - a = N + m - x \le T - x = y$. To justify the matching bounds even when pieces share vertices, restrict any matching to each constituent edge piece; these restrictions partition its edges and their sizes are bounded by the respective matching numbers. If the larger budget was originally y, interchange the two matching-bounded parts. The degree parts stay fixed. This constructs every required certificate at total T.

We now force $\nu(P_k) = b$. Every edge within I was removed, so I is independent in P_k and C is a vertex cover. Every matching has at most b edges, because every edge meets C and disjoint edges use distinct vertices there. If $b = 0$ this gives $\nu(P_k) = 0$. Suppose $b \ge 1$ and, for a contradiction, $m = \nu(P_k) \le b - 1$. Set $T = N + b - 1$. This is nonnegative and at least $N + m$. If $m > 0$, the omission assumption gives $N \ge b - 2$, and hence $T \ge 2b - 3 \ge 2m - 1$. The construction with $i = k$ therefore gives $c_k(F) \le N + b - 1$, contradicting $c_k(F) \ge N + b$. Thus $\nu(P_k) = b$.

The family of matchings in P_k is finite and contains the empty matching, so a maximum exists. In any maximum matching its b edges use at least b distinct vertices of C. There are only b such vertices. Hence every vertex of C is used, and no matching edge uses two vertices of C. Every edge therefore crosses from C to I, and there are no additional matching edges. This also proves the assertion for $b = 0$, when the maximum matching is empty.

Fix v in B. For each $l < k$, the matching M_l removes one edge from v to I and cannot remove any other edge at v. The k removed edges are distinct because the supplied matchings are edge-disjoint. Of the $k + 1$ original edges from v to I, precisely one therefore remains in P_k. Every maximum M_k crosses C to I and saturates v, so it must contain that edge. The set I has no residual internal edge, the independent set B has no internal edge even in H, and all residual edges between B and I are removed by M_k. These three statements exhaust the possible edges within I union B, proving its independence in P_{k+1}. Since M_k has no edge within C, the induced graph on C is unchanged.

Finally, under $N \ge \max(0, \rho_i - 1)$, the total $T = N + \rho_i$ satisfies the certificate construction's conditions with $m = \rho_i$: when $m > 0$, $T \ge 2m - 1$. This proves the asserted certificates and $c_i(F) \le N + \rho_i$ for every stated i. At $i = k$ the additional margin gives $c_k(F) \le N + b$, and the assumed opposite inequality gives equality. □

Remark. A matching smaller than the known cover would lower the original graph's certificate cost. Equality forces every maximum matching across the cover; a layer with one remaining earlier-side edge is then completely cleared.

Depends on. Nothing earlier; this item is self-contained.

Lemma 15 (Restoration of an odd-cycle remainder). *All graphs are finite and simple. Write $V(G)$, $E(G)$, $\nu(G)$, $\Delta(G)$, and $d_G(v)$ for the vertex set, edge set, maximum matching cardinality, maximum degree, and degree of a vertex v in a graph G. An edge set is regarded as a spanning graph, with maximum degree and matching number zero when it is empty. For nonnegative integers A_c, B_c, x, y, an $(A_c, B_c; x, y) - certificate$ of G is a partition*

$E(G) = Q_R$ *dot-union* Q_B *dot-union* P_R *dot-union* P_B

such that $\Delta(Q_R) \le A_c$, $\Delta(Q_B) \le B_c$, $\nu(P_R) \le x$, and $\nu(P_B) \le y$. Define $c_{A_c,B_c}(G)$ to be the least nonnegative integer t such that every nonnegative integer pair x, y with $x + y = t$ has such a certificate, and define $c_i(G) = \max_{A_c+B_c=i} c_{A_c,B_c}(G)$ for every nonnegative integer i.

Let $D \geq 1$, $\ell \geq 2$, and $g \geq 0$ be integers. Let H be an induced subgraph of a graph F, set $S = V(F)$ minus $V(H)$ and $N = |S|$, and prescribe D pairwise edge-disjoint matchings M_1,..., M_D of H, with empty matchings permitted. Put $K = \bigcup_{j=1}^{D} M_j$ and $P = (V(H), E(H) minus K)$. Suppose the entire graph P is a vertex-disjoint union of one cycle C of order 2ell-1, exactly g single-edge components, and any number of isolated vertices. Set $w = \ell - 1$, $L = N + g$, and define $\chi = 0$ when $L \geq \ell - 2$ and $\chi = 1$ when $L < \ell - 2$. Put $T = L + w + \chi$.

For every pair of nonnegative integer capacities $A_c + B_c = D$ and every pair of nonnegative integer budgets $x + y = T$, F has an $(A_c, B_c; x, y) - certificate$ in which Q_R is the union of the first A_c prescribed matchings and Q_B is the union of the others. Every edge outside K, including every original edge incident with S, is in P_R or P_B. In particular,

$c_D(F) \leq T = N + \nu(P) + \chi$.

For fixed D, N, g, ℓ, this T is the least total guaranteed for all such configurations if the union Q_R union Q_B must remain exactly K.

If nonnegative integers r, s, h, β satisfy

$|V(F)| = 2r + s$, $|V(H)| = r + h$, *and* $h = s + \beta + g + \ell$,

then $T = r - \beta - 1 + \chi$. Thus $N + g \geq \ell - 2$ implies $c_D(F) \leq r - \beta - 1$, and such data cannot also have $c_D(F) \geq r - \beta$.

If F has odd order, every F-vertex has degree at least $D + 2$, and $2 \leq \ell \leq 4$, then $N + g \geq 2$. Consequently $\chi = 0$ and the preceding one-unit improvement holds whenever the displayed vertex-count identities hold.

Proof. First, the costs in the statement are finite. Order the vertices of any graph G and assign each edge to its first endpoint in that order. This partitions $E(G)$ into $|V(G)|$ possibly empty stars. For any nonnegative integer split $x + y = |V(G)|$, put x of these stars in P_R and the others in P_B, and leave both degree-bounded parts empty. Every matching uses at most one edge from each assigned star. Thus every capacity pair admits every budget split at this finite total, so its least total exists.

We construct the claimed certificate before using the vertex-count identities. Fix one ordering s_1,..., s_N of S. For $1 \leq j \leq N$ let U_j consist of the F-edges whose earliest endpoint in S is s_j. Each U_j is a possibly empty star centered at s_j. These sets are pairwise edge-disjoint and partition $E(F)$ minus $E(H)$: inducedness of H ensures that an edge outside $E(H)$ has an endpoint in S, and the earliest such endpoint is unique. In particular an edge with both endpoints in S is assigned once. Append the g single edges of P to this list, obtaining U_1,..., U_L. Each piece has matching number at most one. The pieces may share vertices, including vertices of C; their edge sets are disjoint from $E(C)$ and from K. Isolated vertices require no edge assignment.

For given $A_c + B_c = D$, put $Q_R = \bigcup_{j=1}^{A_c} M_j$ and $Q_B = \bigcup_{j=A_c+1}^{D} M_j$. A matching contributes at most one edge at any vertex, so their maximum degrees are at most A_c and B_c. An empty index interval gives an empty part. Regard the omitted vertices as isolated in these two parts.

A cycle of order $2w + 1$ has matching number w: alternating edges avoiding one vertex attain w, and each matching edge uses two distinct vertices. Matching numbers add on vertex-disjoint components, since restricting a matching gives one matching per component and component maximum matchings may be united. Thus $\nu(P) = w + g$.

Suppose first that $\chi = 0$. Then $L \geq w - 1$ and $x + y = L + w \geq 2w - 1$. As x, y are integers, they cannot both be at most w-1. If $x \geq w$, put the whole cycle and the first x-w unit pieces in P_R, and all other unit pieces in P_B. This is defined because $0 \leq x - w \leq L$; the number of remaining pieces is $L - (x - w) = y$. A matching in P_R has at most w cycle edges and at most one edge from each of its other pieces, so its size is at most x. A matching in P_B has at most y edges. If $x < w$, then $y \geq w$; put the first x unit pieces in P_R, and the cycle and all other pieces in P_B. Here $0 \leq x \leq L$ follows from $y \geq w$, and the matching bounds are x and $w + L - x = y$. These arguments do not require the pieces to be vertex-disjoint: a matching restricts to a matching in every edge piece.

For completeness, when $\chi = 1$ we construct every split at total $L + w + 1$. A path of a edges has matching number $\lceil a/2 \rceil$, attained by alternate edges and bounded above by its $a + 1$ vertices. Fix a cyclic ordering of C. For each integer z with $0 \leq z \leq w + 1$ there is a two-part partition of its edges with matching numbers at most z and $w + 1 - z$. At $z = 0$ put the whole cycle in the second part, and at $z = w + 1$ put it in the first part. For $1 \leq z \leq w$ put 2z consecutive cycle edges in the first part and all other edges in the second. These two edge sets are paths of lengths 2z and $2w + 1 - 2z$, with matching numbers z and $w + 1 - z$.

For given $x + y = L + w + 1$, set $a = \min(L, x)$ and $z = x - a$. Then $0 \le a \le L$ and $0 \le z \le w + 1$. Put the first a unit pieces and the first cycle part for z in P_R; put the other pieces and the second cycle part in P_B. Restriction to edge pieces bounds their matching numbers by $a + z = x$ and $(L - a) + (w + 1 - z) = y$.

In both cases all four edge parts are disjoint. The prescribed matchings, the cycle, and the g single edges partition $E(H)$; the N omitted-vertex stars contain precisely all remaining original edges. Hence every F-edge is assigned exactly once. The constructions apply to every integer capacity and budget split with the same prescribed matchings. This proves $c_D(F) \le T$, including $N = 0$, $g = 0$, empty matching classes, and zero endpoint budgets.

We verify the asserted least uniform total under the fixed-union condition. For fixed D, N, g, ℓ construct F as the vertex-disjoint union of C, g single edges, a complete bipartite graph with shores $\{a_0, \dots, a_{D-1}\}$ and $\{b_0, \dots, b_{D-1}\}$, and N additional edges $s_j t_j$ on fresh vertices. Omit the s_j to obtain H. For $1 \le k \le D$ prescribe

$M_k = \{a_i b_(i + k - 1) mod D) : 0 \le i < D\}$.

Each prescribed matching is nonempty, their edge sets are disjoint, and their union K contains exactly all the complete bipartite graph's edges. The entire complement in H is C, the g single edges, and isolates, as required. The graph of all paid edges $E(F)$ minus K is exactly C together with L disjoint single edges and isolates. Its matching number is $w + L$. In any partition of a graph into two edge parts, the sum of their matching numbers is at least the graph's matching number: restrict a maximum matching to the two parts. Thus no total below $w + L$ works for this configuration.

If $\chi = 1$ then $L < w - 1$, so the integer L satisfies $L \le w - 2$. At total $w + L$ choose $x = w - 1$ and $y = L + 1$. Both budgets are less than w. A monochromatic C would violate one budget; therefore both paid colors occur on C. Each color then induces a proper edge subgraph of the cycle, a disjoint union of paths and isolates. The path formula shows that its matching number is at least half its number of edges. The two matching numbers on C consequently sum to an integer at least $(2w + 1)/2$, hence at least $w + 1$. Each of the L disjoint single edges adds one to their sum. That sum exceeds $x + y = w + L$, a contradiction. This proves the natural total fails in this case. Combined with the universal constructions, it proves the least uniform total is precisely $w + L + \chi$. Only the requirement that the degree-bounded union equal K is used for this lower bound.

Now impose the three vertex-count identities. Since H is induced on a subset of $V(F)$, its omitted count is

$N = (2r + s) - (r + h) = r + s - h.$

Substitution gives

$T = N + g + \ell - 1 + \chi$

$= r + s - (s + \beta + g + \ell) + g + \ell - 1 + \chi$

$= r - \beta - 1 + \chi.$

When $\chi = 0$ the certificate already constructed therefore has cost at most r-beta-1 on the same original F. This is incompatible with $c_D(F) \ge r - \beta$. All terms are integers, and $T \ge 0$ also shows that the stated bound has a nonnegative right side whenever these data exist.

Finally assume odd order and minimum degree at least $D + 2$. The graph with edge set K has degree at most D at each vertex, and P has maximum degree at most two. If $N = 0$, then $H = F$ and every vertex v satisfies

$D + 2 \le d_F(v) = d_K(v) + d_P(v) \le D + 2.$

Thus $d_K(v) = D$ at every vertex. Each of the D matchings contributes at most one incident edge, so every prescribed matching covers every vertex. Since $D \ge 1$, a perfect matching exists, forcing $|V(F)|$ even. This contradicts odd order, so $N \ge 1$.

If $N = 1$, simplicity implies $d_H(v) \ge D + 1$ at every retained vertex. Consequently $d_P(v) = d_H(v) - d_K(v) \ge 1$, and P has no isolates. If also $g = 0$, P is exactly the one cycle C on all vertices of H, so $|V(H)|$ is odd. But $|V(F)|$ is odd and one vertex was omitted, so $|V(H)|$ is even, a contradiction. Thus $N = 1$ forces $g \ge 1$. Together with the case $N \ge 2$ this proves $N + g \ge 2$. For $2 \le \ell \le 4$ it implies $N + g \ge \ell - 2$, giving $\chi = 0$ and the asserted improvement. □

Remark. An omitted vertex contributes one separately assignable star; each retained single edge contributes another unit. Their integer allocations let the entire odd cycle fit in one paid part when $N + g \ge \ell -$

2. Otherwise splitting the cycle costs precisely one additional universal unit. The retained-cycle count contributes ℓ to the source identity but its matching rank is ell-1, producing the saving of one.

Depends on. Nothing earlier; this item is self-contained.

Lemma 16 (Two matching boundary construction). *All graphs are finite and simple. For a graph G, write $V(G)$, $E(G)$, $\nu(G)$, and $\Delta(G)$ for its vertex set, edge set, matching number, and maximum degree. For nonnegative integers A, B, define $c_{A,B}(G)$ to be the least nonnegative integer T such that, for every pair of nonnegative integers x, y with $x+y=T$, $E(G)$ has a partition into four sets whose spanning subgraphs have maximum degrees at most A, B and matching numbers at most x, y, respectively. Set $c_2(G)=\max_{A+B=2} c_{A,B}(G)$, with A, B nonnegative integers. These minima exist: assigning each edge separately to one of the last two parts gives feasibility at $T=|E(G)|$.*

Let F be a graph, let W be a subset of $V(F)$, and suppose that $F[W]$ consists of a matching L of m edges and a set Z of isolated vertices, where m is a nonnegative integer. Let P be the set of endpoints of L, so $W = P$ disjoint union Z. Let X be a subset of $V(F)$ minus W such that every vertex of X has exactly two neighbors in W. For Y contained in X, let B_Y be the bipartite graph with parts Y, Z and exactly the F-edges between these parts. Put $K_Y = F[W \text{ union } Y]$, $J_Y = F[Y]$, $N_Y = |V(F) \text{ minus } (W \text{union} Y)|$, and $t_Y = \nu(J_Y)$. Assume that every vertex of P has at most one neighbor in X, and every vertex of Z has at most two neighbors in X.

Let T_1,..., T_d be all connected components of B_X that are paths with both endpoints in X; an isolated X-vertex is included as a path of order one. The integer d is nonnegative. Choose one vertex a_i in X intersect $V(T_i)$ for every integer i with $1 \le i \le d$, independently, and put $A_0 = \{a_1, \dots, a_d\}$ and $Y = X$ minus A_0; when $d=0$ put A_0 empty.

For every such choice there are edge-disjoint matchings M_0, M_1 in K_Y with L contained in M_0, such that M_0 is maximum in K_Y with $|M_0| = m + |Y|$, M_1 is maximum in $K_Y - M_0$ with $|M_1| = |Y|$, and M_0 union M_1 consists of exactly L and all F-edges between W and Y. Consequently $E(K_Y - M_0 - M_1) = E(J_Y)$. Here deleting a matching deletes its edges and retains all vertices. The same conclusions hold for $Y = X$ if and only if $d=0$. More explicitly, under the two assumed neighbor bounds, $d=0$ is equivalent to the existence of two matchings partitioning L and all W-to-X edges with L in the first matching.

If $N_Y \ge t_Y - 1$, these same two matchings extend to a four-part certificate on all of F for every nonnegative integer capacity split $A+B=2$ and every nonnegative integer budget split $x+y = N_Y + t_Y$. In particular $c_2(F) \le N_Y + t_Y$. All edges with an endpoint outside K_Y, including edges with both endpoints outside K_Y, are included in this partition.

Proof. First consider the bipartite graph B_X. Its maximum degree is at most two: each X-vertex has only two neighbors in W, and the bound at Z is assumed. Each nontrivial connected component is a path or a cycle. Indeed, starting at a vertex of degree one produces a path that cannot branch and contains its entire component. If a component has an edge and every vertex has degree two, following consecutive edges closes a cycle, and the degree bound prevents any vertex outside that cycle from attaching. Bipartiteness makes every cycle even.

On a path with both endpoints in X, the numbers of vertices in X and Z differ by one, with X larger. Deleting any one X-vertex splits this path into two possibly empty paths of even order: the deleted vertex occupies an odd position in an alternating ordering starting at an X-endpoint. Each remaining path has equally many vertices in the two parts and has a perfect matching given by alternate edges. On an even cycle or a path with endpoints in different parts, alternate edges give a perfect matching. On a path with both endpoints in Z, leave one Z-endpoint unmatched and take alternate edges on the remaining even path. Isolated Z-vertices require no matching edge.

The components are disjoint. Deleting the chosen vertices A_0 therefore produces a matching Q in B_Y that saturates every vertex of Y. The choices just described also cover every vertex of Z having degree two in B_Y: such a vertex is covered by a perfect matching on its component, or is an internal vertex on a Z-ended path and hence is covered by that path's matching. This construction works for each independent choice of the a_i, including an isolated X-component, whose deletion leaves no vertices.

It follows in particular that B_X has a matching saturating X when $d=0$. When $d>0$, any T_i has $|X \text{ intersect } V(T_i)| = |Z \text{ intersect } V(T_i)| + 1$ and has no edges to other components. A matching cannot saturate its X-side, so no matching of B_X saturates X.

Let G_Y be the spanning subgraph of K_Y whose edges are L and all W-to-Y edges. Set $M_0 = L$ union Q. The vertices used by L lie in P, while Q uses Y and Z, so M_0 is a matching and $|M_0| = m + |Y|$. Every vertex of G_Y has degree at most two. Moreover M_0 covers every vertex of G_Y of degree two. It covers every Y-vertex

through Q, every P-vertex through L, and every degree-two Z-vertex by the chosen property of Q. Hence $G_Y - M_0$ has maximum degree at most one. Its entire edge set is a matching; call it M_1. There are $m + 2|Y|$ edges in G_Y because each vertex of Y has exactly two W-neighbors. Consequently $|M_1| = |Y|$. The edges of K_Y outside G_Y are exactly those of J_Y, proving the residual identity.

Choose one endpoint of each edge of L and call the resulting m-element set U. The set U union Y is a vertex cover of K_Y: it meets every edge of L at its selected endpoint and every other edge at a vertex of Y. A matching uses distinct vertices of any vertex cover on its distinct edges, and therefore has size at most the size of that cover. Thus $\nu(K_Y) \le m + |Y| = |M_0|$, making M_0 maximum. After M_0 is deleted, no edge of L remains. Every remaining edge meets Y, so $\nu(K_Y - M_0) \le |Y| = |M_1|$. This proves maximum status in the stated order on the same retained graph.

For the claimed converse with $Y = X$, suppose two matchings partition G_X with all edges of L in their first member. At every X-vertex the two incident $G_X - edges$ must lie one in each matching. Since every P-vertex is already incident with an L-edge in the first matching, that matching's edge at an X-vertex must go to Z. Its W-to-X edges consequently give a $B_X - matching$ saturating X. The preceding path argument then gives $d = 0$. The construction for $d = 0$ proves the other direction.

Finally fix Y obtained by the construction, and abbreviate $N = N_Y$ and $t = t_Y$. Fix nonnegative integers A, B, x, y with $A + B = 2$ and $x + y = N + t$. Assign A of the two matchings M_0, M_1 to the first degree-bounded part and the other B to the second. Each union has maximum degree at most its number of matchings, including when that number is zero.

Order the N vertices of $V(F)$ minus (W union Y). Every edge of F outside the induced graph K_Y has an omitted endpoint. Assign that edge to its earliest omitted endpoint in this order. This partitions all edges outside K_Y into N labelled stars, allowing empty stars. In particular an edge with two omitted endpoints is assigned exactly once. A matching uses at most one edge of any one of these stars.

If $t = 0$, J_Y is edgeless, and $x + y = N$; place any x labelled stars in the first matching-bounded part and the remaining y in the second. If $t \ge 1$, the hypothesis $N \ge t - 1$ implies $N + t \ge 2t - 1$. Hence at least one of x, y is at least t, since otherwise $x + y \le 2t - 2$. If $x \ge t$, put every edge of J_Y and exactly x-t labelled stars in the first matching-bounded part. This star count is a nonnegative integer at most N because $x \le N + t$. Put the remaining $N - (x - t) = y$ stars in the second matching-bounded part. Every matching in the first part restricts to a matching of size at most t in J_Y and uses at most one edge per assigned star, so has size at most $t + (x - t) = x$. The second part has matching number at most y. If $x < t$, then $y \ge t$; put every $J_Y - edge$ and y-t stars in the second part and the remaining x stars in the first. The same restriction argument gives the two required bounds.

These four edge parts partition $E(F)$. The construction uses the same M_0, M_1 for every capacity and budget split, and includes every omitted edge. Thus $c_{A,B}(F) \le N_Y + t_Y$ for every nonnegative integer $A + B = 2$, proving $c_2(F) \le N_Y + t_Y$. □

Remark. With the original frame matching reserved first, the only obstruction is a bipartite path with one excess selector vertex. Deleting one such vertex repairs that component. Vertex covers certify both matching maxima, and omitted vertices supply stars that extend the same construction to the entire original graph.

Depends on. Nothing earlier; this item is self-contained.

Lemma 17 (Prescribed component ports from a missing matching allocation). *Let F be a finite simple graph and R an induced subgraph. A matching is a set of edges with pairwise disjoint endpoints, $\nu(G)$ is the maximum cardinality of a matching in G, and a perfect matching covers every vertex, including the empty matching in the empty graph. Write $U = V(F)$ minus $V(R)$, $u = |U|$, and $m = \nu(R)$. Suppose every connected component C of R satisfies $\nu(C) = floor(|V(C)|/2)$. For an odd-order component C, call a vertex z exposed if C-z has a perfect matching.*

Let x, y be nonnegative integers with $x + y = u + m - 1$ such that $E(F)$ has no partition into two edge sets whose spanning graphs have matching numbers at most x and y, respectively. Put $j = u - x$, and suppose $j \ge 1$. Then for every $T \subseteq U$ with $h = |T| \le j$,

$\nu(F[V(R) union T]) = m + h.$

Every maximum matching in this induced graph consists of a maximum matching in each component of R together with h edges assigning all vertices of T to exposed vertices in h distinct odd-order components of R.

An independent set is maximal if no vertex can be added and maximum if its cardinality is $\alpha(G)$, the independence number of its graph. Let mathcal P be any family of odd-order components of R. For each C in mathcal P, fix a maximum independent set I_C of C containing every exposed vertex of C. Suppose every exposed vertex of every odd-order component outside mathcal P belongs to some maximum independent set of that component. Put $K = \bigcup_{C\in mathcalP} I_C$, let q be the number of odd-order components outside mathcal P, and suppose $j \geq q+1$. Set $B = \{vinU : N_F(v)intersectKisempty\}$, where $N_F(v)$ is the set of neighbors of v in F. Then $|B| \leq q$ and there is one matching assigning all vertices of B to exposed vertices in distinct odd-order components outside mathcal P, simultaneously compatible with a maximum matching in every component of R. There is a maximum independent set I of R containing K and all these assigned vertices which is maximal independent in F.

If F is well-covered, meaning that every maximal independent set of F is maximum, then $\alpha(R) = \alpha(F)$. Every maximum independent set of R is then maximum in F, and $\alpha(G) = \alpha(F)$ for every induced G with $V(R) \subseteq V(G) \subseteq V(F)$.

In particular, when $\Delta(R) \leq 2$, one may take mathcal P to consist of all odd-order path components, including isolated vertices, and I_C to be the odd-indexed vertices along each such path. The integer q is then exactly the number of odd-cycle components. One may instead take mathcal P to consist only of the isolated vertices, in which case q counts the nonisolated odd-order components. All component counts, graph orders, matching cardinalities, x, y, u, m, h, j, q, and independent-set cardinalities are integers in the stated nonnegative domains. Empty component families and isolated vertices are permitted.

Proof. Fix $T \subseteq U$ with $h = |T| \leq j$, and write $H = F[V(R)$ union T]. Since $u - h \geq u - j = x$, there is an x-element set $Z \subseteq U$ minus T. The graph F is finite, so all maximum matchings and all choices of a largest independent set used below exist.

Place every edge of F incident with Z in the first edge set, and all other edges in the second. A matching in the first set has at most $|Z| = x$ edges: choose for each matching edge an endpoint in Z; those chosen endpoints are distinct. The second set is the edge set of F-Z. Therefore the missing allocation implies $\nu(F - Z) \geq y + 1$.

The set of vertices of F-Z outside H has cardinality u-h-x. Every matching of F-Z has at most $\nu(H)$ edges with both endpoints in H, and at most u-h-x other edges, since the latter use distinct vertices outside H. Consequently

$\nu(H) \geq y + 1 - (u - h - x) = m + h.$

A matching of H has at most m edges wholly in R and at most h edges meeting T, giving the reverse inequality. Thus $\nu(H) = m + h$.

Take any maximum matching of H. Its number of internal R-edges is at most m and its number of edges meeting T is at most h; their sum is $m + h$, so both bounds are equalities. The h edges meeting T have exactly one endpoint in T each and cover all of T. Indeed, each uses at least one distinct T-vertex, and an edge with both endpoints in T would make h such edges use at least $h + 1$ T-vertices.

The m internal edges form a maximum matching of R. The matching number of a disjoint union equals the sum of the component matching numbers: restriction gives the upper bound, and the union of component maxima attains it. The internal edges are therefore maximum in each component C. By hypothesis they cover every vertex of an even component and leave exactly one vertex uncovered in an odd component. Each edge from T must end at such an uncovered vertex. Distinct edges must use different odd components, since only one vertex per component is available. Removing that vertex leaves its component's perfect internal matching. This proves every assertion about H, including the simultaneous compatibility.

Now fix mathcal P and its prescribed sets I_C. An exterior vertex in B has no neighbor at an exposed vertex of any component in mathcal P, because all those vertices belong to K. If $|B| \geq q + 1$, choose a $(q+1) - element$ subset T of B. The bound $q + 1 \leq j$ permits the proved matching assertion. Its $q + 1$ distinct receiving odd components must all be outside mathcal P, but there are only q such components. This contradiction gives $|B| \leq q$.

Apply the same assertion to $T = B$, whose size is at most $q < j$. It supplies one simultaneous assignment of all B-vertices to exposed vertices in distinct odd components outside mathcal P, with internal maximum matchings in all components. In each receiving component choose a maximum independent set containing its assigned vertex; this is possible by the stated exposed-vertex assumption. Only one vertex is prescribed per receiving component. In each C in mathcal P retain the fixed I_C, and in each remaining component choose any maximum independent set. Let I be the union of these choices. Different components have no

edges between them, and R is induced in F, so I is independent in F. Restriction to components and union of component maxima show that $|I| = \alpha(R)$.

Every vertex of R outside I has an I-neighbor, since otherwise its addition would enlarge the selected component independent set. Every vertex of U minus B has a neighbor in $K \subseteq$ I. Every vertex of B has its assigned neighbor in I. Hence every vertex of F outside I has a neighbor in I, proving that I is maximal independent in F. If F is well-covered, $|I| = \alpha(F)$, and therefore $\alpha(R) = \alpha(F)$. Any maximum independent set of R is independent in F with this cardinality and is thus maximum in F. For an induced graph G between R and F, inclusion of independent sets gives $\alpha(R) \leq \alpha(G) \leq \alpha(F)$, so equality holds throughout.

It remains to verify the claimed choices when $\Delta(R) \leq 2$. Every connected component of such a finite simple graph is a path or a cycle. To see this, choose a longest simple path in a component. An internal vertex already has its two possible incident path edges. An endpoint has no neighbor outside the path by maximal length; its only possible extra neighbor on the path is the other endpoint, since internal vertices already have degree two. Connectedness then leaves no vertex outside the path. The component is the path, or the cycle formed by joining its endpoints; a single vertex is allowed.

A path or cycle of order n has matching number $\lfloor n/2 \rfloor$: disjoint consecutive edges attain this number, and distinct matching endpoints give the upper bound. On an odd path of order $2a+1$, with a a nonnegative integer, deleting position i leaves paths of orders i-1 and $2a+1-i$. A path has a perfect matching exactly when its order is even, by parity and consecutive matching edges, including order zero. Thus the exposed vertices of the odd path are exactly its odd-indexed vertices. They form an independent set of size $a+1$. The first 2a vertices can be partitioned into a adjacent pairs, so any independent set has size at most $a+1$. This verifies the required common maximum independent set, including $a = 0$.

For an odd cycle of order $2a+1$, where $a \geq 1$ is an integer, deleting any vertex leaves an even path and hence a perfect matching. Starting with any prescribed vertex, take it and every second vertex in a fixed cyclic direction until a vertices have been selected. They are independent. For any independent cycle set, mapping each selected vertex to its successor in that direction gives distinct unselected vertices, so twice its size is at most $2a+1$. Its size is therefore at most a. Every exposed cycle vertex consequently belongs to a maximum independent set. These calculations verify both choices of mathcal P and the stated values of q. When B or a component family is empty, the corresponding matching, independent union, and sums are empty, and every preceding argument remains valid. □

Remark. A missing matching allocation forces every sufficiently small exterior set to add its full size to the matching number. Equality then provides one compatible assignment to distinct exposed components; odd paths can be dominated in advance by selecting all their exposed vertices.

Depends on. Nothing earlier; this item is self-contained.

Lemma 18 (Cover matchings that improve maximum-degree saturation). *Let $D \geq 3$ be an integer and H a finite simple graph with maximum degree at most D. Let W be any matching of H, let $V(W)$ be its covered vertex set, and let $Q = H - W$ be obtained by deleting only its edges. Put*

$S = \{vinV(H) : deg_H(v) = D\}$, *$I = S$ minus $V(W)$, $R = |I|$,*

and suppose $R \geq 1$. Let C be any vertex cover of Q containing I; a vertex cover meets every edge. Set $C_0 = C$ minus I, $O = V(H)$ minus C, and let B be the bipartite graph consisting of the Q-edges between C and O. Suppose B has a matching saturating C_0 and at least one vertex of I.

Then there are an integer $1 \leq r \leq R$ and a maximum-cardinality matching T of B which saturates C_0, covers exactly r vertices of I, and leaves at most

$R - r + floor(r/(D-1))$

vertices of S uncovered. In particular H has a matching leaving at most R-1 vertices of degree D uncovered. One may choose T among maximum matchings saturating C_0 to maximize the number of covered vertices of S intersect O.

The crossing-matching hypothesis is guaranteed by the following concrete sufficient condition. Let $p, t \geq 1$ be integers, let J be a vertex-disjoint union of t positive-edge stars, and put $E = e(J)$ and $N = p + t - 1$. Suppose $|C| = N$, $E \geq |C_0| + 1$, and Q arrows the ordered pair (pK2,J), where K_2 is one edge. Arrowing means that every red-blue edge coloring contains an entire red matching with p edges or an entire blue copy of J. Copies are ordinary subgraphs, with disjoint components within each target.

Under this sufficient condition, for each t-element subset Z of C, B contains a copy of J using every Z-vertex exactly once and no other C-vertex, with its other E vertices in O. Writing $N_B(X)$ for the set of B-neighbors of $X \subseteq C$, one has $|N_B(X)| \geq \min(|X|, E)$. Thus every $Y \subseteq C$ of size at most $\min(N, E)$ can be saturated by a crossing matching, including C_0 together with any specified vertex of I. The resulting T in the first conclusion need not retain that specified vertex.

All orders, sizes, cardinalities, degrees and matching numbers are integers in their displayed nonnegative or positive domains. The matching W need not have maximum cardinality; C need not be independent or a minimum cover. Unit stars, repeated target components, C_0 empty and isolated vertices are permitted.

Proof. We first construct T assuming the crossing matching in the first hypothesis. If a matching U is smaller than another matching U', their symmetric difference decomposes into alternating paths and even cycles. Since U' has more edges, some path has one more U'-edge than U-edge. Its endpoints are uncovered by U. Switching along this path increases the size of U by one and keeps every previously covered vertex covered. Starting with the supplied crossing matching and repeating this operation finitely many times yields a maximum matching of B still saturating C_0 and at least one vertex of I.

Among all maximum matchings of B saturating C_0 choose T that covers as many vertices of S intersect O as possible. These are nonempty finite choice families. Every edge of B has exactly one endpoint in $C = C_0$ disjoint union I. It follows that each maximum matching saturating C_0 covers exactly

$r = \nu(B) - |C_0|$

vertices of I, where $\nu(B)$ denotes maximum matching cardinality. The preceding extension proves $1 \leq r \leq R$.

A vertex of Q has degree D exactly when it belongs to I: W reduces the degree of every covered vertex by one, and all degrees in H are at most D. In particular every vertex of C_0 has Q-degree at most D-1. Since I is contained in C, every vertex of S intersect O is covered by W and has Q-degree exactly D-1.

Let h be the number of vertices of S intersect O uncovered by T. If $h = 0$, the number of S-vertices uncovered by T is precisely R-r, which proves the claim. Suppose $h \geq 1$. Direct every non-T edge of B from O to C and every T-edge from C to O. Starting from all h uncovered vertices of S intersect O, let O' and C' be the sets reached on the two sides.

Every reached C-vertex is T-covered. Otherwise a shortest directed path to an uncovered C-vertex starts and ends uncovered by T, and switching along it increases the size of T. This contradicts its maximum cardinality. Also every vertex of O' belongs to S. If a vertex z in O' were outside S, a shortest directed path from a starting vertex to z would start with a non-T edge and end with a T-edge. Switching along it preserves cardinality and coverage of every C-vertex; on the O side it covers the starting vertex in S and uncovers only z outside S. This contradicts the maximizing choice of T.

The T-edges pair C' with the matched vertices of O'. All unmatched vertices of O' are precisely the h starting vertices: any other reached O-vertex is first reached along its T-edge. Hence $|O'| = |C'| + h$. The set O is independent in Q because C is a cover. Every Q-neighbor of O' lies in C': non-T edges are traversed towards C, and the T-neighbor of a matched reached O-vertex was already reached.

Each O'-vertex therefore has D-1 Q-edges into C'. A vertex of C' minus I has Q-degree at most D-1, while a vertex of C' intersect I has degree D. Every C'-vertex is T-covered, so $|C' \text{ intersect } I| \leq r$. Counting these crossing edges gives

$(D-1)(|C'| + h) \leq (D-1)|C'| + |C' \text{ intersect } I| \leq (D-1)|C'| + r.$

Consequently $h \leq floor(r/(D-1))$. All S-vertices in C_0 are covered, exactly R-r in I are uncovered, and h in O are uncovered. Their total has the claimed upper bound. For $D \geq 3$ and $r \geq 1$, $floor(r/(D-1)) \leq r-1$, so at most R-1 S-vertices are uncovered. Every B-matching is also an H-matching.

We next verify the sufficient Ramsey condition and its stated witnesses. Fix a t-element subset Z of C and put $A = C$ minus Z; its size is p-1. Color every Q-edge meeting A red and all other edges blue. The red graph has cover A, so a red matching has at most p-1 edges. The arrowing hypothesis forces an entire blue J.

Every vertex of this copied J has a positive incident edge, so none lies in A. Each of its t connected components meets Z, because every edge meets C and its blue edges cannot meet A. Since the components are disjoint and $|Z| = t$, each uses exactly one Z-vertex and they use all of Z. For a nonunit star, the center must be its unique Z-vertex: if the center lay outside C, every one of its at least two leaves would have to lie in Z, which

is impossible. A unit star has one endpoint in Z and one in O. All its edges therefore lie in B. Since J has $E + t$ vertices, the copy has exactly E vertices in O, proving the asserted copy.

For $X \subseteq C$ with $|X| \le t$, extend X to a t-element Z, which is possible since $N \ge t$. Selecting one edge from each of the t components of the corresponding J-copy gives a crossing matching saturating Z and hence $|X|$ distinct neighbors of X. For $|X| \ge t$ choose a t-element Z contained in X instead; all E outside vertices of its copied J are neighbors of X. Since $E \ge t$, these two bounds give $|N_B(X)| \ge \min(|X|, E)$ for every X.

Fix $Y \subseteq C$ with $|Y| \le \min(N, E)$. Each $X \subseteq Y$ consequently has at least $|X|$ neighbors. Choose a maximum matching U in B[Y, O]. If it fails to saturate Y, direct nonmatching edges from Y to O and matching edges from O to Y, and search from all unmatched Y-vertices. No reached O-vertex can be unmatched, since a shortest path to one would augment U. If Y′ and O′ are the reached sets and $k \ge 1$ is the number of unmatched starting vertices, matching edges pair O′ with Y′ minus those k vertices. Every neighbor of Y′ lies in O′. Thus $|N_B(Y')| = |O'| = |Y'| - k < |Y'|$, contradicting the established bound. Therefore U saturates Y.

Take $Y = C_0$ union {x} for any specified x in the nonempty set I. Its size is $|C_0| + 1 \le N$ and, by hypothesis, at most E. The constructed crossing matching saturates Y, proving the first hypothesis and making the earlier construction applicable. None of these steps requires a minimum cover or a maximum original W, and they also apply when C_0 is empty or target stars have size one. □

Remark. A forced full star forest supplies extra outside neighbors of a tight cover. Alternating reachability then charges each remaining uncovered degree-D outside vertex D-1 units against only one extra degree unit per covered exposed root.

Depends on. Nothing earlier; this item is self-contained.

Lemma 19 (Odd-order capacity remnants and profile loss). *All graphs are finite and simple. Let F_1, F_2 be nonempty vertex-disjoint unions of positive-edge stars, with full nonincreasing integer edge-size lists $n_1,\dots, n_s$ and $m_1,\dots, m_t$. Copies are ordinary subgraphs with disjoint target components. Their heads are $A = n_1$ and $B = m_1$. Put $M = s + t - 1$, $\ell_k = max\{n_i + m_j - 1 : 1 \le i \le s, 1 \le j \le t, i + j = k\}$ for $2 \le k \le s + t$, and $P_q = \sum_{k=q+2}^{s+t}(\ell_k - 1)$ for each integer $0 \le q < M$.*

Fix an integer $\mu \ge 3$ and put $v = 2\mu + 1$. For $i = 1, 2$ and integer $j \ge 1$, let $N_{i,j}$ count the full components of F_i of edge size at least j. Let $a_{i,j}$ be the largest cardinality of a subcollection of these components whose total vertex count is at most v. Put $c_{i,j} = \max_{h \ge j}(N_{i,h} - a_{i,h})_+$, where $x_+ = \max(x, 0)$, and let U_i have cumulative component-size counts $c_{i,j}$. Then U_i is a componentwise subforest of F_i, and U_i is empty exactly when $|V(F_i)| \le v$. Superscript β means deletion of β leading whole components from the full sorted list. For every integer $0 \le \beta < c_{i,1}$, and all disjoint graphs C, Q with $|V(C)| \le v$, every copy of F_i^β in C disjoint union Q leaves a copy of U_i^β entirely in Q.

Suppose both U_i are nonempty. Let E_q be the union of all nonempty actual total-q suffix pairs of (U_1, F_2) and (F_1, U_2): a pair (X^α, Y^β) is included when $\alpha, \beta \ge 0$, $\alpha + \beta = q$, and both suffixes are nonempty. The diagonal profile of a pair of positive star forests is defined by the same maximum formula as ℓ_k. Let P_{end} be the sum of the coordinatewise maximum of these profiles after subtracting one from each existing entry, taking missing entries and the maximum of an empty family to be zero. For each integer $z \ge 1$ let Δ_z be the number of entries at least z among $\ell_{q+2} - 1,\dots, \ell_{s+t} - 1$ minus the number of entries at least z in this endpoint envelope. Then $\Delta_z \ge 0$, $\Delta_z \le \mu$ for $z = 1$ and $z \ge 5$, and

$\sum_{z=2}^{4}(\Delta_z - \mu)_+ \le (\mu - 1)/3$.

If $\max(A, B) \ge 3$, the stronger integer bound is

$\sum_{z=2}^{4}(\Delta_z - \mu)_+ \le floor((\mu - 2)/3)$.

Put $D = \ell_{q+2} - 1$, and let R be the initial run length of entries equal to $D + 1$ in the full natural q-tail. If $D \ge 6$, $3 \le \mu \le D$, then

$0 \le P_q - P_{end} \le (D - 1)\mu + R$.

Independently, if $D \ge 6$, $A, B \le D - 1$, $\mu \ge D/2$ and $R = 1$, then

$0 \le P_q - P_{end} \le (D - 1)\mu$.

Both conclusions allow either parity of D and arbitrary legal q, including an empty endpoint family, with every original repetition and unit component retained. All sizes, orders, thresholds, cuts, counts, maxima, D and R are integers in their stated domains.

Proof. All fitting maxima exist because their eligible sets are finite and include the empty set. For one original, abbreviate its counts and maxima by N_j, a_j. We have $0 \le a_j \le N_j$ and $a_j \le floor(v/(j+1))$. The counts $c_j = \max_{h \ge j}(N_h - a_h)_+$ are nonnegative, nonincreasing, eventually zero and at most N_j. Thus they define a star forest componentwise contained in the original: sorting both lists, the k-th required size cannot exceed the k-th original size, since otherwise their counts at that required threshold would violate $c_j \le N_j$. If the full original fits v vertices, every threshold group fits and all c_j vanish. Conversely $c_1 = 0$ implies $a_1 = N_1$, so the full original fits.

In a copy of F_i^β inside C disjoint union Q, every positive star lies wholly in one of those graphs, because it is connected. At most $a_{i,h}$ of the original threshold-h components can lie in C. Hence Q contains at least $(N_{i,h} - a_{i,h} - \beta)_+$ components of size at least h. For every $h \ge j$ these also have size at least j. Maximizing over h gives at least $(c_{i,j} - \beta)_+$ components at threshold j, exactly the threshold counts of U_i^β. Comparing sorted sizes as above gives its copy in Q.

We derive the full envelope formula to include all cut and exhaustion cases. Write $N_x = N_{1,x}$, $T_y = N_{2,y}$. For integer $z \ge 1$ the number of entries $\ell_k - 1$ at least z in the full pair is

$V_z = max\{N_x + T_y - 1 : x + y = z + 2, N_x > 0, T_y > 0\}$, (1)

with maximum zero if there is no pair. Indeed a threshold rectangle $1 \le i \le N_x$, $1 \le j \le T_y$ has every index sum from 2 through $N_x + T_y$. Conversely any sizes $n_i + m_j \ge z + 2$ admit positive thresholds $x \le n_i$, $y \le m_j$ with $x + y = z + 2$. Their union of initial intervals is precisely the initial interval of length V_z. Thus the natural q-tail width is $(V_z - q)_+$.

For a full pair (X, Y), any original indices i, j with $i + j \ge q + 2$ admit cuts $\alpha \le i - 1$, $\beta \le j - 1$ with $\alpha + \beta = q$. To see this, take $\alpha = \min(q, i - 1)$ and $\beta = q - \alpha$. The displayed inequality implies $\beta \le j - 1$. These are legal nonempty cuts, and the local index sum is $i + j - q$. Conversely local indices and such cuts give those full original indices. Therefore the envelope over every actual total-q suffix pair is precisely the natural q-tail. This holds after exhaustion: when q is at least the full component budget, both are empty. For the two endpoint pairs the full superlevel widths are computed by (1) with the corresponding remnant counts; let V'_z be their maximum. The endpoint width is $(V'_z - q)_+$, since taking a maximum commutes with truncation by q. Containment gives $V'_z \le V_z$, so $\Delta_z \ge 0$ and

$P_q - P_{end} = \sum_{z \ge 1} \Delta_z$. (2)

Choose an original maximizing threshold pair x, y in (1). If either remnant threshold survives, using the corresponding endpoint loses at most $N_{i,j} - c_{i,j} \le a_{i,j} \le \mu$ in full width, hence also after q-truncation. Here every component has at least two vertices. If $x = 1$ or $y = 1$, a surviving threshold always exists because both remnants are nonempty. Otherwise, when both thresholds vanish, their whole original threshold groups fit v vertices, and

$\Delta_z \le floor(v/(x+1)) + floor(v/(y+1)) - 1$. (3)

At $z \ge 5$, x, $y \ge 2$ and $x + y = z + 2$. The reciprocal sum is largest at an endpoint $x = 2$ or $y = 2$, as the denominator product is smallest there. It is at most $1/3 + 1/(z+1) \le 1/2$. The bound in (3) is therefore at most $\lfloor v/2 - 1 \rfloor = \mu - 1$, in particular at most μ. At $z = 1$ every pair has a threshold one. This proves the generic level bounds.

Here is a single convexity argument for all three remaining levels. Let h_i be the first vanished remnant threshold on side i. It exists, and $h_i \ge 2$. An excess $\Delta_z > \mu$ at $z = 2$, 3, 4 requires a maximizing pair with $x \ge h_1$, $y \ge h_2$, by the preceding surviving-threshold argument. If $h_1 + h_2 > 6$ there is no such pair. Otherwise the original groups of sizes at least h_i fit v vertices; they are nonempty because $h_i \le 4$ and a positive maximizing pair is under consideration.

Let f_p and g_r be their exact size multiplicities, for $p \ge h_1$ and $r \ge h_2$. They satisfy $\sum_p (p+1) f_p \le v$ and $\sum_r (r+1) g_r \le v$. The sum of positive excesses is at most

$E(f, g) = \sum_{z=2}^{4}$ max(0,max_(x+y=z+2,x>=h_1,y>=h_2)

$[\sum_{p\ge x} f_p + \sum_{r\ge y} g_r - 1 - \mu])$. (4)

If there is no candidate at one level its inner maximum contributes zero. When the actual loss exceeds μ, its original maximizing pair is among these candidates; dropping q can only increase that loss.

The function E is convex, being a sum of maxima of affine functions and zero. Each f is the convex combination of the zero vector and vectors with sole entry $v/(p+1)$ at size p: the coefficient of that vector is $(p+1)f_p/v$ and the remaining coefficient is assigned to zero. The same holds for g. Multiplying their coefficients represents (f, g) as a convex combination of pairs of these vertices. Hence E is bounded by their weighted average of E-values.

When one vector is zero, all relevant expressions are nonpositive, since $v/3 - 1 - \mu < 0$. At a pair supported at sizes p, r, candidates missing either uniform group are likewise nonpositive. A candidate retaining both groups has value

$v/(p+1) + v/(r+1) - 1 - \mu$.

If $p, r \ge 3$ this is at most $v/2 - 1 - \mu < 0$. By symmetry, a positive value thus requires $p = 2$ and $r \ge 2$. It is nonpositive for $r \ge 5$. For $r = 2, 3, 4$, the possible numbers of contributing levels are at most one, two and three, respectively. Their total values are bounded by

$(\mu - 1)/3$, $((2\mu - 5)/6)_+$, $((\mu - 7)/5)_+$, (5)

respectively. The last two are strictly less than $(\mu - 1)/3$ for $\mu \ge 3$. These comparisons include zero values because $(\mu - 1)/3 > 0$. This proves the general low-level bound in the statement.

For the sharper integer bound, if μ is not 1 modulo 3, flooring $(\mu - 1)/3$ already gives $floor((\mu - 2)/3)$. If $\mu = 3t + 1$ with integer $t \ge 1$, the first value in (5) is t and all other vertex values are strictly below t. The only vertex pair that can attain t uses size two on both sides. If either $h_i \ge 3$, that pair is absent. If $h_1 = h_2 = 2$, both full nonunit groups are used in (4). The hypothesis $\max(A, B) \ge 3$ puts positive convex weight on a size at least three in at least one group. Thus positive total product weight lies outside the size-(2,2) pair. In either case $E < t$. The integral actual excess sum is at most $t - 1 = floor((\mu - 2)/3)$, proving the stronger bound.

For the small-order bound assume $D \ge 6$ and $\mu \le D$. The heads satisfy $A + B \ge D + 2$ because the first diagonal dominates the natural leading diagonal. We show there is no low-level excess at all. For a maximizing pair whose two remnant thresholds vanish, reserve the original head in each of its two fitting groups. Then

$N_x \le 1 + (2\mu - A)/(x + 1)$, $T_y \le 1 + (2\mu - B)/(y + 1)$. (6)

Both numerators are nonnegative because these groups fit. At $(x, y) = (2, 2)$ the full width minus μ is at most $1 + (\mu - A - B)/3 \le 1/3$. The loss and μ are integers, so this gives no positive excess. At (2,3) or its exchange the bound is at most

$1 + \mu/6 - (A + B)/4 \le 1/2 - D/12 \le 0$.

At (2,4) or its exchange it is at most

$1 + \mu/15 - (A + B)/5 \le 3/5 - 2D/15 < 0$.

At (3,3) it is at most $1 - (A + B)/4 < 0$.

These are every pair with $x, y \ge 2$ and $x + y$ in {4,5,6}. A surviving threshold already costs at most μ. Thus every level below D costs at most μ. The natural top width is R and no higher natural level exists, so (2) gives $P_q - P_{end} \le (D - 1)\mu + R$. This argument has imposed no individual head bound and no parity condition.

Finally assume A, $B \le D - 1$, $\mu \ge D/2$, $D \ge 6$ and $R = 1$. The head sum $A + B \ge D + 2$ implies both heads are at least three. The top loss Δ_D is at most one. At the penultimate level a maximizing pair has $x + y = D + 1$ and $x, y \ge 2$. A surviving threshold costs at most $\lfloor v/3 \rfloor$. If both vanish, at least one can be increased within its original head since $A + B > D + 1$. Suppose $x + 1$ is available. Its positive threshold count gives

$N_{x+1} + T_y - 1 \le V_D = q + 1$.

Consequently $(N_x + T_y - 1 - q)_+ \le N_x - N_{x+1} + 1 \le N_x \le \lfloor v/3 \rfloor$.

The same argument exchanges sides when y increases. Hence $\Delta_{D-1} \le \lfloor v/3 \rfloor$.

The top two levels are distinct from 2, 3, 4 because $D \geq 6$. The other D-2 levels cost at most (D-2)mu plus their aggregate positive low excess. Using its sharper bound gives

$P_q - P_{end} \leq (D-2)\mu + floor((\mu-2)/3) + floor((2\mu+1)/3) + 1$

$= (D-1)\mu.$

The equality follows on writing $\mu = 3t, 3t+1, 3t+2$: the two floors sum to mu-1 in each case. Nonnegativity follows from (2). This completes both profile bounds with the same full remnants and their every-prefix copy guarantee. □

Remark. A vanished capacity threshold means its entire eligible group fits the odd vertex ∩. Convexity couples the three potentially expensive low levels. A full head above two removes the sole equality case; the resulting integer saving pays the final top-level unit.

Depends on. Nothing earlier; this item is self-contained.

Lemma 20 (Three successive matchings from an induced row). *All graphs are finite and simple. Write $V(G)$, $E(G)$, $\nu(G)$, and $\Delta(G)$ for the vertex set, edge set, maximum matching cardinality, and maximum degree of G. Edge-set quantities refer to the spanning graph, and are zero for the empty edge set. For nonnegative integer capacities A_c, B_c, let $c_{A_c,B_c}(G)$ be the least nonnegative integer T such that, for every pair of nonnegative integers x, y with $x + y = T$, there is an ordered partition $E(G) = Q_R$ disjoint union Q_B disjoint union P_R disjoint union P_B satisfying $\Delta(Q_R) \leq A_c$, $\Delta(Q_B) \leq B_c$, $\nu(P_R) \leq x$, and $\nu(P_B) \leq y$. For every nonnegative integer i put $c_i(G) = max\{c_{A_c,B_c}(G) : A_c + B_c = i\}$, where both capacities are nonnegative integers.*

Let H be an induced subgraph of a graph F, and let A be an induced subgraph of H. Every connected component of A is an isolated vertex, a path on three vertices, a path on $2l$ vertices for an integer $l \geq 1$, or a cycle on j vertices for an integer $j \geq 3$. A whole component means a connected component with all its vertices and edges. Put $r = |V(A)|$, $m = \nu(A)$, let t be the number of even-order path components of A, and let p be its number of odd-cycle components. Put $\beta = m - t$, $h = |V(H)| - r$, and $N = |V(F)| - |V(H)|$. The component conditions imply $0 \leq p \leq \beta$. All these counts and ranks are nonnegative integers. Assume H contains a matching of size $m + h$ and

$N \geq \max(0, \beta + h - 2)$, $c_1(F) \geq N + h + \beta$, $c_2(F) \geq N + h + p$.

Let Rcal consist of all induced subgraphs R of H with r vertices, matching number m, exactly t even-order path components, exactly p odd-cycle components, and only the component types permitted for A. The number of three-vertex path components may vary. This family contains A. Set $P_0 = H$; for $i = 0, 1, 2$ choose any maximum matching M_i of P_i and define P_{i+1} by deleting the edges of M_i and retaining all vertices of H.

Every such sequence exists and has

$|M_0| = \nu(P_0) = m + h$, $|M_1| = \nu(P_1) = \beta + h$, $|M_2| = \nu(P_2) = p + h$.

For every R in Rcal, put $W_R = V(H)$ minus $V(R)$. The same three matchings each pair every vertex of W_R with a vertex of R, have no edge internal to W_R, and are maximum on their current internal row residual. In M_0 the endpoints in R of these crossing edges lie in distinct odd-order whole components and are exposed there: a vertex v of a component C is exposed when $\nu(C - v) = \nu(C)$. In a three-vertex path the exposed vertices are exactly its endpoints.

For every R, the entire first internal residual $R_1 = (V(R), E(R) minus M_0)$ consists of p three-vertex paths, beta-p single-edge components, and isolated vertices. The entire second internal residual $R_2 = (V(R), E(R) minus (M_0 union M_1))$ consists of p single-edge components and isolated vertices. All p edges of R_2 belong to M_2. Thus the prefix removes every original R-edge, preserves every original $H[W_R] - edge$, and uses three distinct edges from each W_R vertex into R.

If w in W_R has exactly one neighbor v that is exposed in an odd-order whole R-component, then wv belongs to M_0. If w has exactly three neighbors in R, the prefix removes all three incident edges into R. Consequently, for every independent set X contained in W_R whose vertices each have exactly three R-neighbors, $V(R)$ union X is independent in P_3 and W_R minus X is a vertex cover of P_3.

For $i = 1, 2$ let L_i be the union of M_0,..., M_{i-1}. For every nonnegative integer capacity split $A_c + B_c = i$ and every nonnegative integer budget split $x + y = N + \nu(P_i)$, there is a certificate of the entire F whose degree-bounded parts have union exactly L_i. Every edge meeting $V(F)$ minus $V(H)$, including an edge with both endpoints there, lies in a matching-bounded part. The restored totals are

$c_1(F) = N + \nu(P_1) = N + h + \beta$, $c_2(F) = N + \nu(P_2) = N + h + p$.

Proof. First, a finite graph with a vertex cover Z admits a two-part edge partition at every nonnegative integer budget split $x + y = |Z|$. Partition Z into Z_R, Z_B of sizes x, y. Assign to the first part every edge with an endpoint in Z_R, and assign all other edges to the second part. Each edge of the second part meets Z_B. A matching in either part has at most as many edges as that part's covering set has vertices. Taking $Z = V(G)$ and the two degree parts empty proves that all the certificate minima in the statement exist.

We record the relevant component calculations. An even path on 2l vertices has a unique perfect matching: the first endpoint forces its incident edge, and repeated removal of the first two vertices forces every later edge. Deleting this matching leaves l-1 disjoint edges and isolated vertices. An even cycle on 2l vertices has matching number l; every perfect matching alternates, and its complement is a matching of size l. An odd cycle on $2l + 1$ vertices has matching number l. Every maximum matching exposes one vertex v and is the unique perfect matching of the even path obtained by deleting v. Its edge complement consists of a three-vertex path centered at v and l-1 disjoint edges. The alternating edges give these matching sizes, and half the vertex order is an upper bound.

A three-vertex path has matching number one; deleting either maximum matching leaves one edge and an isolated vertex. Deleting an endpoint from that path preserves its matching number, whereas deleting its center reduces it to zero. An isolated vertex contributes zero. Matching numbers add over connected components: any matching restricts to a matching in each component, and component maximum matchings can be united. Thus $\beta = m - t \geq p \geq 0$. For every permitted R, deleting a matching that is maximum inside every component leaves exactly p three-vertex paths, beta-p disjoint edges, and isolated vertices. In particular the first internal residual has matching number β and a vertex cover of size β, consisting of each three-vertex path's center and one endpoint of each isolated edge.

Fix an arbitrary sequential choice as in the statement. Maxima exist because each finite graph has finitely many matchings, including the empty one. For any R in Rcal, every H-matching has at most m edges internal to R and at most h other edges, since each of the latter meets W_R. The assumed matching attains $m + h$, so $\nu(H) = m + h$. A maximum M_0 therefore attains both bounds for every R. Its h noninternal edges cannot include an edge inside W_R: such an edge, together with the other h-1 disjoint edges meeting W_R, would require at least $h + 1$ vertices of W_R. Thus all h edges cross, saturating W_R, and the internal restriction is maximum in every R-component.

Every permitted component has matching number half its order rounded down. Its internal maximum matching covers every vertex if the order is even and all but one if the order is odd. The crossing edges must therefore use exposed vertices of distinct odd components. This proves the first matching assertions simultaneously for every R, as well as the stated description of R_1.

Here is an allocation construction used to force the next two ranks. For i in {1,2}, suppose i edge-disjoint matchings have already been removed from H, let P be the remaining spanning graph, and write $w = \nu(P)$. Order $S = V(F)$ minus $V(H)$ as s_1,..., s_N. Assign each original edge meeting S to its earliest endpoint in that order. The resulting sets B_1,..., B_N are possibly empty stars, each of matching number at most one, and partition all edges outside H, including edges with both endpoints in S. At a capacity split $A_c + B_c = i$, place A_c of the matching sets in Q_R and the others in Q_B. A vertex has at most one incident edge from each matching, so the required degree bounds hold.

If $N \geq \max(0, w - 1)$, this yields every certificate at total $N + w$. When $w = 0$, P has no edges; give x of the indexed stars to P_R and the rest to P_B for $x + y = N$. When $w \geq 1$, $N + w \geq 2w - 1$, so at least one of the integer budgets x, y is at least w. If $x \geq w$, assign all of P and exactly x-w indexed stars to P_R and the remaining y stars to P_B. The number x-w lies between zero and N. If $x < w$, use the same instruction with the colors exchanged, since $y \geq w$. Restricting any matching to these edge pieces gives the prescribed upper bound: at most w edges come from P and at most one from each assigned star. Shared vertices do not invalidate this bound. Thus $c_i(F) \leq N + w$ under the indicated inequality.

For the already chosen M_0, any matching in P_1 has at most β edges internal to A and at most h other edges. If $w = \nu(P_1) \leq \beta + h - 1$, then $N \geq \max(0, \beta + h - 2) \geq \max(0, w - 1)$. The construction just given implies

$c_1(F) \leq N + w \leq N + \beta + h - 1$,

contrary to the assumed lower bound. Hence $\nu(P_1) = \beta + h$. For every R, the identical upper bound follows from its R_1. Equality forces every maximum M_1 to saturate W_R into R and to have an internal maximum matching of R_1, by the same endpoint count used for M_0. Such a matching removes all isolated edges of

R_1 and one edge of each three-vertex path. Therefore R_2 consists exactly of p edges and isolated vertices, simultaneously for every R.

Now $\nu(P_2) \le p + h$. If $w = \nu(P_2) \le p + h - 1$, then $\beta \ge p$ and the assumed omission bound imply $N \ge \max(0, w - 1)$. Applying the allocation construction with the two previously removed matchings gives

$c_2(F) \le N + w \le N + p + h - 1$,

again contradicting its lower bound. Thus $\nu(P_2) = p + h$. Equality in the *row/exterior* bound forces every maximum M_2 to saturate W_R into R and to contain every edge of R_2. If either candidate upper rank is zero, a strict smaller matching number is impossible, and the equality follows without the contradiction argument.

For each R, all three matchings cross at every W_R vertex and have no internal W_R edge. They remove every R-edge and preserve every $H[W_R] - edge$. Their edges are distinct because previously selected edges are deleted. A uniquely eligible exposed neighbor must be the endpoint selected by M_0. Three total R-neighbors must be exhausted by the three crossing edges. If X has the stated properties, all R-internal edges and all edges from X into R have been removed, and X had no internal edges. These exhaust the edges within $V(R)$ union X, proving the independent-set and complementary-cover assertions.

It remains to give the claimed complete certificates at the exact ranks, including the boundary $N = \beta + h - 2$. Fix $R = A$. For $i = 1$ choose a cover D_1 of R_1 consisting of each three-vertex path's center and one endpoint of every isolated edge; it has size β. For $i = 2$ choose one endpoint of each of the p edges of R_2, giving a cover D_2 of size p. The sets

$Z_i = S$ union W_A union D_i

are vertex covers of the graph with edge set $E(F)$ minus L_i. Indeed an edge outside H meets S, a retained paid edge outside A meets W_A, and a retained paid edge inside A meets D_i. Their three constituents are disjoint. Hence $|Z_1| = N + h + \beta$ and $|Z_2| = N + h + p$.

For any capacities summing to i, group the i fixed matchings into the degree parts as before. At any budgets $x + y = |Z_i|$, partition Z_i into sets of sizes x, y and apply the first paragraph's cover allocation to the remaining edges. This produces the four edge parts on all of F, with the degree-part union exactly L_i and all omitted-incident edges paid. In view of the forced ranks, the totals are $N + \nu(P_i)$. They give $c_1(F) \le N + h + \beta$ and $c_2(F) \le N + h + p$; the assumed reverse inequalities prove equality. This completes the construction for every sequential choice and every compatible row. □

Remark. A maximum matching attaining the row rank plus the exterior order must cross at every exterior vertex. Exact source costs force the same equality after one and two edge deletions. The remaining row components have matching-sized vertex covers, which allocate every restored budget exactly.

Depends on. Nothing earlier; this item is self-contained.

Lemma 21 (Matching-family edge and cover accounting). *All graphs are finite and simple. Write $e(H)$ for the number of edges, $\Delta(H)$ for maximum degree, and $\tau(H)$ for minimum vertex-cover cardinality, where a vertex cover meets every edge. An edgeless graph has maximum degree zero. Let K_2 denote a single edge. A positive star forest is a vertex-disjoint union of stars $K_{1,m}$ with positive integer edge sizes m. Copies are ordinary subgraphs and different target components have disjoint vertex sets.*

Let $\Gamma = \{(p_x K2, J_x) : x \in calX\}$ be a finite nonempty family, where calX is a finite index set, p_x and t_x are positive integers, and J_x has the complete nonincreasing positive size list m_1^x,..., $m_{t_x}^x$. Put

$N = \max_{x \in calX}(p_x + t_x - 1)$.

For every integer $1 \le i \le N$ define

$\eta_i = max\{m_j^x : x \in calX, 1 \le a \le p_x, 1 \le j \le t_x, a + j = i + 1\}$,

$E = \sum_{i=1}^{N} \eta_i$, *and* $T(b) = \sum_{i=1}^{b} \eta_i$ *for* $0 \le b \le N$.

Each maximum is over a nonempty finite set. The sequence η_1,..., η_N is positive and nonincreasing.

Suppose H arrows every member of Γ separately: every red-blue edge coloring of H has either a red matching with p_x edges or a blue copy of the entire J_x, for each x. Then $\tau(H) \ge N$. For every set Y of b vertices with $0 \le b \le N$,

$e(H-Y) \geq E - T(b) + \tau(H) - N$.

In particular $e(H) \geq E + \tau(H) - N$. For every Y with $b < N$, one also has $\Delta(H-Y) \geq \eta_{b+1}$. Here H-Y is induced vertex deletion; edges are not deleted a second time.

Suppose moreover that r is a nonnegative integer, $e(H) \leq E + r$, and $\tau(H) \geq N + r$. Then $e(H) = E + r$ and $\tau(H) = N + r$. Every sequence of N successive maximum-degree vertex deletions exists, removes exactly η_i edges at its i-th step, and leaves exactly r pairwise vertex-disjoint edges and any number of isolated vertices. For every b-set Y with $b \leq N$, its removed edge count e(H)-e(H-Y) is at most $T(b)$. If that count equals $T(b)$, then

$e(H-Y) = E - T(b) + r$, *and* $\tau(H-Y) = N - b + r$.

Under these same equality hypotheses, put $D = \eta_1$. If $D \geq 2$ and R is the number of entries equal to D, every inclusionwise maximal independent set in the subgraph induced by the vertices of H-degree D has exactly R vertices, including when $R = N$. An independent set contains no edge between its vertices.

Also under the equality hypotheses, let $k \geq 2$ be an integer and let $g = |\{i : \eta_i \geq k\}|$. Let U be any positive star forest with exactly N components whose components of size at least k have exactly the size list η_1,..., η_g, and whose other component sizes are below k. For every copy of U in H, let A be the set of centers of its g components of size at least k. These centers are independent, each has ambient H-degree equal to its copied star size, and its entire neighborhood consists of its copied leaves. Their neighborhoods are pairwise disjoint. Deleting A in nonincreasing copied-size order is a maximum-degree deletion prefix, and

$e(H-A) = \sum_{i=g+1}^{N} \eta_i + r$, $\tau(H-A) = N - g + r$, $\Delta(H-A) \leq k-1$.

Every vertex outside that full copy of U has H-degree at most k-1. If $g = N$, the graph H-A consists of r disjoint edges and isolates. Empty sets A and empty sums are allowed. No assertion that its remaining edges or paths are whole components of the original H is made.

All counts, sizes, ranks, degrees, indices and parameters in this statement are integers in their specified domains.

Proof. A member attaining the maximum N has valid pairs a, j with $a + j = i + 1$ at every integer $1 \leq i \leq N$. Thus each η_i is positive and attained. To compare consecutive entries, take a candidate at $i \geq 2$. At least one of its indices a, j exceeds one. If $a > 1$, decrease a; the same size occurs at i-1. If $a = 1$, then $j > 1$, and decreasing j supplies a size at least as large, by the ordered list. Taking maxima proves monotonicity.

We first prove the degree bound for an arbitrary b-set Y, where $b < N$. Choose a member x and valid indices a, j with $a + j = b + 2$ and $m_j^x = \eta_{b+1}$. Partition Y as Y_R disjoint union Y_B with $|Y_R| = a - 1$ and $|Y_B| = j - 1$. Suppose $\Delta(H - Y) < m_j^x$. Color every edge meeting Y_R red, and all other edges blue. The red graph has vertex cover Y_R, so any red matching has at most $a - 1 < p_x$ edges.

Every blue connected subgraph having an edge and meeting Y must contain a vertex of Y_B: there is no blue edge incident with Y_R. Consequently at most j-1 components of any blue J_x copy could meet Y, because its components are disjoint. Among the first j components of J_x, whose sizes are all at least m_j^x, at least one would therefore lie wholly in H-Y. This would give a vertex of H-Y-degree at least m_j^x, contrary to the supposed degree bound. The coloring avoids both targets of member x, contradicting its arrowing. Hence $\Delta(H - Y) \geq \eta_{b+1} > 0$. This argument includes $b = 0$, $a = 1$, $j = 1$ and unit star sizes, and uses the actual chosen Y.

If H had a vertex cover with $b < N$ vertices, deleting that cover would give an edgeless graph, contradicting this degree bound. Thus $\tau(H) \geq N$.

Fix now any Y with $|Y| = b \leq N$. From H-Y delete N-b vertices successively, each of maximum current degree. Whenever $c < N$ vertices in total have been deleted from H, the degree bound on that actual c-set gives current maximum degree at least $\eta_{c+1} > 0$. Each required step therefore exists. The edge sets removed at different steps are disjoint: an edge with two eventually deleted endpoints is removed at the first of those endpoints. Thus these additional steps remove at least E-T(b) edges from H-Y.

Let Z be the induced terminal graph after all N vertices have been removed. The N deleted vertices together with a minimum vertex cover of Z form a cover of H, so $\tau(Z) \geq \tau(H) - N$. For every finite graph, choosing one endpoint of each edge supplies a cover; hence $e(Z) \geq \tau(Z)$. Adding its edges to the disjoint removed edge sets proves

$e(H-Y) \geq E - T(b) + e(Z) \geq E - T(b) + \tau(H) - N$. (1)

When $b = N$ there are no additional steps and the same proof gives (1). Taking $b = 0$ proves the first edge inequality.

Assume next $e(H) \le E + r$ and $\tau(H) \ge N + r$. Inequality (1) at $b = 0$ gives

$e(H) \ge E + \tau(H) - N \ge E + r$.

It follows that $e(H) = E + r$ and $\tau(H) = N + r$. Take any N-step greedy sequence. Its i-th removed degree is at least η_i, and its terminal graph Z has $e(Z) \ge \tau(Z) \ge r$. Since the degrees plus $e(Z)$ sum exactly to $E + r$, all nonnegative excesses are zero. Every removed degree is η_i and $e(Z) = \tau(Z) = r$. If two edges of Z shared a vertex, that common vertex, together with one endpoint from each of the other r-2 edges, would cover Z with at most r-1 vertices. This contradicts $\tau(Z) = r$. Thus all its edges are disjoint; $r = 0$ and $r = 1$ satisfy the conclusion directly.

Subtracting (1) from $e(H) = E + r$ gives the bound $e(H) - e(H - Y) \le T(b)$ for every b-set. If equality holds, then $e(H - Y) = E - T(b) + r$ and $\tau(H - Y) \ge \tau(H) - b = N - b + r$. Start the same N-b greedy deletions from H-Y. Their removed degrees total at least E-T(b); their terminal graph has at least r edges, because altogether N vertices of H have been removed. The edge equality forces exactly r terminal edges. The N-b additionally deleted vertices and one endpoint per terminal edge form a cover of H-Y of size at most $N - b + r$. This proves its asserted cover equality.

Suppose $D \ge 2$ and let I be inclusionwise maximal independent among the degree-D vertices. First $\Delta(H) = \eta_1 = D$ by the proved exactness of every greedy sequence. While vertices of I are deleted, each other I-vertex retains degree D by independence. Every initial segment of this deletion is therefore greedy.

If $|I| < R$, every degree-D vertex outside I has an I-neighbor by maximality, whereas vertices initially of lower degree cannot gain edges. Thus $\Delta(H - I) \le D - 1$. But $|I| < R \le N$ and the degree bound gives $\Delta(H - I) \ge \eta_{|I|+1} = D$, a contradiction.

If $R < N$ and $|I| > R$, deleting $R + 1$ vertices of I is a greedy prefix with degree D at every step. This contradicts $\eta_{R+1} < D$. If $R = N$ and $|I| > N$, delete N vertices of I. An unchosen vertex of I retains degree D in the terminal graph, whereas that graph consists of disjoint edges and isolates and has maximum degree at most one. This again contradicts $D \ge 2$. Hence $|I| = R$.

Finally fix a copy of U as in the statement and order its high-star centers $z_1,..., z_g$ by the listed copied sizes $\eta_1,..., \eta_g$. We prove sequentially that they form an exact greedy prefix and that each has no neighbors beyond its own copied leaves. Before z_i is removed, earlier centers belong to different copied components, so all η_i edges of its copied star are still present. Its current degree is at least η_i. For $i = 1$ the current maximum degree equals η_1. At later steps, the already established greedy prefix can be continued for N steps, since the degree-forcing assertion supplies every remaining step. Every such greedy sequence has the exact proved profile, so its next maximum degree is η_i. Thus z_i has current degree exactly η_i and no current neighbors beyond its copied leaves.

Each earlier center z_h had only its own copied leaves as neighbors when removed. Since z_i is in a different copied component, it was not one of those leaves. Hence z_i has no earlier-center neighbors either. Its entire ambient neighborhood is therefore its own copied leaf set. This completes the induction, proves independence of the centers, and gives disjoint neighborhoods because the copied components are disjoint.

Deleting A consequently removes exactly $T(g)$ edges. The exact-cost conclusion already proved gives $e(H - A) = E - T(g) + r$ and $\tau(H - A) = N - g + r$. If $g < N$, the next exact greedy degree is $\eta_{g+1} < k$, so $\Delta(H - A) \le k - 1$. If $g = N$, the residual consists of disjoint edges and isolates, which also has maximum degree at most k-1 because $k \ge 2$. A vertex outside the full U copy has no neighbor in A, by the ambient-neighborhood conclusion. Its H-degree equals its degree in H-A and is at most k-1. If $g = 0$ there are no centers to delete, and $\Delta(H) = \eta_1 < k$ gives the same conclusions directly. This proves every assertion for the arbitrarily chosen copy. □

Remark. Every actual deleted set leaves a forced next star degree. Summing those forced degrees and the surviving cover requirement gives the edge bound; equality fixes both the greedy history and any copied centers realizing its large entries.

Depends on. Nothing earlier; this item is self-contained.

Lemma 22 (Replacing complete five-vertex components by stars). *All graphs are finite and simple. Write K_5 for the complete graph on five vertices, $K_{1,h}$ for a star with h edges, and $K_2 = K_{1,1}$. A positive-edge star forest is a vertex-disjoint union of stars $K_{1,h}$ with positive integer h; the empty union is permitted. Copies are ordinary subgraphs and distinct target components must have disjoint vertex sets. Put*

$T = K_{1,5}$ *disjoint union* $K_{1,4}$ *disjoint union* K_2.

For every red-blue edge coloring of T there is an explicit red-blue edge coloring of K_5 such that, for each color, every positive-edge star forest occurring in that color in K_5 occurs in the same color in T. The single constructed coloring of K_5 works simultaneously for every such forest in both colors.

More generally, let $k \geq 0$ be an integer, let G be a finite simple graph with k selected distinct whole connected components isomorphic to K_5, and form G_{star} by replacing each selected component by a separate copy of T, retaining every other component. For every coloring of G_{star} the local prescriptions give one coloring of G agreeing on every unchanged edge such that every monochromatic positive-edge star forest in G occurs in the same color in G_{star}. Thus if G arrows every pair in any prescribed finite family of ordered pairs of nonempty positive-edge star forests, then G_{star} arrows every pair in that same family. Here a graph arrows an ordered pair when every red-blue edge coloring contains its entire first target in red or its entire second target in blue.

Writing $e(H)$ for edge count, $\nu(H)$ for maximum matching cardinality and $\Phi(H) = e(H) - \nu(H)$, the exact changes are

$e(G_{star}) = e(G)$, $\nu(G_{star}) = \nu(G) + k$, $\Phi(G_{star}) = \Phi(G) - k$.

All graph orders, degrees, edge counts, matching sizes, component counts and star sizes are integers in their stated nonnegative or positive domains. Unit stars, repeated sizes, isolated vertices in the unchanged graph, $k = 0$ and an empty family are allowed.

Proof. Label the vertices of K_5 by 1, 2, 3, 4, 5 and index the three components of T by 1, 2, 3 in order of edge sizes 5, 4, 1. Given a coloring of T, let r_i, b_i be its red and blue edge counts in component i. Thus $r_i + b_i$ is respectively 5, 4, 1. For either color, its three capacities are those edge counts: a star of size at most a capacity embeds in that colored component, and forests assigned to different components remain vertex-disjoint.

A positive-edge star forest on at most five vertices has at most two components. A single component has at most four edges. If there are two components of sizes $h_1 \geq h_2 \geq 1$, then $h_1 + h_2 + 2 \leq 5$, so their size list is (1,1) or (2,1). This exhausts the possibilities, together with the empty forest. Consequently a capacity vector with an entry at least four and a different positive entry contains every such forest. If an entry is at least two and a different entry is positive, it contains every such forest whose largest star size is at most two. If all entries are at most one, the colored graph is a matching of size equal to its number of positive entries.

Use the first applicable case below, choosing red first if both colors qualify.

First, suppose a color has all three capacities at most one. Let j in {0,1,2,3} be its number of positive entries, and put $l = \min(j, 2)$. Give that color exactly the first l edges of the list 12, 34 in K_5, and give every other edge the opposite color. Its graph is a matching of size $l \leq j$, so all its star forests occur in its part of T. The opposite-color capacities in components 1 and 2 are at least four and three. They therefore contain every positive-edge star forest on at most five vertices, including every forest appearing in the opposite color of this K_5. This includes $j = 0$.

Second, suppose the first case fails and a color has exactly one positive capacity. That capacity is at least two and occurs in component 1 or 2. Give that color precisely the triangle edges 12, 23, 13 in K_5. A star forest in this triangle has at most one positive component, of size at most two, so it embeds in the unique positive component of that color in T. The opposite color contains the entire other large star, of size at least four, and the whole unit star. These are distinct components and together contain every positive-edge star forest on at most five vertices. Hence the opposite-color forests of K_5 also transfer.

Third, suppose neither case applies. Each color has an entry at least two and at least two positive entries. Color the cycle 12, 23, 34, 45, 15 red and its complement 13, 35, 25, 24, 14 blue. These lists partition all ten K_5 edges, and each color is a five-cycle. Every monochromatic star forest in K_5 therefore has largest star size at most two. The capacity condition and the classification above embed each such forest into the same color of T. This completes the exhaustive local prescription.

For the global construction, fix a labeling of each selected K_5 and the corresponding three replacement stars. Given a coloring of G_{star}, apply the local prescription separately on each selected component and keep every

other edge color. Fix either color and any embedding of an entire positive-edge star forest in the resulting colored G. Each positive target star is connected, so its image lies wholly within one connected component of G. For each selected K_5, group all target components whose images lie in it. Their union is a star forest on at most five vertices, so the local prescription transfers that entire group into the corresponding T in its original color. Keep all remaining target component embeddings unchanged. Different replacement copies and the unchanged graph have disjoint vertex sets; therefore the resulting images are still pairwise disjoint and form the entire original target forest. This works for every forest and both colors for the single constructed coloring, including repeated sizes and unit components.

If G arrows every ordered pair in the stated family, apply its arrowing property to this pulled-back coloring, separately for each pair. In each case its resulting full monochromatic target transfers to G_{star}. Since the original coloring of G_{star} was arbitrary, G_{star} arrows the same family.

Finally $e(K_5) = 10$ and $\nu(K_5) = 2$: the edges 12, 34 attain two, while three matching edges would need six vertices. The three positive stars of T have ten edges in total and matching number three. Edge counts add over disjoint components. Matching numbers also add: restricting any matching to the components proves the upper bound, and uniting component maximum matchings attains it. Each replacement thus keeps e and raises ν by one. Summing over the k selected components proves all three displayed identities. For $k = 0$ the construction is the identity and the same formulas hold. □

Remark. Five vertices allow only a single star, two unit stars, or a two-edge star and a unit. Three cases for the replacement's colored capacities give a matching, a triangle, or complementary five-cycles that preserve all these possibilities simultaneously.

Depends on. Nothing earlier; this item is self-contained.

Lemma 23 (Finite integer representation of certificate costs). *Let $H = (V, E)$ be a finite simple graph, possibly empty. Write $e(H) = |E|$, $\Delta(H)$ for its maximum degree, $\nu(H)$ for its maximum matching cardinality, $\alpha(H)$ for its maximum independent-set cardinality, and $\tau(H)$ for its minimum vertex-cover cardinality. A matching is a set of pairwise vertex-disjoint edges; a vertex cover meets every edge. The maximum degree and matching number of an edge set refer to the spanning graph on V with that edge set; both are zero for an empty edge set. All capacities A, B, matching budgets x, y, totals c, and indices θ below are nonnegative integers.*

Let $C_{A,B}(H; x, y)$ be the set of ordered partitions (Q_R, Q_B, P_R, P_B) of E satisfying $\Delta(Q_R) \le A$, $\Delta(Q_B) \le B$, $\nu(P_R) \le x$, $\nu(P_B) \le y$. The parts may be empty. Define `c_{A,B}(H)=min{c>=0:` *$C_{A,B}(H; x, c - x)$ is nonempty for every integer* `0<=x<=c}`*. For every $x \ge 0$, let $g_{A,B,H}(x) = min\{\nu(P_B) : (Q_R, Q_B, P_R, P_B) partitions E, \Delta(Q_R) \le A, \Delta(Q_B) \le B, \nu(P_R) \le x\}$.*

These minima exist. The exact finite representation is $c_{A,B}(H) = \max(\{0\} union \{x + g_{A,B,H}(x) : 0 \le x < \nu(H), g_{A,B,H}(x) > 0\})$. Every budget pair with total c is feasible if and only if $c \ge c_{A,B}(H)$. Equivalently, the minimum defining $c_{A,B}(H)$ may be restricted to $0 \le c \le \tau(H)$.

Each feasibility test is a binary integer program with one variable $z_{e,L}$ in {0,1} for each e in E and L in $\{Q_R, Q_B, P_R, P_B\}$: $\sum_L z_{e,L} = 1$ for every edge e; $\sum_{e incident with v} z_{e,Q_R} \le A$ and $\sum_{e incident with v} z_{e,Q_B} \le B$ for every vertex v; $\sum_{e \in M} z_{e,P_R} \le x$ and $\sum_{e \in M} z_{e,P_B} \le y$ for every matching M of H. Its feasible family is precisely $C_{A,B}(H; x, y)$, under the edge-indicator correspondence. The program for $g_{A,B,H}(x)$ uses the same binary variables, replaces the last blue bound by $\sum_{e \in M} z_{e,P_B} \le b$ for every matching M, and minimizes the integer b in [0,nu(H)]. Replacing binary variables by real variables in [0,1] defines a separate fractional relaxation; no equality of its feasible set or optimum with these integer problems is asserted.

Define $c_\theta^{\cdot}(H) = max\{c_{A,B}(H) : A + B = \theta\}$, $c_0(H) = c_{0,0}(H)$, $\Sigma(H) = \sum_{\theta \ge 0} c_\theta^{\cdot}(H)$, $ex(H) = c_0(H) - \nu(H)$, $ps(H) = e(H) - \nu(H) - \sum_{\theta \ge 1} c_\theta^{\cdot}(H)$, and $gap(H) = \Sigma(H) - e(H)$. The notation $c_\theta(H)$ is an alias for $c_\theta^{\cdot}(H)$. These quantities are finite integers: $c_\theta(H) = 0$ whenever $\theta \ge max\{1, 2\Delta(H) - 1\}$. Moreover $c_{A,B}(H) = c_{B,A}(H)$; $c_{A,B}(H) \le c_{A',B'}(H)$ if $A' \le A$ and $B' \le B$; $0 \le c_{\theta+1}(H) \le c_\theta(H) \le c_0(H)$; $\nu(H) \le c_0(H) \le \tau(H) = |V| - \alpha(H)$; $0 \le ex(H) \le \tau(H) - \nu(H)$; and $gap(H) = ex(H) - ps(H)$. The quantities ps and gap are defined as signed integers here. Call H LP-good when $gap(H) \le 0$, equivalently $ps(H) \ge ex(H)$.

If $c_0(H) = 0$, define $tr(H) = 0$. Otherwise let $Z_H = \{x \in \{0, \dots, c_0(H) - 1\} : C_{0,0}(H; x, c_0(H) - 1 - x) isempty\}$ and define `tr(H)=max{v-u+1:` *$0 \le u \le v < c_0(H)$, every integer in [u, v] belongs to* `Z_H}`*, where u, v are integers. Then Z_H is nonempty and $1 \le tr(H) \le c_0(H)$. Thus this last maximum has an explicitly nonempty finite feasible set.*

Proof. Fix H, A, B. Putting every edge in P_B shows that the feasible family for $g(x)$ is nonempty for each nonnegative integer x. It is finite because E is finite, so its minimum exists and lies in [0,nu(H)]. Putting every edge in P_R shows $g(x) = 0$ for $x \geq \nu(H)$. For any $x, y \geq 0$, definition and attainment of the minimum give $C_{A,B}(H; x, y)$ nonempty if and only if $y \geq g(x)$.

Put $T = \max(\{0\}union\{x + g(x) : 0 \leq x < \nu(H), g(x) > 0\})$. If $c \geq T$ and $x + y = c$, then $g(x) = 0$ implies feasibility directly. If $g(x) > 0$, then $x < \nu(H)$ and $x + g(x) \leq T \leq c$, giving $y \geq g(x)$, again feasibility. In particular a universally feasible total exists. Universal feasibility is upward closed among integer totals. Suppose all pairs at total c work, and take $x + y = c + 1$. If $x \geq 1$, retain a certificate for (x-1,y); if $x = 0$, retain one for (0,c). Increasing a matching budget preserves its bound. Iteration proves the claim for every larger total. If $T > 0$, choose an integer x attaining T with $g(x) > 0$. At total T-1, the pair (x, g(x)-1) is nonnegative and infeasible. Upward closure shows no smaller total can be universally feasible. If $T = 0$, total zero is feasible and is already the smallest allowed total. This proves the representation and exact threshold characterization.

Let W be a minimum vertex cover and let $x + y = |W|$. Partition W into sets W_R, W_B of cardinalities x, y. Put every edge with an endpoint in W_R in P_R, and put all remaining edges in P_B; take $Q_R = Q_B$ empty. Every P_B edge meets W_B because W covers E. A matching in P_R uses distinct W_R vertices, so has size at most x; a matching in P_B has size at most y. Thus every $total - |W|$ allocation is feasible and $c_{A,B}(H) \leq \tau(H)$. This proves the claimed finite range for its minimum.

Binary variables satisfying the edge equations specify exactly one of the four parts for each edge. The vertex inequalities express precisely the two degree bounds. For any matching M of H, $\sum_{e \in M} z_{e,P_R} = |M$ intersect $P_R|$. If $\nu(P_R) \leq x$, all these sums are at most x. Conversely, every matching of the graph with edge set P_R is a matching of H, so the inequalities imply $\nu(P_R) \leq x$. The same two implications hold for P_B. This proves the feasibility correspondence. With the integer variable b, minimizing the upper bound on all blue matching cardinalities gives exactly $\nu(P_B)$ for each partition. Minimizing across the finite partition family proves the objective assertion for g.

Exchanging the red and blue parts proves $c_{A,B} = c_{B,A}$. A certificate remains valid when A or B is increased, which proves capacity monotonicity. Given $A + B = \theta + 1$, reduce a positive coordinate by one; the resulting capacities have total θ. Monotonicity gives $c_{A,B} \leq c_\theta$, and maximizing over the finitely many pairs gives $c_{\theta+1} \leq c_\theta$. The row-zero alias and nonnegativity yield $0 \leq c_\theta \leq c_0$.

If $\Delta(H) = 0$, every edge set is empty and every cost is zero. If $d = \Delta(H) \geq 1$ and $\theta \geq 2d - 1$, at least one capacity in every pair $A + B = \theta$ is at least d: otherwise both would be at most d-1. Put all edges in that bounded-degree part and the other three parts empty. This certifies total zero for each capacity pair and proves $c_\theta = 0$. Therefore the sums defining Σ, ps, and gap have only finitely many nonzero terms.

When $A = B = 0$ and $x = 0$, a certificate must have $Q_R = Q_B = P_R$ empty, since a nonempty simple edge set has positive maximum degree and a matching of cardinality one. Thus the allocation (0,c) is infeasible if $c < \nu(H)$. This proves $c_0 \geq \nu(H)$; the cover argument proves the upper bound. The complement of a vertex cover is independent and the complement of an independent set is a vertex cover, giving $\tau(H) = |V| - \alpha(H)$. The ex inequalities follow by subtraction. Expanding the definitions gives $ex - ps = c_0 - \nu - e + \nu + \sum_{\theta \geq 1} c_\theta = \Sigma - e = gap$, which also proves the equivalent definitions of LP-goodness.

Finally, if $c_0 > 0$, the total $c_0 - 1$ is not universally feasible by minimality of c_0. Hence Z_H is nonempty. Any member of Z_H supplies a singleton interval in the maximization defining tr, and every interval in its feasible family has cardinality at most c_0. Finiteness gives the asserted existence and bounds. The case $c_0 = 0$ uses its explicit definition and requires no negative total. □

Remark. A fixed red budget leaves a minimum necessary blue budget. The universal total is the largest obstruction height. Indicator variables express actual edge partitions exactly, so integer allocation semantics are preserved.

Depends on. Nothing earlier; this item is self-contained.

Lemma 24 (Marked domination and the exact loss of two trials). *For integers $m \geq 0$ and $j \geq 1$ and a real number p in [0,1], define $q_{m,j}(p) = p + (1-p)Pr(Bin(m,p) \leq j-1)$, where $Bin(m,p)$ is the sum of m independent Bernoulli variables, each equal to one with probability p. Probabilities of values outside {0,...,m} are zero. Let G be a finite simple graph on $n \geq 1$ vertices, let v be a specified vertex, and assume every vertex other than v has degree at least m. There is a set Z of vertices containing v such that every vertex outside Z has at least j neighbors in Z and $|Z| \leq 1 + (n-1)q_{m,j}(p)$. Moreover, for all integers $m \geq 2$ and $j \geq 1$ and every p in [0,1], $q_{m-2,j}(p) - q_{m,j}(p) = (1-p)[p(2-p)Pr(Bin(m-2,p) = j-1) + p^2\ Pr(Bin(m-2,p) = j-2)] \geq 0$. In particular, if H is a connected d-regular finite simple graph, $d \geq 2$ is an even integer, and deletion of every vertex of H leaves a perfect matching, then for every vertex v, every perfect matching M of H-v, every integer $1 \leq j \leq d-1$, every integer $0 \leq m \leq d-1$, and every p in [0,1], the spanning graph $G = H - M$ has such a set Z with $|Z| \leq 1 + (|V(H)|-1)q_{m,j}(p)$.*

Proof. Choose a random subset R of $V(G)$ by including each vertex independently with probability p. Let B consist of all vertices u other than v which do not belong to R and have at most j-1 neighbors in R. Define $Z = \{v\}$ union R union B. This set contains v. If u is outside Z, then u is not v, u is outside R, and u is not in B. The definition of B therefore gives at least j neighbors of u in R, and all those neighbors belong to Z. This verifies the required domination property for every outcome.

For u other than v put $r = deg_G(u)$, so $r \geq m$. Because G is simple, inclusion of u in R is independent of inclusion of its neighbors. Thus $Pr(u \in Z) = p + (1-p)Pr(Bin(r,p) \leq j-1)$. Couple the r trials with their first m trials. Their full sum is at least the sum of the first m, which gives $Pr(Bin(r,p) \leq j-1) \leq Pr(Bin(m,p) \leq j-1)$. Summing the inclusion probabilities yields $E|Z| \leq 1 + (n-1)q_{m,j}(p)$. The distribution has finitely many outcomes, so at least one outcome of positive probability has cardinality at most its expectation. That outcome is the required Z. This proof includes $p = 0$ and $p = 1$ and $n = 1$.

For the identity fix $m \geq 2$. Let X be the sum of m-2 independent Bernoulli trials with success probability p and let Y be the sum of two further such trials, independent of X. Put $F(t) = Pr(X \leq t)$ for every integer t. Conditioning on Y gives $Pr(X+Y \leq j-1) = (1-p)^2\ F(j-1) + 2p(1-p)F(j-2) + p^2\ F(j-3)$. Subtract this equality from $F(j-1)$. Since $1-(1-p)^2 = 2p(1-p) + p^2$, the difference equals $2p(1-p)[F(j-1) - F(j-2)] + p^2[F(j-1) - F(j-3)] = p(2-p)Pr(X = j-1) + p^2\ Pr(X = j-2)$. Multiplying by 1-p proves the displayed identity for q. Every factor on the right is nonnegative. The convention on probabilities covers $j = 1$, $j > m$, and both endpoints of p.

For the last assertion a perfect matching of H-v contains one edge incident with each vertex other than v and none incident with v. Consequently H-M has degree d-1 at each vertex other than v and degree d at v. The first assertion applies because $m \leq d-1$. Its domination property is measured in H-M, with the same vertex set as H, as required. □

Remark. The only possible change after removing two trials occurs at the two boundary values j-1 and j-2. Their masses measure the exact cost of lowering the available degree by two.

Depends on. Nothing earlier; this item is self-contained.

Lemma 25 (Canonical reflection and the exact deficient paths of a quartic frame). *All graphs are finite and simple. Write $\alpha(G)$ and $\nu(G)$ for the independence and matching numbers of a graph G. The symbols P_j and C_j denote a path and a cycle on j vertices. Deleting vertices means taking the induced graph on the remaining vertices.*

Let F be a four-regular graph, let $a = \alpha(F)$, and let m, b be integers with $1 \leq m$ and $0 \leq b \leq m$. Put $q = m - b$. A frame is a pair (S, U), where S is a maximum independent set of F and U is an independent m-element set of vertices of degree three in F-S. Its terminal is $R = F - (S union U)$. Assume the following condition for every frame: $\alpha(R) = a$, and the connected components of R consist of exactly q odd cycles, each C_3, C_5, or C_7; exactly b even-order paths, each P_2, P_4, or P_6; and any number of P_3, C_4, and isolated vertices.

Fix any frame. Put $W = S$ union U, $r = \nu(R)$, and let D be any independent set formed by the following separate choices on all components of R: a maximum independent set on each odd cycle; the center on each P_3; either opposite pair on each C_4; no isolated vertices; and, on each consecutively numbered P_{2s}, where s is an integer in {1,2,3}, either {2,4,...,2s-2} or {3,5,...,2s-1}. Both choices are empty for P_2. Then the following conclusions hold for every such D.

The induced graph $F[W]$ is a matching L of m edges together with $a-m$ isolated vertices. Write P for the endpoints of L and $Z = W$ minus P. One has $|D| = r - b$. There is a frame (J, U') with J union $U' = V(R)$ minus D, and hence

with terminal $R_D = F[W \text{ union } D]$. Every vertex of D has degree two in R_D. Each of its q odd-cycle components and b even-path components contains exactly one edge of L, and these m components contain all edges of L.

Let B_D be the bipartite graph with parts D, Z and all original F-edges between those parts. The path components of B_D having both endpoints in D are exactly the graphs obtained from the q odd-cycle components of R_D by deleting their two P-vertices. An isolated D-vertex counts as such a path. In particular there are exactly q such components, and $\nu(B_D) = |D| - q = r - b - q$. For each of these q paths one may prescribe any one of its D-vertices, independently of the other prescriptions. A maximum matching of B_D exists whose unmatched vertices in D are precisely the prescribed vertices. Every maximum matching of B_D leaves one D-vertex unmatched on each of these paths and none elsewhere in D. When $q = 0$, the prescription is empty and every maximum matching saturates D.

Every choice of one endpoint of each edge of L as a new set U_0 gives another frame $(WminusU_0, U_0)$ with the same terminal. Finally, if p in P has a neighbor x of degree two in R, orient its edge so that p belongs to U_0. The pair $(WminusU_0, (U_0minus\{p\})union\{x\})$ is a frame, with terminal $R - x + p$.

Proof. Every u in U has exactly one neighbor in S, because F is four-regular and u has degree three in F-S. Distinct vertices u, v in U cannot share that neighbor s: the set (S minus {s}) union {u, v} would be independent and would have $a + 1$ vertices. The independence of S and U therefore makes F[W] precisely m disjoint edges and a-m isolated vertices. In particular $m \leq a$.

The component choices defining D are independent. On an odd cycle of order $2s + 1$ they select $s = \nu(C_{2s+1})$ vertices. On P_3 and C_4 they select respectively one and two vertices, their matching numbers. On a path P_{2s} they select $s - 1 = \nu(P_{2s}) - 1$ vertices. Matching numbers add over components: a matching restricts to one in each component, and a union of component matchings is a matching. Since there are b even paths, $|D| = r - b$. Every selected vertex has degree two in R.

We construct J and U′. On an odd cycle O of order $2s + 1$, a maximum independent set D intersect $V(O)$ has s vertices. It has 2s incident edges, all distinct because it is independent. The cycle has $2s + 1$ edges, so O-D has exactly one edge. Call its endpoints x_O, y_O, choose either endpoint as x_O, put x_O in U′, and put $V(O)$ minus $(Dunion\{x_O\})$ in J. This is an independent s-set, and x_O has exactly one neighbor in it, namely y_O. On a path P_{2s}, when D selects the even internal vertices, put the odd-position vertices in J and put vertex 2s in U′. When D selects the odd internal vertices, put the even-position vertices in J and put vertex 1 in U′. In either case J takes a maximum independent set of the path, and the selected endpoint has exactly one neighbor in J. These instructions include P_2.

On P_3 put both leaves in J; on C_4 put the opposite pair complementary to D in J; put every isolate in J. Add no U′-vertex on these components. Thus J takes a maximum independent set on every component of R. The assumed equality $\alpha(R) = a$ gives $|J| = a$, so J is also a maximum independent set of F. There is one U′-vertex on each of the $q + b = m$ designated components, and none on the others. These vertices lie in different components of R, so U′ is independent. Each has exactly one J-neighbor. Four-regularity makes its degree in F-J equal to three. Hence (J, U′) is a frame.

The construction partitions $V(R)$ into J, U′, D. Thus its terminal is $R_D = F[W \text{ union D}]$. The uniform terminal hypothesis applies to this actual frame. A vertex of D has two neighbors in R, neither selected into D, and consequently exactly two neighbors in W. Its degree in R_D is two.

There are no D-to-D edges in R_D, and its only W-to-W edges are L. On any cycle in R_D the number of D-to-W edges is twice the number of D-vertices on that cycle. An odd cycle must therefore contain an odd number of L-edges, at least one. On an even path of order 2s in R_D, no endpoint lies in D, since D-vertices have degree two. If t is its number of L-edges, its edge count gives $2s - 1 = 2|D \text{ intersect } V(P_{2s})| + t$, so t is again odd and positive. The terminal has q odd cycles and b even paths, a total of m designated components. There are exactly m edges of L, and components are vertex-disjoint. Each designated component therefore contains exactly one L-edge and there is no L-edge in another component.

Each P-vertex is incident with an L-edge. It belongs to the designated component containing that edge. Since the maximum degree of R_D is two, each P-vertex has at most one neighbor in D; each Z-vertex has at most two.

Deleting P from R_D gives exactly B_D: Z has no internal edges, D is independent, and all remaining edges join the two sets. On an odd-cycle component of order $2s + 1$, remove the endpoints of its unique L-edge. Every other edge alternates between D and Z. What remains is a path with s vertices in D and s-1 in Z, with both endpoints in D. This includes the isolated D-vertex when $s = 1$. Its maximum matching size is s-1, bounded by its Z-side and attained by alternate edges.

On an even-path component, remove the endpoints of its unique L-edge. Every nonempty remaining segment begins next to a removed endpoint with a D-vertex. At its other end it reaches an original path endpoint in Z: an endpoint in D would contradict degree two for D, and an endpoint in P would require a second L-edge in that component. Thus each remaining segment is a balanced alternating path and has a perfect matching. Empty segments cause no difficulty.

A component of R_D containing no L-edge has no P-vertex. The remaining component types are P_3, C_4, and isolates. The center of such a P_3 lies in D and its leaves in Z, since its leaves cannot lie in D. It has a matching covering its sole D-vertex. A C_4 alternates between D and Z and has a perfect matching. Every isolate belongs to Z. This proves the complete claimed description of the D-ended path components of B_D.

On a D-ended path, a prescribed D-vertex occupies an odd position in an alternating order that begins in D. Removing it leaves two possibly empty even-order paths. Match alternate edges in each. This leaves precisely the prescribed D-vertex unmatched and attains the maximum s-1 for that original component. Use these matchings independently on the q paths, and match all D-vertices on every other component by the constructions just given. The resulting size is $|D| - q$. Any matching has at least one unmatched D-vertex in each of the q D-heavy components, so this size is maximum. A maximum matching must attain the maximum size in each component, since otherwise its restriction could be replaced by a larger component matching. Every component matching maximum has the stated number of unmatched D-vertices. The rank formula and all prescription claims follow.

For the orientation assertion, let U_0 contain one endpoint from each edge of L. It is independent, and W minus U_0 is independent of size $|W| - m = a$, hence maximum in F. Each $U_0 - vertex$ has precisely its mate as a neighbor in W minus U_0. It therefore has degree three in $F - (WminusU_0)$. This gives the asserted frame.

Finally let p in U_0 be adjacent to x in R with degree two in R, and put $S_0 = W$ minus U_0. The vertex x has exactly two neighbors in W. One is p. Since S_0 is maximum independent, it dominates every outside vertex: otherwise that vertex could be added. Hence the other W-neighbor of x lies in S_0. Thus x has exactly one $S_0 - neighbor$, has degree three in $F - S_0$, and has no neighbor in U_0 minus {p}. Replacing p by x preserves independence and the size m of the cubic set. Its terminal is the induced graph on (V(R) minus {x}) union {p}, as asserted. □

Remark. A componentwise complementary independent set realizes the reflection inside the same graph. Parity assigns one old frame edge to each designated component; deleting its endpoints identifies exactly where a selector matching must leave a vertex unmatched.

Depends on. Nothing earlier; this item is self-contained.

Lemma 26 (Matchings and cut vertices at independence number at most two). *Let H be a finite simple graph with independence number $\alpha(H) \leq 2$, where $\alpha(H)$ is the largest cardinality of an independent vertex set. A matching consists of edges with pairwise disjoint endpoints; it is maximal if no edge can be added, perfect if it covers every vertex, and near-perfect if it covers all but one vertex.*

If H is connected and has positive even order, then H has a perfect matching. If H has odd order, then it has a near-perfect matching and every maximal matching is near-perfect.

Suppose now that H is connected and has odd order at least three. A graph is 2-connected if it has at least three vertices and deleting any vertex leaves it connected. If H is 2-connected, it is factor-critical: deleting any vertex leaves a perfect matching. If H is not 2-connected, there is a cut vertex c such that $H-c$ has exactly two nonempty components with vertex sets O_1, O_2, both induced cliques; at least one of O_1 union $\{c\}$ and O_2 union $\{c\}$ also induces a clique; and $H-x$ has a perfect matching for every vertex x different from c. A cut vertex is one whose deletion disconnects the graph. All graph orders, independent-set sizes, matching cardinalities and component counts are integers in the stated domains, and every vertex deletion is induced.

Proof. The vertices uncovered by a maximal matching form an independent set: an edge between two uncovered vertices could be added. Hence any maximal matching of H leaves at most two vertices uncovered. Such a matching exists by repeatedly adding disjoint edges; finiteness makes this process terminate.

First suppose H is connected of positive even order. Choose a maximum matching M. If M is not perfect, its uncovered set has positive even cardinality and at most two members, so it is {u, v} for distinct nonadjacent

vertices u, v. Every other vertex neighbors at least one of u, v, since otherwise it would complete an independent triple.

No vertex a can neighbor both u and v. Its mate b under M is distinct from u, v. If b neighbors v, replacing ab by ua and bv increases the matching size. If b neighbors u, replacing ab by va and bu does so. Thus b would neighbor neither u nor v, contradicting the independent-triple condition. Consequently the vertices other than u, v partition into S, adjacent to u but not v, and T, adjacent to v but not u.

Every edge of M has both endpoints in S or both endpoints in T. Otherwise an M-edge ab with a in S and b in T could be replaced by ua and bv. There is therefore no M-edge between these sets. The two nonempty sets {u} union S and {v} union T partition $V(H)$. Connectedness forces an edge between them. Its endpoints cannot be u or v, by their specified neighborhoods and nonadjacency. It is an edge ab with a in S and b in T. Let a′ and b′ be their respective mates under M; these lie in S and T. All six vertices u, a′, a, b, b′, v are distinct. Removing aa′ and bb′ from M and adding ua′, ab and b′v gives a larger matching. This final contradiction proves that M is perfect.

If H has odd order, every matching leaves an odd number of vertices uncovered. A maximal matching leaves at most two and cannot cover the odd order, so it leaves exactly one. Its existence proves both odd-order assertions. If H is also 2-connected, deleting any vertex gives a connected graph of positive even order with independence number at most two. The even-order assertion just proved gives a perfect matching after every such deletion, proving factor-criticality.

It remains to treat connected odd-order H that is not 2-connected. Choose a cut vertex c. Its deletion has at least two components and cannot have three: choosing one vertex from each of three components would give an independent triple. Let A, B be the vertex sets of its two nonempty components. Both induce cliques, since a nonadjacent pair in one and a vertex in the other would be an independent triple. The vertex c is complete to at least one of A, B, for otherwise a missed vertex from each together with c would form such a triple. Rename the sets so that c is complete to A. Connectedness gives at least one neighbor of c in B. The positive integers $|A|$ and $|B|$ have the same parity because their sum is $|V(H)| - 1$, which is even.

If both orders are odd, consider x in A. Perfectly match the even clique A minus {x}, match c to one of its neighbors b in B, and perfectly match the even clique B minus {b}. The union is a perfect matching of H-x. For x in B, perfectly match B minus {x}, match c to any vertex a in A, and perfectly match A minus {a}. This treats every x different from c. Taking $O1 = A$ and $O2 = B$ gives all the required cut properties.

If both orders are even and c has at least two neighbors in B, consider x in A. Choose a in A minus {x}, possible because $|A| \geq 2$. Use ca and perfect matchings of A minus {x, a} and of B. For x in B, choose a neighbor b of c in B minus {x}, possible by the two-neighbor condition. Use cb and perfect matchings of B minus {x, b} and of A. Empty cliques have the empty perfect matching, so these constructions include the smallest part sizes. Again every x different from c is covered and $O1 = A$, $O2 = B$ satisfy the statement.

Finally suppose both orders are even and c has exactly one neighbor b in B. Use b as the new cut vertex. The components of H-b have vertex sets B minus {b} and A union {c}. Both are nonempty odd cliques. There is no edge between them, since A and B were different components of H-c and b was the only B-neighbor of c. The vertex b is complete to B minus {b} and has a neighbor c in A union {c}. Thus the new cut and its two components satisfy precisely the odd-part hypotheses of the preceding construction. Applying that explicit construction gives a perfect matching of H-x for every x different from b. With this new cut vertex and these two component sets, every required conclusion follows. The parity and adjacency cases exhaust the possibilities. □

Remark. Two uncovered vertices leave no room for a third independent vertex. Their neighbors either yield a short augmenting path or disconnect the graph. In odd order, a cut can only split the graph into two cliques, and parity identifies the one potentially exceptional vertex deletion.

Depends on. Nothing earlier; this item is self-contained.

Lemma 27 (Even-path attachments cover the degree-three vertices). *All graphs are finite and simple. For a positive integer n, let P_n be the path on n vertices. A matching is a set of edges with pairwise disjoint endpoints; $V(W)$ is the set of endpoints of a matching W. For a graph Q, let $\nu_2(Q)$ be the maximum number of vertex-disjoint ordinary copies of P_3 in Q. Empty graphs have $\nu_2 = 0$. For a vertex set I in G, write $N_G(I)$ for the union of the open neighborhoods of its vertices.*

Let G have maximum degree at most three, put $S = \{vinV(G) : deg_G(v) = 3\}$, and let

$\delta = min\{|SminusV(W')| : W'isamatchingofG\}$.

Let W be a maximum-cardinality matching of G attaining this minimum δ. Put $I = S$ minus $V(W)$, and let $H = (G - W) - I$, where G-W deletes only the edges of W and the subsequent deletion removes the vertices of I and all incident edges.

Suppose every vertex outside I has at most one G-neighbor in I, every connected component of H belongs to $\{P_1, P_2, P_3, P_4, P_6\}$, and

$\nu_2(H - z) = \nu_2(H)$ *for every z in $N_G(I)$.*

Then $\delta = 0$. In particular the specified matching W covers every degree-three vertex of G. All cardinalities, degrees, matching sizes and packing numbers are nonnegative integers, and the empty graph and empty sets are allowed.

Proof. There are finitely many matchings of G and the empty matching is one of them, so the minimum defining δ exists. If $\delta = 0$ the conclusion holds. Suppose instead that $\delta > 0$. No two vertices missed by the maximum-cardinality matching W are adjacent: adjoining their edge would give a larger matching. Therefore I is independent, and every vertex of $N_G(I)$ lies outside I and is a vertex of H. No W-edge meets I.

The maximum $P_3 - packing$ numbers of P_1, P_2, P_3, P_4, P_6 are respectively 0, 0, 1, 1, 2. For P_1 and P_2 the order is too small for a P_3. The middle two paths contain a P_3 and have fewer than six vertices. The path P_6 consists of two disjoint consecutive three-vertex subpaths, and its order bounds the packing number by two. Packing numbers add over connected components: each P_3 lies in one component, and disjoint choices in distinct components can be combined.

Deleting a vertex of P_3 leaves two vertices and reduces its packing number to zero. Deleting any vertex of P_6 leaves only five vertices, so its packing number becomes at most one. Deleting an internal vertex of P_4 leaves an edge and an isolated vertex, whereas deleting an endpoint leaves P_3. Thus the hypothesis $\nu_2(H - z) = \nu_2(H)$ forces every attachment z in $N_G(I)$ to be isolated in H, to belong to a P_2 component, or to be an endpoint of a P_4 component.

Choose a matching M within H that is perfect on each P_2, P_4, P_6 component, uses one edge incident with the center of each P_3 component, and has no edge at any isolated vertex. These choices form a matching because the components are disjoint. It covers every vertex of H-degree two, and every vertex in a P_2 or P_4 component.

We claim that M covers S minus I. Such a vertex v is covered by W, so it has degree two in G-W. At most one edge incident with v is deleted on removing I, by the assumed neighbor bound. Its H-degree is therefore two or one. In the first case M covers it. In the second case exactly one incident edge was removed, so v belongs to $N_G(I)$. Its H-degree is one, and the attachment restriction puts it in P_2 or at an endpoint of P_4, where M again covers it. This proves the claim. The matching M misses all of I and no other vertex of S, so it attains the same minimum deficiency δ. It need not have maximum cardinality; the definition of δ applies to every matching of G.

If an attachment z were isolated in H, both z and an adjacent root x in I would be missed by M. The matching M union {xz} would cover x and keep every previously covered vertex covered. Its uncovered S-count would be less than δ, a contradiction. Consequently every attachment is an endpoint of a P_2 or P_4 component.

Fix x in I and any G-neighbor z of x. Let C be the P_2 or P_4 component of H containing z, and let w be its other endpoint. Starting with xz and then traversing C from z to w gives a simple path alternating between edges outside M and edges in M, beginning outside M and ending in M. Here M restricts to the perfect matching on the even path C. Replace the M-edges on this path by its other edges. The result is a matching of the same cardinality: it covers x, uncovers w, and preserves coverage of all other vertices. If w were not in S, this would reduce the number of uncovered S-vertices, contrary to the definition of δ. Therefore w is in S.

Since w lies in H, it is not in I and is covered by W. Its degree in G-W is consequently two. Its degree in H is one, so deleting I removed exactly one incident edge. There is a unique neighbor y of w in I. If y differs from

x, append wy to the preceding alternating path. The extended path is simple, because y lies outside H and is distinct from x. Its two endpoints x, y are missed by M. Replacing its M-edges by its other edges now covers both x and y and keeps every other vertex's coverage. It is a matching with one additional edge and with at least two fewer uncovered S-vertices. This again contradicts minimality of δ. Hence $y = x$.

We have shown that, for every neighbor z of x, the other endpoint w of its even path component is also a neighbor of x. Every neighbor of x is an endpoint of one of these components, and different components have disjoint endpoints. The set $N_G(\{x\})$ is therefore a disjoint union of two-element endpoint sets. Its cardinality is even. But x belongs to S and has degree three in the simple graph G, so its neighborhood has exactly three vertices. This contradiction excludes $\delta > 0$ and proves the theorem, including cases with empty H or isolated vertices. □

Depends on. Nothing earlier; this item is self-contained.

Lemma 28 (Exact residual targets for a disjoint star pool). *All graphs are finite and simple, all copies are ordinary subgraphs, and every star component has a positive integer number of edges. A star forest may be empty, and the empty forest is contained in every graph. Write $K_{1,a}$ for the star with a edges. Let F have labelled component sizes $a_1,\dots, a_s$, where $s \geq 0$ is an integer, and let $c = (c_1, \dots, c_N)$ with integers $N \geq 0$ and $c_i \geq 0$. On the positive component-size multiset define D_b, for each integer $b \geq 0$, to delete one largest remaining size at most b, or to do nothing if no such size exists. Define $res_c(F)$ by applying $D_{c_1},\dots, D_{c_N}$, and let $P(c)$ be the vertex-disjoint union of K_{1,c_i} over the indices with $c_i > 0$.*

The operators D_b commute. Thus $res_c(F)$, up to isomorphism, is independent of the capacity order and of choices among equal component sizes.

Define the finite nonempty feasible set

`I_c(F)={I` $\subseteq$ *{1,...,s}: there is an injection* $f : I \to \{1, \dots, N\}$ *with* $a_j \leq c_{(}f(j))$ *for every j in I}.*

The empty subset is feasible. For I in $I_c(F)$, let F-I be the forest of the components with labels outside I. The forest $res_c(F)$ is one of these feasible complements and embeds in every one of them. Consequently it is their greatest common star subforest: a star forest U embeds in every F-I if and only if U embeds in $res_c(F)$. This universal property determines it up to isomorphism. It is nonempty exactly when $P(c)$ contains no copy of F.

For every graph J vertex-disjoint from $P(c)$,

J disjoint union $P(c)$ contains F if and only if J contains $res_c(F)$.

Both implications have actual witnesses. From any full copy, successive exchanges of its pool assignments produce a copy of $res_c(F)$ in the same J; from a residual copy, the greedy pool assignment reconstructs a full copy.

Let F_R, F_B be nonempty star forests, let Q be a vertex-disjoint union of N stars with positive integer sizes $d_1,\dots, d_N$, and let H be any graph disjoint from Q. Put $G = H$ disjoint union Q. For each integer vector $r = (r_1, \dots, r_N)$ with $0 \leq r_i \leq d_i$, put

$F_R(r) = res_r(F_R)$, $F_B(r) = res_{d-r}(F_B)$.

For every fixed coloring of H, and every coloring of Q having exactly r_i red edges on its ith star, G contains a red F_R exactly when the red graph of H contains $F_R(r)$, and contains a blue F_B exactly when the blue graph of H contains $F_B(r)$.

Writing $G \to (A, B)$ for the assertion that every red-blue edge coloring contains a red A or a blue B, one has

$G \to (F_R, F_B)$

if and only if

$H \to (F_R(r), F_B(r))$ *for every such r with both residuals nonempty.*

If there are no such vectors, Q itself arrows (F_R, F_B). Equivalently, a full avoiding coloring of G exists exactly when some vector with two nonempty residuals has a core coloring avoiding its residual pair. Every such coloring extends by choosing r_i arbitrary edges of the ith pool star red and all its other edges blue. Empty capacity lists, zero capacities, repeated sizes, and all unit components are included.

Proof. Fix two capacities $b \le d$ and a positive integer multiset. If no entry is at most b, D_b does nothing before or after D_d. Otherwise let x be the largest entry at most b and y the largest entry at most d. If $y > x$, then $y > b$. Deleting x does not affect the largest entry y fitting d, and deleting y does not affect the largest entry x fitting b. Both orders therefore delete one x and one y. If $y = x$, no entry belongs to (b, d], so both operations act on the same submultiset of entries at most b. Their two orders remove its two largest entries, or its only entry when it has size one. Thus they commute in all cases, including $b = 0$. Adjacent swaps generate every permutation of a finite list, proving order independence. Equal sizes are indistinguishable in the final multiset.

A connected positive-edge subgraph of a disjoint union of stars lies in one star. Every edge of a host star meets its center, so two vertex-disjoint positive-edge target components cannot occupy the same host star. Hence a subforest of F embeds in $P(c)$ exactly when its labels form a feasible set in $I_c(F)$. Necessity assigns each component to its containing pool star; sufficiency selects the required number of edges in each distinct assigned star. The set $I_c(F)$ is finite and contains the empty set, including when $s = 0$ or $N = 0$.

We prove the exchange assertion while keeping the actual core locations. Begin with any copy of F in J disjoint union $P(c)$. Label its components by those of F. Some components are assigned to distinct pool positions and every other component has an actual vertex-disjoint copy in J. Process positions $k = 1,..., N$, leaving all previously processed assignments fixed. Choose a remaining component g of largest size fitting c_k, if one exists. If none exists, position k cannot currently contain a remaining component and is left unused.

If g is already assigned to position k, fix it there. Otherwise let h be the component currently assigned to k, when that position is used. Its size is at most the size of g. If g is assigned to a later position l, put g at k and put h at l when h exists; h fits c_l because its size is at most that of g. When h is absent, leave l unused. This changes no core demand. If g instead occupies a core location in J, put g at k. When h exists, replace the old core copy of g by a copy of h obtained by retaining the required number of its edges. When h is absent, discard that core copy. These operations preserve all other core copies and their vertex disjointness, keep all assignments feasible, and do not alter earlier positions.

After N steps the unassigned component multiset is $res_c(F)$, so J contains that forest. The greedy choices themselves assign distinct components to distinct fitting positions. Their complement is therefore feasible and is exactly $res_c(F)$. Conversely, given a copy of $res_c(F)$ in J, assign all other target components greedily to the pool and select the required number of edges of each assigned star. These copies are mutually disjoint and disjoint from J. Their union with the specified residual copy is a full copy of F.

For any I in $I_c(F)$, take J to be a fresh copy of F-I and combine it with its feasible assignment to the pool. The exchange argument gives $res_c(F)$ inside this very F-I. Because $res_c(F)$ is itself a feasible complement, a forest U that embeds in all feasible complements embeds in it. Conversely, an embedding of U in $res_c(F)$ composes with its embeddings in all complements. This proves the universal property. Any two forests with this property embed in each other. Mutual subgraph embeddings of finite graphs imply equal vertex counts and edge counts; either embedding is consequently an isomorphism. Taking J empty in the containment equivalence shows that $P(c)$ contains F exactly when the residual is empty: a nonempty positive-edge forest cannot embed in an empty graph. This also covers F empty, for which the residual and all complements are empty. Zero capacities make no greedy deletions.

For the colorwise assertion, fix r and a coloring of Q with these red edge counts. Its red graph, after isolated vertices are disregarded, is $P(r)$; its blue graph is $P(d - r)$. Both remain disjoint from H. Isolated vertices cannot participate in a positive-edge star component, so they do not change any containment in question. Apply the one-color equivalence to the red graph of H and $P(r)$, then to the blue graph of H and $P(d - r)$. The two equivalences concern the same fixed coloring of G and give the claimed exact containment statements. The red and blue pool subgraphs may share their centers; each equivalence only compares copies within its own color.

Suppose G arrows the originals. For any vector having two nonempty residuals, extend an arbitrary coloring of H by choosing r_i red edges on each pool star. One original target must appear, and the colorwise equivalence gives its corresponding residual in H. Thus H arrows every specified residual pair. Conversely, suppose those residual pairs are all arrowed by H and fix any coloring of G. Read its vector r from Q. If either residual is empty, its corresponding full target already lies in the pool by the one-color equivalence with empty core. If both are nonempty, H's residual arrowing supplies one original target by the colorwise equivalence. This proves the arrowing equivalence.

If no vector has two nonempty residuals, the preceding empty-residual argument applies to every coloring of Q itself. Finally a coloring of G avoiding both originals must have two nonempty residuals and restrict to

an avoiding coloring of their core pair. An avoiding core coloring for such a vector extends by any indicated pool choices, and the exact colorwise equivalence guarantees avoidance of both originals. This proves the existence equivalence and supplies its witness for every boundary case stated. □

Depends on. Nothing earlier; this item is self-contained.

Lemma 29 (Star profiles and normalization). *All graphs are finite and simple. Write $V(G)$, $E(G)$, and $e(G) = |E(G)|$ for the vertex set, edge set, and edge count of a graph G. A copy means an ordinary subgraph isomorphic to the specified graph; it need not be induced. For a positive integer h, the star $K_{1,h}$ has one center, h distinct leaves, and exactly the h edges from the center to its leaves. A positive star forest is a nonempty vertex-disjoint union of such stars. Thus it has neither isolated vertices nor components with zero edges.*

A profile is a nonincreasing list of positive integers. Fix positive integers s, t and profiles $n_1 \geq \cdots \geq n_s \geq 1$ and $m_1 \geq \cdots \geq m_t \geq 1$. Let F_1 and F_2 be the vertex-disjoint unions of the stars K_{1,n_i}, $1 \leq i \leq s$, and K_{1,m_j}, $1 \leq j \leq t$, respectively. Write $G \to (F_1, F_2)$ when every function $E(G) \to \{red, blue\}$ yields a red copy of F_1 or a blue copy of F_2. The size Ramsey number $\hat{R}(F_1, F_2)$ is the least nonnegative integer $e(G)$ among such hosts, whenever such a host exists.

For every integer k with $2 \leq k \leq s + t$, define

$$\ell_k = max\{n_i + m_j - 1 : 1 \leq i \leq s, 1 \leq j \leq t, i + j = k\}.$$

Put $M = s + t - 1$ and $L = \sum_{k=2}^{s+t} \ell_k$. For integers $0 \leq a < s$ and $0 \leq b < t$, let F_1^a be the actual suffix consisting of components $a + 1$,..., s of F_1, and let F_2^b consist of components $b + 1$,..., t of F_2. For each integer q with $0 \leq q < M$, define the family of ordered suffix pairs

$$A_q = \{(F_1^a, F_2^b) : 0 \leq a < s, 0 \leq b < t, a + b = q\}$$

and the tail budget $L_q = \sum_{k=q+2}^{s+t} \ell_k$. All cuts and indices in these definitions are integers.

Every positive star forest has a unique profile, obtained by sorting its component edge counts with multiplicity. Replacing an arbitrary enumeration of its components by this profile preserves its isomorphism type. Consequently, sorting the component lists preserves the arrowing problem, its set of possible host edge counts, and the budget L computed from the sorted profiles. No occurrence of size one or any repeated size is removed.

Every maximum defining ℓ_k is over a nonempty finite set, and

$$\ell_2 \geq \ell_3 \geq \cdots \geq \ell_{s+t} \geq 1,$$

$$\ell_2 = n_1 + m_1 - 1, \text{ and } \ell_{s+t} = n_s + m_t - 1.$$

Every A_q in the stated range is nonempty and contains only pairs of nonempty forests. Its members have a total of $s + t - q$ components, and its tail budget is a positive integer; in particular $A_0 = \{(F_1, F_2)\}$ and $L_0 = L$.

If $G^c irc$ is obtained by deleting all isolated vertices of an arbitrary finite simple host G, then $e(G^c irc) = e(G)$ and $G^c irc \to (F_1, F_2)$ if and only if $G \to (F_1, F_2)$. Exchanging the two target profiles preserves each ℓ_k and L, exchanges the entries of every pair in A_q, and preserves the set of possible host edge counts. Thus sorting, exchanging colors, and removing host isolates preserve the target problem and its budget.

Proof. The connected components of a graph are determined by adjacency. Each component of a positive star forest is a star with a positive integer edge count. Two stars with the same edge count are isomorphic: map a chosen center to a chosen center and choose any bijection between their leaves. This also applies when the edge count is one, where either endpoint may be chosen as center. An isomorphism between two forests maps connected components to connected components and preserves their edge counts. Therefore the multiset of component edge counts determines, and is determined by, the isomorphism type. A finite multiset of positive integers has a unique nonincreasing enumeration. This proves the profile assertion, including repeated sizes and size one.

Composing a monochromatic subgraph embedding with a target isomorphism gives a monochromatic embedding of the isomorphic target in the same coloring. This works in both directions. Sorting either list therefore preserves arrowing for every host, and hence preserves its set of possible host edge counts. The two sorted lists themselves do not depend on the original component enumerations, so the displayed expression for L has the same value after any such enumeration and sorting.

Fix an integer k with $2 \le k \le s+t$. Set $i = \max(1, k-t)$ and $j = k-i$. The inequalities $k-t \le s$ and $1 \le s$ give $i \le s$. Both 1 and k-t are at most k-1, so $i \le k-1$ and $j \ge 1$. The inequality $i \ge k-t$ gives $j \le t$. Thus $1 \le i \le s$, $1 \le j \le t$, and $i+j = k$. The defining set for ℓ_k is nonempty, finite, and consists of positive integers. In particular its maximum exists and is positive.

For $2 \le k < s+t$ choose indices i, j attaining ℓ_{k+1}. They satisfy $i+j = k+1 \ge 3$. If $i \ge 2$, the valid pair (i-1,j) has sum k and $n_{i-1} \ge n_i$, so $\ell_k \ge n_{i-1} + m_j - 1 \ge \ell_{k+1}$. If $i = 1$, then $j \ge 2$; the valid pair (1,j-1) has sum k and $m_{j-1} \ge m_j$, again giving $\ell_k \ge \ell_{k+1}$. At sum two the only valid pair is (1,1), and at sum $s+t$ the only valid pair is (s, t). This proves the endpoint identities and monotonicity. When $s = t = 1$ there are no monotonicity comparisons and the two endpoint identities coincide.

Fix an integer q with $0 \le q < M$, and put $a = \max(0, q-(t-1))$ and $b = q-a$. Then $0 \le a \le s-1$ because $q \le s+t-2$, and $0 \le b \le t-1$ by the definition of a. These are legal cuts with $a+b = q$. Thus A_q is nonempty. For every legal cut, both remaining lengths s-a, t-b are positive and their sum is $s+t-q$. The sum defining L_q contains exactly $s+t-q-1 = M-q \ge 1$ positive integer terms. At $q = 0$ the only cuts are $a = b = 0$, proving the asserted identities.

Deleting host isolates removes no edges. Hence restriction and extension give a bijection between the *red/blue* colorings of G and $G^c irc$. Every vertex of either target is incident to at least one target edge. Its image in any target copy is therefore incident to a host edge and cannot be an isolated vertex of G. Thus the copies in any fixed coloring of G are precisely the copies in its restriction to $G^c irc$. This proves the equivalence of arrowing.

Finally, for a fixed host, interchange red and blue on every edge. This operation is its own inverse, and the assertion that the new coloring contains a red F_2 or a blue F_1 is exactly the assertion that the old coloring contains a blue F_2 or a red F_1. Hence $G \to (F_1, F_2)$ if and only if $G \to (F_2, F_1)$. After exchanging the lists, the bijection $(i,j) \to (j,i)$ preserves the index sum and the value $n_i + m_j - 1$, so it preserves every ℓ_k and their sum L. The bijection $(a,b) \to (b,a)$ preserves q and exchanges the two entries of the suffix pairs. All normalization operations consequently have the claimed invariance. □

Remark. The target depends only on the multiset of positive component sizes. Sorting chooses a canonical representation without changing embeddings or the diagonal budget.

Depends on. Nothing earlier; this item is self-contained.

Lemma 30 (Small dominating sets from octic terminal blocks). *Let H be a finite simple graph in which every vertex has degree eight. Write $n = |V(H)|$ and let $\alpha(H)$ be the maximum cardinality of an independent vertex set. A set Z contained in $V(H)$ is dominating if every vertex outside Z has a neighbor in Z. Write C_3 for a triangle, K_2 for a single edge and K_1 for an isolated vertex; sums of these graphs denote vertex-disjoint unions with no edges between the summands. In each row of the following table, if n and the displayed independence bound hold and H has an induced subgraph isomorphic to R, then H has a dominating set of cardinality at most z:*

n | *independence bound* | R | z

25 | $\alpha(H) \le 6$ | $3C3 + K2 + 2K1$ | 5

27 | $\alpha(H) \le 6$ | $4C3 + 2K1$ | 5

27 | $\alpha(H) \le 7$ | $3C3 + K2 + 3K1$ | 6.

The conclusions require no connectedness or matching hypothesis. All vertex orders, degrees, independent-set sizes, block multiplicities and the bound z are nonnegative integers with the displayed values.

Proof. Fix one of the rows and an induced subgraph R of the indicated form. Put $W = V(H)$ minus $V(R)$, and let H[W] denote the induced graph on W. For vertex sets U, V with no common vertex, write $e(U,V)$ for the number of edges joining them; write $e(J)$ for the number of edges of a graph J. Summing degrees gives $e(H) = 4n$ and $e(V(R), W) = 8|V(R)| - 2e(R)$. Therefore

$e(H[W]) = 4n - 8|V(R)| + e(R)$.

In the three rows respectively, the triples $(|V(R)|, |W|, e(H[W]))$ are (13,12,6), (14,13,8), (14,13,6).

A vertex cover is a set meeting every edge, and its complement is independent. Thus every vertex cover of H[W] has size at least $|W| - \alpha(H)$. In the first and third rows this lower bound is six. If two of the six edges of H[W] shared an endpoint, that endpoint together with one endpoint of each of the other four edges would be a cover of size at most five. Consequently those six edges are disjoint. In the first row they cover W; in the third row they leave exactly one isolated vertex.

In the second row every cover has size at least seven. A vertex of degree d in H[W], together with one endpoint of each edge not incident with it, gives a cover of size at most $1 + 8 - d$. It follows that $d \leq 2$. Each component of H[W] is therefore a path or a cycle. For a path on m vertices the minimum cover size is $\lfloor m/2 \rfloor$, and for a cycle it is $\lceil m/2 \rceil$. The path value follows from alternate vertices and the disjoint consecutive edges giving the opposite bound. For the cycle, an independent set has at most $\lfloor m/2 \rfloor$ vertices, since the cyclic successors of its vertices are distinct vertices outside the set; alternating choices attain this size. Taking complements gives the stated cover value.

Minimum cover sizes add over components. Since there are eight edges and cover size at least seven, the sum over components of edge count minus minimum cover size is at most one. An isolate or an edge contributes zero. The only other components contributing at most one are a three-vertex path, a four-vertex path, and a triangle; each contributes one. Every longer path or other cycle contributes at least two. There must be a component of positive contribution, since eight disjoint edges would use sixteen vertices. A three-vertex path and the remaining six disjoint edges would use fifteen vertices, and a four-vertex path and the remaining five edges would use fourteen. Thus H[W] is exactly one triangle and five disjoint edges, using all thirteen vertices.

We use the following finite counting observation in every case. Select one vertex from each of four complete blocks of R, with block sizes either (3,3,3,2) or (3,3,3,3). For a fixed exterior vertex w, let a_j be the number of its nonneighbors in block j. The number of selections containing no neighbor of w is exactly the product of the four a_j. If any factor is zero, this number is zero. Otherwise $1 \leq a_j \leq 3$, and $a_j \leq 2^{a_j - 1}$, as checked at the values 1, 2, 3. Hence the number is at most $2^{\sum_j a_j - 4}$. To find one selection that succeeds for every remaining exterior vertex, it is enough that the sum of their failure counts is smaller than the total number of selections: a union of finite failure sets has cardinality at most the sum of their cardinalities.

First row. Let u, v be the two isolates of R. Their eight-element neighborhoods lie in the twelve-element set W, so they have a common neighbor x. Let x′ be the partner of x in the matching H[W]. Select one vertex from each of the three triangles and one endpoint of the edge of R. There are $3^3 \cdot 2 = 54$ selections U. For every such U, U union {x} dominates R, since the complete blocks have representatives and x neighbors both isolates. It also dominates x, x′. Each remaining w in W has one W-neighbor and hence seven R-neighbors, at most two of which are u, v. It has at least five neighbors in the eleven vertices of the four complete blocks. Thus $\sum_j a_j \leq 6$ and its failure count is at most four. The ten remaining exterior vertices forbid at most forty selections, which is less than fifty-four. One selection U succeeds at all of them. The set U union {x} is the required dominating set of size five.

Second row. Again write u, v for the two isolates of R, and let T be the triangle of H[W]. The neighborhoods of u, v in W intersect because $8 + 8 > 13$. We will choose a common neighbor x and one representative from each of the four triangles of R. Every one of the $3^4 = 81$ representative selections U, together with x, dominates R.

If a common neighbor x belongs to T, then x dominates T. Each of the ten remaining W-vertices is an endpoint of a W-edge and has seven R-neighbors, at most two in {u, v}. It has at least five neighbors among the twelve triangle vertices, so $\sum_j a_j \leq 7$. Its failure count is at most eight. The total at most eighty is smaller than eighty-one, so a successful U exists.

Otherwise no vertex of T neighbors both u and v. Choose any common neighbor x; it belongs to a W-edge with other endpoint x′. Put $j = |N_H(x') \text{ intersect } \{u, v\}|$ and $t = \sum_{w \in T} |N_H(w) \text{ intersect } \{u, v\}|$. Here $N_H(w)$ denotes the set of neighbors of w in H. One has $0 \leq j \leq 2$ and $0 \leq t \leq 3$. The selected vertex x dominates itself and x′. Eight endpoints of the other W-edges and the three vertices of T remain.

For any remaining W-edge endpoint w put $i(w) = |N_H(w) \text{ intersect } \{u, v\}|$. Its seven R-neighbors leave $\sum_j a_j \leq 5 + i(w)$, so it forbids at most $2^{(1 + i(w))} \leq 2 + 3i(w)$ selections; the inequality holds for $i(w) = 0, 1, 2$ by the values 2, 4, 8. For a vertex w of T, one has $i(w) \leq 1$ and six R-neighbors, so the failure count is at most $2^{(2 + i(w))} = 4 + 4i(w)$. The sixteen incidences from u, v to W include two at x, j at x′, and t on T. Therefore

the eight remaining W-edge endpoints have total $i(w) = 14 - j - t$. Their failure counts and those of T sum to at most

$8 \cdot 2 + 3(14 - j - t) + 3 \cdot 4 + 4t = 70 - 3j + t \leq 73 < 81.$

Again some U succeeds at every remaining vertex. In both cases U union {x} is a dominating set of size five.

Third row. Let I be the three isolates of R, and let $w0$ be the single isolate of H[W]. There are twenty-four incidences from I to W, and at most three meet $w0$. Give each of the six W-edges the weight equal to the total number of I-neighbors of its two endpoints. These weights sum to at least twenty-one, so some W-edge P has weight at least four. One endpoint x of P has at least two neighbors in I. If x misses an element of I, choose that element u; otherwise choose any u in I. Since u has eight W-neighbors, one of them, denoted y, lies outside the three vertices in $V(P)$ union {w0}. This y is an endpoint of a second W-edge P′. Together x, y dominate all three vertices of I, and the weights of P and P′ sum to at least five.

There are fifty-four selections U of one vertex from each of the three R-triangles and one endpoint of the R-edge. Each U union {x, y} dominates R and the four vertices of P and P′. For any of the nine remaining W-vertices w, put $i(w) = |N_H(w)$ intersect $I|$. Its W-degree is at most one, so at least seven of its neighbors lie in R and at least 7-i(w) lie in the eleven vertices of the four complete blocks. Thus $\sum_j a_j \leq 4 + i(w)$, and its failure count is at most $2^{i}(w)$. The four vertices of P and P′ account for at least five incidences from I, so $\sum_w i(w) \leq 19$ over the nine remaining vertices. For the four values $i = 0, 1, 2, 3$, the inequality $2^i \leq 1 + 7i/3$ follows respectively from $1 \leq 1$, $2 \leq 10/3$, $4 \leq 17/3$ and $8 \leq 8$. The total failure count is therefore at most

$9 + (7/3) \cdot 19 = 160/3 < 54.$

Some U succeeds at every remaining W-vertex. Its four vertices and the distinct exterior vertices x, y form a dominating set of size six.

In all three constructions the selected exterior vertices lie outside R, so the asserted cardinalities count distinct vertices. The complete-block representatives dominate their entire blocks, the chosen exterior vertices dominate all isolates of R and the specified exterior blocks, and the counting argument covers every other exterior vertex. These are all vertices of H, completing each row. □

Remark. Regularity and the independence bound force a sparse exterior. One or two exterior vertices cover the terminal isolates; a finite count then selects complete-block representatives that dominate all remaining vertices.

Depends on. Nothing earlier; this item is self-contained.

Lemma 31 (Saturation of quartic frames from the terminal identities). *All graphs are finite and simple. Write $\alpha(G)$ for the independence number of G, and G-A for the subgraph induced by $V(G)$ minus A. A graph is well-covered if every maximal independent set is maximum. The symbols P_j, C_j denote paths and cycles on j vertices.*

Let F be a four-regular well-covered graph and put $a = \alpha(F)$. For each maximum independent set S of F, let $C(S)$ be the subgraph induced by the vertices of degree three in F-S. Define the nonnegative integer

$h = max\{\alpha(C(S)) : Sismaximumindependent \in F\}.$

Let m, b be integers with $m \geq 1$ and $b \geq 0$. Assume $h \leq m$, put $d = m - h$, and assume that $d > 0$ implies $d \geq b$.

An attaining frame is a pair (S, U), where S is maximum independent in F and U is an independent h-element subset of V(C(S)). Its terminal is $R = F - (SunionU)$. Assume that for every attaining frame:

(i) $\alpha(R) = \alpha(F - S) = a - d$;

(ii) R has exactly $q = h + 2d - b$ whole odd-cycle components, exactly b whole even-order path components, and any number of components P_3, C_4, and isolated vertices. Here q is a nonnegative integer, every odd cycle has order at least three, and every even-order path has order at least two.

Then $h = m$. In particular an attaining frame exists with m roots, its terminal has independence number a and exactly m-b odd-cycle components, and F has two vertex-disjoint maximum independent sets.

More precisely, if $d > 0$, assumptions (i) and (ii), well-coveredness, and four-regularity imply $33d \leq 28b$ even without the assumption $d \geq b$.

Proof. The finite graph has at least one maximum independent set. The finite maximum defining h is attained, and a maximum independent set in the attaining $C(S)$ supplies U. Fix such a frame, put $W = S$ union U and $R = F - W$. We prove that $d > 0$ is impossible. Throughout that argument, d is a positive integer and all counts introduced below are nonnegative integers.

1. Orientations and exact extensions.

Each u in U has exactly one S-neighbor: it has degree four in F and degree three in F-S. Two distinct vertices of U cannot share an S-neighbor s, since replacing s in S by those two independent vertices would yield an independent set of size $a + 1$. Thus F[W] consists of h disjoint edges and a-h isolates. Let X be its set of isolates. In particular X is independent, and every neighbor of a vertex of X lies in R.

Orient each of those h edges arbitrarily by choosing one endpoint for S', put all isolates in S', and put the opposite endpoints in U'. Then S' is independent of size a, U' is independent of size h, and every U'-vertex has precisely its mate as its S'-neighbor. Hence U' lies in $C(S')$ and (S',U') is an attaining frame with the same W, R.

Let A be any maximum independent set of R. In any orientation, assumption (i) makes A maximum in F-S'. Consequently every U'-vertex has a neighbor in A, since otherwise it could be added. Either endpoint of any frame edge can lie in U', so every endpoint of every frame edge has a neighbor in A.

Put $Z(A) = \{xinX : N_F(x)intersectAisempty\}$, where N_F denotes open neighborhood. The set A union $Z(A)$ is independent. It dominates R, since A is maximum in R. It dominates W minus $Z(A)$, by the endpoint assertion and the definition of $Z(A)$. Hence it is maximal independent in F, and well-coveredness makes it maximum. Therefore

$|Z(A)| = a - \alpha(R) = d$

for every maximum independent A of R.

2. Choosing residual edges without a positive common neighbor.

Call x in X positive if x belongs to $Z(A)$ for at least one maximum independent set A of R. On an odd cycle O of order $2s + 1$, a maximum independent set has s vertices and its unselected vertices induce exactly one edge. Indeed, the s selected vertices have disjoint incident-edge sets of total size 2s, leaving exactly one of the cycle's $2s + 1$ edges. A maximum independent set leaving any prescribed cycle edge is obtained by alternating along the remaining path.

Suppose every edge e of one whole odd-cycle component O has a common positive neighbor x_e in X. A positive vertex cannot own two distinct cycle edges: a maximum independent set of R avoiding it would leave both those edges unselected, contrary to the unique-edge description. Thus the $2s + 1$ chosen vertices x_e are distinct. Each O-vertex already has the distinct owners of its two incident cycle edges as neighbors, as well as its two cycle neighbors. Four-regularity exhausts its neighborhood. Consequently $N_F(x_e)$ intersect $V(O)$ is exactly e, and every X-vertex other than the chosen owners has no O-neighbor.

Fix a maximum independent set K of R-V(O). For each e, let $g_e(K)$ be one when x_e has no K-neighbor, and zero otherwise. Choosing a maximum independent O-set with residual edge e makes x_e the only possible owner in its extension $Z(A)$. All other X-vertices have extension membership independent of this choice on O. Constancy of $|Z(A)|$ therefore makes all values $g_e(K)$ equal, for each K. Positivity of any owner supplies one maximum independent K for which its value is one; all values for that K are then one.

Fix this K, choose a maximum independent set I of O, and put $A = K$ union I. Let $Z = Z(A)$ and $Y = \{x_e : einE(O)\}$. Exactly one member of Y belongs to Z. The set K union Z union Y is independent: X is independent, Z avoids K, and every owner avoids K by its chosen common value. Its size is

$|A \text{ union } Z| - s + (2s + 1) - 1 = |A \text{ union } Z| + s,$

contradicting the maximal cardinality of A union Z. Thus every odd-cycle component has an edge whose endpoints have no common positive neighbor in X.

Choose such an edge on each odd cycle and choose a maximum independent set leaving that edge. On each even path choose either full alternating maximum independent set; on each P_3 choose its leaves; on each C_4 choose an opposite pair; include every isolate. Their union J is maximum independent in R. Put $Z = Z(J)$ and $T = J$ union Z. Thus $|Z| = d$ and T is maximum independent in F.

3. Killed blocks and one-vertex exchanges.

On each odd cycle take the two endpoints of its chosen residual edge as a block. On each even path take its unique unselected endpoint as a singleton block. There are $q + b = h + 2d$ blocks, in distinct components of R. Every block vertex has exactly one J-neighbor. Call a block killed when each of its vertices has a Z-neighbor, and let k_c, k_p be the numbers of killed cycle and path blocks.

For any block vertex v with a Z-neighbor, let j be its unique J-neighbor and put $J_v = (Jminus\{j\})$ union {v}. This is independent of the same size as J and hence maximum in R. Write $Z_v = Z(J_v)$. The set

$(JintersectJ_v)$ union Z union Z_v

is independent: its exterior vertices lie in X and avoid the common terminal part. Its size is $a - 1 + |Z_v$ minus $Z|$, so $|Z_v$ minus $Z| \leq 1$. Both extension sets have size d, and

Z minus $Z_v = Z$ intersect $N_F(v)$.

The latter is nonempty by the choice of v, so both set differences have size one. In particular v has a unique Z-neighbor. The unique vertex y_v in Z_v minus Z has j as its unique J-neighbor. It is positive, and has no Z-neighbor because X is independent. Hence its degree in F-T is three.

Choose one v_B from each killed block and let Y be the set of the resulting y_{v_B}. Distinct blocks lie in distinct components, so their unique J-neighbors are distinct. The uniqueness of a candidate's J-neighbor therefore makes all candidates distinct. Consequently Y is an independent set of cubic vertices in F-T with

$|Y| = k_c + k_p$.

Every candidate has at most three neighbors outside J and, by positivity, cannot meet both endpoints of any chosen cycle block.

The surviving vertices of blocks, namely those with no Z-neighbor, are also cubic in F-T. Choosing one from each surviving block gives an independent set of size $q + b - k_c - k_p$. Since $\alpha(C(T)) \leq h$ by the definition of h, the integer

$\eta = k_c + k_p - 2d$

is nonnegative.

4. Disjoint pairs on clean killed cycles.

Call a killed cycle bad if Z has a neighbor among its unselected vertices outside the residual edge; call it good otherwise. Let r_{bad} and $g = k_c - r_{bad}$ be the numbers of bad and good killed cycles.

For a good killed cycle O, write its residual endpoints u_O, v_O and let $K_O = V(O)$ minus $(Junion\{u_O, v_O\})$. The unique-edge description shows that K_O is independent, has $s_O - 1$ vertices for $|V(O)| = 2s_O + 1$, and is anticomplete to both endpoints. Both endpoints have unique Z-neighbors by step 3, and these neighbors are different by the protected-edge choice.

We show that the resulting two-element Z-pairs for distinct good cycles are disjoint. Fix two good cycles O_1, O_2. Put $K_\cdot = K_{O_1}$ union K_{O_2}, $J_{out} = J$ minus $(V(O_1)unionV(O_2))$, and

$X_\cdot = \{x \in X : N_F(x)avoidsJ_{out}unionK_\cdot\}$, $P_\cdot = X_\cdot$ minus Z.

Goodness gives $Z \subseteq X_\cdot$. The set $P_\cdot$ union $K_\cdot$ is independent and disjoint from the maximum independent set T. Its neighbors in T lie only in J intersect $(V(O_1)unionV(O_2))$.

For any independent Q disjoint from a maximum independent set M, replacing $N_F(Q)$ intersect M in M by Q proves $|Q| \leq |N_F(Q)$ intersect $M|$. Applying this inequality here yields

$|P_\cdot| + (s_{O_1} - 1) + (s_{O_2} - 1) \leq s_{O_1} + s_{O_2}$,

so $|P_\cdot| \leq 2$.

Choose one residual endpoint on each of O_1, O_2. Together with J_{out} union $K_\cdot$ these endpoints form a maximum independent set of R. All X-vertices outside $X_\cdot$ already meet J_{out} union $K_\cdot$, so the extension law says the two chosen endpoints hit exactly $|P_\cdot|$ members of $X_\cdot$. One can choose their unique Z-owners distinct, since each cycle has two distinct owners. Such a choice hits at least two vertices, giving $|P_\cdot| = 2$. Moreover, either endpoint on either cycle can be paired with an endpoint on the other having a different owner. In each such choice exactly two members of Z are hit and no member of $P_\cdot$ is hit. Thus $P_\cdot$ avoids all four residual endpoints.

If the two owner pairs shared a vertex, choosing both endpoints owned by that vertex would hit only one member of Z and none of $P.$, contradicting the required two hits. The pairs are therefore disjoint. Since the two cycles were arbitrary, the g good cycles use 2g distinct Z-vertices. Thus $z_0 = d - 2g$ is nonnegative.

5. The exact incidence count.

Every killed cycle contributes two distinct actual edges from Z, one to each residual endpoint. Every killed path contributes one such edge. Every bad killed cycle contributes at least one further edge to an unselected vertex outside the residual block. Choose exactly one edge for each required incidence. These are $2k_c + k_p + r_{bad}$ distinct edges incident with Z. All neighbors of Z lie in R, and four-regularity gives exactly 4d such edges. Therefore

$\tau = 4d - 2k_c - k_p - r_{bad} \geq 0.$

Every Z-edge to a surviving odd cycle is outside this chosen collection, so there are at most τ such edges.

Put $u = b - k_p$, which is nonnegative. Since $k_c = 2d - k_p + \eta$ and $r_{bad} = k_c - g$, direct substitution using $2g = d - z_0$ gives

$4b - 3d = 4u + 6\eta + z_0 + 2\tau.$ (1)

Let B_{sur} be all block vertices with no Z-neighbor. Its induced graph in $C(T)$ consists of edges and isolates. Each surviving cycle block contributes an edge if neither endpoint has a Z-neighbor, and an isolate if exactly one does. Each surviving path block contributes an isolate. Let f count the cycles of the latter singleton type, let p count these edge components, and let I be the isolate set. Each of the f cycles is surviving and has a Z-edge, so $f \leq \tau$ by the preceding incidence bound. Therefore

$|I| = f + u,\ p + |I| = q + b - k_c - k_p = h - \eta.$ (2)

Let Y_{hit} consist of the candidates in Y having an I-neighbor. A cycle singleton has degree two in R and hence at most two neighbors in $Y \subseteq W$. A path singleton has degree one in R and hence at most three neighbors in Y. Counting the actual incidences from I to Y gives

$|Y_{hit}| \leq 2f + 3u \leq 2\tau + 3u.$ (3)

6. Averaging and saturation.

Put $Y_0 = Y$ minus Y_{hit}. Select one endpoint uniformly and independently from each of the p edges of $C(T)[B_{sur}]$, and include all vertices of I. This is a finite distribution on 2^p choices, with one choice if $p = 0$. Each resulting set Q is independent of size $p + |I| = h - \eta$.

A vertex y in Y_0 has no I-neighbor, has at most three neighbors among the selected matching's endpoints, and meets at most one endpoint of each matching edge. The probability that Q avoids every neighbor of y is therefore at least $1/8$. Let Y_Q be the subset of Y_0 avoided by Q. Since Y is independent, Q union Y_Q is independent in $C(T)$. Thus $|Y_Q| \leq \eta$ for every choice, because $\alpha(C(T)) \leq h$. Averaging and adding the individual occurrence probabilities gives

$|Y_0|/8 \leq average\ |Y_Q| \leq \eta.$

No independence between different candidates is needed for this sum.

Combining $|Y| = 2d + \eta$ with (3) yields

$2d + \eta \leq 8\eta + 2\tau + 3u,$

and hence $2d \leq 7\eta + 2\tau + 3u$. The nonnegative quantities in (1) satisfy

$6(7\eta + 2\tau + 3u) = 42\eta + 12\tau + 18u$

$\leq 42\eta + 14\tau + 28u + 7z_0$

$= 7(4b - 3d).$

It follows that $12d \leq 28b - 21d$, or $33d \leq 28b$. This proves the more precise assertion without using $d \geq b$.

Under all the theorem's assumptions, $d > 0$ also gives $d \geq b$. Thus $33d \leq 28d$, impossible for positive d. Consequently $d = 0$ and $h = m$. An attaining frame exists by the finite maximization at the start. Assumptions (i) and (ii) now give $\alpha(R) = a$ and $q = m - b$. Any maximum independent set of this R is a maximum independent set of F and is disjoint from S. This proves the asserted existence and all conclusions. □

Remark. An attaining terminal already has maximum independence in every orientation's complement, forcing an exact extension law. Protected residual edges create distinct cubic replacements when killed. Disjoint clean owner pairs and the full incidence count leave too little room for those replacements to avoid increasing the global cubic maximum.

Depends on. Nothing earlier; this item is self-contained.

Lemma 32 (Density of a complete natural profile). *Let S, T be positive integers and let $n_1 \geq \cdots \geq n_S \geq 1$ and $m_1 \geq \cdots \geq m_T \geq 1$ be integer lists. Let F, J denote the vertex-disjoint unions of stars with these complete edge-size lists, and write $e(F) = \sum_i n_i$ and $e(J) = \sum_j m_j$. Put $M = S + T - 1$ and $\ell_l = max\{n_i + m_j - 1 : 1 \leq i \leq S, 1 \leq j \leq T, i + j = l\}$ for every integer $2 \leq l \leq S + T$. Let q be an integer with $0 \leq q < M$, and put $K = M - q$, $L_q = \sum_{l=q+2}^{S+T} \ell_l$, $P_q = L_q - K$, $D = \ell_{q+2} - 1$, and $Z = \max(e(F), e(J))$. Assume $D \geq 3$ and $n_1, m_1 \leq D$. Let R be the length of the initial $D + 1$ plateau of the sequence ℓ_{q+2},..., ℓ_{S+T}.*

Orient the two complete lists so that $a = e(F) - S \leq b = e(J) - T$. Let $u = |\{i : n_i = 1\}|$. Define $\omega = 0$ when $m_1 < D$; when $m_1 = D$, put $f = |\{j : m_j = D\}|$ and $\omega = \max(u + f - T, 0)$. Then $P_q \leq (D-1)(Z-q-1) + R + \omega$. In particular the uncorrected bound $P_q \leq (D-1)(Z-q-1) + R$ holds if $m_1 < D$, or if $m_1 = D$ and $u + f \leq T$.

More precisely, for positive integer thresholds x, y define $N_x = |\{i : n_i \geq x\}|$ and $B_y = |\{j : m_j \geq y\}|$. For every positive integer z let $V_z = \max(\{N_x + B_y - 1 : x, y \geq 1 integers, x + y = z + 2, N_x > 0, B_y > 0\} union \{0\})$. Then $P_q = \sum_{z=1}^{D}(V_z - q)$, $V_D = q + R$, and

$V_z \leq Z - 1 + \max(B_{z+1} - a, 0)$ *for* $1 \leq z \leq D - 2$,

$\sum_{z=1}^{D-2} \max(B_{z+1} - a, 0) \leq b - a$,

$V_{D-1} \leq Z - 1 - (b - a) + \omega$.

All counts, indices, sizes, thresholds, a, b, D, R and ω are integers in their stated nonnegative or positive domains. Unit entries and all repeated or intermediate entries are retained. No graph arrowing, matching, degree, or component hypothesis is imposed.

Proof. The set defining every ℓ_l is nonempty. The sequence is nonincreasing: for a candidate $n_i + m_j - 1$ on a diagonal $l \geq 3$, at least one index exceeds one. Decreasing that index produces a valid candidate on diagonal l-1 with value at least as large. Taking the maxima proves $\ell_{l-1} \geq \ell_l$. The integers ℓ_l are positive, so the stated initial plateau exists and $1 \leq R \leq K$.

For any $z \geq 1$, a diagonal has $\ell_l - 1 \geq z$ precisely when it contains indices in a rectangle $1 \leq i \leq N_x$, $1 \leq j \leq B_y$ for positive integers x, y with $x + y = z + 2$ and both counts nonzero. In one direction entries in that rectangle have size sum at least $z + 2$. Conversely, if $n_i + m_j \geq z + 2$, positive integers $x \leq n_i$ and $y \leq m_j$ with $x + y = z + 2$ exist: choose $x = \max(1, z + 2 - m_j)$ and $y = z + 2 - x$. The positive index sums in a nonempty rectangle fill all integers from 2 through $N_x + B_y$. Therefore the number of full-profile entries with $\ell_l - 1 \geq z$ is V_z.

After removal of the first q positions of this nonincreasing sequence, the number remaining at level z is $\max(V_z - q, 0)$. Its largest entry is exactly D. Thus $V_z > q$ for $1 \leq z \leq D$, and $V_z \leq q$ for $z > D$. Summing the threshold incidences of every remaining nonnegative integer $\ell_l - 1$ gives $P_q = \sum_{z=1}^{D}(V_z - q)$. Exactly the first R tail entries have excess D, so $V_D - q = R$.

The full edge excesses satisfy $a = \sum_{x \geq 2} N_x$ and $b = \sum_{y \geq 2} B_y$. Both sums are finite and a, b are nonnegative integers. Fix $1 \leq z \leq D - 2$. A candidate with $x = 1$ has width $S + B_{z+1} - 1 = e(F) - 1 + B_{z+1} - a \leq Z - 1 + \max(B_{z+1} - a, 0)$. A candidate with $x \geq 2$ has $N_x \leq a$ and $B_y \leq T$, hence width at most $a + T - 1$. Because $a \leq b$, this is at most $b + T - 1 = e(J) - 1 \leq Z - 1$. The zero included in the definition of V_z also satisfies the claimed bound, since $Z \geq 1$. Maximizing proves the displayed estimate for V_z.

Let I be the set of integers y with $2 \leq y \leq D - 1$ and $B_y > a$. If I is empty, the sum of the positive errors is zero, at most b-a. Otherwise $\sum_{y \in I}(B_y - a) \leq b - |I|a \leq b - a$, because $\sum_{y \in I} B_y \leq b$, $|I| \geq 1$ and $a \geq 0$. This proves the full positive-error bound, including $a = 0$.

At level $z = D - 1$, the threshold pairs satisfy $x + y = D + 1$. Every pair with $x \geq 2$ again has width at most $a + T - 1$. The only other possible pair is $x = 1$, $y = D$. It is absent if $m_1 < D$. If $m_1 = D$ it has width $S + f - 1$, and $S = u + N_2$ with $N_2 \leq a$. Therefore $S + f - 1 \leq a + u + f - 1 \leq a + T - 1 + \omega$. This last upper bound is nonnegative because $T \geq 1$ and a, $\omega \geq 0$. Hence it bounds the included zero as well. In both cases $V_{D-1} \leq a + T - 1 + \omega = e(J) - 1 - (b - a) + \omega \leq Z - 1 - (b - a) + \omega$.

Sum the bounds for levels 1 through D-2 and then the bound for D-1. The positive errors total at most b-a, and the last bound subtracts exactly b-a. Consequently $\sum_{z=1}^{D-1} V_z \leq (D - 1)(Z - 1) + \omega$. Subtract (D-1)q and add $V_D - q = R$. The threshold expression for P_q then yields $P_q \leq (D - 1)(Z - q - 1) + R + \omega$. The two asserted uncorrected cases are exactly the cases $\omega = 0$. All sums and maxima used the complete lists, so no lower entry or multiplicity was discarded. □

Remark. Lower levels can exceed the full edge budget only on one threshold axis. Their total error is bounded by the difference of the two full edge excesses; the next-to-top level returns that difference, leaving only the explicit unit-versus-head correction.

Depends on. Nothing earlier; this item is self-contained.

Lemma 33 (Monochromatic exhaustion of actual suffix families). *Let $s, t \geq 1$ be integers and let $n_1 \geq \cdots \geq n_s \geq 1$ and $m_1 \geq \cdots \geq m_t \geq 1$ be integer lists. Let F_1 and F_2 be the vertex-disjoint unions of stars K_{1,n_i} and K_{1,m_j}, respectively; the first target is prescribed red and the second blue. All graphs are finite and simple, and target copies are ordinary subgraphs with vertex-disjoint components. A graph arrows an ordered pair if every red-blue edge coloring contains its entire prescribed red or blue target.*

Put $M = s + t - 1$. For an integer $0 \leq q < M$ let A_q consist of all pairs $(F_1^{\alpha}, F_2^{\beta})$ with integer cuts $0 \leq \alpha < s$, $0 \leq \beta < t$ and $\alpha + \beta = q$. The superscript deletes exactly the indicated number of leading whole components. Put $K = M - q$. For a graph H let $\nu(H)$ be its maximum matching cardinality and $\Delta(H)$ its maximum degree, with value zero when H has no edges.

If H arrows every member of A_q, then

$\nu(H) \geq \min(s, K)$ *and* $\nu(H) \geq \min(t, K)$.

If in addition the integer $\delta = K - \nu(H)$ is positive, then

$q + \delta \leq \min(s, t) - 1$, $q \leq \min(s, t) - 2$, *and* $\nu(H) \geq \max(s, t)$.

The same H contains each entire original target F_1 and F_2 as an uncolored subgraph, and $n_1, m_1 \leq \Delta(H)$.

Without assuming arrowing, suppose $\delta = K - \nu(H) \geq 1$ and $q + \delta \geq \min(s, t)$. Then the following explicit monochromatic coloring avoids both targets of the specified legal nonempty pair in A_q:

if $s \leq t$, take $\alpha = \min(q, s - 1)$, $\beta = q - \alpha$, and color all H-edges blue;

if $t < s$, take $\beta = \min(q, t - 1)$, $\alpha = q - \beta$, and color all H-edges red.

No edge-budget or degree hypothesis is needed. Empty H, isolated vertices, unit stars, $q = 0$, $s = 1$ and $t = 1$ are included.

Proof. Each positive star contains an edge. Selecting one edge from each component of any target copy gives a matching with as many edges as that target has components.

First suppose H arrows every A_q member. Take $\alpha = \max(0, q - t + 1)$ and $\beta = q - \alpha$. If $q \leq t - 1$, these cuts are $\alpha = 0$ and $\beta = q$; the red suffix has s components, and $K = s + t - 1 - q \geq s$. If $q \geq t$, the cuts are $\alpha = q - t + 1$ and $\beta = t - 1$; $q \leq s + t - 2$ implies $\alpha \leq s - 1$, and the red suffix has $s - \alpha = K$ components. Thus in both cases the cuts are legal and the red suffix has $\min(s, K)$ components. Color all H-edges red. Its positive blue target is absent, so the red suffix must occur and yields $\nu(H) \geq \min(s, K)$.

For the other inequality take $\beta = \max(0, q - s + 1)$, $\alpha = q - \beta$. If $q \leq s - 1$, $\beta = 0$ and $\alpha = q$, and the blue suffix has t components with $K \geq t$. If $q \geq s$, $\beta = q - s + 1$ and $\alpha = s - 1$, and $q \leq s + t - 2$ implies $\beta \leq t - 1$; the blue suffix has $t - \beta = K$ components. The cuts are again legal. An all-blue coloring has no positive red target, so arrowing forces a blue copy and gives $\nu(H) \geq \min(t, K)$.

Now suppose $\delta = K - \nu(H) \geq 1$. Since $\nu(H) < K$, neither of these two minima can equal K. Therefore $s < K$, $t < K$ and $\nu(H) \geq s, t$. Substitution gives

$q + \delta = M - \nu(H) \leq s + t - 1 - \max(s,t) = \min(s,t) - 1.$

Because $\delta \geq 1$, also $q \leq \min(s,t) - 2$. Both cuts $(\alpha, \beta) = (q, 0)$ and (0,q) are legal. At the first cut, color every H-edge blue; arrowing forces the entire original F_2. At the second cut, color every H-edge red; arrowing forces the entire original F_1. Forgetting these respective colors proves the two uncolored containment assertions. Each largest copied star has a vertex incident with its n_1 or m_1 edges, respectively, so $n_1, m_1 \leq \Delta(H)$.

For the constructive avoidance assertion no arrowing is assumed. Suppose first $s \leq t$. If $q \geq s - 1$, the prescribed cuts are $\alpha = s - 1$ and $\beta = q - s + 1$. They satisfy $\beta \geq 0$ and $\beta \leq t - 1$ because $q \leq s + t - 2$. Their blue target has $t - \beta = K > \nu(H)$ positive components. It cannot occur: one edge per component would form a matching larger than $\nu(H)$. The red suffix is nonempty and has a positive edge, so it is absent in the all-blue coloring. If $q < s - 1$, the cuts are $\alpha = q$ and $\beta = 0$, and both suffixes are nonempty. The assumption $q + \delta \geq s$ gives

$\nu(H) = s + t - 1 - q - \delta \leq t - 1.$

The entire original blue target has t components and is absent by the matching bound, while the positive red suffix is absent because every edge is blue.

Finally suppose $t < s$. If $q \geq t - 1$, choose $\beta = t - 1$ and $\alpha = q - t + 1$. Then $0 \leq \alpha \leq s - 1$, and the red target has $s - \alpha = K > \nu(H)$ components. It is absent in the all-red coloring by the matching bound; the positive blue suffix is absent because there are no blue edges. If $q < t - 1$, choose $\beta = q$ and $\alpha = 0$. The inequality $q + \delta \geq t$ gives $\nu(H) \leq s - 1$, so the entire original red target with s components is absent. The positive blue suffix is again absent in the all-red coloring. These four cases prove the stated witness and handle every endpoint without introducing an empty target. □

Remark. Monochromatic corner colorings turn the number of target components into a matching lower bound. A positive deficit keeps both actual suffix corners away from exhaustion and makes them retain the whole opposite target.

Depends on. Nothing earlier; this item is self-contained.

Lemma 34 (Vertex and matching deletion for whole star forests). *All graphs are finite and simple. A positive star forest is a nonempty vertex-disjoint union of stars $K_{1,d}$ with positive integer edge sizes d. Its component count is $c(F)$, its number of components with at least two edges is $r(F)$, and its once-shortened forest F^- is obtained by replacing each size $d \geq 2$ by d-1 and omitting each size-one component. The empty forest has an empty copy in every graph. Write $K_2 = K_{1,1}$ and $e(G) = |E(G)|$. Copies are ordinary subgraphs, and components of a target copy are vertex-disjoint. The statement $G \to (F, J)$ means that every red-blue coloring of $E(G)$ contains a red F or a blue J, with the same convention for an empty target.*

Let F, J have positive nonincreasing integer component sizes $n_1,\ldots, n_s$ and $m_1,\ldots, m_t$, respectively, where $s, t \geq 1$. For integers $0 \leq a < s$ and $0 \leq b < t$, let $F_{>a}$ and $J_{>b}$ be the forests remaining after omitting the a largest F-components and b largest J-components. For every $X \subseteq V(G)$ with $|X| = a + b$, vertex deletion retains the implication

$G \to (F, J)$ *implies* $G\text{-}X \to (F_{>a}, J_{>b})$.

More precisely, partition $X = X_R$ disjoint union X_B with $|X_R| = a$ and $|X_B| = b$. Any coloring of $G\text{-}X$ avoiding this suffix pair extends to a coloring of G avoiding (F, J) by making every edge incident with X_R red and every remaining edge incident with X_B blue.

Let F, J be positive star forests and let W be any matching of a graph G. Write $G\text{-}W$ for the graph obtained by deleting exactly the edges of W and retaining every vertex. If $G \to (F, J)$, then

$G\text{-}W \to (F^-, J)$ *and* $G\text{-}W \to (F, J^-)$.

Consequently,

$G\text{-}W \to (r(F)K_2, J)$ *whenever* $r(F) \geq 1$,

and $G\text{-}W \to (c(F)K_2, J^-)$ *whenever J^- is nonempty.*

The two analogous conclusions after interchanging F, J and their prescribed colors also hold. These conclusions apply simultaneously, with the same fixed W, to every pair in any family of positive star forests arrowed by G.

Each implication has a color-preserving contrapositive construction. An avoiding coloring of G-W for (F^-, J), or for $(r(F)K_2, J)$ when $r(F) \geq 1$, extends to an avoiding coloring for (F, J) by coloring all W-edges red. An avoiding coloring of G-W for (F, J^-), or for $(c(F)K_2, J^-)$ when J^- is nonempty, extends by coloring all W-edges blue. Every edge outside W retains its color.

In addition, suppose $n_s \geq 2$. Put $M = s + t - 1$ and, for each integer $2 \leq k \leq s + t$, define

$$\ell_k = max\{n_i + m_j - 1 : 1 \leq i \leq s, 1 \leq j \leq t, i + j = k\}.$$

Put $L = \sum_{k=2}^{s+t} \ell_k$. The corresponding diagonal values for the profiles $(n_1 - 1, \ldots, n_s - 1)$ and $(m_1, \ldots, m_t)$ are exactly $\ell_k - 1$, and their sum is L-M. If every finite simple graph Q satisfying $Q \to (F^-, J)$ has at least L-M edges, then every G satisfying $G \to$ (F, J) and admitting a matching W with $|W| \geq M$ has at least L edges. The same assertion holds with the two original colors interchanged. All matching sizes, profile indices, component counts and edge counts are nonnegative integers within their stated domains; W may be empty.

Proof. First consider vertex deletion and the displayed partition of X. In the extension, every edge incident with X_R is red. Hence a blue copy of J avoids X_R. At most b of its vertex-disjoint components meet X_B, so at least t-b whole blue component stars lie in G-X. List any t-b such components in their inherited nonincreasing order. Their h-th size is at least m_{b+h} for $1 \leq h \leq t - b$: among the first $b + h$ components of J, omitting at most b leaves at least h. Trimming extra leaves gives a blue $J_{>b}$ in G-X.

For the red color, any component star of a red F meeting X_B also meets X_R, since every red edge at a vertex of X_B has its other endpoint in X_R. Thus every component of that F that meets X meets X_R. At most a of its vertex-disjoint components meet X_R, so at least s-a whole red stars lie in G-X. Their h-th size is at least n_{a+h} for $1 \leq h \leq s - a$, by the same count of retained components among the first $a + h$ positions. Trimming gives a red $F_{>a}$. Either monochromatic original target would therefore contradict avoidance of the suffix pair. This proves the stated extension and its arrowing contrapositive, including $a = b = 0$.

Fix an arbitrary coloring of G-W, and color every edge of W red. A blue copy of J in this extension uses no edge of W and is already present in G-W. A red copy of F has at most one W-edge in each of its component stars: all edges of a star share a center, whereas distinct matching edges have no common endpoint. For each component of size $d \geq 2$, deleting W therefore leaves at least d-1 red spokes at that center. Keeping exactly d-1 such spokes and discarding every size-one component gives a red copy of F^- in G-W. The chosen stars remain vertex-disjoint because they lie in the corresponding vertex-disjoint original components. If there is no component of size at least two, F^- is empty and is present by convention. Since G arrows (F, J), these alternatives prove G-W $\to (F^-, J)$. Equivalently, if the original coloring avoids both F^- and J, then the red extension avoids F and J. This is the first asserted construction.

Coloring every W-edge blue gives the other construction. A red F then lies entirely in G-W. Any blue J loses at most one spoke from each component upon deletion of W, so shortening its nonunit components and discarding units gives a blue J^- in G-W. This proves G-W $\to (F, J^-)$ and the corresponding avoiding extension, including the empty-shortening convention.

When $r(F) \geq 1$, the forest F^- contains a matching of size $r(F)$: select one edge from each of its $r(F)$ nonempty components. Therefore any red F^- contains a red $r(F)K_2$, and G-W $\to (F^-, J)$ implies G-W $\to (r(F)K_2, J)$. If a coloring of G-W avoids this matching and J, it in particular avoids F^- and J; the first construction extends it as claimed.

The positive forest F contains a matching of size $c(F)$, obtained by choosing one edge in each original component, including every unit component. Therefore G-W $\to (F, J^-)$ implies G-W $\to (c(F)K_2, J^-)$. Avoidance of this matching and J^- implies avoidance of F and J^-; the blue extension then avoids F and J. This projection is stated when J^- is nonempty so that both projected targets are positive. If J^- is empty, the unprojected assertion remains valid but is automatically true.

Interchanging the words red and blue in these constructions interchanges the two target positions and yields all reversed-color assertions. The argument used no property of W beyond being a matching and made no choice of W depending on a target pair. It can therefore be applied to every member of an arrowed family while retaining the same W. This asserts each of the resulting arrowing statements, without requiring different avoiding colorings to be compatible.

For the diagonal assertion, the condition $n_s \geq 2$ ensures that all s numbers $n_i - 1$ remain positive and in nonincreasing order. Thus the valid index set for each diagonal is unchanged. It is nonempty: for $2 \leq k \leq s+t$, choose an integer i between $\max(1, k-t)$ and $\min(s, k-1)$, and put $j = k - i$. Both indices are valid. Subtracting one from every candidate in this finite set gives

$max\{(n_i - 1) + m_j - 1 : i + j = k\} = \ell_k - 1$.

There are exactly $s + t - 1 = M$ diagonals, so their sum is L-M.

Under the conditional lower-bound hypothesis, the first matching-deletion implication gives $Q = G - W \to (F^-, J)$, hence $e(G - W) \geq L - M$. Deleting W removes exactly $|W|$ distinct edges. Consequently $e(G) = e(G - W) + |W| \geq L - M + M = L$ whenever $|W| \geq M$. The profile condition ensures the shortened target is positive, as required by that conditional lower bound. Color exchange proves the symmetric version. No step requires W to be maximum or nonempty. □

Remark. Coloring deleted vertices by assigned sides removes only a bounded number of whole components. Coloring a deleted matching in one color removes at most one spoke from each star center, preserving the complete opposite target and every surviving component.

Depends on. Nothing earlier; this item is self-contained.

Lemma 35 (Covering matchings and degree splittings). *Let $D \geq 1$ be an integer and let G be a finite simple graph with maximum degree $\Delta(G) \leq D$. Write $d_G(v)$ for the degree of v and put $S_D = \{vinV(G) : d_G(v) = D\}$. A matching is an edge set with pairwise disjoint endpoints, and $V(M)$ is the set of endpoints of a matching M. The following conditions are equivalent:*

(a) There is a matching M of G with $S_D \subseteq V(M)$.

(b) For every pair of nonnegative integers A, B with $A + B = D$, the edge set $E(G)$ has a partition Q_R disjoint union Q_B such that the spanning graphs with these edge sets have maximum degrees at most A and B, respectively.

The construction in (a) implies (b) has the following simultaneous form. For every specified matching M satisfying (a), form two labeled copies G^0, G^1 of G and add $D - d_G(v)$ distinct edges between the corresponding vertices v^0, v^1, for each v in $V(G)$. The resulting finite loopless multigraph K is D-regular and has a perfect matching N whose restriction to either copy is exactly the specified M. This one graph K and this one matching N can be fixed before choosing A, B. For every integer A with $0 \leq A \leq D$, K has a spanning subgraph F_A with degree exactly A at every vertex, and restriction of F_A and its complement to G^0 gives the partition in (b). When $A = 1$ one may choose $F_A = N$, so the red edge set in G^0 is exactly M. Empty graphs and S_D empty are permitted; maximum degree of an edgeless graph is zero.

Proof. We give the factor construction, including the matching argument it needs. Throughout the auxiliary constructions, parallel edges are distinct instances. A pseudograph may also have loops, and each loop contributes two incidences to its vertex degree.

First consider a finite 2r-regular pseudograph P, where $r \geq 0$ is an integer. We construct a spanning subgraph of degree 2k for every integer $0 \leq k \leq r$. For $r = 0$ or $k = 0$ choose no edges. Otherwise construct an Euler circuit in each connected component. To justify this construction, a trail following unused edges cannot stop away from its initial vertex: its terminal vertex would have an odd number of used incidences and even total degree, leaving an unused incident edge. Hence it closes. Removing the used edges leaves even degrees. If edges of the connected component remain unused, connectedness supplies a vertex of the current closed trail incident with an unused edge. Build a closed trail of unused edges there and splice it into the previous trail. Each splice adds edges, so finiteness gives a circuit using all edges. This reasoning includes loops and parallel edges.

Orient the edges along these circuits. Every vertex has r incoming and r outgoing incidences. Construct a bipartite multigraph J with an outgoing and an incoming copy of each vertex of P. An oriented edge uv becomes an edge from the outgoing copy of u to the incoming copy of v. A loop in P gives an edge between two different copies of the same vertex. Thus J is loopless and r-regular, with equally many vertices in its two parts.

For every subset X of the left part, counting edge instances gives $r|X| \leq r|N_J(X)|$, where $N_J(X)$ is its set of right neighbors. Since $r > 0$, $|N_J(X)| \geq |X|$. Here is a direct proof that these inequalities yield a perfect

matching. Choose a matching of maximum cardinality. If a left vertex x is uncovered, consider vertices reachable from x by paths directed along nonmatching edges from left to right and matching edges from right to left. An uncovered reachable right vertex would give, along a shortest such path, an alternating path whose edge exchange increases the matching size. Thus every reached right vertex is covered. Write X′ and Y′ for the reached left and right sets. Their matching edges give a bijection from Y′ to X′ minus {x}; all neighbors of X′ belong to Y′, since nonmatching edges can be traversed and a matched left vertex's matching neighbor was reached immediately before it. Consequently $|N_J(X')| = |Y'| = |X'| - 1$, a contradiction. The matching covers the left part and therefore both parts. Remove it and repeat with the remaining (r-1)-regular bipartite multigraph, stopping at degree zero. This partitions J into r perfect matchings. Taking k of them and returning to P selects k incoming and k outgoing incidences at each vertex, and hence gives the required degree-2k subgraph.

We next establish the particular interpolation construction used for K. Let the edge instances of any finite pseudograph H be partitioned into E_0, E_1. Let $d_0(v)$, $d_1(v)$ be their degree contributions, and suppose $d_1(v) - d_0(v) = 2r$ at every vertex, for a nonnegative integer r. We construct an edge subset of degree $d_0(v) + 2k$ at each vertex for every integer $0 \le k \le r$. If $r = 0$, take E_0. For $r > 0$, at each vertex pair every E_0 incidence with a different E_1 incidence. Exactly 2r incidences remain unpaired there, all belonging to E_1.

Make a bipartite auxiliary multigraph L whose vertices are the individual edge instances in E_0 and E_1. Every paired pair of incidences creates an edge in L between their two original edge instances. Every E_0 vertex of L has degree two and every E_1 vertex has degree at most two. Thus each component of L is a cycle, a path with its two ends in E_1, or a single E_1 vertex. The classification follows by tracing the unique continuation at degree-two vertices; a component without a degree-one vertex is a cycle unless it is isolated. An isolated E_1 vertex is treated as a path having two unpaired original incidences.

Each path describes an alternating trail of distinct edge instances in H, starting and ending with E_1 edges. Successive instances join at the vertex of their paired incidences. Its two end incidences are unpaired. A trail may revisit a vertex or have equal endpoints. Distinct path components use disjoint original edge instances. Replace each path by one edge between the original vertices of its unpaired end incidences. The resulting pseudograph P on $V(H)$ is 2r-regular, since each vertex has exactly 2r unpaired incidences. Equal endpoints give a loop. Choose a degree-2k spanning subgraph of P by the preceding construction. Starting from E_0, reverse membership on the alternating trail represented by each chosen edge of P. The operations are consistent because their edge instances are disjoint. At every paired incidence, one E_0 incidence is removed and one E_1 incidence is added, with total change zero. Every chosen unpaired end incidence increases the degree by one. There are exactly 2k chosen end incidences at each vertex, counting both ends of a loop. The final edge subset therefore has degree $d_0(v) + 2k$ at every vertex. Cycle components of L are left unchanged.

Now fix a matching M satisfying (a), and construct K as in the statement. Every vertex of K has degree D. Put both copies of M into N. For each v not covered by M, add one of the new edges v^0v^1 to N. Such an edge exists: v is not in S_D, so $D - d_G(v) \ge 1$. These added matching edges join distinct pairs of uncovered corresponding vertices. Hence N is a perfect matching of K and restricts to M in both copies. Its construction has made no choice of A or B.

Fix an integer A with $0 \le A \le D$. If D is even and A is even, apply the interpolation construction to $H = K$, E_0 empty, $E_1 = E(K)$, with $r = D/2$ and $k = A/2$. If D is even and A is odd, then $1 \le A \le D - 1$; take $H = K$, $E_0 = N$, $E_1 = E(K)$ minus N, $r = (D-2)/2$, and $k = (A-1)/2$. The two input degrees are 1 and D-1, and the output degree is A.

If D is odd, K-N is (D-1)-regular. For even A, use $H = K - N$, E_0 empty, $E_1 = E(K)$ minus N, $r = (D-1)/2$, and $k = A/2$. For odd A, first construct in K-N the even degree D-A subgraph, with $k = (D-A)/2$, and take its complement in K. In every case the displayed k is an integer in [0,r]. Cases with $r = 0$ use E_0 as specified in the interpolation construction. This includes $D = 1$. We have therefore constructed F_A for every required A. For $A = 1$, N itself can be used instead.

Color F_A red and the remaining edges of K blue. The degrees in K are exactly A and D-A. Restrict both colors to the original edge instances of G^0. Restriction can only decrease color degrees, proving (b) with $B = D - A$, while the previously fixed K and N remain the same for every capacity choice.

Conversely, assume (b) and select $A = 1$ and $B = D - 1$. The red edge set is a matching. At a vertex of G-degree D, blue degree at most D-1 forces a red incident edge. This matching covers S_D, proving (a). If G has no vertices, every construction above uses empty sets and the assertions hold. If G has vertices but no edges, the same degree and matching constructions apply because D is positive. □

Remark. Complete the given covering matching to a perfect matching after doubling the graph and filling its degree deficits. Alternating trails allow interpolation between degree prescriptions of the same parity, and regular bipartite matching decomposition selects the required number of trail endpoints.

Depends on. Nothing earlier; this item is self-contained.

Lemma 36 (A largest-star block forced by a tight vertex cover). *All graphs are finite and simple. A star $K_{1,j}$ has j edges, and K_2 is one edge. A positive-edge star forest is a vertex-disjoint union of stars of positive integer sizes. Copies are ordinary subgraphs, with different target components vertex-disjoint. Write Q arrows(F, J) if every red-blue coloring of $E(Q)$ contains a red copy of the entire F or a blue copy of the entire J. A matching has disjoint endpoints; $\nu(Q)$ is its maximum cardinality. A vertex cover meets every edge; $\tau(Q)$ is its minimum cardinality. An independent set has no adjacent pair. Write $d_Q(v)$, $\Delta(Q)$, $N_Q(X)$, and $N_Q[X] = X$ union $N_Q(X)$ for vertex degree, maximum degree, the vertices with a neighbor in X, and closed neighborhood. Vertex deletion is induced.*

Let $d \geq 3$, $p \geq 1$, $r \geq 1$ and $t_0 \geq 0$ be integers. Let J_0 be any positive-edge star forest with exactly t_0 components, all of size less than d; when $t_0 = 0$ it is the empty graph. Put $J = rK_{1,d}$ disjoint union J_0, $R = p + r - 1$, and $N = p + r + t_0 - 1$. Suppose $\Delta(Q) \leq d$, Q arrows(pK_2, J), and every independent set of degree-d vertices of Q has size at most R. Fix any vertex cover C of Q with $|C| = N$ and define

$A = \{vinC : d_Q(v) = dandN_Q(\{v\})intersectCisempty\}$, $K = Q - A$.

Then A is independent, $|A| = R$, and its closed Q-neighborhoods are pairwise disjoint. Also $\Delta(K) \leq d - 1$ and $\nu(K) = \tau(K) = t_0$. For every r-element subset X of A, the graph $Q - (AunionN_Q(X))$ contains the entire J_0, with every one of its components retained. For every vertex cover C' of K with $|C'| = t_0$, no Q-edge joins A to C'.

If, in addition, every vertex of K-degree d-1 belongs to some $t_0 - element$ vertex cover of K, then A is exactly the entire set of vertices of Q-degree d. This extra condition holds whenever every sequence that repeatedly deletes a current maximum-degree vertex of K, stopping when no edge remains, has exactly t_0 positive-degree deletions. Such sequences are finite, and any vertex of K-degree d-1 may be chosen first when one exists.

All degrees, matching and cover sizes, component counts, set cardinalities and deletion lengths are nonnegative integers in the displayed domains. The smaller target J_0, its lower graph K, and the relevant lower covers may be empty. No edge-count, matching-deficit, minimum-cover, or lower-profile assumption on Q is imposed.

Proof. Put $t = r + t_0$. Thus J has exactly t positive-edge components and $N = p + t - 1$. We first prove $|A| = R$. Let $a = |A|$ and suppose $a < R$. Choose a set C_R of p-1 vertices of C containing $\min(a, p - 1)$ members of A. There are enough vertices for this choice because $|C| = N \geq p - 1$. Color every edge meeting C_R red and every remaining edge blue. A red matching has at most p-1 edges, since C_R covers the red graph.

The blue graph is covered by $Z = C$ minus C_R, which has $N - p + 1 = t$ vertices. Any blue copy of the full J consists of t vertex-disjoint positive-edge components. Each uses at least one vertex of Z, so each must use exactly one and together they use all of Z.

Consider a size-d star in such a copy. Its center cannot lie outside Z: a blue-active center outside Z lies outside C and all its neighbors lie in Z, so the d distinct leaves would use at least three cover vertices. Vertices of C_R have no blue edge. If the center lies in Z, attaining blue degree d under $\Delta(Q) \leq d$ forces its degree in Q to be d and every incident Q-edge to be blue. It has no neighbor in C_R, and it cannot have a neighbor in Z, since that would make its star use two cover vertices. Hence its center must lie in A minus C_R. There are exactly $\max(0, a - p + 1) < r$ such vertices, too few for the r largest stars. This contradicts arrowing and proves $a \geq R$.

No vertex of A has a neighbor in C, so A is independent. Its vertices all have degree d. The assumed bound on independent degree-d sets gives $a \leq R$, proving equality.

Put $C_0 = C$ minus A. It has $N - R = t_0$ vertices and covers $K = Q - A$. There are no Q-edges between A and C_0. If a vertex z of K had K-degree d, then $\Delta(Q) \leq d$ would imply that it has Q-degree d and no neighbor in A. The set A union {z} would be an independent degree-d set of size $R + 1$, a contradiction. Thus $\Delta(K) \leq d - 1$. In particular, vertices of C_0 have Q-degree at most d-1 because they have no neighbor in A.

Fix X contained in A with $|X| = r$. Color every edge meeting A minus X red and every other edge blue. The red graph has the cover A minus X, of size $R - r = p - 1$, so it has no p-edge matching. The blue graph has

the cover $Z = C_0$ union X of size $t_0 + r = t$. A forced full blue copy of J must again use exactly one Z vertex in each component.

No size-d center can lie outside Z, because then its d leaves all lie in Z; a vertex of A minus X is blue-isolated. No center can lie in C_0 because its Q-degree is at most d-1. Thus the r largest stars must be centered at all r vertices of X. Each such star uses the entire Q-neighborhood of its center. The vertices of A minus X are blue-isolated and cannot belong to any positive-edge target component. By disjointness of the components in the copy, the entire smaller forest J_0 lies in $Q - (A union N_Q(X))$.

Taking one edge from each of its t_0 components yields a matching of size t_0 in K. The cover C_0 of K has size t_0. Since every matching uses distinct vertices of any vertex cover, $\nu(K) \le \tau(K) \le |C_0| = t_0$. Together these inequalities give $\nu(K) = \tau(K) = t_0$, including $t_0 = 0$.

We next prove disjointness of the closed neighborhoods of A. Its vertices are already independent, so it suffices to exclude a common neighbor of two distinct vertices. If $p = 1$, then $R = r$ and the preceding argument with $X = A$ forces all r largest stars at A simultaneously. Their vertex-disjointness proves the claim.

Suppose $p \ge 2$ and distinct x, y in A have a common neighbor z. Choose p-2 vertices of A other than x, y and call their set B. This is possible because $|A| = p + r - 1 \ge p$. Color all edges meeting B and the two edges xz, yz red, and every other edge blue. Since A is independent, z lies outside A and hence outside C: vertices of A have no neighbor in C. The red graph is covered by B union {z}, a set of p-1 vertices, so it avoids pK_2.

The blue graph is covered by $Z = C$ minus B, of size $N - p + 2 = t + 1$. In a full blue copy of J, a size-d star centered outside Z would use $d \ge 3$ vertices of Z; each of the other t-1 positive-edge components would require another Z vertex. This would require at least $t + 2$ distinct cover vertices, more than $|Z|$. A center in B has no blue edges. A largest-star center must therefore lie in Z.

Vertices of C_0 have Q-degree at most d-1. The vertices x, y each lose at least one incident blue edge, and vertices in B are blue-isolated. Only the $R - (p - 2) - 2 = r - 1$ vertices of A minus (B union {x, y}) could remain as size-d centers. They are insufficient for the forced full J. The contradiction excludes a common neighbor, proving pairwise disjoint closed neighborhoods.

Now let C′ be any $t_0 - element$ vertex cover of K. Suppose some x in A has a neighbor in C′. Choose p-1 vertices of A other than x, possible because $R - 1 \ge p - 1$, and denote this set by B. Color every edge meeting B red and all others blue. Red is covered by B and avoids pK_2. The blue graph is covered by $Z = C'$ union (A minus B), because C′ covers all edges in K and the other retained edges meet A minus B. Its size is $t_0 + r = t$.

A forced full blue copy of J uses exactly one Z vertex per component. A size-d center outside Z would have all d leaves in Z, so is impossible. A center in C′ has K-degree at most d-1; if its blue degree is d, it must have a blue neighbor in A minus B, and its star then uses two Z vertices. Hence no vertex of C′ can be a usable largest-star center. The only possible centers are the r vertices of A minus B. But the star at x would use its whole Q-neighborhood and thus its neighbor in C′, again using two Z vertices. At most r-1 centers remain, a contradiction. Therefore no edge joins A to C′.

Finally suppose every vertex of K-degree d-1 belongs to a $t_0 - element$ cover of K. The conclusion just proved means that no such vertex has a neighbor in A. Pairwise disjoint closed neighborhoods of A imply that each vertex outside A has at most one neighbor in A. If such a vertex y had Q-degree d, the bound $\Delta(K) \le d - 1$ would force it to have precisely one neighbor in A and K-degree d-1. This contradicts the preceding absence of an A-neighbor. Consequently every degree-d vertex belongs to A.

For the asserted sufficient greedy condition, a sequence of current positive maximum-degree deletions exists until K becomes edgeless, because at each nonterminal step there is an edge and hence a vertex of positive maximum degree. Deleting distinct vertices gives termination after at most $|V(K)|$ steps. The deleted vertices form a cover of K: every original edge has a deleted endpoint when the remaining graph is edgeless. If z has K-degree d-1, then it is a maximum-degree vertex by $\Delta(K) \le d - 1$ and may be selected first. Extend this to any such finite sequence. Under the displayed greedy hypothesis, its deleted set has exactly t_0 vertices, covers K, and contains z. This proves membership in a $t_0 - element$ cover for every such z. If K is edgeless there are no vertices of degree d-1, since $d \ge 3$, and this membership requirement is vacuous. Every stated empty case is included. □

Remark. A full target with as many positive components as a surviving cover must use each cover vertex exactly once. This forces all largest stars onto a fixed block, separates their neighborhoods, and prevents that block from touching any smallest cover of the lower graph.

Depends on. Nothing earlier; this item is self-contained.

Lemma 37 (Finite-order bounds for marked domination). *Let n, d be integers with $d \geq 1$ and $n \geq d+1$. Let G be a finite simple graph on n vertices, let v be a specified vertex with degree at least d, and suppose every vertex other than v has degree at least d-1. Put $r_0 = n-1-d$ and, while $r_i > 0$, define $r_{i+1} = \max(0, r_i - ceil(dr_i/(n-1-i)))$. This recurrence reaches zero with positive denominators; let s be its first zero index. There is a set Z containing v with $|Z| \leq 1+s$ such that every vertex outside Z has a neighbor in Z. In addition, for every integer j with $1 \leq j \leq d-1$, every vertex set Z containing v and having exactly $n-d+j$ vertices has at least j neighbors in Z from every vertex outside Z. All sets in these assertions belong to the supplied graph.*

Proof. We first justify the recurrence and its monotonicity. For positive integers N, d and nonnegative integers r define $\phi_N(r) = \max(0, r - \lceil dr/N \rceil)$. If $d \geq N$, then $\lceil dr/N \rceil \geq r$ and ϕ_N is identically zero. If $d < N$, then $ceil(d(r+1)/N) \leq \lceil dr/N \rceil + 1$, so $(r+1) - ceil(d(r+1)/N) \geq r - \lceil dr/N \rceil$. Taking maxima with zero shows that ϕ_N is nondecreasing on the nonnegative integers in both cases. If $r_0 = 0$, the recurrence has $s = 0$ and uses no denominator. Otherwise, as long as the recurrence is defined with $r_i > 0$, its ceiling is at least one, hence $r_{i+1} \leq r_i - 1$. Inductively $r_i \leq r_0 - i$. Thus $r_i > 0$ implies $i < r_0 = n-1-d$ and $n-1-i > d \geq 1$. The next denominator is therefore positive. Each positive step strictly lowers a nonnegative integer, proving termination and $s \leq r_0$.

Start with $Z_0 = \{v\}$. For any current Z_i containing v with $|Z_i| = i+1$, let U_i consist of the vertices outside Z_i with no neighbor in Z_i. The degree assumption on v gives $|U_0| = n-1-deg_G(v) \leq n-1-d = r_0$. Suppose U_i is nonempty. For every u in U_i its closed neighborhood $N_G[u]$, consisting of u and its neighbors, is disjoint from Z_i. Also u is not v, so this neighborhood has at least d vertices. Count pairs (w, u) with u in U_i and w in $N_G[u]$. There are at least $d|U_i|$ such pairs, and every first coordinate belongs to $V(G)$ minus Z_i, which has n-1-i vertices. Some w outside Z_i therefore occurs in at least $ceil(d|U_i|/(n-1-i))$ pairs. Adjoin this w to Z_i. A vertex of U_i ceases to be undominated precisely when it is w or a neighbor of w, while no previously dominated vertex becomes undominated. Consequently $|U_{i+1}| \leq \phi_{n-1-i}(|U_i|)$. Induction and the proved monotonicity give $|U_i| \leq r_i$ at every step before stopping. At or before step s the undominated set is empty. Its corresponding Z has size at most $s+1$, contains v, and dominates every outside vertex. If already U_0 is empty, this is the set {v}.

For the second assertion put $L = n-d+j$. The assumptions imply $1 \leq L \leq n-1$, so sets of that size containing v exist. Let Z be any such set and let u be outside it. Then u differs from v and has at least d-1 neighbors among the n-1 other vertices. At most n-d of those other vertices are nonneighbors of u. Since all L vertices of Z are among them, u has at least $L-(n-d) = j$ neighbors in Z. This proves the assertion for every such Z, independently of the greedy construction. □

Remark. Counting closed neighborhoods gives a guaranteed greedy step at every finite order. At the smallest orders, simply choosing more vertices than any outside vertex can miss gives a second useful bound.

Depends on. Nothing earlier; this item is self-contained.

Lemma 38 (Fixed-cardinality marked domination). *Let n, d, j, r be integers with $2 \leq d \leq n-1$, $1 \leq j \leq d-1$, and $0 \leq r \leq n-1$. Let J be a finite simple graph with n vertices and a specified vertex v such that v has degree d and every other vertex has degree d-1. A set Z containing v is called a marked j-dominating set if every vertex outside Z has at least j neighbors in Z.*

For nonnegative integers a define $binom(a,b)$ to be the binomial coefficient when $0 \leq b \leq a$ and zero for all other integer b. An empty sum is zero. Put

$A = \sum_{i=0}^{j-2} binom(d-2,i)\ binom(n-d, r-i)$,

$B = \sum_{i=0}^{j-1} binom(d-1,i)\ binom(n-d-1, r-i)$,

$R = (dA + (n-1-d)B)/binom(n-1,r)$.

Then J has a marked j-dominating set Z with

$|Z| \le \min(n, 1 + r + \lfloor R \rfloor)$.

In particular, let H be a finite simple d-regular graph of odd order n, let v be a vertex, and let M be a perfect matching of H-v. Deleting the edges of M from H gives a graph J to which the asserted bound applies. The set Z is constructed in this particular graph H-M, for the specified vertex and matching.

Proof. Choose uniformly an r-element subset U of V(J)-{v}, and start with $Z_0 = \{v\}$ union U. A vertex u outside Z_0 is deficient when it has fewer than j neighbors in Z_0. Let $W(U)$ be the set of all deficient vertices and put $Z(U) = Z_0$ union $W(U)$. Each vertex outside $Z(U)$ already has at least j neighbors in Z_0, hence also in $Z(U)$. Thus $Z(U)$ is a marked j-dominating set and

$|Z(U)| = 1 + r + |W(U)|$.

There are $binom(n-1, r) > 0$ equally likely choices of U. Fix first a vertex u adjacent to v. There are d such vertices. For u to be deficient it must not belong to U. Among the n-2 possible elements remaining after v and u are excluded, exactly d-2 are neighbors of u: its degree is d-1 and one of its neighbors is v. The other n-d vertices are not neighbors of u. Since v already supplies one neighbor in Z_0, deficiency is equivalent to choosing at most j-2 of those d-2 neighbors. Consequently precisely A choices of U make this u deficient. When $j = 1$, it cannot be deficient, in agreement with $A = 0$.

Next fix a vertex u different from v and not adjacent to v. There are n-1-d such vertices. After u is excluded from U, the n-2 available vertices consist of d-1 neighbors of u and n-d-1 nonneighbors. Here v supplies no neighbor, so deficiency is equivalent to choosing at most j-1 of the d-1 neighbors. Precisely B choices of U make this u deficient.

Summing these indicator counts over all vertices gives

$E[|W(U)|] = (dA + (n-1-d)B)/binom(n-1, r) = R$.

This is a finite average. Some choice of U therefore has $|W(U)| \le R$, and integrality gives $|W(U)| \le \lfloor R \rfloor$. For that choice $|Z(U)| \le 1 + r + \lfloor R \rfloor$. Since $Z(U)$ is a subset of the n vertices, it also has size at most n. All upper arguments of binomial coefficients are nonnegative under the stated integer domains; the zero convention handles impossible choices and the endpoints $r = 0$ and $r = n - 1$.

Finally a perfect matching M of H-v is incident once with every vertex other than v and is not incident with v. Deleting its edges therefore leaves degree d at v and degree d-1 at every other vertex. The construction just proved applies directly to $J = H - M$ without changing the specified vertex or matching. □

Remark. Sampling a fixed number of vertices avoids the variance in the initial set size. Exact finite counting pays only for deficient vertices and respects the exposed vertex and its actual matching.

Depends on. Nothing earlier; this item is self-contained.

Lemma 39 (The saving from a forced dominating vertex). *For integers $r \ge 0$ and $j \ge 1$ and real p in [0,1], let $Bin(r, p)$ denote the sum of r independent Bernoulli variables with success probability p, and define $q_{r,j}(p) = p + (1-p)Pr(Bin(r, p) \le j - 1)$. Probabilities outside the support {0,...,r} are zero. Let G be a finite simple graph on $n \ge 1$ vertices, let v be a specified vertex, put $a = deg_G(v)$, and suppose every vertex other than v has degree at least an integer $m \ge 1$. Then for every integer $j \ge 1$ and real p in [0,1] there is a vertex set Z containing v such that every vertex outside Z has at least j neighbors in Z and $|Z| \le 1 + (n-1)q_{m,j}(p) - a(1-p)^2\ Pr(Bin(m-1, p) = j - 1)$. In particular, let $d \ge 2$ be an even integer and let H be any nonempty connected d-regular finite simple graph such that deletion of each vertex leaves a perfect matching. For every specified vertex v, every perfect matching M of H-v, every integer $1 \le j \le d - 1$, and every real p in [0,1], the spanning graph $G = H - M$ has a marked j-dominating set Z with $|Z| \le 1 + (|V(H)| - 1)q_{d-1,j}(p) - d(1-p)^2\ Pr(Bin(d-2, p) = j - 1)$, where marked j-dominating means containing v and giving at least j neighbors to each vertex outside Z.*

Proof. Sample a subset R of $V(G)$ minus {v} by including each vertex independently with probability p, and put $S = \{v\}$ union R. Let B consist of the vertices outside S with fewer than j neighbors in S, and put $Z = S$ union B. For every outcome, Z contains v. A vertex outside Z is outside S and not in B, hence has at least j neighbors in S and therefore in Z.

For u different from v, let $b = 1$ when u is adjacent to v and $b = 0$ otherwise, and let $r = deg_G(u) \geq m$. Given that u is not sampled, the number of its neighbors in S is $b + Bin(r - b, p)$. Simplicity makes inclusion of u independent of all these neighbor trials. Consequently $Pr(u \in Z) = p + (1 - p)Pr(Bin(r - b, p) \leq j - 1 - b)$. For $b = 0$, coupling the first m of the r trials bounds this probability by $q_{m,j}(p)$. For $b = 1$, the remaining trial count r-1 is at least m-1, and coupling bounds it by $q^- = p + (1 - p)Pr(Bin(m - 1, p) \leq j - 2)$. The conventions on support give this expression also when $j = 1$.

To compute the difference, let X have distribution $Bin(m - 1, p)$ and let Y be one further independent Bernoulli variable with success probability p. Then $Pr(X + Y \leq j - 1) = (1 - p)Pr(X \leq j - 1) + p\ Pr(X \leq j - 2)$. Subtract $Pr(X \leq j - 2)$ and multiply by 1-p to obtain $q_{m,j}(p) - q^- = (1 - p)^2\ Pr(X = j - 1)$. There are exactly a vertices adjacent to v and n-1-a unmarked vertices not adjacent to it. Summing their inclusion probabilities, and the certain inclusion of v, gives $E|Z| \leq 1 + a\ q^- + (n - 1 - a)q_{m,j}(p) = 1 + (n - 1)q_{m,j}(p) - a(1 - p)^2\ Pr(Bin(m - 1, p) = j - 1)$. The probability space has finitely many outcomes. At least one outcome of positive probability has size at most this expected-size upper bound, so its actual vertex set satisfies both required properties. The argument includes $p = 0$, $p = 1$, $m = 1$, $j = 1$, j beyond the trial support and $n = 1$.

For the last assertion, deleting the specified perfect matching M lowers the degree of every vertex other than v from d to d-1 and leaves the degree of v equal to d. Apply the first result to this graph with $a = d$ and $m = d - 1 \geq 1$. Its set Z has the required neighbor property in that same spanning graph H-M, and substitution gives the displayed bound. No compatibility between separately chosen sets for different j or p is required. □

Remark. Every neighbor of the forced vertex already has one required neighbor. The binomial mass at the boundary measures precisely the savings that this guaranteed neighbor contributes.

Depends on. Nothing earlier; this item is self-contained.

Lemma 40 (Missing matching allocations in a connected graph). *For a finite simple graph G, write $V(G)$, $E(G)$, $\Delta(G)$, and $\nu(G)$ for its vertex set, edge set, maximum degree, and maximum matching cardinality. A matching is an edge set with disjoint endpoints. The matching number of an edge subset means that of its spanning graph. Call a pair of nonnegative integers (x, y) feasible in G if $E(G)$ can be partitioned into two parts whose matching numbers are at most x and y, respectively. Define $c_0(G)$ to be the least nonnegative integer c such that every pair (x, y) of nonnegative integers with $x + y = c$ is feasible.*

Let $k \geq 1$ and $D \geq 3$ be integers. Let H be a connected finite simple graph on $2k + 1$ vertices with $\nu(H) = k$ and $\Delta(H) \leq D$. The number $c_0(H)$ exists. Put $E = c_0(H) - k$, and suppose $E \geq 1$. Put $v = \max(0, floor((2E - 3)/(D - 2)))$. Then $E \leq \lceil k/3 \rceil$. Every integer x with $0 \leq x \leq k + E - 1$ for which $(x, k + E - 1 - x)$ is infeasible satisfies

$$2E - 1 + v \leq x \leq k - E - v.$$

There is at least one such x. Define $tr(H)$ to be the maximum length of a consecutive run of these infeasible integer values of x. Then

$$1 \leq tr(H) \leq k - 3E + 2 - 2v$$

and $k \geq 3E - 1 + 2v$.

For every integer B, if $tr(H) + E \geq B + 1$, then $k \geq B + 2E - 1 + 2v$. Both coordinates of every infeasible pair obey the same interval. No regularity or minimality assumption is imposed.

Proof. We first give the two constructions used to color edges. Write $n = 2k + 1$.

If a set W of at most x vertices leaves at least $n - |W| - 2y - 1$ isolated vertices in H-W, assign all edges incident with W to the first part and all other edges to the second. A matching in the first part has at most $|W| \leq x$ edges, since its edges have distinct endpoints in W. The second part has at most $2y + 1$ nonisolated vertices, so its matching number is at most y. This proves feasibility. In particular, if $x \leq n$ and $n - x \leq 2y + 1$, any x-vertex W works, even when no isolated vertices remain.

For the second construction, fix any initial vertex of a connected graph on $n \geq 2$ vertices with maximum degree at most D. Begin with its singleton independent set I. If I does not dominate the graph, connectedness

gives an edge from its closed neighborhood I union $N(I)$ to a vertex z outside that neighborhood. The endpoint inside the neighborhood belongs to $N(I)$, since z has no neighbor in I. Thus z shares a neighbor with I. Adding z preserves independence and adds at most D-1 new vertices to the open neighborhood $N(I)$. Starting with at most D neighbors, induction gives

$|N(I)| \le (D-1)|I| + 1$

at every step. Each step adds a vertex, so this finite construction terminates in an independent dominating set of size r. Independence and domination imply $n = r + |N(I)| \le Dr + 1$, hence $r \ge ceil((n-1)/D)$. Consequently for every integer $1 \le s \le ceil((n-1)/D)$ a prefix gives an independent s-set with at most $(D-1)s + 1$ neighbors.

We next bound $c_0(H)$ without using connectedness. Put $e = \lceil k/3 \rceil$ and $c = k + e$. Consider any nonnegative integer pair of total c, and exchange its coordinates if needed to arrange $x \le y$. Since $e \le k$, $x \le \lfloor c/2 \rfloor \le k < n$. Also $c \ge 4k/3$ implies

$x \le c/2 \le 2c - 2k$.

Therefore $n - x \le 2(c - x) + 1 = 2y + 1$, and the first construction proves feasibility. This holds for every pair of total c. The set of permissible totals is thus a nonempty set of nonnegative integers and has a least element $c_0(H)$. At a permissible total c′, the pair (0,c′) forces $c' \ge \nu(H) = k$. Hence E is a nonnegative integer and $E \le \lceil k/3 \rceil$.

Set $\ell = k + E - 1$ and $L = 2E - 2 + v$. We prove that every nonnegative integer pair (x, y) with $x + y = \ell$ and $\min(x, y) \le L$ is feasible. Exchange the coordinates so that $x \le y$ and $x \le L$. We have $x \le \lfloor \ell/2 \rfloor < n$ because $E \le \lceil k/3 \rceil \le k$.

If $x \le 2E - 2$, the identity

$n - x - 2y - 1 = x - 2E + 2$

has a nonpositive right side. The first construction with any x-vertex set proves feasibility.

For the remaining case put $s = x - 2E + 2$. Then $1 \le s \le v$, so $v \ge 1$ and $E \ge 2$. The definition of v yields $(D-2)s \le 2E - 3$; therefore

$(D-1)s + 1 \le s + 2E - 2 = x$.

We check the size range for the independent-set construction. Since $E \le \lceil k/3 \rceil \le (k+2)/3$,

$v \le (2E-3)/(D-2) \le (2k-5)/(3(D-2)) < 2k/D$.

The final inequality follows after multiplication by the positive denominators from

$6k(D-2) - D(2k-5) = 4k(D-3) + 5D > 0$.

Thus $s \le \lceil 2k/D \rceil$. The second construction supplies an independent s-set I with $|N(I)| \le (D-1)s + 1 \le x$. Its neighborhood is disjoint from I. Enlarge $N(I)$ to an x-vertex set W avoiding I. There are enough vertices because

$n - s = x + 2y + 1 \ge x$.

Every vertex of I is isolated in H-W. Since $s = n - x - 2y - 1$, the first construction again proves feasibility.

We have proved feasibility whenever the smaller coordinate is at most L. Thus each coordinate of an infeasible pair at total ℓ is at least $L + 1 = 2E - 1 + v$. Subtracting this lower bound for y from ℓ gives $x \le k - E - v$, proving the interval.

Since $E \ge 1$, the integer $c_0(H) = k + E$ is positive. At least one pair at total $c_0(H) - 1$ is infeasible; otherwise that smaller nonnegative total would contradict the definition of $c_0(H)$. Therefore the finite set of infeasible x values is nonempty and has a longest consecutive run of positive integer length $tr(H)$. All such values lie in the integer interval just proved, whose length is $k - 3E + 2 - 2v$. This proves both run bounds and $k \ge 3E - 1 + 2v$. Finally $tr(H) + E \le k - 2E + 2 - 2v$. Combining this with $tr(H) + E \ge B + 1$ and rearranging proves the last assertion. □

Depends on. Nothing earlier; this item is self-contained.

Lemma 41 (Canonical pools and finite potential reduction). *Let s, t be positive integers and let $n_1 \geq \cdots \geq n_s \geq 1$ and $m_1 \geq \cdots \geq m_t \geq 1$ be integer lists. Let F_1 and F_2 be the vertex-disjoint unions of K_{1,n_i} and K_{1,m_j}, respectively. All copies are ordinary subgraph copies, with components of each target vertex-disjoint; F_1 is prescribed red and F_2 blue. Write $J \to (F_1, F_2)$ when every red-blue edge coloring of the finite simple graph J contains one of these full prescribed-color targets. Put $M = s + t - 1$, $\ell_k = max\{n_i + m_j - 1 : 1 \leq i \leq s, 1 \leq j \leq t, i + j = k\}$ for every integer $2 \leq k \leq s + t$, and $L = \sum_{k=2}^{s+t} \ell_k$. For each integer $0 \leq q \leq M$ put $T_q = \sum_{k=2}^{q+1} \ell_k$, $K_q = M - q$, $L_q = L - T_q$, $P = L - M$, and $P_q = L_q - K_q$; empty sums are zero. For $0 \leq q < M$, let A_q contain every ordered pair (F_1^α, F_2^β) with integers $0 \leq \alpha < s$, $0 \leq \beta < t$, $\alpha + \beta = q$, where each superscript deletes that many leading whole components. Let S_q be a disjoint union of q stars with edge counts ℓ_2,..., ℓ_{q+1}; S_0 is empty. For any finite simple graph J, let $e(J)$ be its edge count, $\nu(J)$ its maximum matching cardinality, $\Delta(J)$ its maximum degree, and $\Phi(J) = e(J) - \nu(J)$, with all these quantities zero for the empty graph.*

(I) For every integer $0 \leq q < M$ and finite simple H disjoint from S_q, H disjoint union $S_q \to (F_1, F_2)$ if and only if H arrows every member of A_q. Every coloring of H disjoint union S_q avoiding both full targets restricts to a coloring of H avoiding both targets of some member of A_q. Conversely every such coloring of H extends to an avoiding coloring of the full originals with all pool stars monochromatic.

(II) Let G be any finite simple graph and X any vertex set of size $q < M$. Every coloring of $H = G - X$ avoiding a member (F_1^α, F_2^β) of A_q extends to an avoiding coloring of G preserving all H-edge colors. More precisely, partition $X = X_R$ disjoint union X_B with $|X_R| = \alpha$ and $|X_B| = \beta$; color edges inside X_R and from X_R to H red, edges inside X_B and from X_B to H blue, and each $X_R - -X_B$ edge arbitrarily. Put $c_X = e(G) - e(H)$, $h_X = \nu(G) - \nu(H)$, and $G' = H$ disjoint union S_q. Then $0 \leq h_X \leq q$, $G \to (F_1, F_2)$ implies $G' \to (F_1, F_2)$, and

$$e(G') = e(G) - c_X + T_q,$$

$$\nu(G') = \nu(G) - h_X + q,$$

$$\Phi(G') = \Phi(G) - c_X + T_q + h_X - q.$$

(III) For every finite simple G with $e(G) < L$ there is a finite deletion procedure producing distinct original vertices v_1,..., v_q with $0 \leq q < M$, graphs $G_0 = G$ and $G_i = G_{i-1} - v_i$, and $H = G_q$, such that $d_i := d_{G_{i-1}}(v_i) \geq \ell_{i+1}$ for $1 \leq i \leq q$ and $\Delta(H) \leq \ell_{q+2} - 1$. Put $X = \{v_1, \ldots, v_q\}$, $\epsilon = \sum_{i=1}^{q} (d_i - \ell_{i+1})$, and $j = q - \nu(G) + \nu(H)$. Then ϵ, j are nonnegative integers,

$$e(H) = e(G) - T_q - \epsilon < L_q,$$

$$\Phi(H) - P_q = \Phi(G) - P - \epsilon - j,$$

$$e(H disjoint union S_q) = e(G) - \epsilon,$$

$$\nu(H disjoint union S_q) = \nu(G) + j.$$

If G arrows the original pair, H arrows every entire A_q member, and every avoiding final H-coloring has the extension in (II) to that same original G. Every original edge incident with X is charged once, at its earliest deleted endpoint. The procedure makes at most M-1 deletions; it also covers $q = 0$, $M = 1$, unit components, and edgeless G.

Proof. First note that every ℓ_k is defined and positive: for $2 \leq k \leq s + t$ there are valid positive indices i, j summing to k, and $n_i + m_j - 1 \geq 1$. For $q < M$, the integer interval $[\max(0, q - t + 1), \min(q, s - 1)]$ is nonempty because $q \leq s + t - 2$. Choosing α there and $\beta = q - \alpha$ shows that A_q is nonempty.

We will repeatedly use the following direct observation. If at most α components are removed from a copy of F_1, list the surviving original indices as $i_1 < i_2 <$.... For $1 \leq a \leq s - \alpha$ one has $i_a \leq a + \alpha$, so $n_{i_a} \geq n_{a+\alpha}$. Keeping the first s-alpha surviving stars and removing surplus spokes produces a copy of F_1^α. The same argument applies to F_2 and β.

For (I), start with an avoiding coloring of H disjoint union S_q. Scan the pool stars in order. Before its rth star, $1 \leq r \leq q$, suppose α red heads and β blue heads have been placed in the preceding stars, with $\alpha + \beta = r - 1$. Neither target has been exhausted, since the placed stars would already be a full target in the pool. Thus the next sizes $n_{\alpha+1}$ and $m_{\beta+1}$ exist, and their index sum is $r + 1$. The current star has $\ell_{r+1} \geq n_{\alpha+1} + m_{\beta+1} - 1$ edges. If it had at most $n_{\alpha+1} - 1$ red edges and at most $m_{\beta+1} - 1$ blue edges, the sum would be too small. Choose a color with enough edges and place its next full head in this pool star. The chosen target stars in

different pool components are vertex-disjoint. Avoidance prevents exhaustion at every step. After the qth step, the final integers satisfy $\alpha + \beta = q$, $\alpha < s$ and $\beta < t$. A red F_1^α in H, together with the chosen α red heads in the pool, would be a full red F_1. The blue assertion is identical after exchanging the two lists and colors. Thus the unchanged H-coloring avoids this A_q member.

Conversely, take a coloring of H avoiding a fixed member with cuts α, β. Color any α whole pool stars red and the other β blue. A collection of vertex-disjoint positive-edge stars inside one host star contains at most one component, since all host edges share its center. Hence a hypothetical full red F_1 uses at most α pool components; every other component lies wholly in H, since the host is a disjoint union. The observation in the second paragraph supplies a forbidden red F_1^α in H. A hypothetical blue F_2 similarly supplies the forbidden blue suffix. This proves the avoiding extension. Taking contrapositives of the two constructions proves the arrowing equivalence. The s can is empty and the constructions are identities when $q = 0$.

For (II), the stated rule colors every edge outside H exactly once except cross edges, whose colors are chosen separately. Any nontrivial connected red subgraph meeting X contains a vertex of X_R. Indeed, if it meets only X_B initially, a vertex of X_B in that connected subgraph is incident with a red edge of the subgraph, and every such edge has its other endpoint in X_R. Every nontrivial connected blue subgraph meeting X likewise contains X_B. Consequently a red target copy has at most α components meeting X, because its components are vertex-disjoint. Removing these components and applying the observation gives a red F_1^α in H, a contradiction. The blue argument gives a blue F_2^β in H. This also handles intersections at target leaves and arbitrary cross-edge colors. If G arrows the originals, H must therefore arrow every A_q member, and (I) implies that G′ arrows the originals.

Restriction of a matching of G to H loses at most q edges, since its edges have distinct endpoints in X; conversely every matching of H is a matching of G. Thus $0 \le \nu(G) - \nu(H) \le q$. The disjoint pool has T_q edges and matching number q, since all q star sizes are positive. Every matching in a disjoint union splits into matchings in its components, and maximum matchings in distinct components can be united; matching number is therefore additive here. It follows that $e(G') = e(H) + T_q$ and $\nu(G') = \nu(H) + q$. Substituting the definitions of c_X, h_X and subtracting these two identities proves every displayed formula in (II).

For (III), start at G_0. At a stage $i < M$, delete any current vertex of degree at least ℓ_{i+2} if one exists, calling it v_{i+1}; otherwise stop with $q = i$. All deleted vertices are distinct original vertices. The process cannot perform M deletions. In that event the disjoint sets $B_i = E(G_{i-1})$ minus $E(G_i)$, $1 \le i \le M$, would have sizes $d_i \ge \ell_{i+1}$, and $e(G) \ge \sum_{i=1}^{M} d_i \ge L$, contrary to the hypothesis. It must therefore stop after $q \le M - 1$ deletions, and absence of a further eligible vertex gives the integer degree bound $\Delta(H) \le \ell_{q+2} - 1$. Each edge with one or two endpoints in X lies in exactly the B_i associated with the earliest deleted endpoint; all remaining edges are precisely $E(H)$. Consequently $c_X = \sum_{i=1}^{q} d_i = T_q + \epsilon$, where $\epsilon \ge 0$. This yields $e(H) = e(G) - T_q - \epsilon < L_q$. The bound $h_X \le q$ from (II) gives $j = q - h_X \ge 0$, and both are integers.

Finally, $\Phi(H) = \Phi(G) - c_X + h_X = \Phi(G) - T_q - \epsilon + q - j$, while $P_q = P - T_q + q$. Subtracting gives $\Phi(H) - P_q = \Phi(G) - P - \epsilon - j$. Applying (II) to this actual X gives $e(Hdisjointunion S_q) = e(G) - \epsilon$ and $\nu(Hdisjointunion S_q) = \nu(G) + j$, as well as all arrowing and avoiding-color transfers. For $M = 1$ no deletion can occur because an eligible first degree would be at least $\ell_2 = L > e(G)$. Edgeless graphs also stop immediately. Every argument uses positive star sizes and therefore includes size one; no empty target convention was used. □

Depends on. Nothing earlier; this item is self-contained.

Lemma 42 (Unit-dominated heads and natural-profile density). *All graphs are finite and simple. Let F, J be vertex-disjoint unions of stars with complete nonincreasing positive integer edge-size lists $n_1,\dots, n_S$ and $m_1,\dots, m_T$, where S, T are positive integers. Copies are ordinary prescribed-color subgraphs with vertex-disjoint components; F is red and J is blue. Write e, ν and Δ for edge count, maximum matching cardinality and maximum degree. Put $M = S + T - 1$ and $\ell_l = max\{n_i + m_j - 1 : 1 \le i \le S, 1 \le j \le T, i + j = l\}$ for every integer $2 \le l \le S + T$. Fix an integer $0 \le q < M$, and put $K = M - q$, $L_q = \sum_{l=q+2}^{S+T} \ell_l$, $P_q = L_q - K$, and $D = \ell_{q+2} - 1$. Assume $D \ge 3$, and let R be the length of the initial $D + 1$ plateau of this complete tail.*

Let A_q consist of every ordered pair (F^α, J^β) with nonnegative integer cuts $\alpha < S$, $\beta < T$ and $\alpha + \beta = q$, where superscripts delete that many leading whole components.

Suppose $q < S$ and $m_1 = D$. Put $f = |\{j : m_j = D\}|$ and $u = |\{i : n_i = 1\}|$. If $u \geq T - f$, then, writing $r = S - q$ and K_2 for one edge,

$L_q - R = (r-1)D + e(J)$.

Every graph H with $e(H) \leq L_q - R$ and $\nu(H) < K$ has a coloring avoiding a red rK2 and a blue entire J, and therefore avoiding both entire targets of (F^q, J). In particular this holds if, for a nonnegative integer λ,

$e(H) - \nu(H) \leq P_q + \lambda$ *and* $K - \nu(H) \geq R + \lambda$.

Every coloring of H avoiding (F^q, J) extends, preserving every H-edge color, both over the disjoint union of q stars with sizes ℓ_2,..., ℓ_{q+1} and across any specified q-vertex set X in any graph G with $G - X = H$, to avoidance of the entire originals F, J.

Now suppose both full original heads are at most D, $q < \min(S, T)$, and H arrows every entire member of A_q with $e(H) \leq L_q - R$ and $\nu(H) < K$. Put $Z = \max(e(F), e(J))$. Then

$P_q \leq (D-1)(Z - q - 1) + R$.

More precisely orient the complete originals and their prescribed colors so that $a = e(F) - S \leq b = e(J) - T$. If the second head is D, then its multiplicity f and the first list's unit count u satisfy $u < T - f$. If the second head is below D, no numerical condition is needed. In particular the density bound applies when $R = 1$ and $e(H) \leq L_q - 1$.

All sizes, counts, cuts, indices, matching numbers, degrees, allowances and widths are integers in the stated domains. Isolated host vertices, unit target stars, repeated sizes and $q = 0$ are allowed. No maximum-degree bound on H is assumed.

Proof. We first prove the equality property needed for the matching relaxation. Fix a complete positive star forest J with largest size $D \geq 3$ and f largest components. For a positive integer p put $E_p = (p-1)D + e(J)$. Applying Lemma 21 to the singleton family {(pK2,J)} gives $e(G) \geq E_p$ for every graph G arrowing this pair. Indeed its complete profile is D at positions 1,..., p and then m_2,..., m_T, so its sum is E_p and its component budget is $p + T - 1$. If $e(G) = E_p$, the equality assertion of that fact with surplus zero gives $\Delta(G) = D$.

We claim that every such equality host, after isolated vertices are removed, consists of p copies of the D-edge star and one copy of each star of size m_2,..., m_T. We prove the claim by induction on p, with J fixed. For $p = 1$ the all-blue coloring forces an entire J copy. This copy uses all $e(G) = e(J)$ edges, so every nonisolated component of G is exactly a component of that copy.

Let $p \geq 2$ and assume the claim for p-1. For every vertex w, the graph G-w arrows ((p-1)K2,J). If it had an avoiding coloring, color all edges incident with w red. The red matching number increases by at most one, to at most p-1, and the blue graph gains no edge. This would avoid (pK2,J) in G. Choose v of degree D, which exists by the equality conclusion above. Then

$e(G - v) = E_p - D = E_{p-1}$,

so induction identifies all nonisolated components of G-v.

Let I_0 be the centers of its D-edge stars, of which there are $p + f - 2$, and let W be the union of all its smaller star components. Each $I_0 - vertex$ already has degree D in G-v. Since $\Delta(G) = D$, no such vertex is adjacent to v. The set I_0 union {v} is therefore independent and all its vertices have degree D in G.

Choose any p-1 centers in I_0, color every edge at those centers red, and color every other edge blue. The red graph has a cover of p-1 vertices and therefore no p-edge matching. The blue edges are exactly the D edges at v, the f-1 retained old D-stars, and all edges of W. These constituent stars have disjoint edge sets and their total edge count is $e(J)$. Their number is T. If any two of them met in a vertex, their connected union would have fewer than T nonisolated connected components. A blue J copy, because it would use all $e(J)$ blue edges, would force exactly T nonisolated components. Thus any meeting would make both targets absent, contradicting arrowing. For every choice of the red centers, all the constituent blue stars must therefore have disjoint vertex sets.

In particular the whole star at v is disjoint from W, which was retained in every choice. If $f \geq 2$, any specified old D-star can also be retained: there are $p + f - 3 \geq p - 1$ other old centers from which to choose the red ones. The star at v is consequently disjoint from every old nonisolated component. All its neighbors were

isolated in G-v, and adding v has formed one additional disjoint D-star, proving the induction step in this case.

It remains to treat $f = 1$. Now $I = I_0$ union {v} has p vertices. The proved disjointness from W means that W remains a union of whole components of G. Every other edge of G has exactly one endpoint in I, and every vertex of I has degree D. Suppose two distinct centers c_1, c_2 in I had a common neighbor z. Color all edges at I minus $\{c_1, c_2\}$, together with c_1z and c_2z, red; color every other edge blue. The red graph has vertex cover $(Iminus\{c_1, c_2\})$ union {z}, of size p-1. The two remaining centers have blue degree D-1; the other I-vertices have blue degree zero. Every vertex outside I and W has blue degree at most two, because its only possible blue neighbors are c_1, c_2. Every vertex of W has degree at most D-1. Since $D \geq 3$, the maximum blue degree is at most D-1, so there is no blue J. This is again an avoiding coloring. Hence the I-centers have pairwise disjoint neighborhoods. Their stars are disjoint from one another and from W, proving the induction claim.

The canonical graph in this claim has exactly $p + T - 1$ nonisolated star components. Each star has matching number one, and maximum matchings on disjoint components unite. Thus every equality host has matching number $p + T - 1$. We used only necessity of the canonical description, not an unproved converse.

Return to the first hypotheses in the statement, with $q < S$, $m_1 = D$ and $u \geq T - f$. For positive integer thresholds x, y put $N_x = |\{i : n_i \geq x\}|$ and $B_y = |\{j : m_j \geq y\}|$. For each integer $z \geq 1$, the number V_z of entries in the full profile ℓ whose excess over one is at least z equals

$\max(\{N_x + B_y - 1 : x, y \geq 1, x + y = z + 2, N_x > 0, B_y > 0\} union \{0\})$. (1)

To check (1), a threshold rectangle contains all original index sums from 2 through $N_x + B_y$. Its entries have size sum at least $z + 2$. Conversely any pair of positive sizes with sum at least $z + 2$ dominates positive thresholds x, y with sum exactly $z + 2$: take $x = \max(1, z + 2 - m_j)$ and $y = z + 2 - x$. The qualifying diagonals therefore form the union of the corresponding initial intervals, whose length is the maximum in (1). The full sequence ℓ is nonincreasing, since decreasing a noninitial index on any diagonal weakly increases its size. Hence its q-tail width at excess level z is $\max(V_z - q, 0)$.

For $1 \leq z < D$, the candidate $x = 1$, $y = z + 1$ has width $S + B_{z+1} - 1$. Every candidate with $x \geq 2$ has width at most

$S - u + T - 1 \leq S + f - 1 \leq S + B_{z+1} - 1$.

The first inequality is the unit-count hypothesis, and the second uses $B_{z+1} \geq f$ because $z + 1 \leq D$. The first candidate is present. Thus $V_z = S + B_{z+1} - 1$, and $V_z - q$ is positive since $q < S$ and $B_{z+1} \geq 1$. At level D the natural tail has exactly R entries, and above D it has none. Counting each nonnegative integer $\ell_l - 1$ once at each positive level gives

$P_q = R + (D - 1)(S - q - 1) + \sum_{y=2}^{D} B_y$

$= R + (D - 1)(S - q - 1) + e(J) - T$.

Adding $K = S - q + T - 1$ proves $L_q - R = D(S - q - 1) + e(J)$, the asserted identity.

Put $r = S - q$. If a graph H with the stated edge and matching bounds arrowed (rK2,J), the singleton-family lower bound would give $e(H) \geq E_r = L_q - R$. Its assumed upper bound forces equality. The equality induction just proved then gives $\nu(H) = r + T - 1 = K$, contrary to $\nu(H) < K$. Therefore H has a coloring avoiding that pair. An entire red F^q would contain an r-edge matching, by selecting one edge from each positive vertex-disjoint component, so the same coloring avoids (F^q, J).

Under the stated potential criterion, let $\delta = K - \nu(H)$. The condition $\delta \geq R + \lambda$ implies $\delta \geq 1$ because $R \geq 1$. Also

$e(H) \leq P_q + \lambda + \nu(H) = L_q + \lambda - \delta \leq L_q - R$.

Thus the preceding coloring conclusion applies. Its actual cuts are (q,0), legal because $q < S$ and the opposite target is untouched. Parts (I) and (II) of Lemma 41 extend every actual-pair avoiding coloring both over the canonical disjoint q-star pool and across any specified actual q-set in any outer graph, retaining the core colors. Those statements handle unit stars and target components meeting the deleted set at leaves.

Finally suppose the density hypotheses hold. Simultaneously exchange the two complete lists and prescribed colors if necessary so that $a = e(F) - S \leq b = e(J) - T$. The full diagonal profile, K, L_q, P_q, D, R, Z and the

family-arrowing assertion are unchanged; $q < \min(S, T)$ remains true. The pure density theorem Lemma 32 gives

$P_q \le (D-1)(Z-q-1) + R + \omega,$

where $\omega = 0$ if $m_1 < D$ and $\omega = \max(u + f - T, 0)$ if $m_1 = D$. In the latter case, $u \ge T - f$ would satisfy all the first part's hypotheses: $q < S$, $D = m_1 \ge 3$, $e(H) \le L_q - R$ and $\nu(H) < K$. Its avoiding coloring would avoid the genuine member (F^q, J) of A_q, contradicting common arrowing. Consequently $u < T - f$ and $\omega = 0$. If $m_1 < D$, ω was already zero. Substituting proves the claimed density bound. At $R = 1$ the edge condition is exactly $e(H) \le L_q - 1$. Every induction and selection was finite, and no bound on the host maximum degree was assumed outside the equality argument where it was proved. □

Remark. At matching equality, deleting a largest-degree vertex leaves another equality host. Any collision among its restored stars creates an avoiding coloring, so the matching number reaches the full component budget. Unit domination makes the natural profile exceed that matching budget by exactly the initial plateau, removing the sole correction in the density estimate.

Depends on. Lemma 21, Lemma 32, Lemma 41.

Lemma 43 (Endpoint projections for three-versus-four profiles). *Use the positive star forests, ordinary copies, sorted profiles, diagonal values ℓ_k, component budget M, and actual suffix families A_q of Lemma 29. For a positive integer d, write $K_{1,d}$ for a d-edge star and $K_2 = K_{1,1}$. A superscript on a profile entry denotes its repetition count; a zero repetition count omits that entry.*

Let a, r, p, b, u, v, h be nonnegative integers satisfying

$a \ge 1,\ r \ge 1,\ p > b,\ u > v,\ h \ge 3a + 2p - b,$ *and* $u \ge 3r + b.$

Take the original profiles

$F_1 = (3^a, 2^p, 1^u),\ F_2 = (4^r, 3^b, 2^h, 1^v),$

with $s = a + p + u$, $t = r + b + h + v$, and $M = s + t - 1$. Let q be any integer satisfying

$a + r + b - 1 \le q \le a + p + r - 2.$

Put $K = M - q$, $\lambda_i = \ell_{q+i+1} - 1$ for $1 \le i \le K$, and $P_q = \sum_{i=1}^{K} \lambda_i$. Define

$E = a + p + r - 1,\ R = E - q,$

$A = \max(u, b + h - p),\ B = b + \max(u, h, h + v - p),\ N = R + u + b + h.$

Then $0 \le q < \min(s, r + b + h) \le M$, $R \ge 1$, $A \ge 1$, and the complete potential tail is

$(\lambda_1, \dots, \lambda_K) = (4^R, 3^A, 2^{B-A}, 1^{u+b+h-B}, 0^v).$

All repetition counts are nonnegative integers. In particular exactly N entries are positive, exactly R equal four, exactly A equal three, $K = N + v$, and

$P_q = 4R + A + B + u + b + h.$

For a star forest F, let $c(F)$ be its component count, let $r(F)$ be the number of components with at least two edges, and let F^- be obtained by shortening each such component by one edge and discarding all one-edge components. The empty forest is allowed as F^- and has $c(F^-) = 0$. Form a family C_q as follows. For each actual pair in A_q, take both orders (F, J) of its two entries; include $(r(F)K_2, J)$ when $r(F) \ge 1$, and include $(c(F)K_2, J^-)$ when J^- is nonempty. Every included matching size and every opposite forest is positive. The component budget of a pair (mK_2, J) is $m + c(J) - 1$.

Every member of C_q has component budget at most N. The two pairs

$((s - q)K_2, F_2^-)$ *and* $((r + b + h - q)K_2, F_1)$

belong to C_q and both have component budget N. There is a member (mK_2, J) of C_q with component budget N whose largest component size is three, with a positive multiplicity f satisfying

$m + f - 1 = R + A.$

One may choose the first displayed pair if $u \geq b + h - p$, and the second otherwise. No potential or arrowing property of a host graph is assumed or concluded.

Proof. The definitions and monotonicity of the diagonal sequence are provided by Lemma 29. The integer assumptions give $p \geq 1$, $u \geq 1$, and $h \geq 3a + 2p - b > 0$. Write $B_2 = r + b + h$ for the blue nonunit count. Since $q \leq a + p + r - 2$,

$s - q \geq u - r + 2 \geq 2r + b + 2 > 0$,

$B_2 - q \geq b + h - a - p + 2 \geq 2a + p + 2 > 0$.

Also $q \geq a + r + b - 1 \geq 1$. Consequently $q < s$ and $q < B_2 \leq t$, so both cuts (q,0) and (0,q) are legal, both resulting forests are nonempty, and $0 \leq q < M$. Since $B_2 \leq t$ and $s \geq 1$, one also has $\min(s, B_2) \leq M$. The definition of E gives $R = E - q \geq 1$.

Index the full potential profile by $j = 1,..., M$:

$\pi_j = \ell_{j+1} - 1 = max\{n_i + m_k - 2 : i + k - 1 = j\}$.

It is a nonincreasing sequence of nonnegative integers by the cited monotonicity. For a positive threshold z, the number of its entries at least z equals the largest index attaining that threshold, provided it is attained. If a condition on the two sizes is expressed by an index rectangle $1 \leq i \leq x$ and $1 \leq k \leq y$ with positive x, y, its largest coordinate is $x + y - 1$, attained at (x, y). For a union of finitely many such rectangles, the largest coordinate is the maximum of these values.

Here the maximum potential is $3 + 4 - 2 = 5$. Potential at least five requires a red size three and a blue size four. Its last coordinate is $a + r - 1$, which is at most q. Thus none of the retained entries $\pi_{q+1},..., \pi_M$ exceeds four.

Potential at least four means either red size at least three and blue size at least three, or red size at least two and blue size at least four. The last coordinates of these nonempty rectangles are $a + r + b - 1$ and $a + p + r - 1$. Since $p > b$, their maximum is E. Therefore exactly $E - q = R$ retained entries are at least four, and all of them equal four.

Potential at least three means one of the following: red size at least three and blue size at least two; red size at least two and blue size at least three; red size at least one and blue size at least four. Their last coordinates are

$a + B_2 - 1 = E + b + h - p$,

$a + p + r + b - 1 = E + b$,

$s + r - 1 = E + u$.

All rectangles are nonempty because a, p, r, h are positive. Since $u \geq 3r + b > b$, the largest coordinate is $E + \max(u, b + h - p) = E + A$. This is greater than E, since $A \geq u \geq 1$. Hence the number of retained entries at least three is $R + A$, and the number equal to three is A.

Potential at least two means one of the following: red size at least three and any blue size; red size at least two and blue size at least two; any red size and blue size at least three. Their last coordinates are

$a + t - 1 = E + b + h + v - p$,

$a + p + B_2 - 1 = E + b + h$,

$s + r + b - 1 = E + u + b$.

The maximum is $E + B$, where $B = b + \max(u, h, h + v - p)$. It is attained within 1,..., M. The nesting of threshold conditions gives $E + B \geq E + A$. Thus the number of retained entries equal to two is $(R + B) - (R + A) = B - A \geq 0$.

Potential is positive exactly when at least one selected size is nonunit. The two last coordinates are

$a + p + t - 1 = E + b + h + v$,

$s + B_2 - 1 = E + b + h + u$.

Since $u > v$, the largest is $E + b + h + u = M - v$. Consequently the number of positive retained entries is $M - v - q = R + u + b + h = N$. Threshold nesting gives $N \geq R + B$, so there are $u + b + h - B$ nonnegative-

integer entries equal to one. The remaining $K - N = v$ entries equal zero. Nonincreasing order now gives the entire displayed tail. Its sum is

$4R + 3A + 2(B - A) + (u + b + h - B) = 4R + A + B + u + b + h.$

It remains to verify the projection family, including the original unit components. Let (X, Y) be the pair produced by any legal cuts α, β with $\alpha + \beta = q$, where X is the red suffix and Y the blue suffix. Then

$c(X) = s - \alpha, c(Y) = t - \beta,$

$r(X) = \max(a + p - \alpha, 0), r(Y) = B_2 - \beta.$

The last equality is positive because $\beta \le q < B_2$. Shortening a forest preserves exactly one positive component for each of its nonunit components, so $c(X^-) = r(X)$ and $c(Y^-) = r(Y)$.

For the order (X, Y), the budget of $(c(X)K_2, Y^-)$ is $c(X) + r(Y) - 1$. The order (Y, X) provides $(r(Y)K_2, X)$ with the same budget. Both pairs are always included, and their common budget is

$s - \alpha + B_2 - \beta - 1 = s + B_2 - q - 1 = E + u + b + h - q = N.$

The two other possible projections are included only if $r(X) \ge 1$. In that case $\alpha < a + p$, and each has budget

$r(X) + c(Y) - 1 = a + p - \alpha + t - \beta - 1$

$= a + p + t - q - 1 = N - u + v < N.$

These are all possibilities from the two orders and the two projection rules. Hence every member has budget at most N.

Use the legal cut (q,0) and the projection $(c(X)K_2, Y^-)$. It gives $((s - q)K_2, F_2^-)$, where F_2^- has profile $(3^r, 2^b, 1^h)$. Its matching size s-q is positive, its opposite component count is B_2, and its largest size is three with multiplicity $r \ge 1$. It has budget N and

$(s - q) + r - 1 = E + u - q = R + u.$

Use the other legal cut (0,q) and the reversed order (Y, X), with projection $(r(Y)K_2, X)$. It gives $((B_2 - q)K_2, F_1)$. Its matching size is positive and its entire opposite profile is $(3^a, 2^p, 1^u)$, retaining all red units. Its largest size is three with multiplicity $a \ge 1$. Its budget is N, and

$(B_2 - q) + a - 1 = E + b + h - p - q = R + b + h - p.$

The larger of these two values is $R + \max(u, b + h - p) = R + A$. Choosing the first at a tie gives exactly the selection stated. Every object used belongs to the actual endpoint family; no single endpoint is asserted to attain the whole diagonal profile. □

Depends on. Lemma 29.

Lemma 44 (Path structure at a packed two-level envelope). *All graphs are finite and simple. Write $e(H)$, $\nu(H)$, $\Delta(H)$ for the edge count, maximum matching cardinality, and maximum degree of H. For every positive integer n let P_n be the path on n vertices, and for $n \ge 3$ let C_n be the cycle on n vertices. Put $K_2 = P_2$ and let $\nu_2(H)$ be the maximum number of vertex-disjoint ordinary copies of P_3 in H. The empty graph has $\nu = \nu_2 = 0$. Arrowing is for ordinary prescribed red-blue subgraph copies; copied forest components have disjoint vertex sets.*

Let h, u be nonnegative integers with $N = h + u \ge 1$. Let calA be a finite nonempty label set. For each γ in calA let p_γ, $t_\gamma \ge 1$ be integers and let J_γ be the vertex-disjoint union of positive-edge stars with integer sizes $d_1^\gamma \ge \cdots \ge d_{t_\gamma}^\gamma \ge 1$. Suppose $\max_\gamma(p_\gamma + t_\gamma - 1) = N$. For each integer $2 \le k \le N + 1$ define

$\ell_k = max\{d_j^\gamma : \gamma \in calA, 1 \le i \le p_\gamma, 1 \le j \le t_\gamma, i + j = k\}.$

Assume $(\ell_2, \ldots, \ell_{N+1})$ consists of h entries two followed by u entries one.

Suppose H arrows $(p_\gamma K_2, J_\gamma)$ for every γ, $e(H) = 2h + u$, and H contains h vertex-disjoint copies of P_3. Then every nonisolated connected component of H belongs to P_2, P_3, P_4, P_6. If a, b, c, d count its whole P_3, P_4, P_6, P_2 components, respectively, then

$h = a + b + 2c, u = b + c + d, \nu(H) = h + u,$ *and* $\nu_2(H) = h.$

For every vertex z of H, the graph H-z contains h vertex-disjoint P_3 copies if and only if z is isolated, belongs to a whole P_2 component, or is an endpoint of a whole P_4 component.

In addition, every sequence that repeatedly deletes any current maximum-positive-degree vertex until no edges remain has exactly h deletions of degree two followed by exactly u deletions of degree one. All counts, orders, degrees, indices, and cardinalities are integers in the displayed domains. Cases $h = 0$ or $u = 0$, repeated target sizes, unit target components, and isolated host vertices are included.

Proof. Use the vertex-deletion implication of Lemma 34 with its explicit convention that both retained target forests are nonempty.

Fix any greedy sequence, and suppose its first q vertices form X, where $0 \le q < N$. Choose a label γ and valid indices i, j attaining ℓ_{q+2}. These exist: a label attaining the maximum component budget N has valid indices for every sum from two to $N+1$. The cuts i-1 and j-1 sum to q and are smaller than p_γ and t_γ. The deletion implication says that H-X arrows the matching of size $p_\gamma - i + 1$ against the suffix of J_γ beginning with size d_j^γ. Coloring all remaining edges blue forces that suffix to occur, since the red graph cannot contain the nonempty matching target. Hence H-X has maximum degree at least $d_j^\gamma = \ell_{q+2}$, and in particular has an edge. Thus the next greedy vertex exists, even if the sequence is continued through $q = N - 1$.

Let D_{q+1} be its degree when deleted. Each original edge is removed at its earliest deleted endpoint, or survives all N steps, so

$e(H) = \sum_{q=0}^{N-1} D_{q+1} + e(H - X_N)$.

Every $D_{q+1} \ge \ell_{q+2}$, and the sum of the latter numbers is $2h + u = e(H)$. All these inequalities are therefore equalities: $D_{q+1} = \ell_{q+2}$, and the final graph is edgeless. This proves the asserted greedy profile for every choice sequence. In particular $\Delta(H) \le 2$.

We now classify its whole components using only this profile. Every connected finite graph of maximum degree at most two is a path or a cycle. Indeed, following successive unused edges from a degree-one vertex gives the entire path: an earlier return would create degree at least three, and an additional unvisited vertex would contradict connectedness. If the component has an edge and no degree-one vertex, all its degrees are two; following edges to the first return gives a cycle with no possible additional vertex. An isolated vertex is P_1.

For a path or cycle, let q_2 count the degree-two steps in a greedy deletion sequence. Degree-two steps precede degree-one steps, and the latter count is $e - 2q_2$. Consequently the sequence of positive deleted degrees is fixed exactly when q_2 is fixed. If a whole component permits two different q_2 values, perform either run's degree-two steps in that component and fixed degree-two runs in the others, before deleting degree-one vertices. All these degree-two choices are legal global maximum-degree choices. The two global profiles then differ. The same argument applies after any fixed greedy degree-two prefix. Hence every whole component, and every component exposed by such a prefix, must have a fixed q_2.

The path P_5 does not: deleting its middle vertex leaves two edges, giving positive profile (2,1,1), while deleting its second vertex leaves an isolate and P_3, giving (2,2). For every $n \ge 7$, deleting vertex six of P_n first leaves a whole P_5 on vertices one through five. Vertex six has degree two and is a legal first choice, so this also forbids P_n. The remaining paths P_1, P_2, P_3, P_4, P_6 have fixed positive profiles (), (1), (2), (2,1), and (2,2,1), respectively. For P_6, up to reflection the first choice is vertex two or three, leaving P_1 plus P_4 or P_2 plus P_3; both give the stated profile. This checks every possible first choice. In a cycle C_n every first vertex deletion leaves P_{n-1}. Therefore precisely C_3, C_4, C_5, C_7 are allowed cycles, with profiles (2,1), (2,2), (2,2,1), and (2,2,2,1).

The resulting local counts are

T P_1 P_2 P_3 P_4 P_6 C_3 C_4 C_5 C_7

q_2 0 0 1 1 2 1 2 2 3

q_1 0 1 0 1 1 1 0 1 1

$\nu(T)$ 0 1 1 2 3 1 2 2 3

$\nu_2(T)$ 0 0 1 1 2 1 1 1 2.

Here q_1 is the degree-one step count. The matching and packing rows follow from the upper bounds $\lfloor n/2 \rfloor$ and $\lfloor n/3 \rfloor$, since each matching edge uses two vertices and each P_3 uses three. Consecutive disjoint pairs or triples attain these bounds on every listed path or cycle.

Both matching and $P_3 -$ *packing* numbers add over whole components: each edge or connected P_3 is contained in one component, and choices in distinct components may be united. The q_2 values also add by the legal interleaving of component degree-two runs. Their sum is h. In the table $\nu_2(T) \le q_2$, with a deficit of one precisely for C_4, C_5, C_7. The assumed packing of h copies implies $\nu_2(H) \ge h$, so $\nu_2(H) = h$ and all three deficient cycle types are absent. Only the five listed paths and triangles remain.

Every path or triangle T permits every exact matching split at total $\nu(T)$. Here is a coloring for each integer r with $0 \le r \le \nu(T)$. On an even path P_{2k}, for $1 \le r \le k$ color its first 2r-1 edges red and the remaining edges blue; their matching numbers are r and k-r. For $r = 0$ color all edges blue. On an odd path P_{2k+1}, color its first 2r edges red and the rest blue for $0 \le r \le k$; the two matching numbers are again r and k-r. These statements follow by counting consecutive pairs on the respective colored path segments; empty segments have matching number zero. A triangle has matching number one, and either all-blue or all-red coloring realizes its two quotas.

Put $K = \nu(H)$. For any integer $0 \le r \le K$, assign integers r_T between zero and $\nu(T)$ to its finitely many components with sum r. Fill these finite capacities in any order until r is reached. Combining the local colorings gives matching numbers exactly r and K-r, because distinct original components have disjoint vertex sets.

Choose a family member (pK_2, J) with t components in J and $p + t - 1 = N$. If $K < N$ and $K < p$, the all-red coloring avoids both this matching and the nonempty blue J. If instead $p \le K < N$, use the exact split $r = p - 1$. The red matching number is p-1, and the blue matching number $K - p + 1$ is less than t. Every J contains a matching of t edges, obtained by selecting one edge from each positive component. Thus this split also avoids both targets. Both possibilities contradict the assumed arrowing, proving $K \ge N$.

Let z_3 be the number of triangle components. For each remaining path the table gives $\nu(T) = q_2 + q_1$; for a triangle it gives $\nu(T) = q_2 + q_1 - 1$. Adding yields $K = h + u - z_3 = N - z_3$. The inequality $K \ge N$ forces $z_3 = 0$ and $K = N$. The path list now follows. Summing its q_2 and q_1 rows gives exactly $h = a + b + 2c$ and $u = b + c + d$.

Finally $\nu_2(H) = h$ and additivity show that deleting z preserves h copies exactly when it preserves the local packing number in its whole component. Deleting any vertex of P_3 leaves at most two vertices, and deleting any vertex of P_6 leaves five, so their local maxima decrease. Deleting an internal vertex of P_4 leaves an edge and an isolate, whereas deleting an endpoint leaves P_3. A P_2 and an isolate have local packing number zero before and after deletion. These exhaust the component types and prove the stated equivalence. □

Remark. Every possible degree-two deletion is forced to use the same number of slots. A full packing uses all of those slots, eliminating the inefficient cycles. Exact matching allocations on the remaining paths and triangles then rule out triangles through a maximum-budget target pair.

Depends on. Lemma 34.

Lemma 45 (Missing allocations and independent-star bounds). *Use the finite simple graph certificate costs $c_0(G)$, matching number $\nu(G)$, excess $ex(G) = c_0(G) - \nu(G)$, and top missing-allocation run $tr(G)$ of Lemma 23. A nonnegative integer pair (x, y) is feasible in G when $E(G)$ has a partition into two parts with matching numbers at most x and y. At zero degree capacities this is exactly the certificate definition. Vertex deletion is induced, and all intervals below contain integers.*

For every finite simple graph G, every independent vertex set S, and every integer x with $0 \le x \le |S|$, the pair $(x, |S| + \nu(G - S) - x)$ is feasible in G.

Let $k \ge 1$ and $D \ge 3$ be integers. Let H be connected, have order $2k + 1$, satisfy $\nu(H) = k$ and $\Delta(H) \le D$, and have $E = ex(H) \ge 1$. Put $v = \max(0, floor((2E - 3)/(D - 2)))$. Then $E \le \lceil k/3 \rceil$. Every infeasible pair $(x, k + E - 1 - x)$ at total $k + E - 1$ satisfies

$$2E - 1 + v \le x \le k - E - v.$$

There is at least one such pair; both coordinates obey the displayed interval. Consequently

$1 \le tr(H) \le k - 3E + 2 - 2v$, *and* $k \ge 3E - 1 + 2v$.

For every signed integer B, *if* $tr(H) + E \ge B + 1$, *then* $k \ge B + 2E - 1 + 2v$.

Every independent set S *of this same* H *satisfying* $|S| \ge k - E - v + 1$ *obeys*

$|S| + \nu(H - S) \ge k + E$.

When $D = 4$ *and* $E \ge 2$, *its size condition is* $|S| \ge k - 2E + 3$. *All orders, degrees, cardinalities, capacities, matching budgets, costs, excesses,* k, D, v *and run lengths are integers in the specified nonnegative or positive domains. No regularity, factor-criticality, or minimality assumption is imposed.*

Proof. At zero capacities the bounded-degree parts contain no edge, so feasibility is exactly the two-part condition in the statement, by Lemma 23.

For the independent-set assertion choose an x-element subset S_1 of S. Give the first part every edge meeting S_1 and the second part every other edge. A first-part matching has at most x edges. In a second-part matching, edges internal to G-S number at most $\nu(G - S)$. Every other edge meets S minus S_1, because S is independent and no remaining edge meets S_1. These edges use distinct vertices of S minus S_1, so they number at most $|S| - x$. This proves the asserted feasible pair, including S empty and $x = 0$ or $x = |S|$.

We first give the two constructions used to color edges. Write $n = 2k + 1$.

If a set W of at most x vertices leaves at least $n - |W| - 2y - 1$ isolated vertices in H-W, assign all edges incident with W to the first part and all other edges to the second. A matching in the first part has at most $|W| \le x$ edges, since its edges have distinct endpoints in W. The second part has at most $2y + 1$ nonisolated vertices, so its matching number is at most y. This proves feasibility. In particular, if $x \le n$ and $n - x \le 2y + 1$, any x-vertex W works, even when no isolated vertices remain.

For the second construction, fix any initial vertex of a connected graph on $n \ge 2$ vertices with maximum degree at most D. Begin with its singleton independent set I. If I does not dominate the graph, connectedness gives an edge from its closed neighborhood I union $N(I)$ to a vertex z outside that neighborhood. The endpoint inside the neighborhood belongs to $N(I)$, since z has no neighbor in I. Thus z shares a neighbor with I. Adding z preserves independence and adds at most D-1 new vertices to the open neighborhood $N(I)$. Starting with at most D neighbors, induction gives

$|N(I)| \le (D - 1)|I| + 1$

at every step. Each step adds a vertex, so this finite construction terminates in an independent dominating set of size r. Independence and domination imply $n = r + |N(I)| \le Dr + 1$, hence $r \ge ceil((n - 1)/D)$. Consequently for every integer $1 \le s \le ceil((n - 1)/D)$ a prefix gives an independent s-set with at most $(D - 1)s + 1$ neighbors.

We next bound $c_0(H)$ without using connectedness. Put $e = \lceil k/3 \rceil$ and $c = k + e$. Consider any nonnegative integer pair of total c, and exchange its coordinates if needed to arrange $x \le y$. Since $e \le k$, $x \le \lfloor c/2 \rfloor \le k < n$. Also $c \ge 4k/3$ implies

$x \le c/2 \le 2c - 2k$.

Therefore $n - x \le 2(c - x) + 1 = 2y + 1$, and the first construction proves feasibility. This holds for every pair of total c. The set of permissible totals is thus a nonempty set of nonnegative integers and has a least element $c_0(H)$. At a permissible total c', the pair (0,c') forces $c' \ge \nu(H) = k$. Hence E is a nonnegative integer and $E \le \lceil k/3 \rceil$.

Set $\ell = k + E - 1$ and $L = 2E - 2 + v$. We prove that every nonnegative integer pair (x, y) with $x + y = \ell$ and $\min(x, y) \le L$ is feasible. Exchange the coordinates so that $x \le y$ and $x \le L$. We have $x \le \lfloor \ell/2 \rfloor < n$ because $E \le \lceil k/3 \rceil \le k$.

If $x \le 2E - 2$, the identity

$n - x - 2y - 1 = x - 2E + 2$

has a nonpositive right side. The first construction with any x-vertex set proves feasibility.

For the remaining case put $s = x - 2E + 2$. Then $1 \le s \le v$, so $v \ge 1$ and $E \ge 2$. The definition of v yields $(D - 2)s \le 2E - 3$; therefore

$(D-1)s+1 \le s+2E-2 = x$.

We check the size range for the independent-set construction. Since $E \le \lceil k/3 \rceil \le (k+2)/3$,

$v \le (2E-3)/(D-2) \le (2k-5)/(3(D-2)) < 2k/D$.

The final inequality follows after multiplication by the positive denominators from

$6k(D-2) - D(2k-5) = 4k(D-3) + 5D > 0$.

Thus $s \le \lceil 2k/D \rceil$. The second construction supplies an independent s-set I with $|N(I)| \le (D-1)s+1 \le x$. Its neighborhood is disjoint from I. Enlarge $N(I)$ to an x-vertex set W avoiding I. There are enough vertices because

$n - s = x + 2y + 1 \ge x$.

Every vertex of I is isolated in H-W. Since $s = n - x - 2y - 1$, the first construction again proves feasibility.

We have proved feasibility whenever the smaller coordinate is at most L. Thus each coordinate of an infeasible pair at total ℓ is at least $L+1 = 2E-1+v$. Subtracting this lower bound for y from ℓ gives $x \le k-E-v$, proving the interval.

Since $E \ge 1$, the integer $c_0(H) = k+E$ is positive. At least one pair at total $c_0(H)-1$ is infeasible; otherwise that smaller nonnegative total would contradict the definition of $c_0(H)$. Therefore the finite set of infeasible x values is nonempty and has a longest consecutive run of positive integer length $tr(H)$. All such values lie in the integer interval just proved, whose length is $k-3E+2-2v$. This proves both run bounds and $k \ge 3E-1+2v$. Finally $tr(H)+E \le k-2E+2-2v$. Combining this with $tr(H)+E \ge B+1$ and rearranging proves the last assertion.

Finally fix an independent set S in H of size $a \ge k-E-v+1$. Suppose $a+\nu(H-S) \le k+E-1$ and put $\ell = k+E-1$. For every integer x in [0,a], the independent-set construction gives a partition with matching bounds x and $a+\nu(H-S)-x \le \ell - x$. Our supposition implies $a \le \ell$, so all these pairs are nonnegative. But every infeasible first budget at total ℓ is at most $k-E-v < a$, by the interval already proved. It would therefore belong to [0,a], where the same budget is feasible. This contradicts the existence of an infeasible pair at total ℓ. Hence $a+\nu(H-S) \ge k+E$. If $D=4$ and $E \ge 2$, then $v = floor((2E-3)/2) = E-2$, giving the displayed specialized threshold. □

Remark. An independent vertex pool gives all small red budgets while retaining a bound on the blue matching number. Connected growth finds small-neighborhood independent sets, which isolate vertices after a controlled deletion and push every missing top budget into an interior interval. A large independent set would cover that entire interval unless its complementary matching number is correspondingly large.

Depends on. Lemma 23.

Lemma 46 (Exact prescribed matching deficiency). *Let H be a finite simple graph, possibly empty or containing isolated vertices, and let U be any subset of its vertex set $V(H)$. A matching is a set of pairwise vertex-disjoint edges. For a matching M, let $V(M)$ be its covered vertex set, and let $\nu(H)$ be the maximum cardinality of a matching. For $S \subseteq V(H)$, let H-S be induced vertex deletion and let $o_U(H-S)$ count the odd-order connected components of H-S whose vertex sets are contained in U. Define $\eta_U(H) = \min_M |U \text{ minus } V(M)|$, where the minimum is over all matchings of H. Then $\eta_U(H) = \max_{S \subseteq V(H)}(o_U(H-S) - |S|)$. Both extrema are attained, their common value is a nonnegative integer, and some matching of cardinality $\nu(H)$ attains the minimum. More precisely, for every matching M_0 attaining the minimum there is a maximum matching M with $V(M_0) \subseteq V(M)$ and $|U \text{ minus } V(M)| = \eta_U(H)$.*

For $W \subseteq V(H)$, let H[W] be the induced subgraph, let $o(H[W])$ count all its odd-order components, and let $N_{ext}(W)$ be the set of vertices in $V(H)$ minus W adjacent to a vertex of W. An equivalent formula is $\eta_U(H) = \max_{W \subseteq U}(o(H[W]) - |N_{ext}(W)|)$.

For each nonnegative integer k, H has a matching leaving at most k vertices of U uncovered if and only if $o_U(H-S) \le |S| + k$ for every $S \subseteq V(H)$; when these conditions hold, such a matching can be chosen maximum. For $U = V(H)$, write $o(H-S)$ for the number of all odd-order components. The identity specializes to $|V(H)| - 2\nu(H) = \max_{S \subseteq V(H)}(o(H-S) - |S|)$, and equivalently $\nu(H) = \min_{S \subseteq V(H)}[|S| + \sum_C floor(|V(C)|/2)]$, where C runs over all connected components C of H-S, including isolated vertices. All cardinalities, matching sizes, allowances, and extrema in these assertions are integers in the stated nonnegative domains.

Proof. The families of vertex subsets and matchings are finite and nonempty, so all the stated minima and maxima are attained. Put $d = \max_S(o_U(H - S) - |S|)$. Taking S empty gives $d \geq 0$.

Fix a matching M and a vertex set S. Among the $o_U(H - S)$ counted components, at most $|S|$ have an M-edge to S, since such edges have distinct endpoints in S. Each remaining counted component has no M-edge leaving it: every edge from a component of H-S to its complement has its other endpoint in S. Its internal matching edges cover an even number of vertices, so its odd vertex set includes an uncovered vertex of U. Different components give different uncovered vertices. Thus $|U$ minus $V(M)| \geq o_U(H - S) - |S|$. Maximize over S and minimize over M to obtain $\eta_U(H) \geq d$.

Adjoin a set Z of d new vertices to H. Join each vertex of Z to every vertex of U, and add no other new edge. Let J denote the resulting finite simple graph, with U still its prescribed set. We check the saturation criterion of Lemma 10. Given any $T \subseteq V(J)$, put $S = T$ intersect $V(H)$ and $Y = T$ intersect Z. If some z in Z minus Y remains, every surviving vertex of U is adjacent to z. Any component of J-T containing a vertex of U therefore also contains a vertex outside U. The count $o_U(J - T)$ is zero in this case. If all vertices of Z are deleted, then $J - T = H - S$ and $o_U(J - T) = o_U(H - S) \leq |S| + d = |T|$. These cases exhaust every T, including $d = 0$, when only the second case occurs. The cited criterion provides a matching of J covering U.

Remove its edges incident with Z. At most d edges are removed because they have distinct endpoints in Z. Each removed edge has exactly one endpoint in U. The remaining edges form a matching of H that leaves at most d vertices of U uncovered. Hence $\eta_U(H) \leq d$, proving equality.

Given any matching M_0 attaining $\eta_U(H)$, apply the augmentation conclusion of Lemma 10. It provides a maximum matching M with $V(M_0) \subseteq V(M)$. Consequently $|U$ minus $V(M)| \leq \eta_U(H)$, and the reverse inequality follows from the definition of the minimum. This proves maximum-cardinality attainment with the required vertex inclusion. The criterion with allowance k is equivalent to $d \leq k$, which by the proved identity is $\eta_U(H) \leq k$. This also proves its maximum-matching qualification.

For the neighborhood formula, fix $W \subseteq U$ and delete $S = N_{ext}(W)$. No edge joins W to a vertex outside W union S. Each component of H[W] is therefore an entire component of H-S. It follows that $o(H[W]) - |N_{ext}(W)| \leq o_U(H - S) - |S| \leq d$. Conversely, given any S, let W be the union of the vertex sets of all the odd components of H-S contained in U. Then $o(H[W]) = o_U(H - S)$ and $N_{ext}(W) \subseteq S$. Consequently $o_U(H - S) - |S| \leq o(H[W]) - |N_{ext}(W)|$. Taking the two finite maxima proves their equality.

For $U = V(H)$, every matching M leaves $|V(H)| - 2|M|$ vertices uncovered, so $\eta_U(H) = |V(H)| - 2\nu(H)$. Every component of H-S is contained in U. The first specialized identity follows. If $n = |V(H)|$, summing component orders gives $\sum_C floor(|V(C)|/2) = (n - |S| - o(H - S))/2$. Thus $|S|$ plus this sum is $(n - (o(H - S) - |S|))/2$. Minimizing it over S gives $(n - (n - 2\nu(H)))/2 = \nu(H)$, which is the second specialized identity.

When U is empty, $\eta_U(H) = 0$ and every o_U count is zero. Prescribed isolated vertices are counted as odd components when they survive deletion, and the auxiliary construction joins them to all added vertices as required. When H is empty all the displayed sums and extrema equal zero. No exception to the argument is needed in these cases. □

Remark. Required odd components in excess of separator vertices force uncovered prescribed vertices. Adding exactly that many auxiliary neighbors of the prescribed set makes the zero-deficiency criterion applicable.

Depends on. Lemma 10.

Lemma 47 (A second-row certificate from common neighbors). *Use the finite integer certificate costs $c_{A,B}(H)$ and $c_2(H) = \max_{A+B=2} c_{A,B}(H)$ from Lemma 23. All capacities and matching budgets are nonnegative integers. Let H be a finite simple eight-regular graph on 27 vertices.*

If two distinct vertices have at least four common neighbors, then $c_2(H) \leq 16$. The same conclusion holds if H has an independent four-element set I whose open neighborhood $N_H(I)$, the union of the neighborhoods of its vertices, has size at most fourteen. In particular it holds whenever H contains an induced subgraph consisting of three disjoint triangles and four isolated vertices, with no edges between those seven components. No connectedness, factor-criticality, slack, gap or minimality hypothesis is required.

Proof. Fix any nonnegative integer capacity pair $A + B = 2$. We construct the four edge parts of Lemma 23 for every nonnegative integer budget pair $x + y = 16$. Interchanging the exceptional parts leaves the degree parts unchanged, so it suffices to handle $x \leq y$.

If $y \geq 10$, then $x \leq 6$. Choose any x-element vertex set W. Put all edges meeting W in the exceptional part of budget x and all other edges in the part of budget y. A matching in the first part has at most $|W| = x$ edges. The second has at most $floor((27 - x)/2)$ edges, which is at most $16 - x = y$ because $x \leq 6$. Both degree parts are empty.

For $(x, y) = (7, 9)$, choose any vertex p and a seven-element set W avoiding p and containing six of its eight neighbors. Such a set exists among the other 26 vertices. Put all edges meeting W in the exceptional part of budget seven. The edges avoiding W and meeting p form a star with at most two edges. Put this star into the two degree parts: for $(A, B) = (2, 0)$ or (0,2), put the whole star in the degree-two part; for $(A, B) = (1, 1)$, put its two edges in different parts, with the empty-edge cases treated by restriction. Every remaining edge has both ends outside W union {p}, a set with nineteen vertices, and so all remaining edges together have matching number at most nine. Put them in the other exceptional part.

It remains to realize (8,8). Choose distinct vertices p, q with $c = |N_H(p)$ intersect $N_H(q)| \geq 4$, and let ϵ be one if pq is an edge and zero otherwise. Simplicity and degree eight give $c \leq 8 - \epsilon$. The common-neighbor set C excludes p, q and has cardinality c. The two private-neighbor sets $P = N_H(p)$ minus (C union {q}) and $Q = N_H(q)$ minus (C union {p}) are disjoint and each has cardinality $r = 8 - \epsilon - c$. Let $l = \min(r, 2 - \epsilon)$. Choose l vertices to retain from each private-neighbor set. Define W_0 to contain C and all the private neighbors not retained. It avoids p, q and has size $|W_0| = c + 2(r - l)$. If $r \geq 2 - \epsilon$, this is $12 - c \leq 8$. If $r < 2 - \epsilon$, it is $c \leq 8$. Pad W_0 to a set W of exactly eight vertices avoiding p, q; there are 25 vertices available. Padding may remove some of the chosen retained neighbors, which only deletes edges from the bounded-degree graph described next.

Put all edges meeting W in the first exceptional part. Its matching number is at most eight. Among the remaining edges, let L consist of precisely those meeting p or q. Every common neighbor was deleted. If $\epsilon = 0$, L is a disjoint union of two stars with at most two edges each. If $\epsilon = 1$, L is a subgraph of the path on four vertices whose middle edge is pq and whose two possible end edges go to the retained private neighbors. In both cases L has maximum degree at most two and its edges partition into two matchings: alternate along each path component, with either choice on a single edge. Hence at $(A, B) = (1, 1)$ those two matchings are the degree parts, and at (2,0) or (0,2) all of L belongs to the degree-two part.

Every edge still unassigned has both ends outside W union {p, q}. These seventeen vertices support a matching of at most eight edges. Put all their edges in the second exceptional part. The four parts are disjoint and exhaust the original edges, and all displayed capacities and budgets hold. Together the three budget cases exhaust $x + y = 16$ up to exchanging x and y. Since the capacity pair was arbitrary, the definition gives $c_2(H) \leq 16$.

Now let I be independent of cardinality four and suppose $|N_H(I)| \leq 14$. Index fourteen positions containing its neighborhood, padding with zero-incidence positions if necessary, and let s_w be the number of neighbors in I at position w. The graph is eight-regular and I is independent, so $\sum_w s_w = 32$, with each s_w in {0,1,2,3,4}. For these five integers, $binomial(s_w, 2) \geq 2s_w - 3$, because the difference is $(s_w - 2)(s_w - 3)/2 \geq 0$. Therefore $\sum_w binomial(s_w, 2) \geq 64 - 42 = 22$. The sum counts common neighbors over the six unordered vertex pairs in I. If every such pair had at most three common neighbors it would be at most eighteen. Some pair therefore has at least four, and the already proved certificate applies. Finally, the four isolates of the specified thirteen-vertex induced subgraph have all their neighbors among the fourteen exterior vertices. They supply exactly the independent set required by this argument. □

Remark. Omit the common neighbors and all but a few private neighbors. The edges left at the chosen pair split into two matchings, while the remaining seventeen vertices fit the other matching budget.

Depends on. Lemma 23.

Lemma 48 (Order-sensitive absorption of a paid edge block). *Use the finite integer certificate families $C_{A,B}(G;x,y)$, costs $c_{A,B}(G)$, and row maxima $c_D(G)$ of Lemma 23. Thus a certificate partitions every edge into Q_R, Q_B, P_R, P_B with respective maximum-degree bounds A, B and matching-number bounds x, y. All graphs are finite and simple. Write $\nu(P)$ for the matching number of an edge set P, viewed as a spanning graph, and write $V(G)$, $E(G)$ for vertex and edge sets. All capacities, budgets, indices, cardinalities, matching numbers, and totals below are nonnegative integers.*

First, if a graph J has n vertices and x, y are nonnegative integers satisfying

$n \le x + y + \max(x,y) + 1,$

then $E(J)$ has a partition into two parts of matching numbers at most x, y. For an edge set P with $r = \nu(P)$, suppose some vertex set W contains every endpoint of every P-edge and $|W| \le 2r+1$. For each nonnegative integer δ put $l = r + \delta$ and let $Y_l(P)$ be the integers z in [0,l] for which P has a partition with matching bounds z, l-z. Then 0, l belong to $Y_l(P)$, and

[0,l] minus $Y_l(P)$ is contained in $[2\delta+1, r-\delta-1]$.

All intervals in this statement consist of integers and are empty when their lower endpoint exceeds their upper endpoint. In particular every consecutive missing run has length at most $\max(0, r-3\delta-1)$.

Let F be a graph, X a subset of $V(F)$, $a = |X|$, and $H = F - X$ the induced graph. Suppose

$E(H) = K$ dot-union P dot-union B_1 dot-union... dot-union B_m

is an edge partition, where $m \ge 0$. Empty parts and arbitrary shared vertices are allowed. Put $r = \nu(P)$, $r_j = \nu(B_j)$, and assume the endpoint-set condition on P just stated. Fix integers b, δ with $0 \le b \le m$ and $\delta \ge 0$, and put $L = a + \sum_{j=1}^{b} r_j$. Suppose

$r_j \le a + \sum_{h<j} r_h + 1$ *for* $1 \le j \le b$,

$\max(0, r-3\delta-1) \le L$,

$r_j \le L + r + \delta + \sum_{b<h<j} r_h + 1$ *for* $b < j \le m$.

Empty sums are zero. Put $T = a + r + \delta + \sum_{j=1}^{m} r_j$.

Fix capacities A, B and any partition $K = Q_R$ dot-union Q_B with $\Delta(Q_R) \le A$ and $\Delta(Q_B) \le B$. For every integer budget split $x + y = T$, this same partition extends to a member of $C_{A,B}(F;x,y)$. Every B_j is wholly in one matching-bounded part; P may be divided between those parts; every edge incident with X, including edges inside X, is matching-bounded. More explicitly, there are integers t, z and a subset I of {1,...,m} such that

$0 \le t \le a$, *z in* $Y_{r+\delta}(P)$,

$x = t + z + \sum_{j \in I} r_j$,

$y = (a-t) + (r+\delta-z) + \sum_{jnot \in I} r_j$.

The choices of t, z, I and of the partition of P may be fixed before choosing the capacities or their permitted partition of K. Consequently $c_{A,B}(F) \le T$ whenever K has the stated partition. For a nonnegative integer D, if K admits such a partition for every nonnegative integer capacity split $A + B = D$, then $c_D(F) \le T$.

In particular, let Q_1,..., Q_p be all components of F-X, including isolates, and put $R = \sum_i \nu(Q_i)$. If every component has matching number at most $a+1$, then $c_0(F) \le a + R$. If all but one component Q have matching number at most $a+1$ and $|V(Q)| \le 2\nu(Q)+1$, put $r = \nu(Q)$ and $L = a + \sum_{Q_i different from Q} \nu(Q_i)$. Then for every nonnegative integer δ with $3\delta \ge r - L - 1$,

$c_0(F) \le a + R + \delta$.

Thus $\delta = \max(0, ceil((r-L-1)/3))$ always suffices. These assertions require no independence of X, factor-criticality, matching-additivity identity, or source hypothesis.

Proof. We use Lemma 23 for the finite certificate definitions and their integer threshold interpretation. The constructions below supply actual edge partitions.

For the first assertion suppose $x \geq y$. Choose a set Z of $\min(n, 2x+1)$ vertices of J. Put all edges inside Z in the first part, and all other edges in the second part. A matching inside Z has at most $\lfloor |Z|/2 \rfloor \leq x$ edges. Every edge of the other part meets $V(J)$ minus Z. A matching there therefore has at most

$|V(J) \text{ minus } Z| = \max(0, n-2x-1) \leq y$

edges; the last inequality is the assumed bound $n \leq 2x+y+1$. If $y > x$, use the construction with x and y interchanged and interchange its two edge parts. This includes zero budgets and empty graphs.

Apply that construction to the graph with vertex set W and edge set P. Adding or removing isolated vertices does not change matching bounds. At total $l = r+\delta$, let z be the smaller budget. If $z \leq 2\delta$, then the larger budget is l-z, and

$l + (l-z) + 1 = 2r + 2\delta - z + 1 \geq 2r + 1 \geq |W|$.

The construction realizes that split. Therefore an infeasible red budget must satisfy both $z \geq 2\delta+1$ and $l - z \geq 2\delta + 1$, where here z denotes the red budget, not necessarily the smaller one. Equivalently every infeasible red budget lies in $[2\delta+1, r-\delta-1]$. Its number of integers, if positive, is r-3delta-1. The endpoint budgets 0 and l are feasible by putting all P-edges in the part of budget l, since $l \geq \nu(P) = r$. This proves all local allocation claims.

We next establish the integer interval rules used to choose the allocations. If all red budgets in [0,u] are available and a new whole edge block has matching number v, putting it wholly blue or wholly red supplies [0,u] and $[v, u+v]$. These integer intervals have no gap when $v \leq u+1$. Their union is then $[0, u+v]$. This statement also holds for $u = 0$ or $v = 0$.

More generally, let Y be a subset of [0,l] containing both endpoints, with no consecutive missing run longer than u. List its distinct members as $y_0 < \cdots < y_k$. If $l = 0$ this list consists only of zero. Otherwise $y_0 = 0$, $y_k = l$, and every successive difference is at most $u+1$: the integers strictly between consecutive members form a missing run of length $y_{i+1} - y_i - 1$. Hence the intervals $[y_i, y_i + u]$ have no intervening missing integer. Their union is $[0, l+u]$. Thus every integer in that interval can be written $y + t$ with y in Y and $0 \leq t \leq u$. The singleton case $l = 0$ has the same conclusion.

Order X as v_1,..., v_a. Assign each F-edge having an endpoint in X to its earliest endpoint in this order, obtaining a labelled star U_i centered at v_i. Each edge with two endpoints in X is assigned only to its earlier endpoint. Because $H = F - X$ is induced, the U_i partition exactly $E(F)$ minus $E(H)$. Each U_i has matching number at most one and may be empty. Assigning any t of them red and the rest blue supplies upper budgets t and a-t for every integer t in [0,a]. These are allowed upper bounds even when a labelled star is empty.

Starting from this interval [0,a], add B_1,..., B_b in the prescribed order, using only the wholly red and wholly blue allocations of B_j. The first set of inequalities in the statement and the first interval rule show by induction that every integer in [0,L] is represented as $t + \sum_{j \in I_0} r_j$, where $0 \leq t \leq a$ and I_0 is a subset of {1,...,b}. No restriction on incidences between edge blocks is involved in this arithmetic representation.

The local result for P bounds each missing run of $Y_{r+\delta}(P)$ by $\max(0, r-3\delta-1) \leq L$. The second interval rule therefore supplies every integer in $[0, L+r+\delta]$ by adding an attainable red budget of P to an available red budget of the earlier pieces. Append B_{b+1},..., B_m in order. At the step for B_j, the current complete interval has length $L + r + \delta + \sum_{b<h<j} r_h$. The last set of inequalities permits the first interval rule again. At termination the available red budgets fill [0,T].

Unwinding these finite interval choices for any prescribed x gives the stated t, z, I. The complementary blue sum equals $T - x = y$ by subtracting the red sum from the definition of T. Choose any partition $P = P'_R$ dot-union P'_B with matching bounds z, $r + \delta - z$; this exists by z in $Y_{r+\delta}(P)$. Fix these choices solely from x and the supplied edge blocks, without using A, B or the partition of K.

Keep the given Q_R, Q_B on K. Put in P_R the first t omitted-vertex stars, P'_R, and all B_j with j in I. Put in P_B the other omitted-vertex stars, P'_B, and all remaining B_j. The edge pieces are disjoint: the stars are exactly outside H, and the displayed edge partition is exactly inside H. They therefore assign every original F-edge once, including every edge internal to X. The degree bounds on Q_R, Q_B are unchanged. A matching in P_R restricts to a matching in each of its edge pieces. It has at most one edge in each of its t star pieces, at most z edges in P'_R, and at most r_j edges in each selected B_j. Its cardinality is consequently at most x. The identical restriction argument with the complementary pieces gives the blue bound y. Shared vertices can only restrict

which piece matchings coexist; the upper bounds do not require vertex-disjointness. This proves feasibility independently of the cost conclusion.

Every integer budget split at T has now been realized with the same degree-bounded partition. The definition of $c_{A,B}$ gives $c_{A,B}(F) \le T$. If every capacity pair of total D has a permitted partition of K, repeat this same construction for each pair, keeping the numerical choices and exceptional edge parts fixed. Taking the maximum over the finitely many capacity pairs gives $c_D(F) \le T$.

For the component conclusions, first suppose every component has matching number at most $a+1$. Set K and P empty, take W empty, $r = \delta = 0$, use the components as the B_j, and set $b = m = p$. Each ordered inequality holds because the current interval length is at least a. The middle condition is $zero \le L$. The construction gives $c_0(F) \le a + R$.

If Q is the distinguished component, let $P = E(Q)$, $W = V(Q)$, K empty, and list all other components as B_j before P, so $b = m$. Their ranks are at most $a+1$ and hence satisfy the first ordered inequalities. The condition $3\delta \ge r - L - 1$ is equivalent to $r - 3\delta - 1 \le L$; since $L \ge 0$ it implies $\max(0, r - 3\delta - 1) \le L$. There are no later blocks. The construction yields total $a + R + \delta$. The displayed ceiling is a nonnegative integer satisfying this condition. No matching-additivity assertion for F is used: R is simply the sum of the component matching numbers. All empty-part and zero-budget cases have already been included. □

Remark. Vertex order makes the extreme matching allocations available in a paid block. Each additional budget unit removes three possible positions from the remaining missing interval. The omitted vertices and earlier whole blocks supply an interval wide enough to bridge that interval, while each boundary edge is paid exactly once.

Depends on. Lemma 23.

Lemma 49 (Certificates from domination after a matching deletion). *Use the integer certificate costs and families of Lemma 23. Let $D \ge 1$ and $0 \le j \le D-1$ be integers. Let H be a finite simple graph with $\Delta(H) \le D$, let M be any matching of H, and let Z be a subset of $V(H)$ containing every vertex not covered by M. Suppose every vertex outside Z has at least j neighbors in Z in the spanning graph H-M obtained by deleting the edges of M. Then $\Delta(H - Z) \le D - j$, and the matching M intersect $E(H - Z)$ covers every vertex of degree D-j in H-Z. For every pair of nonnegative integers A, B with $A + B = D - j$ and every pair of nonnegative integers x, y with $x + y = |Z|$, there is an $(A, B; x, y)$-certificate of H whose bounded-degree parts partition $E(H - Z)$ and whose exceptional parts partition every edge meeting Z. One may first fix H-Z and its retained matching, independently of A, B, x, y. Consequently $c_{D-j}(H) \le |Z|$. In particular, if H is a connected d-regular finite simple graph for an even integer $d \ge 2$ and deletion of each vertex leaves a perfect matching, then for every vertex v, every perfect matching M_v of H-v, and every integer $1 \le j \le d-1$, any set Z containing v which has at least j neighbors in Z from every vertex outside it in $H - M_v$ satisfies $c_{d-j}(H) \le |Z|$. Taking $Z = \{v\}$ and $j = 0$ also gives $c_d(H) \le 1$.*

Proof. Put $Q = H - Z$. Every vertex u outside Z has H-degree at most D and at least j H-neighbors in Z, since deleting matching edges cannot create neighbors. Therefore $deg_Q(u) \le D - j$. If equality holds, both inequalities must be equalities: $deg_H(u) = D$ and u has exactly j H-neighbors in Z. The hypothesis gives at least j neighbors in Z in H-M, so no H-edge from u to Z can belong to M. The vertex u is covered by M because Z contains all uncovered vertices. Its unique matching partner must therefore lie outside Z, and its matching edge belongs to Q. This proves the required covering property, including $j = 0$.

The integer D-j is positive. Apply Lemma 35 to Q with the retained matching M intersect $E(Q)$. For every nonnegative integer $A + B = D - j$ it supplies a partition $E(Q) = Q_R$ disjoint union Q_B with maximum degrees at most A and B. The graph Q and the retained matching are fixed before these capacity choices. Fix nonnegative integers x, y with $x + y = |Z|$, and choose a partition $Z = Z_R$ disjoint union Z_B with $|Z_R| = x$ and $|Z_B| = y$. Such a partition exists because x is an integer in $[0, |Z|]$. Assign every H-edge meeting Z_R to P_R; assign every remaining H-edge meeting Z to P_B. Each edge of P_B has an endpoint in Z_B, since it meets Z and has no endpoint in Z_R. The four parts are disjoint and cover $E(H)$, because the edges not meeting Z are exactly $E(Q)$. A matching in P_R has at most $|Z_R|$ edges: assign each of its edges an endpoint in Z_R, which must be distinct across disjoint edges. The same argument bounds matchings in P_B by $|Z_B|$. Thus the four parts give the stated certificate. The definition in Lemma 23 gives $c_{A,B}(H) \le |Z|$ for every $A + B = D - j$, and maximizing over these capacities gives $c_{D-j}(H) \le |Z|$.

For the particular case, a perfect matching M_v of H-v is a matching of H with exactly v uncovered. The hypotheses on Z are therefore precisely those of the general construction with $D = d$ and $M = M_v$. For $j = 0$ the empty neighbor requirement is satisfied by $Z = \{v\}$; applying the same construction gives $c_d(H) \le 1$. □

Remark. A vertex attaining the residual degree bound has lost exactly its required neighbors to Z. None of those lost edges can be its matching edge, so its partner stays in the residual and certifies a degree split.

Depends on. Lemma 23, Lemma 35.

Lemma 50 (Connected isolation bounds for certificate excess and top runs). *Use the finite integer certificate cost $c_0(H)$, matching number $\nu(H)$, excess $ex(H) = c_0(H) - \nu(H)$, and longest missing-allocation run $tr(H)$ from Lemma 23. Let $D \ge 3$ and $k \ge 1$ be integers, and let H be a connected finite simple graph on $2k+1$ vertices with $\Delta(H) \le D$. For a vertex set I, let $N_H(I)$ be the union of its vertices' open neighborhoods. For every vertex v and every integer t with $1 \le t \le \lceil 2k/D \rceil$, there is an independent set I of cardinality t containing v and satisfying $|N_H(I)| \le (D-1)t+1$. These sets can be chosen nested as t increases.*

One has $c_0(H) \le k + \min(floor((k+1)/3), ceil(((D-2)k+2)/(3D-2)))$. If also $\nu(H) = k$ and $E := ex(H) \ge 1$, put $u = \max(0, floor((2E-3)/(D-2)))$. Every missing red budget at total $c_0(H) - 1 = k + E - 1$ lies in the integer interval $[2E-1+u, k-E-u]$. Consequently $1 \le tr(H) \le k - 3E + 2 - 2u$ and $k \ge 3E - 1 + 2u$. For every integer B with $tr(H) + E \ge B + 1$, one has $k \ge B + 2E - 1 + 2u$. In particular $tr(H) + E \ge \max(4, \lfloor D/2 \rfloor + 1) + 1$ implies $E \le floor((k - \max(4, \lfloor D/2 \rfloor + 1) + 1 - 2u)/2)$. All orders, cardinalities, degrees, budgets, costs, interval endpoints, k, D, E, u, B and t are integers in the displayed domains; an interval with lower endpoint above its upper endpoint is empty.

Proof. Fix a vertex v. Start with $I_1 = \{v\}$, which is independent and has at most $D = (D-1)+1$ neighbors. Suppose an independent set I_t containing v has been constructed but does not dominate H. Put $B_t = I_t$ union $N_H(I_t)$. This is a nonempty proper vertex set. Connectedness supplies an edge wz with w in B_t and z outside B_t: take the first edge leaving B_t along a path from v to an outside vertex. The vertex w is not in I_t, since z has no neighbor in I_t. Thus w belongs to $N_H(I_t)$, and z shares this neighbor with a member of I_t. The set $I_{t+1} = I_t$ union {z} is independent and its neighborhood has size at most $|N_H(I_t)| + D - 1$. Induction gives $|N_H(I_t)| \le (D-1)t+1$ for every constructed prefix. The construction increases cardinality at every step, so it stops at a dominating independent set I_r. At that point $2k+1 = r + |N_H(I_r)| \le Dr + 1$, hence $r \ge \lceil 2k/D \rceil$. This proves the nested-set assertion, including every indicated endpoint.

We describe explicitly how to turn such a set into a zero-capacity certificate. Let x, y be nonnegative integers. If a vertex set W has cardinality x and its deletion leaves at least t isolated vertices, where $2k+1-x-t \le 2y+1$, put every edge meeting W in the red exceptional part and every other edge in the blue exceptional part, leaving both bounded-degree parts empty. Every red matching uses distinct W-vertices, so has size at most x. The blue edges are supported on at most $2y+1$ nonisolated vertices, so every blue matching has size at most y. Thus (x, y) is feasible. In particular, if I is independent of size t and $|N_H(I)| \le x \le 2k+1-t$, enlarge $N_H(I)$ to a set W of size x avoiding I; all vertices of I become isolated in H-W. For $t \le 0$ the same argument needs no isolated vertices and any set W of size x suffices whenever the displayed order inequality holds.

First prove the vertex-count bound $c_0(H) \le k + e_0$ for $e_0 = floor((k+1)/3)$. More generally, a total c is universally feasible whenever $c + \lceil c/2 \rceil + 1 \ge 2k+1$. For a split $x + y = c$ arrange $x \le y$ by interchanging the names of the exceptional parts. If $y \ge k$, all edges can be placed in the blue part. Otherwise choose x vertices W; since $x + 2y = c + y \ge c + \lceil c/2 \rceil$, the residual order is at most $2y+1$ and the preceding construction applies with $t = 0$. At $c = k + e_0$, the quantity $c + \lceil c/2 \rceil + 1$ equals $6h+1$, $6h+3$, $6h+6$ when $k = 3h$, $3h+1$, $3h+2$, respectively, while $2k+1$ equals $6h+1$, $6h+3$, $6h+5$. Thus this total is universally feasible in all three residue classes. Its least feasible total is $c_0(H)$ by Lemma 23.

For the connected improvement put $e_1 = ceil(((D-2)k+2)/(3D-2))$, $c = k + e_1$, $x_\cdot = \lfloor c/2 \rfloor$, and $t_\cdot = x_\cdot - 2e_1$. Fix any split $x + y = c$ and arrange $x \le y$, so $x \le x_\cdot$. Again $y \ge k$ is immediate. Suppose $y \le k-1$. Then $0 \le x \le y < k$, so sets of size x can be chosen. Put $t = x - 2e_1$. Direct substitution gives $t = 2k+1-x-2y-1$. If $t \le 0$ the vertex-count construction applies. Otherwise $t_\cdot > 0$ and the defining inequality $(3D-2)e_1 \ge (D-2)k+2$ gives

$$(D-2)x_\cdot + 1 \le ((D-2)(k+e_1)+2)/2 \le 2(D-1)e_1.$$

Since $D \ge 3$ and $x \le x_\cdot$, this also holds with x in place of $x_\cdot$. Rearrangement gives $(D-1)t + 1 \le x$. Furthermore

$t \le t. \le (k-3e_1)/2 \le (2k-3)/(3D-2) < 2k/D$.

For the strict last inequality, the difference between 2k(3D-2) and $D(2k-3)$ is $4k(D-1)+3D > 0$. Thus $t \le \lceil 2k/D \rceil$. The nested construction supplies an independent t-set I with neighborhood size at most x. Since $2k+1-t = x+2y+1 \ge x$, that neighborhood can be enlarged to x vertices while avoiding I. The isolation certificate makes the split feasible. Every split at total c is therefore feasible, proving $c_0(H) \le k+e_1$. Combining the two totals proves the stated minimum bound.

Now assume $\nu(H) = k$ and $E = ex(H) \ge 1$. Put $\ell = k+E-1 = c_0(H)-1$ and $L = 2E-2+u$. We prove that every split $x+y = \ell$ with $\min(x,y) \le L$ is feasible. Arrange $x \le y$ and $x \le L$. If $x \le 2E-2$, then $x+2y = 2\ell - x \ge 2k$, so the vertex-count construction applies; if $y \ge k$ it is already feasible by placing all edges in the blue part. These arguments cover all x in that first range.

In the remaining case $t := x-2E+2$ is an integer with $1 \le t \le u$. Hence $u \ge 1$. From $(D-2)u \le 2E-3$ we obtain $(D-1)t+1 \le t+2E-2 = x$. The proved vertex-count bound gives $E \le floor((k+1)/3) \le (k+1)/3$, and therefore

$u \le (2E-3)/(D-2) \le (2k-7)/(3(D-2)) < 2k/D$.

The final inequality follows because $6k(D-2) - D(2k-7) = 4k(D-3)+7D > 0$. Thus $t \le \lceil 2k/D \rceil$, as needed for the nested independent-set construction. Take its t-set I and enlarge its neighborhood to a set W of size x avoiding I. This is possible because, at the present total ℓ, $t = 2k+1-x-2y-1$ and therefore $2k+1-t = x+2y+1 \ge x$. The isolation certificate proves feasibility of this split also.

Consequently every missing split has both its budgets at least $L+1 = 2E-1+u$. Its first budget x is therefore in $[2E-1+u, \ell-(2E-1+u)] = [2E-1+u, k-E-u]$. Since $c_0(H) = k+E > 0$, the defining minimality in Lemma 23 ensures that at least one split at total ℓ is missing and $tr(H) \ge 1$. Every missing run lies in this interval, whose integer length is $k-3E+2-2u$. This proves both the run bound and $k \ge 3E-1+2u$. Adding E to the run bound gives $tr(H)+E \le k-2E+2-2u$. Comparing it with any assumed lower bound $B+1$ gives $k \ge B+2E-1+2u$. Substitute $B = \max(4, \lfloor D/2 \rfloor + 1)$ and divide the resulting integer inequality by two to obtain the last assertion. □

Remark. Connectedness permits independent vertices to be added at distance two, sharing one neighbor at each step. Deleting their small common neighborhood supplies actual matching-budget certificates and excludes long missing runs near the endpoints.

Depends on. Lemma 23.

Lemma 51 (Ordered absorption of whole edge blocks). *Use the certificate families $C_{A,B}$, integer costs $c_{A,B}$, and rows c_θ of Lemma 23. Let H be a finite simple graph, let X be any subset of $V(H)$, and put $a = |X|$. Suppose the edges of the induced graph H-X are partitioned into spanning graphs J, B_1,..., B_m on $V(H)$ minus X, where m is a nonnegative integer. Empty edge parts are allowed and the graphs may share vertices. Put $r_j = \nu(B_j)$ for integers $1 \le j \le m$.*

Fix nonnegative integers A, B, c with $c \ge c_{A,B}(J)$. Suppose the given ordering satisfies

$r_j \le a + c + \sum_{h<j} r_h + 1$ *for every integer* $1 \le j \le m$.

Put $T = a + c + \sum_{j=1}^{m} r_j$. For every pair of nonnegative integers x, y with $x+y = T$, there exist integers t, u and a subset I of {1,...,m} such that

$0 \le t \le a$, $0 \le u \le c$,

$x = t + u + \sum_{j \in I} r_j$,

$y = (a-t) + (c-u) + \sum_{jnot \in I} r_j$.

Every certificate in $C_{A,B}(J; u, c-u)$ extends to a certificate in $C_{A,B}(H; x, y)$ with exactly the same bounded-degree parts, with every B_j wholly in the red exceptional part for j in I and wholly in the blue exceptional part for j not in I. All edges meeting X are exceptional. For any t-element subset X_R of X, assign such an edge red if it meets X_R and blue otherwise; this includes edges with both endpoints in X.

Consequently $c_{A,B}(H) \le T$.

For a nonnegative integer θ, if the ordered inequalities hold with $c = c_\theta(J)$, then

$c_\theta(H) \le a + c_\theta(J) + \sum_j r_j$.

The same integers t, u and subset I may be fixed for a given x before choosing the capacity split $A + B = \theta$; only the chosen certificate of J may depend on that split. Thus a finite table of actual four-part partitions is supplied for every capacity split and every integer budget allocation at this total. If $a \ge 1$ and every $r_j \le 2$, these conclusions hold for every $\theta \ge 0$. For $m = 0$ the ordered inequalities are empty and the conclusion still holds.

In particular, suppose $q \ge 0$ is an integer, M_1,..., M_q are pairwise edge-disjoint matchings in H-X, J is their spanning edge union, and P is the spanning graph of all other edges of H-X. Put $\rho = \nu(P)$. If $a \ge \max(0, \rho - 1)$, then

$c_q(H) \le a + \rho$.

For every capacity split $A + B = q$ and allocation $x + y = a + \rho$, the constructed certificate has bounded-degree parts whose union is exactly $E(J)$; the whole edge set $E(P)$ is in one exceptional color, and all edges meeting X are exceptional. The supplied matchings need not be maximum or saturate any specified vertices. The cases $q = 0$, $\rho = 0$, X empty, and empty graphs are included.

Proof. For integers $L, r \ge 0$ with $r \le L + 1$, the integer intervals [0,L] and $[r, L + r]$ have union $[0, L + r]$. If $r = 0$ this is equality with one interval. If $r > 0$, their possible separating gap contains no integer because $r \le L + 1$. Equivalently, every integer z in $[0, L + r]$ either lies in [0,L] or has z-r in [0,L]. More explicitly, choose the first option when $z \le L$. Otherwise $z \ge L + 1 \ge r$ and $z - r \le L$, so the second option is valid.

Define $L_0 = a + c$ and $L_j = L_0 + \sum_{h=1}^{j} r_h$ for $1 \le j \le m$. The assumed inequality is exactly $r_j \le L_{j-1} + 1$. Starting with $z = x$ in $[0, L_m]$, work backwards from $j = m$ to $j = 1$. If $z \le L_{j-1}$, omit j from I and leave z unchanged. If $z > L_{j-1}$, put j in I and replace z by $z - r_j$. The preceding paragraph proves after this step that the new z lies in $[0, L_{j-1}]$. The procedure terminates after m steps with an integer z in $[0, a + c]$ and $x = z + \sum_{j \in I} r_j$. Put $t = \min(a, z)$ and $u = z - t$. Then $0 \le t \le a$. If $z \le a$, $u = 0$; if $z > a$, $u = z - a \le c$. Thus $0 \le u \le c$ and the required red identity holds. Subtracting it from T gives the blue identity and its nonnegative terms. When $m = 0$ this construction starts directly with $z = x$.

By the threshold characterization in Lemma 23, the family $C_{A,B}(J; u, c - u)$ is nonempty. Fix any member and retain its bounded-degree parts. Add every edge of B_j to the red exceptional part if j is in I, and to the blue exceptional part if j is not in I. These assignments partition all edges of H-X because its supplied edge parts were disjoint. A matching in their red union restricts to a matching in the original red exceptional part and one in each red B_j; consequently its cardinality is at most $u + \sum_{j \in I} r_j$. For the same reason the blue bound is $c - u + \sum_{jnot \in I} r_j$. Shared vertices between edge parts can reduce these matching numbers and cannot invalidate the upper bounds.

Choose any t-element subset X_R of X. Let E_R be all H-edges meeting X_R, and let E_B be the other H-edges meeting X. Every edge in E_B meets X minus X_R. Assign E_R to the red exceptional part and E_B to the blue exceptional part. The two sets are disjoint, and together they contain exactly the edges not already assigned on H-X. Every matching in E_R has at most t edges, since its edges meet distinct vertices of X_R. Every matching in E_B has at most a-t edges for the corresponding reason. Restricting a full exceptional matching between its edges on H-X and its edges meeting X therefore gives the total bounds x and y from the proved identities. No bounded-degree edge was added, so the degree bounds remain A and B. Every edge with both endpoints in X has been assigned once by this rule. This verifies every part of the asserted extension and proves $c_{A,B}(H) \le T$.

For fixed θ and $c = c_\theta(J)$, every pair $A + B = \theta$ satisfies $c \ge c_{A,B}(J)$ by Lemma 23. The backward integer procedure depends on a, c, r_1,..., r_m, x and uses no capacity choice. Apply the extension for each capacity pair, taking a certificate of J at the same budgets u, c-u. There are $\theta + 1$ capacity pairs and $T + 1$ allocations, both finite; selecting the supplied certificate for each pair gives the stated finite table. Taking the maximum of the fixed-capacity cost bounds proves the row bound. If $a \ge 1$ and every $r_j \le 2$, then $a + c_\theta(J) + \sum_{h<j} r_h + 1 \ge 2$, so every required ordered inequality holds.

For the final assertion, fix any $A + B = q$. Put M_1,..., M_A in Q_R and the remaining B supplied matchings in Q_B. At each vertex each matching contributes at most one incident edge, so these two parts have maximum degrees at most A and B. They are disjoint and together form exactly $E(J)$. Empty exceptional parts complete an (A, B;0,0)-certificate of J. This works for every capacity split, so $c_q(J) = 0$. Apply the proved construction

with one block $B_1 = P$, with $c = 0$ and $r_1 = \rho$. Its inequality is $\rho \le a + 1$, which is equivalent, for $a \ge 0$, to $a \ge \max(0, \rho - 1)$. The row total becomes $a + \rho$, and $u = 0$ forces the retained J-certificate to have empty exceptional parts. Hence the bounded-degree union is exactly $E(J)$, P is wholly in the selected exceptional color, and the cover rule assigns every edge meeting X. If $q = 0$, J is edgeless; if $\rho = 0$, P is edgeless because a nonempty simple edge set contains a matching of size one. All zero and empty cases therefore satisfy the same construction. □

Remark. A whole edge block offers either zero or its matching rank to a color. The ordered bound ensures these choices fill the integer budget interval already supplied by the deleted vertices and retained certificate.

Depends on. Lemma 23.

Lemma 52 (Balanced matching allocations on short equal-cycle rows). *All graphs are finite and simple. Write $V(G)$, $E(G)$, and $\nu(G)$ for the vertex set, edge set, and maximum matching cardinality of a graph G. For an edge set, its matching number is that of its spanning graph. A whole component means an entire connected component; cycle orders count vertices.*

Let $p \ge 2$ be an integer and let w belong to {2,3}. Put

$d = 2p + 2$, $k = (3p + 2)w$, $b = (2p + 1)w$, *and* $z = (p + 2)w - p$.

Let F be d-regular with $2k + 1$ vertices. Suppose F contains an induced subgraph R whose entire component list is p cycles, each of order $2w + 1$, and z isolated vertices.

Then there is a partition $E(F) = P_R$ dot-union P_B with

$\nu(P_R) \le b$ *and* $\nu(P_B) \le b$.

All counts, orders, degrees, budgets, and matching cardinalities are nonnegative integers, with $p \ge 2$ and w in the stated two-element domain. No hypothesis on the edges induced outside R is imposed.

Proof. We use Lemma 7.

Suppose the asserted partition does not exist. A cycle of order $2w + 1$ has matching number w: alternating edges avoiding one vertex attain w, and counting two endpoints per matching edge proves the upper bound. Matching numbers add across disjoint components by restricting a matching and by taking unions of componentwise maximum matchings. Thus

$|V(R)| = p(2w + 1) + z = k$ and $\nu(R) = pw$.

Every component of R has odd order and matching number half its order rounded down.

Put $U = V(F)$ minus $V(R)$, so $|U| = k + 1$, and put $j = (p + 1)w + 1$. The supposed missing pair is (b, b). Its total plus one satisfies

$2b + 1 = pw + (k + 1)$,

and $|U| - b = j$. Apply Lemma 7 with row vertex set $V(R)$. For every subset W of U with $|W| = j$, it gives actual F-edges assigning all j vertices of W to distinct whole components of R. Its exterior-size condition holds with equality, and $|U| \ge b$ since $j > 0$.

Let Z be the set of z isolated vertices of R. For each a in Z, define S_a to be its F-neighborhood. Since a is isolated in the induced graph R, the set S_a is contained in U. Regularity and simplicity give $|S_a| = d$.

For any w distinct vertices a_1,..., a_w of Z, we claim

$|S_{a_1}$ union... union $S_{a_w}| \ge b + 1$. (1)

If their union had at most b vertices, U would have at least $|U| - b = j$ vertices avoiding every one of these neighborhoods. Choose W of size j among them. None could be assigned to any of these w isolated components. The total number of R-components is $p + z = (p + 2)w$, leaving only $(p + 1)w = j - 1$ other components. This contradicts the component-distinct assignment of all j members of W. Such a choice of w isolates is permitted because $z - w = (p + 1)w - p \ge 0$.

First take $w = 3$. Then

$z = 2p + 6$, $d = 2p + 2$, $|U| = 9p + 7$, and $b + 1 = 6p + 4 = 3d - 2$.

For a three-element subset A of Z, put

$D_A = 3d - |\bigcup_{a \in A} S_a|$.

By (1), $0 \le D_A \le 2$. For each v in U let $t_v = |\{a in Z : v in S_a\}|$, an integer between zero and z. Count the contribution of v to the sum of D_A over all three-element A. It is counted with multiplicity t_v $binom(z-1,2)$ in the sum of individual neighborhood cardinalities, and with multiplicity $binom(z,3) - binom(z - t_v, 3)$ in the union cardinalities. Therefore

$\sum_{A \subseteq Z, |A|=3} D_A = \sum_{v \in U} g_z(t_v)$,

where

$g_z(t) = t\ binom(z-1,2) - binom(z,3) + binom(z-t,3)$.

Here $binom(n, j)$ counts j-element subsets of an n-element set and is zero for integers $0 \le n < j$.

For every integer $0 \le t \le z$ one has the identity

$g_z(t) - (2z-5)t + (3z-8)$

$= (t-2)(t-3)(3z-8-t)/6$.

The product (t-2)(t-3) is nonnegative at every integer t. Also $z \ge 10$ and $t \le z$ give $3z - 8 - t \ge 2z - 8 > 0$. Hence

$g_z(t) \ge (2z-5)t - (3z-8)$. (2)

Because every S_a has d members, counting all incidences (a, v) gives $\sum_{v \in U} t_v = zd$. Summing (2) and using $D_A \le 2$ yields

$(2z-5)zd - (3z-8)|U| \le 2\ binom(z,3)$. (3)

After substituting the values of z, d, $|U|$ above, the left side minus the right side of (3) is

$(40p^3 + 54p^2 - 64p - 78)/3$.

Putting $q = p - 2 \ge 0$, its numerator is

$40q^3 + 294q^2 + 632q + 330$,

which is positive. This contradicts (3).

Now take $w = 2$. Then $z = p + 4 \ge 6$, $|U| = 6p + 5$, and $b + 1 = 4p + 3 = 2d - 1$. By (1), every two distinct neighborhoods satisfy

$|S_a$ intersect $S_{a'}| = 2d - |S_a$ union $S_{a'}| \le 1$.

Choose four distinct members of Z. Add their neighborhoods in any order. The second meets the first in at most one point; the third meets their union in at most two points; the fourth meets the previous union in at most three points. Their union consequently has size at least

$d + (d-1) + (d-2) + (d-3) = 4d - 6 = 8p + 2$.

This exceeds $|U| = 6p + 5$, since $2p - 3 > 0$ for $p \ge 2$. This is again a contradiction.

Both permitted values of w contradict the supposed missing partition. Its negation is exactly the asserted two-part edge partition with matching bounds b, b. □

Remark. Missingness would make every sufficiently large exterior subset match into distinct row components. Avoiding a few isolated components then forces their neighborhoods to have very large unions. Pairwise counting handles pentagons, while integer incidence counting over triples handles heptagons.

Depends on. Lemma 7.

Lemma 53 (Weighted marked domination and top-cost certificates). *Use the finite integer certificate families $C_{A,B}(F;x,y)$ and costs $c_i(F)$ of Lemma 23. A certificate partitions every edge into two parts of maximum degrees at most the nonnegative integer capacities A, B and two parts of matching numbers at most the nonnegative integer budgets x, y. The cost c_i is the maximum over $A+B=i$ of the least integer total $x+y$ for which every budget split is feasible. Write $V(G)$, $E(G)$, $d_G(v)$, $\Delta(G)$, $\nu(G)$, and $N_G[v]$ for the vertex set, edge set, vertex degree, maximum degree, matching number, and closed neighborhood.*

Let $d \geq 4$ be an even integer and put $D = d-1$. Let $a > 0$, $\sigma > 0$ and b_0,..., b_D be real numbers satisfying

$b_0 = 0, 0 \leq b_D - b_{D-1} \leq \cdots \leq b_2 - b_1 \leq b_1, b_D \leq a,$

$a + D\, b_1 \geq \sigma, (D+1)a - D\, b_{D-1} \geq \sigma,$

$j\, a - (j-1)b_{j-2} + j(D+1-j)(b_j - b_{j-1}) \geq \sigma$ *for every integer* $2 \leq j \leq D$.

Let F be a connected finite simple d-regular graph of order $n = 2k+1$, where $k \geq 1$ is an integer. Assume F is factor-critical: F-z has a perfect matching for every vertex z. Fix any vertex z, any perfect matching M of F-z, and any ordering of $V(F)$. Form the spanning graph $J = (V(F), E(F) minus M)$, put $Q = J - z$ by induced vertex deletion, and put $S = V(F)$ minus $N_J[z]$.

Construct Y in Q as follows. Initially Y is empty and $W = S$. While W is nonempty, select a vertex of Q maximizing $|N_Q[v]$ intersect $W|$, p refer a vertex in W among maximizers, and use the fixed ordering to break remaining ties. Add that vertex to Y and remove its closed neighborhood from W. Put $Z = \{z\}$ union Y and $R = F - Z$.

This procedure terminates after at most $|S|$ selections. The set Z dominates J in closed neighborhoods, contains z, and satisfies

$|Z| \leq 1 + floor((a(n-d-1) + db_D)/\sigma).$

One has $\Delta(R) \leq d-1$, and M intersect $E(R)$ covers every vertex of R-degree d-1. The same Z, R and retained matching are fixed before any capacities or budgets. For every nonnegative integer $A+B = d-1$ and $x+y = |Z|$, a certificate in $C_{A,B}(F;x,y)$ has its degree-bounded parts partitioning $E(R)$ and its matching-bounded parts partitioning every original edge meeting Z, including edges with both endpoints in Z. Consequently

$c_{d-1}(F) \leq 1 + floor((a(n-d-1) + db_D)/\sigma).$

For each d in {6,8,10,12}, explicit weight choices give the following bounds for this same construction, for every z, M and ordering:

$c_5(F) \leq |Z| \leq floor((3450n - 806)/10000)$ *when* $d = 6$;

$c_7(F) \leq |Z| \leq floor((400n - 580)/1300)$ *when* $d = 8$;

$c_9(F) \leq |Z| \leq floor((280n - 710)/1000)$ *when* $d = 10$;

$c_{11}(F) \leq |Z| \leq floor((250n - 726)/1000)$ *when* $d = 12$.

The procedure deletes vertices only in forming the explicitly induced graphs; deleting a matching always deletes its edges and retains all vertices. All graph orders, degrees, matching numbers, indices, capacities, budgets, totals and selected-set sizes are nonnegative integers in the stated domains. Empty S and Y are allowed. No excess, slack, source-minimality or independence-number hypothesis is imposed.

Proof. We use Lemma 1 for weighted greedy domination, Lemma 35 for splitting a graph with a specified covering matching, and Lemma 23 for the integer certificate definition.

1. Marked domination.

The matching M covers precisely the vertices other than z. Thus in $J = F - M$ the degree of z is d and every other degree is d-1. In $Q = J - z$ the maximum degree is at most $D = d-1$. The set S has n-d-1 vertices, each of Q-degree D, because its vertices have no J-edge to z. Moreover $|V(Q)| - |S| = d$. Simplicity and d-regularity give $n \geq d+1$.

The weights satisfy exactly the assumptions of Lemma 1 at degree $\cap\, D$, and S is a permitted prescribed subset of the degree-D vertices. Its greedy construction is precisely the stated rule with the induced ordering on Q. It terminates after at most $|S|$ distinct selections and gives

$|Y| \leq floor((a|S| + b_D(|V(Q)| - |S|))/\sigma)$

$= floor((a(n-d-1)+db_D)/\sigma)$.

Since z is not a vertex of Q, $|Z| = 1 + |Y|$. The set Y dominates S in Q and hence in J, while z dominates all of $N_J[z]$. These two subsets cover $V(F)$. Thus Z dominates J, contains z, and has the asserted size. If S is empty, the procedure selects no vertices, so Y is empty and $Z = \{z\}$. The same inequality holds since $b_D \geq 0$.

2. Covering and partitioning the same residual.

Every vertex v outside Z has at least one neighbor in Z in J, hence also in F. Therefore its degree in $R = F - Z$ is at most d-1. If it equals d-1, exactly one F-edge joins v to Z. The required J-neighbor in Z uses that same edge, so the edge does not belong to M. The vertex v is covered by M because z is its only uncovered vertex and z belongs to Z. Its matching partner consequently lies outside Z. Its matching edge therefore belongs to R. This proves that the retained matching M intersect $E(R)$ covers every degree-(d-1) vertex of R.

The graph R and this matching were determined by F, z, M and the vertex ordering. Apply Lemma 35 with its positive degree bound d-1. For every nonnegative integer $A + B = d - 1$ it partitions $E(R)$ into Q_R, Q_B with maximum degrees at most A, B. For any nonnegative integers x, y with $x + y = |Z|$, choose a partition $Z = Z_R$ dot-union Z_B with $|Z_R| = x$ and $|Z_B| = y$. Such a partition exists because x is an integer in $[0, |Z|]$. Let P_R consist of all F-edges meeting Z_R, and let P_B consist of all remaining F-edges meeting Z. Every P_B edge has an endpoint in Z_B. A matching in P_R or P_B uses distinct vertices of its corresponding set, so its size is at most x or y, respectively. These four edge parts are disjoint and cover every original edge: the edges not meeting Z are precisely $E(R)$. In particular all edges internal to Z are included.

This constructs a member of $C_{A,B}(F; x, y)$ for every indicated pair of splits. Its graph and retained matching do not depend on those splits. The definition in Lemma 23 gives $c_{d-1}(F) \leq |Z|$ and then the general displayed bound.

3. Four explicit weight tables.

For clarity use the names a, σ, b_j from the statement and keep $D = d - 1$. In each case below the sufficient gain at closed uncovered count one is $a + D\ b_1$, at count $D + 1$ it is $(D+1)a - D\ b_{D-1}$, and for integers $2 \leq j \leq D$ it is

$j\,a - (j-1)b_{j-2} + j(D+1-j)(b_j - b_{j-1})$.

We list all these gains in increasing order of the closed uncovered count.

At $D = 5$ take

$a = 3450$, $\sigma = 10000$,

$(b_0, b_1, b_2, b_3, b_4, b_5) = (0, 1310, 1698, 1951, 2113, 2224)$.

The differences are 1310, 388, 253, 162, 111. They are nonnegative and nonincreasing, and $b_5 \leq a$. The six gains are

10000, 10004, 10007, 10002, 10001, 10135.

All are at least σ.

At $D = 7$ take

$a = 400$, $\sigma = 1300$,

$(b_0, b_1, b_2, b_3, b_4, b_5, b_6, b_7) = (0, 129, 171, 195, 209, 215, 215, 215)$.

The differences are 129, 42, 24, 14, 6, 0, 0. They are nonnegative and nonincreasing, and $b_7 \leq a$. The eight gains are

1303, 1304, 1302, 1311, 1310, 1355, 1510, 1695.

All are at least σ.

At $D = 9$ take

$a = 280$, $\sigma = 1000$,

$(b_0, b_1, b_2, b_3, b_4, b_5, b_6, b_7, b_8, b_9)$

$= (0, 80, 108, 124, 133, 137, 137, 137, 137, 137)$.

The differences are 80, 28, 16, 9, 4, 0, 0, 0, 0. They are nonnegative and nonincreasing, and $b_9 \le a$. The ten gains are

1000, 1008, 1016, 1012, 1004, 1015, 1138, 1281, 1424, 1567.

All are at least σ.

At $D = 11$ take

$a = 250$, $\sigma = 1000$,

$(b_0, b_1, b_2, b_3, b_4, b_5, b_6, b_7, b_8, b_9, b_{10}, b_{11})$

$= (0, 69, 94, 109, 118, 124, 127, 127, 127, 127, 127, 127)$.

The differences are 69, 25, 15, 9, 6, 3, 0, 0, 0, 0, 0. They are nonnegative and nonincreasing, and $b_{11} \le a$. The twelve gains are

1009, 1000, 1017, 1006, 1024, 1018, 1006, 1111, 1234, 1357, 1480, 1603.

All are at least σ.

Thus each table satisfies every weight and gain condition. Finally,

$1 + floor((a(n-d-1) + db_D)/\sigma)$

$= floor((an + \sigma - a(d+1) + db_D)/\sigma)$.

For $d = 6, 8, 10, 12$ the constant $\sigma - a(d+1) + d\,b_D$ is respectively -806, -580, -710, -726. Substitution gives the four asserted bounds. The greedy rule depends only on the prescribed graph, set and ordering, so using these tables changes the proved bound without changing the selected set. Every certificate from step 2 therefore uses that same Z. This proves the four numerical applications and all stated boundaries. □

Remark. The fixed marked greedy set dominates the matching-deleted graph. Its residual retains a covering matching and supports every capacity and budget split. Both transcribed numerical lists are restored to the saved draft.

Depends on. Lemma 1, Lemma 23, Lemma 35.

Lemma 54 (Averaged isolation with private neighbors). *Use the finite integer certificate costs $c_\theta(H)$ of Lemma 23. Let $d \ge 2$ and $k \ge 1$ be integers, put $n = 2k+1$, and let H be a d-regular finite simple graph on n vertices. Let S be an independent vertex set of cardinality a with $1 \le a < n$. Put $b = n - a$, $q = \lfloor da/b \rfloor$, and $r = da - qb$. For integers A, t with $A \ge 0$ and $t \ge 0$ use $binom(A,t) = 0$ if $t > A$. For each integer $1 \le t \le a$ define $f_t(j) = 1 - binom(a-j,t)/binom(a,t)$ for $0 \le j \le a$, and $U_t = (b-r)f_t(q) + r\,f_t(q+1)$, omitting the second term when $r = 0$. For integers $0 \le \theta \le d$ and $1 \le t \le a$ define $W_t(0) = \lfloor U_t \rfloor$. When $\theta \ge 1$ put $m_0 = 0$, $m_l = \max(d - \theta + 1, ceil(l(d-\theta+1)/2))$ for $1 \le l \le t$, and $W_t(\theta) = \max_{0 \le l \le t}[\min(\lfloor U_t \rfloor, dt - m_l) - \theta(t-l)]$. Let $e \ge 0$ be an integer, put $x. = floor((k+e)/2)$ and $t. = x. - 2e$. If $t. \le 0$, or if $1 \le t. \le a$ and $W_t(\theta) \le t + 2e$ for every integer $1 \le t \le t.$, then $c_\theta(H) \le k + e$. In particular, for integers θ, e with $1 \le \theta \le d$, $3\theta \le d+1$ and $e \ge 0$, one has $c_\theta(H) \le k + e$ whenever $t. \le 0$, or whenever $1 \le t. \le \alpha(H)$ and $(d-\theta)t. \le x.$. Here $\alpha(H)$ is the largest independent-set cardinality. All graph orders, degrees, costs, capacities, matching budgets, cardinalities and parameters d, k, n, a, b, q, r, e, $x.$, $t.$, θ, t, l, m_l are integers in their displayed domains; $t.$ may be signed, and $f_t(j)$, U_t are rational numbers. Under the respective hypotheses the certificates cover every original edge and exist at every nonnegative integer capacity split of total θ and every nonnegative integer matching-budget split of total $k + e$. Connectedness and matching hypotheses are not required.*

Proof. All neighbors of S lie outside S, so the numbers $s_w = |N_H(w)$ intersect $S|$ for w outside S satisfy $0 \le s_w \le a$ and $\sum_w s_w = da$. Thus $0 \le q \le a$, $0 \le r < b$, and $q + 1 \le a$ whenever $r > 0$. The definitions of U_t are therefore valid.

Fix $1 \le t \le a$ and choose a uniformly random t-element subset I of S. For each w outside S the probability that I contains a neighbor of w is $f_t(s_w)$. Linearity of expectation gives $E[|N_H(I)|] = \sum_w f_t(s_w)$, where $N_H(I)$ is the external neighborhood. The function f_t on the integers 0,..., a has nonincreasing forward differences, because $f_t(j+1) - f_t(j) = binom(a-j-1, t-1)/binom(a,t)$. If two entries of the list (s_w) differ by at least two, lower the larger one by one and raise the smaller one by one. The sum of the entries is preserved, all

entries stay in [0,a], and the sum of their $f_t - values$ does not decrease by the forward-difference formula. The sum of their squares strictly decreases. Repeating this finite operation terminates at a list whose entries differ by at most one; its fixed sum forces b-r entries equal to q and r entries equal to $q+1$. Hence $\sum_w f_t(s_w) \le U_t$. Some t-element subset I of S has $|N_H(I)| \le U_t$, and its integer neighborhood cardinality is therefore at most $\lfloor U_t \rfloor$.

Suppose first $\theta \ge 1$ and fix one such I. For u in I, let p_u be the number of its private neighbors, meaning vertices w with $N_H(w)$ intersect $I = \{u\}$. Let $L = \{u \in I : p_u < \theta\}$ and $l = |L|$. For every u choose $\min(\theta, p_u)$ of its private neighbors; call this set R_u, and put $R = \bigcup_u R_u$. These sets are pairwise disjoint. The edges between I and R form vertex-disjoint stars centered in I, with at most θ edges per star. For w in $N_H(I)$ put $h_w = |N_H(w)$ intersect $I|$. Since I is independent and H is regular, $\sum_w h_w = dt$, and $dt - |N_H(I)| = \sum_w (h_w - 1)$. If $l \ge 1$, every u in L has at least $d - \theta + 1$ neighbors with $h_w \ge 2$. Hence there are at least $d - \theta + 1$ such shared neighbors, each contributing at least one to the sum of $h_w - 1$. Also $\sum_w (h_w - 1) \ge one - half \sum_{w:h_w \ge 2} h_w \ge l(d - \theta + 1)/2$. The first bound and integrality give $dt - |N_H(I)| \ge m_l$. For $l = 0$ the same statement holds with $m_0 = 0$. Consequently

$$|N_H(I) \text{ minus } R| \le |N_H(I)| - \theta(t-l) \le \min(\lfloor U_t \rfloor, dt - m_l) - \theta(t-l) \le W_t(\theta). \quad (1)$$

For $\theta = 0$ use R empty and the previously obtained $|N_H(I)| \le W_t(0)$; no private-neighbor argument is needed.

We now construct the certificate at each required budget split. Fix nonnegative capacities $A + B = \theta$ and nonnegative budgets $x + y = k + e$. Interchanging the two exceptional colors if necessary, it suffices to produce matching bounds x, y when $x \le y$; the degree-bounded colors retain the specified capacities A, B. If $y \ge k$, put all edges in its exceptional part: every graph on $2k+1$ vertices has matching number at most k. Otherwise $x \le y < k$ and $x \le x.$. Put $t = x - 2e$. If $t \le 0$, choose any vertex set W of cardinality x, put every edge meeting W into the smaller exceptional part, and all remaining edges in the other exceptional part. A matching in the former has at most x edges. Since $n - x \le 2y + 1$ when $t \le 0$, a matching in the latter has at most y edges.

Suppose $t \ge 1$. Then $t \le t.$, so the hypotheses supply the preceding I, R construction with $|N_H(I)$ minus $R| \le t + 2e = x$. Enlarge $N_H(I)$ minus R to a vertex set W of exactly x vertices avoiding I. This is possible because $n - t = x + 2y + 1 \ge x$. Let Q consist of all H-edges meeting I and avoiding W. The set Q is contained in the edges between I and R, so it is a disjoint union of stars of at most θ edges, with empty stars allowed. For each star with $s \le \theta$ edges, assign $\min(A, s)$ edges to the first degree-bounded part and the remaining $s - \min(A, s) \le B$ to the second. Both degree bounds hold at centers and leaves. For $\theta = 0$, Q is empty and both bounded-degree parts are empty. Put all edges meeting W in the smaller exceptional part, and all edges inside $V(H)$ $minus(IunionW)$ in the larger exceptional part. These two edge sets and the two parts of Q partition $E(H)$. Their matching numbers are at most x and y: W is a vertex cover of the first, and the last is supported on exactly $n - x - t = 2y + 1$ vertices. The construction therefore works for every original capacity and budget split. The threshold definition in Lemma 23 proves $c_\theta(H) \le k + e$. When $t. \le 0$ every split was already treated without I.

For the last assertion choose a maximum independent set S; it has $1 \le a < n$ because $d \ge 2$. If $3\theta \le d + 1$, then for every $l \ge 1$, $m_l \ge l(d - \theta + 1)/2 \ge \theta \cdot l$. Thus every entry in the maximum defining $W_t(\theta)$ is at most (d-theta)t, including $l = 0$. Write $D = d - \theta$. If $t. \ge 1$ and $Dt. \le x.$, then $D \ge 1$ and $(D-1)x. \le 2De$. For every smaller budget $x \le x.$ with $t = x - 2e \ge 1$ this gives $Dt = D(x - 2e) \le x$. Equivalently $W_t(\theta) \le (d - \theta)t \le t + 2e$ for all $1 \le t \le t.$. Since $t. \le a$, the general criterion applies. This proves the final assertion and completes the construction. □

Depends on. Lemma 23.

Lemma 55 (Factor-critical components of a maximal marked barrier). *Let G be a finite simple graph and S a subset of $V(G)$. A matching is a set of edges with pairwise disjoint endpoints, and a perfect matching covers every vertex; the empty graph has its empty perfect matching. For $X \subseteq V(G)$, let G-X denote induced vertex deletion, and let $q_S(G - X)$ count the connected components K of G-X whose orders are odd and whose vertex sets are contained in S. Define $\delta_S(G) = \max_{X \subseteq V(G)} (q_S(G - X) - |X|)$.*

Choose an inclusionwise maximal set X among the sets attaining this maximum: no strictly larger set containing X is another maximizer. Such a choice exists. Every odd component K of G-X whose vertices all belong to S is factor-critical, meaning that for every w in $V(K)$, the graph K-w has a perfect matching.

In particular, suppose G is connected, $\delta_S(G) > 0$, and the empty set is an inclusionwise maximal maximizer. Then $S = V(G)$, $|V(G)|$ is odd, $\delta_S(G) = 1$, and G-w has a perfect matching for every w in $V(G)$. Consequently every vertex is exposed by a maximum matching of G, where exposed means incident with no matching edge and maximum means of largest cardinality. All orders, component counts, matching cardinalities and values of the displayed maximum are integers.

Proof. The finite family of vertex subsets contains a maximizing set and hence an inclusionwise maximal maximizing set. Fix such a set X. Suppose that a marked odd component K is not factor-critical. There is a vertex w for which $H = K - w$ has no perfect matching. The graph H has even order. It cannot have order zero, because the empty matching would then be perfect.

Apply Lemma 10 to H with prescribed set $U = V(H)$. Absence of a perfect matching gives $Y \subseteq V(H)$ with $o(H - Y) > |Y|$, where o counts all odd-order connected components. The sum of the component orders shows that $o(H - Y)$ is congruent to $|V(H)| - |Y|$ modulo two. Since $|V(H)|$ is even, $o(H - Y) - |Y|$ is an even positive integer and is at least two.

Put $X' = X$ union {w} union Y. The sets X, {w}, and Y are pairwise disjoint. Deleting {w} union Y from the component K leaves all other components of G-X unchanged. Every odd component of H-Y has all its vertices in $V(K) \subseteq S$, so it is counted by q_S. Therefore

$q_S(G - X') \geq q_S(G - X) - 1 + |Y| + 2$

$= |X| + \delta_S(G) + |Y| + 1$

$= |X'| + \delta_S(G)$.

By definition of $\delta_S(G)$ the reverse bound holds, so X′ is another maximizing set. It strictly contains X because w is outside X. This contradicts the chosen inclusionwise maximality. Thus every marked odd component is factor-critical.

For the final assertion, X empty attaining the maximum gives $q_S(G) = \delta_S(G) > 0$. Connectedness means that G itself is the sole component counted: its order is odd, all its vertices belong to S, and $q_S(G) = 1$. The first part with X empty gives a perfect matching of G-w for every w. Such a matching has $(|V(G)| - 1)/2$ edges. No matching of an odd-order graph has more edges, because its distinct edges cover twice its cardinality in vertices. This matching is therefore maximum in G and exposes w, as required. □

Remark. A failed vertex-deletion matching inside a marked odd component supplies enough additional odd components to enlarge the barrier without reducing its value.

Depends on. Lemma 10.

Lemma 56 (Finite-family diagonal potential and its deletion invariant). *Let calA be a finite nonempty set of labels. For every γ in calA let s_γ, $t_\gamma \geq 1$ be integers and let $n_1^\gamma \geq \cdots \geq n_{s_\gamma}^\gamma \geq 1$ and $m_1^\gamma \geq \cdots \geq m_{t_\gamma}^\gamma \geq 1$ be integer lists. Let F_1^γ and F_2^γ be the corresponding vertex-disjoint unions of stars K_{1,n_i^γ} and K_{1,m_j^γ}, prescribed red and blue. Copies are ordinary subgraph copies. Put $N = \max_\gamma(s_\gamma + t_\gamma - 1)$. For each integer $2 \leq k \leq N + 1$ let ℓ_k be the maximum of $n_i^\gamma + m_j^\gamma - 1$ over all labels and valid indices with $i + j = k$. Define $P = \sum_{k=2}^{N+1}(\ell_k - 1)$. All these maxima are attained.*

For a finite simple graph J, let $e(J)$, $\nu(J)$, $\Delta(J)$ be its edge count, maximum matching cardinality and maximum degree, and put $\Phi(J) = e(J) - \nu(J)$. Empty graphs have all three counts zero. If a finite simple graph G arrows every ordered pair (F_1^γ, F_2^γ), then

$\Phi(G) \geq P$.

More precisely, there are N distinct original vertices v_1,..., v_N with $H_0 = G$ and $H_l = H_{l-1} - v_l$ induced, such that v_l has maximum degree d_l in H_{l-1}. Put $h_l = \nu(H_{l-1}) - \nu(H_l)$ and $\eta_l = d_l - h_l - (\ell_{l+1} - 1)$. For every $1 \leq l \leq N$, these integers satisfy h_l in {0,1}, $\eta_l \geq 0$, and

$\Phi(G) - P = \Phi(H_N) + \sum_{l=1}^{N} \eta_l$.

The edges deleted at different steps are disjoint: every original edge with a deleted endpoint is counted at its earliest such endpoint. The construction always uses exactly N steps; all the required vertices exist under the arrowing hypothesis.

If $\Phi(G) = P$, then every $\eta_l = 0$ and the entire edge set of H_N is a matching. For each l one has $d_l = \ell_{l+1} - 1 + h_l$, and the number of indices with $h_l = 0$ is exactly $N - \nu(G) + \nu(H_N)$. These conclusions hold for every execution of the selection rule in the proof.

For one label, put $M = s + t - 1$ and $L = \sum_{k=2}^{s+t} \ell_k$. The bound becomes $e(G) \geq L + \nu(G) - M$. Hence $\nu(G) \geq M$ implies $e(G) \geq L$, and $e(G) < L$ implies $1 \leq L - e(G) \leq M - \nu(G)$. For the general family, $\Phi(G) < P$ implies the existence of a coloring avoiding both prescribed targets of at least one member of calA. No positive-deficit avoidance conclusion at $\Phi(G) = P$ is asserted. All list lengths, indices, degrees, edge counts and matching cardinalities are nonnegative integers, list entries are positive integers, and the finite construction permits unit stars and repeated sizes.

Proof. Choose a label attaining N. Its two positive index intervals have every possible sum from 2 to $N + 1$, so each maximum defining ℓ_k has at least one candidate. The full candidate sets are finite, proving attainment. Every $\ell_k \geq 1$ and $N \geq 1$.

We first establish two ingredients for the finite construction. For any vertex v of a finite simple graph J, restriction of a maximum matching to J-v removes at most one edge, while every matching of J-v is a matching of J. Thus $h = \nu(J) - \nu(J - v)$ belongs to {0,1} and

$\Phi(J) - \Phi(J - v) = deg_J(v) - h.$ (1)

Also $\Phi(J) \geq 0$ because a matching is an edge subset.

Fix nonnegative integers A, B and put $D = A + B$. Suppose the edges of J do not admit a partition into parts of maximum degrees at most A and B. The graph has an edge, since an empty edge set admits that partition. We claim that one can choose a maximum-degree vertex v with $\Phi(J) - \Phi(J - v) \geq D$. If $\Delta(J) \geq D + 1$, choose any maximum-degree vertex and apply (1), since $h \leq 1$. If $D = 0$, the presence of an edge already puts us in this case. Suppose $D \geq 1$. If $\Delta(J) < D$, the empty matching covers every vertex of degree D, of which there are none; the covering-splitting theorem Lemma 35 gives the prohibited partition. Therefore the only remaining case is $\Delta(J) = D$. By the same theorem, a matching covering every degree-D vertex would again give that partition. No such matching exists. Choose any maximum matching and any degree-D vertex v that it misses; both choices exist because the graph is finite. This matching lies in J-v, so $h = 0$ and (1) gives exactly D. This proves the claim and its selection rule.

The second ingredient is actual-vertex persistence of tails. Fix one ordered pair of lists of lengths s, t and integers $0 \leq \alpha < s$, $0 \leq \beta < t$. For any set X of $\alpha + \beta$ vertices, an avoiding coloring of J-X for the two suffixes after these cuts extends to avoidance of the two original forests. To prove this, partition X into U, W with sizes α, β. Keep every color in J-X. Color every remaining edge meeting U red and every other remaining edge blue. Each connected red subgraph having an edge and meeting X contains a vertex of U: a red edge at a vertex of W can only go to U. Each connected blue subgraph having an edge and meeting X contains a vertex of W, since no blue edge meets U. Thus at most α vertex-disjoint red target components, and at most β blue ones, can meet X. If a full red target existed, at least s-alpha of its components would lie entirely in J-X. The j-th largest of any remaining s-alpha original sizes is at least $n_{\alpha+j}$: deleting at most α entries cannot remove all of the first $\alpha + j$ entries before that position. Trimming each remaining star to its corresponding suffix size would therefore give the prohibited red suffix. The same reasoning applies to blue. This proves the extension and, by contraposition, shows that arrowing a full pair makes J-X arrow these suffixes. It includes units, empty U or W, and components meeting X only at a leaf.

Now suppose G arrows all the given pairs. We construct the vertices recursively. At stage q, where $0 \leq q < N$, let X_q be the q previously deleted vertices and $H_q = G - X_q$. Choose any label γ and valid indices i, j attaining ℓ_{q+2}, so $i + j = q + 2$. The integers $\alpha = i - 1$, $\beta = j - 1$ are valid cuts of that label and total q. By actual-vertex persistence, H_q arrows its two nonempty suffix forests. A partition of $E(H_q)$ into maximum-degree bounds $A = n_i^\gamma - 1$ and $B = m_j^\gamma - 1$ would avoid both suffixes, because it avoids their respective leading stars. Such a partition is therefore impossible. The first ingredient, with $D = A + B = \ell_{q+2} - 1$, supplies a maximum-degree vertex v_{q+1} whose Φ drop is at least $\ell_{q+2} - 1$. Delete that vertex to form H_{q+1}.

The vertex lies outside X_q by construction. The absence of the degree partition also guarantees H_q has an edge. Hence the recursive step is valid at every $q < N$, producing N distinct original vertices. There is no need for one label or one pair of indices to attain multiple diagonals. Each h_l belongs to {0,1} by (1), and each η_l is nonnegative by the selection rule. Summing (1) over exactly these N steps yields

$\Phi(G) - \Phi(H_N) = \sum_{l=1}^{N} (d_l - h_l)$

$= P + \sum_{l=1}^{N} \eta_l$.

This is the exact asserted identity. Its right-hand terms $\Phi(H_N)$ and η_l are nonnegative, so $\Phi(G) \geq P$. An edge removed during this induced sequence belongs to exactly the first deletion of either of its endpoints; edges with no deleted endpoint are precisely $E(H_N)$. This proves the disjoint edge accounting.

If $\Phi(G) = P$, the identity is a sum of nonnegative integers equal to zero. Thus $\Phi(H_N) = 0$ and every $\eta_l = 0$. A maximum matching of H_N then has $\nu(H_N) = e(H_N)$ edges and so comprises every edge of H_N, proving that its edge set is a matching. The equalities $\eta_l = 0$ give $d_l = \ell_{l+1} - 1 + h_l$. Finally telescoping matching losses gives $\sum_l h_l = \nu(G) - \nu(H_N)$. Each h_l is zero or one, so the number of zero values is $N - \sum_l h_l = N - \nu(G) + \nu(H_N)$. Every permitted execution satisfies the same step inequalities and identities, which proves the equality assertion with its stated quantifier.

For a single label, $N = M$ and $P = L - M$; rearrange $\Phi(G) \geq P$ to obtain the claimed edge bound. If $\nu(G) \geq M$ it yields $e(G) \geq L$. If $e(G) < L$, the positive integer L-e(G) is at most M-nu(G), giving exactly the final inequality. For a general family, if $\Phi(G) < P$ then G cannot arrow every pair, by the theorem proved. Negating this finite conjunction gives at least one member and one coloring avoiding its red and blue targets. This uses no boundary assertion at equality. □

Depends on. Lemma 35.

Lemma 57 (Certificate rows and zero slack in graphs of maximum degree two). *Use the finite integer certificate costs $c_{A,B}(R)$, row maxima $c_i(R)$, matching number $\nu(R)$, independence number $\alpha(R)$, edge count $e(R)$, and signed slack $ps(R) = e(R) - \nu(R) - \sum_{i\geq 1} c_i(R)$ of Lemma 23. In particular $c_{A,B}(R)$ is the least nonnegative integer total t such that every nonnegative integer split $x + y = t$ admits an edge partition into parts Q_R, Q_B, P_R, P_B with maximum-degree bounds A, B and matching-number bounds x, y, respectively. All graphs are finite and simple, and all capacities, budgets, totals and row indices are nonnegative integers.*

Let R have maximum degree at most two, allowing the empty graph and isolated vertices. Write its path components as P_{a_1},..., P_{a_u} and its cycle components as C_{b_1},..., C_{b_v}, where P_n is a path on n vertices, C_n is a cycle on n vertices, u, v are nonnegative integers, $a_j \geq 1$, and $b_j \geq 3$ are integers. All these components are whole connected components and are vertex-disjoint. Let p be the number of odd integers among b_1,..., b_v. Empty sums and empty component lists are permitted. Then

$c_1(R) = \sum_{j=1}^{u} \lfloor a_j/3 \rfloor + \sum_{j=1}^{v} \lceil b_j/3 \rceil$,

$c_2(R) = p$,

$c_i(R) = 0$ *for every integer* $i \geq 3$.

Consequently $ps(R)$ is the sum over its path components of

$a_j - 1 - \lfloor a_j/2 \rfloor - \lfloor a_j/3 \rfloor$

and over its cycle components of

$b_j - \lfloor b_j/2 \rfloor - \lceil b_j/3 \rceil - 1_{b_j odd}$,

where the final symbol is one for an odd cycle order and zero otherwise. Every summand is nonnegative. Thus $ps(R) = 0$ if and only if every component of R belongs to

P_1, P_2, P_3, P_4, P_6, C_3, C_4, C_5, C_7.

When $ps(R) = 0$, let ϵ be the number of positive even-order path components. Then

$c_1(R) = \nu(R) - \epsilon$, $c_2(R) = p$,

$e(R) = 2\nu(R) - \epsilon + p$,

$\alpha(R) = |V(R)| - \nu(R) - p$.

Proof. We use only the certificate definitions, color symmetry, and finite row bounds in Lemma 23.

A connected graph of maximum degree at most two is a path or a cycle. An isolated vertex is P_1. If there is a degree-one vertex, start there and continue along the unused next edge. An internal repeated vertex

would have three incident edges, and returning to the first vertex would contradict its degree one. Finiteness therefore forces another endpoint. No vertex can attach to this path without violating the degree bound or extending an endpoint, so connectedness makes it the entire component. If all degrees are two, following edges until the first return gives a cycle, and the same degree bound and connectedness leave no further vertices.

First consider capacity pair (1,0). We give a universal construction. Suppose a vertex set S in a graph J has the property that the edges with both endpoints outside S form a matching. For any nonnegative integer x, y with $x + y = |S|$, partition S as S_R, S_B with these respective cardinalities. Put all edges outside S into Q_R and take Q_B empty. Among the remaining edges put those having an endpoint in S_R into P_R, and put the rest into P_B. Every P_B edge has an endpoint in S_B. A matching in either exceptional part uses distinct vertices of its assigned set, so the bounds x, y hold. Every edge is assigned exactly once. This gives every allocation at total $|S|$, including zero budgets.

For a path on vertices 1,..., n in order, put $q = \lfloor n/3 \rfloor$. If $n = 3q$ or $n = 3q + 1$, take $S = \{2, 5, \ldots, 3q - 1\}$, interpreted as empty when $q = 0$. If $n = 3q + 2$, take $S = \{3, 6, \ldots, 3q\}$, again allowing $q = 0$. Each set has q vertices. Every consecutive run outside S has at most two vertices, so the edges outside S form a matching. For a cycle on vertices 1,..., n in cyclic order, take $S = \{1 + 3j : 0 \le j < \lceil n/3 \rceil\}$. Consecutive selected vertices have cyclic distances at most three, including the last-to-first distance, so again the edges outside S form a matching. Taking the union of these selected sets over all components of R gives a set of cardinality

$t = \sum_j \lfloor a_j/3 \rfloor + \sum_j \lceil b_j/3 \rceil$

and a matching outside it. The construction proves $c_{1,0}(R) \le t$; color symmetry gives $c_{0,1}(R) \le t$ and hence $c_1(R) \le t$.

For the reverse bound, any certificate at capacities (1,0) and budgets (0,c) has $Q_B = P_R$ empty. Thus Q_R is a matching M and P_B is exactly $E(R)$ minus M. Its matching number must be at most c. On a path P_n, partition its first $3floor(n/3)$ vertices into consecutive triples. The two consecutive edges in each triple cannot both belong to M. Choose one remaining edge in each triple; their vertex sets are disjoint, giving at least $\lfloor n/3 \rfloor$ edges of a matching in $P_n - M$.

On a cycle C_n, if M uses no edge, the remaining matching number is $\lfloor n/2 \rfloor$, which is at least $\lceil n/3 \rceil$ because $\lfloor n/2 \rfloor \ge (n-1)/2 \ge n/3$ for $n \ge 3$. If M uses an edge, its deletion breaks the cycle into paths. Every vertex retains at least one cycle edge, since M is a matching, so each resulting path has order $h \ge 2$. A path on h vertices has matching number $\lfloor h/2 \rfloor$: pairing consecutive vertices attains the upper bound from its disjoint endpoint pairs. For $h \ge 2$, $\lfloor h/2 \rfloor \ge h/3$; at $h = 2$ this follows from $1 \ge 2/3$, and for $h \ge 3$ it follows from $(h-1)/2 \ge h/3$. Adding over the remaining paths gives matching number at least $n/3$, and hence at least $\lceil n/3 \rceil$. Matching numbers add over vertex-disjoint components because a matching restricts to each component and component matchings can be united. Therefore $\nu(R - M) \ge t$ for every M, and $c \ge t$ is necessary in the extreme allocation. By the definition of the least universal total, $c_{1,0}(R) \ge t$. This proves the exact first row.

For the second row, pairs (2,0) and (0,2) permit all edges in their degree-two part, so their costs are zero. At capacities (1,1), choose one edge from each of the p odd cycles. Removing these edges leaves only paths and even cycles; alternately assigning consecutive edges on each such component puts all remaining edges into two matchings Q_R, Q_B. For any $x + y = p$, place x of the chosen edges in P_R and the other y in P_B. The chosen edges lie in distinct components, so their matching numbers are x and y. This gives $c_{1,1}(R) \le p$. Conversely, at budgets (0,c), P_R is empty and Q_R, Q_B are matchings. Each odd cycle must have an edge in P_B: otherwise consecutive cycle edges would alternate between Q_R and Q_B, which cannot close around an odd number of edges. One P_B edge from each odd-cycle component gives a matching of size p. Hence $c \ge p$, proving $c_2(R) = p$. For $i \ge 3$, every split $A + B = i$ has $A \ge 2$ or $B \ge 2$. Putting all edges in that degree-bounded part certifies total zero, so $c_i(R) = 0$.

A path or cycle on n vertices has matching number $\lfloor n/2 \rfloor$, by the consecutive-pair construction and the endpoint bound. Its edge count is n-1 for a path and n for a cycle. Substituting these values and the two row formulas into ps gives the displayed component contributions; their sum is exact by edge and matching additivity and the proved row formulas. For paths of orders 1, 2, 3, 4, 5, 6 these contributions are respectively 0, 0, 0, 0, 1, 0. Replacing a path order n by $n + 6$ increases its contribution by $6 - 3 - 2 = 1$. Thus all path contributions are nonnegative, and the zero ones have orders 1, 2, 3, 4, 6. For cycles of orders 3, 4, 5, 6, 7, 8 the contributions are respectively 0, 0, 0, 1, 0, 1. Replacing a cycle order by its value plus six preserves parity and

increases its contribution by one. This proves nonnegativity and the four zero cycle orders 3, 4, 5, 7. A sum of these nonnegative integers is zero exactly when each component contributes zero, including the empty sum.

For the final identities, a positive even-order path has edge count 2nu-1; an odd-order path or even cycle has edge count 2nu; and an odd cycle has edge count $2\nu + 1$. Therefore $e(R) = 2\nu(R) - \epsilon + p$ for every degree-two graph. When $ps(R) = 0$, the definition of slack and $c_2(R) = p$ give $c_1(R) = e(R) - \nu(R) - p = \nu(R) - \epsilon$. On a path P_n, alternating vertices give an independent set of size $\lceil n/2 \rceil$, and disjoint consecutive pairs bound every independent set by that size. On a cycle C_n, sending each selected independent vertex to its following vertex gives an injection into the unselected vertices, so every independent set has size at most $\lfloor n/2 \rfloor$; alternating vertices attain that bound. Independence numbers add across components. Subtracting these component formulas from their orders gives $\alpha(R) = |V(R)| - \nu(R) - p$. All constructions and identities permit isolates, empty graphs, and empty edge parts. □

Remark. At the first row, selected vertices absorb the exceptional edges while the rest form a matching. Disjoint path triples and odd-cycle fragments prove the matching lower bounds. The second row counts precisely the odd cycles, leaving an explicit additive slack formula.

Depends on. Lemma 23.

Lemma 58 (Quartic frame contacts from matching and first-cost identities). *All graphs are finite and simple. Write $\alpha(G)$, $\nu(G)$, $\Delta(G)$ for independence number, matching number, and maximum degree. Vertex deletion is induced. A graph is well-covered if every maximal independent set is maximum. For nonnegative integers A, B, let $c_{A,B}(G)$ be the least nonnegative integer T such that every nonnegative integer split $x + y = T$ permits an edge partition into four parts with maximum degrees at most A, B and matching numbers at most x, y, respectively. Put $c_1(G) = \max_{A+B=1} c_{A,B}(G)$, with nonnegative integer capacities. The symbols P_j, C_j denote paths and cycles on j vertices.*

Let F be connected, four-regular, and well-covered, and put $a = \alpha(F) \geq 4$. Let m, b be integers with $m \geq 1$ and $0 \leq b < m$, and set $q = m - b$. A frame consists of a maximum independent set S of F and an independent m-element set U of degree-three vertices in F-S. Put W = S union U, $R = F - W$, and $r = \nu(R)$. Assume that for every frame:

(i) $\alpha(R) = a$, and the whole components of R are exactly q odd cycles from C_3, C_5, C_7, exactly b even paths from P_2, P_4, P_6, and any number of P_3, C_4, and isolates.

(ii) For every endpoint p of the matching F[W], the induced graph $K_p = F[V(R)$ union $\{p\}]$ satisfies $\nu(K_p) = r + 1$ and $c_1(K_p) = r + 1 - b$.

(iii) There are nonnegative integers x, y with $x + y = |W| + r - 1$ for which $E(F)$ has no partition into two parts of matching numbers at most x, y, and $|W| - x \geq q + 2$.

Then every frame has all the following properties. No whole P_6 occurs in its terminal. Every frame-matching endpoint has at most one degree-two terminal neighbor and cannot meet both the center and a leaf of one whole P_3. Every odd-cycle component has at most one edge to frame-matching endpoints. If such an edge px occurs on an odd cycle O, the entire terminal neighborhood of p is x together with the two endpoints of one whole even path.

For every $T \subseteq W$ with $|T| \leq q + 2$ there is one matching assigning each t in T to a vertex v_t in a distinct whole odd terminal component C_t such that $C_t - v_t$ has a perfect matching. The assignment is simultaneously compatible with a maximum matching in every terminal component. No existence of an initial frame is assumed or concluded.

Proof. The terminal hypotheses imply those of Lemma 25. Thus in every frame F[W] is m disjoint edges and a-m isolates; write P, Z for their endpoints and isolates. Canonical sets D select maximum independent sets on odd cycles, each P_3 center, opposite C_4 pairs, the stated alternating internal choices on even paths, and nothing on isolates. The reflected graph $R_D = F[W$ union D] is an actual frame terminal with complementary frame union $V(R)$ minus D. Every D-vertex has degree two in R_D, and every odd-cycle or even-path component of R_D contains exactly one old frame edge. Those components exhaust all old frame edges. In particular every p in P has at most one D-neighbor, since it also has its mate as a neighbor in the degree-at-most-two graph R_D.

Every frame edge may be oriented arbitrarily. If p in P contacts a degree-two terminal vertex v, the root exchange from Lemma 25 gives the actual terminal $R - v + p$. All hypotheses of the present theorem apply to every such new frame. Every p has exactly three neighbors in R.

1. Port assignments and elimination of P_6.

Each path or cycle C has $\nu(C) = floor(|V(C)|/2)$, by alternate edges and the distinct-endpoint upper bound. Thus Lemma 17 applies with exterior W and the missing allocation in (iii). It supplies the asserted simultaneous assignments for all $|T| \leq q+2$, including compatible maximum internal matchings. Its exposed vertices are precisely those whose deletion leaves a perfect matching in their odd component.

On an odd path, the exposed vertices are its odd-indexed positions from an endpoint; on an odd cycle every vertex is exposed, as proved in Lemma 17. Protect all odd-path components, including isolates, by selecting all their exposed vertices. There are q unprotected odd cycles and the range is at least $q+2 > q$. The same fact supplies one assignment of all exterior vertices avoiding these fixed odd-path choices to distinct odd cycles.

Let I select the fixed odd-path sets and maximum independent sets on all odd cycles containing their assigned vertices. Select no vertices on even components. Every vertex of W is dominated by I, either by a fixed odd-path choice or by its assigned cycle vertex. Every odd terminal component is also dominated, because its chosen set is maximum there. If R_{even} is the union of the even-order components, no vertex of I is adjacent to R_{even}, since these are different whole R-components. Therefore $F - N_F[I] = R_{even}$, where $N_F[I]$ is I together with its neighbors.

Every maximal independent set L of R_{even} extends to the maximal independent set I union L of F. Well-coveredness makes all such L have the same size. Each whole component of R_{even} is consequently well-covered: otherwise fix maximal independent sets on the other components and vary this one's cardinality. But P_6, numbered consecutively, has a maximal independent set {2,5} of size two and a maximum independent set {1,3,5} of size three. The latter is maximum by the perfect matching 12, 34, 56. Thus no P_6 occurs. All terminals from now on have the eight-type list in (i) with P_6 removed.

2. Elementary first-cost bounds.

We record explicit edge partitions for later comparisons. A free matching together with k disjoint edge pieces, each of matching number at most one, proves $c_1 \leq k$: put the free matching in the degree-one part for either capacity split, and assign any x of the k pieces to one paid part and the remaining k-x to the other. Empty pieces are allowed. Componentwise such partitions combine, since a union of matchings on different components is a matching.

For P_2 or P_4 take consecutive disjoint edges as the free matching and each intervening edge as a separate paid piece. This gives $c_1 \leq \nu - 1$ on every allowed even path. For P_3 take the whole graph as one paid piece, and for C_3 take the whole triangle as one paid piece. For C_4 use one opposite edge pair as the free matching and the other two edges as separate paid pieces. For C_5 or C_7, consecutively label the cycle 1,..., $2s+1$ with $s = 2$ or 3; take edges 12, 34,..., (2s-1)(2s) as the free matching. The remainder consists of s-1 separate edges and a P_3 through vertex $2s+1$, giving s paid pieces. An isolate uses none. Thus every other allowed component has $c_1 \leq \nu$. All these are certificates for every integer paid-budget split.

3. No double internal contact on a P_4.

Suppose p in P neighbors both internal vertices 2, 3 of a whole path 1-2-3-4. Its third terminal neighbor h has degree at most one and lies outside the internal pair. Indeed, every high vertex on another component can be selected canonically together with 2, which would give p two D-neighbors.

If h is an endpoint of this same P_4, the component of K_p containing p has five vertices and matching number at most two. All other components are unchanged, so $\nu(K_p) \leq r$, contrary to (ii). If h is an endpoint of another even path, the joined component has odd order $4 + |V(L)| + 1$ and matching number at most $2 + |V(L)|/2$, again yielding $\nu(K_p) \leq r$.

The remaining possibilities are that h is an isolate or a P_3 leaf. If h is an isolate, the modified component C has the perfect matching 12, 34, ph. Removing its edges leaves exactly the triangle p-2-3-p. Hence $\nu(C) = 3$ and $c_1(C) \leq 1 = \nu(C) - 2$. If h is a leaf on h-l-t, add lt to this free matching. It is now perfect in the eight-vertex C, and its removal leaves that triangle and the single edge hl. Thus $\nu(C) = 4$ and $c_1(C) \leq 2 = \nu(C) - 2$.

In either case the other b-1 even paths are unchanged. Combining the explicit partitions of step 2 with this partition of C gives

$$c_1(K_p) \leq \nu(K_p) - 2 - (b-1) = r - b,$$

contrary to (ii), which says $c_1(K_p) = r + 1 - b$. Since a P_4 was present, $b \geq 1$. No endpoint has two internal neighbors on any whole even path, as P_2 has none and P_6 was excluded.

4. Odd-cycle contacts and the K_5 alternative.

Suppose p in P meets an odd cycle O. A canonical set can select any high vertex on another component independently of a selected O-neighbor, so p has no high neighbor outside O. On C_5 every nonadjacent pair is a maximum independent set. On C_7 any nonadjacent pair extends to an independent triple: the two closed neighborhoods have at most six vertices in total. Thus a canonical set could select two nonadjacent O-neighbors of p, which is impossible. Its O-neighbors are consequently one vertex, two adjacent vertices, or all three vertices of a triangle.

If p meets exactly two adjacent O-vertices, exchange p for one of them as a root. The old cycle becomes an even path and restored p joins one end of it to a low vertex on another component. This creates no cycle, decreasing the number q of odd cycles by one, contrary to (i).

If p has exactly one O-neighbor x, the exchange $R - x + p$ destroys precisely that odd cycle. The other two neighbors of p are low and outside O. Adding p creates a cycle only if they are the endpoints of one path component. To restore the number q of odd cycles, that path must have even order. Thus $N_R(p)$ is x and both endpoints of one whole P_2 or P_4.

It remains to exclude $N_R(p) = V(O)$ for O a triangle. Write $O = \{t, u, v\}$, let p' be p's frame mate, and choose a canonical D containing t. In R_D, the old edge pp' belongs to a designated odd cycle or even path, p is adjacent to t, and both p, t have degree two. The unselected edge uv belongs to the reflected frame, whose two endpoints each neighbor p and t.

If pp' lies on an even path, p and t are two adjacent internal vertices of that path. This contradicts step 3 in the reflected frame. If pp' lies on an odd cycle O', apply the preceding preliminary odd-cycle classification in that reflected frame to u and v, each of which has two O'-neighbors p, t. The only surviving possibility is that O' is a triangle and both u, v neighbor every vertex of O'. The triangle O' must have vertices p, p', t. Together with uv, these edges form a K_5 on {p, p',t, u, v}. Every vertex of this set already has its four neighbors there. Connectedness makes F exactly K_5, contradicting $\alpha(F) \geq 4$. The triple-contact alternative is impossible.

We have proved the stated complete neighborhood for every odd-cycle contact. Its only exposed odd-component neighbor is x on O. If two distinct frame endpoints met the same O, the two-element assignment in step 1 would require two distinct receiving odd components even though both endpoints have only O available. Thus every odd cycle has at most one edge to P.

5. At most one high neighbor at every endpoint.

A frame endpoint cannot have high neighbors in different components, since both could be selected in one canonical D. It has at most one on an odd cycle by step 4 and at most one on an even path by step 3. A P_3 has only one high vertex, and an isolate has none.

Only C_4 remains. A canonical set can select either opposite pair. Thus, if p has two high neighbors on a C_4, they must be adjacent, and it has no third neighbor there. Label the square $x_1 - x_2 - x_3 - x_4 - x_1$ so that p meets x_1, x_2. Its third terminal neighbor h is low and on another component, since a high neighbor there would be simultaneously selectable with x_1. Exchange p for x_1. The old square minus x_1 is the whole three-vertex path $x_2 - x_3 - x_4$. Restored p joins its end x_2 to h, which is an endpoint of a path of order one, two, three, or four. The resulting whole path has order between five and eight, outside the eight-type terminal list. This contradiction excludes the adjacent-square contact. Every endpoint therefore has at most one high neighbor.

6. Reserved P_3 rigidity and the own-leaf exclusion.

We prove a small consequence of the assignments in step 1. Suppose all exposed odd-component neighbors of p lie in one P_3 C with center h and leaves L. Let K select h, the leaves of every other P_3, all terminal isolates, and arbitrary maximum independent sets in even components, but no odd-cycle vertices. Suppose a choice of K dominates p, and let B be the W-vertices not dominated by K.

Then p is not in B. Every odd exposed neighbor of a B-vertex lies on an odd cycle or on the reserved leaf pair L. If $|B| \geq q + 1$, assigning $q + 1$ of its members together with p by step 1 would require $q + 2$ distinct components from only q odd cycles and C. Thus $|B| \leq q$. Assign B union {p}; p must be assigned to C, and the B-vertices to distinct odd cycles. Choose maximum independent sets on the odd cycles containing their assigned vertices. Together with K they form a maximal independent set of F: K and the assigned cycle neighbors dominate W, and the choices dominate every R-component. Its size is $\alpha(R) - 1 = a - 1$, because C uses its center in place of its two leaves. This contradicts well-coveredness.

It follows that no such K can dominate p. In particular p cannot meet h or any even-component vertex, since each even-component vertex belongs to some maximum independent choice. Its odd exposed-neighbor restriction also excludes isolates, odd cycles, and other P_3 leaves. Its remaining possible neighbors are the two members of L and centers of other P_3 components. Step 5 and the three terminal neighbors of p force $N_R(p)$ to be both members of L and the center of a different P_3.

Now suppose p meets both the center h and a leaf l of a whole P_3 with other leaf l′. Its third terminal neighbor z is low by step 5. If z is l′ or an even-path endpoint, all its exposed odd neighbors lie in that one P_3, contradicting the just-proved rigidity. If z is a leaf of a different P_3, exchange p for h. The leaf l becomes an isolate on deleting h, and restored p joins it to the unchanged P_3 through z, forming a whole P_5. This is forbidden.

The final possibility is that z is an original isolate. In K_p the changed whole component is the five-vertex graph with triangle p-h-l-p and pendant edges hl′ and pz. Its matching number is two, attained by those two pendant edges and bounded by its five vertices. Use those pendant edges as the free matching and the entire triangle as one paid piece. Hence its first cost is at most one, a saving of one from its matching number. All b even terminal paths are unchanged and each saves one more by step 2; every other unchanged component costs at most its matching number. Combining these certificates yields

$$c_1(K_p) \le \nu(K_p) - b - 1 = r - b,$$

again contradicting $c_1(K_p) = r + 1 - b$. This excludes the last case.

The terminal catalogue exhausts all possible low vertices, so no endpoint has an own-center-and-leaf contact. Together with steps 1, 4, and 5 this proves every assertion. All component changes, matchings, and edge partitions occur in the same original graph F, and every use of a hypothesis is on an actual frame. □

Remark. A missing allocation forces simultaneous exposed-component assignments. These remove P_6 by well-coveredness and permit reserved-P3 comparisons. Canonical reflection reduces bad contacts to two small components whose explicit first-cost savings contradict the same one-root cost identity, giving exactly the endpoint rules needed for the final cycle exclusions.

Depends on. Lemma 17, Lemma 25.

Theorem 59 (The antidiagonal star-union upper bound). *Let s, t be positive integers and let $n_1 \ge \cdots \ge n_s \ge 1$ and $m_1 \ge \cdots \ge m_t \ge 1$ be nonincreasing lists of positive integers. For a positive integer h let $K_{1,h}$ be the finite simple star with one center, h leaves, and h edges. Let F_1 be the vertex-disjoint union of K_{1,n_i}, $1 \le i \le s$, and let F_2 be the vertex-disjoint union of K_{1,m_j}, $1 \le j \le t$. For each integer k with $2 \le k \le s + t$ put*

$$\ell_k = max\{n_i + m_j - 1 : 1 \le i \le s, 1 \le j \le t, i + j = k\},$$

and put $L = \sum_{k=2}^{s+t} \ell_k$.

Let $G^{\cdot}$ be the vertex-disjoint union of K_{1,ℓ_k}, one component for each integer k from 2 through $s + t$. Every red/blue coloring of $E(G^{\cdot})$ contains either a red ordinary subgraph isomorphic to F_1 or a blue ordinary subgraph isomorphic to F_2, and $e(G^{\cdot}) = L$. Consequently the set of edge counts of finite simple arrowing hosts is nonempty and its least value $\hat{R}(F_1, F_2)$ satisfies $\hat{R}(F_1, F_2) \le L$. This includes $s = 1$, $t = 1$, repeated component sizes, and stars of size one.

Proof. Use the star, profile, arrowing, and size Ramsey definitions of Lemma 29. That fact proves that each ℓ_k is a positive integer defined by a nonempty finite maximum. Thus the stipulated vertex-disjoint union $G^{\cdot}$ is a finite simple graph, and its edge count is the sum L of the component edge counts.

Fix any *red/blue* coloring of its edges. We select monochromatic target stars one component at a time. At the start set $i = j = 1$. Maintain the following invariant until one target is complete: the previously selected red stars have sizes n_1,..., n_{i-1}, the previously selected blue stars have sizes m_1,..., m_{j-1}, all selected stars lie in distinct host components, and precisely the host components indexed 2,..., $i + j - 1$ have been used. At the initial state both lists and the set of used components are empty, so this invariant holds.

Suppose both targets are incomplete. Then $1 \le i \le s$ and $1 \le j \le t$, and the next unused component has index $k = i + j$, where $2 \le k \le s + t$. Let r_k and u_k be the nonnegative integer numbers of red and blue edges in that component. They satisfy $r_k + u_k = \ell_k \ge n_i + m_j - 1$. If $r_k \ge n_i$, choose n_i red edges at its center and

their endpoints; these form a red K_{1,n_i}. Add this star to the selected red list and increase i by one. If $r_k < n_i$, integrality gives $r_k \le n_i - 1$ and therefore $u_k = \ell_k - r_k \ge m_j$. Choose m_j blue edges at its center, add the resulting blue K_{1,m_j} to the selected blue list, and increase j by one.

In either case exactly one component and one target entry have been used. The next value of $i + j$ has increased by one. Vertex-disjointness of the host components ensures that the new selected star is disjoint from every previously selected star, and the used components now have exactly the required index interval. Thus the invariant is preserved whenever both targets remain incomplete. If i becomes $s + 1$, the selected red stars give a red F_1; if j becomes $t + 1$, the selected blue stars give a blue F_2.

The process must reach one of these two terminal states. If it completed $s + t - 1$ selections without reaching either, the counters would satisfy $i \le s$ and $j \le t$ but also $i + j = 2 + (s + t - 1) = s + t + 1$, a contradiction. Before termination every required component has an allowed index, as checked in the preceding step. Hence every coloring contains a target. The possible edge counts of arrowing finite simple hosts form a nonempty set of nonnegative integers containing L; the least-element property of the nonnegative integers gives $\hat{R}(F_1, F_2) \le L$. The argument used only positivity and the specified index ranges, so the stated boundary cases need no separate restriction. □

Remark. Each host star forces the next required red star or the next required blue star. After at most $s + t - 1$ such choices one entire target has been selected in distinct components.

Depends on. Lemma 29.

Lemma 60 (Successive matchings from two rows of unequal orders). *Use the finite integer certificate costs $c_{A_c,B_c}(G)$ and $c_i(G)$ of Lemma 14. A certificate partitions every edge into parts of maximum degrees at most the nonnegative integer capacities A_c, B_c and parts of matching numbers at most the nonnegative integer budgets x, y. The cost c_{A_c,B_c} is the least nonnegative integer total $x + y$ feasible for every such budget split, and c_i is its maximum over $A_c + B_c = i$. Write $V(G)$, $E(G)$, $d_G(v)$, and $\nu(G)$ for the vertex set, edge set, vertex degree, and matching number.*

Let F be a finite simple six-regular graph with a vertex partition $V(F) = V(A)$ dot-union $V(R)$ dot-union X, where A and R are the full induced graphs on their vertex sets. Fix nonnegative integers r, s, τ with $0 \le \tau \le s$ and

$|V(A)| = r$, $|V(R)| = r + \tau$, $|X| = s - \tau$.

Every whole component of either row is an isolated vertex, a path on three vertices, a path on $2j$ vertices for an integer $j \ge 1$, or a cycle on j vertices for an integer $j \ge 3$. Whole means an entire connected component. Put $m = \nu(A)$, let e_A count its even-order path components, let p count its odd-cycle components, and put $\beta = m - e_A$. Assume $\nu(R) = m + \tau$, R has exactly e_A even-order path components and $p + \tau$ odd-cycle components. Assume X is independent in F and every vertex of X has exactly three neighbors in R.

Suppose nonnegative integers b_R, b_B satisfy $b_R + b_B + 1 = r + s + m$ and there is no partition of $E(F)$ into two parts of matching numbers at most b_R, b_B. Assume

$c_1(F) \ge r + s + \beta$, $c_2(F) \ge r + s + p$, $c_3(F) \ge r$.

Choose any subset Y of $\{vinV(A) : d_A(v) = 2\}$, put $b = |Y|$, $N = r - b$, and assume

$\min(b_R, b_B) \le N$, $N \ge \max(0, s + \beta + b - 2)$.

Let $H = F[V(R)$ union X union $Y]$, $S = V(A)$ minus Y, and $I = V(R)$ union X. Thus $H = F - S$ and $|S| = N$. Set $P_0 = H$. For each integer $i = 0, 1, 2, 3$, choose any maximum matching M_i of P_i and form the spanning graph P_{i+1} by deleting precisely the edges of M_i and retaining every vertex of H. Every such sequence exists, and every such sequence satisfies

$|M_0| = m + s + b$, $|M_1| = \beta + s + b$, $|M_2| = p + s + b$, $|M_3| = b$.

Each of the first three matchings sends every vertex of X union Y to R, and together they remove every R-edge and every X-to-R edge. In P_3, I is independent, Y is a vertex cover, and each y in Y has exactly one remaining neighbor $\phi(y)$ in I. The map ϕ is injective. The unique maximum matching of P_3 is $M_3 = \{y\phi(y) : y \in Y\}$. The entire edge set of P_4 is $E(A[Y])$; every vertex of I is isolated. No matching in the sequence removes any $A[Y]$-edge.

For each integer $i = 1, 2, 3$ put $L_i = \bigcup_{0 \le j < i} M_j$. Every nonnegative integer capacity split $A_c + B_c = i$ and budget split $x + y = N + \nu(P_i)$ has a certificate of the entire F whose degree-bounded parts have union exactly L_i. Every

original edge with one or both endpoints in S belongs to a matching-bounded part. In particular $c_1(F) = r + s + \beta$, $c_2(F) = r + s + p$, and $c_3(F) = r$.

If an odd cycle C of order 2ell-1, for an integer $\ell \geq 2$, satisfies $E(A[Y]) = E(C)$, then P_4 consists of that same cycle and isolated vertices and $\nu(P_4) = \ell - 1$. All orders, cardinalities, component counts, capacities, budgets, totals, degrees, indices, and matching sizes are integers in the stated nonnegative or positive ranges. Empty X, empty Y, $\tau = 0$, and $\tau = s$ are allowed whenever the hypotheses hold.

Proof. We use the missing-allocation matching bound Lemma 7, the three-matching construction Lemma 20, and the cover transition Lemma 14. Their definitions of the certificate costs agree with those in the statement.

1. The initial matching with R as its row.

Put $u_R = |V(F)$ minus $V(R)| = r + s - \tau$, $h_R = |X$ union $Y| = s - \tau + b$, $m_R = m + \tau$, $\beta_R = \beta + \tau$, and $p_R = p + \tau$. These are nonnegative integers: the component calculation in Lemma 20 gives $\beta \geq p \geq 0$, and the assumed matching and component counts for R give $\beta_R = \nu(R) - e_A$. Also

$u_R - h_R = r - b = N$,

$u_R + m_R = r + s + m = b_R + b_B + 1$.

Writing $q = \min(b_R, b_B)$, the reserve $N \geq q$ implies both $u_R \geq q$ and $h_R \leq u_R - q$. Apply Lemma 7 with its initial row $V(R)$ and its retained exterior set X union Y. Every premise is now checked, including that H is the full induced graph on these vertices. The conclusion is

$\nu(H) = m_R + h_R = m + s + b$.

In particular the initial matching required by the three-matching construction exists.

2. Three matchings in their actual residuals.

The numerical hypotheses of Lemma 20, applied with initial row R, hold because

$\beta_R + h_R = s + \beta + b$,

$N + h_R + \beta_R = r + s + \beta$,

$N + h_R + p_R = r + s + p$.

The actual row order is $r + \tau$ and the actual omitted order is N. Its permitted component domain is precisely the one assumed here. Thus that theorem applies to every successive choice of M_0, M_1, M_2 on the same H, and gives their stated sizes, their crossing saturation of X union Y into R, and deletion of every R-edge. It also supplies the complete certificates at $i = 1, 2$ and the displayed equalities for $c_1(F)$, $c_2(F)$.

Since X is independent and each of its vertices has exactly three R-neighbors, the three distinct crossing edges at each such vertex exhaust those neighbors. Hence the first three matchings remove every edge within $I = V(R)$ union X. None has an edge with both endpoints in X union Y, because every vertex there is already used by a crossing edge to R. In particular every original A[Y]-edge remains.

For y in Y, its degree in A is two, while its degree in F is six. It has exactly four original neighbors outside A, all in I by the displayed vertex partition. Every such edge lies in H. Each of M_0, M_1, M_2 removes exactly one of these four edges, with endpoint in R. The three edges are distinct and no one of these matchings can remove another edge at y. Precisely one y-to-I edge remains in P_3. Since I is independent there, the entire edge set of P_3 is $E(A[Y])$ together with these b boundary edges. This description includes $b = 0$.

3. The fourth matching and every remaining edge.

The cover Y has size b. Its rank lower bound is

$c_3(F) \geq r = N + b$.

The margin $N \geq \max(0, s + \beta + b - 2)$ implies $N \geq \max(0, b - 2)$, since s and β are nonnegative. Apply Lemma 14 with its prefix length $k = 3$, independent side I, cover Y, and next layer B empty. Its prefix is the three actual pairwise edge-disjoint matchings just constructed. Every edge of H[I] belongs to that prefix, and the empty-layer conditions impose no additional requirements. The theorem gives $\nu(P_3) = b$ and says that every maximum matching saturates Y using only edges from Y to I.

Each y has exactly one such remaining edge, so every maximum matching consists of all b of these edges. A maximum matching exists; its disjoint endpoints prove that their endpoints $\phi(y)$ in I are distinct. This proves uniqueness, including the empty maximum matching when $b = 0$. Deleting precisely these edges leaves $E(A[Y])$ as the entire edge set, and all I-vertices are isolated. The fourth matching has no edge inside Y, so every original A[Y]-edge survives the whole sequence.

For completeness the certificate at $i = 3$ follows at the stated weak margin by a cover construction. The set S union Y has $N + b = r$ vertices and covers every edge of F not in L_3: an edge outside H meets S, and any remaining edge of H belongs to P_3 and meets Y. For any capacities $A_c + B_c = 3$, place A_c of the first three matching edge sets in the first degree-bounded part and the other B_c in the second. A vertex meets at most one edge of each matching, so these bounds hold and their union is L_3.

For any budgets $x + y = r$, partition S union Y into sets Z_R, Z_B of sizes x, y. Assign every remaining edge having an endpoint in Z_R to the first matching-bounded part and every other remaining edge to the second. The latter part is covered by Z_B because S union Y covers the remaining graph. Each matching uses distinct cover vertices, so the two matching numbers are at most x, y. These are disjoint edge parts exhausting F. Every edge meeting S is outside L_3 and is assigned exactly once, including an edge with both endpoints in S. Therefore $c_3(F) \le r$; the assumed lower bound proves equality. Since $\nu(P_3) = b$, the constructed budget total equals $N + \nu(P_3)$, as asserted.

Finally, if $E(A[Y]) = E(C)$ for the stated odd cycle, all remaining noncycle vertices are isolated. An odd cycle on 2ell-1 vertices has matching number at most ell-1 by counting endpoints. Delete one cycle vertex and take alternate edges of the resulting even-order path to attain ell-1. Thus $\nu(P_4) = \ell - 1$. At every step the choice was a maximum matching of the residual from the preceding choices. Finiteness gives such a maximum even for an edgeless residual, so existence and all conclusions concern one compatible sequence and hold for every such sequence. □

Remark. Use the larger row for the initial matchings and keep the smaller row as the final cover. The row-size difference cancels from all three initial ranks, and the final cover has exactly the cost-forced size.

Depends on. Lemma 7, Lemma 14, Lemma 20.

Lemma 61 (Nonnegative positive-level slack). *Use the finite integer certificate costs $c_{A,B}(H)$, their row maxima $c_i(H)$, matching number $\nu(H)$, edge count $e(H)$, maximum degree $\Delta(H)$, and quantities $\Sigma(H)$, $ex(H)$, $ps(H)$, $gap(H)$, $\tau(H)$ of Lemma 23. Thus $c_{A,B}(H)$ is the least nonnegative integer total c for which every nonnegative integer $x + y = c$ admits an edge partition (Q_R, Q_B, P_R, P_B) with maximum-degree bounds A, B and matching-number bounds x, y; $c_i(H) = \max_{A+B=i} c_{A,B}(H)$, $ps(H) = e(H) - \nu(H) - \sum_{i\ge1} c_i(H)$, and $gap(H) = ex(H) - ps(H)$. All graphs are finite and simple, all capacities, costs, degrees and matching cardinalities are nonnegative integers, and empty graphs are allowed.*

For every such H, one has $c_i(H) = 0$ for every integer $i \ge \Delta(H) + 1$, $ps(H) \ge 0$, and $\Sigma(H) \le e(H) + ex(H)$. If $\tau(H) = \nu(H)$, then $\Sigma(H) \le e(H)$. If $D \ge 1$ is an integer, $\Delta(H) \le D$, and a matching covers every vertex of H-degree D, then $c_D(H) = 0$.

More precisely, if H has an edge, v has degree $D = \Delta(H)$, $h = \nu(H) - \nu(H - v)$, and $a_i = c_i(H) - c_i(H - v)$ for integers $1 \le i \le D$, then h and every a_i belong to {0,1}, and

$ps(H) - ps(H - v) = D - h - \sum_{i=1}^{D} a_i$.

This difference is nonnegative whenever $h = 0$ or a matching covers every degree-D vertex. In the latter case $a_D = 0$ and the difference is at least 1-h. Every H has a sequence of maximum-degree vertex deletions, stopping at an edgeless graph after at most $|V(H)|$ deletions, along which ps never increases upon deletion. Consequently the asserted nonnegativity follows from a finite induction, including $D = 1$, isolated vertices, and the empty graph.

Proof. The costs, their integer threshold property, and $gap = ex - ps$ are supplied by Lemma 23. First establish the restriction and extension estimates used in the induction. Restricting each of the four parts of a certificate to any subgraph preserves all degree and matching bounds. Hence $c_{A,B}(J) \le c_{A,B}(H)$ for a subgraph J of H, and the same inequality holds for every row maximum.

For a vertex v of H put $c = c_{A,B}(H - v)$. Given nonnegative integers $x + y = c + 1$, if $x \ge 1$ take a certificate of H-v with budgets (x-1,y) and put every edge incident with v into P_R. These added edges form a star, so

any matching in their union with the old P_R has at most one more edge than an old P_R matching. The new red matching bound is therefore x; the other three bounds are unchanged. If $x = 0$, take a certificate with budgets (0,c) and instead put all v-edges into P_B. The new blue bound is $c + 1 = y$. Every edge of H has been assigned exactly once, and both constructions use the integer threshold property in Lemma 23. Thus $c_{A,B}(H) \le c_{A,B}(H - v) + 1$. Maximizing over capacities of a fixed total and combining with restriction yields $0 \le c_i(H) - c_i(H - v) \le 1$ for every nonnegative integer i. Since costs are integers, these differences are zero or one.

If $D \ge 1$, $\Delta(H) \le D$ and a matching covers all degree-D vertices, Lemma 35 gives, for every nonnegative integer $A + B = D$, an edge partition into maximum-degree bounds A and B. Taking the two exceptional matching parts empty certifies total zero, so $c_D(H) = 0$. For any $i \ge \Delta(H) + 1$, the set of degree-i vertices is empty; the empty matching covers it. Apply the same statement with $D = i$ to obtain $c_i(H) = 0$. If H is edgeless, every cost is zero directly from the empty partition, including row zero.

Now let H have an edge, let $D = \Delta(H) \ge 1$ and let v have degree D. Restricting a maximum matching to H-v loses at most the unique edge incident with v, while every matching of H-v is a matching of H. Therefore $h = \nu(H) - \nu(H - v)$ belongs to {0,1}. Both H and H-v have zero rows above D by the preceding paragraph, and $e(H) - e(H - v) = D$. Subtracting the two definitions of ps gives the displayed exact identity. If $h = 0$, its right side is nonnegative because the D summands a_i are at most one. If a matching covers all degree-D vertices, then $c_D(H) = 0$. Restriction and nonnegativity give $c_D(H - v) = 0$ as well. Thus $a_D = 0$ and the sum is at most D-1, giving the lower bound 1-h.

It remains to show that an eligible deletion always exists. If some maximum matching misses a vertex v of maximum degree, that matching lies in H-v; hence $h = 0$ and the exact identity permits deletion of this v. If no such matching and vertex exist, every maximum matching covers every maximum-degree vertex. A maximum matching exists because H is finite. It is then a covering matching, so every maximum-degree vertex v is eligible by the preceding paragraph. Delete any eligible v while an edge remains. Each step removes one vertex, and the process stops after at most $|V(H)|$ steps. The terminal graph is edgeless and has $ps = 0$. The inequalities $ps(parent) \ge ps(child)$, composed over the finite sequence, prove $ps(H) \ge 0$ for every H. An initially edgeless graph uses an empty sequence.

Finally Lemma 23 gives $gap = ex - ps$ and $0 \le ex \le \tau - \nu$. Thus $ps \ge 0$ implies $\Sigma - e = gap \le ex$. If $\tau = \nu$, those bounds give $ex = 0$ and hence $\Sigma - e \le 0$. These are exactly the remaining asserted inequalities. □

Depends on. Lemma 23, Lemma 35.

Lemma 62 (An actual binary suffix attaining the complete profile). *Use the positive star forests, ordinary prescribed-color arrowing, profiles, diagonal values ℓ_k, and actual suffix families A_q of Lemma 29. Write $K_2 = K_{1,1}$ and $P_3 = K_{1,2}$. Let $p, t \ge 1$ and $u \ge 0$ be integers. Let F be the vertex-disjoint union of p copies of P_3 and u copies of K_2, and let J have a positive nonincreasing integer profile $m_1,\dots, m_t$. Put $s = p + u$, let $n_1,\dots, n_s$ be p twos followed by u ones, and define $M = s + t - 1$ and ℓ_k using these complete profiles.*

Fix an integer $0 \le q < M$. Let H be a finite simple graph that arrows every member of A_q for (F, J), and suppose its maximum degree satisfies $\Delta(H) \le \ell_{q+2} - 1$. Put

$\alpha = \min(q, p - 1)$, $\beta = q - \alpha$.

Then $0 \le \alpha < s$, $0 \le \beta < t$. The actual pair

$F' = F^{\alpha} = (p - \alpha)P_3$ *disjoint union* uK_2, $J' = J^{\beta}$

belongs to A_q and is arrowed by H. Its component budget is $M' = M - q$, and its complete diagonal profile is

$(\ell_{q+2}, \ell_{q+3}, \dots, \ell_{M+1})$.

Equivalently, for every integer $2 \le d \le M - q + 1$,

$max\{n_{\alpha+i} + m_{\beta+j} - 1 : 1 \le i \le s - \alpha, 1 \le j \le t - \beta, i + j = d\} = \ell_{q+d}$.

Moreover $\Delta(H) = m_{\beta+1}$.

When $u \ge 1$ and $q \ge p$, put $j_0 = q - p + 2$ and $b_0 = \max(0, q - p - u + 1)$. Then $j_0 \le t$ and

$m_{b_0+1} = m_{b_0+2} = \cdots = m_{j_0}$.

In particular all these conclusions apply whenever a finite simple graph G arrows the full pair (F, J), X is any set of q vertices, $H = G - X$ is the induced vertex deletion, and $\Delta(H) \le \ell_{q+2} - 1$. The original unit components are retained in F′. All displayed indices, degrees, component counts, cuts and budgets are integers, and no host matching-number or potential hypothesis is imposed.

Proof. The definitions of actual suffixes and diagonal profiles are those of Lemma 29. Each candidate on local diagonal d of a legal q-suffix comes from the corresponding original diagonal $q + d$. Thus every such suffix value is at most ℓ_{q+d}. We prove equality for the specified cuts.

First suppose $0 \le q \le p - 1$. The cuts $\alpha = q$, $\beta = 0$ are legal, and $p - q \ge 1$ twos remain in the red suffix. Consider any original candidate $n_i + m_j - 1$ on diagonal $q + d$. If $i \ge q + 1$, it is already a candidate in the selected suffix. If $i \le q$, its red size is two. Replace its indices by $i' = q + 1$ and $j' = d - 1$. Then $1 \le i' \le p$, and $1 \le j' \le j \le t$ because $j = q + d - i \ge d > j'$. The index sum is still $q + d$. The red size remains two, and $m_{j'} \ge m_j$. This replacement gives a selected-suffix candidate at least as large. Every original candidate is therefore bounded by a selected-suffix candidate, proving equality on each diagonal. At the leading diagonal, $(q + 1, 1)$ is valid and has value $m_1 + 1$, which is also an upper bound on every candidate. Hence $\ell_{q+2} = m_1 + 1$. The degree assumption gives $\Delta(H) \le m_1$. The all-blue coloring of H cannot contain the nonempty red target F′, so arrowing of (F′,J′) forces a blue J′, which contains an $m_1 - edge$ star. Therefore $\Delta(H) \ge m_1$, proving equality.

Now suppose $q \ge p$ and $u = 0$. Define $j_0 = q - p + 2$. Since $q < M = p + t - 1$, one has $1 \le j_0 \le t$. The proposed cuts are $\alpha = p - 1$ and $\beta = j_0 - 1$, so they are legal and leave a single red P_3. Every red size is two. On original diagonal $q + d$ the smallest available blue index is $q + d - p = j_0 + d - 2$, obtained at red index p. This index belongs to [1,t] throughout $2 \le d \le M - q + 1$. The largest candidate is therefore $m_{j_0+d-2} + 1$. The selected pair, consisting of a single P_3 and the suffix starting at j_0, has exactly these diagonal entries. At $d = 2$ its value is $m_{j_0} + 1$. The degree upper bound and the all-blue coloring of this actual pair give $\Delta(H) = m_{j_0} = m_{\beta+1}$.

It remains to consider $u \ge 1$ and $q \ge p$. Define

$b_0 = \max(0, q - s + 1)$, $a_0 = q - b_0$, $j_0 = q - p + 2$.

The inequalities $q \le s + t - 2$ and $s - 1 \ge p$ give $0 \le b_0 < t$ and $p \le a_0 \le s - 1$. Indeed $a_0 = \min(q, s - 1)$. Thus (a_0, b_0) is a legal actual cut whose red suffix is a nonempty matching. The blue suffix has largest size $c = m_{b_0+1}$. Since H arrows this pair, its all-blue coloring gives

$\Delta(H) \ge c$. (1)

On original diagonal $q + 2$, the largest candidate from a red unit has value c. To see this, its smallest possible blue index is $\max(1, q + 2 - s) = b_0 + 1$. The corresponding red index is between $p + 1$ and s because $q \ge p$ and $u \ge 1$. A candidate from a red two exists exactly when $j_0 \le t$; when it exists, its largest value is $m_{j_0} + 1$, attained at red index p. Also $b_0 + 1 \le j_0$, so $m_{j_0} \le c$ whenever this entry exists.

If $j_0 > t$, only the unit candidates exist and $\ell_{q+2} = c$. This contradicts (1) and $\Delta(H) \le \ell_{q+2} - 1$. Hence $j_0 \le t$. Now

$c \le \Delta(H) \le \max(c, m_{j_0} + 1) - 1 \le c$.

Equality throughout forces $m_{j_0} = c$ and $\Delta(H) = c$. Monotonicity of the blue list implies $m_j = c$ for every $b_0 + 1 \le j \le j_0$. The proposed cuts $\alpha = p - 1$, $\beta = j_0 - 1$ are consequently legal. They leave one P_3, all u red units, and the blue suffix starting at j_0.

We compare every later diagonal. For $2 \le d \le M - q + 1$, the largest original red-two candidate is $m_{j_0+d-2} + 1$ when $j_0 + d - 2 \le t$, and no red-two candidate exists otherwise. This is exactly the red-two candidate of the selected pair. An original red-unit candidate exists throughout the displayed range: $q + d \ge p + 2$ and $q + d \le s + t$. Its largest value is $m_{(}b(d))$, where

$b(d) = \max(1, j_0 + d - u - 2)$.

For $d \ge 3$, the selected suffix also has a red-unit candidate, whose largest value is $m_{(}b'(d))$, where

$b'(d) = \max(j_0, j_0 + d - u - 2)$.

Both indices are at most t on these diagonals. If $b(d) \geq j_0$, the two indices coincide. If $b(d) < j_0$, then

$b(d) \geq b(2) = \max(1, j_0 - u) = b_0 + 1$

and $b'(d) = j_0$. The constant blue run just proved gives equal values again. At $d = 2$ the selected suffix has no red-unit candidate, but its red-two candidate is $c + 1$ and dominates the original unit value c. Taking maxima therefore yields the same value on every diagonal. The retained component counts are $u + 1$ and $t - j_0 + 1$, whose sum minus one is $u + t - j_0 + 1 = M - q$. This verifies both the complete range of diagonals and the component budget. All three cases also give the asserted degree and actual arrowing conclusions.

For the final original-host assertion, we prove the needed vertex-deletion implication. Suppose G arrows (F, J) and X has q vertices. Fix any legal cuts a, b with $a + b = q$. Partition X into sets X_R, X_B of sizes a, b. If $H = G - X$ had a coloring avoiding red F^a and blue J^b, extend it to G by coloring every edge meeting X_R red, and every remaining edge meeting X_B blue. Edges of H keep their colors.

Any red component of a full red F-copy that is not wholly inside H must meet X_R: an edge at a vertex of X_B but not X_R was colored blue, and each target vertex has positive degree. Since target components are vertex-disjoint, at most a red components meet X_R. At least s-a whole components of that F-copy would therefore lie in H. From any such collection one can obtain F^a: list their sizes in decreasing order; its kth size is at least n_{a+k}, because removing at most a entries from the sorted full list cannot place its kth retained entry after index $a + k$. Select s-a components if more are available, and delete excess leaves as needed. Their vertex sets remain disjoint, producing a red F^a in H, a contradiction. Blue edges never meet X_R, so every blue J-component outside H must meet X_B. The same counting and leaf deletion produce a blue J^b in H from a full blue J-copy, again a contradiction. The extension thus avoids both original targets, contrary to G arrowing.

Therefore H arrows every legal member of A_q. Applying the already proved common-host result gives all the final conclusions for the same H and the same original lists. This argument includes $a = 0$ or $b = 0$ and isolated vertices in X; it uses only the positive edge sizes of the target components. □

Depends on. Lemma 29.

Lemma 63 (Perfect matchings forced by failed absorption). *Use the finite integer certificate costs of Lemma 23. In particular, $c_{A,B}(F)$ is the least nonnegative integer total T such that every pair of nonnegative integers x, y with $x + y = T$ permits an edge partition into two parts of maximum degrees at most A, B and two parts of matching numbers at most x, y. Write $c_0 = c_{0,0}$, and write $\nu(G)$ for matching number. All graphs are finite and simple.*

Let F be a graph, X a subset of $V(F)$, $a = |X|$, and $H = F - X$ the induced graph. Let Q be a graph with vertex set W contained in $V(H)$, with $|W| = 2k + 1$ and $\nu(Q) = k$ for a nonnegative integer k. Suppose

$E(H) = K$ *dot-union* $E(Q)$ *dot-union* B_1 *dot-union... dot-union* B_m,

where m is a nonnegative integer and the parts are pairwise disjoint edge sets. The graphs defined by different parts may share vertices. For integers $1 \leq j \leq m$ put $r_j = \nu(B_j)$, viewing B_j as a spanning graph, and assume

$r_j \leq a + \sum_{h<j} r_h + 1$.

Fix nonnegative integers A, B and a partition $K = Q_R$ dot-union Q_B with $\Delta(Q_R) \leq A$ and $\Delta(Q_B) \leq B$. Fix a nonnegative integer δ, and put

$L = a + \sum_{j=1}^{m} r_j$, $l = k + \delta$, $T = L + l$.

Empty lists and sums are allowed. For each nonnegative integer b, let $Y_b(Q)$ be the integers z in [0,b] for which $E(Q)$ has a two-part partition with matching numbers at most z and b-z. Every interval below contains integers only.

If $c_{A,B}(F) > T$, then the complement of $Y_l(Q)$ in [0,l] contains a consecutive run of at least $L + 1$ integers, every such missing integer belongs to $[2\delta + 1, k - \delta - 1]$, and

$L \leq k - 3\delta - 2$.

If equality $L = k - 3\delta - 2$ holds, then

$Y_l(Q) = [0, 2\delta]$ *union* $[k - \delta, k + \delta]$.

In this equality case, the following conclusions hold on this same Q:

1. For every subset Z of W with $|Z| = 2\delta + 1$, the induced graph Q-Z has a perfect matching.

2. Every even-order induced subgraph of Q with at least 2k-2delta vertices has a perfect matching. Every odd-order induced subgraph with at least $2k - 2\delta + 1$ vertices is factor-critical, meaning that each of its single-vertex deletions has a perfect matching.

3. If M is any matching of Q with $|M| \leq \delta$ and v is any vertex not incident with M, there is a maximum matching of Q that contains every edge of M and exposes v, meaning that v is not incident with any matching edge.

4. Every vertex of Q has degree at least $2\delta + 2$.

No minimality, regularity, independence of X, or matching-additivity identity for F is assumed. All capacities, budgets, totals, orders, indices and matching cardinalities have the integer domains stated above.

Proof. We use Lemma 23 only for certificate definitions and the exact integer threshold interpretation. We first show how every available allocation extends to the original F, then infer the missing interval, and finally construct the asserted matchings.

Order X as v_1,..., v_a. Assign each F-edge meeting X to its first endpoint in this order. This gives a labelled, possibly empty star U_i centered at v_i for every integer $1 \leq i \leq a$. An edge with both endpoints in X is assigned once. Because H is induced, the stars partition exactly $E(F)$ minus $E(H)$. Each star has matching number at most one. Coloring any t of these stars red and all the others blue gives matching upper budgets t, a-t for every integer $0 \leq t \leq a$.

Starting with these choices, add B_1,..., B_m in order, always coloring an entire B_j in one color. Suppose the available red budgets fill [0,u], at total u. A block of matching number r_j supplies [0,u] if colored blue and $[r_j, u + r_j]$ if colored red. The two integer intervals have no intervening missing integer when $r_j \leq u + 1$. The assumed ordered inequalities therefore prove inductively that the stars and all B_j supply every red budget in [0,L] at total L. For each such budget s there is a specific assignment whose red matching bound is s and whose blue matching bound is L-s.

For any z in $Y_l(Q)$, choose its defining two-part partition of $E(Q)$. Combine it with any of the preceding assignments of budget s, and retain the prescribed Q_R, Q_B on K. All four edge parts are disjoint and cover $E(F)$: the stars cover exactly the omitted edges and the displayed partition covers $E(H)$. A matching in a combined exceptional part restricts to a matching in each of its disjoint edge pieces. It consequently has cardinality at most the sum of the assigned upper bounds. Thus this is a certificate of F with matching budgets $s + z$ and $(L - s) + (l - z)$. Shared vertices between pieces cannot invalidate an upper bound obtained by restricting a matching. In particular, if

$$[0, L] + Y_l(Q) = [0, T],$$

where the plus sign denotes addition of integer sets, then every budget split at total T has an actual certificate with the given degree-bounded parts. That would imply $c_{A,B}(F) \leq T$.

The endpoints 0 and l belong to $Y_l(Q)$, because $l \geq \nu(Q) = k$ and the entire edge set can be put in either color. List the elements of $Y_l(Q)$ increasingly. If every missing run has length at most L, consecutive listed elements differ by at most $L + 1$. Translating [0,L] by these listed elements gives integer intervals with no intervening missing integer, starting at zero and ending at $L + l$. This also covers $l = 0$, when the list is the singleton zero. Under $c_{A,B}(F) > T$, the equality of integer sets above must therefore fail, and there is a missing run of length at least $L + 1$.

We locate all missing budgets by an explicit vertex-order construction. For nonnegative integers p, q and any graph on n vertices, a two-part partition with matching bounds p, q exists whenever $n \leq p + q + \max(p, q) + 1$. To prove this, suppose $p \geq q$, choose $\min(n, 2p + 1)$ vertices, color their internal edges with the first color, and color all remaining edges with the second. The first matching number is at most p by its endpoint count. Every edge of the second part meets the complementary vertex set, of size $\max(0, n - 2p - 1) \leq q$, so its matching number is at most q. If $q > p$, exchange the two colors.

Apply this construction to Q at total $l = k + \delta$. If the smaller budget z satisfies $z \leq 2\delta$, the larger is l-z, and

$$l + (l - z) + 1 = 2k + 2\delta - z + 1 \geq 2k + 1 = |W|.$$

Hence this allocation is feasible. Every missing red budget must therefore be at least $2\delta + 1$ and have complementary budget at least $2\delta + 1$. All missing budgets lie in $[2\delta + 1, k - \delta - 1]$. Since a missing run

of length at least $L+1$ exists, this interval is nonempty and its cardinality k-3delta-1 is at least $L+1$. This proves $L \le k-3\delta-2$. It also gives $k \ge 3\delta+2$.

Now suppose $L = k-3\delta-2$. The containing interval has exactly $L+1$ integers. A missing run of at least $L+1$ integers contained in it must be the entire interval. All its integers are therefore missing, and every integer outside it in [0,l] is feasible by the preceding construction. This proves the asserted exact description of $Y_i(Q)$. In particular, the allocation

$(2\delta+1, k-\delta-1)$

is not feasible on Q; both entries are nonnegative since $k \ge 3\delta+2$.

Fix any Z contained in W of cardinality $2\delta+1$. The graph Q-Z has exactly 2k-2delta vertices. If it had no perfect matching, its matching number would be at most k-delta-1: a matching on an even number 2k-2delta of vertices has at most k-delta edges, and equality covers all its vertices. Color every Q-edge meeting Z red and every other Q-edge blue. The red matching number is at most $|Z| = 2\delta+1$ because each red matching edge uses a distinct vertex of Z. The blue graph is Q-Z with added isolates and would have matching number at most k-delta-1. This would realize the forbidden allocation. Therefore every such Q-Z has a perfect matching.

Put $n_0 = 2k-2\delta$. We have proved that every induced subgraph of Q on n_0 vertices has a perfect matching. This even integer is positive because $k \ge 3\delta+2$. Induct on even integers n with $n_0 \le n \le |W|$. At the initial value the assertion is proved. For $n > n_0$, let J be any induced subgraph on n vertices. It contains an induced $n_0 - vertex$ subgraph with a nonempty perfect matching, so J has an edge uv. The induced graph J-{u, v} has $n-2 \ge n_0$ vertices and has a perfect matching by the induction hypothesis. Add uv to it to obtain a perfect matching of J. For an odd-order induced subgraph of order at least n_0+1, deleting any vertex leaves an even-order induced subgraph of order at least n_0, which has a perfect matching. This proves the second conclusion.

For the third conclusion, write $t = |M| \le \delta$ and delete v together with all 2t endpoints of M. The remaining induced graph has even order $2k-2t \ge n_0$, so the second conclusion supplies a perfect matching in it. Its union with M is a matching with $(k-t)+t = k$ edges, uses no edge incident with v, and contains M. Since $\nu(Q) = k$, it is a maximum matching.

Finally suppose a vertex v of Q has at most $2\delta+1$ neighbors. There are 2k vertices different from v, and $2k \ge 2\delta+1$. Choose a set Z of exactly $2\delta+1$ vertices, all different from v, containing every neighbor of v. Then Q-Z has even positive order n_0 and contains the isolated vertex v, so it has no perfect matching. This contradicts the first conclusion. Thus every vertex has degree at least $2\delta+2$. All constructions concern the same original edge sets and vertex sets throughout. □

Remark. The omitted vertices and small edge blocks supply an integer interval of allocations. At the last possible failing size, the odd block must miss its entire central interval. Coloring all edges at a chosen deletion set then turns any failure of a perfect matching into one of those forbidden allocations.

Depends on. Lemma 23.

Lemma 64 (Actual suffixes and certificate row bounds). *Use the star profiles and ordinary prescribed-color arrowing of Lemma 29, and the finite integer certificate costs $c_{A,B}$, row maxima c_i, and quantities Σ, ν, ex, ps, gap of Lemma 23. Explicitly, s, t are positive integers, $n_1 \ge \cdots \ge n_s \ge 1$ and $m_1 \ge \cdots \ge m_t \ge 1$ are integer lists, and F_1, F_2 are their vertex-disjoint star unions, prescribed red and blue respectively. Put $M = s+t-1$ and $\ell_k = max\{n_i + m_j - 1 : 1 \le i \le s, 1 \le j \le t, i+j = k\}$ for every integer $2 \le k \le s+t$. Fix an integer $0 \le q < M$, put $K = M-q$, $L_q = \sum_{k=q+2}^{s+t} \ell_k$ and $P_q = L_q - K$, and let A_q contain all ordered actual suffix pairs $\lambda = (F_1^{\alpha}, F_2^{\beta})$ with integer cuts $0 \le \alpha < s$, $0 \le \beta < t$, $\alpha+\beta = q$. Superscripts delete that many leading whole components.*

For every nonnegative integer θ define $\rho_\theta^q = |\{k : q+2 \le k \le s+t, \ell_k \ge \theta+1\}|$. For λ as above and nonnegative integers A, B define $r_\lambda(A) = |\{i : \alpha < i \le s, n_i > A\}|$ and $s_\lambda(B) = |\{j : \beta < j \le t, m_j > B\}|$. Then

$$\rho_\theta^q = max\{r_\lambda(A) + s_\lambda(B) - 1 : \lambda \in A_q, A+B = \theta, r_\lambda(A) > 0, s_\lambda(B) > 0\},$$

where A, B are nonnegative integers and an empty maximum is zero. Every positive maximum is attained by actual legal cuts. Moreover $\rho_0^q = K$ and $\sum_{\theta \ge 0} \rho_\theta^q = L_q$.

Let H be any finite simple graph, possibly empty. Set $\Phi(H) = e(H) - \nu(H)$, $\delta = K - \nu(H)$, $\sigma_\theta = c_\theta(H) - \rho_\theta^q$, and $S = \sum_{\theta \geq 0} \sigma_\theta$. These quantities are integers, δ, σ_θ and S may be signed, the sum has finite support, and

$gap(H) = L_q - e(H) + S$,

$ex(H) = \delta + \sigma_0$,

$ps(H) = \Phi(H) - P_q - \sum_{\theta \geq 1} \sigma_\theta$.

If $e(H) = L_q - 1$, then $gap(H) = 1 + S$ and $\Phi(H) - P_q = \delta - 1$.

If H arrows every entire prescribed-color member of A_q, then for every λ and A, B having positive counts $r_\lambda(A)$, $s_\lambda(B)$,

$c_{A,B}(H) \geq r_\lambda(A) + s_\lambda(B) - 1$.

Consequently $\sigma_\theta \geq 0$ for every θ, $\Sigma(H) \geq L_q$, $ex(H) \geq \delta$, and $gap(H) \geq L_q - e(H)$. In particular every such host with $e(H) < L_q$ has positive gap. Conversely, any θ with $\sigma_\theta < 0$ yields some maximizing λ, A, B and a certificate with exceptional matching budgets $r_\lambda(A) - 1$, $s_\lambda(B) - 1$; coloring its two red parts red and its two blue parts blue avoids both entire targets of that actual member. The maximizing suffix may depend on θ. All components, including repeated sizes and units, are retained.

Proof. The definitions, positive integer diagonals and their monotonicity $\ell_2 \geq \cdots \geq \ell_{s+t}$ come from Lemma 29. Fix a nonnegative integer θ. Consider a legal tuple λ, A, B in the displayed maximum, and abbreviate $r = r_\lambda(A) > 0$ and $u = s_\lambda(B) > 0$. Because the lists are sorted, $n_{\alpha+r} > A$ and $m_{\beta+u} > B$. The two valid original indices $\alpha + r$ and $\beta + u$ have sum $k = q + r + u$. Thus $\ell_k \geq n_{\alpha+r} + m_{\beta+u} - 1 \geq A + B + 1 = \theta + 1$. Monotonicity implies that every diagonal from $q + 2$ through k qualifies in ρ_θ^q. There are $k - q - 1 = r + u - 1$ of them, proving that every term in the maximum is at most ρ_θ^q.

If $\rho_\theta^q = 0$, no positive term is possible, and the empty-maximum convention gives the desired equality. Suppose $\rho = \rho_\theta^q > 0$. Monotonicity says the last qualifying suffix diagonal has index $k = q + \rho + 1$. Choose valid indices i, j attaining ℓ_k. Then $i + j = k$ and $n_i + m_j \geq \theta + 2$. Set $A = \max(0, \theta - m_j + 1)$, $B = \theta - A$. These are nonnegative integers with sum θ, $A \leq n_i - 1$ and $B \leq m_j - 1$. Indeed $\theta - m_j + 1 \leq n_i - 1$ by the last inequality; if $A = 0$ then $\theta \leq m_j - 1$, while if $A > 0$ then $B = m_j - 1$.

Set $\alpha = \max(0, q - j + 1)$ and $\beta = q - \alpha$. These are nonnegative integers. One has $\beta \leq j - 1$ by the choice of α, and $\alpha \leq i - 1$ because $q \leq i + j - 2$ follows from $\rho \geq 1$. Also $\alpha \leq q$ since $j \geq 1$. Therefore $0 \leq \alpha \leq i - 1 < s$ and $0 \leq \beta \leq j - 1 < t$, so these are legal cuts in A_q. The surviving counts satisfy $r_\lambda(A) \geq i - \alpha \geq 1$ and $s_\lambda(B) \geq j - \beta \geq 1$. Their sum minus one is at least $i + j - q - 1 = \rho$. The upper bound already proved forces equality, establishing the maximum and its attainment. At $\theta = 0$ every positive suffix diagonal qualifies, so $\rho_0^q = M - q = K$. Each positive integer ℓ_k contributes one to ρ_θ^q for exactly $\theta = 0,..., \ell_k - 1$. Summing this finite set of incidences gives $\sum_\theta \rho_\theta^q = L_q$.

Now fix H. The finite support of $c_\theta(H)$ is part of Lemma 23; ρ_θ^q is zero for $\theta \geq \ell_{q+2}$, so every indicated sum has finite support. Their difference sums to $S = \Sigma(H) - L_q$, yielding $gap(H) = L_q - e(H) + S$. Since $c_0 = c_{0,0}$ and $\rho_0^q = K$, subtraction of $\nu(H)$ gives $ex(H) = \sigma_0 + \delta$. The positive ρ levels sum to $L_q - K = P_q$. Substitute $c_\theta = \rho_\theta^q + \sigma_\theta$ in the definition of ps to obtain $ps(H) = \Phi(H) - P_q - \sum_{\theta \geq 1} \sigma_\theta$. When $e(H) = L_q - 1$, these identities give $gap(H) = 1 + S$ and $\Phi(H) - P_q = L_q - 1 - \nu(H) - (L_q - K) = \delta - 1$. None of these identities uses arrowing.

We give the avoiding coloring that proves the cost bound. For any fixed legal λ, A, B with $r = r_\lambda(A) > 0$ and $u = s_\lambda(B) > 0$, suppose there is a certificate with exceptional matching budgets r-1, u-1. Color Q_R union P_R red and Q_B union P_B blue. A red copy of the entire actual suffix F_1^α would have r vertex-disjoint star components with more than A edges each. At the center of each such star at most A of its incident edges can belong to Q_R, so at least one edge of each of the r stars belongs to P_R. These chosen edges have disjoint endpoints because the target components are vertex-disjoint. They form an r-edge matching in P_R, contradicting its bound r-1. A blue copy of the entire suffix F_2^β has u vertex-disjoint stars with more than B edges, and choosing a P_B edge from each contradicts $\nu(P_B) \leq u - 1$. Thus the stated actual coloring avoids both whole suffixes. This argument includes stars of one edge whenever the corresponding capacity is zero.

If H arrows every A_q member, this certificate cannot exist. If $c_{A,B}(H) \le r+u-2$, the integer threshold property of Lemma 23 would supply a certificate at the particular nonnegative allocation (r-1,u-1), because its total is $r+u-2$. Hence $c_{A,B}(H) \ge r+u-1$. At a positive envelope maximum this implies $c_\theta(H) \ge \rho_\theta^q$; if $\rho_\theta^q = 0$ the same bound is just nonnegativity of the cost. Thus every σ_θ is nonnegative. Summing gives $\Sigma \ge L_q$, and the already proved identities give $ex \ge \delta$ and $gap \ge L_q - e(H)$. Strictly below budget, the latter right side is a positive integer.

Finally suppose $\sigma_\theta < 0$. Nonnegativity of c_θ forces $\rho_\theta^q > 0$. Choose its attaining λ, A, B, with positive counts r, u. Then $c_{A,B}(H) \le c_\theta(H) \le \rho_\theta^q - 1 = r+u-2$, where the middle inequality uses integrality. The threshold property supplies the certificate at (r-1,u-1), and the preceding construction gives the asserted avoiding coloring. A separate choice is allowed for each θ, so no simultaneous maximizing suffix was assumed. □

Depends on. Lemma 23, Lemma 29.

(printed in plain notation)

Lemma 65 (Deficit-preserving deletion inside a minimum cover). `Deficit-preserving deletion inside a minimum cover.`

`All graphs are finite and simple. Write e(H), nu(H), and tau(H) for edge count, maximum matching cardinality, and minimum vertex-cover cardinality. A matching consists of edges with disjoint endpoints, and a vertex cover meets every edge. Let F and J be the entire vertex-disjoint unions of stars with complete nonincreasing positive integer edge-size lists n_1,...,n_s and m_1,...,m_t, where s,t>=1. Copies are ordinary prescribed-color subgraphs, with disjoint components within each target.`

`Put M=s+t-1. For every integer 0<=a<M let A_a contain every ordered pair (F^alpha,J^beta) with nonnegative integer cuts alpha<s, beta<t, alpha+beta=a, where a superscript deletes that many leading whole components. Suppose 0<=q<M and H arrows every entire A_q member. Put K=M-q, delta=K-nu(H), E_*=max(e(F),e(J)), and assume the integer delta>=1. Then q+delta<=min(s,t)-1, and tau(H)>=max(K,E_*+2delta-1).`

`More precisely, for every minimum vertex cover C of H, the induced graph H[C] contains a matching of at least delta edges. Choose any matching W in H[C] with exactly delta-1 edges, and put Y=V(W), B=H-Y, r=q+2delta-2, K'=M-r, and h=nu(H)-nu(B). Then |Y|=2delta-2, delta-1<=h<=2delta-2, tau(B)=tau(H)-2delta+2, r<M, and K'-nu(B)>=1. The same actual induced graph B arrows every entire A_r member. In particular r<=min(s,t)-2, so both extreme pairs (F^r,J) and (F,J^r) are legal and arrowed by B.`

`Every choice of C and W specified above has these properties. At delta=1 the matching and deleted set are empty and B=H. There is no host-degree, edge-budget, connectivity, target-head, unit-count, or induction hypothesis, and no assertion that the endpoint cut removes any prescribed number of edges. All indices, orders, cardinalities, sizes, deficits, and matching losses are integers in the specified domains.`

Proof. `We first establish two matching facts by finite alternating-path arguments.`

`Let Q be a bipartite graph with parts L and O and let U be a maximum matching. Direct edges outside U from L to O and matching edges from O to L. Starting at all unmatched L-vertices, let X subseteq L and Z subseteq O be the reached sets. No unmatched O-vertex is reached: a shortest directed path to one would be a simple alternating path with unmatched endpoints, and replacing its matching edges by its other edges would increase the matching by one. Thus every reached O-vertex is matched, and its matching partner is reached in L. Conversely every matched reached L-vertex has its partner in Z, since it can first be reached only along its matching edge. Also N_Q(X)=Z. A nonmatching edge out of X reaches its O-end, and a matching edge at a reached L-vertex leads to its already reached partner. All unmatched L-vertices belong to X. Therefore U pairs Z with exactly the matched vertices of X and pairs every vertex of L minus X with a vertex of O minus Z. These assertions include the case that every L-vertex is matched, when X and Z are empty.`

`If L itself is a minimum vertex cover of Q, then U saturates L. Otherwise |X|>|Z| by the preceding pairing and the existence of an unmatched L-vertex. But (L minus X) union Z is a vertex cover, by N_Q(X)=Z, and has size smaller than |L|, a contradiction. Thus in this case nu(Q)=|L|.`

Now let C be any minimum cover of a graph G. For every Y subseteq C,

tau(G-Y)=tau(G)-|Y|. (1)

Indeed C minus Y is a cover of G-Y, while Y together with any cover of G-Y covers G. Choose a maximum matching W_0 in G[C], write k_0=|W_0| and Y_0=V(W_0). The set C minus Y_0 is independent, because an edge on two of its vertices could be added to W_0. The set O=V(G) minus C is also independent. Thus G-Y_0 is bipartite with parts C minus Y_0 and O. By (1), its first part is a minimum cover. The preceding saturation argument supplies a matching there of size tau(G)-2k_0. Unite it with W_0 to obtain a matching in G of size tau(G)-k_0. Consequently

nu(G[C])>=tau(G)-nu(G). (2)

For any matching W in G[C] with k edges and endpoint set Y, unite W with a maximum matching of G-Y to get nu(G)-nu(G-Y)>=k. Restrict a maximum matching of G to G-Y; at most 2k matching edges are lost because Y has 2k vertices. Hence

k<=nu(G)-nu(G-Y)<=2k. (3)

All choices exist because the graphs and their matching and cover families are finite.

We next prove a target-dependent cover bound. Let U_0 be an entire union of t_0>=1 positive stars, let p>=1, and put K_0=p+t_0-1 and E_0=e(U_0). Suppose a graph G arrows (pK2,U_0), has a vertex cover C_0 of size K_0+b with integer b>=0, and nu(G)<K_0. We claim

nu(G)>=E_0-b. (4)

Here K2 is one edge. Put O_0=V(G) minus C_0 and take a maximum matching V in the bipartite crossing graph between C_0 and O_0. Apply the alternating construction already proved, with reached sets X subseteq C_0 and Z subseteq O_0. Put l=|X|. Some C_0-vertex is unmatched, since |V|<=nu(G)<K_0<=|C_0|. The construction shows that Z is the crossing neighborhood of X, that all Z-vertices are matched into X, and that exactly |C_0|-l edges of V lie outside X union Z.

For any set T subseteq C_0 with |T|=t_0+b, color every edge incident with C_0 minus T red and the others blue. The red graph has a cover of size p-1, so a full blue U_0 must occur. All vertices of this copy lie in T union O_0: no vertex of C_0 minus T has a blue incident edge, and every target vertex has positive degree. Every copied component meets T, because O_0 is independent.

If l<=t_0+b, choose T containing X. At most |T minus X|=t_0+b-l copied components meet T minus X. At least l-b copied components therefore avoid T minus X. Every outside vertex of these components has a copied neighbor in X, so belongs to Z. Selecting one edge per such component gives a matching inside X union Z of size at least l-b; when l-b<=0 the empty matching suffices for this lower bound. Unite it with the |C_0|-l fixed edges of V outside X union Z. The result has at least |C_0|-b=K_0 edges, contrary to nu(G)<K_0.

Thus l>t_0+b. Choose instead T subseteq X of size t_0+b. Its forced entire blue U_0 copy has E_0+t_0 vertices, at most t_0+b of which are in C_0. At least E_0-b are outside C_0, and each has a neighbor in T subseteq X, so lies in Z. Hence |Z|>=E_0-b. Since V matches every Z-vertex, nu(G)>=|V|>=|Z|, proving (4).

For later use, consider any ordered pair of positive star forests with component counts a_0,b_0 and budget L_0=a_0+b_0-1. Every graph arrowing that pair has cover number at least L_0. A cover of size at most L_0-1 could be partitioned into parts of sizes at most a_0-1 and b_0-1. Coloring edges incident with the first part red and every other edge blue gives a red cover and a blue cover of these respective sizes. Every entire target contains a matching of its component count, so neither target occurs.

If such a graph also has nu(G)<L_0, relax each entire target in turn to a matching by selecting one edge from each positive star. It arrows both (a_0 K2,second target) and, after exchanging color names, (b_0 K2,first target). Its minimum cover has size L_0+b for b>=0. Apply (4) to each relaxation to get

tau(G)+nu(G)>=L_0+max(edge counts of the two entire targets). (5)

The color exchange is a bijection on all edge colorings, and does not require the witnesses for the two relaxed pairs to be the same.

Return to the full actual family A_q in the statement. It is nonempty. Its member with alpha=max(0,q-t+1) has exactly min(s,K) red components, and its member with

beta=max(0,q-s+1) has exactly min(t,K) blue components. These cuts are legal since q<=s+t-2. In the all-red or all-blue coloring, respectively, each full target copy yields a matching of its component count. Thus nu(H)>=min(s,K), and nu(H)>=min(t,K). Since nu(H)=K-delta<K, neither minimum can equal K. Therefore nu(H)>=s,t, and substitution of K=s+t-1-q gives

q+delta<=s-1, q+delta<=t-1. (6)

The cover argument preceding (5), applied to any A_q member, also gives tau(H)>=K.

Fix an arbitrary minimum cover C of H. By (2), nu(H[C])>=tau(H)-nu(H)>=delta. Choose any of its matchings W with k=delta-1 edges, set Y=V(W) and B=H-Y. Formulas (1) and (3) give the stated exact cover loss and k<=h<=2k. From (6), s,t>=q+delta+1, so K>=q+2delta+1. With r=q+2k and K'=K-2k, it follows that K'>=q+3>0 and r<M. Moreover

K'-nu(B)=delta-2k+h>=delta-k=1. (7)

We verify simultaneous suffix persistence on this same B using the actual-vertex extension in Lemma 41 (II). Fix any legal cuts alpha',beta' totaling r. Choose alpha=min(q,alpha') and beta=q-alpha. If alpha=q, then beta=0<=beta'. Otherwise alpha=alpha'<q and beta=q-alpha'<=r-alpha'=beta'. Thus alpha<=alpha'<s, beta<=beta'<t, and alpha+beta=q. The graph H arrows (F^alpha,J^beta). The additional cuts alpha'-alpha,beta'-beta are nonnegative, sum to |Y|=2k, and leave both targets nonempty. An avoiding coloring on H-Y would extend across precisely Y to avoidance of (F^alpha,J^beta), by the cited extension. This is impossible. Hence B arrows (F^alpha',J^beta'), for every legal final allocation.

Apply the elementary count argument (6) to the original lists at depth r and graph B, whose deficit in (7) is positive. It gives r<=min(s,t)-2. Both extreme pairs (F^r,J) and (F,J^r) are therefore legal, have common budget K', and are arrowed by B. Apply (5) separately to them. Since nu(B)<K', one obtains tau(B)+nu(B)>=K'+e(J), and tau(B)+nu(B)>=K'+e(F). Thus tau(B)+nu(B)>=K'+E_*. Substitute tau(B)=tau(H)-2k, nu(B)=nu(H)-h, and K'=K-2k. The result is tau(H)+nu(H)>=K+E_*+h. Using nu(H)=K-delta and h>=k=delta-1 gives tau(H)>=E_*+2delta-1. Together with tau(H)>=K this proves the required bound. The choices of C and W were arbitrary, and every formula remains valid when k=0, establishing all witness and endpoint assertions.

□

Depends on. Lemma 41.

Lemma 66 (Quartic components with exact matching-complement covers). *Let $\mu \geq 2$ be an integer and let C be a finite connected simple four-regular graph of order $2\mu + 1$. Suppose C is factor-critical: deleting any vertex leaves a perfect matching. A matching is a set of edges with pairwise disjoint endpoints, and it is perfect if it covers every vertex. Write $\nu(J)$ for the maximum matching cardinality and $\tau(J)$ for the minimum vertex-cover cardinality of a finite graph J. A vertex cover meets every edge. For x in $V(C)$, let $N_C(x)$ be its open neighborhood and $N_C[x] = N_C(x)$ union $\{x\}$.*

Suppose that for every vertex x and every perfect matching W of C-x, the graph $U = (C - W) - x$ satisfies

$\tau(U) = \mu$,

and $(C - N_C[x]) - W$ has a perfect matching.

Here subtracting a matching deletes its edges only, every vertex deletion is induced, and the empty graph has a perfect matching.

If C is equimatchable, meaning that every inclusionwise maximal matching has maximum cardinality, then C is isomorphic to K_5, the complete graph on five vertices. Equivalently, every C satisfying the preceding hypotheses and not isomorphic to K_5 has an inclusionwise maximal matching of size at most nu(C)-1. All orders, degrees, matching and cover cardinalities, and parameters are integers in the stated domains. No arrowing or upper-order hypothesis is imposed.

Proof. Assume C is equimatchable. Factor-criticality and odd order give $\nu(C) = \mu$. Every matching in a finite graph extends, by adding disjoint edges, to an inclusionwise maximal one. Here every such extension therefore has μ edges and leaves exactly one vertex uncovered.

We first prove that C is 2-connected, meaning that it has at least three vertices and has no cut vertex. Its order is at least five by regularity and simplicity. Suppose v is a cut vertex. Every component of C-v has even order,

since a perfect matching of C-v exists. Select distinct components B, B′, and choose a neighbor w of v in B′. Fix a perfect matching F of C-w. Its edge incident with v must go into B′: otherwise the odd set B′ minus {w} has no matching edge leaving it and cannot be covered. All other components of C-v are matched internally by F.

Let M consist of the F-edges having at least one endpoint in $N_C(w)$. All neighbors of w lie in B′ union {v}, and the preceding description of F shows that every endpoint of M also lies in B′ union {v}. Moreover v is covered by M, since v is a neighbor of w. The matching M lies in C-w and covers $N_C(w)$. Apply the first assertion of Lemma 6 to C, w, M. The set B remains a whole component of its residual, so C[B] is an even clique or a balanced complete bipartite graph. In either case it has a common internal degree a. Each of its vertices has total degree four and has only v as a possible outside neighbor. Connectedness gives at least one edge to v; thus $4 - a = 1$ and every vertex of B neighbors v. An even clique with internal degree three has four vertices, and a balanced complete bipartite graph with internal degree three has six vertices. In both cases v has at least four neighbors in B, as well as w in B′, contradicting degree four. Hence C is 2-connected.

For every vertex x, the hypotheses imply that $C - N_C[x]$ has a perfect matching Y, since they give one after additional edge deletions. It has mu-2 edges. If three neighbors a, b, c of x were independent, let d be its fourth neighbor. Then Y union {xd} is a matching of size mu-1 whose uncovered vertices are exactly a, b, c. Such a matching is maximal: an edge could be added only between two uncovered vertices. This contradicts equimatchability. Thus C is claw-free, meaning that no vertex has three independent neighbors.

There is a vertex x for which the four-vertex graph $C[N_C(x)]$ has a perfect matching. To prove this, a four-vertex graph with independence number at most two and no perfect matching must be a triangle and an isolated vertex. Indeed, choose an edge ab and call the other vertices c, d. They are nonadjacent, since otherwise ab, cd would be a perfect matching. Each of a, b has a neighbor in {c, d}, or that vertex together with c, d would be independent. The two nonempty neighbor sets cannot admit distinct representatives, which would give a perfect matching. They are therefore the same singleton, say {c}. This gives exactly the triangle abc and the isolated vertex d.

If every neighborhood in C had this form, each vertex x would belong to a unique four-clique, namely x and the triangle in its neighborhood. For every vertex y of that clique, its other three clique vertices form the triangle in $N_C(y)$, so y has the same unique four-clique. These cliques consequently partition $V(C)$, making its order divisible by four. This contradicts odd order and proves the existence of x.

Fix such x, put $A = N_C(x)$, and choose a perfect matching M_A of C[A]. The matching M_A lies in C-x and covers $N_C(x)$; deleting either of its two edges uncovers two neighbors of x. It is therefore inclusion-minimal with this property. By Lemma 6, using the proved 2-connectivity, the induced graph $B = C - N_C[x]$ is empty, an even complete graph, or a balanced complete bipartite graph. An even clique has at most four vertices because its internal degree is at most four. A balanced complete bipartite B has both parts of size at most two: a vertex and three vertices in the other part would otherwise induce a claw in C. Thus $|V(B)| \le 4$ and $|V(C)|$ is five, seven or nine.

The order-seven case is impossible. Write $\alpha(C)$ for the maximum independent-set cardinality. Choose a maximum independent set I and a vertex y in it. All vertices of I minus {y} lie in $C - N_C[y]$. A perfect matching of that graph implies that an independent set there has size at most half its order, namely mu-2. Hence $\alpha(C) \le \mu - 1$. At order seven, $\mu = 3$ and $\alpha(C) \le 2$. The graph complement of C is two-regular. Each component of a finite simple two-regular graph is a cycle of length at least three: following a path until a vertex repeats gives a cycle, and degree two leaves no edge out of it. Its possible cycle lengths at order seven are seven or three and four. A three-cycle in the complement would be an independent triple in C, so the complement is a seven-cycle.

Label its vertices 0,..., 6 in cyclic order. Take $y = 0$ and $W = \{14, 25, 36\}$. Each listed edge joins vertices at cyclic distance three, so is present in C, and these edges form a perfect matching of C-y. If $U = (C - W) - y$ had an independent triple T, all C-edges within T would belong to W. Thus each vertex of T would have at most one C-neighbor within T and at least one neighbor there in the seven-cycle. An induced three-vertex subgraph of a seven-cycle with no isolated vertex is a path on three vertices. Its endpoints are at cyclic distance two, so their C-edge would have to belong to W, whose edges all have cyclic distance three. This is impossible. Hence U has independence number at most two. But a vertex cover and its complementary independent set partition $V(U)$, and $\tau(U) = \mu = 3$ on six vertices would give an independent triple. This excludes order seven.

Order five makes $C = K_5$ by degree four. It remains to exclude order nine. Here $\mu = 4$, A has four vertices and B has four vertices, and the residual classification gives $B = K_4$ or $B = K_{2,2}$, a four-cycle.

First suppose $B = K_4$. Every B-vertex has exactly one A-neighbor: it has three neighbors in B and none equal to x. There are four A–B edges. Since each A-vertex neighbors x, summing its remaining three degrees gives $12 = 2e(C[A]) + 4$, so $e(C[A]) = 4$. A four-vertex graph with four edges is a four-cycle or a triangle with a pendant vertex, as its complement consists of two disjoint edges or two edges with a common endpoint.

If C[A] is a four-cycle, each A-vertex has exactly one B-neighbor. Its two A-neighbors are nonadjacent, and its B-neighbor has no other A-neighbor. These three vertices are independent neighbors of that A-vertex, contrary to claw-freeness. Thus name the triangle $a1$, $a2$, $a3$ and the pendant vertex $a4$, attached to $a3$. Their respective B-degrees are one, one, zero and two. Name $B = \{b1, b2, b3, b4\}$ so that the four cross edges are a1b1, a2b2, a4b3, a4b4. The two sets {a1,a2,x, a3} and {b1,b2,b3,b4} are four-cliques, their only cross edges are a1b1, a2b2, and the remaining vertex $a4$ neighbors x, $a3$, $b3$, $b4$.

Now suppose B is the cycle b1b2b3b4b1. Every B-vertex has two A-neighbors, giving eight A–B edges and $12 = 2e(C[A]) + 8$. Hence C[A] has two edges. They are disjoint: if they shared an endpoint, their two other endpoints and the fourth A-vertex would be three independent neighbors of x. Every A-vertex consequently has exactly two B-neighbors. These neighbors are adjacent on the cycle, since otherwise they and x would be three independent neighbors of that A-vertex.

Let $N_A(b)$ denote the A-neighbors of b. The two sets $N_A(b2)$, $N_A(b4)$, each of size two, partition A: every adjacent pair on the cycle contains exactly one of $b2$, $b4$. No A-edge ac crosses this partition. If it did, with a adjacent to $b2$ and c to $b4$, then the edges b2a, b4c and the other edge of C[A] would form a matching leaving exactly x, $b1$, $b3$ uncovered. This is an independent triple and would make the matching maximal of size three, contradicting $\nu(C) = 4$. Thus $N_A(b2)$ and $N_A(b4)$ are precisely the endpoint sets of the two A-edges. The same argument using the opposite pair $b1$, $b3$ and the uncovered triple x, $b2$, $b4$ shows that $N_A(b1)$, $N_A(b3)$ are also those two endpoint sets.

Every B-vertex therefore neighbors both endpoints of one A-edge. The endpoints of each A-edge have the same two B-neighbors, which are adjacent. Rotate the cycle labels so that one A-edge is completely joined to {b1,b2} and the other to {b3,b4}. These give two four-cliques. The only cross edges are b2b3, b4b1. The remaining vertex x is adjacent precisely to the two A-edges.

In both nine-vertex cases, C has the following common description: two disjoint cliques $P = \{p1, p2, p3, p4\}$ and $Q = \{q1, q2, q3, q4\}$, cross edges exactly p1q1, p2q2, and one additional vertex z adjacent exactly to $p3$, $p4$, $q3$, $q4$. In the first case take $P = (a1, a2, x, a3)$, $Q = (b1, b2, b3, b4)$, $z = a4$. In the second case take $p1 = b2$, $p2 = b1$, $q1 = b3$, $q2 = b4$, take $p3$, $p4$ to be the first A-edge and $q3$, $q4$ the second, and take $z = x$. The preceding degree counts and edge descriptions verify that these are the complete edge lists.

For this common graph choose root $p3$ and perfect matching

$W = \{zp4, p1p2, q1q3, q2q4\}$

of C-p3. All four edges are present and cover its eight vertices. The graph $U = (C - W) - p3$ has the path p1p4p2 on the three remaining P-vertices, the cycle q1q2q3q4q1, edges zq3, zq4, and cross edges p1q1, p2q2.

Every independent set of U has at most three vertices. If it contains z, it avoids $q3$, $q4$ and has at most one of the adjacent $q1$, $q2$. Its P-part has at most two vertices; if it has two, they must be $p1$, $p2$, which forbid both $q1$, $q2$. Its total size is at most three. If it omits z and its P-part has two vertices, these are $p1$, $p2$; the Q-part then lies in the adjacent pair $q3$, $q4$ and has size at most one. If its P-part has size at most one, its Q-part has size at most two since that part is a four-cycle. Again the total is at most three. The set {p1,p2,z} attains three. Thus $\tau(U) = 8 - 3 = 5$, contradicting the required $\tau(U) = \mu = 4$.

All possible orders except five have been excluded, so $C = K_5$. For the equivalent formulation, a finite graph has a maximal matching. If C is not K_5, the implication just proved says that not every maximal matching is maximum. Some has integer size strictly below $\nu(C)$, hence at most nu(C)-1, as asserted. □

Remark. Far-neighborhood perfect matchings forbid claws under equimatchability. Minimal isolation then leaves at most four vertices, and explicit matchings exclude both remaining nonclique orders.

Depends on. Lemma 6.

Lemma 67 (Connectivity and an odd-clique exclusion in regular graphs). *All graphs are finite and simple, and vertex deletion is induced. A graph is 2-connected if it has at least three vertices, is connected, and remains connected after deleting any vertex. It is factor-critical if deleting any vertex leaves a perfect matching, and equimatchable if every inclusion-maximal matching has maximum cardinality. An independent set contains no adjacent pair of vertices, and is maximum when its cardinality is largest. A perfect matching covers every vertex; the empty graph has an empty perfect matching. For a vertex h, $N_G(h)$ denotes its open neighborhood. A graph is D-regular if every vertex has degree D.*

Let $D \geq 6$ be an even integer, and let G be a connected factor-critical equimatchable D-regular graph. Then G is 2-connected.

Furthermore, let I and I' be disjoint maximum independent sets of G, both of size $\alpha \geq 3$, and let u be any vertex outside I union I'. Suppose G-(I union I' union $\{u\}$) has a perfect matching, and every vertex h of this graph satisfies $|N_G(h)$ intersect $I'| \leq 1$.

Then there is no vertex set X disjoint from I union $\{u\}$ such that $|X|$ is odd, $|X| \geq 3$, $G[X]$ is complete, $|X$ intersect $I'| = 1$, and every neighbor of a vertex of X outside X belongs to I union $\{u\}$.

All graph orders, degrees, matching and set cardinalities are nonnegative integers, subject to the displayed positive and parity restrictions. No restriction is imposed on the cardinality of the perfectly matchable residual, which may be empty. The assertion applies to every choice of I, I', u satisfying the same hypotheses.

Proof. First we prove 2-connectivity. Simplicity and degree D give $|V(G)| \geq D + 1 \geq 3$. Suppose v were a cut vertex. Each component of G-v has even order, since factor-criticality gives a perfect matching of G-v. Fix distinct components C, C′, and choose a vertex w in C′ adjacent to v, using connectedness of G.

Take a perfect matching F of G-w. The vertex v must be matched into C′ minus{w}. Indeed, if it were matched into another component of G-v, that even component would lose just one vertex to the edge at v. An odd number would remain to be matched internally, since there is no other available exit from that component. This contradicts perfection of F. Thus every vertex in C′ union {v} other than w has its F-partner in that same set.

Let M be the subset of F consisting of the edges incident with $N_G(w)$. Every neighbor of w lies in C′ union {v}, so $V(M)$ is contained in C′ union {v}. The matching M lies in G-w and covers $N_G(w)$. It covers v because vw is an edge. Consequently C is a whole connected component of G-(V(M) union {w}). Apply the first assertion of Lemma 6, which requires neither 2-connectivity nor inclusion-minimality of M. It shows that C is an even complete graph or a balanced complete bipartite graph.

In either case C has a common internal degree, say a. Its only possible neighbor outside C is v. Every vertex of C has degree D in G, so each has exactly D-a neighbors outside C. At least one vertex has the neighbor v by connectedness, and thus $D - a = 1$ and every vertex of C neighbors v. If C is complete then $|V(C)| = a + 1 = D$. If C is balanced complete bipartite then $|V(C)| = 2a = 2(D - 1) \geq D$. These at least D vertices, together with w in C′, give v at least $D + 1$ neighbors. This contradicts regularity. No cut vertex exists, proving 2-connectivity.

Now fix I, I′, u under the further hypotheses.

Suppose such X exists. Let $x0$ be its unique vertex in I′, put $Y = V(G)\ minus(IunionI'unionXunion\{u\})$, and write $N_I(z) = N_G(z)$ intersect I for a vertex z. Since I is maximum, every vertex outside I has an I-neighbor. The same assertion holds for I′. The boundary assumption on X gives no edge from X to Y or to I′ minus{x0}. The assumed perfect matching of G-(I union I′ union {u}) therefore restricts to a perfect matching of G[Y]: the remaining part of that residual is X minus{x0}, with no edge to Y. Fix such a matching on Y, allowing it to be empty.

There is no three-edge matching consisting of uz, x1z1, x2z2, where $x1$, $x2$ are distinct vertices of X and z, $z1$, $z2$ are distinct vertices of I. If it existed, choose any $x3$ in X minus{x1,x2}, match the even clique X minus{x1,x2,x3} perfectly, and add the fixed perfect matching of Y. Together with the three specified edges this is a matching of G. Its remaining vertex sets are I minus{z, z1,z2}, of size alpha-3, and (I′ minus{x0}) union {x3}, of size α. Each is independent, the second because $x3$ has no neighbor in I′ minus{x0}. They comprise all remaining vertices. Every extension can match vertices only across these two sets, so it leaves at least three vertices uncovered. This contradicts factor-criticality and equimatchability: every matching extends in the finite graph to a maximal one, and each maximal matching must have size $(|V(G)| - 1)/2$. We call this the three-edge restriction.

Every vertex of X has an I-neighbor. Since X is an odd clique and G has even degree D, this gives $|X| \leq D-1$. If $|X| \leq D-3$, each vertex of X has at least three I-neighbors: its internal degree is at most D-4 and it has at most one possible other neighbor u. Choose uz with z in $N_I(u)$, which is nonempty. Two distinct vertices $x1$, $x2$ of X can then be assigned different I-neighbors avoiding z, choosing the first and then the second from sets of size at least three. This violates the three-edge restriction. Therefore $|X| = D-1$. Each vertex x of X now has precisely $2 - 1_(xuinE(G))$ neighbors in I, because its internal degree is D-2 and its only possible neighbors outside X lie in I union {u}.

We record a consequence of the one-neighbor bound for Y. If a proper subset I_0 of I has the property that either G-(X union I0) or G-(X union I0 union {u}) is a nonempty balanced complete bipartite graph, then Y is empty. Indeed, every y in Y has a neighbor in I' by maximality of I', and none of these neighbors is $x0$ because X has no edge to Y. Thus y neighbors a vertex of I' minus{x0}. That entire independent set remains in the complete bipartite graph. Its size is $\alpha - 1 \geq 2$, and a nonempty independent set in a complete bipartite graph lies wholly in one part. The vertex y must then be in the other part and adjacent to all alpha-1 of its vertices. This contradicts the assumed bound $|N_G(y)$ intersect $I'| \leq 1$. The vertices of Y and all of I' minus{x0} were retained in either of the displayed deletions, so the argument applies exactly as stated.

Suppose u has at most D-3 neighbors in X. There are two distinct vertices a, b in X not adjacent to u, and $|N_I(a)| = |N_I(b)| = 2$. The three-edge restriction implies $N_I(a) = N_I(b) =: I1$ and $N_I(u) \subseteq I_1$. To verify this small selection implication, if the two two-element sets have union of size at least three, then for any z in $N_I(u)$, after avoiding z one can choose distinct representatives from them: a failure could occur only if both remaining sets were the same singleton, which would make the original two sets equal. If their union has size two but z lies outside it, choose its two different elements. Either situation would produce the forbidden three-edge matching. Thus equality and containment hold. Moreover $N_I(x) \subseteq I_1$ for every x in X. Otherwise take an I-neighbor z' outside I_1 of such x, choose z in $N_I(u)$, and use either a or b different from x with the other element of I_1 as its partner. The three distinct partners again violate the restriction.

Write $I1 = \{z1, z2\}$. Choose c in X minus{a, b}. It has an I-neighbor in I_1; name that neighbor $z1$, exchanging the two names if needed. The edges bz2 and cz 1, together with a perfect matching of the even clique X minus{a, b, c}, form a matching whose endpoint set is exactly (X minus{a}) union I_1. This is precisely $N_G(a)$, since a is not adjacent to u and has I-neighborhood I_1. It is therefore an inclusion-minimal matching isolating a: every matching edge has endpoints in $N_G(a)$, and deleting any of its edges uncovers a neighbor of a. The earlier residual theorem Lemma 6 applies to this graph and matching. Its residual G-(X union I1) is empty, an even clique, or a balanced complete bipartite graph. It contains the independent set (I minus I1) union {u}, of size $\alpha - 1 \geq 2$, since $N_I(u) \subseteq I_1$. Hence it is nonempty and is not complete, so it is balanced complete bipartite.

The preceding consequence, with the proper set I_1 ($\alpha \geq 3$), now gives Y empty. The residual is partitioned into the two independent sets (I minus I1) union {u} and I' minus{x0}, each of size alpha-1. In a complete bipartite graph these are its two parts, so u has at least alpha-1 neighbors in I' minus{x0}, and also at least one neighbor in I. Therefore $D \geq \alpha$. On the other hand, any z in the nonempty set I minus I_1 has no neighbor in X union {u}; it has none in I, and Y is empty. Its neighbors all lie in I' minus{x0}, so $D \leq \alpha - 1$, a contradiction. This proves $|N_G(u)$ intersect $X| \geq D-2$.

Put $X0 = N_G(u)$ intersect X. Its size is at least $D - 2 \geq 4$, and each of its vertices has exactly one I-neighbor. At least two distinct vertices a, b of X_0 have the same I-neighbor z. Otherwise all their I-neighbors would be distinct; choosing any z_u in $N_I(u)$, at least two of the at least four distinct neighbors would differ from z_u, giving a forbidden three-edge matching using u and their two X0-vertices. Choose c in X_0 minus{a, b}. The edges bz and cu, together with a perfect matching of the even clique X minus{a, b, c}, form a matching on exactly (X minus{a}) union $\{u, z\} = N_G(a)$. It lies in G-a and covers every neighbor of a; each of its edges has endpoints in $N_G(a)$, so deleting any edge uncovers a neighbor and proves inclusion-minimality. By Lemma 6, its residual G-(X union {u, z}) is empty, complete even, or balanced complete bipartite. It contains the independent set I minus{z}, of size $\alpha - 1 \geq 2$, so it is balanced complete bipartite. The established consequence with $I0 = \{z\}$ gives Y empty.

This last residual is partitioned into I minus{z} and I' minus{x0}, both of size alpha-1. It is therefore $K_{\alpha-1,\alpha-1}$. Its vertices have degree at least alpha-1 in G, so $\alpha - 1 \leq D$. Equality $\alpha - 1 = D$ would saturate every degree of this nonempty residual within itself, making it a whole component disconnected from the nonempty removed set X union {u, z}. Since G is connected, equality is impossible, and $\alpha \leq D$.

The vertex u has at least D-2 neighbors in X and at least one in I. It therefore has at most one neighbor in I' minus{x0}, which has size $\alpha - 1 \geq 2$. Choose w in I' minus{x0} not adjacent to u. This vertex has no neighbor

in X by the boundary condition, none in I′ by independence, and none in Y since Y is empty. Thus every neighbor of w lies in I, giving $D \le \alpha$. Combined with the opposite bound, $\alpha = D$.

We finish with an explicit neighborhood count. Let S be the set of the unique I-neighbors of vertices in X_0. The three-edge restriction gives $|S| \le 2$: if S had at least three values, choose z_u in $N_I(u)$ and two distinct values of S different from z_u, with their distinct X0-vertices. If $|S| = 2$, that same restriction gives $N_I(u) \subseteq S$, since an I-neighbor of u outside S together with one representative of each value of S would again give the three forbidden edges. If $|S| = 1$, the inequality $|X0| \ge D - 2$ implies $|N_I(u)| \le 2$ by degree D. Consequently in all cases $|S$ union $N_I(u)| \le 3$. At most one vertex of X lies outside X_0; when it exists, its I-neighborhood has size exactly two by the earlier degree formula. Hence at most five vertices of I neighbor X union {u}. Since $|I| = \alpha = D \ge 6$, choose z′ in I with no such neighbor. Its remaining possible neighbors are only I′ minus{x0}, because I is independent and Y is empty. This gives $D \le \alpha - 1$, contradicting $\alpha = D$. The contradiction proves the statement. □

Remark. Three independent cross edges would leave too many vertices unmatched. Excluding those edges forces the clique to meet almost all neighbors of the exceptional vertex. Two explicit isolating matchings then expose balanced complete bipartite residuals, and degree counts give the contradiction.

Depends on. Lemma 6.

Lemma 68 (Top-run contraction under matching-additive deletion). *Use the finite simple graph certificate quantities $c_0(G)$, $\nu(G)$, $ex(G) = c_0(G) - \nu(G)$, and $tr(G)$ of Lemma 23. Thus $c_0(G)$ is the least nonnegative integer t such that every pair of nonnegative integers x, y with $x + y = t$ admits a partition of $E(G)$ into two parts of matching numbers at most x, y. For each nonnegative integer t, write $Y_t(G)$ for the set of integers x in [0,t] admitting such a partition with bounds x, t-x. If $c_0(G) > 0$, $tr(G)$ is the longest consecutive integer run missing from $Y_(c_0(G) - 1)(G)$; if $c_0(G) = 0$, it is zero.*

Let H be any finite simple graph and X any subset of $V(H)$. Write H-X for the induced graph after deleting X, and let Q_1,..., Q_p be all its connected components, including isolated vertices, where p is a nonnegative integer. Put $b = |X|$ and $N = \sum_{i=1}^{p} \nu(Q_i)$, with an empty sum equal to zero. Suppose $\nu(H) = b + N$. Put $m = \max(\{0\} union \{ex(Q_i) : 1 \le i \le p\})$. Then $ex(H) \le m$.

If in addition $ex(H) = m \ge 1$, the set of components with excess m is nonempty and $tr(H) + b \le max\{tr(Q_i) : 1 \le i \le p, ex(Q_i) = m\}$. In particular, for X empty, excess is at most the largest component excess; if this excess equals the positive largest component excess, the top run is at most the largest top run among components attaining it. No connectedness, regularity, positive-gap, or hereditary hypothesis is imposed. All displayed counts, matching numbers, excesses, totals, and run lengths are nonnegative integers.

Proof. The zero-capacity degree parts in the certificate definition are empty, so its row-zero feasibility is exactly the two-part edge partition in the statement. Lemma 23 gives nonnegative integer excess and universal feasibility at every total at least c_0.

We first record a coloring extension used at two totals. Suppose the residual graph H-X has a partition with matching bounds z, t-z. For any integer a with $0 \le a \le b$ choose a subset X_R of X of size a. Retain all residual colors. Color every edge with an endpoint in X_R red and every remaining edge incident with X blue. These rules assign every edge outside H-X exactly once. A matching has at most a red edges outside H-X, because they meet distinct vertices of X_R. Every blue edge outside H-X meets X minus X_R, so a matching has at most b-a such edges. The extended partition has matching bounds $z + a$, $t - z + b - a$. Consequently $Y_t(H - X) + [0, b] \subseteq Y_{t+b}(H)$, where the interval and the sumset consist of integers.

Put $w_i = \nu(Q_i)$, $W_0 = 0$, and $W_j = \sum_{i=1}^{j} w_i$ for $1 \le j \le p$. For any nonnegative integer l and any integer $0 \le z < N$, choose the first j with $W_j > z$ and put $r = z - W_{j-1}$. Then $w_j > 0$ and $0 \le r < w_j$. If r belongs to $Y_{w_j+l}(Q_j)$, color every component before Q_j wholly red, every component after it wholly blue, and Q_j according to this feasible split. Matching numbers add across disjoint components: restricting any matching gives the upper bound by the sum, and taking maximum matchings in the separate components attains that sum. Thus the constructed residual split has red bound $W_{j-1} + r = z$ and blue bound $N - W_j + w_j + l - r = N + l - z$. For $N \le z \le N + l$, coloring the whole residual graph red also gives a split with bounds z, $N + l - z$.

Take $l = m$. For every component, $c_0(Q_i) = w_i + ex(Q_i) \le w_i + m$. Hence every local budget needed in the preceding construction is feasible. It follows that $Y_{N+m}(H - X)$ contains every integer in $[0, N + m]$. This remains valid when $N = 0$ or $p = 0$, since the whole-red construction covers that interval. The extension then gives every integer in $[0, N + m + b]$ on H: for a given x in this interval, choose $a = \min(b, x)$ and $z = x - a$. These choices satisfy $0 \le a \le b$ and $0 \le z \le N + m$. Therefore $c_0(H) \le N + m + b = \nu(H) + m$, proving $ex(H) \le m$.

Now suppose $ex(H) = m \ge 1$. Since m is positive and is a maximum over a finite family together with zero, at least one Q_i has excess m. Its c_0 is at least one, so its top run is defined by a nonempty missing set. Write $L = max\{tr(Q_i) : ex(Q_i) = m\}$ and $l = m - 1 \ge 0$. The matching-additivity and excess equalities give $c_0(H) - 1 = N + b + l$.

Construct a set A of sufficient residual budgets at total $N + l$. Include every integer in $[N, N + l]$. For $0 \le z < N$ use the first index j with $W_j > z$ and put $r = z - W_{j-1}$, as before; include z in A exactly when r belongs to $Y_{w_j+l}(Q_j)$. The component coloring already constructed proves A $\subseteq Y_{N+l}(H - X)$.

Every distinct prefix boundary W_j lies in A. At a boundary smaller than N, the next component of positive matching number has local first budget zero, which is feasible by coloring that component wholly blue, since $w_i + l \ge w_i$. The boundary N belongs to the included terminal interval. This also proves 0 and $N + l$ belong to A, including the case $N = 0$. If $ex(Q_j) < m$, integrality gives $c_0(Q_j) \le w_j + m - 1 = w_j + l$, so its entire local interval has feasible budgets. If $ex(Q_j) = m$, the local total $w_j + l$ equals $c_0(Q_j) - 1$, and every consecutive missing local run has length at most $tr(Q_j) \le L$. A missing run of A cannot cross a prefix boundary, all of which belong to A; components with $w_i = 0$ create no interval. Translating a missing run inside one positive-length component interval to its local first budgets therefore shows that every missing run of A has length at most L.

The extension gives $A + [0, b] \subseteq Y_{N+l+b}(H)$. Let z,..., $z + r - 1$ be any nonempty consecutive run missing from the latter set. Because A contains 0 and $N + l$, the sumset contains [0,b] and $[N + l, N + l + b]$. Thus $z > b$ and $z + r - 1 < N + l$. Every integer u in $[z - b, z + r - 1]$ must be missing from A: if u belonged to A, the integer intervals $[u, u + b]$ and $[z, z + r - 1]$ would intersect, giving an element of the missing run in $A + [0, b]$, a contradiction. This produces a missing run of A of length $r + b$, so $r + b \le L$. Lemma 23 guarantees a nonempty missing run at total $c_0(H) - 1 = N + l + b$ because $c_0(H) > 0$. Taking a longest such run gives $tr(H) + b \le L$.

When X is empty, $b = 0$ and $\nu(H) = \sum_i \nu(Q_i)$ by disjoint-component additivity. Both stated component consequences follow from the inequalities just proved. □

Remark. Order the components and split only the component straddling a chosen matching budget. Restoring b vertices expands every attainable residual budget by all integer shifts from zero through b, shortening each missing run by at least b.

Depends on. Lemma 23.

Lemma 69 (Triangle exclusion by a complete quartic frame certificate). *All graphs are finite and simple. Write $\alpha(G)$, $\nu(G)$, $\Delta(G)$ for the independence number, matching number, and maximum degree of G. Vertex deletion takes an induced subgraph. A graph is well-covered if every maximal independent set is maximum. For nonnegative integers A, B, define $c_{A,B}(G)$ as the least nonnegative integer T such that every nonnegative integer split $x + y = T$ permits a partition of $E(G)$ into four parts with maximum degrees at most A, B and matching numbers at most x, y, respectively. Set $c_2(G) = \max_{A+B=2} c_{A,B}(G)$, with nonnegative integer capacities. Paths P_j and cycles C_j have j vertices.*

Let F be four-regular and well-covered, let $a = \alpha(F)$, and let m, b be integers with $m \ge 1$ and $0 \le b < m$. Put $q = m - b$. A frame consists of a maximum independent set S of F and an independent m-element set U of degree-three vertices in F-S. Put $W = S$ union U and $R = F - W$. Assume that every frame has the following properties:

(i) $\alpha(R) = a$, and its whole components are exactly q odd cycles from C_3, C_5, C_7, exactly b even paths from P_2, P_4, and any number of P_3, C_4, and isolates.

(ii) Let P, Z be the endpoints and isolates of the matching $F[W]$. Every p in P has at most one neighbor of degree two in R, and no p is adjacent to both the center and a leaf of the same whole P_3.

(iii) For every $T \subseteq W$ with $|T| \le q+2$, there are vertices v_t in R, one for each t in T, such that tv_t is an edge, the whole R-components C_t containing v_t have odd order and are pairwise distinct, and $\nu(C_t - v_t) = \nu(C_t)$.

Assume $c_2(F) = a + 2m - b$. Then no frame terminal contains C_3. Moreover, in every frame no degree-two vertex on a whole P_4 or C_4 has its two W-neighbors equal to the endpoints of one frame edge.

The strict certificate used in the proof has a fixed edge partition into a bipartite graph of maximum degree at most two and exactly $a + 2m - b - 1$ labelled graphs of matching number at most one, allowing empty labelled graphs. It therefore handles every nonnegative integer capacity split $A + B = 2$ and every nonnegative integer budget split $x + y = a + 2m - b - 1$ on the entire original graph F.

Proof. Fix any frame. The hypotheses imply those of Lemma 25. Its conclusions apply in every original, reflected, or exchanged frame below. Thus F[W] consists of m disjoint edges and a-m isolates. Every canonical selection D takes a maximum independent set on each odd cycle, each P_3 center, an opposite C_4 pair, one internal P_4 vertex, and nothing on P_2 or isolates. The induced graph $R_D = F[W \text{ union } D]$ is an actual frame terminal, with complementary frame union $V(R)$ minus D. Every D-vertex has degree two in R_D. Each odd-cycle or even-path component of R_D contains exactly one old frame edge, and these components exhaust the m old frame edges. Every orientation of the frame edges is a frame, and if p in P contacts a degree-two vertex x of R, exchanging p for x as a root gives the actual terminal $R - x + p$.

We call a degree-two terminal vertex high and all other terminal vertices low. A high vertex is full if its two W-neighbors form a frame edge. Every p in P has exactly three terminal neighbors, because its only W-neighbor is its mate.

1. Odd-component assignments and one reserved P_3.

An exposed vertex of an odd component C means a vertex v satisfying $\nu(C - v) = \nu(C)$. On an odd cycle every vertex is exposed: deleting it leaves an even path with a perfect matching. In a P_3 precisely the two leaves are exposed; in an isolate its sole vertex is exposed. These statements follow by taking alternate path or cycle edges and using the bound of one matching edge per two vertices. These are all odd component types in (i). Assumption (iii), applied to a singleton, consequently gives an exposed odd-component neighbor to every frame vertex.

We first prove that if all exposed odd-component neighbors of p in P lie in one P_3 with leaf pair L and center h, then

$N_R(p) = L$ union {g}

for the center g of a different P_3.

Let K select h on this reserved P_3, both leaves on every other P_3, all isolates, and arbitrary maximum independent sets on every even component. It selects nothing on odd cycles. Suppose these choices can be made so that K has a neighbor of p. Let B be the vertices of W with no K-neighbor; then p is not in B. Every exposed odd-component neighbor of a B-vertex lies either on an odd cycle or among the reserved pair L, because K contains all other odd-path exposed vertices.

If $|B| \ge q+1$, choose $q+1$ members of B and adjoin p. By (iii) this $(q+2) - set$ must be assigned to distinct odd components. Its available components are only the q odd cycles and the reserved P_3, a total of $q+1$, contradiction. Hence $|B| \le q$. Apply (iii) to B union {p}. The vertex p must be assigned to the reserved P_3, so all B-vertices are assigned to distinct odd cycles. On each assigned cycle choose a maximum independent set containing its assigned vertex, and on other odd cycles choose any maximum independent set. Such choices exist by alternating around the cycle starting at the specified vertex.

Unite these cycle choices with K. The resulting independent set dominates R component by component, and dominates W by K and the assigned cycle neighbors of B. It is therefore maximal independent in F. Its cardinality is $\alpha(R) - 1 = a - 1$, since precisely the reserved P_3 uses one vertex instead of its maximum two. This contradicts well-coveredness. Thus no such K can dominate p.

In particular p cannot neighbor h or any even-component vertex: h always belongs to K, and any specified even-component vertex belongs to some maximum choice on its P_2, P_4, or C_4. The assumed restriction on its exposed neighbors also excludes isolates, odd cycles, and other P_3 leaves. Its remaining neighbors are in L or are centers of other P_3 components. It has three distinct terminal neighbors and at most one high neighbor by (ii), so it must use both members of L and exactly one other P_3 center. This proves the assertion.

2. Low neighbors at a binary child and odd-cycle contacts.

Suppose p in P has a high neighbor x on a whole P_4 or C_4, which we call its binary component. Its two other terminal neighbors are low by (ii). They cannot be both leaves of one P_3, since its only exposed odd neighbors would then lie in that P_3 and the preceding assertion forbids the binary third neighbor.

If one low neighbor l were a P_3 leaf, the other low neighbor s would be outside that P_3. Exchange p for x as a root. In R-x the whole P_3 through l is unchanged. The component of s is a path of order one, two, three, or four, with s an endpoint. If s lies on the binary component, that component was P_4 and deleting its internal x leaves s on an isolate or an edge; otherwise its component is unchanged. Adding p joins these two paths through l and s, creating a whole path of order between five and eight. No such path is in (i). Thus p has no $P_3 - leaf$ neighbor.

The singleton case of (iii) supplies an exposed odd neighbor of p. This cannot be an odd-cycle vertex, since x is already its unique high neighbor, and cannot be a P_3 leaf by the preceding paragraph. It must be an isolate i. The remaining low neighbor is another isolate or an even-path endpoint. If it were an endpoint of a P_4 different from x's component, the exchange $R - x + p$ would join that unchanged P_4 to i through p and create a whole P_6. Thus each binary-child parent has an isolate neighbor, no $P_3 - leaf$ neighbor, and its other low neighbor is an isolate, a P_2 endpoint, or an endpoint of the same P_4 containing x.

We also need at most one P-to-cycle edge on each odd cycle. Suppose p contacts x on an odd cycle O. Its other two terminal neighbors are low and hence outside O. In $R - x + p$ the old O becomes an even path, so one odd cycle is lost. Adding p to its two low neighbors can create a cycle only when those neighbors are the two endpoints of the same path component; restoring the fixed number q of odd cycles requires this path to have even order. Thus $N_R(p)$ consists of x and both endpoints of one whole P_2 or P_4. In particular its only exposed odd-component neighbor is x on O. If two different members of P contacted O, assumption (iii) on that two-element set would force both assignments into O, contrary to distinct components. Each individual p contacts at most one O-vertex by (ii). Hence every whole odd cycle has at most one edge to P, in every frame.

3. A full vertex with disjoint low pairs has a strict certificate.

Let x be any high vertex with full parent edge pp′. Suppose its two complete low pairs

$L_p = N_R(p)$ minus {x}, $L_{p'} = N_R(p')$ minus {x}

are disjoint, each contains a terminal isolate, and every other member is an endpoint of a whole even path. We construct a forbidden certificate directly on F.

First order the a-m vertices of Z and assign every edge incident with Z to its earliest such endpoint. This gives a-m labelled stars, possibly empty, and removes every edge with an endpoint in Z. Next order P minus {p, p′} and assign each remaining edge incident with this set to its earliest endpoint in the set. This gives 2m-2 labelled stars. Among the remaining edges take the triangle pp′x as one labelled piece. It has matching number one. Take one edge from each of the q whole odd cycles of R as q further labelled pieces. These edge sets are disjoint: the triangle and the second collection of stars have frame endpoints, while the q selected edges lie in R. The triangle has no endpoint in Z or in P minus {p, p′}.

The number of labelled pieces is

$(a - m) + (2m - 2) + 1 + q = a + 2m - b - 1.$

Each has matching number at most one. All remaining edges form precisely R with one edge removed from each odd cycle, together with the low edges from p and p′. The R-part has maximum degree at most two and every cycle in it is even. A terminal isolate receives at most one new low edge because the low pairs are disjoint; an even-path endpoint also receives at most one. Thus the entire remaining graph has maximum degree at most two. Each of p, p′ is adjacent to an original isolate of degree one in this remaining graph, so neither lies on a cycle. Every remaining cycle is consequently an even cycle already in the R-part. The remaining graph is a union of paths and even cycles, hence is bipartite.

For capacities (2,0) or (0,2), put this remaining graph in the part of capacity two. For capacities (1,1), alternate its path and even-cycle edges between the two degree-bounded parts. For any nonnegative integer budgets $x_0 + y_0 = a + 2m - b - 1$, assign any x_0 labelled pieces to the first matching-bounded part and the remaining y_0 to the second. A matching in either union uses at most one edge per piece, so its size is at most its budget. All edges of F are assigned exactly once. This proves $c_2(F) \le a + 2m - b - 1$, contradicting the hypothesis. The displayed full-vertex configuration is therefore impossible.

4. Full binary vertices are impossible.

Suppose a high x on P_4 or C_4 has full parents p, p'. Step 2 gives the stated isolate-containing low menu for each parent. It remains to prove the low pairs disjoint.

Suppose first they share an even-path endpoint e. If e still has degree one in R-x, exchange p for x. The restored p has degree two, and e now also has degree two because it acquired the edge ep. The remaining frame endpoint p', now mated to x, neighbors both p and e, contradicting (ii). Otherwise e becomes isolated in R-x, so x was its neighbor on its original P_4. The other low neighbor of p is an original isolate by step 2. The exchange then makes p the center of a whole P_3 with leaf e. The endpoint p' neighbors this center and its leaf e, again violating (ii).

Now suppose the shared low is an original isolate i. Write the other low neighbors as l_p and $l_{p'}$. Each has degree zero or one in R-x, by step 2. If l_p has degree zero there, exchanging p for x creates a whole P_3 centered at p with leaf i, while p' neighbors both p and i, forbidden by (ii). Interchanging the roles of p, p' gives the same contradiction if $l_{p'}$ has degree zero. Therefore both have degree one in R-x.

Exchange p for x. The restored p is joined to i and l_p. Its whole component is a path of order at least four; by (i) it must be P_4. The endpoint p', still mated to x, has terminal neighbors p, i, $l_{p'}$. In this new terminal i has degree one, and $l_{p'}$ retains at least one edge from R-x. Thus p' has no terminal-isolate neighbor. It is a parent of the binary high vertex p, contrary to the universal isolate conclusion in step 2. If $l_{p'}$ has acquired degree two, (ii) already gives a contradiction. All shared-low possibilities have been excluded.

The low pairs are consequently disjoint and satisfy step 3. Its strict certificate contradicts $c_2(F) = a + 2m - b$. Hence no full binary vertex occurs in any frame.

5. A triangle forces private isolate parents.

Suppose a whole triangle $T = \{t, u, v\}$ occurs in a frame terminal. We first exclude repeated W-parents. Suppose w in W neighbors two triangle vertices, say t and u. Both are high, so (ii) gives w in Z. If w had a high neighbor y outside T, choose a canonical D containing t and y; these choices are independent on different components. In R_D, both t and w have degree two. The unselected triangle edge uv is a frame edge of the reflected frame. Its endpoint u neighbors both t and w, contrary to (ii) in that frame. Thus w has no high neighbor outside T.

Choose a canonical D containing t. Its only neighbor at w is t, since D selects no other T-vertex and only high vertices outside T. Hence w is a leaf in R_D. The vertex t has degree two, and its two reflected frame parents are u, v, which form the reflected frame edge uv. The component beginning w-t has at least three vertices. With a leaf and the terminal list (i), it is P_3 or P_4. The latter would make t a full binary vertex in the reflected frame, excluded in step 4. It is therefore a whole P_3 centered at t. The reflected frame endpoint u neighbors both t and its leaf w, violating (ii). No repeated parent exists. Thus the triangle's six W-parents are distinct.

For any canonical D selecting t, its reflected parents remain u, v, so t is full and has degree two in R_D. If its component were an odd cycle, that component would receive both edges ut and vt from reflected frame endpoints, contradicting the at-most-one cycle contact proved in step 2. Its component cannot be P_4 or C_4 by step 4. The component list and its degree leave only P_3, centered at t.

Both original W-parents of t must belong to Z: an original frame endpoint among them would have its frame-mate edge as well as its edge to t in R_D, whereas a leaf of that whole P_3 has degree one. Since t was arbitrary, all six triangle parents belong to Z.

Each such parent w has its T-child as its only high neighbor in the original terminal. It has no other T-child by distinctness. If it had a high neighbor y on another component, a canonical choice containing its T-child and y would give it degree at least two, contradicting its degree one as a leaf of the just-proved reflected P_3. Such a canonical choice exists for every high y: it is the P_3 center, one appropriate P_4 internal vertex, an opposite C_4 pair, or a maximum set on an odd cycle containing y.

6. Reflection yields the final strict certificate.

Take a canonical D selecting t on T, and consider its reflected frame. Its terminal R_D has a whole P_3 centered at t, full parents u, v, and frame edge uv. Let L_u, L_v be the original two-element W-parent sets of u, v. These pairs are disjoint, consist of original Z-vertices, and avoid the pair belonging to t, by step 5.

Every vertex of L_u or L_v is isolated in R_D. It has no old W-neighbor because it lies in Z. Its unique original high neighbor is respectively u or v, neither of which belongs to D. Since D contains only high vertices, it has no D-neighbor either. Moreover the complete terminal neighborhoods of the reflected frame endpoints are

$N_{R_D}(u) = \{t\}$ union L_u, $N_{R_D}(v) = \{t\}$ union L_v:

the fourth neighbor of each is its mate in the reflected frame.

Thus in this actual reflected frame the full high vertex t has two disjoint complete low pairs, each consisting of two terminal isolates. Step 3 applies to this same F and this frame, producing exactly the stated edge partition at total $a + 2m - b - 1$. The contradiction excludes the hypothetical whole triangle. Every reflection and root exchange used the original graph, and the final certificate includes all its edges. This proves all assertions. □

Remark. A simultaneous port assignment rules out a reserved P_3 center that would produce a smaller maximal independent set. This forces enough private isolate neighbors for a full parent pair to save one paid unit. Canonical reflection turns every hypothetical terminal triangle into exactly such a full parent pair, while the explicit certificate covers the original graph.

Depends on. Lemma 25.

Lemma 70 (A tight matching block containing all odd row paths). *Let F be a finite simple well-covered graph: every maximal independent set has maximum cardinality $\alpha(F)$, where independent means containing no adjacent pair and maximal means inclusion-maximal. Let R be an induced subgraph with maximum degree $\Delta(R) \leq 2$. A matching is a set of edges with pairwise disjoint endpoints, $\nu(G)$ is the largest matching cardinality of G, and a perfect matching covers every vertex. The empty graph has its empty perfect matching and independence number zero. Paths and cycles are counted by vertices; an isolated vertex is a path of order one.*

Put $U = V(F)$ minus $V(R)$, $u = |U|$, and $m = \nu(R)$. Let ℓ be the number of odd-order R-components with at least three vertices. Suppose x, y are nonnegative integers with $x + y = u + m - 1$ for which $E(F)$ has no partition into two edge sets whose spanning graphs have matching numbers at most x and y. Put $j = u - x$ and assume $j \geq \ell + 1$.

Let I_0 be the isolated vertices of R and let $B0 = \{vinU : N_F(v)intersectI0isempty\}$, where $N_F(v)$ is the open neighborhood. Let Ocal be the family of odd-order R-components with at least three vertices. For C in Ocal put `P(C)={z` *in V(C):C-z has a perfect matching}. Form the bipartite graph Γ from B_0 to Ocal, with an edge bC exactly when an F-edge joins b to a vertex of $P(C)$. Then $\alpha(F) = \alpha(R)$, $|B0| \leq \ell$, and Γ has a matching saturating B_0. Every such matching covers every nontrivial odd-order path component of R.*

Fix any such matching M. Direct its edges from Ocal to B_0 and all other Gamma-edges from B_0 to Ocal. Start directed reachability at all nontrivial odd-order path components, and write Hcal and B for the reached subsets of Ocal and B_0. Then Hcal contains all those paths, every component of Hcal is covered by M, M pairs B bijectively with Hcal, and

$N_\Gamma(B) = Hcal$, $|B| = |Hcal| \leq \ell$,

where $N_\Gamma(B)$ is the component-side neighborhood. Let Z be the vertex union of the even-order R-components and put

$Y = Z$ union $\bigcup_{C \in Hcal} V(C)$, $W = Y$ union B.

Every F-edge from B to $V(R)$ ends in Y. There exists an independent I contained in $V(R)$ minus Y, maximum in every discarded row component, such that

$V(F)$ minus $N_F[I] = W$,

where $N_F[I]$ is I together with all its F-neighbors. The graph $F[W]$ is well-covered and has a perfect matching formed from the M-edges between B and Hcal, lifted to arbitrary actual exposed ports, together with perfect matchings after those ports are removed and perfect matchings of the even components.

Let h count all nontrivial odd-order path components of R, let p_H count the odd-cycle components in Hcal, and put $t = \nu(F[Y])$. Then

$\nu(F[W]) = t + |Hcal|$, $\alpha(F[W]) = t + h$,

$\nu(F[W]) - \alpha(F[W]) = p_H$.

The same I extends every maximal independent set of $F[W]$ to a maximum independent set of F. For every $T \subseteq U$ minus B with $|T| \leq j - |B|$, there is one assignment of all vertices of B union T to exposed ports in distinct odd R-components that sends B into Hcal and T outside Hcal. It is simultaneously compatible with maximum internal matchings in every R-component and gives a maximum matching of $F[V(R)$ union B union $T]$ of size $m + |B| + |T|$. This new assignment need not preserve the previously fixed matching M. When $h = 0$, the reachability sets B and Hcal are empty. All orders,

degrees, budgets, component counts, matching sizes, and independent-set sizes are nonnegative integers; j is a positive integer. Empty component families, chosen sets, and retained graphs are allowed.

Proof. We use only the prescribed component-port theorem Lemma 17. We first check its hypotheses and the component calculations needed here.

A connected finite simple graph of maximum degree at most two is an isolated vertex, a path, or a cycle. Indeed, take a longest simple path in the component. An internal path vertex already has its two possible neighbors. An endpoint cannot have a neighbor outside the path by maximal length; its only possible extra neighbor on the path is the other endpoint, since internal vertices already have degree two. Connectedness leaves no vertex outside the path. The possible edge between its endpoints makes a cycle.

Consecutive disjoint edges give a matching of size $\lfloor n/2 \rfloor$ in each path or cycle of order n. No larger matching exists because matching edges use two distinct vertices. Every even component therefore has a perfect matching. On an odd path, the vertices whose deletion leaves a perfect matching are exactly the odd positions in a consecutive labeling: deleting position i leaves paths of orders i-1 and n-i, both even exactly when i is odd. These odd positions form a maximum independent set. On an odd cycle every vertex is a port, since its deletion leaves an even path; each such vertex belongs to a maximum independent set obtained by taking alternate vertices until $\lfloor n/2 \rfloor$ have been selected. The upper bound for cycle independence follows by pairing every selected vertex with its cyclic successor, which is unselected and distinct. On paths, disjoint adjacent pairs and the possible final vertex give the corresponding upper bound. Thus $\alpha = \nu$ in every even component and odd cycle, and $\alpha = \nu + 1$ in every odd path.

Apply Lemma 17 with the protected components precisely the isolates of R and with their unique maximum independent sets. Its parameter q is ℓ and its j is the present j. All its domain, missing-allocation, component-matching, and exposed-vertex conditions have just been checked. Since F is well-covered and $j \geq \ell + 1$, it gives $\alpha(F) = \alpha(R)$, $|B0| \leq \ell$, and an assignment of B_0 to exposed ports in distinct members of Ocal. This is a Gamma-matching saturating B_0. The same cited result also gives, for every $T_0 \subseteq U$ with $|T0| \leq j$, a maximum matching of F[V(R) union T0] of size $m + |T0|$ whose crossing edges assign T_0 to exposed ports in distinct odd R-components and whose internal edges are maximum in each component.

We next prove mandatory coverage. Suppose some Gamma-matching saturating B_0 leaves a nontrivial odd path C unmatched. In every other nontrivial odd component choose a maximum independent set containing its assigned port when it is matched; in unmatched components choose any maximum independent set. Add I_0 and choose maximum independent sets in all even components. Let L be their union, omitting C entirely. It is independent because R is induced and its components are disjoint. Every exterior vertex in B_0 has its assigned neighbor in L, and every exterior vertex outside B_0 has an I0-neighbor. All row vertices outside C are dominated by L by component maximality, while no vertex of C has an L-neighbor. Hence $F - N_F[L] = C$ exactly.

For each maximal independent set K of C, L union K is maximal independent in F. But if C has order $2a + 1$ with integer $a \geq 1$, its odd-indexed and even-indexed vertices are maximal independent sets of sizes $a + 1$ and a. Both dominate the path because every vertex of one parity has a neighbor of the other parity. Their two extensions contradict well-coveredness of F. Thus every saturating Gamma-matching covers every nontrivial odd path.

Fix M and the directed reachability in the statement. If an unmatched component were reached, a shortest directed path from a starting odd path to an unmatched component would be simple and alternating. Its first edge belongs to M and its last edge does not. Switching along that path leaves every B0-vertex covered, covers its final component, and uncovers its starting odd path. This contradicts mandatory coverage. Therefore every reached component is M-covered.

Every reached component has its M-partner in B, by following its directed matched edge. Every reached B0-vertex is reached along its unique matched edge from a component in Hcal. This proves the bijection between B and Hcal. Each non matched neighbor of B is reachable along its directed edge, while each matched neighbor is already in Hcal. Conversely every component in Hcal has its partner in B. Hence $N_\Gamma(B) = Hcal$ and $|B| = |Hcal| \leq |B0| \leq \ell$. Empty starting sets give empty reachability sets.

Every odd component outside Hcal other than an isolate is a cycle, since all nontrivial odd paths are reached. Every vertex of an odd cycle is a port, as checked above. Consequently $N_\Gamma(B) = Hcal$ says that B has no neighbor anywhere in any of those discarded cycles. The vertices of B avoid I_0 by definition. All even components are in Y, so every F-edge from B to R ends in Y.

In each discarded odd cycle choose a maximum independent set containing the port assigned by M to its matched vertex, if one exists, and otherwise choose any maximum independent set. Add all isolates I_0 and call the union I. Matching bijectivity implies that B_0 minus B is assigned outside Hcal. Hence every exterior vertex outside B is dominated by I: vertices outside B_0 see I_0, and vertices in B_0 minus B see their selected ports. Every discarded row component is dominated by its chosen maximum independent set. The set I is anticomplete to Y because these are different R-components, and anticomplete to B by the preceding full-neighborhood argument. It follows that $V(F)$ minus $N_F[I] = W$ exactly.

Any maximal independent K of F[W] combines with I to give a maximal independent set of F, using the exact neighborhood identity. Thus F[W] is well-covered with $\alpha(F[W]) = \alpha(F) - |I|$, and the same I gives all claimed maximum extensions. The component selections make $|I| = \alpha(R) - \alpha(F[Y])$. Since $\alpha(F) = \alpha(R)$, we obtain $\alpha(F[W]) = \alpha(F[Y])$. The graph F[Y] is precisely the disjoint union of its retained R-components. The component calculations above give $\alpha(F[Y]) = \nu(F[Y]) + h = t + h$.

Lift each M-edge between B and Hcal to any actual adjacent exposed port. In the corresponding odd component, choose a perfect matching after removal of that port. In each even component of Y choose a perfect matching. Distinct incidence edges use distinct B-vertices and distinct components, and all internal choices avoid their used ports. Their union is a perfect matching of F[W]. Its internal contribution is t and its crossing contribution is $|B| = |Hcal|$, giving $\nu(F[W]) = t + |Hcal|$ by the perfect-matching order bound. Hcal consists of its h odd paths and p_H odd cycles, so $|Hcal| = h + p_H$, and subtraction gives the displayed difference p_H.

Finally fix T as in the statement. Its disjoint union with B has size at most j. Apply the earlier consequence of Lemma 17 to $T0 = B$ union T. It supplies one maximum matching with an assignment of that whole set to distinct odd R-components. A B-vertex has no isolate neighbor, and all its other port neighbors lie in Hcal. Thus the $|B|$ vertices of B occupy all $|Hcal|$ components of Hcal, forcing every T-vertex into a component outside Hcal. The same matching is internally maximum in every row component and has size $m + |B| + |T|$. It is newly chosen for the union, so no preservation of the previously fixed M is required or inferred. This proves every assertion, including the empty cases. □

Remark. Well-coveredness makes every odd path essential to an exterior-saturating assignment. Its alternating closure gives a tight set whose full row boundary is retained; exposed ports produce one perfect matching and a common independent exterior.

Depends on. Lemma 17.

Lemma 71 (Exclusion of pentagons and heptagons under the quartic frame hypotheses). *All graphs are finite and simple. For a graph G, write $\alpha(G)$, $\nu(G)$, and $\Delta(G)$ for its independence number, matching number, and maximum degree. For nonnegative integers A, B, let $c_{A,B}(G)$ be the least nonnegative integer T such that every nonnegative integer split $x + y = T$ admits a partition of $E(G)$ into four parts with maximum degrees at most A, B and matching numbers at most x, y, respectively. Define $c_2(G) = \max_{A+B=2} c_{A,B}(G)$, with nonnegative integer capacities. The symbols P_j, C_j denote paths and cycles on j vertices.*

Let F be four-regular, put $a = \alpha(F)$, and let m, b be integers with $a \geq 4$, $m \geq 1$, and $0 \leq b < m$. Set $q = m - b$. Define a frame to be a maximum independent set S of F together with an independent m-element set U of vertices of degree three in F-S. Write $W = S$ union U and $R = F - W$. Assume the following properties for every frame:

(1) $\alpha(R) = a$. The connected components of R are exactly q odd cycles, each C_3, C_5, or C_7; exactly b even paths, each P_2 or P_4; and any number of P_3, C_4, and isolated vertices.

(2) Writing P for the endpoints and Z for the isolated vertices of the induced matching $F[W]$, each vertex of P has at most one neighbor of degree two in R. No vertex of P is adjacent to both the center and a leaf of the same P_3 component of R.

(3) Each odd-cycle component of R has at most one edge to P.

Assume also $c_2(F) = a + 2m - b$. Then no frame terminal has a component C_5 or C_7.

More precisely, the contradiction for either hypothetical component $C = C_{2s+1}$, with s in {2,3}, uses the following single construction in the original graph. A canonical set D selects a maximum independent set on every odd cycle, the center of every P_3, either opposite pair on every C_4, either internal vertex on every P_4, and nothing on P_2 components

or isolates. Let B_D consist of all original edges between D and Z. Its q path components with both endpoints in D are denoted T_1,..., T_q, including isolated D-vertices.

For every canonical D, none of these paths meets C. Choose any vertex d_i in D intersect $V(T_i)$, independently for each i, and put $A_0 = \{d_1, \dots, d_q\}$, $Y = (DminusA_0)$ union $V(C)$, $K = F[W$ union $Y]$, and $N = |V(F)$ minus $V(K)|$. There are edge-disjoint matchings M_0, M_1, maximum in K and $K - M_0$ respectively, whose union is exactly $E(F[W])$ together with all W-to-Y edges. Their sizes are $m + |Y|$ and $|Y|$, and $K - M_0 - M_1$ has exactly the edges of C. Moreover,

$N = a + 2m - b - s - 1 \geq s - 1.$

For this same pair of matchings, every nonnegative integer capacity split $A + B = 2$ and every nonnegative integer budget split $x + y = N + s$ has the defining four-part certificate on all of F. Thus $c_2(F) \leq N + s = c_2(F) - 1$, a contradiction. Matching deletion here removes edges and retains vertices.

Proof. The frame hypotheses imply those of Lemma 25: the permitted even paths are a subset of its list. We use its canonical construction on every frame, including every frame subsequently obtained by reflection or root exchange. It gives $F[W] = mK_2 + (a - m)K_1$, the sets P, Z, and the actual reflected terminal $R_D = F[W$ union D]. Every D-vertex has degree two in R_D. Every odd cycle and every even path in R_D contains exactly one old frame edge, and these components exhaust the m old frame edges. It also identifies exactly q D-ended path components of B_D, one from each reflected odd cycle. All these assertions concern the same graph F.

1. Heptagon contacts and pentagon contacts.

No frame endpoint can meet a heptagon in any frame. Indeed, suppose p in P meets x on a whole C_7. Property (2) makes x its sole neighbor of degree two in R. The other two R-neighbors of p are therefore outside that cycle. Orient the frame edge at p so that p is a root and replace p by x, using the final exchange assertion of Lemma 25. The actual terminal is $R - x + p$. The six surviving vertices of the old C_7 form a whole P_6: p has no remaining neighbor among them, and the old cycle had no terminal edges outside itself. Property (1) forbids this component. Thus heptagon vertices have both frame neighbors in Z, in every frame.

We need a separate assertion for a possible pentagon contact. For each edge e of F[W] and each canonical D, let $I_e(D)$ be one if both endpoints of e have a D-neighbor, and zero otherwise. In R_D, an old frame edge lies either on an odd cycle or on P_2 or P_4. In an odd cycle both endpoints have a D-neighbor. On P_2 neither does. In a P_4 the old frame edge cannot be the middle edge, since the two path endpoints would then lie in D and have degree one. Thus it is an end edge, and exactly one of its endpoints has a D-neighbor. Consequently

$\sum_{e \in E(F[W])} I_e(D) = q$

for every canonical D.

Suppose px is the possible P-to-pentagon edge, let p′ be p's frame mate, and let z be the other W-neighbor of x. Property (3) gives z in Z. If p′ had a degree-two terminal neighbor y, then y would be outside the pentagon, again by (3). Choose a canonical set selecting y and fix all component choices outside the pentagon. On a cyclic labeling with $x = 0$, the pentagon choices {0,2} and {1,3} contain and omit x, respectively. Property (2) says x is p's sole degree-two terminal neighbor. Thus this change decreases $I_{pp'}$ by one, while every other I_e is unchanged because there is no other P-to-pentagon edge. This contradicts the constant sum q. Therefore p′ has no degree-two terminal neighbor.

For every canonical choice containing x, the reflected terminal contains the path p′-p-x-z. Its first vertex has degree one. Its component contains the old frame edge pp′, so it is a designated component; the only possible paths are P_2 and P_4. It must be exactly this whole P_4. In particular no other selected vertex can neighbor p or z.

2. The two distinct local reflection choices.

Fix a hypothetical cycle C of order five or seven, and a vertex x on C with both parents in Z. Suppose a canonical choice D′ contains x and some neighbor w of x on C is an endpoint of the unique edge of C-D′. That edge is an actual frame edge in the reflected frame constructed by Lemma 25.

Consider the component of x in $R_{D'}$. The vertex x has degree two, so the component is not an isolate or P_2. It is not P_4: a P_4 would need its unique old frame edge, requiring the two old P-endpoints in addition to x and its two distinct Z-parents, a total of five vertices. An odd reflected cycle of order $2j + 1$ with one old frame edge has $j + 1$ W-vertices and j D′-vertices, since every other edge alternates between W and D′. For C_3 or C_5

it has at most three W-vertices; the two Z-parents of x and the endpoints of its old frame edge require four. Finally the component cannot be C_7, since it is met at x by the reflected frame endpoint w, contrary to the heptagon-contact exclusion in step 1. The exhaustive list in (1) leaves a whole P_3 centered at x or a whole C_4.

For a pentagon, label $x = 0$ and choose {0,2}; the residual edge is 3-4 and x is adjacent to its endpoint 4. For a heptagon, label $x = 0$ and choose {0,2,4}; the residual edge is 5-6 and x is adjacent to endpoint 6. Each choice can be combined with any fixed choices outside C. The heptagon argument chooses this triple anew for each tested x; it does not assume that all three selected vertices touch the residual edge in one common triple.

There is also a choice useful for an adjacent pair $x = 0$, $w = 1$. On C_5 use {0,3}, and on C_7 use {0,3,5}. In both cases the unique residual edge is 1-2, so the preceding component conclusion applies to x with w itself a reflected frame endpoint.

3. Parent pairs within the cycle.

Write $A_x = N_F(x)$ intersect W for each x in C; it has two elements by four-regularity. Adjacent cycle vertices cannot share a parent. If x, w were adjacent and shared one, property (3) ensures that at least one, say x, has both parents in Z, and that a shared parent z belongs to Z. Use the adjacent-pair choice in step 2. The reflected frame endpoint w neighbors both x and z. If x lies in a reflected P_3, then x is its center and z is a leaf, violating the second clause of (2) in that reflected frame. If x lies in a reflected C_4, then x, z both have reflected degree two, violating the first clause of (2). This proves disjointness for adjacent vertices.

If x is the possible P-contacted pentagon vertex, its other parent z has no other neighbor on C. Adjacent neighbors have just been excluded. For any nonadjacent y, select the maximum independent pair {x, y} on C and any choices elsewhere. The whole P_4 p′-p-x-z from step 1 precludes the edge zy. The unique-contact rule already precludes another cycle neighbor of p. Thus this parent pair is disjoint from all other parent pairs on C.

Now let x, y be nonadjacent cycle vertices with two Z-parents and a shared parent z. They can be selected together in a maximum independent set on C. This is immediate for C_5. For C_7, their two closed neighborhoods have at most six vertices in total, so a seventh vertex outside their union extends the pair to an independent triple. Choose such a set as part of D′. If their other parents z_x, z_y were distinct, $R_{D'}$ would contain the five-vertex path $z_x - x - z - y - z_y$. No allowed path component has five vertices. A cycle containing this path cannot have order three or four. A C_5 would require an edge between z_x and z_y, impossible in the independent set Z. A C_7 would require a three-edge return path from z_y to z_x. Its first and last internal vertices would both belong to D′, since z_y and z_x have no W-neighbors; those two vertices would then be adjacent, contrary to independence of D′. The component list is exhausted. Hence $z_x = z_y$ and $A_x = A_y$.

Thus the parent pairs of C are equal or disjoint. An equality class is independent by the adjacent-vertex result. Its size is at most two on C_5. On C_7 its size is at most three, since four independent cycle vertices would need at least four intervening unselected vertices. If a class had three vertices, they would be a canonical maximum independent triple, making each common parent have degree at least three in the reflected terminal. Property (1) forbids that. Every equality class therefore has size one or two, and any two-vertex class is nonadjacent.

4. Compatibility with all fixed outside choices.

Fix any original canonical D and retain all its choices $D_{out} = D$ minus $V(C)$ outside C. Let x in C have two Z-parents and let y in D_{out} share a parent with x. Apply the appropriate pentagon or heptagon choice of step 2 for this particular x, keeping exactly D_{out} outside C. The component of x is P_3 or C_4. Its shared parent has the two selected neighbors x, y, so it cannot be a leaf of a P_3 centered at x. The component is a whole C_4, necessarily alternating between x, y and their two common Z-parents. Thus $A_y = A_x$ is contained in Z. For the possible contacted pentagon vertex, its whole reflected P_4 in step 1 instead forbids every $D_{out} - neighbor$ of either parent.

For each distinct two-Z parent pair Q occurring on C, let k_Q be its number of occurrences on C and l_Q its number among the fixed D_{out}. We proved that k_Q is one or two, and that every $D_{out} - vertex$ adjacent to Q has its entire parent pair equal to Q. Choose a maximum independent C-set containing the entire $k_Q - class$, extending a singleton to a maximum set or using the nonadjacent-pair extension just proved. Keep all outside choices fixed. Each parent in Q then has at least $k_Q + l_Q$ neighbors in the resulting reflected terminal, so $k_Q + l_Q \leq 2$.

5. Keeping the whole cycle preserves the deficient paths.

Put $X = D$ union $V(C)$ and let B_X have parts X, Z with all original edges between them. For each pair Q from step 4, its k_Q cycle vertices and l_Q outside vertices together with Q form a whole component $K_{k_Q+l_Q,2}$ of B_X. Every neighbor of either parent is counted, every listed X-vertex has both parents in Q, and different pairs are disjoint. Its left size is one or two, so it is a Z-ended P_3 or a C_4. The possible contacted pentagon vertex instead gives a whole single edge xz, by steps 3 and 4. Each of these components has a matching saturating its X-side.

The restrictions of these components to B_D also have matchings saturating their D-side: they have zero, one, or two left vertices against their two Z-vertices, or are restrictions of the single contacted edge. None is a D-ended path. Every new vertex in X-D lies on C, so every new edge has a Z-end in one of the displayed components. Outside them B_X and B_D are identical. Consequently their deficient path components are precisely the same q actual paths T_1,..., T_q, and these avoid C.

The load bounds required by Lemma 16 hold for X. Each Z-parent appearing on C has load $k_Q + l_Q \le 2$, or load one at the contacted vertex; every other Z-vertex retains its original D-load at most two in R_D. A P-vertex without a C-neighbor retains its D-load at most one. At the possible contacted P-vertex, step 4 forbids outside D-neighbors, leaving exactly its single C-neighbor. All vertices of X have exactly two W-neighbors because they had degree two in the original terminal.

6. Ordered matchings and the strict cost inequality.

Delete any one D-vertex d_i from each T_i, independently, and form A_0, Y, K, N as in the statement. By the path construction of Lemma 16, there are the asserted two sequential maximum matchings of sizes $m + |Y|$ and $|Y|$, clearing exactly the frame edges and the W-to-Y edges. Since C is a whole component of R and D is independent, F[Y] consists of C and isolates. Its matching number is s: s alternating cycle edges attain this, and its $2s + 1$ nonisolated vertices bound it above.

No vertex of C was deleted from D, so $|Y| = |D| - q + s + 1$. The reflected-frame construction of Lemma 25 gives $|V(R)| - |D| = a + m$, since $V(R)$ minus D is the disjoint union of a maximum independent set and m reflected roots. Hence

$N = |V(R)| - |Y| = a + m + q - s - 1 = a + 2m - b - s - 1.$

The assumptions $a \ge 4$, $m \ge 1$, $q \ge 1$ give $N \ge 5 - s$, which is at least s-1 for both $s = 2$ and $s = 3$. Apply the full-extension part of Lemma 16 to the same pair of matchings. It provides every required integer capacity and budget split at total $N + s$, including all original edges with omitted endpoints. Thus

$c_2(F) \le N + s = a + 2m - b - 1 = c_2(F) - 1.$

This contradiction excludes both hypothetical cycle orders. Every auxiliary reflection, local exchange, deficient path, matching, and edge partition used above is realized in the original F. □

Remark. Local reflected frames force each long-cycle parent pair to support a complete small bipartite component. Retaining the entire cycle then leaves the original deficient paths untouched, so one compatible matching construction lowers the same full-graph certificate cost by one.

Depends on. Lemma 16, Lemma 25.

Lemma 72 (Restriction and extension of certificate partitions). *Use the certificate families and integer costs of Lemma 23. All capacities, budgets, totals, indices, and cardinalities below are nonnegative integers. Let H be a finite simple graph and K any subgraph, with arbitrary vertex and edge deletions allowed. Regard K as a spanning subgraph by adding isolated vertices of H when needed, and put $D = (V(H), E(H) minus E(K))$. Then every certificate of H restricts to one of K, and every $(0, 0; x_0, y_0) - certificate$ of D and $(A, B; x_1, y_1) - certificate$ of K combine to an $(A, B; x_0 + x_1, y_0 + y_1) - certificate$ of H whose bounded-degree parts are exactly those on K. Consequently*

$0 \le c_{A,B}(H) - c_{A,B}(K) \le c_0(D) \le \tau(D),$

$0 \le c_\theta(H) - c_\theta(K) \le c_0(D) \le \tau(D).$

If W is any vertex cover of D, a concrete extension with total increase $|W|$ assigns each added edge to an exceptional part using a partition $W = W_R$ disjoint union W_B.

In particular, for any $X \subseteq V(H)$, with H-X the induced graph on $V(H)$ minus X,

$0 \le c_{A,B}(H) - c_{A,B}(H - X) \le |X|$,

$0 \le c_\theta(H) - c_\theta(H - X) \le |X|$.

For any $B_0 \subseteq E(H)$, with $H - B_0$ retaining all vertices,

$0 \le c_\theta(H) - c_\theta(H - B_0) \le c_0((V(H), B_0)) \le \tau((V(H), B_0))$.

Thus deletion of a vertex, or of any set of edges meeting one vertex, changes each individual cost by either zero or one.

More generally, suppose $m \ge 0$ and the edges of H-X are partitioned into spanning graphs K, D_1,..., D_m. Then

$c_{A,B}(H) \le |X| + c_{A,B}(K) + \sum_{j=1}^{m} c_0(D_j)$,

$c_\theta(H) \le |X| + c_\theta(K) + \sum_{j=1}^{m} c_0(D_j)$.

Each allocation at the displayed total admits a certificate with bounded-degree parts wholly in K, and every edge meeting X, including edges inside X, belongs to an exceptional part.

Finally, if H is the vertex-disjoint union of finite simple graphs H_1,..., H_m, with no edges between them, then

$c_{A,B}(H) \le \sum_j c_{A,B}(H_j)$, $c_\theta(H) \le \sum_j c_\theta(H_j)$,

$ps(H) \ge \sum_j ps(H_j)$.

The empty union is allowed.

Proof. Adding isolated vertices changes neither the edge-partition family nor any degree or matching bound, so it changes no certificate cost. Intersecting the four parts of a certificate of H with $E(K)$ preserves both degree and matching bounds, proving restriction and cost monotonicity by Lemma 23.

Write a supplied certificate on K as (Q_R, Q_B, P_R^1, P_B^1). The degree-zero parts in a certificate on D must be empty, so write $E(D) = P_R^0$ disjoint union P_B^0. Since $E(D)$ and $E(K)$ are disjoint, the four edge sets

Q_R, Q_B, P_R^1 union P_R^0, P_B^1 union P_B^0

partition $E(H)$. Their bounded-degree parts are unchanged. Every matching in a union of edge-disjoint graphs restricts to a matching in each graph, so its size is at most the sum of their matching numbers. Hence the exceptional matching bounds are $x_0 + x_1$ and $y_0 + y_1$.

Let $C_0 = c_0(D)$, $C_1 = c_{A,B}(K)$ and $x + y = C_0 + C_1$. Choose $x_0 = \min(x, C_0)$, $y_0 = C_0 - x_0$, $x_1 = x - x_0$, and $y_1 = C_1 - x_1$. These four integers are nonnegative: if $x \le C_0$ then $x_1 = 0$, $y_1 = C_1$; if $x \ge C_0$ then $x_1 = x - C_0 \le C_1$. Also $y_0 + y_1 = y$. The threshold characterization in Lemma 23 supplies both certificates. Their union proves $c_{A,B}(H) \le C_0 + C_1$. Taking the maximum over $A + B = \theta$ gives the row inequality; its cover bound is $c_0(D) \le \tau(D)$ from that same fact.

For a direct cover witness at an increment $|W|$, split $x + y = |W| + c_{A,B}(K)$ into a pair of total $|W|$ and a pair of total $c_{A,B}(K)$ by the preceding minimum rule. Choose $W_R \subseteq W$ with cardinality equal to the first red budget, and put $W_B = W$ minus W_R. Assign a D-edge red when it meets W_R and blue otherwise. Every blue D-edge meets W_B. Thus each matching in the new red edges has size at most $|W_R|$ and each matching in the new blue edges has size at most $|W_B|$. Combine these with the residual certificate. This includes all edges having two endpoints in W and gives each edge exactly one color.

For $K = H - X$, the set X covers all missing edges. The cover witness proves the vertex-set bounds. For $K = H - B_0$, the missing graph is precisely $(V(H), B_0)$, giving the edge-deletion bounds. A single vertex covers every edge in a nonempty set of edges meeting that vertex; the empty deleted set changes no costs. Integrality therefore gives the zero-or-one assertion.

For a finite list of totals t_0,..., t_m and any x in $[0, \sum_j t_j]$, choose successive red contributions by taking the minimum of the remaining red budget and each next total. Each contribution lies between zero and that total; after the last choice the remaining budget is zero since $x \le \sum_j t_j$. Set each blue contribution equal to its total minus its red contribution. Apply this rule to $|X|$, $c_{A,B}(K)$, and the $c_0(D_j)$. Choose the corresponding certificates and assign edges meeting X by a partition of X as in the cover construction. Union all exceptional parts. Matching restriction bounds their cardinalities by the summed budgets despite arbitrary overlap of vertices. Only K supplies bounded-degree edges. This proves the fixed-capacity finite-family bound.

Alternatively choose total $c_\theta(K)$ for K; it is at least $c_{A,B}(K)$ for every $A+B=\theta$, so the same construction proves the row bound. The cases X empty or $m=0$ require no omitted assignments.

For vertex-disjoint graphs, apply the same budget decomposition to $c_{A,B}(H_j)$ and union all four parts. Each vertex belongs to only one graph, so the bounded-degree maxima remain at most A and B. Matching numbers add over vertex-disjoint graphs: restriction proves one inequality, and the union of maximum matchings proves the reverse. The exceptional parts therefore meet the summed budgets. Taking row maxima gives row subadditivity. Edge counts and matching numbers add over these graphs, while the finite positive-row sum is subadditive. Subtracting that sum from e(H)-nu(H) yields $ps(H) \geq \sum_j ps(H_j)$. Empty graphs contribute zero throughout, by Lemma 23. □

Remark. An added edge part can be paid at its row-zero cost while preserving the retained graph's bounded-degree edge sets. A vertex cover gives a concrete universal budget for all added edges.

Depends on. Lemma 23.

Lemma 73 (Certificates from small components of odd maximum degree). *Use the finite integer certificate families $C_{A,B}(F;x,y)$ and costs $c_i(F)$ of Lemma 23. A certificate partitions every edge of F into Q_R, Q_B, P_R, P_B with maximum degrees at most A, B in the first two parts and matching numbers at most x, y in the last two. The cost c_i is the maximum over nonnegative integer $A+B=i$ of the least total $x+y$ for which every nonnegative integer budget split is feasible.*

Let $D \geq 1$ be an odd integer. Let F be a finite simple graph, let Z be any subset of $V(F)$, and put $R = F - Z$ by induced vertex deletion. Assume $\Delta(R) \leq D$ and that every connected component of R has at most $2D+4$ vertices.

There exists a maximum matching M of R covering every vertex of R-degree D. Fix any such M. For every nonnegative integer capacity pair $A+B=D$ and every nonnegative integer budget pair $x+y=|Z|$, there is a certificate in $C_{A,B}(F;x,y)$ whose degree-bounded parts partition exactly $E(R)$ and whose matching-bounded parts partition exactly all original F-edges meeting Z, including those with both endpoints in Z. The same Z, R, M are fixed before all capacity and budget choices. Consequently $c_D(F) \leq |Z|$.

In particular, let $d \geq 2$ be an even integer, let j be an odd integer with $1 \leq j \leq d-1$, and assume $\Delta(F) \leq d$. If every vertex outside Z has at least j neighbors in Z and every component of F-Z has at most $2(d-j)+4$ vertices, then all the preceding conclusions hold with $D = d-j$. The component-order hypothesis holds whenever $|V(F)| - |Z| \leq 2(d-j)+4$. For $j=1$ this gives $c_{d-1}(F) \leq |Z|$ for every dominating set Z whose induced complement has order at most $2d+2$; a set is dominating when every vertex outside it has a neighbor in it.

All orders, degrees, cardinalities, capacities, budgets and matching sizes are integers in their displayed nonnegative or positive domains. Empty F, empty Z, empty R, isolated components and zero budgets are allowed. There is no regularity, factor-criticality, independence or minimality hypothesis.

Proof. We use the prescribed-saturation criterion of Lemma 10, the covering-matching splitting theorem Lemma 35, and the integer certificate definitions of Lemma 23.

First let K be one connected component of R and put $U = \{vinV(K) : d_K(v) = D\}$. Degrees in K equal degrees in R. If $D=1$, every component of K is an isolated vertex or a single edge: a vertex incident with an edge has no other neighbor. Thus $E(K)$ is itself a maximum matching covering U.

Suppose $D \geq 3$ and no matching of K covers U. By Lemma 10, there is a subset X of $V(K)$ such that K-X has $q \geq |X|+1$ odd-order components whose vertex sets are contained in U. These are components of the induced deletion K-X. Denote their family by L, and put $b = |X|$. For C in L let $t_C = |V(C)|$ and let a_C be the number of K-edges between $V(C)$ and X. Every vertex of C has K-degree D and has no K-neighbor outside $V(C)$ union X. Counting degrees within C therefore gives

$a_C = D\,t_C - 2e(C)$.

Both D and t_C are odd, so a_C is odd and in particular $a_C \geq 1$. Since L is nonempty, this also proves $b \geq 1$.

Simplicity gives $2e(C) \leq t_C(t_C-1)$, and hence

$a_C \geq t_C(D+1-t_C)$.

Call C small if $t_C \le D$ and large otherwise, and let l be the number of large members of L. For a small member, $t_C \ge 1$ and

$t_C(D+1-t_C) - D = (t_C-1)(D-t_C) \ge 0,$

so $a_C \ge D$. A large member has odd order $t_C > D$ with D odd, and therefore $t_C \ge D+2$; its boundary satisfies $a_C \ge 1$. Distinct members of L have disjoint vertex sets. Counting their edges at the X endpoints, and using $\Delta(K) \le D$, gives

$D(q-l) + l \le \sum_{C \in L} a_C \le Db.$

Since $q \ge b+1$, it follows that

$D - (D-1)l \le 0.$

Thus $l \ge D/(D-1) > 1$, and the integer l is at least two. Consequently

$|V(K)| \ge b + 2(D+2) \ge 2D+5.$

This contradicts the component-order assumption. Every component K therefore has a matching covering its vertices of degree D. The maximum-matching conclusion of Lemma 10 permits choosing such a matching of cardinality $\nu(K)$.

Take the union of these chosen matchings over the components of R. They have disjoint endpoints, and their union M covers every vertex of R-degree D. Every matching in R restricts to a matching in each component, so its size is at most the sum of the component matching numbers. The union attains this sum and is therefore maximum in R. If R is empty the empty union is its empty maximum matching. This proves the matching assertion in all cases.

Now fix any maximum matching M with the asserted coverage. For each nonnegative integer $A + B = D$, apply Lemma 35 to the same graph R and the specified matching M. It supplies a partition $E(R) = Q_R$ dot-union Q_B with maximum degrees at most A and B. Regard both edge sets as spanning graphs on $V(F)$; this adds only isolated vertices and leaves their maximum degrees unchanged.

For any nonnegative integer $x + y = |Z|$, choose a partition $Z = Z_R$ dot-union Z_B with $|Z_R| = x$ and $|Z_B| = y$. Let P_R be all F-edges with an endpoint in Z_R. Let P_B be all remaining F-edges that meet Z. Every edge of P_B has an endpoint in Z_B. A matching in either part uses distinct vertices of its corresponding set Z_R or Z_B, so $\nu(P_R) \le x$ and $\nu(P_B) \le y$. The two parts are disjoint by construction, contain every edge meeting Z, and contain no edge of R. Thus Q_R, Q_B, P_R, P_B partition all original edges of F, including every edge internal to Z, and form the required certificate. All choices of capacities and budgets use the same previously fixed Z, R, M. By Lemma 23, universal feasibility at total $|Z|$ for every capacity split gives $c_D(F) \le |Z|$. Empty parts and zero budgets obey the same construction.

For the stated specialization put $D = d - j$. The parity and range of d, j give a positive odd integer D. Every vertex v outside Z has at most d total neighbors in F and at least j neighbors in Z, so $d_{F-Z}(v) \le d - j = D$. The assumed component bounds are exactly those used above, giving the matching and certificate conclusions. A bound on the entire order of F-Z bounds each component order. At $j = 1$, the neighbor hypothesis is domination and $2(d-j)+4 = 2d+2$. This proves every claimed specialization. □

Remark. An odd-degree coverage failure requires two large odd pieces behind a nonempty separator. Small residual components exclude that obstruction and permit every original-edge certificate split.

Depends on. Lemma 10, Lemma 23, Lemma 35.

Lemma 74 (Independent covers at equality for matching families). *All graphs are finite and simple. A matching has pairwise disjoint endpoints; $\nu(Q)$ denotes its maximum cardinality. A vertex cover meets every edge; $\tau(Q)$ denotes its minimum cardinality. An independent set contains no adjacent pair. For a positive integer d, $K_{1,d}$ is the star with d edges, and $K_2 = K_{1,1}$. Copies are ordinary subgraphs, and different components of a target copy have disjoint vertex sets.*

Let calA be a finite nonempty index set. For every a in calA let p_a, $t_a \ge 1$ be integers with the same value $p_a + t_a - 1 = N$. Let $m_1^a \ge \cdots \ge m_{t_a}^a \ge 1$ be integers and let J_a be the vertex-disjoint union of stars with these complete edge-size lists. For every integer $1 \le k \le N$ put

$\lambda_k = max\{m_j^a : a \in calA, 1 \le i \le p_a, 1 \le j \le t_a, i + j = k + 1\}$,

and put $B = \sum_{k=1}^{N} \lambda_k$.

Suppose Q arrows (p_aK_2, J_a) *for every a: every red-blue coloring of Q contains a red* $p_a -$ *edge matching or a blue copy of the whole* J_a. *Suppose* $e(Q) = B$.

Then Q has an independent vertex cover I of cardinality N whose degrees in Q, counted with multiplicity, are exactly λ_1,..., λ_N. *Consequently Q is bipartite and* $\nu(Q) = \tau(Q) = N$. *For each integer* $d \ge 3$, *distinct vertices of I of degree d have disjoint open neighborhoods in Q. Put*

$h = |\{d \ge 3 : disanintegerand\lambda_k = dforsome1 \le k \le N\}|$,

$c_0 = \max(\{\lambda_k : 1 \le k \le N, \lambda_k \le 2\}union\{0\})$.

Every vertex outside I has Q-degree at most $h + c_0$.

The common component budget N is required. Every original unit component and repeated size is retained. All degrees, orders, indices, cardinalities and budgets are integers in their displayed nonnegative or positive domains, and Q may have isolated vertices.

Proof. We use the actual vertex-deletion implication of Lemma 34. If a host arrows a positive star-forest pair, deleting any b vertices leaves a host for every legal pair of actual suffixes omitting a total of b leading components.

First record the profile consequences needed for the induction. The profile of (pK_2, J), when J has sorted sizes m_1,..., m_t, is m_1 repeated p times followed by m_2,..., m_t. Indeed, at coordinate k the smallest valid blue index is $\max(1, k + 1 - p)$, and its blue size is the maximum. Every profile here has length N. Their coordinatewise maximum λ_1,..., λ_N is therefore positive and nonincreasing.

For $0 \le b < N$, take all legal total-b suffixes of all the family members. Every resulting pair has positive targets, its first target is a matching, and its component budget is N-b. This family is nonempty: for a member with lengths p, t, choose cuts $a' = \min(b, p - 1)$, $b' = b - a'$; the inequality $b \le p + t - 2$ gives $0 \le b' < t$. Its entire envelope is λ_{b+1},..., λ_N. One inequality holds because every suffix candidate comes from the corresponding original diagonal. Conversely, an original candidate with indices i, j on coordinate $b + k$ satisfies $i + j = b + k + 1$ and $i + j - 2 \ge b$. Choose $a' = \min(b, i - 1)$ and $b' = b - a'$. These cuts satisfy $a' \le i - 1$, $b' \le j - 1$ and $a' + b' = b$. They are legal and place the same candidate on suffix coordinate k. This proves the reverse inequality for every $1 \le k \le N - b$.

Consequently, after any b vertex deletions with $b < N$, the remaining graph arrows this suffix family by Lemma 34. Its all-blue coloring must contain each blue target, so its maximum degree is at least λ_{b+1}, the largest suffix head. Starting with Q, successive maximum-degree deletions can therefore continue for N steps: premature exhaustion would contradict this positive degree lower bound. If their successive degrees are d_1,..., d_N and their terminal graph is Q_N, then

$B = e(Q) = \sum_{k=1}^{N} d_k + e(Q_N), d_k \ge \lambda_k$.

Since $B = sum\ \lambda_k$ and $e(Q_N) \ge 0$, equality holds term by term: $d_k = \lambda_k$ for every k and Q_N is edgeless. This argument applies to every maximum-degree deletion sequence. In particular $\Delta(Q) = \lambda_1$.

We now prove the asserted independent cover and its degree and neighborhood properties by induction on the positive integer N. This induction also applies to every finite family with common budget smaller than N. Put $D = \lambda_1$. If $D = 1$, then $\Delta(Q) = 1$ and $e(Q) = N$, so Q consists of N disjoint single edges and isolates. Selecting one endpoint of each edge gives the required independent cover. All outside degrees are at most one, $h = 0$, $c_0 = 1$, and there are no cover vertices of degree at least three.

Assume $D \ge 2$. Let R be the number of initial entries equal to D, where $1 \le R \le N$. Choose the first R vertices of a maximum-degree deletion sequence and let A be their set. Each is deleted with degree D, and $\Delta(Q) = D$. No later chosen vertex can have a neighbor among earlier chosen vertices, since that would make its original degree greater than its current degree D. Hence A is independent and each of its vertices has original Q-degree D. Put $Q_0 = Q - A$.

If $R = N$, the deletion conclusion makes Q_0 edgeless; put C empty. If $R < N$, form the family of all legal total-R suffixes. It has common budget N-R, is arrowed by Q_0, and has envelope λ_{R+1},..., λ_N. The exact deletion degrees give

$e(Q_0) = \sum_{k=R+1}^{N} \lambda_k$.

The induction hypothesis supplies an independent cover C of Q_0 of cardinality N-R with precisely those degrees, disjoint neighborhoods among its equal-degree vertices of degree at least three, and the asserted outside-degree bound. Every vertex of C has $Q_0 - degree$ at most D-1. These same assertions hold with C empty when $R = N$.

Choose one original member whose profile attains D at coordinate R. Write its parameters as p, t and let r be the number of its blue D-stars. Its blue head is D, since every blue head is at most the full envelope head D. Its leading D plateau has length $p + r - 1$. This length is at least R. It is at most R, because the full envelope has no further D entry (and has total length N if $R = N$). Thus $p + r - 1 = R$. Its full blue target is $rK_{1,D}$ disjoint union J_0, where J_0 has $t - r = N - R$ components of size below D. In particular $p \le R$ and $r \ge 1$.

There is no edge between A and C. Suppose instead that x in A has a neighbor in C. Choose p-1 vertices of A other than x; this is possible because $p - 1 \le R - 1$. Let A_{red} be that set. Color every edge meeting A_{red} red and all other edges blue. The red graph has the cover A_{red} and thus no p-edge matching. The blue graph has cover

$Z = (AminusA_{red})$ union C,

of size $R - p + 1 + N - R = N - p + 1 = t$. A full blue target has t vertex-disjoint positive components, each meeting Z, so each must meet Z exactly once.

A blue D-star centered outside Z has all its D leaves in Z because Z is a cover. As $D \ge 2$, it cannot be a component of such a full copy. A blue D-star centered in C needs a neighbor in A minus A_{red} because its degree in Q_0 is at most D-1; its center and that neighbor both lie in Z, again impossible. Each vertex of A minus A_{red} retains all its D edges in blue because A is independent. Its D-star uses its entire neighborhood, and it is unusable if that neighborhood meets C. The vertex x is therefore unusable. Only the other at most r-1 vertices of A minus A_{red} could supply the required r D-stars. This rules out the full blue target and contradicts the chosen member's arrowing.

It follows that $I = A$ union C is an independent cover of Q. It covers every edge since C covers Q-A. It has cardinality N. Vertices of C gain no incident edge on restoring A, so their degrees retain the tail list, while all vertices of A have degree D. This proves the required complete degree multiset.

Suppose $D \ge 3$ and distinct x, y in A share a neighbor z. If $p = 1$, arrowing forces the all-blue Q to contain the entire J. Here $r = R$ and $t = N$. Its N components would meet the size-N independent cover I exactly once each. A D-star centered outside I uses D cover vertices and is impossible. Vertices of C have degree below D. Hence all R required D-stars must use every vertex of A as a center and its full neighborhood. The shared neighbor z prevents their vertex-disjointness, a contradiction.

If $p \ge 2$, choose p-2 vertices of A distinct from x, y and call the set A_{red}; this is possible since $p \le R$. Color all edges meeting A_{red} and also xz, yz red, and color the other edges blue. The red graph has the cover A_{red} union {z}, of size p-1: z lies outside A because A is independent. Thus it has no red p-edge matching. The blue graph is covered by $Z = (AminusA_{red})$ union C, of size $N - p + 2 = t + 1$. A D-star centered outside Z uses $D \ge 3$ distinct cover vertices, and each of the other t-1 required components uses at least one, totaling at least $t + 2 > |Z|$. Such a D-star cannot occur in a full blue target. Vertices of C have degree below D, and x, y now have blue degree D-1. Vertices of A_{red} have blue degree zero. At most $R - (p - 2) - 2 = r - 1$ remaining vertices of A can center a blue D-star. Again the full target is impossible. This proves disjoint neighborhoods for vertices of A. For equal-degree vertices in C with degree at least three, the induction hypothesis persists because their neighborhoods do not change when A is restored. Since every degree in C is below D, all asserted degree classes are covered.

For the outside-degree bound, if $D = 2$ the inequality follows directly from $\Delta(Q) = 2$, $h = 0$ and $c_0 = 2$. If $D \ge 3$, each vertex outside I has at most one neighbor in A by the just-proved disjointness. When $R < N$, the residual envelope has exactly h-1 distinct values at least three and the same c_0. Its inductive outside-degree bound gives Q-degree at most $1 + (h - 1) + c_0 = h + c_0$. When $R = N$, Q_0 is edgeless and $h = 1$, $c_0 = 0$, so the bound is again one. Together with the $D = 1$ case, this completes the induction.

It remains to verify the matching and minimum-cover conclusions. Any cover K of Q with $|K| < N$ can be partitioned into K_{red}, K_{blue} with $|K_{red}| \le p - 1$ and $|K_{blue}| \le t - 1$ for any fixed member satisfying $p + t - 1 = N$. Color edges meeting K_{red} red and all remaining edges blue. The red matching number is at most p-1. The blue graph has cover K_{blue} and hence matching number at most t-1; it cannot contain the whole J, from whose t positive disjoint components one could select a t-edge matching. This contradicts arrowing.

Thus every cover has size at least N. The constructed I proves $\tau(Q) = N$; its independent complement makes Q bipartite, and any matching has at most N edges.

For completeness, choose a maximum matching M of Q and suppose it misses a vertex of I. From all missed vertices in I, follow alternating paths which leave I on edges outside M and return on edges in M. Let U be the reached vertices in I and V the reached vertices outside I. Every reached vertex in V is matched, since otherwise an alternating path to it would augment M. Its partner lies in U. Conversely, every matched vertex in U was reached via its matching edge from V. Therefore $|V|$ is the number of matched vertices in U and is strictly less than $|U|$. Moreover V is the full neighborhood of U: every nonmatching edge from U reaches its other endpoint, and the matching neighbor of a reached matched vertex of U is already reached. Then (I minus U) union V is a cover: each edge either meets I minus U or has its endpoint in I inside U and its other endpoint in V. Its size $N - |U| + |V|$ is less than N, a contradiction. Hence M saturates I and $\nu(Q) = N$. Isolates affect none of these constructions. □

Remark. At exact edge budget every greedy maximum-degree deletion matches its envelope entry. The original member attaining the end of the largest-degree plateau forces each full blue target component to use exactly one cover vertex, ruling out attachments between the new centers and the recursive cover.

Depends on. Lemma 34.

Lemma 75 (Allocation-set composition and component absorption). *Use the finite certificate families and integer costs of Lemma 23. All graphs below are finite and simple, and all capacities, totals, indices, and cardinalities are nonnegative integers. For a graph Q and capacities A, B define*

$Y_l^{A,B}(Q) = \{x \in \{0,\dots,l\} : C_{A,B}(Q; x, l-x) is nonempty\}.$

For a subset Y of {0,...,l}, let $run_l(Y)$ be the largest cardinality of a nonempty consecutive integer interval contained in {0,...,l} minus Y, or zero if that complement is empty. These sets are finite, so this definition is attained whenever it is positive. Intervals in this statement contain integers only. For integer sets U, V write $U + V = \{u + v : u in U, v in V\}$.

If $l \geq c_{A,B}(Q)$, then $Y_l^{A,B}(Q) = \{0,\dots,l\}$. If $l \geq \nu(Q)$, then 0, l belong to $Y_l^{A,B}(Q)$, every missing integer lies in $[l - \nu(Q) + 1, \nu(Q) - 1]$, and

$run_l(Y_l^{A,B}(Q)) \leq \max(0, 2\nu(Q) - 1 - l).$

For any $Y \subseteq$ {0,...,l} containing 0, l, and any $L \geq 0$,

$\{0,\dots,L\} + Y = \{0,\dots,L+l\}$ *if and only if* $run_l(Y) \leq L$.

For vertex-disjoint Q_1, Q_2 with no edges between them,

$Y_{l_1+l_2}^{A,B}(Q_1 disjointunion Q_2)$ *contains*

$Y_{l_1}^{A,B}(Q_1) + Y_{l_2}^{A,B}(Q_2).$

For any graph H, subset X of $V(H)$, and $b = |X|$,

$Y_{l+b}^{A,B}(H)$ *contains* $Y_l^{A,B}(H - X) + \{0,\dots,b\}$.

The first inclusion is witnessed by unions of corresponding edge parts. For the second, given x in $Y_l^{A,B}(H - X)$, an integer t in [0,b], and any t-element subset X_R of X, retain the certificate on H-X, put all edges meeting X_R into its red exceptional part, and put every remaining edge meeting X into its blue exceptional part.

In particular, for vertex-disjoint Q_1, Q_2, if $L \geq c_{A,B}(Q_1)$, $l \geq \nu(Q_2)$, and $run_l(Y_l^{A,B}(Q_2)) \leq L$, then

$c_{A,B}(Q_1 disjointunion Q_2) \leq L + l.$

Consequently

$c_{A,B}(Q_1 disjointunion Q_2) \leq \max(c_{A,B}(Q_1), \nu(Q_2) - 1) + \nu(Q_2).$

More generally let H-X be the vertex-disjoint union Q_0,..., Q_m, with $m \geq 0$, no edges between different pieces, and pieces allowed to be empty. Put $b = |X|$, $c_j = c_{A,B}(Q_j)$, and $r_j = \nu(Q_j)$. Let $I = \{j \in \{1, \ldots, m\} : c_j > r_j\}$, list its elements as j_1,..., j_p in any order, and put

$L = b + c_0 + \sum_{1 \leq j \leq m, j not \in I} c_j$.

If $r_{j_t} \leq L + \sum_{s<t} r_{j_s} + 1$ for each integer $1 \leq t \leq p$, then

$c_{A,B}(H) \leq b + c_0 + \sum_{j=1}^{m} \min(c_j, r_j)$.

Every allocation at this total has an actual four-part certificate on H; all edges meeting X are exceptional.

Finally fix $\theta \geq 0$. If every j in {1,...,m} with $c_\theta(Q_j) > \nu(Q_j)$ satisfies

$\nu(Q_j) \leq 1 + b + c_{\theta,0}(Q_0)$,

then

$c_\theta(H) \leq b + c_\theta(Q_0) + \sum_{j=1}^{m} \min(c_\theta(Q_j), \nu(Q_j))$.

Here $c_{\theta,0}(Q_0)$ is the fixed-capacity cost with $A = \theta$, $B = 0$, not the row maximum. It satisfies $c_{\theta,0}(Q) \leq c_{A,B}(Q)$ whenever $A + B = \theta$. At $\theta = 0$ the condition is imposed exactly on pieces with $ex(Q_j) > 0$, and the conclusion is

$c_0(H) \leq b + c_0(Q_0) + \sum_{j=1}^{m} \nu(Q_j)$.

When $p = 0$ or $m = 0$ the respective list of inequalities is empty. Empty graphs, zero totals, and zero matching numbers are included.

Proof. The threshold characterization in Lemma 23 gives the first assertion about $l \geq c_{A,B}(Q)$. If $l \geq \nu(Q)$ and $x \geq \nu(Q)$, assign all edges to the red exceptional part. If $l - x \geq \nu(Q)$, assign them all to the blue exceptional part. Each assignment has empty other parts and satisfies the stated allocation. It follows that 0, l are feasible and any infeasible x must satisfy both $x \leq \nu(Q) - 1$ and $x \geq l - \nu(Q) + 1$. The number of integers in this last interval is max(0,2nu(Q)-1-l), giving the run bound, also when $\nu(Q) = 0$.

To prove the interval criterion, suppose $run_l(Y) \leq L$ and take an integer z in $[0, L + l]$. If $z \leq L$, write $z = z + 0$. If $z \geq l$, write $z = (z - l) + l$; the two summands lie in [0,L] and Y. In the remaining case $L < z < l$, failure of a representation would mean that every integer in [z-L, z] is missing from Y. This interval lies in [0,l] and has $L + 1$ elements, contradicting the run bound. All pairwise sums lie in $[0, L + l]$, proving equality. Conversely, if $run_l(Y) > L$, choose $L + 1$ consecutive missing integers h, $h + 1$,..., $h + L$ in [0,l]. No y in Y lies in $[h, h + L]$, so $z = h + L$ cannot equal $t + y$ with $0 \leq t \leq L$: any such y would lie in that interval. Thus the sumset fails to contain z, which belongs to $[0, L + l]$. This proves both directions, including $L = 0$ and $l = 0$.

For the disjoint-union inclusion, choose certificates at budgets x_i, $l_i - x_i$ on Q_i, for $i = 1, 2$. Union the two red bounded-degree parts, the two blue bounded-degree parts, and the exceptional parts by color. At any vertex only one graph contributes an edge, so the bounded degrees remain at most A and B. A matching in an exceptional union restricts to a matching in each part, so the red and blue matching bounds are $x_1 + x_2$ and $l_1 + l_2 - x_1 - x_2$. All edges have been assigned exactly once. This proves the displayed inclusion and proves its version for any finite list by iteration.

For the vertex-set inclusion, choose the supplied certificate on H-X and a t-element subset X_R of X. Each new red edge meets X_R. Each new blue edge meets X minus X_R, because it meets X but not X_R. These two edge sets partition exactly the edges omitted from H-X, including edges inside X. Every matching in their respective sets has cardinality at most t or b-t, by assigning distinct matching edges to distinct vertices in those sets. Restricting a matching between new and old edges gives exceptional bounds $x + t$ and $l - x + b - t$. The bounded-degree parts are unchanged. This proves the inclusion with the asserted actual extension for every chosen X_R.

In the two-piece absorption assertion, the first graph supplies the full interval [0,L]. The second supplies a set containing 0, l whose missing runs have length at most L. The interval criterion and disjoint-union inclusion therefore make every allocation at total $L + l$ feasible. The threshold definition gives the cost bound. Set $l = \nu(Q_2)$ and $L = \max(c_{A,B}(Q_1), \nu(Q_2) - 1)$. This L is nonnegative; the proved run bound is at most $\max(0, \nu(Q_2) - 1) \leq L$. The stated consequence follows.

For the ordered construction put

$T = b + c_0 + \sum_{jnot\in I, 1\le j\le m} c_j + \sum_{j\in I} r_j$.

Each piece Q_0 and each piece with j not in I supplies the entire interval of budgets at its cost c_j. Their intervals, together with [0,b] for X, sum to [0,L]. To see this directly for any number of intervals, split a required integer red budget successively by taking the minimum of the remaining budget and the next interval length. Every chosen value is in its interval, and the remaining budget is zero at the end because the requested value does not exceed the sum of the lengths.

For a piece indexed by j_t in I, its allocation set at total r_{j_t} contains 0, r_{j_t} and has missing runs of length at most $\max(0, r_{j_t} - 1)$. The ordered hypothesis makes this at most $L + \sum_{s<t} r_{j_s}$. Repeated application of the interval criterion thus proves

$[0, L] + \sum_{t=1}^{p} Y_{r_{j_t}}^{A,B}(Q_{j_t}) = [0, T]$.

Every integer in this sumset has, by definition of a finite sumset, a decomposition into a pool contribution and a feasible red budget for each component. Choose corresponding component certificates, union them by parts, and restore edges meeting X by the proved vertex-set extension. This is an actual certificate for each allocation at T, with all external edges exceptional. If $p = 0$, only the full intervals occur. The total T equals $b + c_0 + \sum_{j=1}^{m} \min(c_j, r_j)$, proving the ordered bound and its empty cases.

For $A + B = \theta$, any certificate on a graph Q may replace its two bounded-degree parts by their union in the red bounded-degree part and an empty blue bounded-degree part. The degree of that union at each vertex is at most $A + B = \theta$; both exceptional parts are unchanged. Hence every allocation feasible for capacities A, B is feasible for θ, 0, and Lemma 23 gives $c_{\theta,0}(Q) \le c_{A,B}(Q)$.

Now fix an arbitrary $A + B = \theta$. If $c_{A,B}(Q_j) > \nu(Q_j)$, then $c_\theta(Q_j) > \nu(Q_j)$, so the final hypothesis gives

$r_j \le 1 + b + c_{\theta,0}(Q_0) \le 1 + b + c_{A,B}(Q_0) \le 1 + L$.

Thus the ordered construction applies in any order to all its required indices. Its cost bound is at most

$b + c_\theta(Q_0) + \sum_j \min(c_\theta(Q_j), \nu(Q_j))$,

since $c_{A,B}(Q_j) \le c_\theta(Q_j)$ and the minimum is nondecreasing in its first argument. This upper bound is independent of A, B, so taking the row maximum proves the final assertion. When $\theta = 0$, the only capacity pair is (0,0); Lemma 23 gives $c_0(Q_j) \ge \nu(Q_j)$, so the relevant pieces are exactly those with positive excess and each minimum is $\nu(Q_j)$. No assumption on connectedness or nonzero order of a piece was used. □

Remark. A component at its certificate cost supplies every budget in an interval. Such an interval fills missing budgets of the other components, so those components can sometimes be charged only their matching numbers.

Depends on. Lemma 23.

Lemma 76 (Independent private neighbors give a threshold-one certificate). *Use the finite integer certificate costs $c_{A,B}(H)$ and $c_i(H)$, and the independence number $\alpha(H)$, from Lemma 23. Let H be a finite simple graph, possibly empty, on n vertices. Let S be a maximum independent set of H, and let U be an independent subset of $V(H)$ minus S such that every vertex of U has exactly one neighbor in S. Then $c_1(H) \le n - \alpha(H) - |U|$. More explicitly, for each pair of nonnegative integer capacities $A + B = 1$ and each pair of nonnegative integer matching budgets $x + y = n - \alpha(H) - |U|$, there is a certificate whose bounded-degree edges are precisely the edges of H[S union U], all placed in the part of capacity one, and whose remaining edges are assigned to the exceptional parts using a partition of $V(H)$ minus(SunionU) of cardinalities x, y. Consequently, if $d \ge 1$ is an integer and H is d-regular, then for every maximum independent set S and every independent set U of vertices having degree d-1 in the induced graph H-S, one has $|U| \le n - \alpha(H) - c_1(H)$. All orders, cardinalities, degrees, capacities, budgets and costs are integers in the displayed domains; S and U may be empty.*

Proof. For each u in U, let $s(u)$ be its unique neighbor in S. This map is injective. Otherwise distinct u, v in U would have $s(u) = s(v) = s$. The set (S minus{s}) union {u, v} would be independent: S is independent, u and v are nonadjacent because U is independent, and neither has a neighbor in S minus{s}. Its cardinality is $|S| + 1$, contradicting maximal cardinality of S. As both S and U are independent and the edges between them are exactly the pairs $u\ s(u)$, the induced graph H[S union U] is a matching with isolated vertices.

Put $W = V(H)\ minus(SunionU)$, and $w = |W| = n - \alpha(H) - |U|$. Fix nonnegative integers x, y with $x + y = w$, and partition W into W_R, W_B with $|W_R| = x$ and $|W_B| = y$. Such a partition exists also when W or one part is empty. First take $capacities(A, B) = (1, 0)$. Let Q_R consist of all edges of H[S union U], and set Q_B empty. Among the remaining edges put every edge meeting W_R in P_R and all others in P_B. Every remaining edge meets W because Q_R contains the entire induced edge set on its complement. Consequently every edge assigned to P_B meets W_B. In a matching of P_R, its distinct edges meet distinct vertices of W_R; thus $\nu(P_R) \le x$. The same argument gives $\nu(P_B) \le y$. Also $\Delta(Q_R) \le 1$ and $\Delta(Q_B) = 0$. The four disjoint parts cover all edges, so this is the required certificate in the sense of Lemma 23. For $capacities(0, 1)$, place the same induced matching in Q_B instead, keep Q_R empty, and use the identical exceptional assignment to P_R, P_B. These are the only nonnegative integer capacity pairs of sum one. Every budget pair of total w has therefore been certified at both pairs of capacities, proving $c_1(H) \le w$ by definition.

For the stated regular-graph consequence, a vertex u outside S has $d_H(u) = d$. Its degree in H-S is d-1 if and only if it has exactly one neighbor in S. An independent set U among these vertices satisfies the just-proved hypotheses. Rearranging $c_1(H) \le n - \alpha(H) - |U|$ gives the displayed bound. This argument also covers $d = 1$ and an empty U; no nonempty graph is required. □

Remark. A maximum independent set forbids two independent private neighbors from sharing their only neighbor. The resulting induced matching can be placed entirely in the degree-one part, and every other edge is paid for by a remaining vertex.

Depends on. Lemma 23.

Lemma 77 (Matching partners of a maximum independent set). *All graphs are finite and simple. For a graph G, let $V(G)$ be its vertex set, $N_G(x)$ its open neighborhood at x, $\alpha(G)$ its largest independent-set cardinality, and $\nu(G)$ its largest matching cardinality. A matching has pairwise disjoint endpoints, and $V(M)$ denotes its covered vertex set. It is maximal if no edge can be added and perfect if it covers every vertex. A graph is factor-critical if it is nonempty and every one-vertex deletion has a perfect matching; it is equimatchable if every maximal matching is maximum. All vertex deletions and all subgraphs specified by vertex sets are induced.*

Let $D \ge 6$ be an even integer. Let G be a connected D-regular factor-critical equimatchable graph of order n, with $\alpha = \alpha(G) \ge 3$. Let I be any maximum independent set, v any vertex of I, and Z any perfect matching of G-v. Let M_1 be the edges of Z meeting I, $W = V(M1)$ minus I, $M2 = Z$ minus M_1, and $T = V(M2)$. For $i = 0, 1$, let $M2i$ consist of edges xy in M_2 for which $\max(|N_G(x)intersectW|, |N_G(y)intersectW|) = i$. Put $M22 = M2$ minus $(M20$ union $M21)$ and $T'' = V(M20)$. For w in W, let T_w consist of w and the endpoints of all edges in M_{21} having an endpoint adjacent to w.

Then $\alpha \le (n-1)/2$, and for every matching J of G,

$$\alpha(G - V(J)) \le (n - 2|J| + 1)/2.$$

The matching M_2 is maximum in G-I; W is independent of size alpha-1; and G[T union U] has no perfect matching for any $U \subseteq W$ with $|U| \ge 2$.

At most one vertex of T has at least two neighbors in W. The mate in M_2 of any such vertex has no W-neighbor. In particular $|M22| \le 1$. The vertices of T having no W-neighbor form a clique; T'' is an even clique with $|T''| \le D$, allowing the empty clique.

Every M_{21} edge belongs to exactly one T_w. These sets are pairwise disjoint connected odd sets, each anticomplete to W minus $\{w\}$, and $\alpha(G[T_w unionT'']) \le 2$. For distinct $w1$, $w2$, neither $G[T_w1$ union $T_w2]$ nor $G[T_w1$ union T_w2 union $T'']$ has a perfect matching. At most one $G[T_w]$ is noncomplete, and any two distinct complete blocks $G[T_w]$ are anticomplete.

All degrees, orders, matching and independence cardinalities, and indices are integers in their displayed domains. Empty M_{20}, M_{21} or M_{22} and singleton T_w are permitted. No assertion that M_{21} union M_{22} is nonempty is included.

Proof. Use Lemma 26 only for its matching and cut conclusions for connected odd graphs of independence number at most two.

Factor-criticality gives the perfect matching Z, odd n and $\nu(G) = (n-1)/2$. Every matching extends by adding disjoint edges to a maximal matching, hence to a near-perfect matching by equimatchability. For any independent set A, its $D|A|$ incident edges go to its complement, whose vertices have total degree $D(n - |A|)$.

Thus $|A| \le n/2$, and odd n gives $\alpha \le (n-1)/2$. Extend any matching J to a near-perfect matching. Its added edges form a near-perfect matching of G-V(J). An independent set of that residual has at most one endpoint from each added edge and at most its one uncovered vertex. This proves $\alpha(G - V(J)) \le (n - 2|J| + 1)/2$.

Since I is independent, Z pairs every vertex of I minus {v} to a distinct vertex outside I. Thus $|M1| = |W| = \alpha - 1$, and $n = 2\alpha - 1 + 2|M2|$. If a matching J in G-I were larger than M_2, then G-V(J) would retain I and have at most 2alpha-3 vertices. The preceding inequality would bound its independence number by alpha-1, a contradiction. Therefore M_2 is maximum in G-I. An edge inside W could be added to M_2, so W is independent. A perfect matching on T union U, for $U \subseteq W$ with $|U| \ge 2$, would have more than $|M2|$ edges in G-I and is likewise impossible. In particular there is no alternating path relative to M_2 with distinct endpoints in W and with its first and last edges outside M_2: reversing its edges would enlarge M_2.

If xy belongs to M_2 and x, y have distinct respective neighbors w, w' in W, then w, x, y, w' is such an augmenting path. Hence if both endpoints have W-neighbors, each has the same unique W-neighbor. The mate of an endpoint having at least two W-neighbors therefore has none.

Suppose distinct endpoints $x1$, $x2$ each have at least two W-neighbors. They have distinct mates $y1$, $y2$, neither of which has a W-neighbor. Choose distinct $w1$, $w2$ in W adjacent to $x1$, $x2$ respectively. If y1y2 were an edge, the path $w1$, $x1$, $y1$, $y2$, $x2$, $w2$ would augment M_2. Therefore y1y2 is absent. Then W union {y1,y2} is an independent set of size $\alpha + 1$, a contradiction. This proves uniqueness of the possible endpoint and $|M22| \le 1$.

Any two nonadjacent vertices of T without W-neighbors could similarly be adjoined to W to exceed α. These vertices form a clique. In particular T'' is a clique of even order, since it consists of the endpoint sets of matching edges. Every vertex outside I has an I-neighbor, since otherwise it could be adjoined to the maximum independent set I. Thus any vertex of a nonempty T'' has $|T''| - 1$ neighbors inside it and at least one outside, giving $|T''| \le D$. The empty set satisfies the bound as well.

Each pair xy in M_{21} has a W-neighbor and at most one at each endpoint. The preceding three-edge augmentation shows that all such neighbors are the same vertex w. Thus each pair is assigned to exactly one T_w and is anticomplete to W minus {w}. Distinct pairs have disjoint endpoints. Consequently the T_w are pairwise disjoint, have odd order, and are connected: each assigned pair has its matching edge and an edge to its root w. The vertices of T'' have no W-neighbor. An independent triple in T_w union T'' could therefore be adjoined to W minus {w}, whose size is alpha-2, to exceed α. This proves its independence bound.

A perfect matching of T_w1 union T_w2 would replace the M_2 edges inside these blocks and cover their two previously uncovered roots. Keeping all other M_2 edges would enlarge M_2 by one edge. The same replacement applies when T'' is included in the union, replacing its M_{20} edges too. Thus neither union has a perfect matching. If two complete blocks had an edge between them, use that edge and perfectly match the two even cliques obtained by removing its endpoints. This would perfectly match their union. Distinct complete blocks are therefore anticomplete.

Finally suppose $H1 = G[T_w1]$ and $H2 = G[T_w2]$ are both noncomplete. They are connected odd graphs of independence number exactly two. Their union has independence number at most three: four independent vertices in it, together with W minus {w1,w2} of size alpha-3, would exceed α. In particular an edge joins the two blocks, since otherwise one independent pair from each would violate this bound.

By Lemma 26, a 2-connected H_i is factor-critical. If H_i is not 2-connected, that fact supplies a cut vertex c_i such that $H_i - x$ has a perfect matching for every x different from c_i and $H_i - c_i$ consists of two nonempty anticomplete cliques. If both H_i are 2-connected, any interedge and perfect matchings after deleting its endpoints give a forbidden perfect matching of their union. If only H_1 is not 2-connected, every interedge must be incident to $c1$; an edge avoiding $c1$ would give the same perfect matching. After deleting $c1$ there are therefore no edges between H1-c1 and H_2. Each has an independent pair, contradicting the union independence bound of three. If both are not 2-connected, every interedge must meet $c1$ or $c2$, by the same perfect-matching argument. After deleting $c1$, $c2$ the two remaining graphs are anticomplete and each has an independent pair, the same contradiction. Interchanging the block labels handles the case that only H_2 is not 2-connected. Thus at most one block is noncomplete, completing every conclusion. □

Remark. A near-perfect extension bounds independence in every matching residual. The partner set is then exposed by a maximum matching, and forbidden augmentations restrict all remaining pairs to odd rooted blocks and at most one endpoint with several partner neighbors.

Depends on. Lemma 26.

(printed in plain notation)

Lemma 78 (Exact cover costs after a matching deletion). Exact cover costs after a matching deletion.

Let D>=3 be an integer and H a finite simple graph with maximum degree at most D. A matching is an edge set with disjoint endpoints. Write S={v:deg_H(v)=D}. Choose a matching W minimizing |S minus V(W)| over all matchings, where V(W) is its endpoint set. Put I=S minus V(W), rho=|I| and Q=H-W, deleting only edges. For every integer x>=0 define

g_D(x)=x-floor(x/(D-1)).

For a nonempty star forest J with positive integer component edge sizes m_1,...,m_t and every integer c>=0 define

r_J(c)=max{|B|:B subseteq {1,...,t}, sum_(j in B)m_j<=c}.

The empty index set is feasible. Write K2 for one edge. All target components are vertex-disjoint, copies are ordinary subgraphs, and arrowing prescribes the displayed matching red and the full forest blue.

First, suppose Q arrows (pK2,J), with integer p>=1. For every vertex cover C of Q containing I, put N_J=p+t-1 and c=|C|-N_J. Then c is a nonnegative integer and

g_D(max(0,rho-p+1))<=c+r_J(c)<=2c.

If every component of J has at least h edges, with integer h>=1, the last upper bound can be replaced by c+floor(c/h).

Now let calF be a finite nonempty indexed family of pairs (p_a K2,J_a), with positive integers p_a,t_a and positive nonincreasing integer sizes m_1^a,...,m_(t_a)^a for J_a. Suppose Q arrows every family member. Define

N=max_a(p_a+t_a-1),

lambda_k=max{m_j^a:a in calF, 1<=i<=p_a, 1<=j<=t_a, i+j=k+1}

for integers 1<=k<=N, L=sum_(k=1)^N lambda_k, and sigma=e(Q)-L, where e denotes edge count. For every integer r>=0 put

A_D(r)=sum_(k=1)^min(r,N)(D-lambda_k)+(D-1)max(r-N,0).

All displayed maxima have nonempty finite feasible sets. Then

sigma>=A_D(rho)>=0,

and there is an actual vertex cover C of Q containing I such that

N<=|C|<=N+sigma-A_D(rho).

For each selected member (pK2,J) with t components put g=N-(p+t-1) and s=sigma-A_D(rho). These are nonnegative integers, and

g_D(max(0,rho-p+1))<=s+g+r_J(s+g)<=2(s+g).

If all sizes of J are at least an integer h>=1, the last upper bound can instead be s+g+floor((s+g)/h). In particular a maximum-budget member has the weaker bound sigma+r_J(sigma)<=2sigma on the right, and the same constructed cover has size at most N+sigma.

The cover is constructed by choosing the exposed vertices first in a greedy maximum-degree deletion when rho<=N, followed by one endpoint of each remaining edge. If rho>N, use all of I followed by one endpoint of each edge of Q-I. W need not have maximum cardinality. If it does, then e(Q)=e(H)-nu(H), where nu is maximum matching cardinality. Unit stars, repeated components, rho=0, zero surplus, unequal family budgets and arbitrary host components are included. All parameters, degrees, indices, cardinalities, packing values and surplus quantities in the conclusions are integers.

Proof. The function r_J is nondecreasing because its finite feasible sets increase with c. Each selected component costs at least one edge, so r_J(c)<=c; if every size is at least

h then r_J(c)<=floor(c/h). The integer function g_D is nondecreasing, and increasing its argument by one increases its value by zero or one.

We first prove a crossing-matching bound for any cover C containing I. Put C_0=C minus I, O=V(H) minus C, and let B be the bipartite graph of Q-edges between C and O. Suppose a matching U of B covers k vertices of I and all but z vertices of C_0. We claim g_D(k)<=z.

Augment U to maximum cardinality in B preserving all already covered vertices. This is possible because the symmetric difference of a nonmaximum matching with a larger one has an alternating path whose endpoints are uncovered by the smaller matching; switching along it increases size and preserves prior saturation. Repeating terminates. Among maximum matchings covering all the C-vertices initially covered by U, choose T maximizing coverage of S intersect O. This is a nonempty finite choice set. Write r=|I intersect V(T)|>=k and let h count the T-uncovered vertices of S intersect O. At most z C_0-vertices are uncovered.

Every S-vertex in O is W-covered, so its Q-degree is D-1. Every C_0-vertex has Q-degree at most D-1, while each I-vertex has Q-degree D. Indeed the degree-D vertices of Q are exactly I. If h>0, direct non-T edges of B from O to C and T-edges from C to O. Start reachability at all h uncovered S-vertices in O, and denote the reached sets by O' and C'.

Every reached C-vertex is T-covered, since otherwise a shortest directed path to one would augment T. Every reached O-vertex is in S: a shortest path from an uncovered start to an O-vertex outside S would end with its matching edge; switching would preserve cardinality and all covered C-vertices, cover the start and uncover only that outside-S endpoint. This would improve the defining objective of T.

The T-edges pair C' with O' minus the h starts, giving |O'|=|C'|+h. Every Q-neighbor of O' is in C': O is independent because C covers Q; nonmatching edges are traversed, and each matching neighbor was already reached. Counting these edges gives

(D-1)(|C'|+h)<=(D-1)|C'|+|C' intersect I|<=(D-1)|C'|+r.

Here every reached C-vertex is matched, so |C' intersect I|<=r. Thus h<=floor(r/(D-1)), also valid when h=0. The matching T leaves at most rho-r+z+floor(r/(D-1)) vertices of S uncovered. Since W minimizes this count among all H-matchings and T is an H-matching, rho is at most that number. Consequently g_D(r)<=z, and monotonicity with r>=k proves the claim.

We next prove the first assertion for a full pair (pK2,J). Any cover C of Q has size at least p+t-1. Otherwise partition it into C_R,C_B with |C_R|<=p-1 and |C_B|<=t-1. Color all edges meeting C_R red and all others blue. Red matchings have at most p-1 edges; every blue edge meets C_B, so blue matchings have at most t-1 edges. The full forest J contains t disjoint edges, contradicting arrowing. This proves c>=0.

Put m=max(0,rho-p+1). If m=0 the required inequality follows from nonnegativity. If m>0, choose A subseteq I of size p-1 and color all Q-edges meeting A red and all others blue. Arrowing supplies a full blue J-copy, since the red graph has cover A. This copy avoids every vertex of A, as its vertices all have positive blue degree.

Put Z=C minus A, so |Z|=t+c. For each copied star let q_j be its number of C-vertices. Every copied star meets C, and the components are disjoint and avoid A, so q_j>=1 and sum_j(q_j-1)<=c. A copied star having no edge between C and O lies entirely in C: it is connected, has an edge and meets C. Such a star j uses q_j=m_j+1 C-vertices. The sum of m_j over all these internal stars is at most c, so at most r_J(c) copied stars are internal.

Choose one crossing edge from each noninternal star. Their union U is a matching of B, since the copied components are disjoint. It covers one vertex of Z per noninternal star and leaves at most c+r_J(c) vertices of Z uncovered. Let ell be the number of uncovered vertices in I minus A and z the number uncovered in C_0. These sets partition Z, so ell+z<=c+r_J(c). The matching U covers k=m-ell vertices of I and all but z vertices of C_0. The crossing bound gives g_D(k)<=z. Since k<=m and m=k+ell,

g_D(m)<=g_D(k)+ell<=z+ell<=c+r_J(c).

The initial packing bounds prove the other single-pair conclusions. This argument used an arbitrary cover containing I.

For the family, a member attaining N has valid index sums at every coordinate 1 through N, so each lambda_k is defined and attained. The all-blue coloring of Q contains every full J_a; hence all its component sizes are at most Delta(Q)<=D. Thus 1<=lambda_k<=D and A_D(r)>=0 for every integer r>=0. Applying the preceding cover lower bound to a member attaining N proves that every cover of Q has size at least N.

We verify the degree requirement for the greedy construction. After any j<N vertices have been deleted from Q, select a family member a and indices i,l attaining lambda_(j+1), so i+l=j+2. The legal cuts i-1 and l-1 total j and leave two nonempty targets. Apply the vertex-deletion assertion of Lemma 34 to this pair and this actual j-vertex set. The residual arrows its full suffix pair. In its all-blue coloring it contains the blue suffix with leading size m_l^a=lambda_(j+1), so its maximum degree is at least this positive integer. Thus every next greedy step exists through N deletions, independently of earlier choices.

The set I is independent in H: an edge between two of its vertices could be added to W to cover two more S-vertices, contradicting exposure minimality. Every I-vertex has Q-degree D and every other Q-vertex has degree at most D-1. Deleting I-vertices does not lower the degrees of the remaining I-vertices.

If rho<=N, choose all rho vertices of I first, then continue by maximum-degree choices to N vertices. These first rho choices are greedy, and the subsequent degrees are at least lambda_(rho+1),...,lambda_N. If Y is the resulting N-element set, disjoint counting of edges at their earliest deleted endpoint gives

e(Q-Y)<=e(Q)-D*rho-sum_(k=rho+1)^N lambda_k

=sigma-A_D(rho).

Adjoin to Y one endpoint of each residual edge. The resulting cover contains I and has size at most N+sigma-A_D(rho); repeated endpoint choices only decrease its size.

If rho>N, take I and one endpoint of each edge of Q-I. Independence gives e(Q-I)=e(Q)-D*rho. The resulting cover contains I and has size at most

rho+e(Q-I)=e(Q)-(D-1)rho=N+sigma-A_D(rho),

using A_D(rho)=DN-L+(D-1)(rho-N). In either case every cover has size at least N, proving sigma>=A_D(rho) and the claimed cover interval, including rho=N and rho=0.

Fix any member of budget N-g. Then g>=0 by the definition of N. For the constructed cover write c=|C|-(N-g), which lies between g and s+g where s=sigma-A_D(rho)>=0. The single-pair result gives

g_D(max(0,rho-p+1))<=c+r_J(c)<=s+g+r_J(s+g).

The second inequality is monotonicity; the stated packing bounds then follow. Since A_D(rho)>=0, one has s<=sigma. For g=0 this also gives the weaker right side sigma+r_J(sigma), and every constructed cover has size at most N+sigma. Finally, if W is maximum, deletion of exactly its |W|=nu(H) edges gives e(Q)=e(H)-nu(H). Every construction retained the originally specified W.

□

Remark.

First pay the exact degree cost of the exposed roots. The remaining cover surplus can hide one uncovered crossing endpoint per extra cover vertex, plus one for each full internal star whose entire edge size fits that surplus.

Depends on. Lemma 34.

Lemma 79 (Exclusion of the three-triangle octic frame). *Use the integer certificate costs $c_\theta(H)$ of Lemma 23. There is no finite simple 8-regular graph H on 29 vertices satisfying all the following conditions. First, $c_2(H) \geq 18$ and every maximal independent set of H has exactly eight vertices, where independent means containing no adjacent pair and maximal means not properly contained in another independent set. Second, $V(H)$ has a partition R disjoint union {x} disjoint union A such that each of H[R] and H[A] is the disjoint union of three triangles and five isolated vertices, and x has exactly four neighbors in each of R and A. A triangle has three vertices and all three possible edges;*

the indicated isolated vertices have no neighbors within their own side. Third, for every independent set S of eight vertices, every complete sequence that starts with H-S and repeatedly deletes a vertex of current maximum degree while that maximum degree is at least three has at least six deletions, and the degrees of its first six deleted vertices are, in order, 7, 7, 7, 6, 6, 6. Complete means that the sequence stops precisely when the remaining maximum degree is at most two, with the empty graph assigned maximum degree zero. All orders, cardinalities, degrees and sequence lengths are integers.

Proof. Suppose such H exists. Every independent set extends to a maximal independent set by repeatedly adjoining a vertex not adjacent to any already chosen vertex. Finiteness makes this procedure terminate. The maximal-set hypothesis therefore implies that every independent set has size at most eight and every such set extends to one of size eight. Also $c_1(H) \geq c_2(H) \geq 18$ by the monotonicity in Lemma 23.

We first prove that distinct nonadjacent vertices u, v have $|N_H(u)$ union $N_H(v)| \geq 11$, where N_H denotes open neighborhood. Suppose their union has size $m \leq 10$. Since both degrees are eight, $8 \leq m \leq 10$. If $m = 8$, take an empty matching M. If $m = 9$ or 10, each of the two neighborhood differences is nonempty. Choose p in $N_H(u)$ minus $N_H(v)$ and q in $N_H(v)$ minus $N_H(u)$, and put $M = \{up, vq\}$. Its edges are disjoint, since the differences are disjoint and neither contains u or v. In H-M the neighborhood union of u, v has size $m - 2 \leq 8$; for $m = 8$ the same conclusion holds without deleting edges. Enlarge it to a set W of exactly eight vertices avoiding u, v. Put M in the degree-one part Q_R, leave Q_B empty, put the remaining edges incident with W in P_R and all other edges in P_B. Then $\nu(P_R) \leq 8$. The vertices u, v are isolated in the graph with edge set P_B, whose edges are supported on at most $29 - 8 - 2 = 19$ vertices, so $\nu(P_B) \leq 9$. Swapping P_R, P_B gives both allocations (8,9) and (9,8) at capacities (1,0).

The allocations (7,10), (10,7) are always available. Choose an edge vz, put $M = \{vz\}$ and $W = N_H(v)$ minus {z}, of size seven. Put M in Q_R and partition all remaining edges between those meeting W and those not meeting W. Their matching numbers are at most seven and $floor((29 - 7 - 1)/2) = 10$, because v is isolated in the latter part. Swapping the exceptional parts gives both orders. For any other allocation of total 17, the larger budget b is at least 11 and the smaller budget a satisfies $a + b = 17$. Put all edges inside a chosen set of $\min(29, 2b + 1)$ vertices in the part with budget b and all remaining edges in the part with budget a. Their matching numbers are at most b and $\max(0, 29 - 2b - 1) \leq a$, respectively; the last inequality follows from $29 \leq a + 2b + 1 = 18 + b$. The bounded-degree parts may be empty. Thus every allocation of total 17 has a (1,0)-certificate. Capacity symmetry gives the same for (0,1), so $c_1(H) \leq 17$, a contradiction. We have proved the neighborhood-union bound.

Fix the supplied partition R, {x}, A. Take any transversal P consisting of one vertex from each R-triangle. In each triangle designate its two remaining vertices as one S-vertex and one T-vertex, and put all five R-isolates in S. Then S is independent of size eight. The three T-vertices are pairwise nonadjacent, and each has exactly one S-neighbor. Because S is maximal, every vertex outside S has at least one S-neighbor, so H-S has maximum degree at most seven. The T-vertices can therefore be deleted successively as its first three maximum-degree vertices, each of degree seven. The remaining graph is $J = H[A$ union {x} union P]. Each P-vertex has degree six, each A-vertex has degree at most $2 + 1 + 3 = 6$, and x has degree $4 + r$, where $r = |N_H(x)$ intersect $P|$. If $r = 3$, x supplies a fourth maximum-degree deletion of degree seven, contrary to the required first six values. If $r = 2$, delete x at maximum degree six. Two P-vertices now have degree five and the third still has degree six; each A-vertex now has degree at most $2 + 3 = 5$. Delete that remaining degree-six P-vertex. The maximum degree is now at most five, contradicting the required sixth deletion of degree six. Any partial deletion sequence used here can be extended to a complete one by the stated finite greedy rule. Therefore $r \leq 1$ for every transversal P. All triangle neighbors of x in R consequently lie in at most one R-triangle: neighbors in two triangles could both be selected in one transversal.

Let t be the number of R-triangle neighbors of x. Then $0 \leq t \leq 3$. We exclude $t = 2$ using c_2. In that case x is joined to two vertices a, b of one triangle abc and to two R-isolates l_1, l_2. The component L containing x in H[R union {x}] has the seven edges ab, ac, bc, xa, xb, xl_1, xl_2. It has $c_2(L) \leq 1$. For capacities (2,0) or (0,2), put the triangle in the degree-two part and all four x-edges in the exceptional part that has its unit of matching budget. For capacities (1,1), take $Q_R = \{xl_1, ac\}$, $Q_B = \{xl_2, bc\}$; each is a matching. Put the remaining triangle {xa, xb, ab} in whichever exceptional part has budget one. This covers both allocations at total one for every capacity split of total two. Each of the other two triangle components has $c_2 \leq 1$: for capacities (2,0) or (0,2) its edges fit in the degree-two part, and for (1,1) put one edge in each degree-one part and the last edge in the exceptional part with budget one. Isolates have cost zero. Subadditivity and vertex extension in Lemma 72 therefore give $c_2(H[Runion\{x\}]) \leq 3$ and $c_2(H) \leq 3 + |A| = 17$, contradicting $c_2(H) \geq 18$. It remains that t is 0, 1 or 3.

Call a vertex of A available to an independent set I if it has no neighbor in I. In the next three cases the chosen I dominates R union {x}, meaning each vertex there outside I has a neighbor in I. Every extension of I to a maximal independent set must therefore add its vertices in A, all available to I. Since each maximal independent set has size eight, this gives a lower bound of $8 - |I|$ on the number of available vertices for every indicated choice.

If $t = 0$, let z be the unique R-isolate not adjacent to x. For every one of the 27 transversals P of the three R-triangles, $I = \{x, z\}$ union P is independent of size five and dominates R union {x}. At least three A-vertices are available. They all belong to $B = A$ minus $(N_H(x) union N_H(z))$. All eight neighbors of z lie in A, so $|B| \leq 6$. A vertex v of B avoids x, z and has internal A-degree zero or two, so it has at least six R-neighbors. At most four are R-isolates other than z; thus at least two are triangle vertices. Such a vertex avoids at most 12 of the 27 transversals: two forbidden vertices in distinct triangles leave at most $2 \cdot 2 \cdot 3 = 12$ choices, and two in the same triangle leave at most $1 \cdot 3 \cdot 3 = 9$. Additional forbidden vertices cannot increase this count. Counting pairs consisting of a transversal and an available vertex gives at most $6 \cdot 12 = 72$ pairs, whereas the required three available vertices per choice give at least $27 \cdot 3 = 81$, a contradiction.

If $t = 1$, let a be the triangle neighbor of x and let z, w be the two R-isolates not adjacent to x. Select P using a vertex other than a in its triangle and one representative from each other R-triangle. There are 18 choices. Each $I = \{x, z, w\}$ union P is independent of size six and dominates R union {x}, so it has at least two available A-vertices. These belong to $B = A$ minus $(N_H(x) union N_H(z) union N_H(w))$. The nonadjacent z, w have all their neighbors in A, and the proved neighborhood-union bound gives $|B| \leq 14 - 11 = 3$. Each v in B has at least six R-neighbors, avoids z, w, and has at most four R-neighbors outside the eight selectable triangle vertices: the other three R-isolates and a. Thus at least two selectable vertices are forbidden. The three counts of allowable representatives have respective caps 2, 3, 3 and sum at most six; their product is at most eight. To check this, a zero count gives zero; a first count one leaves the product of the last two at most six, and a first count two leaves their sum at most four and product at most four. Each available vertex therefore covers at most eight choices. The total is at most $3 \cdot 8 = 24$, less than $18 \cdot 2 = 36$, a contradiction.

Finally let $t = 3$. The vertex x meets one whole R-triangle and one R-isolate y. Let Z be the other four R-isolates. All their neighborhoods lie in A and have size eight. Their union has at least 12 vertices. Indeed, if its size were at most 11, the neighborhood-union bound for each nonadjacent pair would force the union to have size 11 and every pair of these four neighborhoods to have the whole union. Their complements inside that 11-set would be four pairwise disjoint sets of size three, requiring 12 vertices, a contradiction. Choose one representative from each of the two R-triangles not adjacent to x, with nine choices P. Each $I = \{x\}$ union Z union P is independent of size seven and dominates R union {x}, so some A-vertex must be available. All possible such vertices lie in $B = A$ minus $(N_H(x) union N_H(Z))$, with $|B| \leq 2$. A vertex v in B has at least six R-neighbors and avoids Z. Outside the six selectable vertices it can meet only the triangle adjacent to x and y, at most four vertices. Thus it has at least two neighbors among the selectable vertices. The allowable counts in the two triangles have caps three and sum at most four, so their product is at most four. Each v is available for at most four choices. There are at most $2 \cdot 4 = 8$ available-vertex incidences, less than the required nine. This contradiction excludes the final case and proves that H does not exist. □

Remark. The deletion invariant restricts the center's triangle contacts. Small certificate partitions exclude the two-contact case, and counting extensions of independent transversals excludes the remaining contacts.

Depends on. Lemma 23, Lemma 72.

Lemma 80 (Count-based binary remnants and component avoidance). *All graphs are finite and simple. Let S, $T \geq 1$ be integers, and let F, J be star forests with complete nonincreasing positive integer edge-size lists $n_1,..., n_S$ and $m_1,..., m_T$. Copies are ordinary subgraphs with vertex-disjoint target components; F is prescribed red and J blue. Their heads are $B = n_1 \geq 4$ and $A = m_1 \geq 4$. Write e for edge count and ν for maximum matching cardinality. Put*

$U = e(F) - S - (B - 1)$, $W = \sum_{j=1}^{T} \max(m_j - 2, 0)$.

Let μ be an integer satisfying

$\mu \geq S + B + U$, $T \geq \mu + 3$, $W \leq \mu - B - U - 4$.

Define

$h = \max(A, B)$, $r = \max(A - B, 0)$, $a = \min(A, B)$,

$\Gamma = (r+1)U + (r+2)B + r + a + 6$,

$Q = |\{j : m_j \geq 2\}|$, $v = T - Q$,

$b_2 = floor((2\mu+1)/3)$, $\rho = \mu - b_2$,

$b_0 = \min(\mu, floor((2\mu+1+v)/3))$.

Write $K2 = K_{1,1}$. Define the full remnant $J.$ by

$J. = (T - b_0)K2$ *if* $Q \leq b_2$,

and $J. = (Q - b_2)K_{1,2}$ *disjoint union max(v-rho,0)K2 if* $Q > b_2$.

For any nonempty positive star forests X, Y with complete sorted sizes x_1,..., x_s and y_1,..., y_t, define

$\Pi(X,Y) = \sum_{k=2}^{s+t} max\{x_i + y_j - 2 : 1 \leq i \leq s, 1 \leq j \leq t, i+j = k\}$.

Then $J.$ is nonempty and componentwise contained in J, and

$0 \leq \Pi(F,J) - \Pi(F,J.) \leq (h+1)\mu + b_2 - \Gamma$.

Superscript β means deletion of β leading whole components. For every nonnegative integer β smaller than the component count of $J.$, and all disjoint graphs C, Q_0 with $|V(C)| \leq 2\mu+1$ and $\nu(C) \leq \mu$, every copy of J^{β} in C disjoint union Q_0 leaves a copy of $J.^{\beta}$ entirely in Q_0.

Put $M = S + T - 1$ and $\ell_k = max\{n_i + m_j - 1 : 1 \leq i \leq S, 1 \leq j \leq T, i+j = k\}$. For every integer $0 \leq q < M$ let $P_q = \sum_{k=q+2}^{S+T}(\ell_k - 1)$. Define $P.q$ by the same natural-tail formula for the new full pair $(F, J.)$, taking $P.q = 0$ if q is at least its component budget. Then

$0 \leq P_q - P.q \leq (h+1)\mu + b_2 - \Gamma$.

Let $D \geq 2$ be even and $z \geq 0$ be integers. Suppose H has a whole connected odd D-regular component C with a near-perfect matching and $\nu(C) = \mu$. Put $\Phi(H) = e(H) - \nu(H)$. If

$\Phi(H) \leq P_q + z$,

$b_2 \leq (D - h - 2)\mu + D/2 + \Gamma - z - 1$,

then H has a coloring avoiding both targets of a legal actual suffix pair (F^{α}, J^{β}), with $\alpha + \beta = q$, $0 \leq \alpha < S$ and $0 \leq \beta < T$. This coloring can be chosen with C wholly blue. It extends over the disjoint canonical q-star pool of sizes ℓ_2,..., ℓ_{q+1}, and across every specified actual q-vertex set in any outer graph whose deletion gives H, preserving every H-edge color and avoiding both full originals. No degree bound outside C is required. All sizes, orders, counts, thresholds, cuts and allowances are integers in their displayed domains; all original units and repetitions are retained.

Proof. The hypotheses give $U \geq 0$, $\mu \geq S + B + U \geq 5$, and $0 \leq b_2 \leq \mu$. Thus $\rho \geq 0$, $0 \leq b_0 \leq \mu$, and $b_0 \geq b_2$. All defined counts are integers. In the matching case $T - b_0 \geq 3$. In the other case $k = Q - b_2 \geq 1$ and $l = \max(v - \rho, 0) \geq 0$; moreover $k \leq Q$ and $k + l \leq T$. These observations prove nonemptiness and componentwise containment: choose k distinct original nonunits and trim them to two edges, then choose l further components and trim them to one edge. A matching remnant is obtained by choosing its required number of distinct original components and one edge in each.

We first prove the every-prefix property for this same remnant. A positive star in the disjoint union C and Q_0 lies wholly in one part because it is connected. The components lying in C number at most μ, since one edge from each gives a matching, and their nonunits number at most b_2, since each such component occupies at least three vertices. In the matching case a copy of J^{β} has at most v unit components. If c of its components lie in C, their vertex count is at least 3c-v. Hence $c \leq floor((2\mu+1+v)/3)$ and $c \leq \mu$, so $c \leq b_0$. At least $T - \beta - b_0$ components remain outside. Selecting one edge from each of the required components gives $J.^{\beta}$.

In the binary case use $k = Q - b_2$ and $l = \max(v - \rho, 0)$. If $\beta \leq k$, then $\beta \leq Q$ and J^{β} has Q-beta nonunits. At least k-beta lie outside C, and at least T-beta-mu total components lie outside. When $l > 0$, $k + l = T - \mu$, so these two bounds supply k-beta nonunits and l additional distinct components. Trim them to the required stars. When $l = 0$, only the nonunit bound is needed. If $\beta > k$ and $J.^{\beta}$ is nonempty, then $l > 0$ and that suffix

is a matching of $k+l-\beta = T-\mu-\beta$ edges. The total-component bound supplies it. The case $\beta = k$ is included. This proves the assertion for every admissible cut and every permitted pair C, Q_0.

We now prove the complete-profile estimate. For nonnegative integer thresholds x, z define $R_x = |\{i : n_i - 1 \geq x\}|$ and $B_z = |\{j : m_j - 1 \geq z\}|$; thus $R_0 = S$ and $B_0 = T$. At a positive excess level w the number of complete diagonal entries at least w is

$V_w = max\{R_x + B_z - 1 : x, z \geq 0 integers, x + z = w, R_x B_z > 0\}$,

with maximum zero if there are no candidates. A positive threshold rectangle realizes every index sum from two through $R_x + B_z$. Conversely an entry with $n_i + m_j - 2 \geq w$ dominates thresholds $x \leq n_i - 1, z \leq m_j - 1$ summing to w: take $x = \min(n_i - 1, w)$, $z = w - x$. The union of all these initial intervals gives the claimed width. Counting every nonnegative diagonal value by its positive integer levels yields $\Pi(F, J) = \sum_{w \geq 1} V_w$.

For $1 \leq w \leq B-1$ let f_w count red components of size w, let $F_w = \sum_{j=1}^{w} f_j$, and let $d_w = T - B_w$ count blue components of size at most w. Then $d_1 = v$, $R_w \geq 1$, $R_{w-1} - R_w = f_w$, and $\sum_{w=1}^{B-1}(R_w - 1) = U$. A candidate with $x \geq 1$, $z \geq 2$ has $R_x \leq U + 1$ and $B_z \leq W$, so its width is at most $U + W \leq \mu - B - 4 < T$. At these levels it is dominated by $R_w + T - 1 \geq T$. The only remaining candidates use $z = 0$, $z = 1$, or $x = 0$. Their maximum is exactly

$V_w = T + R_w - 1 + \Psi_w$,

$\Psi_w = \max(0, f_w - v, F_w - d_w)$.

If $w > A - 1$ the $x = 0$ candidate is absent, but $F_w - d_w = F_w - T < 0$, so its inclusion in this maximum is harmless. Put $\Psi = \sum_{w=1}^{B-1} \Psi_w$ and $\Omega(l) = \sum_{w=1}^{B-1} \max(f_w - l, 0)$ for integers $l \geq 0$.

For $0 \leq l \leq v$, at $w = 1$ one has $\Psi_1 = \max(f_1 - v, 0) \leq \max(f_1 - l, 0)$. At $w \geq 2$, the inequality $F_w \leq S - 1 < T$ gives

$\Psi_w \leq \max(f_w - l, 0) + \max(B_w - 1, 0)$.

Indeed $f_w - v \leq f_w - l$, while $F_w - d_w = F_w - T + B_w \leq B_w - 1$. The extra positive term is possible only for $2 \leq w \leq \min(A-1, B-1)$; every B_w in that range is positive, and $\sum_{w=2}^{A-1} B_w = W$. Consequently

$\Psi - \Omega(l) \leq W - (a - 2)$. (1)

Also $\Omega(v) \leq \max(S - 1 - v, 0)$: if any terms are positive, summing the corresponding f_w subtracts v at least once and their total f_w is at most S-1. Therefore

$\Psi \leq \max(S - 1 - v, 0) + W - (a - 2)$. (2)

Let H_{hi} be the sum of widths at levels B,..., $A + B - 2$, writing a level as $B - 1 + w$ for $1 \leq w \leq A - 1$. Any candidate can be indexed by an integer j with $0 \leq j \leq \min(B - 1, A - w - 1)$: its red size threshold is B-j and its blue size threshold is $w + j + 1$. Its width is at most

$B_w + |\{i : n_i \geq \max(1, B - A + w + 1)\}| - 1$.

Remove one fixed B-head in the red count. Each remaining component of size n_i contributes for exactly $\max(A - B + n_i - 1, 0)$ choices of w. It follows that

$H_{hi} \leq Q + W + R_{hi}$,

$R_{hi} = \sum_{i=2}^{S} \max(A - B + n_i - 1, 0) \leq U + r(S - 1)$. (3)

Here $\sum_{w=1}^{A-1} B_w = Q + W$. The last inequality uses $\max(n_i - 1 + A - B, 0) \leq n_i - 1 + r$. The complete potential is

$\Pi(F, J) = (B-1)T + U + \Psi + H_{hi}$. (4)

For any binary forest $kK_{1,2}$ disjoint union lK2 with $k \geq 1$, the same width formula gives

$\Pi(F, kK_{1,2} disjoint union lK2)$

$= (B-1)(k+l) + U + k + (R_{B-1} - 1) + \Omega(l)$. (5)

To verify this, the low widths are $k + l + R_w - 1 + \max(f_w - l, 0)$ for $1 \leq w \leq B - 1$. The sole higher positive width is $R_{B-1} + k - 1$, at level B. For a nonempty matching lK2 the width formula instead gives

$\Pi(F, lK2) = (B-1)l + U$. (6)

If $Q > b_2$, use $k = Q - b_2$, $l = \max(v - \rho, 0)$. We have $0 \le l \le v$, $Q - k = b_2$ and $T - k - l \le \mu$. Subtract (5) from (4), apply (1) and (3), and discard the nonnegative term $R_{B-1} - 1$. This gives

$\Pi(F, J) - \Pi(F, J_\cdot) \le (B-1)\mu + b_2 + 2W + R_{hi} - (a-2)$. (7)

If $Q \le b_2$, subtract (6) with $l = T - b_0$ and use (2)-(4). The same bound (7) follows once

$(B-1)b_0 + \max(S - 1 - v, 0) \le (B-1)\mu$ (8)

is checked. If $v \ge \mu - 1$, its positive part vanishes because $S \le \mu - B - U$, and $b_0 \le \mu$ proves (8). Otherwise $x = \mu - v \ge 2$ is an integer, and the definition gives $\mu - b_0 = ceil((x-1)/3)$. Since $B - 1 \ge 3$,

$(B-1)(\mu - b_0) \ge x - 1$,

while $\max(S - 1 - v, 0) \le \max(x - B - U - 1, 0) \le x - 1$. This proves (8).

Substitute $W \le \mu - B - U - 4$, $R_{hi} \le U + r(S-1)$, and $S - 1 \le \mu - B - U - 1$ in (7). Since $h = B + r$, the resulting expression is

$(B-1)\mu + b_2 + 2(\mu - B - U - 4) + U + r(\mu - B - U - 1) - (a-2)$

$= (h+1)\mu + b_2 - [(r+1)U + (r+2)B + r + a + 6]$.

This is the claimed upper bound. Componentwise containment orders the sorted remnant sizes below their corresponding original sizes. Thus every new diagonal value, at the same index, is at most the original value, proving the lower bound.

For completeness the same estimate holds at every actual natural depth q. The full positive superlevel sets just proved are initial intervals. Removing q leading positions changes a width V to $\max(V - q, 0)$. For original width V and remnant width $V_\cdot$ with $0 \le V_\cdot \le V$, their truncated difference lies between zero and $V - V_\cdot$. Summing proves the stated natural-tail estimate.

These natural tails are also the coordinatewise envelopes of the complete actual total-q suffix families. Every local candidate with cuts $\alpha + \beta = q$ comes from full indices with index sum increased by q. Conversely any full indices i, j with $i + j \ge q + 2$ admit cuts $\alpha = \min(q, i-1)$, $\beta = q - \alpha \le j - 1$. They retain both components and yield that local index sum. Thus both directions of the envelope identity hold. When q is at least the full component budget no such indices or legal nonempty pair exists, and every truncated width is zero. This proves the assertion including exhaustion of the new family.

Finally impose the graph hypotheses. The selected component has $|V(C)| = 2\mu + 1$ and $e(C) = Dmu + D/2$, so $\Phi(C) = (D-1)\mu + D/2$. Matching number and edge count add over whole component unions. The scalar premise and the natural-tail bound give

$P_q - P_q^{\cdot} \le (h+1)\mu + b_2 - \Gamma$

$\le (D-1)\mu + D/2 - z - 1 = \Phi(C) - z - 1$.

Hence, for $Q_0 = H - V(C)$,

$\Phi(Q_0) = \Phi(H) - \Phi(C) \le P_q^{\cdot} - 1$.

Every graph has nonnegative potential because each matching is an edge subset. Thus $P_q^{\cdot} > 0$ and the new actual q-family is nonempty. Its envelope potential is $P_q^{\cdot}$, by the proved identity. The finite-family theorem Lemma 56 gives a coloring of Q_0 avoiding both targets of one new member $(F^\alpha, J_\cdot^\beta)$. Color C blue. A blue copy of J^β would leave a forbidden $J_\cdot^\beta$ outside C by the every-prefix property; every red edge lies outside C, where F^α is absent. These cuts are legal for the originals because $J_\cdot$ is componentwise contained in J.

We have obtained the asserted original actual-family avoidance with all C edges blue. The two extension conclusions follow from Lemma 41 (I), (II), using precisely these cuts and keeping every H-edge color. The first adjoins the stated disjoint canonical pool; the second applies to any specified actual set in an outer graph. Both include $q = 0$. All constructions and counts used the full original lists. □

Remark. Vertex and matching capacities give one remnant that survives every prefix. Exact low profile corrections and a single high profile count bound its cost. Subtracting the same regular component pays that cost and leaves a strict finite-family potential deficit.

Depends on. Lemma 41, Lemma 56.

Lemma 81 (Extremal components under matching-additive deletion). *Use the integer certificate costs $c_{A,B}$, row maxima c_i, matching number ν, excess $ex = c_0 - \nu$, and top missing-allocation run tr from Lemma 23. For nonnegative integer l write $Y_l(Q) = Y_l^{0,0}(Q)$ and run_l for the feasible allocation set and missing-run function of Lemma 75.*

Let H be any finite simple graph and X any vertex subset such that H-X has at least one vertex and $\nu(H) = |X| + \sum_Q \nu(Q)$, where Q ranges over all connected components of H-X, including isolated vertices. Put $b = |X|$, $m = ex(H)$, and $k = \max_Q \nu(Q)$. All cardinalities, capacities, costs, matching numbers, excesses and run lengths are nonnegative integers.

Every component Z with $\nu(Z) = k$ satisfies $ex(Z) \geq m$. If a component W has $ex(W) < m$, it is different from every such Z, and $k \geq b + \nu(W) + m + 1$. If a maximum-matching-number component Z satisfies $ex(Z) = m \geq 1$, then $tr(H) + b \leq tr(Z)$. No maximum-excess hypothesis on Z or the other components is required.

For any integer $T \geq 1$ define $w_T(Q) = |\{i : 1 \leq i \leq T, c_i(Q) > \nu(Q)\}|$. There exists a component Z maximizing $\nu(Z)$ and, among those attaining k, maximizing $w_T(Z)$. Fix any such Z. Simultaneously for every integer j with $1 \leq j \leq T$, if a component R satisfies $c_j(R) > \nu(R)$, $r := \nu(R) \geq b + c_j(Z) + 2$, then $r \leq k - 1$ and $c_j(Z) \leq k - b - 3$. If additionally $ex(R) < m$, then the stronger bound $c_j(Z) \leq k - 2b - m - 3$ holds. The same fixed Z satisfies every asserted threshold implication.

Proof. The graph H-X has a finite nonempty component family. Each component has a nonnegative integer matching number, so the maximum k exists. The subfamily attaining k is finite and nonempty, and w_T is an integer in [0,T], so its maximum on that subfamily exists. This proves existence of every selection used below. By Lemma 23 the sequence $c_i(Q)$ is nonincreasing, and $ex(Q) \geq 0$.

We use Lemma 75 throughout for its allocation sumsets, missing-run criterion, and actual restoration of all edges meeting X. Thus a sum of feasible component allocation sets, together with [0,b], supplies feasible allocations on H at the sum of their budgets. Internal component certificates have disjoint vertex supports. The X contribution is realized by taking an appropriate subset of X for the red exceptional edges and assigning the remaining incident edges blue. This assigns also edges inside X and does not change the internal bounded-degree parts. All intervals and sets of allocations below contain integers only.

Fix any Z with $\nu(Z) = k$. At row zero Z supplies the full interval $[0, c_0(Z)]$. Each other component Q supplies both endpoints $0, \nu(Q)$ at total $\nu(Q)$, by placing all its edges in one exceptional part. Together with [0,b], the initial full interval has length $L = b + c_0(Z) \geq k$. Every other $\nu(Q) \leq k \leq L$. Adding endpoints {0,nu(Q)} to [0,L] gives the union [0,L] union $[\nu(Q), L + \nu(Q)]$, which is $[0, L + \nu(Q)]$. This can be repeated for all components. Consequently $c_0(H) \leq b + c_0(Z) + \sum_{Q \neq Z} \nu(Q)$. Subtracting the assumed matching identity proves $ex(H) \leq ex(Z)$.

Now let W have rank $r = \nu(W)$ and excess $z = ex(W) < m$. Integrality gives $m \geq z + 1 \geq 1$. Excess dominance proves $W \neq Z$ for every rank-k component Z. Suppose, for a contradiction, that $k \leq b + r + m$. The graph W supplies a full allocation interval at total $c_0(W) = r + z$. Together with X this gives [0,L] with $L = b + r + z$. Put $l = k + m - 1 - z$, which is at least k. The endpoint and run estimates of Lemma 75 give $0, l$ in $Y_l(Z)$ and $run_l(Y_l(Z)) \leq \max(0, 2k - 1 - l) = \max(0, k - m + z) \leq L$. Its sumset criterion implies $[0, L] + Y_l(Z) = [0, L + l]$. This interval has length at least $l \geq k$, so every remaining component Q can be included at its endpoint budget $\nu(Q) \leq k$ by the preceding interval argument. The final total is $L + l + \sum_{Q \neq W,Z} \nu(Q) = b + r + k + m - 1 + \sum_{Q \neq W,Z} \nu(Q) = \nu(H) + m - 1 = c_0(H) - 1$. The sumset constructions give an actual certificate for every allocation at this total, contradicting minimality of $c_0(H)$ in Lemma 23. Hence $k > b + r + m$ and integrality proves the strict-excess rank separation.

We prove the asserted run contraction for a maximum-rank Z with $ex(Z) = m \geq 1$. Two elementary run estimates will be used. If A is an integer subset of [0,a] containing both endpoints and $b \geq 0$ is an integer, then $run_{a+b}(A + [0, b]) = \max(0, run_a(A) - b)$. To check this, list the elements of A in increasing order. The sumset is the union of intervals $[u, u + b]$ over this list. For consecutive elements $u < v$, the intervening missing interval has length $\max(0, v - u - b - 1)$, proving the equality on taking a maximum. When A has one element, necessarily $a = 0$ and both sides are zero. Next, for any integer $0 \leq w \leq a + 1$, $run_{a+w}(A + \{0, w\}) \leq run_a(A)$. For $w = 0$ the sets coincide. For $w > 0$ the integer a belongs to $A + \{0, w\}$, so no missing interval crosses a. A missing interval to its left lies in [0,a] and misses A. One to its right lies

in $[a+1, a+w]$, hence in $[w, a+w]$, and after translation by -w lies in [0,a] and misses A. Both lengths are bounded as claimed.

Set $l = c_0(Z) - 1 = k + m - 1 \geq k$ and $A = Y_l(Z)$. The endpoints 0, l are feasible by Lemma 75, and $run_l(A) = tr(Z)$ by definition. After adding [0,b], the first estimate bounds the missing run by max(0,tr(Z)-b). Add the endpoint sets {0,nu(Q)} for all $Q \neq Z$. Their shifts are at most k, while the current interval length is initially $l + b \geq k$ and never decreases. The second estimate applies at every step. The final set is therefore contained in $Y_{c_0(H)-1}(H)$, by the component and vertex-set inclusions, has the same interval endpoints, and has missing run at most max(0,tr(Z)-b). Here its total is $l + b + \sum_{Q \neq Z} \nu(Q) = \nu(H) + m - 1 = c_0(H) - 1$. Associativity and commutativity of sumset addition allow the X interval to be added first for the run estimates and last for the actual vertex-set extension. Taking a superset cannot enlarge missing runs on the same interval. Thus $tr(H) \leq \max(0, tr(Z) - b)$. Because $m \geq 1$, $c_0(H) > 0$ and Lemma 23 gives $tr(H) \geq 1$. The maximum on the right is positive, yielding $tr(H) + b \leq tr(Z)$.

Finally fix T and a component Z selected by the stated two successive maxima. Monotonicity of $c_i(Q)$ implies that the indices counted by $w_T(Q)$ form an initial interval. For an R satisfying the threshold hypotheses, maximal rank gives $r \leq k$. If $r = k$, then $c_j(R) > r$ gives $w_T(R) \geq j$. But $r \geq b + c_j(Z) + 2$ gives $c_j(Z) \leq k - b - 2 < k$, so $w_T(Z) \leq j - 1$. Replacing Z by R keeps the maximum rank k and strictly increases w_T, contrary to selection. Therefore $r \leq k - 1$, and $c_j(Z) \leq r - b - 2 \leq k - b - 3$. If $ex(R) < m$, strict-excess rank separation gives $r \leq k - b - m - 1$, which improves the same inequality to $c_j(Z) \leq k - 2b - m - 3$. The argument uses the one fixed Z for arbitrary j, so the implications are simultaneous. □

Remark. The largest matching component supplies enough allocation range to absorb every smaller component. A component of smaller excess must be much smaller in matching number; at equal excess, the separator's allocation interval shortens the top missing run.

Depends on. Lemma 23, Lemma 75.

Lemma 82 (Density loss along covered vertex deletions). *Use the finite integer certificate costs c_i, matching number ν, independence number α, edge count e, maximum degree Δ and slack $ps = e - \nu - \sum_{i \geq 1} c_i$ of Lemma 61. All graphs are finite and simple, and vertex deletion is induced. A graph of positive maximum degree is covered if a matching covers every vertex of that degree. Fix an integer $D \geq 4$. For a graph J put*

$$B_D(J) = (3 - D)|V(J)| + (2D - 3)\nu(J) + (D - 3)\alpha(J),$$

$$E_D(J) = e(J) - B_D(J), \text{ and } f(J) = |V(J)| - 2\nu(J).$$

Let H have $\Delta(H) \leq D$. A complete covered process is a sequence $H_0 = H, H_1, \ldots, H_t = R$ such that, for each integer $1 \leq i \leq t$, H_{i-1} is covered, v_i has degree $d_i = \Delta(H_{i-1}) \geq 3$, $H_i = H_{i-1} - v_i$, and $\Delta(R) \leq 2$. Such a process is a hypothesis in the first assertion. Put $s = ps(H)$, and let m count its deletions that preserve matching number. Then $0 \leq m \leq s$ and

$$(2D - 3)s - E_D(H)$$

$$= (2D - 3)(s - m) + \textstyle\sum_{i=1}^{t}(D - d_i)$$

$$+(D - 3)(\alpha(H) - \alpha(R)) + B_D(R) - e(R). \text{ (1)}$$

Every summand on the right is nonnegative. In particular

$$E_D(H) \leq (2D - 3)ps(H).$$

Writing P_j and C_j for a path and a cycle on j vertices, the contribution $B_D(C) - e(C)$ of each whole terminal component C is

$(D - 2)h + 1$ *on P_{2h}, $h \geq 1$;*

(D-2)h on P_{2h+1}, $h \geq 0$;

(D-2)h on C_{2h}, $h \geq 2$;

(D-2)(h-1) on C_{2h+1}, $h \geq 1$.

If $s = 1$ and $E = E_D(H) \geq D+1$, put $a = \alpha(H)$, $k = \nu(H)$, $f = f(H)$ and $L = 2D-3-E$. Then $0 \leq L \leq D-4$. Every complete covered process has exactly one matching-preserving deletion, preserves independence number at every step, and terminates in a disjoint union of triangles and isolated vertices. More precisely,

$t = a + 2 - f$, $\sum_{i=1}^{t}(D - d_i) = L$,

$c_1(H) = k + 1$,

$c_{D-1}(H) = |\{i : d_i = D\}| \geq a - f + E - 2D + 5$,

$E \leq f + 2D - 5$. *(2)*

Every vertex deleted at degree D has original H-degree D, and these vertices are pairwise nonadjacent. At each deletion, every cost row with integer index $1 \leq j < d_i$ drops by exactly one, while all rows with $j \geq d_i$ vanish on both parent and child.

There is also a criterion supplying the required processes. Suppose $ps(H) = 1$ and that every induced subgraph J of H, including H itself, satisfies both following conditions:

(a) if $ps(J) = 0$ and $\Delta(J) \geq 3$, then J is covered;

(b) if $ps(J) = 1$, $\Delta(J) \geq 3$ and J is not covered, then $E_D(J) \leq D$.

Then $E_D(H) \leq 2D - 3$. If $E_D(H) \geq D + 1$, every process deleting an arbitrary current maximum-degree vertex while that degree is at least three is a complete covered process, and all conclusions (2) hold for every such process.

All orders, degrees, costs, cardinalities, counts, indices, matching losses and parameters D, s, m, t, a, k, f, L are integers in the displayed domains; E_D may be signed. Empty processes, empty graphs and isolated vertices are permitted in (1). The two induced-subgraph conditions in the last assertion are explicit hypotheses.

Proof. The nonnegative slack, covered-degree vanishing, zero tails and one-vertex row bounds used below are supplied by Lemma 61. For each step of a complete covered process define

$h_i = \nu(H_{i-1}) - \nu(H_i)$, $u_i = \alpha(H_{i-1}) - \alpha(H_i)$,

$a_{i,j} = c_j(H_{i-1}) - c_j(H_i)$ for $1 \leq j < d_i$,

and $z_i = \sum_{j=1}^{d_i - 1}(1 - a_{i,j})$.

The matching and independence losses are zero or one: restriction of an optimal matching or independent set loses at most one member, and each object on the child is also one on the parent. The cited row bounds give $a_{i,j}$ in {0,1}. Covered-degree vanishing and restriction give zero rows on parent and child at every index at least d_i. Subtracting the slack definitions therefore gives

$ps(H_{i-1}) - ps(H_i) = 1 - h_i + z_i$. (3)

All terms on the right are nonnegative integers. Since $m = \sum_i (1 - h_i)$, telescoping yields

$s = ps(R) + m + \sum_i z_i$.

Nonnegative slack implies $m \leq s$.

Direct subtraction in the definition of E_D gives

$E_D(H_{i-1}) - E_D(H_i)$

$= (2D-3)(1-h_i) - (D - d_i) - (D-3)u_i$.

Sum over all steps and use $\sum_i u_i = \alpha(H) - \alpha(R)$. Rearranging proves (1).

For completeness, every finite simple graph of maximum degree at most two is a disjoint union of paths, cycles and isolated vertices: in a connected component, following successive unused edges either visits every vertex as a path or closes a cycle whose vertices already have their two neighbors. A path of order j has j-1 edges, matching number $\lfloor j/2 \rfloor$, and independence number $\lceil j/2 \rceil$. Consecutive disjoint edges and alternating vertices attain the latter two values; disjoint consecutive vertex pairs and the possible unpaired endpoint prove the independence upper bound. A cycle of order j has j edges and matching and independence numbers $\lfloor j/2 \rfloor$. The matching upper bound counts endpoints; for the independence upper bound, the cyclic successors of selected vertices are distinct unselected vertices. Alternating choices attain the bounds.

Orders, edge counts, matching numbers and independence numbers add over whole components: optimal objects restrict to component objects and unions of component optima attain the sums. Substituting the preceding path and cycle values into $B_D - e$ gives exactly the four component values in the statement. They are nonnegative. They vanish only on an isolated vertex or a triangle, and every other component contributes at least D-2. Also $d_i \le D$, $s - m \ge 0$ and $\alpha(H) \ge \alpha(R)$. Thus every right-side contribution in (1) is nonnegative, proving the density bound.

Now let $s = 1$ and $E \ge D + 1$. The proved bound gives $L = 2D - 3 - E \ge 0$, and the lower bound on E gives $L \le D - 4$. In (1), a positive integer 1-m would contribute at least $2D - 3 > L$, which is impossible. Hence $m = 1$. A positive independence loss would contribute at least $D - 3 > L$, and a terminal component other than a triangle or isolate would contribute at least $D - 2 > L$. Neither can occur. Therefore $\alpha(R) = a$, R consists only of triangles and isolates, and $\sum_i (D - d_i) = L$. Every u_i is nonnegative, so their zero sum also proves independence preservation at every step.

The terminal order and matching number are $|V(R)| = |V(H)| - t$ and $\nu(R) = k - t + 1$. A union of triangles and isolates satisfies $|V(R)| = 2\nu(R) + \alpha(R)$. Substitution yields $t = a + 2 - f$. Since $s = m = 1$, (3) gives $ps(R) = 0$ and $z_i = 0$ for every i. Every $a_{i,j}$ equals one. This proves the complete row-drop assertion. Lemma 57, applied to this actual triangle-and-isolate terminal, gives $c_1(R) = \nu(R)$ and $c_j(R) = 0$ for all $j \ge 3$. Every deletion degree is at least three, so the first row drops once per step. Hence $c_1(H) = t + \nu(R) = k + 1$. The row with index D-1 drops exactly at degree-D deletions; at a lower deletion degree that row is already zero. Since $D - 1 \ge 3$, its terminal value is zero, proving the exact count for $c_{D-1}(H)$.

Every deletion with $d_i < D$ contributes at least one to $\sum_i (D - d_i) = L$. Consequently at least $t - L = a - f + E - 2D + 5$ of the deletions have degree D. Each such vertex has original degree D because $\Delta(H) \le D$ and induced deletion cannot increase degree. Two of these vertices cannot be adjacent: deletion of the earlier one would leave the later one with degree at most D-1 thereafter. Their set is independent, of size $c_{D-1}(H) \le a$. Combining this with the lower bound yields $E \le f + 2D - 5$. This proves (2).

It remains to prove the criterion from the two local hypotheses. Every zero-slack induced subgraph J of H admits a complete covered process: if its current maximum degree is at least three, condition (a) supplies coverage, and (3) and nonnegative slack force the next graph to have slack zero. Vertex count decreases, so this procedure terminates. Formula (1), with $s = 0$, gives $E_D(J) \le 0$ for every such J.

We prove $E_D(J) \le 2D - 3$ for every slack-one induced subgraph J by induction on its order. At maximum degree at most two the component calculation gives $E_D(J) \le 0$. If J is not covered, condition (b) gives $E_D(J) \le D \le 2D - 3$. Otherwise choose any maximum-degree vertex v of degree $d \ge 3$ and set $K = J - v$, $h = \nu(J) - \nu(K)$, $u = \alpha(J) - \alpha(K)$. Equation (3) gives $ps(K) \le h$. The exact error change is

$$E_D(J) - E_D(K) = (2D - 3)(1 - h) - (D - d) - (D - 3)u. \quad (4)$$

If $h = 0$, K has slack zero and $E_D(K) \le 0$, so $E_D(J) \le 2D - 3$. If $h = 1$, (4) gives $E_D(J) \le E_D(K)$. A zero-slack K again has nonpositive error; a slack-one K has smaller order and the induction hypothesis applies. This proves the bound in all cases.

Finally suppose $E_D(H) \ge D + 1$ and start any allowed maximum-degree deletion process. At a slack-one stage whose error is at least $D + 1$, the graph cannot have maximum degree at most two, by the component calculation, and cannot be uncovered, by (b). If the chosen deletion has $h = 1$, (4) makes the child error at least the parent error. The child cannot have slack zero, whose error would be nonpositive; hence it still has slack one and error at least $D + 1$. Such steps cannot continue indefinitely because the order decreases and a stage of degree at most two cannot have this error. Therefore a step with $h = 0$ occurs. Its child has slack zero, and condition (a) supplies coverage throughout the remaining process, with slack staying zero by (3). Thus the entire arbitrary process is covered. The already proved conclusions (2) apply to it. This verifies every quantifier in the final assertion. □

Remark. Each matching-preserving deletion consumes a slack unit. After charging those units, the density discrepancy is a sum of degree, independence and terminal-component losses; a small discrepancy forces exact row counts.

Depends on. Lemma 57, Lemma 61.

Lemma 83 (Marked barriers after a dominating vertex deletion). *Use the finite integer certificate costs $c_{A,B}$ and c_i of Lemma 23: $c_{A,B}(H)$ is the least nonnegative integer T such that every nonnegative integer split $x+y=T$ admits a partition of $E(H)$ into four parts with maximum degrees at most A, B and matching numbers at most x, y, respectively; $c_i(H)=\max_{A+B=i} c_{A,B}(H)$. All graphs are finite and simple, vertex deletion is induced, and a matching is a set of edges with pairwise disjoint endpoints.*

Let $d \geq 4$ be an even integer, let F be a connected d-regular graph, and let j be an integer with $1 \leq j \leq d-2$. Let $Z \subseteq V(F)$ satisfy $|N_F(v)$ intersect $Z| \geq j$ for every v outside Z, where $N_F(v)$ denotes the neighbors of v. Put $G=F-Z$ and $D=d-j$, and suppose $\Delta(G)=D$. Let $U=\{vinV(G) : d_G(v)=D\}$; equivalently U consists of vertices outside Z having exactly j neighbors in Z. Define $\kappa=\min_M |U$ minus $V(M)|$ over all matchings M of G, where $V(M)$ is the set of covered vertices.

Then $c_D(F) \leq |Z|+\kappa$. For every nonnegative integer capacity pair $A+B=D$ and every nonnegative integer budget pair $x+y=|Z|+\kappa$, a certificate attaining these bounds exists on all original edges of F.

For $X \subseteq V(G)$, let $o_U(G-X)$ count the odd-order connected components of G-X wholly contained in U. There is a set X with $o_U(G-X)-|X|=\kappa$. Fix any such X, let L be the family of components counted by $o_U(G-X)$, and put $q=|L|=|X|+\kappa$. For each C in L define $\delta(C)=e_G(V(C),X)$, the number of edges between $V(C)$ and X. Then

$\delta(C)$ congruent to D modulo 2, and $\sum_{C\in L}\delta(C) \leq D|X|$.

Consequently the subfamily $L_0=\{C \in L : \delta(C) \leq D-2\}$ has at least κ members. Every C in L_0 has odd order at least $D+1$.

No edge of F joins two different components in L. For every integer J with $0 \leq J \leq |L_0|$ and every choice of J distinct members C_1,..., C_J of L_0, put $K=\bigcup_{i=1}^{J} V(C_i)$ and $M=\sum_{i=1}^{J}|V(C_i)|$. Let $N_F(K)$ be the vertices outside K with a neighbor in K. Then

$|N_F(K)| \leq jM+(D-2)J$.

This includes $\kappa=0$ and the empty selected family $J=0$, when K is empty and $M=0$. All degrees, orders, capacities, budgets, costs, cardinalities, matching sizes, and displayed parameters d, j, D, κ, q, J, M are integers in their stated positive or nonnegative domains.

Proof. By Lemma 46, the minimum defining κ equals $\max_X(o_U(G-X)-|X|)$; both extrema are attained, and a maximum matching of G attains the minimum. Fix such a maximum matching M_0 and let $B_0=U$ minus $V(M_0)$, so $|B_0|=\kappa$. Define $H=G-B_0$. No edge of M_0 has an endpoint in B_0, so the same edge set M_0 is a matching of H. If a vertex has degree D in H, it had degree D in G, belongs to U, and is outside B_0. It is therefore covered by M_0. Also $\Delta(H) \leq D$.

For every nonnegative integer pair $A+B=D$, apply Lemma 35 to H and this specified matching. It partitions $E(H)$ into parts of maximum degrees at most A and B. The two matching-bounded parts can be empty, so $c_D(H)=0$ by the definitions in Lemma 23. The omitted vertex set $O=Z$ union B_0 is disjointly partitioned by Z and B_0 and has cardinality $t=|Z|+\kappa$. Moreover $H=F-O$. The vertex-deletion extension in Lemma 72 gives $c_D(F) \leq t$.

Here is the actual extension for every budget pair $x+y=t$. Order O and assign each F-edge meeting O to its earliest endpoint in O. This partitions all edges outside $E(H)$ into t labeled stars, with empty labels allowed. Give any x of the labels to the first matching-bounded part and the other y labels to the second. Each matching in a union of x stars has at most x edges: its restriction to each star has at most one edge. The corresponding bound is y in the other part. Edges internal to O are assigned once by the ordering. Together with the two degree-bounded parts on H this partitions every original edge of F and meets all four bounds. The construction includes $t=0$ and either budget equal to zero.

Now fix any X attaining the displayed deficiency and form L. Each C in L is a whole connected component of G-X, and every vertex of C lies in U and hence has G-degree D. All G-edges leaving C go to X. Degree summation gives

$D|V(C)|=2e(C)+\delta(C)$.

Since $|V(C)|$ is odd, $\delta(C)$ has the same parity as D. The edges counted in $\delta(C)$ for distinct C are disjoint. Counting their endpoints in X therefore gives

$\sum_{C\in L}\delta(C) \leq \sum_{v\in X} d_G(v) \leq D|X|$.

Let h be the number of members of L with $\delta(C) \geq D$. The preceding sum implies $Dh \leq D|X|$, and $D \geq 2$ gives $h \leq |X|$. At least $q - |X| = \kappa$ members consequently have $\delta(C) < D$. Their parity forces $\delta(C) \leq D - 2$, so they belong to L_0.

Fix C in L_0 and put $\ell = |V(C)|$. If $\ell \leq D$, simplicity gives at most ell-1 internal neighbors at each vertex. Each of its ℓ vertices thus has at least $D + 1 - \ell$ neighbors in X. It follows that

$\delta(C) \geq \ell(D + 1 - \ell) = D + (\ell - 1)(D - \ell) \geq D,$

contradicting $\delta(C) \leq D - 2$. Hence $\ell \geq D + 1$. Its odd order is part of the definition of L.

Any edge of F joining different members of L would have both endpoints outside Z and X. It would then be an edge of G-X between different connected components, which is impossible. For the selected family C_1,..., C_J, every vertex in K has exactly j neighbors in Z, because it belongs to U. Its other neighbors lie either in its own component or in X. Therefore every vertex of $N_F(K)$ lies in Z or X. At most jM distinct Z-vertices occur, by counting the jM incident edges from K to Z. At most $\sum_{i=1}^{J} \delta(C_i) \leq (D - 2)J$ distinct X-vertices occur, by counting the corresponding edges. Since X and Z are disjoint, adding these bounds proves the assertion. For $J = 0$ both neighbor sets and both edge counts are empty. □

Remark. A matching deficiency can be paid by deleting its uncovered prescribed vertices. A barrier attaining that deficiency has too many odd components for all of them to use a full residual degree of boundary edges.

Depends on. Lemma 23, Lemma 35, Lemma 46, Lemma 72.

Lemma 84 (Degree-restricted remnants at a unique plateau). *All graphs are finite and simple. Let F_1, F_2 be nonempty star forests with full nonincreasing positive integer sizes n_1,..., n_s and m_1,..., m_t, where $s, t \geq 1$ are integers. Copies are ordinary subgraphs with disjoint target components; F_1 is prescribed red and F_2 blue. Put $M = s + t - 1$ and $\ell_k = max\{n_i + m_j - 1 : 1 \leq i \leq s, 1 \leq j \leq t, i + j = k\}$. Fix an integer $0 \leq q < M$ and put $P_q = \sum_{k=q+2}^{s+t}(\ell_k - 1)$. Suppose $D = \ell_{q+2} - 1 \geq 6$ is even and every later natural entry is at most D. Let $\mu \geq D/2$ be an integer and put $n = 2\mu + 1$.*

For i in {1,2} and integer $j \geq 1$, let $N_{i,j}$ count all original F_i components of size at least j. Let $a_{i,j}$ be the maximum cardinality of a subcollection of original components with sizes in [j, D] whose total vertex count is at most n. Set

$c_{i,j} = \max_{z \geq j} \max(N_{i,z} - a_{i,z}, 0).$

Let U_i have exactly $c_{i,j} - c_{i,j+1}$ components of size j. Then U_i is a possibly empty componentwise subforest of F_i. It is empty exactly when the full original has head at most D and order at most n. At every $j > D$ one has $c_{i,j} = N_{i,j}$. Superscript β denotes deletion of β leading whole components. For every integer $0 \leq \beta < c_{i,1}$, every copy of F_i^{β} in C disjoint union Q, where C, Q are disjoint, $|V(C)| \leq n$ and $\Delta(C) \leq D$, leaves a copy of U_i^{β} entirely in Q.

For any full nonempty positive star pair (X, Y), write $P_q(X, Y)$ for its own complete natural q-tail potential, defined by the same diagonal formula and set to zero if q is at least its component budget, the sum of its two component counts minus one. Let $A_q(X, Y)$ be its full family of nonempty actual suffix pairs after nonnegative leading cuts totaling q. For a finite family E of such pairs, write each member's full lists as x_i, y_j. Its envelope potential is the sum over integers $k \geq 2$ of max({x_i+y_j-2: a member of E and valid indices `i+j=k`} union {0}). The sum is finite, and an empty family has potential zero.

If both U_i are nonempty, let $E_q = A_q(U_1, F_2)$ union $A_q(F_1, U_2)$ and let P_{end} be its envelope potential. Then

$0 \leq P_q - P_{end} \leq (D - 1)\mu + 1.$

Independently orient the two originals as F, J, let S be the full component count and B the head of F, and let U, c_j be the displayed remnant and cumulative counts of J. Put $h = \max(1, D + 1 - B)$. If $S \leq \mu$ and $c_h > 0$, then

$0 \leq P_q - P_q(F, U) \leq (D - 1)\mu + 1.$

This same one-sided estimate holds without a count or retained-threshold assumption if the full head of J exceeds D.

Let H have a whole connected odd D-regular component C with a matching missing exactly one vertex and with $\nu(C) = \mu$, where ν denotes maximum matching cardinality. Let $z \geq 0$ be an integer with $z + 1 < D/2$, and suppose

$\Phi(H) := e(H) - \nu(H) \le P_q + z$. *If either both U_i are nonempty, or one of the one-sided alternatives holds, then H has a coloring avoiding both targets of some original actual suffix pair with total cut q. The component C can be colored wholly in the prescribed color of the shortened target. Every such coloring extends over the original canonical q-star pool of sizes ℓ_2,..., ℓ_{q+1}, and across every specified actual q-vertex set in any outer graph whose deletion gives H, preserving all H-edge colors and avoiding both full originals.*

Consequently, under these graph hypotheses, if H arrows every original actual total-q suffix pair, both full original heads are at most D and at least one full original has order at most n. For each orientation F, J with the full count S of F at most μ and head B, the entire group of original J components of sizes at least $\max(1, D + 1 - B)$ *has at most n vertices. No maximum-degree bound outside C, positive matching-deficit assumption, or original-head bound is imposed in the hypotheses. Every order, degree, size, count, maximum, cut and allowance is an integer in its stated domain. All original units and repetitions are retained.*

Proof. Every fitting maximum exists because the eligible subcollections form a finite family containing the empty subcollection. A selected component at threshold j occupies at least $j + 1$ vertices, so

$0 \le a_{i,j} \le \min(N_{i,j}, floor(n/(j+1))) \le \mu$.

There are no eligible components when $j > D$. The $c_{i,j}$ are nonnegative nonincreasing integers, at most $N_{i,j}$, and zero above the original head. Thus they define a finite star forest. Comparing sorted sizes proves its componentwise containment: if its k-th size exceeded the k-th original size, the cumulative count at that size would exceed the original count, contradicting $c_{i,j} \le N_{i,j}$.

If the entire original has head at most D and order at most n, every threshold group is eligible and fits, giving $c_{i,j} = 0$ for all j. Conversely, $c_{i,1} = 0$ gives $a_{i,1} = N_{i,1}$, so the full original is an eligible fitting subcollection and satisfies both bounds. When $j > D$, every $a_{i,z}$ with $z \ge j$ is zero. Since the original counts are nonincreasing, $c_{i,j} = \max_{z \ge j} N_{i,z} = N_{i,j}$.

Fix a legal β and a copy of F_i^β in C disjoint union Q. Its positive connected components lie wholly in one part. Any copied component lying in C has edge size at most D, by the degree bound. At threshold z, the components lying in C form an eligible subcollection of original labels with total vertex count at most n, and hence number at most $a_{i,z}$. The outside copy has at least $\max(N_{i,z} - a_{i,z} - \beta, 0)$ components at threshold z. For $z \ge j$ all these components also have size at least j. Maximizing over z gives at least $\max(c_{i,j} - \beta, 0)$, precisely the threshold-j count of U_i^β. Sorting and trimming the outside components gives that suffix. The argument applies to the same U_i after every cut, without requiring compatible maximizing subcollections.

We establish the profile identities needed for both estimates. Write N_x, T_y for the full original size-at-least-x and size-at-least-y counts. At positive integer excess level w the full width is

$V_w = max\{N_x + T_y - 1 : x, y \ge 1 integers, x + y = w + 2, N_x T_y > 0\}$,

with value zero for an empty candidate set. Each positive threshold rectangle realizes all diagonal index sums from two through $N_x + T_y$. Conversely any original sizes whose sum is at least $w + 2$ dominate positive thresholds with sum exactly $w + 2$: take $x = \max(1, w + 2 - m_j)$ and $y = w + 2 - x$. This proves the full-width identity. The full diagonal sequence is nonincreasing: decreasing a positive index that exceeds one gives a candidate on the preceding diagonal at least as large. Thus the natural width at level w is $\max(V_w - q, 0)$. The unique initial plateau means $V_D = q + 1$, and all natural widths above D vanish.

For any full pair, its natural q-tail is the coordinatewise envelope of its entire actual total-q suffix family. A local candidate after cuts α, β comes from full indices with sum larger by q. Conversely full indices i, j with $i + j \ge q + 2$ admit cuts $\alpha = \min(q, i - 1)$, $\beta = q - \alpha \le j - 1$. Both retained suffixes are nonempty and produce the required local candidate. If q is at least the component budget there are no such full indices or legal cuts, and all natural widths vanish. Hence this identity includes exhaustion. For a union of two such families, its superlevel width is the maximum of their widths, because each positive superlevel is an initial interval. Summing the widths gives the defined envelope potential. Componentwise containment makes each endpoint width at most the original width, proving all asserted lower bounds.

At a maximizing original threshold pair x, y, suppose the threshold of a shortened side survives in its remnant. The endpoint retaining that threshold loses at most $N_{i,j} - c_{i,j} \le a_{i,j} \le \mu$ in full width. The map V to $\max(V - q, 0)$ cannot increase a nonnegative width difference, so it loses at most μ in natural width as well. This comparison needs only one surviving shortened threshold, with the other side unchanged.

First assume both U_i are nonempty. If both full heads are at most D-1, the degree restriction excludes no original component. The remnants are precisely those of Lemma 19; its unique-plateau estimate, applicable at $\mu \geq D/2$, gives the stronger bound $P_q - P_{end} \leq (D-1)\mu$. Suppose instead that one full head is at least D, and exchange the two complete lists for this calculation so that $N_D > 0$. If $T_2 > 0$, the threshold pair (D,2) gives

$q + 1 = V_D \geq N_D + T_2 - 1,$

so $q \geq T_2 - 1$. If $T_2 = 0$ this last inequality holds because $q \geq 0$.

Fix a positive natural level and choose an original maximizing pair x, y. If either shortened threshold survives, the preceding $\cap$ comparison bounds the endpoint-envelope loss by μ. Otherwise both thresholds vanish. Since both remnants are nonempty, each has positive threshold-one count, so $x, y \geq 2$. In particular $T_y \leq T_2$ and

$\max(N_x + T_y - 1 - q, 0) \leq N_x.$

The vanishing red threshold implies $N_x - a_{1,x} = 0$. Thus its entire original threshold group is eligible and fits n vertices; hence $N_x \leq floor(n/(x+1)) \leq \mu$. Even complete loss of this natural level therefore costs at most μ. There are at most D positive natural levels and the top width is one. The total loss is at most $(D-1)\mu + 1$. This proves the two-sided bound without a full-head restriction.

For the one-sided assertion, orient the full lists as F, J and abbreviate the J-remnant counts by c_j. Suppose $S \leq \mu$ and $c_h > 0$, where $h = \max(1, D + 1 - B)$. Because $B \geq 1$, one has $1 \leq h \leq D$. Set $x_0 = D + 1 - h$; then $1 \leq x_0 \leq B$. If $T_{h+1} > 0$, the positive original pair $(x_0, h+1)$ lies at level D, giving

$q + 1 = V_D \geq N_{x_0} + T_{h+1} - 1,$

and thus $q \geq T_{h+1} - 1$. If that blue count is zero, the latter inequality follows from $q \geq 0$.

At a maximizing original pair x, y, a surviving blue remnant threshold bounds the loss by μ. If that threshold vanishes, monotonicity and $c_h > 0$ force $y \geq h + 1$. The original natural width is then at most

$\max(N_x + T_y - 1 - q, 0) \leq N_x \leq S \leq \mu.$

Each positive natural level therefore loses at most μ, and the top loses at most one. Summation proves the first one-sided bound. Its remnant is nonempty because $c_h > 0$.

If the head of J exceeds D, its remnant retains a positive count at threshold $D + 1$ and hence at every $y \leq D + 1$. Every maximizing pair at a positive natural level $w \leq D$ has $y \leq w + 1 \leq D + 1$. Thus the shortened blue threshold always survives. The $\cap$ comparison gives loss at most μ at each level, and the top again loses at most one. This proves the other one-sided bound without any restriction on S. Such a remnant is nonempty.

Now impose the graph hypotheses and any of the stated alternatives. The whole odd near-perfect component has order $2\mu + 1$ and, by summing its degrees, $e(C) = Dmu + D/2$. Thus $\Phi(C) = (D-1)\mu + D/2$. Let $P.$ denote the applicable endpoint-envelope potential or one-sided natural potential. Matching number and edge count add over whole component unions, so

$\Phi(H - V(C)) = \Phi(H) - \Phi(C)$

$\leq P. + z + 1 - D/2 < P..$

A graph's potential is nonnegative because a matching is an edge subset. Therefore $P. > 0$ and the relevant endpoint family is nonempty. Its exact envelope potential is $P.$, by the proved identity. The finite-family bound Lemma 56 supplies a coloring of H-V(C) avoiding both targets of some endpoint member.

Color all C edges in the prescribed color of the shortened target. A copy of the corresponding original suffix would leave the prohibited remnant suffix outside C by the every-prefix property, since $\Delta(C) = D$ and $|V(C)| = n$. All edges of the other color lie outside C, where the unchanged opposite suffix is absent. These are legal original cuts because every remnant is componentwise contained in its original. We have obtained the asserted original actual-family avoidance. The extensions over the canonical pool and over any specified actual outer q-set are Lemma 41 (I), (II), with the same cuts and every H-edge color retained.

Finally suppose H arrows every original actual suffix pair. A full head exceeding D would give the just-proved one-sided avoidance, so both heads are at most D. If neither full original had order at most n, both remnants would be nonempty by the emptiness criterion, again giving avoidance. Thus one original fits. Fix

any orientation F, J whose full count S is at most μ, and put $h = \max(1, D + 1 - B)$. If the whole J-group of sizes at least h had more than n vertices, then it could not all be selected by the fitting maximum at h. Thus $a_h < N_h$ and $c_h > 0$, contradicting the one-sided avoidance. The group therefore has at most n vertices. This conclusion uses only the stated count threshold, not an unstated ordering of the edge totals or a matching-deficit premise. □

Remark. A surviving shortened threshold pays by its fitting capacity. When it vanishes, the unique top rectangle has already removed enough of the opposite threshold group to make the remaining width small. Stars above the component degree survive automatically.

Depends on. Lemma 19, Lemma 41, Lemma 56.

Lemma 85 (Graphs attaining the minimum-degree matching bound). *All graphs are finite and simple. For a graph J, let $\nu(J)$ be its maximum matching cardinality; a matching consists of edges with disjoint endpoints. Write K_a for the complete graph on a positive integer a vertices. An independent set has no edge between its vertices. A perfect matching covers every vertex, and the empty graph has its empty perfect matching.*

Every graph on an integer $n \geq 1$ vertices whose minimum degree is at least a nonnegative integer d has $\nu(J) \geq \min(d, \lfloor n/2 \rfloor)$.

Let $r \geq 0$ and $N \geq 2r + 2$ be integers, and let J have order N, matching number r, and minimum degree at least r. Then at least one of the following holds:

(a) There is a partition $V(J) = X$ disjoint union I with $|X| = r$ and $|I| = N - r$ such that I is independent and every edge between X and I is present. Edges within X are unrestricted.

(b) The integer r is even, $N = 2r + 2$, and J is the vertex-disjoint union of two copies of K_{r+1}.

Consequently let $h \geq 1$ be an integer and G a graph of order $2h + 1$ and minimum degree at least h. If G is not factor-critical, meaning that G-v has no perfect matching for at least one vertex v, then at least one of the following holds:

(c) There is a partition $V(G) = Z$ disjoint union I with $|Z| = h$ and $|I| = h + 1$ such that I is independent and every edge between Z and I is present. Edges within Z are unrestricted.

(d) The integer h is odd, and G consists of two vertex-disjoint copies of K_h and one additional vertex adjacent to every vertex of those copies, with no edge between the copies.

All orders, degrees, cardinalities, and matching sizes are integers in the displayed domains. The alternatives include $r = 0$ and $h = 1$; they are not required to be disjoint.

Proof. We first prove the matching lower bound. Take a maximum matching M with k edges. If $k = \lfloor n/2 \rfloor$, the bound holds. Otherwise at least two vertices u, v are uncovered. No two uncovered vertices are adjacent, since adding their edge would enlarge M. Thus every neighbor of u or v is an endpoint of M. At most two neighbor incidences from {u, v} meet the endpoints of a given matching edge ab: if at least three of ua, ub, va, vb existed, there would be two disjoint edges matching u and v to a and b. Replacing ab by those edges would enlarge M. Summing over its k edges gives $deg(u) + deg(v) \leq 2k$. Both degrees are at least d, hence $k \geq d$. This proves the lower bound in every case, including $d = 0$ and $n = 1$.

For the classification, apply the ordinary matching-deficiency identity of Lemma 46 to J. Among all sets X attaining

$o(J - X) - |X| = N - 2r$,

choose one of maximum cardinality. Here o counts odd-order connected components, and attainment and finiteness are supplied by that fact. Put $b = |X|$ and $\rho = N - 2r \geq 2$.

There is no even-order component C of J-X. If such a component existed, choose a vertex v in C. The graph C-v has odd total order, so the sum of its component orders forces at least one of those components to be odd. All other components of J-X are unchanged by deleting v. Thus $o(J - (Xunion\{v\})) \geq o(J - X) + 1$. Subtracting $|X| + 1$ makes the new objective at least ρ. Since ρ is its maximum, equality holds. This gives a larger attaining set, contradicting the choice of X.

Let the p components of J-X therefore have odd orders $2s_1+1$,..., $2s_p+1$, where each s_i is a nonnegative integer. Their count satisfies $p = b+\rho \geq b+2$. Summing their orders gives

$N = b + 2sum_i\ s_i + p = 2b + 2sum_i\ s_i + \rho$,

so $r = b + \sum_i s_i$. Put $W = \sum_i s_i = r - b \geq 0$. Every vertex in component i has at most $2s_i$ neighbors in that component and at most b outside it, because different components of J-X have no edge between them. The minimum-degree hypothesis therefore gives

$r \leq b + 2s_i$, hence $2s_i \geq W$

for every i.

If $p \geq 3$, summing these inequalities gives $2W \geq pW$. Because $W \geq 0$ and $p \geq 3$, this forces $W = 0$. Each s_i is then zero, so every component of J-X is a singleton and $b = r$. Put $I = V(J)$ minus X. It is independent. Each of its vertices has at least r neighbors and can have neighbors only in the r-element set X. It is adjacent to all of X. This is alternative (a), with no restriction on edges within X.

The only other possibility is $p = 2$, since $p \geq 2$. The identity $p = b+\rho$ and $\rho \geq 2$ give $b = 0$ and $\rho = 2$. Thus the two components are whole components of J, $N = 2r+2$, and $s_1 + s_2 = r$. Their inequalities $2s_i \geq r$ force $2s_1 = 2s_2 = r$. Hence r is even, and each component has order $r+1$. Its vertices have degree at least r and no neighbors in the other component. Simplicity makes its degree at most r, so every two distinct vertices in it are adjacent. Both components are K_{r+1}, proving (b). This argument includes $r = 0$, when both components in this second case are singletons.

Finally let G have the stated odd order and minimum degree. The initial matching bound gives $\nu(G) \geq h$, while the order bound gives $\nu(G) \leq h$. Thus $\nu(G) = h$. Choose v such that $J = G - v$ has no perfect matching. Deleting one vertex from a maximum matching loses at most one edge, so $\nu(J) \geq h-1$. A matching of h edges in its 2h vertices would be perfect; therefore $\nu(J) = h-1$. Its minimum degree is at least h-1. Apply the proved classification with $r = h-1$ and $N = 2h = 2r+2$.

In alternative (a), J has a set X of size h-1 and an independent set I of size $h+1$, with all X-I edges present. Every I-vertex has degree exactly h-1 in J. Its G-degree is at least h, so it must be adjacent to the restored vertex v. Put $Z = X$ union {v}. Then every Z-I edge is present and $|Z| = h$, giving (c).

In alternative (b), h-1 is even, so h is odd, and J is two disjoint copies of K_h. Every vertex of J has degree h-1 there. Again its degree in G is at least h, forcing adjacency to v. There are no other new edges on restoring one vertex. Thus G has precisely the form in (d). All conclusions also apply at $h = 1$, where each K_h is a single vertex. □

Depends on. Lemma 46.

Lemma 86 (Degree-threshold exposure and vertex correction). *Use the integer certificate costs $c_r(G)$ of Lemma 23. For an integer $r \geq 1$, call a finite simple graph r-good if its maximum degree is at most r and it has a matching covering every vertex of degree r. For any finite simple graph G define $\delta_r(G)$ as the minimum cardinality of a vertex set U for which the induced graph G-U is r-good. Empty graphs have maximum degree zero and their empty matching is permitted.*

Let J have maximum degree at most r, let $S = \{vinV(J) : d_J(v) = r\}$, and let κ be the minimum number of vertices of S left uncovered by a matching of J. Then

$\delta_r(J) \leq \kappa \leq floor(|S|/(r+1))$.

If $r \geq 3$ and J has no connected component that is r-regular and has odd order, put $\epsilon_r = 2$ and $b_r = r+1$ when r is even, and $\epsilon_r = 1$ and $b_r = r+2$ when r is odd. If $\kappa \geq 1$, then

$|S| \geq b_r\ ceil(r\kappa/(r-\epsilon_r)) \geq 2b_r$.

Consequently $|S| \leq 2b_r - 1$ implies that J is r-good under this component exclusion. For odd r the component exclusion is automatic.

Now let $D \geq 2$ be an integer and let G be a finite simple graph of order $n \geq 0$ with maximum degree at most D. Let N_j be the number of its vertices of degree j and put $W = N_D + N_{D-1}/D$. For every real p in [0,1],

$\delta_{D-1}(G) \leq floor(np + W(1-p)^D)$.

If $n \geq D+1$, then for every integer s with $0 \leq s \leq n$,

$\delta_{D-1}(G) \leq s + floor(W binomial(n-D,s)/binomial(n,s))$.

Here $binomial(a,b) = 0$ for integers $a \geq 0$ and b outside [0,a]. In particular, $W\, binomial(n-D,s) < binomial(n,s)$ guarantees an induced (D-1)-good graph on at least n-s vertices.

For every integer $r \geq 1$ and every finite simple G, $c_r(G) \leq \delta_r(G)$. More precisely, any fixed U for which G-U is r-good supplies, for every nonnegative integer pair $A+B=r$ and every nonnegative integer pair $x+y=|U|$, a certificate whose bounded-degree parts partition $E(G-U)$ and whose exceptional parts partition all original G-edges meeting U. The set U and a covering matching of G-U can be fixed before both pairs; for each fixed x, y, its exceptional edge sets can also be fixed before A, B. Thus all the displayed deletion bounds also bound the corresponding certificate costs.

Proof. The minimum defining $\delta_r(G)$ exists: there are finitely many vertex subsets, and deleting every vertex leaves an r-good graph. The minimum defining κ also exists because the matching family is finite and contains the empty matching.

Apply Lemma 46 to J with prescribed set S. Choose a vertex set X attaining $\kappa = q_S(J-X) - |X|$, where q_S counts odd components whose vertex sets are wholly in S. Let calQ be this family of components. For Q in calQ put $m_Q = |V(Q)|$ and $d_Q = e_J(V(Q), X)$, the number of J-edges between $V(Q)$ and X. Every edge leaving Q ends in X, and every vertex of Q has J-degree r. Therefore $d_Q = r\, m_Q - 2e(Q)$. Since the degree sum over X is at most $r|X|$, one has $\sum_Q d_Q \leq r|X|$. It follows that

$\sum_Q (r - d_Q) \geq r(|calQ| - |X|) = r\,\kappa$.

Let calP consist of the components Q with $d_Q < r$. Discarding the other, nonpositive, summands gives $\sum_{Q \in calP}(r-d_Q) \geq r\,\kappa$. Every remaining summand is at most r, so $|calP| \geq \kappa$. If a member had $1 \leq m_Q \leq r$, simplicity would give

$d_Q \geq m_Q(r - m_Q + 1) = r + (m_Q - 1)(r - m_Q) \geq r$,

a contradiction. Thus its order is at least $r+1$. These components have disjoint vertex sets contained in S, so $|S| \geq (r+1)|calP| \geq (r+1)\kappa$. Integrality gives the stated floor bound.

Choose a matching M attaining κ, and let U′ be the vertices of S that it misses. No edge of M meets U′. In J-U′ every vertex still of degree r was originally in S and is outside U′, so the same M covers it. Deletion does not increase degree. Thus J-U′ is r-good and $\delta_r(J) \leq |U'| = \kappa$.

Under the additional component exclusion, every Q in calQ has $d_Q > 0$. Otherwise Q itself is a connected component of J, of odd order and regular degree r. The identity $d_Q = r\, m_Q - 2e(Q)$, with m_Q odd, gives d_Q congruent to r modulo two. Hence $d_Q \geq \epsilon_r$. The preceding positive-summand inequality now yields $|calP|(r - \epsilon_r) \geq r\,\kappa$. The denominator $r - \epsilon_r$ is positive for the stated $r \geq 3$. Each member of calP has odd order greater than r, so has at least b_r vertices. This proves the ceiling bound. If $\kappa \geq 1$, then $r\,\kappa/(r-\epsilon_r) > 1$ and its ceiling is at least two. The covering assertion follows by contradiction, since $\kappa = 0$ is exactly the covering requirement. When r is odd, an odd-order r-regular component would have odd degree sum, contradicting the identity between degree sum and twice the edge count.

For the deletion construction, fix any vertex set U of G. Let B consist of original degree-D vertices outside U with no neighbor in U. Put $J = G - (U union B)$. Every original degree-D vertex retained in J has a neighbor in U, so $\Delta(J) \leq D-1$. Let T consist of the original degree-D vertices outside U with exactly one neighbor in U, together with the original degree-(D-1) vertices outside U with no neighbor in U. Every vertex of J-degree D-1 is in T: a degree-D vertex with at least two deleted neighbors already has degree at most D-2, as does a degree-(D-1) vertex with a neighbor in U; smaller original degrees cannot attain D-1. Deletion of B only reduces degrees. Apply the first part to J with $r = D-1$. At most $|T|/D$ further vertex deletions suffice to leave an r-good graph. Consequently some actual deletion set containing U union B has integer cardinality at most

$F(U) = |U| + |B| + |T|/D$.

Select every vertex independently for U with probability p, and write $q = 1-p$. For an original degree-D vertex, the probabilities of belonging to B and T are respectively q^{D+1} and Dp q^D. For an original degree-(D-1) vertex, the probability of belonging to T is q^D. All these formulas hold at $p=0$ and $p=1$. By summing indicator expectations,

$E\,F(U) = np + N_D\,q^{D+1} + (DN_D pq^D + N_{D-1}q^D)/D$

$= np + (N_D + N_{D-1}/D)q^D.$

The distribution has finite support and total probability one, so at least one set in its support has $F(U)$ at most this expectation. The associated deletion set has integer size, proving the first probabilistic bound, including $n = 0$.

Suppose $n \geq D+1$ and select U uniformly from the s-element subsets. An original degree-D vertex belongs to B with probability $binomial(n-D-1,s)/binomial(n,s)$. It belongs to T with probability $D\ binomial(n-D-1,s-1)/binomial(n,s)$: choose exactly one of its D neighbors and avoid the vertex itself and the other neighbors. An original degree-(D-1) vertex belongs to T with probability $binomial(n-D,s)/binomial(n,s)$. The denominator is positive for $0 \leq s \leq n$, and the zero-binomial convention includes $s = 0$ and $s = n$. Pascal's identity combines the two degree-D contributions to give

$E(|B| + |T|/D) = W\ binomial(n-D,s)/binomial(n,s).$

Here $|U| = s$ deterministically. Finite averaging and integer cardinality give the second bound. If the ratio is less than one, its floor is zero, leaving at most s deleted vertices.

Finally fix any r-good induced residual G-U and one of its covering matchings. By Lemma 35 its same edge set has a degree split for every $A + B = r$. With empty exceptional parts this proves $c_r(G-U) = 0$ in the definitions of Lemma 23. The vertex extension theorem Lemma 72 gives $c_r(G) \leq |U|$ and restores every edge meeting U. Its concrete construction is to partition U into U_R, U_B of cardinalities x, y, put every missing edge meeting U_R in the red exceptional set, and put all remaining missing edges in the blue exceptional set. Every blue missing edge meets U_B. A matching in either exceptional set has size at most the respective part cardinality. These sets are chosen independently of the capacities, while the same covering matching supplies each residual degree split. Minimizing over U proves $c_r(G) \leq \delta_r(G)$ and the full compatibility statement. □

Remark. An uncovered maximum-degree vertex forces a large odd matching obstruction. After deleting a chosen set, remove vertices that still exceed the target degree, then repair only the remaining tight vertices. Averaging this concrete construction yields bounds that retain an actual common residual.

Depends on. Lemma 23, Lemma 35, Lemma 46, Lemma 72.

Lemma 87 (Induced partitions and complementary certificate levels). *Use the integer certificate costs $c_\theta(G) = \max_{A+B=\theta} c_{A,B}(G)$ of Lemma 23. Let G be any finite simple graph with n vertices. For every integer $m \geq 1$ and nonnegative integers $a_1,\ldots, a_m$ with $\sum_i (a_i+1) \geq \Delta(G)+1$, there is a partition $V(G) = V_1$ disjoint union... disjoint union V_m, with empty parts allowed, such that $\Delta(G[V_i]) \leq a_i$ for every i. Consequently, for positive integers $\theta_1,\ldots,\theta_m$ with $\sum_i \theta_i \geq \Delta(G)+1$, one has $\sum_i c_{\theta_i}(G) \leq (m-1)n$. In particular $c_\theta(G) = 0$ for every integer $\theta \geq \Delta(G)+1$, and $c_\theta(G) + c_\eta(G) \leq n$ for positive integers θ, η with $\theta + \eta \geq \Delta(G)+1$. Now let $D \geq 1$ be an integer with $\Delta(G) \leq D$, let v be a vertex of G, and fix any perfect matching M of G-v. For all positive integers θ, η with $\theta + \eta \geq D$, one has $c_\theta(G) + c_\eta(G) \leq n+1$. If $deg_G(v) \leq D-1$, the bound improves to n. In both cases $c_D(G) \leq 1$. If G is factor-critical, meaning it has at least one vertex and deleting any vertex leaves a perfect matching, then n is odd, $\nu(G) = (n-1)/2$, and $c_\theta(G) \leq (n+1)/2$ for every positive integer θ with $2\theta \geq D$. If such a G has a vertex of degree at most D-1, this last bound improves to $c_\theta(G) \leq \nu(G)$. All graph orders, capacities, indices, and matching cardinalities are integers. An empty graph has maximum degree zero.*

Proof. We first record how an induced graph supplies a certificate bound. Suppose $\theta \geq 1$ is an integer, X is a vertex set, $Q = G - X$ has maximum degree at most θ, and a matching of Q covers every vertex whose Q-degree is θ. The covering-matching implication of Lemma 35, with $D = \theta$, partitions $E(Q)$ into degree-bounded parts of capacities A, B for every nonnegative integer $A + B = \theta$. With empty exceptional parts this gives $c_\theta(Q) = 0$ by Lemma 23. The vertex-set extension bound of Lemma 72 then gives $c_\theta(G) \leq |X|$. In particular this applies when $\Delta(Q) \leq \theta - 1$, because there is no vertex of degree θ and the empty matching suffices.

We prove the asserted partition by minimizing a finite potential. There are m^n ordered partitions of the vertex set; this is a positive integer, including for $n = 0$, so the candidate family is nonempty and finite. For a partition $P = (V_1,\ldots,V_m)$, put $\Phi(P) = \sum_i e(G[V_i])/(a_i+1)$. Its denominators are positive, and a minimum exists in the finite family. Choose a minimizing partition. Suppose u in V_i has $d_i \geq a_i + 1$ neighbors in V_i. For every j let d_j be the number of its neighbors in V_j. Then $\sum_{j\neq i} d_j = deg_G(u) - d_i \leq \Delta(G) - (a_i+1) \leq \sum_{j\neq i}(a_j+1) - 1$.

If $m = 1$ this is impossible because its left side is zero and the right side is -1. Otherwise some $j \neq i$ has $d_j < a_j + 1$, hence $d_j \leq a_j$. Move u from V_i to V_j. This remains an ordered partition, since empty parts are allowed. The potential changes by exactly $-d_i/(a_i+1) + d_j/(a_j+1) \leq -1 + a_j/(a_j+1) < 0$, contradicting its minimality. Thus each induced maximum degree is at most its prescribed a_i.

For the first certificate sum, set $a_i = \theta_i - 1$. The partition just constructed satisfies $\Delta(G[V_i]) \leq \theta_i - 1$. Apply the induced-graph certificate bound with $X = V(G)$ minus V_i to obtain $c_{\theta_i}(G) \leq n - |V_i|$. Summing over all i gives $\sum_i c_{\theta_i}(G) \leq mn - n$. The zero-tail assertion is the case $m = 1$, and the pair assertion is the case $m = 2$.

Fix D, v, M for the near-perfect-matching assertion. Delete v and all edges of M, leaving a graph J on $V(G)$ minus {v}. Each of its vertices loses its unique matching edge, so $\Delta(J) \leq D - 1$. Use the proved partition theorem on J with bounds theta-1 and eta-1. Their sum plus two is $\theta + \eta \geq D \geq \Delta(J) + 1$. We obtain parts V_1, V_2 of $V(G)$ minus {v} for which the J-degrees inside the respective parts are at most theta-1 and eta-1. In $G[V_1]$, an internal matching edge increases each degree by at most one. A vertex can have $G[V_1]-$*degree* θ only if its M-partner belongs to V_1. Hence $\Delta(G[V_1]) \leq \theta$ and M intersect $E(G[V_1])$ covers all vertices of that degree. The induced-graph certificate bound gives $c_\theta(G) \leq n - |V_1| = |V_2| + 1$. In $G[V_2]$, a vertex of degree η must likewise have its M-partner inside V_2, giving $c_\eta(G) \leq |V_1| + 1$. Addition proves the bound $n + 1$.

If $deg_G(v) \leq D - 1$, instead retain v and delete only M. The resulting spanning graph J has maximum degree at most D-1: each vertex different from v loses one matching edge, and v already has the required degree. Partition all vertices of J with the same two bounds. A vertex different from v that attains degree θ in $G[V_1]$ has its matching partner inside V_1; if v lies in V_1, its degree is at most theta-1 because it has no matching edge. Thus the restricted matching covers every degree-theta vertex, and $c_\theta(G) \leq |V_2|$. The same argument in V_2 gives $c_\eta(G) \leq |V_1|$, proving the improved total n. Separately G-v has maximum degree at most D and its perfect matching covers every vertex. The induced-graph certificate bound with $X = \{v\}$ therefore gives $c_D(G) \leq 1$.

Finally, if G is factor-critical, a perfect matching of G-v has $(n-1)/2$ edges, so n is odd and $\nu(G) \geq (n-1)/2$. Every matching uses two distinct vertices per edge, giving $\nu(G) \leq \lfloor n/2 \rfloor = (n-1)/2$. Apply the near-perfect pair bound with $\eta = \theta$ to obtain $2c_\theta(G) \leq n + 1$. If a vertex of degree at most D-1 exists, use that vertex in the improved bound to obtain $2c_\theta(G) \leq n$. The costs are integers by Lemma 23, so in the latter case $c_\theta(G) \leq \lfloor n/2 \rfloor = \nu(G)$. □

Remark. A low-degree induced part costs only the vertices outside it. Removing a near-perfect matching before partitioning allows one additional unit of degree inside each part, because the retained matching covers every vertex attaining that unit.

Depends on. Lemma 23, Lemma 35, Lemma 72.

Lemma 88 (Well-covered rows from a sharp second certificate cost). *All graphs are finite and simple. Write $V(G)$, $E(G)$, $\nu(G)$, and $\alpha(G)$ for vertex set, edge set, maximum matching cardinality, and maximum independent-set cardinality. A graph is well-covered if all its inclusionwise maximal independent sets have size $\alpha(G)$. For a vertex set S, write $N_G(S)$ for its open neighborhood and $N_G[S] = S$ union $N_G(S)$. Write P_i and C_i for the path and cycle on i vertices, and K_1 for an isolated vertex. A whole component means an entire connected component.*

For nonnegative integer capacities A, B, let $c_{A,B}(G)$ be the least nonnegative integer T such that every nonnegative integer pair $x + y = T$ admits an ordered edge partition into four parts having maximum degrees at most A, B and matching numbers at most x, y, respectively. Put $c_2(G) = \max_{A+B=2} c_{A,B}(G)$.

Let F be well-covered and let R be an induced subgraph whose whole components belong to

K_1, P_2, P_3, P_4, P_6, C_3, C_4, C_5, C_7.

Put $U = V(F)$ minus $V(R)$, $u = |U|$, and $m = \nu(R)$. Let p count the odd-cycle components of R, and let ℓ count its nonisolated odd-order components. Suppose nonnegative integers x, y satisfy $x + y = u + m - 1$ and $E(F)$ has no partition into two parts with matching numbers at most x, y. Suppose $j = u - x \geq \ell + 1$ and $c_2(F) \geq u + p$.

Then R has no whole P_3 or P_6 component. It is well-covered, $\alpha(R) = \alpha(F)$, and every maximal independent set of R is a maximum independent set of F. In particular its entire component list lies in

K_1, P_2, P_4, C_3, C_4, C_5, C_7.

All counts, orders, cardinalities, matching numbers, capacities, budgets, and costs are nonnegative integers; j is positive. No regularity, minimum degree, source-minimality, or deletion-process assumption is imposed.

Proof. Use Lemma 70 for component assignments and Lemma 72 for full certificate restoration. Put $a = \alpha(F)$, let Z be the isolates of R, and put $B0 = \{binU : N_F(b) intersect Z is empty\}$. Every listed component has maximum degree at most two, so all the hypotheses of Lemma 70 are exactly the displayed missing-allocation and j hypotheses. It gives $\alpha(R) = a$ and a nonempty finite family of matchings saturating B_0 in the bipartite graph Γ of incidence with exposed vertices in distinct nonisolated odd components. A vertex of an odd component is exposed when its deletion leaves a perfect matching. Every such matching covers every whole P_3, and its actual assigned vertex there is an endpoint. Every vertex of an odd cycle is exposed.

For any independent set S in F, every maximal independent set J in $F - N_F[S]$ makes S union J maximal in F. Indeed vertices outside this remainder are in S or adjacent to S, and maximality of J handles the remainder. Hence $F - N_F[S]$ is well-covered with independence $a - |S|$. We use this exact remainder observation throughout.

1. Localizing the even components.

Let D be the union of all even-order row components. Fix any B0-saturating Gamma-matching and realize all its incidences by actual edges to exposed vertices. Choose a maximum independent set in every odd component, containing its assigned vertex when assigned, and include all Z. These choices are possible: the endpoints of P_3 form its maximum independent set, and alternate vertices on an odd cycle give a maximum independent set containing any specified vertex. Their union I is independent and dominates every odd row component. It dominates every exterior vertex, because B_0 vertices have assigned selected neighbors and other exterior vertices meet Z. It is anticomplete to D since R is induced and D is a union of its other components. Thus $F - N_F[I] = D$ exactly, and D is well-covered.

Each whole component of a well-covered disjoint union is well-covered: unite two maximal independent sets of different sizes in that component with fixed maximum independent sets in all other components to contradict the equality of total sizes. A path P_6, consecutively labeled 1,..., 6, has maximal independent sets {1,3,5} and {2,5}, of sizes three and two. Thus D has no P_6. Its components are now P_2, P_4, C_4, each bipartite with a perfect matching. Fix its two bipartition classes D_0, D_1; both have size $\nu(D)$ and are maximum independent sets.

2. The companions of three-vertex paths.

Suppose there are $h \geq 1$ whole P_3 components. Fix any saturating matching M in Γ and any actual exposed vertices realizing its assignments. For each assigned odd cycle C write t_C for its chosen vertex, and choose any maximum independent set containing t_C; on an unassigned cycle choose any maximum independent set. Denote their union by T. Let C_{mid} contain the centers of all h paths. For $i = 0, 1$ put $S_i = Z$ union T union C_{mid} union D_i.

This set is independent, dominates R, and has size a-h, because only the h paths contribute one center in place of two endpoints. A maximal independent extension in F must therefore add h exterior vertices. Any available exterior vertex avoids Z and T, so it belongs to B_0 and cannot be assigned to a cycle. The possible available vertices thus lie in the set of M-assignees of P_3 components, which has exactly h members by the cited result's mandatory coverage. Call this fixed set B. Every extension must add all B. Consequently B is independent, avoids C_{mid} and both D_i, and avoids every permitted T. This holds for every choice of cycle sets and every realization of M. Membership in B_0 also gives avoidance of Z.

We prove that a member of B contacts no other path. Fix one permitted T and let $I = Z$ union T union D_0. Every exterior vertex outside B has a neighbor in Z or T. The set I dominates every row component except the P_3 components, and B avoids I. Hence $F - N_F[I]$ is exactly the graph H induced by B and all h paths. It is well-covered with independence 2h, because $|I| = a - 2h$. For each b in B write its assigned path as $l_b - c_b - r_b$, with l_b the endpoint used by the actual assigned edge. The 2h edges $b\ l_b$ and $c_b\ r_b$ form a perfect matching N of H.

Every maximum independent set of H contains exactly one endpoint of each N-edge: it contains at most one, and its total size is $|N| = 2h$. For an N-edge vw, there is no common neighbor of v, w, since extending that neighbor to a maximal independent set would miss both endpoints. Moreover every vertex z in $N_H(v)$ minus {w} is adjacent to every vertex z′ in $N_H(w)$ minus {v}: these sets are disjoint by the common-neighbor assertion, and if z, z′ were nonadjacent, a maximal independent extension of {z, z′} would again miss both endpoints of vw.

Apply this property to b l_b. Every neighbor of b other than l_b must be adjacent to c_b. The center c_b has only its two path neighbors in H, since B avoids all centers and different row components have no edges between them. Thus the only possible other neighbor of b in H is r_b. Each b consequently contacts a nonempty subset of the endpoints of its own path and no other noncycle row vertex.

An unassigned odd cycle has no B-neighbor: every vertex of it belongs to some permitted maximum independent set. For an assigned cycle C the union of its maximum independent sets containing t_C is $V(C)$ minus $N_C(t_C)$. To verify this for order $2q+1$, label the vertices t_C, c_1,..., c_{2q} cyclically. For $q = 1$ the only set is $\{t_C\}$. For $q \geq 2$ the two maximum sets

$\{t_C, c_2, c_4, \ldots, c_{2q-2}\}$, $\{t_C, c_3, c_5, \ldots, c_{2q-1}\}$

together cover exactly every vertex other than c_1, c_{2q}; those two vertices cannot be selected with t_C. The upper bound q on independence follows by pairing each selected cycle vertex with its distinct unselected cyclic successor. Since B avoids every permitted choice, for all b in B

$N_F(b)$ intersect $V(C)$ is contained in $N_C(t_C)$.

These conclusions hold for every realization of every saturating M.

3. A forced pair of long-cycle contacts.

Fix b in B. Suppose it does not contact both neighbors of t_C on any assigned C_5 or C_7. On every odd cycle C its contacts can then be contained in the endpoints of one actual cycle edge e_C: it has no contact on unassigned cycles, at most one on assigned long cycles, and at most two, necessarily adjacent, on a triangle. Let L_C consist of e_C and every edge from b to C. Its matching number is one; its nonisolated graph is an edge, a two-edge path, or a triangle.

In the full induced graph $J = F[V(R)$ union {b}], delete all p edge sets L_C. The remaining graph Q is bipartite of maximum degree at most two. Each odd cycle has become a path and has no remaining edge to b. The assigned P_3 together with b is P_4 or C_4, according as b meets one or both endpoints; all other P_3 components, D, and isolates are unchanged. These exhaust all edges since the entire noncycle neighborhood of b was proved above.

Alternate along each path and even cycle to partition Q into two matchings. For capacity split (1,1) use them as the bounded-degree parts; for (2,0) or (0,2) put all Q in the capacity-two part. For every matching-budget split of total p, assign that many of the p whole sets L_C to the respective exceptional parts. Any matching in a union restricts to a matching in each edge piece, so its size is at most the number of assigned pieces, including when pieces share b. Thus $c_2(J) \leq p$.

Since b is exterior, $u \geq 1$. Restore the other u-1 vertices by Lemma 72, with all original incident edges including edges among those vertices. This gives $c_2(F) \leq u + p - 1$, contrary to the assumed lower bound. Therefore each b in B meets both neighbors of the assigned vertex of some matched C_5 or C_7. By the preceding containment its entire neighborhood on that cycle is exactly that pair.

4. Maximizing over the complete assignment family.

Call a matched C_5 or C_7 marked when its assignee has both neighbors of some cycle vertex as neighbors in F. Let $\mu(M)$ be the number of marked cycles. This nonnegative integer function on the finite nonempty family of all B0-saturating Gamma-matchings has a maximum. Choose a maximizing M, then realize each incidence by any actual exposed vertex. No constraint on a partial assignment or extension of such a partial assignment is used.

If C is marked, its assignee contacts both cycle neighbors v, w of some c. Realize the same incidence first at v and then at w, keeping M fixed. For every b in the same path-assignee set B, the containment proved above then gives

$N_F(b)$ intersect $V(C) \subseteq N_C(v)$ intersect $N_C(w) = \{c\}$.

The equality holds on C_5 and C_7: in cyclic labels with $c = 0$ and v, $w = 1$, L-1, the intersection of {0,2} and {L-2,0} is {0}. Thus the compulsory pair for any b from the previous paragraph must occur on an unmarked matched long cycle.

Choose one path Q with assignee b, and choose such an unmarked C with assignee w and actual assigned vertex t_C. Then b has exactly the two contacts $N_C(t_C)$ on C, and b, w are distinct. Construct S in the row by taking the center of Q, both endpoints of every other P_3, all isolates, any maximum independent set in each

D-component, and on every odd cycle other than C a maximum independent set containing its assigned vertex when assigned. On C choose a maximum independent set containing both neighbors of t_C. In the labels above these sets are $\{c_1, c_4\}$ on C_5 and $\{c_1, c_3, c_6\}$ on C_7.

The set S is independent and dominates R, with size a-1. Every exterior vertex outside B_0 meets its selected isolates. Every M-assignee other than b, w is dominated by its selected assigned vertex. The vertex b is dominated by its pair of C-contacts. Therefore w is the only possible vertex to add to S. Well-coveredness forces a maximal extension of S to have size a, so w avoids S.

The vertex w has an endpoint neighbor in Q. Otherwise replacing its selected center in S by both endpoints would yield an independent set of size a that still avoids w; adjoining w would give size $a+1$, a contradiction. Replace the M-assignments bQ and wC by wQ and bC. The first uses this actual endpoint contact, which is exposed in P_3, and the second uses either actual C-contact, all cycle vertices being exposed. Distinct assignees and distinct receiving components are preserved, and every other assignment is unchanged. Thus the modified matching still saturates B_0. The formerly unmarked cycle C is now marked, since b meets both neighbors of t_C; no other long-cycle assignee changed. Its μ value is exactly $\mu(M)+1$, a contradiction. Hence $h = 0$.

We have eliminated both P_3 and P_6. Each remaining component is well-covered. This follows directly for K_1, P_2, C_3. For P_4, C_4, C_5 the independence number is two and a singleton closed neighborhood does not cover the component, so every maximal independent set has size two. For C_7 the independence number is three, while at most two vertices have a union of closed neighborhoods of size at most six and cannot dominate it. Thus every maximal independent set has size three. Unions of maximal independent sets over disjoint components are exactly the maximal independent sets of R, proving that R is well-covered. Since $\alpha(R) = a$, every such set is maximum in F. This proves all assertions. □

Remark. All path companions are forced into one independent set by the exact loss from replacing endpoint pairs by centers. The sharp second cost forces a long-cycle pair; exchanging its assignee with the path companion strictly improves a finite maximum.

Depends on. Lemma 70, Lemma 72.

Lemma 89 (Full-target half-order matching capacity). *All graphs are finite and simple. A matching has pairwise disjoint endpoints, and $\nu(G)$ is its largest possible cardinality. Let F, J be vertex-disjoint unions of positive-edge stars with full nonincreasing positive integer sizes $n_1,\dots, n_s$ and $m_1,\dots, m_t$, where $s, t \geq 1$ are integers. The red target is F and the blue target is J; copies are ordinary subgraphs with disjoint target components. Put $M = s + t - 1$, $T(F) = \sum_i floor((n_i+1)/2)$, $T(J) = \sum_j floor((m_j+1)/2)$, $W_F = T(F) - s$, and $W_J = T(J) - t$. Thus $W_F = \sum_i floor((n_i-1)/2)$ and $W_J = \sum_j floor((m_j-1)/2)$.*

Every graph G arrowing (F, J) satisfies

$\nu(G) \geq \min(M, \max(T(J) + floor((s-1)/2), T(F) + floor((t-1)/2)))$.

In particular $W_J \geq ceil((s-1)/2)$ or $W_F \geq ceil((t-1)/2)$ implies $\nu(G) \geq M$.

More precisely suppose $\delta = M - \nu(G) \geq 1$. Call S an attaining separator if $\nu(G) = |S| + \sum_C floor(|V(C)|/2)$, where C ranges over all connected components of $G-S$, including isolated vertices. Such a set exists. For every attaining separator S, put $h = |S|$. Then

$h \leq \min(s,t) - 2$,

$\nu(G) \geq T(J) + floor((s-1+h)/2)$,

$\nu(G) \geq T(F) + floor((t-1+h)/2)$,

and consequently $s \geq 2W_J + 2\delta + h$, $t \geq 2W_F + 2\delta + h$.

For every integer $0 \leq q < M$ let A_q be all ordered nonempty actual suffix pairs (F^α, J^β), where α, β are nonnegative integers with $\alpha < s$, $\beta < t$, $\alpha + \beta = q$, and superscripts delete leading whole components. If H arrows every A_q member and $\delta = M - q - \nu(H) \geq 1$, then $q < \min(s,t)$ and, for every attaining separator S of H with $h = |S|$,

$q + 2\delta + h \leq \min(s - 2W_J, t - 2W_F)$.

All sizes, counts, indices, separator cardinalities and matching numbers are integers in their displayed domains. Unit stars, repeated sizes, isolated host vertices and the empty separator are permitted. No maximum-degree, edge-count, connectivity or smaller-target hypothesis is imposed.

Proof. We use the attained separator formula of Lemma 46, with the prescribed set equal to the entire vertex set. It supplies every existence assertion about attaining separators made here.

First suppose G arrows (F, J) and put $v = \nu(G) < M$. The all-red coloring gives a copy of F, and selecting one edge from each positive component gives $v \geq s$. The all-blue coloring gives $v \geq t$. Thus $s, t \geq 2$, since $s = 1$ would give $v \geq t = M$ and $t = 1$ would give $v \geq s = M$.

Every vertex set X with $|X| \leq s - 1$ leaves an entire copy of J in G-X. To prove this, color every edge incident with X red and all other edges blue. A red matching has size at most $|X|$, because its edges have distinct endpoints in the cover X. A red F would supply an s-edge matching and is impossible. A blue J therefore occurs. Every vertex of J has an incident target edge, so every vertex of that copy avoids X. Interchanging F with J and red with blue gives the corresponding statement for F after at most t-1 vertex deletions.

Fix any attaining separator S and write $h = |S|$. The separator identity gives $h \leq v < M$. Suppose $h \geq s - 1$. Choose $X \subseteq S$ with $|X| = s - 1$. A copy of J in G-X exists by the preceding coloring argument. At most $h - s + 1$ of its t components meet S minus X, since distinct copied components are vertex-disjoint. At least $t - (h - s + 1) = M - h \geq 1$ copied components avoid S entirely. Each such component lies in one connected component C of G-S and contains an edge. Taking one edge from each yields a matching in G-S with at least M-h edges. Any matching inside C has at most $floor(|V(C)|/2)$ edges. Summing these bounds and using attainment gives $M - h \leq \sum_C floor(|V(C)|/2) = v - h$, contrary to $v < M$. Hence $h \leq s - 2$. Apply this argument to the interchanged pair, with the same G and S, to obtain $h \leq t - 2$.

Put $r = floor((s - 1 - h)/2)$, a nonnegative integer. Within each component C of G-S, partition any $2floor(|V(C)|/2)$ of its vertices into disjoint unordered pairs. These pairs need not be graph edges. Across all components there are exactly v-h pairs. Since $v \geq s$, this number is at least s-h and hence at least r. Select any r of these pairs, let r_C be the number selected within C, and let Y be their combined endpoint set. Then $|Y| = 2r$ and $\sum_C r_C = r$. The set $X = S$ union Y has cardinality $h + 2r \leq s - 1$, so G-X contains an entire J.

Assign each copied star of J to the unique component C of G-S containing it, and let I_C be the set of its target indices. These sets partition {1,...,t}. Their vertex-disjointness gives $\sum_{j \in I_C}(m_j + 1) \leq |V(C)| - 2r_C$. For nonnegative integers a_j, the sum of $\lfloor a_j/2 \rfloor$ is at most $\lfloor \sum_j a_j/2 \rfloor$: write each a_j as twice an integer plus a remainder in {0,1}. Apply this to the displayed bound to get $\sum_{j \in I_C} floor((m_j + 1)/2) \leq floor(|V(C)|/2) - r_C$. The inequality includes empty I_C. Summing over C yields $T(J) \leq v - h - r$. Therefore $v \geq T(J) + h + floor((s - 1 - h)/2) = T(J) + floor((s - 1 + h)/2)$. Apply the same argument after interchanging the entire targets and their colors, while keeping the same attaining separator S, to obtain $v \geq T(F) + floor((t - 1 + h)/2)$. No compatibility between the two target copies is required: both inequalities concern this fixed v and h.

Now $\delta = M - v \geq 1$. Substitute $T(J) = t + W_J$ and $v = s + t - 1 - \delta$ in the first inequality. This gives $s - 1 - \delta \geq W_J + floor((s - 1 + h)/2)$. Since $(s - 1) - floor((s - 1 + h)/2) = floor((s - h)/2)$, it follows that $floor((s - h)/2) \geq W_J + \delta$. All quantities are integers, so $s - h \geq 2W_J + 2\delta$. The interchanged inequality gives $t - h \geq 2W_F + 2\delta$. These are the asserted separator-strengthened capacity bounds.

Because $h \geq 0$, the two lower bounds for v imply the stated maximum without h whenever $v < M$. If instead $v \geq M$, the outer minimum is automatically at most v. This proves the unconditional matching inequality. Under $W_J \geq ceil((s - 1)/2)$, its first inner expression is $t + W_J + floor((s - 1)/2) \geq t + s - 1 = M$. The outer minimum is therefore M. The other sufficient condition follows by substituting the second inner expression.

Finally suppose H arrows all A_q and $\delta = M - q - \nu(H) \geq 1$. Put $K = M - q$. The family is nonempty because $q \leq s + t - 2$. Choose $\alpha = \max(0, q - t + 1)$, $\beta = q - \alpha$. These cuts are legal and retain exactly $\min(s, K)$ red components. Its all-red coloring forces $\nu(H) \geq \min(s, K)$. Choosing instead $\beta = \max(0, q - s + 1)$, $\alpha = q - \beta$ gives $\nu(H) \geq \min(t, K)$ from the all-blue coloring. Since $\nu(H) = K - \delta < K$, neither minimum can be K. Thus $\nu(H) \geq s, t$, and substituting $K = s + t - 1 - q$ yields $q + \delta \leq \min(s, t) - 1$. In particular $q < \min(s, t)$.

The two pairs (F^q, J) and (F, J^q) are now legal and arrowed by H, with the same budget K and the same deficit δ. Fix any attaining separator S of H and let $h = |S|$. Apply the proved bound for the first pair to its red component count s-q and the unchanged complete blue target J. It gives $s - q \geq 2W_J + 2\delta + h$. Apply the proved bound for the second pair to its blue component count t-q and the unchanged complete red

target F. It gives $t - q \geq 2W_F + 2\delta + h$. Combining proves the stated family inequality for every such S. All constructions were finite vertex selections and all copies retained their unit components. □

Remark. Deleting pairs of vertices inside an attaining separator decomposition reduces the available half-order capacity, while the entire opposite target must still fit. Retaining the separator size strengthens both full-original capacity inequalities.

Depends on. Lemma 46.

Lemma 90 (Exact binary remnants below the critical degree). *All graphs are finite and simple. Let $D \geq 6$ be even, and let F, J be nonempty star forests with complete nonincreasing positive integer edge-size lists n_1,..., n_S and m_1,..., m_T, where S, $T \geq 1$ are integers. Copies are ordinary subgraphs with vertex-disjoint target components; F is prescribed red and J blue. Their heads $B = n_1$ and $A = m_1$ satisfy $4 \leq A$, $B \leq D - 1$. Put*

$U = e(F) - S - (B - 1)$, $Wplus = 2sum_{j=1}^{T} floor((m_j - 1)/2)$.

Suppose an integer μ satisfies $\mu \geq e(F) + 1 = S + B + U$, $T \geq \mu + 3$ and $Wplus \leq S - 4$. Put $M = S + T - 1$ and $\ell_k = max\{n_i + m_j - 1 : 1 \leq i \leq S, 1 \leq j \leq T, i + j = k\}$ for $2 \leq k \leq S + T$. Fix an integer $0 \leq q < M$ with $q + \mu < M$, $\ell_{q+2} = D + 1$, and every later $\ell_k \leq D$. Write $L_q = \sum_{k=q+2}^{S+T} \ell_k$ and $P_q = L_q - (M - q)$.

For each positive integer j let T_j count the original J components of size at least j, let a_j be the maximum cardinality of a subcollection of these components using at most $2\mu + 1$ vertices, and put $c_j = T_j - a_j$. Set $a = a_1$ and $\zeta = \mu - a$. Let J_{bin} be the union of c_2 two-edge stars and $c_1 - c_2$ one-edge stars. If $B = D - 1$, assume additionally that the entire group of nonunit components of J uses at most $2\mu + 1$ vertices.

Then J_{bin} is a nonempty componentwise subforest of J, has $T - a \geq 3$ components, and $1 \leq a \leq \mu$. For every integer $0 \leq \beta < T - a$ and every pair of disjoint graphs C, Q with $|V(C)| \leq 2\mu + 1$, every copy of J^β in C disjoint union Q leaves a copy of J_bin^β entirely in Q. Superscripts delete that many leading whole components from the respective complete sorted lists.

Let $M_{bin} = M - a$ and define $L_q^b in$ and $P_q^b in$ from the new full pair (F, J_{bin}) by the same natural-tail formulas, through its own final diagonal. These tails exist because $q < M_{bin}$. All their corresponding losses are nonnegative. More precisely:

(i) If $B \leq D - 2$ and $c_2 > 0$, then $P_q - P_q^b in \leq (D - 1)\mu - B - 2$.

(ii) If $B \leq D - 2$ and $c_2 = 0$, then $L_q - L_q^b in \leq Dmu - B - 2$.

(iii) If $B = D - 1$, then $c_2 = 0$, $P_q - P_q^b in \leq (D - 1)\mu + \zeta + 2$ and $L_q - L_q^b in \leq Dmu + 2$.

Consequently, if H satisfies $e(H) \leq L_q - 1$, $\Phi(H) := e(H) - \nu(H) \leq P_q + 2$, and has a whole connected odd D-regular component C with a near-perfect matching and $\nu(C) = \mu$, then H has a coloring avoiding both entire targets of some actual pair (F^α, J^β), where α, $\beta \geq 0$ are integers, $\alpha < S$, $\beta < T$ and $\alpha + \beta = q$. Here ν is maximum matching cardinality, and a near-perfect matching misses exactly one vertex. The coloring can be chosen with C wholly blue. It extends both over the disjoint canonical q-star pool of sizes ℓ_2,..., ℓ_{q+1} and across every specified actual q-vertex set in any outer graph whose deletion gives H, retaining all H-edge colors and avoiding both full originals. No degree bound outside C or smaller-target lower-bound hypothesis is imposed.

Proof. Write $W = \sum_j (m_j - 2)_+$. For every positive integer m, $(m - 2)_+ \leq 2floor((m - 1)/2)$, so $W \leq Wplus \leq S - 4$. Also $U \geq 0$. The capacities a_j are finite maxima over families containing the empty subcollection. Removing all size-j components from a fitting subcollection gives

$a_j \leq a_{j+1} + T_j - T_{j+1}$.

Thus c_j is nonnegative and nonincreasing, is at most T_j and vanishes above A. It defines a componentwise subforest of J: otherwise a sorted size larger than its corresponding original size would contradict the threshold-count bound. Truncating its sizes at two gives J_{bin}. Since each positive component uses at least two vertices, $a \leq \mu$ and $T - a \geq 3$. The inequality $W \geq A - 2$ gives $S \geq A + 2$, hence $\mu \geq S + B > A$; at least one J component therefore fits and $a \geq 1$. Since $a \leq \mu$ and $q + \mu < M$, $q < M - a = M_{bin}$.

A copy of J^{β} distributed between two disjoint graphs places every connected positive component wholly in one part. At threshold j, at most a_j copied components can lie in the part of order at most $2\mu+1$. Outside it there remain at least $(T_j - a_j - \beta)_+ = (c_j - \beta)_+$ components of size at least j. Sorting and trimming supplies the beta-suffix of the forest with cumulative counts c_j, and then its truncation $J_b in^{\beta}$. This proves the same remnant works for every permitted β.

We establish the numerical estimates from full threshold widths. Put $N_x = |\{i : n_i \geq x\}|$ for positive integers x, and use the already defined T_y. At positive integer excess level z the full diagonal width is

$V_z = \max(\{N_x + T_y - 1 : x, y \geq 1, x + y = z + 2, N_x T_y > 0\} union \{0\})$.

A nonempty threshold rectangle realizes every index sum from two through $N_x + T_y$. Conversely, sizes whose sum is at least $z+2$ dominate positive thresholds summing to $z+2$. This proves the width formula. The full profile is nonincreasing, because decreasing an index greater than one on a noninitial diagonal cannot decrease its candidate. Thus the natural width is $w_z = (V_z - q)_+$. Define $w_z^b in$ for the new pair in the same way and $\Delta_z = w_z - w_z^b in$. Componentwise containment gives $\Delta_z \geq 0$. The hypotheses give $w_D = 1$ and $w_z = 0$ for $z > D$. Counting each integer excess by its positive levels yields

$P_q - P_q^b in = \sum_z \Delta_z$,

$L_q - L_q^b in = a + \sum_z \Delta_z$. (1)

In particular both losses are nonnegative.

The same width identity identifies each natural tail with the envelope of its entire actual suffix family. A local candidate after cuts α, β restores to full indices with their sum increased by q. Conversely a full pair i, j with $i + j \geq q + 2$ admits cuts $\alpha = \min(q, i-1)$, $\beta = q - \alpha \leq j - 1$. The retained components realize that local candidate. This proves both directions and applies to the new pair because $q < M_{bin}$.

We first treat $B \leq D - 2$. Put $r = D - B \geq 2$, $h = \mu - S \geq B + U$, $t = T - \mu \geq 3$, $g = T - S + 1 = t + h + 1$, $Q = T_2$, and $v = T - Q$. For $x \geq 2$ and $y \geq 3$,

$N_x \leq U + 1$, $T_y \leq W$, $N_x + T_y - 1 \leq U + W \leq \mu - B - 4 < T$. (2)

Every level-D candidate has $x \geq 3$ and $y \geq r + 2$ because $A \leq D - 1$ and $B = D - r$. Since $V_D = q + 1$, (2) gives $q < T$. Moreover $A + B \geq D + 2$, so the threshold pair $(B, r+2)$ is present. Its width is at most $q+1$, whence

$q \geq N_B - 1$ and $T_{r+2} \leq q + 1$. (3)

For $1 \leq z < B$ put $f_z = N_z - N_{z+1}$, $F_z = S - N_{z+1}$, $d_z = T - T_{z+1}$, and

$\psi_z = \max(0, f_z - v, F_z - d_z)$.

The candidate with blue threshold one is $T + N_{z+1} - 1 \geq T$ and dominates (2). The only remaining candidates have blue threshold two or red threshold one. Their differences from this baseline are $f_z - v$ and $F_z - d_z$. If the latter candidate is absent, $d_z = T$ and $F_z - d_z < 0$, so including it in the maximum changes nothing. Thus

$V_z = T + N_{z+1} - 1 + \psi_z$. (4)

Define

$E_{low} = \sum_{y=3}^{B} (T_y - g)_+$,

$E_{high} = \sum_{y=B+1}^{r+1} (T_y - h - q - 1)_+$,

with an empty sum zero.

If $c_2 > 0$, put $u_{bin} = c_1 - c_2 = v - a + a_2$; then $0 \leq u_{bin} \leq v$. The exact new low width is

$V_z^b in = T - a + N_{z+1} - 1 + (f_z - u_{bin})_+$.

For $z = 1$, $\psi_1 = (f_1 - v)_+ \leq (f_1 - u_{bin})_+$. For $2 \leq z < B$,

$\psi_z \leq (f_z - u_{bin})_+ + (F_z - d_z)_+$

$\leq (f_z - u_{bin})_+ + (T_{z+1} - g)_+$.

The last step uses $N_{z+1} \geq 1$. Positive-part truncation by q cannot enlarge a nonnegative difference. Hence

$\sum_{z=1}^{B-1} \Delta_z \leq (B-1)a + E_{low}$. (5)

For use when $c_2 = 0$, the retained threshold-one candidate instead gives $\Delta_z \leq a + \psi_z$. We show

$\sum_{z=1}^{B-1} \psi_z \leq \kappa + E_{low}$,

where $\kappa = (S-1-v)_+$.

As a function of the nonnegative vector $(f_1, \ldots, f_{B-1})$ with sum at most S-1, the left side is convex, being a sum of maxima of affine functions. If $S-1 > 0$, each such vector is the convex combination of zero and the vectors with S-1 in one coordinate, with coefficients $f_j/(S-1)$ and the remaining coefficient at zero. If $S-1 = 0$ only zero occurs. At the vertex with its sole nonzero coordinate j, the value is

$(S-1-v)_+ + \sum_{z=j+1}^{B-1} (S-1-d_z)_+$,

since $d_j \geq v$. This is at most $\sum_{z=1}^{B-1} (S-1-d_z)_+ = \kappa + E_{low}$. The value at zero is zero. Thus

$a + \sum_{z=1}^{B-1} \Delta_z \leq Ba + \kappa + E_{low}$. (6)

There are r-1 levels $B \leq z \leq D-2$. At each, a blue threshold-one candidate is absent. Candidates with $x \geq 2$, $y \geq 3$ have full width below μ by (2). A blue threshold-two candidate occurs only at $z = B$. If $c_2 > 0$ it is retained and loses at most $a_2 \leq \mu$; if $c_2 = 0$, then $Q = a_2 \leq \mu$, and (3) bounds its old natural width by Q. The remaining candidate has $x = 1$, $y = z+1$ and natural width at most $\mu + (T_y - h - q - 1)_+$. This extra term vanishes for $y \geq r+2$ by (3). Taking the maximum of the candidates, and using the retained candidate in the case where it has width exceeding μ, proves

$\sum_{z=B}^{D-2} \Delta_z \leq (r-1)\mu + E_{high}$. (7)

Let $b = T_{r+1} - T_{r+2}$ and $p = \max(b, f_2, \ldots, f_{B-1})$. At level D-1 a present pair has $2 \leq x \leq B$ and $y = D+1-x$. If $x < B$, increasing x by one gives a present top pair and bounds its natural width by $f_x + 1$. If $x = B$, increasing y gives the present pair in (3) and bounds its width by $b+1$. Therefore $w_{D-1} \leq p+1$. The new binary widths at D-1 and D vanish because $B \leq D-2$. Every f_x occurring in p is at most U.

Put $Z = \sum_{y=3}^{B} \min(T_y, g)$. The summation ranges in E_{low}, E_{high} are disjoint, and $W = \sum_{y=3}^{A} T_y$, so

$E_{low} + E_{high} \leq W - Z$, $Z \geq \min(b, g)$. (8)

The latter follows from $T_3 \geq b$ and $B \geq 4$.

Suppose $c_2 > 0$. We claim

$E_{low} + E_{high} + p - (B-1)\zeta \leq S - 4 + U$. (9)

If $p \leq U$, this follows from (8) and $W \leq S-4$. Otherwise $p = b$. If B is even, $r+1$ is odd and at least three. The identity $Wplus = W + |\{j : m_j \geq 3 isodd\}|$ gives $W + b \leq Wplus \leq S-4$ and proves (9). If B is odd, then $B \geq 5$. When $b \leq g$, (8) gives $Z \geq b$ and proves (9). When $b > g$, the inequality $b \leq T - v$ gives $v < S-1$, so $\kappa = S-1-v > 0$.

A fitting subcollection of a components uses at least

$2a + (a-v)_+ + (b-(T-a))_+$

vertices. The extra terms count nonunits and selected $size - (r+1)$ components; each of the latter has at least four vertices, so these contributions can be added. Since $T - a = t + \zeta$ and $a - v = \kappa + h + 1 - \zeta$, its vertex bound gives

$\kappa \leq 3\zeta - h - (b - t - \zeta)_+$. (10)

If $b \leq t + \zeta$, then $b - g \leq \zeta - h - 1 \leq (B-1)\zeta$. If $b > t + \zeta$, (10) gives

$b - g \leq 4\zeta - 2h - \kappa - 1 \leq 4\zeta \leq (B-1)\zeta$.

Thus $Z \geq g$ and $b - Z \leq (B-1)\zeta$, proving (9).

Combine (5), (7), $w_{D-1} \leq p+1$ and $w_D = 1$. Since $a = \mu - \zeta$ and $B + r = D$, (9) yields

$P_q - P_q^b in$

$\le (D-2)\mu + E_{low} + E_{high} + p + 2 - (B-1)\zeta$

$\le (D-2)\mu + S - 4 + U + 2$

$= (D-1)\mu - h + U - 2 \le (D-1)\mu - B - 2.$

This proves (i).

Now suppose $c_2 = 0$. We claim

$\kappa + p - Z - Bzeta \le U$. (11)

If $\kappa > 0$, counting only nonunits in a fitting a-subcollection gives $\kappa \le 3\zeta - h$; hence $\kappa \le 3\zeta$ in all cases. If $p \le U$, (11) follows because $B \ge 4$. Otherwise $p = b$. If $\kappa = 0$, then $v \ge S - 1$ and $g = v + Q - S + 1 \ge Q \ge b$, so $Z \ge b$. If $\kappa > 0$, (10) applies again, without any parity restriction. Using $Z \ge \min(b, g) \ge \min(b, t)$ and $B \ge 4$ gives

$Bzeta - \kappa + Z$

$\ge \zeta + h + (b - t - \zeta)_+ + \min(b, t)$

$\ge h + (b - t)_+ + \min(b, t) = h + b.$

Here $\zeta + (b - t - \zeta)_+ \ge (b - t)_+$. This proves (11).

Equations (1), (6), (7), (8) and (11) now give

$L_q - L_q^b in$

$\le (D-1)\mu - Bzeta + \kappa + E_{low} + E_{high} + p + 2$

$\le (D-1)\mu + W + U + 2$

$\le Dmu - h + U - 2 \le Dmu - B - 2.$

This proves (ii).

It remains to treat $B = D - 1$. Now $A \le B$ and $B \ge 5$. The assumed fitting of the entire nonunit group gives $Q = T_2 \le a$, $a_2 = Q$ and $c_2 = 0$. Set $k = T_3$. Counting its vertices gives $Q \le (2\mu + 1)/3$ and $k \le floor((2\mu + 1)/4)$.

Define f_z, F_z, d_z, ψ_z as in (4), for $1 \le z < B$. The proof of (2) and (4) remains valid, using only $W \le S - 4$ and $\mu \ge S + B + U$. The new matching forest has $l = T - a$ positive components, so its low full width is $l + N_{z+1} - 1$. Consequently its total low natural-width loss is at most $(B-1)a + \Psi$, where $\Psi = \sum_{z=1}^{B-1} \psi_z$.

Write $u = N_B$ and let p count the red nonunit components below B. The present top threshold pair (B,3) gives $q \ge u + k - 2$. At level B, a maximizing pair has $x, y \ge 2$. For $y = 2$, its natural width is at most $u + Q - 1 - q \le Q - k + 1 \le Q$. For $y \ge 3$, its full width is at most $u + p + k - 1$, so its natural width is at most $p + 1$. Hence $w_B \le \max(Q, p + 1)$. The matching-remnant widths at B, D vanish and $w_D = 1$.

We prove

$\Psi + \max(Q, p + 1) \le \mu + Bzeta + 1$. (12)

Remove one specified red B-head and let g_j be the remaining exact size-j counts, $1 \le j \le B$. Then $\sum_j jg_j = e(F) - B \le \mu - B - 1 =: L_0$. The left side of (12) is a convex function of this nonnegative vector. If $L_0 > 0$, the vector is the convex combination of zero and the vectors having sole nonzero coordinate $g_j = L_0/j$, with coefficients jg_j/L_0 and the remaining coefficient at zero. If $L_0 = 0$ the vector is zero. We check these vertices with all blue data fixed.

At zero and the size-B vertex, $\Psi = 0$ and $\max(Q, p + 1) = Q \le \mu$. At a size-j vertex, $2 \le j < B$, put $r_0 = L_0/j$. Then $r_0 \le \mu/2$ and $r_0 + 1 \le \mu$. For $z \ge 2$ one has $d_z \ge T - k > \mu/2$. Thus $\Psi = (r_0 - v)_+$ and $p = r_0$. If $r_0 \le v$ the expression is at most μ. If $r_0 > v$ and the maximum is $r_0 + 1$, it is $2r_0 + 1 - v \le \mu - B - v \le \mu$. If that maximum is Q, use $v = T - Q \ge \mu + 3 - Q$ to bound it by

$r_0 + 2Q - \mu - 3$

$\le (\mu - B - 1)/2 + 2(2\mu + 1)/3 - \mu - 3 < \mu$.

At the size-one vertex $r_0 = L_0$ is an integer less than T, $p = 0$ and $\Psi = \sum_{z=1}^{B-1}(r_0 - d_z)_+$. This is the excess above one of the r_0 smallest original blue components, by counting those components at each threshold. The a smallest blue components realize the maximum fitting cardinality a: among any a components they minimize total vertices. Their excess is at most $2\mu + 1 - 2a = 2\zeta + 1$. If $r_0 \le a$, then $\Psi + Q \le 2\zeta + 1 + a = \mu + \zeta + 1$. If $r_0 > a$, each added component has excess at most B-1, so

$\Psi \le 2\zeta + 1 + (B - 1)(r_0 - a) = (B + 1)\zeta + 2 - B^2$.

Adding $Q \le a$ gives at most $\mu + Bzeta + 2 - B^2 \le \mu + Bzeta + 1$. This proves (12) at all vertices and hence throughout the simplex.

It follows that

$P_q - P_q^b in \le (B - 1)a + \Psi + \max(Q, p + 1) + 1$

$\le Bmu + \zeta + 2$.

Adding $a = \mu - \zeta$ proves $L_q - L_q^b in \le Dmu + 2$. This establishes (iii).

Finally let H, C satisfy the graph hypotheses and put $Q_0 = H - V(C)$. Since C is a whole odd regular near-perfect component, $|V(C)| = 2\mu + 1$, $e(C) = Dmu + D/2$ and $\Phi(C) = (D - 1)\mu + D/2$. Edge count and maximum matching cardinality add across whole components.

When $c_2 > 0$, only case (i) can occur, and

$\Phi(Q_0) \le P_q^b in + 2 - B - 2 - D/2 < P_q^b in$.

The new actual family is finite and nonempty, and its envelope is $P_q^b in$ by the proved identity. Apply Lemma 56 to obtain a coloring of Q_0 avoiding an entire member $(F^\alpha, J_b in^\beta)$.

When $c_2 = 0$, cases (ii), (iii) and $e(H) \le L_q - 1$ give, respectively,

$e(Q_0) \le L_q^b in - B - D/2 - 3 < L_q^b in$,

or $e(Q_0) \le L_q^b in + 1 - D/2 < L_q^b in$.

Write $l = T - a \ge 3$ and choose $\alpha = \max(0, q - (l - 1))$, $\beta = q - \alpha$. Since $q < S + l - 1$, these are legal cuts with $\alpha < S$ and $\beta < l$. The full diagonal profile of (F, lK2) is B repeated l times followed by n_2,..., n_S. If $q \le l - 1$, the selected pair is (F,(l-q)K2); if $q > l - 1$, it is $(F^\alpha, K2)$. In either case its full edge budget is exactly $L_q^b in$. The singleton matching-family lower bound in Lemma 21, exchanging prescribed colors, therefore supplies a coloring of Q_0 avoiding this pair.

In either case color C wholly blue. A blue copy of J^β would leave a forbidden $J_b in^\beta$ in Q_0 by the every-prefix property; every red edge lies in Q_0, where F^α is absent. The cuts are legal for the originals because J_{bin} is componentwise contained in J. This gives the claimed actual-family avoidance with C blue. Apply Lemma 41 (I), (II) to those same cuts to extend over the stated canonical pool and across any specified actual q-set. Both extensions preserve every H-edge color and treat all positive components, including units. □

Remark. Exact fitting capacities retain a common binary remnant after every prefix. Low-width errors and the exceptional high widths are paid by disjoint pieces of the blue excess or by unused fitting capacity. A matching remnant is paid through its exact edge budget, while a surviving two-star is paid through potential.

Depends on. Lemma 21, Lemma 41, Lemma 56.

Lemma 91 (Whole-cycle selection with the required matching margins). *All graphs are finite and simple. Write $V(G)$, $E(G)$, $d_G(v)$, $\Delta(G)$, $\nu(G)$, and $\alpha(G)$ for the vertex set, edge set, vertex degree, maximum degree, maximum matching cardinality, and maximum independent-set cardinality. An edge subset has the matching number of its spanning graph. A whole component means an entire connected component; path and cycle orders count vertices.*

Let d belong to {6,8,10,12}. Let k, E, λ, δ, ϵ be integers satisfying

$k \geq 1$, $E \geq d/2+1$, $0 \leq \epsilon \leq \delta \leq \lambda+1$.

Define the integers

$p = E-2-\lambda$, $m = p+\delta$, $\beta = m-\epsilon$,

$r = k-\lambda-1+\delta$, $s = 2\lambda+3-2\delta$, $u = r+s$,

$v = floor((2E-3)/(d-2))$.

Assume $r \geq 0$ *and* $p \geq (d-2)/2$.

Let F be d-regular of order $2k+1$. *Let R be an induced subgraph of F with* $|V(R)| = r$, $\Delta(R) \leq 2$, $\nu(R) = m$, *exactly p whole odd-cycle components, and exactly* ϵ *whole path components of positive even order. Every whole odd cycle of R has order at most seven. Other path and even-cycle orders are unrestricted.*

Let Bcal consist of all nonnegative integer pairs (b_R, b_B) *with* $b_R + b_B = u+m-1$ *for which* $E(F)$ *has no partition into two edge parts having matching numbers at most* b_R, b_B. *Suppose Bcal is nonempty and every pair in it satisfies*

$u - b_R \geq s+m+v+1$, $u - b_B \geq s+m+v+1$.

Put D2 $= \{zinV(R) : d_R(z) = 2\}$. *Choose any maximum independent set* T_0 *of R[D2] and any shortest whole odd-cycle component C of R. These choices exist. Put*

$\ell = (|V(C)|+1)/2$, $Y = T0$ *union* $V(C)$,

$H = F[(V(F)minusV(R))$ *union* $Y]$,

$N = |V(F)$ *minus* $V(H)|$, $h = |V(H)| - r$.

Then

$|T0| = \beta$, $|Y| = \beta+\ell$, $h = s+\beta+\ell$, $N = r-\beta-\ell$,

$E(R[Y]) = E(C)$, $\nu(R[Y]) = \ell-1$,

$N + \nu(R[Y]) = r-\beta-1$, $2 \leq \ell \leq 4$.

All vertices of R[Y] outside C are isolated. Every y in Y has exactly d-2 F-neighbors outside $V(R)$, *and all those vertices and edges belong to H.*

For every pair (b_R, b_B) *in Bcal, the same H satisfies*

$\min(b_R, b_B) \leq N$,

$N \geq \max(0, \beta+h-2)$, $N \geq \ell-2$.

Thus the choices of T_0 *and C can be made before choosing the missing pair. No positivity assumption on* ϵ *is imposed. All cardinalities, degrees, matching and independence numbers, budgets and counts are nonnegative integers; the displayed differences are integers. In particular s is positive odd and* $\beta \geq p \geq 2$.

Proof. We use the balanced-allocation theorem Lemma 52. All remaining arguments concern the explicitly given degree-two row and integer parameters.

1. The row and its selected cycle.

The parameter inequalities give $\lambda \geq -1$, $s \geq 1$, $m \leq E-1$, $\beta \geq p \geq 2$, and $2r+s = 2k+1$. The excess bound $E \geq d/2+1$ implies $v \geq 1$. Since R is induced, its complement has order $2k+1-r = r+s = u$.

A connected finite graph of maximum degree at most two is an isolated vertex, a path, or a cycle: following edges without repetition either stops at an endpoint or closes a cycle, and the degree bound permits no additional branch. A path on n vertices has matching number $\lfloor n/2 \rfloor$ and independence number $\lceil n/2 \rceil$. Consecutive disjoint vertex pairs give the matching and independent-set upper bounds, attained by alternating edges and alternating vertices. A cycle of order n has matching and independence numbers $\lfloor n/2 \rfloor$. The matching bound follows by counting endpoints. For an independent set on a cycle, sending each chosen vertex to its cyclic successor injects it into the complement; alternating vertices attain the bound. These quantities add across disjoint components by restricting any candidate and by uniting componentwise maximizing candidates.

In an even-order path on 2a vertices, $a \geq 1$, the degree-two vertices induce a path on 2a-2 vertices, with independence number a-1, one less than the original matching number a. In an odd-order path on $2a+1$ vertices, $a \geq 1$, they induce a path on 2a-1 vertices, with independence number a, equal to the original matching number. An isolate contributes zero, and a cycle remains unchanged. Therefore $\alpha(R[D2]) = m - \epsilon = \beta$.

Finiteness guarantees maximum independent sets. There are $p \geq 2$ odd cycles, so shortest ones exist. Every maximum T_0 restricts to a maximum independent set in each whole component of R[D2]; otherwise replacing one restriction would enlarge T_0. On C it has ell-1 vertices. Adding all 2ell-1 vertices of C adds ℓ vertices, proving $|Y| = \beta + \ell$. The other selected vertices are mutually nonadjacent and have no edge to the whole component C. Thus the entire edge set of R[Y] is $E(C)$, its matching number is ell-1, and all its other vertices are isolated. The hypothesis on odd-cycle orders gives $2 \leq \ell \leq 4$.

Because H contains the entire complement of R and exactly Y from R, its omitted count is $N = r - \beta - \ell$ and $h = s + \beta + \ell$. This proves the displayed order and cost identities. Every selected vertex has R-degree two; regularity and inducedness give exactly d-2 neighbors outside R, all retained in H.

2. The usual shortest-cycle bound.

Sum the two bounds for any pair in the nonempty Bcal, using $b_R + b_B = u + m - 1$. This gives

$2u - (u + m - 1) \geq 2(s + m + v + 1)$,

or

$r \geq s + 3m + 2v + 1$. (1)

This bound is independent of the chosen pair.

Put $w = \ell - 1$. Every odd cycle has matching number at least w, and each positive even-order path has matching number at least one. Hence

$pw + \epsilon \leq m \leq E - 1$. (2)

If $d - 2 < 2p$, these even integers differ by at least two. Thus

$2E - 3 \geq 2pw + 2\epsilon - 1 \geq (d-2)w$,

since $(2p - (d-2))w + 2\epsilon - 1 \geq 1$. Taking floors gives $v \geq w$, or $\ell \leq v + 1$.

More generally suppose $\ell \leq v + 1$, including any case with $d - 2 = 2p$ where this inequality holds. For either coordinate b of any pair in Bcal,

$u - b - h \geq m - \beta + v + 1 - \ell = \epsilon + v + 1 - \ell \geq 0$.

As $u - h = N$, each coordinate is at most N. Also (1) yields

$N - (\beta + h - 2) = r - s - 3\beta - 2\ell + 2$

$\geq 3\epsilon + 2(v + 1 - \ell) + 1 \geq 1$. (3)

The count N is nonnegative, and $\beta + h - 2 = s + 2\beta + \ell - 2 \geq \ell - 2$. All asserted margins follow in this case.

3. The exceptional shortest-cycle case.

It remains to treat $\ell > v + 1$. Step 2 and the assumption $2p \geq d - 2$ force $d - 2 = 2p$. Since $w > v = floor((2E-3)/(2p))$, one has $2E - 3 < 2pw$ and therefore $E \leq pw + 1$. Together with (2) this forces

$E = pw + 1$, $m = pw$, $\epsilon = 0$.

Equality in (2) means every odd cycle has matching number exactly w and every other component has matching number zero. Thus R consists of exactly p cycles of order $2w + 1$ and isolated vertices. The parameter definitions give

$\lambda = pw - p - 1$, $\delta = pw - p$, $\beta = pw$,

$r = k$, $s = 1$, $u = k + 1$, $v = w - 1$.

The bound $E \geq d/2 + 1 = p + 2$ excludes $w = 1$. Since the cycle order is at most seven, w belongs to {2,3}.

Write $k_0 = (3p+2)w$. Bound (1) becomes $k \geq k_0$. For every pair in Bcal its total and coordinate bounds become

$b_R + b_B = k + pw$, $b_R \leq k - (p+1)w$, $b_B \leq k - (p+1)w$. (4)

If $k = k_0$, these conditions force

$b_R = b_B = (2p+1)w$.

The number of isolated vertices of R is then

$z = k_0 - p(2w+1) = (p+2)w - p$.

The same graph F satisfies every hypothesis of Lemma 52: it is $(2p+2) - regular$ of order $2k_0+1$, its specified induced row consists of these p cycles and z isolates, $p \geq 2$, and w belongs to {2,3}. That theorem partitions every F-edge into two parts of matching numbers at most $(2p+1)w$. This contradicts the missing pair forced by (4), which exists because Bcal is nonempty. Therefore the integer k satisfies $k \geq k_0 + 1$.

In this case

$h = (p+1)w + 2$, $N = k - (p+1)w - 1$,

$N - (\beta + h - 2) = k - k_0 - 1 \geq 0$. (5)

Furthermore $\beta + h - 2 = (2p+1)w \geq \ell - 2$, so (5) gives both lower bounds on N.

Fix any pair in Bcal. By (4), both coordinates are at most $N+1$. If both exceeded N, integrality would give $b_R = b_B = N+1 = k - (p+1)w$. Substituting into their sum in (4) would force $k = k_0$, already excluded. Hence $\min(b_R, b_B) \leq N$. This conclusion holds for every pair on the same graph H.

Steps 2 and 3 exhaust all possible values of the shortest cycle. The only choices used were an arbitrary maximum T_0 and an arbitrary tied shortest C. Neither choice depends on the missing pair. This proves all conclusions with the stated nonnegative integer and zero-deficit boundaries. □

Remark. Only the smaller missing budget and a weak omission margin are required by the matching construction. At the one-unit failure of the usual shortest-cycle reserve, failure of either required margin would force an exactly balanced allocation on a short equal-cycle row; that allocation is feasible by the earlier lemma.

Depends on. Lemma 52.

Lemma 92 (Parity corrections for domination certificates). *Use the integer certificate costs $c_{A,B}(G)$ and $c_\theta(G)$ of Lemma 23. For a finite simple graph G and an integer $\theta \geq 2$, put $T_\theta(G) = \{v \in V(G) : deg_G(v) = \theta\}$, and let $m_\theta = \theta + 1$ when θ is even and $m_\theta = \theta + 2$ when θ is odd. If $\Delta(G) \leq \theta$, then $c_\theta(G) \leq floor(|T_\theta(G)|/m_\theta)$. For every finite simple graph H and vertex set Z with $\Delta(H - Z) \leq \theta$, one has $c_\theta(H) \leq |Z| + floor(|T_\theta(H - Z)|/m_\theta)$. For integers $d \geq 3$ and $1 \leq j \leq d-2$, put $\theta = d - j$. For a real number p in [0,1], let $Bin(d,p)$ be the sum of d independent Bernoulli variables of success probability p and define $Q_{d,j}(p) = p + (1-p)Pr(Bin(d,p) \leq j-1) + (1-p)Pr(Bin(d,p) = j)/m_\theta$. Every d-regular finite simple graph H on $n \geq 0$ vertices satisfies $c_{d-j}(H) \leq floor(nQ_{d,j}(p))$. These are bounds for the actual integer certificate costs, valid simultaneously for all nonnegative integer capacity pairs of total d-j. In particular, every 16-regular finite simple H satisfies $c_{14}(H) \leq floor(30429|V(H)|/100000)$, $c_{13}(H) \leq floor(37639|V(H)|/100000)$. Empty graphs and residuals, $p = 0$, and $p = 1$ are included. Every graph order, degree, capacity, budget, component count, cost, d, j, θ and m_θ is an integer in its displayed nonnegative or positive domain.*

Proof. Fix $\theta \geq 2$ and $\Delta(G) \leq \theta$, and write $T = T_\theta(G)$. Let κ be the least number of T-vertices uncovered by a matching of G. The finite minimum and its attainment follow from Lemma 46. Choose an attaining matching M and put $U = T$ minus $V(M)$, so $|U| = \kappa$ and no M-edge meets U. In G-U every vertex of degree θ also had degree θ in G, belongs to T minus U, and remains covered by the retained matching M. Thus Lemma 35 applied to G-U with degree bound θ supplies every degree splitting of total θ. With empty exceptional parts these are certificates of total zero. The vertex-extension bound Lemma 72 therefore gives $c_\theta(G) \leq |U| = \kappa$.

The exact prescribed-deficiency formula Lemma 46 gives a vertex set X such that $\kappa = q_T(G - X) - |X|$, where q_T counts the odd components whose vertices all lie in T. Let K be the family of those components,

so $|K| = |X| + \kappa$. For each component C in K, every vertex has G-degree θ and every edge leaving C goes to X. Its boundary size is $b(C) = \theta|V(C)| - 2e(C)$. Distinct components have disjoint boundary edges, so $\sum_{C \in K} b(C) \leq \theta|X|$. Consequently at most $|X|$ members of K have $b(C) \geq \theta$. At least κ members have $b(C) \leq \theta - 1$.

For any such member let $s = |V(C)|$. If $1 \leq s \leq \theta$, simplicity would give $b(C) \geq \theta\, s - s(s-1) = s(\theta + 1 - s) \geq \theta$; the final inequality follows from $(s-1)(\theta - s) \geq 0$. This contradicts $b(C) \leq \theta - 1$. Hence $s \geq \theta + 1$. Its order is odd, so $s \geq m_\theta$ in both parity cases. These components are vertex-disjoint subsets of T; therefore $\kappa\, m_\theta \leq |T|$ and $\kappa \leq \lfloor |T|/m_\theta \rfloor$. This proves the first inequality, also when $\kappa = 0$ or T is empty. Applying it to $G = H - Z$ and then applying Lemma 72 proves the deletion inequality.

Now fix d, j, p and a d-regular graph H with n vertices. Choose a random set Z_0 by independently including every vertex with probability p. For each vertex v let $X_v = |N_H(v)$ intersect $Z_0|$. Simplicity implies that X_v has distribution $Bin(d, p)$ and is independent of whether v belongs to Z_0. Define $D_0 = \{voutside Z_0 : X_v \leq j - 1\}$, $Z = Z_0$ union D_0, and $T' = \{voutside Z_0 : X_v = j\}$. Every vertex outside Z has at least j neighbors in Z_0, so $\Delta(H - Z) \leq d - j = \theta$. If it has degree exactly θ in H-Z, exactly j of its H-neighbors lie in Z. Since Z_0 is contained in Z and $X_v \geq j$, it follows that $X_v = j$. Thus $T_\theta(H - Z)$ is a subset of T′.

For every outcome the deletion inequality gives

$c_\theta(H) \leq |Z_0| + |D_0| + \lfloor |T'|/m_\theta \rfloor$

$= \lfloor |Z_0| + |D_0| + |T'|/m_\theta \rfloor$.

Linearity of expectation and the independence just proved give

$E[|Z_0| + |D_0| + |T'|/m_\theta]$

$= n\, p + n(1-p)Pr(Bin(d,p) \leq j-1) + n(1-p)Pr(Bin(d,p) = j)/m_\theta$

$= n\, Q_{d,j}(p)$.

There are finitely many outcomes, including at the probability endpoints. Some outcome of positive probability has the quantity inside the floor at most its expectation. Its actual vertex set Z proves $c_\theta(H) \leq floor(nQ_{d,j}(p))$. The certificate threshold definition in Lemma 23 means that every capacity pair and every budget split at that total is certified. The same Z supplies the deletion bound for all these choices, although the edge partitions may depend on them. If $n = 0$, all quantities and the unique outcome are empty or zero.

For the displayed constants at $d = 16$, take $j = 2$, $p = 1/4$, so $\theta = 14$ and $m_\theta = 15$. Expanding the binomial probabilities gives

$Q_{16,2}(1/4) = 1/4 + (65/16)(3/4)^{15}$

$= 5227646251/17179869184$.

Indeed the coefficient $65/16$ is $(3/4)^2 + 16(1/4)(3/4) + 8(1/4)^2$. The exact comparison

$30429 \cdot 17179869184 - 100000 \cdot 5227646251 = 1614299936 > 0$

proves $Q_{16,2}(1/4) < 30429/100000$.

For $j = 3$, $p = 3/10$, $\theta = 13$ and $m_\theta = 15$. Expansion gives

$Q_{16,3}(3/10) = 3/10 + (11263/1000)(7/10)^{14}$.

Here the four coefficient numerators with denominator 1000 are 343, 2352, 7560, 1008, from $(1-p)^3$, $16p(1-p)^2$, $120p^2(1-p)$, $(560/15)p^3$, and they sum to 11263. The exact comparison

$7639 \cdot 10^{12} - 11263 \cdot 7^{14} = 173530501713 > 0$

proves $Q_{16,3}(3/10) < 37639/100000$. Substitution in the proved general bound and monotonicity of the floor give both degree-sixteen conclusions. □

Remark. An uncovered full-degree vertex is charged to a disjoint odd component that must contain at least $\theta + 1$ or $\theta + 2$ vertices. This makes the boundary binomial mass much cheaper than deleting every tight residual vertex.

Depends on. Lemma 23, Lemma 35, Lemma 46, Lemma 72.

Lemma 93 (Certificates from separated odd vertex sets). *Use the finite integer certificate costs c_θ of Lemma 23. Let F be a connected finite simple graph on $n = 2N+1$ vertices, with integers $N \geq 1$ and $D \geq 3$ and maximum degree at most D. Let Q be any subset of $E(F)$ and let H be the spanning graph F-Q. Let e be an integer with $1 \leq e \leq N$, and put $c = N + e$. Let $J \geq 1$ be an integer, and let K_1,..., K_J be disjoint nonempty vertex sets of odd cardinalities m_1,..., m_J, with no H-edge between different sets. Put $K = \bigcup_i K_i$, $M = \sum_i m_i$, `W={v outside K:v has an H-neighbor in K}`, $B = |W|$, and $H' = H - (KunionW)$. All vertex deletions are induced. Put $T = \sum_C \max(1, ceil((|V(C)|-1)/D))$, $t_0 = floor((2e-1)/(D-2))$, $t_1 = \min(T, floor((2e+J-B)/(D-2)))$, $L = B + (M-J)/2$, where the sum defining T is over the connected components of H' and is zero for an empty H'. Assume $t_0 \leq ceil((n-1)/D)$, $B \leq 2e+J$, $(M-J)/2 \leq 2e+t_0+1$. Assume also either $L \geq \lfloor c/2 \rfloor$, or both $2e + t_1 + (M+J)/2 \geq \lfloor c/2 \rfloor$ and $\lfloor c/2 \rfloor + (M+J)/2 + t_1 \leq n$. Then $c_0(H) \leq c$.*

The conclusion also holds under the following sufficient numerical hypotheses, in place of those involving B, T, t_1, L: there is an integer $\beta \geq B$ such that $t_0 \leq ceil((n-1)/D)$, $\beta \leq 2e+J$, $(M-J)/2 \leq 2e+t_0+1$, $2e + \min(ceil((n-M-\beta)/(D+1)), floor((2e+J-\beta)/(D-2))) + (M+J)/2 \geq \lfloor c/2 \rfloor$, and $\lfloor c/2 \rfloor + (M+J)/2 + floor((2e+J)/(D-2)) \leq n$.

In either version, if Q is the union of l edge-disjoint matchings, for an integer $l \geq 0$, then $c_l(F) \leq c$. The case $l = 0$ means Q is empty. Every capacity, matching budget, index, order, degree, cardinality, and defined parameter is an integer in its indicated domain; n-M-beta is allowed to be signed. Empty edge sets, empty residual graphs, singleton odd sets, and zero matching budgets are allowed.

Proof. All statements about costs will be justified by actual edge partitions at every required budget split. Fix $x + y = c$ with x, y nonnegative integers, and interchange the names of the exceptional parts if necessary so that $x \leq y$. Then $x \leq \lfloor c/2 \rfloor \leq N$. We first partition $E(H)$ into two exceptional parts of matching numbers at most x and y.

If $x \leq 2e$, choose any set U of x vertices. Put all edges meeting U into the first part and all others into the second. A matching in the first part has at most x edges. The other edges are supported on n-x vertices, and $n - x \leq 2y + 1$ is equivalent to $x \leq 2e$. Thus their matching number is at most y.

If $2e < x \leq 2e + t_0$, put $t = x - 2e$. By the connected independent-set assertion of Lemma 50, applied to F with half-order N and degree bound D, there is an independent set I of size t whose neighborhood in F has at most $(D-1)t + 1$ vertices. The hypothesis on t_0 checks the allowed size. Its H-neighborhood is no larger. Moreover $(D-2)t \leq 2e - 1$, so $(D-1)t + 1 \leq x$. Since $x + t \leq 2x \leq c \leq n$, enlarge that neighborhood to a set U of exactly x vertices avoiding I. Put the edges meeting U into the first part. In the remaining graph I is isolated, so the other edges are supported on at most $n - x - t = 2y + 1$ vertices. This gives the required partition.

We now treat $x > 2e + t_0$. Write $U_0 = 2e + (M+J)/2$. Both L and U_0 are integers, since every m_i is odd, and $U_0 - L = 2e + J - B \geq 0$. Suppose first $x < L$, and put $r = L - x$. The lower bound $x \geq 2e + t_0 + 1 \geq (M-J)/2$ implies $1 \leq r \leq B$. Choose r vertices R from W and put $W'' = W$ minus R. Color first all H-edges meeting W'', together with all H-edges internal to a set K_i. Color second all other edges. The first matching number is at most $|W''| + \sum_i (m_i - 1)/2 = B - r + (M-J)/2 = x$. Every second-colored edge either meets R or has both endpoints in H': edges internal to K_i were colored first, no edge joins different K_i, and every edge from K goes to W. The second matching number is therefore at most $r + floor((n - M - B)/2)$. The inequality $B \leq 2e + J$ gives $n - M - B \leq 2(c - L) + 1$, so this bound is at most $r + c - L = c - x = y$. The same construction with $r = 0$ works when $x = L$, with R empty.

It remains to treat $x > L$. Necessarily $L < \lfloor c/2 \rfloor$, so the two hypotheses involving t_1 hold. Put $t = \max(0, x - U_0)$, an integer. The first such hypothesis gives $t \leq t_1$. If $t = 0$, then $x \geq L + (D-1)t$. If $t \geq 1$, then $x = U_0 + t$, and $(D-2)t \leq (D-2)t_1 \leq 2e + J - B = U_0 - L$ gives $x \geq L + (D-1)t$ in this case as well.

We construct an independent t-set I in H' with $|N_H(I)$ minus $W| \leq (D-1)t$. First consider one nonempty component C of H'. Since K is nonempty and F is connected, an F-edge vu leaves $V(C)$. If that edge belongs to Q, then v has at most D-1 H-neighbors. If it does not, it is an H-edge and its other endpoint belongs to W: it cannot belong to K by the definition of W, and it cannot belong to another component of the induced graph H'. In either case v has at most D-1 neighbors within C.

Start an independent set in C with v. Whenever the current independent set does not dominate C, connectedness supplies a vertex at distance two from it, which can be added while sharing at least one neighbor with

the current set. Every addition increases its size by one and its neighborhood size by at most D-1. Thus an s-vertex prefix has at most (D-1)s neighbors in C. On termination, a dominating independent set of size r satisfies $|V(C)| \le Dr$, so $r \ge ceil(|V(C)|/D) \ge \max(1, ceil((|V(C)|-1)/D))$. We may therefore use a prefix of every size from zero through the summand defining T. As $t \le T$, allocate t among these finite component capacities and take the union of the corresponding prefixes. Components are pairwise nonadjacent, so this union I is independent. Its neighborhood outside W lies in H', has cardinality at most (D-1)t, and is disjoint from K.

Choose a vertex set U containing W union $N_H(I)$, avoiding K union I, and with $|U| = x - (M-J)/2$. This is possible: the required initial set has at most $B + (D-1)t \le x - (M-J)/2$ vertices, and $x - (M-J)/2 \le n - M - t$ follows from $x + (M+J)/2 + t \le \lfloor c/2 \rfloor + (M+J)/2 + t_1 \le n$. Put all edges meeting U and all edges internal to each K_i into the first part. Its matching number is at most $|U| + (M-J)/2 = x$. In the second part every vertex of U, K and I is isolated. These sets are disjoint. Its edges are supported on at most $n - |U| - M - t = 2N + 1 - x - (M+J)/2 - t$ vertices. This number is at most $2y+1$, because $x \le U_0 + t$. The second matching number is at most y.

The cases cover every $x \le \lfloor c/2 \rfloor$, and swapping exceptional parts covers every original ordered budget split. With the two degree-bounded parts empty, the definition in Lemma 23 proves $c_0(H) \le c$.

For the sufficient numerical conditions, $\beta \ge B$ implies $B \le 2e + J$. Each component C of H' satisfies $\max(1, ceil((|V(C)|-1)/D)) \ge |V(C)|/(D+1)$. For $|V(C)| \le D+1$ the left side is at least one; for $|V(C)| \ge D+1$ use $(|V(C)|-1)/D \ge |V(C)|/(D+1)$. Thus, since T is an integer, $T \ge ceil((n-M-B)/(D+1)) \ge ceil((n-M-\beta)/(D+1))$. The other entry in the minimum defining t_1 is at least $floor((2e+J-\beta)/(D-2))$. The numerical lower bound therefore implies $2e + t_1 + (M+J)/2 \ge \lfloor c/2 \rfloor$. Also $t_1 \le floor((2e+J)/(D-2))$, so the numerical upper bound implies $\lfloor c/2 \rfloor + (M+J)/2 + t_1 \le n$. All hypotheses of the first version hold.

Finally suppose Q is the union of l edge-disjoint matchings Q_1,..., Q_l. For any nonnegative capacities $A + B = l$, assign the first A matchings to the first degree-bounded part and the others to the second. Their degrees are at most A and B. For every matching-budget split at total c, use the proved two-part partition of $E(H)$ for the exceptional parts. Since Q and $E(H)$ are disjoint and exhaust $E(F)$, this is a certificate of F at every capacity and budget split. Definition Lemma 23 gives $c_l(F) \le c$. At $l = 0$ the matching union and both bounded-degree parts are empty, and the same construction applies. □

Remark. The intermediate budgets are filled by leaving some boundary vertices available to the second matching. Once the whole boundary fits, independent sets in the residual components pad the construction through the balanced budget.

Depends on. Lemma 23, Lemma 50.

Lemma 94 (Matching-rank boundary under a failed run bound). *Use the finite integer zero-capacity certificate cost $c_0(F)$, matching number $\nu(F)$, excess $ex(F) = c_0(F) - \nu(F)$, and top missing-allocation run $tr(F)$ of Lemma 45. All graphs are finite and simple, and vertex deletion is induced. Let F be connected, have order $2k+1$, satisfy $\nu(F) = k$, and have actual maximum degree $D \ge 3$, where $k \ge 1$ and D are integers. Put $E = ex(F)$ and suppose the integer $E \ge 1$.*

If $k = 3E - 1$, then $tr(F) = 1$ and $E \le \lfloor D/2 \rfloor$. Consequently, if

$tr(F) + E > \max(4, \lfloor D/2 \rfloor + 1)$,

then $k \ge 3E$.

Under this last strict inequality, let S be any vertex subset, put $a = |S|$ and $G = F - S$, and suppose there is a nonnegative integer m such that

$|V(G)| = 2m + 1$, $\nu(G) = m$, *and* $c_0(F) = a + m + 1$.

Then a is even, $E = a/2 + 1$, and $m \ge a + 3$. The set S need not be independent, and no regularity, factor-criticality, or gap assumption on G is imposed. All costs, orders, degrees, matching numbers, subset cardinalities, and run lengths are integers in their stated nonnegative or positive domains.

Proof. Apply Lemma 45 to F with its actual degree bound D. Its connectedness, odd-order, matching-number and positive-excess hypotheses hold by assumption. Put $v = \max(0, floor((2E-3)/(D-2)))$. That fact gives

$k \geq 3E - 1 + 2v$,

$1 \leq tr(F) \leq k - 3E + 2 - 2v$.

If $k = 3E - 1$, the first inequality forces $v = 0$. The second then gives $tr(F) = 1$. The equality $v = 0$ implies $floor((2E-3)/(D-2)) \leq 0$. Since D-2 is positive, it follows that $2E - 3 < D - 2$. Both D and E are integers, so $2E \leq D$ and $E \leq \lfloor D/2 \rfloor$. Therefore

$tr(F) + E \leq \lfloor D/2 \rfloor + 1 \leq \max(4, \lfloor D/2 \rfloor + 1)$.

This is incompatible with the strict inequality in the statement. Since the same cited result already gives $k \geq 3E - 1$ and k is an integer, that strict inequality forces $k \geq 3E$.

Now impose the stated conditions on S and G. The vertex partition $V(F) = S$ disjoint union $V(G)$ gives

$2k + 1 = a + 2m + 1$,

so $a = 2(k - m)$ is even and $k = m + a/2$. Subtract this matching number from the actual cost identity $c_0(F) = a + m + 1$ to obtain

$E = c_0(F) - k = a/2 + 1$.

Substituting both identities in $k \geq 3E$ yields

$m + a/2 \geq 3a/2 + 3$,

which is exactly $m \geq a + 3$. Every step uses the order and cost of this same graph F and this same induced complement G. No matching compatibility or independent-set property is required. □

Depends on. Lemma 45.

Lemma 95 (Exact greedy deletion with a fixed maximum matching). *Let calA be a finite nonempty set of labels. For each γ in calA let s_γ, $t_\gamma \geq 1$ be integers, and let $n_1^\gamma \geq \cdots \geq n_{s_\gamma}^\gamma \geq 1$ and $m_1^\gamma \geq \cdots \geq m_{t_\gamma}^\gamma \geq 1$ be integer lists. Let F_1^γ, F_2^γ be the corresponding vertex-disjoint unions of stars with those edge sizes, prescribed red and blue. Copies are ordinary subgraphs. Put $N = \max_\gamma(s_\gamma + t_\gamma - 1)$. For each integer $2 \leq k \leq N + 1$ let ℓ_k be the maximum of $n_i^\gamma + m_j^\gamma - 1$ over all labels and valid indices with $i + j = k$. For $0 \leq q \leq N$ put $P_q = \sum_{k=q+2}^{N+1}(\ell_k - 1)$, with empty sums zero.*

For a finite simple graph J write $e(J)$, $\nu(J)$, $\Delta(J)$ for edge count, maximum matching cardinality and maximum degree, and put $\Phi(J) = e(J) - \nu(J)$. An edgeless graph has maximum degree zero. Suppose a finite simple graph G arrows every ordered pair in calA and $\Phi(G) = P_0$. Fix any maximum matching W of G. Deleting W means deleting its edges and retaining every vertex; vertex deletion is induced.

For every vertex set X of size q with $0 \leq q \leq N$, put $H_X = G - X$, $W_X = \{abinW : a, bnotinX\}$, and $J_X = (G - W) - X$. Then

$\Phi(H_X) \geq P_q$, $e(J_X) \geq P_q$.

Equality $e(J_X) = P_q$ holds if and only if both $\Phi(H_X) = P_q$ and W_X is a maximum matching of H_X.

Starting with X_0 empty, for each integer $0 \leq q < N$ choose any maximum-degree vertex v_{q+1} in $J_q = (G - W) - X_q$, and set $X_{q+1} = X_q$ union $\{v_{q+1}\}$. Such a vertex exists at every step. Put $H_q = G - X_q$ and $W_q = \{abinW : a, bnotinX_q\}$. For every sequence of these choices and every $0 \leq q \leq N$,

W_q is a maximum matching of H_q, and $e(J_q) = \Phi(H_q) = P_q$.

For $0 \leq q < N$ one has

$\Delta(J_q) = deg_{J_q}(v_{q+1}) = \ell_{q+2} - 1$.

Let ϵ_q be one if v_{q+1} is covered by W_q and zero otherwise. Then

$\nu(H_q) - \nu(H_{q+1}) = \epsilon_q$,

$deg_{H_q}(v_{q+1}) = \ell_{q+2} - 1 + \epsilon_q$.

If, whenever possible, a W_q − covered vertex is chosen among the maximum-degree vertices of J_q, then every chosen vertex also has maximum degree in H_q. Finally $E(H_N) = W_N$ and

$N - \nu(G) = |\{q : 0 \le q < N, \epsilon_q = 0\}| - |W_N|$.

Thus the entire sequence retains restrictions of the initially prescribed matching, without replacing its edges. Unit stars, repeated sizes, isolated host vertices, zero entries $\ell_k - 1$, and empty terminal graphs are included. All indices, degrees, edge counts, matching sizes, set cardinalities and potentials are nonnegative integers in their displayed domains; N-nu(G) may be signed.

Proof. The full finite family and its envelope have precisely the domains of Lemma 56, which guarantees all displayed maxima are attained. We first establish a residual potential bound for every vertex set, independent of the fixed matching.

Fix $q < N$ and any q-element set X. For each label γ include all its nonempty actual suffix pairs after integer cuts α, β with $\alpha + \beta = q$, $0 \le \alpha < s_\gamma$ and $0 \le \beta < t_\gamma$. The union of these suffix families is nonempty: a label attaining N admits such cuts whenever $q < N$. Every pair in the union has component budget $s_\gamma + t_\gamma - 1 - q$, so its largest component budget is N-q. By the actual-vertex extension in Lemma 41 (II), $H_X = G - X$ arrows each of these pairs.

Their diagonal envelope is ℓ_{q+2},..., ℓ_{N+1}. To check this at an integer local diagonal k with $2 \le k \le N - q + 1$, a suffix candidate with local indices i, j and cuts α, β has original indices $I = \alpha + i$ and $J = \beta + j$. They satisfy $I + J = q + k$, so its size sum minus one is a candidate for ℓ_{q+k}. Conversely, take any original candidate at indices I, J with $I + J = q + k$. Set $\alpha = \min(q, I - 1)$ and $\beta = q - \alpha$. If $\alpha = q$, then $\beta = 0$. Otherwise $\beta = q - I + 1 \le J - 1$ because $k \ge 2$. Thus both cuts are nonnegative, $\alpha \le I - 1 < s_\gamma$, $\beta \le J - 1 < t_\gamma$, and they total q. The resulting local positive indices I-alpha, J-beta total k and recover that candidate. This proves equality of the candidate sets and hence of their maxima.

Apply Lemma 56 to this finite nonempty family on H_X. Its potential bound is exactly $\Phi(H_X) \ge P_q$. For $q = N$, the same inequality follows from $P_N = 0$ and $\nu(H_X) \le e(H_X)$, since a matching is an edge subset. Thus the bound holds for every displayed X, including $q = N$.

The edges of H_X partition into W_X and $E(J_X)$. Since W_X is a matching in H_X,

$e(J_X) = \Phi(H_X) + \nu(H_X) - |W_X| \ge P_q$.

Both terms $\Phi(H_X) - P_q$ and $\nu(H_X) - |W_X|$ are nonnegative. Their sum is zero exactly when both are zero, proving the stated characterization of equality.

At X_0 empty, W is maximum and $\Phi(G) = P_0$, so $e(J_0) = P_0$. Suppose an arbitrary greedy prefix has reached $q < N$ with $e(J_q) = P_q$. The equality characterization gives $\Phi(H_q) = P_q$ and that W_q is maximum in H_q. For every vertex v of H_q, apply the general bound to X_q union {v}. Since deleting v removes precisely $deg_{J_q}(v)$ edges from J_q, it gives

$deg_{J_q}(v) \le P_q - P_{q+1} = \ell_{q+2} - 1$.

Write $D = \ell_{q+2} - 1 \ge 0$.

Choose a label γ and original indices i, j attaining ℓ_{q+2}. Their cuts i-1, j-1 are legal and total q. The graph H_q arrows the corresponding nonempty suffix pair by Lemma 41 (II). Both targets have a positive-edge component, so an edgeless H_q would avoid them. Hence H_q has an edge and in particular has a vertex. Its vertex set is also the vertex set of J_q. If $D = 0$, the upper degree bound already gives $\Delta(J_q) = 0$ and a maximum-degree vertex exists.

Suppose $D \ge 1$ and $\Delta(J_q) < D$. Every vertex has $H_q - degree$ at most its $J_q - degree$ plus one, so $\Delta(H_q) \le D$. Any vertex of $H_q - degree\ D$ must be covered by W_q, since an uncovered vertex has unchanged degree at most D-1. Apply Lemma 35 with this covering matching and with capacities $A = n_i^\gamma - 1$, $B = m_j^\gamma - 1$. These are nonnegative integers with $A + B = D$. The resulting red-blue degree split of H_q omits both leading stars

of the chosen suffix pair and therefore avoids both entire targets. This contradicts its arrowing. Consequently $\Delta(J_q) = D$.

Every maximum-degree choice v_{q+1} thus removes exactly D edges from J_q and leaves $e(J_{q+1}) = P_q - D = P_{q+1}$. The equality characterization again gives $\Phi(H_{q+1}) = P_{q+1}$ and that W_{q+1} is maximum there. This proves the induction for every possible greedy choice, and the preceding existence argument permits all N steps.

The restriction W_{q+1} loses the unique $W_q - edge$ incident with v_{q+1} exactly when $\epsilon_q = 1$. Since both restrictions are maximum, their cardinality difference is $\nu(H_q) - \nu(H_{q+1}) = \epsilon_q$. The partition $E(H_q) = E(J_q)$ disjoint union W_q also gives the displayed $H_q - degree$ identity.

For the optional tie rule, if a $J_q - degree - D$ vertex is $W_q - covered$, selecting it gives $H_q - degree\ D + 1$, which is maximum because every $H_q - degree$ is at most $D + 1$. If no such vertex is covered, a vertex of $H_q - degree\ D + 1$ would have to be covered and have $J_q - degree\ D$, a contradiction. Every $H_q - degree$ is then at most D, and any selected $J_q - degree - D$ vertex attains D in H_q. This proves the rule in both cases.

At $q = N$, $e(J_N) = P_N = 0$, so $E(H_N) = W_N$. Telescoping the exact matching losses gives $\nu(G) = |W_N| + \sum_{q=0}^{N-1} \epsilon_q$. There are N summands, each zero or one. Rearranging this equality proves the last identity, with no sign assumption on N-nu(G). All arguments permit isolated vertices and zero complement degree, so the stated boundary cases are included. □

Remark. The matching-complement edge count is the residual potential plus the unused matching capacity. Equality forces both terms to be exact. A smaller complement degree would make the same matching cover every critical-degree vertex and yield a forbidden coloring.

Depends on. Lemma 35, Lemma 41, Lemma 56.

Lemma 96 (Exact deficiency for interval degree prescriptions). *Let G be a finite simple graph, and let g, f: V(G) → Z be integer functions satisfying $0 \le g(v) \le f(v) \le d_G(v)$ for every vertex v. For a spanning subgraph R of G, define*

$dev_{g,f}(R) = \sum_{v \in V(G)} (\max(g(v) - d_R(v), 0) + \max(d_R(v) - f(v), 0))$.

For disjoint subsets S, T of $V(G)$, let $q_{g,f}(S,T)$ count the connected components C of G-(S union T) such that $g(v) = f(v)$ for every v in C and $\sum_{v \in C} f(v) + e_G(C,T)$ is odd. Here $e_G(A,B)$ denotes the number of edges with one endpoint in each of the disjoint vertex sets A, B. Then

$\min_{R spanning subgraph of G}\ dev_{g,f}(R)$

$= \max_{S,T disjoint subsets of V(G)}\ [q_{g,f}(S,T) + e_G(S,T) - \sum_{v \in S} f(v) - \sum_{v \in T}(d_G(v) - g(v))]$.

Both extrema are attained. In particular, a spanning subgraph with $g(v) \le d_R(v) \le f(v)$ for every vertex exists if and only if every expression in the displayed maximum is nonpositive. The statement includes the empty graph, isolated vertices, zero bounds, and bounds equal to the ambient degree.

Proof. Write $d_v = d_G(v)$. We first represent the left side by a prescribed matching problem. Construct a finite simple graph K. For every vertex v of G, introduce a set I_v of d_v incidence vertices, one for each original edge incident with v, and a set B_v of $d_v - g(v)$ additional vertices. Mark every incidence vertex and exactly $d_v - f(v)$ vertices of B_v; its remaining f(v)-g(v) vertices are unmarked. Join every vertex of I_v to every vertex of B_v. For each edge uv of G, join its incidence vertex in I_u to its incidence vertex in I_v. These latter edges are called cross edges. All vertex sets used in distinct blocks are disjoint. Let W be the marked set.

Fix a spanning subgraph R of G and include its cross edges in a matching of K. The unused incidence vertices in block v number $h_v = d_v - d_R(v)$. Match as many of them as possible to B_v, using marked vertices of B_v first. The number of marked vertices left uncovered in this block is

$\max(h_v - (d_v - g(v)), 0) + \max((d_v - f(v)) - h_v, 0)$

$= \max(g(v) - d_R(v), 0) + \max(d_R(v) - f(v), 0)$.

The first term counts unused incidence vertices, and the second counts unused marked vertices of B_v; at most one term is positive. The complete bipartite local edges realize these choices independently in every block.

Conversely, let a matching of K be given and let R consist of its cross edges. There remain $h_v = d_v - d_R(v)$ incidence vertices in block v. At most $d_v - g(v)$ can be matched locally, so at least $\max(g(v) - d_R(v), 0)$ marked incidence vertices remain uncovered. At most h_v marked vertices of B_v can be matched, so at least $\max(d_R(v) - f(v), 0)$ marked vertices there remain uncovered. It follows that the minimum number $\eta_W(K)$ of uncovered marked vertices over all matchings of K is exactly $\min_R dev_{g,f}(R)$.

By fact Lemma 46, this prescribed matching deficiency is

$\eta_W(K) = \max_{X \subset of V(K)}(o_W(K - X) - |X|)$,

where o_W counts components of odd order whose vertices all belong to W. Among the maximizing sets X choose one of minimum cardinality. All choices are possible because the graphs are finite. We put this separator into a form indexed by original vertices.

The following restoration observations concern this fixed separator. If a deleted marked vertex has neighbors in at most two components of K-X, restoring it reduces o_W by at most one. Indeed, if two formerly counted components are joined, their odd orders together with the restored marked vertex have odd total order, and every vertex remains marked, so the resulting component is counted. If at most one of the touched components was counted, at most one count can be lost. The cases of zero or one touched component obey the same bound. If a deleted unmarked vertex has neighbors in at most one component, restoring it likewise loses at most one count. Each restoration reduces $|X|$ by one, so it cannot reduce $o_W - |X|$. Maximality of that expression and minimality of $|X|$ therefore prohibit every such restoration.

Suppose a block has at least one surviving incidence vertex and at least one surviving vertex of B_v. Its surviving local vertices all lie in one component. A deleted incidence vertex would touch this component and at most the component of its unique cross-edge neighbor; it is marked and hence could be restored. A deleted vertex of B_v would touch just the local component and could also be restored. Thus this block has no deleted vertices.

If all incidence vertices of a nonempty I_v are deleted, no vertex of B_v can be deleted: restoring such a vertex makes an isolated vertex, contradicting the same minimality argument. If all vertices of a nonempty B_v are deleted, no incidence vertex can be deleted: it could touch only the component of its unique cross-edge neighbor when restored. Finally, if B_v is empty, every incidence vertex has only its cross-edge neighbor, so no incidence vertex is deleted. These statements also show that both nonempty groups of a block cannot be deleted simultaneously.

Define S to be the set of vertices v with $d_v > 0$ for which all of I_v is deleted. Define T to consist of the vertices with B_v nonempty and entirely deleted, together with every vertex whose B_v is empty. These sets are disjoint. Isolated vertices belong to T and contribute no gadget vertices and no costs. Every block indexed by $U = V(G) - (SunionT)$ has both local groups nonempty and entirely present. We have

$X = (\bigcup_{v \in S} I_v)$ union $(\bigcup_{v \in T} B_v)$.

We count the odd entirely marked components of K-X. For v in S the surviving vertices of B_v are isolated; exactly $d_v - f(v)$ of them are marked. For v in T its incidence vertices have no local neighbors. An original edge from S to T leaves one isolated marked incidence vertex, contributing one count. An edge within T gives a component with two marked vertices and contributes zero. An edge from T to U gives a marked leaf attached to the U block.

The connected components of G[U] correspond to the remaining connected components of K-X, with their attached leaves from T. This follows because every U block is connected and each original edge between U blocks becomes a cross edge. For a component C of G[U], the corresponding gadget component is entirely marked exactly when $g(v) = f(v)$ for every v in C: the only possible unmarked vertices are the f(v)-g(v) vertices of B_v. When this equality holds, its order is

$\sum_{v \in C}(2d_v - g(v)) + e_G(C, T)$,

which has the same parity as $\sum_{v \in C} f(v) + e_G(C, T)$. Consequently

$o_W(K - X) = \sum_{v \in S}(d_v - f(v)) + e_G(S, T) + q_{g,f}(S, T)$.

Subtracting $|X| = \sum_{v \in S} d_v + \sum_{v \in T}(d_v - g(v))$ gives

$\min_R dev_{g,f}(R)$

$= q_{g,f}(S,T) + e_G(S,T) - \sum_{v\in S} f(v) - \sum_{v\in T}(d_v - g(v))$.

In particular the minimum deviation is at most the maximum asserted in the statement.

For the reverse inequality, let arbitrary disjoint S, T and arbitrary R be given, and again put $U = V(G) - (SunionT)$. Denote the summand of $dev_{g,f}(R)$ at v by a_v. Set

$P = \sum_{v\in S} d_R(v) + \sum_{v\in T}(d_v - d_R(v))$.

The degree inequalities $d_R(v) \le f(v) + a_v$ and $d_v - d_R(v) \le d_v - g(v) + a_v$ give

$P \le \sum_{v\in S} f(v) + \sum_{v\in T}(d_v - g(v)) + \sum_{v\in SunionT} a_v$.

Every edge from S to T contributes exactly one to P, either as an R edge at its S endpoint or as a non-R edge at its T endpoint.

Consider a component C counted by $q_{g,f}(S,T)$ for which $a_v = 0$ at every vertex. Since $g = f$ on C, each $d_R(v) = f(v)$ there. Taking parity in the degree sum over C gives

$\sum_{v\in C} f(v) + e_G(C,T)$

congruent to $e_R(C,S) + e_{G-R}(C,T)$ modulo 2.

The left side is odd, so the right side is a positive integer. These boundary edges contribute to P. Different components use disjoint boundary edges, none of which are S-T edges. Since every a_v is a nonnegative integer, the number of counted components on which some a_v is positive is at most $\sum_{v\in U} a_v$. We therefore obtain

$P \ge e_G(S,T) + q_{g,f}(S,T) - \sum_{v\in U} a_v$.

Combining the two bounds on P proves that $dev_{g,f}(R)$ is at least the barrier expression for these S, T. Taking the minimum over R and the maximum over S, T proves equality.

There are finitely many spanning subgraphs and ordered disjoint pairs S, T, so both extrema are attained. Every deviation is nonnegative and vanishes exactly when all interval constraints hold. Thus the maximum is nonnegative, and interval feasibility is equivalent to every barrier expression being nonpositive. The construction and counting used empty groups and zero sums without exceptions, which covers the stated boundary cases. □

Remark. A local incidence gadget converts interval deviation exactly to uncovered required vertices of a matching. A minimum-cardinality maximizing separator has only two kinds of deleted blocks; restoring vertices rules out every partial block. The resulting matching barrier is the asserted degree barrier.

Depends on. Lemma 46.

(printed in plain notation)

Lemma 97 (Edge deletion in a regular factor-critical graph with hereditary goodness).

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Edge deletion in a regular factor-critical graph with hereditary goodness.

Use the finite integer certificate costs c_i(G), sum Sigma(G)=sum_(i>=0)c_i(G), matching
number nu(G), excess ex(G)=c_0(G)-nu(G), and signed gap gap(G)=Sigma(G)-e(G) from Lemma
23 . A graph is LP-good when gap<=0. A graph is factor-critical if deleting any one
vertex leaves a perfect matching. An edge-deleted graph retains all vertices.

Let d>=1 and b0>=0 be integers. Let F be a finite simple d-regular factor-critical graph
with gap(F)=1. Put E=ex(F), and assume E>=b0. Suppose every proper subgraph J of F with
ex(J)>=b0 is LP-good; proper subgraphs allow arbitrary vertex deletions, edge deletions,
or both.

Then d is even, and for an integer k>=1 one has |V(F)|=2k+1, nu(F)=k, and c_d(F)=1. For
every edge e of F and every integer i>=0 define r_i(e)=c_i(F)-c_i(F-e), and put
R(e)=sum_(i=0)^d r_i(e). These quantities satisfy

r_i(e) in {0,1}, r_i(e)=0 for i>=d+1, r_d(e)=1,
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nu(F-e)=k, ex(F-e)=E-r_0(e), R(e)>=2,

and gap(F-e)=2-R(e)<=0.

In particular every edge strictly lowers at least one row with index in {0,...,d-1}, where the index may depend on the edge. All cardinalities, degrees, costs, capacities, matching budgets and row indices are nonnegative integers; gap and differences not explicitly bounded may be signed integers.

Proof. An empty graph has every certificate cost zero by Lemma 23 , so the hypothesis gap(F)=1 implies F is nonempty. Since d>=1, it has an edge and at least two vertices. For any vertex u, the perfect matching of F-u shows |V(F)|-1 is even. Write |V(F)|=2k+1 with k>=1. That perfect matching has k edges, and no matching on 2k+1 vertices has more than k edges, proving nu(F)=k. The degree sum d(2k+1)=2e(F) makes d even.

Fix any vertex u. The perfect matching of F-u covers every vertex of that graph, in particular every vertex whose F-u degree is d. Its maximum degree is at most d. The covering-matching conclusion of Lemma 61 therefore gives c_d(F-u)=0. The vertex extension bound of Lemma 72 gives c_d(F)<=1.

To prove the reverse inequality, take degree capacities (1,d-1). If total matching budget zero were feasible, both exceptional parts would be empty: a nonempty edge set contains a matching of size one. The remaining edge parts Q_R,Q_B would have maximum degrees at most 1,d-1. At every vertex v of the d-regular graph F, d_Q_R(v)+d_Q_B(v)=d, so d_Q_R(v)>=1 and hence equals one. Thus Q_R would be a perfect matching of the odd-order graph F, impossible because its edges cover vertices in pairs. Total zero is therefore infeasible at this capacity pair. By the definition of the nonnegative integer cost in Lemma 23 , c_{1,d-1}(F)>=1 and c_d(F)>=1. Together with the upper bound, this gives c_d(F)=1.

Now fix any edge e=uv and put J=F-e, retaining all vertices. Every edge of the perfect matching of F-u belongs to J. It has k edges, so nu(J)>=k. Since J has 2k+1 vertices, nu(J)<=k and therefore nu(J)=k. The vertices u and v have J-degree d-1. Every vertex of J-degree d is different from u and v and is covered by that same perfect matching of F-u. Applying the covering-matching conclusion of Lemma 61 with degree bound d gives c_d(J)=0. Consequently r_d(e)=1.

A single vertex covers the graph consisting of the one removed edge. Restriction and edge restoration from Lemma 72 imply 0<=r_i(e)<=1 for every i>=0; integrality makes each drop zero or one. The strong tail of Lemma 61 gives c_i(F)=c_i(J)=0 for i>=d+1, so all those drops are zero. The matching identity already proved gives ex(J)=c_0(J)-k=E-r_0(e). Subtracting the finite cost sums and using e(J)=e(F)-1 yields

gap(J)=Sigma(F)-R(e)-(e(F)-1)=gap(F)+1-R(e)=2-R(e).

The nonnegative drops and r_d(e)=1 give R(e)>=1. Suppose R(e)=1. Every r_i(e) below d must then be zero, including r_0(e). Thus ex(J)=E>=b0, and J is a proper subgraph because exactly one existing edge was deleted. The hereditary goodness hypothesis gives gap(J)<=0, whereas the exact identity gives gap(J)=1. This contradiction proves R(e)>=2. The same identity then gives gap(J)<=0. Since the top row contributes exactly one to R(e), at least one index below d has drop one. The edge e was arbitrary, proving every stated quantifier.

□

Depends on. Lemma 23, Lemma 61, Lemma 72.

Lemma 98 (Exclusion of a single odd block beside a maximum independent set). *All graphs are finite and simple, and every vertex deletion is induced. Let $D \geq 6$ be an even integer. There is no connected D-regular factor-critical equimatchable graph G with a partition $V(G) = I$ disjoint union X disjoint union W_0 having the following properties. The set I is a maximum independent set of size $\alpha \geq D$. The set W_0 is independent of size alpha-2 and is anticomplete to X. The induced graph G[X] is connected, has odd order at least three, and has independence number at most two. Exactly 2D edges join X to I. If G[X] is 2-connected, it is K_3 with $D = 6$, K_{D-1}, or K_{D-1} with one edge deleted.*

Here K_j is the complete graph on j vertices. A graph is 2-connected if it has at least three vertices and remains connected after deleting any vertex. Factor-critical means that deleting any vertex leaves a perfect matching; equimatchable means every inclusion-maximal matching has maximum cardinality. A perfect matching covers every vertex and the empty

graph has an empty perfect matching. All orders, degrees, matching and independence cardinalities, and parameters are integers in the specified domains.

Proof. The graph G is 2-connected by Lemma 67. Every vertex outside the maximum independent set I has an I-neighbor, since otherwise it could be adjoined to I. The stated partition and anticompleteness make every neighbor of a vertex in W_0 belong to I. Write $b(X) = 2D$ for the number of edges between X and I.

We first show G[X] is 2-connected. If not, Lemma 26 gives a cut vertex c such that X minus{c} is partitioned into two nonempty clique components A, B, with c complete to A and adjacent to at least one vertex of B. Write $a = |A|$ and $b = |B|$. Each clique has size at most D-1, because any vertex in it adjacent to c also has an I-neighbor. Moreover $a \leq D - 2$, since c has all a neighbors in A, a neighbor in B, and an I-neighbor. Each vertex of A has exactly D-a I-neighbors. The total number of I-edges from B is at least $b(D - b)$, since its internal degree in X is at most b. The vertex c contributes at least one further I-edge. If $2 \leq a \leq D - 2$, this gives $b(X) \geq 2D - 4 + (D - 1) + 1 = 3D - 4 > 2D$. If $a = 1$ and $2 \leq b \leq D - 2$, it gives the same contradiction. If $a = b = 1$, X is a three-vertex path and $b(X) = (D - 1) + (D - 1) + (D - 2) = 3D - 4 > 2D$. These inequalities use $t(D - t) \geq D - 1$ for $1 \leq t \leq D - 1$ and $t(D - t) \geq 2D - 4$ for $2 \leq t \leq D - 2$, verified by the concavity of that quadratic and its endpoint values.

The only remaining orders are $a = 1$, $b = D - 1$. Let $k = |N_G(c)$ intersect $B|$. The vertex c has D-1-k I-neighbors, the unique vertex $y0$ in A has D-1, and B contributes 2(D-1)-k. Hence $b(X) = 4(D - 1) - 2k$. The equality $b(X) = 2D$ forces $k = D - 2$. Let $x0$ be the unique vertex of B not adjacent to c. Set $I' = W0$ union {x0,y0}. This is independent, disjoint from I, and of size α. The vertex c is outside both sets. The residual G-(I union I' union {c}) is the even clique B minus{x0}, which has a perfect matching. Each residual vertex has exactly the I'-neighbor $x0$: it has none in W_0 and none in A. The odd clique B has order $D - 1 \geq 5$, meets I' exactly in $x0$, and has all external neighbors in I union {c}. This is forbidden by Lemma 67. The cut case is excluded.

Therefore G[X] is 2-connected. By hypothesis it is K_3 with $D = 6$, K_{D-1}, or K_{D-1} with one edge deleted. The graph K_{D-1} has $boundary(D - 1)(D - (D - 2)) = 2D - 2$, contradicting $b(X) = 2D$.

We give a finite matching principle for the remaining arguments. In a finite bipartite graph with specified left set L, if no matching covers L, some nonempty $S \subseteq L$ has fewer than $|S|$ neighbors. Indeed choose a maximum matching and start alternating paths at all unmatched left vertices, using nonmatching edges left to right and matching edges right to left. Every reached right vertex is matched, since otherwise an augmenting path would increase the matching. Its partner is reached on the left. Conversely every reached matched left vertex was reached through its matched partner. Thus for the reached sets S on the left and R on the right, $|R| < |S|$. Every neighbor of S is in R: nonmatching edges are followed, and matching edges point to already reached partners. Therefore R is the full neighbor set of S. Contrapositively, if every $S \subseteq L$ has at least $|S|$ neighbors, a matching covering L exists. This argument also applies to any selected subset of left vertices by taking its induced bipartite incidence graph.

Suppose G[X] is K_{D-1} with the one edge x1x2 deleted. The vertices $x1$, $x2$ have exactly three I-neighbors, and every other vertex of X has exactly two I-neighbors, because all edges out of X go to I. There is no matching of four edges between X and I. To prove this, suppose J is such a matching. The remaining subgraph on X minus $V(J)$ has odd order $D - 5 \geq 1$ and is a clique with at most one missing edge. Select a vertex x' from it so that deleting x' leaves an even clique: if both endpoints of the missing edge remain, select one of them; otherwise any vertex works. Perfectly match that even clique and adjoin this matching to J. The remaining vertices form two independent sets, I minus $V(J)$ of size alpha-4 and W_0 union {x'} of size alpha-1. The latter is independent because X is anticomplete to W_0. Every extension leaves at least three vertices uncovered, contradicting factor-criticality and equimatchability of G. This proves the four-edge restriction.

The I-neighbor sets of $x1$ and $x2$ must be identical triples, and must contain the I-neighborhood of every other X-vertex. Here are all selection details. Choose any two distinct ordinary vertices $x3$, $x4$ in X minus{x1,x2}; there are $D - 3 \geq 3$ ordinary vertices. For the four left vertices $x1$, $x2$, $x3$, $x4$, any single vertex has at least two I-neighbors and any pair has at least two. Any triple includes at least one of $x1$, $x2$ and consequently has at least three neighbors. If the union of all four neighbor sets had size at least four, the finite matching principle would give a four-edge matching, prohibited above. The union therefore has size at most three. Since each of $x1$, $x2$ has three neighbors, their neighborhoods coincide in a triple I_0 and the neighborhoods of $x3$, $x4$ are contained in I_0. Every ordinary vertex can be included in such a pair, so every vertex of X has its I-neighborhood contained in I_0.

There is no three-edge matching from I_0 to three ordinary vertices of X. Suppose J is one. The vertices of X minus{x1,x2} not used by J form an even clique of order D-6, allowing a perfect matching J'. The endpoint set of J union J' is exactly I_0 $union(Xminus\{x1, x2\})$, which is precisely the open neighborhood of $x1$. Thus J union J' is an inclusion-minimal matching isolating $x1$: every selected matching edge covers a neighbor of $x1$, and all its neighbors are covered. Lemma 6 says the residual G-(V(J union J') union {x1}) is empty, an even clique, or a balanced complete bipartite graph. Its vertices partition into the two independent sets I minus I_0 of size alpha-3 and W_0 union {x2} of size alpha-1. Both are nonempty, the second has size at least 5 since $\alpha \geq D \geq 6$, and the graph is therefore nonempty and noncomplete. If it were a balanced complete bipartite graph, these two independent sets covering its vertices would be its two bipartition parts, yet their sizes differ by two. This is impossible, proving the restriction.

Each ordinary vertex has a two-element neighborhood inside the triple I_0. They all have the same pair of neighbors: otherwise choose two ordinary vertices with different pairs and any third ordinary vertex. The three pairs have union I_0 of size three. Every one-vertex subset has at least two neighbors, every two-vertex subset at least two, and all three together have three. The finite matching principle would give the just-prohibited three-edge matching. Name the common pair {z1,z2}, and write $I0 = \{z1, z2, z3\}$.

Choose any ordinary vertex $x3$. Match $x1$ to $z1$ and $x2$ to $z2$, and perfectly match the even clique X minus{x1,x2,x3}. This matching has endpoint set exactly (X minus{x3}) union {z1,z2}, which is the full neighborhood of $x3$; hence it is inclusion-minimal isolating $x3$. Its Lemma 6 residual is G-(X union {z1,z2}). It is partitioned into the independent sets I minus{z1,z2} and W_0, each of size $\alpha - 2 \geq 4$, so must be $K_{\alpha-2,\alpha-2}$. The vertex $z3$ therefore neighbors all W_0. Its only neighbors in X are $x1$, $x2$, and it has none in I, so $D = d_G(z3) = |W0| + 2 = \alpha$. Since $\alpha \geq 6$ there is a vertex z in I minus I_0. It has no neighbor in X and none in I, hence at most $|W0| = \alpha - 2 < D$ neighbors, a contradiction. This excludes K_{D-1} minus an edge.

We have reached the final possibility $D = 6$ and X a triangle. Name its vertices $w0$, $x1$, $y1$. Let $I0 = N_G(w0)$ intersect I, a set of four vertices. The graph G-X is bipartite with independent parts I and W_0. There is no matching in it saturating I_0. If J saturated I_0, keep just its four edges incident with I_0. Together with the edge x1y1 it forms an inclusion-minimal matching isolating $w0$, since it covers all four I-neighbors of $w0$ and its two triangle neighbors, and each matching edge is necessary to cover a neighbor. The Lemma 6 residual has independent parts I minus I_0, of size $\alpha - 4 \geq 2$, and W_0 minus $V(J)$, of size alpha-6. It cannot be empty or a clique because the first part is independent of size at least two. It also cannot be nonempty balanced complete bipartite: these two sets cover all its vertices and are independent, and either one is empty, leaving no edge, or they are its unequal bipartition parts. Their sizes differ by two. Thus such J cannot exist.

Every z in I_0 has at least three neighbors in W_0, since its total degree is six and X has only three vertices. By the finite matching principle, failure of a matching saturating the four-element set I_0 gives a nonempty subset S of I_0 with fewer than $|S|$ W0-neighbors. Each nonempty subset has at least three neighbors, so S must be all of I_0, and its full W0-neighbor set, denoted W_1, has size exactly three. Each z in I_0 neighbors all three vertices of W_1 and, to reach degree six, all three vertices of X; it has no other neighbor. Write $I0 = \{z1, z2, z3, z4\}$ and $W1 = \{w1, w2, w3\}$.

The four edges x1z1, y1z2, w1z3, w2z4 form an inclusion-minimal matching isolating $w0$. Each edge covers a distinct neighbor of $w0$ and all six neighbors of $w0$ lie among the matching endpoints. Its Lemma 6 residual is partitioned into $I1 = I$ minus I_0 and $W2 = W0$ minus{w1,w2}, each of size $\alpha - 4 \geq 2$. It must be $K_{\alpha-4,\alpha-4}$. The vertex $w3$ lies in W_2. Its neighbors are all in I, it has all four I0-neighbors, and the complete bipartite residual makes it adjacent to all alpha-4 vertices of I_1. Thus $6 = d_G(w3) = 4 + (\alpha - 4) = \alpha$. Hence $|W0| = \alpha - 2 = 4$ and $|I1| = 2$. Choose w' in the nonempty set W_0 minus W_1. It has no neighbor in I_0 by the definition of W_1, and has no neighbor outside I since it lies in W_0. It has at most the two neighbors of I_1, contradicting its degree six.

All possible shapes of G[X] have been contradicted. Therefore no graph and partition satisfying the hypotheses exist. □

Remark. A single odd block exhausts only twice the regular degree in boundary edges. Cut structure leaves one exceptional shape; the remaining almost-clique and triangle shapes yield minimal isolating matchings whose residual parts have incompatible sizes or degrees.

Depends on. Lemma 6, Lemma 26, Lemma 67.

Lemma 99 (Exposure and maximal-matching loss in regular components). *Let calA be a finite nonempty set of labels. For each γ in calA let s_γ, t_γ be positive integers, and let $n_1^\gamma \geq \cdots \geq n_{s_\gamma}^\gamma \geq 1$ and $m_1^\gamma \geq \cdots \geq m_{t_\gamma}^\gamma \geq 1$ be integer lists. Let F_1^γ, F_2^γ be their vertex-disjoint star unions, prescribed red and blue. Copies are ordinary subgraphs with vertex-disjoint target components. Put $N = \max_\gamma(s_\gamma + t_\gamma - 1)$. For each integer $2 \leq j \leq N+1$ let ℓ_j be the maximum of $n_i^\gamma + m_h^\gamma - 1$ over every label and valid indices with $i + h = j$. Put $D = \ell_2 - 1$, $P = \sum_{j=2}^{N+1}(\ell_j - 1)$, and suppose $D \geq 3$. Let R be the number of initial entries of ℓ equal to $D+1$.*

Let H be a finite simple graph with maximum degree at most D, arrowing every pair of calA separately. Write $e(G)$, $\nu(G)$, and $\Phi(G) = e(G) - \nu(G)$ for edge count, maximum matching cardinality and potential. Put $\sigma = \Phi(H) - P$ and $S_D = \{vinV(H) : deg_H(v) = D\}$. Let ρ be the minimum of $|S_D$ minus $V(W)|$ over all matchings W of H, where $V(W)$ is the endpoint set. Then σ is a nonnegative integer,

$R \leq \rho \leq R + \sigma$,

and every maximum matching leaves between R and $R + \sigma$ vertices of S_D uncovered.

More generally, let I be any independent subset of S_D, put $k = |I|$ and $b = \nu(H) - \nu(H - I)$, and suppose $k \geq R$. Then

$\sigma \geq k - R - b + (D-1)\max(k - N, 0)$.

Vertex deletion is induced. This assertion imposes no condition that I arise from a matching.

Let Q be the union of any selected whole connected D-regular components of H, allowing Q to be empty, and put $J = H - V(Q)$. Choose any inclusionwise maximal matching W of Q and any maximum matching Z of J. Put

$b_Q = \nu(Q) - |W|$,

$k_0 = |V(Q)| - 2\nu(Q) + |\{v \in V(J) : deg_H(v) = Dandvnot \in V(Z)\}|$,

$k = k_0 + 2b_Q$.

Then $k_0 \geq R$ and

$\sigma \geq k_0 - R + b_Q + (D-1)\max(k - N, 0)$.

In particular $b_Q \leq \sigma$.

For each whole connected D-regular component C define

`g(C)=max{nu(C)-|W|:W` *is an inclusionwise maximal matching of C}.*

Put $\zeta = \sum_C g(C)$ and $h_{reg} = \sum_C(|V(C)| - 2\nu(C))$, summing over all such components. Then $\zeta \leq \sigma$. Every maximum matching of the complement of their union leaves at most $R + \sigma - h_{reg} - \zeta$ vertices of ambient H-degree D uncovered. At $\sigma = 0$ every whole D-regular component is equimatchable, meaning all its inclusionwise maximal matchings are maximum. At $\sigma = 1$ at most one such component is not equimatchable, and every maximal matching of that component is short of maximum by at most one edge.

All graph counts, sizes, capacities, indices, matching numbers, losses, exposure counts, and extrema are integers in their stated nonnegative or positive domains. Empty selected unions, empty complementary graphs, isolated vertices, unit targets and repeated target sizes are permitted. No half-order matching, factor-criticality, edge-budget or positive matching-deficit assumption is imposed.

Proof. A label attaining N supplies candidates on every diagonal from 2 through $N+1$, so all ℓ_j are attained and positive. The envelope is nonincreasing: on any diagonal beyond the first, one of the two positive indices exceeds one; decreasing it gives a candidate on the preceding diagonal at least as large. Thus $1 \leq R \leq N$, the first R entries of the potential profile $\ell_j - 1$ equal D, and every later one is at most D-1. In particular

$P \leq (D-1)N + R$. (1)

The finite-family potential theorem Lemma 56 gives $\Phi(H) \geq P$, hence $\sigma \geq 0$. All matching extrema exist because the graph and its matching family are finite.

We prove that every matching of H leaves at least R members of S_D uncovered. Choose a terminal maximizing label and valid indices i, j with $i + j = R + 1$ and $n_i^\gamma + m_j^\gamma = D + 2$. Put $A = n_i^\gamma - 1$ and $B = m_j^\gamma - 1$, nonnegative integers with $A + B = D$. Suppose a matching W left $r < R$ members of S_D uncovered. For each

of these r vertices choose one incident edge and let E_0 be their union. Such edges exist because $D \geq 3$. None belongs to W, and $|E_0| \leq r \leq R-1 = i+j-2$.

In the spanning graph $H - E_0$, every vertex still of degree D was covered by W in H, and W remains a matching of that graph. The covering-splitting theorem Lemma 35 gives an edge partition of $H - E_0$ into red and blue parts of maximum degrees at most A, B. Color $\min(i-1, |E_0|)$ of the edges of E_0 red and the remainder blue. The number of added red edges is at most i-1 and the number of added blue edges is at most j-1, by $|E_0| \leq i+j-2$.

Any red copy of the entire F_1^γ would have i vertex-disjoint stars of size at least $n_i^\gamma = A+1$. Each would need an added red edge, since the red degree in $H - E_0$ is at most A. These edges would be distinct, and even form a matching, contradicting the count i-1. The first j components of a blue F_2^γ similarly require j distinct added blue edges. Thus this coloring avoids both entire targets of that label, contradicting arrowing. This proves $\rho \geq R$ and the same lower bound for every matching, without requiring it to be maximum.

We next establish the residual potential bound for any actual vertex set Y with integer size $l \leq N$. Put

$P_l = \sum_{j=l+2}^{N+1} (\ell_j - 1)$,

where $P_N = 0$. If $l = N$, $\Phi(H - Y) \geq 0 = P_N$ because a matching is a subset of the edge set. Suppose $l < N$. For each original label retain all its nonempty actual suffix pairs after legal leading cuts α, β with $\alpha + \beta = l$. The resulting family is nonempty, since a label of budget N allows such cuts, and its maximum component budget is N-l. By the actual-vertex extension in Lemma 41 (II), H-Y arrows every retained pair: any avoiding coloring would extend across the same Y to avoid its original label.

The new family envelope at local diagonal h is ℓ_{l+h}, for every integer $2 \leq h \leq N - l + 1$. A retained candidate has this original index sum. Conversely take any valid original i, j with $i + j = l + h$. Set $\alpha = \max(0, l - j + 1)$, $\beta = l - \alpha$. Then α, β are nonnegative integers, total l, with $\alpha \leq i - 1$ and $\beta \leq j - 1$. They are legal cuts leaving both corresponding components. Their local indices i-alpha, j-beta sum to h. Thus every original candidate on that diagonal occurs in the retained family. Applying Lemma 56 to this complete retained family gives

$\Phi(H - Y) \geq P_l$. (2)

No suffix with an empty target has been used.

Now let I, k, b be as in the independent-set assertion. Independence and degree D give $e(H - I) = e(H) - Dk$: no edge is counted at two deleted endpoints. By the definition of b,

$\Phi(H - I) = P + \sigma - Dk + b$. (3)

Restriction of a maximum matching to H-I shows $0 \leq b \leq k$, and all quantities are integers.

If $k \leq N$, apply (2) and use $k \geq R$ to bound the first k potential entries:

$P - P_k \leq RD + (k - R)(D - 1) = Dk - (k - R)$.

Combining with (3) yields $\sigma \geq k - R - b$. Here $\max(k - N, 0) = 0$. If $k > N$, use nonnegative residual potential and (1) in (3) to obtain

$\sigma \geq Dk - b - P \geq k - R - b + (D - 1)(k - N)$.

These are precisely the two cases of the independent-set inequality, including $k = N$.

Fix any maximum matching of H and take I to be its uncovered $S_D - vertices$. Uncovered vertices of a maximum matching are independent: an edge joining two could be added to increase the matching. All its matching edges survive deletion of I, so $\nu(H - I) = \nu(H)$ and $b = 0$. The lower exposure bound already proved gives $k \geq R$. The independent-set inequality implies $\sigma \geq k - R$, proving $k \leq R + \sigma$ for every maximum matching. Since ρ is the minimum over all matchings, it is no larger than such a k, completing the asserted interval for ρ.

Consider now a selected whole regular union Q and W, Z as in the statement. Matching numbers add across whole component unions: every matching splits over the components, and componentwise maximum matchings unite. A maximum matching of Q together with Z is therefore maximum in H. Every vertex of Q has ambient degree D, so its uncovered S_D count is exactly k_0. The lower exposure bound gives $k_0 \geq R$.

Let I contain every Q-vertex uncovered by W, together with every ambient degree-D vertex of J uncovered by Z. The W-uncovered vertices are independent because W is inclusionwise maximal; the Z-uncovered

vertices are independent because Z is maximum. There are no edges between Q and J, so I is independent in H and lies in S_D. Its size is $k = k_0 + 2b_Q$.

The matching W is perfect on Q-I: precisely its uncovered vertices were deleted, and all its edges remain. Consequently $\nu(Q - I) = |W|$. The matching Z survives in J-I and still has size $\nu(J)$; the opposite bound holds by subgraph inclusion, so $\nu(J - I) = \nu(J)$. Additivity gives $\nu(H - I) = \nu(H) - b_Q$. The independent-set inequality with $b = b_Q$ now yields

$\sigma \geq k_0 - R + b_Q + (D-1)\max(k - N, 0)$.

All terms except $k_0 - R$ and b_Q are nonnegative and $k_0 \geq R$, proving $b_Q \leq \sigma$. If Q is empty, its unique matching is empty and all the same identities remain valid.

Finally choose Q to be the union of all whole D-regular components. In each such C choose a maximal matching attaining the nonnegative integer $g(C)$, which exists by finiteness. Their union is maximal in Q and has $b_Q = \zeta$. For any maximum matching Z of the complementary J, write u for its uncovered ambient degree-D count. Then $k_0 = h_{reg} + u$. The proved inequality, after dropping its nonnegative last term, gives

$\sigma \geq h_{reg} + u - R + \zeta$.

Thus $u \leq R + \sigma - h_{reg} - \zeta$ and $\zeta \leq \sigma$. The latter also follows directly from $b_Q \leq \sigma$. Both conclusions permit Q empty. Since $g(C) = 0$ exactly when C is equimatchable, $\sigma = 0$ forces every $g(C) = 0$. At $\sigma = 1$, the nonnegative integer summands have total at most one, so at most one is positive and it equals one. This is the stated bound on every maximal matching in the sole possible exceptional component. All constructions and sums concern the supplied finite graph. □

Remark. Uncovered maximum-degree vertices pay their complete degrees upon deletion. A maximal matching in a whole regular component leaves two extra independent vertices per lost matching edge, while its surviving matching is perfect there. Exact tail potential then charges every such matching loss to the same surplus.

Depends on. Lemma 35, Lemma 41, Lemma 56.

Lemma 100 (Binary remnants and component deletion). *All graphs are finite and simple. Write $e(H)$, $\nu(H)$, and $\Phi(H) = e(H) - \nu(H)$ for edge count, maximum matching cardinality, and potential. A positive star forest is a nonempty vertex-disjoint union of stars $K_{1,d}$ with positive integer edge sizes d. Copies are ordinary subgraphs, and distinct target components have disjoint vertex sets. Arrowing an ordered pair means that every red-blue edge coloring contains its whole first target in red or its whole second target in blue.*

Let F, J have full nonincreasing positive integer component sizes $n_1,\dots, n_S$ and $m_1,\dots, m_T$, with integer counts $S, T \geq 1$ and heads $B = n_1$, $A = m_1$. Put $M = S + T - 1$ and

$\ell_k = max\{n_i + m_j - 1 : 1 \leq i \leq S, 1 \leq j \leq T, i + j = k\}$

for every integer $2 \leq k \leq S + T$. For integers $0 \leq z \leq M$ put

$L_z = \sum_{k=z+2}^{S+T} \ell_k$, $P_z = L_z - (M - z)$,

with empty sums zero. For $z < M$, the actual total-z suffix family consists of every pair (F^α, J^β) with integer cuts $0 \leq \alpha < S$, $0 \leq \beta < T$ and $\alpha + \beta = z$; a superscript deletes exactly that many leading whole components.

Fix integers $D \geq 5$ and $0 \leq q < \min(S, T)$. Suppose $\ell_{q+2} = D + 1$ and $\ell_k \leq D$ for every $q + 3 \leq k \leq S + T$. Assume $3 \leq B \leq D$ and $2 \leq A \leq D$. Put $W = \sum_{j=1}^{T} \max(m_j - 2, 0)$. Let μ, p be integers satisfying

$p \geq 1$, $\mu \geq \max(D, e(F) + 1, S + W)$, $T \geq \mu + 2$, $W \leq S$.

If $B = 3$, assume in addition $2\mu \geq 3S + 2p - 3$.

For $j = 1, 2$, let T_j count J-components of size at least j, and put

`a_j=max{|U|:` $U \subseteq \{i : m_i \geq j\}$, $|U| \leq \mu$,

sum_(i in U)(m_i+1)<=2mu+p}.

Define $c_j = T_j - a_j$, $a = a_1$, $\delta_0 = \mu - a$, $h = \mu - S$, and $\epsilon = \max(p - 1 - h, 0)$. Then $c_1 \geq c_2 \geq 0$ and $c_1 \geq 2$. Let J_{bin} consist of c_2 two-edge stars and $c_1 - c_2$ unit stars. It is a nonempty componentwise subforest of J. Define $M_{bin} = S + c_1 - 1$ and its full-pair tails $L_z^b in$, $P_z^b in$ by the same diagonal formulas for (F, J_{bin}). Its actual q and $q+1$ suffix families are nonempty, and

$$0 \leq P_q - P_q^b in \leq (D-1)\mu + 1 + \epsilon,$$

$$0 \leq L_q - L_q^b in \leq Dmu + 1 + \epsilon - \delta_0.$$

For every integer $0 \leq \beta < c_1$ and disjoint finite simple graphs C_0, Q with $|V(C_0)| \leq 2\mu + p$ and $\nu(C_0) \leq \mu$, every copy of J^β in C_0 disjoint union Q leaves a copy of $J_b in^\beta$ entirely in Q.

Let $r \geq 0$ be an integer and let H satisfy $\Phi(H) \leq P_q + r$. Each of the following two conditions implies that H has a coloring avoiding both full targets of some original actual total-q suffix pair.

(i) C is a union of whole components of H, with $|V(C)| = 2\mu + p$, $\nu(C) \leq \mu$, and every vertex of C has degree D in H, and $r < Dp/2 - 1 - \epsilon$.

(ii) y is an actual vertex of H, and C is a union of whole components of H-y, with $|V(C)| = 2\mu + p$, $\nu(C) \leq \mu$, and every vertex of C has degree D in H. Put $b = e_H(C, \{y\})$, $v_0 = d_H(y) - b/2$, and $\Theta = Dp/2 + v_0 - B - 2 - \epsilon$. Assume $r < \Theta$. Here b counts boundary edges and $d_H(y)$ is the degree of y; v_0 is nonnegative and may be a half-integer, while Θ is an integer. If $p \geq 4$, then $\Theta \geq D - 2$, so $0 \leq r \leq D - 3$ suffices.

Every resulting coloring extends, preserving all H-edge colors, over the disjoint canonical q-star pool of sizes ℓ_2,..., ℓ_{q+1} to avoidance of F, J. It also extends across any specified actual q-vertex set X in any finite simple G with $H = G - X$ to avoidance of F, J. Both target positions and prescribed colors may be exchanged. All profile entries, indices, cuts, component counts, fitting maxima, costs, degrees and matching cardinalities are integers in their stated domains.

In particular, for $D = 5$ and $p = 4$ one has $\epsilon = 0$, the two loss bounds are $4\mu + 1$ and $5\mu + 1 - \delta_0$, and (ii) holds for every integer $0 \leq r \leq 2$. The additional hypothesis when $B = 3$ is then exactly $2\mu \geq 3S + 5$. Repeated sizes and all original unit components are retained throughout.

Proof. The maxima a_j exist because their finite sets of candidate subsets include the empty subset. Thus $0 \leq a_j \leq \min(\mu, T_j)$. Removing the original units from an $a_1 - attaining$ subset leaves an eligible subset for a_2 with at least $a_1 - (T_1 - T_2)$ members. Hence $c_1 \geq c_2 \geq 0$. A largest J-component fits: $A \leq D \leq \mu$ and $A + 1 \leq 2\mu + p$. Therefore $a \geq 1$. Also $c_1 = T - a \geq T - \mu \geq 2$. Choose c_2 distinct nonunit J-components and trim them to two edges, then choose $c_1 - c_2$ additional components and trim them to units. The bounds $c_2 \leq T_2$ and $c_1 \leq T$ make this possible. This proves componentwise containment. Since $M_{bin} = M - a \geq S + 1$ and $q + 1 \leq S$, both q and $q + 1$ are below M_{bin}. Their suffix families are nonempty: any integer $z < M_{bin}$ has legal cuts $\alpha = \min(z, S - 1)$, $\beta = z - \alpha$.

Fix an integer $0 \leq \beta < c_1$ and a copy of J^β in C_0 disjoint union Q as in the statement. Each copied component is connected and lies wholly in one part. For $j = 1, 2$, the copied components of size at least j lying in C_0 number at most a_j. Selecting one edge from each bounds their number by $\nu(C_0) \leq \mu$, and their disjoint vertex sets satisfy the same vertex $\cap$ as the defining subsets. The full suffix has $\max(T_j - \beta, 0)$ components of size at least j. Therefore Q retains at least $\max(c_j - \beta, 0)$ such disjoint components. These are precisely the two threshold counts of $J_b in^\beta$. Choose and trim the required nonunits first, then choose enough additional components and trim them to units. The resulting copy lies in Q. This argument uses the same binary forest for every prefix, without requiring compatible subsets attaining a_1 and a_2.

For the numerical bounds put $N_x = |\{i : n_i \geq x\}|$ and extend $T_y = |\{j : m_j \geq y\}|$ to all positive integer thresholds. Counts beyond a head are zero. The full excess profile has

$$V_z = max\{N_x + T_y - 1 : x + y = z + 2, x, y \geq 1, N_x T_y > 0\} \quad (1)$$

entries at least a positive integer z, with maximum zero if there is no candidate. A threshold rectangle realizes every index sum from two through $N_x + T_y$, giving an initial interval of $N_x + T_y - 1$ profile positions. Conversely, if $n_i + m_j \geq z + 2$, choose $x = \max(1, z + 2 - m_j)$ and $y = z + 2 - x$. These are positive, have sum $z + 2$, and satisfy $x \leq n_i$, $y \leq m_j$. Thus the union of those initial intervals is exactly the superlevel set, proving (1). Its size after q initial positions is $\max(V_z - q, 0)$. Summing these sizes over positive z gives P_q. The same argument applies to the binary pair.

Let Δ_z be the old natural superlevel size minus the new one. The sorted sizes of J_{bin} are bounded by the corresponding initial sorted sizes of J, by componentwise containment. Thus each new diagonal is bounded by the old diagonal, and $\Delta_z \geq 0$. The old natural excess profile has maximum D occurring exactly once, so

$P_q - P_q^b in = \sum_{z=1}^{D} \Delta_z$, $\Delta_D \leq 1$. (2)

Put $U = e(F) - S - (B-1)$, $v = T - T_2$, and $\kappa = \max(S - 1 - v, 0)$.

The definitions give $U \geq 0$ and $h = \mu - S \geq B + U \geq B$. For $x \geq 2$, one has $N_x \leq U + 1$ because each counted component other than one fixed head contributes at least one to U. For $y \geq 3$, $T_y \leq W$. Consequently a rectangle with $x \geq 2$, $y \geq 3$ has width at most $U + W \leq U + S \leq \mu - B < \mu$; a rectangle with $x = 1$, $y \geq 3$ has width at most $S + W - 1 < \mu$.

For $1 \leq z \leq B - 1$, the blue-unit rectangle has width $T + N_{z+1} - 1 \geq T > \mu$, so it dominates all candidates with blue threshold at least three. The only competitor has thresholds (z,2), and $T_2 > 0$ since $A \geq 2$. Write $f_z = N_z - N_{z+1}$. Formula (1) gives

$V_z = T + N_{z+1} - 1 + \max(f_z - v, 0)$.

This width exceeds q. The new unit rectangle gives natural size at least $T - a + N_{z+1} - 1 - q$, even if that expression is negative. Hence $\Delta_z \leq a + \max(f_z - v, 0)$. Since $\sum_{z=1}^{B-1} f_z = S - N_B \leq S - 1$, the sum of all positive corrections is at most κ: if any are positive their sum subtracts v at least once, and otherwise it is zero. Thus

$\sum_{z=1}^{B-1} \Delta_z \leq (B-1)a + \kappa$. (3)

At heights $B \leq z \leq D - 1$ there is no blue-unit rectangle. All candidates with blue threshold at least three have full width below μ by the preceding bounds. The sole possible blue-threshold-two candidate occurs at $z = B \leq D - 1$ and has red threshold B. The original head sum is at least $D + 2$ because $\ell_{q+2} = D + 1$. The threshold pair $(B, D + 2 - B)$ therefore exists, and its blue threshold is at least three. At height D its width is at most $q + 1$, since the natural height-D width is one. Thus $N_B + T_{D+2-B} - 1 \leq q + 1$, and $T_{D+2-B} \geq 1$ gives $q \geq N_B - 1$.

If $c_2 > 0$, the binary pair retains the candidate (B,2). Its natural width loses at most $a_2 \leq \mu$, since the map w to $\max(w - q, 0)$ does not increase a nonnegative width difference. If $c_2 = 0$, then $T_2 = a_2 \leq \mu$ and the old candidate has natural size $\max(N_B + T_2 - 1 - q, 0) \leq \mu$. Every other candidate has natural size below μ. Taking their maximum proves $\Delta_z \leq \mu$ at all these middle heights. When $B = D$ there is no middle height and no use of that top threshold pair is needed. Equations (2), (3), and $a = \mu - \delta_0$ now give

$P_q - P_q^b in \leq (D-1)\mu + 1 + \kappa - (B-1)\delta_0$. (4)

If $\kappa = 0$, the last correction is at most zero and hence at most ϵ. Otherwise $v = S - 1 - \kappa$. An a-component subset attaining a_1 contains at most v units and uses at least $2a + \max(a - v, 0)$ vertices. Its vertex $\cap$ implies $a - v \leq 2\delta_0 + p$. Substitution of $a = \mu - \delta_0$ yields

$\kappa \leq 3\delta_0 - h + p - 1$. (5)

For $B \geq 4$, this gives

$\kappa - (B-1)\delta_0 \leq (4 - B)\delta_0 - h + p - 1 \leq \epsilon$.

For $B = 3$, the additional hypothesis implies $2h \geq S + 2p - 3$, so $h - p + 1 \geq (S-1)/2 \geq \kappa/2$. Equation (5) gives $3\delta_0 \geq \kappa + h - p + 1 \geq 3\kappa/2$, hence $\kappa \leq 2\delta_0$. The correction in (4) is again nonpositive. Thus (4) proves the potential bound in all cases. The component budgets differ by a, so

$L_q - L_q^b in = P_q - P_q^b in + a \leq Dmu + 1 + \epsilon - \delta_0$.

Both losses are nonnegative by diagonal containment and nonnegative excess entries.

We record how the full tails describe the two finite families used below. For $u = q$ or $q + 1$, the complete diagonal envelope of all actual total-u suffix pairs of (F, J_{bin}) is its full profile with the first u positions removed. A local candidate at position k comes from an original candidate at position $u + k$. Conversely, for original indices $i + j = u + k + 1$, take $\alpha = \min(u, i - 1)$ and $\beta = u - \alpha$. The inequality $(i-1) + (j-1) \geq u$ ensures $\beta \leq j - 1$, and the original index bounds make both cuts legal. Thus that candidate occurs at the

specified local position. Every family member has positive common budget $M_{bin} - u$, so this accounts for its entire envelope and gives potential $P_u^b in$.

Consider (i) and put $Q = H - V(C)$. Since C is a union of whole components and every one of its vertices has degree D,

$e(C) = Dmu + Dp/2, \nu(H) - \nu(Q) = \nu(C) \leq \mu.$

These identities in particular force Dp even in this case. Hence

$\Phi(Q) \leq P_q + r - (D-1)\mu - Dp/2.$

Using the proved loss bound yields

$P_q^b in - \Phi(Q) \geq Dp/2 - 1 - \epsilon - r > 0.$ (6)

Apply Lemma 56 to Q and the nonempty actual q-suffix family of the binary pair. Its strict-potential contrapositive supplies a coloring avoiding a pair $(F^\alpha, J_b in^\beta)$ with $\alpha + \beta = q$ and legal cuts. Color all edges of C blue. There are no edges from C to Q. A red F^α would therefore lie in Q and is absent. A blue J^β would leave $J_b in^\beta$ in Q by the proved copy-remnant property, again a contradiction. This proves original actual q-suffix avoidance.

For (ii) put $Q = H - (V(C) union \{y\})$. Every edge leaving C goes to y. With $b = e_H(C, \{y\})$, the degree sum gives

$e(C) = Dmu + Dp/2 - b/2.$

Deleting C and y removes exactly $Dmu + Dp/2 + v_0$ edges, because this count adds all $d_H(y)$ edges at y to $e(C)$. One has $b \leq d_H(y)$, so $v_0 \geq 0$. Also Dp-b is even; hence $Dp/2 + v_0 = Dp/2 + d_H(y) - b/2$ is an integer, proving the stated integrality of Θ. Any matching has at most μ internal edges in C and at most one edge at y. Its restriction to Q consequently gives

$\Phi(Q) \leq P_q + r - (D-1)\mu - Dp/2 - v_0 + 1.$ (7)

Since the blue head of J_{bin} is at most two, every new excess entry is at most B. Thus

$P_{q+1}^b in \geq P_q^b in - B \geq P_q - (D-1)\mu - B - 1 - \epsilon.$ (8)

Subtracting (7) from (8) gives $P_{q+1}^b in - \Phi(Q) \geq \Theta - r > 0$. By Lemma 56 and the actual-family envelope identity, Q has a coloring avoiding some pair $(F^\alpha, J_b in^\beta)$ with $\alpha + \beta = q + 1$ and legal cuts. Color C blue. The same copy-remnant argument proves avoidance of the original suffix pair (F^α, J^β) on H-y.

At least one cut is positive because $\alpha + \beta = q + 1$. Decrease such a cut by one. The resulting pair is a legal original total-q pair, and the avoided pair on H-y is its one-vertex suffix. Apply Lemma 41 (II) to that pair and the actual vertex y. This extends the coloring to H, with every previous edge color retained, proving (ii).

For its sufficient numerical condition, $h \geq B$. If $p \geq 4$ and $\epsilon = 0$, then $\Theta \geq 2D - B - 2 \geq D - 2$. If $\epsilon > 0$, substitution gives

$\Theta = p(D/2 - 1) + (h - B) + v_0 - 1 \geq 2D - 5 \geq D - 2,$

where the last inequality uses $D \geq 5$. Therefore $r \leq D - 3$ guarantees $r < \Theta$ in every allowed case.

Each coloring obtained in (i) or (ii) avoids an original actual total-q pair. Parts (I) and (II) of Lemma 41 respectively extend it over the original canonical q-star pool and across every specified actual q-set X, without changing any H-edge color. These statements include $q = 0$ and allow all original units and repeated sizes. Interchanging the two complete targets and their color names gives the claimed symmetry.

Finally take $D = 5$, $p = 4$. Since $h \geq B \geq 3$, $\epsilon = \max(3 - h, 0) = 0$. The two loss formulas specialize to $4\mu + 1$ and $5\mu + 1 - \delta_0$. The condition at $B = 3$ becomes $2\mu \geq 3S + 5$, and the sufficient range $r \leq D - 3$ is $r \leq 2$. This proves all the stated degree-five specializations directly. □

Remark. A single unused-capacity correction controls the profile loss for every vertex allowance. The same forced binary remainder then transfers strict potential savings from either whole regular components or components behind one actual vertex, while preserving all original suffixes.

Depends on. Lemma 41, Lemma 56.

Lemma 101 (A certificate from two isolated row contacts). *All graphs are finite and simple. Write $V(G)$, $E(G)$, $\nu(G)$, and $\alpha(G)$ for vertex set, edge set, maximum matching cardinality, and maximum independent-set cardinality. A graph is well-covered if every inclusionwise maximal independent set has size $\alpha(G)$. Write $N_G(S)$ for the vertices adjacent to at least one member of S and $N_G[S] = S$ union $N_G(S)$.*

For nonnegative integer capacities A, B, let $c_{A,B}(G)$ be the least nonnegative integer T such that every nonnegative integer split $x + y = T$ permits an ordered partition of $E(G)$ into four parts having maximum degrees at most A, B and matching numbers at most x, y, respectively. Put $c_2(G) = \max_{A+B=2} c_{A,B}(G)$.

Let F be well-covered. Let R be an induced disjoint union of p odd cycles, a bipartite graph D of maximum degree at most two with a perfect matching of size m_D, and z isolated vertices Z, where $p \geq 0$, $m_D \geq 1$, $z \geq 2$ are integers. Every listed cycle is a whole connected component of R and has odd order at least three. Put $U = V(F)$ minus $V(R)$ and $u = |U|$. Suppose nonnegative integers b_R, b_B satisfy $b_R + b_B = u + \nu(R) - 1$, $E(F)$ has no two-part partition with matching numbers at most b_R, b_B, and $j = u - b_R \geq p + 3$.

Let $I = \{i_1, i_2\}$ be a two-element subset of Z, let r belong to $V(D)$, and let x belong to U with entire row neighborhood

$N_F(x)$ intersect $V(R) = I$ union $\{r\}$.

Then $\alpha(R) = \alpha(F)$, and there is y in U minus $\{x\}$ such that xy is not an edge, y has a neighbor in I and none in Z minus I, and

$N_F(y)$ intersect $V(D)$ is contained in $\{r\}$.

The same y has at most one neighbor in each odd row cycle of order at least five and has its neighborhood in each row triangle contained in a proper subset of that triangle.

For $K = F[V(R)$ union $\{x, y\}]$ one has $c_2(K) \leq p + 1$ and $c_2(F) \leq u + p - 1$. For every nonnegative integer capacity split $A + B = 2$ and every nonnegative integer matching-budget split of total $u + p - 1$, there is a certificate of F with both degree-bounded parts contained in $E(K)$; every edge meeting $V(F)$ minus $V(K)$, including edges with both endpoints there, belongs to a matching-bounded part.

All counts, orders, cardinalities, capacities, budgets, costs, and optimization values are integers in their displayed nonnegative domains. In particular $u \geq 2$. The vertex r may be internal on a path or lie on an even cycle.

Proof. Use Lemma 17 for simultaneous assignments from the missing allocation and Lemma 72 for full certificate restoration.

1. Complete feasible assignments.

Each component of D has a perfect matching, by restriction of the prescribed one, so its order is even and its matching number is half its order. On an odd cycle of order $2k + 1$, where $k \geq 1$ is an integer, consecutive pairs give a matching of size k. No matching is larger by vertex count. Every cycle vertex is exposed, meaning that deleting it leaves a perfect matching, since the deletion leaves an even path. Alternate vertices give a maximum independent set of size k containing any prescribed vertex; the upper bound follows by mapping each selected vertex to its distinct unselected cyclic successor. Isolates are exposed components of order one.

Thus Lemma 17 applies with the stated missing pair and gives, for every subset of U of size at most j, one actual matching assigning its vertices to exposed vertices in distinct odd R-components. Protecting all isolates in the second part of that fact gives $\alpha(R) = \alpha(F)$, since its number of unprotected odd components is p and $j \geq p + 3$.

Put `B0={b` in $U : N_F(b)$ intersect (Z minus I) is empty}. If $|B0| \geq p + 3$, applying the assignment to any $p + 3$ of its members would require $p + 3$ distinct odd components among only the p cycles and two isolates I. Hence $|B0| \leq p + 2 < j$, and one simultaneous assignment exists for all of B_0.

Form the finite bipartite incidence graph with parts B_0 and the p cycles together with the two singleton components I. An incidence bC means that b has an actual F-neighbor in that component. Its family of matchings saturating all of B_0 is nonempty. For each such matching M let $\Phi(M)$ count its assigned cycle incidences whose exterior vertex has at least two neighbors on that cycle. Choose M maximizing this nonnegative integer over the entire family. Each M can be realized by actual edges, together with compatible maximum internal row matchings, because its receiving components are distinct and all their vertices are exposed.

Let B_I be the vertices assigned to I by M. Then $|B_I| \leq 2$, and x belongs to B_I because x has no odd-cycle neighbor.

2. The companion and its entire even-row neighborhood.

In every assigned cycle choose any maximum independent set containing any actual contact of its assignee; in an unassigned cycle choose any maximum independent set. Their union is denoted T. Choose a maximum independent set J_D of D avoiding r: both bipartition classes of each component have half its order, since a perfect matching pairs them, and a class not containing r may be selected in its component. Their union has size m_D and is maximum, since the perfect matching bounds independence by m_D.

Put $S = (ZminusI)$ union T union J_D. It is independent, has size alpha(F)-2, and dominates every row vertex outside I. Every exterior vertex outside B_0 meets Z minus I, and every cycle assignee meets its selected contact. Thus the graph $F - N_F[S]$ has vertex set I union B, where B is a subset of B_I. The pivot x survives because J_D avoids r.

For any independent set S in a well-covered graph, every maximal independent set L of $F - N_F[S]$ makes S union L maximal in F. Vertices outside this residual are in S or adjacent to S, and maximality of L handles the residual. Consequently the residual is well-covered with independence $\alpha(F) - |S|$. Here its independence is two. A maximal independent extension of {x} cannot contain either vertex of I, because x meets both. Since at most two exterior vertices survive, this forces $B = B_I = \{x, y\}$, with xy absent.

This same pair survives every permitted choice of T and J_D for the fixed M. The assigned isolate is an actual neighbor of y, and membership in B_0 makes y avoid Z minus I.

Omit J_D and use $S0 = (ZminusI)$ union T. Every exterior vertex outside B_I is still dominated, and both members of B_I avoid S_0. Different row components are anticomplete. Therefore

$F - N_F[S0] = F[V(D)$ union I union {x, y}]

as full induced graphs. This graph is well-covered with independence $m_D + 2$. It has a perfect matching formed from any perfect matching of D and the actual edges $x\ i_x$, $y\ i_y$, where i_x and i_y are the distinct isolates assigned by M.

Suppose y has a D-neighbor v other than r. Then {x, v} is independent, since r is x's only D-neighbor. A maximal independent extension of {x, v} in this residual misses both endpoints of the matching edge $y\ i_y$: v excludes y, and x excludes i_y. It has at most one endpoint of each of the other $m_D + 1$ matching edges, so its size is at most $m_D + 1$, contradicting well-coveredness. This proves $N_F(y)$ intersect $V(D)$ is contained in {r}.

3. A legal strict exchange controls every cycle.

An unassigned cycle has no y-neighbor, since every cycle vertex belongs to some permitted maximum independent set and y survives every such choice. If a cycle C is assigned to w and a is any contact of w on C, then

$N_F(y)$ intersect $V(C)$ is contained in $N_C(a)$.

Indeed the union of maximum independent sets of C containing a is $V(C)$ minus $N_C(a)$. To check this, label the cycle a, c_1,..., c_{2k} cyclically. For $k = 1$ the only such set is {a}. For $k \geq 2$ the two maximum sets

$\{a, c_2, c_4, \ldots, c_{2k-2}\}$ and $\{a, c_3, c_5, \ldots, c_{2k-1}\}$

together cover exactly all vertices other than c_1 and c_{2k}. Those two neighbors cannot be selected with a. Since y survives all the choices, the asserted containment follows. It holds for every contact a of the same assignee w. On a triangle it already gives a proper subset.

Suppose C has odd order at least five and y has two C-neighbors. They equal $N_C(a)$. The assignee w has only the contact a: a second contact would have the same two cycle neighbors, and two distinct vertices with the same two neighbors would make a four-cycle be the whole connected degree-two component, contrary to the order of C. Thus wC contributes zero to Φ.

Replace only the chosen set on C by

$\{c_1, c_3, \ldots, c_{2k-3}, c_{2k}\}$,

a maximum independent set containing both neighbors of a. Keep all other cycle choices and J_D. The resulting S′ has size alpha(F)-2. Both I and x survive in $F - N_F[S']$, y is dominated, and every exterior vertex

other than x and possibly w remains dominated by unchanged choices. No D-vertex survives since J_D is maximal in D, and every row cycle is dominated by its selected maximum set. Thus the residual is contained in I union {x, w}, contains I union {x}, and is well-covered with independence two.

A maximal independent extension of {x} forces w to survive with xw absent. The vertex w has an I-neighbor, since otherwise I union {w} would be an independent set of size three. Assign w to an actual neighbor in I and x to the other isolate, which x also meets. Assign y to C through either of its two actual contacts. Keep every other cycle assignment.

This modified incidence matching still saturates exactly B_0, uses distinct receiving components, and has actual contacts for every incidence. Exactly one cycle assignee changes, from w with one contact to y with two. Hence Φ increases by one, contradicting its maximum. Therefore the same y has at most one neighbor on every odd row cycle of order at least five.

4. The full certificate.

Take K as in the statement. Let P_0 contain every D-edge incident with r, together with rx and ry if ry is present. This is a star centered at r and has matching number one. Put all other D-edges and all edges from x or y to I into an edge set Q. The latter edges form a subgraph of $K_{2,2}$ on {x, y} and I, disjoint from the remaining D-edges. The vertex r is isolated in the D-remainder. There are no other root-to-D edges, and xy is absent.

For each row triangle choose one further exceptional edge set P_C. With no y-contact take the whole triangle. With sole contact a take ya and both triangle edges at a. With contacts a, b take ya, yb, ab. These sets are respectively a triangle, a star, and a triangle, of matching number one; their remaining triangle edges form a path or an empty graph.

For a longer odd cycle with no y-contact take any one cycle edge as P_C. With sole contact a take ya and both cycle edges at a. These are respectively an edge and a star of matching number one. The remaining cycle edges form paths, with possible isolates. Add all remaining cycle edges to Q.

These sets partition $E(K)$. They include every contact of x and y, every D-edge, every cycle edge, and no edge to an isolate outside I. The graph Q is bipartite with maximum degree at most two: it is the disjoint union of the even-row remainder, the root-isolate subgraph of $K_{2,2}$, and the cycle remnants. Each finite connected degree-two graph is a path or cycle, and bipartiteness makes every cycle even. Alternating along these paths and cycles partitions Q into two matchings.

For capacities (1,1), take these two matchings as the bounded-degree parts. For (2,0) or (0,2), put all Q in the capacity-two part. For any nonnegative integer matching budgets of total $p+1$, assign the indicated number of the $p+1$ whole exceptional pieces P_0 and P_C to each exceptional part. A matching in their union restricts to a matching in each piece, so its cardinality is at most the number of pieces assigned there, even if pieces share vertices. This proves every required certificate of K and $c_2(K) \le p+1$.

The companion gives $u \ge 2$. Restore the other u-2 vertices by Lemma 72. It supplies every capacity and budget split at total $(u-2)+(p+1) = u+p-1$, retaining both bounded-degree parts inside K and assigning every omitted-incident edge, including edges with two omitted endpoints, to an exceptional part. Thus $c_2(F) \le u+p-1$, with all the asserted edge-placement guarantees. □

Remark. A complete maximum assignment forces one companion and confines its even-row contacts through a perfect-matching count. A strict exchange limits its cycle contacts, after which one star and one piece per odd cycle pay for the full extension.

Depends on. Lemma 17, Lemma 72.

Lemma 102 (Matching-size and component-order bounds for excess). *Use the integer certificate cost $c_0(H)$, matching number $\nu(H)$, and excess $ex(H) = c_0(H) - \nu(H)$ of Lemma 23. Thus $c_0(H)$ is the least nonnegative integer t such that every pair of nonnegative integers x, y with $x+y=t$ admits a partition of $E(H)$ into two parts of matching numbers at most x, y. All graphs are finite and simple, including the empty graph and isolated vertices. Every budget, order, matching number, excess, and parameter below is an integer in its specified nonnegative or positive domain.*

Fix an integer $l \ge 0$. Let X be a vertex subset of H such that

$\nu(H) = |X| + \sum_Q floor(|V(Q)|/2)$,

where Q ranges over all connected components of the induced graph H-X. Such a set X exists. If each odd-order component of H-X has at most $6l+3$ vertices and each even-order component has at most $6l+6$ vertices, then $c_0(H) \le \nu(H) + l$.

Consequently $ex(H) \le l$ whenever either every connected component of H has order at most $6l+4$ or $\nu(H) \le 3l+1$. For every integer $m \ge 1$, $ex(H) \ge m$ implies $\nu(H) \ge 3m-1$ and H has a connected component of order at least 6m-1. In particular $ex(H) \le floor((\nu(H)+1)/3)$.

These bounds are sharp: for every integer $l \ge 0$, the complete graph K_{6l+5} has matching number $3l+2$, cost $c_0 = 4l+3$, and excess $l+1$.

Proof. We first give a coloring on a graph Q with q vertices and prescribed nonnegative integer budgets x, y. Suppose $q \le x+y+1+\max(x,y)$. If $x \ge y$, choose a subset A of $\min(q, 2x+1)$ vertices. Color all edges within A red and every other edge blue. A red matching has at most x edges, and the complement of A covers every blue edge. Its cardinality is $\max(0, q-2x-1) \le y$, so a blue matching has at most y edges. If $y > x$, use the same construction with the colors and budgets exchanged. Thus the displayed vertex-count condition supplies a partition on every edge of Q.

Let $q = 2w+1$ be odd with $q \le 6l+3$, or let $q = 2w$ be even with $q \le 6l+6$. For every integer a in $[0, w+l]$, we claim that Q has a coloring with matching budgets a, $w+l-a$. Indeed, $\max(a, w+l-a) \ge ceil((w+l)/2)$, so the preceding construction applies provided

$q \le w+l+1+ceil((w+l)/2)$. (1)

For odd q, one has $w \le 3l+1$. If $w \le 3l$, then $w-l \le (w+l)/2$, and if $w = 3l+1$, then $w-l = ceil((w+l)/2) = 2l+1$. Both cases prove (1). For even q, $w \le 3l+3$. If $w \le 3l+2$, then $w-l-1 \le (w+l)/2$, and if $w = 3l+3$, then $w-l-1 = ceil((w+l)/2) = 2l+2$. This proves (1) in that case too.

By Lemma 46 there is a set X attaining the stated matching identity. Now use any such X satisfying the component-size hypotheses. Put $b = |X|$, list the components of H-X as Q_1,..., Q_p, and put $w_i = floor(|V(Q_i)|/2)$. Empty lists and zero weights are allowed. Their sum satisfies $b + \sum_i w_i = \nu(H)$.

Fix any nonnegative integers x, y with $x+y = \nu(H)+l$. If $x \ge \nu(H)$, color all edges red; this meets both matching budgets. Suppose $x < \nu(H)$.

If $x < b$, choose x vertices of X, call their set X_R, and color every edge meeting X_R red and every remaining edge blue. The set X_R covers the red edges, so the red matching number is at most x. In a blue matching, at most b-x edges meet X minus X_R, and every other edge is inside a component Q_i. There are at most w_i matching edges in that component by its order. Hence its total size is at most $b - x + \sum_i w_i = \nu(H) - x \le y$.

If $x \ge b$, choose the least index j with $b + \sum_{i=1}^{j} w_i > x$; such an index exists because $x < \nu(H)$. Set $a = x - b - \sum_{i<j} w_i$. Then $0 \le a < w_j$. Color all edges meeting X red, all internal edges of Q_i red for $i < j$, and all internal edges of Q_i blue for $i > j$. On Q_j use the coloring of the preceding vertex-count construction with budgets a, $w_j + l - a$. There are no edges between different components of H-X, so every edge has been assigned exactly once. A red matching has at most b edges meeting X, at most $\sum_{i<j} w_i$ edges in earlier components, and at most a edges in Q_j; the total is at most x. A blue matching has at most $w_j + l - a + \sum_{i>j} w_i = \nu(H) + l - x = y$ edges. This proves every allocation at total $\nu(H)+l$, so the definition in Lemma 23 gives $c_0(H) \le \nu(H) + l$.

If every connected component of H has order at most $6l+4$, each component of H-X also has order at most $6l+4$, since it lies within one original component. Its odd orders are at most $6l+3$, and its even orders are at most $6l+4 \le 6l+6$. Thus the first assertion applies to an attaining X. If $\nu(H) \le 3l+1$, its matching identity gives $w_i \le 3l+1$ for every component, hence $|V(Q_i)| \le 2w_i + 1 \le 6l+3$. The same assertion applies. In each case subtract $\nu(H)$ to get $ex(H) \le l$.

For $m \ge 1$ apply both contrapositives with $l = m-1$. The inequality $ex(H) \ge m$ excludes $\nu(H) \le 3m-2$, so $\nu(H) \ge 3m-1$; it also excludes all component orders at most 6m-2, so a component has order at least 6m-1. The nonnegativity of $ex(H)$ comes from Lemma 23. Taking $m = ex(H)$ when this is positive yields $ex(H) \le floor((\nu(H)+1)/3)$; when it is zero that bound follows because $\nu(H) \ge 0$.

We finish by proving sharpness. We will use the following elementary matching count, proved here. A complete multipartite graph on $N \ge 1$ vertices, with nonempty independent parts and largest part size a, has a matching of size $\min(\lfloor N/2 \rfloor, N-a)$. The two expressions are upper bounds, since a matching covers two

vertices per edge and every edge meets the complement of the largest part. If $a \geq \lceil N/2 \rceil$, match each vertex outside that part to a distinct vertex inside it, giving N-a edges. Otherwise $a < \lceil N/2 \rceil$. For even $N = 2t$ the assertion of a perfect matching whenever all parts have size at most t follows by induction on t. At $t = 1$ there are two singleton parts. For $t > 1$ choose vertices in two largest parts and pair them. The remaining graph has 2t-2 vertices and largest part at most t-1: if a part had size t before removal, it was among the two chosen, since at most two parts can have that size; every other part was already smaller. Remove any empty parts and apply induction. For odd $N = 2t + 1$ in the present case, $a \leq t$; delete any one vertex to obtain 2t vertices in parts of size at most t, and use the even case to obtain t matching edges. This proves the count in every case.

Put $N = 6l + 5$ and let $H = K_N$. Its matching number is $\lfloor N/2 \rfloor = 3l + 2$, because disjoint pairs of vertices attain the order bound. For $c = 4l + 3$, every split $x + y = c$ satisfies

$x + y + 1 + \max(x, y) \geq c + 1 + \lceil c/2 \rceil = 6l + 6 \geq N$.

The initial coloring construction gives $c_0(H) \leq 4l + 3$.

For the reverse inequality suppose its edges admit a partition into red and blue spanning graphs R, B, both with matching number at most $2l + 1$. Apply Lemma 46 to R. There is a vertex set Y for which

$o(R - Y) - |Y| = N - 2\nu(R) \geq 2l + 3$,

where o counts odd connected components. Put $b = |Y|$ and let p count all connected components of R-Y. Then $p \geq b + 2l + 3$. Their vertex sets partition N-b vertices, so $b + 2l + 3 \leq N - b$ and $b \leq 2l + 1$. Every edge joining two different such components is blue, since H is complete and a red edge would join them into one component. Thus B contains the complete multipartite graph with those component sets as parts, on $N' = N - b$ vertices. Its largest part has size $a \leq N' - (p - 1)$. The proved matching count gives

$\nu(B) \geq \min(\lfloor N'/2 \rfloor, N' - a) \geq \min(floor((N - b)/2), p - 1) \geq 2l + 2$.

Here $b \leq 2l + 1$ gives $N - b \geq 4l + 4$, and $p - 1 \geq b + 2l + 2 \geq 2l + 2$. This contradicts $\nu(B) \leq 2l + 1$.

Therefore the allocation $(2l + 1, 2l + 1)$, whose total is $4l + 2$, is missing for K_N. The integer threshold characterization in Lemma 23 implies $c_0(K_N) \geq 4l + 3$. Together with the upper bound this proves the exact cost and $ex(K_N) = (4l + 3) - (3l + 2) = l + 1$, as asserted. □

Remark. A single component absorbs the entire extra matching allowance while all other barrier pieces are colored wholly red or blue. A blue multipartite matching proves the complete graphs attain the first excluded size.

Depends on. Lemma 23, Lemma 46.

Lemma 103 (Exact matching allocations in graphs of maximum degree two). *Use the integer certificate costs $c_i(G)$, matching number $\nu(G)$, signed slack $ps(G)$, excess $ex(G) = c_0(G) - \nu(G)$, signed $gap(G) = ex(G) - ps(G)$, and top missing-allocation run $tr(G)$ of Lemma 23. Paths P_n and cycles C_n are indexed by their vertex orders, with integers $n \geq 1$ and $n \geq 3$ respectively. All graphs below are finite and simple. Every total, budget, order, component count, index and matching number is a nonnegative integer unless its positive lower bound is stated.*

Let G have maximum degree at most two. List its whole odd-cycle components of order at least five as $C_{2h_1+1}, \ldots, C_{2h_m+1}$, where $m \geq 0$ and $2 \leq h_1 \leq \cdots \leq h_m$ are integers. Let Z be the induced union of all other whole components, including isolated vertices, and put $r = \nu(Z)$ and $N = r + \sum_{i=1}^{m} h_i = \nu(G)$. Define

$D = \max(\{0\} union \{h_i - 1 - r - \sum_{j=1}^{i-1} h_j : 1 \leq i \leq m\})$.

Empty lists and sums are allowed, so $D = 0$ when $m = 0$.

For every nonnegative integer T, let $Y_T(G)$ be the set of integers x in [0,T] for which $E(G)$ partitions into two parts with matching numbers at most x and T-x. Every such partition assigns each original edge exactly once. Then $c_0(Z) = r$, and

$Y_T(G)$ is empty for $T < N$;

$Y_N(G) = \{s + \sum_{i \in I} h_i : I \subseteq \{1, \ldots, m\}, 0 \leq s \leq r, saninteger\}$;

$Y_T(G) = [0, T]$ for $T \geq N + 1$, with integer intervals understood.

Consequently $c_0(G) = N$ when $D = 0$, and $c_0(G) = N + 1$ when $D > 0$. If $D = 0$ then $ex(G) = 0$ and $tr(G) = N$, including $tr(G) = 0$ when $N = 0$. If $D > 0$ then $ex(G) = 1$ and $tr(G) = D$.

Furthermore $gap(G) > 0$ holds if and only if both of the following conditions hold:

(a) every whole component belongs to P_1, P_2, P_3, P_4, P_6, C_3, C_4, C_5, C_7;

(b) if q and b count the whole C_5 and C_7 components, respectively, then $q + b \geq 1$ and either $r = 0$ or $(q = 0 and r = 1)$.

Here Z and r retain their original definitions. In every positive-gap case $gap(G) = 1$, and $tr(G) = 1$ if $q \geq 1$, whereas $tr(G) = 2 - r$ if $q = 0$. Thus every G of maximum degree at most two satisfies

$gap(G) > 0$ implies $gap(G) = 1$ and $tr(G) + ex(G) \leq 3$.

In particular, if G is the disjoint union of q five-cycles, b seven-cycles, r single edges and z isolated vertices, where q, b, r, $z \geq 0$ are integers, then $N = 2q + 3b + r$, $e(G) = 5q + 7b + r$, $ps(G) = 0$,

$c_1(G) = 2q + 3b$, $c_2(G) = q + b$, and $c_i(G) = 0$ for all integers $i \geq 3$.

Its exact row-zero cost and allocation sets are the formulas above, including the case $q + b = 0$.

Proof. We use the certificate definitions and integer threshold monotonicity from Lemma 23, and the positive-row formulas and zero-slack classification from Lemma 57. In particular, the latter fact classifies all connected degree-two graphs as paths or cycles, gives $ps(G) \geq 0$, and gives exactly the component list in condition (a) when $ps(G) = 0$. Neither a source assumption nor a literature result is needed.

We first establish the component matching allocations. A path or cycle on n vertices has matching number $\lfloor n/2 \rfloor$: disjoint endpoint pairs give the upper bound, and consecutive pairs give a matching attaining it. In any edge partition of any graph J into two parts, a maximum matching of J restricts to a matching in each part. The sum of the two part matching numbers is therefore at least $\nu(J)$. Matching numbers add across whole components, since both restricting a matching and uniting component matchings preserve disjoint endpoints.

Every path P_n has all allocations at total $\nu(P_n)$. Label its edges consecutively. For $n = 2k$ and an integer $1 \leq a \leq k$, put the first 2a-1 edges in the first part, and the remaining 2k-2a edges in the second. The parts are paths, with matching numbers a and k-a. For $a = 0$ put all edges in the second part. For $n = 2k + 1$ and $0 \leq a \leq k$, put the first 2a edges in the first part and the remaining 2(k-a) edges in the second. Their matching numbers are again a and k-a. Empty edge parts and the isolate $k = 0$ are included.

An even cycle C_{2k}, where $k \geq 2$, has all allocations at total k. The two endpoint allocations use a monochromatic cycle. For $1 \leq a \leq k - 1$, put 2a consecutive edges in the first part and the remaining 2(k-a) consecutive edges in the second. These are paths with matching numbers a and k-a. A triangle has matching number one and its two allocations at total one are obtained by putting all edges in either part. Thus every component of Z has all allocations at its matching number.

For any integer s in [0,r], assign to the whole components of Z integers between zero and their matching numbers with sum s. Such a choice exists by successively using the smaller of the remaining quota and the next component matching number; the total available quota is r. Choose the component partitions just constructed and unite their first and second parts. Component additivity gives matching numbers exactly s and r-s. Hence all total-r allocations exist. The general matching lower bound excludes any total below r, so $c_0(Z) = r$. This also proves the assertion when Z or its edge set is empty.

Consider an odd cycle C_{2h+1} with $h \geq 2$. Its only allocations at total h are the two monochromatic allocations, of first budget zero or h. Indeed, a partition using both colors has a cyclic sequence of nonempty maximal monochromatic runs. Each run is a path with a positive integer number l of edges, hence matching number $\lceil l/2 \rceil$. Distinct runs of the same color have disjoint vertices, because a nonempty run of the other color separates their endpoints. The sum of the two color matching numbers is consequently the sum of these ceilings. The sum of the run lengths is the odd integer $2h + 1$, so at least one run has odd length. Therefore the sum of the ceilings is at least $h + 1$, excluding total h for a non monochromatic partition.

The same cycle has every allocation at total $h + 1$. Budgets zero and $h + 1$ are handled by monochromatic partitions, with unused budget permitted. For $1 \leq a \leq h$, assign 2a-1 consecutive edges to the first part and the remaining $2h + 2 - 2a$ edges to the second. These are paths of matching numbers a and $h + 1 - a$. This explicitly supplies every required partition.

Now consider a partition at total N on G. On each of the m listed cycles and on the block Z, let a_J, b_J be the two actual part matching numbers. The lower bounds $a_J + b_J \geq \nu(J)$ sum to N. The two global budget upper bounds also sum to N. Thus equality holds in every block, and the first actual matching numbers sum exactly to the prescribed first budget. Each listed cycle contributes either zero or h_i, and Z contributes an integer s in [0,r]. This proves necessity of the displayed formula for $Y_N(G)$. For sufficiency, choose the indicated cycles monochromatically in the first part and the other listed cycles monochromatically in the second, and choose on Z the partition with actual matching numbers s, r-s. Uniting the block partitions gives the prescribed allocation on the original graph. The general matching lower bound already proved shows that every $Y_T(G)$ with $T < N$ is empty.

To prove full feasibility at total $N+1$, first take $m = 0$. The established $c_0(Z) = N$ and integer threshold monotonicity give every allocation at $N+1$ and above. If $m \geq 1$, select the cycle with parameter h_m. It supplies the integer interval $[0, h_m + 1]$ of first budgets at total $h_m + 1$. Keep all other listed cycles monochromatic, so each supplies a first-budget add end from $\{0, h_i\}$. If an available budget set is [0,L] and $h_i \leq L + 1$, adding $\{0, h_i\}$ gives the union of [0,L] and $[h_i, L + h_i]$, which is the full integer interval $[0, L + h_i]$. Initially $L = h_m + 1$ and every remaining $h_i \leq h_m$, so this property persists as all cycles are added. The neutral block Z then adds the full interval [0,r]. The resulting interval is $[0, N + 1]$. These additions express actual independent choices of block partitions, so every point of the interval is attained by an original-edge partition. Threshold monotonicity gives all totals above $N + 1$.

We determine the gaps in the finite set $Y_N(G)$. For $0 \leq i \leq m$ put $L_i = r + \sum_{j=1}^{i} h_j$ and

$S_i = \{s + \sum_{j \in I} h_j : I \subseteq \{1, \ldots, i\}, 0 \leq s \leq r, saninteger\}$.

Thus $S_0 = [0, r]$, each S_i contains 0 and L_i, and $S_i = S_{i-1}$ union $(h_i + S_{i-1})$. For a finite set containing both endpoints of [0,L], define its largest missing-run length to be zero if the set fills the interval, and otherwise the largest number of consecutive missing integers in that interval.

If $h_i > L_{i-1}$, the two interval hulls of these copies are disjoint or adjacent. The only new possible missing run between them has length $\max(0, h_i - L_{i-1} - 1)$; all other missing runs lie within a copy. If $h_i \leq L_{i-1}$, the hulls overlap. Any missing run that meets $[0, L_{i-1}]$ lies wholly there, because L_{i-1} is present; any remaining run lies in the second hull $[h_i, h_i + L_{i-1}]$. Its length is therefore bounded by the largest missing run of the preceding set. Induction shows that the largest missing run of S_m is at most D.

For the reverse bound, whenever $h_i - L_{i-1} - 1 > 0$, every integer strictly between L_{i-1} and h_i is absent from S_m. A subset using only weights with index less than i has sum, including its neutral contribution, at most L_{i-1}. A subset using an index at least i has sum at least h_i, because the weights are sorted and all contributions are nonnegative. Thus the entire interval of that positive length remains missing. Maximizing proves that the largest missing run of S_m is exactly D. Since its endpoints are present, $D = 0$ is equivalent to $S_m = [0, N]$.

The allocation formulas now give $c_0(G) = N$ if $D = 0$, and $c_0(G) = N + 1$ otherwise. In the latter case its top allocation set is $Y_N(G)$, so $tr(G) = D$. If $D = 0$ and $N > 0$, the allocation set $Y_{N-1}(G)$ is empty, making its entire N-element domain missing and giving $tr(G) = N$. If $N = 0$, $c_0 = 0$ and the definition gives $tr = 0$. Subtracting $\nu(G) = N$ proves the stated excess values.

Since $ps(G) \geq 0$ by Lemma 57 and $ex(G)$ is either zero or one, the integer identity $gap = ex - ps$ shows that $gap(G) > 0$ is equivalent to $ex(G) = 1$ and $ps(G) = 0$; the gap then equals one. The same cited result says $ps(G) = 0$ precisely when condition (a) holds. Under that condition, the listed odd cycles have weights two and three, with multiplicities q and b. If $q = b = 0$, then $D = 0$. If $q \geq 1$, the first weight two gives $h_1 - 1 - r = 1 - r$. Every later weight two gives a smaller value. At the first weight three the expression is $2 - r - 2q \leq 0$, and later weights three give still smaller values. Therefore $D = \max(0, 1 - r)$. If $q = 0$ and $b \geq 1$, the first weight three gives 2-r and every later weight gives a smaller value, so $D = \max(0, 2 - r)$. This proves exactly condition (b), the positive-case trace formulas, and $tr + ex \leq 3$.

Finally, in the specified union of five-cycles, seven-cycles, edges and isolates, matching and edge additivity give $N = 2q + 3b + r$ and $e = 5q + 7b + r$. The exact positive-row formulas of Lemma 57 give $c_1 = 2q + 3b$, $c_2 = q + b$ and all later rows zero. Substitution into $ps = e - \nu - \sum_{i \geq 1} c_i$ gives $ps = 0$. All preceding allocation arguments apply with Z consisting of its r edges and z isolates, including $q + b = 0$. This proves the final specialization. □

Remark. At the matching-number total every odd cycle must remain monochromatic; all other components permit every split. One extra unit makes a largest odd cycle absorb every missing allocation. Sorted subset sums identify the exact missing intervals.

Depends on. Lemma 23, Lemma 57.

Lemma 104 (Boundary accounting for component credit absorption). *Use the finite integer certificate costs $c_{A,B}$, row maxima c_i, matching number ν, and quantities ex, ps, gap from Lemma 23. Let H be a connected finite simple graph with maximum degree $D \geq 2$. Let S be its degree-D vertices and let $\kappa \geq 1$ be the minimum number of S-vertices left uncovered by a matching. Supply a nonempty vertex set X for which $q_S(H - X) - |X| = \kappa$, where q_S counts odd-order connected components whose entire vertex sets belong to S. Write $b = |X|$, let calQ be all components of H-X, let calK be the components counted by q_S, and let Y be the union of vertices of calQ minus calK. For K in calK set $d(K) = e_H(V(K), X)$, and put $e_{inc} = e(H) - e(H - X)$. Here $e_H(U, W)$ counts edges between disjoint vertex sets U, W.*

Then $|calK| = b + \kappa$, $\nu(H) = b + \sum_{Q \in calQ} \nu(Q)$, $c_D(H) \leq \kappa$, every $d(K)$ is positive and congruent to D modulo two, and

$e_{inc} = e(H[X]) + e_H(X, Y) + \sum_{K \in calK} d(K)$,

$Db + \kappa - e_{inc} = \sum_{K \in calK}(D - d(K)) - (D-1)\kappa - e(H[X]) - e_H(X, Y)$.

Define the nonnegative integer $\zeta = \kappa - c_D(H) + \sum_{Q \in calQ} c_D(Q)$.

Fix any component Z in calQ. For K in calK minus {Z}, set $\delta_K = \max(D - d(K) - ps(K), 0)$. Supply nonnegative integers $a_i(Q)$, for $1 \leq i \leq D-1$ and Q in calQ, such that $a_i(Q) = 0$ if $Q = Z$ or Q is not in calK,

$a_i(Q) \leq \max(c_i(Q) - \nu(Q), 0)$,

and $\sum_{i=1}^{D-1} a_i(K) = \delta_K$ for every K in calK minus {Z}.

Put $\mu_i(Q) = c_i(Q) - a_i(Q)$ and $M_i = \max_{Q \in calQ} a_i(Q)$.

For each i perform the following finite procedure, with arbitrary choices among eligible components. Start with $L = b + \sum_{Q : a_i(Q) = 0} c_i(Q)$. The remaining components have $a_i(Q) > 0$. A component Q is eligible if $\nu(Q) \leq L + 1$; on absorbing it remove it from the remaining family and increase L by $\mu_i(Q)$. Stop when no remaining component is eligible. Let I be the set of indices whose chosen procedures absorb every remaining component, and set $E = \sum_{inot \in I} M_i$, with i ranging from 1 through D-1. Every procedure uses at most $|calQ|$ absorptions. At every index,

$c_i(H) \leq b + \sum_Q \mu_i(Q) + M_i$,

and if i belongs to I the stronger bound $c_i(H) \leq b + \sum_Q \mu_i(Q)$ holds.

Consequently, with $\eta = ex(Z) - ex(H)$, which may be a signed integer,

$ps(H) \geq \sum_Q ps(Q) + \sum_{K \in calKminus\{Z\}} \delta_K + e_{inc} - Db - \kappa - E + \zeta$,

and

$gap(H) \leq gap(Z) - \eta + E - (D-1)\kappa + 1_{Z \in calK}(D - d(Z))$

$-e(H[X]) - e_H(X, Y) - \zeta$

$-\sum_{Qnot \in calK, Q \neq Z} ps(Q)$

$-\sum_{K \in calK, K \neq Z} \max(ps(K) - (D - d(K)), 0)$.

The indicator term is zero when Z is not in calK, so $d(Z)$ is only used when defined. Empty sums are zero.

If every procedure finishes, $gap(Z) \leq 1$, and $ex(Z) \geq ex(H)$, then even D implies $gap(H) \leq 0$. For odd D, if $gap(H) \geq 1$, then $\kappa = 1$, Z belongs to calK, $d(Z) = 1$, $gap(H) = gap(Z) = 1$, $ex(H) = ex(Z)$, and every nonnegative quantity subtracted after the indicator term in the displayed gap bound is zero. All cardinalities, credits, capacities, matching budgets and iteration indices are integers in their specified nonnegative domains; D, b, κ are positive, and gap and η may be signed.

Proof. All cost thresholds and $gap = ex - ps$ are provided by Lemma 23. We use Lemma 61 for $ps \geq 0$, the zero tail above maximum degree, and the vanishing of row D when a matching covers all degree-D vertices. Restriction, union and vertex-set extension of certificates are supplied by Lemma 72. Feasible allocation sets and their integer sumset criterion are supplied by Lemma 75.

First prove the matching identity. Choose a matching M leaving exactly κ vertices of S uncovered; the finite nonempty family of matchings attains this minimum. For K in calK let u_K count its uncovered vertices and let t_K count M-edges from K to X. There are no edges from K to another component of H-X. Since K has odd order, $u_K + t_K$ is an odd positive integer, and is at least one. Also $\sum_K u_K \leq \kappa$ because all vertices of K are in S, and $\sum_K t_K \leq b$ because these matching edges use distinct vertices of X. Thus

$\kappa \geq \sum_K u_K \geq |calK| - \sum_K t_K \geq b + \kappa - b = \kappa.$

All inequalities are equalities. Every vertex of X is matched to calK, each $u_K + t_K = 1$, and each K contains exactly $(|V(K)| - 1)/2$ internal matching edges, a maximum matching of K. No M-edge joins X to Y or has both endpoints in X. Replacing the internal matching of each component outside calK by any of its maximum matchings therefore constructs a matching of H of size $b + \sum_Q \nu(Q)$. Conversely any matching of H has at most b edges meeting X and at most $\nu(Q)$ edges inside each Q. This proves the identity.

For the top cost, use the chosen M and select one incident edge at each uncovered S-vertex. Let F be the set of selected edges, so $|F| \leq \kappa$. These edges exist since $D \geq 2$, and none belongs to M. In H-F, every vertex still having degree D is covered by the retained matching M. Hence $c_D(H - F) = 0$ by Lemma 61. For any $A + B = D$, take its degree-bounded edge partition. Given $x + y = \kappa$, put $\min(x, |F|)$ selected edges into the red exceptional part and the rest into the blue exceptional part. The latter number is at most y because $|F| \leq x + y$. Matching cardinalities are at most the corresponding edge counts. Thus every such allocation is feasible and $c_D(H) \leq \kappa$. Nonnegativity of costs gives $\zeta \geq 0$.

Every edge leaving K in calK ends in X, and every vertex of K has H-degree D. Therefore $D|V(K)| = 2e(K) + d(K)$. Odd order gives $d(K)$ congruent to D modulo two. Connectedness and X nonempty give $d(K) > 0$. Edges meeting X lie either inside X, between X and Y, or between X and one of the K. These three classes are disjoint and exhaustive, proving the identity for e_{inc}. Substituting $|calK| = b + \kappa$ into its right side gives the displayed identity for $Db + \kappa - e_{inc}$.

Fix an index i and capacities $A + B = i$. Enumerate calQ as Q_1,..., Q_m, and write $C_j = c_i(Q_j)$, $a_j = a_i(Q_j)$, $\mu_j = C_j - a_j$, and $M' = \max_j a_j$. Both endpoint allocations $(\mu_j, 0)$ and $(0, \mu_j)$ admit (A, B)-certificates of Q_j. Indeed, if $a_j = 0$ then $\mu_j = C_j \geq c_{A,B}(Q_j)$, so the integer threshold property supplies them. If $a_j > 0$, its bound gives $C_j > \nu(Q_j)$ and $\mu_j \geq \nu(Q_j)$; assigning every edge of Q_j to the chosen exceptional part supplies the endpoints.

Put $W = \sum_j \mu_j$. Given $x + y = W + M'$, if $x \geq W$ use red endpoints in every Q_j. If $x < W$, let j be the first index with $\sum_{h \leq j} \mu_h > x$, and put $r = x - \sum_{h<j} \mu_h$. Then $0 \leq r < \mu_j$ and $\mu_j + M' \geq C_j \geq c_{A,B}(Q_j)$. Use red endpoints before j, blue endpoints after j, and a certificate of Q_j at $(r, \mu_j + M' - r)$. Partwise union over vertex-disjoint components preserves degree bounds and adds the matching bounds. In this case the resulting budget totals are exactly x, y; in the first case any unused budget is allowed. This includes $W = 0$ in the first case. Thus $c_i(H - X) \leq \sum_Q \mu_i(Q) + M_i$ after maximizing over $A + B = i$. Vertex extension over X costs at most b by Lemma 72, giving the first row bound.

For the stronger bound use $Y_l^{A,B}(Q)$, the feasible red-budget set at total l, and run_l from Lemma 75. The zero-credit components supply full intervals at totals C_j. Their sum with [0,b], reserved for X, is initially [0,L]. A remaining component Q has $\mu = \mu_i(Q) \geq r = \nu(Q)$. The same fact gives endpoints 0, μ in $Y_\mu^{A,B}(Q)$ and

$run_\mu(Y_\mu^{A,B}(Q)) \leq \max(0, 2r - 1 - \mu) \leq \max(0, r - 1).$

Eligibility $r \leq L + 1$ makes the last bound at most L. The integer sumset criterion therefore gives $[0, L] + Y_\mu^{A,B}(Q) = [0, L + \mu]$. This is the induction invariant: every integer at the current total decomposes into a reserved X-budget and feasible budgets for the components already absorbed or assigned full cost. Each absorption removes one remaining component, so there are at most m absorptions. If the procedure finishes, component union and vertex-set restoration in Lemma 75 turn each decomposition into an actual certificate on H at total $b + \sum_Q \mu_i(Q)$; all edges meeting X, including those inside X, are exceptional. The argument applies to every $A + B = i$, establishing the stronger row bound.

Sum the applicable row bounds over $i = 1,...,$ D-1. The total amount subtracted from component costs is $\sum_{K \neq Z} \delta_K$, and the additional amount is E. All rows above D vanish in H and every Q by Lemma 61. Using the matching identity and $e(H) = \sum_Q e(Q) + e_{inc}$, direct subtraction in $ps(H)$ gives

$ps(H) \geq \sum_Q ps(Q) + \sum_{K \neq Z} \delta_K + e_{inc} - Db - E$

$+ \sum_Q c_D(Q) - c_D(H).$

The last two top-level terms are zeta-kappa by definition. This proves the asserted ps inequality without discarding them.

Subtract this inequality from $ex(H)$, replace $Db + \kappa - e_{inc}$ by its proved edge identity, and separate the term of Z. One has $ex(H) - ps(Z) = gap(Z) - \eta$. For each K in calK minus {Z}, its contribution is

$D - d(K) - ps(K) - \delta_K = -\max(ps(K) - (D - d(K)), 0).$

A nonrequired component other than Z contributes -ps(Q). These equalities give exactly the displayed gap bound.

Finally assume every procedure finishes, so $E = 0$, with $gap(Z) \leq 1$ and $\eta \geq 0$. All terms explicitly subtracted after the indicator term are nonnegative by Lemma 61 and the definitions. Put $r_D = 1$ for odd D and $r_D = 2$ for even D; every $d(K) \geq r_D$ by positivity and parity. If Z is not required, the upper bound is at most $1 - (D-1)\kappa \leq 0$. If Z is required and $\kappa \geq 2$, it is at most $1 - 2(D-1) + D - r_D = 3 - D - r_D \leq 0$ for every $D \geq 2$. Therefore positive gap requires $\kappa = 1$ and Z required, and then $gap(H) \leq 2 - d(Z)$. At even D this is at most zero. At odd D positive integer gap forces $d(Z) = 1$ and $gap(H) = 1$. Returning to the full inequality, equality forces $gap(Z) = 1$, $\eta = 0$, and every nonnegative subtracted quantity to be zero. These are all of the asserted final conclusions. □

Remark. An exact marked barrier fixes the matching budget. Each absorbed component saves a specified integer credit from its row cost, and the exact incident-edge identity converts those savings into the stated global gap bound. Unfinished rows retain their maximum-credit error.

Depends on. Lemma 23, Lemma 61, Lemma 72, Lemma 75.

Lemma 105 (Terminal complement slack and independence). *Use the finite integer certificate costs c_i, matching number ν, independence number α, and slack $ps = e - \nu - \sum_{i \geq 1} c_i$ of Lemma 23. All graphs are finite and simple. Orders, degrees, cardinalities, indices and matching budgets are nonnegative integers; the parameters defined as differences below are initially integers. Let $d \geq 6$ be an even integer and $k \geq 1$ an integer. Let F be a connected d-regular graph of order $2k+1$ with $\nu(F) = k$. Put*

$a = \alpha(F)$, $E = c_0(F) - k$,

$\lambda = 2E - 3 - (k - a)$, $\delta = c_0(F) - c_2(F)$, $\epsilon = c_0(F) - c_1(F)$,

and assume $E \geq 1$.

Let R be an induced subgraph with $\Delta(R) \leq 2$ and $ps(R) = 0$. Put $U = F - V(R)$, $u = |V(U)|$, and assume the three exact identities

$c_0(F) = u + \nu(R)$, $c_1(F) = u + c_1(R)$, $c_2(F) = u + c_2(R)$.

Write $h = a - \alpha(R)$, $t = h + \delta$, $\mu = E + \lambda + 1 - t$, and

$v = \max(0, floor((2E-3)/(d-2)))$.

Then $h \geq 0$, $0 \leq \epsilon \leq \delta$, $\lambda \leq E - 2$, $k \geq 3E - 1 + 2v$, and

$u = k + \lambda + 2 - t$, $\nu(R) = E - 2 - \lambda + t$,

$c_1(R) = \nu(R) - \epsilon$, $c_2(R) = \nu(R) - \delta$.

Moreover

$\nu(U) \geq \mu$, $c_1(U) \geq \mu - \epsilon$, $c_2(U) \geq \mu - \delta$,

and the exact identity

$(\nu(U)-\mu)+(c_1(U)-\mu+\epsilon)+(c_2(U)-\mu+\delta)$

$+ps(U)+\sum_{i\ge 3} c_i(U)$

$=((d-6)/2)(2\lambda+3-2t)$

has only nonnegative terms on its left side.

If $d=6$, then $\Delta(U)\le 3$, $ps(U)=0$, $c_i(U)=0$ for every integer $i\ge 3$, $\nu(U)=\mu$, $c_1(U)=\mu-\epsilon$, and $c_2(U)=\mu-\delta$.

If $d\ge 8$, then $h+\delta\le\lambda+1$. Let z be the number of nonisolated odd-order components of R. For every missing allocation $x\le y$ of total $c_0(F)-1$, one has $u-x\ge z+v+2$. Here missing means that $E(F)$ admits no partition into two edge sets with matching numbers at most x and y. If, in addition, F is well-covered, meaning that every maximal independent set has size a, then $\alpha(R)=a$. Every maximum independent set of R is then maximum in F, and every induced graph H with $V(R)$ contained in $V(H)$ contained in $V(F)$ satisfies $\alpha(H)=a$. The well-coveredness assumption is used only for this last independence conclusion. No source-minimality or deletion-process assumption is imposed.

In particular, suppose $d\ge 8$ and F is well-covered. Choose a maximum independent set S and any complete process from F-S deleting current maximum-degree vertices while that degree is at least three. Assume $c_i(F-S)=c_i(F)-a$ for $i=1,2$ and $c_0(F)=a+\nu(F-S)+ps(F-S)$. Assume every deletion parent of current maximum degree D has a matching covering all its degree-D vertices, every chosen deletion lowers each c_i with $1\le i<D$ by one, and the terminal has zero slack. Then the terminal and every induced prefix remainder have independence number a. The process may have zero deletions.

Proof. The definitions, row monotonicity, and vertex-cover bound are from Lemma 23. The restoration bound is from Lemma 72, nonnegative slack from Lemma 61, and the degree-two identities from Lemma 57. We use the missing-allocation bounds of Lemma 40, the induced matching bound of Lemma 7, and finally the well-covered domination theorem Lemma 5.

Inducedness gives $h\ge 0$. Row monotonicity gives $0\le\epsilon\le\delta$, hence $t\ge 0$. Since $c_0(F)\le|V(F)|-\alpha(F)$, one has $k+E\le 2k+1-a$. Thus $k-a\ge E-1$ and $\lambda\le E-2$. The hypotheses on connectedness, order, matching number and degree allow Lemma 40 with maximum-degree bound d. It gives $k\ge 3E-1+2v$ and a missing allocation of total $k+E-1$. By exchanging colors, write any such allocation as $x\le y$. Then $x\le(k+E-1)/2$.

The assumed row identities give $\nu(R)=k+E-u$, $c_1(R)=\nu(R)-\epsilon$, and $c_2(R)=\nu(R)-\delta$. The degree-two, zero-slack identities from Lemma 57 give

$\alpha(R)=|V(R)|-\nu(R)-c_2(R)$

$=(2k+1-u)-2(k+E-u)+\delta$

$=u+1-2E+\delta.$

Since $\alpha(R)=a-h$ and $a=k-2E+3+\lambda$, this rearranges to

$u=k+\lambda+2-t$, $\nu(R)=E-2-\lambda+t$.

In particular $|V(R)|=k-\lambda-1+t$. The lower bound on k and the upper bound on λ show that

$2|V(R)|-(k+E-1)$

$=k-E-2\lambda-1+2t$

$\ge 2+2v+2t\ge 0.$

Consequently $|V(R)|\ge x$. This verifies the complement-size hypothesis needed for the induced matching bound; no independence-preserving terminal is being assumed.

Apply Lemma 7 to the missing allocation (x, y) and the vertex set $V(U)$, whose complement has at least $x=\min(x,y)$ vertices. The bound gives

$\nu(U)\ge u+c_0(F)-|V(F)|$

$=u-k+E-1$

$=E+\lambda+1-t=\mu.$

Apply the vertex-restoration inequality of Lemma 72 to F and its induced subgraph U. For $i = 1, 2$ it gives $c_i(U) \geq c_i(F) - |V(R)|$. Substitution yields $c_1(U) \geq \mu - \epsilon$ and $c_2(U) \geq \mu - \delta$. Thus the first three terms in the asserted identity are nonnegative, even before any sign for μ is deduced. Every row cost is nonnegative and Lemma 61 gives $ps(U) \geq 0$, proving the same assertion for the other terms.

We compute the identity on the full induced complement, retaining every cross edge. The degree-two formulas and $ps(R) = 0$ imply

$e(R) = \nu(R) + c_1(R) + c_2(R) = 3(k + E - u) - \epsilon - \delta.$

As F is d-regular, its total edge count is $d(2k+1)/2$, and its edges from R to U number $d|V(R)| - 2e(R)$. Subtracting $e(R)$ and these cross edges from $e(F)$ gives

$e(U) = d(2k+1)/2 - d|V(R)| + e(R)$

$= (d-3)u - d(2k+1)/2 + 3(k+E) - \epsilon - \delta.$

Using $u = k + \lambda + 2 - t$ and $\mu = E + \lambda + 1 - t$, this becomes

$e(U) = 3\mu - \epsilon - \delta + ((d-6)/2)(2\lambda + 3 - 2t).$

On the other hand, by the definition of slack,

$e(U) = \nu(U) + c_1(U) + c_2(U) + \sum_{i \geq 3} c_i(U) + ps(U).$

Subtracting 3mu-epsilon-delta gives precisely the asserted identity. Its sum is finite by Lemma 23.

When $d = 6$ the right side is zero. Every nonnegative left term is therefore zero. This proves the displayed matching, first-row, second-row and slack equalities, and $c_i(U) = 0$ for every $i \geq 3$. In particular $c_3(U) = 0$. The cost at capacities (3,0) is at most $c_3(U) = 0$, so the finite certificate definition supplies a certificate with both matching budgets zero. Its degree-zero part and its matching-number-zero parts contain no edges; every edge lies in its maximum-degree-three part. Hence $\Delta(U) \leq 3$.

When $d \geq 8$, the factor $(d-6)/2$ is positive. Nonnegativity of the left side gives $2\lambda + 3 - 2t \geq 0$. As t and λ are integers, $t \leq \lambda + 1$.

Each nonisolated odd-order path or cycle component of R has matching number at least one. Matching numbers add over disjoint components, so $z \leq \nu(R)$. With the smaller coordinate x of any missing allocation,

$u - x - z \geq u - x - \nu(R)$

$\geq u - (k + E - 1)/2 - (k + E - u)$

$= (k - 3E + 1)/2 + 2\lambda + 4 - 2t$

$\geq v + 2.$

The last inequality uses $k \geq 3E - 1 + 2v$ and $t \leq \lambda + 1$. This proves $u - x \geq z + v + 2$ for every stated missing allocation.

Finally assume F is well-covered. All components of R are paths or cycles, including isolated vertices, because $\Delta(R) \leq 2$. The component matching and exposed-vertex conditions of Lemma 5 therefore hold. The identity $c_0(F) = u + \nu(R)$ gives $x + y = u + \nu(R) - 1$, and the inequality just proved gives $u - x \geq z + 1$. The same missing allocation thus satisfies every hypothesis of that theorem. It supplies a maximum independent set of R that is maximal in F and proves $\alpha(R) = a$. Its further conclusions give the asserted statement about all maximum independent sets of R and all induced intermediate graphs. This invocation is confined to $d \geq 8$; the sextic conclusions above required no well-coveredness.

For the process conclusion, let H be a deletion parent, H′ its child, and D its current maximum degree. By its covered matching, Lemma 61 gives $c_D(H) = 0$, and restriction gives $c_D(H') = 0$. Every higher row on both graphs is zero by row monotonicity. Each of the D-1 lower positive rows loses one by assumption. Thus the local slack identity in that fact gives

$ps(H) - ps(H') = 1 - (\nu(H) - \nu(H')),$

or equivalently

$\nu(H) + ps(H) = 1 + \nu(H') + ps(H').$

If the complete process has b roots and terminal R, summing gives $c_0(F) = a + b + \nu(R) + ps(R) = a + b + \nu(R)$. The two lower rows likewise satisfy $c_i(F) = a + b + c_i(R)$ for $i = 1, 2$. The full complement of R consists of S and the b distinct roots, so its order is $u = a + b$. These are exactly the three terminal identities required above, with $\Delta(R) \leq 2$ and $ps(R) = 0$. Applying the proved result yields $\alpha(R) = a$, and every induced prefix remainder contains R and lies in F, giving the same independence number. When $b = 0$ the sums are empty and the assumed initial identities supply these equations directly. □

Remark. A missing allocation forces a strong matching lower bound on the terminal complement. Together with the first two restored costs, it uses all available complement edges in degree six and bounds terminal loss in higher degree. That bound makes the terminal's maximum independent set dominate a well-covered host.

Depends on. Lemma 5, Lemma 7, Lemma 23, Lemma 40, Lemma 57, Lemma 61, Lemma 72.

Lemma 106 (Top-cost bounds and cycle-count floors in degrees six and eight). *Use the finite integer certificate costs $c_{A,B}(G)$ and $c_i(G)$ of Lemma 49. Thus $c_{A,B}(G)$ is the least nonnegative integer total T such that every nonnegative integer split $x + y = T$ admits a partition of all G-edges into parts with maximum degrees at most A, B and parts with matching numbers at most x, y; $c_i(G)$ is the maximum over nonnegative integer capacities $A + B = i$. Write $V(G)$, $E(G)$, $d_G(v)$, $\nu(G)$, and $\alpha(G)$ for the vertex set, edge set, degree, matching number, and independence number.*

Let d belong to {6,8}. Let F be a connected finite simple d-regular graph which is factor-critical, meaning that F-v has a perfect matching for every vertex v. Suppose $|V(F)| = n = 2k + 1$ for an integer $k \geq 1$. Define the nonnegative integer

$T_6(n) = floor((3450n - 806)/10000)$ *when* $d = 6$,

$T_8(n) = floor((400n - 580)/1300)$ *when* $d = 8$.

Then $c_{d-1}(F) \leq T_d(n)$.

More precisely, for every vertex v, every perfect matching M of F-v, and every fixed ordering of the vertices, there is an explicitly constructed set Z containing v with $|Z| \leq T_d(n)$ which dominates the entire spanning graph (V(F), E(F) minus M) in closed neighborhoods. For every nonnegative integer capacity split $A + B = d - 1$ and budget split $x + y = |Z|$, the same induced graph F-Z and the same retained matching M intersect $E(F - Z)$ give an (A, B; x, y)-certificate on the entire F. Its degree-bounded parts partition $E(F - Z)$, and its matching-bounded parts partition every edge meeting Z, including edges with both endpoints in Z. The set Z is fixed before the capacity and budget splits.

Assume in addition $c_{d-1}(F) \geq \alpha(F)$. Put $g = k - \alpha(F)$, $E = c_0(F) - k$, and $p = g - E + 1$; these are integers, initially allowed to be signed. Then

$3100k \leq 10000g + 2644$ *when* $d = 6$,

$25k \leq 65g - 9$ *when* $d = 8$.

If $d = 6$ and $E \geq 5$, then $p \geq 2$. If $d = 8$ and $E \geq 8$, then $p \geq 4$; more precisely, with $v_8 = floor((2E - 3)/6)$,

$65p \geq 10E + 50v_8 + 49$.

Under the parameter definition $\lambda = 2E - 3 - g$, the same p equals E-2-lambda. Hence whenever p is the number of whole odd-cycle connected components of a specified induced subgraph of F, the displayed lower bound applies to that same count.

All capacities, budgets, totals, orders, degrees, matching and independence cardinalities, step numbers, and set cardinalities are nonnegative integers in the displayed domains. The differences g, E, p, λ are signed integers unless a stated hypothesis or conclusion gives a stronger bound. A matching deletion removes its edges and keeps every vertex; a vertex deletion takes the full induced graph. No source-minimality or deletion-process assumption is included.

Proof. We use Lemma 1 for the weighted greedy construction, Lemma 49 for certificates obtained from domination after deleting a matching, and Lemma 40 for the missing-allocation interval on a connected graph of odd order. The first part of the proof has no excess or independence-cost hypothesis.

1. Two weight choices.

For the weighted construction, use its degree $\cap D$ and denote its uncovered weight by A_0, its required gain by σ, and its outside weights by b_j.

At $D = 5$ take

$A_0 = 3450, \sigma = 10000,$

$(b_0, b_1, b_2, b_3, b_4, b_5) = (0, 1310, 1698, 1951, 2113, 2224).$

The successive differences are 1310, 388, 253, 162, 111. They are nonnegative and nonincreasing, and $b_5 \leq 3450$. For maximum closed uncovered counts $k' = 1, 2, 3, 4, 5, 6$, the corresponding sufficient gains in Lemma 1 are respectively

10000, 10004, 10007, 10002, 10001, 10135.

These values follow by substituting the displayed weights into the endpoint gains $A_0 + D\, b_1$, $(D+1)A_0 - D\, b_{D-1}$, and into

$k'A_0 - (k'-1)b_{k'-2} + k'(D+1-k')(b_{k'} - b_{k'-1})$

for $2 \leq k' \leq D$. Thus every required gain is at least σ. If a graph Q has maximum degree at most five and S is any subset of its degree-five vertices, the construction gives Y dominating S with

$|Y| \leq floor((3450|S| + 2224(|V(Q)| - |S|))/10000).$ (1)

At $D = 7$ take

$A_0 = 400, \sigma = 1300,$

$(b_0, b_1, b_2, b_3, b_4, b_5, b_6, b_7) = (0, 129, 171, 195, 209, 215, 215, 215).$

The differences are 129, 42, 24, 14, 6, 0, 0 and satisfy the same conditions, with $b_7 \leq 400$. For $k' = 1, 2, 3, 4, 5, 6, 7, 8$, the sufficient gains are

1303, 1304, 1302, 1311, 1310, 1355, 1510, 1695.

Each is at least 1300. Consequently, when Q has maximum degree at most seven and S consists of any of its degree-seven vertices, the same construction gives

$|Y| \leq floor((400|S| + 215(|V(Q)| - |S|))/1300).$ (2)

Both applications permit the empty prescribed set and every ordering used to resolve the remaining ties.

2. Domination in the original matching residual.

Fix v in $V(F)$ and a perfect matching M of F-v. Form the spanning graph $J = (V(F), E(F) minus M)$, deleting only the matching edges, and let $Q = J - v$ be the full induced deletion of v. The vertex v has J-degree d, because it is uncovered by M. Every other vertex has J-degree d-1, because it is incident with exactly one edge of M. Thus $\Delta(Q) \leq d - 1$.

Let $S = V(F)$ minus $N_J[v]$, where $N_J[v]$ is the closed neighborhood of v in J. Its order is n-d-1. Every vertex in S has Q-degree d-1, since it has no edge to v. The graph Q has exactly d vertices outside S. Apply (1) when $d = 6$ and (2) when $d = 8$, using the ordering on Q induced by the prescribed ordering of $V(F)$. Put $Z = \{v\}$ union Y.

For $d = 6$, the bound becomes

$|Z| \leq 1 + floor((3450(n-7) + 2224 \cdot 6)/10000)$

$= floor((3450n - 806)/10000) = T_6(n).$

For $d = 8$, it becomes

$|Z| \leq 1 + floor((400(n-9) + 215 \cdot 8)/1300)$

$= floor((400n - 580)/1300) = T_8(n).$

Since a simple d-regular graph has $n \geq d + 1$, both displayed $T_d(n)$ are nonnegative.

The vertex v dominates its entire closed neighborhood in J. The set Y dominates S within Q, so it also dominates S within J. Hence Z dominates every vertex of J and contains v. Every vertex outside Z therefore has a J-neighbor in Z. The only vertex uncovered by M is v, which belongs to Z. Apply Lemma 49 to F, M, Z

with its degree bound d and parameter $j = 1$. It gives precisely the asserted certificates on F, for every split $A + B = d - 1$ and $x + y = |Z|$, with the same induced residual and retained matching fixed first. It gives $c_{d-1}(F) \le |Z| \le T_d(n)$, including all edges meeting the omitted set. This proves the constructive bound for every initial choice of v and M.

3. Compare the top cost with missing allocations.

Now assume $c_{d-1}(F) \ge \alpha(F)$. When $d = 6$, the upper bound in step 2 and $n = 2k + 1$ give

$10000\,\alpha(F) \le 3450(2k + 1) - 806 = 6900k + 2644.$

Substitute $\alpha(F) = k - g$ and rearrange to obtain

$3100k \le 10000g + 2644.$ (3)

When $d = 8$, the same argument gives

$1300\,\alpha(F) \le 400(2k + 1) - 580 = 800k - 180,$

so $500k \le 1300g - 180$, equivalently

$25k \le 65g - 9.$ (4)

Factor-criticality and the specified odd order give $\nu(F) = k$: a perfect matching of F-v has k edges, while no matching on $2k + 1$ vertices has more than k edges. Thus, whenever $E \ge 1$, the graph F satisfies every hypothesis of Lemma 40 with degree bound d. That theorem gives

$k \ge 3E - 1 + 2max(0, floor((2E - 3)/(d - 2)))$. (5)

The row-zero certificate in the current notation is exactly its two-part matching allocation, because a degree-zero edge part must be empty.

For $d = 6$ suppose $E \ge 5$ and, toward a contradiction, $p \le 1$. The identity $p = g - E + 1$ gives $g = E + p - 1 \le E$. If $E = 2m$ is even, twice $floor((2E - 3)/4)$ equals E-2. If $E = 2m + 1$ is odd, the same expression equals E-3. Hence (5) gives $k \ge 4E - 4$ in both cases. Combining this with (3) yields

$12400E - 12400 \le 3100k \le 10000E + 2644,$

and therefore $2400E \le 15044$. Since $2400 \cdot 7 = 16800 > 15044$, the integer $E \ge 5$ belongs to {5,6}. At $E = 6$, the exact bound (5) is $k \ge 17 + 2floor(9/4) = 21$, whereas (3) gives $3100k \le 62644$. This is impossible because $3100 \cdot 21 = 65100 > 62644$.

At $E = 5$, (5) gives $k \ge 14 + 2floor(7/4) = 16$, while $g \le E = 5$ and (3) give $3100k \le 52644 < 52700 = 3100 \cdot 17$. Hence $k = 16$, $n = 33$, and $\alpha(F) = k - g \ge 11$. Since $c_0(F) = k + E = 21$, fact Lemma 40 supplies at least one missing allocation at total twenty. Its interval is

$2E - 1 + floor((2E - 3)/4) \le x \le k - E - floor((2E - 3)/4),$

which here has both endpoints equal to ten. Thus (10,10) is missing.

Suppose two distinct nonadjacent vertices z, z′ have at least two common neighbors. Their open-neighborhood union has size at most $6 + 6 - 2 = 10$ and avoids both z, z′. Enlarge this union to a set W of exactly ten vertices avoiding z, z′, which is possible since there are thirty-one vertices available. Assign every F-edge meeting W to the first part and all remaining edges to the second. The first part has matching number at most ten because W covers it. The second is the induced F-W on twenty-three vertices, of which z, z′ are isolated. Its matching number is at most $\lfloor 21/2 \rfloor = 10$. This contradicts the missing allocation. Therefore every pair of nonadjacent vertices has at most one common neighbor.

Choose an independent set J of exactly eleven vertices, possible because $\alpha(F) \ge 11$. For each of the twenty-two vertices v outside J put $a_v = |N_F(v)$ intersect $J|$, where $N_F(v)$ is the open neighborhood. Regularity and independence give $\sum_v a_v = 66$ by counting edges from J. Counting unordered pairs of J through their common neighbors gives $\sum_v binom(a_v, 2) \le binom(11, 2) = 55$. For each nonnegative integer a, the identity

$binom(a, 2) - (2a - 3) = (a - 2)(a - 3)/2 \ge 0$

holds: the factors have the same sign except strictly between the consecutive integers two and three, which contains no integer. Summing this inequality instead gives $\sum_v binom(a_v, 2) \ge 2 \cdot 66 - 3 \cdot 22 = 66$. The resulting $66 \le 55$ is impossible. Both values of E have now been excluded, and $p \ge 2$ follows for every $E \ge 5$.

For $d = 8$ suppose $E \geq 8$, and put $v_8 = floor((2E-3)/6) \geq 2$. Multiply (5) by 25 and combine with (4), using $g = E + p - 1$:

$75E - 25 + 50v_8 \leq 25k \leq 65E + 65p - 74$.

Rearrangement proves $65p \geq 10E + 50v_8 + 49$. Its right side is at least $80 + 100 + 49 = 229$, which exceeds $65 \cdot 3 = 195$. Since p is an integer, $p \geq 4$.

Finally $\lambda = 2E - 3 - g$ implies $E - 2 - \lambda = g - E + 1 = p$. This is a parameter identity on the same graph, so identifying p with an existing row's whole odd-cycle count requires no change of object or witness. The proof has not assumed such a row exists. □

Remark. Explicit weights give small dominating sets in a near-perfect matching residual. The last sextic endpoint would force a missing allocation whose common-neighbor restriction contradicts the independent set required by the same cost bound.

Depends on. Lemma 1, Lemma 40, Lemma 49.

Lemma 107 (Compatible maximum matchings at a marked barrier). *Let G be a finite simple graph, let U be any subset of $V(G)$, and let $\nu(G)$ be its maximum matching cardinality. A matching is a set of pairwise vertex-disjoint edges, $V(M)$ is the set of its covered vertices, and a perfect matching covers every vertex of its graph. The empty graph has its empty perfect matching. Define $\eta_U(G) = \min_M |U \text{ minus } V(M)|$ over all matchings M of G. For $X \subseteq V(G)$, let G-X be induced vertex deletion and let $o_U(G - X)$ count the odd-order connected components of G-X whose vertex sets are contained in U.*

Choose an inclusionwise maximal set X among the maximizers of $o_U(G - X) - |X|$; thus no proper superset of X attains that maximum. Let O be the family of all components counted by $o_U(G - X)$, let $q = |O|$, and let R be the subgraph induced by $V(G)$ minus $(X \text{union} \bigcup_{C \in O} V(C))$. Form a bipartite graph B with parts X and O, where xC is an edge exactly when x has a G-neighbor in $V(C)$. Then $q = |X| + \eta_U(G)$, B has a matching saturating X, each C in O is factor-critical (C-v has a perfect matching for every v in $V(C)$), and R has a maximum matching saturating U intersect $V(R)$. Moreover,

$\nu(G) = |X| + \sum_{C \in O}(|V(C)| - 1)/2 + \nu(R)$.

The following choices can be made independently and in every case give one maximum matching of G attaining $\eta_U(G)$. Choose any matching L in B saturating X. For each xC in L choose any actual edge xv_x of G with v_x in $V(C)$, and choose any perfect matching of $C - v_x$. For each C in O not used by L, prescribe any vertex z_C in $V(C)$ and choose any perfect matching of $C - z_C$. Choose any maximum matching M_R of R saturating U intersect $V(R)$. Take the union of M_R, the chosen edges xv_x, and all the chosen internal perfect matchings. Its vertices of U left uncovered are exactly the $\eta_U(G)$ prescribed vertices z_C of the components not used by L.

Conversely every maximum matching M of G with $|U \text{ minus } V(M)| = \eta_U(G)$ is obtained by this construction for suitable choices. More generally, every matching attaining $\eta_U(G)$, regardless of its cardinality, matches all vertices of X to distinct members of O, has exactly one such edge to each used component, has exactly $(|V(C)| - 1)/2$ internal edges in each C in O, and covers all marked vertices outside the unused components. It restricts to a matching of R saturating U intersect $V(R)$. All orders, component counts, cardinalities, matching sizes, and values of η_U are nonnegative integers. Empty X, empty O, empty U, isolated vertices, and empty G are permitted.

Proof. By Lemma 46, the finite maximum of $o_U(G - X) - |X|$ equals $\eta_U(G)$ and is attained. There is an inclusionwise maximal maximizer because the family of vertex subsets is finite. The identity $q = |X| + \eta_U(G)$ follows from its definition. By Lemma 55, applied with marked set U, every C in O is factor-critical.

Fix any matching M attaining $\eta_U(G)$, and let l be the number of components in O having an M-edge to X. Distinct such components require distinct matching endpoints in X, so $l \leq |X|$. Each of the q-l other components has no matching edge leaving it: all graph edges from that component to its complement have their other endpoints in X. Internal matching edges cover vertices in pairs, so each such odd component has at least one uncovered vertex of U. Therefore

$\eta_U(G) \geq q - l \geq q - |X| = \eta_U(G)$.

Every inequality is an equality. Thus $l = |X|$, each of the q-l unused components has exactly one uncovered vertex, and no vertex of U outside those components is uncovered.

There are $|X|$ used components, each requiring an edge to X, and only $|X|$ possible matching endpoints in X. Consequently each used component receives exactly one M-edge from X and every vertex of X is on one of these edges. These edges define a matching in B saturating X. In a used component C, all its vertices except its one endpoint of the edge to X are covered internally, giving $(|V(C)|-1)/2$ internal edges. In an unused component C, exactly one vertex is uncovered and every matching edge meeting C is internal, giving the same number of internal edges. The components of G-X have no edges between them; furthermore all vertices of X are already matched to O. Hence no M-edge joins R to its complement. The restriction of M to R covers U intersect $V(R)$, because no marked vertex outside the unused components is uncovered. This proves the structural assertion for every deficiency-attaining matching.

By Lemma 46, one can choose such an M, denoted $M^{\cdot}$, with cardinality $\nu(G)$. Its restriction to R is maximum in R. If a larger matching of R existed, replacing the restriction by that larger matching would preserve disjointness from all edges of $M^{\cdot}$ outside R and would increase the cardinality in G, a contradiction. We have proved existence of the claimed maximum marked-covering matching of R. Counting the edges of $M^{\cdot}$ in its disjoint parts gives

$\nu(G) = |X| + \sum_{C\in O}(|V(C)|-1)/2 + \nu(R)$.

Now make any of the choices in the construction. Its incidence matching L exists by the matching extracted above. Every edge xC of B has at least one actual realizing edge by definition of B. The components assigned to distinct vertices x are distinct, so the chosen realizing edges have pairwise disjoint endpoints. In an assigned component C, factor-criticality gives a perfect matching after its chosen port v_x is deleted; these internal edges avoid the realizing edge. In an unassigned component C, factor-criticality gives a perfect matching after any chosen z_C is deleted. Internal matchings in different components have disjoint vertex sets. The vertices of R lie in none of these components and outside X, so the chosen M_R is disjoint from all other chosen edges. Their union is therefore a matching of G.

It has one edge for each x in X, $(|V(C)|-1)/2$ internal edges for every C in O, and $\nu(R)$ edges in R. The rank identity just proved makes its cardinality $\nu(G)$. Every X vertex is covered, every vertex of an assigned component is covered, every unassigned component has precisely its prescribed z_C uncovered, and M_R covers all marked vertices in R. There are exactly $q - |X| = \eta_U(G)$ unassigned components. Thus the uncovered marked vertices are exactly the prescribed z_C and their count is $\eta_U(G)$. This verifies simultaneously all the choices and all matching and exposure requirements.

Finally take any maximum M attaining $\eta_U(G)$. The proved structural assertion supplies its incidence matching L, its actual port in every assigned component, and its unique uncovered vertex z_C in every unassigned component. Its internal edges are perfect matchings of the respective vertex-deleted components. The same replacement argument used for $M^{\cdot}$ shows that its restriction to R is maximum in R, and it covers the marked vertices there. Taking precisely these choices in the construction recovers M. This proves the converse. For a singleton component, its vertex deletion is empty and the internal matching is empty. Empty families contribute no edges and zero sums throughout; hence every stated boundary case satisfies the same construction. □

Remark. Equality in the prescribed odd-component bound exhausts all separator endpoints. The resulting disjoint pieces permit arbitrary internal exposed vertices and compatible actual port edges without changing maximum cardinality.

Depends on. Lemma 46, Lemma 55.

Lemma 108 (A certificate from three isolated row contacts). *All graphs are finite and simple. Write $V(G)$, $E(G)$, $\nu(G)$, and $\alpha(G)$ for vertex set, edge set, maximum matching cardinality, and maximum independent-set cardinality. Write $N_G(S)$ for the vertices adjacent to at least one member of S and $N_G[S] = S$ union $N_G(S)$. A graph is well-covered if every inclusionwise maximal independent set has size $\alpha(G)$.*

For nonnegative integer capacities A, B, define $c_{A,B}(G)$ as the least nonnegative integer T such that every nonnegative integer pair $x + y = T$ admits an ordered edge partition into four parts with maximum degrees at most A, B and matching numbers at most x, y, respectively. Put $c_i(G) = \max_{A+B=i} c_{A,B}(G)$, with nonnegative integer capacities and index.

Let F be well-covered and let R be an induced subgraph that is the disjoint union of p whole odd cycles, a bipartite graph D of maximum degree at most two with a perfect matching of size m_D, and a set Z of z isolated vertices, where p,

$m_D \geq 0$ and $z \geq 3$ are integers. The graph D may be empty. Bipartite means that the vertices admit a partition into two independent sets. Cycle orders count vertices and are at least three. A whole component includes all its vertices and edges. Put $U = V(F)$ minus $V(R)$ and $u = |U|$. Assume $\alpha(R) = \alpha(F)$ and $c_0(F) = u + \nu(R)$. Suppose there are nonnegative integers b_R, b_B with $b_R + b_B = c_0(F) - 1$ for which $E(F)$ has no partition into two parts with matching numbers at most b_R, b_B, and suppose $j = u - b_R \geq p + 4$.

Let I be a three-element subset of Z and let x belong to U with $N_F(x)$ intersect $V(R) = I$.

Then there is a two-element set Y contained in U minus {x} such that {x} union Y is independent, every member of Y has a neighbor in I and none in $V(D)$ union (Z minus I), and the combined neighborhood $N_F(Y)$ in each whole R-cycle of order at least five has size at most one. Its combined neighborhood in each whole R-triangle is a proper subset of that triangle. For every y in this same Y, the full induced graph $K = F[V(R)$ union $\{x, y\}]$ satisfies $c_2(K) \leq p + 1$, and

$c_2(F) \leq u + p - 1$.

For every nonnegative integer capacity split $A + B = 2$ and every nonnegative integer budget split of total $u + p - 1$, a certificate of F attains these bounds with both degree-bounded parts contained in $E(K)$. Every edge meeting $V(F)$ minus $V(K)$, including edges with both endpoints there, belongs to a matching-bounded part. All orders, counts, cardinalities, costs, capacities, budgets, indices, and optimization values in this statement are nonnegative integers.

Proof. We use Lemma 7 for assignments to actual odd components and Lemma 72 for certificate restoration.

1. A finite family of complete assignments.

Put `B0={b` in $U : N_F(b)$ intersect (Z minus I) is empty}. Every vertex of an odd cycle C is exposed: deleting it leaves an even path with a perfect matching. A cycle of order $2k + 1$ has matching and independence numbers k, attained by consecutive pairs and alternate vertices; the upper bounds follow respectively from vertex count and from injecting an independent set into its complement by taking cyclic successors. Isolates are exposed components of order one. Each connected component of D has a perfect matching, obtained by restricting the given one, and hence has even order and matching number half its order. In a bipartite component with a perfect matching the two bipartition classes have equal size, and each is a maximum independent set: any independent set uses at most one endpoint of each matching edge.

The hypotheses of Lemma 7 apply with row R and the given missing pair. Indeed the missing total plus one equals $u + \nu(R)$, and $j > 0$ implies $u \geq \min(b_R, b_B)$. Every exterior subset of size at most j is consequently assigned through actual edges to distinct odd row components.

If $|B0| \geq p + 4$, apply this to any $p + 4$ of its vertices. They avoid every isolate outside I and therefore have only $p + 3$ available components, a contradiction. Thus $|B0| \leq p + 3 < j$, and the same fact assigns all of B_0 to distinct components among the p cycles and the three singleton components in I.

Construct the finite bipartite incidence graph with parts B_0 and these $p + 3$ components; an incidence means that the original graph F has an edge between the vertex and component. Its family of matchings saturating B_0 is nonempty by the preceding assignment. Among this entire finite family choose M maximizing the integer $\Phi(M)$, the number of assigned cycle incidences bC for which b has at least two actual neighbors in C. Each matching lifts to actual edges with compatible internal maximum row matchings, because all receiving components are distinct and all their vertices are exposed.

Let B_I be the vertices assigned to isolates by M. The vertex x belongs to B_I, since its entire row neighborhood is I. Initially $|B_I| \leq 3$.

2. Companions forced by well-coveredness.

On every assigned cycle choose a maximum independent set containing any actual contact of its assignee; on each unassigned cycle choose any maximum independent set. Such a set containing any specified cycle vertex is obtained by starting the alternating choice at that vertex. In each component of D choose either bipartition class, and let J_D be their union. Let S be the union of J_D, all these cycle sets, and Z minus I. Then S is independent, dominates all row vertices outside I, and has size alpha(F)-3. Every vertex outside B_0 meets Z minus I, and every cycle assignee meets its chosen port. Thus $W = F - N_F[S]$ has vertex set I union B with B contained in B_I. The pivot x survives because S avoids its entire row neighborhood.

For any independent S in a well-covered graph, every maximal independent set J of $F - N_F[S]$ makes S union J maximal in F: all vertices outside that residual are in S or adjacent to S, while maximality of J handles the residual. Hence W is well-covered of independence number three. A maximal independent extension of {x} in W cannot contain a vertex of I, since x meets all three. It must therefore use three vertices of B. This

forces $B = B_I$, $|B_I| = 3$, and independence of B_I. The argument applies for every permitted choice of cycle ports, maximum independent cycle sets, and component bipartition classes forming J_D, while M is fixed.

Put $Y = B_I$ minus {x}. It has two members, and each has an actual I-neighbor by its matching assignment and no neighbor in Z minus I by membership in B_0. Varying the chosen bipartition class in each D-component makes the union of permitted J_D equal $V(D)$. Since every companion survives every such choice, Y has no neighbor in D.

An unassigned cycle has no Y-neighbor, since every one of its vertices belongs to an allowed maximum independent set. Suppose a cycle C is assigned to w and a is any contact of w on C. The union of maximum independent sets of C containing a is $V(C)$ minus $N_C(a)$. To check this, label its vertices a, c_1,..., c_{2k} cyclically. For $k = 1$ the only such set is {a}. For $k \geq 2$ the sets

$\{a, c_2, c_4, \ldots, c_{2k-2}\}$ and $\{a, c_3, c_5, \ldots, c_{2k-1}\}$

are maximum independent sets and together contain exactly all vertices other than c_1, c_{2k}. No independent set containing a can include those two neighbors. Since all of Y survives every allowed choice, $N_F(Y)$ intersect $V(C)$ is contained in $N_C(a)$ for every contact a of w. On a triangle this is a proper subset, and on a longer odd cycle it has size at most two.

3. A legal strict improvement rules out two long-cycle contacts.

Suppose the combined Y-neighborhood on some cycle C of order at least five has size two. It must equal $N_C(a)$ for every contact a of its assignee w. Distinct vertices of such a cycle have distinct open neighborhoods: a second common neighbor of the two neighbors of a would form a whole four-cycle in this connected degree-two graph. Thus w has only the contact a on C and contributes zero to Φ.

Choose instead on C a maximum independent set containing both neighbors of a. With the labels above it is

$\{c_1, c_3, \ldots, c_{2k-3}, c_{2k}\}$.

Keep every other chosen cycle set and J_D. The resulting S′ still has size alpha(F)-3, so $W' = F - N_F[S']$ is well-covered of independence number three. The vertices I and x survive.

Let $t_{hit} \geq 1$ count the members of B_I meeting C. All such members are dominated by this new choice, since all their contacts lie in $N_C(a)$. Every vertex outside B_I union {w} remains dominated by an unchanged part of S′. Hence at most $4 - t_{hit}$ exterior vertices survive. A maximal independent extension of {x} again needs three exterior vertices. Thus $t_{hit} = 1$, w survives, and precisely

$B' = (B_I minus \{y\})$ union {w}

survives outside I, where y is the unique hit companion. Moreover B′ is independent. Therefore W′ is a bipartite graph with independent classes I and B′, each of size three.

For each b in B_I let i_b be its old assigned isolate. The two old edges $b\ i_b$ for b different from y form a matching M_0 between B′ and I, leaving only w and i_y uncovered. Search from w along alternating paths, using nonmatching edges from B′ to I and matching edges from I to B′. If i_y is reached, a shortest such path has no repeated vertex, and interchanging its matching and nonmatching edges gives a perfect matching M_1.

If i_y is not reached, let $B.$ and $I.$ be the sets reached when the finite search closes. Every neighbor in I of $B.$ belongs to $I.$: nonmatching edges were searched, and the matching neighbor of a reached matched vertex was already reached. Every vertex of $I.$ is matched with a different vertex of $B.$ minus {w}, and these matching partners are all of $B.$ minus {w}. Hence $|B.| = |I.| + 1$. Since both classes of W′ are independent and $B.$ has no neighbor in I minus $I.$, the set $B.$ union $(I minus I.)$ would be independent of size four, contrary to $\alpha(W') = 3$. The perfect matching M_1 therefore exists.

Replace the incidence wC in M by yC, and replace all its old isolate assignments by the actual matching M_1 between B′ and I. Keep every other cycle assignment. This still saturates exactly B_0 with distinct receiving components. The new incidence yC has two contacts, whereas wC had one. No other cycle incidence changed, so Φ increases by exactly one, contradicting its maximum. This proves the required bound of one combined contact on every long odd cycle.

4. A certificate on the full induced retained graph.

Fix any y in Y and take K as in the statement. The edge xy is absent. Choose i in $N_F(y)$ intersect I. Let P_0 consist of ξ and yi, a star of matching number one. Put all remaining edges from x or y to I into an edge set Q; these form a subgraph of $K_{2,2}$ on {x, y} and I minus {i}. Put every D-edge into Q; both x and y avoid D.

For each whole row cycle C choose one more exceptional edge set P_C. On a triangle with no y-contact take the whole triangle. With sole contact a take ya and the two triangle edges at a. With contacts a, b take ya, yb, ab. These are respectively a triangle, a star, and a triangle, all of matching number one; each remaining triangle edge set is a path or empty.

On a longer odd cycle with no y-contact take any one cycle edge. With sole contact a take ya and the two cycle edges at a. These are respectively one edge and a star; their remaining cycle edges form a path, with possible isolated vertices. Add all remaining cycle edges to Q.

All K-edges have now been assigned exactly once. Indeed x has only its I-neighbors in R, y has no neighbor in Z minus I, every y-to-cycle edge has been included in its P_C, and there are no edges between distinct whole row components. The graph Q is bipartite of maximum degree at most two: its cycle remnants are paths, its D-part has those two properties by assumption, and its separate root-isolate part is a subgraph of $K_{2,2}$. Alternate edges on its paths and even cycles to split Q into two matchings. They serve capacities (1,1); for capacities (2,0) or (0,2), put all Q into the capacity-two part.

There are $p+1$ exceptional pieces, namely P_0 and all P_C, each of matching number one. For any nonnegative integer budgets of total $p+1$ assign that many pieces to the respective matching-bounded parts. A matching in their union restricts to a matching in each piece, so its size is at most the number of pieces assigned there, even though pieces may share vertices. This constructs every required certificate and proves $c_2(K) \le p+1$.

The omitted set $V(F)$ minus $V(K)$ has u-2 vertices. The vertex-restoration construction of Lemma 72 now produces every capacity and budget split at total $(u-2)+(p+1) = u+p-1$, with bounded-degree parts contained in K and all other edges in the matching-bounded parts, including edges inside the omitted set. Thus $c_2(F) \le u+p-1$, as asserted. □

Remark. The complete component assignment forces compatible companions by well-coveredness. Its strict swap aligns their odd-cycle contacts, while varying bipartition classes forces them away from the entire even row part.

Depends on. Lemma 7, Lemma 72.

Lemma 109 (Positive slack in a factor-critical graph of small defect). *Use the finite integer certificate rows $c_i(G)$, matching number $\nu(G)$, edge count $e(G)$, maximum degree $\Delta(G)$, and positive-level slack*

$ps(G) = e(G) - \nu(G) - \sum_{i\ge 1} c_i(G)$

of Lemma 61. Let $D \ge 2$ be an integer and let K be a nonempty finite simple factor-critical graph: deleting any one vertex leaves a perfect matching. Suppose $\Delta(K) \le D$ and the integer

$d = \sum_{v\in V(K)} (D - deg_K(v))$

satisfies $0 < d < D$.

Then K is connected and nonregular, and, for integers $k \ge 1$ and $u \ge 1$,

$|V(K)| = 2k+1 > D$, $\nu(K) = k$, $\Delta(K) = D$,

d is congruent to D modulo two,

$u = (D-d)/2$, *and* $e(K) = Dk + u$.

For every vertex z with $deg_K(z) < D$, every perfect matching of K-z is a matching of K covering all vertices of degree D. Such a vertex z exists. Consequently $c_i(K) = 0$ for every integer $i \ge D$. For every vertex v of degree D one has

$\nu(K-v) = \nu(K) = k$,

$ps(K) \ge ps(K-v) + 1 \ge 1$.

In particular,

$ps(K) = (D-1)k + u - \sum_{i=1}^{D-1} c_i(K) \ge 1$.

All orders, degrees, cardinalities, costs and indices are integers in the displayed domains. No hypothesis about the gaps of induced subgraphs is imposed.

Proof. Write $n = |V(K)|$. Since K is nonempty, choose any vertex. A perfect matching after its deletion shows that n-1 is even. Thus $n = 2k + 1$ for a nonnegative integer k. The supplied perfect matching has k edges and is also a matching of K. No matching on $2k + 1$ vertices can have more than k edges, so $\nu(K) = k$, and $\nu(K - v) = k$ for every vertex v.

The graph is connected. Otherwise choose vertices x and y from distinct connected components, and let C be the component containing y. A perfect matching of K-x restricts to a perfect matching of C, so $|V(C)|$ is even. A perfect matching of K-y restricts to a perfect matching of C-y, so $|V(C)| - 1$ is even. These two parity conclusions contradict one another.

Suppose $1 \le n \le D$. Simplicity gives $deg_K(v) \le n - 1$ for each vertex, whence

$d \ge n(D - n + 1) = D + (n - 1)(D - n) \ge D,$

contrary to $d < D$. Therefore $n > D \ge 2$, which also implies $k \ge 1$. If every vertex had degree at most D-1, the sum defining d would have n terms at least one, so $d \ge n > D$. This is impossible. Hence some vertex has degree D and $\Delta(K) = D$. Since $d > 0$ and every summand defining d is nonnegative, some vertex z has degree less than D. Together these two facts show that K is nonregular.

Fix any such z and any perfect matching M of K-z. In K it covers every vertex other than z, and z has degree less than D. Thus M covers every degree-D vertex of K. The covering-matching conclusion of Lemma 61 gives $c_D(K) = 0$. The same fact gives $c_i(K) = 0$ for $i \ge \Delta(K) + 1 = D + 1$, so all rows with $i \ge D$ vanish.

Now fix any vertex v of degree D. The matching equality already proved makes the loss $h = \nu(K) - \nu(K - v)$ equal to zero. Apply the covered maximum-degree deletion inequality of Lemma 61 to K and v, using the covering matching M in K. It gives

$ps(K) - ps(K - v) \ge 1 - h = 1.$

That cited result also proves $ps(K - v) \ge 0$. These inequalities hold for every degree-D vertex v and give the asserted positive slack.

Finally count edge endpoints:

$d = Dn - 2e(K) = D(2k + 1) - 2e(K).$

It follows that D-d is even. Since $0 < d < D$, the number $u = (D - d)/2$ is a positive integer, and rearranging gives $e(K) = Dk + u$. Substitute this equality, $\nu(K) = k$, and the zero rows with $i \ge D$ into the definition of $ps(K)$. This yields

$ps(K) = (D - 1)k + u - \sum_{i=1}^{D-1} c_i(K),$

whose lower bound one was just established. Every claimed count and inequality follows. □

Remark. A perfect matching exposing a deficient vertex covers the entire maximum-degree set. Deleting a maximum-degree vertex then preserves the matching number but saves a full unit in the slack ledger.

Depends on. Lemma 61.

Lemma 110 (Zero slack and odd-clique attachments). *Use the finite integer certificate costs $c_i(G)$, matching number $\nu(G)$, independence number $\alpha(G)$, and positive-level slack $ps(G) = e(G) - \nu(G) - \sum_{i \ge 1} c_i(G)$ of Lemma 23. All graphs are finite and simple. Write K_n for the complete graph on n vertices.*

For every integer $k \ge 2$, $ps(K_{2k+1}) \ge 1$. More precisely, for every integer $1 \le i \le 2k$,

$c_i(K_{2k+1}) \le 2k + 1 - i - 1_{iodd},$

where 1_{iodd} equals one for odd i and zero for even i. The sum of these upper bounds is $2k^2$. At $k \ge 3$ the bound on c_1 improves to $c_1(K_{2k+1}) \le 2k - 2$; at $k = 2$ the bound on c_2 improves to $c_2(K_5) \le 2$. All rows with $i \ge 2k + 1$ vanish.

Let G be any finite simple graph, let x be a vertex, and suppose the connected components of the induced graph G-x are precisely the odd complete graphs $T_1, \ldots, T_p$, where $p \ge 1$ is an integer. If $\alpha(G) \le p$ and $ps(G) = 0$, then $|V(T_j)|$ belongs to $\{1,3\}$ for every integer $1 \le j \le p$. Edges from x to the components are arbitrary; some components may be anticomplete to x. No connectedness, regularity, hereditary-gap, or source hypothesis is imposed on G.

All orders, degrees, costs, indices, matching cardinalities, capacity bounds, and matching budgets are integers in the displayed positive or nonnegative domains.

Proof. Use Lemma 61 for nonnegative slack, vanishing rows above the maximum degree, and the zero row at a degree threshold covered by a matching. Use Lemma 72 for vertex restoration and slack superadditivity over components. All certificate costs have the finite integer meaning of Lemma 23.

First put $Q = K_{2k+1}$, with $k \geq 2$. For odd i in [1,2k], retain any $i+1$ vertices. The induced clique on these vertices has degree i and even order, and pairing its vertices gives a perfect matching. Its row i therefore vanishes by Lemma 61. Restoring the other 2k-i vertices gives $c_i(Q) \leq 2k - i$. For even i in this interval, retain any i vertices. This clique has maximum degree i-1, so its row i vanishes, and restoration gives $c_i(Q) \leq 2k + 1 - i$. These are the stated bounds. At the pair of indices $i = 2j - 1$, 2j, with $1 \leq j \leq k$, both bounds equal $2k + 1 - 2j$. Their sum over all j is $2k^2$. The rows from $2k + 1$ onwards vanish because $\Delta(Q) = 2k$.

For $k \geq 3$, fix any nonnegative integers $u + v = 2k - 2$. Their larger coordinate is at least k-1, and hence

$u + v + \max(u, v) + 1 \geq 3k - 2 \geq 2k + 1$.

The first construction of Lemma 48 partitions all edges of Q into parts of matching numbers at most u, v. This holds for every such pair, so $c_0(Q) \leq 2k - 2$. Row monotonicity gives $c_1(Q) \leq 2k - 2$, improving the preceding c_1 bound by one.

For $k = 2$, label the five vertices 1, 2, 3, 4, 5. We construct every certificate at matching total two and capacity total two. The endpoint matching-budget pairs (0,2) and (2,0) are obtained by placing all edges in the exceptional part with budget two; every matching on five vertices has at most two edges. It remains to treat matching budgets (1,1).

For capacities (2,0), let the red exceptional part consist of all edges meeting vertex 1, let the blue exceptional part consist of 23, 24, 25, and let the red degree-bounded part be the triangle on 3, 4, 5. These disjoint parts partition all ten edges. Each exceptional part is a star, and the degree-bounded part has maximum degree two. For capacities (0,2), put that same triangle in the blue degree-bounded part and keep the two exceptional parts unchanged. For capacities (1,1), use the triangle on 1, 2, 5 as the red exceptional part and the triangle on 3, 4, 5 as the blue exceptional part. These triangles have disjoint edge sets, and each has matching number one. The four remaining edges 13, 14, 23, 24 partition into the two matchings {13,24} and {14,23}, which are the two degree-bounded parts. This treats all nonnegative integer capacity pairs of total two and all matching-budget pairs of total two. Thus $c_2(K_5) \leq 2$, improving its preceding bound by one.

The complete graph Q has $e(Q) = k(2k + 1)$ and $\nu(Q) = k$: pairing all but one vertex attains the largest possible matching. Therefore $e(Q) - \nu(Q) = 2k^2$. In each case the positive cost sum is at most $2k^2 - 1$, so $ps(Q) \geq 1$.

Now impose the attachment hypotheses. Put $n_j = |V(T_j)|$, $n = \max_j n_j$, $h = deg_G(x)$, and $R = \sum_j (n_j - 1)/2$. If every T_j contained a vertex not adjacent to x, choosing one such vertex in each component together with x would give an independent set of size $p + 1$. Thus x is adjacent to every vertex of at least one T_j, and $h \geq 1$.

Every matching uses at most one edge at x. If that edge joins x to T_j, the remaining vertices of T_j have even order $n_j - 1$ and permit at most $(n_j - 1)/2$ matching edges; the other cliques have the same respective upper bounds. Matchings avoiding x have at most R edges. Conversely, match x to any neighbor in a component T_j, perfectly match the other vertices of T_j, and match all but one vertex in every other odd clique. This constructs a matching of size $1 + R$. Consequently

$\nu(G) = 1 + R$, and $\nu(G - x) = R$.

Suppose $n \geq 5$. If an order-n clique has no neighbor of x, it is a whole component of G. The clique bound just proved, slack superadditivity, and nonnegative slack of all other components would give $ps(G) \geq 1$. Hence every order-n clique has a neighbor of x.

First suppose $h \geq n$. Every vertex of T_j has degree $n_j - 1$ or n_j, according to whether it is nonadjacent or adjacent to x. Therefore $\Delta(G) = h$; degree-h vertices other than x can occur only in an order-h clique and must be adjacent to x. There is a matching covering every degree-h vertex, as follows. If an order-h clique is entirely adjacent to x, these h vertices are all neighbors of x. Match x into that clique, perfectly match the remaining vertices there, and take near-perfect matchings in all other cliques. If no order-h clique is entirely

adjacent to x, match x into any neighbor clique and perfectly match its remaining vertices. In every other order-h clique omit a nonneighbor of x from a near-perfect matching; such a nonneighbor exists by this case assumption. Use arbitrary near-perfect matchings in all smaller cliques. In both cases every degree-h vertex is covered.

Apply the covered maximum-degree deletion bound of Lemma 61 to x. The matching-number loss is one, as calculated, so $ps(G) \geq ps(G - x)$. But G-x has an order-n clique with $n \geq 5$ as a whole component. Slack superadditivity and the proved clique bound give $ps(G - x) \geq 1$, a contradiction.

It remains to consider $h < n$. Because a largest clique meets x, $\Delta(G) = n$, and the vertices of degree n are exactly the neighbors of x lying in order-n cliques. Every order-n clique has a nonneighbor of x, because x has only $h < n$ neighbors in total. A clique T_j entirely adjacent to x exists by the independence argument, and its order is at most $h < n$. Match x into this smaller clique and perfectly match its remaining vertices. In each largest clique choose a near-perfect matching omitting a nonneighbor of x; complete by arbitrary near-perfect matchings in the other cliques. This covers every degree-n vertex.

Choose a vertex v adjacent to x in a largest clique. It has degree n. There is also a matching of size $1 + R$ that omits v: match x into the same smaller clique, omit v from a near-perfect matching of its largest clique, and use the indicated perfect or near-perfect matchings in all remaining cliques. Thus $\nu(G - v) = \nu(G)$. The covering matching and this maximum matching need not be the same. The deletion bound of Lemma 61 uses their two properties and yields

$ps(G) \geq ps(G - v) + 1 \geq 1.$

This again contradicts $ps(G) = 0$. We have excluded $n \geq 5$. Every n_j is positive and odd, so each n_j is one or three, as required. □

Remark. Two elementary vertex-retention bounds nearly account for every clique edge; one extra certificate saves a unit. In an attachment of odd cliques, the independence hypothesis forces one fully adjacent clique. It then supplies a maximum-degree deletion that preserves enough matching rank to retain this positive slack.

Depends on. Lemma 23, Lemma 48, Lemma 61, Lemma 72.

Lemma 111 (Exclusion of saturated quartic frames with zero first deficit). *Use the finite integer first-row certificate cost c_1 of Lemma 57: at each of the capacity pairs (1,0), (0,1), every nonnegative integer budget split of total $c_1(G)$ permits an edge partition into two degree-bounded parts and two matching-bounded parts. Write $\alpha(G)$, $\nu(G)$ for independence and matching numbers. All graphs are finite and simple, and vertex deletion is induced. A graph is well-covered when every maximal independent set has maximum cardinality. Paths P_j and cycles C_j have j vertices.*

Let F be a connected four-regular well-covered graph, let $a = \alpha(F) \geq 4$, and let $m \geq 1$ be an integer. For every maximum independent set S, suppose that the graph induced by the degree-three vertices of F-S has independence number at most m. A frame is a pair (S, U) with S maximum independent and U an independent m-set of degree-three vertices of F-S. For every frame put W = S union U and R = F − W, and assume:

(i) $\alpha(R) = a$, and R consists of exactly m whole odd cycles from C_3, C_5, C_7 and any number of P_3, C_4, and isolated components;

(ii) for every endpoint p of an edge of F[W], the induced graph $K_p = F[V(R) \text{ union } \{p\}]$ satisfies $\nu(K_p) = \nu(R) + 1$ and $c_1(K_p) = \nu(R) + 1$.

Then no frame exists. All degree, order, cardinality, component-count and budget parameters have the stated nonnegative integer domains. The conclusion includes $m = 1$ and empty families of any optional components.

Proof. Suppose a frame exists. We use the canonical reflection theorem Lemma 25 with its parameters m and $b = 0$. Its uniform terminal hypotheses are exactly (i). It gives $F[W] = m$ disjoint edges and a-m isolates, arbitrary orientations of its edges as actual frames, and the following reflection. A canonical independent set D of R takes a maximum independent set on each odd cycle, the center of every P_3, an opposite pair on every C_4, and no isolates. The graph F[W union D] is the terminal of another frame. Every D-vertex has degree two there. Its m odd cycles each contain exactly one old W-edge and exhaust those edges. Hence every endpoint of a W-edge has exactly one neighbor in every canonical D. No degree-two vertex of R can neighbor endpoints on two different W-edges. These conclusions apply each time a new frame is constructed below.

1. Constructing the whole components around the roots.

Fix an endpoint z of a W-edge. It has exactly three R-neighbors. Canonical choices on distinct R-components vary independently. Therefore the number of z-neighbors selected on each component is constant over its canonical choices, and the nonnegative component counts sum to one.

On an odd cycle of order $2s+1$, the maximum independent sets with cyclic labels $\{i, i+2, \dots, i+2s-2\}$ and $\{i+2, i+4, \dots, i+2s\}$ differ by replacing i with i-1. Equal intersection counts force the indicator of membership in $N_F(z)$ to be constant around the cycle. Thus z meets either none or every cycle vertex. With three R-neighbors, a nonzero possibility is precisely all of a triangle. On a C_4, a constant count of one means one neighbor in each opposite pair, hence an adjacent pair. A P_3 contributes one precisely when its center is a neighbor. The vertices in no canonical D are exactly P_3 leaves and isolates. It follows that $N_F(z)$ intersect R is a whole triangle; an adjacent C_4 pair and one leaf or isolate; or a P_3 center and two leaves or isolates.

The whole-triangle case is impossible. Let p be the mate of z and T its triangle. For any t in T, reflect with t in D. The old edge zp lies on an odd terminal cycle containing t. The other two vertices of T form a new frame edge, and both meet t. Applied to that reflected frame, the preceding menu forbids an endpoint from meeting a C_5 or C_7. Thus the cycle containing z, p, t is a triangle, giving pt. Doing this for all t makes {z, p} union T a K_5. All its vertices already have degree four, so connectedness gives $F = K_5$, contrary to $a \geq 4$. Every frame endpoint therefore avoids all terminal odd cycles.

The $adjacent - C_4$ case is also impossible. Orient z into U and let b, c be its adjacent square neighbors and x its remaining neighbor. The vertex b has its two R-neighbors and z in F-S. Since F-S is subcubic, b has no neighbor in U minus {z}; its degree there is exactly three. Replacing z by b in U is another frame. Its terminal $R - b + z$ contains a whole P_5 if x was an isolate and a whole P_7 if x was a P_3 leaf: deleting b leaves a P_3 ending at c, and z joins c to x. These are forbidden by (i). Thus each endpoint has neighborhood {t} union Y, with t a P_3 center and Y a two-element set of P_3 leaves or isolates.

Write L_t for t's two leaves and I for the R-isolates. Replacing z by t is again a legal root swap by the same subcubic argument. If Y contains a leaf of a different P_3, the new component containing z is P_5 when the other neighbor becomes isolated, P_7 when the two leaves belong to different foreign P_3s, or C_4 when they are the two leaves of one foreign P_3. The first two contradict (i). The last contradicts the just-proved square prohibition, because the new frame endpoint t meets z on that square. Therefore Y is contained in L_t union I.

Endpoints on different W-edges have different centers t, by the reflection prohibition on a common degree-two neighbor. Their sets Y are disjoint. Indeed, a common member must be an old isolate i. Orient both endpoints as roots and swap the first root z with its center t. The old isolate i becomes a leaf of the new P_3 centered at z, whereas the other center and its P_3 remain unchanged. The second root still meets i, now a foreign P_3 leaf, contradicting the locality just proved.

Put $G = F - S$. For each selected root z its whole G-component is {z, t} union L_t union Y. Distinct roots use distinct centers, no foreign leaves and disjoint isolate pairs, so no other root attaches to this component. Original R-components are whole, so there is no further edge leaving it in G. If $Y = L_t$ it is a diamond, meaning K_4 minus an edge, with adjacent cubic centers z, t. If Y consists of two isolates it is the six-vertex tree D_6 with two adjacent cubic centers and two leaves at each. The only other possibility has one leaf from L_t and one isolate; it is a triangle with a pendant edge at each of two adjacent triangle vertices.

Exclude this third possibility using (ii). In $K_z = R + z$ this component has matching number two. Put its two disjoint pendant edges in the degree-one free part, and its triangle in whichever matching-bounded part receives the unit budget. This proves its c_1 is at most one for both capacity pairs and both splits of total one. Every unchanged R-component has c_1 equal to its matching number by Lemma 57 and (i). For disjoint components, these universal totals add: distribute a specified first budget successively among the component totals, give each component the remainder of its total as second budget, and unite the component certificates. The free degree bound persists because the components are disjoint, and their exceptional matching bounds add. Hence $c_1(K_z) \leq \nu(K_z) - 1 = \nu(R)$, contradicting (ii).

We have proved, for every existing frame, that G consists of m whole components Q_1,..., Q_m, each D_6 or diamond, m whole odd cycles, and otherwise P_3, C_4, or isolates. Each endpoint's low pair is either its center's two leaves or two isolates. In particular every cubic vertex of G is one of the two centers of a Q_i. Either center from each Q_i can be selected to give another frame with the same S.

2. Restrictions on the parents of the odd cycles.

For a vertex v of a whole odd cycle of G, put $P_v = N_F(v)$ intersect S. This set has two elements. The unique S-neighbor of any cubic vertex of G avoids every such odd cycle. To prove this, select that vertex and one center in each other Q_i. These m roots give a frame whose terminal still contains the old whole odd cycles. The selected vertex's S-parent is a frame endpoint, and step 1 says it avoids them.

Choose a maximum independent set J of G by taking all leaves in every D_6, both outer vertices in each diamond, all P_3 leaves, an opposite pair in each C_4, all isolates, and a maximum independent set on each odd cycle. We have $|J| = a$: the original R is induced in G in F and $\alpha(R) = \alpha(F) = a$, so $\alpha(G) = a$; the displayed component choices are maximum. Every old odd cycle minus J has one residual edge. Both endpoints have exactly one J-neighbor and are cubic in F-J. Select one endpoint per cycle. They form an independent m-set, hence a new frame.

For a residual edge bc, with b selected, the new terminal neighborhood of c is exactly P_c. Vertex b is a frame endpoint meeting the degree-two vertex c. By step 1, c is its terminal P_3 center and P_c is its leaf pair. Centers obtained from distinct old cycles have disjoint leaf pairs. Any prescribed vertex of an odd cycle can be made such a residual endpoint by fixing an incident residual edge. Choices on different cycles are independent. Therefore P_v and P_w are disjoint whenever v, w lie on different old cycles.

For a triangle with vertices t, b, c, choose t in J and b as the root. Vertices t, b are the endpoints of one new frame edge and both meet c, whose P_3 leaf pair is P_c. Their other terminal neighbors are P_t, P_b. Step 1 says each of those pairs is either P_c or a pair of terminal isolates disjoint from P_c. Every triangle vertex can serve as c, so its three parent pairs are pairwise equal or disjoint. We now exclude equality.

For any triangle vertex t with $P_t = \{p, q\}$, the independent set (S minus {p, q}) union {t} has size a-1 and is not maximal by well-coveredness. A vertex x that can be added lies in G, avoids t, and has its nonempty S-neighborhood contained in {p, q}. It cannot be cubic by the first paragraph of this step. It cannot have G-degree at most one, since that would give at least three S-neighbors. It therefore has degree two in G and has exactly the parents {p, q}. It is not on t's triangle, whose other vertices meet t, or another odd cycle, whose parent pairs are disjoint. Thus x is a P_3 center, a C_4 vertex, or an outer vertex of a diamond.

Suppose exactly two triangle vertices t, b have pair {p, q}; the third c has a disjoint pair. Set $S' = (Sminus\{p, q\})$ union {t, x}, a maximum independent set of F. In $G' = F - S'$, vertex b has precisely the neighbors c, p, q and is cubic. Vertex c has degree one there. Each of p, q has degree two: its four original neighbors include t, b, x, and deleting t, x leaves two. Thus b has no cubic neighbor.

Nevertheless b belongs to an independent cubic m-set in G′. If x lies in a diamond Q_i, choose one center from each of the other m-1 Q_i; otherwise choose centers from any m-1 of the m root components. Their S-parents are not p, q because p, q meet an odd cycle. Their original components avoid t, x, so they remain cubic and are independent of b and of each other. For $m = 1$ this is just {b}. The resulting frame must have the structure proved in step 1, in which every selected cubic root has a cubic center neighbor. This contradicts the property of b.

Suppose all three triangle vertices have pair {p, q}. Choose x as before. Degree four gives $N_F(p) = N_F(q) = T$ union {x}. Put $M = T$ union {p, q}. The only edges from M to its complement are px, qx. Vertex x has a G-neighbor y outside M, since it has G-degree two in a different component from T. Extend {y} to a maximal independent set J_0 of F-M. Then x is not selected. For any t in T, J_0 union {t} is maximal independent in F, since J_0 dominates F-M and t dominates M. Thus $|J_0| + 1 = a$. But J_0 union {p, q} is independent: their only neighbor outside M is x, which is absent. Its size is $a + 1$, a contradiction. We have proved that all parent pairs within a terminal triangle are disjoint as well.

3. Excluding diamond components.

Suppose some Q_i is a diamond Q with centers u, v and outer vertices b_1, b_2. Choose {u} on Q, the maximum choices described in step 2 on the other noncycle components, and independently choose maximum independent sets on all odd cycles. Let J′ be their union. Each choice is maximal on its whole G-component, and the diamond choice loses exactly one vertex, so J′ is maximal in G with size a-1. The vertices Z of S having no J′-neighbor extend J′ to a maximal independent set of F. Well-coveredness gives $|Z| = 1$. Write $Z = \{z\}$ and $T' = J'$ union {z}, a maximum independent set.

Let q be the unique S-neighbor of v. We claim q always meets J′. If not, $S_v = (Sminus\{q\})$ union {v} is maximum independent. Its complement is $G - v + q$. Vertex q is cubic there. Select also one center from every Q_i other than Q. None has parent q: if one did, it and v could replace q in S, increasing independence.

These centers remain cubic and together with q form an independent m-set. By step 1, q is the center of a whole D_6 or diamond in $G - v + q$.

Deleting a center of D_6 leaves a P_3 and two isolates, and that center neighbors both isolates. The only isolates of G-v are the original G-isolates, all in J′. Deleting a center of a diamond leaves a P_3 and the center had all three as neighbors. The P_3 components of G-v are the old P_3s and the one centered at u with leaves b_1, b_2. Every old P_3 has its leaves in J′, and the new one has u in J′. Both alternatives force q to meet J′, a contradiction. Thus $q \neq z$.

Before deleting z, vertex v is cubic in F-J′, with neighbors b_1, b_2, q, and stays cubic in F-T′. Both b_i are also cubic before deleting z, having lost only u. The two endpoints of every residual odd-cycle edge are cubic in F-J′. If z met at most one endpoint on each such edge, select a surviving cubic endpoint on every old cycle. Together with v these would be an independent cubic $(m+1) - set$ in F-T′, contradicting the uniform bound. Thus z meets both endpoints of some residual cycle edge. Step 2 says z meets at most one old cycle and cannot meet two vertices on a triangle. The affected cycle consequently has length five or seven; all other residual cycle endpoints remain cubic.

If z missed both b_1, b_2, these two nonadjacent cubic vertices and one endpoint from each of the other m-1 residual cycle edges would give another independent cubic $(m+1) - set$. Therefore z belongs to

$C = (N_F(b_1) intersect S)$ union $(N_F(b_2) intersect S)$,

which has at most four elements.

Choose the independent cycle sets uniformly and independently by their residual edges; there are exactly l choices on a cycle of odd order l, one for each edge. This is a finite product distribution. Call w in C eligible if it avoids J′ and meets both endpoints of some residual cycle edge. Step 2 permits it to meet at most one old cycle and forbids eligibility on a triangle. On its possible cycle of length l in {5,7}, at most one residual-edge choice makes it eligible. Indeed, if two choices worked, the fixed neighbor set of w on the cycle would contain two distinct edges but avoid the selected independent set in either choice; the complement of that set induces exactly one edge. Thus each w has eligibility probability at most $1/5$. Summing these probabilities gives expected eligible count at most $|C|/5 \leq 4/5$. Yet for every choice the unique extender z just constructed is an eligible member of C. The count is always at least one, a contradiction. Hence no diamond occurs.

4. Counting the cycles after a deficient independent choice.

The whole components Q_i are now all D_6. First no vertex of S meets both endpoints of an edge bc of any old odd cycle. To prove this, choose its maximum independent set leaving bc residual, and use maximum component choices elsewhere as in step 2. Their union J has size a. Both b, c are cubic in F-J. Select b and one residual endpoint on each other cycle, giving a frame. Step 1 and step 3, which apply to every frame, say that all cubic components of F-J are D_6. Adjacent cubic vertices b, c must be the two centers of one D_6, and such centers have no common neighbor. All vertices of S remain in F-J, so no S-vertex can neighbor both b and c.

Choose one of the m D_6 components, with centers u, v and corresponding leaf pairs L_u, L_v. Select {u} union L_v there, a maximal independent three-set in place of its maximum four-set. Use maximum choices on all other G-components, obtaining a maximal independent set J′ of G of size a-1. Exactly one vertex z of S avoids J′, by the extension and well-coveredness argument in step 3. The set $T' = J'$ union {z} is maximum independent in F.

The preceding common-neighbor prohibition means at least one endpoint of each residual old odd-cycle edge remains cubic in F-T′. One from each gives an independent cubic m-set, and hence a frame. Consequently F-T′ has m whole D_6 components and m whole odd cycles by steps 1 and 3. Every vertex on each of these new whole odd cycles has degree two in F-T′.

Partition $V(F - T')$ into $A = S$ minus {z} and $B = V(G)$ minus J′. The set A is independent. The only edges induced by B are the m residual old odd-cycle edges and the m-1 center edges from the D_6 components other than the selected one. The selected deficient three-set removes every edge of its D_6; maximum choices in P_3, C_4, and isolates leave no edges.

Every new odd cycle needs an edge inside B, since without such an edge it would alternate between the independent parts A and B and have even length. It cannot use an old residual cycle edge, each of which has a cubic endpoint in F-T′, while a whole cycle component has only degree-two vertices. It must therefore use an old D_6 center edge. The m new cycles are vertex-disjoint and require distinct such edges, but only m-1 are

available. This contradiction also applies when $m = 1$. It rules out the assumed frame and completes the proof. □

Remark. Zero first deficit rigidifies every root component into a double star or diamond. Maximum-independent exchanges separate cycle parents; a finite averaging argument removes diamonds. A deficient independent choice in one double star then leaves too few internal edges to support the required odd cycles.

Depends on. Lemma 25, Lemma 57.

Lemma 112 (Bipartite degree-two parts of a four-regular graph). *Use the finite integer certificate costs $c_{A,B}(H)$, row maxima $c_i(H)$, and matching number $\nu(H)$ of Lemma 23. Let H be a finite simple connected four-regular graph other than the complete graph K_5 on five vertices. Put $n = |V(H)|$. There is a vertex partition $V(H) = W_1$ disjoint union W_2 such that each induced graph $H[W_i]$ is bipartite and has maximum degree at most two.*

Consequently there is a subset U of $V(H)$ with $|U| \le \lfloor n/2 \rfloor$ such that H-U is a disjoint union of paths and even cycles, with isolated vertices permitted. For every nonnegative integer capacity pair $A + B = 2$ and every nonnegative integer budget pair $x + y = |U|$, the original edge set $E(H)$ has a certificate with bounded-degree parts contained in $E(H - U)$ and exceptional parts consisting of all edges meeting U. In particular $c_2(H) \le \lfloor n/2 \rfloor$. If H is factor-critical, meaning that H-v has a perfect matching for every v in $V(H)$, then n is odd and $c_2(H) \le \nu(H) = (n-1)/2$. All orders, degrees, cardinalities, costs, capacities, and matching budgets are integers in the displayed domains.

Proof. We use the classical Brooks coloring theorem in its list form as proved by Jan Hladky, Daniel Kral and Uwe Schauz, Algebraic proof of Brooks' theorem, arXiv:0905.3475v2 (2009), Corollary 6: a connected graph of maximum degree Δ that is neither a complete graph nor an odd cycle is list Delta-colorable. Here list Delta-colorable means that every assignment of a set of at least Δ available colors to each vertex admits a choice of one available color at every vertex with different choices at adjacent vertices. This is the coloring assertion of that corollary; no assertion about its painting game is used.

All hypotheses hold for H. It is connected, finite and simple by assumption, and $\Delta = 4$. A four-regular complete graph must have five vertices and is K_5, which was excluded. An odd cycle has degree two at every vertex, so H is not an odd cycle. Give every vertex the same list {1,2,3,4}. The cited coloring theorem supplies a proper coloring of H with these four colors.

There are finitely many such proper colorings, and at least one. For a coloring let C_i be its color-i class, put $W_1 = C_1$ union C_2 and $W_2 = C_3$ union C_4, and set $\Phi = e(H[W_1]) + e(H[W_2])$. Choose a coloring minimizing the nonnegative integer Φ. Suppose some vertex v has at least three neighbors in its own part W_i. Since its total degree is four, at most one neighbor lies in the other part. That other part uses two colors, so at least one of those colors is absent among all neighbors of v. Recolor v with that color. This is still a proper four-coloring. At least three formerly internal edges at v become crossing edges, and at most one formerly crossing edge becomes internal. Every edge not incident with v keeps its status. Thus Φ decreases by at least two, contradicting its minimality.

Every vertex therefore has at most two neighbors within its part. Each $H[W_i]$ is bipartite: its two original color classes are independent and every internal edge goes between them. This proves the partition assertion.

Let U be whichever of W_1, W_2 has smaller cardinality, choosing either in a tie. Then $|U| \le \lfloor n/2 \rfloor$. Its complement induces a finite bipartite graph of maximum degree at most two. Each connected component of a finite graph of maximum degree at most two is a path or a cycle: beginning at a degree-one vertex, a walk following the only unused incident edge cannot branch and exhausts its component; if all vertices of a nontrivial component have degree two, following edges closes a cycle, and connectedness leaves no vertex outside it. An isolated vertex is a path of order one. Bipartiteness forces each cycle to have even length, because walking around it alternates the two parts. This proves the asserted form of H-U.

Alternating edges along every path and even cycle partitions $E(H - U)$ into two matchings. At capacities (1,1), use these as the two bounded-degree parts. At capacities (2,0), put all edges of H-U in the first bounded-degree part, and at capacities (0,2), put them in the second. These are all nonnegative integer capacity pairs of total two. Each gives a certificate on H-U with empty exceptional parts.

For any nonnegative integer $x + y = |U|$, choose $U_R \subseteq U$ of size x and let $U_B = U$ minus U_R. Assign every original edge meeting U_R to the red exceptional part, and every remaining edge meeting U to the blue exceptional part. The new red edges meet U_R and the new blue edges meet U_B. A matching in either part

uses distinct vertices of the corresponding set, so the matching bounds are x and y. Every edge internal to U is assigned exactly once by the red-priority rule; all other edges outside H-U are assigned as well. The bounded-degree parts on H-U remain unchanged. This is the concrete vertex restoration of Lemma 72, and it proves every claimed certificate on H. The threshold definition in Lemma 23 gives $c_2(H) \le |U| \le \lfloor n/2 \rfloor$.

Finally suppose H is factor-critical. A perfect matching of H-v covers n-1 vertices, so n is odd and this matching has $(n-1)/2$ edges. It is also a matching of H. No matching of H can have more than $\lfloor n/2 \rfloor = (n-1)/2$ edges. Thus $\nu(H) = (n-1)/2$, and the already proved cost bound becomes $c_2(H) \le \nu(H)$. □

Remark. A proper four-coloring pairs the color classes into two bipartite graphs. Minimizing the edges inside the pairs prevents any vertex from having more than two neighbors in its pair. Deleting the smaller pair leaves paths and even cycles, whose edges split into two matchings.

Depends on. Lemma 23, Lemma 72.

Lemma 113 (Component control of the independent-deletion potential). *Use the finite integer certificate families $C_{A,B}(H;x,y)$, costs $c_{A,B}(H)$, row maxima $c_i(H)$, and quantities $\nu(H)$, $e(H)$, $\Delta(H)$, $ex(H)$, $ps(H)$, and $gap(H)$ of Lemma 23. Thus $c_0(H) = c_{0,0}(H)$, $ex(H) = c_0(H) - \nu(H)$, $ps(H) = e(H) - \nu(H) - \sum_{i\ge1} c_i(H)$, and $gap(H) = ex(H) - ps(H)$. A graph is LP-good if $gap(H) \le 0$. All graphs are finite and simple. All capacities, matching budgets, row indices, orders, degrees, cardinalities, and costs below are nonnegative integers. Empty sums are zero.*

Let H be any such graph, let X be a subset of $V(H)$, put $a = |X|$ and $G = H - X$, and let P range over the connected components of G, including isolated vertices. Define the finite family

$B = \{P : Pisacomponentof Gandfloor(|V(P)|/2) \ge a+2\}$

and the integer $m = \max(\{0\}union\{ex(P) : P \in B\})$. Then

$c_0(H) \le a + \nu(G) + m.$

For every pair of nonnegative integers x, y with $x + y = a + \nu(G) + m$, an actual partition of $E(H)$ into parts of matching numbers at most x, y exists. In this construction all edges meeting X, including edges with both endpoints in X, are assigned exactly once. This assertion includes X empty, G empty, and B empty.

Now let $d \ge 2$ be an even integer, and suppose H is connected, d-regular, of odd order, and H-v has a perfect matching for every v in $V(H)$. Suppose also that H is not the complete graph on $d+1$ vertices. Let S be any maximal independent set of H, where maximal means inclusionwise maximal, and use the preceding definitions with $X = S$. Then

$a \ge 2$, $\Delta(G) \le d-1$, $c_d(H) = 1$,

$c_i(H) = 0$ *for every integer $i > d$, and* $c_i(G) = 0$ *for every integer $i \ge d$.*

The following potential bounds hold:

$gap(H) \le c_0(H) + 1 - a - \nu(G) - ps(G) \le 1 + m - ps(G).$

If $m = 0$, this implies

$gap(H) \le 1 - \sum_{Pcomponentof G} ps(P).$

If $m \ge 1$, then for every $P^{\cdot}$ in B with $ex(P^{\cdot}) = m$,

`gap(H)<=1+gap(P*)-sum_{P` *component of G, P different from* `P*}ps(P)`

$\le 1 + \max(0, gap(P^{\cdot})).$

In particular, if every P in B with $ex(P) > 0$ is LP-good, then $gap(H) \le 1$. If additionally $gap(H) \ge 1$, all of the following equalities hold:

$gap(H) = 1$,

$c_i(H) = a + c_i(G)$ *for every integer* $1 \le i < d$,

$c_0(H) = a + \nu(G) + m$,

$ps(G) = m = \sum_{Pcomponentof G} ps(P).$

When $m = 0$, every component of G has zero ps. When $m \geq 1$, every choice of $P^{\cdot}$ in B with $ex(P^{\cdot}) = m$ has $gap(P^{\cdot}) = 0$, $ps(P^{\cdot}) = m$, and every other component has zero ps. No minimality, bound on the order, or bound on the independence number is assumed.

Proof. First prove the arbitrary-deletion assertion. Let Q_0 be the vertex-disjoint union of the components in B, taking Q_0 to be the empty graph when B is empty. List the remaining components as Q_1,..., Q_s, where s is a nonnegative integer. There are no edges between different pieces of this list. For each integer $1 \leq j \leq s$, put $r_j = \nu(Q_j)$. Since a matching uses two different vertices for each edge,

$r_j \leq floor(|V(Q_j)|/2) \leq a + 1.$

Matching numbers add over vertex-disjoint graphs: restriction of a matching proves the upper bound by the sum, and taking the union of maximum matchings proves the reverse bound. Hence

$\nu(G) = \nu(Q_0) + \sum_{j=1}^{s} r_j.$

Apply Lemma 68 to Q_0 with its deleted set empty. Its actual components are precisely the members of B, and the required matching-additivity identity is the one just proved. It gives $ex(Q_0) \leq m$, including the empty case, so

$c_0(Q_0) \leq \nu(Q_0) + m.$

For every j with $ex(Q_j) > 0$ we have

$r_j \leq a + 1 \leq 1 + a + c_0(Q_0).$

The $\theta = 0$ assertion of Lemma 75 therefore applies to H, X and the vertex-disjoint pieces Q_0,..., Q_s. Its conclusion is

$c_0(H) \leq a + c_0(Q_0) + \sum_{j=1}^{s} r_j \leq a + \nu(G) + m.$

For completeness, the same application supplies an actual certificate at total $T_0 = a + c_0(Q_0) + \sum_j r_j$, retaining component partitions and restoring every edge meeting X by a partition of X. In that restoration an edge is red if it meets the chosen red subset of X, and blue otherwise; every new blue edge meets the complementary subset. Thus internal edges of X are also assigned exactly once and the added matching budgets have their specified bounds. Put $T = a + \nu(G) + m \geq T_0$. Given $x + y = T$, set $x' = \min(x, T_0)$ and $y' = T_0 - x'$. Then $0 \leq x' \leq x$ and $0 \leq y' \leq y$. Indeed, if $x \geq T_0$ then $y' = 0$, while if $x < T_0$ then $y' = T_0 - x \leq T - x = y$. The certificate supplied at (x',y') is also a certificate at (x, y). At zero degree capacities its two bounded-degree parts are empty, so this is the claimed two-part partition. All lists and constructions remain valid when a piece or the whole residual graph is empty.

Now impose the regularity, matching, and independent-set hypotheses. Every vertex outside S has a neighbor in S, since otherwise it could be added to S. It consequently loses at least one of its d incident edges, giving $\Delta(G) \leq d - 1$. A maximal independent set is nonempty because H has vertices. If $S = \{v\}$, then v is adjacent to all other vertices, so $|V(H)| = d + 1$. Each vertex of the simple d-regular graph H would then be adjacent to every other vertex, contradicting the excluded complete graph. Thus $a \geq 2$.

For any v in $V(H)$, the perfect matching of H-v covers all of its vertices and therefore all vertices of degree d. The covered-degree assertion of Lemma 61 gives $c_d(H - v) = 0$, and Lemma 72 gives $c_d(H) \leq 1$. On the other hand, a certificate at total zero for capacities (d-1,1) would partition $E(H)$ into a graph of maximum degree at most d-1 and a graph of maximum degree at most one. Since every H-vertex has degree d, the latter part would have degree exactly one at each vertex and would be a perfect matching of H. Its odd order makes this impossible: a matching covers an even number of vertices. Thus total zero is infeasible for that capacity pair, and the integer cost characterization in Lemma 23 gives $c_d(H) \geq 1$. We have proved $c_d(H) = 1$. The zero-tail conclusion in Lemma 61, applied to H and G with their respective maximum-degree bounds, gives all the asserted higher-row zeros.

By Lemma 72, for every integer $1 \leq i < d$ one has $c_i(H) \leq a + c_i(G)$. Since S is independent, no edge has two endpoints in S. Summing the degree d at its a vertices counts exactly the deleted edges, so $e(H) - e(G) = da$. The top-row identity and the zero tails now give

$gap(H) = c_0(H) + \sum_{i=1}^{d-1} c_i(H) + 1 - e(H)$

$\leq c_0(H) + (d-1)a + \sum_{i=1}^{d-1} c_i(G) + 1 - e(G) - da$

$= c_0(H) + 1 - a - \nu(G) - ps(G)$.

The final equality is the definition of $ps(G)$, whose rows at $i \geq d$ vanish. Inserting the arbitrary-deletion bound already proved gives the second potential inequality.

Every component has nonnegative ps by Lemma 61, and Lemma 72 gives

$ps(G) \geq \sum_{P component of G} ps(P)$.

If $m = 0$, substituting this inequality gives the stated bound by one minus that sum. If $m \geq 1$, the finite definition of m supplies $P^{\cdot}$ in B with $ex(P^{\cdot}) = m$. For every such choice,

$1 + m - ps(G) \leq 1 + m - ps(P^{\cdot}) - \sum_{P different from P^{\cdot}} ps(P)$

$= 1 + gap(P^{\cdot}) - \sum_{P different from P^{\cdot}} ps(P)$.

Here $ex(P^{\cdot}) - ps(P^{\cdot}) = gap(P^{\cdot})$ by Lemma 23. The sum subtracted in the last expression is nonnegative, so that expression is at most $1 + \max(0, gap(P^{\cdot}))$. This proves all component potential bounds.

Suppose every member of B having positive excess is LP-good. If $m = 0$, nonnegative $ps(G)$ and the potential bound give $gap(H) \leq 1$. If $m \geq 1$, the chosen $P^{\cdot}$ has $gap(P^{\cdot}) \leq 0$, so the last component bound gives the same conclusion.

Finally also suppose $gap(H) \geq 1$. Then $gap(H) = 1$. When $m = 0$, the chain

$1 = gap(H) \leq c_0(H) + 1 - a - \nu(G) - ps(G) \leq 1 - ps(G) \leq 1$

consists of equalities. It gives $ps(G) = 0$ and $c_0(H) = a + \nu(G)$. Superadditivity and nonnegativity then give $ps(P) = 0$ for every component P.

When $m \geq 1$, fix any $P^{\cdot}$ in B with $ex(P^{\cdot}) = m$. The chain

$1 = gap(H) \leq c_0(H) + 1 - a - \nu(G) - ps(G)$

$\leq 1 + m - ps(G)$

$\leq 1 + gap(P^{\cdot}) - \sum_{P different from P^{\cdot}} ps(P) \leq 1$

also consists of equalities. The last equality and the signs $gap(P^{\cdot}) \leq 0$ and $ps(P) \geq 0$ force $gap(P^{\cdot}) = 0$ and $ps(P) = 0$ for every other component. Thus $ps(P^{\cdot}) = ex(P^{\cdot}) = m$. Equality at the preceding steps gives $ps(G) = m$ and $c_0(H) = a + \nu(G) + m$. It also proves that $ps(G)$ is the sum of all component slacks. The reasoning applied to each allowed choice of $P^{\cdot}$, which proves the stated quantifier.

In both cases, equality holds in the first potential inequality. The difference between its right and left sides is exactly

$\sum_{i=1}^{d-1} (a + c_i(G) - c_i(H))$.

Each summand is nonnegative by vertex restoration. A finite sum of nonnegative numbers is zero only when each term is zero, proving $c_i(H) = a + c_i(G)$ for every integer $1 \leq i < d$. These are all of the asserted equalities. □

Remark. The deleted vertices absorb every component whose matching number fits their budget interval. The other components cost their total matching number plus only their largest excess. Equality in the resulting potential bound forces equality in every positive row.

Depends on. Lemma 23, Lemma 61, Lemma 68, Lemma 72, Lemma 75.

Lemma 114 (Two-component residual profiles and boundary restoration). *Let $s, t \geq 1$ be integers and let F_1, F_2 be the star forests with positive nonincreasing integer edge-size lists $n_1,\ldots, n_s$ and $m_1,\ldots, m_t$, prescribed red and blue. Copies are ordinary subgraphs with vertex-disjoint target components. Put $M = s + t - 1$, $\ell_k = max\{n_i + m_j - 1 : 1 \leq i \leq s, 1 \leq j \leq t, i + j = k\}$ for $2 \leq k \leq s + t$, and $P_q = \sum_{k=q+2}^{s+t} (\ell_k - 1)$. For $0 \leq q < M$ let A_q consist of every ordered nonempty actual suffix pair after α red and β blue leading cuts, where $\alpha + \beta = q$.*

Fix an integer $0 \leq q < M$, let $D = \ell_{q+2} - 1$, and suppose the natural q-tail has exactly one initial entry $D + 1$. Let H be finite simple, write ν for maximum matching cardinality and $\Phi(H) = e(H) - \nu(H)$, and suppose

$\Phi(H) \le P_q + r$ for an integer $r \ge 0$. Let y be an actual vertex with $d_H(y) \le D-1$. Suppose H-y has four distinct odd connected components C_1,..., C_4, each adjacent to y and wholly of ambient H-degree D. Order their half-orders $\mu_i = (|V(C_i)|-1)/2$ increasingly, and put $C = C_1$ disjoint union C_2, $\mu = \mu_1 + \mu_2$, $n = 2\mu + 2$, $b_i = e_H(V(C_i), \{y\})$, $v_0 = d_H(y) - (b_1 + b_2)/2$. For $1 \le h \le \mu$ put $d_h = \min(D, \lfloor n/h \rfloor - 1)$. In each full target apply these capacities by deleting a largest remaining component of size at most the capacity whenever possible. Let U_1, U_2 be the resulting actual residual forests, and assume both are nonempty. Put $E = \{(U_1, F_2), (F_1, U_2)\}$. For every integer $z \ge 0$ let Ψ_z be the complete diagonal-envelope potential of all nonempty actual total-z suffix pairs from E; it is zero when that family is empty. For any nonempty family this potential means: on each attained diagonal $i + j = k$, maximize the two component sizes minus one over all pairs and valid indices, subtract one from that maximum, and sum over all attained diagonals.

The following two regimes have the stated guarantees.

(I) Suppose $D \ge 7$, both full original heads belong to [4,D-1], and $0 \le r \le D$. Put $\Gamma = \lceil n/20 \rceil$ when $D = 7$ and $\Gamma = ceil((\mu+1)(4/15 - 2/D))$ when $D \ge 8$. Then $\Gamma \ge 1$ and $0 \le P_q - \Psi_q \le (D-1)\mu + 1 - \Gamma \le (D-1)\mu$. If $v_0 + \Gamma \ge r + 2$, then H has a coloring avoiding both entire targets of some member of A_q. In particular $v_0 \ge r + 1$ suffices. No factor-criticality or near-perfect matching is assumed.

(II) Suppose $D \ge 6$, both full original heads are at most D, and each C_i is factor-critical, meaning that deleting any vertex leaves a perfect matching. Suppose either a full original head equals D, or $D \ge 7$, both full heads are at least three, $\mu \ge 8$, and $s, t > 3\mu/2$. Then $0 \le P_q - \Psi_q \le (D-1)\mu + 1$. If $v_0 \ge r + 1$, then H has a coloring avoiding both entire targets of some member of A_q. In particular this holds for $0 \le r \le 2$, since $v_0 \ge 3$. The full-head alternative imposes no lower bound on the other head; neither alternative imposes an upper bound on r.

Every asserted avoiding coloring extends over the original canonical q-star pool with capacities ℓ_2,..., ℓ_{q+1}, and across any specified actual outer q-set X when $H = G - X$, to avoidance of the unchanged original targets, preserving all H-edge colors. Neither regime assumes a matching deficit, equimatchability, or a degree bound outside the four components and y. Units, repeated sizes, all complete suffixes, and simultaneous exchange of target roles and colors are included. All list lengths, indices, degrees, half-orders, capacities, cuts and surpluses are integers in the stated domains.

Proof. Each boundary b_i is positive and at most D-1. For $v = |V(C_i)|$, the ambient degree sum and simplicity give $b_i \ge v(D+1-v)$. If $1 \le v \le D$, then $v(D+1-v) - D = (v-1)(D-v) \ge 0$, a contradiction. Hence $v > D$ and $\mu_i \ge \lceil D/2 \rceil$; thus $\mu \ge D$. Also $v_0 \ge (b_1 + b_2)/2 + b_3 + b_4 \ge 3$. All displayed capacities are positive because $n/h \ge 2 + 2/\mu > 2$ for $h \le \mu$. The first capacity equals D.

Define $N_x = |\{i : n_i \ge x\}|$ and $T_z = |\{j : m_j \ge z\}|$ for positive integer thresholds, and define N'_x, T'_z from U_1, U_2. Let $a_R = s - |U_1|$ and $a_B = t - |U_2|$, where the bars here count components. At most μ operations remove a component, so a_R, $a_B \le \mu$ and N'_1, $T'_1 \ge 1$. For $j \ge 2$ put $k_j = floor(n/(j+1)) \le \mu$. A component of size at least j can be removed only by a capacity position $h \le k_j$, whence $N_j - N'_j \le k_j$, $T_j - T'_j \le k_j$. (1)

For $h \ge 1$ let V_h be the maximum of $N_x + T_z - 1$ over positive counts with $x + z = h + 2$, or zero if no such thresholds exist. The number of full profile positions with $\ell_k - 1 \ge h$ equals V_h. Indeed, the inequality $n_i + m_j \ge h + 2$ is equivalent to the existence of positive thresholds $x \le n_i$, $z \le m_j$ with $x + z = h + 2$. For fixed thresholds the valid indices form a rectangle with diagonal sums exactly 2,..., $N_x + T_z$. Their union is an initial interval, of length V_h.

The corresponding natural width is $W_h = (V_h - q)_+$, where $u_+ = \max(u, 0)$. To check the entire suffix operation, a candidate at original indices $i + j = q + k$ with $k \ge 2$ survives cuts $\alpha = \max(0, q - j + 1)$, $\beta = q - \alpha$; these are nonnegative and satisfy $\alpha \le i - 1$, $\beta \le j - 1$. Conversely any suffix candidate has that original index sum. Apply this reasoning to each endpoint and take their envelope, calling its natural width W'_h. Endpoints with no surviving q-family contribute zero. Their components are contained in the original lists, so $\Delta_h = W_h - W'_h \ge 0$, and $P_q - \Psi_q = \sum_{h=1}^{D} \Delta_h$. (2) The plateau assumption gives $W_D = 1$ and $W_h = 0$ for $h > D$.

For a height with $W_h > 0$, choose maximizing thresholds (x, z). If $x = 1$, the endpoint (U_1, F_2) retains that threshold and loses at most $a_R \le \mu$ in width. If $z = 1$ use (F_1, U_2). If $x, z \ge 2$ and $N'_x > 0$, the first endpoint loses at most $k_x \le \mu$; if $T'_z > 0$, the second loses at most $k_z \le \mu$. Taking positive parts does not enlarge any loss. Therefore $\Delta_h > \mu$ requires a mixed pair with $N'_x = T'_z = 0$. In that case (1) gives $\Delta_h \le k_x + k_z - 1$. (3)

In the first alternative of regime (II), exchange target roles and colors so $m_1 = D$. A natural entry $D + 1$ implies the full head sum is at least $D + 2$, so $N_2 > 0$. The top threshold pair (2,D) gives $N_2 + T_D - 1 \le q + 1$,

hence $N_2 \le q+1$. In a disappearing mixed case below height D, $W_h = N_x + T_z - 1 - q \le T_z \le k_z \le \mu$. All other cases were already bounded by μ. The top loses at most one, proving (II)'s profile bound through (2).

For its second alternative, some full head is at least five because the head sum is at least $D+2 \ge 9$. Orient it red. If $h \ge 5$ and both mixed thresholds disappear, their reciprocal sum is at most $1/3 + 1/(h+1) \le 1/2$, so (3) gives $\Delta_h \le n/2 - 1 = \mu$. Height one has only axis pairs. At $h = 2, 3, 4$, $N_{h+1} \ge 1$. The endpoint (F_1, U_2) gives $W'_h \ge N_{h+1} + t - a_B - 1 - q$. If $\Delta_h > \mu$, subtraction from $W_h = N_x + T_z - 1 - q$ yields $t < N_x + T_z$. The mixed thresholds both disappear, so $t \le k_x + k_z \le 2floor(n/3) \le 4(\mu+1)/3 \le 3\mu/2$, using $\mu \ge 8$. This contradicts the assumed $t > 3\mu/2$. Hence every height below D loses at most μ, proving the same profile bound.

We now prove the sharper bound in regime (I). Put $\rho = \sum_{i=1}^{4} \mu_i$ and $b_{all} = \sum_{i=1}^{4} b_i$. The induced graph on y and the four pieces has $D\,\rho + 2D + b_{all}/2$ edges and matching number at most $\rho + 1$. Thus its potential is at least $(D-1)\rho + 2D - 1 + b_{all}/2$. Since $\rho \ge 2\mu$ and $b_{all} \ge 4$, this is at least $(D-1)n + 3$. Potential cannot decrease on restoring a vertex: its change is d-h with $0 \le h \le 1$, and $h = 0$ when $d = 0$. Consequently $\Phi(H) \ge (D-1)n + 3$. (4) Only an upper bound on component matching size was used.

Suppose $\Delta_2 > \mu$. Its maximizing pair must be (2,2) with $N'_2 = T'_2 = 0$. Both full heads are at least four, so N_3, $T_3 > 0$. Endpoint unit-axis candidates give $W'_2 \ge s - a_R + T_3 - 1 - q$, $W'_2 \ge N_3 + t - a_B - 1 - q$, even if these right sides are negative. Subtract from $W_2 = N_2 + T_2 - 1 - q$ and use $\Delta_2 > \mu$ to get s, $t \le N_2 + T_2 \le 2k_2$. Every original nonunit threshold count is at most k_2. All unit-axis widths are at most $3k_2 - 1 \le n - 1$ and all mixed widths at most $2k_2 - 1 \le n - 1$. Hence $P_q \le (D-1)(n-1) + 1$. Its sum with r is below (4), by at least $D + 1 - r \ge 1$, a contradiction. Thus $\Delta_2 \le \mu$.

Suppose $\Delta_3 > \mu$. After exchanging targets its maximizing pair is (2,3), with $N'_2 = T'_3 = 0$. The positive counts N_4, T_4 permit the same unit-axis comparison, giving s, $t \le N_2 + T_3 \le k_2 + k_3$. Every N_j with $j \ge 2$ is at most k_j, and every T_j with $j \ge 3$ is at most k_j. Widths at heights one and two are at most $2(k_2 + k_3) - 1 \le 7n/6 - 1$. At any height $h \ge 3$ a unit-axis candidate is at most $k_2 + 2k_3 - 1$: its nonunit count is at most k_3 and its other count at most $k_2 + k_3$. A mixed pair with blue threshold at least three is at most $k_2 + k_3 - 1$. If its blue threshold is two, its red threshold is at least three and its width at most $k_3 + t - 1 \le k_2 + 2k_3 - 1$. Thus all such heights are at most $5n/6 - 1$. Using top width one, $P_q \le 2(7n/6 - 1) + (D-3)(5n/6 - 1) + 1 = (5D-1)n/6 - D + 2$. The lower bound (4) exceeds this plus r by $(D-5)n/6 + D + 1 - r > 0$. This contradiction proves $\Delta_3 \le \mu$.

At height one only unit-axis pairs occur, giving $\Delta_1 \le \mu$. At height four the disappearing mixed pairs are (2,4), (3,3), (4,2). Their losses in (3) are at most $8n/15 - 1$ or $n/2 - 1$, respectively. Therefore $\Delta_4 \le \mu + (\mu+1)/15$. For $h \ge 5$ a mixed pair satisfies $1/(x+1) + 1/(z+1) \le 1/3 + 1/(h+1) \le 1/2$, because $(x+1)(z+1) \ge 3(h+1)$ when $x, z \ge 2$ and $x + z = h + 2$. Thus its loss in (3) is at most $n/2 - 1 = \mu$. The surviving and axis cases have the same bound.

When $D \ge 8$, at height D-1 the two thresholds are at least two, since both full heads are at most D-1. Their reciprocal sum is at most $1/3 + 1/D$. A disappearing pair loses at most $n(1/3 + 1/D) - 1$. A surviving pair loses at most $n/3$, which is smaller because $n/D > 1$. These heights are distinct from the first four. Summing, and using $\Delta_D \le 1$, gives $P_q - \Psi_q \le (D-2)\mu + (\mu+1)/15 + n(1/3 + 1/D) = (D-1)\mu + 1 - (\mu+1)(4/15 - 2/D)$. The left side is an integer, so rounding subtracts precisely Γ. Its defining coefficient is positive for $D \ge 8$, so $\Gamma \ge 1$.

For $D = 7$, each odd piece has order at least nine, hence $\mu \ge 8$ and $n \ge 18$. The preceding proofs of Δ_2, $\Delta_3 \le \mu$ remain valid: their contradiction margins are $D + 1 - r$ and $(D-5)n/6 + D + 1 - r$, both positive at $D = 7$ and $r \le 7$. At height six the possible pairs are (2,6), (3,5), (4,4), (5,3), (6,2). For (2,6), $N_3 \ge 1$ and the top rectangle (3,6) gives $T_6 \le q + 1$. If $N'_2 > 0$ its loss is at most k_2; if $N'_2 = 0$, its old natural width is at most $N_2 \le k_2$. This also handles (6,2) after exchanging colors. Each other pair, if a threshold survives, loses at most $n/4$. If both disappear its loss is at most $5n/12 - 1$ for (3,5), (5,3), or $2n/5 - 1$ for (4,4). Since $n \ge 18$, all these bounds, including $n/3$, are at most $5n/12 - 1$. Thus $\Delta_6 \le 5n/12 - 1$. Combining Δ_1, Δ_2, Δ_3, $\Delta_5 \le \mu$, $\Delta_4 \le 8n/15 - 1$ and $\Delta_7 \le 1$ gives $P_q - \Psi_q \le 4\mu + (8n/15 - 1) + (5n/12 - 1) + 1 = 6\mu + 1 - n/20$. Integer rounding yields the stated $\Gamma = \lceil n/20 \rceil \ge 1$. This completes (I).

It remains to convert the bounds into actual avoidance. Put $b = b_1 + b_2$ and $Q = H - (V(C)union\{y\})$. The ambient degree sum gives $e(C) = Dmu + D - b/2$. No edges leave C except those to y. Deleting C and y removes exactly $Dmu + D + v_0$ edges. Every matching has at most μ internal edges of C and at most one edge at y, so its deletion loses at most $\mu + 1$ matching edges. Thus $\Phi(Q) \le P_q + r - (D-1)\mu - D + 1 - v_0$. (5)

By Lemma 28, capacity deletions commute; the first capacity D deletes an original head. Hence the i-th residual size is at most the $(i+1)-st$ original size. Every existing first potential entry of the endpoint total-q family is at most $\ell_{q+3}-1 \le D-1$. That original index exists whenever the endpoint family exists, since each endpoint has fewer total components. Profiles are nonincreasing: lower any noninitial index of a candidate on a later diagonal to get a candidate on the preceding diagonal of at least equal size. This also holds for an envelope; append zero after its last potential entry. The suffix identity already proved gives, for every integer $j \ge 0$, $\Psi_{q+j} \ge \Psi_q - j(D-1)$. (6)

In regime (I), (6) and the profile bound give $\Psi_{q+1} \ge P_q - (D-1)\mu - D + \Gamma$. If $v_0 + \Gamma \ge r+2$, this strictly exceeds the right side of (5). Since $\Phi(Q) \ge 0$, the endpoint $total-(q+1)$ family is nonempty. By Lemma 56 there is a coloring of Q avoiding one entire pair in that family.

In regime (II), (6) instead gives $\Psi_{q+2} \ge P_q - (D-1)\mu - 2D + 1$. The component C_3 has order greater than D and at most $b_3 \le D-1$ neighbors of y. Choose w in C_3 not adjacent to y. It has degree D in Q. Factor-criticality gives a maximum matching of C_3 exposing w, and C_3 is a whole component of Q. Therefore $\nu(Q-w) = \nu(Q)$ and $\Phi(Q-w) = \Phi(Q) - D$. If $v_0 \ge r+1$, (5) yields $\Phi(Q-w) \le P_q - (D-1)\mu - 2D < \Psi_{q+2}$. The endpoint $total-(q+2)$ family is nonempty by nonnegativity of graph potential, and Lemma 56 supplies a coloring avoiding one entire member. Decrease a positive cut by one and use the actual one-vertex extension Lemma 41 to restore this same w. This produces an avoiding coloring on Q for that endpoint's $total-(q+1)$ suffix pair.

In either regime we now have such a Q-coloring, with an identified originating endpoint and cuts α, β summing to $q+1$. Every disjoint star collection in C has at most μ components: selecting one edge from each gives a matching, and each odd piece has matching number at most its half-order. If its sizes are $c_1 \ge c_2 \ge ...$, its first h stars use at least $h(c_h+1)$ vertices. Thus $c_h \le \lfloor n/h \rfloor - 1$ and $c_h \le D$. Assign its h-th star to the h-th star of a fresh disjoint pool of capacities d_h; this is a feasible embedding. A target copy in C disjoint union Q can therefore be moved into this pool disjoint union Q without changing its part in Q. The residual containment property of Lemma 28 then leaves its actual greedy residual in Q. Every leading cut is another capacity-D operation, since the full original heads are at most D; commutation proves this same containment after the specified cuts.

Color all edges of C in the color of the originating residual endpoint. There are no C–Q edges in H-y. A copy of that color's full original suffix would leave its prohibited residual in Q; the other color's positive-edge target lies entirely in Q. Hence this coloring avoids the corresponding original $total-(q+1)$ suffix pair on H-y. Both original suffixes are nonempty because the endpoint suffixes are nonempty. Decrease a positive cut by one and use Lemma 41 at exactly y. The result avoids an entire A_q member on H. The same fact extends it over the canonical pool and through every specified actual outer q-set, preserving the H-colors. All operations concern full actual target components, so units and repeated sizes are retained. □

Depends on. Lemma 28, Lemma 41, Lemma 56.

Lemma 115 (Exact independent deletion in a minimal gap-one failure). *Use the finite integer certificate costs $c_i(G)$, matching number $\nu(G)$, edge count $e(G)$, maximum degree $\Delta(G)$, excess $ex(G) = c_0(G) - \nu(G)$, slack $ps(G) = e(G) - \nu(G) - \sum_{i\ge1} c_i(G)$, signed gap $gap(G) = ex(G) - ps(G)$, and top missing-allocation run $tr(G)$ of Lemma 23. For each nonnegative integer t let $Y_t(G)$ be the integers x in [0,t] for which $E(G)$ has a partition into two parts whose matching numbers are at most x and t-x. Intervals consist of integers.*

Let h be a nondecreasing integer-valued function on the nonnegative integers. Say that G satisfies T_h if $gap(G) > 0$ implies both $gap(G) = 1$ and $tr(G) + ex(G) \le h(\Delta(G))$. Let $d \ge 4$ be an even integer. Let F be a finite simple connected d-regular factor-critical graph, meaning that deleting any one vertex leaves a perfect matching. Suppose $gap(F) = 1$, F violates T_h, and every proper induced subgraph of F satisfies T_h, including the empty induced subgraph. Put $k = \nu(F)$ and $E = ex(F)$. Then $|V(F)| = 2k+1$ and $E \ge 1$.

For every nonempty independent vertex set I, put $j = |I|$, $J = F - I$, $b(I) = j + \nu(J) - k$, and $z(I) = gap(J)$. Then

$c_i(J) = c_i(F) - j$ for every integer $1 \le i < d$,

$c_d(F) = 1$, $c_d(J) = 0$,

$c_i(F) = c_i(J) = 0$ for every integer $i > d$,

$z(I)$ in {0,1}, $c_0(J) = c_0(F) - j + z(I)$,

$ex(J) = E - b(I) + z(I)$, $ps(J) = E - b(I)$,

$1 \le b(I) \le \min(E, \lceil j/2 \rceil)$,

$c_0(F) = j + \nu(J) + ps(J)$.

For nonempty independent sets I contained in I′, one has $z(I) \le z(I')$. Thus when $z(I) = 0$ every row of integer index $0 \le i < d$ drops by exactly j. All orders, degrees, capacities, budgets, costs, row indices, matching numbers, cardinalities, and displayed parameters are integers in the stated domains; gaps may be signed on other graphs. No excess floor beyond $E \ge 1$ is assumed.

Proof. Factor-criticality gives $|V(F)| = 2k + 1$. Indeed, deletion of a vertex leaves a perfect matching with $(|V(F)| - 1)/2$ edges, and an odd-order graph has no larger matching. The slack nonnegativity in Lemma 61 and $gap(F) = 1$ give $E = ps(F) + 1 \ge 1$.

For any vertex v, a perfect matching of F-v covers every vertex of degree d in F-v. Lemma 61 gives $c_d(F - v) = 0$. The extension inequality in Lemma 72 gives $c_d(F) \le 1$. If $c_d(F) = 0$, the capacity pair (1,d-1) at budgets (0,0) would partition $E(F)$ into a matching and a graph of maximum degree at most d-1. Since F is d-regular, each vertex would meet that matching. It would be a perfect matching of the odd-order graph F, which is impossible. Thus $c_d(F) = 1$. Lemma 61 makes all rows of F above d zero.

Fix a nonempty independent I, and write $j = |I|$ and $J = F - I$. Choose v in I. Since J is a subgraph of F-v, restriction in Lemma 72 gives $c_d(J) = 0$. All rows of J above d vanish as well. For integers $0 \le i < d$ define

$a_i = j - [c_i(F) - c_i(J)]$.

Restriction and extension in Lemma 72 give $0 \le a_i \le j$. Since I is independent and F is d-regular, deleting I removes exactly dj edges. Subtracting the finite sums defining gap, including the decrease of one in row d, yields

$gap(J) = gap(F) - \sum_{i \ge 0}[c_i(F) - c_i(J)] + dj$

$= 1 - [dj - \sum_{i=0}^{d-1} a_i + 1] + dj$

$= \sum_{i=0}^{d-1} a_i$.

The proper induced graph J satisfies T_h. Hence this nonnegative integer sum is either zero or one.

Suppose some a_i is positive with $1 \le i < d$. The sum must then be one, $a_0 = 0$, and $gap(J) = 1$. In particular $c_0(J) = c_0(F) - j$. Slack nonnegativity gives $ex(J) \ge 1$. Therefore the integer $t = c_0(J) - 1$ is nonnegative and satisfies $t \ge \nu(J)$. Assigning all edges of J to either one part shows that both 0 and t belong to $Y_t(J)$.

We compare this top allocation set to that of F using the actual deleted vertices. If x belongs to $Y_t(J)$ and r is an integer in [0,j], choose an r-element subset I_R of I. Retain the given edge partition of J, put every edge meeting I_R in its first part, and put every remaining edge meeting I in its second part. The added first part has vertex cover I_R and matching number at most r. The added second part has vertex cover I minus I_R and matching number at most j-r. Matching numbers of unions are at most the sums of their matching numbers, since a matching partitions into matchings on the two edge sets. Thus $x + r$ belongs to $Y_{t+j}(F)$. Equivalently,

$Y_t(J) + [0, j]$ is contained in $Y_{t+j}(F)$,

where the sum of integer sets consists of sums of their members. This is also the concrete extension of Lemma 72.

Here $t + j = c_0(F) - 1$. List the distinct members of $Y_t(J)$ in increasing order. The endpoints are 0 and t. The sum with [0,j] is the union of the intervals $[y, y + j]$ over these members y. A run of r missing integers between consecutive members becomes a run of $\max(0, r - j)$ missing integers between the corresponding intervals. There are no endpoint gaps. If $Y_t(J)$ has one member, then $t = 0$ and its sum with [0,j] is the whole interval $[0, t + j]$, so the same bound holds. Consequently the longest missing run in the sum is max(0,tr(J)-j). The displayed containment implies

$tr(F) \le \max(0, tr(J) - j)$.

The definition in Lemma 23 gives $tr(F) \ge 1$, since $c_0(F) = k + E > 0$. Therefore $tr(J) \ge tr(F) + j$.

Using the zero-row equality, we also have

$ex(J) = ex(F) - j + \nu(F) - \nu(J)$.

Restriction of matchings gives $\nu(J) \le \nu(F)$. Combining the last two relations and then the hypotheses on F gives

$tr(J) + ex(J) \ge tr(F) + ex(F) + \nu(F) - \nu(J)$

$\ge tr(F) + ex(F) > h(d)$.

On the other hand, J satisfies T_h and has $gap(J) = 1$, so

$tr(J) + ex(J) \le h(\Delta(J)) \le h(d)$,

because h is nondecreasing and $\Delta(J) \le d$. This contradiction proves $a_i = 0$ for every integer $1 \le i < d$.

We now have all the positive-row identities. The gap subtraction above gives $z(I) = a_0$ in {0,1}, so $c_0(J) = c_0(F) - j + z(I)$. Writing $\nu(J) = k - j + b(I)$ in the excess identity gives $ex(J) = E - b(I) + z(I)$, and subtracting $gap(J) = z(I)$ gives $ps(J) = E - b(I)$. Hence

$j + \nu(J) + ps(J) = j + (k - j + b(I)) + (E - b(I)) = c_0(F)$.

To bound $b(I)$, restrict a perfect matching of F-v to J. Each of the j-1 vertices in I minus {v} is matched to a vertex outside I, because I is independent. They meet distinct matching edges. Exactly j-1 of the k matching edges are deleted, leaving $k - j + 1$ edges in J. This proves $b(I) \ge 1$. The order bound $\nu(J) \le floor((2k + 1 - j)/2)$ gives $b(I) \le \lceil j/2 \rceil$, and $ps(J) \ge 0$ gives $b(I) \le E$.

Finally let I contained in I′ be nonempty independent sets and put $q = |I'| - |I|$. The zero-row extension inequality gives

$c_0(F - I) - c_0(F - I') \le q$.

Applying the two zero-row formulas already proved yields

$z(I') - z(I) = q - [c_0(F - I) - c_0(F - I')] \ge 0$.

This proves the monotonicity and the stated zero-gap specialization. □

Remark. All possible failures of the independent-cut row identities fit in a single unit of gap. A failure in a positive row forces exact restoration in row zero, whose missing intervals then contract too much to remain a minimal threshold failure.

Depends on. Lemma 23, Lemma 61, Lemma 72.

Lemma 116 (Matching-assisted deletion in graphs of odd order). *Use the integer certificate costs $c_{A,B}(G)$ and row maxima $c_i(G)$ from Lemma 23. A closed neighborhood of a vertex set consists of that set and all vertices adjacent to it. For every integer $t \ge 2$ and every finite simple graph Q of odd order $2\mu + 1$, where $\mu \ge 0$ is an integer, there exists a vertex set U with $|U| \le \max(0, \mu - t)$ whose closed neighborhood contains every vertex of Q of degree at least $2t + 1$.*

Consequently let $D \ge 6$ and $k \ge 0$ be integers, let K be any finite simple graph on $2k + 1$ vertices with maximum degree at most D, let v be a vertex with $deg_K(v) \le D - 1$, and supply any perfect matching M of K-v. Put $a = floor((D - 2)/2)$. There is a vertex set U with $|U| \le \max(0, k - a)$ such that K-U has maximum degree at most D-1 and the surviving edges of the same M cover every vertex of K-U of degree D-1. In particular $c_{D-1}(K) \le \max(0, k - a)$.

The set U and that residual covering matching can be fixed before choosing any nonnegative integer capacities $A + B = D - 1$ or matching budgets $x + y = |U|$. For every such pair of pairs, there is a certificate of K whose two bounded-degree parts partition $E(K - U)$, and whose exceptional parts partition all edges of K meeting U, including edges inside U. Empty U and empty residuals are allowed.

Proof. Fix $t \ge 2$ and Q of odd order $n = 2\mu + 1$, and let S be its vertices of degree at least $2t + 1$. Put $b = \mu - t$, so $n = 2t + 2b + 1$. If $b \le 0$, then every degree is at most $n - 1 \le 2t$ and S is empty; the empty set works. If S is empty for any other parameters the same choice works. We may therefore assume $b \ge 1$ and S nonempty. Every probability space below is finite; existence follows by averaging over that finite space.

First suppose $b = 1$, so $n = 2t + 3$ and the degree threshold is n-2. If Q has a vertex adjacent to every other vertex, its singleton dominates S. Otherwise every vertex of S has degree exactly n-2 and hence has exactly one neighbor in the complement graph. If some s in S has that complement-neighbor outside S, then s is

adjacent in Q to every other member of S and {s} dominates S. If this does not occur, each complement-neighbor of a member of S also belongs to S. Thus the complement induced by S is a disjoint union of edges and $|S|$ is even. Since n is odd, a vertex z outside S exists. There is no complement edge from S to its outside, so z is adjacent in Q to every member of S. The singleton {z} works. This proves $b = 1$.

Next suppose $2 \le b \le t$. Choose a b-element vertex set U uniformly. For any vertex s in S, at most $n - (2t + 2) = 2b - 1$ vertices lie outside its closed neighborhood. Thus the expected number of members of S not dominated by U is at most $n\ binom(2b - 1, b)/binom(n, b)$. Here and below $binom(N, j)$ counts j-element subsets of an N-element set. We have $n \ge 4b + 1$. For fixed $b \ge 2$, the quantity $n/binom(n, b) = b/binom(n - 1, b - 1)$ is nonincreasing in n. The preceding expectation is therefore at most $(4b + 1)\ binom(2b - 1, b)/binom(4b + 1, b)$. For $b = 2, 3, 4$ these values are respectively $3/4$, $5/11$, $1/4$. For $b \ge 5$, write the binomial ratio as the product, for $0 \le j < b$, of $(2b - 1 - j)/(4b + 1 - j)$. Each factor is less than $1/2$, since $4b - 2 - 2j < 4b + 1 - j$. Hence the expectation is less than $(4b + 1)/2^b$. This is at most $21/32$ for $b \ge 5$: at $b = 5$ it equals $21/32$ and its successive ratio is $(4b + 5)/(2(4b + 1)) < 1$. All these expectations are below one. A nonnegative integer random variable with expectation below one has an outcome zero, so a suitable U of size b exists.

It remains to consider $b \ge t + 1$, so $n \ge 4t + 3$. Select every vertex independently with probability p, call the selected set U_0, and let W consist of the members of S outside the closed neighborhood of U_0. The set $U = U_0$ union W dominates S. The two parts are disjoint and each s in S has closed-neighborhood size at least $2t + 2$, giving $E|U| \le n[p + (1 - p)^{2t+2}]$. An expectation strictly below the integer $b + 1$ guarantees an outcome with $|U| \le b$.

For $t \ge 4$ take $p = 1/4$ and write $z_t = (3/4)^{2t+2}$. The inequality $n(1/4 + z_t) < (n - 2t + 1)/2 = b + 1$ is equivalent to $(1/2 - 2z_t)n > 2t - 1$. The coefficient on n is positive because $z_t \le z_4 < 1/16$; the last strict inequality follows from $59049 < 65536$. At the smallest allowed $n = 4t + 3$ the difference between the two sides is $5/2 - 2(4t + 3)z_t$. The sequence $(4t + 3)z_t$ decreases for $t \ge 4$ because its successive ratio is $9(4t + 7)/(16(4t + 3)) < 1$. At $t = 4$ it is $19z_4 < 19/16 < 5/4$. Thus the difference is positive at the smallest n and stays positive as n grows.

For $t = 3$ the remaining odd orders start with $n = 15$. At $n = 15$ one has $b = 4$; a uniformly chosen four-element set leaves an expected number of undominated required vertices at most $15\ binom(7, 4)/binom(15, 4) = 5/13 < 1$, so one such set works. For $n \ge 17$ choose $p = 1/4$. The coefficient is $1/4 + (3/4)^8 = 22945/65536 < 1/2$. At $n = 17$ its product is $390065/65536 < 6 = (17 - 5)/2$. Since the coefficient is less than $1/2$, the inequality $22945n/65536 < (n - 5)/2 = b + 1$ holds for every larger n.

For $t = 2$ the remaining small orders are $n = 11, 13, 15, 17, 19$ with $b = (n - 5)/2$. A uniform b-element set leaves an expected number of undominated required vertices at most $n\ binom(n - 6, b)/binom(n, b)$. At those five orders the ratios are respectively $2/3$, $7/11$, $90/143$, $33/52$, $11/17$, each less than one. For $n \ge 21$ choose $p = 1/3$. The expectation of the completed set is at most $307n/729$, since $1/3 + (2/3)^6 = 307/729$. The desired strict inequality $307n/729 < (n - 3)/2 = b + 1$ is equivalent to $115n > 2187$, which holds at $n = 21$ and for larger n. This exhausts every permitted t and odd order, proving the domination assertion.

For the matching-assisted construction, retain every vertex of K and remove the edges of the supplied matching M, obtaining a spanning graph F. Every vertex other than v loses exactly one incident edge, and v already has degree at most D-1. Therefore $\Delta(F) \le D - 1$. Put $a = floor((D - 2)/2) \ge 2$. Apply the domination assertion to F with $t = a$ and $\mu = k$. It gives U of size at most $\max(0, k - a)$ dominating every vertex of F of degree at least $2a + 1$. If D is even then $2a + 1 = D - 1$; if D is odd then $2a + 1 = D - 2$. In either case every vertex of F-degree D-1 is dominated. Each such vertex retained outside U has a neighbor in U and loses that edge. Vertices of smaller F-degree cannot acquire edges on deletion. Hence $\Delta(F - U) \le D - 2$.

The graph K-U is the union of F-U and the edges of M with both endpoints outside U. Its maximum degree is therefore at most D-1. Every vertex attaining D-1 must have a surviving M-edge, since its degree in F-U is at most D-2. These surviving edges form a matching and cover every degree-(D-1) vertex. This proves the construction with one fixed U and the specified retained matching.

By Lemma 35, for every nonnegative integer $A + B = D - 1$ the same residual K-U has an edge partition with maximum-degree bounds A, B. With empty exceptional parts this is a zero-total certificate in the definitions of Lemma 23. Given any nonnegative integers $x + y = |U|$, choose an x-element subset U_R of U. Put every original edge meeting U_R into the red exceptional part and every remaining original edge meeting U into the blue exceptional part. The latter edges all meet U minus U_R. Matchings in these two sets have cardinalities at most x and y because their distinct edges meet distinct vertices of the respective covering

sets. All edges meeting U, including those with both endpoints in U, are assigned exactly once. Equivalently this is the actual vertex-set extension in Lemma 72. The construction supplies every allocation at total $|U|$, so the threshold property gives $c_{D-1}(K) \le |U| \le \max(0, k-a)$. The choices of U and its residual covering matching precede both pairs of capacities and budgets, as asserted. □

Remark. Dominate only the high-degree vertices after removing the prescribed matching. Restoring its surviving edges raises degrees by at most one and certifies that every vertex reaching the new degree threshold is covered.

Depends on. Lemma 23, Lemma 35, Lemma 72.

Lemma 117 (Completing a dominating set by pairing uncovered vertices). *Let $\delta \ge 2$ be an integer, and let F be a finite simple graph on $n \ge 0$ vertices such that every vertex has degree at least δ. For w in $V(F)$, let $N_F[w]$ be its closed neighborhood, and for a set Z let $N_F[Z]$ be the union of these closed neighborhoods. A set D dominates F if $N_F[D] = V(F)$; all graph vertices may be chosen. Fix any set Z, put $W = V(F)$ minus $N_F[Z]$, $m = |W|$ and $N = n - |Z|$, and suppose $|N_F[x]$ intersect $W| \le 2$ for every vertex x outside Z. Then there is a set Y contained in $V(F)$ minus Z that dominates W and satisfies $|Y| \le \min(m, floor((\delta \cdot m + 2N)/(3\delta + 2)))$. More generally, for every set A of cardinality a and every nonnegative integer m_0 with $|V(F)$ minus $N_F[A]| \le m_0$, there exists a dominating set D containing this same A such that $|D| \le floor((3\delta \cdot a + \delta \cdot m_0 + 2n)/(3\delta + 2))$. All orders, degrees, cardinalities, and parameters δ, n, m, N, a, m_0 are integers in their indicated domains. Empty graphs, empty prefix sets, and an already dominating prefix are included.*

Proof. First assume the additional hypothesis on Z. If W is empty, take Y empty. Suppose W is nonempty. For each w in W, its closed neighborhood avoids Z, since otherwise w would belong to $N_F[Z]$. Also F[W] has maximum degree at most one: if w had two neighbors in W, then $|N_F[w]$ intersect $W|$ would be at least three, contrary to the hypothesis applied to w, which lies outside Z.

Define a finite simple auxiliary graph L with vertex set W, joining distinct w, w' if $N_F[w]$ intersect $N_F[w']$ is nonempty. Apply Lemma 46 to L with every vertex prescribed. It gives a vertex set X such that, writing $b = |X|$ and q for the number of odd components of L-X,

$\nu(L) = (m + b - q)/2$. (1)

This is an attained separator, not an assumed extremum. Let l be the number of components of L-X with exactly one vertex.

For every component C of L-X put $A_C = \bigcup_(w \in V(C))\ N_F[w]$, and write $c = |V(C)|$. The sets A_C for distinct components are disjoint: a common member would belong to the two closed neighborhoods of vertices in different components and would join them by an L-edge. Every A_C avoids Z. Count incidences (y, w) with w in $V(C)$ and y in $N_F[w]$. Each w contributes at least $\delta + 1$ incidences. Each y in A_C contributes at most two, because membership y in $N_F[w]$ is equivalent to w in $N_F[y]$, and the hypothesis applies to y outside Z. Thus $2|A_C| \ge (\delta + 1)c$.

If c is odd, F[V(C)] has maximum degree at most one and an odd number of vertices. Its edges cover an even number of vertices, so some vertex w_0 has no neighbor in $V(C)$. It belongs to $N_F[w]$ for w in $V(C)$ only when $w = w_0$. Thus at least one element of A_C contributes just one incidence, and $2|A_C| - 1 \ge (\delta + 1)c$. If $c = 1$, the stronger direct bound $|A_C| \ge \delta + 1$ holds. Summing these three bounds over the disjoint component sets gives

$N \ge (\delta + 1)(m - b)/2 + q/2 + \delta \cdot l/2$. (2)

Here the first term accounts for all components, the second adds one half for every odd component, and the last corrects the singleton bound from $(\delta + 2)/2$ to $\delta + 1$.

Every odd component other than a singleton has at least three vertices, so $m - b \ge l + 3(q - l)$. Hence $l \ge (3q - m + b)/2$. Substitution in (2) yields

$N \ge (\delta + 2)(m - b)/4 + (3\delta + 2)q/4$.

By (1), $q = m + b - 2\nu(L)$, so

$N \ge (\delta + 1)m + \delta \cdot b/2 - (3\delta + 2)\nu(L)/2$

$\geq (\delta+1)m-(3\delta+2)\nu(L)/2$.

Rearranging with the positive denominator $3\delta+2$ gives

$m-\nu(L)\leq(\delta\cdot m+2N)/(3\delta+2)$. (3)

Choose a maximum matching of L. For every matched pair w, w′ choose one element of $N_F[w]$ intersect $N_F[w']$; such an element exists by the definition of L. For every unmatched vertex w choose w itself. Let Y be the set of chosen vertices, deleting repetitions if they occur. All chosen vertices lie outside Z. Every w in W is in the closed neighborhood of its chosen representative, so Y dominates W. There are at most $\nu(L)+(m-2\nu(L))=m-\nu(L)$ chosen vertices. This is at most m, and (3) together with integral cardinality gives the asserted floor bound.

For the general assertion, start with the specified A and repeatedly adjoin a vertex outside the current selected set whose closed neighborhood contains at least three currently undominated vertices, whenever one exists. Every step adds a new selected vertex and reduces the undominated count by at least three. The process is finite and stops after t steps, with a selected set Z of size $a+t$ and an undominated set W of size $m\leq m_0-3t$. In particular $t\leq m_0/3$. At termination every vertex outside Z covers at most two vertices of W, so the first assertion supplies Y outside Z dominating W, of size at most $(\delta\cdot m+2(n-a-t))/(3\delta+2)$. The set $D=Z$ union Y dominates F and contains the original A. Its size is at most

$a+t+(\delta\cdot(m_0-3t)+2(n-a-t))/(3\delta+2)$

$=(3\delta\cdot a+\delta\cdot m_0+2n)/(3\delta+2)$.

The t terms cancel exactly. Taking the floor proves the result for the actual integer cardinality of D. If the initial set already dominates, the process has $t=0$ and Y empty, and the same bound holds because $n\geq a$ and $m_0\geq 0$. The empty graph has $A=Z=W=D$ empty and satisfies every assertion. □

Remark. When no vertex covers three remaining vertices, a matching pairs vertices that can share a dominator. A matching-deficiency separator forces disjoint neighborhood unions, and the extra incidence saving in each odd component bounds the number of unpaired vertices.

Depends on. Lemma 46.

Lemma 118 (Terminal absorption and exact covered-process accounting). *Use the finite integer certificate costs $c_i(G)$, matching number $\nu(G)$, independence number $\alpha(G)$, edge count $e(G)$, maximum degree $\Delta(G)$, and signed slack $ps(G)=e(G)-\nu(G)-\sum_{i\geq1}c_i(G)$ of Lemma 23. All graphs are finite and simple. Vertex deletion is induced. All orders, degrees, indices, cardinalities, costs, matching budgets and process lengths are nonnegative integers.*

Let S be a maximum independent set of F, put $a=|S|=\alpha(F)$ and $H_0=F-S$, and assume

$c_0(F)=a+\nu(H_0)+ps(H_0)$.

Fix any integer $t\geq0$ and complete process $H_0, H_1,\ldots, H_t=R$ in which, for every integer $1\leq j\leq t$, a vertex v_j of current maximum degree $d_j=\Delta(H_{j-1})\geq3$ is deleted, and $\Delta(R)\leq2$. Put

$\mu_j=\nu(H_{j-1})-\nu(H_j)$,

$A_{j,i}=c_i(H_{j-1})-c_i(H_j)$ *for* $1\leq i\leq d_j$,

$Z=\sum_{j=1}^{t}\sum_{i=1}^{d_j-1}(1-A_{j,i})$,

$Q=\sum_{j=1}^{t}A_{j,d_j}$,

$W_j=S$ *union* $\{v_1,\ldots,v_j\}$, $p=ps(H_0)$, $P=ps(R)$.

The quantities μ_j and $A_{j,i}$ belong to {0,1}, and p, P, Z, Q are nonnegative integers. Then every nonnegative integer allocation $x+y=a+t+\nu(R)$ admits a partition of all edges of F into two parts of matching numbers at most x, y. Moreover

$c_0(F)=a+t+\nu(R)+P+Z-Q$,

$Z+P\leq Q$.

If, additionally, every deletion parent has a matching covering all its vertices of degree d_j, then $Q = Z = P = 0$. Every row of index at least d_j is zero on both H_{j-1} and H_j, and $A_{j,i} = 1$ for every $1 \le i < d_j$. For every integer $0 \le j \le t$ and every integer $i \ge 1$,

$c_i(H_j) = c_i(H_0) - |\{l : 1 \le l \le j, d_l > i\}|$,

$ps(H_j) = p - |\{l : 1 \le l \le j, \mu_l = 0\}|$,

$c_0(F) = |W_j| + \nu(H_j) + ps(H_j)$.

Exactly p of the t deletions preserve matching number, and $\nu(R) = c_0(F) - a - t$. The covering matching may depend on the parent. No zero-initial-slack assumption, graph regularity, source minimality, or existence of a covered process is imposed. The result applies to every specified process meeting the hypotheses; empty graphs, empty sums, and $t = 0$ are allowed.

Proof. Use Lemma 23 for the certificate meaning and finite integer thresholds, and Lemma 61 for nonnegative slack, the strong cost tail, the covered-degree zero cost, and the exact local slack identity. These are the only mathematical cited results.

First construct the asserted row-zero certificates on the entire graph. Put $W = W_t$ and $w = |W| = a + t$. Every whole connected component C of R is a path or a cycle, with isolated vertices included as paths of order one. To see this classification, a component with an endpoint can be traversed as a path until the other endpoint; an additional vertex attached to it would violate the degree bound or extend it. A component with all degrees two is a cycle, since following successive edges must return and no further vertex can attach. An isolated vertex is the remaining possibility.

For a path on m vertices, consecutive disjoint pairs show that its matching number is $\lfloor m/2 \rfloor$, and alternating vertices give an independent set of size $\lceil m/2 \rceil$. For a cycle on m vertices, consecutive pairs after omitting a vertex when necessary give matching number $\lfloor m/2 \rfloor$; alternating vertices give an independent set of that size. Thus in every component,

$\nu(C) \le \alpha(C) \le \alpha(F) = a \le w$.

The middle inequality holds because C is induced in F. Matching numbers add across whole components: every matching restricts to each, and the union of maximum component matchings attains their sum. Write the component matching numbers as m_1,..., m_b, where b is the number of components and each m_l lies in [0,w]. In particular $\sum_l m_l = \nu(R)$.

We claim that every integer x in $[0, w + \sum_l m_l]$ has a representation

$x = z + \sum_{l \in B} m_l$

with an integer $0 \le z \le w$ and a subset B of {1,...,b}. Start with all integers in [0,w]. If the first q components have produced every integer in [0,L], adjoining the choice to add m_{q+1} or zero produces [0,L] union $[m_{q+1}, L + m_{q+1}]$. Since $m_{q+1} \le w \le L$, these integer intervals cover $[0, L + m_{q+1}]$. Finite induction proves the representation, including zero weights and an empty component list.

Given nonnegative $x + y = w + \nu(R)$, choose such a representation. Split the actual vertex set W into W_R, W_B of cardinalities z, w-z. Put every edge of R-components indexed by B in a red edge part and all other R-component edges in a blue edge part. An edge of F not internal to R has an endpoint in W, because $R = F - W$ is induced. Assign each such edge red when it meets W_R, and blue otherwise. Every remaining blue edge meets W_B. This gives every edge exactly one color, including edges with both endpoints in W. A matching in the red part has at most $\sum_{l \in B} m_l$ edges internal to R and at most z other edges, since those other matching edges use distinct W_R vertices. Thus its size is at most x. The blue part is bounded by $\sum_{l not \in B} m_l + w - z = y$ in the same way. The two bounded-degree parts at capacities (0,0) are empty. We have proved the universal certificate assertion and therefore

$c_0(F) \le w + \nu(R)$.

Next compute the exact slack identity without a covering assumption. At step j, put

$Z_j = \sum_{i=1}^{d_j - 1} (1 - A_{j,i})$, $Q_j = A_{j,d_j}$.

The local statement of Lemma 61 gives μ_j, $A_{j,i}$ in {0,1}; it also gives

$ps(H_{j-1}) - ps(H_j)$

$= d_j - \mu_j - \sum_{i=1}^{d_j} A_{j,i}$

$= 1 - \mu_j + Z_j - Q_j.$

The nonnegative quantities Z_j and Q_j sum to Z and Q. Summing the identities and using $\sum_j \mu_j = \nu(H_0) - \nu(R)$ gives

$\nu(H_0) + ps(H_0) = t + \nu(R) + P + Z - Q.$

Insert the assumed initial identity to obtain

$c_0(F) = a + t + \nu(R) + P + Z - Q.$

The certificate upper bound just proved now gives $Z + P \le Q$. Nonnegative slack from Lemma 61 gives p, $P \ge 0$. Every sum here is finite, and integrality follows from the finite cost definitions.

Now impose the displayed covering hypothesis for every parent. The covered-degree zero-cost conclusion of Lemma 61 gives $c_{d_j}(H_{j-1}) = 0$. Its restriction property gives that same row zero on H_j. The strong tail makes every row above d_j zero on both graphs. Consequently $Q_j = 0$ for every j, so $Q = 0$. Since Z, P are nonnegative and $Z + P \le Q$, both vanish. Each summand $1 - A_{j,i}$ of Z is nonnegative; hence every such summand is zero. This proves all unit lower-row losses.

For a fixed positive index i, step j changes c_i by one when $i < d_j$ and by zero when $i \ge d_j$. Summing these changes through any prefix proves the stated $c_i(H_j)$ formula. With $Z_j = Q_j = 0$, the local slack identity reduces to $ps(H_{j-1}) - ps(H_j) = 1 - \mu_j$. Each μ_j is zero or one, so summing gives the displayed count of matching-preserving deletions. At the terminal $P = 0$, this count equals p.

Finally, at each covered step the matching-number loss plus the slack loss is $\mu_j + (1 - \mu_j) = 1$, whereas $|W_j|$ increases by one. The quantity $|W_j| + \nu(H_j) + ps(H_j)$ is therefore unchanged. At $j = 0$ it equals $c_0(F)$ by hypothesis, proving every prefix identity. At $j = t$, use $ps(R) = 0$ and $|W_t| = a + t$ to obtain $\nu(R) = c_0(F) - a - t$. With $t = 0$ all local statements are vacuous, and the initial identity together with the terminal certificate bound forces $p = P = 0$ as claimed. The proof asserts no existence of a covering matching unless that matching is one of the supplied process hypotheses. □

Remark. Whole terminal components can each be colored entirely red or blue; the actually deleted vertices supply a long enough integer budget interval to fill every gap. Comparing this concrete certificate with the local slack ledger forces exactness whenever every parent is covered.

Depends on. Lemma 23, Lemma 61.

Lemma 119 (Optimal configurations for complementary certificate levels). *Use the finite integer certificate costs $c_\theta(H)$ of Lemma 23. Let $d \ge 2$ be an even integer, let H be a nonempty connected d-regular finite simple graph such that H-v has a perfect matching for every vertex v, and let $\theta_1, \theta_2 \ge 1$ be integers with $\theta_1 + \theta_2 = d$. Write $n = |V(H)|$. A configuration consists of a near-perfect matching M, whose unique uncovered vertex is x, and an ordered partition (V_1, V_2) of $V(H)$, with empty parts allowed. Its value is $\Phi(M, V_1, V_2) = e((H - M)[V_1])/\theta_1 + e((H - M)[V_2])/\theta_2$. Here H-M retains all vertices and deletes precisely the matching edges. Call a configuration optimal if it minimizes Φ over all configurations and balanced if x has exactly θ_i neighbors in V_i for each $i = 1, 2$. Optimal configurations exist, and $c_{\theta_1}(H) + c_{\theta_2}(H) \le n + 1$. If an optimal configuration is not balanced, this sum is at most n. Suppose the sum equals $n + 1$. Then every optimal configuration is balanced, all its matching edges are internal to its two parts, and every vertex is uncovered in some optimal configuration. For any optimum, let V_s contain x and let V_o be the other part, with thresholds t and t', respectively. Put $J = H[V_s]$, and define $D_J = \{u \in V_s : \nu(J - u) = \nu(J)\}$, where ν is maximum matching cardinality. Then $|V_s|$ is odd, $|V_o|$ is even, $\Delta(J) \le t$, and $c_t(H) = n - |V_s| + 1$, $c_{t'}(H) = |V_s|$. Every u in D_J has degree exactly t in J. In particular, both complementary costs are odd integers. Consequently if integers U_1, U_2 satisfy $c_{\theta_i}(H) \le U_i$ for $i = 1, 2$ and $U_1 + U_2 = n + 1$ with U_1 even, then $c_{\theta_1}(H) + c_{\theta_2}(H) \le n$. All graph orders, degrees, thresholds, costs, capacities, budgets, matching sizes and cardinalities are integers in their indicated positive or nonnegative domains.*

Proof. Deleting any vertex of H leaves a perfect matching, so n is odd and near-perfect matchings exist. There are finitely many matchings and ordered partitions and at least one configuration. Each denominator in Φ is positive. Hence the finite family has an optimum.

If W is a vertex subset, $\Delta(H[W]) \leq a$ for an integer $a \geq 1$, and a matching in H[W] covers every vertex of degree a, then Lemma 35 provides every degree splitting of total a. Empty exceptional parts give $c_a(H[W]) = 0$ by Lemma 23. Vertex extension Lemma 72 therefore gives $c_a(H) \leq n - |W|$.

Fix an optimal configuration, put $F = H - M$, and for a vertex in a part of threshold a let b be the other threshold, own its number of F-neighbors in its own part, and cross its number in the other part. Moving this vertex changes Φ by $-own/a + cross/b$, which must be nonnegative. If the vertex differs from x, then $own + cross = d - 1 = a + b - 1$. Therefore $own \geq a$ would give $cross \leq b - 1$ and a strictly negative change. Thus $own \leq a - 1$. At x the degree is d, and $own \geq a + 1$ would give $cross \leq b - 1$, also a negative change. Hence $own \leq a$. If equality holds, $cross = b$ and the configuration is balanced.

Let x belong to V_s, with the notation of the statement. In $H[V_o]$, restoring the matching edges raises each vertex degree by at most one. Every vertex attaining degree t′ must have its matching partner inside V_o, so the restricted matching covers all such vertices. The same statement with threshold t holds in $H[V_s] - x$: its vertices had internal F-degree at most t-1, and restoration of an internal matching edge is the only way to reach t. The preceding induced-subgraph bound gives

$c_{t'}(H) \leq |V_s|, c_t(H) \leq |V_o| + 1.$ (1)

This proves the universal sum bound $n + 1$. If the optimum is unbalanced, x has internal F-degree at most t-1. Since x has no matching edge, it also has $H[V_s] - degree$ at most t-1. The restricted matching now covers all degree-t vertices of the whole $H[V_s]$. Thus $c_t(H) \leq |V_o|$, and the sum in (1) improves to n.

Suppose from now on that the sum equals $n + 1$. Every optimum is balanced by the preceding paragraph. Moving the uncovered vertex to the other part preserves Φ: its t edges to its old part and t′ edges to its other part change the value by $-t/t + t'/t' = 0$. The resulting configuration is again optimal.

There is a second permitted change. Suppose y is an H-neighbor of x in the same part, and z is the M-partner of y. Replace yz in M by xy. The new matching is near-perfect and uncovers z. In H-M the operation removes the internal edge xy and restores yz. If yz crosses the partition, Φ falls by $1/t$, contradicting optimality. Hence yz is internal to the same part, the value is unchanged, and the new configuration is optimal.

Fix any original optimum (M, P) uncovering v, and any u different from v. Choose a perfect matching M_u of H-u. In M symmetric-difference M_u, only v and u have degree one, all other degrees are zero or two, and the edges alternate between the two matchings. The component starting at v is therefore a simple even-length path

$x_0 = v, x_1, \ldots, x_{2m} = u$

for an integer $m \geq 1$, with edges $x_{2i}x_{2i+1}$ in M_u and $x_{2i+1}x_{2i+2}$ in M for $0 \leq i < m$.

Carry the uncovered vertex along this path. Before step i, the current matching has reversed membership on its first 2i edges, uncovers x_{2i}, and its optimal partition differs from P only at some of $x_0, x_2, \ldots, x_{2i-2}$. This is true initially. If x_{2i+1} is in the other part, move the current uncovered vertex across by the first permitted change. Now apply the second change to replace $x_{2i+1}x_{2i+2}$ by $x_{2i}x_{2i+1}$. The next configuration is optimal and uncovers x_{2i+2}, establishing the invariant. The removed original matching edge is internal in the current partition. Neither of its endpoints has ever moved, because the path is simple and only previous even-indexed vertices have moved. Thus that edge was internal in P. At the final step it is the original M-edge incident with u. Since u was arbitrary outside v, all original M-edges are internal. The final configuration also shows that each u can be uncovered by an optimum. The starting optimum was arbitrary, proving both universal assertions.

For every optimum, its internal matching covers all of V_o and all of V_s except x. Hence $|V_o|$ is even and $|V_s|$ is odd. The earlier internal-degree bound shows $\Delta(H[V_s]) \leq t$. The two costs in (1) have sum $n + 1$, so both inequalities there are equalities. Their right sides $|V_o| + 1$ and $|V_s|$ are odd, proving the asserted parity of both costs.

Finally $J = H[V_s]$ has a matching missing just x. Its odd order implies $\nu(J) = (|V_s| - 1)/2$. For u in D_J, a maximum matching N of J-u has this same cardinality, so it covers every vertex of J except u. Combine N with M restricted to V_o. This is a near-perfect matching of H exposing u, all of whose edges are internal to the unchanged partition. The number of removed matching edges in each part is the same as for M, so the value

Φ is unchanged. The configuration is optimal and therefore balanced. Its uncovered vertex u has exactly t neighbors in V_s, proving $deg_J(u) = t$.

For the final consequence, the universal sum bound and integrality leave only the values at most n or equal to $n+1$. Equality, together with the supplied upper bounds whose sum is $n+1$, would force each cost to equal its U_i. The first cost would then be even, contradicting the parity just proved. Thus the sum is at most n. □

Remark. A minimum weighted partition supplies two covered residual graphs. Tightness forces every exposed vertex to balance the partition, allowing alternating-path switches that make all matching edges internal and force both exact costs to be odd.

Depends on. Lemma 23, Lemma 35, Lemma 72.

Lemma 120 (Exact row losses in a covered zero-slack deletion process). *Use the integer certificate costs $c_i(J)$, matching number $\nu(J)$, independence number $\alpha(J)$, edge count $e(J)$, and slack $ps(J) = e(J) - \nu(J) - \sum_{i\geq 1} c_i(J)$ of Lemma 23. All graphs are finite and simple. All orders, degrees, row indices, matching and independence cardinalities, and process lengths are nonnegative integers.*

Let H have $ps(H) = 0$. Fix an integer $t \geq 0$ and a sequence $H_0 = H$, H_1,..., $H_t = R$ of induced graphs. For each integer $1 \leq j \leq t$, let v_j be a vertex of H_{j-1}, put $d_j = \Delta(H_{j-1}) \geq 3$, require $d_{H_{j-1}}(v_j) = d_j$, and set $H_j = H_{j-1} - v_j$. Assume that H_{j-1} has a matching covering every vertex of degree d_j, for each such j, and that $\Delta(R) \leq 2$. A covering matching may be chosen separately at each parent; no common matching is assumed.

Then for every integer $0 \leq j \leq t$,

$ps(H_j) = 0$,

$|V(H_j)| = |V(H)| - j$,

$e(H_j) = e(H) - \sum_{h=1}^{j} d_h$,

$\nu(H_j) = \nu(H) - j$,

and, for every integer $i \geq 1$,

$c_i(H_j) = c_i(H) - |\{h : 1 \leq h \leq j, d_h > i\}|$.

In particular $\nu(H_j) - c_1(H_j)$ and $\nu(H_j) - c_2(H_j)$ do not depend on j.

Put $b = \nu(H) - c_1(H)$ and $p = c_2(H) - t$. These are nonnegative integers. The terminal R has exactly b whole positive even-order path components and exactly p whole odd-cycle components. All its whole components belong to

P_1, P_2, P_3, P_4, P_6, C_3, C_4, C_5, C_7,

where P_a is the path on $a \geq 1$ vertices and C_a the cycle on $a \geq 3$ vertices. Moreover

$\alpha(R) = |V(H)| - \nu(H) - c_2(H) + t$,

$0 \leq t \leq min\{\nu(H), c_2(H), \alpha(H) - |V(H)| + \nu(H) + c_2(H)\}$.

Every asserted formula holds for any complete sequence satisfying the displayed covering hypothesis. Empty graphs, isolates and $t = 0$ are included.

Proof. The only permitted operation is deletion of a current maximum-degree vertex at a parent whose maximum-degree vertices admit a covering matching. We prove its exact effect first.

Let J be one such parent, of maximum degree $d \geq 3$, with $ps(J) = 0$, and let v have degree d. Put $J' = J - v$, $h = \nu(J) - \nu(J')$, and $a_i = c_i(J) - c_i(J')$ for integers $1 \leq i \leq d$. By Lemma 61, h and every a_i belong to {0,1}, $a_d = 0$, and

$ps(J) - ps(J') = d - h - \sum_{i=1}^{d} a_i \geq 1 - h$.

The same fact gives $ps(J') \geq 0$. The left side is therefore nonpositive, while 1-h is nonnegative. Both sides must be zero: $h = 1$ and $ps(J') = 0$. Substituting $h = 1$ and $a_d = 0$ into the identity gives

$\sum_{i=1}^{d-1} a_i = d - 1$.

There are d-1 integer summands, each at most one and at least zero. Hence $a_i = 1$ for every $1 \le i < d$. For every $i > d$, both graphs have zero row i by the tail bound in Lemma 61, since their maximum degrees are at most d. Their row-d costs are also zero: the parent's cost vanishes by coverage, and $a_d = 0$ gives the child's value. Thus deletion lowers exactly the positive rows whose indices are less than d, and it lowers the matching number by one.

Apply this conclusion successively to the given sequence. Initially $ps(H_0) = 0$. If $ps(H_{j-1}) = 0$, the supplied covering matching makes the preceding argument applicable and gives $ps(H_j) = 0$, matching loss one, and the specified row losses. Finite induction proves these assertions for every prefix. Summing the matching losses gives $\nu(H_j) = \nu(H) - j$. Summing the losses in a fixed row i gives precisely the count of indices $h \le j$ with $d_h > i$. Because every $d_h \ge 3$, the losses in rows one and two are both j, proving the two invariant differences. Each operation deletes exactly one vertex and d_h current edges. An edge removed during the process is counted exactly once, at its first removed endpoint. This proves the order and edge identities, including the empty prefix.

At the terminal, $\Delta(R) \le 2$ and $ps(R) = 0$. Lemma 57 gives the displayed complete component list and says that $c_2(R)$ is the number of its odd-cycle components. Since each of the t operations lowers row two by one, this count is $p = c_2(H) - t$. The same fact says $c_1(R) = \nu(R) - b_R$, where b_R is the number of positive even-order path components. The prefix identities give

$b_R = \nu(R) - c_1(R) = (\nu(H) - t) - (c_1(H) - t) = b$.

Both b and p are therefore nonnegative integers defined by actual component counts.

The terminal independence identity in Lemma 57 is $\alpha(R) = |V(R)| - \nu(R) - p$. Substitute $|V(R)| = |V(H)| - t$, $\nu(R) = \nu(H) - t$, and $p = c_2(H) - t$. This yields

$\alpha(R) = |V(H)| - \nu(H) - c_2(H) + t$.

Every independent set in the induced graph R is independent in H, so $\alpha(R) \le \alpha(H)$. Rearranging gives the third upper bound on t. Nonnegativity of $\nu(R)$ and p gives the first two upper bounds. The input already gives $t \ge 0$. All arguments apply when $t = 0$; then $R = H$ has maximum degree at most two and the covering premises form an empty list. Empty graphs and isolates have no edges and are covered by the same component formulas. □

Remark. At zero slack, the lower bound on a covered deletion's slack loss must be an equality. It forces one matching unit and every eligible positive row to be lost together; the degree-two terminal turns their differences into component counts.

Depends on. Lemma 23, Lemma 57, Lemma 61.

Lemma 121 (Connectedness and total criticality of minimal failures). *Use the certificate costs $c_i(G)$, matching number $\nu(G)$, maximum degree $\Delta(G)$, excess $ex(G)$, signed gap $gap(G)$, positive-level slack $ps(G)$, and top missing-allocation run $tr(G)$ of Lemma 23. In particular $c_0(G) = c_{0,0}(G)$, $ex(G) = c_0(G) - \nu(G)$, $gap(G) = \sum_{i \ge 0} c_i(G) - e(G) = ex(G) - ps(G)$, and $tr(G) = 0$ when $c_0(G) = 0$. All graphs are finite and simple, with maximum degree zero for an edgeless graph. All indices, capacities, cardinalities and matching numbers are nonnegative integers.*

Let h be any nondecreasing integer-valued function on the nonnegative integers. Say that G satisfies T_h if

$gap(G) > 0$ *implies* $[gap(G) = 1$ *and* $tr(G) + ex(G) \le h(\Delta(G))]$.

Let F violate T_h and suppose every proper induced subgraph of F satisfies T_h. A proper induced subgraph is $F[U]$ for a proper subset U of $V(F)$, including the empty subset.

Then F is connected, has an edge, and has $ex(F) \ge gap(F) \ge 1$. Write $D = \Delta(F) \ge 1$. For every vertex v of degree D and every integer i with $1 \le i \le D$,

$c_i(F - v) = c_i(F) - 1$.

Thus F is totally critical, meaning that deletion of each maximum-degree vertex strictly lowers every positive row through its maximum degree. No matching of F covers all vertices of degree D.

In particular these conclusions hold when F violates T_h while every proper subgraph satisfies T_h, where proper subgraphs allow arbitrary vertex deletions, edge deletions, or both, including spanning edge deletions. Taking $h(D) = \max(4, \lfloor D/2 \rfloor + 1)$ gives the same conclusions for every proper-subgraph-minimal T_- failure, with $T_- = T_h$ for this specified h. No regularity or factor-criticality conclusion is included here.

Proof. Since F violates T_h, it has positive integer gap. Lemma 61 gives $ps(F) \geq 0$, and Lemma 23 gives $gap(F) = ex(F) - ps(F)$, so $ex(F) \geq gap(F) \geq 1$. An edgeless graph has every certificate cost zero by the empty partition and hence gap zero. Thus F has an edge and $D = \Delta(F) \geq 1$.

Suppose F is disconnected. Among its components choose Z of largest excess, and among those choose one with largest top run. By Lemma 68 with X empty, the largest component excess $m = ex(Z)$ satisfies $m \geq ex(F) \geq 1$. By Lemma 72, $ps(F)$ is at least the sum of the component slacks, and each such slack is nonnegative by Lemma 61. Consequently $ps(F) \geq ps(Z)$ and

$$gap(Z) - gap(F) = (m - ex(F)) + (ps(F) - ps(Z)) \geq 0.$$

The component Z is a proper induced subgraph. If $gap(Z) \geq 2$, it violates T_h, contradicting the hypothesis. Otherwise the positive integer gaps of Z and F both equal one. Equality in the displayed formula forces $m = ex(F) = ex(Z)$, so the positive-equality assertion of Lemma 68 gives $tr(Z) \geq tr(F)$, by the choice of Z. Since F fails T_h at gap one, $tr(F) + ex(F) > h(D)$. Monotonicity of h and $\Delta(Z) \leq D$ give

$$tr(Z) + ex(Z) \geq tr(F) + ex(F) > h(D) \geq h(\Delta(Z)),$$

so Z again violates T_h. Both alternatives contradict the hypothesis, proving connectedness.

Fix a vertex v of degree D and put $J = F - v$. For each nonnegative integer i let $a_i = c_i(F) - c_i(J)$, and let $\epsilon = \nu(F) - \nu(J)$. The restriction and single-vertex extension bounds in Lemma 72 give a_i in {0,1}. Restricting a maximum matching loses at most the edge incident with v, while each matching of J is a matching of F; hence ϵ is in {0,1}. By Lemma 61, both graphs have zero rows above D. If g is the number of indices i in {1,...,D} for which $a_i = 0$, then

$$\sum_{i=1}^{D} a_i = D - g.$$

Exactly D edges are deleted with v. Subtracting the finite cost sums and edge counts therefore gives

$$gap(F) - gap(J) = a_0 - g,$$

and the definition of excess gives

$$ex(J) = ex(F) - a_0 + \epsilon.$$

Suppose $g \geq 1$. Then $gap(J) \geq gap(F) \geq 1$ since $a_0 \leq 1$. The graph J is a proper induced subgraph and satisfies T_h, so $gap(J) = 1$. Thus $gap(F) = gap(J) = 1$ and $a_0 = g$. With a_0 in {0,1} and $g \geq 1$ this forces $a_0 = g = 1$, in particular $c_0(J) = c_0(F) - 1$ and $ex(J) \geq ex(F) - 1$.

We check the needed run increase explicitly. Put $c = c_0(F)$ and $n = \nu(F)$. The graph F has an edge and $ex(F) \geq 1$, so $n \geq 1$ and $c = n + ex(F) \geq 2$. At total c-1, any integer first budget x with $x \leq ex(F) - 1$ is feasible by coloring all F-edges blue, since $c - 1 - x \geq n$. Any $x \geq n$ is feasible by coloring all edges red. Hence every missing first budget at that total lies in the integer interval [ex(F), n-1]. There is a longest nonempty missing run by Lemma 23; write it as x_0,..., $x_0 + R - 1$, where $R = tr(F) \geq 1$. Its endpoints satisfy $x_0 \geq 1$ and $x_0 + R - 1 \leq n - 1 \leq c - 2$.

If J has a two-part partition with matching bounds u, c-2-u, inserting all edges incident with v into the blue part gives bounds u, c-1-u on F; inserting them into the red part gives bounds $u + 1$, c-2-u. Each matching uses at most one of the inserted edges. Therefore feasibility of u on J at total c-2 implies feasibility of both u and $u + 1$ on F at total c-1. For every integer x in the chosen missing F-run, both x and x-1 are consequently missing on J at total c-2. The full integer interval $[x_0 - 1, x_0 + R - 1]$ is thus missing; it lies in [0,c-2] and has length $R + 1$. Since $c_0(J) = c - 1$, this is the level defining $tr(J)$, so $tr(J) \geq tr(F) + 1$.

Combining the run and excess inequalities gives

$$tr(J) + ex(J) \geq tr(F) + ex(F).$$

Since J satisfies T_h and has gap one, monotonicity of h yields

$$tr(F) + ex(F) \leq tr(J) + ex(J) \leq h(\Delta(J)) \leq h(D).$$

Together with $gap(F) = 1$ this says F satisfies T_h, a contradiction. Thus $g = 0$. Since v was arbitrary, every maximum-degree vertex deletion drops each row from 1 through D by exactly one.

Finally, if a matching covered every degree-D vertex, Lemma 61 would give $c_D(F) = 0$. Restriction and nonnegativity would then give $c_D(F - v) = 0$ for every maximum-degree v, contradicting the strict decrease just proved. No such matching exists.

Every proper induced subgraph is a proper subgraph under the stated broader relation, so proper-subgraph minimality implies the hypothesis used in the proof. The integer function $h(D) = \max(4, \lfloor D/2 \rfloor + 1)$ is nondecreasing on nonnegative integer degrees. Substituting it gives the claimed T_- specialization with no change to the allowed subgraphs. □

Depends on. Lemma 23, Lemma 61, Lemma 68, Lemma 72.

Lemma 122 (Complementary phases and physical reversal). *All graphs are finite and simple. Use the integer certificate costs $c_{A,B}(G)$, their row maxima $c_i(G)$, and $ps(G) = e(G) - \nu(G) - \sum_{i\geq 1} c_i(G)$ defined in Lemma 72 and Lemma 61. Thus $c_{A,B}$ is the least nonnegative integer total for which every matching-budget split permits a partition of all edges into two parts of maximum degrees at most A, B and two parts with the prescribed matching bounds; c_i maximizes over nonnegative integer $A + B = i$. Write $V(G)$, $E(G)$, $e(G)$, $\nu(G)$, $\Delta(G)$, $N_G(v)$, and $d_G(v)$ for vertices, edges, edge count, matching number, maximum degree, neighborhood, and degree. Vertex deletion always takes an induced graph.*

Let $D \geq 6$ be an even integer and let r, s, m, β, p be nonnegative integers. Put $u = r + s$. Let F be D-regular of order $2r + s$ and let Rcal be a nonempty family of induced subgraphs A of F satisfying

$|V(A)| = r$, $\Delta(A) \leq 2$, $\nu(A) = m$, $c_1(A) = \beta$, $c_2(A) = p$, $ps(A) = 0$.

Assume $c_1(F) = u + \beta$ and $c_2(F) = u + p$. Suppose there are nonnegative integers x, y with $x + y + 1 = u + m$ and $\min(x, y) \leq r$ such that $E(F)$ has no two-part partition with matching numbers at most x, y.

For every A in Rcal assume $\Delta(F - V(A)) \leq D - 3$. A complete process from F-V(A) repeatedly deletes any vertex of current maximum degree while that maximum exceeds two. Assume that every such process ends at a member of Rcal. For each such process let n_j be its number of deletions at degree j, for integers $3 \leq j \leq D - 3$, and assume

$c_i(F) = r + \sum_{j=i+1}^{D-3} n_j$ *for every integer $3 \leq i \leq D - 3$.*

These identities are required for every initial row and every process.

Fix any A in Rcal and any such process. Write its ordered deleted vertices as z_1,..., z_s, their deletion degrees as d_1,..., d_s, its terminal row as R, its full degree-j phase as T_j, and $X = \bigcup_{j=3}^{D-3} T_j$. Then:

(i) The counts $n_j = |T_j|$ depend only on F and satisfy $n_j = n_{D-j}$ for $3 \leq j \leq D - 3$. The sets $V(A)$, X, $V(R)$ partition $V(F)$. Every T_j is independent. Starting from F-V(R), deleting the same physical vertices in exactly the reverse order z_s,..., z_1 is a legal complete process ending at A. The reverse deletion degree of z_l is $D - d_l$. Each T_j is the complete degree-(D-j) phase in this reverse process, and any order within any phase is permitted. One has $d_l + d_{s+1-l} = D$ for $1 \leq l \leq s$. If s is odd, then $n_{D/2}$ is positive and odd.

(ii) For $3 \leq j \leq D - 3$ put

$I_j = V(A)$ *union* $\bigcup_{j<k\leq D-3} T_k$,

$J_j = V(R)$ *union* $\bigcup_{3\leq k<j} T_k$.

For every v in T_j its entire neighborhood is partitioned between I_j and J_j, with $|N_F(v)$ intersect $I_j| = D - j$ and $|N_F(v)$ intersect $J_j| = j$. Moreover

$\sum_{j<k\leq h} |N_F(v)$ *intersect* $T_k| \leq h - j$ *for* $j \leq h \leq D - 3$,

$\sum_{h\leq k<j} |N_F(v)$ *intersect* $T_k| \leq j - h$ *for* $3 \leq h \leq j$.

Consequently, for distinct integers i, j in {3,...,D-3}, all edges between T_i and T_j form a bipartite graph of maximum degree at most $|i - j|$; consecutive phases have a matching as their entire edge set.

(iii) For any $Y \subseteq V(R)$, put $W = X$ union Y, $H = F[V(A)$ union $W]$, $N = |V(F)$ minus $V(H)|$, and $h = |W|$. Then $h = s + |Y|$, $N = r - |Y|$. Every earlier side I_j is retained in H with all its original incident edges from T_j. Every y in Y has exactly $D - d_R(y)$ neighbors in $V(A)$ union X, all retained. For $3 \le i \le D - 3$ define

$C_i = W$ *minus* $\bigcup_{D-i \le j \le D-3} T_j$.

Then

$|C_i| = |Y| + \sum_{j=3}^{D-i-1} n_j$,

$c_i(F) = N + |C_i|$.

In particular $C_{D-3} = Y$ and $c_{D-3}(F) = r$. These are set and cost identities; no matching deletion is part of the definition of C_i.

All indices, orders, counts, capacities, budgets, and matching numbers are integers in the stated domains. Empty phases, $s = 0$, empty sums, and Y empty are allowed. The differences defining ps may be signed, although ps is nonnegative by the cited theorem.

Proof. We use Lemma 72 for restriction and vertex-extension inequalities, Lemma 61 for nonnegative slack and the zero cost tail, and Lemma 7 for the matching bound from a missing allocation. All refer to the same finite integer certificate costs.

1. Common counts and restored prefixes.

Every process exists and is finite: whenever the current maximum exceeds two, choose a vertex attaining it and delete that vertex. The number of vertices decreases. Its initial order is $r + s$ and its terminal order is r, so exactly s vertices are deleted. Their degrees lie between three and D-3. Maximum degree never increases under vertex deletion, so equal-degree deletions occur consecutively. At the beginning of a degree-j phase every remaining vertex has degree at most j. If two vertices selected at degree j were adjacent, the later one would lose a neighbor at the earlier deletion and thereafter have degree at most j-1. Therefore each phase T_j is independent.

Taking differences of the assumed cost identities gives

$n_j = c_{j-1}(F) - c_j(F)$ for $4 \le j \le D - 3$,

$n_3 = s - c_3(F) + r$.

The second identity uses the fact that all phase counts sum to s. These formulas prove that all counts are common to every row and process, including $D = 6$ when the first range is empty.

This common-count property also controls every actual process boundary. In a remaining graph of maximum degree at most j, let Q be its vertices of degree j. Any maximal independent set Z of the induced graph on Q is a full degree-j phase in every ordering. Its members do not lower one another's degrees; every other degree-j vertex has a neighbor in Z and drops below j after Z is removed; degrees initially below j stay below j. Such a set exists because Q is finite, including Z empty if Q is empty. Extend this phase by a complete process. Whenever the preceding phases were whole actual phases from a row complement, the extension has the common counts. Thus every such Z has cardinality n_j, and every independent subset of Q has cardinality at most n_j by extension to a maximal one. An independent subset of cardinality n_j is itself maximal, since a larger maximal extension would have the same cardinality n_j.

Fix the stated forward process. For any subset Z of $V(F)$ minus $V(A)$ with $l = |Z| \le s$, let $G = F[V(A)$ union $Z]$. Set $q = \min(x, y)$. The missing allocation has total $x + y + 1 = u + m$, and $q \le r = u - s$ implies $l \le u - q$. Apply Lemma 7 with its retained row vertex set $V(A)$ and retained exterior Z. It gives $\nu(G) = m + l$. Vertex extension from A gives $c_1(G) \le \beta + l$ and $c_2(G) \le p + l$. Restoring the other u-l vertices to reach F, the same extension inequalities give

$c_1(G) \ge c_1(F) - (u - l) = \beta + l$,

$c_2(G) \ge c_2(F) - (u - l) = p + l$.

Hence both are equalities. For every positive integer i, extension also gives $c_i(G) \ge c_i(F) - (u - l)$.

For an ordered forward prefix $Z = \{z_1, \dots, z_l\}$, denote G by G_l. At its forward deletion z_a has d_a neighbors among later roots and $V(R)$. D-regularity therefore gives exactly $D - d_a$ neighbors among $V(A)$ and the earlier roots. Counting an edge when its later endpoint is inserted proves

$e(G_l) = e(A) + \sum_{a=1}^{l}(D - d_a)$.

Since $\Delta(A) \le 2$, Lemma 61 gives $c_i(A) = 0$ for $i \ge 3$. With $ps(A) = 0$ this implies $e(A) = m + \beta + p$. Subtracting the just-proved matching and first two cost identities from the definition of ps gives

$ps(G_l) + \sum_{i\ge 3} c_i(G_l) = \sum_{a=1}^{l}(D - 3 - d_a)$. (1)

Every summand on the left is nonnegative, and the sum is finite by Lemma 61. No assertion that G_l is already a reverse-process boundary has been used.

2. Paired counts and legality of the first reverse phases.

We prove successively for $t = 3$,..., $D/2 - 1$ that $n_t = n_{D-t}$ and that deleting T_3, T_4,..., T_t, in that order, gives complete phases of the opposite process from F-V(R). At the induction entry, assume the claims for the smaller t. For $t = 3$ this is an empty assumption; F-V(R) has maximum degree at most D-3 since R belongs to Rcal.

Write $a_k = n_{D-k}$ for $3 \le k \le t$, and let Z_t be the forward prefix consisting of all phases of degrees at least D-t. Its length is $l_t = \sum_{k=3}^{t} a_k$. Independence of each phase means that any internal order in the chosen forward process gives the same induced graph $G_t = F[V(A) \text{ union } Z_t]$. Formula (1) says its nonnegative total is

$ps(G_t) + \sum_{i\ge 3} c_i(G_t) = \sum_{k=3}^{t}(k-3)a_k$. (2)

For $3 \le i < t$, the cost identity in the statement can be written as $c_i(F) = u - \sum_{k=3}^{i} n_k$. The restoration lower bound from step 1 and the induction equalities $n_k = a_k$ for $k < t$ yield

$c_i(G_t) \ge l_t - \sum_{k=3}^{i} n_k = \sum_{k=i+1}^{t} a_k$.

Summing these lower bounds for $i = 3$,..., t-1 gives exactly the right side of (2), because a_k is counted k-3 times. Therefore nonnegativity forces $ps(G_t) = 0$ and $c_i(G_t) = 0$ for every $i \ge t$. At $t = 3$ the right side of (2) is zero, so the same conclusions follow directly from nonnegativity.

Restoring $u - l_t$ vertices to F now gives $c_t(F) \le u - l_t$. Using $c_t(F) = u - \sum_{k=3}^{t} n_k$ and the previously paired counts implies

$a_t \le n_t$. (3)

In the actual opposite process the phases T_3,..., T_{t-1} have already been removed by induction. Its remaining maximum degree is at most D-t, since all larger-degree phases were complete; this statement also covers skipped empty phases. A vertex v of the original T_t had exactly t forward neighbors in $V(R)$ and in the smaller-degree phases, and none in T_t. Those vertices are precisely its neighbors that have now been deleted in the opposite process. Thus its current opposite degree is D-t. The set T_t is independent and lies in the current degree-(D-t) set. The common-count argument in step 1, applied at this actual opposite boundary, bounds its size by $n_{D-t} = a_t$. Together with (3) this proves equality. It also proves that T_t is itself a maximal independent set of the current maximum-degree set and hence a complete legal degree-(D-t) phase, in every ordering. This closes the induction without a positive-phase assumption.

3. Finish the physical reversal.

After step 2, the opposite process has maximum degree at most $D/2$. This is also the initial bound for $D = 6$, when step 2 has no iterations. The original $T_{D/2}$ is independent and every one of its vertices currently has degree $D/2$ by the complementary-neighbor calculation. Its size is already the common count $n_{D/2}$. The boundary argument in step 1 therefore makes it a complete phase in any order.

Proceed with $t = D/2 + 1$,..., D-3. All original phases of smaller degree have now been removed in the opposite process, whose maximum degree is at most D-t. Every original T_t vertex has current degree D-t and T_t is independent. Its size n_t equals n_{D-t} by the pairs from step 2. Hence it is the complete degree-(D-t) phase by the same common-count argument. After the last phase the remaining graph is exactly A, of maximum degree at most two. Thus this is a complete opposite process, with the same physical roots and any internal phase orders. In particular the exact reversal z_s,..., z_1 is legal, and its deletion degrees are $D - d_s$,..., $D - d_1$.

Both complete processes have the common counts, and their degrees are nonincreasing. The common ordered degree list is uniquely determined by those counts. Therefore $d_l = D - d_{s+1-l}$ for every $1 \le l \le s$. If s is odd, the equality

$s = n_{D/2} + 2sum_{j=3}^{D/2-1} n_j$

shows that $n_{D/2}$ is odd, and its nonnegativity makes it positive. These arguments also include $s = 0$, when all lists and phases are empty. This proves (i).

4. Physical boundaries and retained costs.

Fix j and v in T_j. Phase independence and the order of the forward process show that its remaining neighbors at deletion are exactly its neighbors in J_j. Thus it has j of them and D-j neighbors in I_j; these two sets contain its entire neighborhood.

For $j \le h \le D-3$ the graph on $V(R)$ together with all T_k for $k \le h$ is the actual boundary after all higher-degree phases. Its maximum degree is at most h. The degree of v there is $j + \sum_{j<k\le h} |N_F(v)$ intersect $T_k|$, proving the first inequality of (ii). Apply that inequality to the proved reverse process, where the same physical T_k has degree label D-k. For $3 \le h \le j$ it gives the second inequality with right side $(D-h) - (D-j) = j-h$. The first inequality bounds neighbors in a higher phase and the second bounds neighbors in a lower phase; they bound both shores of every stated bipartite graph. A bipartite graph of maximum degree at most one has pairwise vertex-disjoint edges, proving the consecutive-phase assertion.

The full vertex partition gives $h = s + |Y|$ and the omitted set exactly $V(R)$ minus Y, of size $N = r - |Y|$. Every I_j is wholly retained and H is induced, so all its relevant original edges survive. Since F is D-regular and R induced, a vertex y in Y has exactly $D - d_R(y)$ original neighbors outside R, all in the retained $V(A)$ union X.

For $3 \le i \le D-3$, the complement of the removed phases inside W is exactly Y together with T_j for $3 \le j \le D-i-1$. This proves the displayed cardinality of C_i. Replacing each j by D-j in its phase-count sum and using $n_j = n_{D-j}$ gives

$N + |C_i| = r + \sum_{j=3}^{D-i-1} n_j$

$= r + \sum_{j=i+1}^{D-3} n_j = c_i(F).$

The change of index remains valid for empty sums. At $i = D-3$ it gives $C_i = Y$ and $c_i(F) = r$. This proves (iii) and completes the theorem. □

Remark. Restoring a high-degree root prefix forces matching and two low-cost equalities. The remaining nonnegative slack budget forces one inequality between complementary phase lengths; the opposite process gives the reverse inequality and turns each physical layer into a legal reverse phase.

Depends on. Lemma 7, Lemma 61, Lemma 72.

Lemma 123 (Residual potential after maximum-degree deletions). *Use the finite integer certificate costs $c_i(H)$, matching number $\nu(H)$, independence number $\alpha(H)$, maximum degree $\Delta(H)$, edge count $e(H)$, excess $ex(H) = c_0(H) - \nu(H)$, and slack $ps(H) = e(H) - \nu(H) - \sum_{i\ge1} c_i(H)$ of Lemma 23. Put $\Phi(H) = e(H) - \nu(H)$. All graphs are finite and simple.*

For every graph G,

$\Phi(G) \ge \max(0, 2|V(G)| - 4\alpha(G))$.

A complete maximum-degree deletion sequence of H is a sequence $H_0 = H$, H_1,..., H_T such that, for every integer $1 \le j \le T$, the vertex v_j has positive maximum degree $d_j = \Delta(H_{j-1})$ in H_{j-1}, $H_j = H_{j-1} - v_j$ is induced, and H_T is edgeless. Stage j is covered when H_{j-1} has a matching covering all its degree $- d_j$ vertices, and uncovered otherwise. Call the sequence economical when $\nu(H_j) = \nu(H_{j-1})$ at every uncovered stage.

An economical sequence exists and has $c_0(H) \le T \le |V(H)|$. Fix any such sequence and any integer $c_0(H) \le k \le T$. Put $R = H_k$. Let z_k count the covered stages $j \le k$ for which $\nu(H_j) = \nu(H_{j-1})$. Let β_k count uncovered stages $j \le k$ having some earlier covered stage $l < j$ with $d_l = d_j$. Then

$ps(H) \geq z_k + \beta_k + \Phi(R)$

$\geq z_k + \beta_k + \max(0, 2(|V(H)| - k) - 4\alpha(H))$.

Consequently every finite simple H satisfying

$2(|V(H)| - c_0(H)) \geq ex(H) + 4\alpha(H)$

has $gap(H) \leq 0$, *where* $gap(H) = ex(H) - ps(H)$.

All sequence lengths, degrees, row indices, matching cardinalities, cutoff indices and counts are integers in the displayed nonnegative domains. The empty graph, isolated vertices, $T = 0$, $k = 0$ *and an edgeless residual are included.*

Proof. Use Lemma 23 for nonnegative integer costs, $c_i \leq c_0 \leq \tau$, and $gap = ex - ps$, where τ is minimum vertex-cover cardinality. Use Lemma 61 for nonnegative slack, zero cost at a covered degree threshold, the zero tail above maximum degree, and the fact that deleting one vertex changes each cost and the matching number by either zero or one.

First prove the bound on Φ. Fix a maximum matching M of G. Independently for each edge of M choose exactly one endpoint, with the two endpoints equally likely. Independently choose each unmatched vertex with probability one half. Let A be the selected vertex set. Each vertex has selection probability one half, so the average of $|A|$ over this finite probability space is $|V(G)|/2$. No edge of M lies inside A. The endpoints of an edge outside M belong to distinct choice blocks: simplicity ensures that the only edge inside a matching-edge block is that matching edge itself. Their selections are independent and each has probability one half. The average of $e(G[A])$ is therefore $(e(G) - |M|)/4 = \Phi(G)/4$.

For every outcome A, choose one endpoint of each edge of G[A] and delete the union of these chosen endpoints. At most $e(G[A])$ vertices are deleted, and the remaining set is independent. Hence $\alpha(G) \geq |A| - e(G[A])$ for every outcome. Taking the finite average gives $\alpha(G) \geq |V(G)|/2 - \Phi(G)/4$. Rearranging yields $\Phi(G) \geq 2|V(G)| - 4\alpha(G)$. Also $e(G) \geq \nu(G)$, because a matching is an edge subset, so $\Phi(G) \geq 0$.

Construct an economical sequence as follows. At a covered positive-degree stage, choose any maximum-degree vertex to delete. At an uncovered stage, choose a maximum matching. It cannot cover all maximum-degree vertices, by the definition of uncovered, so choose an uncovered maximum-degree vertex v. This matching lies in the graph after deleting v; restriction and inclusion imply that the matching number is unchanged. Every deletion removes one vertex, so this procedure ends at an edgeless induced graph in at most $|V(H)|$ steps. The deleted vertices form a vertex cover: an edge avoiding all of them would remain in the terminal graph. Thus $T \geq \tau(H) \geq c_0(H)$. If H is edgeless, the empty sequence has $T = c_0(H) = 0$.

Fix the sequence and cutoff in the statement. Its degrees d_j are positive and nonincreasing, and $\sum_{j=1}^{T} d_j = e(H)$. Indeed vertex deletion cannot increase maximum degree, and each edge is removed exactly when its first endpoint is deleted.

For every integer $i \geq 1$ define

$m_i = |\{j : 1 \leq j \leq Tandd_j > i\}|$.

Among the consecutive stages of degree i, let w_i count the uncovered stages before the first covered stage of that degree. If none of its stages is covered, let w_i count the entire degree-i block; if that degree never occurs, put $w_i = 0$. Set $N_i = m_i + w_i$ and $W = \sum_{i \geq 1} w_i$. All these sums are finite.

The residual after the first N_i deletions has cost c_i equal to zero. If a first covered degree-i stage exists, this residual is its parent and the covering condition gives the zero cost. Otherwise all stages of degree at least i have been completed, so its maximum degree is less than i; the empty matching covers its empty set of degree-i vertices. Restoring the first N_i vertices, one at a time, costs at most N_i in row i. Thus

$c_i(H) \leq N_i$.

Also $c_i(H) \leq c_0(H) \leq k$. Therefore $c_i(H) \leq \min(k, N_i)$.

Each deletion of degree d_j contributes $d_j - 1$ to $\sum_i m_i$. Consequently

$\sum_i N_i = e(H) - T + W$.

Define $C(k) = \sum_{i \geq 1} \max(N_i - k, 0)$. Substituting the preceding row bounds into the definition of slack gives

$ps(H) \geq e(H) - \nu(H) - \sum_i \min(k, N_i)$

$= T - \nu(H) - W + C(k)$. (1)

Let ℓ_i be the number of the initial uncovered degree-i stages that occur after the cutoff. Their indices are precisely $m_i + 1$,..., $m_i + w_i$. Counting those greater than k gives

$\ell_i = \max(N_i - k, 0) - \max(m_i - k, 0)$.

Put $L = \sum_i \ell_i$. Counting pairs (i, j) with $j > k$ and $d_j > i$ gives

$\sum_i \max(m_i - k, 0) = \sum_{j=k+1}^{T} (d_j - 1)$

$= e(R) - (T - k)$.

Hence

$C(k) = e(R) - (T - k) + L$. (2)

Let U be the total number of uncovered stages, U_{tail} their number after k, z_{all} the total number of covered stages preserving matching number, and z_{tail} their number after k. Every vertex deletion changes matching number by zero or one. Its total decrease along the full sequence is $\nu(H)$, since the terminal graph is edgeless. The economical condition therefore gives

$T - \nu(H) = U + z_{all}$. (3)

The uncovered stages outside the initial uncovered block of their degree are exactly those with an earlier covered stage at that same degree. Of the W initial uncovered stages, L lie after k. Thus counting uncovered stages before and after the cutoff yields

$U - W = \beta_k + U_{tail} - L$. (4)

Insert (2), (3), and (4) into (1):

$ps(H) \geq z_{all} + \beta_k + U_{tail} + e(R) - (T - k)$. (5)

The tail sequence starts at R and ends edgeless. Its total matching-number decrease is $\nu(R)$, so

$z_{tail} + U_{tail} = (T - k) - \nu(R)$.

Also $z_{all} = z_k + z_{tail}$. These two equalities transform (5) into

$ps(H) \geq z_k + \beta_k + e(R) - \nu(R)$,

which is the first required cutoff bound. The Φ inequality already proved, $|V(R)| = |V(H)| - k$, and $\alpha(R) \leq \alpha(H)$ give the second bound. These counts remain valid for empty blocks, $k = T$ and $T = 0$.

Finally choose an economical sequence and $k = c_0(H)$, which is permitted by $T \geq c_0(H)$. The displayed numerical hypothesis and $ex(H) \geq 0$ imply

$ps(H) \geq \max(0, 2(|V(H)| - c_0(H)) - 4\alpha(H)) \geq ex(H)$.

Thus $gap(H) = ex(H) - ps(H) \leq 0$. This argument also includes every edgeless graph. □

Remark. Each cost row becomes zero either at its first covered stage or when its degree level ends. Truncating these restoration bounds at the cutoff records the later uncovered stages exactly. Their contribution cancels with the tail matching loss, leaving the full residual edge-minus-matching potential.

Depends on. Lemma 23, Lemma 61.

Lemma 124 (Constructive top-cost bounds in degrees ten and twelve). *Use the finite integer certificate costs $c_{A,B}(G)$ and $c_i(G)$ of Lemma 49. Thus $c_{A,B}(G)$ is the least nonnegative integer total T for which every nonnegative integer split $x + y = T$ admits a partition of all G-edges into parts of maximum degrees at most A, B and parts of matching numbers at most x, y; $c_i(G)$ is the maximum over nonnegative integer capacities $A + B = i$. Write $V(G)$, $E(G)$, $\nu(G)$, $\alpha(G)$, and $d_G(v)$ for the vertex set, edge set, matching number, independence number, and vertex degree.*

Let d belong to {10,12}. Let F be a connected finite simple d-regular factor-critical graph, meaning that deleting any vertex leaves a perfect matching. Suppose $|V(F)| = n = 2k + 1$ for an integer $k \geq 1$. Define

$T_{10}(n) = floor((280n - 710)/1000)$, $T_{12}(n) = floor((250n - 726)/1000)$.

Then $c_{d-1}(F) \le T_d(n)$.

More precisely, for every vertex z, every perfect matching M of F-z, and every ordering of $V(F)$*, there is an explicitly constructed set Z containing z with* $|Z| \le T_d(n)$ *that dominates in closed neighborhoods the spanning graph* $J = (V(F), E(F) minus M)$*. The same set Z, induced graph F-Z, and retained matching M intersect* $E(F - Z)$ *are fixed before any capacity or budget split. For every nonnegative integer* $A + B = d - 1$ *and* $x + y = |Z|$ *they give a certificate on the entire F whose degree-bounded parts partition* $E(F - Z)$ *and whose matching-bounded parts partition every edge with at least one endpoint in Z, including edges inside Z.*

Assume in addition $c_{d-1}(F) \ge \alpha(F)$*. Define the integers*

$g = k - \alpha(F)$, $E = c_0(F) - k$, $p = g - E + 1$, $\lambda = 2E - 3 - g$.

Then

$44k \le 100g - 43$ *when* $d = 10$,

$125k \le 250g - 119$ *when* $d = 12$.

Whenever $E \ge 1$*, put* $v_d = \max(0, floor((2E - 3)/(d - 2)))$*. One has* $k \ge 3E - 1 + 2v_d$ *and*

$100p \ge 32E + 88v_d + 99$ *when* $d = 10$,

$250p \ge 125E + 250v_d + 244$ *when* $d = 12$.

Consequently, at degree ten $p \ge 4$ *for* $E \ge 6$ *and* $p \ge 5$ *for* $E \ge 7$*; at degree twelve* $p \ge 6$ *for* $E \ge 7$*. The identity* $p = E - 2 - \lambda$ *holds, so these bounds apply whenever this p is the whole odd-cycle component count of a specified induced subgraph.*

All capacities, budgets, totals, orders, degrees, indices, matching and independence cardinalities, and selected-set sizes are nonnegative integers in the displayed domains. The differences g, E, p, λ *are signed integers unless a hypothesis or conclusion gives a stronger bound. Deleting a matching removes its edges and retains all vertices; deleting vertices takes the full induced graph. An empty prescribed set in the construction is allowed. No source minimality or deletion-process hypothesis is assumed.*

Proof. We use the weighted construction of Lemma 1, the domination certificate of Lemma 49, and the missing-allocation interval of Lemma 40. The first two steps require no excess or top-cost lower bound.

1. Explicit admissible weights.

In Lemma 1 denote the maximum-degree bound by D, the uncovered weight by A_0, and the required gain by σ. For $D = 9$ set $A_0 = 280$, $\sigma = 1000$, and

$(b_0, b_1, b_2, b_3, b_4, b_5, b_6, b_7, b_8, b_9)$

$= (0, 80, 108, 124, 133, 137, 137, 137, 137, 137)$.

The successive differences are 80, 28, 16, 9, 4, 0, 0, 0, 0. They are nonnegative and nonincreasing, and $b_9 = 137 \le 280$. For maximum closed uncovered counts $j = 1, 2,..., 10$, the sufficient gains are respectively

1000, 1008, 1016, 1012, 1004, 1015, 1138, 1281, 1424, 1567.

The first and last entries are $A_0 + D\, b_1$ and $(D + 1)A_0 - D\, b_{D-1}$. For $2 \le j \le D$ the entries are obtained by substitution in

$j\, A_0 - (j - 1)b_{j-2} + j(D + 1 - j)(b_j - b_{j-1})$.

Thus all gains are at least σ, and every weight hypothesis holds.

For $D = 11$ set $A_0 = 250$, $\sigma = 1000$, and

$(b_0, b_1, b_2, b_3, b_4, b_5, b_6, b_7, b_8, b_9, b_{10}, b_{11})$

$= (0, 69, 94, 109, 118, 124, 127, 127, 127, 127, 127, 127)$.

The successive differences are 69, 25, 15, 9, 6, 3, 0, 0, 0, 0, 0. They are nonnegative and nonincreasing, and $b_{11} = 127 \le 250$. For $j = 1, 2,..., 12$ the sufficient gains, using precisely the endpoint and interior formulas just displayed, are

1009, 1000, 1017, 1006, 1024, 1018, 1006, 1111, 1234, 1357, 1480, 1603.

Each is at least 1000. This checks every required inequality for the second table.

Consequently, if Q has maximum degree at most D and S is any subset of its degree-D vertices, the fixed-order greedy rule in Lemma 1 constructs a set Y dominating S in closed neighborhoods, with

$|Y| \le floor((A_0|S| + b_D(|V(Q)| - |S|))/1000)$. (1)

The rule maximizes the number of currently uncovered vertices in a closed neighborhood, p refers an uncovered vertex among maximizers, and resolves remaining ties by the prescribed ordering. It terminates for every such ordering and also when S is empty.

2. The same marked set supplies every certificate.

Fix the vertex z, perfect matching M of F-z, and ordering in the statement. Form the spanning graph $J = F - M$ by deleting only M's edges. The vertex z has J-degree d, while every other vertex has degree d-1. Let $Q = J - z$ be the full induced graph and put $S = V(F)$ minus $N_J[z]$, where $N_J[z]$ is the closed neighborhood of z. The graph Q has maximum degree at most d-1. The set S has n-d-1 vertices, each of Q-degree exactly d-1. There are exactly d vertices in $V(Q)$ minus S.

Apply (1) with $D = d - 1$ and the induced vertex ordering on Q, and set $Z = \{z\}$ union Y. For $d = 10$ the bound is

$|Z| \le 1 + floor((280(n - 11) + 137 \cdot 10)/1000)$

$= floor((280n - 710)/1000) = T_{10}(n)$.

For $d = 12$ it is

$|Z| \le 1 + floor((250(n - 13) + 127 \cdot 12)/1000)$

$= floor((250n - 726)/1000) = T_{12}(n)$.

The simple d-regular graph has $n \ge d + 1$. Both bounds are nonnegative throughout this range. If S is empty, the greedy rule selects no vertex and $Z = \{z\}$; the same displayed inequalities remain valid.

The vertex z dominates its closed neighborhood in J. The set Y dominates S within Q and therefore within J. These two sets of vertices cover $V(J)$, so Z dominates J. The only vertex uncovered by M is z, and it belongs to Z. Thus the hypotheses of Lemma 49 hold on the same graph F with matching M, omitted set Z, degree bound d, and neighbor parameter one. That fact gives the stated certificate for every capacity split $A + B = d - 1$ and every matching-budget split $x + y = |Z|$. Its induced remainder and retained matching were fixed before either split; it partitions all original edges, including those inside Z. It also gives $c_{d-1}(F) \le |Z| \le T_d(n)$.

3. Integer cycle-count consequences.

Assume now $c_{d-1}(F) \ge \alpha(F)$. At degree ten the preceding bound implies

$1000\,\alpha(F) \le 280(2k + 1) - 710 = 560k - 430$.

Substituting $\alpha(F) = k - g$ and dividing by ten gives $44k \le 100g - 43$. At degree twelve it gives

$1000\,\alpha(F) \le 250(2k + 1) - 726 = 500k - 476$,

which rearranges, after dividing by four, to $125k \le 250g - 119$.

Factor-criticality gives $\nu(F) = k$: a perfect matching of F-z has k edges, while no matching on $2k + 1$ vertices has more. The zero-capacity certificate cost $c_0(F)$ equals the two-part threshold in Lemma 40 because degree-zero edge parts are empty. For $E \ge 1$ that theorem therefore applies to the connected F with maximum-degree bound d and yields $k \ge 3E - 1 + 2v_d$.

Since $g = E + p - 1$, at degree ten the two bounds give

$44(3E - 1 + 2v_d) \le 44k \le 100(E + p - 1) - 43$.

Rearrangement yields $100p \ge 32E + 88v_d + 99$. If $E \ge 6$ then $v_d = floor((2E - 3)/8) \ge 1$, so $100p \ge 379 > 300$ and $p \ge 4$. If $E \ge 7$ the right side is at least $411 > 400$, so $p \ge 5$.

At degree twelve they give

$125(3E-1+2v_d) \le 125k \le 250(E+p-1)-119$.

Rearrangement yields $250p \ge 125E + 250v_d + 244$. If $E \ge 7$ then $v_d = floor((2E-3)/10) \ge 1$, so $250p \ge 1369 > 1250$. As p is an integer, $p \ge 6$.

Finally $E - 2 - \lambda = E - 2 - (2E - 3 - g) = g - E + 1 = p$. Hence identifying this integer with a specified induced subgraph's odd-cycle component count transfers each proved bound without changing the graph or any certificate. This last assertion does not assume or construct such a subgraph. □

Remark. Simple integer weight tables control domination of the full-degree vertices left after deleting a near-perfect matching. Marking the exposed vertex turns that set into an actual certificate, and the missing-allocation order bound forces the cycle counts at small excess.

Depends on. Lemma 1, Lemma 40, Lemma 49.

Lemma 125 (Robust induced matchings at order twenty-nine). *Use the finite integer zero-capacity certificate cost $c_0(H)$, matching number $\nu(H)$, excess $ex(H) = c_0(H) - \nu(H)$, and top missing-allocation run $tr(H)$ of Lemma 45. A pair of nonnegative integer budgets (x, y) is feasible if all edges admit a partition into two parts whose matching numbers are at most x, y. All graphs are finite and simple, and vertex deletion is induced. A nonempty graph is factor-critical if deleting any vertex leaves a perfect matching.*

Let H be connected, have 29 vertices, satisfy $\nu(H) = 14$ and $\Delta(H) \le 12$, and have $ex(H) = 5$. Then $c_0(H) = 19$, the unique infeasible allocation of total 18 is (9,9), and $tr(H) = 1$. Every induced subgraph of H of even order at least 20 has a perfect matching. Every induced subgraph of odd order at least 21 is factor-critical. Consequently every induced subgraph J of order at least 20 satisfies $\nu(J) = floor(|V(J)|/2)$.

For every vertex set S of cardinality a with $0 \le a \le 9$, one has $\nu(H - S) = 14 - \lfloor a/2 \rfloor$. In particular, if S is an independent set of five vertices, then H-S has a perfect matching with 12 edges and $\nu(H - S) - (\nu(H) + 1 - |S|) = 2$. Regularity, positivity of gap, and any hereditary condition are unnecessary. All orders, degrees, costs, budgets, matching cardinalities, indices and set sizes are nonnegative integers in their stated ranges.

Proof. Apply Lemma 45 with $k = 14$, $D = 12$ and $E = 5$. Its integer v equals $\max(0, \lfloor 7/10 \rfloor) = 0$. Since $c_0(H) = \nu(H) + ex(H) = 19$, that fact gives a nonempty set of infeasible first budgets at total 18, all lying in the integer interval $[2E - 1 + v, k - E - v] = [9, 9]$. Thus the only infeasible allocation at that total is (9,9), and its longest consecutive missing run has length one.

Fix any twenty-element vertex subset L of H. If H[L] had no perfect matching, then $\nu(H[L]) \le 9$: a matching of ten edges would cover all twenty vertices. Its complement W has nine vertices. Assign every original edge with at least one endpoint in W to the first part and all edges of H[L] to the second part. A matching in the first part has at most nine edges, since each edge meets W and its edges have distinct endpoints. The second part has matching number at most nine by supposition. This partitions all edges, including edges inside W, and realizes (9,9), a contradiction. Therefore every such H[L] has a perfect matching.

We extend this assertion by induction on even induced order m, in steps of two. The base $m = 20$ was proved. Let $m \ge 22$ be even and let J be an induced m-vertex subgraph. Any twenty vertices in J induce a graph with a perfect matching, so J has an edge uv. The induced graph J-{u, v} has even order $m - 2 \ge 20$ and has a perfect matching by induction. Its union with the edge uv is a perfect matching of J, because its vertex set excludes both u and v. This proves the assertion for all even induced orders available in H.

If J is an induced subgraph of odd order at least 21, then for every vertex v in J the graph J-v is induced in H and has even order at least 20. The result just proved supplies a perfect matching of J-v, so J is factor-critical. Such a matching has $(|V(J)| - 1)/2$ edges, and no matching in J can be larger. The even case has already given $|V(J)|/2$ edges, proving the common floor-half formula.

Finally, a set S of size $a \le 9$ leaves an induced graph of order $29 - a \ge 20$. Hence $\nu(H - S) = floor((29 - a)/2) = 14 - \lfloor a/2 \rfloor$. For $a = 5$ this is 12, and subtracting $\nu(H) + 1 - |S| = 14 + 1 - 5 = 10$ gives the asserted gain two. Independence is needed only to describe that final application, not for the matching formula. □

Remark. The only missing budget split is balanced. Any twenty-vertex imperfect matching would realize it with the complementary nine-vertex pool, and an elementary two-vertex induction extends perfect matchings to every larger even order.

Depends on. Lemma 45.

Lemma 126 (Cubic deletion at zero slack). *Use the finite integer certificate costs $c_i(G)$, matching number $\nu(G)$, independence number $\alpha(G)$, edge count $e(G)$, and signed slack $ps(G) = e(G) - \nu(G) - \sum_{i\geq 1} c_i(G)$ of Lemma 23. All graphs are finite and simple. All orders, degrees, matching and independence cardinalities, row indices, and process lengths are nonnegative integers. Let G satisfy $\Delta(G) \leq 3$, $ps(G) = 0$, and $c_3(G) = 0$.*

A complete cubic deletion process is a sequence $G_0 = G, G_1, \ldots, G_z = R$ in which $G_{j+1} = G_j - v_j$ for a vertex v_j of current degree three, for every integer $0 \leq j < z$, and $\Delta(R) \leq 2$. Vertex deletion removes all incident edges and retains the full induced graph on the remaining vertices. Such a process always exists, including $z = 0$ when $\Delta(G) \leq 2$.

For every complete cubic deletion process and every integer $0 \leq j \leq z$,

$ps(G_j) = 0$, $c_i(G_j) = 0$ *for every integer* $i \geq 3$,

$|V(G_j)| = |V(G)| - j$, $e(G_j) = e(G) - 3j$,

$\nu(G_j) = \nu(G) - j$, $c_1(G_j) = c_1(G) - j$, $c_2(G_j) = c_2(G) - j$.

Its terminal satisfies

$\alpha(R) = |V(G)| - \nu(G) - c_2(G) + z$.

In particular

$0 \leq z \leq min\{c_2(G), \alpha(G) - |V(G)| + \nu(G) + c_2(G)\}$.

Consequently $\alpha(G) \geq |V(G)| - \nu(G) - c_2(G)$, with equality if and only if $\Delta(G) \leq 2$.

Proof. The definitions and monotonicity in the row index are supplied by Lemma 23. The zero-or-one vertex-deletion changes, nonnegative slack on every finite simple graph, and exact local deletion identity are supplied by Lemma 61. We use the degree-two terminal formula from Lemma 57.

Starting from G, delete a degree-three vertex whenever one exists. Vertex count strictly decreases, so this procedure stops after at most $|V(G)|$ steps. Every intermediate graph is induced in G and has maximum degree at most three. At termination its maximum degree is at most two. This proves existence and applies to every sequence of the prescribed choices.

We prove the displayed identities by induction on j. Since $c_3(G) = 0$ and the row costs are nonnegative and nonincreasing in their indices, every row of G at index at least three is zero. The assertions at $j = 0$ follow. Suppose they hold for an integer $j < z$. Put $H = G_j$, $H' = G_{j+1}$, $h = \nu(H) - \nu(H')$, and $a_i = c_i(H) - c_i(H')$ for $i = 1, 2, 3$. Each h, a_i is either zero or one by Lemma 61. Restriction gives $0 \leq c_3(H') \leq c_3(H) = 0$, so $a_3 = 0$, and index monotonicity makes every higher row of H′ zero. Because the deleted vertex has degree three, the exact identity in that fact reads

$ps(H) - ps(H') = 3 - h - a_1 - a_2$.

The left side is $-ps(H') \leq 0$, since $ps(H) = 0$ by the induction hypothesis and $ps(H') \geq 0$. The right side is nonnegative, since h, a_1, $a_2 \leq 1$. Both sides must therefore be zero. This yields $ps(H') = 0$ and $h = a_1 = a_2 = 1$. Vertex count drops by one and edge count by three. These conclusions give all induction assertions at $j + 1$.

At the terminal, Lemma 57 applies because $\Delta(R) \leq 2$ and $ps(R) = 0$. Its independence identity is $\alpha(R) = |V(R)| - \nu(R) - c_2(R)$. Substitution of the three established process formulas gives

$\alpha(R) = (|V(G)| - z) - (\nu(G) - z) - (c_2(G) - z)$

$= |V(G)| - \nu(G) - c_2(G) + z$.

Since $c_2(R) = c_2(G) - z \geq 0$, one has $z \leq c_2(G)$. Every independent set of the induced graph R is independent in G, so $\alpha(R) \leq \alpha(G)$; substituting the terminal identity proves the other stated upper bound on z. A complete process exists and has $z \geq 0$, proving the lower bound on $\alpha(G)$. If $\Delta(G) = 3$, every complete process must have $z \geq 1$, so that lower bound is strict. If $\Delta(G) \leq 2$, apply the same degree-two formula directly to G to obtain equality. This proves both directions, including the empty graph and isolated vertices. □

Remark. With zero third row, the three units of degree at a cubic deletion must all be accounted for by matching and the first two rows. The degree-two terminal then turns the number of cubic roots into a strict independence surplus.

Depends on. Lemma 23, Lemma 57, Lemma 61.

Lemma 127 (Concentration when matching endpoints have at most one partner neighbor). *Use the graph, matching and rooted-block notation of Lemma 77. Thus $D \geq 6$ is an even integer, G is a finite connected simple D-regular factor-critical equimatchable graph, I is a maximum independent set of size $\alpha \geq 3$, v belongs to I, Z is a perfect matching of G-v, W consists of the Z-mates of I minus{v}, M_2 is the part of Z disjoint from I, T″ consists of the endpoints of M_2 edges having no W-neighbor, and T_w consists of w and the endpoints of the M_2 edges whose maximum endpoint W-degree is one and whose W-neighbor is w. Assume every vertex outside I union W has at most one neighbor in W.*

There is exactly one w0 in W for which $|T_w 0| \geq 3$. Put $X = T_w 0$ union T″. The sets I, X, W minus{w0} partition $V(G)$. The induced graph G[X] is connected, of odd order at least three, and has independence number at most two. The set X is anticomplete to W minus{w0}; precisely 2D edges join X to I; every vertex of W minus{w0} has all its neighbors in I; and $\alpha \geq D$. If G[X] is 2-connected, then it is K_3 with $D = 6$, K_{D-1}, or K_{D-1} with one edge deleted. A graph is 2-connected if it has at least three vertices and remains connected after deleting any vertex; K_j denotes a complete graph on j vertices. All orders, degrees, matching and independence cardinalities are integers, and all vertex deletions are induced.

Proof. Use the notation and conclusions of Lemma 77. Lemma 67 gives 2-connectivity of G from the assumed connected even-regular factor-critical equimatchable hypotheses. Our assumption says M_{22} is empty. We first prove $W' = \{w \in W : |T_w| \geq 3\}$ is nonempty. Otherwise $M2 = M20$ and all its endpoints form the even clique T″. This clique is nonempty: if it were empty, $n = 2\alpha - 1$ would contradict $\alpha \leq (n-1)/2$ from Lemma 77. Write $q = |T''|$, so $2 \leq q \leq D$. Each vertex of W has all its D neighbors in I; T″ has no W-neighbor and each of its vertices has $D - q + 1$ neighbors in I. Counting edges from I gives $D\ \alpha = D(\alpha - 1) + q(D - q + 1)$, hence $(q-1)(D-q) = 0$ and $q = D$. Choose distinct u, u' in T″ and set $I' = W$ union {u'}, $X = T''$ minus{u}. The independent sets I, I' are disjoint and both have size α; u lies outside them. The residual G-(I union I' union {u}) is the even clique T″ minus{u, u'}, with a perfect matching and exactly one I'-neighbor u' at every vertex. The set X is an odd clique of order $D - 1 \geq 5$, meets I' just in u', and has every external neighbor in I union {u}. Lemma 67 excludes it. Therefore W' is nonempty. The sets T_w partition $V(G)\ minus(IunionT'')$. They are disjoint connected odd blocks, T″ is an even clique, and each T_w union T″ has independence number at most two and is anticomplete to W minus{w}. At most one T_w is noncomplete. Distinct complete T_w blocks are anticomplete. For different roots, the union of their two blocks, with or without T″, has no perfect matching. Write $b(Y)$ for the number of edges of G having exactly one endpoint in a vertex set Y. Every vertex outside I has an I-neighbor; otherwise it could be adjoined to I.

Choose $w0$ in W' to be the noncomplete block root if one exists, and otherwise to maximize the number of edges between T_w and T″. Put $X0 = T_w 0$ union T″. All other blocks are cliques. We first show that G[X0] is connected. If it were disconnected, the connected graph $G[T_w 0]$ and the clique T″ would be nonempty anticomplete subgraphs. Since their union has independence number at most two, $T_w 0$ would be a clique. All blocks would then be cliques. The maximizing choice of $w0$ would force every block to be anticomplete to T″; this already holds for singleton blocks by definition. All blocks would also be pairwise anticomplete. A nontrivial odd clique T_w has order m between 3 and D-1, because each vertex has an I-neighbor and D is even. Its boundary is $m(D - m + 1) \geq 2D - 2$; the inequality follows by minimizing the concave quadratic on [3,D-1], whose endpoint values are 3D-6 and 2D-2. The nonempty even clique T″, of order q between 2 and D, has boundary $q(D - q + 1) \geq D$ by the same endpoint argument. Singleton blocks have boundary D. Set $k = |W'| \geq 1$. Counting all edges between I and its complement, using $|W| = \alpha - 1$, gives $D\ \alpha \geq D + k(2D - 2) + (\alpha - 1 - k)D = D\ \alpha + k(D - 2)$, a contradiction. Thus G[X0] is connected.

We record boundary estimates for any connected odd set Y of order at least 3 contained outside I, with independence number at most two. If $|Y| \geq D + 1$, then $b(Y) \geq |Y| \geq D + 1$ because each vertex has an I-neighbor. If $3 \leq m = |Y| \leq D - 1$, simplicity gives $b(Y) = Dm - 2e(G[Y]) \geq m(D - m + 1) \geq 2D - 2$. In particular $b(Y) \geq D + 1$ in either case. If additionally $m \leq D - 1$ and $b(Y) \leq 2D$, then G[Y] is one of the three graphs in the assertion. Here is the numerical and edge check. If $3 \leq m \leq D - 3$ and $D \geq 8$, concavity gives $m(D - m + 1) \geq \min(3D - 6, 4D - 12) > 2D$, impossible. If $D = 6$ this interval has only the odd value

$m = 3$; $b(Y) \le 12$ then forces all three edges, so Y is K_3. Otherwise $m = D - 1$, and if h denotes the number of missing edges from the clique on Y, $b(Y) = 2D - 2 + 2h \le 2D$, so $h \le 1$.

If such G[Y] is not 2-connected, Lemma 26 gives a cut vertex c whose removal leaves two nonempty cliques A, B; c is complete to one of them, and G[Y]-x has a perfect matching for every x different from c. Both clique orders are at most D-1. Indeed, c has a neighbor in each, and that neighbor has an I-neighbor, so a clique of order at least D would give degree at least $(D-1)+2 > D$. For $1 \le a = |A|$, $b = |B| \le D-1$, vertices of A have internal degree at most a, vertices of B at most b, and c has at least one I-neighbor. Thus the edge boundary of Y is at least $a(D-a) + b(D-b) + 1 \ge 2D - 1$, since $t(D-t) \ge D-1$ for $1 \le t \le D-1$.

Suppose X_0 is anticomplete to every other nontrivial block. The other blocks are cliques and are mutually anticomplete; X_0 is anticomplete to the other singleton blocks from the decomposition. Boundary counting then gives $D\ \alpha = b(X0) + \sum_{w \in W' minus \{w0\}} b(T_w) + (\alpha - 1 - k)D$. The lower bounds $b(T_w) \ge 2D - 2$ imply $(D-2)k \le 3D - 2 - b(X0)$. Since $b(X0) \ge D + 1$, $k \le 2$. If G[X0] is not 2-connected, its stronger boundary bound 2D-1 gives $k \le 1$.

First suppose G[X0] is not 2-connected and, for contradiction, $k \ge 2$. Choose the cut vertex c and cliques A, B just supplied. Any edge from X_0 to another block T_w must be incident to c on the X_0 side. Otherwise its endpoint x different from c could be deleted from X_0 leaving a perfect matching, while deletion of the other endpoint leaves a perfect matching of the odd clique T_w. Together with the interedge these would perfectly match X_0 union T_w, prohibited by Lemma 77. There must be an edge from c to another nontrivial block, since otherwise the preceding boundary count gives $k \le 1$. The orders of A and B have the same parity. If both were even, they would perfectly match X0-c, and an edge from c to an odd clique block, with a perfect matching of that block off its endpoint, would again perfectly match their union. Thus both A and B are odd.

Choose $y1$ in A and $y2$ in B and set $I' = (W minus \{w0\})$ union {y1,y2}. This is independent: A and B are anticomplete, and X_0 is anticomplete to W minus{w0}. It has $size(\alpha - 2) + 2 = \alpha$ and is disjoint from I; c lies outside both. The residual G-(I union I' union {c}) has a perfect matching, formed by perfect matchings of the even cliques A-y1 and B-y2 and the original M_2 edges outside X_0. Every residual vertex has at most one I'-neighbor. Inside X_0 it is in one of A-y1, B-y2, so can neighbor only the corresponding y_i and none of W minus{w0}. Outside X_0 it has no neighbor in A union B by the interedge restriction and has at most one W-neighbor by our hypothesis. Choose any nontrivial T_w with w different from $w0$. This is an odd clique of size at least 3, meets I' exactly in w, and has external neighbors only in I union {c}: other clique blocks are anticomplete and its only possible X0-neighbor is c. This contradicts Lemma 67. Hence $k = 1$ in the case that G[X0] is not 2-connected.

Now suppose G[X0] is 2-connected. Its odd order and independence number at most two make it factor-critical by Lemma 26. Any edge between it and another clique block would perfectly match their union by the same deletion argument, so no such edge exists. The boundary count therefore gives $k \le 2$. If $k = 1$, $b(X0) = 2D$. If $k = 2$, write $X1 = T_w1$ for the other nontrivial block. Then $b(X0) + b(X1) = 3D$, and $b(X1) \ge 2D - 2$ gives $b(X0) \le D + 2$. In all cases $b(X0) \le 2D$. If $|X0| \le D - 1$, the boundary estimates classify X_0 as one of the three claimed graphs. Also k cannot be 2, since then $b(X0) + b(X1) \ge 4D - 4 > 3D$. Thus in this size range $k = 1$ and the assertion follows.

It remains to exclude $|X0| \ge D + 1$. If X_1 exists, it must be K_{D-1}. To prove this, its odd order is between 3 and D-1. If it is at most D-3, its clique boundary is at least 2D: for $D = 6$ it is K_3 with boundary 12, and for $D \ge 8$ the concave-quadratic estimate above is larger than 2D. But $b(X0) \ge D + 1$ would then give $b(X0) + b(X1) > 3D$, impossible. Thus $|X1| = D - 1$.

We show that no z in I has all its neighbors in X_0. Suppose such z exists. Since $|X0| > D$, select u in X_0 not adjacent to z. Factor-criticality of G[X0] gives a perfect matching of X0-u. Take the submatching consisting precisely of the edges that meet $N_G(z)$. It covers every neighbor of z, and each selected edge covers at least one such neighbor that no other matching edge covers. It is therefore an inclusion-minimal matching isolating z. Apply Lemma 6 to the whole graph G and this matching. Its residual after deleting z and all matching endpoints is empty, an even clique, or a balanced complete bipartite graph. It contains I minus{z}, an independent set of size $\alpha - 1 \ge 2$, so must be a nonempty balanced complete bipartite graph. It also contains u, which has an I-neighbor different from z because u was not adjacent to z. In a complete bipartite graph the nonempty independent set I minus{z} lies entirely in one part; hence u lies in the other and is adjacent to all alpha-1 of its vertices. Since G[X0] is 2-connected, u has at least two neighbors in X_0. Thus $D \ge \alpha + 1$. If X_1 exists, it is a retained clique of order $D - 1 \ge 5$ in this bipartite residual, impossible. Hence $k = 1$. There is a vertex w in W minus{w0}, since $\alpha \ge 3$. Its neighbors all lie in I: there are no other nontrivial

blocks and X_0 is anticomplete to it. Therefore $D \le \alpha$, contradicting $D \ge \alpha + 1$. This proves that every z in I has some neighbor outside X_0.

Every vertex of X_0 has at least one I-neighbor, and $b(X0) \le 2D$. Because $|X0| \ge D + 1$ and X_0 is anticomplete to all other blocks, some $x0$ in X_0 has exactly one I-neighbor $z0$. By the preceding result choose a neighbor $y0$ of $z0$ outside X_0. The vertex $y0$ lies outside I because I is independent. Let Z_0 be a perfect matching of X0-x0, and let J_0 consist of its edges meeting $N_G(x0)$ intersect X_0. The matching J_0 union {z0y0} is inclusion-minimal isolating $x0$. All internal neighbors of $x0$ are covered by J_0; its only external neighbor is $z0$ and is covered by the added edge. Each matching edge covers a neighbor of $x0$ and is indispensable in this matching. Lemma 6 again makes the residual $H = G - (V(J0) union \{x0, z0, y0\})$ a nonempty balanced complete bipartite graph, since it contains the independent set $I1 = I$ minus{z0} of size $\alpha - 1 \ge 2$.

If X_1 exists, all of X_1 except possibly $y0$ remains in H. This is a clique of order at least $D - 2 \ge 4$, contradicting bipartiteness. Thus $k = 1$, and $y0$ belongs to W minus{w0}. Put $W1 = W$ minus{w0,y0} and $R = X0$ minus(V(J0) union {x0}). The vertex sets I_1, W_1, R partition H. The set R has a perfect matching given by Z_0 minus J_0, so it contributes the same number of vertices to both parts of the bipartition of H. The nonempty independent set I_1 lies in one part and has size alpha-1. If W_1 is nonempty it also lies wholly in one part, since H is complete bipartite, and its size is alpha-3. If it lies opposite I_1 the two parts therefore differ in size by 2; if on the same side they differ by $2\alpha - 4 > 0$. If W_1 is empty, then $\alpha = 3$ and the difference is 2. In all cases the parts are unequal, contradicting balanced bipartiteness. The case $|X0| \ge D + 1$ is excluded. Since $|X0|$ is odd, the alternatives considered exhaust its possible orders.

We have proved $k = 1$ in both connectivity cases. Set $X = X0$ and let $w0$ be its unique nontrivial root. The decomposition supplies the claimed partition and anticompleteness. Connectedness and independence number at most two were established above. Since every vertex of W minus{w0} now has all D neighbors in I, counting the boundary at I gives $D\,\alpha = b(X) + D(\alpha - 2)$, whence $b(X) = 2D$. The set W minus{w0} is nonempty, so one of its vertices also gives $\alpha \ge D$. If G[X] is 2-connected the earlier classification applies, proving all assertions. □

Remark. The degree boundary of the maximum independent set supports at most two substantial odd blocks. Cut and isolation arguments remove the second block and any large first block, leaving one small block with boundary exactly twice the regular degree.

Depends on. Lemma 6, Lemma 26, Lemma 67, Lemma 77.

Lemma 128 (Two units of slack in a factor-critical cubic component with one defect). *Use the finite simple graph certificate costs $c_i(G)$, matching number $\nu(G)$, independence number $\alpha(G)$, edge count $e(G)$, and slack $ps(G) = e(G) - \nu(G) - \sum_{i \ge 1} c_i(G)$ of Lemma 23. A graph is factor-critical if deleting any one vertex leaves a perfect matching; the empty graph has its empty perfect matching. A vertex set is independent if it contains no edge, and a graph is bipartite if its vertices admit a partition into two independent sets.*

Let K be a finite simple factor-critical graph with a vertex u of degree two and with every other vertex of degree three. Then K is connected and, for an integer $k \ge 2$, has

$|V(K)| = 2k + 1$, $\nu(K) = k$, $e(K) = 3k + 1$, $\alpha(K) \le k$.

There exists a subset U of $V(K)$ with $|U| \le k - 1$ such that the induced graph K-U is bipartite and has maximum degree at most two. In particular it is a disjoint union of paths and even cycles, with isolated vertices permitted.

The same graph K satisfies

$c_1(K) \le k$, $c_2(K) \le k - 1$, $c_i(K) = 0$ *for every integer* $i \ge 3$,

and hence $ps(K) \ge 2$. These bounds concern the universal integer allocation costs on all original edges of K. All cardinalities, degrees, costs, matching budgets and cost indices are nonnegative integers.

Proof. Since K-u has a perfect matching, $|V(K)|$ is odd; write it as $2k + 1$. The graph K has at least three vertices because u has degree two. It cannot have three vertices, since either of its other vertices would then have degree at most two. Thus $k \ge 2$. A perfect matching of K-u has k edges, and no matching of K can have more than k edges, proving $\nu(K) = k$. The degree sum is $2 + 3(2k) = 6k + 2$, so $e(K) = 3k + 1$.

The graph K is connected. Otherwise, since its total order is odd, it has an odd-order component C and another component containing a vertex v. A perfect matching of K-v would restrict to a perfect matching

of C, which is impossible at odd order. If an independent set I had at least $k+1$ vertices, there would be a vertex v outside it because K has edges and I cannot be all of $V(K)$. In K-v the at least $k+1$ independent vertices would have to be matched to distinct vertices outside I and v, of which there are at most k-1. This contradicts factor-criticality, so $\alpha(K) \leq k$.

Construct a proper coloring of K with three colors as follows. Order the vertices by nonincreasing distance from u, with arbitrary order within each distance, and color greedily in that order. Every vertex other than u has a neighbor strictly closer to u that appears later; thus at most two of its neighbors have already been colored. The final vertex u has degree two. In either case one of the three colors is available, and assigning it keeps every edge between already colored vertices properly colored. Relabel the colors so that u does not have color three.

Let S_0 be the vertices of color three. This is an independent subset of the degree-three vertices, and $K - S_0$ is bipartite. Extend S_0 to an inclusionwise maximal independent subset S of the degree-three vertices by adding an eligible vertex while one exists. Each addition increases cardinality, so the finite procedure terminates. Every degree-three vertex outside S then has a neighbor in S. Consequently K-S has maximum degree at most two, and it is bipartite as an induced subgraph of $K - S_0$. The independence bound gives $|S| \leq k$.

If $|S| \leq k-1$, choose $U = S$. Suppose instead $|S| = k$ and put $T = V(K)$ minus S. The vertex u belongs to T. Since S is independent and all its vertices have degree three, exactly 3k edges meet S, and $e(K) = 3k+1$ leaves exactly one edge ab inside T. Choose an endpoint w of this edge different from u. Then w has degree three, exactly one neighbor in T, and two distinct neighbors s, s′ in S. Define $U = \{w\}$ union (S minus {s, s′}), which has cardinality k-1. The vertices of K-U are (T minus {w}) union {s, s′}. Both displayed sets are independent: deleting w removes the only edge of K[T], and S was independent. A vertex in T minus {w} has neighbors in K-U only among s, s′, so its degree there is at most two. Each of s, s′ had all three neighbors in T, including w, and hence also has at most two neighbors left. This proves both required properties of U in the remaining case.

For the row-one bound we construct a needed bipartite vertex cover. In any finite bipartite graph B with parts L, R, fix a maximum matching M. Starting at unmatched vertices of L, traverse nonmatching edges from L to R and matching edges from R to L. Let Z_L, Z_R be the reachable sets. No unmatched vertex of R is reachable: a shortest such traversal is an alternating path whose first and last vertices are unmatched, and replacing its matching edges by its nonmatching edges would produce a matching of size $|M|+1$. Every vertex of Z_R is therefore matched to a vertex of Z_L. Conversely each matched vertex of Z_L is reached from its unique matching partner in Z_R. Thus matched vertices in Z_L and vertices in Z_R correspond bijectively under M. All unmatched vertices of L belong to Z_L.

The set $W = (Lminus Z_L)$ union Z_R covers every edge of B. If an edge from a vertex of Z_L to a vertex outside Z_R were nonmatching, the traversal would reach its right endpoint. If it were matching, its reachable left endpoint would have its matching partner in Z_R by the preceding correspondence. Both are impossible. Finally L minus Z_L consists only of matched vertices, and together its matched vertices and those in Z_L account for the $|M|$ left endpoints of M. Therefore $|W| = |L \text{ minus } Z_L| + |Z_R| = |M| = \nu(B)$.

Among the finitely many partitions $V(K) = L$ disjoint union R, choose one maximizing the number of crossing edges. Every vertex has at most one neighbor in its own part: a vertex with at least two such neighbors has at most one across, since its total degree is at most three, and moving it would strictly increase the crossing count. Hence the internal edge set Q has maximum degree at most one, and the crossing graph B is bipartite. The preceding construction gives a cover W of B with $|W| = \nu(B) \leq \nu(K) = k$. For each pair of nonnegative integers x, y with $x + y = |W|$, choose $W_R \subseteq W$ of size x and put $W_B = W$ minus W_R. Put every B-edge meeting W_R in P_R, and every other B-edge in P_B. Every P_B edge meets W_B; hence their matching numbers are at most x and y, respectively. Taking Q as the degree-one part and the other degree part empty gives a certificate at capacities (1,0) for every such allocation. Lemma 23 supplies symmetry and the universal threshold characterization, so $c_1(K) = c_{1,0}(K) \leq |W| \leq k$.

For the row-two bound, a finite graph of maximum degree at most two has components that are paths or cycles: tracing edges in a component cannot branch, and either terminates or closes a cycle containing that component. Bipartiteness forces each cycle to have even order. Thus the edges of K-U split into two matchings by alternating along each path and each even cycle. This gives a zero-cost certificate at capacities (1,1); putting all these edges in one degree-two part gives zero cost at capacities (2,0) and (0,2). Therefore $c_2(K - U) = 0$. The vertex-set extension inequality of Lemma 72 gives $c_2(K) \leq |U| \leq k-1$; it assigns every edge incident with U to an exceptional part for each allocation and preserves all other edges.

The perfect matching of K-u covers every degree-three vertex of K. The covering-matching conclusion of Lemma 61 gives $c_3(K) = 0$, and its strong tail gives $c_i(K) = 0$ for $i \geq 4$. Substituting the established bounds in the definition of ps yields

$ps(K) = 3k + 1 - k - c_1(K) - c_2(K) \geq 3k + 1 - k - k - (k - 1) = 2.$

Every construction used the same graph K, and the certificate steps covered every integer allocation at their stated totals. □

Remark. A greedy three-coloring removes an independent set to leave paths and even cycles. If that set is largest possible, the single edge in its complement permits an exchange saving one deleted vertex. This saving supplies the second unit of slack.

Depends on. Lemma 23, Lemma 61, Lemma 72.

Lemma 129 (Closed lumps of degree at least twenty-six are LP-good). *Use the integer certificate costs $c_\theta(H)$, the matching number $\nu(H)$, the excess $ex(H) = c_0(H) - \nu(H)$, and $gap(H) = \sum_{\theta \geq 0} c_\theta(H) - e(H)$ from Lemma 23. A closed d-lump is a nonempty connected d-regular finite simple graph such that deleting any vertex leaves a perfect matching. For every even integer $d \geq 26$ and every closed d-lump H, one has $gap(H) \leq 0$. More precisely, write $|V(H)| = 2k + 1$, so $k = \nu(H)$ is an integer and $k \geq d/2 \geq 13$. If $k = 13$ then $gap(H) < -41/14$; if $k \geq 14$ then $gap(H) < 2 - k/14$. Thus every such H is LP-good. For an even integer $a \geq 2$, let $P(a)$ mean that every closed d-lump with even integer $d \geq a$ is LP-good. The theorem establishes $P(26)$, and hence $P(28)$ and $P(30)$. No upper bound on the order of H is assumed.*

Proof. Let H and d satisfy the hypotheses, and put $n = |V(H)|$. A perfect matching after a vertex deletion shows that n is odd and $\nu(H) = (n - 1)/2$; write $n = 2k + 1$. Simplicity and regularity give $d \leq n - 1 = 2k$, hence $k \geq 13$. The degree sum gives $e(H) = dn/2$. Put $E = ex(H)$, a nonnegative integer by Lemma 23.

We first prove the vertex-count estimate $E \leq floor((k + 1)/3)$ directly. For any graph on n vertices, suppose x, y are nonnegative integers and $n \leq x + y + 1 + \max(x, y)$. If $x \geq y$, take a set S of $\min(n, 2x + 1)$ vertices. Put its internal edges in P_R and all remaining edges in P_B, with both bounded-degree parts empty. A red matching has at most x edges because it uses vertices of S. Every blue edge meets $V(H)$ minus S, whose size is $\max(0, n - 2x - 1) \leq y$, so a blue matching has at most y edges. If $y > x$, exchange the two colors in this construction. Therefore every allocation with total c is feasible at zero capacities whenever $c + \lceil c/2 \rceil + 1 \geq n$, since $\max(x, y) \geq \lceil c/2 \rceil$. Let $e_0 = floor((k + 1)/3)$ and $c = k + e_0$. Writing k as 3h, $3h + 1$, or $3h + 2$ gives $e_0 = h$, h, or $h + 1$ and $c + \lceil c/2 \rceil + 1 = 6h + 1$, $6h + 3$, or $6h + 6$, respectively. These numbers are at least $n = 6h + 1$, $6h + 3$, or $6h + 5$. Hence $c_0(H) \leq c$ by the cost definition, proving $E \leq e_0$.

The complementary-level theorem Lemma 87 gives $c_\theta(H) = 0$ for $\theta \geq d + 1$, $c_d(H) \leq 1$, $c_i(H) + c_{d+1-i}(H) \leq n$ for $1 \leq i \leq d/2$, and $c_i(H) + c_{d-i}(H) \leq n + 1$ for $1 \leq i \leq d - 1$. The definition fact Lemma 23 gives $c_0(H) = k + E$ and $0 \leq c_i(H) \leq k + E$. Thus the integer sequence $c_0(H)$,..., $c_d(H)$ satisfies every hypothesis of Lemma 12. Its g is $gap(H)$ because the higher rows vanish and $e(H) = d(2k + 1)/2$. With $t = 4$, any bounds $b_j \geq c_{d-j}(H)$ therefore give

$gap(H) \leq 6E - 4k - 4 + \sum_{j=1}^{4} b_j.$

The value $t = 4$ is permitted since $d \geq 26$.

Fix any vertex v and any perfect matching M_v of H-v. For $j = 1, 2, 3, 4$, use the probabilities $p_j = 1/8, 1/6, 2/9, 3/11$, respectively, and put $q_j = q_{25,j}(p_j)$ in the notation of Lemma 24. Its trial count 25 is permitted because $25 \leq d - 1$, and every j is at most d-1. That theorem provides, for each j, a set Z containing v which dominates every outside vertex by at least j neighbors in $H - M_v$ and has size at most $1 + 2kq_j$. The construction Lemma 49 then gives $c_{d-j}(H) \leq 1 + 2kq_j$. The sets for different j need not agree; each bounds a separate scalar cost of the same graph. Taking these as b_j yields

$gap(H) \leq 6E - 4k + 2k \sum_{j=1}^{4} q_j.$

Here are explicit rational bounds for their sum. Expanding the binomial probabilities in Lemma 24 gives

$q_1 = 1/8 + (7/8)^{26},$

$q_2 = 1/6 + 5(5/6)^{25}$,

$q_3 = 2/9 + (1599/81)(7/9)^{24}$,

$q_4 = 3/11 + (89012/1331)(8/11)^{23}$.

The coefficients use $binom(25,1) = 25$, $binom(25,2) = 300$, and $binom(25,3) = 2300$. In particular $1599 = 49 + 350 + 1200$ and $89012 = 512 + 4800 + 21600 + 62100$. The following four integer inequalities give strict upper bounds on the four powers:

$10000 \cdot 7^{26} = 938748033764775430564900000 < 939939824750374183333405184 = 311 \cdot 8^{26}$;

$100000 \cdot 5^{25} = 29802322387695312500000 < 29823372143396256743424 = 1049 \cdot 6^{25}$;

$1000000 \cdot 7^{24} = 191581231380566414401000000 < 191598996270647768691793122 = 2402 \cdot 9^{24}$;

$1000000 \cdot 8^{23} = 590295810358705651712000000 < 590983960548456665682710460 = 660 \cdot 11^{23}$.

Substitution gives $q_j < Q_j$, where

$Q_1 = 1561/10000$,

$Q_2 = 1/6 + 1049/20000$,

$Q_3 = 2/9 + 640133/13500000$,

$Q_4 = 3/11 + 6069/137500$.

Their sum is $Q = 71407829/74250000$. This is less than $27/28$ because $28 \cdot 71407829 = 1999419212 < 2004750000 = 27 \cdot 74250000$. Hence

$gap(H) < 6E - 4k + 2kQ$.

If $k \geq 14$, then $E \leq floor((k+1)/3) \leq (k+1)/3$, giving $gap(H) < 2 - 2k(1-Q) < 2 - k/14 \leq 1$. The gap is an integer by Lemma 23, so $gap(H) \leq 0$. If $k = 13$, the stronger integer estimate $E \leq \lfloor 14/3 \rfloor = 4$ gives $gap(H) < 24 - 52 + 26 \cdot (27/28) = -41/14 < 0$. These cases exhaust $k \geq 13$ and establish the stated bounds. Finally every even degree at least 28 or 30 is at least 26, so $P(26)$ implies both $P(28)$ and $P(30)$ by their definitions. □

Remark. Four small domination certificates pay for the lower certificate levels. The exact integer vertex-count bound and the coefficient sum below $27/28$ cover every legal order, so neither a dense-graph reduction nor a finite order table is needed.

Depends on. Lemma 12, Lemma 23, Lemma 24, Lemma 49, Lemma 87.

Lemma 130 (Strict cubic potential from a full smaller-forest packing). *All graphs are finite and simple. A star $K_{1,d}$ has d edges; write $P_3 = K_{1,2}$ and $K_2 = K_{1,1}$. Let p, r be positive integers and let h, u, v be nonnegative integers satisfying $h > u > v$. Form the entire ordered target forests*

$F = pP_3$ *disjoint union* uK_2,

$J = rK_{1,3}$ *disjoint union* hP_3 *disjoint union* vK_2.

Every displayed union is vertex-disjoint. Let $s = p + u$ and $t = r + h + v$, and let $n_1 \geq \cdots \geq n_s$ and $m_1 \geq \cdots \geq m_t$ be their positive component edge-size lists. Put $M = s + t - 1$, $R = p + r - 1$, and

$\ell_k = max\{n_i + m_j - 1 : 1 \leq i \leq s, 1 \leq j \leq t, i + j = k\}$

for all integers $2 \leq k \leq s + t$. Put $L = \sum_{k=2}^{s+t} \ell_k$ and $P = L - M$. Then $P = 3R + 2h + u$.

For a graph G write $e(G)$, $\nu(G)$, $\Delta(G)$ for its edge count, maximum matching cardinality, and maximum degree, and put $\Phi(G) = e(G) - \nu(G)$. If $\Delta(G) \leq 3$ and every red-blue edge coloring of G contains a red copy of F or a blue copy of J, then

$\Phi(G) \geq P + 1 = 3(p + r - 1) + 2h + u + 1$.

Copies are ordinary subgraphs with vertex-disjoint target components. Equivalently, every such degree-capped G with $\Phi(G) \leq P$ has an edge coloring avoiding both entire prescribed targets. The same assertion holds with the two forests

and their prescribed colors interchanged. All indices, multiplicities, graph quantities, matching sizes and cardinalities are integers in the displayed domains. No matching-deficit, connectedness, regularity, or minimum-degree assumption is imposed.

Proof. The complete original diagonal list is

(4 repeated R times,3 repeated h times,2 repeated u times,1 repeated v times).

To check it, pairs of a red two with a blue three give precisely all index sums from two through $p+r$, with value four. After this interval, a value at least three can occur only from a red two with a blue nonunit or a red unit with a blue three. Their largest possible index sums are $p+r+h$ and $p+u+r$; the first is larger because $h>u$. Every index sum k from $p+r+1$ through $p+r+h$ attains three using $i=p$ and $j=k-p$. Next, a value at least two requires a red two or a blue nonunit. The largest possible sums are $p+r+h+v$ and $p+u+r+h$, with the latter larger because $u>v$. For k from $p+r+h+1$ through $p+r+h+u$ take $j=r+h$ and $i=k-j$ to attain two. All later candidates have both entries one. Their number is v. Subtracting one from each of the M diagonal entries gives $P=3R+2h+u$.

Suppose G arrows the displayed pair and $\Delta(G)\le 3$. The finite-family potential theorem Lemma 56, applied to this one label, gives $\Phi(G)\ge P$. Assume equality for a contradiction. Fix a maximum matching W, and write $Q=G-W$ for deletion of its edges only. Then $e(Q)=P$.

Let S be the set of degree-three vertices of G, and let I be the set of degree-three vertices of Q. Since a covered vertex loses one incident edge and an uncovered vertex loses none, $I=S$ minus $V(W)$. The set I is independent: an edge between two uncovered vertices could be added to W, contradicting maximal cardinality. The exact greedy theorem Lemma 95 applies to the same single label and the initially fixed W. It says that every greedy maximum-degree deletion in Q has degrees

(3 repeated R times,2 repeated h times,1 repeated u times,0 repeated v times).

An initially lower-degree vertex cannot gain degree. Deleting vertices of the independent set I leaves the degrees of its other vertices unchanged. Therefore the first R greedy choices delete exactly all of I, and $|I|=R$. This conclusion holds for every maximum matching, because W was arbitrary.

The minimum number of uncovered S-vertices over all matchings is also R. For completeness, any non-maximum matching can be enlarged without uncovering any vertex it already covered. Compare it with a matching of greater cardinality. Their symmetric difference has components that are alternating paths and even cycles, since every vertex has at most one incident edge from each matching. The greater total cardinality gives a path component with one more edge from the larger matching. Its endpoints are uncovered by the smaller matching. Interchanging the two edge sets on this path increases cardinality by one and retains coverage of every previously covered vertex. Repeating terminates at a maximum matching, because cardinality increases and is bounded by $floor(|V(G)|/2)$. Thus a matching leaving fewer than R vertices of S uncovered would extend to a maximum matching with the same property, contradicting the preceding paragraph.

Matching deletion Lemma 34 gives, simultaneously on this Q,

$Q\ arrows(pK_2, J)$,

$Q\ arrows((p+u)K_2, rP_3 disjointunion hK_2)$. (1)

We next prove the required full-star separation directly from the first pair. For $p=1$, the all-blue coloring forces J. Its r largest stars must be centered at all $R=r$ vertices of I and use their entire neighborhoods; the disjoint copied components therefore give pairwise disjoint closed neighborhoods. For $p\ge 2$, suppose two distinct roots x, y in I share a neighbor z. Independence gives z outside I. Choose p-2 other roots, possible because $|I|-2=p+r-3\ge p-2$, and call their set C. Color every edge meeting C and both edges xz, yz red, and the other edges blue. The red graph has a cover C union {z} of size p-1 and hence no p-edge matching. Only I can supply blue three-star centers. The vertices in C are blue-isolated, and x, y each lose an incident edge, leaving at most $R-(p-2)-2=r-1$ possible centers. This excludes the full blue J, contradicting (1). Thus no two roots share a neighbor. Together with independence, this proves disjoint closed Q-neighborhoods for every p.

For every A $\subseteq$ I with $|A|=r$, color all edges meeting I minus A red and all others blue. The red graph has a cover of size $R-r=p-1$, so (1) forces a blue J. The r largest stars must be centered at all vertices of A and

use their complete neighborhoods. Every vertex of I minus A is blue-isolated. Therefore, writing $H = Q - I$ and $N_Q(A)$ for the union of the Q-neighborhoods of A, the graph

$H - N_Q(A)$ contains the entire hP_3 disjoint union vK_2. (2)

All copied smaller components lie outside the removed vertices by vertex-disjointness and their positive edge sizes. Since no W-edge meets I, Q- and G-neighborhoods of roots agree. The disjointness just proved implies that every vertex outside I has at most one G-neighbor in I. Also $e(H) = e(Q) - 3R = 2h + u$.

We verify the exact finite family needed for Lemma 44. A matching target of size k against a sorted positive forest profile has its largest opposite size repeated k times, then the remaining opposite sizes: on diagonal b the smallest valid opposite index is $\max(1, b - k)$, and nonincreasing order attains the maximum there. Thus the two pairs in (1) have diagonal lists

(3 repeated R times,2 repeated h times,1 repeated v times),

$(2repeatedR + utimes, 1repeatedhtimes)$.

Padding missing coordinates by zero and using $h > u > v$, their maximum is

(3 repeated R times,2 repeated h times,1 repeated u times).

The first component budget is $R + h + v$ and the second is $R + u + h$, both greater than R.

Let calB consist of all pairs obtained from either member of (1) by deleting a leading matching prefix of length a and a leading opposite-forest prefix of length b, where a, b are nonnegative integers with $a + b = R$ and both retained targets are nonempty. Such cuts exist for each original pair because its budget exceeds R. Every retained first target is a positive matching, the largest remaining component budget is $h + u$, and Lemma 34 applied to the actual vertex set I makes H arrow every member.

The full calB envelope is exactly the previous envelope after omitting its first R entries. Indeed, a candidate on local diagonal k of an R-cut pair comes from its original diagonal $k + R$. Conversely, an original candidate with indices i, j and $i + j = k + R$, $k \geq 2$, is retained by the cuts $a = \min(R, i - 1)$, $b = R - a$. They satisfy $a \leq i - 1$, $b \leq j - 1$, and retain nonempty targets; the candidate has local index sum k. Hence both inclusions of candidate sets hold on every diagonal. The resulting full envelope is h twos followed by u ones.

Since $R \geq r$, choose an r-element A in (2); this gives h disjoint P_3s in H. All hypotheses of Lemma 44 now hold: calB is finite and nonempty, its maximum budget is $h + u$, its envelope is $(2^h, 1^u)$, H arrows it, and $e(H) = 2h + u$. Consequently every component of H belongs to P_1, P_2, P_3, P_4, P_6 and $\nu_2(H) = h$, where ν_2 counts disjoint P_3s.

For any z in $N_G(I)$, let x in I be its adjacent root. Choose an r-element subset A of I containing x, possible because $R \geq r$. The packing supplied by (2) avoids z, so $\nu_2(H - z) \geq h$. Deleting a vertex cannot increase a packing number, because every remaining packing was also present before deletion. Thus $\nu_2(H - z) = \nu_2(H)$ for every such z.

Apply Lemma 27 to this original G and the same W. It has maximum degree at most three. The matching W has maximum cardinality and attains the minimum R of uncovered degree-three vertices. Its exposed set is I, every outside vertex has at most one neighbor in I, its residual (G-W)-I is exactly H with the required path components, and every root attachment preserves $\nu_2(H)$. The lemma concludes that this minimum is zero. This contradicts $R = p + r - 1 \geq 1$. Equality $\Phi(G) = P$ is therefore impossible. The lower bound $\Phi(G) \geq P$ and integer values give $\Phi(G) \geq P + 1$.

Negating arrowing yields the stated avoiding coloring whenever $\Phi(G) \leq P$. Exchanging red and blue in each definition and coloring preserves the entire argument and proves the reversed prescribed-color assertion. □

Remark. Equality fixes the number of exposed cubic roots. Their whole neighborhoods must be disjoint, and every selected group leaves the entire smaller blue forest outside it. The resulting full P_3 packing fills all degree-two slots, so only paths remain. The path attachment parity theorem then covers every cubic root, contradicting their positive prescribed number.

Depends on. Lemma 27, Lemma 34, Lemma 44, Lemma 56, Lemma 95.

Lemma 131 (Degree-four residuals and positive-defect payment). *Use the integer certificate costs $c_i(G)$, matching number $\nu(G)$, and slack $ps(G)$ of Lemma 61, and r-goodness and the induced deletion number $\delta_r(G)$ of Lemma 86. Define*

$p_{cap}(G) = e(G) - \nu(G) - \sum_{i\geq 1} \min(c_i(G), \nu(G))$.

The sum is finite by the zero-tail statement in Lemma 61.

Let K be a nonempty connected finite simple graph of odd order $2k+1$, where $k \geq 0$ is an integer. Assume $\Delta(K) \leq 4$ and K is not 4-regular, meaning that some vertex has degree different from four. Then $\delta_2(K) \leq k$ and $c_2(K) \leq k$. If $e(K) = 4k+1$, then $k \geq 2$, $\delta_3(K) \leq k-1$, and $c_3(K) \leq k-1$. Each deletion bound supplies an actual induced residual and a matching covering every vertex of its threshold degree; the deleted set and matching can be fixed before choosing any certificate capacities or budgets.

If the graph with $e(K) = 4k+1$ is also factor-critical, meaning that deleting any one vertex leaves a perfect matching, then $\nu(K) = k$, a matching covers every degree-four vertex of K, $c_i(K) = 0$ for every integer $i \geq 4$, and $ps(K) \geq 1$. If $ps(K) = 1$, then $c_1(K) \geq k+1$.

More generally, for every nonempty connected factor-critical finite simple graph K with $\Delta(K) \leq 4$ and positive integer total degree defect

$d = \sum_{v\in V(K)}(4 - d_K(v))$,

one has $p_{cap}(K) \geq 4 - d$.

All graph orders, degrees, matching cardinalities, cost indices, costs and deletion counts are integers. The empty matching and empty deletion set are allowed.

Proof. We first construct a proper colouring of K using four colours. Every nonempty induced subgraph K[W] has a vertex of degree at most three in that subgraph. Otherwise all its vertices would have degree four in K[W] and in K, by $\Delta(K) \leq 4$. No edge would leave W. Connectedness would force $W = V(K)$, making K 4-regular contrary to the assumption. Delete such a vertex repeatedly until no vertices remain. Reinsert vertices in reverse order; each has at most three previously inserted neighbours, so one of the four colours is absent from those neighbours and can be assigned to it. This constructs a proper colouring.

For each proper four-colouring let W_1 be the union of its first two colour classes and W_2 the union of its last two classes. Among the finite nonempty set of such colourings, choose one minimizing

$\Phi = e(K[W_1]) + e(K[W_2])$.

If a vertex had at least three neighbours in its own W-part, it would have at most one neighbour in the other part. One of the other part's two colours would be absent from its neighbours. Recolouring the vertex with that colour preserves properness, removes at least three edges from the count Φ, and adds at most one. Thus Φ would strictly decrease, a contradiction. Both induced parts therefore have maximum degree at most two, and each is bipartite because it is the union of two independent colour classes.

Each component of either part is a path, an isolated vertex, or an even cycle. Indeed, a component containing a cycle cannot have further vertices at maximum degree two, and a component without a cycle has no branches and is a path; bipartiteness makes every cycle even. On a path v_1,..., v_l choose the edges v_1v_2, v_3v_4,..., leaving at most the last endpoint uncovered. This covers every degree-two vertex of the path. Alternating edges form a perfect matching of an even cycle. Taking these matchings componentwise shows that both induced parts are 2-good.

Relabel them so $|W_1| \leq |W_2|$. Since their orders sum to $2k+1$, $|W_1| \leq k$. Deleting W_1 leaves a 2-good graph, proving $\delta_2(K) \leq k$. The implication $c_r \leq \delta_r$ and its full certificate compatibility are provided by Lemma 86.

Now suppose $e(K) = 4k+1$. If $k = 0$ or $k = 1$, this would require respectively one edge on one vertex or five edges on three vertices, both impossible in a simple graph. Thus $k \geq 2$. If $|W_1| \leq k-1$, its deletion leaves maximum degree at most two and hence a 3-good graph, because there are no degree-three vertices to cover. It remains to consider $|W_1| = k$ and $|W_2| = k+1$.

We will use the following specialization of Lemma 86: every graph of maximum degree at most three with at most four degree-three vertices is 3-good. The component exclusion in that fact is automatic for odd $r = 3$, and its threshold $2b_r - 1$ is nine, so the stated specialization applies.

If some v in W_1 has at most three neighbours in W_2, let $J = K[W_2 \text{ union } \{v\}]$. Its maximum degree is at most three. Every degree-three vertex is either v or one of its at most three neighbours in W_2, so J has at most four such vertices. The preceding specialization makes J 3-good. This deletes W_1 minus {v}, whose size is k-1.

In the remaining case every v in W_1 has at least four neighbours in W_2. The maximum-degree bound forces exactly four cross-neighbours and no neighbour in W_1. Thus W_1 is independent, exactly 4k edges cross the partition, and $e(K[W_2]) = 1$. Since $4k > k + 1$, some w in W_2 has distinct neighbours u, v in W_1. Let

$J = K[(W_2 minus \{w\}) \text{ union } \{u, v\}]$.

The vertices u, v each have exactly three neighbours in J and are nonadjacent. Every other retained vertex has at most two neighbours among u, v and at most one within W_2 minus {w}, since that part has at most one edge. Thus $\Delta(J) \leq 3$. Any degree-three vertex outside {u, v} must be an endpoint of the sole possible edge within W_2 minus {w}. There are at most two such endpoints, so J again has at most four degree-three vertices and is 3-good. Its deleted set is $(W_1 minus \{u, v\}) \text{ union } \{w\}$, of size k-1. This proves $\delta_3(K) \leq k - 1$ and hence $c_3(K) \leq k - 1$ in every case.

Suppose in addition that K is factor-critical. A perfect matching of K-v has k edges for every vertex v, while a matching on $2k + 1$ vertices cannot have more than k edges. Thus $\nu(K) = k$. The average degree $4 - 2/(2k + 1)$ lies strictly between three and four, so there is a vertex of degree four and also a vertex z of degree at most three. A perfect matching of K-z covers every degree-four vertex. Lemma 61 gives $c_4(K) = 0$ and zero costs at all higher indices. Choose a degree-four vertex w. Factor-criticality gives $\nu(K - w) = k$, so the matching loss h of that fact is zero. Its covered-vertex deletion inequality and its nonnegative slack conclusion give

$ps(K) \geq ps(K - w) + 1 \geq 1$.

If $ps(K) = 1$, the definition of ps and the zero tail give

$c_1(K) + c_2(K) + c_3(K) = 3k$.

The two proved cost bounds imply $c_1(K) \geq 3k - k - (k - 1) = k + 1$.

Finally let K satisfy the hypotheses of the positive-defect assertion. Nonemptiness and factor-criticality force odd order $2k + 1$ for an integer $k \geq 0$ and $\nu(K) = k$. The integer d is even because $d = 4|V(K)| - 2e(K)$, and it is positive by assumption. For every graph, $\min(c_i, \nu) \leq c_i$ at each positive index, so Lemma 61 implies $p_{cap} \geq ps \geq 0$. Consequently $d \geq 4$ gives $p_{cap}(K) \geq 0 \geq 4 - d$.

The only remaining possibility is $d = 2$. Degree summation now gives $e(K) = 4k + 1$, and $d > 0$ ensures that K is not 4-regular. All preceding conclusions apply. The costs at indices at least four vanish, while the capped costs at indices one and two are at most k and the cost at index three is at most k-1. Therefore

$p_{cap}(K) \geq (4k + 1) - k - [k + k + (k - 1)] = 2 = 4 - d$.

This proves the last assertion, including the possible one-vertex graph in the $d \geq 4$ case. □

Remark. A proper four-colouring can be chosen so that paired colour classes have maximum degree two. When the total defect is two, moving at most two vertices between the retained and deleted parts leaves only four cubic vertices, which a matching must cover. The resulting cost saving pays the entire degree-four defect requirement.

Depends on. Lemma 61, Lemma 86.

Lemma 132 (Sparse triples and the potential of regular component collections). *Let s, $t \geq 1$ be integers, and let $n_1 \geq \cdots \geq n_s \geq 1$ and $m_1 \geq \cdots \geq m_t \geq 1$ be integer lists. Let F_1, F_2 be the respective vertex-disjoint unions of stars with these edge sizes, prescribed red and blue. Copies are ordinary subgraphs with vertex-disjoint target components. A graph arrows an ordered pair when every red-blue edge coloring contains the entire first target in red or the entire second target in blue.*

Put $M = s + t - 1$ and

$\ell_j = max\{n_i + m_l - 1 : 1 \leq i \leq s, 1 \leq l \leq t, i + l = j\}$

for every integer $2 \le j \le s+t$. For each integer $0 \le z \le M$ put $P_z = \sum_{j=z+2}^{s+t}(\ell_j - 1)$, with empty sum zero. For $0 \le z < M$ let A_z be the family of every ordered nonempty actual suffix pair $(F_1^{\alpha}, F_2^{\beta})$ with integers $0 \le \alpha < s$, $0 \le \beta < t$, $\alpha + \beta = z$; superscripts delete the specified number of leading whole components.

Fix an integer $0 \le q < M$. Put $D = \ell_{q+2} - 1$ and assume $D \ge 6$ is even. Let R be the largest integer r with $1 \le r \le M - q$ such that ℓ_{q+2},..., ℓ_{q+r+1} all equal $D+1$. For a finite simple graph J, let $e(J)$ and $\nu(J)$ be its edge count and maximum matching cardinality, and put $\Phi(J) = e(J) - \nu(J)$. Let H be finite and simple with $\delta = M - q - \nu(H) \ge 1$. Select $\kappa \ge 1$ distinct whole connected components C_1,..., C_κ of H, each D-regular, of odd order, possessing a near-perfect matching, and not isomorphic to K_{D+1}. A near-perfect matching leaves exactly one vertex uncovered. Let λ be the number of selected components satisfying $\nu(C_i) > D$. Suppose an integer k satisfies

$\lambda \le k < \kappa$, $q + 3k + 1 \ge D$, *and* $R \le 3\kappa$.

If H arrows every member of A_q, then

$\Phi(H) - P_q \ge (D/2 - 1)(\kappa - k) + \kappa - R$.

Consequently, strict failure of this inequality supplies a coloring of H avoiding both entire prescribed targets of some member of A_q. This coloring extends to avoidance of F_1, F_2 on the disjoint union of H with the q canonical stars of sizes ℓ_2,..., ℓ_{q+1}, and across every actual q-vertex set X in any finite simple graph G with $G - X = H$, preserving all H-edge colors.

There is no degree restriction on the other components of H and no upper bound on the matching numbers of the k selected components reserved to include all λ large components. All indices, component counts, orders, matching cardinalities, degrees, and deficits are integers in the stated domains. Unit stars and repeated sizes are included.

Proof. We use the matching theorem Lemma 107, the family potential theorem Lemma 56, and the canonical-pool and actual-vertex extensions Lemma 41. The proof constructs the three-vertex deletions, verifies all suffix allocations on the same complementary graph, and then compares their potentials.

First we prove the local deletion assertion needed for a selected component. Let C be connected, D-regular of odd order $2m+1$, with a near-perfect matching, and not K_{D+1}; here $D \ge 4$ is even. We claim that some three-vertex set Z_C satisfies

$\Phi(C) - \Phi(C - Z_C) \ge 3D - 2$. (1)

If an independent triple Z_C contains a vertex x exposed by a maximum matching W, then W has m edges and deleting the other two vertices removes at most two of them. The residual has 2m-2 vertices, so

$1 \le \nu(C) - \nu(C - Z_C) \le 2$.

Exactly 3D edges meet the independent triple. Subtracting the matching loss from this edge loss proves (1).

Suppose no independent triple contains a vertex exposed by any maximum matching of C. Apply Lemma 107 with every vertex marked and an inclusionwise maximal attaining set S. Such a set exists by finiteness. Since C has a near-perfect matching, its minimum full-vertex exposure is one. Thus C-S has $|S| + 1$ odd components, each factor-critical, and every maximum matching covers S and exposes its unique uncovered vertex x in one of those odd components.

If $|S| \ge 2$, take one vertex from each of two other odd components. Together with x they form a prohibited independent triple. If $|S| = 1$, let B be the other odd component. It is not a clique: if it were a clique of odd order b, every vertex of B would have degree $D = b - 1 + \epsilon$, where ϵ is zero or one according as it meets the unique vertex of S. Both D and b-1 are even, forcing $\epsilon = 0$ at every vertex. Then B would be a whole component of C, contrary to connectedness. Two nonadjacent vertices of B together with x again give the prohibited triple. Hence S is empty. Connectedness and Lemma 107 now imply that C is factor-critical. Every vertex can therefore be exposed by a maximum matching. Our assumption implies that C has no independent triple.

Write $n = |V(C)|$. Regularity and noncompleteness give $n > D + 1$. Since n is odd and D even, every vertex u has at least two nonneighbors v, w. They are adjacent, since u, v, w cannot be independent. The triple $Z_C = \{u, v, w\}$ spans exactly one edge. We show that $J = C - Z_C$ has a perfect matching.

If it did not, the full-vertex instance of Lemma 107 would supply an attaining set T with

$o(J - T) - |T| = |V(J)| - 2\nu(J) \ge 2$,

where o counts odd connected components. Representatives from distinct such components are independent in C, so $o(J-T) \le 2$. Consequently T is empty and J has exactly two odd components. There is no additional component, since three component representatives would be independent. Each of these two components is a clique, since a nonadjacent pair in one together with a vertex in the other would also be an independent triple.

The nonneighbors of any vertex of C form a clique of order n-1-D. A clique in a graph of maximum degree D has at most $D+1$ vertices, so $n \le 2D+2$ and, by odd parity, $n \le 2D+1$. Each of the two cliques in J has minimum degree at least D-3, because only three vertices were deleted. Its odd order is therefore at least D-1. Their orders sum to n-3, forcing $n = 2D+1$ and both orders D-1. Each vertex of J then has exactly two neighbors in Z_C. Counting edges between J and Z_C gives 4D-4 from J and 3D-2 from Z_C, since its induced triple has exactly one edge. Equality would imply $D = 2$, a contradiction. Thus J has a perfect matching. The matching loss is one and the edge loss is 3D-1, proving (1). This completes the local assertion without a factor-criticality hypothesis.

Now assume H arrows every A_q member and put $h = D/2$. Each selected C_i has order $2\nu(C_i)+1$. Simplicity gives order at least $D+1$, and equality would make it K_{D+1}. Therefore

$\nu(C_i) \ge h+1$. (2)

Reserve any k selected components including all λ whose matching number exceeds D. This is possible because $\lambda \le k < \kappa$. Let Q be the union of the remaining kappa-k components and let $\mu = \nu(Q)$. This union is nonempty, and each of its component matching numbers lies in $[h+1, D]$.

In each reserved component choose the triple supplied by (1), and let Z be their union; take Z empty if $k = 0$. The selected components are whole and pairwise disjoint. Edges and matching numbers are additive over whole-component unions, so

$|Z| = 3k$, $\Phi(H) - \Phi(H-Z) \ge k(3D-2)$. (3)

Indeed, each component outside the reserved collection is unchanged, and the matching and edge losses inside different reserved components add.

Set $q' = q+3k$ and $p = q'+\mu$. Additivity, (2), and the positive deficit give

$M - q = \nu(H) + \delta \ge \mu + k(h+1) + \delta$,

and hence

$M - p \ge k(h-2) + \delta > 0$. (4)

Thus $q' < M$ and $p < M$. These inequalities also ensure the nonempty actual suffix families needed below.

By Lemma 41, the canonical completion G_c consisting of H and the q disjoint canonical stars arrows the original pair. Let Y be the set of those q star centers. Deleting Y union Z leaves H-Z and isolated pool leaves. Denote this residual by H′ and let $J = H' - V(Q)$. The components of Q remain whole in H′, and

$\Phi(J) = \Phi(H-Z) - \Phi(Q)$. (5)

Only the actual $q' = q+3k$ vertices of Y union Z have been deleted.

We prove that J arrows every A_p member. List the matching numbers of the components of Q as w_1,..., w_l, where $l = \kappa - k$ and w_1 is largest. Each $w_i \le D \le q'+1$. If a subcollection of these components is colored entirely red and the others entirely blue, then the two matching numbers on Q are w and mu-w, where w is the sum of the selected w_i.

Every integer in $[0, q'+\mu]$ is the sum of an integer in [0,q′] and such a subset sum. To verify this, start with [0,q′] and add the choice 0 or w_1. The two integer intervals [0,q′] and $[w_1, q'+w_1]$ meet or are consecutive, since $w_1 \le q'+1$. Their union is the complete integer interval $[0, q'+w_1]$. If the weights through w_{i-1} have produced [0,A], adjoining 0 or w_i gives [0,A] union $[w_i, A+w_i]$; these meet or are consecutive because $A \ge q'+w_1 \ge w_i$. Finite induction proves the assertion, including $l = 1$ and $q' = 0$.

Fix legal cuts α, β with $\alpha+\beta = p$ and suppose a coloring of J avoids (F_1^α, F_2^β). Choose a representation $\alpha = a+w$ with $0 \le a \le q'$ and w a subset sum, and put $b = q'-a$. The identities $\alpha = a+w$ and $\beta = b+(\mu-w)$ imply $a \le \alpha < s$ and $b \le \beta < t$. Color the selected components of Q red and the others blue. In $H' = J$ disjoint union Q, a red copy of F_1^a would use at most w components in Q: one edge from each

such vertex-disjoint positive-edge component would form a red matching in Q. Its other components lie entirely in J. Removing at most w original components from F_1^a leaves stars that contain its suffix $F_1^{a+w} = F_1^{\alpha}$, by the nonincreasing sizes and trimming surplus spokes. This contradicts the J-coloring. The same argument, with matching quota mu-w, excludes blue F_2^b. Hence H' avoids this $A_{q'}$ member. Apply the actual-vertex extension Lemma 41 across precisely Y union Z, with cuts a, b. It gives an avoiding coloring of G_c, contrary to its arrowing. This proves that J arrows every A_p member.

The full coordinatewise envelope of A_p is exactly the natural p-tail. For completeness, a local candidate with index sum r corresponds to original indices with sum $p + r$. Conversely, given valid original indices i, j with $i + j = p + r$, $r \geq 2$, choose integers $\alpha \leq i - 1$, $\beta \leq j - 1$ with α, $\beta \geq 0$ and $\alpha + \beta = p$. Such a choice exists because $(i - 1) + (j - 1) = p + r - 2 \geq p$; both cuts are legal, and the retained local indices sum to r. The members all have component budget M-p. Thus the finite-family bound Lemma 56 applies on J and gives

$\Phi(H - Z) - \Phi(Q) = \Phi(J) \geq P_p$. (6)

Isolated pool leaves contribute neither edges nor matching edges.

Each component of Q with matching number w has $2w + 1$ vertices and $D(2w + 1)/2$ edges. Therefore

$\Phi(Q) = (D - 1)\mu + h(\kappa - k)$. (7)

The complete diagonal profile is nonincreasing: for a candidate on any diagonal after the first, at least one index exceeds one; decreasing that index gives a candidate on the preceding diagonal with at least as large a value. The first R entries of the natural q-potential tail are D, and every later entry is at most D-1. By (2),

$p - q = \mu + 3k \geq (h + 1)(\kappa - k) + 3k \geq 3\kappa \geq R$.

Together with $p < M$, this shows that all the R top entries, and only valid further entries, are removed between P_q and P_p. Consequently

$P_q - P_p \leq (D - 1)(\mu + 3k) + R$. (8)

Combining (3), (6), (7), and (8) yields

$\Phi(H) - P_q$

$\geq k(3D - 2) + (D - 1)\mu + h(\kappa - k) - [(D - 1)(\mu + 3k) + R]$

$= (h - 1)(\kappa - k) + \kappa - R$,

as asserted. The reserved component matching numbers entered only the lower bound (2).

If the displayed conclusion fails strictly, H cannot arrow every A_q member. Negating this finite conjunction supplies a particular actual member and an avoiding coloring of H. The two extension assertions follow from Lemma 41, using its canonical pool for the first and the specified actual set X for the second. Both preserve the H-edge colors and handle copies meeting deleted vertices through leaves. The $q = 0$ extensions are identities. Every extraction and extension used the entire sorted suffix forests, so unit stars and repeated sizes remain included. □

Remark. Sparse triples in a few whole components provide real deletion vertices that bridge all gaps between the matching quotas of the remaining components. Exact suffix coverage converts the extra deletion saving into a natural-potential lower bound.

Depends on. Lemma 41, Lemma 56, Lemma 107.

Lemma 133 (Degree splittings with even capacities or sparse odd cycles). *Let A, B be nonnegative integers, put $D = A + B$, and let G be a finite simple graph with maximum degree at most D. The maximum degree of an edgeless graph is zero. Put $S_D = \{vinV(G) : deg_G(v) = D\}$, and write $G[S_D]$ for its induced subgraph. Suppose either A and B are both even, or D is odd and $G[S_D]$ contains no two vertex-disjoint odd cycles. Then the edge set of G has a red-blue coloring in which every red degree is at most A and every blue degree is at most B.*

In the second case there exists a maximum matching of G covering every vertex of S_D. A matching consists of edges with disjoint endpoints, and maximum refers to its cardinality. The first case needs no covering-matching assumption. The conclusion includes $A = 0$, $B = 0$, $D = 0$, empty graphs and isolated vertices. All degrees, capacities, orders and matching cardinalities are integers in the indicated nonnegative domains.

Proof. First suppose A and B are both even. If $D = 0$ then G is edgeless and the empty coloring works. Assume $D > 0$ and put $r = D/2$, a positive integer. Form a finite loopless multigraph K from two labeled copies G^0, G^1 of G by adding $D - deg_G(v)$ distinct parallel edges between the corresponding vertices v^0, v^1 for each v in $V(G)$. Every vertex of K has degree D. Parallel edge instances are distinguished throughout. This construction is finite and contains G^0 as a subgraph. If G has no vertices the assertion already holds, so empty auxiliary graphs may be omitted.

We construct a circuit using every edge of each connected component of K. Start along unused edges, never repeating an edge. A trail cannot stop away from its starting vertex: at another terminal vertex, its used incidences have odd cardinality, whereas the total degree is even, so another unused edge is available. Thus it closes. Removing its edges leaves even degrees. If an edge of the component remains unused, connectedness gives a vertex of the current trail incident with an unused edge. One justification is to take a path from an unused edge to the trail and use its first intersection with the trail. Starting there in the unused graph produces another closed trail, which can be spliced into the first. Each splice adds at least one edge. Finiteness therefore yields a circuit through every edge of that component, and the same construction applies to all components.

Orient every edge along its circuit. Since the degree in K is 2r, every vertex has r incoming and r outgoing edges. Form a bipartite multigraph L with an outgoing copy and an incoming copy of each K-vertex. Every oriented edge xy of K becomes an edge from the outgoing copy of x to the incoming copy of y. All vertices of L have degree r, and its two parts have equal size.

We prove directly that every finite positive-degree regular bipartite multigraph has a perfect matching. In an r-regular bipartite graph, every set U of left vertices satisfies $|N(U)| \geq |U|$ by counting its $r|U|$ incident edge instances at the right neighbors, each of degree r. Choose a maximum matching. If a left vertex x is unmatched, follow nonmatching edges from left to right and matching edges from right to left, starting at x. If an unmatched right vertex were reached, a shortest such alternating path would have distinct vertices; reversing membership along it would increase the matching size. Thus every reached right vertex is matched. Let U and W be the reached left and right sets. Matching edges pair W with U minus {x}: every reached left vertex other than x is first reached by its matching edge, and the matching partner of every reached right vertex is reached. Every neighbor of U lies in W, because nonmatching edges extend reachability and matching partners are already reached. Hence $|N(U)| = |W| = |U| - 1$, contradicting the counting inequality. The maximum matching therefore covers the left part and, since the parts have equal size, both parts.

Remove a perfect matching from L. The remaining bipartite multigraph is (r-1)-regular; repeat this construction until degree zero. This partitions all L-edges into r perfect matchings. Select any $A/2$ of these classes. Color a K-edge red when its corresponding L-edge lies in a selected class, and blue otherwise. At every K-vertex each class has exactly one outgoing and one incoming edge. These are distinct edge instances because K has no loops. Thus its red degree is A and its blue degree is $D - A = B$. Restrict the coloring to G^0. Restriction cannot increase any color degree, proving the first case. This also handles $A = 0$ and $B = 0$ by selecting no classes or all classes, respectively.

Now suppose D is odd and $G[S_D]$ contains no two vertex-disjoint odd cycles. In particular $D \geq 1$. Let ρ be the minimum number of $S_D - vertices$ uncovered by a matching of G. Suppose for contradiction that $\rho \geq 1$. Apply Lemma 107 to G with marked set S_D, and choose its inclusionwise maximal maximizing separator X. It gives a family O of $q = |X| + \rho$ odd connected components of G-X, all of whose vertices lie in S_D, and each such component is factor-critical: deleting any one of its vertices leaves a perfect matching.

A nontrivial connected factor-critical graph contains an odd cycle. To verify this, a graph without an odd cycle is bipartite: choose a root in each component and assign vertices by parity of path length from that root. Two paths to the same vertex with different parity, or an edge joining equal-parity classes, would give an odd closed walk. A shortest odd closed walk has no repeated internal vertex, since splitting at a repetition leaves a shorter odd closed walk in one of the two parts. It is therefore an odd cycle. Thus the parity assignment is well-defined and every edge crosses its two classes. In a nontrivial connected bipartite graph both classes are nonempty. Deleting a vertex in the first class can leave a perfect matching only if the first class originally had one more vertex than the second. Deleting a vertex in the second class requires the reverse equality. Both cannot hold, proving the assertion about factor-critical graphs.

Every odd cycle in a member of O lies in $G[S_D]$, and distinct members have disjoint vertex sets. Therefore at most one member of O is nontrivial; all other members are singletons. A singleton in O has ambient degree D and all its neighbors in X, since it is a component of G-X. It sends exactly D edges to X.

If every member is a singleton, their boundary edges total Dq. Since $\rho \geq 1$ gives $q > |X|$, this exceeds $D|X|$. But all these edges have an endpoint in X, whose total degree is at most $D|X|$, a contradiction.

Otherwise let C be the unique nontrivial member. Its order is odd and every vertex has ambient G-degree D. Its boundary count β, which counts edges from C to X, satisfies

$\beta = D|V(C)| - 2e(C)$.

All edges leaving C go to X, because C is a component of G-X. Both D and $|V(C)|$ are odd, so β is an odd nonnegative integer and hence $\beta \geq 1$. The q-1 singleton members together with C consequently send at least $D(q-1)+1$ edges to X. Since $q = |X| + \rho \geq |X| + 1$, this is at least $D|X| + 1$, again impossible. This includes X empty.

It follows that $\rho = 0$. The maximum-matching construction in Lemma 107 supplies a matching of cardinality $\nu(G)$ attaining ρ, so this maximum matching covers all of S_D. Apply Lemma 35 to this same G, D and covering matching. Its degree-splitting implication gives a coloring with the required bounds for every nonnegative $A + B = D$, including zero capacities. Every matching, separator and auxiliary graph used above was selected from or constructed over a finite graph, and the edgeless cases have the indicated empty colorings and matchings. □

Remark. Even capacities are obtained by grouping 2-regular classes in a doubled regular completion. At odd total degree, any marked matching obstruction would need more required components than separator vertices; without two disjoint odd cycles, singleton degree counts and one odd boundary already exceed the separator's degree capacity.

Depends on. Lemma 35, Lemma 107.

Lemma 134 (Closed lumps of degree twenty-two are LP-good). *Use the integer certificate costs $c_\theta(H)$, matching number $\nu(H)$, excess $ex(H) = c_0(H) - \nu(H)$, and $gap(H) = \sum_{\theta \geq 0} c_\theta(H) - e(H)$ of Lemma 23. Let H be any nonempty connected 22-regular finite simple graph such that deleting each vertex leaves a perfect matching. Then $gap(H) \leq 0$. More precisely, write $|V(H)| = 2k+1$. The integer k equals $\nu(H)$ and satisfies $k \geq 11$. If $k \geq 31$, then $gap(H) < 217/30 - 31k/150 < 1$; if $11 \leq k \leq 30$, the explicit integer bounds in the proof are nonpositive. No hereditary or minimality hypothesis and no upper bound on $|V(H)|$ is imposed.*

Proof. Put $n = |V(H)|$. A perfect matching after any vertex deletion gives odd $n = 2k+1$ and $\nu(H) = k$. Simplicity gives $22 \leq 2k$, so $k \geq 11$, and regularity gives $e(H) = 11(2k+1)$. Set $E = ex(H) \geq 0$, an integer by Lemma 23.

We first prove $E \leq e_0 := floor((k+1)/3)$. For nonnegative integer budgets $x \geq y$, if $n \leq 2x + y + 1$, choose $\min(n, 2x+1)$ vertices, put their internal edges in the red exceptional part and all other edges in the blue exceptional part, and leave both bounded-degree parts empty. Every red matching has size at most x. Every blue edge meets the complement, whose size is at most y, so every blue matching has size at most y. For $y > x$ interchange the colors. Thus all allocations of total c are feasible at zero capacities whenever $c + \lceil c/2 \rceil + 1 \geq n$. Put $c = k + e_0$. For $k = 3h, 3h+1, 3h+2$, the left side equals $6h+1, 6h+3, 6h+6$ and the corresponding order is $6h+1, 6h+3, 6h+5$. The required inequality holds in all three residue classes, proving $c_0(H) \leq k + e_0$ and $E \leq e_0$.

The theorem Lemma 87 gives zero costs above 22, $c_{22}(H) \leq 1$, and $c_i(H) + c_{23-i}(H) \leq 2k+1$ and $c_i(H) + c_{22-i}(H) \leq 2k+2$ in the required positive-index ranges. Together with $0 \leq c_i(H) \leq c_0(H) = k + E$ from Lemma 23, these are all hypotheses of Lemma 12 at $d = 22$. Its sequence gap equals $gap(H)$. With $t = 3$ it gives, for any real bounds $b_j \geq c_{22-j}(H)$,

$gap(H) \leq -3k + b_1 + b_2 + b_3 + \min(5E - 3, 4E + 5)$. (1)

Fix any vertex v and perfect matching M of H-v. The graph H-M has degree 22 at v and 21 elsewhere. Apply Lemma 39 with $p_1 = 1/7$, $p_2 = 1/5$, $p_3 = 1/4$ and $j = 1, 2, 3$. Put $q_j = q_{21,j}(p_j)$ and $C_j = 22(1-p_j)^2 \, Pr(Bin(20, p_j) = j-1)$. It gives, for each j, a set containing v and j neighbors from every outside vertex in H-M, of size at most $1 + 2kq_j - C_j$. The theorem Lemma 49 with $D = 22$ converts this actual set into $c_{22-j}(H) \leq 1 + 2kq_j - C_j$. Every set belongs to the same supplied graph and may use the same v, M.

Put $(Q_1, Q_2, Q_3) = (1770, 2465, 3065)/10000$. The binomial expansions are

$q_1 = 1/7 + 6^{22}/7^{22}$,

$q_2 = 1/5 + 25 \cdot 4^{21}/5^{22}$,

$q_3 = 1/4 + 282 \cdot 3^{20}/4^{22}$.

Here $25 = 4 + 21$ and $282 = 9 + 21 \cdot 3 + 210$, using $binom(21, 2) = 210$. The differences $Q_j - p_j$ are $239/7000$, $93/2000$, $113/2000$. Thus $q_j < Q_j$ follows from the three explicit integer comparisons

$7000 \cdot 6^{22} = 921351926895869952000 < 934447230611334143711 = 239 \cdot 7^{22}$;

$2000 \cdot 25 \cdot 4^{21} = 219902325555200000 < 221729278564453125 = 93 \cdot 5^{22}$;

$2000 \cdot 282 \cdot 3^{20} = 1966546402164000 < 1987917023019008 = 113 \cdot 4^{22}$.

The corrections expand to $C_1 = 22 \cdot 6^{22}/7^{22}$, $C_2 = 88 \cdot 4^{21}/5^{21}$, and $C_3 = 4180 \cdot 3^{20}/4^{22}$. Each is greater than $7/10$ because

$220 \cdot 6^{22} = 28956774845298769920 > 27368747340080916343 = 7^{23}$;

$880 \cdot 4^{21} = 3870280929771520 > 3337860107421875 = 7 \cdot 5^{21}$;

$41800 \cdot 3^{20} = 145747587961800 > 123145302310912 = 7 \cdot 4^{22}$.

Consequently $c_{22-j}(H) < 3/10 + 2kQ_j$ for every $j = 1, 2, 3$. Their integer values are therefore at most $\lfloor 3/10 + 2kQ_j \rfloor$.

Since $\sum_j Q_j = 73/100$, use these real bounds and $E \le (k+1)/3$ in the second expression of (1) to obtain

$gap(H) < -3k + 9/10 + 2k \cdot (73/100) + 4(k+1)/3 + 5 = 217/30 - 31k/150$.

At $k = 31$ this is $124/150 < 1$ and it decreases with k. Since $gap(H)$ is an integer, it is nonpositive for every $k \ge 31$.

For $11 \le k \le 30$ apply Lemma 37 to H-M with $d = 22$, $n = 2k + 1$. Its recurrence starts at $r_0 = 2k - 22$ and has updates $r_{i+1} = \max(0, r_i - ceil(22r_i/(2k - i)))$. If s is its first zero index, its marked dominating set and Lemma 49 give $c_{21}(H) \le 1 + s$. Its second construction and the same certificate theorem give $c_{22-j}(H) \le 2k - 21 + j$ for $j = 2, 3$. Thus we can use $b_1 = 1 + s$ and $b_j = \min(\lfloor 3/10 + 2kQ_j \rfloor, 2k - 21 + j)$ for $j = 2, 3$. Replace E by e_0 in (1); both expressions increase with E. The table gives k; e_0; (b_1, b_2, b_3); the full recurrence through zero; and $U = -3k + sum\ b_j + \min(5e_0 - 3, 4e_0 + 5)$. Every recurrence entry follows from the displayed integer formula.

11;4;(1,3,4);(0);-8

12;4;(2,5,6);(2,0);-6

13;4;(2,6,8);(4,0);-6

14;5;(3,7,8);(6,1,0);-2

15;5;(3,7,9);(8,2,0);-4

16;5;(3,8,10);(10,3,0);-5

17;6;(4,8,10);(12,4,1,0);-2

18;6;(4,9,11);(14,5,1,0);-3

19;6;(4,9,11);(16,6,2,0);-6

20;7;(5,10,12);(18,8,3,1,0);-1

21;7;(5,10,13);(20,9,4,1,0);-3

22;7;(5,11,13);(22,11,5,2,0);-5

23;8;(6,11,14);(24,12,6,3,1,0);-1

24;8;(6,12,15);(26,14,7,3,1,0);-2

25;8;(7,12,15);(28,15,8,4,2,1,0);-4

26;9;(7,13,16);(30,17,9,5,2,1,0);-1

27;9;(7,13,16);(32,18,10,5,2,1,0);-4

28;9;(8,14,17);(34,20,12,7,4,2,1,0);-4

29;10;(8,14,18);(36,22,13,7,4,2,1,0);-2

30;10;(8,15,18);(38,24,15,9,5,3,1,0);-4.

Every last entry is nonpositive, so (1) proves the claim for every remaining integer k. □

Remark. The forced marked vertex supplies enough additional saving in the three top rows to control the former exceptional small orders without a dense-residual certificate.

Depends on. Lemma 12, Lemma 23, Lemma 37, Lemma 39, Lemma 49, Lemma 87.

Lemma 135 (Near-regular profiles and supported boundary credits). *Use the finite integer certificate costs $c_i(K)$, matching number $\nu(K)$, excess $ex(K) = c_0(K) - \nu(K)$, and slack $ps(K) = e(K) - \nu(K) - \sum_{i\geq 1} c_i(K)$ from Lemma 23; put $gap(K) = ex(K) - ps(K)$. A graph is factor-critical if it is nonempty and deletion of every vertex leaves a perfect matching. Let $D \geq 4$ be an integer and let K be a finite simple factor-critical graph with maximum degree at most D. Suppose its total degree defect $d = \sum_(v \in V(K))(D - deg_K(v))$ satisfies $0 < d < D$. Put $r = \nu(K)$, $u = (D-d)/2$, $T = floor((D-1)/2)$, and $E = ex(K)$.*

Then r, u are positive integers, D and d have the same parity, $|V(K)| = 2r+1 > D$, $\Delta(K) = D$, and $e(K) = Dr + u$. There is a matching covering every degree-D vertex. One has $ps(K) \geq 1$, $c_i(K) = 0$ for all integer $i \geq D$, $c_i(K) \leq r$ for all integer $i > T$, and

$c_i(K) + c_{D-i}(K) \leq 2r+1$ *for every integer* $1 \leq i < D$.

Define $p_{cap}(K) = e(K) - r - \sum_{i\geq 1} \min(c_i(K), r)$. Then

$p_{cap}(K) = u + \sum_{i=1}^{D-1}(r - \min(c_i(K), r))$.

If $D \geq 6$, then $c_{D-1}(K) \leq r - floor((D-2)/2)$. If $D \geq 6$ and either D is even or $d \geq 3$, then $r - c_{D-1}(K) \geq u$ and $p_{cap}(K) \geq D - d$.

Suppose additionally $gap(K) \leq 1$ and impose the following condition: for D in {4,5}, $p_{cap}(K) \geq D - d$; for odd $D \geq 7$ with $d = 1$, $c_{D-1}(K) \leq r - u$. No additional condition is imposed in the other cases. Write $\psi = ps(K)$, $h_i = \max(c_i(K) - r, 0)$, and $\delta = \max(D - d - \psi, 0)$. Define integers $a_1, \ldots, a_{D-1}$ by $a_1 = \min(\delta, h_1)$, by $a_i = 1$ when $2 \leq i \leq T$ and $i \leq 1 + \delta - a_1$, and by $a_i = 0$ at every other integer index $2 \leq i < D$. Then

$0 \leq a_i \leq h_i$, $\sum_{i=1}^{D-1} a_i = \delta$, $a_1 \leq u$, *and* $a_i \leq 1$ *for* $i \geq 2$.

Every active index i, meaning $a_i > 0$, satisfies $E \leq D - d - i + 1$. If also $E \geq 4$ implies $gap(K) \leq 0$, the stronger bound $E \leq \max(3, D - d - i)$ holds at every active index. For every $2 \leq j \leq T$ with $a_j > 0$,

$0 \leq r - c_{D-1}(K) \leq u + T - j$.

Moreover $\psi \geq \max(1, ceil((D - d - T)/2))$, and the nonnegative reduced costs $\mu_i = c_i(K) - a_i$ satisfy $\mu_i \geq \min(c_i(K), r)$. These assertions include $\delta = 0$ and empty index intervals. All degrees, defects, orders, ranks, costs, capacities, indices, credits, and matching budgets are integers in the stated positive or nonnegative domains; gap may be signed.

Proof. The definitions, finiteness, monotonicity, and identity $gap = ex - ps$ are those of Lemma 23. Since K is nonempty and factor-critical, its order is odd, say $2r+1$: a perfect matching of any vertex deletion has exactly half the remaining order, and no matching of the original odd-order graph can have more edges. Thus this r equals $\nu(K)$. Let $n = 2r+1$. If $n \leq D$, simplicity gives

$d \geq n(D - n + 1) = D + (n-1)(D-n) \geq D$,

contrary to the hypothesis. Hence $n > D$. If $\Delta(K) \leq D - 1$, every vertex contributes at least one to d and $d \geq n > D$, also impossible. Thus $\Delta(K) = D$. The degree-sum identity gives $d = Dn - 2e(K)$, so d has the parity of D, u is a positive integer, and $e(K) = Dr + u$.

Since $d > 0$, choose z of degree at most D-1, and choose a perfect matching M of K-z. This matching covers every degree-D vertex. The covering and zero-tail clauses of Lemma 61 give $c_i(K) = 0$ for all $i \geq D$. Choose w of degree D. A perfect matching of K-w has r edges, so $\nu(K - w) = \nu(K)$. The exact deletion identity of the same fact, in its covered case with matching loss zero, gives $ps(K) \geq ps(K - w) + 1 \geq 1$. In particular $r \geq 1$.

We prove the paired cost estimates by one finite induced-partition construction. Let J be the spanning graph obtained from K by removing exactly M. Every vertex except z loses one edge and z already has degree at most D-1; hence $\Delta(J) \leq D - 1$. Fix positive integers θ, η with $\theta + \eta = D$. Among all ordered bipartitions (U, W) of $V(J)$, choose one minimizing

$e(J[U])/\theta + e(J[W])/\eta$.

There are 2^n such bipartitions, a nonempty finite number, and the denominators are positive, so a minimizer exists. If a vertex in U has at least θ neighbors in U, it has at most eta-1 in W because its total J-degree is at most D-1. Moving that vertex to W changes the potential by at most $-1 + (\eta - 1)/\eta < 0$, a contradiction. The same argument for a vertex in W proves $\Delta(J[U]) \leq \theta - 1$ and $\Delta(J[W]) \leq \eta - 1$.

Restoring the edges of M inside U raises each degree by at most one. A vertex attaining degree θ in K[U] must be incident with a restored edge, so M intersect $E(K[U])$ covers every such vertex. This includes z if present, because z has no M-edge and its degree remains below θ. Thus Lemma 35 supplies every bounded-degree edge split of K[U] with capacity sum θ. Its exceptional parts may be empty, giving $c_\theta(K[U]) = 0$. Extending over W by Lemma 72 gives $c_\theta(K) \leq |W|$; this assigns every edge meeting W, including its internal edges, to the exceptional parts for every prescribed integer split of total $|W|$. The analogous construction gives $c_\eta(K) \leq |U|$. Addition proves $c_\theta(K) + c_\eta(K) \leq n = 2r + 1$ for every $\theta + \eta = D$. If $i > T$ and $i < D$, then $i \geq D - i$. Monotonicity gives $2c_i(K) \leq c_i(K) + c_{D-i}(K) \leq 2r + 1$, so integrality yields $c_i(K) \leq r$. For $i \geq D$ the costs are already zero. This proves every paired and upper-half bound, on the actual original graph.

Substitute $e(K) = Dr + u$ and the zero rows at $i \geq D$ in the definition of p_{cap}. Direct subtraction gives its displayed identity; each summand is nonnegative. When $D \geq 6$ apply Lemma 116 to K, the already fixed low-degree vertex z, and the already fixed matching M. Its actual deletion construction gives

$c_{D-1}(K) \leq \max(0, r - floor((D-2)/2))$.

The second argument is positive because $2r + 1 > D$; thus the maximum can be removed. For even D, positivity and parity give $d \geq 2$, and $floor((D-2)/2) = (D-2)/2 \geq u$. For odd D with $d \geq 3$, the floor is $(D-3)/2 \geq u$. In these cases the tail saving is at least u, and the p_{cap} identity gives $p_{cap} \geq 2u = D - d$. In the remaining odd one-defect case the extra hypothesis stated for the credit construction gives this same tail saving; no such improvement has been assumed in the unconditional part.

We now apply the numerical credit theorem Lemma 2 to the actual entries $c_0(K)$,..., $c_{D-1}(K)$, with its r, E, D, d unchanged. The sequence is nonincreasing and nonnegative by Lemma 23, $c_0 = r + E$, and the upper-half and paired bounds were just proved. Since the higher rows vanish,

$\psi = (D-1)r + u - \sum_{i=1}^{D-1} c_i(K)$.

We proved $\psi \geq 1$, and $gap \leq 1$ gives $E \leq \psi + 1$. For D_4 or D_5, direct subtraction shows $p_{cap} = \psi + \sum_{i=1}^{T} h_i$, so the explicitly imposed capped bound is exactly the additional numerical hypothesis. For $D \geq 6$, the established tail saving, or the explicit odd one-defect hypothesis, gives $r - c_{D-1} \geq u$. All hypotheses of the numerical theorem are therefore met. It yields the displayed credit construction, exact total and support, first-coordinate $\cap$, active-index bound $E \leq D - d - i + 1$, tail upper bound, slack lower bound, and reduced-cost inequalities.

For completeness, the strengthened active-index estimate under the last gap implication uses its actual quantifier. That implication and $gap \leq 1$ give $E \leq \max(3, \psi)$: if $E \geq 4$, $gap \leq 0$ says $E \leq \psi$; if $E \leq 3$ the inequality already holds. If $a_1 > 0$, then $\delta > 0$ and $\psi \leq D - d - 1$. If $a_i > 0$ for $i \geq 2$, the explicit credit formula gives $\delta > h_1$ and $\delta - h_1 \geq i - 1$. Its support and monotonicity give $h_1 \geq h_i \geq 1$, so $\delta \geq i$ and $\psi = D - d - \delta \leq D - d - i$. In both cases $E \leq \max(3, D - d - i)$, as asserted. Zero credits require no active-index conclusion, and every empty sum in the construction is already permitted by Lemma 2. □

Depends on. Lemma 2, Lemma 23, Lemma 35, Lemma 61, Lemma 72, Lemma 116.

Lemma 136 (Low-degree hosts of actual suffix families). *Use the positive star profiles, ordinary prescribed-color copies and arrowing of Lemma 29. All graphs are finite and simple. Write $e(G)$, $\Delta(G)$ and $\nu(G)$ for the edge count, maximum degree and maximum matching cardinality of G. An edgeless graph has maximum degree zero. A path P_j and a cycle C_j have j vertices; put $K2 = P_2$ and $P3 = P_3 = K_{1,2}$. Components in a copied forest have disjoint vertex sets.*

Let a, b be positive integers and u, v nonnegative integers. If a graph G of maximum degree at most two arrows

(aP3 disjoint union uK2, bP3 disjoint union vK2),

then

$e(G) \geq 3(a+b-1) + 2max(u,v) + \min(u,v)$.

More generally let s, t be positive integers, let $n_1 \geq \cdots \geq n_s \geq 1$ and $m_1 \geq \cdots \geq m_t \geq 1$ be integer profiles, and let F_1, F_2 be their vertex-disjoint star unions, prescribed red and blue. Put $M = s + t - 1$ and

$\ell_k = max\{n_i + m_j - 1 : 1 \leq i \leq s, 1 \leq j \leq t, i + j = k\}$

for every integer $2 \leq k \leq s + t$. Fix an integer $0 \leq q < M$ and put

$L_q = \sum_{k=q+2}^{s+t} \ell_k$, $D = \ell_{q+2} - 1$.

Let A_q be all ordered pairs $(F_1^{\alpha}, F_2^{\beta})$ with integer cuts $0 \leq \alpha < s$, $0 \leq \beta < t$ and $\alpha + \beta = q$, where each superscript deletes exactly that many leading whole components.

Suppose $0 \leq D \leq 2$. Every graph H with $\Delta(H) \leq D$ that arrows every entire member of A_q satisfies $e(H) \geq L_q$. Equivalently, every such degree-capped H with $e(H) < L_q$ has one red-blue edge coloring avoiding both entire targets of some actual member of A_q. In fact no common arrowing H under this degree $\cap$ exists when $D = 0$ or $D = 1$.

All orders, multiplicities, indices, cuts, edge counts and matching cardinalities are integers in the stated domains. Empty hosts, isolated host vertices, repeated target sizes and all unit components are included. The first assertion requires both targets to contain a P_3; the actual-suffix assertion retains every positive profile allowed above.

Proof. We use Lemma 103 in precisely this form: every finite simple graph of maximum degree at most two has every matching-budget allocation at any integer total at least its matching number plus one. Such an allocation is a partition of its original edges into red and blue parts whose matching numbers are bounded by the two specified nonnegative integer budgets. All profile and suffix definitions are those of Lemma 29.

First prove the assertion for the two displayed forests. Exchanging targets and colors, assume $u \geq v$. Put $r = a + b - 1$ and $L = 3r + 2u + v$, and suppose G arrows this pair with $e(G) < L$.

We construct a maximum matching Q of G. Each whole connected component is a path or a cycle: starting at an endpoint in a component of degree at most two visits every vertex as a path, while a component with every degree two is a cycle. Isolates are paths of order one. Label each component consecutively. On P_{2k}, P_{2k+1}, C_{2k} or C_{2k+1}, choose the edges (1,2), (3,4),..., (2k-1,2k). The path cases allow $k = 0$ when the order is one; cycle orders are at least three. These matchings have $\lfloor j/2 \rfloor$ edges on a component of order j, which is maximum since matching edges have disjoint endpoints. Their union Q is therefore maximum.

Let $J = G - E(Q)$, retaining all vertices. On an even path P_{2k}, $k \geq 1$, J has k-1 single-edge components. On an odd path P_{2k+1}, it has k single-edge components. On an even cycle C_{2k}, it has k single-edge components. On an odd cycle C_{2k+1}, it has one P_3, using the two edges incident with vertex $2k + 1$, and k-1 single-edge components. Other vertices in these descriptions are isolated. Let c be the number of whole odd cycles of G, let p be the number of whole even-order paths with at least one edge, and let h be the number of positive-edge components of J. Thus J is a vertex-disjoint union of c copies of P_3 and h-c single edges, together with isolates. The displayed component counts give

$\nu(G) = h + p$, $e(J) = h + c$, $e(G) = 2h + c + p$.

The graph J arrows (aP3 disjoint union uK2,bK2). Otherwise color its edges to avoid that pair and restore all Q-edges in blue. A blue copy of bP3 disjoint union vK2 in G would have, in each of its b disjoint P3s, an edge outside Q, because two adjacent edges cannot both belong to a matching. One such edge from each P_3 would give b disjoint blue edges in J, a contradiction. The red target is unchanged when only blue edges are restored. This proves the asserted arrowing by J.

Consequently $c \geq r$ and $h \geq r + u$. Indeed, if $c \leq a + b - 2$, color $\min(c, a - 1)$ of the P_3 components of J red and its remaining P_3 components blue, and color every single-edge component red. There are at most a-1 red P_3 components and at most b-1 blue positive-edge components. Every P_3 in J lies in one of its P_3 components, so the red target is absent; the blue matching number is at most b-1. This contradicts the preceding arrowing. If instead $h \leq a + b + u - 2$, color $\min(h, a + u - 1)$ positive-edge components red and all other components blue. Each positive-edge component has matching number one. The red and blue matching numbers are at most $a + u - 1$ and b-1, respectively, excluding both targets and giving the second contradiction.

Write $c = r + x$ and $h = r + u + y$, with nonnegative integers x, y. From the edge identity and $e(G) \leq L - 1$,

$x + 2y + p \leq v - 1$.

Put $A = a + u - 1$ and $B = b + v - 1$, nonnegative integers. Then

$A + B - \nu(G) = v - 1 - y - p \geq x + y \geq 0$.

If $A + B - \nu(G) \geq 1$, the stated consequence of Lemma 103 supplies an original-edge coloring with red matching number at most A and blue matching number at most B. The red target contains a matching of $a + u = A + 1$ edges and the blue target one of $b + v = B + 1$ edges. That coloring avoids both, a contradiction. Thus $A + B - \nu(G) = 0$. The displayed inequalities force $x = y = 0$ and $p = v - 1$. In particular $v \geq 1$, $c = r$ and $\nu(G) = r + u + p$.

Choose b distinct whole odd cycles of G, possible since $c = r = a + b - 1 \geq b$. In a chosen triangle color all edges blue. In any other chosen odd cycle color exactly three consecutive edges blue; they form a P_4. In every one of the p positive even-order path components of G, color one end edge blue. Color all remaining original edges red.

The positive-edge blue components are b blocks, each a triangle or a P_4, and p single edges. Each block contains at most one member of a family of vertex-disjoint P3s, since it has at most four vertices. Hence b disjoint blue P3s must use one from each block. After removing their vertices, every block has at most one vertex and contains no edge. Only the $p = v - 1$ blue single edges remain available, which cannot supply v additional disjoint blue edges. Thus the entire blue target is absent.

On a chosen triangle the red matching number decreases from one to zero. On a chosen C_{2k+1} with $k \geq 2$, removing the three consecutive blue edges leaves, apart from isolates, a path with 2k-2 edges and matching number k-1, again a decrease of one. On an even path P_{2k}, deleting its blue end edge leaves an isolate and a P_{2k-1}, decreasing its matching number from k to k-1. All other original components are wholly red. Matching numbers add over whole components, since matchings restrict to components and component matchings can be united. Therefore the red matching number is

$\nu(G) - b - p = r + u - b = a + u - 1$,

excluding the entire red target. This avoiding coloring is the final contradiction and proves $e(G) \geq L$.

Now consider the actual suffix family in the statement, and assume that H arrows every member with $\Delta(H) \leq D \leq 2$. A legal suffix member whose red head exceeds D is avoided by coloring all H-edges red: that head cannot embed and the blue graph has no edge. The same argument with blue applies if its blue head exceeds D. Thus every head in every actual suffix member is at most D.

Choose indices i_0, j_0 attaining ℓ_{q+2}, with $i_0 + j_0 = q + 2$. The cuts $\alpha = i_0 - 1$ and $\beta = j_0 - 1$ are legal and sum to q. If $D = 0$, both nonempty targets contain an edge while H is edgeless, so no arrowing is possible. If $D = 1$, then $n_{i_0} + m_{j_0} = 3$, and one of these heads exceeds one, again impossible. It remains that $D = 2$. The same head bound and $n_{i_0} + m_{j_0} = 4$ force $n_{i_0} = m_{j_0} = 2$.

Every entry in a pair of original indices i, j with $i + j \geq q + 2$ is at most two. To check this, choose $\alpha' = \min(q, i - 1)$ and $\beta' = q - \alpha'$. These are nonnegative, sum to q, and satisfy $\alpha' \leq i - 1 < s$ and $\beta' \leq j - 1 < t$. The last inequality follows either from $\beta' = 0$ or from $q - (i - 1) \leq j - 1$. These legal suffixes begin no later than i, j, and their heads are at most two, so $n_i, m_j \leq 2$.

Let A_2 be the number of original n-entries at least two and B_2 the number of original m-entries at least two, and put $u = s - A_2$, $v = t - B_2$. The chosen suffix member has exactly

$a = A_2 - i_0 + 1 \geq 1$, $b = B_2 - j_0 + 1 \geq 1$

two-edge stars and retains all u and v original unit stars. Its value $r = a + b - 1$ equals $A_2 + B_2 - q - 1$.

The natural tail consists of r entries three, then $\max(u,v)$ entries two, then $\min(u,v)$ entries one. Here is a direct check that also handles original entries larger than two. For every k from $q+2$ through A_2+B_2, increase the coordinates of (i_0,j_0) within their respective intervals ending at A_2, B_2 until their sum is k. Both corresponding entries are at least two, and the preceding paragraph makes them at most two. Their value is therefore three, the largest possible tail value. If $k > A_2+B_2$, both entries cannot be at least two. If $v \geq u$, every k from A_2+B_2+1 through A_2+B_2+v is attained by $i=A_2$, $j=k-A_2$, giving one entry two and one entry one. If $u>v$, use $i=k-B_2$, $j=B_2$ over the analogous interval through A_2+B_2+u. These give precisely the asserted entries two. For $k > \max(A_2+t, s+B_2) = A_2+B_2+\max(u,v)$, any valid pair has $i > A_2$ and $j > B_2$, and hence both entries one. This accounts for all tail indices through $s+t$.

It follows that $L_q = 3(a+b-1)+2max(u,v)+\min(u,v)$. The same H arrows the selected actual member (aP3 disjoint union uK2,bP3 disjoint union vK2), and has maximum degree at most two. The first part of the proof gives $e(H) \geq L_q$. This proves the second assertion. Negating common arrowing gives an actual member and a coloring avoiding both of its entire targets, which establishes the equivalent below-budget formulation. Every construction assigns the original edges and allows isolated vertices and empty edge parts. □

Remark. A maximum matching leaves only two-edge stars and single edges. The resulting arrowing constraints almost force the edge bound; at equality, short blue paths inside selected odd cycles give a coloring that preserves too few disjoint blue units. A legal binary suffix then owns the whole natural tail at critical degree two.

Depends on. Lemma 29, Lemma 103.

Lemma 137 (Excluding an odd clique at an even terminal rectangle). *Let $s, t \geq 1$ be integers and let $n_1 \geq \cdots \geq n_s \geq 1$ and $m_1 \geq \cdots \geq m_t \geq 1$ be integer lists. Let F_1, F_2 be their respective vertex-disjoint unions of stars K_{1,n_i}, K_{1,m_j}, prescribed red and blue. All graphs are finite and simple. Copies are ordinary subgraphs with vertex-disjoint target components. Arrowing an ordered pair means that every red-blue edge coloring contains its entire first target in red or its entire second target in blue.*

Put $M = s+t-1$ and $\ell_k = max\{n_i+m_j-1 : 1 \leq i \leq s, 1 \leq j \leq t, i+j=k\}$ for every integer $2 \leq k \leq s+t$. Fix an integer $0 \leq q < M$, put $K = M-q$ and $P_q = \sum_{k=q+2}^{s+t}(\ell_k-1)$, and suppose $D = \ell_{q+2}-1$ is even and $D \geq 6$.

Let R be the number of initial entries equal to $D+1$ in the tail ℓ_{q+2},..., ℓ_{s+t}. Let A_q contain every pair (F_1^α, F_2^β) with integers $0 \leq \alpha < s$, $0 \leq \beta < t$ and $\alpha+\beta = q$, where a superscript deletes that many leading whole components.

Suppose integers $a, b \geq 2$ satisfy $a+b = D+2$. Put $N_a = |\{i : n_i \geq a\}|$ and $T_b = |\{j : m_j \geq b\}|$, and assume N_a, $T_b \geq 1$ and $N_a+T_b-1 = q+R$. Define

$\mu = D/2$, $n = D+1$, $\pi = floor(n/(a+1)) + floor(n/(b+1)) - 1$.

Let $\sigma \geq 0$ be an integer satisfying $\sigma+\pi < \mu$. If H has a whole connected component isomorphic to the complete graph K_{D+1}, and

$\Phi(H) := e(H) - \nu(H) \leq P_q + \sigma$,

$\delta := K - \nu(H) \geq 1$,

then H has a coloring avoiding both entire prescribed targets of some member of A_q. Here e and ν are edge count and maximum matching cardinality.

In particular the conclusion holds whenever $4 \leq a < b$ are even and $0 \leq \sigma \leq a-2$. Every resulting avoiding coloring extends, preserving all H-edge colors, to avoidance of F_1, F_2 on H disjoint union S_q, where S_q consists of stars of sizes ℓ_2,..., ℓ_{q+1}. It also extends across every specified q-vertex set X in any finite simple G with $G-X = H$. There is no degree restriction outside the selected clique, original-head bound, short-prefix assumption or smaller-target induction hypothesis. All sizes, counts, degrees, indices, matching cardinalities, potentials and parameters have the displayed integer domains. All unit stars and repeated components are retained; $q=0$ is allowed.

Proof. Assume for contradiction that H arrows every member of A_q. Let C be a selected whole K_{D+1} component and $Q = H - V(C)$. Since D is even, a matching pairing D vertices of C is maximum, so $|V(C)| = 2\mu+1 = n$ and $\nu(C) = \mu$.

We first show that both full originals have more than n vertices. Write $A = n_1$ and $B = m_1$ for their heads. The family bound in Lemma 89 applies with the present positive deficit. It gives

$s \geq q + 2\delta + 2sum_j\ floor((m_j - 1)/2) \geq B,$

$t \geq q + 2\delta + 2sum_i\ floor((n_i - 1)/2) \geq A.$

Indeed $q \geq 0$, $\delta \geq 1$, and $2floor((B-1)/2) \geq B - 2$; the other inequality uses A in the same way. Also $A + B \geq D + 2$ because the head sum is at least the sum of any pair of original sizes attaining ℓ_{q+2}. A star forest with s positive components and head A has at least $A + 2s - 1$ vertices. Consequently

$|V(F_1)| \geq A + 2s - 1 \geq A + 2B - 1 \geq D + B + 1 > D + 1.$

Interchanging A, B and s, t gives $|V(F_2)| > D + 1$. This reasoning concerns the complete original targets.

We construct residual forests for the common vertex capacity n. For i in {1,2} and integer $j \geq 1$, let $N_i(j)$ count the components of F_i having size at least j. Let $k_i(j)$ be the largest cardinality of a subcollection of those components using at most n vertices; the empty subcollection is permitted. Set

$c_i(j) = \max_{z \geq j}(N_i(z) - k_i(z)),$

where all quantities are zero above the corresponding original head. These maxima are attained over a finite set of relevant thresholds. Since $0 \leq k_i(j) \leq N_i(j)$, the numbers $c_i(j)$ are nonnegative integers, nonincreasing in j, at most $N_i(j)$, and eventually zero. They therefore define a star forest U_i with $c_i(j)$ components of size at least j. The inequalities $c_i(j) \leq N_i(j)$ imply that U_i embeds componentwise in F_i: after sorting, every required size is at most the size in the same original position. Both U_i are nonempty. If $c_i(1) = 0$, then $k_i(1) = N_i(1)$, so all components of F_i fit within n vertices, contrary to the preceding paragraph.

These forests have the following property for every integer leading cut $0 \leq \beta < N_i(1)$. Suppose a copy of the suffix F_i^{β} lies in a disjoint union of an n-vertex graph and another graph. Each copied star, being connected and having an edge, lies wholly in one of these two graphs. At threshold z, at most $k_i(z)$ copied components can be in the n-vertex graph. Thus the second graph contains at least $(N_i(z) - \beta - k_i(z))_+$ components of size at least z, where $x_+ = \max(x, 0)$. For every j, its count of components of size at least j is at least the maximum of these bounds over $z \geq j$, namely $(c_i(j) - \beta)_+$. These are the threshold counts of the suffix of U_i obtained by deleting its β leading components, with all counts zero if $\beta \geq c_i(1)$. Comparison of sorted sizes, followed by trimming surplus spokes, gives a copy of U_i^{β} whenever that suffix is nonempty.

Let E_q be the union of all nonempty actual total-q suffix pairs of (U_1, F_2) and (F_1, U_2). It is a finite family, possibly empty. Let P_{end} be its complete diagonal-envelope potential, defined as the sum of every envelope entry minus one, up to its maximum component budget, and set $P_{end} = 0$ if the family is empty. We prove

$0 \leq P_q - P_{end} \leq (D-1)\mu + \pi.$ (1)

For a positive integer h, let W_h be the number of entries of the full original profile ℓ_2,..., ℓ_{s+t} that are at least $h + 1$. Its threshold formula is

$W_h = \max_{(}\ x + y = h + 2, N_1(x), N_2(y) > 0)(N_1(x) + N_2(y) - 1),$

with maximum zero if there is no eligible pair. To justify it, the rectangle of indices whose red size is at least x and whose blue size is at least y supplies every diagonal position through $N_1(x) + N_2(y) - 1$ at excess at least h. Conversely, if original sizes have sum at least $h + 2$, choose positive integers x, y below these sizes with $x + y = h + 2$. This places their indices in such a rectangle. The full diagonal profile is nonincreasing: lowering an index greater than one in a later-diagonal candidate gives an at-least-as-large earlier candidate.

The corresponding width for all actual total-q suffixes is $(W_h - q)_+$. Here is the precise cut correspondence. A local candidate with indices u, v after cuts α, β has original indices $\alpha + u$, $\beta + v$. Conversely, if original indices u, v satisfy $u + v = q + l$ with $l \geq 2$, take $\alpha = \min(q, u - 1)$, $\beta = q - \alpha$. Then $\beta \leq v - 1$, both cuts are legal, and the same sizes occur on local diagonal l. This proves the complete envelope shift. The argument also applies to either endpoint pair when its full component budget exceeds q. If that budget is at most q, no nonempty total-q suffix pair exists and all its truncated widths are zero.

Let W_h' be the maximum of the two full widths obtained by replacing one original by its U_i. The width of E_q is $(W_h' - q)_+$. Componentwise embedding gives $W_h' \leq W_h$. Define

$\Delta_h = (W_h - q)_+ - (W_h' - q)_+ \geq 0.$

There are no positive original natural widths above D. Summing the integer superlevel widths therefore gives $P_q - P_{end} = \sum_{h=1}^{D} \Delta_h$, including an empty E_q.

We claim $\Delta_h \le \mu$ for every $1 \le h \le D$. There is nothing to prove if the original natural width is zero. Otherwise choose x, y attaining W_h. If $x = 1$, use the endpoint (U_1, F_2). Its first threshold survives because U_1 is nonempty, and its count loss is at most

$N_1(1) - c_1(1) \le k_1(1) \le \lfloor n/2 \rfloor = \mu$.

The positive-part truncation by q cannot increase this loss. If $y = 1$ use (F_1, U_2) with the same argument. If $x, y \ge 2$ and either $c_1(x)$ or $c_2(y)$ is positive, use that surviving endpoint threshold: its loss is at most $k_i(j) \le floor(n/(j+1)) \le \mu$.

It remains that $x, y \ge 2$ and both corresponding remnant counts vanish. Each entire eligible original group then fits in n vertices, since $k_1(x) = N_1(x)$ and $k_2(y) = N_2(y)$. Each group contains its own full head. Hence A, $B \le 2\mu$, and counting its head separately yields

$N_1(x) \le 1 + (2\mu - A)/(x+1)$,

$N_2(y) \le 1 + (2\mu - B)/(y+1)$.

The numerators are nonnegative and $x+1, y+1 \ge 3$. Thus

$\Delta_h \le W_h = N_1(x) + N_2(y) - 1$

$\le 1 + (4\mu - A - B)/3 \le (2\mu + 1)/3 \le \mu$.

We used $A + B \ge D + 2 = 2\mu + 2$ and $\mu \ge 3$. This establishes the uniform claim without restricting either original head.

At height D there is a better bound. Put $k_a = floor(n/(a+1))$ and $k_b = floor(n/(b+1))$. Both are at least one because a, $b \ge 2$ and $a + b = D + 2$ imply a, $b \le D$. The specified rectangle has width $N_a + T_b - 1 = q + R = W_D$. If $c_1(a) > 0$, the endpoint using U_1 loses at most k_a at this rectangle. If $c_2(b) > 0$, its endpoint loses at most k_b. Each of these losses is at most $k_a + k_b - 1 = \pi$. If both vanish, the two complete threshold groups fit, so $N_a \le k_a$ and $T_b \le k_b$. Then $W_D \le \pi$, and $\Delta_D \le W_D \le \pi$. In every case $\Delta_D \le \pi$. Summing it and the other D-1 bounds proves (1).

Matching number and edge count add over whole components: every matching restricts componentwise and unions of maximum component matchings attain the resulting sum. Since $e(C) = D(D+1)/2$, we have $\Phi(C) = Dmu = (D-1)\mu + \mu$. It follows from (1) that

$\Phi(Q) = \Phi(H) - \Phi(C)$

$\le P_q + \sigma - (D-1)\mu - \mu$

$\le P_{end} + \sigma + \pi - \mu < P_{end}$. (2)

Every graph has nonnegative potential because a matching is an edge subset. Thus (2) also proves E_q is nonempty. Apply Lemma 56 to this finite family: Q has a coloring avoiding both full targets of one E_q member.

Suppose this member comes from (U_1, F_2), with cuts α, β. Color every edge of C red and retain the coloring of Q. A red F_1^{α} would leave a red U_1^{α} in Q by the proved n-vertex residual property, contradicting the selected coloring. Every blue edge lies in Q, so the full blue F_2^{β} is also absent. The selected cuts are legal for F_1, F_2 because U_1 embeds componentwise in F_1. If the member instead comes from (F_1, U_2), color C blue; the same residual property excludes F_2^{β}, while every red edge lies in Q. In either case H avoids a member of A_q, contradicting the assumed simultaneous arrowing. This proves the main assertion.

For its stated specialization, write $b = a + 2w$ with integer $w \ge 1$. Then $\mu = a + w - 1$ and $n = 2a + 2w - 1$. We have $b + 1 \le n < 2(b+1)$, so $k_b = 1$. Also

$(w+1)(a+1) - n = (w-1)(a-1) + 1 > 0$,

so $k_a \le w$ and $\pi = k_a \le w$. Therefore $\sigma + \pi \le a - 2 + w = \mu - 1 < \mu$, as required.

Finally apply Lemma 41 (I), (II) to the avoiding actual A_q member. It gives the extension to the specified canonical pool and to every actual q-vertex set, preserving the entire coloring of H, including target intersec-

tions at leaves. All component capacities, residual comparisons and selected endpoint families retained every original unit and repeated size. □

Remark. Removing the clique costs more potential than the two possible target remnants can lose. The full matching-capacity bound ensures that both remnants survive. At the top threshold, the unequal even terminal sizes make the two fitting capacities small enough to pay the entire allowed surplus.

Depends on. Lemma 41, Lemma 56, Lemma 89.

Lemma 138 (Strict potential from a full matching projection). *Let $s, t \geq 1$ be integers and $n_1 \geq \cdots \geq n_s \geq 1$, $m_1 \geq \cdots \geq m_t \geq 1$ be integer lists. Let F_1, F_2 be the vertex-disjoint unions of stars with these edge sizes, prescribed red and blue. Assume each forest has at least one component of size at least two. Let u, v be their respective numbers of one-edge components and put $U = \min(u, v)$, $M = s + t - 1$,*

$\ell_k = max\{n_i + m_j - 1 : 1 \leq i \leq s, 1 \leq j \leq t, i + j = k\}$ *for* $2 \leq k \leq s + t$.

Fix an integer $0 \leq q < M$. Let A_q contain every ordered pair of nonempty actual suffix forests obtained by deleting α and β leading whole components, with integers $0 \leq \alpha < s$, $0 \leq \beta < t$ and $\alpha + \beta = q$. Put $K = M - q$,

$P_q = \sum_{k=q+2}^{s+t} (\ell_k - 1)$, $D = \ell_{q+2} - 1$, $N = K - U$,

and assume $D \geq 3$. Let R be the length of the initial run of entries equal to $D + 1$ in $\ell_{q+2}, \ldots, \ell_{s+t}$.

Suppose some member of A_q has a suffix F in a color whose original unit count equals U, and F contains $p \geq 1$ components of size at least two. Write J for its entire opposite suffix, $c(F)$ and $c(J)$ for their positive component counts, and F^- for the forest obtained by shortening each nonunit component of F by one edge and discarding its unit components.

If

$\max(e(J), e(F) - c(F)) \geq N - R + 1$,

then every finite simple graph H with maximum degree at most D which arrows every entire prescribed-color pair in A_q satisfies

$e(H) - \nu(H) \geq P_q + 1$,

where ν denotes maximum matching cardinality and copies are ordinary subgraphs with vertex-disjoint components. The first alternative $e(J) \geq N - R + 1$ is equivalently $p \leq R + e(J) - c(J)$.

All indices, counts, sizes and graph quantities are integers; in this domain $1 \leq R \leq N \leq K$, $c(F) = p + U$, $p + c(J) - 1 = N$ and $e(F^-) = e(F) - c(F)$. The assertion permits both original unit counts to be zero, repeated sizes, arbitrary numbers of remaining units and arbitrary host components. It imposes no matching-deficit condition or equality between an individual suffix pair's diagonal profile and the natural tail. If a host of maximum degree at most D instead has $e(H) - \nu(H) \leq P_q$, some member of A_q admits a coloring avoiding both of its entire targets.

Proof. The diagonal profile is positive and nonincreasing: decreasing an index greater than one in a candidate from a later diagonal produces a no-smaller candidate on the preceding diagonal. A positive potential candidate $n_i + m_j - 2$ requires a nonunit on at least one side. The red nonunit rectangle gives precisely an initial interval of M-u positive coordinates, and the blue nonunit rectangle gives an initial interval of M-v positive coordinates, where coordinate k-1 corresponds to diagonal k. Both rectangles are nonempty by hypothesis. Their union has M-U coordinates. Thus the q-tail has exactly $N = M - q - U$ positive potential entries. Since its first value is $D \geq 3$, one has $1 \leq R \leq N \leq K$.

The chosen F retains a nonunit, so its leading deletion has removed no original units. It therefore contains exactly U units and p nonunits. The component budget of its actual pair is $K = c(F) + c(J) - 1$. Consequently $c(F) = p + U$ and $p + c(J) - 1 = N$. Shortening each nonunit once and discarding the units gives a positive p-component forest with $e(F^-) = e(F) - c(F)$. Substituting $N = p + c(J) - 1$ proves the stated equivalence of numerical tests.

We check the complete family envelope before applying the two finite-family cited results. Each member of A_q has component budget K, and the family is finite and nonempty: α ranges over the nonempty integer

interval $[\max(0, q-t+1), \min(q, s-1)]$ and $\beta = q - \alpha$. At local diagonal h, with $2 \le h \le K+1$, every suffix candidate has original index sum $q+h$ and is therefore a candidate for ℓ_{q+h}. Conversely take original indices i, j with $i + j = q + h$. Set $\alpha = \min(q, i-1)$, $\beta = q - \alpha$. These are nonnegative, $\alpha \le i - 1 < s$ and $\beta \le j - 1 < t$, because $h \ge 2$; they sum to q. The original candidate is realized on local indices i-alpha, j-beta of this legal suffix pair. Hence the complete envelope of A_q is exactly the natural q-tail, with potential sum P_q.

By Lemma 56 applied to this same family, any common arrowing H has $\Phi(H) := e(H) - \nu(H) \ge P_q$. Suppose equality holds. Fix any maximum matching W of H and put $Q = H - W$, retaining every vertex. Apply Lemma 95 to the full family A_q, with its maximum component budget K and the verified tail envelope. It says that every sequence of maximum-degree vertex deletions in Q removes, in order, precisely the potential degrees

$\lambda_i = \ell_{q+i+1} - 1$ for $1 \le i \le K$,

and that after any l such steps the remaining Q-edge count is $\sum_{i=l+1}^{K} \lambda_i$.

Let I be the degree-D vertices of H uncovered by W, and put $k = |I|$. The set I is independent, since an edge between two W-uncovered vertices could be added to this maximum matching. The Q-vertices of degree D are exactly I: covered vertices lose one edge, all H-degrees are at most D, and uncovered vertices keep their degrees. Deleting vertices of I does not lower the Q-degree of any other vertex in I.

It follows that $k = R$. If $k < R$, choose all k vertices of I first, which are maximum-degree choices while any remain. The residual Q has maximum degree at most D-1, whereas the next degree prescribed by the theorem is $\lambda_{k+1} = D$. This contradiction also covers $k = 0$. If $k > R$, choose R vertices of I first. Another I-vertex still has degree D. If $R < K$ the theorem prescribes the next maximum degree $\lambda_{R+1} < D$; if $R = K$ its residual edge count is zero. Both cases are impossible. Thus every maximum matching of H exposes exactly R degree-D vertices.

Every matching of H can be enlarged to maximum cardinality preserving all already covered vertices. Indeed symmetric difference with a larger matching contains an alternating path with endpoints uncovered by the smaller matching; switching along it increases size and preserves prior saturation. Repeating finitely many times proves the assertion. Therefore every matching of H exposes at least R degree-D vertices, since a matching exposing fewer would extend to a maximum one exposing fewer.

Choose the greedy Q-sequence with all R vertices of I first and continue through its first N steps. Its N degrees are positive and all later λ_i vanish. Thus the N chosen vertices form a cover C of Q. They consist of I and a disjoint set C_0 of N-R further vertices.

We now verify the matching projections on this very Q. Orient the colors so the selected actual pair is (F, J). Take any coloring of Q and give all W-edges the color of F. If H contains the full opposite J, that copy lies in Q. Otherwise it contains a full monochromatic F. At most one edge in each copied star belongs to W because W is a matching. Keeping the required shortened number of edges at each nonunit center and omitting the unit components produces a copy of F^- in Q with its components still disjoint. Consequently Q arrows (F^-, J). Selecting one edge from each positive copied component proves that Q arrows (pK2,J), and, after exchanging color names and weakening J instead, it arrows $(c(J)K2, F^-)$. Both projected pairs have component budget $p + c(J) - 1 = N$.

At least one of their full forest targets has at least $N - R + 1$ edges, by the hypothesis and $e(F^-) = e(F) - c(F)$. Apply Lemma 18 to H, W, I, C and the corresponding one of these two projected pairs. Its hypotheses hold: $D \ge 3$, $|I| = R \ge 1$, C covers Q and contains I, $|C| = N$ equals the selected component budget, and the opposite forest has at least $|C_0| + 1 = N - R + 1$ edges. That theorem constructs a crossing matching saturating C_0 and an exposed root, then an H-matching exposing at most R-1 degree-D vertices. This contradicts the lower bound R just proved.

Thus equality $\Phi(H) = P_q$ is impossible. Since the non-strict lower bound and all potentials are integral, $\Phi(H) \ge P_q + 1$. Finally any host satisfying $\Phi(H) \le P_q$ cannot arrow the entire finite nonempty A_q family. Negating that conjunction supplies an actual member and an edge coloring avoiding both entire targets, as stated. □

Remark. At exact potential the matching complement has a forced greedy cover. Either full forest in the two matching projections supplies enough crossing neighbors to cover one more critical-degree root, contradicting the forced exposure count.

Depends on. Lemma 18, Lemma 56, Lemma 95.

Lemma 139 (Regular remainders and internal edges of the deleted set). *Use the finite integer certificate costs $c_i(G)$, matching number $\nu(G)$, edge count $e(G)$, excess $ex(G) = c_0(G) - \nu(G)$, slack $ps(G) = e(G) - \nu(G) - \sum_{i\ge 1} c_i(G)$, and signed gap $gap(G) = ex(G) - ps(G)$ of Lemma 23. All graphs are finite and simple. A nonempty graph is factor-critical if deleting any vertex leaves a perfect matching. Vertex deletion is induced.*

Let d belong to {8,10,12} and put $D = d - 2$. There do not exist a connected d-regular factor-critical graph F, a set $W \subseteq V(F)$, and a connected component C of $R = F - W$ satisfying the following conditions. Write $P = R - V(C)$, $k = \nu(F)$, $E = ex(F)$, $w = |W|$, $\ell = e(F[W])$, $m = \nu(C)$, and $t = \nu(P)$. Thus C is a whole component and P is its whole exterior in R, with no edges between them. Require

$gap(F) = 1$,

$c_{d-1}(F) + \ell \le w$,

C is D-regular and factor-critical,

$ps(C) = s$ *for a positive integer* s,

$gap(C)$ *belongs to {0,1},*

$c_D(P) = 0$,

$c_0(F) = w + m + t + s$,

$m \ge w + t + 3s - 1$.

The exterior P may be empty. No independent set, deletion process, or source minimality is assumed. All degrees, orders, cardinalities, matching numbers, costs, and indices are nonnegative integers; k, m, s are positive integers, E is nonnegative, and generic gaps may be signed.

Proof. Suppose such a tuple exists. We use the finite costs and their monotonicity from Lemma 23. The complementary-level bounds, top-row bound, and zero rows above maximum degree come from Lemma 87. Component subadditivity and restoration of W come from Lemma 72. The numerical local top-row estimates come from Lemma 53; equality-parity steps use Lemma 119.

Put $\gamma = gap(C)$, so γ is zero or one and $ex(C) = s + \gamma$. Factor-criticality gives $|V(F)| = 2k + 1$ and $|V(C)| = 2m + 1$: a perfect matching after one deletion has half the remaining order and is maximum in the original odd-order graph. Matching numbers add over C and P, and $|V(P)| \ge 2t$. Comparing orders gives $2k \ge w + 2m + 2t$. Substituting $k + E = w + m + t + s$ yields

$w \ge 2(E - s)$. (1)

The integer ℓ is nonnegative because it counts actual edges. Also $c_D(P) = 0$ implies $\Delta(P) \le D$: the certificate definition at capacities zero and D, with both matching budgets zero, puts all edges of P in the degree-D part.

On C the top-row bound gives $c_D(C) \le 1$. Subadditivity and restoration therefore give

$c_{d-2}(F) \le w + c_D(C) + c_D(P) \le w + 1$. (2)

The other assumed top-row bound is $c_{d-1}(F) \le w - \ell$.

For indices $3 \le i \le d - 3$, complementary levels on F satisfy $c_i(F) + c_{d-i}(F) \le 2k + 2$. Pairing distinct indices and using $c_{d/2}(F) \le k + 1$ at the sole fixed index gives

$\sum_{i=3}^{d-3} c_i(F) \le (d - 5)(k + 1)$.

Also $c_1(F), c_2(F) \le k + E$, $c_d(F) \le 1$, and all rows above d vanish. Since $gap(F) = 1$ and $e(F) = dk + d/2$, it follows that

$dk + d/2 + 1$

$= \sum_{i\ge 0} c_i(F)$

$\le 3(k + E) + (d - 5)(k + 1) + (w + 1) + (w - \ell) + 1$.

Rearranging and substituting the same cost identity gives

$2m + 2t + 2s \le 5E - \ell + d/2 - 4$. (3)

The rank hypothesis and (1) imply $m \ge 2E + t + s - 1$, so $2E \le m - t - s + 1$. Double (3) and substitute:

$4m + 4t + 4s \le 10E - 2\ell + d - 8$

$\le 5m - 5t - 5s - 2\ell + d - 3.$

Consequently

$m \ge 9s + 9t + 2\ell + 3 - d$. (4)

On C the indices 2,..., D-2 contribute at most $(D-3)(m+1)$, including the fixed index $D/2$. Since $c_0(C) = m + s + \gamma$, $c_1(C) \le c_0(C)$, $c_D(C) \le 1$, and $gap(C) = \gamma$, its row sum gives

$Dm + D/2 + \gamma$

$\le 2(m + s + \gamma) + (D-3)(m+1) + c_{D-1}(C) + 1.$

Thus

$m \le 2s + \gamma + D/2 - 2 + c_{D-1}(C)$. (5)

The connected even-regular factor-critical graph C satisfies the hypotheses of Lemma 53 at degree D and order $2m + 1$. For $D = 6, 8, 10$, respectively, it gives

$c_5(C) \le floor((6900m + 2644)/10000)$,

$c_7(C) \le floor((800m - 180)/1300)$,

$c_9(C) \le floor((560m - 430)/1000)$.

Substituting the real arguments of these floors into (5) yields

$3100m \le 20000s + 10000\gamma + 12644$,

$500m \le 2600s + 1300\gamma + 2420$,

$440m \le 2000s + 1000\gamma + 2570$.

Combine these with (4). For $d = 8, 10, 12$, respectively, we obtain

$7900s + 27900t + 6200\ell \le 28144 + 10000\gamma$, (6)

$1900s + 4500t + 1000\ell \le 5920 + 1300\gamma$, (7)

$1960s + 3960t + 880\ell \le 6530 + 1000\gamma$. (8)

We next prove $t = 0$. Suppose $t \ge 1$. Since $s \ge 1$, $\ell \ge 0$, and $\gamma \le 1$, each of (6)-(8) forces $t = 1$ and $s = 1$. Indeed, their left sides at $s = 1$, $t = 2$ are respectively 63700, 10900, 9880, exceeding the largest possible right sides 38144, 7220, 7530. At $s = 2$, $t = 1$ the corresponding left sides are 43700, 8300, 7880, again too large. Equations (6) and (7) further force $\ell = 0$. The local real bounds give $m \le 13, 12, 12$ in the three degree cases. The rank assumption gives $w \le m - 3$.

We will use two elementary consequences of these bounds. First, if m is odd, then $c_{D/2}(C) \le m$. The complementary upper bound is initially $m + 1$. If equality held, the sum of the two equal middle costs would be $2m + 2 = |V(C)| + 1$. Lemma 119 would make each cost odd, whereas $m + 1$ is even. This proves the improvement, and reduces the right side of (5) by one.

Second, a graph of matching number one is a nonempty star or a triangle, together with isolated vertices. To prove this, fix an edge uv. All edges meet {u, v}, since otherwise two disjoint edges exist. If every edge meets u, the nonisolated graph is a star; the same applies to v. Otherwise choose edges ux and vy with x, y outside {u, v}. They must intersect, so $x = y$. The triangle uv x is then present. An edge using a vertex outside that triangle would be disjoint from the opposite triangle edge, so no such edge is possible. This proves the classification.

If $d = 8$, (4) now gives $m \ge 13$, hence $m = 13$. The middle-row improvement and the local bound $c_5(C) \le \lfloor 92344/10000 \rfloor = 9$ give

$m \le 2 + 1 + 3 - 3 + 9 = 12,$

a contradiction.

If $d = 10$, (4) gives $11 \le m \le 12$. When $m = 11$, the middle-row improvement and $c_7(C) \le \lfloor 8620/1300 \rfloor = 6$ give $m \le 2 + 1 + 4 - 3 + 6 = 10$, a contradiction. Hence $m = 12$ and $w \le 9$. Write $n_P = |V(P)|$ and $j = e(P)$. Since $\nu(P) = 1$ and $\Delta(P) \le 8$, the star-or-triangle classification gives $1 \le j \le 8$. The exact order and cost identities give

$w - n_P = 2(E - s - t)$.

Count edges between W and R. The degree sum in W is dw-2ell. Each C-vertex has two neighbors in W, while the total contribution from P is $d\, n_P - 2j$. Thus

$dw - 2\ell = 2(2m + 1) + d\, n_P - 2j$,

and consequently

$d(E - s - t) = 2m + \ell + 1 - j$.

Here this becomes $10(E - 2) = 25 - j$. The right side lies in [17,24], so integrality forces $E = 4$ and $j = 5$. The classification makes P a star with five edges together with isolates, so $n_P \ge 6$. It follows that $w = 2(E - s - t) + n_P = 4 + n_P \ge 10$, contradicting $w \le 9$.

If $d = 12$, the bounds give $w \le m - 3 \le 9$. The nonisolated component of P is a star or triangle. A leaf of a star has one neighbor in P and none in C, so it would need eleven neighbors in W to have F-degree twelve. A triangle vertex has two neighbors in P and none in C, so it would need ten neighbors in W. Both are impossible when $w \le 9$. This completes the proof that $t = 0$.

Now P is edgeless; write $z = |V(P)|$. The exact order and cost identities give

$w - z = 2(E - s)$. (9)

Count edges between W and R again. Every C-vertex has two neighbors in W, and every vertex of the edgeless P has all its d neighbors in W. Thus

$dw - 2\ell = 2(2m + 1) + dz$.

Using (9) and dividing by two yields

$d(E - s) = 2m + \ell + 1$. (10)

Because d is even, ℓ is odd; hence $\ell \ge 1$.

Substituting (10) into (3), with $t = 0$, gives

$(d - 5)E \le (d - 2)s + d/2 - 3$. (11)

The rank bound and (1) give $m \ge 2E + s - 1$. Substitution into (10), together with $\ell \ge 1$, gives

$(d - 4)E \ge (d + 2)s$. (12)

Multiply (11) by the positive d-4, multiply (12) by the positive d-5, and compare. This gives

$3(d - 6)s \le (d - 4)(d - 6)/2$.

Since $d - 6 > 0$, we conclude $6s \le d - 4$. For $d = 8$ this contradicts $s \ge 1$. For $d = 10$ or 12 it forces $s = 1$. At $d = 12$, (12) gives $8E \ge 14$, hence $E \ge 2$, while (11) gives $7E \le 13$, hence $E \le 1$, again impossible.

It remains that $d = 10$ and $s = 1$. Equations (11) and (12) force $E = 2$. Equation (10) reads $10 = 2m + \ell + 1$. Simplicity and the degree eight of C give $|V(C)| \ge 9$, hence $m \ge 4$. Together with $\ell \ge 1$ this forces $m = 4$ and $\ell = 1$. The rank gives $w \le 2$ and (1) gives $w \ge 2$, so $w = 2$. Equation (9) gives $z = 0$, and the cost identity gives $k = 5$. Therefore F has order eleven and $e(F) = 55$. Its costs satisfy

$c_0(F) = 7$, $c_1(F)$, $c_2(F) \le 7$,

$c_8(F) \le 3$, $c_9(F) \le 1$, $c_{10}(F) \le 1$.

Complementary levels 3 and 7, and levels 4 and 6, each sum to at most twelve. At the middle index $c_5(F) \le 6$. If $c_5(F) = 6$, the two complementary costs sum to twelve, equal to $|V(F)| + 1$. Lemma 119 would make both costs odd, contradicting $c_5(F) = 6$. Hence $c_5(F) \le 5$. The sum of all costs on F is consequently at most

$3 \cdot 7 + 12 + 12 + 5 + 3 + 1 + 1 = 55 = e(F)$,

contradicting $gap(F) = 1$. All cases have been excluded. □

Remark. The cost sum on the host forces a large regular remainder, while the local top cost permits only a small one. Parity and the actual cut across the deleted set exclude all small-slack exceptions.

Depends on. Lemma 23, Lemma 53, Lemma 72, Lemma 87, Lemma 119.

Lemma 140 (Matching gain after a maximum independent cut). *Use the finite integer certificate cost $c_0(H)$, matching number $\nu(H)$, independence number $\alpha(H)$, and excess $ex(H) = c_0(H) - \nu(H)$ of Lemma 23. All graphs are finite and simple, and vertex deletion is induced. A nonempty graph is factor-critical if deleting any vertex leaves a perfect matching.*

Let $d \geq 4$ be an even integer and let H be a nonempty connected d-regular factor-critical graph. Write $|V(H)| = 2k+1$ and $E = ex(H)$, so $k = \nu(H)$. Choose any maximum independent set S, and put $s = |S| = \alpha(H)$, $G = H - S$, and $\eta = \nu(G) - (k+1-s)$. For a graph J let $o(J)$ be the number of its odd-order connected components. Choose any X of maximum cardinality among the sets maximizing $o(G-X) - |X|$, and put $b = |X|$.

Then $\eta \geq 0$, $b \leq 1 + 2\eta$, and G-X consists of $p = s + b - 1 - 2\eta \geq 1$ odd factor-critical components. Each such component Q satisfies $\alpha(Q) \leq 2\eta + 2 - b$ and $|V(Q)| \leq (d-1)(2\eta + 2 - b)$.

If $E \geq \eta + 2$, then at least one such Q has $|V(Q)| \geq 2s + 2b + 5$, and necessarily $2s + (d+1)b \leq 2(d-1)(\eta+1) - 5$. Consequently $s \geq (d-1)(\eta+1) - 2$ implies $E \leq \eta + 1$.

In particular: (i) If $d = 6$, $s \geq 3$, and $E \geq 2$, then $\eta \geq 1$. (ii) If $d = 6$, $s \geq 3$, $E \geq 3$, and $\eta = 1$, then either $b = 0$ and $s \leq 7$, or $b = 1$ and $s \leq 4$. (iii) If $d = 8$, $E \geq 2$, and $\eta = 0$, then $b = 0$ and $s \leq 4$. (iv) If $d = 8$, $s \geq 3$, $E \geq 3$, and $\eta = 1$, then either $b = 0$ and $s \leq 11$, or $b = 1$ and $s \leq 7$.

All conclusions apply to every permitted S and X. No gap, slack, source-minimality, total-criticality, or hereditary hypothesis is imposed. All degrees, orders, matching and independence cardinalities, costs, excesses, parameters, and component counts are integers in their displayed nonnegative or positive domains.

Proof. Factor-criticality gives odd order and $\nu(H) = k$: a perfect matching of H-v has k edges, while no matching in $2k+1$ vertices has more than k. The set S is nonempty. Choose v in S and a perfect matching of H-v. Removing the other s-1 vertices of S removes at most s-1 of its edges, leaving a matching of G of size at least $k + 1 - s$. Thus $\eta \geq 0$.

The maximizing sets X form a nonempty finite family because there are finitely many vertex subsets, including the empty set. The chosen maximum-cardinality member is inclusionwise maximal. Apply Lemma 107 to G with every vertex marked. Every odd component of G-X is factor-critical. There can be no even-order component C of G-X: choose w in $V(C)$. The graph C-w has odd total order, so at least one of its components has odd order. Passing from X to X union {w} therefore increases $o(G - X)$ by at least one and $|X|$ by exactly one. The maximized difference cannot decrease; by maximality of its value it must stay equal. This contradicts the maximum cardinality of X. Hence every component of G-X is odd.

The residual outside the counted components in Lemma 107 is therefore empty, and that fact gives $\nu(G) = b + \sum_Q (|V(Q)| - 1)/2$. Equivalently the number p of components is

$p = |V(G)| - 2\nu(G) + b$

$= (2k + 1 - s) - 2(k + 1 - s + \eta) + b$

$= s + b - 1 - 2\eta.$

This family is nonempty. Otherwise the same matching identity would give $\nu(G) = b = |V(G)|$, whereas $\nu(G) \leq |V(G)|/2$, forcing G empty. But G cannot be empty: then $S = V(H)$ would make H edgeless, contrary to $d \geq 4$.

Independent sets from distinct components of G-X have no edges between them in H, since G-X is induced. Their union is an independent set of H, so $\sum_Q \alpha(Q) \leq s$. Each of the p components is nonempty, giving $p \leq s$ and hence $b \leq 1 + 2\eta$. For any specified component Q, the other p-1 components each contribute at least one independent vertex. Thus

$\alpha(Q) \leq s - (p - 1) = 2\eta + 2 - b.$ (1)

Every vertex outside S has a neighbor in S; otherwise adjoining it would enlarge the maximum independent set. Consequently $\Delta(G) \leq d - 1$, and the same bound holds in each Q. Put $D = d - 1$, which is odd. A

connected odd-order graph Q of maximum degree at most D has a vertex w of degree at most D-1. Indeed, if all its degrees were D, its degree sum $D|V(Q)|$ would be odd, contradicting its equality to twice the number of edges.

Such a Q admits a proper coloring with D colors. Grow a spanning tree from w by repeatedly adding an edge to a vertex not yet reached; connectedness ensures such an edge until all vertices are reached. Order the vertices by nonincreasing distance from w in this tree, with arbitrary order among equal distances. Every vertex other than w has its tree parent later in the order, so at most D-1 of its neighbors have already received colors. The root w also has at most D-1 neighbors in total. Greedily give each vertex a color absent from its earlier neighbors, using the fixed palette of D colors. At each step a color is available. Each color class is independent and has size at most $\alpha(Q)$, whence

$|V(Q)| \le D\,\alpha(Q) \le (d-1)(2\eta+2-b)$. (2)

This argument includes a singleton component.

Suppose $E \ge \eta+2$. Assume for contradiction that every component Q of G-X has rank $r_Q = (|V(Q)|-1)/2$ at most $s+b+1$. Set $U = S$ union X, of cardinality $s+b$. The induced graph H-U is precisely G-X. Apply Lemma 51 with deleted set U, both degree capacities zero, retained spanning graph edgeless, retained cost zero, and the separate edge sets of the components Q as the whole blocks. Add isolated vertices to each block to give the common spanning vertex set; this does not change its matching number, which is r_Q by factor-criticality. In any ordering,

$r_Q \le s+b+1 \le |U| + sum$ of the earlier block $ranks + 1$.

Every required ordered inequality holds. That fact supplies all integer allocations at total

$|U| + \sum_Q r_Q = s+b+(\nu(G)-b) = s+\nu(G) = k+1+\eta$

on every original edge of H, including all edges meeting U. Thus $c_0(H) \le k+1+\eta$ by the definition in Lemma 23, contrary to $E = c_0(H) - k \ge \eta+2$.

There is therefore a component of rank at least $s+b+2$, and its odd order is at least $2s+2b+5$. Combining this with (2) gives $2s+2b+5 \le (d-1)(2\eta+2-b)$, which rearranges to

$2s+(d+1)b \le 2(d-1)(\eta+1)-5$. (3)

If $s \ge (d-1)(\eta+1)-2$, then 2s exceeds the right side of (3), even at $b=0$. Hence $E \ge \eta+2$ is impossible, and integrality gives $E \le \eta+1$.

For (i), $\eta=0$ would make (3) read $2s+7b \le 5$, contradicting $s \ge 3$. For (ii), (3) reads $2s+7b \le 15$. At $s \ge 3$ this forces $b \le 1$; $b=0$ gives $s \le 7$ and $b=1$ gives $s \le 4$. For (iii), (3) reads $2s+9b \le 9$. Since $s \ge 1$, it forces $b=0$ and then $s \le 4$. For (iv), (3) reads $2s+9b \le 23$. At $s \ge 3$ this forces $b \le 1$, giving $s \le 11$ at $b=0$ and $s \le 7$ at $b=1$. These prove all stated consequences. □

Remark. A small matching gain forces many odd components with small independent sets. Their odd order permits a short greedy coloring. If every component rank were small, the deleted vertices would absorb all components into a cheaper row-zero certificate; a large rank then forces the stated inequality.

Depends on. Lemma 23, Lemma 51, Lemma 107.

Lemma 141 (Cubic independence and preserving deletion). *Use the finite integer certificate costs c_i, matching number ν, independence number α, and slack $ps = e - \nu - \sum_{i\ge1} c_i$ of Lemma 126. All graphs are finite and simple; all orders, degrees, cardinalities, row indices and process lengths are nonnegative integers. Let G satisfy $\Delta(G) \le 3$, $ps(G) = 0$, and $c_3(G) = 0$. Put $n = |V(G)|$, $X = \{vinV(G) : deg_G(v) = 3\}$, and $q = \alpha(G[X])$, where G[X] is the induced graph on X.*

A complete cubic deletion process repeatedly deletes a current degree-three vertex and stops at maximum degree at most two. Let T be its set of deleted vertices and $R = G - T$ its full induced terminal. A set T arises this way if and only if T is a maximal independent set of $G[X]$; every ordering of such a T is a complete cubic deletion process. Here maximal means that no further vertex of X can be added while retaining independence, whereas maximum means having cardinality q. One has

$\alpha(G) = n - \nu(G) - c_2(G) + q$.

For every complete cubic deletion process,

$\alpha(G) - \alpha(R) = q - |T|$,

$\nu(R) = \nu(G) - |T|$, $c_1(R) = c_1(G) - |T|$, $c_2(R) = c_2(G) - |T|$.

The process preserves $\alpha(G)$ at every intermediate graph if and only if T is a maximum independent set of G[X]. Consequently, whenever G has a cubic vertex, some cubic vertex v satisfies $\alpha(G - v) = \alpha(G)$. There exists a complete process preserving independence at every step, and it has exactly q roots. Its intermediates have zero slack and zero rows at every index at least three. Its terminal has the same independence number as G and every whole component belongs to P_1, P_2, P_3, P_4, P_6, C_3, C_4, C_5, C_7, where P_j and C_j denote the path and cycle on j vertices. Empty graphs, X empty, T empty, and zero-length processes are included. No assertion that every maximal independent set of G[X] is maximum is made.

Proof. We use the arbitrary-process identities in Lemma 126 and the path, cycle and terminal formulas of Lemma 57. In particular every permitted cubic deletion preserves $ps = 0$ and $c_3 = 0$, and loses one unit from each of ν, c_1, c_2.

First characterize the root sets. A vertex selected at current degree three had original degree three, since deletion cannot increase degree and $\Delta(G) \leq 3$. None of its original neighbors was previously deleted, since its degree remains three. Thus every root belongs to X and no two roots are adjacent. At the end, every unselected vertex of X has lost a neighbor; otherwise it would still have degree three. Hence every vertex of X minus T has a neighbor in T, which is exactly maximality of the independent set T in G[X]. Conversely, if T is maximal independent in G[X], every ordering deletes its vertices at degree three because no earlier root is adjacent to a later one. Every remaining original cubic vertex has a neighbor in T, and all other original vertices already had degree at most two. Thus the terminal has maximum degree at most two. This proves both directions and the assertion about every ordering.

We next prove that some cubic vertex preserves independence whenever a cubic vertex exists. Call a vertex essential when every maximum independent set contains it. Suppose, to the contrary, that every vertex of the nonempty set X is essential. Put $t = |X|$ and $Q = G - X$. The set X is independent, since every maximum independent set contains it. Its deletion in any order is therefore a complete cubic process, and Lemma 126 gives

$\nu(G) = \nu(Q) + t$.

Every vertex adjacent to X has degree at most two in G: a degree-three neighbor would belong to X and contradict independence of X.

Choose a maximum matching M of G. At most t edges of M meet X, while at most $\nu(Q)$ lie entirely in Q. Equality in the displayed matching identity forces exactly t edges to meet X and forces $M_Q = M$ intersect $E(Q)$ to be maximum in Q. Since X is independent, the t edges meeting it have t distinct endpoints in Q, each unmatched by M_Q. An endpoint so obtained has a neighbor in X and degree at most two in G, hence degree at most one in Q. It is therefore an endpoint of a whole path component of Q, allowing a one-vertex path.

The restriction of M_Q to each component of Q is maximum there: replacing any nonmaximum restriction by a larger matching would improve M_Q. Every path or cycle component on l vertices has matching number $\lfloor l/2 \rfloor$, by Lemma 57. An even-order path consequently has no unmatched vertex in this restriction, and an odd-order path has exactly one. The t distinct endpoints used by the cross edges thus lie in t distinct odd-order path components of Q.

Let N be the set of all neighbors of X in Q. Every vertex of N is a path endpoint in Q by the same degree argument. The preceding t odd paths all meet N. On an odd path of order $2j + 1 \geq 3$, independence is $j + 1$; deleting either endpoint or both endpoints lowers it to j. These values follow from the path formula in Lemma 57. Deleting the sole vertex of a one-vertex path lowers independence from one to zero. Hence deleting N lowers independence by one on each odd path it meets. On every other component independence cannot increase under induced deletion. Independence numbers add over disjoint components, so

$\alpha(Q) - \alpha(Q - N) \geq t$.

Every maximum independent set of G contains X and avoids N. Conversely, X united with any independent set of Q-N is independent in G. Therefore

$\alpha(G) = t + \alpha(Q - N) \leq \alpha(Q) \leq \alpha(G)$.

A maximum independent set of Q is now maximum in G and avoids the nonempty set X, contradicting its essentiality. This proves the existence of a nonessential cubic vertex v. A maximum independent set omitting v gives $\alpha(G - v) = \alpha(G)$.

Delete such a vertex whenever a cubic vertex remains. The zero-slack and zero-third-row hypotheses persist by Lemma 126, so the argument just proved applies to every intermediate graph. The vertex count decreases at each step, ensuring termination. This constructs a complete independence-preserving process, with root set T_0.

Put $b = n - \nu(G) - c_2(G)$. The terminal formula of Lemma 126 gives $\alpha(G - T) = b + |T|$ for every complete root set T. A maximum independent set of G[X] exists because X is finite. It is maximal, so taking it as T gives $b + q \leq \alpha(G)$ by inducedness. For the constructed preserving set T_0 the same identity gives $\alpha(G) = b + |T_0| \leq b + q$. The two bounds prove $\alpha(G) = b + q$, including X empty. Subtraction gives the exact loss $q - |T|$ for every process.

Every intermediate graph contains its terminal and is contained in G as an induced graph. Thus $\alpha(R) \leq \alpha(H) \leq \alpha(G)$ for each intermediate H. It follows that independence is preserved at every step exactly when $\alpha(R) = \alpha(G)$, equivalently $|T| = q$. This is precisely the maximum-independent-set condition. The remaining matching and cost identities and the zero-slack intermediate assertions are those of Lemma 126. At any terminal the degree bound and zero slack allow Lemma 57, which gives exactly the listed whole components. If G has no cubic vertices the empty root set is maximum in G[X], all formulas reduce to the degree-two identities, and the unique zero-length process satisfies the assertions. □

Remark. Exact matching loss rules out all cubic vertices being unavoidable in maximum independent sets. A preserving deletion process then turns the terminal independence formula into an exact independence formula on the original cubic vertex set.

Depends on. Lemma 57, Lemma 126.

Lemma 142 (Covered deletions forced by induced perfect matchings). *Use the finite integer costs $c_0(H)$, matching number $\nu(H)$, independence number $\alpha(H)$, slack $ps(H)$, and complete maximum-degree deletion sequences of Lemma 123. A perfect matching covers every vertex. A graph of odd order is factor-critical if deleting any vertex leaves a perfect matching. A graph is noncomplete when some two distinct vertices are nonadjacent.*

Let H be a noncomplete finite simple graph on n vertices, and let q be an even integer with $2 \leq q \leq n$. Suppose every induced q-vertex subgraph of H has a perfect matching. Then every induced even-order subgraph of H of order at least q has a perfect matching, and every induced odd-order subgraph of order at least $q + 1$ is factor-critical.

Every complete maximum-degree deletion sequence of H has at least n-q positive-degree deletions. Among its first n-q stages, at least

$b = floor(ceil((n - q)/2)/2)$

are covered stages whose chosen deletion preserves matching number. Covered means that the parent has a matching covering every vertex of its current maximum degree. The choices of deleted vertices are arbitrary among maximum-degree vertices; the sequence need not be economical.

If $c_0(H) \geq n - q$, then

$ps(H) \geq b + \max(0, 2(n - c_0(H)) - 4\alpha(H))$.

The perfect-matching hypothesis is satisfied whenever the minimum degree of H is at least $n - q/2$. All orders, degrees, matching cardinalities, sequence lengths, indices and counts n, q, b are integers in the displayed nonnegative or positive domains. The cases $q = n$ and $b = 0$ are included.

Proof. The only matching bound imported is Lemma 85: a graph on $m \geq 1$ vertices with minimum degree at least $d \geq 0$ has matching number at least $\min(d, \lfloor m/2 \rfloor)$. All cost and deletion-sequence notation is supplied by Lemma 123.

First prove inheritance of perfect matchings. Induct on an even integer $m \geq q$. At $m = q$ the hypothesis gives the conclusion for every induced m-vertex graph. Suppose $m > q$ and let J be induced of order m. Some

q-vertex induced subgraph has a perfect matching and therefore an edge, since $q \geq 2$. Choose any such edge uv. The induced graph J-{u, v} has even order $m - 2 \geq q$ and a perfect matching by induction. Add uv to this matching. The union is a perfect matching of J. This proves the even-order assertion. If J is induced of odd order at least $q + 1$, deleting any vertex leaves an induced even graph of order at least q, which has a perfect matching. Thus J is factor-critical.

We prove two properties of maximum-degree deletions. First, noncompleteness persists whenever a positive maximum-degree vertex is deleted. Suppose Q is noncomplete, v has positive maximum degree, and Q-v is complete. Since all nonedges of Q must then meet v, noncompleteness gives a nonneighbor of v and $deg_Q(v) \leq |V(Q)| - 2$. Positive degree gives a neighbor u of v. The vertex u is adjacent to every other vertex of Q-v and also to v, so $deg_Q(u) = |V(Q)| - 1$. This contradicts maximality of the degree of v.

Second, two successive maximum-degree deletions from a positive-degree noncomplete regular graph of odd order cannot leave a regular graph. Let Q have odd order N and positive regular degree r. Its degree sum shows that r is even. Because Q is noncomplete and $r > 0$, a chosen vertex u has both a neighbor and a nonneighbor. In Q-u, the neighbors of u have degree r-1 and its nonneighbors have degree r. Therefore Q-u is nonregular, its maximum degree is r, and a next maximum-degree vertex v is nonadjacent to u in Q.

Suppose Q-{u, v} were p-regular. Its order N-2 is odd, so p is even. Every remaining vertex has lost r-p neighbors among u, v; this common loss lies in {0,1,2}. Evenness of r and p leaves only loss zero or two. At loss zero, no remaining vertex is adjacent to u, and v is also nonadjacent to u. This gives $deg_Q(u) = 0$, contrary to $r > 0$. At loss two, u is adjacent to all N-2 remaining vertices and not to v, giving $r = N - 2$, which is odd. This contradicts evenness of r. Thus the residual is nonregular.

Fix any complete maximum-degree deletion sequence of H. Every induced graph of order at least q contains an edge, since any q-subset has a perfect matching. Thus the sequence makes at least n-q deletions before its order becomes q, and all its first n-q stages have positive maximum degree.

Among the parent graphs of those first n-q stages, exactly $K = ceil((n - q)/2)$ have odd order. They have order at least $q + 1$, and hence are factor-critical by the proved inheritance. If such a parent Q has order $2r + 1$, a perfect matching after deleting any vertex has r edges. The order bound gives $\nu(Q) \leq r$, so every chosen vertex deletion preserves its matching number r.

If this Q is nonregular, choose a vertex w of degree below its maximum. A perfect matching of Q-w covers every maximum-degree vertex of Q, so this parent is covered. If Q is regular, no matching covers every vertex because Q has odd order; the parent is uncovered. Thus among the listed odd parents, being nonregular is exactly the condition of being covered.

All parents remain noncomplete by the first deletion property. Two consecutive parents in the list of odd-order stages are separated by exactly two maximum-degree deletions. By the second property they cannot both be regular. Pair the first and second listed parents, the third and fourth, and so on. Each of the $\lfloor K/2 \rfloor$ disjoint pairs contains a nonregular parent. Each supplies a covered matching-preserving deletion among the first n-q stages. This proves the required count $b = \lfloor K/2 \rfloor$; an empty list gives $b = 0$.

For the slack assertion choose an economical sequence, whose existence and length $T \geq c_0(H)$ are guaranteed by Lemma 123. Under $c_0(H) \geq n - q$, the first $c_0(H)$ stages include the prefix just counted, so their covered matching-preserving count is at least b. The cutoff bound of that fact with $k = c_0(H)$, after discarding its nonnegative β_k term, gives

$ps(H) \geq b + \max(0, 2(n - c_0(H)) - 4\alpha(H))$.

All counts occur in one fixed sequence and within this cutoff.

Finally suppose the minimum degree of H is at least $n - q/2$. In any induced graph J of order q, every vertex loses at most n-q neighbors, so its degree in J is at least $q/2$. The imported matching bound with $m = q$ and $d = q/2$ gives $\nu(J) \geq q/2$. A matching cannot have more than $q/2$ edges on q vertices; equality means it is perfect. This establishes the claimed sufficient condition on minimum degree. Every argument also applies when $q = n$, for which the counted prefix is empty. □

Remark. Perfect matchings on every fixed-size induced subgraph force larger odd induced graphs to be factor-critical. Along maximum-degree deletion, parity prevents consecutive odd regular parents. Each forced nonregular parent is covered and preserves matching number, producing a slack contribution before the residual cutoff.

Depends on. Lemma 85, Lemma 123.

Lemma 143 (Exact color overflow and the star-forest lower bound). *Let G be a finite simple graph. For nonnegative integers A, B, define $\mu_{A,B}(G)$ to be the minimum over red-blue edge colorings of G of*

$\sum_{v\in V(G)}[\max(d_{red}(v)-A,0)+\max(d_{blue}(v)-B,0)]$,

where $d_{red}(v)$, $d_{blue}(v)$ are the two color degrees. For positive A, B put $D=A+B$,

$C_D(G)=\sum_v \max(d_G(v)-D,0)$,

$a_v=\max(\min(A,d_G(v)),d_G(v)-B)$, *and*

$b_v=\max(\min(B,d_G(v)),d_G(v)-A)$.

For disjoint S, $T\subseteq V(G)$, let $q_{D,A}(S,T)$ count the connected components C of $G-(S\cup T)$ whose every vertex has degree D in G and for which $A|V(C)|+e_G(C,T)$ is odd. For disjoint vertex sets X, Y, $e_G(X,Y)$ counts their joining edges. Then

$\mu_{A,B}(G)=C_D(G)+\max_{S,T disjoint}[q_{D,A}(S,T)+e_G(S,T)-\sum_{v\in S}a_v-\sum_{v\in T}b_v]$.

If $\Delta(G)\le D$, then $C_D(G)=0$, $a_v=\min(A,d_G(v))$, $b_v=\min(B,d_G(v))$, and the same maximum also has the constant-cost form

$\mu_{A,B}(G)=\max_{S,T disjoint}[q_{D,A}(S,T)+e_G(S,T)-A|S|-B|T|]$.

For every integer $D\ge 2$ and every G with $\Delta(G)\le D$, let $\rho_D(G)$ be the minimum number of degree-D vertices of G uncovered by a matching. Then

$\mu_{1,D-1}(G)=\mu_{D-1,1}(G)=\rho_D(G)$.

Let F_1 and F_2 be nonempty star forests with positive nonincreasing integer component edge sizes $n_1,\dots,n_s$ and $m_1,\dots,m_t$. Arrowing $G\to(F_1,F_2)$ means that every red-blue coloring of $E(G)$ contains a red copy of the entire F_1 or a blue copy of the entire F_2, as ordinary subgraphs. If $G\to(F_1,F_2)$, then for every $1\le i\le s$ and $1\le j\le t$,

$\mu_{n_i-1,m_j-1}(G)\ge i+j-1$.

In particular, when n_i, $m_j\ge 2$ the first maximum with $A=n_i-1$, $B=m_j-1$ is at least $i+j-1-C_D(G)$. If additionally $\Delta(G)\le n_i+m_j-2$, an attaining disjoint pair S, T satisfies both the clipped-cost and constant-cost lower bound $i+j-1$. All extrema are attained. Empty G, isolated vertices and unit target stars are included wherever their stated hypotheses allow them.

Proof. First assume A, $B\ge 1$, and write $d_v=d_G(v)$, $p_v=\min(A,d_v)$ and $t_v=\max(0,d_v-B)$. Put $l(v)=\min(p_v,t_v)$ and $u(v)=\max(p_v,t_v)$. These are integer bounds with $0\le l(v)\le u(v)\le d_v$. For any integer r between 0 and d_v,

$\max(r-A,0)=\max(r-p_v,0)$, and

$\max(d_v-r-B,0)=\max(t_v-r,0)$.

If $t_v\le p_v$, then $d_v\le D$, and the sum of these two terms is the deviation of r from [l(v), u(v)]. If $t_v>p_v$, then $d_v>D$, $p_v=A$ and $t_v=d_v-B$; the identity

$\max(r-p_v,0)+\max(t_v-r,0)$

$=(t_v-p_v)+\max(p_v-r,0)+\max(r-t_v,0)$

holds in each of the ranges $r\le p_v$, $p_v\le r\le t_v$, and $r\ge t_v$. Therefore in every case

$\max(r-A,0)+\max(d_v-r-B,0)$

$=\max(d_v-D,0)+\max(l(v)-r,0)+\max(r-u(v),0)$.

The red edge set ranges over every spanning subgraph R of G. Thus the coloring minimum equals $C_D(G)$ plus the minimum interval deviation for [l, u].

Apply the interval-deficiency theorem Lemma 96 to these bounds. Its costs are $u(v)$ and $d_v-l(v)$. The first equals a_v because replacing t_v by d_v-B inside $\max(p_v,t_v)$ does not change that maximum. Also

$d_v - l(v) = \max(d_v - p_v, d_v - t_v)$

$= \max(\max(0, d_v - A), \min(B, d_v)) = b_v.$

Positivity of A and B implies $l(v) = u(v)$ exactly when $d_v = 0$ or $d_v = D$. To check this, when $0 < d_v < D$ the strict inequality $t_v < p_v$ follows if $d_v \le A$ from $B > 0$, and if $d_v > A$ from $d_v - B < A$; when $d_v > D$ we have $t_v > p_v$; the cases $d_v = 0, D$ give equality directly. A component containing a vertex of ambient degree zero is that isolated vertex; its interval upper bound and its edge count into T both vanish, so it is never an odd counted component. Every other equality component consists of ambient degree-D vertices and has $u(v) = A$ throughout. The interval parity count is therefore exactly $q_{D,A}$. This proves the uncapped formula.

If $\Delta(G) \le D$, then $C_D(G) = 0$ and $d_v - B \le A$, $d_v - A \le B$, giving the two clipped costs in the statement. We next remove their clipping. Among pairs maximizing the clipped expression choose S, T with $|S| + |T|$ minimum, and put $U = V(G) - (SunionT)$. Suppose v in S has $d_v \le A$. Because $B \ge 1$, this vertex has $d_v < D$. Move v from S to U. A previously counted component can be lost only if adjacent to v; at most $e_G(\{v\}, U)$ counted components are lost. Components now containing v cannot be counted, since its ambient degree is below D. The cross-edge count decreases by $e_G(\{v\}, T)$, and the deleted S cost is d_v. Hence the barrier change is at least

$d_v - e_G(\{v\}, U) - e_G(\{v\}, T) = e_G(\{v\}, Sminus\{v\}) \ge 0.$

An increase contradicts maximality, and equality contradicts minimality of $|S| + |T|$. Thus $d_v > A$ for every v in S. Moving a vertex v in T with $d_v \le B$ to U gives the same argument with S and T interchanged; the parity counts of untouched components are unchanged, while every component newly containing v has ambient degree below D. Thus $d_v > B$ for every v in T. At this maximizing pair all clipped costs equal their constants A, B. Every constant-cost expression is at most its clipped-cost expression, because $\min(A, d_v) \le A$ and $\min(B, d_v) \le B$. Their maxima are equal. In particular the selected pair attains both.

For the endpoint identity assume $D \ge 2$ and $\Delta(G) \le D$. For a coloring with red degree r_v the overflow at v for capacities 1, D-1 is

$\max(r_v - 1, 0) + 1_{d_v = D and r_v = 0}.$

Indeed the blue degree exceeds D-1 exactly in the indicated case and then by one. If some vertex has red degree at least two, recolor an incident red edge blue. The first term at that vertex decreases by one and the second remains zero. At the other endpoint the first term cannot increase, and the second increases by at most one. Hence total overflow does not increase. Each step reduces the number of red edges, so finitely many steps produce a red matching. Its overflow is exactly the number of degree-D vertices it leaves uncovered, which is at least $\rho_D(G)$. Thus every initial coloring has overflow at least $\rho_D(G)$. Conversely color a matching attaining $\rho_D(G)$ red and all other edges blue; this has overflow exactly $\rho_D(G)$. Interchanging red and blue proves the other endpoint identity.

For the star-forest assertion fix i, j, set $A = n_i - 1$ and $B = m_j - 1$, and now allow these capacities to be zero. Suppose some coloring has total overflow $k \le i + j - 2$. In its red graph, while a vertex has degree greater than A, delete one incident red edge. The sum of positive red degree excesses decreases by at least one at each deletion. This yields a red base graph Q_R of maximum degree at most A after at most the initial red excess number of deletions. Apply the corresponding operation in the blue graph with capacity B, obtaining a blue base graph Q_B. The combined set P of deleted edges has cardinality at most k. The edge sets $E(Q_R)$, $E(Q_B)$, P partition $E(G)$.

Partition P into P_R and P_B with $|P_R| \le i - 1$ and $|P_B| \le j - 1$: put $\min(i - 1, |P|)$ of its edges in P_R and the rest in P_B. Color Q_R and P_R red, and Q_B and P_B blue. In a red copy of F_1, its first i star components would each have at least $n_i = A + 1$ edges. Since Q_R has maximum degree at most A, each of these components must contain an edge of P_R. These components are vertex-disjoint, so their required P_R edges are distinct. This contradicts $|P_R| \le i - 1$. The same reasoning for the first j components of F_2 and the base Q_B contradicts any blue copy. The recoloring thus avoids both entire forests, contrary to arrowing. This proves $\mu_{n_i - 1, m_j - 1}(G) \ge i + j - 1$, including $A = 0$ or $B = 0$.

Substitution into the uncapped formula gives the first final barrier inequality when $n_i, m_j \ge 2$. Under the degree $\cap$, the minimum-cardinality maximizing pair constructed above attains both cost versions and gives both lower bounds. All choices range over finite sets, including the colorings and matchings. Their extrema are attained and all deletion procedures terminate; zero degrees and empty sets introduce no exception. □

Remark. Total color excess is a fixed ambient excess plus distance from an interval of red degrees. Canonicalizing an attaining interval barrier removes clipped costs. Deleting excess edges makes two bounded-degree color classes; too few deleted edges cannot support all required large target stars.

Depends on. Lemma 96.

Lemma 144 (A regular remainder four degrees below its host). *Use the finite integer certificate costs $c_i(G)$, matching number $\nu(G)$, edge count $e(G)$, excess $ex(G) = c_0(G) - \nu(G)$, slack $ps(G) = e(G) - \nu(G) - \sum_{i\geq 1} c_i(G)$, and signed gap $gap(G) = ex(G) - ps(G)$ of Lemma 23. All graphs are finite and simple. A nonempty graph is factor-critical if deleting any one vertex leaves a perfect matching. Vertex deletion is induced.*

Let d belong to {10,12} and put $D = d - 4$. There do not exist a connected d-regular factor-critical graph F, a set $W \subseteq V(F)$, and a connected component C of $R = F - W$ satisfying all the following conditions.

Write $P = R - V(C)$, $k = \nu(F)$, $E = ex(F)$, $w = |W|$, $m = \nu(C)$, and $t = \nu(P)$. Require

$gap(F) = 1$,

C is D-regular and factor-critical,

$ps(C) = s$ *for an integer* $s \geq 1$, $gap(C) \leq 1$,

$c_D(P) = 0$,

$c_0(F) = w + m + t + s$,

$m \geq w + t + 3s - 1$.

Thus C and P are whole complementary parts of R with no edges between them. The set W may contain internal edges, and P may be empty or have isolated vertices. No independent starting set, deletion process, source-minimality, independence-number bound, sign condition on $gap(C)$, or additional top-row bound on F is assumed. All degrees, orders, cardinalities, matching numbers, costs and indices are nonnegative integers; k, m are positive integers and the displayed gap may be a signed integer.

Proof. Suppose such a tuple exists. We use Lemma 23 for row monotonicity and the identity $\sum_{i\geq 0} c_i(G) = e(G) + gap(G)$, Lemma 61 for the zero tail and the vanishing top cost under a covering matching, Lemma 72 for restriction and extension, Lemma 119 for complementary rows and their equality parity, and Lemma 53 for local top-cost bounds.

Factor-criticality gives $|V(F)| = 2k + 1$ and $|V(C)| = 2m + 1$. A perfect matching after deleting a vertex achieves half the remaining order, and odd order rules out a larger matching. Whole-component additivity gives $\nu(R) = m + t$, while $|V(P)| \geq 2t$. The actual orders therefore give

$2k + 1 = w + 2m + 1 + |V(P)| \geq w + 2m + 1 + 2t$.

Substituting $k + E = w + m + t + s$ yields

$w \geq 2(E - s)$.

Together with $m \geq w + t + 3s - 1$ this implies

$2E \leq m - t - s + 1$. (1)

We next control the four highest nontrivial rows on F. For any vertex v of C, the perfect matching of C-v covers all its vertices, so Lemma 61 gives $c_D(C - v) = 0$. Restoring v by Lemma 72 gives $c_D(C) \leq 1$. The same argument on F gives $c_d(F) \leq 1$. The zero-tail theorem makes $c_i(C) = 0$ for $i \geq D + 1$ and $c_i(F) = 0$ for $i \geq d + 1$. By monotonicity, $c_D(P) = 0$ implies $c_i(P) = 0$ for every $i \geq D$. Component subadditivity and restoration of the actual set W therefore give

$c_D(F) \leq w + 1$, and $c_i(F) \leq w$ for $D + 1 \leq i \leq d - 1$. (2)

The vertex extension includes every edge inside W. There are exactly four indices D,..., d-1 in (2), whose costs sum to at most $4w + 1$.

Apply Lemma 119 to F. For each integer $1 \leq i < d$ it gives $c_i(F) + c_{d-i}(F) \leq 2k + 2$. The interval 5,..., d-5 is nonempty for both allowed d and has d-9 indices. Pairing its indices around $d/2$, with the midpoint bound $c_{d/2}(F) \leq k + 1$, gives

$\sum_{i=5}^{d-5} c_i(F) \le (d-9)(k+1)$. (3)

At $d = 10$ this is the single midpoint; at $d = 12$ it has three indices.

Each of $c_0(F)$,..., $c_4(F)$ is at most $c_0(F) = k + E$. The intervals 0,..., 4; 5,..., d-5; D,..., d-1; and {d} partition all potentially nonzero indices. Since F is d-regular and has gap one, (2)-(3) give

$dk + d/2 + 1$

$= \sum_{i \ge 0} c_i(F)$

$\le 5(k+E) + (d-9)(k+1) + 4w + 2.$

Rearranging gives $4k \le 5E + 4w + d/2 - 8$. Substituting $k = w + m + t + s - E$ yields

$4(m+t+s) \le 9E + d/2 - 8$. (4)

Double (4), and multiply (1) by nine, to obtain

$8(m+t+s) \le 18E + d - 16 \le 9(m - t - s + 1) + d - 16.$

Hence

$m \ge 17(s+t) + 7 - d$. (5)

Put $g = gap(C) \le 1$ and $q = c_{D-1}(C)$. These are integers, with $q \ge 0$, and $c_0(C) = m + s + g$. Apply Lemma 119 to C and pair the indices 2,..., D-2, including its midpoint. The sum of these D-3 rows is at most $(D-3)(m+1)$. Use $c_1(C) \le c_0(C)$, $c_D(C) \le 1$, and the zero tail. Regularity then gives

$Dm + D/2 + g$

$= \sum_{i \ge 0} c_i(C)$

$\le 2(m+s+g) + (D-3)(m+1) + q + 1.$

Consequently

$m \le 2s + g + D/2 - 2 + q$. (6)

No lower bound on g has been used.

If $d = 10$, then $D = 6$. The degree-six bound in Lemma 53, at the actual order $2m + 1$ of C, gives

$q \le floor((6900m + 2644)/10000)$.

Replace the floor by its argument in (6), multiply by 10000, and use $g \le 1$. The result is

$3100m \le 20000s + 22644$.

On the other hand, (5) gives $m \ge 17(s+t) - 3$. Combining yields

$32700s + 52700t \le 31944$.

This is impossible for the integers $s \ge 1$ and $t \ge 0$.

It remains to consider $d = 12$, $D = 8$. The degree-eight bound in Lemma 53 gives

$q \le floor((800m - 180)/1300)$.

Equation (6) and $g \le 1$ imply

$500m \le 2600s + 3720$.

Together with (5), namely $m \ge 17(s+t) - 5$, this yields

$5900s + 8500t \le 6220$.

The integer domains force $s = 1$ and $t = 0$. Equation (5) now gives $m \ge 12$, while $500m \le 6320$ gives $m \le 12$. Thus $m = 12$.

Equation (1) gives $E \le 6$. Equation (4), with $m = 12$, $t = 0$, $s = 1$, $d = 12$, gives $52 \le 9E - 2$, so $E \ge 6$. Thus $E = 6$ and the original cost identity yields $k = w + 7$. The assumed rank bound gives $12 \ge w + 2$, hence $w \le 10$. Comparing the actual vertex counts again gives

$|V(P)| = 2k - w - 2m = w - 10 \geq 0$.

Therefore $w = 10$, P is empty, $k = 17$, and $c_0(F) = 23$.

At these values (2) bounds the sum of rows 8, 9, 10, 11 by 41. Rows 0,..., 4 sum to at most 115, and $c_{12}(F) \leq 1$. The complementary bound gives $c_5(F) + c_7(F) \leq 36$ and $c_6(F) \leq 18$. In fact $c_6(F) \leq 17$: if $c_6(F) = 18$, apply the equality clause of Lemma 119 to the thresholds $\theta_1 = \theta_2 = 6$ on F. Their sum is $d = 12$, and their two costs have sum $36 = |V(F)| + 1$. That clause makes both costs odd, contradicting $c_6(F) = 18$. Integrality proves the stronger bound.

Summing all rows now gives

$\sum_{i \geq 0} c_i(F) \leq 115 + (36 + 17) + 41 + 1 = 210$.

But $|V(F)| = 35$, so regularity and $gap(F) = 1$ require this sum to equal $e(F) + 1 = 12 \cdot 35/2 + 1 = 211$. This contradiction excludes the second degree pair. Both permitted pairs have been excluded on the same actual graphs and costs, proving the theorem. □

Depends on. Lemma 23, Lemma 53, Lemma 61, Lemma 72, Lemma 119.

Lemma 145 (The exceptional sextic parameters below two odd cycles). *All graphs are finite and simple. Write $V(G)$, $E(G)$, $\nu(G)$, and $\alpha(G)$ for the vertex set, edge set, maximum matching cardinality, and maximum independent-set cardinality. For nonnegative integer capacities A, B, let $c_{A,B}(G)$ be the least nonnegative integer T such that every nonnegative integer pair $x + y = T$ admits an ordered edge partition into four parts with maximum degrees at most A, B and matching numbers at most x, y, respectively. Put $c_i(G) = \max_{A+B=i} c_{A,B}(G)$, over nonnegative integer capacities and indices. A matching number of an edge subset is that of its spanning graph.*

Let F be a connected six-regular graph such that deleting any vertex leaves a perfect matching. Write $|V(F)| = 2k + 1$ with integer $k \geq 1$. Assume $c_5(F) \geq \alpha(F)$, and define the integers

$E = c_0(F) - k$, $g = k - \alpha(F)$, $p = g - E + 1$, $\lambda = 2E - 3 - g$.

Suppose $E \geq 4$. Then either $p \geq 2$, or all the following hold:

$E = 4$, $k = 13$, $g = 4$, $p = 1$, $\lambda = 1$, $|V(F)| = 27$,

$\alpha(F) = c_5(F) = 9$, $c_0(F) = 17$.

In the latter alternative the entire set of nonnegative integer pairs (b_R, b_B) of total sixteen for which $E(F)$ has no partition into two parts of matching numbers at most b_R, b_B is exactly {(8,8)}. Every two distinct nonadjacent vertices of F have at most three common neighbors.

The second alternative is a necessary parameter description, not an existence assertion. No well-coveredness, source-minimality, induced-row, or deletion-process hypothesis is imposed. All degrees, orders, matching and independence cardinalities, capacities, budgets, costs, and indices are nonnegative integers. The expressions g, p, λ are initially signed integers. If p is separately identified with the odd-cycle component count of an actual induced row, the same alternative applies to that count.

Proof. Use Lemma 106 for the top-cost bound and its excess-at-least-five cycle-count consequence, and Lemma 40 for missing matching allocations.

Deleting a vertex of F leaves a perfect matching with k edges. No matching on $2k + 1$ vertices has more than k edges, so $\nu(F) = k$. The graph is connected and has maximum degree six. Thus Lemma 40 applies with $D = 6$ whenever $E \geq 1$. Its zero-capacity cost is the present $c_0(F)$, because both degree-zero edge parts must be empty. It gives

$k \geq 3E - 1 + 2max(0, floor((2E - 3)/4))$.

All hypotheses of Lemma 106 also hold, including $c_5(F) \geq \alpha(F)$. Its same-graph inequality is

$3100k \leq 10000g + 2644$. (1)

If $E \geq 5$, that fact gives $p \geq 2$, as desired.

It remains to consider $E = 4$ and $p \leq 1$. The identity $g = E + p - 1$ gives $g \leq 4$. The missing-allocation order bound yields $k \geq 12 - 1 + 2 = 13$. Inequality (1) gives

$3100k \le 42644 < 43400 = 3100 \cdot 14$.

Integrality forces $k = 13$. Returning to (1), one has

$10000g \ge 3100 \cdot 13 - 2644 = 37656$,

so the integer g is at least four. Hence $g = 4$ and $p = 1$. The definitions now give $\lambda = 8 - 3 - 4 = 1$, $\alpha(F) = 13 - 4 = 9$, $|V(F)| = 27$, and $c_0(F) = 13 + 4 = 17$. The top-cost bound in Lemma 106 is

$c_5(F) \le floor((3450 \cdot 27 - 806)/10000)$

$= \lfloor 92344/10000 \rfloor = 9$.

The assumed opposite bound $c_5(F) \ge \alpha(F) = 9$ proves equality.

At this parameter point, Lemma 40 gives $v = floor((2E - 3)/4) = 1$ and bounds every missing first budget at total $c_0(F) - 1 = 16$ between

$2E - 1 + v = 8$ and $k - E - v = 8$.

It also proves that at least one pair at this total is missing. Therefore the complete missing-pair set is exactly {(8,8)}.

Suppose distinct nonadjacent vertices z, z′ have at least four common neighbors. Six-regularity and the elementary union identity give

$|N_F(z)$ union $N_F(z')| \le 6 + 6 - 4 = 8$,

where $N_F(z)$ denotes the full open neighborhood. Since z, z′ are nonadjacent, this union avoids both. Enlarge it to an eight-element set W still avoiding z, z′; this is possible because there are twenty-five vertices other than z, z′. Put every F-edge meeting W into the first edge part and all remaining edges into the second.

Each edge in a matching of the first part uses a distinct vertex of W, so its matching number is at most eight. The second part is the full induced graph F-W on nineteen vertices, and z, z′ are isolated there because their entire neighborhoods lie in W. Its nonisolated vertices number at most seventeen, so its matching number is at most $\lfloor 17/2 \rfloor = 8$. This realizes the missing pair (8,8), a contradiction. Thus every nonadjacent pair has at most three common neighbors.

The cases $E \ge 5$ and $E = 4$ exhaust the given integer domain. All conclusions concern the original graph F. Identifying the already defined integer p with an actual row count changes no object or hypothesis and gives the final conditional assertion. □

Remark. The existing top-cost inequality and connected-graph allocation interval meet at a single integer point when $E = 4$ and $p < 2$. Missingness at that point imposes a concrete common-neighbor restriction.

Depends on. Lemma 40, Lemma 106.

Lemma 146 (Exact density loss along covered zero-slack deletion). *Use the finite simple graph quantities c_i, ps, ν and α defined in Lemma 23. Let $D \ge 3$ be an integer. Let H be a finite simple graph with $\Delta(H) \le D$ and $ps(H) = 0$. Fix a complete sequence $H_0 = H$, H_1,..., $H_t = R$ as follows: for each integer $1 \le j \le t$, delete a vertex v_j of current maximum degree $d_j = \Delta(H_{j-1}) \ge 3$ to obtain H_j, and assume that H_{j-1} has a matching covering every degree $- d_j$ vertex. Require $\Delta(R) \le 2$. Vertex deletion is induced deletion. All degrees, orders, indices, cardinalities and t are integers in their stated nonnegative ranges.*

Put $n = |V(H)|$, $k = \nu(H)$, $a = \alpha(H)$, $b = k - c_1(H)$, and, for any finite simple graph J, define

$B_D(J) = (3 - D)|V(J)| + (2D - 3)\nu(J) + (D - 3)\alpha(J)$.

For a whole component C of R put

$F_D(C) = D|V(C)| - e(C) - (2D - 3)(|V(C)| - \nu(C)) + (D - 3)\alpha(C)$.

Let $\epsilon(C)$ be one when C is a positive even-order path and zero otherwise. Then b is the number of those path components and

$B_D(H) - (D - 1)b - e(H)$

$= \sum_{j=1}^{t}(D-d_j) + (D-3)(a-\alpha(R))$

$+ \sum_C (F_D(C) - (D-1)\epsilon(C))$.

Every summand on the right is nonnegative. The last summand has the following values, where paths and cycles are indexed by vertex order:

(D-2)(h-1) on P_{2h}, $h \geq 1$;

(D-2)h on P_{2h+1}, $h \geq 0$;

(D-2)h on C_{2h}, $h \geq 2$;

(D-2)(h-1) on C_{2h+1}, $h \geq 1$.

In particular $e(H) \leq B_D(H) - (D-1)b$.

If $D \geq 4$ and equality holds in this bound, the deleted vertices form an independent set of size $t = 2k - n + a - b$, and each has original and deletion-time degree D. The terminal has exactly b single-edge components, n-k-a triangle components, and $2a - n + k - b$ isolated vertices, and $\alpha(R) = a$.

If $D \geq 4$ and the unrefined deficit $L = B_D(H) - e(H)$ satisfies $0 \leq L \leq D-4$, then $b = 0$, $t = 2k - n + a$, and the terminal has exactly n-k-a triangles and $2a - n + k$ isolates. It satisfies $\alpha(R) = a$ and $\sum_j (D - d_j) = L$. At least t-L of the deleted vertices have original and deletion-time degree D and are pairwise nonadjacent in H. If H is obtained from another finite simple graph G of maximum degree at most D by deleting an integer $q \geq 0$ vertices, each of deletion-time degree D, then $q + t - L \leq \alpha(G)$.

Two explicit consequences retain the supplied covering hypothesis for every process in question.

(i) If $D = 7$, $n = 18$, H has a perfect matching, $a \leq 6$, $e(H) \geq 45$ and $c_1(H) \leq 8$, then $e(H) = 45$, $a = 6$, $c_1(H) = 8$. The process deletes five independent vertices of original and deletion-time degree seven and leaves one single edge, three triangles and two isolates.

(ii) If $D = 7$, $n = 20$, H has a perfect matching, $a \leq 6$ and $e(H) = 53$, then $a = 6$ and $c_1(H) = 10$. The process has exactly six deletions, in degree order (7,7,7,7,7,6), preserves independence number, and leaves four triangles and two isolates.

All conclusions apply to every complete sequence meeting the displayed hypotheses. Empty graphs, isolates, empty sums and $t = 0$ are allowed; the component counts in the conclusions are nonnegative integers.

Proof. Apply Lemma 120 to the supplied covered process. It gives $ps(R) = 0$, $|V(R)| = n - t$, $\nu(R) = k - t$, and $e(H) = \sum_j d_j + e(R)$. It also says that R has exactly $b = k - c_1(H)$ positive even-order path components, so $b \geq 0$. Each component of R is a path or cycle. No covering property of an unlisted graph is used.

We record the elementary component values needed in the identity. On a path with m vertices, consecutive disjoint edges give matching number $\lfloor m/2 \rfloor$, and alternating vertices give an independent set of size $\lceil m/2 \rceil$. Disjoint consecutive vertex pairs, and the possible one remaining endpoint, prove the corresponding independent-set upper bound. On a cycle with m vertices the matching number is $\lfloor m/2 \rfloor$, attained by consecutive disjoint edges and bounded by the number of endpoints. For an independent set in a cycle, send each selected vertex to its cyclic successor. These successors are distinct unselected vertices, so its size is at most $\lfloor m/2 \rfloor$; alternating vertices attain that bound. A path has m-1 edges and a cycle m edges.

Substitution into F_D gives the following four values. For a path P_{2h}, $F_D = (D-2)h + 1$, and subtracting D-1 leaves (D-2)(h-1). For a path P_{2h+1}, $F_D = (D-2)h$. For a cycle C_{2h}, $F_D = (D-2)h$. For a cycle C_{2h+1}, $F_D = (D-2)(h-1)$. The parameters have exactly the ranges in the statement, and the indicator ϵ vanishes in the last three cases. Because $D \geq 3$, all these subtracted values are nonnegative. They vanish only on a single edge, an isolate or a triangle.

Orders, edge counts, matching numbers and independence numbers add over whole components. For matching and independence numbers, restricting an optimal set proves one inequality and uniting optimal component sets proves the reverse. Thus

$\sum_C F_D(C)$

$= D(n-t) - e(R) - (2D-3)(n-k) + (D-3)\alpha(R)$.

Add $\sum_j (D - d_j) = Dt - \sum_j d_j$ and (D-3)(a-alpha(R)). Using the exact edge identity, the result is

$(3-D)n+(2D-3)k+(D-3)a-e(H)=B_D(H)-e(H)$.

Since $\sum_C \epsilon(C)=b$, subtracting (D-1)b proves the displayed refined identity.

For its other nonnegative summands, deletion cannot increase maximum degree, so $d_j \le D$. Also an independent set of the induced graph R is independent in H, giving $\alpha(R) \le a$. The coefficient D-3 is nonnegative. Therefore the right side is a sum of nonnegative terms, proving the density bound.

Suppose $D \ge 4$ and equality holds. Each of those nonnegative terms is zero. Thus all $d_j = D$, $\alpha(R) = a$, and each terminal component is an edge, triangle or isolate by the computed component values. There are exactly b edges. Let y and z be the respective triangle and isolate counts. The identities for order, matching and independence are

$n = t+2b+3y+z$, $k = t+b+y$, $a = b+y+z$.

Solving them gives $y = n-k-a$, $z = 2a-n+k-b$ and $t = 2k-n+a-b$. These are counts of actual sets and components and hence are nonnegative integers.

Every vertex deleted at degree D had original degree D, because $\Delta(H) \le D$ and deletion cannot increase a degree. No two such vertices are adjacent: if one were deleted before an adjacent later one, that later vertex would already have lost a neighbor and its deletion-time degree would be at most D-1. This contradicts degree D. It proves the independence and original-degree conclusions in the equality case.

Now suppose the unrefined deficit L satisfies $0 \le L \le D-4$, with $D \ge 4$. Adding (D-1)b back to the refined identity gives

$L = \sum_j (D-d_j) + (D-3)(a-\alpha(R)) + \sum_C F_D(C)$.

The unsubtracted F_D values are nonnegative. They are zero precisely on triangles and isolates. Every other component contributes at least D-2: a positive even path contributes at least D-1, a nontrivial odd path at least D-2, an even cycle at least 2(D-2), and a nontriangle odd cycle at least D-2. Because $L \le D-4 < D-2$, the terminal can have no such component. In particular $b = 0$. The independence loss is a nonnegative integer, and its coefficient D-3 exceeds L, so that loss is zero. Thus $\alpha(R) = a$. The same three counting identities with $b = 0$ give $t = 2k-n+a$ and the asserted triangle and isolate counts. All component and independence contributions vanish, leaving $\sum_j (D-d_j) = L$.

Each $d_j < D$ contributes a positive integer to that last sum, so at most L of the t deletions have degree below D. At least t-L therefore have degree D, and the earlier-neighbor argument proves that all vertices of degree D at deletion have original degree D and are pairwise nonadjacent. This argument does not require t-L to be positive.

For the optional prefix, prepend the stated q deletions from G to H. The q earlier vertices and every subsequent vertex deleted at degree D all have original G-degree D, since $\Delta(G) \le D$. The same argument concerning an earlier adjacent deletion shows that their union is independent in G. Its size is at least $q+t-L$, so $q+t-L \le \alpha(G)$.

For (i), the perfect matching gives $k = 9$, and $c_1(H) \le 8$ gives $b \ge 1$. The density bound yields

$45 \le e(H) \le 27+4a-6b \le 27+24-6 = 45$.

Every inequality is equality, so $e(H) = 45$, $a = 6$ and $b = 1$, hence $c_1(H) = 8$. The proved equality case gives $t = 5$, three triangles, two isolates and $b = 1$ edge, together with the required independent full-degree roots.

For (ii), the perfect matching gives $k = 10$ and

$53 = e(H) \le 30+4a-6b \le 54-6b$.

If $a \le 5$ the middle bound is at most 50, a contradiction. Hence $a = 6$. Because b is a nonnegative integer, $53 \le 54-6b$ forces $b = 0$ and $c_1(H) = 10$. Now $B_7(H) = 54$, so $L = 1 \le 7-4$. The small-deficit conclusion gives $t = 6$, four triangles, two isolates, and $\sum_j (7-d_j) = 1$. Exactly one of the six deletion degrees is six and the other five are seven. Since the sequence always deletes a vertex of current maximum degree and induced deletion cannot increase that maximum, the degrees are nonincreasing. The degree-six deletion is therefore last. Independence was proved to be preserved.

The process was any one satisfying the stated covering premises. The conclusions therefore hold for each such process separately, including an empty one where admitted. No assertion about arbitrary zero-slack deletion histories without those premises is used. □

Remark. The edge-density deficit is exactly the sum of degree lost before deletion, independence lost at the terminal, and explicit path-or-cycle surpluses. Equality or a small deficit eliminates all but a few terminal forms.

Depends on. Lemma 23, Lemma 120.

Lemma 147 (The run-plus-excess bound in graphs of maximum degree three). *Use the finite integer certificate costs $c_i(G)$, matching number $\nu(G)$, vertex-cover number $\tau(G)$, excess $ex(G) = c_0(G) - \nu(G)$, positive-level slack $ps(G)$, signed gap $gap(G) = ex(G) - ps(G)$, and top missing-allocation run $tr(G)$ of Lemma 23. All graphs are finite and simple, including the empty graph; an edgeless graph has maximum degree zero. All cardinalities, degrees, costs, capacities, matching budgets and cost indices are nonnegative integers; gap may be a signed integer.*

Every finite simple graph G with $\Delta(G) \leq 3$ satisfies

$gap(G) > 0$ *implies* $[gap(G) = 1$ *and* $tr(G) + ex(G) \leq 3]$.

For $\Delta(G) \leq 2$ the stronger conclusions $ex(G) = 1$, $ps(G) = 0$ and $tr(G) \leq 2$ hold whenever $gap(G) > 0$.

In particular every graph of maximum degree at most three satisfies T_-, where

$T_-(G)$ *means* $gap(G) > 0$ *implies* $[gap(G) = 1$ *and* $tr(G) + ex(G) \leq \max(4, floor(\Delta(G)/2) + 1)]$.

Thus every graph violating T_- has maximum degree at least four, including every failure that is minimal under all proper subgraphs, where arbitrary vertex deletions, edge deletions, or both are allowed.

Proof. We first handle $\Delta(G) \leq 2$. Every component is a path or cycle by Lemma 57. A path on $n \geq 1$ vertices has a matching and a vertex cover of size $\lfloor n/2 \rfloor$, obtained from consecutive disjoint pairs and the even-position vertices. An even cycle on $2k \geq 4$ vertices has a matching and a cover of size k, by alternating edges and vertices. An odd cycle on $2k + 1 \geq 3$ vertices has a matching of size k and a cover of size $k + 1$. No matching in a graph on n vertices has more than $\lfloor n/2 \rfloor$ edges. The bound $c_0 \leq \tau$ from Lemma 23 therefore gives excess zero for paths and even cycles, and excess at most one for odd cycles.

Matching numbers add over components: restriction bounds any matching by the sum, and the union of maximum component matchings attains that sum. Apply Lemma 68 with X empty to obtain $ex(G) \leq 1$. Lemma 61 gives $ps(G) \geq 0$. Thus positive integer $gap(G) = ex(G) - ps(G)$ forces $gap(G) = ex(G) = 1$ and $ps(G) = 0$. The empty graph has gap zero and is excluded from this case.

By Lemma 57, $ps(G) = 0$ makes every component one of P_1, P_2, P_3, P_4, P_6, C_3, C_4, C_5, C_7. The maximum component excess equals one: it is at least $ex(G) = 1$ by Lemma 68 and at most one by the preceding component bounds. The positive-equality conclusion of that fact yields a component Z with $ex(Z) = 1$ and $tr(G) \leq tr(Z)$. Such a Z must be one of the odd cycles C_3, C_5, C_7, since paths and even cycles have excess zero.

Put $k = \nu(Z) \leq 3$. Then $c_0(Z) = k + 1$. At the total $c_0(Z) - 1 = k$ defining $tr(Z)$, the first budget zero is feasible by coloring all edges blue, and the first budget k is feasible by coloring all edges red. Hence every missing budget lies in the integer interval [1,k-1], and $tr(Z) \leq k - 1 \leq 2$. The missing set is nonempty by Lemma 23, so these statements also exclude $k = 1$ from the excess-one case. This proves all asserted degree-two conclusions and the run-plus-excess bound at that degree.

For any graph J, write $T_3(J)$ for the implication $gap(J) > 0$ implies $gap(J) = 1$ and $tr(J) + ex(J) \leq 3$. Suppose some graph G with $\Delta(G) \leq 3$ violates T_3. Among its finitely many induced subgraphs violating T_3 choose F with the fewest vertices. This family is nonempty because it contains G. Every proper induced subgraph of F satisfies T_3. Apply Lemma 121 with the nondecreasing integer function $h(D) = 3$ for every nonnegative integer D. It follows that F is connected, $gap(F) \geq 1$, $ex(F) \geq 1$, and no matching covers all maximum-degree vertices. The degree-two case already proved forces $\Delta(F) = 3$.

Let S be the vertices of F-degree three, and let κ be the minimum number of S-vertices left uncovered by a matching of F. Then $\kappa \geq 1$. For $X \subseteq V(F)$, let $q_S(F - X)$ count the odd components of F-X wholly contained in S. Choose a maximum-cardinality set X among those maximizing $q_S(F - X) - |X|$; finiteness gives

such a set, and it is inclusionwise maximal among the maximizers. Lemma 107, with marked set S, gives $q_S(F - X) = |X| + \kappa$ and factor-criticality of every component counted by q_S.

The set X is nonempty. If it were empty, connectedness and $q_S(F) = \kappa \geq 1$ would force F itself to have odd order and every vertex in S. Its degree sum would then be three times an odd integer, whereas every edge contributes two to the degree sum. This is impossible.

Write calQ for all components of F-X and calK for those counted by q_S. For K in calK let $d(K)$ be the number of F-edges between $V(K)$ and X. Apply the preliminary conclusions of Lemma 104 with $H = F$ and $D = 3$. They apply because X is a nonempty exact marked barrier and F is connected and uncovered. They give

$\nu(F) = |X| + \sum_{Q \in calQ} \nu(Q)$

and make every $d(K)$ a positive odd integer. Every Q in calQ is a proper induced subgraph of F, so $T_3(Q)$ holds and $gap(Q) \leq 1$.

If $d(K) = 1$, precisely one vertex of K has internal degree two and every other vertex has internal degree three: all vertices have F-degree three, all edges leaving K end in X, and there is exactly one such edge. The same K is factor-critical by Lemma 107. Hence Lemma 128 gives $ps(K) \geq 2$. If $d(K) \geq 3$, then $ps(K) \geq 0$ by Lemma 61. In both cases $3 - d(K) - ps(K) \leq 0$.

Choose Z in calQ of largest excess. The matching identity and Lemma 68 imply $ex(F) \leq \max(\{0\} union \{ex(Q) : Q \in calQ\})$. Since $ex(F) \geq 1$, this maximum is positive and attained by Z, so $ex(Z) \geq ex(F)$. For every K in calK minus {Z}, the credit required by Lemma 104 is $\delta_K = \max(3 - d(K) - ps(K), 0) = 0$. Supply $a_i(Q) = 0$ for every Q and for $i = 1, 2$. All its credit inequalities and sums hold. No component has positive credit, so both absorption procedures finish immediately, and their error E is zero.

The final odd-degree assertion of Lemma 104 now applies: $gap(Z) \leq 1$, $ex(Z) \geq ex(F)$, $D = 3$, and $gap(F) \geq 1$ were all checked. It forces Z in calK, $d(Z) = 1$, $gap(F) = gap(Z) = 1$ and $ex(F) = ex(Z)$. The one-defect component bound therefore gives $ps(Z) \geq 2$ and $ex(Z) = gap(Z) + ps(Z) \geq 3$. Since Z is a proper induced subgraph, $T_3(Z)$ gives $tr(Z) + ex(Z) \leq 3$. But $c_0(Z) \geq ex(Z) \geq 3$, so Lemma 23 gives $tr(Z) \geq 1$ and hence $tr(Z) + ex(Z) \geq 4$. This contradiction proves T_3 for every graph of maximum degree at most three.

For every integer D with $0 \leq D \leq 3$, $\max(4, \lfloor D/2 \rfloor + 1) = 4 \geq 3$. The proved T_3 bound therefore implies T_- in these degrees. Its contrapositive gives the asserted degree lower bound for every failure. Imposing proper-subgraph minimality does not weaken that conclusion and introduces no restriction on which vertex or edge deletions are permitted. □

Remark. A minimum induced failure is connected and has no matching covering its highest-degree vertices. At degree three, every marked odd component with one boundary edge has two units of slack, so all absorption credits vanish. The exact boundary identity then forces a proper component with excess too large for its inherited run bound.

Depends on. Lemma 23, Lemma 57, Lemma 61, Lemma 68, Lemma 104, Lemma 107, Lemma 121, Lemma 128.

Lemma 148 (Exposure and clique localization from matching remainders). *Let $D \geq 2$ be an integer and let G be a finite simple graph with maximum degree $\Delta(G) \leq D$. A matching is a set of edges with pairwise disjoint endpoints; its covered vertex set is $V(M)$, and $\nu(G)$ is the largest matching cardinality. A perfect matching covers every vertex, and a near-perfect matching covers all but one vertex. A graph is factor-critical if it is nonempty and deleting any vertex leaves a perfect matching. The empty graph has its empty perfect matching.*

Put $S = \{vinV(G) : d_G(v) = D\}$ and $\delta = \min_M |S$ minus $V(M)|$, the minimum over all matchings of G. Let Wcal be the family of matchings of cardinality $\nu(G)$ attaining δ. This family is nonempty. Let κ be the number of whole connected components of G that are odd-order, D-regular, and factor-critical. For every maximum matching M define

$I_M = S$ minus $V(M)$, $A_M = S$ minus $(I_M union N_G(I_M))$,

$T_M = (G - M) - I_M$, $U_M = \{vinV(T_M) : d_{T_M}(v) = D - 1\}$.

Here $N_G(I_M)$ consists of vertices having a neighbor in I_M; G-M removes only the matching edges and retains every vertex; vertex deletion takes the induced graph.

If $(G - M)[A_M]$ is edgeless for every M in Wcal, then $\delta = \kappa$. In particular this conclusion holds if U_M is independent in T_M for every M in Wcal. When D is odd, either sufficient hypothesis therefore implies that a maximum matching covers every vertex of degree D.

Independently, suppose $D \geq 4$ is even. Fix any maximum matching M, without requiring M in Wcal. If U_M is independent in T_M and the closed neighborhoods $N_{T_M}[u] = \{u\}$ union $N_{T_M}(u)$, for u in U_M, are pairwise disjoint, then every whole connected component C of G that is odd-order, D-regular, and admits a near-perfect matching is K_{D+1}. Its unique M-uncovered vertex x belongs to I_M, none of C-x belongs to U_M, and C-x is a whole connected component of $T_M - U_M$ isomorphic to K_D with a perfect matching removed. Here K_j denotes the complete graph on j vertices.

Consequently, when $D \geq 4$ is even and the independence and disjoint-neighborhood conditions hold for every M in Wcal, δ equals the number of whole K_{D+1} components of G. Every M in Wcal leaves exactly one vertex uncovered in each such component and covers every other vertex of S. Empty graphs, empty S, and empty vertex or component families are permitted. All degrees, orders, cardinalities, matching sizes, minima and component counts are nonnegative integers in the displayed domains.

Proof. Use Lemma 107 with marked set S. It guarantees that Wcal is nonempty and that the minimum δ has the compatible marked-barrier construction stated there.

First assume $(G - M)[A_M]$ is edgeless for every M in Wcal. Every whole odd D-regular factor-critical component has all its vertices in S and every matching leaves at least one vertex uncovered in it. These components are disjoint, so $\delta \geq \kappa$. If $\delta = 0$, this inequality gives $\kappa = 0$ and proves the assertion.

Suppose $\delta > 0$. Choose an inclusionwise maximal maximizer X of the marked odd-component deficiency, as allowed by Lemma 107. Let O be its family of odd components of G-X whose vertices lie in S, and put $p = |O|$. The cited fact gives $p = |X| + \delta$, factor-criticality of each C in O, and a matching M_0 in Wcal constructed as follows. Every vertex of X is matched to a distinct member of O; each such component is reached by precisely one edge from X and is fully covered. Every unreached member has exactly one uncovered vertex. All other vertices of S are covered. The uncovered vertices of S therefore lie in unreached members of O and none lie in X.

Fix a member C of O having an edge to X. There is a structured matching M in Wcal reaching C. To prove this, take $M = M_0$ if it already reaches C. Otherwise choose an edge xc with x in X and c in $V(C)$. The matching M_0 matches x to some vertex y of a different member C′ of O. Remove xy and all internal M_0 − *edges* in C. Insert xc and a perfect matching of C-c, which exists by factor-criticality. The removed and inserted internal matchings each have $(|V(C)| - 1)/2$ edges. The new edges meet no unchanged matching edge: C was unreached, and x is freed by removing xy. Thus the new matching has the same cardinality $\nu(G)$.

The new matching covers all of C. In C′ it exposes exactly y, because the unchanged internal matching of C′ covers every other vertex. Hence one uncovered S-vertex is transferred from C to C′, and their total remains δ. The remaining components and the matching outside them are unchanged. This proves the stated structured matching M in Wcal reaching C.

Every vertex of C belongs to A_M. Indeed, its vertices lie in S and are covered, while every vertex of I_M lies in a different component of G-X and none lies in X. There are no edges between distinct components of G-X. Thus no vertex of C has a neighbor in I_M.

The assumed edgelessness on A_M now says that every edge of G[C] belongs to M. The connected graph G[C] consequently has maximum degree at most one, so it has at most two vertices: a connected graph of maximum degree at most one consists of one vertex or one edge. Since C has odd order, it is a singleton.

Let k_0 count the members of O with no edge to X. Each such member is a whole component of G, because it has no edge to X or to another component of G-X. It is odd, factor-critical, and all its vertices have G-degree D. Hence $k_0 \leq \kappa$. Each of the other $p - k_0$ members is a singleton of G-degree D, with all D edges going to X. These edge sets are disjoint, and the total degree of the vertices of X is at most $D|X|$. Therefore

$$D(p - k_0) \leq D|X|.$$

As $D > 0$ and $p = |X| + \delta$, it follows that $\delta \leq k_0 \leq \kappa$. Together with $\delta \geq \kappa$ this proves equality.

For the stated sufficient independence condition, fix M in Wcal and v in A_M. Since v lies in S but not in I_M, it is matched and loses precisely one incident edge on passage to G-M. It has no G-neighbor in I_M, so its degree remains D-1 when those vertices are deleted. Thus A_M is contained in U_M. Its induced graph in T_M equals its induced graph in G-M. Independence of U_M therefore implies the required edgelessness on A_M.

If D is odd, an odd-order D-regular graph would have an odd degree sum, whereas every degree sum is twice an integer edge count. Thus $\kappa = 0$, giving $\delta = 0$; a member of the nonempty family Wcal is the asserted maximum covering matching.

Now prove the independent local assertion for even $D \geq 4$ and the specified maximum matching M. Fix a whole odd D-regular component C that admits a near-perfect matching. The restriction of M to C is maximum in C: replacing it by a larger matching of C would increase the cardinality in G without intersecting the matching outside C. Odd order and the assumed near-perfect matching therefore imply that this restriction exposes exactly one vertex x. All vertices of C lie in S, so I_M intersects $V(C)$ in precisely {x}.

Every vertex z of C other than x loses one edge when M is removed. It then loses one further edge when I_M is deleted exactly if z is adjacent to x. There are no edges from C to other components. Consequently

$d_{T_M}(z) = D - 2$ for z in $N_G(x)$,

$d_{T_M}(z) = D - 1$ for z in $V(C)$ minus $(\{x\} union N_G(x))$.

Put $B = V(C)$ minus $(\{x\} union N_G(x))$. Then $B = U_M$ intersect $V(C)$, and $|N_G(x)| = D$. Every y in B has D-1 neighbors in T_M, all within C-x. Independence of U_M means that none of these neighbors belongs to B, so all belong to $N_G(x)$. Pairwise disjoint closed neighborhoods in T_M imply that these neighbor sets for distinct y are disjoint. Counting them inside the D-element set $N_G(x)$ gives

$(D-1)|B| \leq D$.

For $D \geq 4$ this implies $|B| \leq 1$. On the other hand, $|V(C)| = 1 + D + |B|$ is odd and D is even, so $|B|$ is even. Hence B is empty. It follows that C has $D + 1$ vertices, each of degree D; simplicity then makes C the complete graph K_{D+1}.

The restriction of M to C-x is a perfect matching. In T_M this component's remaining graph is exactly K_D with those matching edges removed. None of its vertices belongs to U_M, since B is empty, so deleting U_M leaves it unchanged. It has no edge to vertices outside C-x. It is connected: two vertices are adjacent unless paired in the removed perfect matching, and such a paired pair has any third vertex as a common neighbor. Such a third vertex exists because $D \geq 4$. Thus C-x is a whole connected component of $T_M - U_M$, as asserted.

Finally impose the independence and disjoint-neighborhood assumptions for every M in Wcal at even $D \geq 4$. The first assertion gives $\delta = \kappa$. Choose any member of the nonempty Wcal and apply the local assertion: every component counted by κ has a near-perfect matching and therefore is K_{D+1}. Conversely every whole K_{D+1} component is odd-order, D-regular, and factor-critical, because deleting any vertex leaves K_D, whose vertices may be paired to give a perfect matching. Thus κ is exactly the number of these clique components. Every maximum matching exposes one vertex in each, by the same componentwise replacement argument. Their exposed vertices already total δ, so a member of Wcal exposes no other S-vertex. Empty families give zero counts throughout. □

Remark. A required odd component with a boundary can be reached by a maximum matching. The independence hypothesis then forces it to be a singleton, whose boundary uses all the available separator degree. The remaining deficiency lies in whole odd regular components, where disjoint neighborhoods and parity force a clique.

Depends on. Lemma 107.

Lemma 149 (Robust induced matchings at excess four and small order). *Use the finite integer zero-capacity certificate cost $c_0(H)$, matching number $\nu(H)$, and excess $ex(H) = c_0(H) - \nu(H)$ of Lemma 23. All graphs are finite and simple, and vertex deletion is induced. A nonempty graph is factor-critical if deleting any vertex leaves a perfect matching.*

Let k be an integer in {11,12,13}. Let H have $2k+1$ vertices, minimum degree at least eight, $\nu(H) = k$, and $ex(H) = 4$. Every induced subgraph of H of even order at least 2k-6 has a perfect matching. Every induced subgraph of odd order at least 2k-5 is factor-critical. In particular every induced subgraph J of order at least 2k-6 has $\nu(J) = floor(|V(J)|/2)$.

For every vertex set S of cardinality a with $1 \leq a \leq 7$, one has

$\nu(H - S) = k - \lfloor a/2 \rfloor$

and $\nu(H - S) - (k + 1 - a) = floor((a-1)/2)$.

For every vertex set S of cardinality $a \geq 7$, one has $\nu(H - S) - (k + 1 - a) \geq 3$.

Connectedness, regularity, positivity of gap, independence of S, and hereditary hypotheses are unnecessary. All orders, degrees, costs, matching numbers, set cardinalities, k and a are integers in their displayed nonnegative or positive domains.

Proof. We have $c_0(H) = k + 4$. By the integer threshold definition in Lemma 23, some allocation of nonnegative matching budgets (x, y) of total $k + 3$ is infeasible at zero degree capacities. Interchange the exceptional colors to arrange $x \leq y$.

If $x \leq 6$, choose any x vertices W. Assign all edges meeting W to the first exceptional part and all other edges to the second. A matching in the first part has at most x edges. The second part is supported on $2k + 1 - x$ vertices, so its matching number is at most $k - \lfloor x/2 \rfloor$. The inequality $\lceil x/2 \rceil \leq 3$ gives $k - \lfloor x/2 \rfloor \leq k + 3 - x = y$. The bounded-degree parts are empty. Thus this allocation would be feasible, a contradiction. Consequently $x \geq 7$, and the only possibilities, up to exchange of coordinates, are

$k = 11$: (7,7);

$k = 12$: (7,8);

$k = 13$: (7,9) or (8,8). (1)

Suppose an allocation (7,k-4) fails. For every seven-element set W, if $\nu(H - W) \leq k - 4$, assigning every edge meeting W to the first part and all remaining edges to the second would realize this allocation. Therefore $\nu(H - W) \geq k - 3$. The graph H-W has exactly 2k-6 vertices, so this matching is perfect. This proves the base even-order assertion except for the case $k = 13$ when the failed allocation chosen in (1) is (8,8).

In that case every induced graph on nineteen vertices has a matching of nine edges. Indeed, otherwise its complementary eight-vertex set, with every incident edge assigned to the first part, would realize (8,8). No matching on nineteen vertices has more than nine edges.

Fix any seven-element set W and put $K = H - W$, so K has twenty vertices. Suppose K has no perfect matching. Each induced nineteen-vertex graph inside K has matching number nine by the preceding paragraph. Thus $\nu(K) \geq 9$, whereas the absence of a perfect matching gives $\nu(K) \leq 9$. Hence $\nu(K) = 9$.

For a graph J let $o(J)$ be its number of odd-order connected components. Choose a set B of maximum cardinality among the maximizers of $o(K - B) - |B|$, and put $b = |B|$. These choices exist because K has finitely many vertex subsets. Apply Lemma 107 to K with every vertex marked. The marked deficiency is $20 - 2\nu(K) = 2$. Its conclusions give $b + 2$ odd factor-critical components counted outside B.

There is no even-order component of K-B. If C were such a component, choose a vertex w of C. The graph C-w has odd total order, and therefore at least one odd-order component. Enlarging B to B union {w} increases the odd-component count by at least one and the set size by exactly one. The maximizing difference cannot decrease; it cannot increase beyond its maximum either. Equality would contradict the maximal cardinality of B. This excludes C.

Thus the counted components are all the components of K-B. The matching identity of Lemma 107 now gives

$9 = b + \sum_C r_C$, where $r_C = (|V(C)| - 1)/2 = \nu(C)$. (2)

If $b \geq 1$, there are $b + 2$ nonempty components, and any specified one has at most

$20 - b - (b + 1) = 19 - 2b$

vertices. Consequently $r_C \leq 9 - b \leq 8 + b$.

If $b = 0$, there are exactly two odd components. Every vertex of K has degree at least one, because H has minimum degree at least eight and W has seven vertices. Each of the two components therefore has at least three vertices, so each has at most seventeen vertices and $r_C \leq 8$. In both cases every component satisfies $r_C \leq 8 + b$.

Apply Lemma 51 on H with deleted vertex set W union B, of size $7 + b$. The remaining whole edge blocks are exactly the components C of K-B, viewed as spanning graphs by adding isolated vertices, and the retained edge graph is empty. Its zero-capacity cost is zero. Every block rank is at most $(7 + b) + 1$, so the ordered absorption inequality holds in any ordering. Equation (2) makes the supplied total

$(7 + b) + \sum_C r_C = 7 + b + 9 - b = 16$.

That theorem realizes every zero-capacity allocation of total sixteen, including (8,8), on all edges of H. This contradicts the chosen failed allocation. Thus K has a perfect matching. Since W was arbitrary, the base assertion at even order 2k-6 holds also in the last case of (1).

Set $m_0 = 2k - 6$. We extend the assertion to larger even induced orders by induction in steps of two. Its base m_0 is proved. Let J be an induced graph of even order $m \geq m_0 + 2$. Choose any m_0 of its vertices. Their induced graph has a perfect matching, so J contains an edge uv since $m_0 \geq 16$. The induced graph J-{u, v} has order m-2 and a perfect matching by the induction hypothesis. Adding uv gives a perfect matching of J.

If J is induced, has odd order at least $m_0 + 1$, and v is any vertex of J, then J-v has even order at least m_0 and a perfect matching by the result just proved. This proves factor-criticality. The order upper bound for a matching and the supplied perfect or near-perfect matchings give $\nu(J) = floor(|V(J)|/2)$ throughout the asserted range.

Now let S be any vertex set of cardinality $1 \leq a \leq 7$. Its complement has order $2k + 1 - a \geq m_0$, so

$\nu(H - S) = floor((2k + 1 - a)/2) = k - \lfloor a/2 \rfloor$.

Subtracting $k + 1 - a$ gives $a - 1 - \lfloor a/2 \rfloor = floor((a - 1)/2)$, proving the exact gain.

If instead $a \geq 7$, choose a seven-element subset I of S. The graph H-I has a perfect matching with k-3 edges. Deleting the other a-7 vertices of S removes at most a-7 edges from this particular matching. The edges left form a matching of H-S of size at least $k - 3 - (a - 7) = k + 4 - a$. This inequality remains valid if its right side is negative. Subtracting $k + 1 - a$ proves the lower bound three. No independence assumption was used in either argument. □

Remark. A failed central allocation forces large induced matchings. An imperfect twenty-vertex residual would split into small odd blocks absorbable by the same omitted seven-vertex pool.

Depends on. Lemma 23, Lemma 51, Lemma 107.

Lemma 150 (An independent exposure witness at a high separator vertex). *Let $D \geq 3$ be an integer and let G be a finite connected simple graph with maximum degree at most D that is not D-regular. Put $A = \{vinV(G) : d_G(v) = D\}$. Let η be the minimum number of A-vertices uncovered by a matching, and write $\nu(G)$ for the maximum matching cardinality. For $Z \subseteq V(G)$, let $q_A(G - Z)$ count the odd-order connected components of the induced deletion G-Z whose vertices all belong to A. Call Z attaining when $q_A(G - Z) - |Z| = \eta$.*

For every attaining set X_0 and every specified vertex y in X_0 intersect A, there exist an attaining superset X of X_0, a matching W of G, and an independent set $I \subseteq A$ with the following properties. Every component counted by $q_A(G - X)$ is factor-critical, meaning that deletion of any one vertex leaves a perfect matching. The matching W has cardinality nu(G)-1, and I is exactly its set of uncovered A-vertices. Moreover $|I| = \eta + 2$, y belongs to I, and the other $\eta + 1$ members of I lie in distinct counted components of G-X and are nonadjacent to y. Consequently

$\nu(G - I) \geq \nu(G) - 1$ *and* $e(G) - e(G - I) = D(\eta + 2)$.

The set X can be chosen inclusionwise maximal among attaining sets. No maximality condition is imposed on X_0 or W. All orders, degrees, cardinalities, matching sizes, and η are nonnegative integers in the stated domains.

Proof. Choose an attaining superset X of X_0 of maximum cardinality among such supersets. This finite nonempty family contains X_0. No proper superset of X can be attaining, since it would also contain X_0. Thus X is inclusionwise maximal among all attaining sets. Apply Lemma 107 with marked set A. Let O be its family of counted odd components, $x = |X|$, and $p = |O| = x + \eta$. Let J be the union of the other components of G-X and put $r_C = (|V(C)| - 1)/2$ for C in O. The cited theorem gives factor-criticality of every C, an incidence matching saturating X, a maximum matching M_J of J covering A intersect $V(J)$, and the exact identity

$\nu(G) = x + \sum_{C \in O} r_C + \nu(J)$. (1)

Here an incidence edge x'C means that x' has an actual G-neighbor in C. The specified y belongs to X, so $x \geq 1$.

Let L be the subfamily of components C in O whose entire vertex set lies in $N_G(y)$, the set of neighbors of y. We prove that L has an incidence matching into X minus {y}. Suppose some nonempty subfamily U of L,

of size h, has a neighborhood Z′ in X minus {y} of size $k < h$. The components are disjoint, nonempty, and contained in the D-element set $N_G(y)$, so

$1 \le h \le D$ and $\sum_{C \in U} |V(C)| \le D$.

Every vertex of C in U has degree D. It has at most $|V(C)| - 1$ internal neighbors, and every external neighbor belongs to Z′ union {y}, because C is a component of G-X. Therefore $|V(C)| \ge D - k$. As $k \le h - 1$,

$D \ge \sum_{C \in U} |V(C)| \ge h(D-k) \ge h(D-h+1) \ge D$. (2)

The final inequality follows from $h(D-h+1) - D = (h-1)(D-h) \ge 0$. Equality holds throughout, so $k = h - 1$ and h is either 1 or D.

If $h = 1$, the sole component has D vertices and has no external neighbor except y. All its vertices are adjacent to y, and their degree D forces every internal pair to be adjacent. These D vertices and y form a K_{D+1}, with no edge leaving it: y already has D neighbors there, as does each of the others. Connectedness makes G this D-regular graph, contrary to hypothesis.

If $h = D$, the components are D singletons and $k = D - 1$. Each singleton has all its D neighbors in the D-element set Z′ union {y}. Thus they and Z′ union {y} form $K_{D,D}$. Every vertex in both parts already has degree D within this graph, so no edge leaves it. Connectedness again makes G D-regular, a contradiction. This proves that every subfamily of L has at least its own size as its neighborhood in X minus {y}.

These inequalities give a matching covering L by the following finite argument. Choose a largest matching in the bipartite incidence graph with parts L and X minus {y}. If it leaves an L-vertex uncovered, explore alternating paths from that vertex, taking nonmatching edges toward X minus {y} and matching edges back toward L. A reached uncovered vertex in the other part gives an augmenting path, contradicting largest cardinality. Otherwise all reached vertices in the other part are matched back into the reached L-set, whose initial vertex is unmatched. Every neighbor of that L-set is reached, so its neighborhood has one fewer vertex than the set. This contradicts the proved inequality. Hence a matching covering L exists, including the empty matching when L is empty.

Now work in the full incidence graph between X minus {y} and O. Deleting the edge at y from an incidence matching saturating X shows that this graph has a matching of cardinality x-1; no larger one exists because its first part has x-1 vertices. Start with the matching covering L and augment until its size is x-1. If a smaller matching had no augmenting path, take its symmetric difference with a matching of size x-1. The components of the symmetric difference are alternating paths and cycles, and the positive total size difference forces a path with one extra edge of the larger matching. This is an augmenting path for the smaller matching. Reversing an augmenting path preserves coverage of every previously covered vertex and covers its two previously uncovered endpoints. Therefore each step keeps all of L covered. Cardinality increases by one per step, so the process terminates at a matching B saturating X minus {y} and covering all of L. Exactly $p - (x-1) = \eta + 1$ components of O are unused by B, and none belongs to L.

Realize every incidence edge of B by any actual edge of G. Their separator endpoints and component endpoints are distinct. In each used component C, take a perfect matching after deleting its chosen endpoint, possible by factor-criticality. In each unused component choose a vertex z_C outside $N_G(y)$, possible because C is not in L, and take a perfect matching of $C - z_C$. Unite these actual edges and internal matchings with M_J. Their vertex sets are disjoint except for the intended endpoints, which are excluded from the internal matchings. The result is a matching W.

By (1), its cardinality is $x - 1 + \sum_C r_C + \nu(J) = \nu(G) - 1$. Every vertex of X except y is covered. Each unused component has exactly its chosen z_C uncovered, all vertices in used components are covered, and M_J covers all marked vertices of J. Since y belongs to A, the entire uncovered A-set is

$I = \{y\}$ union $\{z_C : C in Ounused by B\}$.

It has $\eta + 2$ members. Distinct z_C lie in different components of G-X and therefore are nonadjacent, and each was chosen nonadjacent to y. Thus I is independent. No edge of W meets I, so W survives in G-I and $\nu(G - I) \ge \nu(G) - 1$. Finally every vertex of I has G-degree D and independence ensures that no deleted edge is counted at two of these vertices. Exactly $D|I| = D(\eta + 2)$ edges are removed. This proves every claimed property of the same X, W, I. □

Remark. The only obstruction to matching all components wholly adjacent to the specified high vertex into the other separator vertices would force the entire graph to be regular. Keeping those components used permits nonadjacent exposed roots in every unused component.

Depends on. Lemma 107.

Lemma 151 (Covering the once-dominated vertices by a matching). *Use the finite integer certificate costs $c_i(F)$ and excess $ex(F) = c_0(F) - \nu(F)$ of Lemma 23. Let $d \geq 4$ be an even integer and $k \geq 1$ an integer. Let F be a nonempty connected d-regular finite simple graph on $n = 2k+1$ vertices such that deleting every vertex leaves a perfect matching. Suppose $E = ex(F) \geq 2$, put $e = E - 1$ and $t_0 = floor((2e-1)/(d-2))$, and let Z be a dominating set of F of cardinality z. All vertex deletions below are induced. Assume (i) $d+1 \geq 2floor((k+e)/2) - 4e - 1$, $6e + 2t_0 + 4 \leq n + d$, and $t_0 \leq ceil((n-1)/d)$; (ii) either $2e + 3 \geq z + d$, or $z + d - 3 - 2e \geq 1$ and $d \cdot ceil((d-1)/(z+d-3-2e)) > n + z - 4e - 5$; (iii) $6e + 2t_0 + d + 8 > n$; (iv) $8e + 4t_0 + 11 > n - z$. Then F-Z has a matching covering all its vertices of degree d-1. This includes the case in which no such vertices exist. Consequently $c_{d-1}(F) \leq z$. More precisely, fix Z and one such matching before any capacities or budgets are chosen. For every pair of nonnegative integers $A + B = d - 1$ and every pair of nonnegative integers $x + y = z$, there is a certificate on all original edges of F whose two degree-bounded parts partition $E(F - Z)$ and whose two matching-bounded parts partition all original edges meeting Z. The matching and Z are common to all those choices. Every order, degree, cost, matching size, capacity, budget and parameter d, k, n, E, e, t_0, z is an integer in its stated domain; in particular $e \geq 1$. Empty residual graphs, empty prescribed degree sets and zero budget coordinates are allowed.*

Proof. Factor-criticality gives $\nu(F) = k$. Apply Lemma 50 with $D = d$. It gives $E \leq floor((k+1)/3)$, so $e \geq 1$ and $3e \leq k - 2 < k$. In particular $1 \leq e \leq k$. Put $G = F - Z$. Domination gives $\Delta(G) \leq d - 1$. If $\Delta(G) \leq d - 2$, the empty matching covers its empty degree-(d-1) set; the final certificate conclusion then follows from Lemma 86 with $r = d - 1$ and deletion set Z.

Suppose $\Delta(G) = d - 1$ and put $U = \{vinV(G) : d_G(v) = d - 1\}$. This is exactly the set of vertices outside Z having one neighbor in Z. Let κ be the minimum number of U-vertices left uncovered by a matching of G. Suppose for contradiction that $\kappa \geq 1$. Lemma 83 with $j = 1$ supplies $X \subseteq V(G)$ for which the number q of odd components of G-X wholly contained in U satisfies $q = |X| + \kappa$. Fix any such X and put $x = |X|$. For each counted component C, write $\delta(C) = e_G(V(C), X)$, its number of edges to X. Degree summation gives $\delta(C)$ odd, and that fact gives $\sum_C \delta(C) \leq (d-1)x$.

Call a counted component small-boundary if $\delta(C) \leq d - 3$. Let g count these components and b count the others. Every $\delta(C)$ is a nonnegative odd integer and hence at least one. Each of the other b components has $\delta(C) \geq d - 1$. Thus $g + b = x + \kappa$, $(d-1)b + g \leq (d-1)x$. In particular $x \geq 1$, $b \leq x - 1$, and $g \geq \kappa + 1 \geq 2$. Also

$\sum_{Csmall-boundary} \delta(C) \leq (d-1)(x-b) = (d-1)(g-\kappa) \leq (d-1)(g-1)$. (1)

Every small-boundary component has odd order $m \geq d$ by Lemma 83, so in fact $m \geq d + 1$. Its external F-neighborhood is contained in Z union X and has cardinality at most

$\min(m, z) + \min(\delta(C), x) \leq z + \delta(C)$. (2)

Indeed each vertex of C has exactly one neighbor in Z, all other edges leaving C go to X, and no F-edge joins different counted components.

Put $W = 4e + 2t_0 + 3$, an odd integer. We claim that every small-boundary component C with $m = |V(C)| \leq W$ has external F-neighborhood of cardinality at least $2e + 2$. Otherwise use Lemma 93 with $N = k$, $D = d$, Q empty, $J = 1$, $K_1 = V(C)$, and the integer boundary bound $\beta = 2e + 1$. We check all its sufficient numerical hypotheses. The t_0 bound is part of (i), $\beta = 2e + J$, and $(m-1)/2 \leq 2e + t_0 + 1$ follows from $m \leq W$. Also $n - m - \beta \geq n - 6e - 2t_0 - 4 \geq -d$ by (i). Therefore $ceil((n - m - \beta)/(d+1)) \geq 0$, while $floor((2e + J - \beta)/(d-2)) = 0$. Its lower balanced inequality consequently reduces to $2e + (m+1)/2 \geq floor((k+e)/2)$. This follows from $m \geq d + 1$ and the first inequality in (i). For the upper balanced inequality, use $d - 2 \geq 2$ and $m \leq W$ to obtain

$floor((k+e)/2) + (m+1)/2 + floor((2e+1)/(d-2)) \leq (k+e)/2 + 4e + 2 < n$,

since the difference between n and the right side is $3(k-3e)/2-1>0$. These checks prove $c_0(F)\le k+e$, contradicting $c_0(F)=k+E=k+e+1$. The claim follows. In particular, when such a component exists, (2) gives $x\ge 2e+2-z$. (3)

There is at most one small-boundary component with more than W vertices. Two of them would each have at least $W+2$ vertices, since their orders are odd. Together with $x\ge 1$ they would occupy at least $2(W+2)+1=8e+4t_0+11$ vertices outside Z, contrary to (iv).

First suppose that all g small-boundary components have at most W vertices. Their neighborhood lower bounds and (1), (2) imply $g(2e+2)\le gz+(d-1)(g-1)$, or $g(2e+3-z-d)\le -(d-1)<0$. This is impossible under the first alternative in (ii). Under its second alternative, put $a=z+d-3-2e\ge 1$. The same inequality gives $g\ge ceil((d-1)/a)$. The pairwise disjoint sets Z, X, the g small-boundary components, and the b other counted components have total cardinality at least $z+x+(d+1)g+b$. Since $b=x+\kappa-g\ge x+1-g$, it follows that $dg+2x+1\le n-z$. Using (3), we get $dg\le n+z-4e-5$. This contradicts the strict second inequality in (ii).

The only other possibility is exactly one small-boundary component with more than W vertices. Since $g\ge 2$, another small-boundary component has at most W vertices, so (3) holds. Count X, the large component, and at least one other small-boundary component, all outside Z. Their total cardinality is at least $x+(W+2)+(d+1)\ge(2e+2-z)+(4e+2t_0+5)+(d+1)$. Thus $n\ge 6e+2t_0+d+8$, contradicting (iii).

Every possibility under $\kappa\ge 1$ has been excluded. Hence $\kappa=0$, so a matching of G covers all its degree-(d-1) vertices. Fix any such matching. The final assertion of Lemma 86, applied with $r=d-1$ and this same induced residual F-Z, supplies every stated capacity and budget split, with bounded-degree parts on $E(F-Z)$ and exceptional parts on all edges meeting Z. Its construction also covers edges internal to Z exactly once and permits zero budgets and empty parts. It proves $c_{d-1}(F)\le z$ and the asserted compatibility of choices. □

Depends on. Lemma 23, Lemma 50, Lemma 83, Lemma 86, Lemma 93.

Lemma 152 (Capped certificate payment for degree-five defects). *Use the finite integer certificate costs $c_i(G)$, matching number $\nu(G)$, edge count $e(G)$, maximum degree $\Delta(G)$, and positive-level slack $ps(G)$ of Lemma 23. All graphs are finite and simple. A graph is factor-critical if it has odd order and deleting any vertex leaves a perfect matching. For a graph G put*

$p_{cap}(G)=e(G)-\nu(G)-\sum_{i\ge1}\min(c_i(G),\nu(G))$.

The sum is finite. All degrees, orders, costs, matching budgets, cost indices and matching cardinalities are nonnegative integers.

Let K be any connected factor-critical graph with $\Delta(K)\le 5$ and positive total degree defect

$d=\sum_(\, v\in V(K))(5-d_K(v))$,

where $d_K(v)$ is the degree of v in K. Then

$p_{cap}(K)\ge 5-d$.

More precisely, write $|V(K)|=2k+1$, so $\nu(K)=k$. If d belongs to {1,3}, then $k\ge 3$, $e(K)=5k+(5-d)/2$, a matching covers every degree-five vertex, $ps(K)\ge 1$, $c_i(K)=0$ for every integer $i\ge 5$, $c_3(K)\le k$ and $c_4(K)\le k-1$. If $d=1$, so exactly one vertex has degree four and all others have degree five, then

$c_3(K)+c_4(K)\le 2k-2$.

All these costs are the universal integer allocation costs on the entire original edge set of K.

Proof. We use r-good to mean maximum degree at most the positive integer r with a matching covering every degree-r vertex. Lemma 86 gives the following tools. Every graph of maximum degree three with at most nine degree-three vertices is 3-good. Every graph of maximum degree four with at most nine degree-four vertices and no odd-order 4-regular component is 4-good. In particular, a graph of maximum degree four with at most six degree-four vertices and no complete five-vertex subgraph is 4-good: any odd-order 4-regular component would have at least five and at most six vertices, hence would be complete on five vertices. Also, an induced r-good subgraph obtained by deleting at most t vertices gives $c_r(K)\le t$, retaining certificates on all original edges. We apply these tools with $r=3$ or 4 throughout.

First establish the preliminary bounds at $d = 1$ or 3. Factor-criticality gives $\nu(K) = k$ and the degree sum gives $e(K) = 5k + (5 - d)/2$. If $n = 2k + 1 \leq 5$, simplicity gives

$d \geq n(5 - (n - 1)) = n(6 - n) \geq 5$

for odd n in {1,3,5}, a contradiction. Thus $k \geq 3$ and $n \geq 7$. If every degree were at most four, $d \geq n \geq 7$, again impossible. There is consequently a degree-five vertex and, since $d > 0$, a vertex z of smaller degree. A perfect matching of K-z covers every degree-five vertex. Lemma 61 yields $c_5(K) = 0$; row monotonicity from Lemma 23 gives $c_i(K) = 0$ for $i \geq 5$. For a degree-five vertex v, a perfect matching of K-v has size k, so $\nu(K) - \nu(K - v) = 0$. The covered-deletion inequality in Lemma 61 now gives $ps(K) \geq ps(K - v) + 1 \geq 1$.

Choose a partition $V(K) = S$ disjoint union T maximizing its number of crossing edges, with $|S| \leq |T|$. The finite family of partitions is nonempty, so a maximum exists. Moving a vertex with at least three neighbors in its own part would increase the cut, because its total degree is at most five. Both K[S] and K[T] therefore have maximum degree at most two. Since $|S| \leq k$ and K[T] is 3-good, $c_3(K) \leq k$. If $|S| \leq k - 1$ the same residual is 4-good and $c_4(K) \leq k - 1$.

For the remaining preliminary case, $|S| = k$ and $|T| = k + 1$. Write $N_S(v)$, $N_T(v)$ for the neighbors of v in S, T. If v in S has at most four T-neighbors, the graph K[T union {v}] has maximum degree four and at most one degree-four vertex, namely v. It is 4-good and gives $c_4(K) \leq k - 1$. Otherwise every S-vertex has five T-neighbors, S is independent, and $e(K[T]) = (5 - d)/2 \leq 2$. Since $5k > k + 1$, some w in T has two distinct neighbors u, v in S. In $J = K[(Tminus\{w\})$ union {u, v}], the nonadjacent vertices u, v each have degree four. Any other degree-four vertex needs two internal T-neighbors. A graph with at most two edges has at most one vertex with two neighbors, so J has at most three degree-four vertices and maximum degree four. It is 4-good. Its order is $k + 2$, again giving $c_4(K) \leq k - 1$.

We next prove the extra unit when $d = 1$. Let p be the unique degree-four vertex. Assume, for a contradiction, that $c_3(K) = k$ and $c_4(K) = k - 1$. These are the only simultaneous values that could violate the desired sum bound, by the preliminary bounds and integrality.

Retain a maximum cut (S, T). The preceding deletion bound forces $|S| = k$, $|T| = k + 1$. If some v in S had at most three T-neighbors, K[T union {v}] would be subcubic with at most four degree-three vertices: only v and its at most three T-neighbors can have degree three. It would be 3-good, contradicting $c_3(K) = k$. Hence every S-vertex has at least four T-neighbors and at most one S-neighbor. Thus K[S] is a matching and isolated vertices.

If that matching has two edges, choose u, v from distinct edges. Each has four T-neighbors, and u, v are nonadjacent. The graph $J = K[T$ union {u, v}] has maximum degree four; its degree-four vertices are u, v and at most four common T-neighbors. It contains no complete five-vertex subgraph: one using both u, v would require their absent edge, while one using at most one would require four mutually adjacent T-vertices, contradicting $\Delta(K[T]) \leq 2$. Thus J is 4-good. Its order $k + 3$ gives $c_4(K) \leq k - 2$, a contradiction. Consequently $r = e(K[S])$ belongs to {0,1}; when $r = 1$ write ab for its unique edge. If p is in S, it is isolated in K[S], because it has at least four cross-neighbors and total degree four. Put $\epsilon = 1$ when p is in S and $\epsilon = 0$ otherwise. Counting cross-edges and then all edges gives

$e_K(S, T) = 5k - \epsilon - 2r$, $e(K[T]) = 2 + r + \epsilon$. (1)

We will construct a 4-good induced subgraph on $k + 3$ vertices, contradicting $c_4(K) = k - 1$, except for one final configuration where a 3-good induced subgraph on $k + 2$ vertices contradicts $c_3(K) = k$.

Suppose first that p is in T. Since its internal T-degree is at most two, choose a neighbor u in S. If u is an endpoint of ab, put $W = N_T(u)$, which has four vertices including p, and $J = K[(Sminus\{u\})$ union W]. The S-part of J is independent. Every W-vertex loses neighbor u, so has degree at most four in J, and p has degree at most three. The other side has at most four neighbors in W. At most three W-vertices have degree four. Any degree-four S-vertex is adjacent to every W-vertex, hence to p, so there are at most $|N_S(p)| - 1 \leq 3$ such vertices. A 4-regular component cannot contain p, and hence can use at most three W-vertices. It cannot contain an independent S-vertex, which would need four W-neighbors; nor can it lie in W, whose internal maximum degree is at most two. Thus there is no 4-regular component. The exposure bound makes J 4-good.

If this chosen u is isolated in K[S], it has five T-neighbors including p. There are four subsets W of $N_T(u)$ of size four that contain p. When $r = 1$ exclude $W = N_T(a)$ and $W = N_T(b)$, at most two choices. Also exclude a W containing a triangle of K[T] disjoint from p. Equation (1) gives $e(K[T]) \leq 3$, so there is at most one triangle; together with p it determines at most one excluded W. A choice remains. Again put $J = K[(Sminus\{u\})$ union W]. Every W-vertex loses u and p has degree at most three. An isolated S-vertex has at most four

neighbors in J; each of a, b has at most three W-neighbors, because its four-element T-neighborhood was excluded as a possible W. Hence $\Delta(J) \leq 4$. At most three W-vertices have degree four. Every degree-four isolated S-vertex is a neighbor of p other than u, giving at most three such vertices, and at most two more are a, b. Thus J has at most eight degree-four vertices.

An odd 4-regular component of J cannot contain p, so uses at most three W-vertices. It cannot lie entirely in W. Every S-vertex in such a component would need an S-neighbor to attain degree four, so its S-vertices would be exactly a, b. They would require exactly three W-vertices. Each of those W-vertices would then need its two other neighbors inside W, making them a triangle disjoint from p. This is excluded by the choice of W. Hence J has no odd 4-regular component and is 4-good. Both subcases have order $k+3$ and yield the required contradiction.

Suppose next that p is in S and $r = 0$. By (1), K[T] has three edges. If its maximum degree is at most one, choose one endpoint of each edge to form W with three vertices and put $J = K[S \text{ union } W]$. Its S-vertices have degree at most three, while each selected T-vertex loses its one T-neighbor and has degree four. Thus only three vertices can have degree four, and J is 4-good. If a vertex w in T has internal degree two, it has exactly three S-neighbors. Put $U = N_S(w)$ and $J = K[(Tminus\{w\}) \text{ union } U]$. The set U is independent. Each U-vertex loses w and has degree at most four. There is exactly one remaining internal T-edge, so every retained T-vertex has degree at most four, and only the endpoints of that edge can have degree four. At most five vertices in J have degree four. A complete five-vertex subgraph is impossible because U is independent and the T-part has only one edge. Hence J is 4-good. In each case its order is $k+3$, a contradiction.

It remains that p is in S and $r = 1$. Equation (1) gives $e(K[T]) = 4$. Unless K[T] is a four-cycle and isolated vertices, there is a three-element set W in T each of whose vertices has a T-neighbor outside W. Indeed, if there is a triangle, maximum degree two makes it a component, and the fourth edge is a separate component; take two triangle vertices and one endpoint of that edge. If there is no cycle, the nontrivial components are paths with four edges in total and at least five vertices. In their alternating bipartitions choose a larger side in each path. The union is independent and has at least three nonisolated vertices; any three form W. Every nonisolated selected vertex has a neighbor outside this independent set. These alternatives exhaust graphs of maximum degree two and four edges other than a four-cycle with isolates.

For this W put $J = K[S \text{ union } W]$. Every selected T-vertex loses a T-neighbor, so has degree at most four. Each S-vertex has at most three cross-neighbors in J and at most one internal neighbor. Only a, b and the three W-vertices can have degree four. A complete five-vertex subgraph would require a, b and a triangle on W; vertices of a T-triangle have both their T-neighbors within that triangle, contrary to the outside-neighbor condition defining W. Thus J is 4-good and has order $k+3$, a contradiction.

We are left with a four-cycle in T and isolated T-vertices. Let w, z be any opposite cycle vertices and put $A_w = N_S(w)$, $A_z = N_S(z)$. Both sets have size three. We now describe three possible choices of a three-element set U in S for $J = K[(Tminus\{w\}) \text{ union } U]$. In each choice every U-vertex has degree at most four. The two cycle neighbors of w retain one internal T-neighbor and have at most three U-neighbors; z retains two internal T-neighbors and will have at most two U-neighbors; isolated T-vertices have at most three neighbors in J. Thus $\Delta(J) \leq 4$ and only U and the three retained cycle vertices can have degree four. No complete five-vertex subgraph is possible: K[U] has at most one edge, while the T-part is a path on three vertices and isolates, with no triangle. These facts make each such J 4-good.

If A_w differs from A_z, take $U = A_w$. Every U-vertex loses w, and $|A_w \text{ intersect } A_z| \leq 2$ checks the assertion at z. This contradicts $c_4(K) = k-1$. Thus opposite cycle vertices have identical S-neighborhoods.

If $A_w = A_z$ but p is not in A_w, take U to consist of p and two vertices of A_w. The vertex p has total degree four and is isolated in S; the other two lose w. The vertex z has no neighbor p, so retains at most two U-neighbors. This is another contradiction. Hence p belongs to the S-neighborhood of every cycle vertex.

If an endpoint a of ab is not in A_w, take U to consist of a and two elements of A_w minus {b}; at least two are available. The vertex a does not retain b, so loses its sole S-neighbor and has degree at most four. The other two lose w, and z has at most those two as neighbors in U. This yields the same contradiction. The argument applies separately to either endpoint and either opposite pair. Thus a, b, p belong to every cycle vertex's S-neighborhood and, since that neighborhood has size three, exhaust it.

The seven vertices a, b, p and the cycle have all their degrees saturated within that set: p has four cycle-neighbors; a, b each have the other and four cycle-neighbors; each cycle vertex has its two cycle-neighbors and a, b, p. No edge leaves this set. Connectedness of K makes it all of $V(K)$, so $k = 3$. Keep a, b, p and opposite cycle vertices w, z. The resulting induced graph has degree two at p and degree three at its other

four vertices. The edges aw, bz are a matching covering all four degree-three vertices. This 3-good graph is obtained by deleting two vertices, giving $c_3(K) \le 2 = k-1$, the final contradiction.

All cases exclude $c_3(K) = k$, $c_4(K) = k-1$. The preliminary upper bounds and integer costs therefore imply $c_3(K) + c_4(K) \le 2k-2$ at $d = 1$.

Finally d is odd, because it is five times an odd order minus an even degree sum. The definition and Lemma 61 give $p_{cap}(K) \ge ps(K) \ge 0$. This proves the target bound whenever $d \ge 5$. If $d = 1$ or 3, the already proved vanishing rows and degree sum give the exact identity

$p_{cap}(K) = (5-d)/2 + \sum_{i=1}^{4}(k - \min(c_i(K), k))$.

Every summand is nonnegative. At $d = 3$, $c_4(K) \le k-1$ supplies one unit, giving $p_{cap}(K) \ge 1+1 = 2 = 5-d$. At $d = 1$, $c_3(K)$, $c_4(K) \le k$, so their capped values equal their actual values; the proved sum bound supplies two units, giving $p_{cap}(K) \ge 2+2 = 4 = 5-d$. This covers every positive defect and proves the assertions. □

Remark. Tightness in both high rows forces a maximum cut to have at most one edge on its smaller side. Small vertex exchanges then improve the residual certificate, with a single saturated seven-vertex configuration handled by an explicit covering matching.

Depends on. Lemma 23, Lemma 61, Lemma 86.

Lemma 153 (Full-target localization at an exact matching separator). *Let s, t be positive integers, and let F_1, F_2 be nonempty star forests with s, t positive-edge components, respectively. Their edge sizes are positive integers and their copies are ordinary subgraphs with vertex-disjoint target components. Let G be a finite simple graph arrowing F_1 in red or F_2 in blue, and let ν denote maximum matching cardinality. Suppose $S \subseteq V(G)$ satisfies*

$\nu(G) = |S| + \sum_C \nu(C)$,

where C ranges over the connected components of the induced graph $G-S$. For any selected subcollection of these components, let H be their union, put $\mu = \nu(H)$, and put $p = s - \nu(G) + \mu$. If $p \ge 1$, then H arrows $(pK2, F_2)$, where $pK2$ is a matching of p edges. Every H-coloring avoiding this pair extends to a coloring avoiding the full original pair by coloring every edge outside $E(H)$ red.

Call a vertex set S half-order attaining when

$\nu(G) = |S| + \sum_C floor(|V(C)|/2)$.

Every such S satisfies $\nu(C) = floor(|V(C)|/2)$ for each component C of $G-S$, hence has the matching-additive property above. If additionally $d = s+t-1-\nu(G) \ge 1$, and a selected union H has $\mu \ge t-d$ and o odd-order components, then H-Z contains the entire uncolored F_2 for every $Z \subseteq V(H)$ with $|Z| \le \mu - t + d$. Moreover $e(F_2) \le \mu + o - d$. In particular an odd complementary component C has either $\nu(C) < t-d$ or $\nu(C) \ge e(F_2) + d - 1$; an even complementary component has either $\nu(C) < t-d$ or $\nu(C) \ge e(F_2) + d$.

If $X \subseteq V(G)$ satisfies $\nu(G-X) = \nu(G) - |X|$ and $Y \subseteq V(G-X)$ is half-order attaining in G-X, then X union Y is half-order attaining in G. Thus all preceding original-target conclusions apply to component unions of $(G-X)$-Y with the original s, t and matching deficit. If Y has maximum cardinality among half-order attaining sets of G-X, all these components are odd and factor-critical, meaning that deleting any one vertex leaves a perfect matching. No maximality of X union Y in G is asserted. Every assertion has the color-exchanged version and retains all unit components. All orders, component counts, matching sizes, indices, allowances and deficits are integers in their stated domains; empty selected unions and empty separators are allowed whenever the hypotheses permit them.

Proof. First assume the matching-additive equality for S and fix a selected union H with $p \ge 1$. Keep an H-coloring avoiding red $pK2$ and blue F_2, and color all other G-edges red. A red matching uses at most $|S|$ edges meeting S. Every other edge lies inside a component of $G-S$. Its edges in the unselected components number at most the sum of their matching numbers, and its edges inside H number at most $p-1$. Additivity of matching numbers over disjoint component unions gives $\mu = \sum_{C selected} \nu(C)$. Thus every red matching has cardinality at most

$|S| + \sum_{C unselected} \nu(C) + p - 1 = \nu(G) - \mu + p - 1 = s - 1$.

A red copy of F_1 would contain an s-edge matching obtained by selecting one edge from each positive component, so no red F_1 exists. All blue edges lie in H and contain no F_2. This proves the claimed extension. Since G arrows the original pair, such an H-coloring cannot exist. Hence H arrows $(pK2, F_2)$.

Suppose now that S is half-order attaining. Choose a maximum matching N of G. Partition its edges into those meeting S and those internal to each component C of G-S. These counts are bounded above by $|S|$ and $floor(|V(C)|/2)$, respectively. Their upper bounds sum to $|N|$ by the attaining equality. Therefore every bound is met: if any count were strictly smaller, their sum would be less than $|N|$. The internal edges in C consequently form a matching of size $floor(|V(C)|/2)$. No matching of C can be larger because its endpoints are distinct, proving $\nu(C) = floor(|V(C)|/2)$. The matching-additive equality follows, so the first part applies.

Assume $d \geq 1$ and $\mu \geq t - d$. Then $p = s - \nu(G) + \mu = \mu - t + d + 1$ is a positive integer. For any $Z \subseteq V(H)$ with $|Z| \leq p - 1$, color every H-edge meeting Z red and every other H-edge blue. Every red matching has at most $|Z|$ edges, because its edges have distinct endpoints in the vertex cover Z. There is no red pK2. Arrowing of H therefore gives a whole blue F_2. Every vertex of this target has a positive target degree, so its copy lies entirely outside Z. This proves the stated containment in H-Z for every permitted set Z.

The all-red coloring of H forces a p-edge matching, so $p \leq \mu$. In particular H has at least p-1 vertices, and a set Z of exactly p-1 vertices can be chosen. The half-order equalities in its selected components give $|V(H)| = 2\mu + o$, where o counts their odd orders. A star forest with t positive components has $e(F_2) + t$ vertices. Applying the just proved containment after this exact-size deletion yields

$e(F_2) + t \leq 2\mu + o - (p - 1) = \mu + o + t - d.$

This proves $e(F_2) \leq \mu + o - d$. For a single component, o equals one or zero according to its parity. If its matching number satisfies the threshold $\mu \geq t - d$, rearranging gives the asserted lower bound; otherwise it satisfies the other alternative. The arguments used all original component sizes, including size one. Interchanging the two target and color roles proves each symmetric assertion.

For the lifting assertion, X and Y are disjoint and the components of G-(X union Y) are precisely those of (G-X)-Y. Hence

$|X \text{ union } Y| + \sum_C floor(|V(C)|/2)$

$= |X| + |Y| + \sum_C floor(|V(C)|/2)$

$= |X| + \nu(G - X) = \nu(G).$

Thus X union Y is half-order attaining in the original G. All the proved assertions apply to its components with the unchanged original target counts and matching deficit.

Finally let Y have maximum cardinality among half-order attaining sets in $J = G - X$. For every vertex set Z of J, writing $o(J - Z)$ for its odd-component count gives the identity

$|Z| + \sum_C floor(|V(C)|/2) = (|V(J)| - (o(J - Z) - |Z|))/2.$

The marked-barrier theorem Lemma 107, with all vertices of J marked, identifies the maximum of $o(J - Z) - |Z|$ with $|V(J)| - 2\nu(J)$: the minimum exposure with all vertices marked equals this number. Consequently the half-order attaining sets are exactly its maximizing separators. The present Y is inclusionwise maximal among them. The same theorem makes every odd component of J-Y factor-critical.

There is no even-order component C of J-Y. If one existed, deleting any vertex v in C would leave a graph of odd total order and therefore at least one odd component. In the maximizing objective, replacing Y by Y union {v} would increase the odd-component count by at least one and the separator size by one. The objective could not be smaller. A strict increase contradicts its maximum value, while equality contradicts the maximum cardinality of Y. This excludes even components and proves the final assertion. The argument permits J empty, when the component family is empty. All matching choices, vertex sets and component sums are finite, so their required extrema exist. □

Remark. An exact matching separator fixes the available matching budget outside a selected component union. Coloring that exterior red forces the unchanged blue forest into the union, even after any permitted vertex deletion.

Depends on. Lemma 107.

Lemma 154 (Exact independent cuts in degree ten). *Use the finite integer certificate costs $c_i(H)$, matching number $\nu(H)$, excess $ex(H) = c_0(H) - \nu(H)$, nonnegative slack $ps(H) = e(H) - \nu(H) - \sum_{i\geq 1} c_i(H)$, signed $gap(H) = ex(H) - ps(H)$, and top missing-allocation run $tr(H)$ of Lemma 23. For a nonnegative integer D put $h(D) = \max(4, \lfloor D/2 \rfloor + 1)$. A finite simple graph satisfies T_- when positive gap implies gap one and $tr + ex \leq h(\Delta)$, where Δ is maximum degree. Let H be nonempty, connected, ten-regular, and factor-critical, meaning that deleting any vertex leaves a perfect matching. Assume $gap(H) = 1$, $E = ex(H) \geq 4$, and every proper induced subgraph of H satisfies T_-. Let S be any inclusionwise maximal independent set with $a = |S| \geq 4$, put $G = H - S$ and $k = \nu(H)$, and define $\eta = \nu(G) - (k + 1 - a)$. Call a connected component P of G large if $floor(|V(P)|/2) \geq a + 2$, and put $m = \max(\{0\} union \{ex(P) : P is large\})$. Then $|V(H)| = 2k + 1$, $\Delta(G) \leq 9$, $c_{10}(H) = 1$, $c_i(H) = 0$ for $i \geq 11$, $c_i(G) = 0$ for $i \geq 10$, and*

$c_i(H) = a + c_i(G)$ for every integer $1 \leq i \leq 9$,

$0 \leq \eta \leq \min(E - 1, floor((a - 1)/2))$, $ps(G) = E - 1 - \eta$,

$c_0(H) = a + \nu(G) + ps(G)$.

Furthermore $gap(G)$ belongs to {0,1}, $c_0(G) = c_0(H) - a + gap(G)$, and $ex(G) = ps(G) + gap(G)$. If $m = 0$, then $ps(G) = 0$. If $m \geq 1$, then $m - 1 \leq ps(G) \leq m$; when $ps(G) = m - 1$, every large component P with $ex(P) = m$ has $gap(P) = 1$ and $ps(P) = m - 1$, every other component has slack zero, and $ps(G)$ equals the sum of its component slacks. All these conclusions hold for every specified S meeting the cardinality condition, without a requirement that S be maximum. No failure of T_- by H and no minimality under edge deletion are assumed. All orders, degrees, indices, costs, capacities, budgets, matching cardinalities, excesses, and a, k, E, m, η are integers in their displayed nonnegative or positive domains; gaps may be signed on other graphs.

Proof. Fix S as in the statement. Factor-criticality gives $|V(H)| = 2k + 1$ and $\nu(H) = k$, so $c_0(H) = k + E$ and $ps(H) = E - 1$. Since S has at least four vertices, H is not the complete graph on eleven vertices. Lemma 113 applies with $d = 10$ and gives $\Delta(G) \leq 9$, $c_{10}(H) = 1$, the two zero tails, and

$c_0(H) \leq a + \nu(G) + m$. (1)

Choose v in S and restrict a perfect matching of H-v to G. Every one of the other a-1 vertices of S is matched to a distinct vertex outside S, since S is independent. Deleting those vertices removes exactly a-1 matching edges. Thus G has a matching of $k + 1 - a$ edges and $\eta \geq 0$. The order bound gives $\nu(G) \leq floor((2k + 1 - a)/2)$, hence $\eta \leq floor((a - 1)/2)$.

For each integer $1 \leq i \leq 9$ put $r_i = c_i(H) - c_i(G)$. Lemma 72 gives $0 \leq r_i \leq a$. Because S is independent, its deletion removes exactly 10a edges. Subtracting the definitions of slack, and using the top row and zero tails, gives

$\epsilon := E - 1 - ps(G) - \eta = \sum_{i=1}^{9} (a - r_i) \geq 0$. (2)

The inequality (1) says $E \leq 1 + \eta + m$, so (2) implies

$\epsilon \leq m - ps(G)$. (3)

The slack $ps(G)$ is nonnegative by Lemma 61. If $m = 0$, (3) therefore forces $\epsilon = 0$ and $ps(G) = 0$.

Suppose $m \geq 1$. Choose any large component P with $ex(P) = m$. It is a proper induced subgraph of H, so its T_- property gives $gap(P) \leq 1$ and consequently $ps(P) \geq m - 1$. Slack is superadditive over components by Lemma 72, and each component slack is nonnegative by Lemma 61. Thus $ps(G) \geq m - 1$. Together with (3) this leaves only $\epsilon = 0$ or $\epsilon = 1$.

Assume $\epsilon = 1$. Equalities are then forced in the preceding inequalities:

$ps(G) = ps(P) = m - 1$, $gap(P) = 1$, $E = 1 + \eta + m$,

$c_0(H) = a + \nu(G) + m$,

and every component Q different from P has $ps(Q) = 0$. We will derive a contradiction using actual zero-capacity budget allocations.

For any graph J and nonnegative integer t let $Y_t(J)$ be its feasible first matching budgets at total t and zero degree capacities. These are the sets $Y_t^{0,0}(J)$ from Lemma 75. A companion Q has $ex(Q) = gap(Q)$ because $ps(Q) = 0$. Nonnegative excess and T_- imply $ex(Q)$ is zero or one. If it is zero, $Y_{(}\nu(Q))(Q)$ is the full interval

[0,nu(Q)]. If it is one, this is the top allocation set; since $\Delta(Q) \le 9$, T_- gives $tr(Q) \le h(9) - 1 = 4 \le a$. In both cases its allocation set at total $\nu(Q)$ contains both endpoints and has no missing run longer than a.

Start with the full interval [0,a] supplied by the S-vertices. The interval criterion and composition in Lemma 75 absorb the companions one after another at totals $\nu(Q)$. At every step the full interval has length at least a, so all these absorptions are valid. Write

$L = a + \sum_{Q different from P} \nu(Q)$.

The resulting full interval is [0,L]. Keep P at total $t = c_0(P) - 1 = \nu(P) + m - 1$, which is at least $\nu(P)$. The disjoint component and vertex-pool inclusions in that same fact, restoring all edges meeting S, yield

$[0, L] + Y_t(P)$ contained in $Y_(c_0(H) - 1)(H)$, (4)

since $L + t = a + \nu(G) + m - 1 = c_0(H) - 1$.

Here all intervals and sums are over integers. The set $Y_t(P)$ contains 0 and t. List its consecutive elements. If two consecutive elements leave a missing run of length r between them, adding [0,L] leaves a run of length $\max(0, r - L)$ between their translated intervals; the endpoint intervals leave no missing endpoint segment. Therefore the longest missing run of $[0, L] + Y_t(P)$ is max(0,tr(P)-L). Inclusion (4) shows that $tr(H)$ is at most this value. The definition in Lemma 23 gives $tr(H) \ge 1$, because $c_0(H) = k + E > 0$. Hence

$tr(H) + L \le tr(P)$.

Since $gap(P) = 1$, its T_- property gives $tr(P) + m \le 5$. Combine this with $E = 1 + \eta + m$ and the preceding inequality:

$tr(H) + E \le 6 + \eta - L$

$\le 6 + floor((a - 1)/2) - a$

$= 5 - \lfloor a/2 \rfloor \le 3$.

But $E \ge 4$ and $tr(H) \ge 1$ make the left side at least five. This contradiction rules out $\epsilon = 1$.

We have proved $\epsilon = 0$ in every case. Formula (2) gives $ps(G) = E - 1 - \eta$ and $\eta \le E - 1$. All nine summands $a - r_i$ in (2) are nonnegative integers whose sum is zero, so $r_i = a$ for every $1 \le i \le 9$. The identity $\nu(G) = k + 1 - a + \eta$ now gives

$a + \nu(G) + ps(G) = a + (k + 1 - a + \eta) + (E - 1 - \eta) = k + E = c_0(H)$.

Inequality (1) implies $ps(G) \le m$; the $m = 0$ case and the lower bound m-1 when $m \ge 1$ have already been proved.

If $m \ge 1$ and $ps(G) = m - 1$, take any large component P with $ex(P) = m$. Its slack is at least m-1. Nonnegative component slacks and their superadditivity force $ps(P) = m - 1$, every other component slack zero, and equality between $ps(G)$ and the sum of those slacks. Thus $gap(P) = 1$. This argument applies to every such P.

Finally put $r_0 = c_0(H) - c_0(G)$. The vertex-extension bound gives $r_0 \le a$. The total positive-row decrease from H to G is $9a + 1$, and the edge decrease is 10a. Therefore

$gap(G) = gap(H) - r_0 - (9a + 1) + 10a = a - r_0 \ge 0$.

The graph G is proper induced in H and satisfies T_-, so its gap is zero or one. Rearranging the last equality gives $c_0(G) = c_0(H) - a + gap(G)$; its excess identity follows from $gap = ex - ps$ in Lemma 23. Every construction and count used the initially fixed S, completing the proof. □

Remark. Four independent vertices absorb every possible missing run in a zero-slack companion at degree at most nine. A loss in any positive row would then contract the surviving component run below the required excess.

Depends on. Lemma 23, Lemma 61, Lemma 72, Lemma 75, Lemma 113.

Lemma 155 (Robust induced matchings at three small octic orders). *Use the finite integer certificate costs $c_i(H)$, matching number $\nu(H)$, and excess $ex(H) = c_0(H) - \nu(H)$ of Lemma 23. All graphs are finite and simple, and vertex deletion is induced. A graph is factor-critical if deleting any vertex leaves a perfect matching.*

Let k belong to {11,12,13}, and let H be a nonempty connected eight-regular factor-critical graph on $2k+1$ vertices with $ex(H) = 4$. Then every induced subgraph of H of even order at least 2k-6 has a perfect matching. Every induced subgraph of odd order at least 2k-5 is factor-critical. In particular every induced subgraph J with $|V(J)| \geq 2k-6$ has $\nu(J) = floor(|V(J)|/2)$.

For every nonempty independent set S of cardinality $a \leq 7$, one therefore has

$\nu(H - S) = k - \lfloor a/2 \rfloor$, *and*

$\nu(H - S) - (k + 1 - a) = floor((a-1)/2)$.

For every independent set S of cardinality $a \geq 7$, one has

$\nu(H - S) - (k + 1 - a) \geq 3$.

If k is eleven or twelve, then $c_1(H) \leq k + 3$.

No positive-gap or hereditary-gap hypothesis is imposed. All orders, degrees, costs, capacities, budgets, matching and independence cardinalities, k and a are integers in their displayed nonnegative or positive domains.

Proof. Factor-criticality and the order give $\nu(H) = k$, so $c_0(H) = k + 4$. By Lemma 23 some nonnegative integer allocation (x, y) of total $k + 3$ fails at zero degree capacities. Exchange the two exceptional colors so that $x \leq y$. If $x \leq 6$, choose any x vertices W, put all edges meeting W in the part with matching budget x, and put every other edge in the other part. The latter graph has order $2k + 1 - x$ and matching number at most $k - \lfloor x/2 \rfloor$. Since $x \leq 6$, one has $x - \lfloor x/2 \rfloor = \lceil x/2 \rceil \leq 3$, so $k - \lfloor x/2 \rfloor \leq k + 3 - x = y$. This would realize the allocation. Thus $x \geq 7$.

For $k = 11$, 12, 13 the possibilities with $x + y = k + 3$ and $x \leq y$ are therefore, respectively,

(7,7), (7,8), and either (7,9) or (8,8). (1)

If a pair (7,k-4) fails, every deletion of seven vertices leaves a graph with matching number at least k-3: otherwise color all edges meeting the seven vertices with the first budget and all remaining edges with the second. The remainder has exactly 2k-6 vertices, so a matching of k-3 edges is perfect. This proves the required base matching assertion except when $k = 13$ and the selected failed pair is (8,8).

Suppose that pair fails. Every nineteen-vertex induced graph has matching number at least nine, since its complementary eight-vertex pool would otherwise realize (8,8). Its matching number is consequently exactly nine. Fix any twenty-vertex induced graph $K = H - W$ with $|W| = 7$. If K has no perfect matching, the nineteen-vertex assertion implies $\nu(K) \geq 9$, while its even order and absence of a perfect matching imply $\nu(K) \leq 9$. Thus $\nu(K) = 9$.

Choose a maximum-cardinality set B among the maximizers of the ordinary odd-component deficiency on K, and put $b = |B|$. Such a set exists because the vertex set is finite. Apply Lemma 107 with every vertex marked. Its deficiency is $20 - 2\nu(K) = 2$, so there are $b + 2$ odd components of K-B and

$9 = b + \sum_C (|V(C)| - 1)/2$. (2)

All these components are factor-critical. There is no even component: adjoining one of its vertices to B creates at least one new odd component and costs one separator vertex, so the objective cannot decrease. It cannot increase past its maximum, and equality contradicts the chosen maximal cardinality.

If $b \geq 1$, each component C has at most $20 - b - ((b + 2) - 1) = 19 - 2b$ vertices, since all other components have at least one vertex. Its matching rank $r_C = (|V(C)| - 1)/2$ is therefore at most $9 - b \leq 8 + b$. If $b = 0$, there are two odd components. Every K-vertex has degree at least $8 - 7 = 1$, since H is eight-regular and only W was removed. Each component thus has odd order at least three. Neither has order greater than seventeen, so again $r_C \leq 8 = 8 + b$.

Use the actual pool W union B, of size $7 + b$, in the ordered absorption theorem Lemma 51. The residual edge blocks are precisely these components; the separate retained edge graph is empty and has cost zero. Every block rank is at most $(7 + b) + 1$, so every ordering satisfies the theorem's inequality. Its total exceptional budget is

$(7+b)+\sum_C r_C = 7+b+(9-b) = 16$

by (2). The theorem therefore realizes every zero-capacity allocation of total sixteen on H, including (8,8), restoring every edge meeting the pool. This contradicts its failure. Hence K has a perfect matching. The base matching assertion is now proved in every case of (1).

Put $m_0 = 2k-6$. We extend the assertion to larger even induced orders by induction in steps of two. The base m_0 has just been proved. For an induced graph J of even order $m \geq m_0+2$, choose any m_0 vertices in J. Their induced graph has a perfect matching, and $m_0 \geq 16$, so J has at least one edge uv. The induced graph J-{u, v} has even order m-2 and has a perfect matching by induction. Add uv to it. The result is a perfect matching of J. For any induced odd-order J with $|V(J)| \geq m_0+1$ and any vertex v in it, J-v has even order at least m_0 and therefore a perfect matching. This proves factor-criticality. The exact floor-half matching ranks follow in both parities by the trivial upper bound and the matchings just constructed.

If S is nonempty and $a = |S| \leq 7$, its complement has order $2k+1-a \geq m_0$, so

$\nu(H-S) = floor((2k+1-a)/2) = k - \lfloor a/2 \rfloor$.

Subtracting $k+1-a$ gives $floor((a-1)/2)$. If $a \geq 7$, choose any seven-element subset I of S. The graph H-I has a perfect matching with k-3 edges. Deleting the other a-7 vertices removes at most a-7 edges from this matching. The resulting matching in H-S has at least $k-3-(a-7) = k+4-a$ edges, which proves the lower bound three for the displayed gain. This remains valid when the numerical lower bound on the matching cardinality is negative.

Finally suppose $k = 11$ or 12. Choose any vertex v and a perfect matching M of H-v, and let L be the spanning graph obtained from H by deleting the edges of M. Its degree is eight at v and seven at every other vertex. We claim every zero-capacity allocation of total $k+3$ is feasible on L. A pair whose smaller coordinate is at most six is feasible by the vertex-pool construction in the first paragraph, which used only the graph order. The only remaining pair, up to exchange, is (7,k-4), by (1).

Choose any vertex w different from v, whose L-degree is seven, and take $W = N_L(w)$. Thus $|W| = 7$ and w does not belong to W. Assign all L-edges meeting W to the exceptional part of budget seven. The induced graph L-W has 2k-6 vertices and contains the isolated vertex w. Its matching number is at most $floor((2k-7)/2) = k-4$, so all its edges fit in the other exceptional part. This realizes (7,k-4) and, on exchanging exceptional colors, its reverse. Consequently $c_0(L) \leq k+3$.

For capacity pair (1,0) put the fixed matching M in the degree-one part and use the proved exceptional partitions of L for every budget split of total $k+3$. For pair (0,1), put the same matching in the other degree-one part and again use those partitions. These are the only nonnegative integer capacity pairs of total one. They partition every original H-edge and respect all capacities and budgets, proving $c_1(H) \leq k+3$. All constructions used actual subgraphs and fixed matchings, and no slack condition was required. □

Remark. The few possible missing budget pairs force perfect matchings after seven vertex deletions. In the balanced twenty-seven-vertex case, an imperfect residual decomposes into odd blocks small enough to absorb with the same seven-vertex pool, contradicting the missing allocation.

Depends on. Lemma 23, Lemma 51, Lemma 107.

Lemma 156 (Independent cuts of factor-critical graphs with zero slack). *Use the finite integer certificate costs $c_i(Q)$, matching number $\nu(Q)$, independence number $\alpha(Q)$, and slack $ps(Q) = e(Q) - \nu(Q) - \sum_{i\geq 1} c_i(Q)$ of Lemma 23. A nonempty graph is factor-critical if deleting any one vertex leaves a perfect matching. All graphs are finite and simple. Let Q be factor-critical, with $n \geq 3$ vertices and $ps(Q) = 0$. Then Q is connected and D-regular for a positive even integer D. Put $k = (n-1)/2 = \nu(Q)$ and $s = \alpha(Q)$.*

For every maximum independent set S, put $G = Q - S$. Then

$\Delta(G) \leq D-1$, $ps(G) = 0$, $\nu(G) = k+1-s$,

$c_D(Q) = 1$, $c_i(Q) = s + c_i(G)$ *for* $1 \leq i < D$,

and $c_i(Q) = 0$ *for* $i > D$ *and* $c_i(G) = 0$ *for* $i \geq D$.

For a finite graph J write $o(J)$ for its number of odd-order connected components. Choose any vertex set X of maximum cardinality among the maximizers of $o(G-X) - |X|$. Such a choice exists. Put $b = |X|$. Then b belongs to {0,1};

every component of G-X is odd and factor-critical; and there are exactly $b + s - 1$ such components. For every such S and X, precisely one of the following descriptions applies, where all listed components are whole components of the specified induced graph:

(a) $b = 0$, and G consists of one triangle and isolated vertices, $D = 6$, and $n = 2s + 1$.

(b) $b = 0$, and G consists of two triangles and isolated vertices, $D = 4$, and $n = 2s + 3$.

(c) $b = 1$, say $X = \{x\}$, and G-x is edgeless. Then G is a star with exactly $D/2$ edges together with isolated vertices, and $n = 2s + 1$.

(d) $b = 1$, say $X = \{x\}$, and G-x consists of one triangle and s-1 isolated vertices. Then $D = 4$, $n = 2s + 3$, $deg_G(x) = 3$, and x has exactly one neighbor in S.

Here a triangle is a complete graph on three vertices; isolated vertices are components with one vertex and no edges; and the star in (c) has one center joined to $D/2$ distinct leaves. In particular either $n = 2s + 1$, or $D = 4$ and $n = 2s + 3$. If $s = 1$, then Q is a triangle; if $s = 2$, then Q is the cycle on five vertices. Orders, degrees, costs, indices, matching numbers, cardinalities, k, s, b and component counts are integers in the displayed domains.

Proof. The proof uses the nonnegative-slack and covered-deletion statements of Lemma 61, the restriction, restoration and component inequalities of Lemma 72, the marked-barrier theorem Lemma 107, and the odd-clique result Lemma 110. Costs and matching numbers have the definitions of Lemma 23.

A factor-critical graph of order at least three has odd order and matching number $(n - 1)/2$. It is connected: if distinct components contained vertices u and v, deletion of u would require the component containing v to have even order, while deletion of v would require it to have odd order. It has an edge, because deleting a vertex of an edgeless graph of order at least three leaves vertices unmatched.

If Q were nonregular, choose a vertex w below its maximum degree D. A perfect matching of Q-w covers every degree-D vertex of Q. For every degree-D vertex v, factor-criticality also gives $\nu(Q - v) = \nu(Q)$. The covered-deletion inequality gives $ps(Q) \geq ps(Q - v) + 1 \geq 1$, a contradiction. Thus Q is regular of positive degree D. Odd order and the identity $Dn = 2e(Q)$ imply that D is even. This argument applies to every factor-critical graph of order at least three and zero slack.

Fix a maximum independent set S of size s. It is nonempty. Every vertex outside S has a neighbor in S, since a vertex with no such neighbor could be added to S. Hence $\Delta(G) \leq D - 1$. Choose z in S and a perfect matching of Q-z. Deleting the other s-1 vertices of S destroys at most s-1 matching edges, proving $\nu(G) \geq k + 1 - s$. Write $\eta = \nu(G) - (k + 1 - s) \geq 0$.

The graph Q-v has a perfect matching for any v, so $c_D(Q - v) = 0$ and restoration gives $c_D(Q) \leq 1$. A zero-budget certificate at capacities (D-1,1) in the D-regular graph Q would make its degree-at-most-one part cover every vertex exactly once, yielding a perfect matching on odd order. This is impossible, so the integer cost definition gives $c_D(Q) = 1$. The zero tails asserted in the statement follow from Lemma 61.

For every $1 \leq i < D$ put $r_i = c_i(Q) - c_i(G)$. Restriction and restoration give $0 \leq r_i \leq s$. Independence and regularity give $e(Q) - e(G) = Ds$. Subtracting the slack definitions therefore gives

$ps(Q) = ps(G) + \eta + \sum_{i=1}^{D-1}(s - r_i)$.

Every term on the right is nonnegative. Since $ps(Q) = 0$, all are zero. Thus $\eta = 0$, $ps(G) = 0$, and $r_i = s$ at every indicated index, proving the exact cut identities.

There are finitely many vertex sets of G, so a maximum-cardinality maximizer X of $o(G - X) - |X|$ exists. Apply Lemma 107 with every vertex of G marked. Its deficiency is $|V(G)| - 2\nu(G)$, since a matching of size m leaves exactly $|V(G)| - 2m$ marked vertices. Thus the maximum of $o(G - X) - |X|$ has this value, and every odd component at a maximizer is factor-critical.

No component C of G-X can have even order. If it did, choose v in C. The graph C-v has odd total order and therefore has at least one odd component. Replacing X by X union {v} raises the number of odd components by at least one and raises $|X|$ by exactly one. The objective cannot decrease, so the new set is also a maximizer, contradicting the maximum cardinality of X. Hence every component is odd and factor-critical. The marked-barrier matching identity now reads

$\nu(G) = b + \sum_C (|V(C)| - 1)/2$.

If p is the number of these components, it follows that

$p = |V(G)| + b - 2\nu(G) = b + s - 1$.

One vertex from each component is independent in G, so $p \le \alpha(G) \le s$. Consequently $b \le 1$.

If $s = 1$ then Q is an odd complete graph. Lemma 110 excludes order at least five at zero slack, leaving precisely the triangle. Before analyzing the general component family, we prove the $s = 2$ conclusion without induction.

Let $s = 2$. In case $b = 0$, G is one factor-critical component of order $n - 2 \ge 3$: order $n = 3$ would already force Q to be a triangle of independence one. It has zero slack, and the nonregular exclusion proved above makes it regular of a positive even degree d. Each G-vertex has between one and two neighbors in S, so $1 \le D - d \le 2$. Parity forces $D - d = 2$. Thus every G-vertex is adjacent to both vertices of S. Counting these edges at S gives $D = n - 2$, contradicting even D and odd n.

In case $s = 2$, $b = 1$, write $X = \{x\}$. The graph G-x has two odd components. Their independent sets combine to one of size at most two, so each component is an odd complete graph. Lemma 110 applies to G and x, using $ps(G) = 0$ and $\alpha(G) \le 2$. Both component orders belong to {1,3}. Hence n is 5, 7 or 9.

If $n = 7$, one of the two components is an isolated vertex in G-x. Its neighbors in Q lie among the two vertices of S and x, so $D \le 3$ and the positive even degree is two. A connected finite simple graph with every degree two is a cycle: following consecutive edges first closes a cycle, whose vertices have no other incident edges, and connectedness excludes vertices outside it. Thus Q is a seven-cycle, which has an independent set of size three, a contradiction.

If $n = 9$, put $U = S$ union {x}; its complement consists of two triangles and has six vertices. The number of edges joining that complement to U is 6D-12, at most the total degree 3D at U. Hence $D \le 4$. Degree two would leave each triangle with no edge to U, contradicting connectedness. Degree four makes the crossing edge count equal to 3D, so there is no edge within U; this gives an independent triple, again impossible. Thus $n = 5$. Its positive even regular degree is two or four. Degree four would make Q complete and $s = 1$. Therefore Q is the five-cycle. This proves the $s = 2$ assertion for every graph under the theorem's hypotheses.

Return to arbitrary $s \ge 2$. Suppose $b = 0$. The $p = s - 1$ components of G have slack zero, by component superadditivity and nonnegative slack. Their independence numbers are positive integers summing to at most s, so all are one except possibly a single value two. Components with independence one are odd cliques and, by Lemma 110, are isolated vertices or triangles. A component of independence two is factor-critical with zero slack and is therefore a five-cycle by the conclusion just proved.

Let t be the number of triangle components and let ϵ in {0,1} indicate the presence of a five-cycle. Counting orders and edges gives

$|V(G)| = s - 1 + 2t + 4\epsilon$, $e(G) = 3t + 5\epsilon$,

$n = 2s - 1 + 2t + 4\epsilon$.

The degree sum on G and the Ds edges from G to S give $D(n - 2s) = 2e(G)$. Hence

$D(2t + 4\epsilon - 1) = 6t + 10\epsilon$.

If $\epsilon = 1$, this implies $D = 3 + 1/(2t + 3)$, which is not an integer. Thus $\epsilon = 0$. Then $t = 0$ would force $D = 0$, impossible; for $t \ge 1$,

$D = 3 + 3/(2t - 1)$.

The positive odd integer 2t-1 must divide three. Therefore either $t = 1$, $D = 6$, $n = 2s + 1$, or $t = 2$, $D = 4$, $n = 2s + 3$. These are (a) and (b).

Suppose $b = 1$, with $X = \{x\}$. There are s odd components of G-x. Their independence numbers sum to at most $\alpha(G) \le s$, so every component is an odd clique. Applying Lemma 110 again makes each an isolated vertex or a triangle. Write t for the number of triangles. Then

$|V(G - x)| = s + 2t$, and $n = 2s + 1 + 2t$.

The set $U = S$ union {x} has size $s + 1 > \alpha(Q)$, so $e(Q[U]) \ge 1$. Counting crossing edges between U and its complement in two ways gives

$D(s + 2t) - 6t = D(s + 1) - 2e(Q[U])$,

and therefore

$2e(Q[U]) = D(1-2t) + 6t \geq 2$.

If $t = 0$, G-x is edgeless and $n = 2s+1$. The identity $D(n-2s) = 2e(G)$ yields $e(G) = D/2$. Every G-edge meets x, so G is precisely the star with $D/2$ edges and isolated vertices, as in (c).

If $t \geq 1$, degree $D = 2$ would make any triangle a whole component of Q, contradicting connectedness because U is nonempty. Thus $D \geq 4$. For $t \geq 2$ the right side $D(1-2t)+6t$ is at most $4-2t \leq 0$, contradicting its lower bound two. Hence $t = 1$. The same identity gives $6 - D \geq 2$, forcing $D = 4$ and $e(Q[U]) = 1$. Since S is independent, the unique edge within U joins x to exactly one vertex of S. Consequently $deg_G(x) = 3$. The order is $n = 2s+3$, and G-x has one triangle and s-1 isolates, proving (d). The case $s = 1$, already known to be a triangle, has G a single edge and falls in (c). This proves all alternatives and all their quantified cut identities. □

Remark. Zero slack forces every positive certificate row to lose the full independent-set budget. The remaining matching is then as small as possible, leaving at most one barrier vertex. The clique restriction and a degree count reduce its components to two explicit small configurations.

Depends on. Lemma 23, Lemma 61, Lemma 72, Lemma 107, Lemma 110.

Lemma 157 (Closed lumps of degree twenty-four are LP-good). *Use the integer certificate costs $c_\theta(H)$, matching number $\nu(H)$, excess $ex(H) = c_0(H) - \nu(H)$, and $gap(H) = \sum_{\theta \geq 0} c_\theta(H) - e(H)$ of Lemma 23. Let H be any nonempty connected 24-regular finite simple graph such that deleting each vertex leaves a perfect matching. Then $gap(H) \leq 0$. More precisely, write $|V(H)| = 2k+1$. The integer k equals $\nu(H)$ and satisfies $k \geq 12$. If $k \geq 33$, then $gap(H) < 29/3 - 799k/3000 < 1$; if $12 \leq k \leq 32$, the explicit integer bounds in the proof are nonpositive. No hereditary or minimality hypothesis and no upper bound on $|V(H)|$ is imposed.*

Proof. Put $n = |V(H)|$. A perfect matching after any vertex deletion gives odd $n = 2k+1$ and $\nu(H) = k$. Simplicity gives $24 \leq 2k$, so the integer k is at least 12, and regularity gives $e(H) = 12(2k+1)$. Set $E = ex(H) \geq 0$, an integer by Lemma 23.

We need the exact vertex-count ceiling $E \leq floor((k+1)/3)$. For nonnegative integer budgets $x \geq y$, if $n \leq 2x + y + 1$, take a set of $\min(n, 2x+1)$ vertices, color its internal edges red and every remaining edge blue. Each red matching has at most x edges, and every blue edge meets the complement, whose size is at most y. For $y > x$ interchange the colors. Thus every allocation of total c is feasible at zero capacities when $c + \lceil c/2 \rceil + 1 \geq n$. Put $e_0 = floor((k+1)/3)$ and $c = k + e_0$. For $k = 3h$, $3h+1$, $3h+2$ with nonnegative integer h, the left side is respectively $6h+1$, $6h+3$, $6h+6$, at least the corresponding $n = 6h+1$, $6h+3$, $6h+5$. Therefore $c_0(H) \leq k + e_0$ and $E \leq e_0$.

By Lemma 87 the costs vanish for $\theta \geq 25$, $c_{24}(H) \leq 1$, and the two complementary pair bounds are $c_i(H) + c_{25-i}(H) \leq 2k+1$ and $c_i(H) + c_{24-i}(H) \leq 2k+2$ in the ranges required by Lemma 12. By Lemma 23, all costs are nonnegative integers at most $c_0(H) = k + E$. The scalar theorem Lemma 12 therefore applies with $d = 24$, $t = 4$. For any four real bounds $b_j \geq c_{24-j}(H)$, it yields

$$gap(H) \leq -4k + \sum_{j=1}^{4} b_j + \min(6E - 4, 5E + 4). \quad (1)$$

Indeed its g is $gap(H)$ because the higher rows vanish and the baseline is $e(H)$. In particular the second expression in (1) gives $gap(H) \leq 5E - 4k + 4 + \sum_j b_j$.

Fix a vertex v and a perfect matching M of H-v. The graph H-M has degree 24 at v and 23 at every other vertex. In Lemma 24 take $m = 23$ and $p_1 = 1/8$, $p_2 = 1/6$, $p_3 = 2/9$, $p_4 = 3/11$, and write $q_j = q_{23,j}(p_j)$. For each j the resulting marked j-dominating set, followed by Lemma 49 with $D = 24$, gives $c_{24-j}(H) \leq 1 + 2kq_j$. The marked vertex and matching stay fixed; the four sets may differ because they bound four separate scalar costs.

Define $(Q_1, Q_2, Q_3, Q_4) = (1660, 2375, 2905, 3395)/10000$. We prove $q_j < Q_j$ by explicit rational calculations. Expansion of the binomial expressions gives

$q_1 = 1/8 + 7^{24}/8^{24}$,

$q_2 = 1/6 + 28 \cdot 5^{23}/6^{24}$,

$q_3 = 2/9 + 1383 \cdot 7^{22}/9^{24}$,

$q_4 = 3/11 + 70961 \cdot 8^{21}/11^{24}$.

For q_3 the coefficient is $49 + 46 \cdot 7 + 1012 = 1383$; for q_4 it is $512 + 69 \cdot 64 + 2277 \cdot 8 + 47817 = 70961$. The four positive differences $Q_j - p_j$ are $41/1000$, $17/240$, $1229/18000$, $1469/22000$. The required comparisons are exactly

$1000 \cdot 7^{24} = 191581231380566414401000 < 193617025797655453761536 = 41 \cdot 8^{24}$;

$240 \cdot 28 \cdot 5^{23} = 80108642578125000000 < 80552482751467487232 = 17 \cdot 6^{24}$;

$18000 \cdot 1383 \cdot 7^{22} = 97331085183424904491806000 < 98032958541476314622070669 = 1229 \cdot 9^{24}$;

$22000 \cdot 70961 \cdot 8^{21} = 14398993468359538414452736000 < 14469257300761380698131694429 = 1469 \cdot 11^{24}$.

This proves the comparisons and gives the integer bounds $c_{24-j}(H) \le \lfloor 1 + 2kQ_j \rfloor$. Since $\sum_j Q_j = 2067/2000$, the real bounds and $E \le (k+1)/3$ give

$gap(H) < 5(k+1)/3 - 4k + 4 + 4 + 2k \cdot (2067/2000) = 29/3 - 799k/3000$.

At $k = 33$ this is $2633/3000 < 1$, and it decreases with k. The integral gap is therefore at most zero whenever $k \ge 33$.

It remains to check $12 \le k \le 32$, including every possible smaller order. Apply Lemma 37 to the same graph H-M with $d = 24$, $n = 2k + 1$. Its greedy recurrence starts at $r_0 = 2k - 24$ and has updates $r_{i+1} = \max(0, r_i - ceil(24r_i/(2k - i)))$. If s is its first zero index, the marked dominating set and Lemma 49 give $c_{23}(H) \le 1 + s$. For $j = 2, 3, 4$ the other conclusion of Lemma 37 gives a marked j-dominating set of size $2k - 23 + j$. Again Lemma 49 gives $c_{24-j}(H) \le 2k - 23 + j$. Thus in (1) we may use $b_1 = 1 + s$ and $b_j = \min(\lfloor 1 + 2kQ_j \rfloor, 2k - 23 + j)$ for $j = 2, 3, 4$. Replacing E by e_0 is valid because both affine expressions in E in (1) increase with E.

The following table gives, in order, k; e_0; (b_1, b_2, b_3, b_4); the entire recurrence from r_0 through zero; and $U = -4k + sum\ b_j + \min(6e_0 - 4, 5e_0 + 4)$. Each recurrence step is the displayed integer formula, so all required sizes and inequalities are explicit.

12;4;(1,3,4,5);(0);-15

13;4;(2,5,6,7);(2,0);-12

14;5;(2,7,8,9);(4,0);-4

15;5;(3,8,9,11);(6,1,0);-3

16;5;(3,8,10,11);(8,2,0);-6

17;6;(3,9,10,12);(10,2,0);-2

18;6;(4,9,11,13);(12,4,1,0);-3

19;6;(4,10,12,13);(14,5,1,0);-5

20;7;(4,10,12,14);(16,6,2,0);-2

21;7;(4,10,13,15);(18,7,2,0);-4

22;7;(5,11,13,15);(20,9,3,1,0);-6

23;8;(5,11,14,16);(22,10,4,1,0);-2

24;8;(5,12,14,17);(24,12,5,2,0);-4

25;8;(6,12,15,17);(26,13,6,3,1,0);-6

26;9;(6,13,16,18);(28,15,7,3,1,0);-2

27;9;(7,13,16,19);(30,16,8,4,2,1,0);-4

28;9;(7,14,17,20);(32,18,10,5,2,1,0);-5

29;10;(7,14,17,20);(34,19,11,6,3,1,0);-4

30;10;(8,15,18,21);(36,21,12,7,4,2,1,0);-4

31;10;(8,15,19,22);(38,23,13,7,4,2,1,0);-6

32;11;(8,16,19,22);(40,25,15,9,5,3,1,0);-4.

Every last entry is nonpositive. Formula (1) then proves $gap(H) \leq 0$ for every k in this interval, completing all integer $k \geq 12$. □

Remark. The exact excess ceiling controls all large orders. At the remaining orders, greedy first-level domination and a bound obtained by counting nonneighbors supply all needed certificates.

Depends on. Lemma 12, Lemma 23, Lemma 24, Lemma 37, Lemma 49, Lemma 87.

Lemma 158 (Simultaneous matching losses at exposed neighborhoods). *All graphs are finite and simple. A matching is a set of edges with disjoint endpoints; $V(W)$ denotes its covered vertices, and $\nu(G)$ its maximum cardinality in G. A vertex cover meets every edge, and $\tau(G)$ is its minimum cardinality. A graph is factor-critical if it is nonempty and deleting any vertex leaves a perfect matching. The empty graph has its empty perfect matching. Vertex deletion is induced, whereas G-W deletes only the matching edges.*

Let $D \geq 3$ be an integer and let G have maximum degree at most D. Put $S = \{v : deg_G(v) = D\}$ and $\rho = \min_W |S$ minus $V(W)|$, where the minimum is over all matchings of G. Let Ccal be the family of whole connected components of G which are odd, D-regular and factor-critical. There is a matching W of cardinality $\nu(G)$ attaining ρ for which the following assertion holds simultaneously for every $Z \subseteq I$, where

$I = S$ minus $V(W)$, $Q = G - W$, and $T = Q - I$.

Suppose the open neighborhoods $N_Q(x)$, for x in Z, are pairwise disjoint. Put `Z_0={x` *in Z:x belongs to a member of Ccal}*, *$Z_1 = Z$ minus Z_0, and $N_Q(Z) = \bigcup_{x \in Z} N_Q(x)$. Then*

$\tau(T) - \nu(T - N_Q(Z)) \geq |Z_0|\lceil D/2 \rceil + |Z_1| ceil((D+1)/2)$.

In particular the difference is at least $|Z|\lceil D/2 \rceil$. The matching W is chosen before Z. The assertion permits $\rho = 0$, Z empty, empty G and disconnected G. All orders, degrees, cardinalities and matching and cover sizes are nonnegative integers, subject to $D \geq 3$.

Proof. Remove all components in Ccal and call the remaining whole-component union G_0. Put $S_0 = S$ intersect $V(G_0)$, and let ρ_0 be its minimum uncovered S_0 count. A member C of Ccal has order $2\mu_C + 1$ and matching number μ_C: factor-criticality supplies a matching of that size, and odd order prevents a larger one. Every vertex of C is marked, so its minimum exposure is one. Maximum matching sizes and minimum marked exposure both add over whole components. Indeed restriction bounds any global matching componentwise, and unions of componentwise optimizing matchings attain those bounds. Thus $\rho = |Ccal| + \rho_0$.

First suppose $\rho_0 > 0$. Apply Lemma 107 to G_0 with marked set S_0. Choose a set X of maximum cardinality among the maximizers of the marked odd-component deficiency; it is inclusionwise maximal, as required by that fact. Let C_1,..., C_r be its counted odd components. They are factor-critical, all their vertices have $G_0 - degree\ D$, and $r = |X| + \rho_0$. Each C_i has a neighbor in X. Otherwise it would itself be a whole odd D-regular factor-critical component of G_0 and hence of G, contrary to the removal defining G_0.

In each C_i prescribe a vertex adjacent to X. The compatible construction in Lemma 107 supplies a maximum matching W_0 attaining ρ_0. It matches each vertex of X into a distinct one of exactly $|X|$ counted components, called reached components. In each of the remaining components precisely its prescribed vertex is uncovered. All other marked vertices are covered. Define $I_0 = S_0$ minus $V(W_0)$, $Q_0 = G_0 - W_0$, and $T_0 = Q_0 - I_0$. Each root x in I_0 lies in a distinct unreached component. Write $b_x = |N_G(x)$ intersect $X| \geq 1$ and $k_x = D - b_x$ for its number of internal neighbors; thus $k_x \leq d := D - 1$. The neighborhoods of x in G_0 and Q_0 coincide because W_0 leaves x uncovered.

Write $|V(C_i)| = 2\mu_i + 1$, let $U = X$ union $\bigcup_i V(C_i)$, and put $B = |X| + \sum_i \mu_i$. We prove a bound for an arbitrary vertex cover A of T_0. Put $A_X = A$ intersect X, $U_i = V(C_i)$ minus I_0, $a_i = |A$ intersect $U_i|$ and $O_i = U_i$ minus A. Since A covers T_0, O_i is independent, and all its neighbors outside U_i lie in A_X. Let e_i count its $T_0 - edges$ to A_X.

For reached C_i, every vertex of U_i has degree d in T_0. It lost its matching edge and has no neighbor in I_0, which lies in other components of $G_0 - X$. Counting edges incident with O_i gives $d|O_i| \le d\,a_i + e_i$. Define $t_i = \max(\mu_i + 1 - a_i, 0)$. When $t_i \ge 1$, $|O_i| - a_i = 2t_i - 1$, so $e_i \ge d(2t_i - 1) \ge d\,t_i$. The same final bound holds when $t_i = 0$.

For an unreached component with root x, its $2\mu_i$ retained vertices have degree d, except its k_x internal neighbors, whose degree is d-1. Consequently $d|O_i| - k_x \le d\,a_i + e_i$. Here put $t_i = \max(\mu_i - a_i, 0)$. If $t_i \ge 1$, then $|O_i| - a_i = 2t_i$ and $e_i \ge 2d\,t_i - k_x \ge d\,t_i$, since $k_x \le d$ and $t_i \ge 1$. Again the bound holds for $t_i = 0$.

The edge sets counted by e_i are disjoint and incident with A_X. Every vertex of X is $W_0 -$ *covered* and has degree at most d in T_0. Hence

$d \sum_i t_i \le \sum_i e_i \le d|A_X|$.

As $d > 0$, $\sum_i t_i \le |A_X|$. There are exactly $|X|$ reached components. Summing $a_i \ge \mu_i + 1 - t_i$ on reached components and $a_i \ge \mu_i - t_i$ on the rest proves

$|A$ intersect $U| = |A_X| + \sum_i a_i \ge B$. (1)

Now take any $Z_1 \subseteq I_0$, and write $b_Z = |N_{Q_0}(Z_1)$ intersect $X|$. Let F be any matching of $T_0 - N_{Q_0}(Z_1)$. At most $|X| - b_Z$ of its edges meet X. All its other edges meeting U are internal to counted components, since every edge leaving a component of $G_0 - X$ goes to X. For the component of a tested root x, deletion of that root and its k_x internal neighbors leaves $2\mu_i - k_x$ vertices, and hence at most $\mu_i - \lceil k_x/2 \rceil$ internal matching edges. Every other counted component permits at most μ_i internal matching edges. The tested roots lie in distinct components. Therefore at most

$B - b_Z - \sum_{x \in Z_1} \lceil k_x/2 \rceil$

edges of F meet U.

Every F-edge wholly outside U meets A outside U. Distinct matching edges use distinct such cover vertices, so there are at most $|A$ minus $U|$ of them. Subtract this bound for $|F|$ from (1) plus $|A$ minus $U|$ to obtain

$|A| - |F| \ge b_Z + \sum_{x \in Z_1} ceil((D - b_x)/2)$. (2)

If the tested open neighborhoods are pairwise disjoint, their X-neighborhoods are pairwise disjoint. Thus $b_Z = \sum_{x \in Z_1} b_x$. Each resulting summand is

$b_x + ceil((D - b_x)/2) = ceil((D + b_x)/2) \ge ceil((D + 1)/2)$.

Taking A to be a minimum cover and F a maximum matching yields

$\tau(T_0) - \nu(T_0 - N_{Q_0}(Z_1)) \ge |Z_1| ceil((D + 1)/2)$. (3)

The construction of W_0 preceded the choices of A, F and Z_1.

If $\rho_0 = 0$, use Lemma 107 to choose a maximum matching W_0 covering S_0, and put I_0 empty, $T_0 = G_0 - W_0$. Now Z_1 is necessarily empty. Every matching has size at most every vertex cover, since its disjoint edges require distinct cover vertices. Therefore (3) remains true. This includes empty G_0.

For each C in Ccal choose any vertex x_C and a perfect matching W_C of $C - x_C$. Put $T_C = (C - W_C) - x_C$ and $\mu = \mu_C$. Its 2mu vertices have degree $d = D - 1$ except the D neighbors of x_C, whose degree is d-1. We claim $\tau(T_C) \ge \mu$. If a cover A had $a = |A| \le \mu - 1$, its independent complement O would satisfy $|O| - a = 2\mu - 2a \ge 2$. The total degree deficiency from d among all vertices of T_C is D, so counting edges from O to A gives $d|O| - D \le d$ a. It would follow that $2(D - 1) \le D$, contradicting $D \ge 3$. This proves the claim.

The graph $T_C - N_C(x_C)$ has 2mu-D vertices. Its matching number is at most $floor((2\mu - D)/2) = \mu - \lceil D/2 \rceil$. Consequently

$\tau(T_C) - \nu(T_C - N_C(x_C)) \ge \lceil D/2 \rceil$. (4)

Without that neighborhood deletion one still has $\tau(T_C) - \nu(T_C) \ge 0$. These arguments use no choice of W_C beyond its being a perfect matching of $C - x_C$.

Take W to be the union of W_0 and all W_C. The whole-component additivity proved at the start shows that W is maximum in G and attains ρ. Its exposed set is I_0 together with the roots x_C. Also Q-I is the disjoint

union of T_0 and all T_C. Minimum cover size and maximum matching size add over whole components, since restriction gives one inequality and unions of optimal choices give the reverse one.

For any tested Z as in the statement, its roots in G_0 satisfy the disjoint-neighborhood hypothesis in (3). Each tested root in a component C contributes (4), and every untested such component contributes a nonnegative difference. All these root neighborhoods lie in their respective whole-component parts. Adding their inequalities gives precisely the stated bound. When $\rho = 0$ the only Z is empty and the same proof gives a nonnegative difference. All matching choices were made independently of Z, proving the simultaneous assertion. □

Remark. A marked separator bounds the cover capacity of its required odd components. Deleting disjoint root neighborhoods removes additive matching capacity. Whole regular components satisfy the same estimate by a direct degree count.

Depends on. Lemma 107.

Lemma 159 (Successive matchings along a complete row process). *All graphs are finite and simple. Use the integer certificate costs $c_{A_c,B_c}(G)$, their row maxima $c_i(G)$, and the signed slack $ps(G) = e(G) - \nu(G) - \sum_{i\ge1} c_i(G)$ of Lemma 122. Thus $c_{A_c,B_c}(G)$ is the least nonnegative integer total for which every matching-budget split permits an edge partition into two parts of maximum degrees at most A_c, B_c and two parts with the prescribed matching-number bounds; c_i is the maximum over nonnegative integer $A_c + B_c = i$. Write $V(G)$, $E(G)$, $e(G)$, $\nu(G)$, $\Delta(G)$, and $d_G(v)$ for vertices, edges, edge count, matching number, maximum degree, and vertex degree. Vertex deletion means induced deletion; deleting a matching deletes its edges and retains all vertices.*

Let $D \ge 6$ be an even integer, let r, s, m, β, p be nonnegative integers, and put $u = r + s$. Let F be D-regular of order $2r + s$. Let Rcal be a nonempty family of induced subgraphs A of F satisfying

$|V(A)| = r$, $\Delta(A) \le 2$, $\nu(A) = m$, $c_1(A) = \beta$, $c_2(A) = p$, $ps(A) = 0$.

Assume $c_1(F) = u + \beta$ and $c_2(F) = u + p$. Let x, y be nonnegative integers with $x + y + 1 = u + m$ and $\min(x, y) \le r$ such that $E(F)$ has no two-part partition with matching numbers at most x, y.

For every A in Rcal assume $\Delta(F - V(A)) \le D - 3$. A complete process from F-V(A) repeatedly deletes any vertex of current maximum degree while that degree exceeds two. Assume every such process ends in Rcal and, writing n_j for its number of deletions at degree j, assume

$c_i(F) = r + \sum_{j=i+1}^{D-3} n_j$ *for every integer* $3 \le i \le D - 3$.

These hypotheses apply to every initial row and every complete process.

Fix any A in Rcal and any such process, with terminal row R and full deletion phases T_j for $3 \le j \le D - 3$. Put $X = \bigcup_{j=3}^{D-3} T_j$. Choose any subset Y of $\{vinV(R) : d_R(v) = 2\}$. Define

$W = X$ *union* Y, $H = F[V(A)$ *union* $W]$, $S = V(F)$ *minus* $V(H)$,

$h = |W| = s + |Y|$, $N = |S| = r - |Y|$.

For each integer $3 \le i \le D - 3$ *put*

$I_i = V(A)$ *union* $\bigcup_{D-i\le j\le D-3} T_j$,

$C_i = V(H)$ *minus* I_i,

$b_i = |C_i| = |Y| + \sum_{j=3}^{D-i-1} |T_j|$.

(i) Suppose three pairwise edge-disjoint matchings M_0, M_1, M_2 in H are given, each matching every vertex of W into $V(A)$, and their union contains every edge of A. They need not be maximum. Assume $N \ge \max(0, b_3 - 2)$.

For any matching segment define the spanning graphs

$P_i = (V(H), E(H)minus \bigcup_{0\le a<i} M_a)$,

$L_i = (V(H), \bigcup_{0\le a<i} M_a)$.

Recursively choose any maximum matching M_i of P_i for every integer $3 \le i \le D-3$. Every such sequence exists. At every such index, I_i is independent in P_i, C_i is a vertex cover, $P_i[C_i] = H[C_i]$, and

$|M_i| = \nu(P_i) = b_i = c_i(F) - N$.

Every maximum M_i matches all of C_i into I_i and has no other edges. For $3 \le i \le D-4$, every vertex of T_{D-i-1} has exactly one remaining edge to I_i, and that edge belongs to M_i. No matching in the entire sequence removes an edge of R[Y].

Put $Z = V(A)$ union X. In P_{D-3}, every y in Y has exactly one neighbor $\phi(y)$ in Z, the map $\phi : Y \to Z$ is injective, and the unique maximum matching is

$M_{D-3} = \{y\phi(y) : y \in Y\}$.

The entire edge set of P_{D-2} is $E(R[Y])$; every vertex of Z is isolated. Thus $\nu(P_{D-2}) = \nu(R[Y])$. In particular any chosen whole odd cycle contained in R[Y] survives as the same physical cycle with every original edge.

For every integer $3 \le i \le D-3$ and every nonnegative integer capacity split $A_c + B_c = i$ and budget split $b_R + b_B = N + b_i$, the entire F has a certificate whose degree-bounded parts have union $E(L_i)$. Every edge meeting S, including every edge with both endpoints in S, belongs to a matching-bounded part.

(ii) Instead of supplying the prefix in (i), assume $\min(x, y) \le N$ and $N \ge \max(0, \beta + h - 2)$. Then any three successive maximum matchings in $P_0 = H$, P_1, P_2 supply such a prefix, with

$|M_0| = m + h$, $|M_1| = \beta + h$, $|M_2| = p + h$.

Consequently every sequence of successive maximum matchings M_0,..., M_{D-3} has all conclusions of (i), and every matching is maximum in its own residual graph.

All indices, orders, degrees, cardinalities, phase counts, capacities, budgets, and matching numbers are integers in the displayed domains. Empty phases, $s = 0$, Y empty, and zero matching sizes are allowed. At $D = 6$ the range $3 \le i \le D-4$ is empty and the only appended matching is M_3.

Proof. The five inputs are Lemma 122 for physical phases and their costs, Lemma 14 for a matching transition, Lemma 57 for degree-two rows, Lemma 7 for the initial matching rank, and Lemma 20 for the first three matchings. They use identical finite integer certificate definitions.

1. Physical sets and initialization.

Every hypothesis of Lemma 122 is included in the statement, with the same F, D, r, s, m, β, p, u, Rcal, x, y. Apply it to the fixed A and complete process. It gives the partition $V(A)$, X, $V(R)$, exactly s roots, independent phases, and the displayed values of h, N, b_i. Its complementary phase-count identities give

$c_i(F) = N + b_i$ for $3 \le i \le D-3$. (1)

For $3 \le i \le D-4$, the earlier side for a vertex of T_{D-i-1} is exactly I_i. Every such vertex has $D - (D-i-1) = i+1$ original neighbors in that side, all retained in H. Every T_{D-3} vertex has exactly three A-neighbors. Each y in Y has D-2 original neighbors outside R, all in the retained $Z = V(A)$ union X, because $d_R(y) = 2$ and F is D-regular.

Now suppose the prefix in (i) is supplied. At a vertex of T_{D-3}, its three distinct matching edges into A exhaust all three A-neighbors. The prefix also removes every A-edge, and T_{D-3} is independent. Thus $I_3 = V(A)$ union T_{D-3} is independent in P_3. Each prefix matching saturates W into A and therefore has no edge inside W: its crossing edges already use every W-vertex, so no additional matching edge can meet W. Hence $P_3[C_3] = H[C_3]$. Each vertex of C_3 was matched by each prefix member into A, which is contained in I_3. This establishes initialization without assuming maximality or maximum cardinality of the supplied prefix.

2. The invariant and each transition.

We induct on i from 3 through D-3. The invariant before choosing M_i consists of these three assertions: I_i is independent in P_i; $P_i[C_i] = H[C_i]$; and each vertex of C_i was matched by every earlier M_a, $0 \le a < i$, into I_i. Initialization proves all three at $i = 3$. The sets I_i increase and C_i decrease with i, so $b_i \le b_3$ and

$N \ge \max(0, b_i - 2)$. (2)

For $i \le D-4$ apply Lemma 14 with its index $k = i$, earlier side $I = I_i$, cover $C = C_i$, size $b = b_i$, and next layer $B = T_{D-i-1}$. Its retained graph is exactly $H = F - S$. Its prefix is the already chosen segment M_0,...,

M_{i-1}. All prefix members are pairwise edge-disjoint matchings since each newly chosen matching lies in the current residual. Independence of I_i says every original $H[I_i]$ edge belongs to L_i. Equation (1) supplies the original-graph cost lower bound and (2) supplies the omission bound. The phase B is independent in H, has $i+1$ neighbors per vertex in I_i, and the last invariant supplies all i prior crossing edges.

The transition proves $\nu(P_i) = b_i$ and that every maximum M_i saturates C_i into I_i. It forces the unique remaining $I_i - edge$ at each vertex of B, makes I_i union $B = I_{i+1}$ independent after deletion, and preserves every $C_i - internal$ edge. Since C_{i+1} is contained in C_i, this gives $P_{i+1}[C_{i+1}] = H[C_{i+1}]$. Earlier matchings still send C_{i+1} into the enlarged I_{i+1}, and the new matching sends it into I_i. These statements establish all three invariant clauses at $i+1$ on the same physical graph.

At $i = D-3$ use the identical transition with B empty. The cost and omission bounds remain (1) and (2), and the empty layer requires no neighbor condition. This proves the final appended rank and crossing saturation. Every maximum exists because its finite residual graph has finitely many matchings, including the empty matching. At each step the argument applies to every maximum, so no choices need replacement at later indices.

Throughout this induction Y is contained in every C_i. The prefix has no W-internal edge and every appended matching crosses the current cover into its earlier side. Consequently no edge inside Y is ever removed. Since $H[Y] = R[Y]$, these are exactly the asserted original edges.

3. The entire terminal residual and certificates.

At $i = D-3$ the earlier side is $I_{D-3} = Z$ and the cover is $C_{D-3} = Y$. By the invariant every y in Y has lost exactly one edge into Z in each of M_0,..., M_{D-4}, which are D-3 pairwise edge-disjoint matchings. There were D-2 original such edges. Therefore it has exactly one residual Z-neighbor $\phi(y)$. Every maximum M_{D-3} crosses Y into Z and saturates Y, and must therefore contain all edges $y\,\phi(y)$. Existence makes their Z-endpoints distinct. These edges constitute the entire maximum matching, proving both injectivity and uniqueness.

The three possible edge locations in P_{D-3} are inside Z, between Z and Y, and inside Y. There are no edges of the first kind by the invariant. The edges of the second kind are precisely the forced matching and are all deleted. The third kind is exactly $E(H[Y])$ and remains untouched. Thus $E(P_{D-2}) = E(R[Y])$, with all vertices of Z isolated. This proves the asserted full residual and its matching number, including Y empty.

Fix $3 \le i \le D-3$. The set S union C_i covers the spanning edge graph with edge set $E(F)$ minus $E(L_i)$: every edge outside H meets S, and each remaining edge within H meets C_i because I_i is independent in P_i. These covering sets are disjoint and have total size $N + b_i$. For given nonnegative integer $b_R + b_B = N + b_i$, partition S union C_i into sets of those respective sizes. Assign every paid edge with an endpoint in the first set to the first matching-bounded part; assign every other paid edge to the second. The latter edges all meet the second set. Every matching uses distinct covering vertices, so the two matching bounds hold.

For a capacity split $A_c + B_c = i$, put the first A_c of M_0,..., M_{i-1} in the first degree-bounded part and the others in the second. Each matching contributes at most one incident edge at any vertex; hence the two degree bounds hold. Their union is precisely L_i and is disjoint from the paid edges. All four parts partition $E(F)$. Edges with either or both endpoints in S are in the paid graph and are assigned exactly once. This supplies every printed certificate, including zero capacities and budgets.

4. Construction of the initial prefix.

Under (ii), let $q = \min(x, y)$. The actual omitted count is $N = u - h$, so $q \le N$ gives $h \le u - q$ and $u \ge q$. Apply Lemma 7 with row vertex set $V(A)$, exterior subset W, missing allocation (x, y), and total $x + y + 1 = u + m$. Its conclusion is $\nu(H) = m + h$.

By Lemma 57, a zero-slack degree-two A has only components P_1, P_2, P_3, P_4, P_6, C_3, C_4, C_5, C_7, where P_j, C_j are paths and cycles on j vertices. If ϵ is its number of positive even-order path components, the same theorem gives $\beta = m - \epsilon$ and says p is its number of odd-cycle components. Thus A meets the component and coefficient hypotheses of Lemma 20. The other hypotheses of that theorem are the just-proved rank, the assumed omission bound, and

$c_1(F) = u + \beta = N + h + \beta,$

$c_2(F) = u + p = N + h + p.$

It follows that any first three successive maximum matchings have sizes $m+h$, $\beta+h$, $p+h$, each saturates W into A, and their union removes every A-edge. Hence they supply precisely the common prefix required in (i).

Finally $b_3 = h - |T_{D-3}| \le h$ and $\beta \ge 0$, so

$N \ge \max(0, \beta + h - 2) \ge \max(0, b_3 - 2)$.

Part (i) therefore applies to every such prefix and all its successive maximum continuations. At $D = 6$ no root-layer transition occurs: initialization clears T_3 and the single empty-layer transition gives M_3 and the same terminal conclusion. For $s = 0$ or empty phases the invariant uses the same empty unions and the corresponding empty next layers. All constructions are finite and retain every vertex of H during matching deletions. □

Remark. The reflected phase counts turn exact source costs into cover sizes. A common three-matching prefix clears the initial row, and rank equality then forces every later maximum matching across its current cover. At the final stage the only crossing edges are a forced matching, leaving exactly the original selected terminal subgraph.

Depends on. Lemma 7, Lemma 14, Lemma 20, Lemma 57, Lemma 122.

Lemma 160 (A component run forces a degree bound at gap two). *Use the finite integer certificate costs c_i, matching number ν, excess $ex = c_0 - \nu$, slack ps, and $gap = ex - ps$ of Lemma 113. For a graph Q and nonnegative integer t, let $Y_t(Q)$ be its feasible first matching budgets at total t and let $run_t(Y_t(Q))$ be its longest consecutive missing integer run, as in Lemma 75. Write $tr(Q)$ for the run at total $c_0(Q) - 1$ when $c_0(Q) > 0$ and put $tr(Q) = 0$ otherwise. Put $h(D) = \max(4, \lfloor D/2 \rfloor + 1)$ for nonnegative integers D. Property T_- means that positive gap implies gap one and $tr + ex \le h(\Delta)$.*

Let H be a finite simple connected d-regular graph of odd order $2k+1$ such that every vertex deletion leaves a perfect matching, with even integer $d \ge 4$ and $gap(H) = 2$. Let S be an inclusionwise maximal independent set of cardinality $a \ge 2$, and put $G = H - S$. Suppose every connected component Q of G satisfies T_-, and every such Q with $ps(Q) = 0$ satisfies $run_(\nu(Q))(Y_(\nu(Q))(Q)) \le a$.

Call a component P of G large if $floor(|V(P)|/2) \ge a+2$, and put $m = \max(\{0\} union \{ex(P) : P is large\})$. Then $m \ge 1$. For every large P with $ex(P) = m$, one has

$gap(P) = 1$, $ps(P) = ps(G) = m - 1$,

$ps(Q) = 0$ for every other component Q of G,

$c_0(H) = a + \nu(G) + m$.

Put $L = a + \sum_{Q different from P} \nu(Q)$. Then

$L + 1 \le tr(P)$, *and* $tr(P) + m \le h(d-1) = \max(4, d/2)$.

In particular $a + m + 1 \le \max(4, d/2)$. The integer $\eta = \nu(G) - (k + 1 - a)$ satisfies

$0 \le \eta \le floor((a-1)/2)$, *and* $ex(H) = 1 + m + \eta$.

Also $\Delta(G) \le d - 1$ and every vertex of G has degree at least d-a.

Suppose in addition that every positive-gap component Q of G satisfies the three implications

$ex(Q) = 1 \Rightarrow tr(Q) \le 2$,

$ex(Q) = 2 \Rightarrow tr(Q) \le 4$,

$ex(Q) = 3 \Rightarrow tr(Q) \le 5$.

Then the following necessary bounds hold: if $a = 2$, then $m \ge 2$ and $d \ge 10$; if $a = 3$, then $m \ge 2$ and $d \ge 12$; if $a = 4$, then $m \ge 3$ and $d \ge 16$; if $a \ge 5$, then $m \ge 4$ and $d \ge 20$.

All orders, degrees, costs, budgets, indices, matching numbers, excesses, and displayed cardinalities are integers in the stated domains; generic gaps may be signed. No hereditary assumption on H, no minimality, and no conclusion asserting existence of a graph satisfying these hypotheses is imposed.

Proof. The graph H is not complete because it has the independent set S of size at least two. Thus Lemma 113 applies to this H, S, G and supplies $\Delta(G) \le d-1$ and

$2 = gap(H) \le c_0(H) + 1 - a - \nu(G) - ps(G) \le 1 + m - ps(G).$

Slack is nonnegative by Lemma 61. If $m = 0$ the last expression is at most one, a contradiction. Therefore $m \ge 1$ and a large component P attaining it exists.

Fix any such P. The component inequality in Lemma 113 gives

$2 \le 1 + gap(P) - \sum_{Q different from P} ps(Q) \le 2,$

because $T_{-}(P)$ implies $gap(P) \le 1$ and all component slacks are nonnegative. Every inequality is an equality. Consequently $gap(P) = 1$, $ps(P) = ex(P) - gap(P) = m - 1$, and every other component has slack zero. The chain through $1 + m - ps(G)$, together with $ps(G) \ge \sum_Q ps(Q)$, now gives $ps(G) = m - 1$. Equality in the first displayed chain also gives $c_0(H) = a + \nu(G) + m$. This applies to every indicated P.

For each component Q different from P, the hypotheses give a missing-run bound at total $\nu(Q)$ of at most a. Both endpoints of $Y_{(}\nu(Q))(Q)$ are feasible: placing every edge in either part has matching number at most $\nu(Q)$. Start with the full interval [0,a]. Repeatedly apply the integer interval criterion of Lemma 75 to add these component allocation sets, one at a time. At every step the current interval length is at least a, so the supplied run bound permits the step. The resulting sumset is the full interval [0,L], where $L = a + \sum_{Q different from P} \nu(Q)$.

Put $t = c_0(P) - 1 = \nu(P) + m - 1 \ge \nu(P)$. The set $Y_t(P)$ contains both endpoints, and its missing-run length is $tr(P)$. If $tr(P) \le L$, the same interval criterion would give

$[0, L] + Y_t(P) = [0, L + t].$

Union the actual component partitions and restore all edges meeting S by Lemma 75. This would make every allocation at total

$L + t = a + \nu(G) + m - 1 = c_0(H) - 1$

feasible on H, contradicting the definition of $c_0(H)$. Therefore $tr(P) \ge L + 1$.

The positive-gap component P satisfies T_{-}. Since $\Delta(P) \le \Delta(G) \le d - 1$ and h is nondecreasing, it follows that $tr(P) + m \le h(d-1)$. For even d, $floor((d-1)/2) + 1 = d/2$, so $h(d-1) = \max(4, d/2)$. Combining this with $L \ge a$ proves $a + m + 1 \le \max(4, d/2)$.

A perfect matching of H-v, for any v in S, has k edges. Deleting the remaining a-1 vertices of S destroys at most a-1 matching edges. Hence $\nu(G) \ge k + 1 - a$. Since $|V(G)| = 2k + 1 - a$, the bound by half the order gives $\nu(G) \le floor((2k + 1 - a)/2) = k - \lfloor a/2 \rfloor$. Subtracting $k + 1 - a$ proves the asserted bounds on η. Substitution in $c_0(H) = a + \nu(G) + m$ and $\nu(H) = k$ gives $ex(H) = 1 + m + \eta$. The degree upper bound was already supplied by maximality. Every remaining vertex loses at most a neighbors, proving its degree lower bound d-a.

For the final assertions, the selected P has positive gap. If $a = 2$ or $a = 3$, then $tr(P) \ge a + 1 \ge 3$, so the added excess-one implication excludes $m = 1$. Thus $m \ge 2$. The bound $a + m + 1 \le \max(4, d/2)$ gives $d \ge 10$ when $a = 2$ and $d \ge 12$ when $a = 3$.

If $a = 4$, then $tr(P) \ge 5$. The added implications at excess one and two exclude $m = 1, 2$, so $m \ge 3$ and $\max(4, d/2) \ge 8$, giving $d \ge 16$.

If $a \ge 5$, then $tr(P) \ge 6$. All three added implications exclude $m = 1, 2, 3$, so $m \ge 4$ and $\max(4, d/2) \ge 10$, giving $d \ge 20$. These cases preserve the same actual H, S and component P throughout. □

Remark. Gap two makes the independent-deletion estimate exact. The deleted vertices absorb every zero-slack companion, leaving one component whose top missing run must exceed the whole allocation interval; low-excess run bounds then force a high degree.

Depends on. Lemma 61, Lemma 75, Lemma 113.

Lemma 161 (Certificate bounds from cubic matching deficiency). *Use the finite integer certificate costs $c_i(G)$, matching number $\nu(G)$, edge count $e(G)$, maximum degree $\Delta(G)$, and positive-level slack $ps(G) = e(G) - \nu(G) - \sum_{i\geq 1} c_i(G)$ of Lemma 23. All graphs are finite and simple; vertex deletion is induced. The empty graph and isolated vertices are permitted.*

Let G have maximum degree at most three. Put $U = \{vinV(G) : d_G(v) = 3\}$ and $\kappa = \min_M |U \text{ minus } V(M)|$, where M ranges over all matchings of G and $V(M)$ is its set of covered vertices. Then

$c_1(G) \leq \nu(G)$,

$c_3(G) \leq \kappa \leq floor(ps(G)/2)$,

and $c_i(G) = 0$ for every integer $i \geq 4$.

The minimum defining κ is attained by a maximum matching. If $ps(G) \leq 1$, a maximum matching covers every vertex of degree three and $c_3(G) = 0$. No connectedness, regularity, factor-criticality, or hereditary gap assumption is imposed. Orders, degrees, costs, indices, matching sizes, κ, and $ps(G)$ are nonnegative integers.

Proof. First construct a vertex cover of minimum matching size in a finite bipartite graph B with parts L, R. Choose a maximum matching M. Starting from all uncovered vertices of L, follow nonmatching edges from L to R and matching edges from R to L. Let Z_L, Z_R be the reached vertices in the two parts. No uncovered R-vertex is reached: a shortest directed path to one would alternate between nonmatching and matching edges, begin and end uncovered, and permit an exchange increasing $|M|$ by one.

Every vertex of Z_R therefore has a matching partner in Z_L. Conversely, every covered vertex of Z_L is reached through its matching edge and has its partner in Z_R. Every uncovered vertex of L belongs to Z_L. Thus $W = (LminusZ_L)$ union Z_R has cardinality $|M|$. It covers B: an edge from Z_L to R minus Z_R cannot be nonmatching, because it would be followed; if matching, its left endpoint's partner would be reached. These are the only edges not already incident with W.

Choose a partition $V(G) = L$ disjoint union R maximizing the number of crossing edges. Such a partition exists because $V(G)$ is finite. A vertex with at least two neighbors in its own part has at most one in the other part, since $\Delta(G) \leq 3$. Moving it would increase the number of crossing edges. Hence the internal edge set Q has maximum degree at most one. The crossing graph B is bipartite; the preceding construction gives a vertex cover W of B of size $\nu(B) \leq \nu(G)$.

For each pair of nonnegative integers x, y with $x + y = |W|$, partition W into W_R, W_B of sizes x, y. Put each crossing edge meeting W_R in P_R and all remaining crossing edges in P_B. Every P_B edge meets W_B. A matching in either exceptional part uses distinct vertices of the corresponding cover subset, so its size is at most x or y. Take Q as the degree-one part and the other bounded-degree part empty. This is an actual certificate on every edge of G at capacities (1,0). Symmetry and upward closure of feasible totals in Lemma 23 give $c_1(G) \leq |W| \leq \nu(G)$.

The finite matching family attains κ. By Lemma 107, one can choose a maximum matching M attaining it. Put $Z = U$ minus $V(M)$, so $|Z| = \kappa$. No edge of M meets Z; thus M remains a matching of G-Z. Every degree-three vertex of G-Z belonged to U and is outside Z, so is covered by M. The covered-degree conclusion of Lemma 61 gives $c_3(G - Z) = 0$. Restoring Z with Lemma 72 gives $c_3(G) \leq \kappa$. The same strong-tail theorem gives $c_i(G) = 0$ for $i \geq 4$.

It remains to prove $ps(G) \geq 2\kappa$. If $\kappa = 0$, this is the nonnegativity assertion of Lemma 61. Suppose $\kappa \geq 1$. For a vertex set X let $q_U(G - X)$ count odd-order components of G-X whose vertices all belong to U. Choose an inclusionwise maximal X maximizing $q_U(G - X) - |X|$. Lemma 107 supplies this choice, gives $q_U(G - X) = b + \kappa$ with $b = |X|$, makes every counted component factor-critical, and gives

$\nu(G) = b + \sum_Q \nu(Q)$,

where Q ranges over all components of G-X. In writing this sum, matching numbers add over the uncounted components: a matching restricts to each component and unions of maximum component matchings attain the sum.

Let O be the family of counted components and let R_0 be the induced union of all other components of G-X. For K in O let t_K be the number of G-edges with exactly one endpoint in $V(K)$. Every such edge ends in X. Every vertex of K has G-degree three, and its order is odd, so

$t_K = 3|V(K)| - 2e(K)$

is a positive odd integer. If $t_K = 1$, exactly one K-vertex has internal degree two and every other K-vertex has internal degree three. The factor-criticality just established permits application of Lemma 128, giving $ps(K) \geq 2 = 3 - t_K$. If $t_K \geq 3$, nonnegative slack gives $ps(K) \geq 0 \geq 3 - t_K$. Therefore $ps(K) \geq 3 - t_K$ for every K in O.

Put $e_{inc} = e(G) - \sum_Q e(Q)$. For $i = 1, 2$, vertex restoration and disjoint-union composition in Lemma 72 give $c_i(G) \leq b + \sum_Q c_i(Q)$. The matching identity, these two row bounds and $c_3(G) \leq \kappa$ yield

$ps(G) \geq e_{inc} - 3b - \kappa + \sum_Q [e(Q) - \nu(Q) - c_1(Q) - c_2(Q)]$

$\geq e_{inc} - 3b - \kappa + \sum_Q ps(Q)$.

The last inequality drops the nonnegative terms $c_3(Q)$; all rows above three vanish by Lemma 61.

For disjoint vertex sets A, B write $e_G(A, B)$ for the number of edges between them, and write G[A] for the induced graph. The edges counted by e_{inc} are exactly the edges internal to X, the edges from X to R_0, and the boundary edges of the counted components. Consequently

$e_{inc} = e(G[X]) + e_G(X, V(R_0)) + \sum_{K \in O} t_K$.

Since $|O| = b + \kappa$, substitution in the preceding slack bound gives

$ps(G) \geq 2\kappa + e(G[X]) + e_G(X, V(R_0))$

$+ \sum_{Q not \in O} ps(Q)$

$+ \sum_{K \in O} [ps(K) + t_K - 3]$

$\geq 2\kappa$.

Every term in the last two sums is nonnegative by the proved boundary estimate and Lemma 61. This argument includes disconnected G and isolated uncounted components.

All quantities are integers, so $ps(G) \geq 2\kappa$ implies $\kappa \leq floor(ps(G)/2)$. If $ps(G) \leq 1$, the nonnegative integer κ is zero. The maximum matching attaining it covers U, and $c_3(G) \leq \kappa = 0$ proves the final assertion. □

Remark. A maximum cut pays the first certificate row with a bipartite vertex cover. Each unavoidable exposed cubic vertex corresponds to one surplus marked odd component. Boundary-one components have two units of slack, and exact boundary accounting preserves two units per unmatched cubic vertex in the whole graph.

Depends on. Lemma 23, Lemma 61, Lemma 72, Lemma 107, Lemma 128.

Lemma 162 (Greedy domination with pairing completion). *Let $\delta \geq 2$ and $n \geq 0$ be integers, and let F be a finite simple graph on n vertices with minimum degree at least δ. A set Z dominates F when every vertex outside Z has a neighbor in Z. Let $\gamma(F)$ be the least cardinality of a dominating set, with γ of the empty graph zero. For $n > 0$ put $u_0 = n$ and, as long as $u_i > 0$ and $(\delta + 1)u_i > 2(n - i)$, define $u_{i+1} = \max(0, u_i - ceil((\delta + 1)u_i/(n - i)))$. Let s be the first index with $u_s = 0$ or $(\delta + 1)u_s \leq 2(n - s)$; for $n = 0$ put $s = u_s = 0$. This process terminates with nonnegative integer values and positive denominators whenever a step is taken. Put $P_\delta = product_{m=3}^{\delta+1} \delta \cdot m/(\delta \cdot m + 1)$. Then F has a dominating set of cardinality at most $floor((3\delta \cdot s + \delta \cdot u_s + 2n)/(3\delta + 2))$, and $\gamma(F) \leq [\delta(3\delta + 1)s + (4\delta + 2)n]/[(\delta + 1)(3\delta + 2)]$, $s \leq 1 + (n + 1)(1 - P_\delta)$. For $\delta = 14$, the exact product satisfies $P_{14} > 8789/10000$, and $\gamma(F) \leq floor((654511n + 3374511)/3300000)$. All graph orders, degrees, recurrence indices and values, and cardinalities are nonnegative integers in the displayed domains. Dominating vertices may be chosen anywhere in F. No connectedness or regularity is assumed.*

Proof. For $n = 0$ take the empty dominating set; every displayed bound is nonnegative and holds. Assume $n > 0$. Simplicity and the minimum degree hypothesis give $n \geq \delta + 1$. Set $c = \delta + 1$. While $u_i > 0$, one has $ceil(c \cdot u_i/(n - i)) \geq 1$ whenever the denominator is positive. Starting from $u_0 = n$, induction therefore gives $0 \leq u_i \leq n - i$ and a decrease of at least one at every step with positive u_i. Thus the process stops by index n, and every step before stopping has $n - i \geq 1$.

Construct a greedy prefix A_0 empty, and at stage i choose a vertex outside A_i whose closed neighborhood contains as many currently undominated vertices as possible. If the graph is already dominated, pad

A_i by arbitrary unselected vertices as necessary up to stage s. This is possible since $s \le n$. Write W_i for the undominated set at stage i. Every w in W_i has at least c vertices in its closed neighborhood, all outside A_i. Counting pairs consisting of w in W_i and a vertex in its closed neighborhood shows that some unselected vertex covers at least $ceil(c|W_i|/(n-i))$ vertices of W_i. Thus $|W_{i+1}| \le f_i(|W_i|)$, where $f_i(v) = \max(0, v - ceil(cv/(n-i)))$. This function is nondecreasing on the nonnegative integers. Indeed if $c \le n-i$, the ceiling increases by at most one when v increases by one, so v minus that ceiling is nondecreasing; if $c > n-i$, the function is identically zero. Induction proves $|W_i| \le u_i$ through stage s. The prefix A_s has exactly s vertices and leaves at most u_s undominated. Apply Lemma 117 to this same prefix, with $a = s$ and $m_0 = u_s$. It supplies the required dominating set and the first bound.

The stopping condition gives $u_s \le 2(n-s)/c$, including the case $u_s = 0$. Substituting in the first bound, and dropping the floor, gives $\gamma(F) \le (3\delta \cdot s + 2\delta(n-s)/c + 2n)/(3\delta+2) = [\delta(3\delta+1)s + (4\delta+2)n]/[c(3\delta+2)]$.

It remains to bound the stopping index. If $s \le 1$, the asserted bound follows from $0 < P_\delta < 1$. Assume $s \ge 2$. For $0 \le i < s$ put $N_i = n - i$ and $\phi_i = c \cdot u_i / N_i$. For such i, $\phi_i > 2$, and $\phi_0 = c$. For $0 \le i \le s-2$ the next value is positive and the truncation at zero is inactive. Define $g(\phi) = c \cdot \lceil \phi \rceil - \phi$ for positive real ϕ. Then $D_i := \phi_i - \phi_{i+1} = g(\phi_i)/(N_i - 1) > 0$. The denominator is positive because $N_i - 1 \ge u_{i+1} \ge 1$.

Let $M = \lceil \phi_i \rceil$. On the interval $(\phi_{i+1}, \phi_i]$, if $\lceil \phi \rceil = M$, then $g(\phi) = g(\phi_i) + (\phi_i - \phi) \le g(\phi_i) + D_i$. If instead $\lceil \phi \rceil = M' \le M-1$, then $g(\phi) \le (c-1)M' + 1 \le (c-1)(M-1) + 1 \le (c-1)M \le g(\phi_i)$. Here $c \ge 3$ and $\phi_i \le M$ were used. Since g is positive, these inequalities give $integral_{\phi_{i+1}}^{\phi_i} dphi/g(\phi) \ge D_i/(g(\phi_i) + D_i) = 1/N_i$. The intervals are disjoint and contained in (2,c]. Adding them yields $\sum_{i=0}^{s-2} 1/(n-i) \le integral_2^c dphi/(c \cdot \lceil \phi \rceil - \phi) = \sum_{m=3}^{c} ln((\delta \cdot m + 1)/(\delta \cdot m)) = -ln\ P_\delta$. On the other hand, $1/j \ge integral_j^{j+1} dt/t$ for each positive integer j, so the left side is at least $ln((n+1)/(n-s+2))$. Exponentiating gives $n - s + 2 \ge (n+1)P_\delta$. Rearrangement proves $s \le 1 + (n+1)(1 - P_\delta)$.

For $\delta = 14$ direct multiplication of the thirteen displayed rational factors gives $P_{14} = 597381486941998940160/679661878685656570223$. The exact positive difference $10000 \cdot 597381486941998940160 - 8789 \cdot 679661878685656570223 = 26661765175380591 0053$ proves $P_{14} > 8789/10000$. The previously established estimates give $\gamma(F) \le (301s + 29n)/330$, $s < 1 + 1211(n+1)/10000$, and consequently $\gamma(F) < [301 + 364511(n+1)/10000 + 29n]/330 = (654511n + 3374511)/3300000$. The cardinality $\gamma(F)$ is an integer, so it is at most the floor displayed in the statement. □

Depends on. Lemma 117.

Lemma 163 (Counting odd components after domination). *Use the integer certificate costs c_i and excess $ex = c_0 - \nu$ of Lemma 23. Let $d \ge 8$ be an even integer, and let F be a connected d-regular finite simple graph on $n = 2k+1$ vertices such that deleting any vertex leaves a perfect matching. Assume $E = ex(F) \ge 2$, put $e = E - 1$, and suppose $3e \le k$. Fix j in {1,2}, and put $m_0 = d+1$ if $j = 1$, and $m_0 = d-1$ if $j = 2$; $g(m) = (d-2-2j)m - (d-2j-4)$; $R = (d-2)k + 2(d-3) - (3d-2)e$; $a = (2j+1)/(d-2-2j)$; $b = 2d - 2j - 5 + (2j+1)(d-2j-4)/(d-2-2j)$; $t_0 = floor((2e-1)/(d-2))$; $\tau = \max(0, ceil(R/g(m_0)) - 1)$. Assume either $R \le g(m_0)$, or, with $W = 2R(a + b/g(m_0))$, both $W \le 4e$ and $(d-2)(n - (j+1)W/(2j+1) - (d-j-2)2R/g(m_0)) \ge (d+1)(2e + 2R/g(m_0))$.*

For an odd integer $m \ge m_0$ and a nonnegative integer β, call the following the partial-set test: $\beta \le 2e+1$, $(m-1)/2 \le 2e + t_0 + 1$, $2e + \min(ceil((n-m-\beta)/(d+1)), floor((2e+1-\beta)/(d-2))) + (m+1)/2 \ge floor((k+e)/2)$, and $floor((k+e)/2) + (m+1)/2 + floor((2e+1)/(d-2)) \le n$. Call the following the full-boundary test at m, with $\beta = jm + d - j - 2$: $\beta \le 2e+1$, $2\beta + m - 1 \le 2(2e + t_0 + 1)$, and $(d-2)(n - (j+1)m - (d-j-2)) \ge (d+1)(2e+1)$. For a nonnegative integer ζ, call an odd integer $m. \ge m_0$ admissible if every odd m with $m_0 \le m < m.$ and $g(m) \ge R$ passes either the full-boundary test or the partial-set test with $\beta = \min(jm, \zeta) + d - j - 2$.

Let $Z \subseteq V(F)$ have at least j neighbors at every vertex outside Z, and suppose $|Z| \le \zeta$. Fix any $m.$ admissible for this ζ. Put $G = F - Z$, $D = d - j$, and $U = \{voutsideZ : |N_F(v) intersect Z| = j\}$. If $\Delta(G) < D$ put $\kappa = 0$ and use the empty component family. Otherwise let κ be the minimum number of U-vertices left uncovered by a matching of G, and fix any set X such that $o_U(G - X) - |X| = \kappa$, as supplied by Lemma 83; here o_U counts odd connected components wholly contained in U. In this case call a counted component C good if $\delta(C) := e_G(V(C), X) \le D - 2$. Let t count good components with $g(|V(C)|) < R$, and l count good components with $|V(C)| \ge m.$.

Then $\min(\kappa, t) \le \tau$, no good component of size m with $g(m) \ge R$ passes either stated test with the actual upper bound ζ, and $\kappa \le \tau + l$, $l \cdot m. \le |U| \le n - |Z|$, $c_{d-j}(F) \le |Z| + \tau + \lfloor |U|/m. \rfloor$. For $j = 1$ the stronger bounds hold:

$\kappa \le \tau + \max(0, l-1)$, $c_{d-1}(F) \le |Z| + \tau + \max(0, floor((n-|Z|)/m.) - 1)$. *In particular* $c_{d-1}(F) \le |Z| + \tau$ *if* $2m. > n - |Z|$.

Finally, at $j = 2$, *if* $m.$ *is admissible with* $\zeta = n$, *then for every real* p *in [0,1],* $c_{d-2}(F) \le floor(n[p + (1-p)Pr(Bin(d,p) \le 1) + (1-p)Pr(Bin(d,p) = 2)/m.] + \tau)$, *where* $Bin(d,p)$ *is the sum of* d *independent Bernoulli variables of success probability* p. *All degrees, orders, costs, cardinalities,* k, E, e, j, m, $m.$, m_0, ζ, β, R, t_0, τ *and component counts are integers in the indicated domains;* a, b, W *and* p *are real. All vertex deletions are induced. The conclusions use the supplied* Z *and any of its specified attaining sets* X; *empty residuals and* $\kappa = 0$ *are included.*

Proof. Deleting one vertex leaves a perfect matching of size k, and no matching on $2k+1$ vertices has more than k edges. Thus $\nu(F) = k$ and $c_0(F) = k + E = k + e + 1$. We will contradict this equality whenever a separated-set certificate of cost $k + e$ is constructed.

The denominators d-2-2j are positive. The function g is strictly increasing and $g(m_0) > 0$: its value is $d^2 - 4d + 2$ for $j = 1$ and $d^2 - 8d + 14$ for $j = 2$, positive for $d \ge 8$. Both a and b are positive, and expansion gives $(2j+1)m + (2d - 2j - 5) = a \cdot g(m) + b$. (1) Also $1 \le e \le k/3$, and $t_0 < 2e/(d-2) \le 2k/(3(d-2)) \le 2k/d \le \lceil 2k/d \rceil$. (2)

If $\Delta(G) = D$, apply Lemma 83 to the same Z and attaining X. It gives $c_D(F) \le |Z| + \kappa$, at least κ good components, odd component orders, pairwise separation even in F, and at least $D+1$ vertices in each good component. The least allowed odd size is m_0. The total external neighborhood of J good components of total order M has size at most $jM + (d - j - 2)J$. Each vertex of these components has exactly j neighbors in Z. Consequently a single good component of size m has boundary at most $\min(jm, \zeta) + d - j - 2$. The fact also supplies $\delta(C)$ congruent to D modulo 2 and $\sum_C \delta(C) \le D|X|$.

If $\Delta(G) < D$, the degree-D vertex set of G is empty. Applying Lemma 35 with degree bound D and the empty matching splits $E(G)$ for every capacity sum D. Thus $c_D(G) = 0$ by Lemma 23, and Lemma 72 gives $c_D(F) \le |Z|$. All component-count conclusions then hold for the empty family. Hence it suffices to analyze the case $\Delta(G) = D$.

We first establish the numerical certificate implication used to count small components. Suppose $J \ge 1$ good components, of total order M, satisfy $(2j+1)M + (2d - 2j - 5)J \le 4e$, (3) $(d-2)(n - (j+1)M - (d - j - 2)J) \ge (d+1)(2e + J)$, (4) and $\sum_C g(|V(C)|) \ge R$. (5) Write K for their union, $B = |N_F(K)|$, $H' = F - (KunionN_F(K))$, and T for the component sum in Lemma 93 with degree bound d. Here $B \le jM + (d - j - 2)J$ and $1 \le J \le M$. Since j is 1 or 2 and $d \ge 8$, (3) implies $M \le 4e/3$. It also implies $B + (M - J)/2 \le 2e$, $B \le 2e \le 2e + J$. (6)

For every nonempty component C′ of H′, $\max(1, ceil((|V(C')| - 1)/d)) \ge |V(C')|/(d+1)$: for size at most $d+1$ the term 1 suffices, and for larger size use $(|V(C')| - 1)/d \ge |V(C')|/(d+1)$. It follows from (4) that $T \ge |V(H')|/(d+1) \ge (2e + J)/(d-2)$. Therefore the parameter t_1 of Lemma 93 equals $floor((2e + J - B)/(d-2))$, and $t_1 \ge [2e - jM - (d - j - 3)J - (d-3)]/(d-2)$. (7) The floor inequality used here is $floor(h/(d-2)) \ge (h - (d-3))/(d-2)$ for integer h.

Since the sum in (5) is (d-2-2j)M-(d-2j-4)J, substituting R and rearranging (5) gives $(3d-2)e \ge (d-2)k - (d - 2 - 2j)M + (d - 2j - 4)J + 2(d-3)$. Multiplying (7) by 2(d-2) and using this last inequality proves $2e + t_1 + (M + J)/2 \ge (k + e)/2 \ge floor((k+e)/2)$. (8)

For the required upper size bound, use $M \le 4e/3 \le 4k/9$, $J \le M$ and $d - 2 \ge 6$ to obtain $floor((k+e)/2) + (M + J)/2 + t_1 \le 2k/3 + 4k/9 + (2k/3 + 4k/9)/6 = 35k/27 < 2k + 1 = n$. (9) Together (2), (6), (8), (9) check every hypothesis of Lemma 93 for Q empty and these same components: the weakened fit condition follows from (6), and both balanced conditions are (8), (9). It gives $c_0(F) \le k + e$, a contradiction. Thus no family satisfies (3)–(5).

If $R \le g(m_0)$, there are no good components with g-value less than R, so $t = 0$. Suppose now $R > g(m_0)$, and assume $\min(\kappa, t) \ge \tau + 1 = ceil(R/g(m_0))$. Among at most $\min(\kappa, t)$ small components choose a subfamily whose g-values sum to at least R, with the fewest possible members. Let J′ and M′ denote its size and total order, and S its g-sum. Deleting any member lowers the sum below R, and that member has g-value less than R. Hence $R \le S < 2R$, $J' < 2R/g(m_0)$. By (1), $(2j+1)M' + (2d - 2j - 5)J' = aS + bJ' < W \le 4e$, so (3) holds and $M' < W/(2j+1)$. Both sides of (4) are monotone in the needed directions as M′, J′ increase. The assumed inequality involving W and $2R/g(m_0)$ therefore implies (4) for this family. Its definition gives (5), contrary to the preceding certificate implication. Thus $\min(\kappa, t) \le \tau$.

Next fix a good component of odd size m with $g(m) \geq R$. If its partial-set test holds, use Q empty, $J = 1$, $M = m$, $N = k$, $D = d$ and the displayed integer β in the numerical version of Lemma 93. The actual boundary is at most β. Equation (2) supplies its t_0 condition, and the four lines of the test are precisely its other numerical conditions. It yields $c_0(F) \leq k + e$, which is impossible.

Suppose instead that the full-boundary test holds, and put $\beta = jm + d - j - 2$. The actual boundary B is at most β. It satisfies $B \leq 2e + 1$ and $B + (m-1)/2 \leq 2e + t_0 + 1$. The component sum T on $F - (CunionN_F(C))$ is at least $(n - m - B)/(d+1) \geq (2e+1)/(d-2)$. Thus $t_1 = floor((2e+1-B)/(d-2))$ and $t_1 \geq (2e + 1 - \beta - (d-3))/(d-2)$. The inequality $2e + t_1 + (m+1)/2 \geq (k+e)/2$ follows on multiplying by 2(d-2) from $(3d-2)e \geq (d-2)k - (d-2-2j)m + 3d - 2j - 10$. This is exactly $g(m) \geq R$. To check the remaining upper size condition, $\beta \geq m$ and $2\beta + m - 1 \leq 4e + 2t_0 + 2$ imply $3m \leq 4e + 2t_0 + 3$. Since $d - 2 \geq 6$, this gives $m \leq (14e + 8)/9$. Also $t_1 \leq (2e+1)/6$, so $floor((k+e)/2) + (m+1)/2 + t_1 \leq k/2 + 29e/18 + 10/9 \leq 28k/27 + 10/9 < 2k + 1$. The last inequality holds for $k \geq 1$. Equation (2) and these bounds check Lemma 93 for this singleton family, again contradicting $c_0(F) = k + e + 1$.

Every good component is now either counted by t or has order at least $m.$: if neither held, its odd order m would obey $m_0 \leq m < m.$, $g(m) \geq R$, and admissibility would provide an excluded test. Let h be the number of good components. Then $\kappa \leq h \leq t + l$. If $\kappa \leq t$, the small-component bound gives $\kappa \leq \tau$. If $\kappa > t$, it gives $t \leq \tau$ and hence $\kappa \leq \tau + l$. Disjointness inside U gives $l \cdot m. \leq |U|$. Combining with $c_D(F) \leq |Z| + \kappa$ proves the first cost bound.

For $j = 1$, $D = d - 1$ is odd. Every counted component has positive odd boundary. Let h and b′ count good and other components; $h + b' = |X| + \kappa$. Every other component has boundary at least D, and every good one has boundary at least one. Thus $D\, b' + h \leq \sum_C \delta(C) \leq D|X|$. If $\kappa \geq 1$ then $h \geq \kappa \geq 1$, so $b' \leq |X| - 1$ and $h \geq \kappa + 1$. Since $h \leq t + l$, when $l \leq 1$ we have $t \geq \kappa$ and thus $\kappa \leq \tau$; when $l \geq 2$, both t and κ are at least $\kappa + 1 - l$, so $\kappa + 1 - l \leq \min(\kappa, t) \leq \tau$. These cases prove $\kappa \leq \tau + \max(0, l-1)$. For $\kappa = 0$ the same inequality holds because $\tau \geq 0$. The asserted once-dominated cost bound follows from $l \leq floor((n - |Z|)/m.)$. Its special case uses the strict inequality $2m. > n - |Z|$, which makes $l \leq 1$.

Finally take $j = 2$ and an $m.$ admissible at $\zeta = n$. Select Z_0 by independent vertex choices with probability p. For every vertex v put $X_v = |N_F(v)$ intersect $Z_0|$; let Def consist of vertices outside Z_0 with $X_v \leq 1$, and put $Y = Z_0$ union Def. Every vertex outside Y has at least two neighbors in Y. Let U_Y be those with exactly two neighbors in Y. The preceding cost bound gives, for each outcome, $c_{d-2}(F) \leq |Y| + \tau + |U_Y|/m.$.

For each vertex v, its contribution to $|Y| + |U_Y|/m.$ is one if v is sampled or if it is unsampled and $X_v \leq 1$; it is at most $1/m.$ if unsampled with $X_v = 2$; and it is zero if unsampled with $X_v \geq 3$. The last assertion holds because all its sampled neighbors remain in Y. Regularity and simplicity imply X_v has distribution $Bin(d, p)$, independently of whether v itself is sampled. Summing expected contributions therefore bounds the expectation by the expression inside the final floor minus τ. Some outcome is at most its expectation. The certificate cost is the same integer for every outcome, so adding τ and taking the floor proves the stated bound, including $p = 0$ and $p = 1$. □

Depends on. Lemma 23, Lemma 35, Lemma 72, Lemma 83, Lemma 93.

Lemma 164 (Closed lumps of degree eighteen are LP-good). *Use the finite integer certificate costs $c_i(H)$, matching number $\nu(H)$, excess $ex(H) = c_0(H) - \nu(H)$, and signed gap $gap(H) = \sum_{i \geq 0} c_i(H) - e(H)$ of Lemma 23. A graph is LP-good when its gap is nonpositive.*

Let H be any nonempty connected 18-regular finite simple graph such that deleting every vertex leaves a perfect matching. Then $gap(H) \leq 0$. More precisely, write $|V(H)| = 2k + 1$, where $k = \nu(H) \geq 9$ is an integer. For $k \geq 78$,

$$gap(H) < 2253/260 - 31981k/325000 < 1.$$

For every integer $9 \leq k \leq 77$, the explicit finite integer bounds in the proof are nonpositive.

No hereditary, minimality, excess, top-run, or upper-order hypothesis is imposed. All orders, degrees, costs, matching numbers, sample sizes, capacities and indices are integers in their displayed positive or nonnegative domains; gaps may be signed integers.

Proof. Put $n = |V(H)| = 2k + 1$ and $E = ex(H)$. Factor-criticality gives odd order and $\nu(H) = k$. Simplicity gives $k \geq 9$, and regularity gives $e(H) = 18k + 9$. The connected isolation bound Lemma 50, with $D = 18$, gives

$E \le e_k := \min(floor((k+1)/3), ceil((16k+2)/52))$. (1)

By Lemma 87 the rows above eighteen vanish, $c_{18}(H) \le 1$, $c_9(H) \le k+1$, and

$c_i(H) + c_{18-i}(H) \le 2k+2$ for $1 \le i \le 8$.

The cost definition in Lemma 23 gives $c_i(H) \le c_0(H) = k+E$. For $t = 2$ or $t = 3$, let b_j be any real upper bound for $c_{18-j}(H)$, $1 \le j \le t$. In the eight pairs above, bound the lower row of the first t pairs by $k+E$ and their upper rows by b_j. Bound the other 8-t pairs by $2k+2$. Including rows zero, nine, and eighteen gives

$gap(H) \le (t+1)E - tk + 9 - 2t + \sum_{j=1}^{t} b_j$. (2)

Indeed the pre-subtraction sum is $(t+1)(k+E) + \sum_j b_j + (8-t)(2k+2) + (k+1) + 1$; subtracting $18k+9$ gives (2).

Fix any vertex v and any perfect matching M of H-v. The graph $J = H - M$ has degree eighteen at v and degree seventeen elsewhere. Every marked j-dominating set supplied below is a set in this same J containing v. The certificate construction Lemma 49 converts such a set Z to $c_{18-j}(H) \le |Z|$. Different j may use different sets: they bound separate scalar costs of the same H.

First treat $k \ge 78$. Apply Lemma 39 to J at $j = 1, 2$ with probabilities $p_1 = 1/6$ and $p_2 = 23/100$. With $q_j = q_{17,j}(p_j)$ and $C_j = 18(1-p_j)^2\, Pr(Bin(16, p_j) = j-1)$, that theorem and the certificate construction give

$c_{18-j}(H) \le 1 + 2kq_j - C_j$. (3)

Expansion gives

$q_1 = 1/6 + 5^{18}/6^{18}$, $C_1 = 18 \cdot 5^{18}/6^{18}$,

$q_2 = 23/100 + 468 \cdot 77^{17}/100^{18}$, $C_2 = 6624 \cdot 77^{17}/100^{18}$.

Here $468 = 77 + 17 \cdot 23$ and $6624 = 18 \cdot 16 \cdot 23$. The strict bounds

$q_1 < 20423/100000$, $q_2 < 28503/100000$, $C_1 > 27/40$, $C_2 > 31/40$

follow from these positive integer differences:

$11269 \cdot 6^{18} - 300000 \cdot 5^{18} = 69972008879904$;

$720 \cdot 5^{18} - 27 \cdot 6^{18} = 4463201202768$;

$5503 \cdot 100^{18} - 100000 \cdot 468 \cdot 77^{17} = 1435848510522217403964127404604000000$;

$40 \cdot 6624 \cdot 77^{17} - 31 \cdot 100^{18} = 1546332426894274215312941555617011120$.

Use (2) at $t = 2$, (3), and $E \le ceil((16k+2)/52) < (16k+2)/52 + 1$. As $q_1 + q_2 < 24463/50000$ and $C_1 + C_2 > 29/20$, this gives

$gap(H) < 3((16k+2)/52 + 1) - 2k + 7 - 29/20 + 2k \cdot (24463/50000)$

$= 2253/260 - 31981k/325000$.

At $k = 78$ the last expression is $80433/81250 < 1$; its slope is negative. Since the gap is an integer, $gap(H) \le 0$ for every $k \ge 78$.

For $9 \le k \le 77$ use fixed-cardinality sampling instead. Define

$r_1 = \lfloor k/4 \rfloor$, $r_2 = \lfloor 2k/5 \rfloor$, $r_3 = \lfloor k/2 \rfloor$.

For a nonnegative integer a use $binom(a, b) = 0$ outside $0 \le b \le a$. For $j = 1, 2, 3$ put

$N_{k,j} = 18 \sum_{i=0}^{j-2} binom(16, i) binom(2k-17, r_j - i)$

$+ (2k-18) \sum_{i=0}^{j-1} binom(17, i) binom(2k-18, r_j - i)$,

$D_{k,j} = binom(2k, r_j)$.

An empty sum is zero. All upper binomial arguments are nonnegative because $k \geq 9$, and $D_{k,j} > 0$ because $0 \leq r_j \leq 2k$. The fixed-cardinality construction Lemma 38 with $n = 2k+1$, $d = 18$, $r = r_j$ gives a marked j-dominating set of size at most

$1 + r_j + \lfloor N_{k,j}/D_{k,j} \rfloor$.

Consequently every integer b_j satisfying

$N_{k,j} < (b_j - r_j)D_{k,j}$ (4)

is a valid upper bound for $c_{18-j}(H)$.

The table gives k, e_k from (1), three integer bounds (b_1, b_2, b_3) satisfying (4), and

$U_k = \min(3e_k - 2k + 5 + b_1 + b_2, 4e_k - 3k + 3 + b_1 + b_2 + b_3)$. (5)

The entries in (4) are explicit integer comparisons of the binomial sums just displayed. For example, at $k = 26$ the sample sizes are (6,10,13), and the three fractions $N_{26,j}/D_{26,j}$ reduce to

$336226/149695 < 3, 3916044/1626905 < 3, 285045930/93384347 < 4$.

Thus $(b_1, b_2, b_3) = (9, 13, 17)$ is valid and $U_{26} = \min(27 - 52 + 5 + 22, 36 - 78 + 3 + 39) = 0$. At $k = 9$ all three numerators are zero, giving the first row with the same binomial convention. The following table lists the inequalities (4) and their resulting (5) for every remaining integer in the stated finite range:

9; 3; (3,4,5); 0

10; 3; (3,5,6); -1

11; 4; (3,5,6); 0

12; 4; (4,5,7); -1

13; 4; (4,6,7); -3

14; 5; (4,6,8); -1

15; 5; (4,7,8); -3

16; 5; (5,7,9); -4

17; 6; (5,8,10); -1

18; 6; (5,8,11); -3

19; 6; (6,9,11); -4

20; 7; (6,10,12); -1

21; 7; (7,10,13); -2

22; 7; (7,11,14); -3

23; 8; (8,11,14); -1

24; 8; (8,12,15); -2

25; 8; (8,12,16); -4

26; 9; (9,13,17); 0

27; 9; (9,14,17); -2

28; 9; (10,14,18); -3

29; 9; (10,15,19); -4

30; 10; (10,15,20); -2

31; 10; (11,16,20); -3

32; 10; (11,17,21); -4

33; 11; (12,17,22); -1
34; 11; (12,18,22); -3
35; 11; (13,18,23); -4
36; 12; (13,19,24); -1
37; 12; (13,19,25); -3
38; 12; (14,20,25); -4
39; 13; (14,21,26); -1
40; 13; (15,21,27); -2
41; 13; (15,22,28); -3
42; 13; (16,22,28); -5
43; 14; (16,23,29); -2
44; 14; (16,24,30); -3
45; 14; (17,24,31); -4
46; 15; (17,25,31); -2
47; 15; (18,25,32); -3
48; 15; (18,26,33); -4
49; 16; (18,26,34); -2
50; 16; (19,27,34); -3
51; 16; (20,28,35); -3
52; 17; (20,28,36); -1
53; 17; (20,29,37); -2
54; 17; (21,29,37); -4
55; 17; (21,30,38); -5
56; 18; (21,31,39); -2
57; 18; (22,31,40); -3
58; 18; (22,32,40); -5
59; 19; (23,32,41); -2
60; 19; (23,33,42); -3
61; 19; (24,33,43); -4
62; 20; (24,34,43); -2
63; 20; (25,35,44); -2
64; 20; (25,35,45); -4
65; 21; (25,36,46); -1
66; 21; (26,36,46); -3
67; 21; (26,37,47); -4
68; 21; (26,37,48); -6
69; 22; (27,38,49); -2
70; 22; (28,39,49); -3

71; 22; (28,39,50); -5

72; 23; (28,40,51); -2

73; 23; (29,40,52); -3

74; 23; (29,41,52); -5

75; 24; (30,42,53); -1

76; 24; (30,42,54); -3

77; 24; (30,43,55); -4

Both right sides of (2) increase with E. Thus (1), (2), and the actual certificate bounds (4) give $gap(H) \le U_k$. Every displayed U_k is nonpositive, proving the assertion for all $9 \le k \le 77$. This finite interval and the already proved $k \ge 78$ case exhaust the possible orders. □

Remark. Sampling a prescribed number of vertices provides sharper small-order domination certificates. Three uniform sample-size formulas cover every finite exceptional order, while the two-row Bernoulli estimate controls all larger orders.

Depends on. Lemma 23, Lemma 38, Lemma 39, Lemma 49, Lemma 50, Lemma 87.

Lemma 165 (Weighted cover bounds for exposed maximum-degree vertices). *All graphs are finite and simple. A matching has pairwise disjoint endpoints. For a graph Q let $\nu(Q)$, $\tau(Q)$ and $\alpha(Q)$ denote maximum matching cardinality, minimum vertex-cover cardinality and maximum independent-set cardinality. For an integer $D \ge 5$ put $k = floor((D-1)/2)$. For every Q of maximum degree at most D, set $S_D(Q) = \{vinV(Q) : deg_Q(v) = D\}$ and let $\theta_D(Q)$ be the minimum matching number of a spanning edge subset of Q incident with every vertex of $S_D(Q)$. If $S_D(Q)$ is empty the empty edge subset is permitted. This definition always uses degrees internal to Q. Every odd-order such Q, of order $2\mu+1$ with integer $\mu \ge 0$, satisfies $\theta_D(Q) \le \max(0, \mu - k)$.*

Let H have maximum degree at most D, put $S = S_D(H)$ and $\alpha_S = \alpha(H[S])$, and let ρ be the minimum number of S-vertices left uncovered by a matching of H. For $Y \subseteq V(H)$, let $q_S(H-Y)$ count the odd connected components of the induced graph H-Y whose vertices all belong to S. Choose any inclusionwise maximal set Y among the maximizers of $q_S(H-Y) - |Y|$. Such Y exist.

Let calR be the components counted by $q_S(H-Y)$. For T in calR put $\mu_T = (|V(T)|-1)/2$. Let b count the members T with $\theta_D(T) = \alpha(T)$, and set $\Lambda = \sum_(T \in calR, |V(T)| > 1)(\mu_T - \alpha(T))$. Then

$b \ge \max(0, \theta_D(H) + \rho - \alpha_S)$, $\Lambda \ge k\, b$,

$\tau(H) - \nu(H) \ge \rho + \Lambda \ge \rho + k \max(0, \theta_D(H) + \rho - \alpha_S)$.

In particular,

$k\,\theta_D(H) + (k+1)\rho \le \tau(H) - \nu(H) + k\,\alpha_S$,

$\theta_D(H) + 2\rho \le \tau(H) - \nu(H) + \alpha_S$,

and $\rho \le \tau(H) - \nu(H)$.

More precisely every vertex cover C of H satisfies

$|C| - \nu(H) \ge \rho + \Lambda$ *and* $2e(H[C]) \ge D(\rho + 2kb)$,

where H[C] is induced and e counts edges.

All orders, degrees, counts, matching sizes, cover sizes, independence sizes and extrema have integer domains. Empty S, empty Y, empty calR, isolated vertices, empty H and $\rho = 0$ are permitted. No arrowing, edge-budget, connectivity or whole-component regularity hypothesis is assumed.

Proof. All extrema in the definitions range over finite nonempty families and are attained. The entire edge set is an allowed incidence cover. For any Q of maximum degree at most D, choose an inclusionwise maximal independent vertex set I of Q. Every vertex outside I has a neighbor in I. The edges incident with I therefore cover every internal degree-D vertex: such a vertex in I has positive degree, and one outside I has

an I-neighbor. Every matching in those selected edges has at most $|I|$ edges, since I is a vertex cover. Thus $\theta_D(Q) \le |I| \le \alpha(Q)$.

For Q of order $2\mu + 1$, apply the first assertion of Lemma 116 with its threshold parameter $t = k$. The hypothesis $t \ge 2$ follows from $D \ge 5$, and $D \ge 2k + 1$. The resulting vertex set Z has size at most $\max(0, \mu - k)$ and its closed neighborhood contains all degree-D vertices of Q. Select all edges of Q meeting Z. Each required vertex outside Z has a Z-neighbor; each required vertex inside Z has positive degree D. These edges are consequently an incidence cover and have matching number at most $|Z|$. This proves the odd-order bound. If additionally $\theta_D(Q) = \alpha(Q)$, then $\alpha(Q) \ge 1$ because Q is nonempty. The maximum in that bound cannot be zero, so $\mu - \alpha(Q) \ge k$.

Apply Lemma 107 to H with marked set S and the stipulated inclusionwise maximal maximizing set Y. It gives existence of such a set, $q := |calR| = |Y| + \rho$, factor-criticality of every T in calR, and

$\nu(H) = |Y| + \sum_{T \in calR} \mu_T + \nu(R)$, (1)

where R is the induced union of the other components of H-Y. Factor-criticality means that T-z has a perfect matching for every z in $V(T)$.

For any nontrivial T in calR, one has $\alpha(T) \le \mu_T$. To verify this, take a maximum independent set I of T. Since T is connected and has an edge, I does not contain every vertex; choose z outside I. In a perfect matching of T-z, every I-vertex is matched to a distinct vertex outside I and z. Hence $2|I| \le |V(T)| - 1 = 2\mu_T$. All summands in Λ are therefore nonnegative. A singleton T has $\theta_D(T) = 0$ and $\alpha(T) = 1$, so it is not counted by b. Each component counted by b is nontrivial and satisfies the odd-order equality consequence proved in the preceding paragraph. Each of its summands in Λ is at least k. Thus $\Lambda \ge k\, b$.

We next compare local and ambient incidence covers. For every component Q of H-Y set $S_Q = S$ intersect $V(Q)$, and put $A_0 = \sum_{Q component of H-Y} \alpha(H[S_Q])$. Maximum independent subsets of the pairwise nonadjacent S_Q unite to an independent subset of H[S], so $A_0 \le \alpha_S$. Within each T in calR choose an incidence cover attaining $\theta_D(T)$. If T is counted by b, this rank equals $\alpha(T)$. Otherwise it is at most alpha(T)-1, by the inequality $\theta_D(T) \le \alpha(T)$, integrality and non equality.

Within each other component Q choose a maximal independent subset I_Q of its internal degree-D vertices and select all internal edges at I_Q. This covers those internal degree-D vertices: every unchosen one has an $I_Q - neighbor$ in their induced graph, and a chosen one has positive degree. Its matching number is at most $|I_Q| \le \alpha(H[S_Q])$, because an internal degree-D vertex also has ambient degree D. If there are no such vertices both sets are empty.

Add every edge of H incident with Y to these internal selections. The union covers S. A required vertex in Y has positive degree and a selected incident edge. A required vertex outside Y with a neighbor in Y is covered by the added edges. Any remaining required vertex has no Y-neighbor, so has internal degree D in its component and is covered there. In any matching of this union, at most $|Y|$ edges meet Y and the remaining edges belong to separate component selections. Since $S_T = V(T)$ for T in calR, the total internal bound is $A_0 - (q - b)$. Hence

$\theta_D(H) \le |Y| + A_0 - q + b = A_0 - \rho + b \le \alpha_S - \rho + b.$

Together with $b \ge 0$ this proves the asserted lower bound on b.

Fix any vertex cover C of H and put $C_Y = C$ intersect Y. For each nontrivial T in calR, the set $V(T)$ minus C is independent in T. Thus

$|C \text{ intersect } V(T)| \ge 2\mu_T + 1 - \alpha(T) = \mu_T + 1 + (\mu_T - \alpha(T)).$

Let t_0 be the number of required singleton components whose sole vertices lie outside C. Every other required singleton contributes one vertex to C. Summing gives

$\sum_{T \in calR} |C \text{ intersect } V(T)| \ge \sum_T \mu_T + q + \Lambda - t_0$. (2)

The restriction C intersect $V(R)$ is a cover of R and has size at least $\nu(R)$, because it meets distinct edges of a maximum matching in distinct vertices.

Each of the t_0 excluded singleton vertices has ambient degree D and all its neighbors in Y. All its D edges must be covered at their Y-endpoints, which therefore belong to C_Y. There are $D\, t_0$ distinct such edges, while the degree sum at C_Y is at most $D|C_Y|$. It follows that $t_0 \le |C_Y|$. Add C_Y and the bound on C intersect $V(R)$ to (2), then use (1) and $q = |Y| + \rho$. This yields $|C| \ge \nu(H) + \rho + \Lambda$. Minimizing over C proves the cover-size

chain. It implies $\rho \le \tau(H) - \nu(H)$. Also $\Lambda \ge k\ b$ and $b \ge \max(0, \theta_D(H) + \rho - \alpha_S)$ imply the weighted displayed inequality. Since $k \ge 1$, they also give $\Lambda \ge b$, which yields the unweighted inequality.

For the final edge bound, write $O = V(H)$ minus C, and for every required component T put $C_T = C$ intersect $V(T)$, $O_T = O$ intersect $V(T)$, $e_T = e(H[C_T])$, and $\beta_T = e_H(V(T), Y)$. Here $e_H(U, W)$ counts edges between the disjoint vertex sets U, W. Let $\beta_{C,T}$ and $\beta_{O,T}$ count the boundary edges at C_T and O_T respectively, so $\beta_T = \beta_{C,T} + \beta_{O,T}$. Every vertex in T has ambient degree D, and O is independent. Subtracting the degree sums on C_T and O_T gives

$$D(|C_T| - |O_T|) = 2e_T + \beta_{C,T} - \beta_{O,T} \le 2e_T + \beta_T. \quad (3)$$

For a nontrivial required component, $|O_T| \le \alpha(T) \le \mu_T$, hence $|C_T| - |O_T| \ge 1$. For a component counted by b, $\alpha(T) \le \mu_T - k$ strengthens this difference to at least $1 + 2k$. For a singleton, $e_T = 0$ and $\beta_T = D$, so $2e_T + \beta_T \ge D$ holds regardless of which side contains it. Sum (3) over nontrivial components together with this singleton inequality to get

$$D(q + 2kb) \le 2sum_T\ e_T + \sum_T \beta_T.$$

The first sum is at most $e(H[C])$. Each counted boundary edge has an endpoint in Y, so the second sum is at most $D|Y|$. Substituting $q = |Y| + \rho$ proves $2e(H[C]) \ge D(\rho + 2kb)$. Empty collections have zero sums and all the preceding constructions permit empty selections, so the stated boundary cases are included. □

Remark. An incidence cover can save a rank in every required component except those at equality with independence. Odd-order domination makes every equality component pay a fixed independence defect, and separator incidences pay for the singleton components.

Depends on. Lemma 107, Lemma 116.

Lemma 166 (Component localization of a below-budget host). *Use the finite integer certificate costs c_i, their sum Σ, matching number ν, excess ex, positive-level slack ps, and $gap = ex - ps$ from Lemma 61. Let $s, t \ge 1$ be integers and let $n_1 \ge \cdots \ge n_s \ge 1$ and $m_1 \ge \cdots \ge m_t \ge 1$ be integer lists. Let F_1, F_2 be their vertex-disjoint unions of stars, prescribed red and blue, with copies understood as ordinary subgraphs. Put $M = s + t - 1$, $\ell_k = max\{n_i + m_j - 1 : 1 \le i \le s, 1 \le j \le t, i + j = k\}$ for $2 \le k \le s + t$, $L = \sum_k \ell_k$, and $P = L - M$. For each integer $0 \le q < M$, let A_q be every ordered pair of actual nonempty suffixes obtained by deleting α and β leading whole components, where $0 \le \alpha < s$, $0 \le \beta < t$ and $\alpha + \beta = q$. Put $K = M - q$, $L_q = \sum_{k=q+2}^{s+t} \ell_k$, $P_q = L_q - K$, and $\rho_i = |\{k : q + 2 \le k \le s + t, \ell_k \ge i + 1\}|$ for every integer $i \ge 0$.*

Let H be a finite simple graph that arrows every member of A_q. Define integers $\delta = K - \nu(H)$, $\sigma = e(H) - \nu(H) - P_q$, and $\sigma_i = c_i(H) - \rho_i$. Then $\sigma_i \ge 0$ for every i, the sums below have finite support, and

$$\sigma = ps(H) + \textstyle\sum_{i \ge 1} \sigma_i \ge 0,$$

$$e(H) - L_q = \sigma - \delta,$$

$$ex(H) = \delta + \sigma_0,$$

$$gap(H) = L_q - e(H) + \textstyle\sum_{i \ge 0} \sigma_i.$$

Consequently $\nu(H) \ge K$ implies $e(H) \ge L_q$.

If $e(H) < L_q$, then $\delta \ge 1$ and $0 \le ps(H) \le \sigma \le \delta - 1$. The graph H has an edge. Every connected component C of largest matching number satisfies

$$ex(C) \ge ex(H) \ge \delta,$$

$$gap(C) \ge gap(H) + \textstyle\sum_{Q \ne C} ps(Q) \ge L_q - e(H) \ge 1,$$

$$\Delta(C) \le \Delta(H),$$

where Q ranges over all connected components of H, including isolated vertices. In particular every such C is not LP-good. If additionally $\nu(H) \ge K - 3$, then δ belongs to {1,2,3} and

$$ps(H) + \textstyle\sum_{i \ge 1} \sigma_i = \sigma \le \delta - 1 \le 2.$$

At $\delta = 1$ this forces $ps(H) = 0$ and $c_i(H) = \rho_i$ for every $i \ge 1$.

Now let G be any finite simple graph with $G \to (F_1, F_2)$ and $e(G) < L$. Put $\delta_0 = M - \nu(G)$ and $\sigma_{init} = e(G) - \nu(G) - P$. There is an ordered sequence of $q < M$ original vertices whose induced deletion leaves a graph H arrowing every A_q member, with $e(H) < L_q$ and $\Delta(H) \le \ell_{q+2} - 1$. For that same H and q there are nonnegative integers ϵ, j such that

$\delta = \delta_0 - j \ge 1,$

$\sigma = \sigma_{init} - \epsilon - j,$

$e(H) = e(G) - \sum_{k=2}^{q+1} \ell_k - \epsilon.$

The sequence has at most M-1 steps and charges each deleted edge exactly once at its earliest deleted endpoint. Every coloring of H avoiding an entire member of A_q extends across these same actual deleted vertices to a coloring of G avoiding both original targets while retaining all H-edge colors. If $\delta_0 \le 3$, the literal H satisfies the displayed delta-in-{1,2,3} and slack-at-most-two conclusions. No exclusion of those three remaining deficit values is asserted here. All indices, orders, ranks, costs, degrees, capacities and matching budgets are integers; δ and σ_{init} are initially signed, and all nonnegativity conclusions have the hypotheses stated above.

Proof. The envelope and signed identities of Lemma 64 apply to these exact original lists, legal suffix family and graph H. They give $\sigma_i \ge 0$, $\rho_0 = K$, $\sum_i \rho_i = L_q$, $ex(H) = \delta + \sigma_0$, and the displayed gap identity. Their positive-level identity is $ps(H) = \sigma - \sum_{i \ge 1} \sigma_i$. Rearranging and using $ps(H) \ge 0$ from Lemma 61 proves the first identity and $\sigma \ge 0$. Direct substitution of the definitions gives $e(H) - L_q = \sigma - \delta$. If $\nu(H) \ge K$, then $\delta \le 0$ and $e(H) - L_q \ge 0$, proving that assertion without any degree restriction.

Assume $e(H) < L_q$. Integrality and $\sigma - \delta < 0$ give $\delta \ge \sigma + 1 \ge 1$ and $\sigma \le \delta - 1$. The nonnegative summands in $\sigma = ps(H) + \sum_{i \ge 1} \sigma_i$ give $ps(H) \le \sigma$. The gap identity gives $gap(H) \ge L_q - e(H) \ge 1$. An edgeless graph has every certificate cost zero, since its empty edge partition is feasible at zero budgets and capacities; its gap is therefore zero. Thus H has an edge and a finite nonempty component family.

Matching numbers add over the components: restrictions of a matching bound its size by the sum of the component matching numbers, while the union of component maximum matchings attains that sum. Hence the hypothesis of Lemma 81 holds with the deleted set empty. Its maximum-rank conclusion gives $ex(C) \ge ex(H)$ for every component of largest matching number. The composition inequality of Lemma 72 gives $ps(H) \ge \sum_Q ps(Q)$. Subtracting the component slack and using $ex(C) \ge ex(H)$ yields

$gap(C) = ex(C) - ps(C)$

$\ge ex(H) - ps(C)$

$\ge ex(H) - ps(H) + \sum_{Q \ne C} ps(Q).$

This is the required stronger gap inequality, and each other-component slack is nonnegative by Lemma 61. The excess bound $ex(H) \ge \delta$ and the positive gap lower bound were already proved. A component is an actual subgraph of H, so none of its degrees exceeds $\Delta(H)$. This proves all component assertions simultaneously for every maximum-rank choice.

If $\nu(H) \ge K - 3$, then $\delta \le 3$. Together with $\delta \ge 1$ this gives the stated three integer values. The exact positive-level identity gives the displayed bound by $\delta - 1 \le 2$. When $\delta = 1$, a sum of nonnegative integers is at most zero, so $ps(H) = 0$ and every positive-row surplus $\sigma_i = 0$. This proves the stated row equalities, with no inference that this case is impossible.

For the original host G, apply the finite actual-vertex reduction of Lemma 41. It supplies $q < M$, the ordered deleted set X, literal $H = G - X$, common A_q arrowing, strict $e(H) < L_q$, and the stated degree $\cap$. With d_l the degree of the l-th vertex just before it is deleted, define

$\epsilon = \sum_{l=1}^{q} (d_l - \ell_{l+1}),$

$j = q - \nu(G) + \nu(H).$

That theorem gives ϵ, $j \ge 0$, the exact displayed edge identity, and $\sigma = \sigma_{init} - \epsilon - j$. Directly,

$\delta = K - \nu(H) = M - q - \nu(H) = M - \nu(G) - j = \delta_0 - j.$

Apply the earlier conclusions of the present proof to this same H. They give $\delta \ge 1$; hence $\delta_0 \ge 1$ as well. If $\delta_0 \le 3$, then $1 \le \delta \le \delta_0 \le 3$, and the earlier exact slack bounds apply. The cited result's finite construction

stops after at most M-1 steps because M successful deletions would charge at least all M diagonal entries, contrary to $e(G) < L$. Its earliest-endpoint edge accounting proves the asserted disjoint charging, and its actual-vertex extension preserves every surviving color. All these conclusions refer to the same literal output; no replacement graph is substituted for H. □

Depends on. Lemma 41, Lemma 61, Lemma 64, Lemma 72, Lemma 81.

Lemma 167 (A sharp deletion bound for an odd degree and one unit of defect). *Use r-goodness, the induced vertex-deletion number $\delta_r(G)$, and the certificate costs $c_r(G)$ of Lemma 86. Let $D \geq 7$ be an odd integer and let G be a finite simple graph of odd order n with maximum degree at most D.*

If $n \geq D+6$, then $\delta_{D-1}(G) \leq (n-D)/2$.

If instead the total degree defect satisfies

$\sum_{v \in V(G)} (D - d_G(v)) = 1$,

then $n \geq D+2$ and the same bound $\delta_{D-1}(G) \leq (n-D)/2$ holds, including $n = D+2$ and $n = D+4$.

In each case an actual vertex set of at most $(n-D)/2$ vertices leaves one induced graph of maximum degree at most D-1 with a matching covering all its degree-(D-1) vertices. Consequently $c_{D-1}(G) \leq (n-D)/2$. The graph need not be connected or factor-critical. If the one-defect graph is factor-critical, meaning that deleting any one vertex leaves a perfect matching, put $k = \nu(G)$. Then $n = 2k+1$ and $c_{D-1}(G) \leq \delta_{D-1}(G) \leq k - (D-1)/2$. All orders, degrees, matching cardinalities and deletion counts are integers.

Proof. We first bound the binomial term in Lemma 86. For integers $A \geq 7$ and $t \geq 3$ put

$R_A(t) = (A+2t)\ binomial(2t,t)/binomial(A+2t,t)$.

For fixed t, cancellation gives

$R_{A+1}(t)/R_A(t) = (A+t+1)/(A+2t) \leq 1$.

Moreover $R_7(3) = 10/11$ and

$R_7(t+1)/R_7(t) = 2(2t+1)(t+8)/((2t+7)(2t+8))$.

The denominator minus the numerator in this last fraction equals 40-4t. Therefore $R_7(t) \leq 10/11 < 1$ for every integer $3 \leq t \leq 11$, by successive comparison for $t = 3,...,10$. In particular $R_A(t) < 1$ for $A \geq 7$ and $3 \leq t \leq 7$.

For $8 \leq t \leq A$ the first monotonicity gives

$R_A(t) \leq R_t(t) = 3t\ binomial(2t,t)/binomial(3t,t)$

$= 3t\ product_{i=0}^{t-1}(2t-i)/(3t-i) \leq 3t(2/3)^t$.

The final inequality follows by cross multiplication in each positive factor. At $t = 8$ the last expression is $6144/6561 < 1$. Its ratio at consecutive t is $2(t+1)/(3t) \leq 3/4$ for $t \geq 8$. Thus $R_A(t) < 1$ for every $A \geq 7$ and $3 \leq t \leq A$. We also retain the bound for $A = 7$ and $3 \leq t \leq 11$.

Suppose $n \geq D+6$, and write $n = D+2t$ with integer $t \geq 3$. The weighted count $W = N_D + N_{D-1}/D$ in Lemma 86 is at most n, because $D \geq 1$ and the two degree classes are disjoint. Apply its fixed-size bound with deletion size $s = t$. Its correction term is at most

$n\ binomial(n-D,t)/binomial(n,t) = R_D(t)$.

Wherever the preceding estimates give $R_D(t) < 1$, the integer correction is zero and $\delta_{D-1}(G) \leq t$. This settles $3 \leq t \leq D$ and also $D = 7$ with $3 \leq t \leq 11$.

For $D \geq 9$ and $t \geq D$, use the independent-selection estimate of Lemma 86 with $p = 1/4$. Since $12 \cdot 19683 = 236196 < 262144$, one has $(3/4)^D \leq (3/4)^9 < 1/12$. Therefore

$\delta_{D-1}(G) \leq floor(n[1/4 + (3/4)^D]) < n/3$.

Here $n = D+2t \geq 3D$, which implies $n/3 \leq (n-D)/2 = t$. This proves the required deletion bound in this range.

For $D = 7$ and $t \geq 12$, $n \geq 31$, and the same selection estimate is at most $\lfloor 6283n/16384 \rfloor$. The difference between t and the unfloored expression is

$(n-7)/2 - 6283n/16384 = (1909n - 57344)/16384 > 0,$

since $1909 \cdot 31 = 59179 > 57344$. This gives $\delta_6(G) \leq t$. The ranges treated cover every odd $D \geq 7$ and every $t \geq 3$, proving the first assertion.

Now suppose the total degree defect equals one. Nonnegative integer deficits imply that exactly one vertex p has degree D-1 and all other vertices have degree D. The graph is nonempty. If $1 \leq n \leq D$, simplicity would imply total defect at least

$n(D-n+1) = D + (n-1)(D-n) \geq D,$

contrary to the defect one. Thus $n > D$, and parity of n and D gives $n \geq D+2$. The first assertion applies for $n \geq D+6$. There are two remaining orders.

If $n = D+2$, let C be the complement of G. Its vertex p has degree two and every other vertex has degree one. The two neighbors of p have no other neighbors, so the component of C containing p is a three-vertex path centered at p; every other component is a single edge. Delete a leaf a of that path. On the remaining $D+1$ vertices the complement is a perfect matching, so G-a is the complete graph on those vertices with one perfect matching removed. It is (D-1)-regular. To exhibit a perfect matching in it, label the missing pairs a_i, b_i for $1 \leq i \leq m = (D+1)/2$, and choose $a_i\, b_{i+1}$, with indices modulo m. These edges exist because $m \geq 4$, and they cover every vertex exactly once. Thus G-a is (D-1)-good and one deletion suffices.

Suppose $n = D+4$. In the complement C, p has degree four and all other vertices have degree three. Count triples consisting of an unordered pair {u, v} contained in $V(C)$ minus {p} and one common C-neighbor of that pair. The common neighbor p contributes $binomial(4,2) = 6$ triples. Each of the four neighbors of p contributes $binomial(2,2) = 1$, since p is excluded from the pair. Every other vertex contributes $binomial(3,2) = 3$. The total is $6 + 4 + 3(n-5) = 3n - 5$. For $n \geq 11$,

$binomial(n-1,2) - (3n-5) = (n^2 - 9n + 12)/2 > 0$:

writing $n = 11 + a$ with $a \geq 0$ makes the numerator $a^2 + 13a + 34$ positive. Hence some pair u, v different from p has no common neighbor in C.

Put $J = G - \{u, v\}$. Every retained vertex is adjacent in G to at least one of u, v, because failure would make it a common C-neighbor. Thus $\Delta(J) \leq D-1$ and $d_J(p) \leq D-2$. Since each original degree is at least D-1, every J-degree is at least D-3. Every component of J therefore has at least D-2 vertices. If J had at least two components, it would have at least $2D - 4 > D + 2$ vertices, contradicting $|V(J)| = D+2$ and $D \geq 7$. Thus J is connected. The vertex p has degree below D-1, so J has no whole connected (D-1)-regular component. Its degree-(D-1) vertex set has at most $D+1$ members. Apply the component-exclusion bound of Lemma 86 with even $r = D-1$ and $b_r = D$. A positive exposure would require at least 2D such vertices, whereas $D + 1 < 2D$. The exposure is therefore zero, so J is (D-1)-good. Two deletions suffice, proving the second small-order case.

In every case $\delta_{D-1}(G)$ is attained, by its definition in Lemma 86, which also gives $c_{D-1}(G) \leq \delta_{D-1}(G)$ with an actual common residual and covering matching. Finally, if G is factor-critical, a perfect matching of G-v has $(n-1)/2$ edges, and no matching can have more than $\lfloor n/2 \rfloor$ edges. Since n is odd, $k = \nu(G) = (n-1)/2$. Substitution gives $(n-D)/2 = k - (D-1)/2$ and proves the final bound. □

Remark. Finite averaging handles every larger odd order. At the two smallest one-defect orders, sparse complements permit explicit vertex deletions, after which the exposure count forces a covering matching.

Depends on. Lemma 86.

Lemma 168 (Zero slack of retained cubic extensions). *Use the finite integer certificate costs c_i, matching number ν, independence number α, edge count e, maximum degree Δ, and slack $ps = e - \nu - \sum_{i \geq 1} c_i$ of Lemma 23. All graphs are finite and simple, and graph restriction and vertex deletion are induced. Let R be an induced subgraph of F with $\Delta(R) \leq 2$ and $ps(R) = 0$. Put $U = V(F)$ minus $V(R)$, $u = |U|$, and $m = \nu(R)$. Suppose*

$c_1(F) \geq u + c_1(R),\ c_2(F) \geq u + c_2(R).$

Let x, y be nonnegative integers such that $x + y = u + m - 1$ and $E(F)$ cannot be partitioned into two parts of matching numbers at most x, y. Put $q = \min(x, y)$, and assume $u \geq q$.

For any independent set T contained in U such that every vertex of T has exactly three neighbors in R, put $t = |T|$ and $H = F[V(R) \text{ union } T]$. If $t \leq u - q$, then

$\nu(H) = m + t$, $c_1(H) = c_1(R) + t$, $c_2(H) = c_2(R) + t$,

$ps(H) = 0$, $c_i(H) = 0$ *for every integer* $i \geq 3$, *and* $\Delta(H) \leq 3$.

Every maximum matching of H has m edges within R and t edges from T to R, saturates T, and has no edge within T. In particular, each v in $V(R)$ has at most $3 - deg_R(v)$ neighbors in T.

Deleting T in any order is a complete cubic deletion process on H. More generally, any process deleting a current degree-three vertex until maximum degree is at most two has some integer length $z \geq 0$ and terminal R' satisfying

$\alpha(R') = \alpha(R) - t + z$,

$0 \leq z \leq min\{c_2(R) + t, t + \alpha(H) - \alpha(R)\}$.

Every such process has zero slack and zero rows of index at least three at every prefix, and each deletion lowers ν, c_1, c_2 by one.

If $t = 1$ and $\alpha(H) = \alpha(R)$, every complete cubic deletion process has length exactly one. The vertices of H-degree three form a nonempty clique. Deleting any one of them gives a degree-at-most-two zero-slack graph R' with

$|V(R')| = |V(R)|$, $\nu(R') = \nu(R)$, $c_1(R') = c_1(R)$, $c_2(R') = c_2(R)$, $\alpha(R') = \alpha(R)$.

All orders, degrees, costs, cardinalities, budgets and process lengths are nonnegative integers. Empty R or T and $x = 0$ or $y = 0$ are permitted whenever the hypotheses hold. No regularity, well-coveredness, source minimality or preserving source process is assumed.

Proof. We use the certificate definitions Lemma 23, vertex restoration Lemma 72, nonnegative slack Lemma 61, the degree-two formulas Lemma 57, missing-allocation retention Lemma 7, and cubic deletion Lemma 126.

The missing allocation has total $x + y = u + m - 1$, and $u \geq q$. Apply Lemma 7 with initial vertex set $V(R)$, exterior U, and retained exterior set T. The condition $t \leq u - q$ is precisely its retention hypothesis. It gives $\nu(H) = m + t$ and the asserted structure of every maximum matching. No component assumption is needed for these conclusions.

The omitted vertex set U minus T has u-t vertices. The vertex-restoration bound of Lemma 72 gives

$c_i(F) \leq u - t + c_i(H)$

for $i = 1, 2$. Combining it with the two assumed lower bounds yields

$c_i(H) \geq c_i(R) + t$, $i = 1, 2$. (1)

The degree-two formulas Lemma 57 give all rows of R of index at least three equal to zero. Since $ps(R) = 0$,

$e(R) = m + c_1(R) + c_2(R)$. (2)

The set T is independent and each of its vertices has exactly three neighbors in R. Because H is the full induced graph, its entire edge set consists of $E(R)$ and these 3t distinct crossing edges. Thus $e(H) = e(R) + 3t$.

Nonnegative slack and nonnegative row costs, supplied by Lemma 61 and Lemma 23, now give

$e(H) = \nu(H) + c_1(H) + c_2(H) + \sum_{i \geq 3} c_i(H) + ps(H)$

$\geq m + t + c_1(R) + t + c_2(R) + t = e(R) + 3t = e(H)$.

The difference between the two sides is the sum of the two nonnegative deficits in (1), all the higher row costs, and $ps(H)$. Their sum is zero, so each vanishes. This proves every asserted cost and slack identity.

In particular $c_3(H) = 0$. The capacity pair (3,0) therefore admits a certificate with both matching budgets zero, by Lemma 23. An edge set of matching number zero is empty, since any one edge is a matching. The part of maximum degree zero is also empty. Every H-edge consequently belongs to the part of maximum degree at most three, proving $\Delta(H) \leq 3$. A vertex v of R has degree $deg_R(v) + |N_F(v)$ intersect $T|$ in H, which proves the asserted neighbor bound.

Every vertex of T has H-degree three. Deleting other vertices of T does not alter that degree because T is independent. Consequently any ordering of T deletes current degree-three vertices and ends at precisely R, proving that it is a complete process, including the empty ordering.

For an arbitrary complete cubic process on H, all hypotheses of Lemma 126 now hold: H is subcubic, $ps(H) = 0$, and $c_3(H) = 0$. That fact gives existence, the claimed prefix identities, and

$\alpha(R') = |V(H)| - \nu(H) - c_2(H) + z.$

The degree-two identity from Lemma 57 is $\alpha(R) = |V(R)| - m - c_2(R)$. Substituting $|V(H)| = |V(R)| + t$ and the already proved costs therefore gives $\alpha(R') = \alpha(R) - t + z$. The two upper bounds in Lemma 126 become $z \le c_2(R) + t$ and $z \le t + \alpha(H) - \alpha(R)$, as stated.

Finally suppose $t = 1$ and $\alpha(H) = \alpha(R)$. The added vertex has degree three, so every complete process has at least one deletion. The second upper bound is $z \le 1$, hence $z = 1$ for every process. Any degree-three vertex can be the first deletion, and finite continuation always exists. Thus deleting any such vertex already leaves maximum degree at most two. Two distinct nonadjacent degree-three vertices cannot exist: deleting one would leave the other's degree equal to three and require a second deletion. The degree-three vertices therefore form a clique, nonempty because it contains the added vertex. The prefix identities with $z = 1$ give the order, matching and cost identities for each resulting R'; its independence identity gives $\alpha(R') = \alpha(R)$. All arguments use finite sets, and the displayed identities include every allowed empty-set and zero-budget endpoint. □

Remark. The matching and first two cost lower bounds use every edge of the extension. Nonnegative slack leaves no room for any surplus or higher row, so a local attachment inherits exact cubic accounting.

Depends on. Lemma 7, Lemma 23, Lemma 57, Lemma 61, Lemma 72, Lemma 126.

Lemma 169 (Decomposition by exposable vertices). *Let G be a finite simple graph, and let $\nu(G)$ be its maximum matching cardinality. A matching is an edge set with pairwise disjoint endpoints; a perfect matching covers every vertex, with the empty matching perfect in the empty graph. Define $D_G = \{v \in V(G) : \nu(G - v) = \nu(G)\}$, $A_G = N_G(D_G)$ minus D_G, and $Q_G = G - (D_G union A_G)$, where vertex deletion is induced and $N_G(D_G)$ is the set of vertices adjacent to some member of D_G. Let C_G be the family of connected components of $G[D_G]$. Each component C in C_G has odd order and is factor-critical, meaning C-v has a perfect matching for every v in $V(C)$. The graph Q_G has a perfect matching, and*

$\nu(G) = |A_G| + \sum_{C \in C_G} (|V(C)| - 1)/2 + |V(Q_G)|/2,$

$|C_G| = |A_G| + |V(G)| - 2\nu(G).$

Every maximum matching of G matches all vertices of A_G to vertices in distinct components of C_G, has exactly $(|V(C)| - 1)/2$ internal edges in every C, and is perfect internally on Q_G. Form the bipartite incidence graph with parts A_G and C_G, joining a to C exactly when an actual G-edge joins them. It has a matching saturating A_G; every member of C_G is omitted by at least one such matching. Every A_G − saturating incidence matching, any actual realizing edge for each of its incidences, any perfect matching of each used C after deleting that edge's endpoint, independently prescribed exposed vertices and perfect matchings after their deletion in the unused components, and any perfect matching of Q_G together form one maximum matching of G. Conversely every maximum matching of G has this form. If D_G is empty, A_G is empty and $Q_G = G$ has a perfect matching. Empty graphs and component families and isolated vertices are permitted. All graph orders, degrees, matching cardinalities and component counts are nonnegative integers.

Proof. A vertex v lies in D_G exactly when some maximum matching of G leaves v uncovered. Indeed such a matching is contained in G-v and has size $\nu(G)$; conversely a matching in G-v of size $\nu(G)$ is a maximum matching of G missing v. This proves the equivalence directly from the definition.

If G has a perfect matching, all maximum matchings are perfect, so D_G and A_G are empty and every assertion holds with $Q_G = G$. Suppose G has no perfect matching, and put $\eta = |V(G)| - 2\nu(G) \ge 1$. Apply Lemma 107 with the prescribed set $U = V(G)$. Choose an inclusionwise maximal maximizing separator Y as in that fact. Let O be the family of odd components of G-Y. Every member is factor-critical, and $|O| = |Y| + \eta$. The other components have a perfect matching in their union. In the separator-component incidence graph B choose a matching L_0 saturating Y and use the cited fact to lift it to a maximum matching M_0 of G. The family O_0 of

component vertices omitted by L_0 has exactly η members. The only vertices exposed by M_0 are one chosen vertex in each member of O_0.

In B direct each edge outside L_0 from its O-end to its Y-end, and each edge of L_0 from its Y-end to its O-end. Let $O_\cdot$ and A be the component vertices and separator vertices reachable from O_0 by directed paths, including the starting vertices. Every neighbor in B of a member of $O_\cdot$ lies in A. A nonmatching edge gives immediate reachability; if the component is matched, its matching neighbor was the preceding vertex on any path first reaching it. All vertices of A are matched by L_0, and their partners belong to $O_\cdot$. Every reached component outside O_0 is reached along its unique matching edge. Hence L_0 pairs A bijectively with $O_\cdot$ minus O_0, and $|O_\cdot| = |A| + \eta$.

Write D for the union of the vertex sets of $O_\cdot$, $W = D$ union A, and $Q = G - W$. Distinct members of O are components of G-Y and have no edges between them. All exterior neighbors of each member of $O_\cdot$ therefore lie in Y, and the preceding neighbor property puts them in A. Thus every edge of G is either incident with A, internal to a member of $O_\cdot$, or internal to Q. Every vertex of A has a neighbor in D because its reaching path ends with an incidence from $O_\cdot$.

The matching M_0 has no edge joining W to Q. Its incidence edges meeting A end in $O_\cdot$, its incidence edges at Y minus A end outside $O_\cdot$, and all remaining matching edges are internal to their G-Y components. On W it has $|A|$ incidence edges and $(|V(C)| - 1)/2$ internal edges for each C in $O_\cdot$. All vertices exposed by M_0 belong to W, so M_0 is perfect on Q. Counting its disjoint parts gives

$$\nu(G) = |A| + \sum_{C \in O_\cdot} (|V(C)| - 1)/2 + |V(Q)|/2. \quad (1)$$

Each C in $O_\cdot$ is omitted by some A-saturating incidence matching on A, $O_\cdot$. Choose a shortest directed path from O_0 to C. If it has length zero, L_0 already omits C. Otherwise it is a simple even-length alternating path starting at an $L_0 -$ *unmatched* component and ending at C; reverse matching membership along it. Every separator vertex on it remains matched and C becomes unmatched. The restriction to A, $O_\cdot$ saturates A. Every A-saturating matching on A, $O_\cdot$ can also be combined with the unchanged part of L_0 outside these two sets to give a Y-saturating incidence matching, since those unchanged edges have neither endpoint in the two sets. By Lemma 107 this matching lifts to a maximum matching of G with any specified vertex of any omitted component exposed. Therefore every vertex of D belongs to D_G.

Conversely, let M be any maximum matching of G. Partition its edges into those incident with A, those internal to each C in $O_\cdot$, and those internal to Q. The first part has at most $|A|$ edges, each component part has at most $(|V(C)| - 1)/2$ edges, and the last part has at most $|V(Q)|/2$ edges. Their sum equals $\nu(G)$ by (1), so every bound is attained. Exactly $|A|$ matching edges incident with A force every A-vertex covered and no edge with both endpoints in A: each such edge would use two A-vertices while contributing only one to the attained count. The Q-part is perfect, so no matching edge joins A to Q. Every component part leaves exactly one component vertex available for an edge from A. Hence the A-edges go to distinct components. In particular every vertex outside D is covered by every maximum matching. The first paragraph proves $D_G = D$.

The members of $O_\cdot$ are connected, pairwise anticomplete, and all their exterior neighbors lie in A. Thus they are precisely the components C_G of $G[D_G]$. Since every A-vertex has a neighbor in D and no other vertex outside D does, $A = A_G$ and $Q = Q_G$. Equations (1) and $|O_\cdot| = |A| + \eta$ become the stated formulas.

Finally choose any of the incidence edges, realizing G-edges, internal perfect matchings after the indicated vertex deletions, and perfect matching of Q_G specified in the statement. All required choices exist: the incidence matching was constructed, each realizing edge exists by its definition, and factor-criticality gives each internal perfect matching. Distinct incidence edges use distinct separator vertices and distinct components. Their endpoints avoid the corresponding internal matchings. Different components and Q_G have disjoint vertex sets. Their union is therefore a matching. Its size is the right side of (1), so it is maximum. The structural conclusion for an arbitrary maximum M in the preceding paragraph recovers precisely these choices from M, proving the converse and their simultaneous compatibility. Singleton components have empty internal perfect matchings after deletion, and empty sets contribute zero edges throughout. □

Remark. Alternating reachability removes exactly the separator components whose exposed vertex cannot be moved. The remaining graph is perfectly matched, and equality in one matching-size bound identifies every exposable vertex.

Depends on. Lemma 107.

Lemma 170 (Connected regular equimatchable graphs). *Let $d \geq 2$ be an integer and let G be a finite connected simple d-regular graph. A matching is inclusionwise maximal if no edge can be added to it, and G is equimatchable if every such matching has the maximum possible cardinality $\nu(G)$. Suppose G is equimatchable. Then either G has odd order and is factor-critical, or G is isomorphic to $K_{d,d}$, or d is odd and G is isomorphic to K_{d+1}. Here factor-critical means that deleting any one vertex leaves a perfect matching; $K_{d,d}$ is the complete bipartite graph with two d-element parts, and K_{d+1} is the complete graph on $d+1$ vertices. In particular, for even d the alternatives are odd factor-critical or $K_{d,d}$, and for odd d the alternatives are $K_{d,d}$ or K_{d+1}. No converse for the odd factor-critical alternative is asserted. All orders, degrees and matching cardinalities are integers in their stated domains.*

Proof. Every specified matching extends to a maximum matching: add disjoint edges while possible. Each step increases the edge count, so the finite graph gives termination at an inclusionwise maximal matching; equimatchability makes it maximum. This construction never removes a specified edge.

Apply Lemma 107 with all vertices of G marked, and choose its inclusionwise maximal maximizing separator S. Its marked deficiency is $|V(G)| - 2\nu(G)$, since a matching covers exactly twice its cardinality. No component C of G-S can have even order. Otherwise choose a vertex v in C. The graph C-v has odd total order, so at least one of its components has odd order. Deleting v in addition to S therefore increases the odd-component count by at least one and the separator cardinality by exactly one, without changing any other component. The maximizing objective cannot increase strictly, while equality would contradict inclusionwise maximality of S. This excludes even components.

Thus all components C of G-S are odd and factor-critical by Lemma 107. Put $r_C = (|V(C)| - 1)/2$. The same fact gives

$\nu(G) = |S| + \sum_C r_C$. (1)

Every maximum matching attains the minimum exposure when all vertices are marked. Its structural conclusion therefore says that every maximum matching covers all of S by edges into distinct components of G-S, with exactly r_C internal edges in every C.

The set S is independent. If an edge lay inside S, extend that one-edge matching to a maximum matching; its internal S-edge would contradict the preceding structure. Also no two disjoint edges join S to one component C, because that two-edge matching could likewise be extended to a prohibited maximum matching.

If S is empty, connectedness and the factor-critical component conclusion give the first alternative. Suppose S is nonempty. If a vertex a in S is adjacent to every vertex of some nonsingleton component C of G-S, then no other vertex a' of S has a neighbor c in C: choose c' in C distinct from c, and a'c, ac' would be two disjoint edges into C. Thus every vertex of C has exactly one neighbor outside C, namely a, and its internal degree is d-1. Simplicity gives $|V(C)| \geq d$, whereas $d_G(a) = d$ and its adjacency to all of C give $|V(C)| \leq d$. Hence $|V(C)| = d$, C is complete, and a has no neighbor outside C. All degrees within C union {a} are already d, so no edge leaves it. Connectedness gives $G = K_{d+1}$. Since C has odd order, d is odd.

For the remaining case assume no vertex of S is adjacent to every vertex of a nonsingleton component. Fix a in S and let T be the set of its neighbors that are singleton components of G-S. Then $|T| \leq d$. Each vertex in T has all its d neighbors in S, so exactly d-1 neighbors in S minus {a}. Consider the bipartite graph B with parts T and S minus {a}, retaining these adjacency edges. Every nonempty subset U of T of size at most d-1 has at least d-1 neighbors, by considering even one of its vertices, and therefore at least $|U|$ neighbors.

The only possible further subset has $|U| = d$, forcing $U = T$. If it has fewer than d neighbors in B, its neighborhood has exactly d-1 vertices, and every member of T has the same full d-element G-neighborhood $S_0 = N_B(T)$ union {a}. Each member of S_0 is adjacent to all d vertices of T, exhausting its degree. The same holds for every member of T. These vertices form a whole $K_{d,d}$ component, and connectedness gives $G = K_{d,d}$.

Otherwise every subset of T has at least its own size as its neighborhood in B. We prove that B has a matching covering T. Choose a maximum matching. If it leaves a vertex of T uncovered, follow nonmatching edges from that vertex toward S minus {a}, and matching edges back toward T. A reached uncovered vertex in the second part gives an alternating path whose edge reversal increases the matching cardinality, a contradiction. Thus every reached vertex in the second part is matched back to the reached T-set. The latter set also contains its uncovered starting vertex, and all its neighbors are reached. Its neighborhood then has one fewer vertex than the set, contradicting the subset inequality. Therefore there is a matching covering T. When T is empty choose the empty matching. In either case this matching uses no vertex a.

For every nonsingleton component C of G-S choose z_C nonadjacent to a, possible by the remaining-case assumption, and take a perfect matching of $C - z_C$. Unite all of these matchings with the B-matching covering T. Their vertices are disjoint: the B-matching uses only singleton components and vertices of S minus {a}. The union leaves a uncovered and covers every neighbor of a. A neighbor cannot lie in S, since S is independent; a singleton neighbor lies in T; and every neighbor in a nonsingleton component differs from z_C and is covered internally.

Extend this union by adding edges until it is inclusionwise maximal. The initially covered neighbors of a remain covered, so no added edge can cover a. The final maximal matching therefore leaves a uncovered. It is maximum by equimatchability, contradicting the compulsory coverage of S in every maximum matching. This eliminates the remaining case and proves all three alternatives.

Finally, a d-regular graph of odd order has even degree sum $d|V(G)| = 2e(G)$, so d must be even. Thus the odd factor-critical alternative cannot occur for odd d. The two stated parity-specific conclusions follow. All constructions terminate by finite cardinality, and $d \geq 2$ covers the smallest degree in every counting and matching argument. □

Remark. A maximal matching separator is compulsory in every maximum matching. Equimatchability lets any local matching be extended, so degree constraints force a complete block or allow all neighbors of a compulsory vertex to be covered while leaving it uncovered, a contradiction.

Depends on. Lemma 107.

Lemma 171 (Whole-cycle selection under the failed run bound). *Use the finite integer certificate costs $c_i(G)$ of Lemma 106. Thus $c_i(G)$ is the maximum, over nonnegative integer capacities $A + B = i$, of the least nonnegative integer total T such that every nonnegative integer split $x + y = T$ admits a partition of all G-edges into parts of maximum degrees at most A, B and parts of matching numbers at most x, y. All graphs are finite and simple. Write $V(G)$, $E(G)$, $d_G(v)$, $\Delta(G)$, $\nu(G)$, and $\alpha(G)$ for vertices, edges, vertex degree, maximum degree, maximum matching cardinality, and maximum independent-set cardinality. A whole component is an entire connected component, and path and cycle orders count vertices.*

Let d belong to {8,10,12}. Let F be connected, d-regular, and factor-critical, meaning that F-v has a perfect matching for every vertex v. Write $|V(F)| = 2k + 1$ for an integer $k \geq 1$, put $a_F = \alpha(F)$, and suppose $E = c_0(F) - k \geq 1$. At total $c_0(F) - 1$, call a nonnegative integer pair missing if $E(F)$ has no partition into two parts with its matching-number bounds. Let $tr(F)$ be the greatest length of a consecutive run of missing first coordinates. Suppose

$c_{d-1}(F) \geq a_F$, $tr(F) + E \geq d/2 + 2$.

Define the integers

$\lambda = 2E - 3 - (k - a_F)$, $\delta = c_0(F) - c_2(F)$, $\epsilon = c_0(F) - c_1(F)$,

$p = E - 2 - \lambda$, $m = p + \delta$, $\beta = m - \epsilon$,

$r = k - \lambda - 1 + \delta$, $s = 2\lambda + 3 - 2\delta$, $u = r + s$,

and suppose $0 \leq \epsilon \leq \delta \leq \lambda + 1$.

Let R be an induced subgraph of F with $|V(R)| = r$, $\Delta(R) \leq 2$, $\nu(R) = m$, exactly p whole odd-cycle components, and exactly ϵ whole path components of positive even order. Every whole odd cycle of R has order at most seven; other component orders are unrestricted. Let Bcal be the family of all missing pairs at total $u + m - 1$.

Then Bcal is nonempty and $p \geq 2$. Put $D2 = \{z in V(R) : d_R(z) = 2\}$. Choose any maximum independent set T_0 of R[D2] and any shortest whole odd-cycle component C of R. Such choices exist and can be made before choosing any pair in Bcal. Set

$\ell = (|V(C)| + 1)/2$, $Y = T0$ *union* $V(C)$,

$H = F[(V(F) minus V(R))$ *union* $Y]$,

$N = |V(F)$ *minus* $V(H)|$, $h = |V(H)| - r$.

Then

$|T0| = \beta$, $|Y| = \beta + \ell$, $h = s + \beta + \ell$, $N = r - \beta - \ell$,

$E(R[Y]) = E(C), \nu(R[Y]) = \ell - 1,$

$N + \nu(R[Y]) = r - \beta - 1, 2 \le \ell \le 4.$

All vertices of R[Y] outside C are isolated. Every y in Y has exactly d-2 neighbors outside R, all retained in H. For every pair (b_R, b_B) in Bcal,

$\min(b_R, b_B) \le N, N \ge \max(0, \beta + h - 2), N \ge \ell - 2.$

If $E \le d/2$, every shortest whole odd cycle is a triangle, so $\ell = 2$.

All orders, degrees, capacities, budgets, indices, component counts, matching and independence sizes, and selected-set sizes are integers in their stated nonnegative or positive domains. Defined differences are initially integers. In particular s is positive odd and $\beta \ge p \ge 2$. No well-coveredness or deletion-process hypothesis is imposed, and $\epsilon = 0$ is permitted.

Proof. Use Lemma 40 for missing allocations, Lemma 106 and Lemma 124 for top-cost inequalities, and Lemma 91 for selection at larger excess. Their integer cost definitions agree.

1. Common parameter and missing-pair bounds.

Factor-criticality gives $\nu(F) = k$: a perfect matching of F-v has k edges, and every matching on $2k + 1$ vertices has at most k edges. Therefore Lemma 40 applies to F with maximum-degree bound d. Put

$v = \max(0, floor((2E - 3)/(d - 2))).$

The missing family at total $k + E - 1$ is nonempty, every coordinate b of a missing pair satisfies

$2E - 1 + v \le b \le k - E - v,$ (1)

and

$k \ge 3E - 1 + 2v.$

Its final run inequality applies with $B = d/2 + 1$, since the assumed run bound is $tr(F) + E \ge B + 1$. It gives the additional bound

$k \ge 2E + d/2 + 2v.$ (2)

The parameter definitions give

$2r + s = 2k + 1, u + m = k + E, g := k - a_F = E + p - 1.$

Thus the specified Bcal is precisely this nonempty missing family. Since p counts odd cycles, $p \ge 0$. The bounds on δ give

$\lambda \ge -1, 0 \le \delta \le E - 1 - p, p \le E - 1,$

$s \ge 1$ with s odd, and $\beta \ge p.$ (3)

The row order makes r nonnegative.

The top-cost lower bound and the graph hypotheses permit the unrestricted inequalities of Lemma 106 and Lemma 124:

$25k \le 65g - 9$ when $d = 8$,

$44k \le 100g - 43$ when $d = 10$,

$125k \le 250g - 119$ when $d = 12.$ (4)

These inequalities require no lower bound on E beyond those already imposed.

2. Selection when $E \ge d/2 + 1$.

Here $v = floor((2E - 3)/(d - 2)) \ge 1$. At $d = 8$ combine the first inequality in (4) with $k \ge 3E - 1 + 2v$ and $g = E + p - 1$:

$65p \ge 10E + 50v + 49 \ge 149.$

The last inequality uses $E \ge 5$ and $v \ge 1$. Since p is an integer and $65 \cdot 2 = 130$, it follows that $p \ge 3 = (d - 2)/2$. This uses the unrestricted linear inequality, not its separate excess-eight corollary.

At $d = 10$, $E \geq 6$, and the cycle-count conclusion of Lemma 124 gives $p \geq 4 = (d-2)/2$. At $d = 12$, $E \geq 7$, and the same fact gives $p \geq 6 \geq (d-2)/2$. Its p equals ours by the identity $g - E + 1 = p$.

For every missing coordinate b, (1) and the parameter formulas give

$u - b \geq u - k + E + v = s + m + v + 1$. (5)

All hypotheses of Lemma 91 now hold for this actual F, R and Bcal: the order, degree, component counts and odd-cycle order bound are the stated ones; (3) gives its parameter bounds; this step gives its p and E bounds; and (5) supplies both required coordinate margins. Apply it to any choices T_0, C in the statement. It gives every asserted selection identity, physical edge conclusion, exterior-neighbor count and omission bound in this case, with the same choices before all missing pairs.

3. The small-excess range forces a triangle.

Suppose $1 \leq E \leq d/2$. Then $v = 0$, because $2E - 3 < d - 2$ and the maximum with zero removes any negative floor. By (2),

$k \geq 2E + d/2 \geq 3E$. (6)

At $d = 8$, substitute $k \geq 2E + 4$ and $g = E + p - 1$ into (4):

$50E + 100 \leq 65E + 65p - 74$,

so $65p \geq 174 - 15E \geq 114$. Thus $p \geq 2$. Since $p \leq E - 1 \leq 3$, (3) gives $\delta \leq 1 < p$.

At $d = 10$, the corresponding calculation is

$88E + 220 \leq 100E + 100p - 143$,

so $100p \geq 363 - 12E \geq 303$. Hence $p \geq 4$. Together with $p \leq E - 1 \leq 4$, this forces $E = 5$, $p = 4$ and $\delta = 0$.

At $d = 12$, one obtains

$250E + 750 \leq 250E + 250p - 369$,

so $250p \geq 1119$ and $p \geq 5$. Since $p \leq E - 1 \leq 5$, this forces $E = 6$, $p = 5$ and $\delta = 0$.

In every case $p \geq 2$ and $m = p + \delta < 2p$. If R had no triangle, each of its p odd-cycle components would have order at least five and would contain a matching of size at least two. Uniting these disjoint matchings would give $\nu(R) \geq 2p$, a contradiction. Thus a triangle exists, and every shortest whole odd cycle C is a triangle. In particular $\ell = 2$.

4. All selection margins in the small-excess range.

We give the row calculation directly, so it does not require the larger-excess hypothesis of Lemma 91. A path on n vertices has matching number $\lfloor n/2 \rfloor$ and independence number $\lceil n/2 \rceil$; grouping consecutive vertex pairs gives the upper bounds, attained by alternating edges or vertices. A cycle on n vertices has both quantities $\lfloor n/2 \rfloor$: counting matching endpoints gives one upper bound, and mapping independent vertices to their cyclic successors gives the other; alternating choices attain them.

For a path of order $n \geq 2$, its degree-two vertices induce the internal path on n-2 vertices. Its independence number $ceil((n-2)/2)$ equals the original matching number minus one when n is even, and equals that matching number when n is odd. An isolated vertex contributes zero to both the matching number and the selected degree-two set. A cycle is unchanged. Adding these component contributions proves

$\alpha(R[D2]) = \nu(R) - \epsilon = \beta$.

A maximum independent set T_0 exists by finiteness. Its restriction to the triangle C has exactly one vertex, since each whole component of R[D2] can be optimized independently. Adding all of C therefore adds two vertices. All other selected vertices are independent and have no edge to the whole component C. Hence

$|Y| = \beta + 2$, $E(R[Y]) = E(C)$, $\nu(R[Y]) = 1$.

All vertices of R[Y] outside C are isolated. The entire complement of R is retained, so

$h = s + \beta + 2$, $N = r - \beta - 2 = k - E - 1 + \epsilon$. (7)

This gives $N + \nu(R[Y]) = r - \beta - 1$. Every selected vertex has R-degree two, and inducedness and d-regularity give exactly d-2 neighbors outside R, all lying in H.

For the weaker omission margin, the definitions give

$\beta + h - 2 = s + 2\beta = 2E - 1 - 2\epsilon$.

Consequently (6) and (7) imply

$N - (\beta + h - 2) = k - 3E + 3\epsilon \geq 0$.

The quantity N is the cardinality of the omitted set, so it is nonnegative. This proves $N \geq \max(0, \beta + h - 2)$, and $N \geq \ell - 2$ follows from $\ell = 2$.

Every pair in Bcal has total $L = k + E - 1$. Equation (7) gives

$2N + 1 - L = k - 3E + 2\epsilon \geq 0$.

For integer L, N this implies $\lfloor L/2 \rfloor \leq N$. Therefore

$\min(b_R, b_B) \leq \lfloor L/2 \rfloor \leq N$

for every missing pair. The proof uses no property of a particular maximum T_0 or a particular shortest triangle, so all choices in the statement are allowed before the missing pair. The two excess ranges exhaust the positive integers, completing the proof. □

Remark. The failed-run lower bound supplies the omission reserve at small excess. The top-cost inequalities then force a triangle, so the shortest-cycle selection joins directly to the larger-excess construction without losing the integer equality endpoint.

Depends on. Lemma 40, Lemma 91, Lemma 106, Lemma 124.

Lemma 172 (Independent deletion in an octic graph of excess four). *Use the finite integer certificate costs $c_i(H)$, matching number $\nu(H)$, excess $ex(H) = c_0(H) - \nu(H)$, slack $ps(H)$, signed gap $gap(H) = ex(H) - ps(H)$, and top missing-allocation run $tr(H)$ of Lemma 23. Define $T_-(J)$ to mean that $gap(J) > 0$ implies both $gap(J) = 1$ and $tr(J) + ex(J) \leq \max(4, floor(\Delta(J)/2) + 1)$. Let H be a finite simple connected 8-regular factor-critical graph, where factor-critical means that deleting any one vertex leaves a perfect matching. Suppose $ex(H) = 4$, $gap(H) = 1$, and every proper induced subgraph of H satisfies T_-. Put $k = \nu(H)$. Then $|V(H)| = 2k + 1$ and $k \geq 11$. For every nonempty independent vertex set I, put $j = |I|$, $J = H - I$, $b = j + \nu(J) - k$, and $z = gap(J)$. One has $c_i(J) = c_i(H) - j$ for every integer $1 \leq i \leq 7$, $c_8(H) = 1$, $c_8(J) = 0$, and $c_i(H) = c_i(J) = 0$ for every integer $i \geq 9$. Also z is either zero or one, $c_0(J) = c_0(H) - j + z$, $ex(J) = 4 - b + z$, $ps(J) = 4 - b$, and $1 \leq b \leq \min(4, \lceil j/2 \rceil)$. If I is contained in another nonempty independent set I', then $gap(H - I) \leq gap(H - I')$. More specifically, for every inclusionwise maximal independent set S, put $a = |S|$, $G = H - S$, and $\eta = \nu(G) - (k + 1 - a)$. Call a component P of G large when $floor(|V(P)|/2) \geq a + 2$, and put $m = \max(\{0\} union \{ex(P) : P is large\})$. Then $a \geq 3$, $\Delta(G) \leq 7$, $0 \leq \eta \leq \min(3, floor((a-1)/2))$, $ps(G) = 3 - \eta$, and $c_0(H) = a + \nu(G) + ps(G)$. If $m = 0$ then $ps(G) = 0$; if $m \geq 1$ then $m - 1 \leq ps(G) \leq m$. In the latter case, when $ps(G) = m - 1$, every large component P with $ex(P) = m$ has $gap(P) = 1$ and $ps(P) = m - 1$, every other component of G has slack zero, and $ps(G)$ is the sum of the component slacks. All orders, cardinalities, degrees, costs, capacities, budgets and displayed parameters are integers; signed gaps may be negative on graphs outside the hypotheses. No failure of T_- by H, and no minimality under edge deletion, is assumed.*

Proof. Factor-criticality gives $|V(H)| = 2k + 1$ and $\nu(H) = k$. Since $ex(H) = 4$, $c_0(H) = k + 4$. By the integer threshold definition in Lemma 23 there is an infeasible pair (x, y) of nonnegative budgets with $x + y = k + 3$ at degree capacities zero. Neither budget can be at least k, since placing all edges in that exceptional part would be feasible. Exchange the names of the two budgets so that $x \leq y$. If $x \leq 6$, choose any x vertices W. Put all edges meeting W in the first exceptional part and all other edges in the second. The first matching number is at most x. The second is at most $floor((2k + 1 - x)/2) \leq k + 3 - x = y$, where the inequality follows from $x \leq 6$. This would be a certificate, a contradiction. Thus $x, y \geq 7$, so $k + 3 \geq 14$ and $k \geq 11$.

Fix any maximal independent set S, write $a = |S|$, and put $G = H - S$. Every vertex outside S has a neighbor in S, by inclusionwise maximality. Each closed neighborhood of a vertex of S has nine vertices, so $9a \geq 2k + 1 \geq 23$ and $a \geq 3$. In particular H is not the complete graph on nine vertices. Apply Lemma 113 to obtain $\Delta(G) \leq 7$, $c_8(H) = 1$, zero rows of H above eight and of G from eight onward, and $c_0(H) \leq a + \nu(G) + m$ for the stated large-component parameter m.

Choose v in S and restrict a perfect matching of H-v to G. The a-1 other vertices of S are matched to distinct outside vertices, since S is independent. Their deletion removes exactly a-1 matching edges and leaves $k+1-a$ matching edges in G. Therefore $\eta := \nu(G) - (k+1-a) \geq 0$. The order bound $\nu(G) \leq floor((2k+1-a)/2)$ gives $\eta \leq floor((a-1)/2)$.

For integers $1 \leq i \leq 7$ write $r_i = c_i(H) - c_i(G)$. Restriction and extension in Lemma 72 give $0 \leq r_i \leq a$. Deleting the independent set S removes exactly 8a edges. Since $gap(H) = 1$ and $ex(H) = 4$, $ps(H) = 3$. Subtracting the definitions of ps and using the top row and zero tails gives

$\epsilon := 3 - ps(G) - \eta = \sum_{i=1}^{7}(a - r_i) \geq 0$.

The cost bound $c_0(H) \leq a + \nu(G) + m$ becomes $4 \leq 1 + \eta + m$. Consequently $\epsilon \leq m - ps(G)$. If $m = 0$, the slack nonnegativity from Lemma 61 forces $\epsilon = 0$ and $ps(G) = 0$.

Suppose $m \geq 1$ and choose any large component $P_.$ with $ex(P_.) = m$. This component is proper induced in H, so its T_- property implies $gap(P_.) \leq 1$ and $ps(P_.) \geq m - 1$. Component slack superadditivity in Lemma 72 and nonnegativity from Lemma 61 imply $ps(G) \geq m - 1$. Thus the nonnegative integer ϵ is at most one.

Assume $\epsilon = 1$. Equality must hold in the preceding bounds: $ps(G) = m - 1$, $4 = 1 + \eta + m$, $ps(P_.) = m - 1$, $gap(P_.) = 1$, and every other component Q of G has $ps(Q) = 0$. Also $c_0(H) = a + \nu(G) + m$. To rule out this case define $Y_t(K)$ to be the feasible red budgets at nonnegative integer total t and degree capacities zero on a graph K, as in Lemma 75. Every companion Q satisfies T_- and $\Delta(Q) \leq 7$. Since $ps(Q) = 0$, its nonnegative excess equals $gap(Q)$, so $ex(Q)$ is zero or one. If $ex(Q) = 0$, $Y_{(}\nu(Q))(Q)$ is a full interval. If $ex(Q) = 1$, this is its top allocation set and its longest missing run is $tr(Q) \leq 4 - 1 = 3 \leq a$. In both cases its endpoints are feasible because the total equals $\nu(Q)$.

Start with the actual S-pool interval [0,a]. The interval criterion and composition in Lemma 75 absorb each companion Q at total $\nu(Q)$; the interval length can only increase. This produces a full allocation interval of length $L = a + \sum_{Q different from P_.} \nu(Q)$. Keep $P_.$ separate at total $t = c_0(P_.) - 1 = \nu(P_.) + m - 1$, which is at least $\nu(P_.)$. The same composition lemma, restoring every edge meeting S, yields

$[0, L] + Y_t(P_.)$ contained in $Y_{(}c_0(H) - 1)(H)$,

since $L + t = a + \nu(G) + m - 1 = c_0(H) - 1$. All these intervals consist of integers. Listing consecutive elements of $Y_t(P_.)$, including its endpoints 0, t, shows that adding [0,L] replaces each missing run of length r by one of length $\max(0, r - L)$. The larger set on H therefore has missing runs of length at most $\max(0, tr(P_.) - L)$. Because $c_0(H) > 0$, its top allocation set is not full and $tr(H) \geq 1$ by Lemma 23. Hence $tr(H) + L \leq tr(P_.)$. Apply T_- to the positive-gap component $P_.$ to obtain $tr(P_.) + m \leq 4$. Combining this with $4 = 1 + \eta + m$ yields

$tr(H) + 4 \leq 4 - L + 1 + \eta \leq 4 - a + 1 + floor((a-1)/2) = 4 - \lfloor a/2 \rfloor \leq 3$.

This contradicts $tr(H) \geq 1$. Thus $\epsilon = 0$.

It follows that $ps(G) = 3 - \eta$, so $\eta \leq 3$ by nonnegative slack. Every summand $a - r_i$ in the formula for ϵ is nonnegative, so $r_i = a$ for all $i = 1,..., 7$. The identity $4 = 1 + \eta + ps(G)$ gives $c_0(H) = a + \nu(G) + ps(G)$. The cost bound then implies $ps(G) \leq m$. When $m \geq 1$ its lower bound m-1 was already proved. If $ps(G) = m - 1$, apply component slack superadditivity to any maximizing large P: $ps(P) \geq m - 1$ and all other slacks are nonnegative, so equality forces $ps(P) = m - 1$, $gap(P) = 1$, and zero slack for all companions. This also makes $ps(G)$ the sum of the component slacks. No particular maximizing P was required.

Now let I be any nonempty independent set. Enlarge I to a maximal independent S by adjoining available vertices until none remain; finiteness guarantees termination. Put $j = |I|$ and $J = H - I$. Deleting I lowers each row by at most j; deleting S minus I lowers it by at most a-j, by Lemma 72. For positive rows 1,..., 7 the total loss on deleting S is exactly a. Both bounds must be attained, giving $c_i(J) = c_i(H) - j$.

Choose v in I and restrict a perfect matching of H-v to J. The remaining matching has $k - j + 1$ edges. Every uncovered surviving vertex was formerly matched to a vertex of I, so it has a neighbor in I and degree at most seven in J. The restricted matching therefore covers every degree-eight vertex of J. Lemma 61 gives $c_8(J) = 0$ and the zero rows of both graphs above eight. It also gives nonnegative slack.

Set $r_0 = c_0(H) - c_0(J)$. The extension bound gives $r_0 \leq j$. Exactly 8j edges have been deleted and the total positive-row decrease is $7j + 1$. Thus

$z := gap(J) = 1 - r_0 - (7j + 1) + 8j = j - r_0 \geq 0$.

The proper induced graph J satisfies T_-, so z is zero or one. This proves $c_0(J) = c_0(H) - j + z$. Define $b = j + \nu(J) - k$. Substituting $\nu(J) = k - j + b$ into the definitions gives $ex(J) = 4 - b + z$ and $ps(J) = 4 - b$. The restricted matching gives $b \geq 1$, nonnegative slack gives $b \leq 4$, and the order bound gives

$b \leq j + floor((2k + 1 - j)/2) - k = \lceil j/2 \rceil$.

Finally, if I is contained in a nonempty independent set I′, put $q = |I'| - |I|$. The zero-row extension bound says $c_0(H - I) - c_0(H - I') \leq q$. Using the proved zero-row identities gives

$gap(H - I') - gap(H - I) = q - [c_0(H - I) - c_0(H - I')] \geq 0$.

This proves every claimed assertion. □

Depends on. Lemma 23, Lemma 61, Lemma 72, Lemma 75, Lemma 113.

Lemma 173 (Exact costs and native rows from a complete deletion process). *Use the finite integer certificate costs $c_{A,B}$, row maxima c_i, matching number ν, independence number α, and slack $ps = e - \nu - \sum_{i \geq 1} c_i$ of Lemma 23. All graphs are finite and simple. All capacities, budgets, row indices, orders, degrees, cardinalities, and process lengths are nonnegative integers; differences and the parameters defined below are initially integers.*

Let d belong to {6,8,10,12}. Let F be connected, d-regular, and factor-critical, meaning that F-v has a perfect matching for every vertex v. Write $|V(F)| = 2k + 1$ with integer $k \geq 1$, put $a = \alpha(F)$, and assume $E = c_0(F) - k \geq 1$. Define

$\lambda = 2E - 3 - (k - a)$, $\delta = c_0(F) - c_2(F)$, $\epsilon = c_0(F) - c_1(F)$,

$p = E - 2 - \lambda$, $r = k - \lambda - 1 + \delta$, $s = 2\lambda + 3 - 2\delta$,

$\beta = p + \delta - \epsilon$, $u = 2k + 1 - r$, $M = k + E - r$.

A complete process chooses a maximum independent set S of F, then repeatedly deletes any current maximum-degree vertex from F-S while that degree is at least three, and stops at an induced terminal Q with $\Delta(Q) \leq 2$. Assume the following three conditions.

(a) For every such S and every integer $1 \leq i < d$,

$c_i(F - S) = c_i(F) - a$, *and* $c_0(F) = a + \nu(F - S) + ps(F - S)$.

(b) In every complete process, each deletion parent H, of current maximum degree $D \geq 3$, has a matching covering all its degree-D vertices; deletion of the chosen vertex lowers c_i by exactly one for every integer $1 \leq i < D$. Every terminal Q satisfies $ps(Q) = 0$.

(c) At least one complete process has $\alpha(Q) = a$.

Call an induced subgraph A of F a native row when

$|V(A)| = r$, $\Delta(A) \leq 2$, $ps(A) = 0$,

$\nu(A) = p + \delta$, $c_1(A) = \beta$, $c_2(A) = p$.

Then native rows exist, $p \geq 0$, $0 \leq \epsilon \leq \delta \leq \lambda + 1$, s is positive and odd, $\beta \geq p$, $r = a + 2p + \delta$, $u = r + s$, and $M = p + \delta + s$.

For every complete process, let b_j be the number of its roots deleted at degree j, for integers $3 \leq j \leq d - 1$, and put $t = \sum_{j=3}^{d-1} b_j$ and $h = a - \alpha(Q)$. Then

$h \geq 0$, $t = 2E - 1 - \delta - h$, $|V(Q)| = r + h$,

$\nu(Q) = p + \delta + h$, $c_1(Q) = \beta + h$, $c_2(Q) = p + h$,

$b_{d-1} = p + \epsilon$, $b_{d-2} = \beta$,

$c_i(F) = a + \sum_{j=i+1}^{d-1} b_j$ *for every integer* $3 \leq i \leq d - 1$.

Empty sums are zero. In particular

$c_{d-3}(F) = r$, $c_{d-2}(F) = r - \beta = a + p + \epsilon$,

$c_{d-1}(F) = a$, $c_d(F) = 1$, $c_i(F) = 0$ *for* $i > d$.

For each displayed positive exact cost $K = c_i(F)$, every nonnegative integer capacity split $A_c + B_c = i$ and every nonnegative integer budget split $x + y = K$ admit a certificate on F; some capacity split and some budget split of K-1 admit none.

In every process the induced graph on S and all roots of the first two phases is a native row. Every native row has $\alpha(A) = a$, $e(A) = 3p + 2\delta - \epsilon$, exactly p whole odd-cycle components and ϵ whole positive even-order path components. Its components belong to P_1, P_2, P_3, P_4, P_6, C_3, C_4, C_5, C_7, with paths and cycles indexed by vertex count.

Every native row A has an explicit replay as precisely the initial set and first two full phases of a process on the same F. Label each path consecutively and each cycle cyclically. On components P_1, P_2, P_3, P_4, P_6, C_3, C_4, C_5, C_7, respectively, choose the pairs (S, T_1)

({1},empty), ({1},{2}), ({1,3},empty), ({1,3},{4}), ({1,3,5},{6}),

({1},{2}), ({1,3},empty), ({1,3},{4}), ({1,3,5},{6}).

Take the unions of the component sets and put $T_2 = V(A)$ minus $(SunionT_1)$. Then S is maximum independent in F, every ordering of T_1 is its full degree-(d-1) phase, and every ordering of T_2 is the following full degree-(d-2) phase. Thus $|T_1| = p + \epsilon$ and $|T_2| = \beta$.

For every native row A, its full complement $J = F - V(A)$ has order u, maximum degree at most d-3, and a matching covering all its degree-(d-3) vertices. It satisfies

$\nu(J) = M$, $ps(J) = 0$, $e(J) = 3p + 2\delta - \epsilon + ds/2$,

$c_i(J) = c_i(F) - r$ *for every integer* $1 \le i \le d - 3$.

In particular $c_0(F) = u + p + \delta$, $c_1(F) = u + \beta$, $c_2(F) = u + p$,

$c_1(J) = M - \epsilon$, $c_2(J) = M - \delta$.

Put $v_d = \max(0, floor((2E - 3)/(d - 2)))$. *For every missing nonnegative integer allocation (x, y) of total* $c_0(F) - 1$,

$2E - 1 + v_d \le x, y \le k - E - v_d$,

$s + p + \delta + v_d + 1 \le u - x$, $u - y \le a - \delta - v_d$.

Here missing means no partition into two edge parts with matching-number bounds x, y.

If $Q = R$ is an independence-preserving terminal of a specified process, A is its early native row, and X is its remaining root set, then $V(A)$, X, $V(R)$ partition $V(F)$,

$|V(A)| = |V(R)| = r$, $|X| = s$, $\Delta(F[X]) \le d - 6$,

$e_F(X, V(A)) = e_F(X, V(R)) = ds/2 - e(F[X])$,

where $e_F(U, W)$ counts edges between disjoint vertex sets U, W. These conclusions retain assumptions (a)–(c).

Proof. We use Lemma 23 for the integer certificate thresholds, Lemma 72 for restriction and vertex restoration, Lemma 61 for nonnegative slack and covered-degree zero rows, Lemma 57 for degree-two rows, Lemma 7 for the matching consequence of a missing allocation, and Lemma 40 for its connected odd-order interval. The cubic elimination criterion used below is supplied by Lemma 126.

1. Process accounting and terminal profiles.

Factor-criticality and odd order give $\nu(F) = k$: F-v has a matching of size k, and no matching on $2k + 1$ vertices is larger. A maximum independent S exists because F is finite. It is maximal, so every vertex outside S has a neighbor in S. Hence $\Delta(F - S) \le d - 1$, and every process has only the stated degree indices. Every legal partial process can be completed by choosing a current maximum-degree vertex whenever the degree is at least three; vertex count decreases and forces finite termination.

At a deletion parent H of degree D, its covering matching gives $c_D(H) = 0$ by Lemma 61; all higher rows vanish by index monotonicity, and restriction makes those rows zero after deletion. Write $h_v = \nu(H) - \nu(H - v)$. The local slack identity and assumption (b) yield

$ps(H) - ps(H - v) = D - h_v - (D - 1) = 1 - h_v$.

Thus $\nu + ps$ drops by exactly one. Combining this with (a) proves

$c_0(F) = |W| + \nu(F - W) + ps(F - W)$

for every actual prefix W containing S.

For an integer $3 \le i \le d-1$, stop immediately after all roots of degree greater than i have been deleted. Let W_i be the removed set, including S. The remainder has maximum degree at most i. If its degree equals i, it is a deletion parent and (b) gives a covering matching. If its degree is smaller, its degree-i vertex set is empty and the empty matching covers it. In either case Lemma 61 gives $c_i(F - W_i) = 0$. All prior roots lowered row i by one. Assumption (a) therefore gives

$c_i(F) = |W_i| = a + \sum_{j=i+1}^{d-1} b_j$.

For every integer $1 \le \ell \le i$ it also gives $c_\ell(F - W_i) = c_\ell(F) - |W_i|$. These equalities hold for each actual process, including empty phases.

At any terminal Q, all t roots lowered rows one and two by one, while its slack is zero. Consequently

$k + E = a + t + \nu(Q)$, $c_1(F) = a + t + c_1(Q)$, $c_2(F) = a + t + c_2(Q)$.

The degree-two formulas in Lemma 57 give

$\epsilon = \nu(Q) - c_1(Q) \ge 0$,

$\delta = \nu(Q) - c_2(Q) \ge \epsilon$,

$\alpha(Q) = |V(Q)| - \nu(Q) - c_2(Q)$.

The second inequality uses $c_1(Q) \ge c_2(Q)$. Since Q is induced, $h = a - \alpha(Q) \ge 0$. Substitute $|V(Q)| = 2k + 1 - a - t$ and the preceding cost identities into the α formula. It gives $t = 2E - 1 - \delta - h$; substitution back gives

$|V(Q)| = r + h$, $\nu(Q) = p + \delta + h$, $c_1(Q) = \beta + h$, $c_2(Q) = p + h$.

The process in (c) has $h = 0$, so $p \ge 0$ and its terminal is a native row. For any native row, the same degree-two theorem gives the nine component types and the stated odd-cycle and even-path counts. The parameter identities give $r = a + 2p + \delta$, so its independence is $r - (p + \delta) - p = a$. Its zero slack gives $e(A) = (p + \delta) + \beta + p = 3p + 2\delta - \epsilon$.

2. Universal upper bounds for the first two phases.

The row-zero definition agrees with the two-part definition of Lemma 40, because degree-zero parts are empty. That fact applies to connected F with order $2k + 1$, $\nu(F) = k$, degree $d \ge 3$, and $E \ge 1$. It gives $k \ge 3E - 1$ and a missing allocation (x, y) with $x \le y$ and $x + y = k + E - 1$. Since $p \ge 0$, $\lambda \le E - 2$, whence

$r = k - \lambda - 1 + \delta \ge k - E + 1$,

$x \le floor((k + E - 1)/2) \le k - E + 1 \le r$.

The middle inequality follows from $k \ge 3E - 1$. By Lemma 7, every induced L with at least x exterior vertices satisfies

$\nu(L) \ge |V(L)| + k + E - (2k + 1) = |V(L)| - k + E - 1$.

Fix any process. Let $C = b_{d-1}$, $B = b_{d-2}$, $m = C + B$, and let L be the full induced graph on S and these m roots. Its complement contains its terminal, whose order $r + h$ is at least $r \ge x$. Since $a = k - E + 1 - p$, the matching bound yields $\nu(L) \ge m - p$.

Roots chosen in a single degree phase are independent: an earlier adjacent root would lower the later root's degree below that phase value. Every first-phase root has exactly one neighbor in S, since F is d-regular. Two such roots cannot have the same S-neighbor, because replacing that neighbor in S by both roots would give an independent set larger than a. The initial pool consisting of S and the first phase therefore induces C disjoint edges and a-C isolates; its row-one cost is zero, as all its edges fit in either single degree-one part. Restore all other vertices using Lemma 72. This gives $c_1(F) \le 2k + 1 - a - C$, hence $C \le p + \epsilon$.

Every second-phase root has exactly two neighbors in S together with the first phase, and none in its own phase. Therefore $e(L) = C + 2B = 2m - C$. Vertex restoration also gives

$c_1(L) \ge c_1(F) - (2k + 1 - a - m) = m - p - \epsilon$,

$c_2(L) \ge c_2(F) - (2k + 1 - a - m) = m - p - \delta$.

Nonnegative slack and nonnegative higher costs from Lemma 61 imply

$2m - C = e(L) \geq \nu(L) + c_1(L) + c_2(L) \geq 3m - 3p - \epsilon - \delta.$

Thus $m + C \leq 3p + \epsilon + \delta$. Both phase bounds hold for every complete process.

3. Canonical replay and exact phase lengths.

Take any native row A, whose existence was established in step 1. Use the component sets in the statement. For each path or cycle, the selected S has maximum independent size; their union therefore has size $\alpha(A) = a$ and is maximum in F. The sets T_1 and T_2 are independent inside their respective phases. Each T_1 vertex has exactly one neighbor in S. Each T_2 vertex has exactly two row neighbors, both in S union T_1. These checks follow directly from consecutive path edges and cyclic cycle edges in the nine displayed choices; no edges join different components. There is one T_1 vertex for each odd cycle and for each positive even-order path. Hence

$|T_1| = p + \epsilon, |T_2| = r - a - p - \epsilon = p + \delta - \epsilon = \beta.$

In F-S all T_1 vertices have degree d-1, and their independence preserves those degrees in every ordering. Delete them. If another degree-(d-1) vertex remained, deleting it and continuing greedily would give a complete process with $C > p + \epsilon$, contradicting step 2. Thus T_1 is the full first phase. The T_2 vertices now have degree d-2; their independence permits every ordering. After all are deleted the two-phase total is $2p + \delta$. Any additional degree-(d-2) root would give $m > 2p + \delta$ with $C = p + \epsilon$, contradicting $m + C \leq 3p + \epsilon + \delta$. Thus T_2 is exactly the full second phase, and the graph after them is $J = F - V(A)$.

Applying the prefix formula of step 1 at $i = d - 2$ and $i = d - 3$ gives

$c_{d-2}(F) = a + p + \epsilon = r - \beta,$

$c_{d-3}(F) = a + 2p + \delta = r.$

It also gives $c_{d-1}(F) = a$. These costs are independent of the process. Applying the same prefix formulas to an arbitrary process forces $C = p + \epsilon$ and $C + B = 2p + \delta$, hence $B = \beta$, universally.

Return to its early pool L. Now $e(L) = 3p + 2\delta - \epsilon$. The bounds in step 2 become $\nu(L) \geq p + \delta$, $c_1(L) \geq \beta$, and $c_2(L) \geq p$. Their sum equals $e(L)$. Nonnegative slack forces equality in each bound, $ps(L) = 0$, and every row at index at least three to be zero. In particular, its zero-total certificate at capacities (3,0) has all edges in the degree-three part, so $\Delta(L) \leq 3$. Since L contains S and is induced, $\alpha(L) = a$. The parameter identity gives

$|V(L)| - \nu(L) - c_2(L) = r - (p + \delta) - p = a.$

Lemma 126 therefore implies $\Delta(L) \leq 2$. All native-row conditions now hold for this L.

4. Complement ranks and full certificate meanings.

For every native row A the replay makes it an actual prefix. The prefix identities give $c_i(J) = c_i(F) - r$ for $1 \leq i \leq d - 3$, maximum degree at most d-3, and a matching covering all degree-(d-3) vertices. The row-zero prefix identity gives $\nu(J) + ps(J) = k + E - r = M$.

Choose an x-element set Z inside $V(A)$, possible because $x \leq r$. Missingness of (x, y) gives $\nu(F - Z) \geq y + 1$: otherwise coloring all edges incident with Z red and all other edges blue would realize (x, y). A matching of F-Z has at most $\nu(J)$ edges inside J and at most r-x other edges, since the latter use distinct vertices in $V(A)$ minus Z. Therefore

$\nu(J) \geq y + 1 - r + x = k + E - r = M.$

Nonnegative slack forces $\nu(J) = M$ and $ps(J) = 0$. Substitution gives $c_1(J) = M - \epsilon$ and $c_2(J) = M - \delta$.

For the remaining top cost, a zero-budget certificate at capacities (d-1,1) would partition $E(F)$ into a degree-(d-1) part and a matching meeting every vertex, since F is d-regular. Such a matching would be perfect, impossible on $2k + 1$ vertices. Thus $c_d(F) \geq 1$. For any v, factor-criticality supplies a perfect matching of F-v, which covers every vertex of degree d in that graph. By Lemma 61, $c_d(F - v) = 0$. Restoring v by Lemma 72 gives $c_d(F) \leq 1$. Hence $c_d(F) = 1$, and the strong tail in Lemma 61 gives $c_i(F) = 0$ for $i > d$.

The high-cost upper bounds have concrete witnesses: the corresponding prefix remainder has cost zero, and Lemma 72 restores all deleted vertices for every capacity split and every matching-budget allocation at the asserted total. At row d the same construction uses F-v. For each high row, the exact positive cost K already

proved is the maximum over a finite nonempty set of capacity pairs. Choose a pair attaining K. By the integer threshold statement of Lemma 23, at least one allocation at total K-1 is infeasible for that pair. This establishes both certificate directions without using a fractional relaxation.

5. The physical two-row partition.

Fix an independence-preserving terminal R and its actual process. Its early pool A is a native row, disjoint from R. Let X be all other roots. The partition gives

$|X| = (2E - 1 - \delta) - (2p + \delta) = 2\lambda + 3 - 2\delta = s.$

This is a nonnegative odd integer and hence positive. Thus $\delta \leq \lambda + 1$, $2k + 1 = 2r + s$, $u = r + s$, and $M = p + \delta + s$. Also $\beta = p + \delta - \epsilon \geq p$. Every native row has the common order and edge count already established. In a d-regular graph, deleting A removes dr-e(A) edges, so

$e(J) = d(2k + 1)/2 - dr + e(A) = ds/2 + 3p + 2\delta - \epsilon.$

The algebra $u + p + \delta = k + E$ gives the three low source-cost expressions in the statement.

The bounds on each missing coordinate are those of Lemma 40. Subtracting their upper endpoint from u gives $u - k + E + v_d = s + p + \delta + v_d + 1$, and subtracting their lower endpoint gives $u - 2E + 1 - v_d = a - \delta - v_d$. This proves both missing-coordinate bounds for the same allocation.

Every vertex of X belongs to both row complements F-V(A) and F-V(R), each of maximum degree at most d-3 by step 4. Its F-degree d therefore gives at least three neighbors in each row, leaving at most d-6 neighbors in X. The edges of F-V(A) are precisely the edges of R, those of F[X], and those between X and R. Subtract $e(R) = e(A)$ from the preceding edge count to obtain $e_F(X, V(R)) = ds/2 - e(F[X])$. Applying the same argument with the two rows interchanged gives $e_F(X, V(A)) = ds/2 - e(F[X])$. Every complement and retained graph here is induced on its specified vertex set. This completes the construction. □

Remark. The missing top allocation gives a matching lower bound for every early pool. A componentwise replay of one independence-preserving terminal attains two phase bounds, fixing the high source costs. Zero slack then forces every early pool to have degree at most two, and the same missing allocation fixes each row complement's matching number.

Depends on. Lemma 7, Lemma 23, Lemma 40, Lemma 57, Lemma 61, Lemma 72, Lemma 126.

Lemma 174 (Closed lumps of degree twenty are LP-good). *Use the integer certificate costs $c_\theta(H)$, matching number $\nu(H)$, excess $ex(H) = c_0(H) - \nu(H)$, and $gap(H) = \sum_{\theta \geq 0} c_\theta(H) - e(H)$ of Lemma 23. Let H be any nonempty connected 20-regular finite simple graph such that deleting each vertex leaves a perfect matching. Then $gap(H) \leq 0$. More precisely, write $|V(H)| = 2k + 1$. The integer k equals $\nu(H)$ and satisfies $k \geq 10$. If $k \geq 50$, then $gap(H) < 187/30 - 791k/7500 < 1$; if $10 \leq k \leq 49$, the explicit integer bounds in the proof are nonpositive. No hereditary or minimality hypothesis and no upper bound on $|V(H)|$ is imposed.*

Proof. Put $n = |V(H)|$. A perfect matching after any vertex deletion gives odd $n = 2k + 1$ and $\nu(H) = k$. Simplicity gives $20 \leq 2k$, so $k \geq 10$, and regularity gives $e(H) = 10(2k + 1)$. Set $E = ex(H) \geq 0$, an integer by Lemma 23.

We first prove $E \leq e_0 := floor((k + 1)/3)$. For nonnegative integer budgets $x \geq y$, if $n \leq 2x + y + 1$, choose $\min(n, 2x + 1)$ vertices, put their internal edges in the red exceptional part and all other edges in the blue exceptional part, and leave both bounded-degree parts empty. Every red matching has size at most x. Every blue edge meets the complement, whose size is at most y, so every blue matching has size at most y. For $y > x$ interchange the colors. Thus all allocations of total c are feasible at zero capacities whenever $c + \lceil c/2 \rceil + 1 \geq n$. Put $c = k + e_0$. For $k = 3h, 3h + 1, 3h + 2$, the left side equals $6h + 1, 6h + 3, 6h + 6$ and the corresponding order is $6h + 1, 6h + 3, 6h + 5$. The required inequality holds in all three residue classes, proving $c_0(H) \leq k + e_0$ and $E \leq e_0$.

The theorem Lemma 87 gives zero costs above 20, $c_{20}(H) \leq 1$, and $c_i(H) + c_{21-i}(H) \leq 2k + 1$ and $c_i(H) + c_{20-i}(H) \leq 2k + 2$ in the required positive-index ranges. Together with $0 \leq c_i(H) \leq c_0(H) = k + E$ from Lemma 23, these are all hypotheses of Lemma 12 at $d = 20$. Its sequence gap equals $gap(H)$. With $t = 3$ it gives, for any real bounds $b_j \geq c_{20-j}(H)$,

$gap(H) \leq -3k + b_1 + b_2 + b_3 + \min(5E - 3, 4E + 4).$ (1)

Fix any vertex v and perfect matching M of H-v. The graph H-M has degree 20 at v and 19 elsewhere. Apply Lemma 39 with $p_1 = 1/7$, $p_2 = 3/14$, $p_3 = 3/11$ and $j = 1, 2, 3$. Put $q_j = q_{19,j}(p_j)$ and $C_j = 20(1-p_j)^2 Pr(Bin(18, p_j) = j-1)$. The actual marked dominating sets given by that theorem, followed by Lemma 49 with $D = 20$, give $c_{20-j}(H) \le 1 + 2kq_j - C_j$. The vertex and perfect matching may remain fixed for all three separate cost bounds.

Put $(Q_1, Q_2, Q_3) = (1887, 2640, 3279)/10000$. The binomial expansions are

$q_1 = 1/7 + 6^{20}/7^{20}$,

$q_2 = 3/14 + 68 \cdot 11^{19}/14^{20}$,

$q_3 = 3/11 + 2059 \cdot 8^{18}/11^{20}$.

The coefficients are $68 = 11 + 19 \cdot 3$ and $2059 = 64 + 19 \cdot 3 \cdot 8 + 171 \cdot 9$, using $binom(19, 2) = 171$. The differences $Q_j - p_j$ are $3209/70000$, $87/1750$, $6069/110000$. Thus $q_j < Q_j$ follows from

$70000 \cdot 6^{20} = 255931090804408320000 < 256053382549036911209 = 3209 \cdot 7^{20}$;

$1750 \cdot 68 \cdot 11^{19} = 72779317633613310086290000 < 72791382219997777357701120 = 87 \cdot 14^{20}$;

$110000 \cdot 2059 \cdot 8^{18} = 40800811184125745561600000 < 40829197192457066958408690 = 6069 \cdot 11^{20}$.

The corrections expand to $C_1 = 20 \cdot 6^{20}/7^{20}$, $C_2 = 1080 \cdot 11^{19}/14^{20}$, and $C_3 = 27540 \cdot 8^{18}/11^{20}$. Each exceeds $7/10$ by the integer comparisons

$200 \cdot 6^{20} = 731231688012595200 > 558545864083284007 = 7^{21}$;

$10800 \cdot 11^{19} = 6605181768428770999428000 > 5856777879769936109240320 = 7 \cdot 14^{20}$;

$275400 \cdot 8^{18} = 4961165349511338393600 > 4709249964527920064407 = 7 \cdot 11^{20}$.

Consequently $c_{20-j}(H) < 3/10 + 2kQ_j$, and its integer value is at most $\lfloor 3/10 + 2kQ_j \rfloor$.

Since $\sum_j Q_j = 3903/5000$, the second expression of (1) and $E \le (k+1)/3$ give

$gap(H) < -3k + 9/10 + 2k\cdot(3903/5000) + 4(k+1)/3 + 4 = 187/30 - 791k/7500$.

At $k = 50$ this is $24/25 < 1$, and it decreases with k. The integral gap is nonpositive for every $k \ge 50$.

For $10 \le k \le 49$, apply Lemma 37 to H-M with $d = 20$, $n = 2k+1$. Its recurrence has $r_0 = 2k - 20$ and $r_{i+1} = \max(0, r_i - ceil(20r_i/(2k-i)))$. If s is its first zero index, its marked dominating set and Lemma 49 give $c_{19}(H) \le 1 + s$. Its second conclusion and the same certificate theorem give $c_{20-j}(H) \le 2k - 19 + j$ for $j = 2, 3$. Thus take $b_1 = 1 + s$ and $b_j = \min(\lfloor 3/10 + 2kQ_j \rfloor, 2k - 19 + j)$ for $j = 2, 3$. The two expressions in (1) are increasing in E, so replacing E by e_0 preserves an upper bound. The table lists k; e_0; (b_1, b_2, b_3); the entire recurrence through zero; and $U = -3k + sum\ b_j + \min(5e_0 - 3, 4e_0 + 4)$. All entries are explicit evaluations of these integer formulas.

10;3;(1,3,4);(0);-10

11;4;(2,5,6);(2,0);-3

12;4;(2,6,8);(4,0);-3

13;4;(3,7,8);(6,1,0);-4

14;5;(3,7,9);(8,2,0);-1

15;5;(3,8,10);(10,3,0);-2

16;5;(4,8,10);(12,4,1,0);-4

17;6;(4,9,11);(14,5,1,0);0

18;6;(5,9,12);(16,7,3,1,0);-1

19;6;(5,10,12);(18,8,3,1,0);-3

20;7;(5,10,13);(20,10,4,1,0);0

21;7;(5,11,14);(22,11,5,2,0);-1

22;7;(6,11,14);(24,13,6,3,1,0);-3

23;8;(6,12,15);(26,14,7,3,1,0);0

24;8;(7,12,16);(28,16,9,5,2,1,0);-1

25;8;(7,13,16);(30,18,10,5,2,1,0);-3

26;9;(7,14,17);(32,19,11,6,3,1,0);0

27;9;(8,14,18);(34,21,13,8,4,2,1,0);-1

28;9;(8,15,18);(36,23,14,8,4,2,1,0);-3

29;10;(8,15,19);(38,24,15,9,5,3,1,0);-1

30;10;(9,16,19);(40,26,17,11,7,4,2,1,0);-2

31;10;(9,16,20);(42,28,18,12,7,4,2,1,0);-4

32;11;(9,17,21);(44,30,20,13,8,5,3,1,0);-1

33;11;(10,17,21);(46,32,22,15,10,6,4,2,1,0);-3

34;11;(10,18,22);(48,33,23,16,11,7,4,2,1,0);-4

35;12;(10,18,23);(50,35,24,16,11,7,4,2,1,0);-2

36;12;(11,19,23);(52,37,26,18,12,8,5,3,2,1,0);-3

37;12;(11,19,24);(54,39,28,20,14,10,7,4,2,1,0);-5

38;13;(12,20,25);(56,41,30,21,15,10,7,5,3,2,1,0);-1

39;13;(12,20,25);(58,43,31,22,16,11,7,5,3,2,1,0);-4

40;13;(12,21,26);(60,45,33,24,17,12,8,5,3,2,1,0);-5

41;14;(12,21,27);(62,46,34,25,18,13,9,6,4,2,1,0);-3

42;14;(13,22,27);(64,48,36,27,20,15,11,8,5,3,2,1,0);-4

43;14;(13,23,28);(66,50,38,28,21,15,11,8,5,3,2,1,0);-5

44;15;(13,23,29);(68,52,40,30,22,16,12,9,6,4,2,1,0);-3

45;15;(14,24,29);(70,54,41,31,23,17,13,9,6,4,3,2,1,0);-4

46;15;(14,24,30);(72,56,43,33,25,19,14,10,7,5,3,2,1,0);-6

47;16;(15,25,31);(74,58,45,35,27,21,16,12,9,6,4,3,2,1,0);-2

48;16;(15,25,31);(76,60,47,37,29,22,17,13,10,7,5,3,2,1,0);-5

49;16;(15,26,32);(78,62,49,38,30,23,18,14,10,7,5,3,2,1,0);-6.

Every last entry is nonpositive, so (1) proves the claim for every remaining integer k. □

Remark. The saving from the forced vertex pays for all three top certificate rows. The same elementary excess and greedy bounds cover both the smallest legal orders and the remaining finite interval.

Depends on. Lemma 12, Lemma 23, Lemma 37, Lemma 39, Lemma 49, Lemma 87.

Lemma 175 (Goodness of regular factor-critical graphs with small excess). *Use the finite integer certificate costs $c_i(H)$, matching number $\nu(H)$, excess $ex(H) = c_0(H) - \nu(H)$, and signed gap $gap(H) = \sum_{i\geq 0} c_i(H) - e(H)$ from Lemma 23. A graph is LP-good if $gap(H) \leq 0$. All graphs are finite and simple.*

Let H be nonempty, connected, and d-regular, and suppose deleting any vertex of H leaves a perfect matching. Write $|V(H)| = 2k + 1$, where $k = \nu(H)$, and put $E = ex(H)$. Then:

(i) For every even integer $d \geq 12$, $gap(H) < 4E - 7k/10$. If $E \leq 2$, then H is LP-good.

(ii) For every even integer $d \geq 16$, $gap(H) < 4E - 469k/500$. If $E \leq 3$, then H is LP-good.

These assertions cover every possible order, without a hereditary-gap, minimality, independence-number, or density assumption. The degrees, orders, matching numbers, costs, excesses, capacities, and indices are integers in their displayed positive or nonnegative domains; gaps may be signed.

Proof. Factor-criticality and the matching cardinality bound give odd order $2k+1$ and $\nu(H) = k$. Simplicity gives $d \leq 2k$. Put $n = 2k+1$ and $E = ex(H) \geq 0$. The identities and monotonicity of the costs below use Lemma 23.

We first obtain a bound from two high-index rows. By Lemma 87, all rows above d vanish, $c_d(H) \leq 1$, and

$c_i(H) + c_{d+1-i}(H) \leq 2k+1$,

$c_i(H) + c_{d-i}(H) \leq 2k+2$

in the respective ranges of that theorem. Also $c_i(H) \leq c_0(H) = k + E$. Since $e(H) = d(2k+1)/2$, the sequence $c_0(H)$,..., $c_d(H)$ meets every hypothesis of Lemma 12. Taking $t = 2$ in its first upper envelope gives

$gap(H) \leq 4E - 2k - 2 + c_{d-1}(H) + c_{d-2}(H)$. (1)

The condition $2 \leq d/2 - 1$ holds for every degree used here.

Fix any vertex v and any perfect matching M of H-v. In the spanning graph H-M, v has degree d and every other vertex has degree d-1. Apply Lemma 24 and then Lemma 49 to this same graph, vertex and matching. For $j = 1, 2$, any nonnegative integer $m \leq d-1$ and any real p in [0,1] give

$c_{d-j}(H) \leq 1 + 2k\, q_{m,j}(p)$,

where $q_{m,j}(p) = p + (1-p)Pr(Bin(m,p) \leq j-1)$. The sets obtained for the two values of j may differ; they bound separate costs of H.

For $d \geq 12$ choose $m = 11$, with $p = 1/5$ for $j = 1$ and $p = 2/7$ for $j = 2$. Write the resulting coefficients as q_1, q_2. Expansion gives

$q_1 = 1/5 + (4/5)^{12} = 65605341/244140625$,

$q_2 = 2/7 + (5/7)^{12} + 11(2/7)(5/7)^{11}$

$= 5273012861/13841287201$.

The integer inequalities

$1000 \cdot 65605341 = 65605341000 < 65673828125 = 269 \cdot 244140625$,

$1000 \cdot 5273012861 = 5273012861000 < 5273530423581 = 381 \cdot 13841287201$

give $q_1 < 269/1000$ and $q_2 < 381/1000$, hence $q_1 + q_2 < 13/20$. Equation (1) proves

$gap(H) < 4E - 7k/10$. (2)

Strictness is preserved because $k \geq 6 > 0$.

For $d \geq 16$ instead choose $m = 15$, with $p = 1/6$ for $j = 1$ and $p = 1/4$ for $j = 2$. Now

$q_1 = 1/6 + (5/6)^{16}$,

$q_2 = 1/4 + (9/2)(3/4)^{15}$.

The bounds

$10000 \cdot 5^{16} = 1525878906250000 < 1526220459933696 = 541 \cdot 6^{16}$,

$100000 \cdot 3^{15} = 1434890700000 < 1435592818688 = 1337 \cdot 4^{15}$

show $q_1 < 1/6 + 541/10000$ and $q_2 < 1/4 + 12033/200000$. Their sum is less than

$318559/600000 < 318600/600000 = 531/1000$.

Substitution in (1) gives

$gap(H) < 4E - 469k/500$. (3)

We handle the few ranks not covered by (2) or (3) with explicit row bounds. For any odd $n = 2k + 1$ with $k \geq 2$ and any even integer d with $4 \leq d \leq 2k$, restriction from the complete graph K_n, by Lemma 72, and the clique bounds of Lemma 110 give

$c_{d-1}(H) \leq n - d$, $c_{d-2}(H) \leq n - d + 2$. (4)

Here d-1 is odd and d-2 is even, both within the clique theorem's range.

One further construction treats orders seventeen and twenty-one. Let D be 12 or 16, assume $n = D + 5$ and $d \geq D$, and keep the fixed v and M. Put $J = H - M$ and $N = n - 1 = D + 4$. We prove $c_{d-1}(H) \leq 3$ and $c_{d-2}(H) \leq 4$.

Choose uniformly a two-element subset U of $V(H)$ minus {v}, and put $Z = \{v\}$ union U. Every neighbor of v in J outside Z already has a neighbor in Z. There are at most four other vertices, since $deg_J(v) = d \geq D$ and $N = D + 4$. For any such vertex u, choose D-1 of its J-neighbors, which exist because $deg_J(u) = d - 1 \geq D - 1$. None is v. If u lies outside Z and has no neighbor in Z, U must avoid u and these D-1 specified neighbors. Only $N - 1 - (D - 1) = 4$ choices of individual vertices remain. Thus its failure probability is at most $binom(4, 2)/binom(N, 2)$. The expected number of failing vertices is at most

$24/binom(D + 4, 2)$,

which equals 24/120 or 24/190 and is less than one. This failure count is a nonnegative integer on a finite nonempty sample space, so some sample has no failures. It supplies a set Z of size three containing v and dominating every vertex outside it in J. Lemma 49 gives $c_{d-1}(H) \leq 3$.

For the second row choose uniformly a three-element subset U of the same N vertices and again put $Z = \{v\}$ union U. If u is adjacent to v in J, select D-2 of its other J-neighbors. A failure of two-neighbor domination outside Z requires U to avoid u and these D-2 vertices. There are $N - 1 - (D - 2) = 5$ remaining vertices, so the failure probability is at most $binom(5, 3)/binom(N, 3)$. There are at most N vertices of this type.

If u is not adjacent to v, select D-1 of its J-neighbors. A failure outside Z requires U to avoid u and include at most one of these selected neighbors. Among the remaining vertices there are four outside the selected neighbor set. The number of such samples is at most

$binom(4, 3) + (D - 1)binom(4, 2) = 6D - 2$.

There are at most four vertices of this type. By summing these probability bounds, the expected number of failures is at most

[N $binom(5, 3) + 4(6D - 2)]/binom(N, 3)$

$= (34D + 32)/binom(D + 4, 3)$.

For $D = 12$ this is $440/560 < 1$; for $D = 16$ it is $576/1140 < 1$. Finite averaging again supplies a sample with no failures. Its set Z has size four, contains v, and two-dominates every outside vertex in J. Applying Lemma 49 with $j = 2$ gives $c_{d-2}(H) \leq 4$. Both constructions cover every $d \geq D$ at the indicated order.

Now assume $d \geq 12$ and $E \leq 2$. Since $d \leq 2k$, $k \geq 6$. If $k \geq 11$, (2) gives $gap(H) < 8 - 77/10 = 3/10 < 1$. If $k = 6$ or 7, (4), $d \geq 12$ and (1) give

$gap(H) \leq 8 - 2k - 2 + (2k + 1 - d) + (2k + 3 - d)$

$= 2k + 10 - 2d \leq 2k - 14 \leq 0$.

If $k = 8$, $n = 17 = D + 5$ with $D = 12$, so the construction gives the two row bounds 3 and 4. Equation (1) gives $gap(H) \leq 8 - 16 - 2 + 7 = -3$.

If $k = 9$, the strict coefficient estimates used in (2) imply

$c_{d-1}(H) < 1 + 18 \cdot 269/1000 = 5.842$,

$c_{d-2}(H) < 1 + 18 \cdot 381/1000 = 7.858$.

The two integer costs are therefore at most 5 and 7. Equation (1) gives $gap(H) \leq 8 - 18 - 2 + 12 = 0$.

If $k = 10$, the same estimates give integer row bounds 6 and 8, since $1 + 20 \cdot 269/1000 = 6.38$ and $1 + 20 \cdot 381/1000 = 8.62$. Equation (1) again gives $gap(H) \le 0$. These cases exhaust all $k \ge 6$. Whenever the displayed bound was only strict below one, the integer gap is at most zero. This proves (i).

Finally assume $d \ge 16$ and $E \le 3$, so $k \ge 8$. If $k \ge 12$, (3) gives $gap(H) < 12 - 469 \cdot 12/500 = 93/125 < 1$. For $k = 8$ or 9, (4) and (1) give

$gap(H) \le 12 - 2k - 2 + (2k + 1 - d) + (2k + 3 - d)$

$= 2k + 14 - 2d \le 2k - 18 \le 0.$

If $k = 10$, $n = 21 = D + 5$ with $D = 16$, and the constructed row bounds 3 and 4 give $gap(H) \le 12 - 20 - 2 + 7 = -3$.

If $k = 11$, the strict coefficients used in (3) give

$1 + 22q_1 < 1 + 22(1/6 + 541/10000) < 6,$

$1 + 22q_2 < 1 + 22(1/4 + 12033/200000) < 8.$

For the first final comparison, $22(1/6 + 541/10000) = 145706/30000 < 5$; for the second, $22(1/4 + 12033/200000) = 1364726/200000 < 7$. Thus the integer row costs are at most 5 and 7, and (1) gives $gap(H) \le 12 - 22 - 2 + 12 = 0$. All ranks are covered, and integrality converts the remaining strict bound below one into $gap(H) \le 0$. This proves (ii). □

Remark. Two domination costs pay all sufficiently large orders. Complete-graph row bounds pay the first two ranks, and a fixed-cardinality marked-set count pays the intermediate rank without a general dense-graph theorem.

Depends on. Lemma 12, Lemma 23, Lemma 24, Lemma 49, Lemma 72, Lemma 87, Lemma 110.

Lemma 176 (Regular-component fitting at repeated plateau x). *All graphs are finite and simple. Let $s, t \ge 1$ be integers. Let F_1, F_2 be nonempty star forests with full nonincreasing positive integer edge-size lists $n_1,\dots, n_s$ and $m_1,\dots, m_t$ and heads $A = n_1$, $B = m_1$. Copies are ordinary subgraphs with disjoint target components; F_1 is prescribed red and F_2 blue. Put $M = s + t - 1$ and $\ell_k = max\{n_i + m_j - 1 : 1 \le i \le s, 1 \le j \le t, i + j = k\}$ for every integer $2 \le k \le s + t$. For an integer $0 \le q < M$, let A_q consist of every nonempty actual suffix pair $(F_1^{\alpha}, F_2^{\beta})$, where α, β are nonnegative integers, $\alpha + \beta = q$, $\alpha < s$, $\beta < t$, and superscripts delete leading whole components. Put $K = M - q$, $P_q = \sum_{k=q+2}^{s+t}(\ell_k - 1)$, $D = \ell_{q+2} - 1$, and let R be the initial run length of $D + 1$ in that natural tail. Write $\nu(G)$ for maximum matching cardinality, $e(G)$ for edge count, and $\Phi(G) = e(G) - \nu(G)$.*

Assume $D \ge 6$ is even and $A, B \le D - 1$. Let z be an integer with $0 \le z \le D/2 - 1$. Let H satisfy $\Phi(H) \le P_q + z$ and $\delta = K - \nu(H) \ge 1$, and contain at least R distinct whole connected odd D-regular components with near-perfect matchings, meaning matchings missing exactly one vertex.

If H arrows every member of A_q, then every whole connected odd D-regular near-perfect component C of H, with $\mu = \nu(C)$, has at least one full original target F_i with $|V(F_i)| \le 2\mu + 1$. For every such fitting original, writing c_i for its number of components, one has

$\mu \ge e(F_i) + \delta - 1,$

and C-Y contains the entire uncolored F_i for every vertex set $Y \subseteq V(C)$ with $|Y| \le \mu - c_i + \delta$.

Moreover every such C satisfies

$\mu \ge A + B - 4 + 3\delta \ge D - 2 + 3\delta.$

In particular $\delta \ge 3$ implies $\mu \ge D + 7$.

If $R = 1$ and H arrows every member, the full original edge totals are distinct. Orient them F, J with $e(F) < e(J)$, and let S, T be their component counts. For every such C, exactly F fits its order, and

$e(F) + \delta - 1 \le \mu \le e(J) - q - 1$, $T \ge \mu + \delta + 1.$

If instead $R = 2$, $4 \le A, B \le D - 1$, $\Phi(H) \le P_q + 2$ and $\delta \ge 3$, then H has a coloring avoiding both entire prescribed targets of some member of A_q.

Every avoiding member obtained by the last conclusion, or by failure of a necessary fitting conclusion above, extends over the original disjoint canonical q-star pool of edge sizes ℓ_2,..., ℓ_{q+1}, and across any specified actual q-vertex set in any outer graph whose deletion gives H, preserving all H-edge colors and avoiding both entire originals. No maximum-degree bound is imposed outside the selected regular components. All orders, degrees, counts, cuts, indices, matching numbers and allowances are integers in the stated domains. Unit components, repeated sizes and $q = 0$ are permitted.

Proof. Assume first that H arrows every member of A_q. Fix any whole odd D-regular near-perfect component C, and put $\mu = \nu(C)$. Its order is $2\mu + 1$, and simplicity gives $\mu \geq D/2 \geq 3$. Choose R-1 other such components and, in each, a near-perfect matching and its uncovered vertex. There are enough other components even if C was not among an initially selected collection of R. Let Y_0 consist of these vertices, let $p = R - 1$, $q' = q + p$, and $H' = H - Y_0$. When $R = 1$ these choices are empty.

The set Y_0 is independent, with all its vertices of degree D in H. The chosen matchings, a maximum matching of C, and maximum matchings of all remaining components form a maximum matching of H missing Y_0. Thus $\nu(H') = \nu(H)$, $e(H') = e(H) - Dp$ and $\Phi(H') = \Phi(H) - Dp$. The component C remains unchanged and whole. Since $1 \leq R \leq M - q$, $q' < M$. Deleting the first R-1 entries of the natural q-tail leaves a unique initial $D + 1$ entry. Its potential is $P_{q'} = P_q - Dp$, so $\Phi(H') \leq P_{q'} + z$.

The graph H' arrows every actual $A_{q'}$ member. Indeed, for any legal final cuts alpha', beta' of total q', put $\alpha = \min(q, \alpha')$ and $\beta = q - \alpha$. These legal cuts have total q, $\alpha \leq \alpha'$, $\beta \leq \beta'$. The additional cuts total p. If H' avoided the final pair, the actual-vertex extension of Lemma 41, applied to the pair with the intermediate cuts and the actual set Y_0, would extend it to an avoiding coloring of H for that A_q member. This contradicts common arrowing. This argument asserts no positive matching deficit after the peeling.

Suppose both original targets had more than $2\mu + 1$ vertices. Apply Lemma 19 to their full lists, capacity μ and depth q'. Both full capacity remnants are nonempty. Its unique-plateau estimate gives

$P_{q'} - P_{end} \leq (D - 1)\mu$.

The component has $e(C) = Dmu + D/2$ and $\Phi(C) = (D - 1)\mu + D/2$. Matching number and potential add over whole component unions. Consequently

$\Phi(H' - V(C)) \leq P_{end} + z - D/2 \leq P_{end} - 1$.

Potential is nonnegative since a matching is a subset of the edge set. Hence $P_{end} > 0$, so its endpoint family is nonempty. By Lemma 56 the graph H'-V(C) avoids both targets of one endpoint member. Color every edge of C in the prescribed color of that member's remnant side. The every-prefix guarantee in Lemma 19 shows that a copy of the corresponding original suffix would leave the forbidden remnant suffix in H'-V(C). All edges in the other color lie outside C, so that other original suffix is absent. The endpoint cuts are legal for the originals, because each full remnant is a componentwise subforest. We have contradicted the common $A_{q'}$ arrowing. Thus at least one full original fits the order of C.

Return to the initial H, q and δ. Adjoin the canonical q-star pool to H, obtaining G_c. By Lemma 41 it arrows the full original pair, and $\nu(G_c) = \nu(H) + q = M - \delta$. Every selected component of H remains a whole component of G_c.

We give a localization calculation used below. Let W be a union of o whole odd near-perfect components of G_c, with $w = \nu(W)$ and hence $|V(W)| = 2w + o$. Let F_i have c_i components. If $w \geq c_i - \delta$, the first matching-additive assertion of Lemma 153, applied with empty separator and this selected union, makes W arrow $(p_i K2, F_i)$, after exchanging colors when necessary, where

$p_i = w - c_i + \delta + 1 \geq 1$.

The empty separator is matching-additive because matchings decompose over whole components; no half-order hypothesis on unselected components is required. Coloring all W-edges in the matching color forces $p_i \leq w$. For any $Y \subseteq V(W)$ with $|Y| \leq p_i - 1$, color its incident edges in the matching color and all other edges in $F_i's$ color. The first color has a vertex cover of size at most $p_i - 1$ and cannot contain p_i disjoint edges. Thus W-Y contains the entire F_i. A set of exactly $p_i - 1$ vertices exists, and its use gives

$e(F_i) + c_i \leq 2w + o - (p_i - 1)$,

or $e(F_i) \leq w + o - \delta$. (1)

A fitting original for C has $c_i \le \mu$ since each positive component uses at least two vertices. Thus the localization applies with $W = C$, $w = \mu$, $o = 1$, proving both the asserted robustness and $\mu \ge e(F_i) + \delta - 1$.

Use Lemma 89 on G_c with the same positive deficit δ. If the fitting original has head A_i and its opposite has head B_i, its component count satisfies

$c_i \ge 2\delta + 2sum_{oppositesizesd} floor((d-1)/2)$

$\ge 2\delta + B_i - 2.$

This follows by dropping the nonnegative size of an attaining separator from that theorem's bound; such a separator exists there. Since $e(F_i) \ge c_i + A_i - 1$, equation (1) for C gives

$\mu \ge A_i + B_i - 4 + 3\delta = A + B - 4 + 3\delta.$

Also $A + B \ge D + 2$, since the first original diagonal dominates the natural leading one. This proves the uniform lower bounds.

For the $R = 1$ refinement put $Z = \max(e(F_1), e(F_2))$. The complete-profile density theorem Lemma 32 applies after orienting the full edge excesses over component counts. Both heads are below D, so its correction is zero:

$P_q \le (D-1)(Z-q-1) + 1.$

Nonnegative potential outside C and $z \le D/2 - 1$ give

$(D-1)\mu + D/2 \le \Phi(H) \le P_q + z$

$\le (D-1)(Z-q-1) + D/2.$

Thus $\mu \le Z - q - 1$. If a maximum-edge original of count T satisfied $\mu \ge T - \delta$, equation (1) for C would give $Z \le \mu + 1 - \delta$, contradicting $\mu \le Z - q - 1$ because $q + \delta > 0$. Hence every maximum-edge original has $T \ge \mu + \delta + 1$ and cannot fit $2\mu + 1$ vertices. At least one original does fit. The edge totals are therefore distinct, and precisely the smaller-edge original F fits. Its already proved lower bound and this upper bound are the claimed inequalities, with the unchanged full originals.

It remains to prove the two-component exit. Assume its hypotheses and suppose H arrows every A_q member. Select two of its whole odd D-regular near-perfect components, call them C_1, C_2, and put $\mu_i = \nu(C_i)$, $C = C_1$ disjoint union C_2, $\mu = \mu_1 + \mu_2$. The preceding fitting proof applies because $z = 2 \le D/2 - 1$ and $R = 2$. Each C_i fits a full original. Also $|V(C)| = 2\mu + 2$ and $\Phi(C) = (D-1)\mu + D$.

The same density theorem, now with $R = 2$, and nonnegative potential outside C give

$(D-1)\mu + D \le \Phi(H) \le P_q + 2 \le (D-1)(Z-q-1) + 4.$

Since $D \ge 6$, integrality implies

$\mu \le Z - q - 2.$ (2)

If a maximum-edge original with count T had $\mu \ge T - \delta$, equation (1) applied to C would yield $Z \le \mu + 2 - \delta \le \mu - 1$, contradicting (2). Thus every maximum-edge original has $T \ge \mu + \delta + 1 \ge \mu + 4$ and cannot fit either C_i. Since each C_i fits an original, the two edge totals differ and both components fit the same smaller-edge original F. Write S for its count, B_F for its head, and J for the other original.

The individual fitting bounds give $\mu_i \ge e(F) + \delta - 1$. Consequently

$e(F) \le \mu/2 - \delta + 1$, $S \le e(F) - B_F + 1 \le \mu/2 - B_F - \delta + 2.$

Here $B_F \ge 4$ and $\delta \ge 3$. The full-target capacity bound Lemma 89 on G_c gives

$W_J := \sum_{m \in J} (m-2)_+$

$\le 2sum_{m \in J} floor((m-1)/2) \le S - 2\delta \le S.$

Apply Lemma 4 with the full ordered pair (F, J), component capacity $\lambda = \mu$, extra vertex allowance $l = 2$, and the same original q and $R = 2$. We check every numerical premise. The last bounds give $\mu \ge e(F) + 1$, $2\mu \ge 3S$ and $\mu - S \ge 1 = l - 1$. We have $T \ge \mu + 4$, $W_J \le S$, and both heads lie in [4,D-1]. Thus that theorem gives one nonempty full binary remnant J_{bin} with every-prefix copy guarantee for the literal graph C, whose order is $2\mu + 2$ and matching number is μ. Its natural q-tail potential satisfies

$P_q - P_q^b in \le (D-1)\mu + 3$.

Delete the whole union C. The same witnesses now give

$\Phi(H - V(C)) \le P_q + 2 - [(D-1)\mu + D]$

$\le P_q^b in + 5 - D < P_q^b in$.

If the new actual q-family were exhausted, its potential would be zero by Lemma 4, contradicting nonnegative graph potential. Otherwise Lemma 56 supplies an avoiding member of that full new family on H-V(C). Color C entirely in J's prescribed color. The every-prefix remnant property makes any original J-suffix copy leave its forbidden binary suffix outside C. The other color is wholly outside C and avoids the prescribed F-suffix. These legal original cuts give an avoided A_q member, the required contradiction.

Every argument assumed common A_q arrowing only to derive its necessary conclusions or a contradiction. Failure of a necessary conclusion therefore yields an avoiding member. Both asserted original-target extensions are exactly Lemma 41 (I), (II), applied to this member with its actual legal cuts and unchanged H-edge colors. All constructions kept the full original lists. □

Remark. Peeling exposed vertices removes all but one plateau entry without changing matching number. A full original must then fit every regular component. Original-target localization turns fitting into a strong rank bound; two such components pay for one common binary remnant with a strict potential saving.

Depends on. Lemma 4, Lemma 19, Lemma 32, Lemma 41, Lemma 56, Lemma 89, Lemma 153.

Lemma 177 (Exact critical degree in actual-suffix normalization). *Use the finite integer certificate costs c_i, matching number ν, excess $ex = c_0 - \nu$, positive-level slack ps, $gap = ex - ps$, and top missing-allocation run tr of Lemma 23. Graphs are finite and simple, copies are ordinary subgraphs, and an edgeless graph has maximum degree zero.*

Let s, t be positive integers and let $n_1 \ge \cdots \ge n_s \ge 1$ and $m_1 \ge \cdots \ge m_t \ge 1$ be integer lists. Let F_1, F_2 be the respective vertex-disjoint unions of stars of these edge sizes, prescribed red and blue. Arrowing an ordered pair means that every red-blue edge coloring contains its entire red target or its entire blue target. Put $M = s + t - 1$,

$\ell_k = max\{n_i + m_j - 1 : 1 \le i \le s, 1 \le j \le t, i + j = k\}$ *for every integer* $2 \le k \le s + t$,

$L = \sum_{k=2}^{s+t} \ell_k$, *and* $P = L - M$.

For each integer $0 \le q < M$, let A_q contain every ordered actual suffix pair (F_1^α, F_2^β), where the integer cuts satisfy $0 \le \alpha < s$, $0 \le \beta < t$ and $\alpha + \beta = q$, and each superscript deletes exactly that many leading whole components. Define

$K_q = M - q$, $L_q = \sum_{k=q+2}^{s+t} \ell_k$, $P_q = L_q - K_q$,

$D_q = \ell_{q+2} - 1$, *and* $\rho_i^q = |\{k : q + 2 \le k \le s + t, \ell_k \ge i + 1\}|$ *for every integer* $i \ge 0$.

If a graph H arrows every entire member of A_q, $e(H) < L_q$ and $\Delta(H) \le D_q$, then

$\Delta(H) = D_q \ge 4$, $c_{D_q}(H) \ge \rho_{D_q}^q \ge 1$,

and no matching of H covers every vertex of degree D_q. In particular, when $D_q \le 3$, every graph H with $\Delta(H) \le D_q$ and $e(H) < L_q$ admits a coloring avoiding both entire targets of some actual member of A_q.

Consequently every graph G with $G \to (F_1, F_2)$ and $e(G) < L$ has distinct original vertices v_1,..., v_q, where $0 \le q < M$, whose successive induced deletion leaves $H = G - \{v_1, \ldots, v_q\}$ satisfying all the preceding conclusions and arrowing every A_q member. Write $G_0 = G$, $G_i = G_{i-1} - v_i$, $d_i = deg_{G_{i-1}}(v_i)$ for $1 \le i \le q$, $\delta_0 = M - \nu(G)$, $\sigma_{init} = e(G) - \nu(G) - P$, and

$\epsilon = \sum_{i=1}^{q} (d_i - \ell_{i+1})$, $j = q - \nu(G) + \nu(H)$.

One has $d_i \ge \ell_{i+1}$, ϵ, $j \ge 0$, and, on this same H,

$\delta := K_q - \nu(H) = \delta_0 - j \ge 1$,

$\sigma := e(H) - \nu(H) - P_q = \sigma_{init} - \epsilon - j,$

$e(H) = e(G) - \sum_{k=2}^{q+1} \ell_k - \epsilon < L_q.$

Putting $\sigma_i = c_i(H) - \rho_i^q$ for all integers $i \geq 0$, one has $\sigma_i \geq 0$,

$\sigma = ps(H) + \sum_{i \geq 1} \sigma_i \leq \delta - 1,$

$ex(H) = \delta + \sigma_0$, *and*

$gap(H) = L_q - e(H) + \sum_{i \geq 0} \sigma_i \geq 1.$

All sums of row quantities have finite support.

Every coloring of this H avoiding a member with cuts α, β extends to a coloring of the same original G avoiding both full original targets and retaining every H-edge color. For any partition of the deleted set X into X_R, X_B of sizes α, β, color edges within X_R and from X_R to H red, edges within X_B and from X_B to H blue, and edges between X_R and X_B arbitrarily. The deletion sequence charges every original edge meeting X exactly once, at its earliest deleted endpoint. Empty deleted sets, repeated sizes, unit components, and isolated host vertices are included. All indices, cuts, degrees, counts, matching numbers, costs, ϵ and j are nonnegative integers in their displayed ranges; δ_0, σ_{init} and σ are initially signed integers, and their asserted bounds hold under the given arrowing and strict budget hypotheses.

Proof. Fix q and H under the first assertion. Apply Lemma 166 to these original lists, to the whole legal suffix family A_q and to this same H. Its row inequalities give $c_i(H) \geq \rho_i^q$ for every i, and its gap identity gives $gap(H) \geq L_q - e(H) \geq 1$. The first tail entry is $\ell_{q+2} = D_q + 1$, so $\rho_{D_q}^q \geq 1$. In particular $c_{D_q}(H) \geq 1$.

If $0 \leq D_q \leq 2$, fact Lemma 136 applies to exactly this family and degree $\cap$. It gives $e(H) \geq L_q$, contrary to the strict hypothesis. Thus $D_q \geq 3$.

Suppose $D_q = 3$. Then $\Delta(H) \leq 3$, and Lemma 147 applied to the positive-gap H gives $gap(H) = 1$ and $tr(H) + ex(H) \leq 3$. The definition in Lemma 23 and nonnegative slack in Lemma 61 give $ex(H) = 1 + ps(H) \geq 1$. Hence $c_0(H) \geq ex(H) \geq 1$, so the nonempty missing-allocation set in Lemma 23 gives $tr(H) \geq 1$. Consequently $ex(H) \leq 2$ and $ps(H) = ex(H) - 1 \leq 1$. Now Lemma 161 applies to this subcubic graph without any connectedness or hereditary assumption and yields $c_3(H) = 0$. This contradicts $c_{D_q}(H) \geq 1$. It follows that $D_q \geq 4$.

If $\Delta(H) \leq D_q - 1$, the strong tail in Lemma 61 gives $c_{D_q}(H) = 0$, again a contradiction. The assumed upper bound therefore forces $\Delta(H) = D_q$. If some matching covered every *degree* $- D_q$ vertex, the covered-degree conclusion of that same fact would give $c_{D_q}(H) = 0$. Thus no such matching exists. These arguments prove every conclusion for the given H. When $D_q \leq 3$ they rule out common arrowing. Its negation supplies one member of A_q and one edge coloring avoiding both of its whole targets, as asserted.

Now let G arrow the original pair below L. Apply part (III) of Lemma 41. It supplies one ordered sequence of $q < M$ distinct original vertices, its actual induced remainder H, the inequalities $d_i \geq \ell_{i+1}$, the displayed nonnegative ϵ, j, common A_q arrowing, $e(H) < L_q$ and $\Delta(H) \leq D_q$. The first part of this proof applies to this very output. Thus its degree is exactly $D_q \geq 4$, its $D_q - th$ row is positive, and no matching covers its maximum-degree vertices. No replacement host is needed.

The exact potential and edge identities in Lemma 41 give $\sigma = \sigma_{init} - \epsilon - j$ and the displayed edge equality. The definition of j gives

$\delta = K_q - \nu(H) = M - q - \nu(H) = M - \nu(G) - j = \delta_0 - j.$

Apply the common-host part of Lemma 166 to this H to obtain all displayed row and gap identities. In particular it gives

$\sigma = ps(H) + \sum_{i \geq 1} \sigma_i \geq 0$

and $e(H) - L_q = \sigma - \delta < 0$.

All quantities are integers, so $\delta \geq \sigma + 1 \geq 1$ and $\sigma \leq \delta - 1$. Its finite-support assertion justifies each row sum.

For completeness check the stated extension on the actual deleted vertices. Suppose an H-coloring avoids the suffix pair with cuts α, β, and partition $X = X_R$ disjoint union X_B with those sizes; such a partition exists

because $\alpha + \beta = q = |X|$. Under the prescribed extension, every red edge with an endpoint in X has an endpoint in X_R. Indeed the only edges touching X_B and not X_R lie within X_B or between X_B and H, and they are blue. Therefore any red star component of a copied target which touches X contains a vertex of X_R. Since target components are vertex-disjoint, at most α red components of a red copy of F_1 could touch X. Such a copy would leave at least s-alpha components entirely in H.

Choose s-alpha of those retained components, and list their original profile indices as $i_1 < \cdots < i_{s-\alpha}$. For every integer $1 \le r \le s - \alpha$, one has $i_r \le \alpha + r$: otherwise the final s-alpha-r indices could not fit between $i_r + 1$ and s. Thus $n_{i_r} \ge n_{\alpha+r}$. Trimming leaves in the r-th retained star supplies the r-th star of F_1^α, with all component vertex sets still disjoint and wholly inside H. This contradicts the avoiding H-coloring.

Every blue edge touching X has an endpoint in X_B: edges within X_R or from X_R to H are red, and a crossing edge between the two sets has an X_B endpoint. Thus at most β components of a blue full copy could touch X. Ordering the indices of t-beta retained blue components gives an index bound $j_r \le \beta + r$ for $1 \le r \le t - \beta$. Their edge sizes are at least $m_{\beta+r}$; trimming leaves gives the forbidden blue suffix F_2^β in H. This contradiction proves that the extension avoids both full targets. Unit stars are handled by the same argument, since every target component has a positive number of edges.

Finally an original edge removed during the deletion sequence disappears precisely when its first endpoint in that sequence is deleted. Hence the counts d_i partition all edges meeting X, including those whose two endpoints lie in X. This proves the exact edge charging and completes the construction, also when $q = 0$ and every displayed deletion sum is empty. □

Remark. Positive gap in degree three forces a small excess and therefore a matching covering every cubic vertex. That makes the third certificate row zero, contradicting its positive actual-suffix demand. Literal normalization consequently stops at its exact critical degree, which is at least four.

Depends on. Lemma 23, Lemma 41, Lemma 61, Lemma 136, Lemma 147, Lemma 161, Lemma 166.

Lemma 178 (Residual rank under a uniform independent-cut gain). *Use the finite integer certificate costs $c_i(G)$, matching number $\nu(G)$, independence number $\alpha(G)$, maximum degree $\Delta(G)$, and slack $ps(G) = e(G) - \nu(G) - \sum_{i\ge1} c_i(G)$ of Lemma 23. All graphs are finite and simple, and vertex deletion is induced. A graph is factor-critical if deleting every vertex leaves a perfect matching.*

Let $p \ge 1$ and $d \ge 2p + 4$ be integers, with d even. Let H be a nonempty connected d-regular factor-critical graph with $ps(H) = p$. Suppose every zero-slack induced subgraph J of H with $\Delta(J) \ge 3$ has a matching covering every vertex of degree $\Delta(J)$. Suppose also that for every maximum independent set S of H,

$$\nu(H - S) - (\nu(H) + 1 - |S|) = p.$$

Then, for every maximum independent set S, its actual induced complement $G = H - S$ satisfies

$$ps(G) = 0 \text{ and } \nu(G) \le 3p + 1.$$

Moreover, for every integer i with $1 \le i \le d - 1$,

$$c_i(H) = |S| + c_i(G).$$

In particular $d = 8$ and $p = 2$ give $\nu(H - S) \le 7$ for every maximum independent set.

No hypothesis on the excess or gap of H is imposed. The covering hypothesis is required only at zero-slack induced subgraphs, and the uniform gain hypothesis quantifies over all maximum independent sets, including those chosen later from a deletion terminal. All graph orders, degrees, indices, cardinalities, costs, matching and independence numbers, and parameters p, d are integers in their displayed nonnegative or positive domains.

Proof. Use Lemma 61 for nonnegative slack, covered-degree zero rows, zero tails and the covered deletion inequality, and Lemma 72 for restriction and vertex restoration.

Write $|V(H)| = 2k + 1$ and $a = \alpha(H)$, so factor-criticality gives $\nu(H) = k$. Every maximum independent set is nonempty. The top row is $c_d(H) = 1$. A perfect matching after deleting one vertex covers all remaining degree-d vertices, so the covered-degree zero row and restoration give $c_d(H) \le 1$. If this row were zero, the capacity pair (1,d-1) and zero matching budgets would partition all H-edges into bounded-degree parts.

Since every vertex has degree d, the first part would have degree exactly one at every vertex, giving a perfect matching on odd order. This is impossible.

Fix any maximum independent set S, and put $G = H - S$ and $m = \nu(G)$. Maximality gives $\Delta(G) \le d-1$. For $1 \le i \le d-1$, put $r_i = c_i(H) - c_i(G)$. Restriction and restoration give $0 \le r_i \le a$. Independence removes exactly da edges; rows of H above d and of G at indices at least d vanish. Using $m = k+1-a+p$ from the stated gain, subtraction of the slack definitions gives

$ps(G) = p - p - \sum_{i=1}^{d-1}(a - r_i)$.

The sum and $ps(G)$ are nonnegative, so both are zero. Thus $ps(G) = 0$ and $r_i = a$ at every displayed index, proving the row identities. The order and edge counts are

$|V(G)| = 2m + a - 2p - 1$, and $e(G) = d(m - p - 1/2)$. (1)

Starting from any zero-slack induced subgraph of H, repeatedly delete any current maximum-degree vertex while that degree is at least three. The stated covering hypothesis and the deletion inequality give

$0 \ge ps(J - v) + 1 - (\nu(J) - \nu(J - v)) \ge 0$

at each stage. Thus every child has zero slack and its matching number decreases by one. The process terminates at maximum degree at most two and is covered at every parent. This construction permits any choices among current maximum-degree vertices, including any prescribed eligible initial choices.

Apply Lemma 146 to G with $\cap D = d-1$ and one such process. Let $f = |V(G)| - 2m = a - 2p - 1$. Since $\alpha(G) \le a$, its refined density bound gives, after dropping a nonnegative refinement,

$d(m - p - 1/2) = e(G) \le 3m + (d-4)(\alpha(G) - f)$

$\le 3m + (d-4)(2p+1)$.

Consequently

$m \le 3p + 3/2 + (2p+1)/(2(d-3)) \le 3p + 2$, (2)

because $d - 3 \ge 2p + 1$. If $d > 2p + 4$, the last inequality is strict and integrality gives $m \le 3p + 1$.

It remains to exclude $d = 2p + 4$ and $m = 3p + 2$. Suppose these equalities hold. Every inequality in the density application is then equality. Since $d - 4 > 0$, $\alpha(G) = a$. The nonnegative integer $m - c_1(G)$ in the refined bound must be zero as well, because its coefficient D-1 is positive.

The equality description of Lemma 146 applies to every complete covered process on G. It gives an independent deleted set T of cardinality

$t = 2m - |V(G)| + a = 2p + 1 = D - 2$,

all of whose vertices have original and deletion-time degree D. The terminal R has exactly

$|V(G)| - m - a = p + 1$

triangle components and

$2a - |V(G)| + m = a - p - 1$

isolated vertices. Its independence number is a.

Choose one vertex from every terminal triangle and all terminal isolated vertices, and call the resulting set S′. Since R is induced in H, this is an independent set of H of size a and therefore is maximum. Put $G' = H - S'$. The hypothesis quantified over every maximum independent set gives $\nu(G') = m$, and the initial slack subtraction gives $ps(G') = 0$. Its order and edge count agree with (1), since S′ is independent and has size a. It also has $\Delta(G') \le D$. Repeating (2) and the equality argument gives $\alpha(G') = a$, $c_1(G') = m$, and equality in the same density bound. Therefore every complete covered process on G′ deletes an independent set of exactly $t = D - 2$ vertices, each of original and deletion-time degree D, and ends at maximum degree at most two.

There are $p + 1 \ge 2$ triangles in R. From each of two different triangles take the two vertices not selected in S′. Name the resulting two edges xy and zw. Among these four vertices there are no other edges, since different components of R have no mutual edges and R is induced. Every one of the four vertices has exactly

one neighbor in S′, namely the selected vertex of its own triangle. Their original H-degrees are d, so their G′-degrees are $D = d - 1$.

Start a maximum-degree deletion process on G′ by deleting x and then z. The two chosen vertices are nonadjacent, so z still has degree D after the first deletion and is eligible. Complete the process using the zero-slack coverage construction above. Let T′ be its entire deleted set and B′ its terminal. Density equality requires $|T'| = D - 2$ and every deletion-time degree to be D. The vertex y is adjacent to x, so after the first step its degree is at most D-1 and it can never be deleted in this process. Thus y belongs to B′. It is not adjacent to z, since the four vertices induce only xy and zw. At most $|T'| - 1 = D - 3$ neighbors of y were deleted. Its original G′-degree was D, so its terminal degree is at least three. This contradicts $\Delta(B') \leq 2$.

The endpoint in (2) is impossible. Hence $m \leq 3p + 1$ for the originally arbitrary maximum independent set S. The same proof applies to every such S; zero slack and all row identities were already proved for each separately. Substituting $p = 2$ gives the stated bound seven in degree eight. □

Remark. Density leaves one integer endpoint at $d = 2p + 4$. Its terminal provides another maximum independent set, whose complement contains two disjoint induced edges at maximum degree. Starting the equality process with the opposite endpoints forces a surviving endpoint to retain three neighbors, contradicting the terminal degree bound.

Depends on. Lemma 23, Lemma 61, Lemma 72, Lemma 146.

Lemma 179 (Boundary credits for factor-critical components). *Use the integer certificate rows $c_i(K)$, matching number $\nu(K)$, excess $ex(K) = c_0(K) - \nu(K)$, slack $ps(K) = e(K) - \nu(K) - \sum_{i\geq 1} c_i(K)$, and $gap(K) = ex(K) - ps(K)$ of Lemma 23. All graphs are finite and simple. A nonempty graph is factor-critical if deleting every one of its vertices leaves a perfect matching.*

Let $D \geq 4$ be an integer and let K be a connected factor-critical graph with $\Delta(K) \leq D$ and total degree defect $d = \sum_(v \in V(K))(D - deg_K(v))$ satisfying $0 < d < D$. Put $r = \nu(K)$, $u = (D - d)/2$, $T = floor((D - 1)/2)$, and $E = ex(K)$. Then r, u are positive integers, D and d have the same parity, $|V(K)| = 2r + 1 > D$, $\Delta(K) = D$, and $e(K) = Dr + u$. A matching covers all degree-D vertices,

$ps(K) \geq 1$, $c_i(K) = 0$ for $i \geq D$, $c_i(K) \leq r$ for $i > T$,

and $c_i(K) + c_{D-i}(K) \leq 2r + 1$ for $1 \leq i < D$.

The finite capped sum

$p_{cap}(K) = e(K) - r - \sum_{i\geq 1} \min(c_i(K), r)$

satisfies $p_{cap}(K) \geq D - d$. For $D \geq 6$ one also has $r - c_{D-1}(K) \geq u$.

Suppose in addition that $gap(K) \leq 1$. Define $\psi = ps(K)$, $h_i = \max(c_i(K) - r, 0)$ for $1 \leq i < D$, and $\delta = \max(D - d - \psi, 0)$. Define the integers a_i for $1 \leq i < D$ by

$a_1 = \min(\delta, h_1)$;

$a_i = 1$ *if* $2 \leq i \leq T$ *and* $i \leq 1 + \delta - a_1$;

$a_i = 0$ *at every other index* $2 \leq i < D$.

Then

$0 \leq a_i \leq h_i$, $\sum_{i=1}^{D-1} a_i = \delta$, $a_1 \leq u$,

and $a_i \leq 1$ for $2 \leq i < D$.

Every index i with $a_i > 0$ satisfies $E \leq D - d - i + 1$.

Every index j with $2 \leq j \leq T$ and $a_j > 0$ satisfies

$0 \leq r - c_{D-1}(K) \leq u + T - j$.

Moreover $\psi \geq \max(1, ceil((D - d - T)/2))$, and $\mu_i = c_i(K) - a_i$ are nonnegative integers with $\mu_i \geq \min(c_i(K), r)$. These statements include $\delta = 0$ and empty index intervals; all degrees, defects, orders, costs, credits, ranks and indices have the integer domains displayed above.

Proof. The quantities and their finite integer domains are those of Lemma 23. Apply Lemma 135 to the supplied graph K and integer degree $\cap\, D$. Its hypotheses are met: K is nonempty and factor-critical, $\Delta(K) \le D$, and the same total defect satisfies $0 < d < D$. That theorem gives every asserted order, parity, matching-coverage, edge-count, positive-slack, zero-tail, upper-half and paired-row statement. It also gives

$p_{cap}(K) = u + \sum_{i=1}^{D-1}(r - \min(c_i(K), r))$. (1)

We discharge each extra condition of its credit conclusion on this same graph.

If $D = 4$, apply Lemma 131 to K. Connectedness and factor-criticality are among the present hypotheses, its maximum degree is at most four, and the present d is precisely the positive defect in that theorem. Its final assertion gives $p_{cap}(K) \ge 4 - d = D - d$.

If $D = 5$, apply Lemma 152 to the same K. The connected factor-critical and maximum-degree hypotheses agree, and its defect is again exactly d. It gives $p_{cap}(K) \ge 5 - d = D - d$. Both low-degree payments concern the same universal integer costs used here.

Suppose $D \ge 6$. If D is even, or if $d \ge 3$, the explicitly scoped conclusion of Lemma 135 gives $r - c_{D-1}(K) \ge u$ and $p_{cap}(K) \ge D - d$. The only remaining possibility is odd $D \ge 7$ with $d = 1$, because d has the parity of D. Apply Lemma 167 to K with $n = |V(K)| = 2r + 1$. Its graph has odd order, maximum degree at most this odd D, and total defect one. Its factor-critical specialization gives

$c_{D-1}(K) \le r - (D-1)/2 = r - u$.

This establishes the tail saving in the remaining case. In (1) every summand is nonnegative, and the one at D-1 is at least u, since $c_{D-1}(K) \le r - u$. Hence $p_{cap}(K) \ge u + u = D - d$ there as well.

All degree cases have now been covered. Under the additional hypothesis $gap(K) \le 1$, the conditional credit conclusion of Lemma 135 applies without any further unproved premise: at $D = 4, 5$ its required capped inequality has just been proved, and at odd $D \ge 7$, $d = 1$ its required tail inequality has just been proved. In the remaining cases it imposes no extra condition. Its definitions of ψ, h_i, δ, a_i and μ_i are exactly the displayed definitions, with the same r, E, D, d, T. Therefore it gives their support, exact total, coordinate bounds, active-index excess bound, active tail bound, slack lower bound, and reduced-cost inequalities. Its integer zero-credit and empty-interval cases include those stated here. No assertion about excess-four goodness is needed. □

Depends on. Lemma 23, Lemma 131, Lemma 135, Lemma 152, Lemma 167.

Lemma 180 (Four odd degree-five components at a low vertex). *Let $s, t \ge 1$ be integers and let F_1, F_2 be the star forests with full positive nonincreasing integer edge-size lists $n_1,\ldots, n_s$ and $m_1,\ldots, m_t$, prescribed red and blue. Copies are ordinary subgraphs with vertex-disjoint target components. Put $M = s + t - 1$ and $\ell_k = max\{n_i + m_j - 1 : 1 \le i \le s, 1 \le j \le t, i + j = k\}$ for every integer $2 \le k \le s + t$. Fix an integer $0 \le q < \min(s, t)$, put $K = M - q$, $L_q = \sum_{k=q+2}^{s+t} \ell_k$ and $P_q = L_q - K$, and suppose $\ell_{q+2} = 6$ and every later natural entry is at most five. Let A_q be the entire family of nonempty actual suffix pairs after α red and β blue leading cuts with $\alpha + \beta = q$.*

Assume both full original heads belong to [3,5]. Let H be finite simple, write ν for maximum matching cardinality and $\Phi(H) = e(H) - \nu(H)$, and suppose $\Phi(H) \le P_q + 2$, $\delta = K - \nu(H) \ge 3$. Suppose an actual vertex y has degree at most four, and H-y has four distinct odd connected components $C_1,\ldots, C_4$, each adjacent to y, wholly of ambient H-degree five, and factor-critical. Factor-critical means that deleting any one vertex leaves a perfect matching.

Then H has a coloring avoiding both entire targets of some member of A_q. Every such coloring extends over the canonical q-star pool of capacities $\ell_2,\ldots, \ell_{q+1}$, and across any specified actual outer q-set X with $H = G - X$, to avoidance of the unchanged original targets while preserving every H-edge color. No degree condition outside the four pieces and y, equimatchability, or strict-cut hypothesis is imposed. All units, repeated sizes and full actual suffixes are retained; the assertion is invariant under exchanging full target roles and colors. All sizes, counts, indices, cuts, degrees, matching numbers and deficits are integers in their stated domains.

Proof. Suppose H arrows every member of A_q. Order the four pieces by their half-orders $\mu_i = (|V(C_i)| - 1)/2$ and put $b_i = e_H(V(C_i), \{y\})$. For $v = 2\mu_i + 1$, simplicity and the ambient degree sum give $b_i \ge v(6 - v)$. If $v \le 5$, this is at least five because $v(6 - v) - 5 = (v - 1)(5 - v) \ge 0$, contradicting $b_i \le 4$. Thus $\mu_i \ge 3$, and factor-criticality gives $\nu(C_i) = \mu_i$. The equation $5(2\mu_i + 1) = 2e(C_i) + b_i$ makes every b_i odd. Four positive

boundaries have sum at most four, so $b_i = 1$ for all i, $d_H(y) = 4$. (1) Put $\mu = \mu_1 + \mu_2$, $\lambda = \mu_1 + \mu_2 + \mu_3$, $\rho = \sum_i \mu_i$, $C = C_1$ disjoint union C_2 and $n = 2\mu + 2$. Then $\mu \geq 6$, $\lambda \geq 3\mu/2$, $\rho \geq 2\mu$ and $\rho \geq \lambda + 3$.

The deficit gives $e(H) \leq P_q + 2 + K - \delta \leq L_q - 1$. By Lemma 41, the canonical q-star completion G_c arrows the full originals, with $\nu(G_c) = M - \delta$. Every matching of H has at most one edge at y and at most $\nu(Z)$ edges internal to each component Z of H-y. This bound is attained by matching y to its neighbor in C_1, using a perfect matching in C_1 after removing that neighbor, and using maximum matchings in every other component. Let Y consist of y and the q canonical star centers. The isolated pool leaves contribute zero to the component sum, giving $\nu(G_c) = |Y| + \sum_Z \nu(Z)$. (2)

Let C′ be any union of selected pieces with matching number θ and o pieces. Its order is $2\theta + o$. If an entire original target F has S components and $\theta \geq S - \delta$, the first clause of Lemma 153 applied to (2), with colors exchanged if necessary, gives C′ → (kK2,F), where $k = \theta - S + \delta + 1 \geq 1$. The all-red coloring forces $k \leq \theta$. Choose k-1 vertices of C′ and color incident edges red and all others blue. The red matching number is at most k-1, so the entire blue F lies outside those vertices. Its vertex count gives $e(F) + S \leq 2\theta + o - (k-1)$, or $e(F) \leq \theta + o - \delta$. (3) No half-order condition on unselected components is needed.

Put $Z_0 = \max(e(F_1), e(F_2))$. The density theorem Lemma 42 applies to the full originals: their heads are at most five, $q < \min(s,t)$, the entire family is arrowed, $e(H) \leq L_q - 1$, and $\nu(H) < K$. Thus $P_q \leq 4(Z_0 - q - 1) + 1$. The graph induced by y and all four pieces has $5\rho + 12$ edges and matching number $\rho + 1$, by (1) and the construction above. Its potential is $4\rho + 11$. Potential cannot decrease on restoring a vertex: its increase is d-h with h in {0,1}, and $h = 0$ if $d = 0$. Consequently $\Phi(H) \geq 4\rho + 11 \geq 4n + 3$. Combining bounds gives $Z_0 \geq \rho + q + 3$. (4)

We first exclude an original target F with $S \leq \mu$ components. Let J be the opposite original, with T components. Applying (3) to the first two pieces gives $e(F) \leq \mu + 2 - \delta \leq \mu - 1$. Thus $Z_0 = e(J)$. If $T \leq \rho + \delta$, (3) on all four pieces would give $e(J) \leq \rho + 4 - \delta \leq \rho + 1$, contrary to (4). Therefore $T \geq \rho + \delta + 1 \geq \rho + 4$. For the full list of J put $W = \sum_j (m_j - 2)_+$. The capacity theorem Lemma 89 on G_c gives $W \leq 2sum_j\ floor((m_j - 1)/2) \leq S - 2\delta \leq S$. (5) Hence $\rho \geq \max(5, e(F) + 1, S + W)$, since $\rho \geq 2\mu$. If the head of F is three, then $2\rho - 3S \geq \mu \geq 6 \geq 5$. All hypotheses of the $D = 5$, $p = 4$ boundary clause of Lemma 100 hold, with matching $\cap\ \rho$ and removed union of all four actual pieces. Its surplus allowance two gives an avoiding A_q member, a contradiction. We have proved $s > \mu$ and $t > \mu$. (6)

Use capacities $d_j = \min(5, \lfloor n/j \rfloor - 1)$, $1 \leq j \leq \mu$, and let U_1, U_2 be the actual full-target residuals obtained by successive largest-fitting deletions. Each operation removes at most one component, so (6) ensures both residuals survive. For $z \geq 0$ let Ψ_z be the complete envelope potential of all nonempty actual total-z suffix pairs of $\{(U_1, F_2), (F_1, U_2)\}$, zero when the family is empty.

Write N_x, T_z for the full original positive threshold counts and N'_x, T'_z for the residual counts. Put $a_R = s - |U_1|$, $a_B = t - |U_2|$, where bars count components; both are at most μ. For $j \geq 2$ put $k_j = floor(n/(j+1)) \leq \mu$. A component of size at least j can be removed only in one of the first k_j capacity positions, so $N_j - N'_j \leq k_j$, $T_j - T'_j \leq k_j$. (7) At height $h \geq 1$ let $V_h = max\{N_x + T_z - 1 : x + z = h + 2, N_x T_z > 0\}$, with empty maximum zero. A threshold rectangle supplies all diagonal sums from 2 through $N_x + T_z$. Conversely every size sum at least $h + 2$ dominates positive thresholds summing to $h + 2$. The full excess profile therefore has exactly V_h entries at least h; its natural width is $W_h = (V_h - q)_+$.

The same formula holds for all actual endpoint suffixes. A candidate at original indices $i + j = q + k$ permits cuts $\alpha = \max(0, q - j + 1)$, $\beta = q - \alpha$, which satisfy $\alpha \leq i - 1$, $\beta \leq j - 1$; conversely a suffix candidate has that original index sum. Thus the whole-family envelope deletes the first q diagonal positions even when endpoint lengths differ. Call its width W'_h and put $\Delta_h = W_h - W'_h \geq 0$. Containment of residual lists in full lists proves nonnegativity. We have $P_q - \Psi_q = \sum_{h=1}^{5} \Delta_h$, $W_5 = 1$. (8)

For a maximizing rectangle at a positive width, a unit threshold survives in its corresponding endpoint and loses at most μ. A surviving mixed residual threshold gives loss at most its $k_j \leq \mu$. Thus $\Delta_h > \mu$ requires both mixed thresholds to disappear, and then $\Delta_h \leq k_x + k_z - 1$. (9) We also use this observation: if $T_z \leq q + 1$, the rectangle loses at most k_x. When $N'_x > 0$ use the first endpoint; when $N'_x = 0$ its old natural width is at most $N_x \leq k_x$. If $N_x \leq q + 1$ exchange the two roles.

If a full head is five, orient it blue. The top rectangle (2,5) is present and has width at most $q + 1$, so $N_2 \leq q + 1$. A disappearing mixed rectangle then has natural width at most $T_z \leq k_z$. All other cases were already bounded by μ, so $\Delta_h \leq \mu$ for $h < 5$.

Otherwise the full heads are (4,4) or (4,3) in some order, because their sum is at least seven. If $\Delta_2 > \mu$ its pair is (2,2), with $N_2' = T_2' = 0$. Both heads are at least three, so N_3, $T_3 \geq 1$. The endpoint unit-axis bounds $W_2' \geq s - a_R + T_3 - 1 - q$, $W_2' \geq N_3 + t - a_B - 1 - q$ and subtraction from $W_2 = N_2 + T_2 - 1 - q$ give s, $t \leq 2k_2$. Every nonunit threshold is at most k_2. Each axis width is at most $3k_2 - 1 \leq n - 1$ and each mixed width at most $2k_2 - 1 \leq n - 1$. Hence $P_q \leq 4(n-1) + 1 = 4n - 3$, contradicting $\Phi(H) \geq 4n + 3$ and $\Phi(H) \leq P_q + 2$. Thus $\Delta_2 \leq \mu$; height one has only axis pairs and $\Delta_1 \leq \mu$.

For heads (4,4), if $\Delta_3 > \mu$, orient its pair as (2,3), so $N_2' = T_3' = 0$. The two unit-axis comparisons with positive N_4, T_4 give s, $t \leq k_2 + k_3$. All N_j for $j \geq 2$ are at most k_j, and all T_j for $j \geq 3$ at most k_j. Heights one and two have widths at most $2(k_2 + k_3) - 1 \leq 7n/6 - 1$. At heights three and four an axis has width at most $k_2 + 2k_3 - 1$; a mixed pair with blue threshold at least three is at most $k_2 + k_3 - 1$; with blue threshold two its red threshold is at least three and its width at most $k_3 + t - 1$. All are at most $5n/6 - 1$. Thus $P_q \leq 2(7n/6 - 1) + 2(5n/6 - 1) + 1 = 4n - 3$, the same contradiction. At height four, the top rectangles (3,4), (4,3) give T_4, $N_4 \leq q + 1$. The extreme pairs (2,4), (4,2) lose at most k_2 by the observation after (9). Pair (3,3) loses at most k_3 if a threshold survives and at most $2k_3 - 1 \leq \mu$ otherwise. This proves all required bounds for heads (4,4).

Orient the remaining heads (4,3) red and blue respectively. The top rectangle (4,3) gives N_4, $T_3 \leq q + 1$. At height four, pairs (3,3), (4,2) lose at most k_3, k_2. At height three pair (2,3) loses at most k_2 and (4,1) is an axis. The only possible $\Delta_3 > \mu$ therefore has pair (3,2), with $N_3' = T_2' = 0$. The second endpoint has width at least $N_4 + t - a_B - 1 - q$. Using $\Delta_3 \geq \mu + 1$ and $a_B \leq \mu$ gives $t \leq N_3 + T_2 - N_4 - 1 \leq k_3 + k_2 - 2$. (10)

Now call the full blue head-three target F, with $S = t$ components, and the full red head-four target J, with $T = s$ components. We have $S \leq k_3 + k_2 - 2 \leq 7n/12 - 2 \leq \lambda$. Equation (3) on the first three pieces gives $e(F) \leq \lambda + 3 - \delta \leq \lambda$. Since $\rho \geq \lambda + 3$, (4) implies $Z_0 = e(J)$. If $T \leq \rho + \delta$, (3) on all pieces contradicts (4) as before. Hence $T \geq \rho + \delta + 1 \geq \rho + 4$.

For this full J its above-two excess is $W = N_3 + N_4$. By (10), $S + W \leq 2N_3 + T_2 - 1 \leq 2k_3 + k_2 - 1 \leq 5n/6 - 1 = (5\mu + 2)/3 \leq 2\mu \leq \rho$. Also $2\rho - 3S \geq 4\mu - 3(k_3 + k_2 - 2) \geq 4\mu - 7n/4 + 6 = (\mu + 5)/2 \geq 5$. Equation (5), with these oriented full targets, still gives $W \leq S$. Finally $\rho \geq 5$ and $\rho \geq e(F) + 1$. The $D = 5$, $p = 4$ boundary clause of Lemma 100 applies to all four pieces and head-three F, including its extra condition $2\rho \geq 3S + 5$. It gives avoidance, a contradiction. Thus $\Delta_3 > \mu$ is impossible here as well. Combining every case with (8) yields $P_q - \Psi_q \leq 4\mu + 1$. (11)

Put $Q = H - (V(C) union \{y\})$. Equation (1) gives $e(C) = 5\mu + 4$; deleting C and y removes $5\mu + 8$ edges and loses at most $\mu + 1$ matching edges. Therefore $\Phi(Q) \leq P_q - 4\mu - 5$. (12) By Lemma 28 the capacity deletions commute. Their first capacity five deletes a full original head, so every i-th residual size is at most its original $(i + 1) - st$ size. Each available first potential entry of the endpoint total-q family is at most $\ell_{q+3} - 1 \leq 4$. Later entries are no greater by the monotonicity obtained by lowering one noninitial candidate index; expired entries are zero. The full suffix-envelope identity and (11) give $\Psi_{q+2} \geq \Psi_q - 8 \geq P_q - 4\mu - 9$. (13)

Choose w in C_3 not adjacent to y; its at least seven vertices and single boundary edge ensure existence. It has degree five in Q. Factor-criticality gives a maximum matching of C_3 exposing w, and C_3 is a whole component of Q, so $\nu(Q - w) = \nu(Q)$. Consequently $\Phi(Q - w) = \Phi(Q) - 5 \leq P_q - 4\mu - 10 < \Psi_{q+2}$. Graph potential is nonnegative, so this endpoint family is nonempty. By Lemma 56, Q-w has a coloring avoiding one entire member of it. Decrease a positive cut and use Lemma 41 to restore this same w. This gives avoidance on Q of that endpoint's $total - (q + 1)$ suffix pair.

Every disjoint star collection in C has at most μ components, by taking one edge per star, and its ordered sizes c_j satisfy $c_j \leq \lfloor n/j \rfloor - 1$ and $c_j \leq 5$. Assign its j-th star to a fresh pool star of capacity d_j. Thus every full target copy divided between C and Q yields a full copy in that pool disjoint union Q, preserving its actual Q-part. The exact residual containment of Lemma 28 leaves its residual in Q. Leading cuts are capacity-five operations and commute with the pool, so the same assertion holds after the chosen cuts. Color C in the originating residual's color. Since H-y has no C–Q edges, this avoids the original $total - (q + 1)$ pair on H-y. Decrease a positive remaining cut and restore exactly y by Lemma 41. The result avoids an entire A_q member, contradicting our assumption.

The same fact gives both asserted original-target extensions, including $q = 0$, while retaining all H-colors. Every copy and cut used complete positive-edge components, including repeated and unit sizes. □

Depends on. Lemma 28, Lemma 41, Lemma 42, Lemma 56, Lemma 89, Lemma 100, Lemma 153.

Lemma 181 (Avoidance from many odd components at a low vertex). *Let $s, t \geq 1$ be integers, and let F_1, F_2 be the star forests with full positive nonincreasing integer edge-size lists $n_1,\ldots, n_s$ and $m_1,\ldots, m_t$, prescribed red and blue. Copies are ordinary subgraphs with vertex-disjoint target components. Put $M = s + t - 1$, $\ell_k = max\{n_i + m_j - 1 : 1 \leq i \leq s, 1 \leq j \leq t, i + j = k\}$ for $2 \leq k \leq s + t$, $L_q = \sum_{k=q+2}^{s+t} \ell_k$, and $P_q = L_q - (M - q)$. For $0 \leq q < M$ let A_q contain every ordered nonempty actual suffix pair after α red and β blue leading cuts, with $\alpha + \beta = q$.*

Fix an integer $0 \leq q < \min(s, t)$, put $K = M - q$ and $D = \ell_{q+2} - 1 \geq 7$, and suppose the natural q-tail has exactly one initial entry $D + 1$. Let $p \geq 4$ and $0 \leq r \leq p - 2$ be integers. Assume either both full original heads belong to [4,D], or $p \geq 6$ and both belong to [3,D]. Let H be finite simple, write ν for maximum matching cardinality and $\Phi(H) = e(H) - \nu(H)$, and suppose $\Phi(H) \leq P_q + r$, $\delta = K - \nu(H) \geq p - 1$. Suppose an actual vertex y has $d_H(y) \leq D - 1$, and H-y has at least p distinct odd connected components, each adjacent to y, wholly of ambient H-degree D, and factor-critical. Here factor-critical means that deleting any vertex leaves a perfect matching.

Then H has a coloring avoiding both entire targets of some member of A_q. In particular, for full heads in [4,D], the conclusion holds for any integer $r \geq 0$ under $\delta \geq \max(3, r + 1)$ and at least $\max(4, r + 2)$ such components. Every asserted coloring extends over the canonical q-star pool with capacities $\ell_2,\ldots, \ell_{q+1}$, and across any specified actual q-set X with $H = G - X$, to avoidance of the unchanged original targets while preserving every H-edge color. There is no strict-cut, equimatchability, or degree condition outside the displayed components and y. All original units, repeated sizes and full legal suffixes are retained. The result is invariant under simultaneously exchanging the full target roles and colors. Every size, count, cut, degree, matching number, deficit and surplus is an integer in the stated domain.

Proof. Select p of the required components and order them as $C_1,\ldots, C_p$ with nondecreasing half-orders $\mu_i = (|V(C_i)| - 1)/2$. Write $b_i = e_H(V(C_i), \{y\})$. Every $b_i \geq 1$ and $\sum_i b_i \leq d_H(y) \leq D - 1$, so $p \leq D - 1$ and $r \leq D - 3$. For $v = 2\mu_i + 1$, simplicity and the ambient degree sum give $b_i \geq v(D + 1 - v)$. If $v \leq D$ this contradicts $b_i < D$, because $v(D + 1 - v) - D = (v - 1)(D - v) \geq 0$. Thus $\mu_i \geq \lceil D/2 \rceil \geq 4$. Factor-criticality gives $\nu(C_i) = \mu_i$.

Put $C = C_1$ disjoint union C_2, $\mu = \mu_1 + \mu_2$, $\lambda = \mu_1 + \mu_2 + \mu_3$, $\rho = \sum_i \mu_i$ and $b_{all} = \sum_i b_i$. Then $\mu \geq \max(D, 8)$, $|V(C)| = 2\mu + 2$ and $\nu(C) = \mu$. Also $\lambda \geq 3\mu/2$, $\rho \geq p\,\mu/2 \geq 2\mu$, $\rho \geq p\,\lambda/3$, and $\rho \geq 4p$. These follow because every later half-order is at least the mean of an initial segment. In particular $p \geq 6$ implies $\rho \geq 2\lambda$. The actual boundary payment is $d_H(y) - (b_1 + b_2)/2 \geq (b_1 + b_2)/2 + \sum_{i=3}^{p} b_i \geq p - 1 \geq r + 1$. (1)

Assume for contradiction that H arrows every A_q member. Since $\delta \geq p - 1 \geq r + 1$, $e(H) \leq P_q + r + K - \delta \leq L_q - 1$. (2) By Lemma 41 its completion G_c by the disjoint canonical q-star pool arrows the full originals, with $\nu(G_c) = \nu(H) + q = M - \delta$.

We claim that a full target F can be chosen with component count $S \leq \theta$, where $\theta = \mu$ if both full heads are at least four, and $\theta = \lambda$ otherwise.

In the first case apply the capacity list $d_h = \min(D, floor((2\mu + 2)/h) - 1)$, $1 \leq h \leq \mu$, to each full original and call its residuals U_1, U_2. If both survive, Lemma 114 applies to $C_1,\ldots, C_4$. With both heads below D use its first regime, including $D = 7$; with a head D use its second. The bound $r \leq D - 3$ and (1) give avoidance, a contradiction. Hence one residual is empty, and its original has at most μ components.

In the second case $p \geq 6$. Suppose both full component counts exceeded λ. The same capacity list leaves both residuals nonempty since it removes at most μ components. The second alternative of regime (II) in Lemma 114 applies: $D \geq 7$, full heads are at least three, $\mu \geq 8$, and both counts exceed $\lambda \geq 3\mu/2$. Equation (1) again yields avoidance, a contradiction. Thus some full original has at most λ components, proving the claim.

Orient this target red and call it F, with S components and head B. Call the other full original J, with T components and head A. Put $o = 2$ when $\theta = \mu$ and $o = 3$ when $\theta = \lambda$. We have $S \leq \theta$, $\rho \geq 2\theta$, $3 \leq B$, $A \leq D$, $q < \min(S, T)$, $p \geq o + 2$. (3)

Every matching of H has at most one edge incident with y and at most $\nu(Z)$ internal edges in each component Z of H-y. This upper bound is attained: choose a neighbor z of y in C_1, match yz, use a perfect matching in $C_1 - z$, and use maximum matchings in all other components. Therefore $\nu(H) = 1 + \sum_Z \nu(Z)$. (4) Let Y consist of y and the q canonical pool centers in G_c. The components of $G_c - Y$ are those of H-y and isolated pool leaves. Equation (4) yields $\nu(G_c) = |Y| + \sum_Z \nu(Z)$. (5)

Let C′ be a union of selected pieces with matching number mu′ and o′ pieces. Its order is $2\mu' + o'$. For either entire original target T_0 with t_0 components and E_0 edges, if $\mu' \geq t_0 - \delta$, the first, matching-additive

clause of Lemma 153 applied to (5), with colors exchanged if necessary, gives $C' \to (kK2, T_0)$, where $k = \mu' - t_0 + \delta + 1 \geq 1$. The all-red coloring forces $k \leq \mu'$. Choose k-1 vertices of C′ and color every edge incident with them red and all other edges blue. The red graph has no k-edge matching, so an entire blue T_0 lies outside those vertices. Counting its $E_0 + t_0$ vertices gives $E_0 + t_0 \leq 2\mu' + o' - (k-1)$, and hence $E_0 \leq \mu' + o' - \delta$. (6) Only the selected pieces need their known half-order matching sizes; no half-order condition on the unselected components or their separator has been used. Apply (6) to the entire F and the first o pieces, which have matching number θ. Equation (3) gives $e(F) \leq \theta + o - \delta \leq \theta - 1$. (7)

Put $Z_0 = \max(e(F), e(J))$. The density conclusion of Lemma 42 applies to these same complete originals and their entire q-family: both heads are at most D, $q < \min(S, T)$, (2) holds, and $\nu(H) = K - \delta < K$. Hence $P_q \leq (D-1)(Z_0 - q - 1) + 1$. (8) The induced graph on y and all p selected pieces has D $\rho + Dp/2 + b_{all}/2$ edges and exactly $\rho + 1$ matching edges, by the construction used for (4). Its potential is therefore $(D-1)\rho + Dp/2 - 1 + b_{all}/2$. Restoring any vertex cannot decrease potential: its change is d-h where $0 \leq h \leq 1$, and $h = 0$ if $d = 0$. This induced potential is at most $\Phi(H)$. Combining with (8) gives $(D-1)(Z_0 - q - 1 - \rho) \geq Dp/2 - 2 + b_{all}/2 - r \geq p(D-1)/2$, where $b_{all} \geq p$ and $r \leq p - 2$. Thus $Z_0 \geq \rho + q + 1 + \lceil p/2 \rceil \geq \rho + q + 3$. (9) By (7) and $\rho \geq 2\theta$, this maximum is $e(J)$.

If $T \leq \rho + \delta$, apply (6) to all p selected pieces and the full original J. It gives $e(J) \leq \rho + p - \delta \leq \rho + 1$, contrary to (9). Therefore $T \geq \rho + \delta + 1 \geq \rho + p$. (10) Put $W = \sum_{j=1}^{T} (m_j - 2)_+$ for the complete size list of J. Since G_c arrows the originals and has matching number $M - \delta < M$, the full-target capacity bound Lemma 89 gives $S \geq 2 \sum_{j=1}^{T} floor((m_j - 1)/2) + 2\delta$. For each positive integer m, $(m-2)_+ \leq 2floor((m-1)/2)$. Consequently $W \leq S - 2\delta \leq S - 6$. (11)

Equations (3), (7), (11), $\mu \geq D$ and $\rho \geq 2\theta$ imply $\rho \geq \max(D, e(F) + 1, S + W)$, $W \leq S$. Equation (10) gives $T \geq \rho + 2$. If $B = 3$, we are in the second head regime, and $S \leq \lambda \leq \rho/2$ together with $\rho \geq 4p$ gives $2\rho - 3S \geq \rho/2 \geq 2p \geq 2p - 3$. This checks the additional head-three inequality $2\rho \geq 3S + 2p - 3$. Apply Lemma 100 with the full oriented originals F, J, matching capacity ρ, vertex excess p, and actual union of all p selected pieces behind the same y. This union has $2\rho + p$ vertices, matching number ρ and ambient degree D at every vertex. Its conditions have all been checked: $D \geq 7$, $3 \leq B \leq D$, $2 \leq A \leq D$, natural plateau one, short prefix, the displayed matching and edge bounds, and the extra inequality when $B = 3$. Its boundary conclusion for $p \geq 4$ and surplus $r \leq D - 3$ supplies an avoiding coloring of an entire A_q member. This contradicts simultaneous arrowing.

Thus H avoids some entire A_q member. For the asserted consequence with full heads at least four, take $p = \max(4, r + 2)$; the given deficit is at least p-1 and $r \leq p - 2$, so the proved theorem applies. Every transfer uses actual components and the same boundary vertex. Lemma 41 extends the resulting coloring over the original canonical pool and through any specified outer q-set, preserving residual colors and accounting for target components meeting deleted vertices through leaves. Full originals or genuine complete suffixes occur throughout, so all units and repetitions remain. □

Depends on. Lemma 41, Lemma 42, Lemma 89, Lemma 100, Lemma 114, Lemma 153.

Lemma 182 (Large closed lumps of degree fourteen are LP-good). *Use the integer certificate rows $c_i(F)$, matching number $\nu(F)$, excess $ex(F) = c_0(F) - \nu(F)$, and signed gap $gap(F) = \sum_{i \geq 0} c_i(F) - e(F)$ of Lemma 23. A closed 14-lump means a nonempty connected 14-regular finite simple graph F such that deleting any vertex leaves a perfect matching. Let $k \geq 148$ be an integer and let F be any closed 14-lump on $n = 2k + 1$ vertices. Then $gap(F) \leq 0$. Equivalently, a closed 14-lump with positive gap has at most 295 vertices. No minimality hypothesis or assumption on proper subgraphs is imposed. All graph orders, degrees, matching numbers, costs, and excesses are integers; gap is a signed integer.*

Proof. Write $E = ex(F)$ and $e = E - 1$, an integer which is allowed to be negative. A perfect matching after one vertex deletion shows $\nu(F) = k$. The connected excess bound in Lemma 50 gives

$E \leq ceil((12k + 2)/40) \leq (12k + 41)/40$,

$e \leq (12k + 1)/40$, $R := 12k + 22 - 40e \geq 21$. (1)

Whenever $e \geq 1$, (1) and $k \geq 148$ imply $3e \leq k$.

The complementary bounds in Lemma 87 give $c_i(F) + c_{14-i}(F) \le 2k+2$ for $i = 3, 4, 5, 6$, $c_7(F) \le k+1$, $c_{14}(F) \le 1$, and $c_i(F) = 0$ for $i \ge 15$. By Lemma 23 the first three rows are at most $k+E$ each. Since $e(F) = 14k+7$, adding these bounds gives

$gap(F) \le 3E - 2k + 3 + c_{13}(F) + c_{12}(F)$. (2)

We first obtain bounds needing no census hypothesis. Fix a vertex v and a perfect matching M of F-v. The spanning graph F-M has minimum degree 13. In Lemma 162 the degree-13 product equals

$P_{13} = product_{m=3}^{14}\, 13m/(13m+1) = 2120125746145771/2423608732701220 > 873/1000$:

the last strict comparison follows from

$1000 \cdot 2120125746145771 - 873 \cdot 2423608732701220 = 4315322497605940 > 0$.

Substituting its stopping-index bound in its affine domination bound at $\delta = 13$ gives

$\gamma(F-M) \le (574 - 520P_{13})n/574 + (1040 - 520P_{13})/574$

$< 3001n/14350 + 2093/2050$.

Add v to a dominating set attaining $\gamma(F-M)$. This same set contains the only vertex exposed by M and dominates F-M, so Lemma 49 with $D = 14$, $j = 1$ gives

$c_{13}(F) < 3 + 21n/100$. (3)

Here $3001/14350 < 21/100$ and $1 + 2093/2050 < 3$.

In Lemma 92 take $d = 14$, $j = 2$, $p = 7/25$, so $m_\theta = 13$. The binomial expression is

$Q = p + (1-p)^{15} + 14p(1-p)^{14} + 7p^2(1-p)^{13}$

$= 311390948480780593783/931322574615478515625 < 33436/100000$.

The positive difference of the last two fractions is

48540541206421611/7450580596923828125000.

Thus

$c_{12}(F) < 33436n/100000$. (4)

Put $e_1 = (198k+363)/731$ and $e_2 = (36k+66)/127$. These are real thresholds, with

$e_2 - e_1 = (1170k+2145)/92837 > 0$.

The small-component hypothesis in Lemma 163 is satisfied at $j = 1$ whenever $e \ge e_1$, and at $j = 2$ whenever $e \ge e_2$. Here is the complete substitution. For $j = 1$, $m_0 = 15$, $g(m) = 10m - 8$, $g(m_0) = 142$ and $W = 66R/71$. Its two sufficient inequalities are

$2924e \ge 792k + 1452$, $24870e \ge 6396k + 13998$.

The first holds at $e = e_1$ with equality. Multiplying the difference in the second by 731 at $e = e_1$ gives $248784k - 1204728 > 0$.

For $j = 2$, $m_0 = 13$, $g(m) = 8m - 6$, $g(m_0) = 98$ and $W = 12R/7$. The two sufficient inequalities are

$508e \ge 144k + 264$, $140610e \ge 38508k + 78438$.

The first holds at $e = e_2$ with equality; multiplying the difference in the second by 127 at $e = e_2$ gives $171444k - 681366 > 0$.

All four left-minus-right expressions increase with e. The displayed differences are positive for $k \ge 148$. This checks the census hypothesis even in the range where its alternative $R \le g(m_0)$ is not available.

At minimum degree 14, Lemma 162 supplies a dominating set Z of F with

$|Z| \le z := floor((1309022k + 4029022)/3300000)$.

Write $z_0 = (1309022k + 4029022)/3300000$. For the following size checks use

$z \le z_0 \le 2k/5 + 5/4$. (5)

Suppose $e \geq e_1$. Then $e > 40$, and $z \leq 2e - 10$: after replacing z by the last upper bound in (5) and e by e_1, the difference multiplied by $20 \cdot 731$ is $2072k - 149955 > 0$.

We verify an admissible odd cutoff for $j = 1$ on this same Z. Put $t_0 = floor((2e-1)/12)$ and $m_1 = 4e + 2t_0 + 5$. Let m be any odd integer with $15 \leq m < m_1$ and $10m - 8 \geq R$, and put $\beta = \min(m, z) + 11$. Then $\beta \leq 2e + 1$ and $(m-1)/2 \leq 2e + t_0 + 1$. Also

$floor((k+e)/2) + (m+1)/2 + floor((2e+1)/12)$

$\leq k/2 + 17e/6 + 2$

$\leq 27k/20 + 497/240 < 2k + 1$. (6)

The upper bound $m \leq 4e + 2t_0 + 3$, (1), and (5) give

$m + \beta \leq (13/3)e + 83/6 + z$

$\leq 17k/10 + 1823/120 < 2k + 15$.

Hence $ceil((n - m - \beta)/15) \geq 0$; also $floor((2e + 1 - \beta)/12) \geq 0$. If $2e + (m+1)/2 \geq floor((k+e)/2)$, these two nonnegative quantities prove the lower balanced inequality in the partial-set test.

Otherwise $m \leq k - 3e$. As $\beta \leq m + 11$ and $e \geq 4$,

$12(n - m - \beta) \geq 12(6e - 10) \geq 15(2e + 1)$.

The minimum in that test therefore equals $floor((2e + 1 - \beta)/12)$, which is at least $(2e - m - 21)/12$. The required inequality follows from

$2e + (2e - m - 21)/12 + (m+1)/2 \geq (k+e)/2$,

which after multiplication by 24 is precisely $10m - 8 \geq R$.

Thus every indicated m passes the partial-set test, and m_1 is admissible with $\zeta = z$.

Moreover $t_0 \geq (2e - 12)/12$, so

$2m_1 \geq (26e + 18)/3 > 2k + 1$,

since at $e = e_1$ the difference is $(762k + 20403)/2193 > 0$.

Apply the $j = 1$ conclusion of Lemma 163 to this Z and cutoff. With $\tau_1 = \max(0, \lceil R/142 \rceil - 1) \leq R/142$, it yields

$c_{13}(F) \leq z + R/142$ whenever $e \geq e_1$. (7)

When $e \geq e_2$, also define m_2 to be the least odd integer satisfying $2m_2 + 10 \geq 2e + 2$. Here $e_2 > 42$ at $k \geq 148$, so $m_2 \geq 13$. We check admissibility for $j = 2$ with $\zeta = n$. If $13 \leq m < m_2$ is odd, then $2m + 10 \leq 2e + 1$ and $m < e$. Since $2m < 2e \leq 2(12k+1)/40 < n$, the partial-set test at $\zeta = n$ has $\beta = \min(2m, n) + 10 = 2m + 10$. The first two size inequalities follow, and the upper balanced inequality follows from (6), whose upper estimates also apply to $m < e$.

Furthermore

$12(n - m - \beta) \geq 12(2k - 9 - 3e) \geq 15(2e + 1)$.

For the last inequality it suffices that $66e \leq 24k - 123$. By (1), the left-minus-right comparison reduces to $84k \geq 2493$, true for $k \geq 148$.

Thus the minimum in the partial-set test is $floor((2e + 1 - \beta)/12) \geq (2e - 20 - 2m)/12$. Its lower balanced inequality follows from

$2e + (2e - 20 - 2m)/12 + (m+1)/2 \geq (k+e)/2$,

equivalent to $8m - 6 \geq R$. This is exactly the gain condition required for admissibility. The cutoff m_2 is therefore admissible at $\zeta = n$.

Use the random $j = 2$ conclusion of Lemma 163 with $p = 13/50$. Write

$f = p + (1-p)^{15} + 14p(1-p)^{14}$

$= 990821173632290681763189 1/305175781250000000000000000$,

$t = 91p^2(1-p)^{13}$

$= 37458510992191144395700463/3051757812500000000000000000.$

The exact differences $32469/100000 - f$ and $12277/100000 - t$ have respective positive numerators 540705083343182368109 and 791967187135604299537 over the displayed common denominator. Also

$m_2 \geq e - 9/2 \geq e_2 - 9/2 = (72k - 1011)/254 > 0,$

$\tau_2 = \max(0, \lceil R/98 \rceil - 1) \leq R/98.$

Therefore, putting $N(k) = 3118358(2k+1)/(100000(72k-1011))$, we obtain

$c_{12}(F) < 32469n/100000 + N(k) + R/98$ whenever $e \geq e_2$. (8)

It remains to combine these bounds in three exhaustive real-threshold ranges.

If $e < e_1$, insert (3), (4) and $E = e + 1$ in (2). The resulting upper bound is strictly less than

$A(k) = 3e_1 + 9 - 2k + 54436(2k+1)/100000.$

This affine function has slope $-901821/9137500$ and $A(148) = -65290837/18275000 < 0$. Hence $gap(F) < 1$ in this range, which includes $E = 0$ and $E = 1$.

If $e_1 \leq e < e_2$, insert (7), (4) in (2). Since

$3e + R/142 = (386/142)e + (12k + 22)/142$

has positive coefficient of e, the resulting upper bound is strictly less than

$B(k) = (386/142)e_2 + (12k+22)/142 + 6 - 2k + z_0 + 33436(2k+1)/100000.$

This affine function has slope $-107602247/1352550000$ and $B(148) = -13148645877/4959350000 < 0$. Again $gap(F) < 1$.

Finally suppose $e \geq e_2$ and insert (7), (8) in (2). Here $R = 12k + 62 - 40E$ and $1/142 + 1/98 = 60/3479$. The coefficient of E in $3E + 60R/3479$ is $8037/3479 > 0$. Bound (1) gives

$3E + 60R/3479 \leq 9k/10 + 478317/139160.$

Consequently

$gap(F) < C(k) := 13092555503/1640100000 - (2023/37500)k + N(k).$ (9)

For $k \geq 148$ the denominator in $N(k)$ is positive, and the function $(2k+1)/(72k-1011)$ strictly decreases: its derivative is $-2094/(72k-1011)^2$. The linear coefficient in (9) is negative. Thus $C(k) \leq C(148) = 1685424335809/1757640500000 < 1$; the positive denominator-minus-numerator difference is 72216164191.

Every case gives $gap(F) < 1$. This quantity is an integer by Lemma 23, so $gap(F) \leq 0$. A closed 14-lump has odd order and $\nu = (n-1)/2$; the equivalent positive-gap order bound follows by contraposition. □

Remark. For small excess, elementary top-row estimates already force nonpositive gap. Larger excess pays for stronger component-count certificates, and the two census thresholds divide the remaining argument into decreasing rational bounds.

Depends on. Lemma 23, Lemma 49, Lemma 50, Lemma 87, Lemma 92, Lemma 162, Lemma 163.

Lemma 183 (Closed lumps of degree at least eighteen are LP-good). *Use the finite integer certificate costs $c_\theta(H)$, edge count $e(H)$, and $gap(H) = \sum_{\theta \geq 0} c_\theta(H) - e(H)$ of Lemma 23. A graph is LP-good when $gap(H) \leq 0$. For a nonnegative integer d, a closed d-lump is a nonempty connected d-regular finite simple graph such that deleting any vertex leaves a perfect matching.*

For every even integer $d \geq 18$, every closed d-lump H is LP-good. Equivalently, $\sum_{\theta \geq 0} c_\theta(H) \leq e(H)$. There is no upper bound on the order of H. All degrees, orders, certificate capacities, budgets, costs, and indices are integers in their nonnegative domains.

Proof. Fix an even integer $d \geq 18$ and a closed d-lump H. It is nonempty, connected, finite and simple, is regular of degree d, and has a perfect matching after each vertex deletion. The finite sum and the equivalence between LP-goodness and its displayed cost inequality are those of Lemma 23.

An even integer $d \geq 18$ either equals 18, 20, 22, 24 or is at least 26. If $d = 18$, fact Lemma 164 applies to this H and gives $gap(H) \leq 0$. If $d = 20$, fact Lemma 174 gives the same conclusion. If $d = 22$, fact Lemma 134 gives it. If $d = 24$, fact Lemma 157 gives it. If $d \geq 26$, fact Lemma 129 applies because its definition of a closed d-lump and its even-degree domain agree with the present ones, and again gives $gap(H) \leq 0$. Each cited theorem covers every order of H. These five cases exhaust the specified integer domain, proving the assertion. □

Remark. The four individual degrees below twenty-six and the uniform higher-degree theorem give one theorem covering the entire even-degree domain beginning at eighteen.

Depends on. Lemma 23, Lemma 129, Lemma 134, Lemma 157, Lemma 164, Lemma 174.

Lemma 184 (Closed degree-fourteen lumps at intermediate orders are LP-good). *Use the finite integer certificate costs $c_i(F)$, matching number $\nu(F)$, excess $ex(F) = c_0(F) - \nu(F)$, and signed gap $gap(F) = \sum_{i \geq 0} c_i(F) - e(F)$ of Lemma 23. Let $77 \leq k \leq 147$ be an integer. If F is a nonempty connected 14-regular finite simple graph on $n = 2k + 1$ vertices such that deleting any vertex leaves a perfect matching, then $gap(F) \leq 0$. No hereditary or minimality hypothesis is imposed. All graph orders, degrees, costs, cardinalities, capacities and matching budgets are integers in their stated domains.*

Proof. Put $E = ex(F)$. Factor-criticality gives $\nu(F) = k$, and regularity gives $e(F) = 14k + 7$. The definitions in Lemma 23 give $E \geq 0$ and $c_i(F) \leq k + E$. The complementary bounds in Lemma 87 give $c_i + c_{14-i} \leq 2k + 2$ for $i = 3, 4, 5, 6$, $c_7 \leq k + 1$, $c_{14} \leq 1$, and $c_i = 0$ for $i \geq 15$. Thus, for any bounds $c_1 \leq k + a_1$ and $c_2 \leq k + a_2$, counting rows 0, 1, 2, 12, 13, 14 separately and the four indicated pairs and row 7 gives

$gap(F) \leq E + a_1 + a_2 - 2k + 3 + c_{13}(F) + c_{12}(F)$. (1)

First we specify the integer excess and low-row bounds. For r in {12,13} and c in {0,1}, define $h(r, c) = ceil(((r - 1)k + 2c)/(3r + 1))$. Set $g(r, c) = h(r, c) - 1$ if $h(r, c) \geq 1$, $k + h(r, c)$ is even, and $(3r + 1)(h(r, c) - 1) \geq (r - 1)(k - 1) + 2c$; otherwise set $g(r, c) = h(r, c)$. If $g = g(r, c)$ and $x = floor((k + g)/2)$, then

$(r - 1)x + c \leq 2rg$. (2)

Indeed, when $g = h$ the left side minus 2rg is at most $((r - 1)k + 2c - (3r + 1)g)/2 \leq 0$. When $g = h - 1$, $k + g$ is odd and that difference equals $((r - 1)(k - 1) + 2c - (3r + 1)g)/2 \leq 0$. Put $E_0 = \min(floor((k + 1)/3), g(13, 1))$, $g_1 = g(13, 0)$, $g_2 = g(12, 0)$. Then

$E \leq E_0$, $c_j(F) \leq k + g_j$ for $j = 1, 2$. (3)

Here are the graph applications proving (3). The floor bound on E is part of Lemma 50. For $g = g(13, 1)$, every smaller budget $x \leq floor((k + g)/2)$, with $t = x - 2g \geq 1$, satisfies $13t + 1 \leq x$ by (2). Also $3x + t \leq 2k$, so $40t + 3 \leq 2k$ and $t \leq \lceil 2k/14 \rceil$. The connected independent-set construction in Lemma 50 supplies an independent t-set I with $|N_F(I)| \leq 13t + 1 \leq x$. Enlarge its neighborhood to an x-set W avoiding I. Such enlargement is possible since $n - t = x + 2y + 1$ for $y = k + g - x$. Put all edges meeting W in the smaller exceptional part, and all edges within the remaining vertices outside I in the other. The former has vertex cover W; the latter has at most $2y + 1$ vertices. Both degree-bounded parts are empty. This covers every edge because I is independent and $N_F(I)$ is in W. For $t \leq 0$, put edges meeting any x-set in the smaller part and the residual edges in the other; $n - x \leq 2y + 1$. If the larger budget is at least k, putting all edges there suffices. These constructions handle every budget split and prove $c_0 \leq k + g$.

For the positive rows, let $r = 14 - j$ and $g = g_j$. Inequality (2) with $c = 0$ gives $rt \leq x$ at $x = floor((k + g)/2)$, $t = x - 2g$. If $t \geq 1$, then $(3r + 1)t \leq 3x + t \leq 2k$, so $t \leq 2k/37 < n/15$. A maximum independent set is maximal, and its closed neighborhoods cover F, each using at most 15 vertices. Hence $\alpha(F) \geq \lceil n/15 \rceil > t$. The final assertion of Lemma 54 applies with $\theta = j$, since $3j \leq 15$. It also covers $t \leq 0$. This proves (3).

For δ in {13,14}, let $u_0(\delta) = n$ and, while $u_i(\delta) > 0$ and $(\delta + 1)u_i(\delta) > 2(n - i)$, put

$u_{i+1}(\delta) = \max(0, u_i(\delta) - ceil((\delta + 1)u_i(\delta)/(n - i)))$.

Let s_δ be the stopping index. Define

$z = floor((42s_{14} + 14u_{s_{14}}(14) + 2n)/44)$,

$b_1 = 1 + floor((39s_{13} + 13u_{s_{13}}(13) + 2n)/41)$,

$b_2 = \lfloor 33358n/100000 \rfloor$.

Then $\gamma(F) \le z$, $c_{13}(F) \le b_1$, $c_{12}(F) \le b_2$. (4)

The first bound is Lemma 162 at minimum degree 14. To prove the second, fix a vertex v and a perfect matching M of F-v. The graph F-M has minimum degree 13. Apply Lemma 162 to it and add v to the resulting dominating set. The same enlarged set contains the only vertex exposed by M and dominates F-M. Lemma 49 with $D = 14$, $j = 1$ proves the bound b_1, including its added unit. For the last bound, apply Lemma 92 with $d = 14$, $j = 2$, $p = 27/100$ and $m_\theta = 13$. Its probability expression Q satisfies

$13 \cdot 100^{15} \cdot Q$

$= 13 \cdot 27 \cdot 100^{14} + 13 \cdot 73^{15} + 13 \cdot 14 \cdot 27 \cdot 73^{14} + 91 \cdot 27^2 \cdot 73^{13}$

$= 433645874178603081029831837 1154$

$< 433654000000000000000000000000$

$= 13 \cdot 33358 \cdot 10^{25}$.

Thus $Q < 33358/100000$, proving the integer bound.

Next we improve the first upper row when E is large. Write $e = E - 1$, $R = 12k + 22 - 40e$, and $L = (198k + 363)/731$, $A = \lceil L \rceil$, $\tau(E) = \max(0, \lceil R/142 \rceil - 1)$. We claim that, whenever $E \ge A + 1$,

$c_{13}(F) \le \gamma(F) + \tau(E) \le z + \tau(E)$. (5)

The connected excess bound in Lemma 50 gives $E \le ceil((12k + 2)/40) \le (12k + 41)/40$, $e \le (12k + 1)/40$, $R \ge 21$. Consequently $3e \le k$. Here $e \ge L > 21$, so all positive-excess hypotheses of Lemma 163 hold. At $d = 14$, $j = 1$ its constants are $m_0 = 15$, $g(m) = 10m - 8$, $g(m_0) = 142$, and $W = 66R/71$. The two sufficient small-component inequalities reduce to $2924e \ge 792k + 1452$, $24870e \ge 6396k + 13998$. The first holds at $e = L$ with equality; the second left-minus-right difference at $e = L$, multiplied by 731, equals $248784k - 1204728 > 0$. Both differences increase with e. This verifies that theorem's small-component hypothesis.

Choose a minimum dominating set Z of F. The product bound in Lemma 162 gives, in addition to $|Z| \le z$,

$|Z| \le z_0 := (1309022k + 4029022)/3300000 \le 2k/5 + 5/4$.

Use $\zeta = \min(z, \lfloor z_0 \rfloor)$, so $|Z| \le \zeta$. From $k \ge 77$ and $e \ge L$ we have $\zeta \le 2e - 10$: the difference between 2L-10 and $2k/5 + 5/4$, multiplied by $20 \cdot 731$, is $2072k - 149955 > 0$. Put $t_0 = floor((2e - 1)/12)$, $m_\cdot = 4e + 2t_0 + 5$. This is odd and at least 15. For every odd m with $15 \le m < m_\cdot$ and $10m - 8 \ge R$, put $\beta = \min(m, \zeta) + 11$. We verify the partial-set test of Lemma 163. First $\beta \le 2e + 1$ and $(m - 1)/2 \le 2e + t_0 + 1$. Also

$floor((k + e)/2) + (m + 1)/2 + floor((2e + 1)/12)$

$\le k/2 + 17e/6 + 2 \le 27k/20 + 497/240 < 2k + 1$.

The bound $m \le 4e + 2t_0 + 3$ gives

$m + \beta \le (13/3)e + 83/6 + \zeta \le 17k/10 + 1823/120 < 2k + 15$.

Thus $ceil((n - m - \beta)/15) \ge 0$, and $floor((2e + 1 - \beta)/12) \ge 0$. If $2e + (m + 1)/2 \ge floor((k + e)/2)$, these nonnegative quantities prove the lower balanced inequality in the test. Otherwise $m \le k - 3e$. Since $\beta \le m + 11$ and $e \ge 4$,

$12(n - m - \beta) \ge 12(6e - 10) \ge 15(2e + 1)$.

Hence the minimum in that test is its floor branch, at least $(2e - m - 21)/12$. The desired lower inequality follows from

$2e + (2e - m - 21)/12 + (m + 1)/2 \ge (k + e)/2$,

which after multiplication by 24 is exactly $10m - 8 \ge R$. Every required m therefore passes. Finally $t_0 \ge (2e - 12)/12$, giving $2m_\cdot \ge (26e + 18)/3 > 2k + 1$, because at $e = L$ the difference is $(762k + 20403)/2193 > 0$.

The stronger $j = 1$ conclusion of Lemma 163 now gives $c_{13}(F) \le |Z| + \tau(E)$, proving (5) with this actual minimum dominating set.

All remaining comparisons are finite integer calculations. Define

$G_A = A + \min(A, g_1) + \min(A, g_2) - 2k + 3 + b_1 + b_2,$

$G_B = E_0 + \min(E_0, g_1) + \min(E_0, g_2) - 2k + 3 + z + b_2 + \tau(E_0).$

The following table gives, in order, k, E_0, g_1, g_2, A, s_{14}, $u_{s_{14}}(14)$, z, s_{13}, $u_{s_{13}}(13)$, b_1, b_2, G_A, G_B. Its entries are exact evaluations of the displayed integer formulas and the two stopping recurrences, with positive denominators at every step.

77, 24, 24, 23, 22, 18, 18, 29, 19, 19, 32, 51, -2, 0

78, 24, 24, 23, 22, 19, 16, 30, 20, 17, 33, 52, -2, 0

79, 24, 24, 24, 22, 19, 17, 30, 19, 20, 33, 53, -3, 0

80, 25, 24, 24, 23, 19, 18, 31, 20, 18, 33, 53, -2, 0

81, 25, 24, 24, 23, 19, 19, 31, 20, 19, 34, 54, -2, -1

82, 25, 25, 25, 23, 20, 17, 32, 21, 18, 34, 55, -3, 1

83, 25, 25, 25, 23, 20, 18, 32, 21, 19, 35, 55, -4, -1

84, 25, 25, 25, 24, 20, 19, 32, 21, 20, 35, 56, -2, -2

85, 26, 26, 26, 24, 20, 20, 33, 21, 21, 35, 57, -3, 1

86, 26, 26, 26, 24, 21, 18, 33, 21, 21, 36, 57, -4, -1

87, 26, 26, 26, 25, 21, 19, 34, 21, 22, 36, 58, -2, -1

88, 27, 27, 27, 25, 21, 20, 34, 22, 20, 36, 59, -3, 1

89, 27, 27, 27, 25, 21, 20, 34, 22, 21, 37, 59, -4, -1

90, 27, 27, 27, 25, 22, 19, 35, 22, 22, 37, 60, -5, -1

91, 28, 28, 28, 26, 22, 20, 35, 22, 23, 38, 61, -2, 1

92, 28, 28, 28, 26, 22, 21, 36, 22, 23, 38, 61, -4, 0

93, 28, 28, 28, 26, 22, 21, 36, 23, 22, 38, 62, -5, -1

94, 29, 29, 28, 26, 22, 22, 36, 23, 23, 39, 63, -5, 0

95, 29, 29, 28, 27, 23, 20, 37, 23, 24, 39, 63, -4, -1

96, 29, 29, 29, 27, 23, 21, 37, 24, 22, 40, 64, -4, -1

97, 30, 30, 29, 27, 24, 20, 38, 24, 23, 40, 65, -5, 1

98, 30, 30, 29, 28, 23, 23, 38, 24, 23, 40, 65, -4, -1

99, 30, 30, 30, 28, 24, 21, 38, 24, 25, 41, 66, -4, -1

100, 31, 30, 30, 28, 24, 22, 39, 25, 23, 41, 67, -5, 0

101, 31, 30, 30, 28, 25, 21, 39, 25, 24, 42, 67, -6, -2

102, 31, 31, 31, 29, 24, 24, 39, 25, 25, 42, 68, -4, -1

103, 31, 31, 31, 29, 25, 22, 40, 25, 26, 43, 69, -4, -1

104, 31, 31, 31, 29, 25, 22, 40, 26, 24, 43, 69, -6, -3

105, 32, 32, 32, 29, 25, 23, 40, 26, 25, 43, 70, -7, -1

106, 32, 32, 32, 30, 25, 24, 41, 26, 26, 44, 71, -4, -1

107, 32, 32, 32, 30, 25, 25, 41, 26, 27, 44, 71, -6, -3

108, 33, 33, 33, 30, 26, 24, 42, 27, 25, 45, 72, -6, 0
109, 33, 33, 33, 31, 26, 25, 42, 27, 26, 45, 73, -4, -1
110, 33, 33, 33, 31, 26, 26, 43, 27, 27, 46, 73, -5, -2
111, 34, 34, 33, 31, 27, 24, 43, 28, 26, 46, 74, -6, -1
112, 34, 34, 33, 31, 27, 24, 43, 28, 27, 47, 75, -6, -2
113, 34, 34, 34, 32, 27, 25, 44, 28, 28, 47, 75, -5, -2
114, 35, 35, 34, 32, 27, 26, 44, 29, 26, 48, 76, -5, -1
115, 35, 35, 34, 32, 27, 27, 44, 28, 29, 48, 77, -6, -2
116, 35, 35, 35, 32, 28, 26, 45, 29, 27, 48, 77, -8, -2
117, 36, 36, 35, 33, 28, 26, 45, 29, 27, 48, 78, -6, -1
118, 36, 36, 35, 33, 28, 27, 46, 29, 29, 49, 79, -6, -1
119, 36, 36, 36, 33, 29, 26, 46, 30, 28, 50, 79, -7, -2
120, 37, 36, 36, 33, 29, 26, 46, 30, 28, 50, 80, -8, -2
121, 37, 36, 36, 34, 29, 28, 47, 30, 29, 50, 81, -6, -2
122, 37, 37, 37, 34, 30, 26, 48, 30, 30, 51, 81, -7, -1
123, 37, 37, 37, 34, 29, 29, 48, 31, 29, 51, 82, -8, -2
124, 37, 37, 37, 35, 30, 27, 48, 31, 30, 52, 83, -5, -3
125, 38, 38, 38, 35, 30, 27, 48, 31, 30, 52, 83, -7, -2
126, 38, 38, 38, 35, 30, 29, 49, 31, 31, 52, 84, -8, -2
127, 38, 38, 38, 35, 31, 28, 50, 32, 30, 53, 85, -8, -2
128, 39, 39, 39, 36, 31, 28, 50, 32, 31, 53, 85, -7, -1
129, 39, 39, 39, 36, 31, 29, 50, 32, 31, 53, 86, -8, -2
130, 39, 39, 39, 36, 31, 30, 51, 32, 32, 54, 87, -8, -2
131, 40, 40, 39, 36, 32, 28, 51, 32, 33, 54, 87, -10, -2
132, 40, 40, 39, 37, 32, 29, 51, 33, 31, 55, 88, -7, -3
133, 40, 40, 40, 37, 32, 30, 52, 33, 33, 55, 89, -8, -2
134, 41, 41, 40, 37, 32, 30, 52, 34, 31, 56, 89, -9, -2
135, 41, 41, 40, 38, 32, 31, 52, 34, 32, 56, 90, -7, -3
136, 41, 41, 41, 38, 32, 32, 53, 34, 32, 56, 91, -8, -2
137, 42, 42, 41, 38, 33, 31, 53, 34, 34, 57, 91, -9, -2
138, 42, 42, 41, 38, 33, 32, 54, 34, 34, 57, 92, -10, -2
139, 42, 42, 42, 39, 33, 32, 54, 34, 35, 58, 93, -7, -2
140, 43, 42, 42, 39, 33, 33, 54, 35, 33, 58, 93, -9, -3
141, 43, 42, 42, 39, 34, 32, 55, 35, 35, 59, 94, -9, -3
142, 43, 43, 43, 39, 34, 33, 55, 36, 33, 59, 95, -10, -2
143, 43, 43, 43, 40, 35, 31, 56, 36, 34, 60, 95, -8, -3
144, 43, 43, 43, 40, 34, 34, 56, 36, 34, 60, 96, -9, -4
145, 44, 44, 44, 40, 35, 32, 56, 36, 35, 60, 97, -10, -2

146, 44, 44, 44, 41, 35, 33, 57, 36, 36, 60, 97, -9, -3

147, 44, 44, 44, 41, 35, 34, 57, 36, 37, 61, 98, -9, -4

To apply the table, first suppose $E \le A$. Equations (1), (3), (4) and row monotonicity give $gap(F) \le G_A \le -2$. If $E \ge A+1$, use (5). The resulting bound $E + \min(E, g_1) + \min(E, g_2) - 2k + 3 + z + b_2 + \tau(E)$ is nondecreasing in integer E: increasing E by one decreases R by 40, so τ falls by at most one; the first E term compensates for this, and the two minima do not decrease. Therefore $gap(F) \le G_B$. Every G_B is nonpositive except at $k = 82, 85, 88, 91, 97$, where it is one. At these five indices $R(E_0)$ respectively equals 46, 42, 38, 34, 26, all at most 102. Thus $\tau(E_0) = \tau(E_0 - 1) = 0$. The first E term makes the bound at $E_0 - 1$ at least one smaller, and monotonicity then excludes every $E < E_0$.

It remains to exclude positive gap in these five cases at $E = E_0$. The upper estimate equal to one forces $c_{13}(F) = z$: reducing this term by one in (1) makes the bound nonpositive. Formula (5), with $\tau(E_0) = 0$, gives $c_{13}(F) \le \gamma(F) \le z$, so $\gamma(F) = z$. Every maximal independent set dominates F, hence $\alpha(F) \ge z$. Choose an independent set S of cardinality $a = z$ and use Lemma 54 with $\theta = 2$. The exact parameters and its finite lists are as follows. Here e′ is its trial excess, $b = n - a$, $q = \lfloor 14a/b \rfloor$, $r = 14a - qb$, $x_\cdot = floor((k + e')/2)$, and $t_\cdot = x_\cdot - 2e'$.

k, E_0, a, e′, b, q, r, $x_\cdot$, $t_\cdot$; $\lfloor U_1 \rfloor$,..., $\lfloor U_{t_\cdot} \rfloor$; $W_1(2)$,..., $W_{t_\cdot}(2)$

82, 25, 32, 24, 133, 3, 49, 53, 5; 14, 26, 38, 49, 59; 12, 22, 32, 41, 51

85, 26, 33, 25, 138, 3, 48, 55, 5; 14, 26, 38, 49, 59; 12, 22, 32, 41, 51

88, 27, 34, 26, 143, 3, 47, 57, 5; 14, 26, 38, 50, 60; 12, 22, 32, 42, 51

91, 28, 35, 27, 148, 3, 46, 59, 5; 14, 27, 39, 50, 60; 12, 23, 33, 42, 51

97, 30, 38, 28, 157, 3, 61, 62, 6; 14, 27, 39, 50, 61, 70; 12, 23, 33, 42, 51, 62

For direct verification of these lists, the numerator of U_t is $(b - r)[binom(a, t) - binom(a - 3, t)] + r[binom(a, t) - binom(a - 4, t)]$, its denominator is $binom(a, t)$, and the formula for W_t uses $m_0 = 0$ and $m_l = \max(13, \lceil 13l/2 \rceil)$. Every displayed W_t is at most $t + 2e'$. For example the last row has comparison list 57, 58, 59, 60, 61, 62; in the other rows all earlier entries are smaller and the last comparison values are 53, 55, 57, 59, respectively. Thus the cited construction proves $c_2(F) \le k + e'$. This saves at least one beyond the c_2 term used for G_B: in the first four cases $e' = E_0 - 1 < \min(E_0, g_2) = E_0$, and in the last case $e' = 28 < \min(30, 29) = 29$. Returning to (1) gives $gap(F) \le 0$ in each case, a contradiction to positive gap. All integer E and all integers $77 \le k \le 147$ have now been treated. Hence $gap(F) \le 0$ throughout the stated class. □

Depends on. Lemma 23, Lemma 49, Lemma 50, Lemma 54, Lemma 87, Lemma 92, Lemma 162, Lemma 163.

Lemma 185 (An actual binary pair at the first matching-preserving deletion). *All graphs are finite and simple. Put $P_3 = K_{1,2}$ and $K_2 = K_{1,1}$, where $K_{1,d}$ is the d-edge star. Let p, t be positive integers and u a nonnegative integer. Let $F = pP_3$ disjoint union uK_2 and let J be the vertex-disjoint union of stars with positive nonincreasing integer edge sizes m_1,..., m_t. Put $s = p + u$, let n_1,..., n_s be p twos followed by u ones, and define $M = s + t - 1$,*

$$\ell_k = max\{n_i + m_j - 1 : 1 \le i \le s, 1 \le j \le t, i + j = k\}$$

for integers $2 \le k \le M + 1$, and $L = \sum_{k=2}^{M+1} \ell_k$. Write e, ν, Δ for edge count, maximum matching cardinality, and maximum degree, and put $\Phi = e - \nu$. Copies are ordinary subgraphs with vertex-disjoint target components; F and J are prescribed red and blue.

Suppose G arrows (F, J). For any vertex set X with $q = |X| < M$, put $H = G - X$ by induced deletion. If $\Delta(H) \le \ell_{q+2} - 1$, define

$$\alpha = \min(q, p - 1),\ \beta = q - \alpha.$$

Then $0 \le \alpha < s$, $0 \le \beta < t$, and H arrows the actual pair

$$F' = (p - \alpha)P_3 \text{ disjoint union } uK_2,$$

$$J' = disjoint \bigcup_{j=\beta+1}^{t} K_{1,m_j}.$$

Its component budget is $M' = M - q$, its complete diagonal list is $(\ell_{q+2}, \dots, \ell_{M+1})$, and $\Delta(H) = m_{\beta+1}$. If $u \geq 1$ and $q \geq p$, additionally $j_0 = q - p + 2 \leq t$, $b_0 = \max(0, q - p - u + 1)$, and $m_{b_0+1} = \cdots = m_{j_0}$.

Assume now that $\Phi(G) = L - M$ and $\delta = M - \nu(G) > 0$. Fix any maximum matching W of G. For $0 \leq j < M$, recursively let $X_j = \{v_1, \dots, v_j\}$, $H_j = G - X_j$, $W_j = \{abinW : a, bnotinX_j\}$, and $Q_j = (G - W) - X_j$. Here G-W removes only matching edges. Choose v_{j+1} of maximum degree in Q_j, preferring a vertex covered by W_j whenever such a maximum-degree vertex exists. Every sequence following this rule exists for all M steps. It has a first index q with $0 \leq q < M$ such that v_{q+1} is not covered by W_q. Put $X = X_q$ and $H = H_q$.

The vertices v_1,..., v_q are successive maximum-degree vertices of the corresponding H_j, and, writing $T_q = \sum_{k=2}^{q+1} \ell_k$ with empty sum zero,

$e(G) - e(H) = T_q$, $\nu(G) - \nu(H) = q$.

The selected matching-preserving step satisfies $\nu(H_{q+1}) = \nu(H)$. The actual pair F', J' defined in the preceding paragraph exists for this same H and satisfies

$L' = L - T_q$, $M' = M - q$, $\Phi(H) = L' - M'$,

$M' - \nu(H) = \delta > 0$, $\Delta(H) = m_{\beta+1} = \ell_{q+2} - 1 \geq 3$,

where L' is its entire diagonal sum. These conclusions hold for every initial maximum matching and every permitted tie choice. All counts, indices, degrees and cardinalities are integers in their stated nonnegative domains; δ is a positive integer. Original unit components remain in F', and J' is an actual unmodified suffix. No original maximum-degree, connectedness, or regularity condition is imposed.

Proof. The original pair and the conditional arbitrary-set conclusion are exactly the domains of Lemma 62. Applying that fact to G, X gives the displayed legal cuts, retained pair, entire diagonal equality, component budget, degree equality, and constant-run clause. This application requires no matching or potential hypothesis.

Now impose the exact potential and positive deficit assumptions, and fix W. Apply Lemma 95 to the single original pair, so its maximum component budget N is the present M. Its potential at index j is

$P_j = \sum_{k=j+2}^{M+1} (\ell_k - 1)$.

It provides all M greedy steps on Q_j, with the exact restricted matching W_j remaining maximum in H_j. At each step put $\epsilon_j = 1$ if v_{j+1} is covered by W_j and zero otherwise. The cited fact gives, for every $j < M$,

$\nu(H_j) - \nu(H_{j+1}) = \epsilon_j$,

$deg_{H_j}(v_{j+1}) = \ell_{j+2} - 1 + \epsilon_j$,

$\Phi(H_j) = P_j$.

It also states that the covered-first tie rule makes each chosen vertex maximum-degree in H_j, and that

$M - \nu(G) = |\{j : 0 \leq j < M, \epsilon_j = 0\}| - |W_M|$.

The left side is $\delta > 0$ and $|W_M| \geq 0$. Therefore there is an index with $\epsilon_j = 0$. Let q be the first. Then $\epsilon_j = 1$ for every $j < q$ and $\epsilon_q = 0$. Telescoping the exact matching losses gives $\nu(H_q) = \nu(G) - q$ and $\nu(H_{q+1}) = \nu(H_q)$.

For $j < q$ the deleted vertex has $H_j - degree\ \ell_{j+2}$. An original edge is deleted exactly when its first selected endpoint is removed. Summing these degrees consequently gives $e(G) - e(H_q) = T_q$. Every such selected vertex has maximum $H_j - degree$ by the tie-rule conclusion. These statements also cover $q = 0$.

At the q-th parent, the chosen vertex is uncovered by W_q. There is no $W_q - covered$ maximum-degree vertex of Q_q, because otherwise the tie rule would have chosen one. The exact degree formula and maximum-degree property in H_q give

$\Delta(H_q) = deg_{H_q}(v_{q+1}) = \ell_{q+2} - 1$.

Thus the conditional arbitrary-set result in the first paragraph applies to this same set X_q and graph $H = H_q$. Its retained pair has full diagonal sum $L' = L - T_q$, component budget $M' = M - q$, and largest blue size $\Delta(H)$.

Using the exact potential at q,

$\Phi(H) = P_q = \sum_{k=q+2}^{M+1}(\ell_k - 1) = L' - M'$.

The matching identity gives

$M' - \nu(H) = (M - q) - (\nu(G) - q) = \delta > 0$.

In particular $e(H) = \Phi(H) + \nu(H) < L'$.

It remains to verify the critical lower bound on the degree. Suppose $D = \Delta(H) = \ell_{q+2} - 1 \le 2$. Apply Lemma 136 to the retained actual pair F′, J′ as its own original pair, with cut zero. This pair consists of nonempty positive star forests by the first paragraph, and H arrows it. Its first diagonal value is ℓ_{q+2}, by the full-profile equality, so its critical degree is exactly D. The cut-zero actual suffix family has only this pair, and $\Delta(H) = D \le 2$. That theorem therefore requires $e(H) \ge L'$, contradicting the strict inequality just proved. Hence $D \ge 3$.

All uses of the fixed-matching theorem were valid for every permitted greedy execution and every initial maximum matching. The construction never replaces an edge of that initially fixed matching; it only restricts it off the chosen vertices. The arbitrary-set profile theorem then uses the very same actual residual, so the stated quantifiers and witness compatibility follow. □

Remark. A positive matching deficit forces a first greedy deletion that does not reduce matching number. At that point the actual graph reaches its critical degree, while preceding deletions preserve the potential surplus and deficit. The actual binary suffix lemma identifies one pair attaining the complete remaining profile, and low-degree avoidance forces critical degree at least three.

Depends on. Lemma 62, Lemma 95, Lemma 136.

Lemma 186 (Avoidance from a small regular component). *All graphs are finite and simple. Write $e(G)$ and $\nu(G)$ for edge count and maximum matching cardinality, and put $\Phi(G) = e(G) - \nu(G)$. Let F_1, F_2 be nonempty star forests with full nonincreasing positive integer edge-size lists $n_1,\dots, n_s$ and $m_1,\dots, m_t$, where $s, t \ge 1$ are integers. Copies are ordinary subgraphs with vertex-disjoint target components, with F_1 prescribed red and F_2 blue. Put $M = s + t - 1$ and*

$\ell_k = max\{n_i + m_j - 1 : 1 \le i \le s, 1 \le j \le t, i + j = k\}$

for every integer $2 \le k \le s + t$. Fix an integer $0 \le q < M$, put

$P_q = \sum_{k=q+2}^{s+t}(\ell_k - 1)$, $D = \ell_{q+2} - 1$,

and suppose $D \ge 6$. Let R be the number of initial entries equal to $D + 1$ in this natural tail. For a forest F, write F^β for deletion of its β leading whole components. Let A_q contain every ordered nonempty pair (F_1^α, F_2^β) with nonnegative integer cuts $\alpha + \beta = q$, $\alpha < s$, $\beta < t$.

Suppose D is even. Let $\sigma \ge 0$ be an integer and let H be a finite simple graph satisfying $\Phi(H) \le P_q + \sigma$ and $\delta := M - q - \nu(H) \ge 1$. Suppose H has a whole connected odd D-regular component C with a near-perfect matching, meaning a matching leaving one vertex uncovered, and $\nu(C) \le D$. If $R + \sigma < D/2$, then H has a red-blue edge coloring avoiding both entire prescribed targets of some member of A_q.

Every such avoiding H-coloring extends to avoidance of both full originals on the disjoint union of H and the canonical pool of q stars of sizes $\ell_2,\dots, \ell_{q+1}$. It also extends across any actual outer set X of q vertices when $H = G - X$, preserving every H-edge color. The conclusion imposes no maximum-degree condition outside C, original-head bound, short-prefix restriction, or smaller-target lower-bound premise. Unit components, repeated sizes and $q = 0$ are permitted. Both entire targets and their prescribed colors may be exchanged together. All orders, degrees, list entries, indices, cuts, matching sizes, deficits and allowances are integers in their stated domains.

Proof. Assume for contradiction that H arrows every member of A_q. Put $\mu = \nu(C)$. The near-perfect matching gives $|V(C)| = 2\mu + 1$. Simplicity and D-regularity give $|V(C)| \ge D + 1$, so $D/2 \le \mu \le D$ and $\mu \ge 3$. The

natural diagonal profile is nonincreasing: for any candidate on a later diagonal, lowering an index greater than one yields a candidate at least as large on the preceding diagonal. Hence $n_1 + m_1 \geq D + 2$.

We first show that neither entire original forest fits $2\mu + 1$ vertices. The full-family conclusion of Lemma 89 gives $q < \min(s, t)$, since $\delta \geq 1$. Suppose one original fits; call it F, with S components, edge total E and head A, and call the opposite original J, with T components and head B. Since each component has at least two vertices, $S \leq \mu$. The pair (J^q, F), with colors exchanged when required, is legal and arrowed by H. Matching numbers add on connected components, so the empty separator is matching-additive. The first clause of Lemma 153, applied to this pair and the selected whole component C, gives

$C \to$ (kK2,F), where $k = (T - q) - \nu(H) + \mu = \mu - S + \delta + 1 \geq 1$.

If $k > \mu$, coloring every C-edge in the matching color contradicts this arrowing. Thus $k \leq \mu$. Select k-1 vertices of C, color every edge meeting them in the matching color, and color all other edges in the F color. The matching color has a vertex cover of size k-1 and contains no k-edge matching. Every positive-edge F-colored component lies outside the selected vertices. Arrowing therefore implies

$E + S \leq 2\mu + 1 - (k - 1)$, hence $\mu \geq E + \delta - 1$. (1)

Write g_1,..., g_T for the full size list of J, so $g_1 = B$. The same full-family capacity theorem, applied with the appropriate orientation and discarding its nonnegative separator and prefix terms, gives

$S \geq 2 \sum_{j=1}^{T} floor((g_j - 1)/2) + 2\delta$

$\geq B - 2 + 2\delta$.

Also $E \geq S + A - 1$ and $A + B = n_1 + m_1 \geq D + 2$. Inserting these inequalities into (1) yields

$\mu \geq A + B - 4 + 3\delta \geq D + 1$,

contradicting $\mu \leq D$. Thus both full originals have more than $2\mu + 1$ vertices. No half-order hypothesis on the other components of H was needed: the localization used only its matching-additive clause.

Apply Lemma 19 at capacity $2\mu + 1$ to the two full originals. It constructs nonempty componentwise remnants U_1, U_2 with the following property: for any actual leading cut β leaving U_i^β nonempty, every copy of F_i^β in C disjoint union Q leaves U_i^β in Q. Let E_q be all nonempty actual total-q suffix pairs of (U_1, F_2) and (F_1, U_2), and let P_{end} be their full diagonal-envelope potential, with value zero for the empty family. The same fact, using $D \geq 6$ and $3 \leq \mu \leq D$, gives

$P_q - P_{end} \leq (D - 1)\mu + R$. (2)

Set $Q = H - V(C)$. The component is whole, so edges and matching numbers add between C and Q. Its regularity gives

$\Phi(C) = D(2\mu + 1)/2 - \mu = (D - 1)\mu + D/2$.

Consequently (2) and the strict allowance bound yield

$\Phi(Q) \leq P_q + \sigma - \Phi(C)$

$\leq P_{end} + R + \sigma - D/2 < P_{end}$.

Every graph has nonnegative Φ, since a matching is an edge subset. Thus E_q is nonempty. The finite-family potential theorem Lemma 56 supplies a Q-coloring avoiding an entire prescribed pair from E_q.

If that pair is (U_1^α, F_2^β), color every edge of C red. A red F_1^α would leave a red U_1^α in Q by the remnant property, which is impossible. A blue F_2^β would lie entirely in Q, also impossible. If the selected endpoint pair is (F_1^α, U_2^β), color C blue and use the color-exchanged argument. The cuts α, β are legal for the full originals because each U_i is componentwise contained in F_i. In either case the obtained H-coloring avoids an A_q member, contradicting the assumed simultaneous arrowing. This proves existence of the required avoiding coloring.

Finally Lemma 41 (I) extends any avoiding A_q coloring over the disjoint canonical q-star pool, and its part (II) extends it across every actual outer q-vertex set with $H = G - X$ while preserving all H-edge colors. At $q = 0$ both extensions are identities. Their hypotheses retain the full original lists and their prescribed colors, so these are the claimed full-target extensions. □

Remark. The regular component pays the entire capacity-remnant loss. A localization and full-target capacity argument first ensures that neither remnant vanishes.

Depends on. Lemma 19, Lemma 41, Lemma 56, Lemma 89, Lemma 153.

Lemma 187 (The octic seven-deletion frame from exact complement rows). *Use the finite integer certificate costs c_i, matching number ν, independence number α, and slack ps of Lemma 23. Let H be a finite simple 8-regular graph on 29 vertices with $\alpha(H) = 8$. For every maximum independent set S put $G = H - S$, and assume $\nu(G) = 10$, $ps(G) = 0$, and $c_i(H) = 8 + c_i(G)$ for every integer $1 \le i \le 7$. A complete process from G repeatedly deletes any vertex of current maximum degree while that degree is at least three, and stops at maximum degree at most two. Assume that at every deletion parent of every such process, a matching covers all vertices of its current maximum degree. Then, for every S and every complete process, $\alpha(G) = 8$, there are exactly seven deletions, their degrees in order are (7,7,7,6,6,6,4), and the terminal induced graph is the disjoint union of three triangles and five isolated vertices. The positive rows $(c_1(H), \dots, c_7(H))$ are (18,18,15,14,14,11,8). Moreover, for every terminal vertex set R arising in any such process there are a vertex x outside R and a set A such that $V(H) = R$ disjoint union $\{x\}$ disjoint union A, both $H[R]$ and $H[A]$ are the disjoint union of three triangles and five isolated vertices, and x has exactly four neighbors in each of R and A. The pair (x, A) is obtained by taking all five isolates and one vertex of each triangle of $H[R]$ as a new maximum independent set, deleting one remaining vertex of each triangle at degree seven and then their three triangle mates at degree six, and finally deleting x at degree four. All orders, degrees, cardinalities, costs, matching sizes and indices are integers. The matching choices at distinct deletion parents are not required to agree.*

Proof. Fix a maximum independent set S and put $G = H - S$. Such sets exist because H is finite. It has $|S| = 8$. Maximality means each outside vertex has an S-neighbor, so $\Delta(G) \le 7$. Regularity gives $e(H) = 8 \cdot 29/2 = 116$, $n(G) = 21$ and $e(G) = 116 - 8 \cdot 8 = 52$. By hypothesis $\nu(G) = 10$ and $ps(G) = 0$. A complete current-maximum-degree deletion process exists: choose an eligible vertex whenever the current maximum is at least three; each deletion removes one vertex and hence the process terminates. The hypothesis supplies a covering matching at each parent of any process so obtained. Fix any such process, write t for its length, R for its terminal induced graph, and $a_G = \alpha(G) \le 8$.

Apply Lemma 120 to this process. It gives $n(R) = 21 - t$, $\nu(R) = 10 - t$, zero slack at every prefix, and the exact positive-row losses. In particular $b := 10 - c_1(G)$ is a nonnegative integer and equals the number of positive even-order path components of R. The source row identity gives $b = 18 - c_1(H)$, independent of S and of the process. The density bound of Lemma 146 with $D = 7$ gives

$$52 \le -4 \cdot 21 + 11 \cdot 10 + 4a_G - 6b = 26 + 4a_G - 6b \le 58 - 6b.$$

Thus $a_G \ge 7$ and $b \le 1$.

If $b = 1$, equality holds in this density bound and $a_G = 8$. Its equality case says that all $t = 2 \cdot 10 - 21 + 8 - 1 = 6$ deleted vertices have original G-degree seven and form an independent set U in G, hence also in H. Since H is 8-regular, each has exactly one neighbor in S. Apply Lemma 76 to the maximum independent set S and U to obtain $c_1(H) \le 29 - 8 - 6 = 15$. But $b = 1$ gives $c_1(H) = 17$, a contradiction. Therefore $b = 0$ and $c_1(H) = 18$.

Let r_7 be the number of degree-seven deletions in any process. Each such vertex had original G-degree seven. No two are adjacent: if adjacent vertices were deleted in that order, the second would already have lost an incident edge and could not be deleted at degree seven. Their set is independent in H, and every member has exactly one S-neighbor. The same private-neighbor bound gives $18 = c_1(H) \le 29 - 8 - r_7$, so $r_7 \le 3$. Also Lemma 120 gives $r_7 = c_6(G)$, because the terminal has maximum degree at most two and hence row six zero. The row identity makes $r_7 = c_6(H) - 8$, so this count has the same value for every maximum independent S and every process. The number r_6 of degree-six deletions is likewise $c_5(G) - c_6(G) = c_5(H) - c_6(H)$, independent of all those choices.

The full density identity in Lemma 146, with $b = 0$, reads

$$4a_G - 26 = \sum_{deleted}(7 - d) + 4\ell + \sum_C F_7(C),$$

where d is the deletion-time degree, $\ell = a_G - \alpha(R) \ge 0$, and C ranges over the terminal components. Every term is nonnegative. There are no positive even-order paths. Among the remaining components, isolates and triangles have $F_7 = 0$; every other component has F_7 at least five. The components contributing exactly five are a three-vertex path P_3 and a five-cycle C_5. All other positive contributions are at least ten, as follows from the component formulas in that fact.

If $a_G = 7$, the left side is two. The small-deficit conclusion of Lemma 146 applies because $2 \le 7 - 4$. It gives $t = 2 \cdot 10 - 21 + 7 = 6$ and at least $t - 2 = 4$ degree-seven roots. This contradicts $r_7 \le 3$. Hence $a_G = 8$, and the left side is six.

Now ℓ is zero or one. If $\ell = 1$, the remaining deficit is two, so all terminal components are triangles and isolates. Let z and i be their counts. Then $z = \nu(R) = 10 - t$, $z + i = \alpha(R) = 7$ and $3z + i = n(R) = 21 - t$. These equations give $t = 6$. The degree deficits sum to two, so at most two deletions have degree below seven and $r_7 \ge 4$, again impossible. Thus $\ell = 0$ and $\alpha(R) = 8$.

If any terminal component has positive F_7, there is exactly one such component and it is P_3 or C_5. All other components are triangles and isolates, and the degree deficits sum to one. A triangle or isolate has value zero for $|V(C)| - 2\nu(C) - \alpha(C)$; each of P_3 and C_5 has value minus one. Adding these values over R gives

$(21 - t) - 2(10 - t) - 8 = t - 7 = -1,$

so $t = 6$. At most one deletion has degree below seven; hence $r_7 \ge 5$, a contradiction. Thus R consists entirely of triangles and isolates. With z, i as their counts, the equations $z = 10 - t$, $z + i = 8$ and $3z + i = 21 - t$ give $t = 7$, $z = 3$ and $i = 5$. All independence and component contributions vanish, so the sum of the seven degree deficits is six. These conclusions were proved for every S and every complete process.

Choose any terminal vertex set R. Let S′ contain all five isolates of H[R] and one vertex from each of its three triangles. It is independent and has eight vertices, so $\alpha(H) = 8$ makes it a maximum independent set. Label the two unchosen vertices of the jth triangle by u_j, v_j for $j = 1, 2, 3$. The vertices u_1, u_2, u_3 are pairwise nonadjacent and each has exactly one neighbor in S′, namely the chosen vertex of its triangle. Thus they have degree seven in H-S′. Since its maximum degree is at most seven, they can be deleted successively as three legal maximum-degree vertices. This partial sequence extends to a complete one by the finite rule. Its r_7 is at least three. As r_7 is independent of S and all deletion choices and was bounded above by three, every process has exactly three degree-seven deletions.

After the u_j are deleted, each v_j has degree six: its only neighbors in R were the two deleted vertices of its triangle, and there are no edges from it to other components of H[R]. The v_j are pairwise nonadjacent. At this point the current maximum cannot exceed six. If it were seven, the partial sequence could be continued by a fourth degree-seven deletion, contradicting the count just proved. Therefore v_1, v_2, v_3 may be deleted successively at current maximum degree six. This proves $r_6 \ge 3$ in the replay process and hence in every process, because r_6 also has a choice-independent expression in the H-rows. Each process has seven deletions, exactly three of degree seven, and degree-deficit sum six. Four degree-six deletions would have deficit sum four, so there are exactly three of degree six. The sole remaining degree has deficit three, and thus equals four. Deletion degrees are nonincreasing because induced deletion cannot increase maximum degree. Every degree list is therefore (7,7,7,6,6,6,4).

The terminal has three triangles and five isolates, so its matching number is three. Lemma 120 gives $c_1(R) = 3$ because $b = 0$ and $c_2(R) = 3$ because there are three odd cycles. For every $i \ge 3$ the maximum-degree-two graph R has row i zero by the tail bound in Lemma 23. The exact process formula adds one to row i for each deletion of degree greater than i. The displayed word therefore gives the G-rows at indices 1,..., 7 as (10,10,7,6,6,3,0). The given complement identities yield the asserted H-rows after adding eight.

Finally start from the chosen arbitrary R and use the replay just described. The new independent set S′ together with the six deleted triangle vertices is exactly R. After those six deletions the remaining induced graph is H[V(H) minus R]. Its process has precisely one remaining deletion, at degree four; call its vertex x, and put $A = V(H)\ minus(Runion\{x\})$. The graph H[A] is the terminal of this complete replay, hence has exactly three triangles and five isolates. The sets R, {x}, A are disjoint and cover $V(H)$. By its current deletion degree, x has four neighbors in A. Its original degree is eight and all its other neighbors lie in R, so it has four neighbors in R as well. This supplies the actual vertex, sets, induced structures and both degree requirements for every original terminal R. □

Remark. The exact density deficit leaves only triangles and isolates. Replaying from a terminal transversal forces three degree-seven and three degree-six roots, so the last root has degree four and exposes the opposite terminal.

Depends on. Lemma 23, Lemma 76, Lemma 120, Lemma 146.

Lemma 188 (Reduction of three small octic orders). *Use the finite integer certificate costs c_i, matching number ν, independence number α, maximum degree Δ, excess $ex = c_0 - \nu$, slack ps, signed $gap = ex - ps$ and top missing-allocation run tr from Lemma 172. A graph satisfies T_- when positive gap implies gap one and $tr + ex \leq \max(4, \lfloor \Delta/2 \rfloor + 1)$. Put $f(J) = |V(J)| - 2\nu(J)$ and $E_7(J) = e(J) + 4|V(J)| - 11\nu(J) - 4\alpha(J)$.*

Let H be a nonempty connected eight-regular finite simple factor-critical graph, meaning that deleting any vertex leaves a perfect matching. Suppose $ex(H) = 4$, $gap(H) = 1$, and every proper induced subgraph satisfies T_-. Put $k = \nu(H)$ and assume k belongs to {11,12,13}. Assume that every proper induced subgraph J of H with $\Delta(J) \leq 7$ has the following two properties: (a) if $ps(J) = 0$ and $\Delta(J) \geq 3$, a matching covers all vertices of degree $\Delta(J)$; (b) if $ps(J) = 1$ and $\alpha(J) - f(J) \leq 5$, then $E_7(J) \leq 11$.

Then $k = 13$ and $\alpha(H) = 7$. The minimum cardinality of a dominating set of H is seven, where a set dominates when every vertex outside it has a neighbor in it. Every maximal independent set has seven vertices. For every such set S, the induced graph $G = H - S$ has $|V(G)| = 20$, $e(G) = 52$, $\nu(G) = 10$, $\Delta(G) \leq 7$, $ps(G) = 0$, and $c_i(H) = 7 + c_i(G)$ for every integer $1 \leq i \leq 7$. Moreover $c_7(G) = 0$ and $c_7(H) = 7$. Thus no H satisfying these hypotheses has order 23 or 25.

The two induced-subgraph conditions are explicit premises. All orders, degrees, cardinalities, costs, matching and independence numbers, indices, and k are integers in their displayed nonnegative or positive domains.

Proof. We use Lemma 61 for nonnegative slack, row restriction and restoration, covered-degree vanishing and the covered deletion inequality. In particular, when a graph J has a matching covering its vertices of actual maximum degree d and v has degree d, put $h = \nu(J) - \nu(J - v)$. Then $ps(J) \geq ps(J - v) + 1 - h$. If equality holds and $d \geq 2$, every positive row below d decreases by one: the exact deletion identity has d-1 possible row losses, each zero or one, whose sum must equal d-1. A nonregular factor-critical graph with an edge is covered at its actual maximum degree, by taking a perfect matching after omitting a vertex below that degree. Deleting any maximum-degree vertex preserves its matching number. The inequality and nonnegative child slack consequently give $ps(J) \geq 1$ for such a graph.

By Lemma 155, every even induced graph of H of order at least 2k-6 has a perfect matching, and every odd induced graph of order at least 2k-5 is factor-critical. Also $c_1(H) \leq k + 3$ when $k = 11$ or 12. Whenever such an induced graph has maximum degree at most seven, a matching covers its degree-seven vertices: use its perfect matching in the even case; in the odd case, degree-sum parity excludes seven-regularity, so omit a vertex below degree seven and match the remainder. If its maximum degree is smaller there are no marked vertices. Its seventh row is therefore zero.

Condition (a) allows a complete covered maximum-degree deletion process on every zero-slack proper induced graph of degree at most seven. Indeed the deletion inequality forces each child to have zero slack and matching loss one. Iterate until maximum degree is at most two. Thus Lemma 146 applies to every such graph J, giving

$e(J) \leq -4|V(J)| + 11\nu(J) + 4\alpha(J) - 6b$, $b = \nu(J) - c_1(J) \geq 0$. (1)

In particular $E_7(J) \leq 0$. Together with (b), this yields

$ps(J) \leq 1$ and $\alpha(J) - f(J) \leq 5$ imply $E_7(J) \leq 11$. (2)

All zero-slack processes used below are of this permitted kind.

We first give dominating sets of H of sizes at most five, six and seven when $k = 11$, 12, 13 respectively. In an eight-regular graph on n vertices, if i vertices have been selected and r remain undominated, the nine-element closed neighborhood of every undominated vertex avoids the selected set. Counting incidences shows that some unselected vertex dominates at least $ceil(9r/(n - i))$ of them. For $n = 25$ the successive residual upper bounds are 25, 16, 10, 6, 3, 1, 0, and for $n = 27$ they are 27, 18, 11, 7, 4, 2, 1, 0. The upper-bound update is nondecreasing in its nonnegative integer input while $n - i \geq 9$, as in these lists: the ceiling increases by at most one per unit increase. The listed chains therefore construct the required sets of sizes six and seven.

At $n = 23$, four greedy choices leave at most two undominated vertices, with residual bounds 23, 14, 8, 4, 2. One additional choice suffices if there is at most one remaining vertex, or if the two remaining closed neighborhoods intersect. Otherwise choose two vertices p, q with disjoint closed neighborhoods. They dominate eighteen vertices, leaving a set W of size five. This W can be dominated by at most three vertices of H. If H[W] has two disjoint edges, choose one endpoint of each and the fifth vertex. If it is a triangle plus isolates, choose one triangle vertex and both isolates. If it is a star with at least two edges plus isolates, choose its center and its isolates. These cases exhaust all graphs except an edgeless graph and one edge plus three

isolates: a family of pairwise intersecting edges is a star or a triangle, because two edges sharing a vertex force any edge avoiding that vertex to join their other endpoints, after which no fourth distinct edge can meet all three.

For one edge plus three isolates, the isolates send twenty-four neighbor incidences outside W union {p, q}, a set of sixteen vertices. One vertex there neighbors at least two isolates. Choose it, an undominated isolate if any, and an endpoint of the edge. For edgeless H[W], forty incidences into those sixteen vertices give a vertex neighboring at least three vertices of W; choose it and the at most two remaining vertices. In all cases at most three extra vertices dominate W, so p, q and those vertices dominate H with at most five vertices.

Pad the dominating sets to exactly five, six or seven vertices in the respective orders. Their complements have orders eighteen, nineteen or twenty and maximum degree at most seven. The robust matching conclusion above makes their seventh rows zero. Restoring the padded set shows

$c_7(H) \le 5, 6, 7$ at $k = 11, 12, 13$ respectively. (3)

Let S be any maximum independent set, put $a = |S|$ and $G = H - S$. The exact independent-deletion result Lemma 172 gives $a \ge 3$, $\Delta(G) \le 7$, and

$\eta = \nu(G) - (k + 1 - a)$ in {0,1,2,3}, $ps(G) = 3 - \eta$,

$c_i(H) = a + c_i(G)$ for $1 \le i \le 7$,

$|V(G)| = 2k + 1 - a$, $e(G) = 8k + 4 - 8a$. (4)

If $a \le 7$, robust matching gives $\eta = floor((a - 1)/2)$ and $c_7(G) = 0$. If $a \ge 7$, the last matching bound of Lemma 155 gives $\eta \ge 3$, so $\eta = 3$ and $ps(G) = 0$. Condition (a) then gives seventh-row vanishing when $\Delta(G) = 7$; lower degree gives it directly. Thus in every case $c_7(H) = a$. Consequently (3) bounds a by five, six or seven, respectively.

The minimum dominating-set size is a. Otherwise pad a smaller dominating set to a-1 vertices. Its complement has order at least 2k-5 and maximum degree at most seven, so robustness again gives zero seventh row. Restoration would yield $c_7(H) \le a - 1$, a contradiction. Every maximal independent set is dominating and has size at most a; hence each has size a. All independent-cut conclusions in (4) apply to every such set.

We exclude $a = 3$ first. Here G has even order 2k-2, matching number k-1, edge count 8k-20 and slack two. Its average degree exceeds six and its maximum degree is at most seven, so a maximum-degree vertex has degree seven. Delete one to obtain J. This odd graph is factor-critical and has order 2k-3 and 8k-27 edges. Its average degree still exceeds six; seven-regularity on odd order is impossible. Thus J is nonregular, covered at maximum degree seven, and has slack at most two by the first deletion. Delete a maximum-degree vertex of J to obtain K. The matching-preserving deletion gives $ps(K) \le 1$. The graph K has even order 2k-4, matching number k-2, exactly 8k-34 edges and independence at most three. Hence $E_7(K) \ge 5k - 40 \ge 15$. Its deficiency $f(K)$ is zero, so (2) applies and gives a contradiction.

For $a = 4$, G is factor-critical of order 2k-3, matching number k-2, with 8k-28 edges and slack two. Its average degree exceeds six, so it is nonregular of maximum degree seven. Delete a maximum-degree vertex to obtain K, which has order 2k-4, matching number k-2, 8k-35 edges and slack at most one. Its independence is at most four and $f(K) = 0$, so $E_7(K) \ge 5k - 45$. At $k = 12$ or 13 this exceeds eleven, contradicting (2).

At $k = 11$, K has eighteen vertices, 53 edges and matching number nine. The bound $E_7(K) \ge 10$ and the zero-slack part of (1) imply $ps(K) = 1$. In (4), $c_1(G) = c_1(H) - 4 \le 10$. The deletion G to K has matching loss zero and slack decrease exactly one, so its first row decreases by one and $c_1(K) \le 9$. Choose a maximum-degree vertex of K, of degree six or seven, and delete it to obtain J′. This seventeen-vertex graph is factor-critical with matching number eight and 46 or 47 edges. It is nonregular because neither 92 nor 94 is divisible by seventeen. The covered deletion gives $ps(J') \le 1$, and nonregular factor-criticality gives $ps(J') \ge 1$. Thus $ps(J') = 1$, and the first row decreases by one. Delete a maximum-degree vertex of J′ to obtain L. Its matching number is still eight, its slack is zero, it has sixteen vertices and at least 39 edges, and again the first row decreases by one. Thus $c_1(L) \le 7$, so $b = \nu(L) - c_1(L) \ge 1$. Its independence is at most four. Bound (1) gives $e(L) \le -64 + 88 + 16 - 6 = 34$, contradicting $e(L) \ge 39$. This completes $a = 4$.

For $a = 5$, G has even order 2k-4, matching number k-2, 8k-36 edges and slack one. At $k = 13$, it has $E_7(G) \ge 5k - 50 = 15$ and $f(G) = 0$, contradicting (2). Suppose $k = 11$ or 12. Its maximum degree is six or seven. Delete a maximum-degree vertex to obtain J. Robustness makes J factor-critical of order 2k-5, with matching number k-3. Its edge count is 45 or 46 at $k = 11$, and 53 or 54 at $k = 12$. The respective doubled edge counts are not divisible by seventeen or nineteen, so J is nonregular. It follows exactly as in the preceding

case that $ps(J) = 1$. Delete a maximum-degree vertex of J to obtain K. Both deletion degrees are at least six and at most seven. The graph K has order 2k-6, matching number k-3, slack zero, and at least 8k-50 edges. Both deletions have first-row loss one, while $c_1(G) = c_1(H) - 5 \le k - 2$. Hence $c_1(K) \le k - 4$ and $b \ge 1$ in (1). Since $\alpha(K) \le 5$,

$8k - 50 \le e(K) \le 3k + 5$. (5)

At $k = 12$ this reads $46 \le 41$, impossible.

At $k = 11$ every inequality in (5) is equality: K has sixteen vertices, 38 edges, matching number eight, independence five and $b = 1$. The equality conclusion of Lemma 146 gives an induced terminal R consisting of three triangles, one edge and one isolate. It has twelve vertices and ten edges. In the original eight-regular H on 23 vertices, its exterior W has eleven vertices, and degree summation gives $e(H[W]) = 4 \cdot 23 - 8 \cdot 12 + 10 = 6$. If two of those six edges meet, their common endpoint and one endpoint of each other edge form a vertex cover of size at most five. Its complement is an independent set of at least six vertices in H, contradicting $a = 5$. If no two edges meet, the six edges require twelve vertices, exceeding $|W| = 11$. This excludes $a = 5$ at $k = 11$ as well.

It remains to consider $a = 6$, possible only at $k = 12$ or 13 by (3). Here G is factor-critical of order 2k-5, matching number k-3, with 8k-44 edges and slack one. Its edge counts are 52 on nineteen vertices or 60 on twenty-one vertices, so it is nonregular. Its maximum degree is six or seven. Deleting a maximum-degree vertex gives a zero-slack graph K of order 2k-6 with a perfect matching of k-3 edges, at least 8k-51 edges, and independence at most six.

At $k = 12$, $c_1(G) \le 15 - 6 = 9$. The matching-preserving deletion has slack decrease one, so $c_1(K) \le 8$. Consequence (i) of Lemma 146 applies to this eighteen-vertex graph with at least 45 edges. Its terminal is three triangles, one edge and two isolates. The first row of Lemma 30 gives a dominating set of H of size at most five, contradicting the already proved minimum dominating-set size $a = 6$.

At $k = 13$, K has twenty vertices, matching number ten, at least 53 edges and independence at most six. Bound (1) forces $\alpha(K) = 6$ and $b = 0$, and gives $e(K) \le 54$. If $e(K) = 53$, consequence (ii) of Lemma 146 gives a terminal of four triangles and two isolates. If $e(K) = 54$, equality holds in (1), and its equality conclusion gives exactly the same terminal: there are $20 - 10 - 6 = 4$ triangles, $12 - 20 + 10 = 2$ isolates and $b = 0$ edges. The second row of Lemma 30 again produces a dominating five-set in H, contradicting $a = 6$.

All possible a at $k = 11$ and 12 have now been excluded, and at $k = 13$ only $a = 7$ remains. Formula (4) and robustness give $\eta = 3$, $ps(G) = 0$, order twenty, edge count 52 and matching number ten for every maximum independent set. The degree and exact row identities were already established, as were $c_7(G) = 0$, $c_7(H) = 7$, the minimum dominating-set size and well-coveredness. These prove all conclusions. □

Remark. Robust matchings turn independent-set deletion into an exact slack calculation. Density removes the small independence cases, and at the endpoint a sparse exterior or a small dominating set contradicts the original graph.

Depends on. Lemma 30, Lemma 61, Lemma 146, Lemma 155, Lemma 172.

Lemma 189 (Strict potential from a cubic matching projection). *Let $s, t \ge 1$ be integers, and let $n_1 \ge \cdots \ge n_s \ge 1$ and $m_1 \ge \cdots \ge m_t \ge 1$ be integer lists. Let F_1, F_2 be the respective vertex-disjoint unions of stars of these edge sizes, prescribed red and blue. All graphs are finite and simple, and copies are ordinary subgraphs with vertex-disjoint target components. A graph arrows an ordered pair when every red-blue edge coloring contains its entire first target in red or its entire second target in blue.*

Put $M = s + t - 1$ and $\ell_j = \max\{n_i + m_b - 1 : 1 \le i \le s, 1 \le b \le t, i + b = j\}$ for every integer $2 \le j \le M + 1$. Fix an integer $0 \le q < M$. Let A_q contain every ordered pair of nonempty actual suffix forests obtained by deleting α leading components of F_1 and β leading components of F_2, where $0 \le \alpha < s$, $0 \le \beta < t$ and $\alpha + \beta = q$ are integers. Put $K = M - q$, $\lambda_i = \ell_{q+i+1} - 1$ for $1 \le i \le K$, and $P_q = \sum_{i=1}^{K} \lambda_i$. Suppose $\lambda_1 = 4$. Let N be the number of positive λ_i, R the number equal to four, and A the number equal to three.

For a positive-edge star forest F let $c(F)$ be its number of components, $r(F)$ its number of components with at least two edges, and F^- the forest obtained by shortening each such component by one edge and discarding all one-edge components. Form a family C_q by taking both orders (F, J_1) of the entries of each A_q member: include $(r(F)K_2, J_1)$

when $r(F) \geq 1$, and include $(c(F)K_2, J_1^-)$ when J_1^- is nonempty. Here K_2 is one edge, and a multiple of K_2 is a matching.

Assume $A \geq 1$ and that C_q contains a pair (pK_2, J), where $p \geq 1$ is an integer, the largest star size of J is three with integer multiplicity $f \geq 1$, and

$p + c(J) - 1 = N$, $p + f - 1 = R + A$.

Then every graph H with $\Delta(H) \leq 4$ which arrows every member of A_q satisfies

$\Phi(H) := e(H) - \nu(H) \geq P_q + 1$,

where $e(H)$ is its edge count and $\nu(H)$ its maximum matching cardinality.

All list lengths, indices, degrees, component counts, cardinalities, matching sizes, and parameters in the displayed domains are integers. Unit stars, repeated sizes, isolated host vertices, $q = 0$, and an empty forest of components of J smaller than three are allowed. No matching-deficit or binary-target hypothesis is imposed.

Proof. The family A_q is finite and nonempty. To see nonemptiness, choose an integer α in $[\max(0, q - t + 1), \min(q, s - 1)]$ and put $\beta = q - \alpha$; the interval is nonempty since $q \leq s + t - 2$. Every member has component budget K. Its full diagonal envelope is ℓ_{q+2},..., ℓ_{M+1}. Indeed, a local candidate with indices i, b and cuts α, β comes from original indices $I = \alpha + i$ and $B = \beta + b$, whose sum is $q + i + b$. Conversely, for original indices I, B with $I + B = q + k$ and $k \geq 2$, choose $\alpha = \min(q, I - 1)$, $\beta = q - \alpha$. Then $0 \leq \alpha \leq I - 1$ and $0 \leq \beta \leq B - 1$, so these are legal cuts recovering that same candidate on local diagonal k. Thus the candidate sets, and their maxima, agree.

The full original profile is nonincreasing: from a candidate on any diagonal after the first, decrease an index greater than one; the corresponding star size cannot decrease. Its tail λ is therefore nonincreasing and nonnegative. Since $\lambda_1 = 4$, its first R entries are four, its next A entries are three, and its first N entries are precisely the positive ones.

Apply Lemma 56 to A_q. It gives $\Phi(H) \geq P_q$. Suppose equality. Fix an arbitrary maximum matching W of H and let $Q = H - W$, removing only its edges. By Lemma 95, applied directly to the same finite family A_q and its envelope just established, every sequence of maximum-degree vertex deletions in Q has successive degrees λ_1,..., λ_K. After its first N steps the graph is edgeless, because the remaining edge count is the sum of the remaining λ entries, which is zero. Before step N each specified degree is positive.

Let S be the set of degree-four vertices of H and put $I = S$ minus $V(W)$. The set I is independent, since an edge between two W-uncovered vertices could be added to W. Its vertices are exactly the degree-four vertices of Q: a W-covered vertex loses one incident edge, and every other vertex keeps its degree. Deleting vertices of I in any order preserves degree four at every remaining vertex of I. After all are deleted the maximum degree is at most three. Comparing these legal greedy choices with the exact profile shows $|I| = R$: exhaustion earlier would contradict the next prescribed degree four, and having any remaining vertex after R deletions would contradict the next prescribed degree three. The latter entry exists since $A \geq 1$. Put $T = Q - I$. Every greedy deletion sequence in T consequently has exactly N-R positive-degree steps, starts with A steps of degree three, and then has degrees at most two. In particular $\Delta(T) = 3$, and the vertices in any complete such sequence form a vertex cover C of T of size N-R.

We next check the projection on this same Q. For either orientation (F, J_1) of an A_q member, color every W-edge red after specifying any coloring of Q. A red copy of F in H loses at most one edge at each of its star centers on deletion of W. Its nonunit components therefore retain a copy of F^-, which contains a matching of $r(F)$ edges; all of its component vertex sets remain disjoint. A blue J_1 lies wholly in Q. This proves Q arrows $(r(F)K_2, J_1)$ whenever $r(F) \geq 1$. Coloring W blue instead proves Q arrows $(c(F)K_2, J_1^-)$ whenever J_1^- is nonempty: a red full F in Q contains $c(F)$ disjoint edges, and a blue full J_1 shortens as described. Exchanging color names justifies both orientations. Thus Q arrows the selected (pK_2, J).

Write $J = fK_{1,3}$ disjoint union J_0, where J_0 has $t_0 = c(J) - f \geq 0$ components, each of size one or two. The hypotheses give

$N = R + A + t_0$.

Set $a = \min(R, p - 1)$ and $b = R - a$. Then $0 \leq a < p$ and $0 \leq b < f$. For the second bound, if $R \leq p - 1$ then $b = 0$; otherwise $b = R - p + 1 = f - A \leq f - 1$. Put $p' = p - a$ and $f' = f - b$. These are positive integers satisfying $p' + f' - 1 = A$.

The graph T arrows $(p'K_2, f'K_{1,3} disjoint union J_0)$. Here is the full deletion argument. If a T-coloring avoided this pair, partition the independent set I into sets of sizes a, b and color every edge incident with the first set red and every edge incident with the second set blue. There are no edges inside I, so the prescriptions are compatible and color every edge of Q outside T. A red p-edge matching would leave at least $p - a = p'$ red edges in T. A full blue J could have at most b of its vertex-disjoint components meeting the second set, and no blue edge meets the first. Removing those at most b components leaves stars dominating the sorted suffix obtained by deleting b largest components of J. Since $b < f$, that suffix is exactly $f'K_{1,3}$ disjoint union J_0. Trimming larger surviving stars gives the forbidden blue target in T. Both alternatives contradict the chosen T-coloring, proving its asserted arrowing.

Every independent set of degree-three vertices of T has size at most A. If it had $A + 1$ vertices, deleting them first would be a legal greedy sequence with $A + 1$ initial degrees three, contrary to the exact inherited profile. Apply Lemma 36 to T with $d = 3$, matching size p', largest-star multiplicity f', smaller forest J_0, and the cover C of size $N - R = A + t_0$. It gives an independent set U of A degree-three vertices, pairwise disjoint closed T-neighborhoods, and a lower graph $B = T - U$ with $\Delta(B) \leq 2$ and $\nu(B) = \tau(B) = t_0$. It also states that no T-edge joins U to any $t_0 - vertex$ cover of B.

Deleting all U vertices first in T is legal because they are independent and all have degree three. Every subsequent greedy sequence in B is therefore a continuation of the same exact profile in Q, and it has exactly $N - R - A = t_0$ positive-degree deletions before becoming edgeless. The last clause of Lemma 36 now applies and says that U is the entire degree-three vertex set of T. Thus for the arbitrary original maximum matching W, the entire degree-three set of (H-W)-I is independent and has pairwise disjoint closed neighborhoods there.

Apply Lemma 148 with $G = H$ and $D = 4$. Its family of maximum matchings attaining the minimum uncovered S-count is nonempty. Every maximum matching has exactly R uncovered S-vertices by the argument above, so that minimum equals R. The independence and disjoint-neighborhood conditions just proved hold for every member of the required family. The cited theorem therefore gives exactly R whole K_5 components of H. Since $R \geq 1$, select one.

Replace that component as in Lemma 22 by disjoint stars of sizes five, four and one. The resulting graph H_{star} still arrows every member of the same A_q and satisfies $\Phi(H_{star}) = \Phi(H) - 1 = P_q - 1$. The finite-family bound Lemma 56 applies without a maximum-degree restriction and gives $\Phi(H_{star}) \geq P_q$. This contradiction excludes equality. Since $\Phi(H)$ and P_q are integers and $\Phi(H) \geq P_q$, the desired bound is $\Phi(H) \geq P_q + 1$. □

Remark. At exact potential, the same fixed matching complement has an exact greedy profile. A full cubic projection turns its next degree layer into disjoint centers for every maximum matching. Clique localization then provides a whole five-vertex component whose simultaneous coloring replacement lowers the potential.

Depends on. Lemma 22, Lemma 36, Lemma 56, Lemma 95, Lemma 148.

Lemma 190 (Size bounds for tight odd complementary levels). *Use the finite integer certificate costs $c_\theta(H)$ and $c_0(H)$ of Lemma 23. Let $d \geq 4$ be an even integer, and let H be a nonempty connected d-regular finite simple graph such that H-v has a perfect matching for every vertex v. Put $n = |V(H)|$. Let $\theta, \theta' \geq 1$ be odd integers with $\theta + \theta' = d$. If $c_\theta(H) + c_{\theta'}(H) = n + 1$, then $\theta, \theta' \geq 3$, $c_{\theta'}(H) \geq 5\theta + 8$, $c_\theta(H) \geq 5\theta' + 8$, and $n \geq 5d + 15$. Consequently the inequality $c_\theta(H) + c_{\theta'}(H) \leq n$ holds if $n \leq 5d + 13$ or if $c_0(H) \leq 5max(\theta, \theta') + 7$. No assumption on gaps, minimality, or proper induced subgraphs is imposed. All graph orders, degrees, thresholds, costs, matching sizes, and cardinalities are integers in their displayed positive or nonnegative domains.*

Proof. Assume the displayed equality. We use the configurations and weighted potential of Lemma 119 for thresholds θ, theta'. Choose any optimal configuration. Let x be its uncovered vertex, S its part of threshold t, and O the other part, of threshold t', so $\{t, t'\} = \{\theta, \theta'\}$. The cited fact gives balance, internal matching edges, odd $|S|$, even $|O|$, and

$c_t(H) = |O| + 1, c_{t'}(H) = |S|$. (1)

Write $J = H[S]$. For a vertex u in either part, define $def(u)$ as its part's threshold minus its number of H-neighbors in that part. All deficiencies are nonnegative: in H-M, moving a covered vertex of threshold a to the other part of threshold b would lower the potential if it had at least a internal neighbors, because its total degree is $a + b - 1$. Its internal H-M degree is therefore at most a-1; the internal matching edge raises this by one. The uncovered vertex has internal degree exactly its threshold by balance.

We first establish structural properties that apply to every such optimum. By Lemma 169, decompose J into its exposable vertex set D, its external neighbor set A, and the remaining vertex set C. The graph J has a matching missing exactly x, so J[D] has exactly $|A|+1$ connected components, each factor-critical, and J[C] has a perfect matching. Every vertex of D has degree t in J by Lemma 119, hence deficiency zero; no J-edge joins D to C.

We will repeatedly exclude a vertex set W of size $|S|$ whose induced graph has maximum degree at most t and has a matching covering every degree-t vertex. Indeed Lemma 35 then supplies every degree splitting of total t on H[W]. Taking the exceptional parts empty gives $c_t(H[W]) = 0$ by Lemma 23. Extension Lemma 72 gives $c_t(H) \le n - |W| = |O|$, contradicting (1). (2)

There is no edge between a deficiency-zero vertex u of S and a deficiency-zero vertex z of O. To prove this, suppose uz is such an edge. If $u = x$, moving x to O changes the potential by zero, since x has t neighbors in S and t' in O. The result is optimal. The vertex z now has t' internal H-M neighbors and t-1 across, so moving z changes the potential by $-1 + (t-1)/t < 0$, a contradiction. If u differs from x, it has t-1 internal H-M neighbors and t' across. Move u to O, changing the potential by $1/t$. Now z has t' internal H-M neighbors and t-1 across, so moving z to S changes it by $-1/t$. The final configuration is optimal. The unchanged matching edge at u now crosses the partition: its partner stayed in S and cannot be z because the original matching edges were internal. This contradicts Lemma 119. (3)

Every deficiency-zero vertex z in O has exactly t neighbors in S, all of positive deficiency and all in C. The number and the positive deficiency follow from regularity and (3), and exclude D. Suppose a neighbor a belongs to A. Choose u_0 in D adjacent to a and a maximum matching N of J missing u_0; such a matching exists by the definition of D. Let b be the N-partner of a. Put $W = (Sminus\{a\})$ union {z} and restrict N by deleting ab. This matching covers all W-vertices except u_0, b, z. All degrees in H[W] are at most t: any vertex gaining a neighbor z had positive deficiency before the change. The degrees of u_0 and z are t-1. The degree of b is at most t-1 as well: it loses its neighbor a; if it gains z, its previous deficiency was at least one, while if it does not gain z, its previous degree was at most t. This contradicts (2), proving the assertion.

Fix any perfect matching N_C of J[C], and write w' for the partner of w in it. For a deficiency-zero z in O and any w in C adjacent to z, we have

$def(w') \le$ [w' is adjacent to z], (4)

where the bracket is 1 when its assertion holds and 0 otherwise. By Lemma 169, extend N_C to a maximum matching N of J exposing x, using any incidence choices that expose x; such choices exist because x lies in D. In $W = (Sminus\{x\})$ union {z}, replace ww' by zw in N. This matching leaves only w' uncovered. All degrees are at most t because the old neighbors of z in S have positive deficiency, while z has t neighbors in W. Since no C-vertex is adjacent to x, the degree of w' in W is $t - def(w') +$ [w' is adjacent to z]. If (4) failed, this degree would be less than t and the matching would cover all degree-t vertices, contradicting (2).

We next count the components of J[D]. For any such component K, all its vertices have J-degree t, so the number b_K of edges from K to A is $b_K = t|V(K)| - 2e(K)$. Both t and $|V(K)|$ are odd, so b_K is an odd nonnegative integer and is therefore at least one. Also $\sum_K b_K \le t|A|$. Let q count components with $b_K \le t-2$ and p the others; since b_K is odd, the others have $b_K \ge t$. The number of components and the edge count give $p + q = |A| + 1$ and $tp + q \le t|A|$. Subtracting gives $t \le (t-1)q$. This excludes $t = 1$; therefore $t \ge 3$ and $q \ge 2$. For any integer r with $1 \le r \le t$, simplicity gives $b_K \ge r(t+1-r) \ge t$ when $|V(K)| = r$, since $r(t+1-r) - t = (r-1)(t-r) \ge 0$. A component of order $t+1$ is impossible because that order is even. Hence each of the q components has at least $t+2$ vertices. Moreover A is nonempty, since otherwise its edge count would be zero despite the positive b_K values. We have proved

$|D| \ge 2t+4$, $|A| \ge 1$, $|S| \ge 2t+5+|C|$. (5)

All these assertions apply to every optimal configuration.

Move x from S to O, which preserves optimality by balance. Its new exposed part is O union {x}, with threshold t'. Let D', A', C' be the corresponding exposable decomposition of $J' = H[O$ union {x}]. The conclusions already proved apply to this configuration with t and t' interchanged. In particular, $t' \ge 3$, J'[D'] has $|A'|+1$ components, at least two of order at least $t'+2$, and $|A'| \ge 1$. Also x belongs to D' and has exactly t' neighbors in J'. Let $N_H[x]$ consist of x and all its H-neighbors.

The set D' minus $N_H[x]$ contains at least $|A'|+2$ pairwise nonadjacent vertices. To check this explicitly, let K_x be the D'-component containing x. Every other D'-component avoids $N_H[x]$, since an edge to x would join it to K_x. A D'-component of order at least $t'+2$ contains two nonadjacent vertices: otherwise its vertices

would have J′-degree at least $t'+1$, although every D′-vertex has degree t′. Choose two large components K_1, K_2. If one of them is K_x, take a nonadjacent pair from the other, one vertex from K_x outside $N_H[x]$, and one vertex from every remaining component. The vertex in K_x exists because that component has at least $t'+2$ vertices but $N_H[x]$ meets J′ in $t'+1$ vertices. This yields $2+1+(|A'|-1)=|A'|+2$ vertices. If neither large component is K_x, take a nonadjacent pair from each, omit K_x, and take one vertex from every other component. This yields $4+(|A'|-2)=|A'|+2$ vertices. The latter case has at least three components, so $|A'|-2$ is nonnegative. Vertices from distinct components are nonadjacent. Choose exactly $m=|A'|+2\geq 3$ vertices z_1,..., z_m in this manner.

We show that their neighborhoods $W_i = N_H(z_i)$ intersect S are pairwise disjoint subsets of C, each of cardinality t. A vertex z in D′ minus $N_H[x]$ has degree t′ in J′ and is not adjacent to x. Thus it lies in O and has exactly t′ neighbors in O, giving deficiency zero in the original configuration. Its t neighbors in S therefore belong to C by the assertion preceding (4).

There is an optimal configuration exposing this z with threshold-t part $S\cdot = (Sminus\{x\})$ union {z} and other part $O\cdot = (Ominus\{z\})$ union {x}. Indeed J′-z has a perfect matching $N\cdot$, since z is exposable in the odd graph J′ which has a near-perfect matching. Combine $N\cdot$ with the original matching on S minus {x}. The result is a near-perfect matching of H exposing z and internal to $S\cdot$, $O\cdot$. The matching-edge counts in the two parts equal their original values. The induced H-edge counts do also: exchanging x and z loses t and gains t edges on the S side, and loses t′ and gains t′ edges on the O side, because xz is not an edge. Thus the weighted potential is unchanged, and the configuration is optimal.

For distinct nonadjacent z, w in the chosen set, w lies in $O\cdot$ and still has deficiency zero there: it is adjacent to neither x nor z. Apply the structural conclusion before (4) to this new optimum. Every neighbor of w in $S\cdot$ lies in its remaining set $C\cdot$. But z is exposable in $H[S\cdot]$, so none of its neighbors in $S\cdot$ lies in $C\cdot$, by the definition of that decomposition's external neighbor set. Consequently z and w have no common neighbor in $S\cdot$. Their possible common neighbors in S are the same, since neither is adjacent to x and z is adjacent to neither w nor itself. This proves the disjointness of all W_i.

Use the fixed perfect matching N_C of J[C]. Each W_i has the odd cardinality t, so at least one vertex w_i in W_i has its partner w_i' outside W_i; otherwise W_i would be partitioned into matching edges and have even cardinality. The partner is not adjacent to z_i by the definition of W_i. Inequality (4) and nonnegative deficiencies imply $def(w_i')=0$. Every vertex of every W_j has positive deficiency, so w_i' lies outside their entire union. The w_i are distinct by disjointness, and their partners are distinct because N_C is a matching. Therefore $|C|\geq mt+m = m(t+1)\geq 3(t+1)$. Combining with (1) and (5) yields $c_{t'}(H)=|S|\geq 2t+5+3(t+1)=5t+8$.

There is an optimum with its exposed vertex in either specified threshold part: moving that vertex across preserves optimality. Apply the last inequality first with $t=\theta$ and then with $t=\theta'$. This proves both claimed cost bounds and θ, $\theta'\geq 3$. Adding the cost bounds to the assumed equality gives $n+1\geq 5(\theta+\theta')+16$, hence $n\geq 5d+15$.

Finally Lemma 119 always gives a complementary sum at most $n+1$. The order n is odd, so $n\leq 5d+13$ excludes the equality case just analyzed. If instead $c_0(H)\leq 5max(\theta,\theta')+7$, tightness would force one complementary cost to be at least $5max(\theta,\theta')+8$, contradicting the monotonic bound $c_r(H)\leq c_0(H)$ from Lemma 23. In either situation the integer sum is at most n. □

Remark. A tight pair forbids an induced residual of the exposed part's size whose top-degree vertices are matchable. This forces two large odd exposable components and three disjoint odd neighbor sets in the perfect remainder. Each such set needs an additional distinct matching partner.

Depends on. Lemma 23, Lemma 35, Lemma 72, Lemma 119, Lemma 169.

Lemma 191 (Quartic terminal identities from independent-complement costs). *Use the finite integer certificate costs c_i, matching number ν, independence number α, and slack $ps = e-\nu-\sum_{i\geq 1} c_i$ of Lemma 126. All graphs are finite and simple. Vertex deletion is induced. A graph is well-covered when every maximal independent set has maximum cardinality. A missing matching allocation (x, y) on F means that $E(F)$ cannot be partitioned into two parts having matching numbers at most x and y.*

Let F be a four-regular well-covered graph, put $a=\alpha(F)$, and let m, b, σ be integers satisfying $a\geq 4$, $m\geq 1$, $0\leq b<m$, and $\sigma\geq b$. Suppose that for every maximum independent set S of F, its complement $G=F-S$ satisfies

$$ps(G)=0,\ c_3(G)=0,$$

$|V(G)| - \nu(G) = a + m - b$,

$c_1(G) = \nu(G) - b$, $c_2(G) = 2m - b$,

$c_0(F) = a + \nu(G)$, $c_2(F) = a + c_2(G)$.

Suppose there are nonnegative integers x, y forming a missing matching allocation with $x + y = c_0(F) - 1$ and $x \le a - \sigma$. For each such S let $C(S)$ be the subgraph induced by the degree-three vertices of F-S, and define

$h = \max_S$ *alpha(C(S))*, $d = m - h$.

Then h exists, $0 \le h \le m$, and d is a nonnegative integer. An attaining frame is a pair (S, U) with S maximum independent and U an independent h-element subset of $C(S)$. Such frames exist. For every attaining frame put $W = S$ union U, $R = F - W$, $r = \nu(R)$, and $q = 2m - b - h = h + 2d - b$. Then

$F[W] = hK_2$ *disjoint union* $(a - h)K_1$,

$\Delta(R) \le 2$, $ps(R) = 0$,

$r = \nu(F - S) - h$, $c_1(R) = r - b$, $c_2(R) = q$,

$\alpha(R) = \alpha(F - S) = a - d$.

Here K_1 is an isolate and K_2 a single edge. The terminal R has exactly q whole odd cycles, chosen from C_3, C_5, C_7, exactly b whole even-order paths, chosen from P_2, P_4, P_6, and otherwise P_3, C_4, or isolates; subscripts count vertices. If $d > 0$ then $2d \ge \sigma + b$, and consequently $d \ge b$.

If $h = m$, all pairs (S, U) with S maximum independent and U an independent m-set of cubic vertices have these assertions with $d = 0$ and $q = m - b$. Every orientation of the m edges of $F[W]$, assigning one endpoint of each edge and every isolate to the independent side, gives another such frame with the same W and R. For every endpoint p of an edge of $F[W]$, the induced graph $K_p = F[V(R)$ union $\{p\}]$ satisfies

$\nu(K_p) = r + 1$, $c_1(K_p) = r + 1 - b$.

For the same fixed missing allocation, put $j = |W| - x$. Then

$x + y = |W| + r - 1$, $j \ge q + b + \sigma$.

In particular $j \ge q + 2$ whenever $b \ge 1$. The exact original-graph second cost is $c_2(F) = a + 2m - b$.

Proof. The graph is finite and therefore has a maximum independent set. A maximum independent set is maximal and dominates every vertex outside it: otherwise an undominated vertex could be added. Four-regularity consequently makes $G = F - S$ subcubic for every such S.

Apply Lemma 141 to G. Its hypotheses are precisely the subcubic bound and the assumed equalities $ps(G) = c_3(G) = 0$. Writing $t = \alpha(C(S))$, its independence identity gives

$\alpha(G) = |V(G)| - \nu(G) - c_2(G) + t$

$= a + m - b - (2m - b) + t = a - m + t$.

Since G is induced in F, $\alpha(G) \le a$, so $t \le m$. The finitely many maximum independent sets S form a nonempty family; hence their finite integer maximum h exists and belongs to [0,m]. A maximum independent set of an attaining $C(S)$ supplies U, establishing frame existence.

Fix an attaining frame. Since $|U| = h$ and $\alpha(C(S)) \le h$, the set U is maximum, and hence maximal, independent in $C(S)$. By Lemma 141, every ordering of U is a complete cubic deletion process, and its terminal R preserves $\alpha(G)$ and satisfies

$\nu(R) = \nu(G) - h$,

$c_1(R) = c_1(G) - h$,

$c_2(R) = c_2(G) - h$.

It has maximum degree at most two and zero slack. The same fact gives $\alpha(G) = a - m + h = a - d$, and preservation gives $\alpha(R) = a - d$. Substitution in the two cost identities yields $c_1(R) = r - b$ and $c_2(R) = 2m - b - h = q$. The degree-two formulas of Lemma 57 identify $c_2(R)$ with the number of odd-cycle components, and $\nu(R) - c_1(R)$ with the number of positive even-order path components. Its zero-slack

classification gives exactly the component list in the statement. This also proves $q \geq 0$ without a separate numerical assumption.

Each u in U has degree three in G and four in F, so it has exactly one neighbor in S. Distinct U-vertices have distinct such neighbors: if u, v shared s in S, then (S minus {s}) union {u, v} would be independent of size $a+1$, since U is independent and neither u nor v has another S-neighbor. This contradicts $\alpha(F) = a$. The sets S and U are individually independent, so these are all edges of F[W]. Thus F[W] is h disjoint edges and a-h isolates; in particular $h \leq a$.

The assumed cost equality and the matching loss give

$c_0(F) = a + \nu(G) = a + h + r = |W| + r.$

Consequently the fixed missing allocation has $x + y = |W| + r - 1$. Its shortage $j = |W| - x$ satisfies

$j \geq a + h - (a - \sigma) = h + \sigma.$

Suppose $d > 0$ and $2d < \sigma + b$. Since $q = h + 2d - b$, the preceding inequality implies

$j - q \geq \sigma + b - 2d \geq 1,$

where the last step uses integrality. Every component of R is a path or a cycle and has matching number equal to the floor of half its order, by Lemma 57. Apply Lemma 17 to this induced R and the same missing allocation, protecting all odd-order path components, including isolates. Its unprotected odd components are exactly the q odd cycles; the inequality just proved gives its required $j \geq q + 1$. Since F is well-covered, that fact implies $\alpha(R) = \alpha(F) = a$. This contradicts $\alpha(R) = a - d < a$. Therefore $d > 0$ implies $2d \geq \sigma + b$. As $\sigma \geq b$, it follows that $2d \geq 2b$ and hence $d \geq b$. No matching allocation is changed in this argument.

Now suppose $h = m$. Every independent m-set of cubic vertices in any maximum-independent complement is maximum there, by the definition of h. All arguments already given apply to every such pair, with $d = 0$. Orient the m edges of F[W] arbitrarily, take one endpoint of each edge and every isolate as S′, and take the other endpoints as U′. There are no further W-edges, so S′ is independent with $|S'| = m + (a - m) = a$ and is maximum in F. The set U′ is independent; each of its vertices has its mate as its unique S′-neighbor, and is therefore cubic in F-S′. Thus (S′,U′) is another frame with terminal R.

Given an endpoint p, choose this orientation with p in U′. Delete the m-1 vertices of U′ minus {p} in any order, leaving K_p. Each deleted vertex still has degree three because U′ is independent. The arbitrary-prefix identities of Lemma 126 apply to this complete cubic process, including the prefix of length m-1 (which is zero when $m = 1$). They give

$\nu(K_p) = \nu(F - S') - (m - 1),$

$c_1(K_p) = c_1(F - S') - (m - 1).$

The complete process ends at the same R, so $r = \nu(F - S') - m$. The uniform input $c_1(F - S') = \nu(F - S') - b$ proves $\nu(K_p) = r + 1$ and $c_1(K_p) = r + 1 - b$.

Finally $|W| = a + m$ and $q = m - b$, so the fixed allocation satisfies

$j = |W| - x \geq m + \sigma = q + b + \sigma.$

If $b \geq 1$, then $\sigma \geq b \geq 1$ and $b + \sigma \geq 2$. This proves $j \geq q + 2$ in exactly that case. The original-graph identity follows directly from the uniform assumption $c_2(F) = a + c_2(G)$ and $c_2(G) = 2m - b$. Every statement uses the same F and, whenever allocations occur, the same nonnegative integer pair (x, y). □

Remark. The exact cubic independence identity turns complement costs into the attaining-frame identities. A single missing allocation rules out a deficit smaller than the even-path count and also supplies the needed port range after positive-deficit saturation.

Depends on. Lemma 17, Lemma 57, Lemma 126, Lemma 141.

Lemma 192 (Whole-triangle contacts at a zero-deficit row). *All graphs are finite and simple. Write $V(G)$, $E(G)$, $\nu(G)$, $\alpha(G)$, and $\Delta(G)$ for vertex set, edge set, matching number, independence number, and maximum degree. A graph is well-covered if every inclusionwise maximal independent set has size $\alpha(G)$. Write $N_F(x)$ for the full set of neighbors of x in F. For nonnegative integer capacities A, B, let $c_{A,B}(G)$ be the least nonnegative integer T such that every nonnegative integer pair $x + y = T$ admits an ordered edge partition into four parts with maximum degrees at most A, B and matching numbers at most x, y, respectively. Put $c_i(G) = \max_{A+B=i} c_{A,B}(G)$ and $ps(G) = |E(G)| - \nu(G) - \sum_{i\geq 1} c_i(G)$. All indices and capacities in these definitions are nonnegative integers.*

Let F be well-covered, and let R be an induced subgraph whose whole components are isolated vertices or cycles of orders three, four, five, or seven. A whole component is an entire connected component; cycle orders count vertices. Put $U = V(F)$ minus $V(R)$, $u = |U|$, $m = \nu(R)$, and let p count its odd-cycle components. Assume

$\alpha(R) = \alpha(F)$, $c_0(F) = u + m$, $c_1(F) \geq u + m$, $c_2(F) \geq u + p$.

Suppose nonnegative integers b_R, b_B satisfy $b_R + b_B = c_0(F) - 1$, $E(F)$ has no two-part partition with matching numbers at most b_R, b_B, and $j = u - b_R \geq p + 4$.

For every x in U having exactly three neighbors in R, those neighbors form one whole triangle T of R. The induced graph $K = F[V(R)$ union $\{x\}]$ consists of the whole four-clique on $V(T)$ union $\{x\}$ and all other original R-components. Every complete process deleting current degree-three vertices from K has exactly one deletion and may delete any one of those four vertices. Each resulting graph is obtained by replacing T by a triangle on the other three vertices; all other row components and their original edges are unchanged.

No regularity, source minimality, whole-source process, or terminal-preservation assumption is imposed. Every order, count, degree, cardinality, capacity, budget, cost, and process length is a nonnegative integer.

Proof. Use Lemma 168 for singleton cubic extensions, Lemma 57 for degree-two costs, Lemma 101 for two-isolate payment, and Lemma 108 for three-isolate payment.

The degree-two formulas of Lemma 57 give $ps(R) = 0$, $c_1(R) = m$, and $c_2(R) = p$, since every component is one of the listed zero-slack cycles or an isolate and no positive even-order path is present. Fix x as in the statement, and put $q = \min(b_R, b_B)$. Since $j = u - b_R \geq p + 4 \geq 4$, one has $u \geq q$ and $1 \leq j \leq u - q$. The singleton {x} is independent and its vertex has exactly three R-neighbors. The cost lower bounds and the missing total $b_R + b_B = u + m - 1$ therefore meet every hypothesis of Lemma 168 with retained set {x}.

That fact gives $\Delta(K) \leq 3$, $\nu(K) = m + 1$, $c_1(K) = m + 1$, $c_2(K) = p + 1$, and $ps(K) = 0$. Since R is induced in K and K is induced in F, $\alpha(R) \leq \alpha(K) \leq \alpha(F) = \alpha(R)$. Its singleton conclusion applies: the vertices of K-degree three form a nonempty clique, and deleting any such vertex leaves a graph R′ with $\Delta(R') \leq 2$, $ps(R') = 0$, $c_1(R') = m$, $\nu(R') = m$, and $c_2(R') = p$.

In particular Lemma 57 says that each such R′ has exactly p odd-cycle components, has no positive even-order path component, and has components only among P_1, P_2, P_3, P_4, P_6, C_3, C_4, C_5, C_7. Here P_i denotes the path on i vertices.

Let t be the number of x's three neighbors that lie on row cycles. Each has row degree two and thus degree three in K. They are pairwise adjacent by the cubic clique property. Hence if $t \geq 2$ they lie on the same cycle; if $t = 3$ that cycle is a whole triangle, giving the desired neighborhood. We exclude the other cases.

If $t = 0$, the entire row neighborhood of x is a set I of three isolates. Decompose R into its p odd cycles, the union D of all its whole four-cycles, and its isolate set Z. The graph D is bipartite of maximum degree at most two and has a perfect matching obtained by pairing opposite edges in each four-cycle; D may be empty. The α equality, c_0 identity, missing pair, and $j \geq p + 4$ are exactly the hypotheses of Lemma 108. It gives $c_2(F) \leq u + p - 1$, contradicting the assumed lower bound.

Suppose $t = 1$. Denote the cycle contact by c on a whole cycle C; the other two neighbors are isolates i_1, i_2. If C is odd, delete the cubic vertex c from K. The cycle C becomes a path, and x together with i_1, i_2 forms a separate whole three-vertex path. All other row components remain unchanged. Thus the resulting graph has exactly p-1 odd-cycle components, contradicting $c_2(R') = p$ and the degree-two odd-cycle formula.

If C is even, it is a whole four-cycle. Use the same decomposition into odd cycles, even part D, and isolates Z. The even graph D is nonempty and has a perfect matching. Apply Lemma 101 with its pivot x, isolate pair $\{i_1, i_2\}$, and arbitrary even-row contact $r = c$. Its missing total is $u + \nu(R) - 1$, and its required reserve $j \geq p + 3$ follows from the stronger given reserve. It gives $c_2(F) \leq u + p - 1$, again a contradiction. This application allows the contact on a whole even cycle, rather than requiring a path endpoint.

Suppose $t = 2$. The two contacts c, d are adjacent on one whole cycle C, and the third neighbor i is a row isolate. Delete the cubic vertex c. The remaining path C-c has d as an endpoint and is extended by d-x-i. It is therefore one whole path on $|V(C)| + 1$ vertices, with all other original row components unchanged. If C is odd, this operation destroys exactly one odd-cycle component and creates none, contradicting the preserved second cost p. If C is even, its order is four, so the resulting graph has a whole P_5 component. The zero-slack component characterization in Lemma 57 excludes P_5, contradicting $ps(R') = 0$.

Thus $t = 3$. The three contacts are pairwise adjacent vertices of a whole cycle, which forces that cycle to have order three. Because x has no other row neighbor and T was a whole row component, T union {x} is a whole four-clique in K, and every other row component is unchanged. Its four vertices are the only cubic vertices of K. Deleting any one leaves the triangle on the remaining three, together with the other unchanged row components, and no cubic vertex remains. Conversely any complete cubic process must start at one of those four vertices. This proves every assertion. □

Remark. Exact singleton accounting makes every cubic vertex exchange legal. Preserved odd-cycle count rules out partial odd-cycle contacts, zero slack rules out the five-vertex path, and the two isolate payments eliminate the remaining masks.

Depends on. Lemma 57, Lemma 101, Lemma 108, Lemma 168.

Lemma 193 (Sextic rows without terminal independence). *Use the finite integer certificate costs c_i, matching number ν, independence number α, and slack $ps = e - \nu - \sum_{i\geq 1} c_i$ of Lemma 23. All graphs are finite and simple, and vertex deletion is induced. Orders, degrees, cardinalities, indices, capacities, budgets and process lengths are nonnegative integers; parameters defined by differences are initially integers.*

Let F be connected, six-regular and factor-critical, meaning that F-v has a perfect matching for every vertex v. Write $|V(F)| = 2k + 1$ with integer $k \geq 1$, put $a = \alpha(F)$, and assume $E = c_0(F) - k \geq 1$. Define

$\lambda = 2E - 3 - (k - a)$, $\delta = c_0(F) - c_2(F)$, $\epsilon = c_0(F) - c_1(F)$,

$p = E - 2 - \lambda$, $r = k - \lambda - 1 + \delta$, $s = 2\lambda + 3 - 2\delta$,

$\beta = p + \delta - \epsilon$, $u = 2k + 1 - r$, $M = k + E - r$.

A complete process chooses a maximum independent set S of F and repeatedly deletes any current maximum-degree vertex from F-S while that degree is at least three, ending at a graph R of maximum degree at most two. Assume:

(a) For every maximum independent S, $c_i(F - S) = c_i(F) - a$ for every integer $1 \leq i \leq 5$, and $c_0(F) = a + \nu(F - S) + ps(F - S)$.

(b) In every complete process, each deletion parent of current maximum degree D has a matching covering all its degree-D vertices; deleting the chosen vertex lowers c_i by one for every integer $1 \leq i < D$. Every terminal has zero slack.

Call an induced A a native row if $|V(A)| = r$, $\Delta(A) \leq 2$, $ps(A) = 0$, $\nu(A) = p + \delta$, $c_1(A) = \beta$ and $c_2(A) = p$. Native rows exist, $p \geq 0$, $0 \leq \epsilon \leq \delta \leq \lambda + 1$, $\beta \geq p$, s is positive odd, $r = a + 2p + \delta$, $u = r + s$ and $M = p + \delta + s$. Every native row has $\alpha(A) = a$, $e(A) = 3p + 2\delta - \epsilon$, exactly p odd-cycle components and ϵ positive even-order path components. Its whole components belong to P_1, P_2, P_3, P_4, P_6, C_3, C_4, C_5, C_7, with subscripts denoting vertex counts.

Every complete process has precisely $p + \epsilon$ roots at degree five and β roots at degree four. Their union with S induces a native row. Conversely every native row admits the following replay on the same F. Label paths consecutively and cycles cyclically. For components P_1, P_2, P_3, P_4, P_6, C_3, C_4, C_5, C_7 choose respectively the pairs (S, T_5)

({1},empty), ({1},{2}), ({1,3},empty), ({1,3},{4}), ({1,3,5},{6}),

({1},{2}), ({1,3},empty), ({1,3},{4}), ({1,3,5},{6}).

Unite the component choices and set $T_4 = V(A)$ minus $(SunionT_5)$. Then S is maximum in F, every ordering of T_5 is the full degree-five phase, and every ordering of T_4 is the next full degree-four phase.

The exact costs on F are

$c_0(F) = u + p + \delta$, $c_1(F) = u + \beta$, $c_2(F) = u + p$,

$c_3(F) = r$, $c_4(F) = r - \beta = a + p + \epsilon$, $c_5(F) = a$, $c_6(F) = 1$,

and $c_i(F) = 0$ for every integer $i \geq 7$.

For each displayed positive cost $K = c_i(F)$, every nonnegative integer capacity split $A_c + B_c = i$ and every nonnegative integer budget split $x + y = K$ admit a certificate on F; some capacity split and some budget split of K-1 admit none.

Every native-row complement $J = F - V(A)$ is subcubic, has $ps(J) = 0$, $\nu(J) = M$, $c_1(J) = M - \epsilon$, $c_2(J) = M - \delta$, and $c_i(J) = 0$ for $i \geq 3$. For any complete process let $h = a - \alpha(R)$, let A be its early native row, and let X be the remaining cubic root set. Then

$0 \leq h \leq s$, $|V(R)| = r + h$, $\nu(R) = p + \delta + h$,

$c_1(R) = \beta + h$, $c_2(R) = p + h$, $|X| = s - h$.

The sets $V(A)$, X, $V(R)$ partition $V(F)$. The set X is independent and each of its vertices has exactly three neighbors in A and exactly three in R. The complement F-V(R) is also subcubic with zero slack, independence number a, and matching and first two costs

$E + \lambda + 1 - h - \delta$,

$E + \lambda + 1 - h - \delta - \epsilon$,

$E + \lambda + 1 - h - 2\delta$,

respectively. Its higher rows vanish. No independence-preserving source terminal, well-coveredness or source-minimality is assumed or concluded.

Proof. We use definitions and finite thresholds from Lemma 23, restoration from Lemma 72, slack and covered-degree identities from Lemma 61, degree-two formulas from Lemma 57, missing-allocation matching bounds from Lemma 7, terminal-complement accounting from Lemma 105, and cubic independence from Lemma 141.

Factor-criticality gives $\nu(F) = k$: a perfect matching of F-v has k edges and no matching on $2k + 1$ vertices has more. All complete processes exist by finite deletion. Since a maximum independent set is maximal, every outside vertex has a neighbor in it, so $\Delta(F - S) \leq 5$.

At a deletion parent H of degree D, coveredness gives $c_D(H) = 0$. Restriction and row monotonicity make that row and all higher rows zero on its child H′. The D-1 lower rows lose one each by (b). The local identity of Lemma 61 therefore gives

$\nu(H) + ps(H) = 1 + \nu(H') + ps(H')$.

Combining this with (a) proves at any prefix with deleted vertex set W, including S,

$c_0(F) = |W| + \nu(F - W) + ps(F - W)$.

For $i = 1, 2$ every root degree is greater than i, so $c_i(F) = |W| + c_i(F - W)$. At a terminal R with b roots after S, put $U = F - V(R)$; then $|U| = a + b$ and

$c_0(F) = |U| + \nu(R)$, $c_i(F) = |U| + c_i(R)$ for $i = 1, 2$.

Apply Lemma 105 at $d = 6$ to this actual R. With $h = a - \alpha(R) \geq 0$ it gives

$|U| = k + \lambda + 2 - h - \delta$,

$\nu(R) = p + \delta + h$, $c_1(R) = \beta + h$, $c_2(R) = p + h$,

$\Delta(U) \leq 3$, $ps(U) = 0$, $c_i(U) = 0$ for $i \geq 3$,

$\nu(U) = E + \lambda + 1 - h - \delta$,

$c_1(U) = \nu(U) - \epsilon$, $c_2(U) = \nu(U) - \delta$.

It also gives $0 \leq \epsilon \leq \delta$, $\lambda \leq E - 2$ and $k \geq 3E - 1$. Because U contains S and is induced in F, $\alpha(U) = a$. The terminal order is $|R| = r + h$, and the root count is $b = 2E - 1 - \delta - h$.

By Lemma 141, U has a complete cubic process preserving $\alpha(U)$. Its root count q equals

$\alpha(U) - |U| + \nu(U) + c_2(U) = 2\lambda + 3 - h - 2\delta = s - h$.

Its terminal A therefore has

$|A| = |U| - q = r$, $\nu(A) = \nu(U) - q = p + \delta$,

$c_1(A) = \beta$, $c_2(A) = p$, $ps(A) = 0$, $\Delta(A) \le 2$.

Thus a native row exists and $p \ge 0$. Since $q \ge 0$, $h \ge 0$ and s is odd, $s \ge 1$; equivalently $\delta \le \lambda + 1$. The remaining parameter identities in the statement follow by substitution. In any native row the degree-two formulas give $\alpha(A) = r - (p + \delta) - p = a$, the stated edge count, component types and component counts. In particular $\beta \ge p$ follows from $\epsilon \le \delta$. This construction did not require the original terminal to preserve independence.

For any process let C and B count its entire degree-five and degree-four phases, and put $m = C + B$. The identities

$c_5(F) = a$, $c_4(F) = a + C$, $c_3(F) = a + m$

hold before their values are determined. Indeed after S, the degree-five row is zero: if the actual degree is five, (b) covers it, and if smaller the strong tail in Lemma 61 gives zero. After the full degree-five phase the same argument makes row four zero. After the full degree-four phase it makes row three zero, including a degree-two terminal. Summing the assumed lower-row losses from S to these boundaries gives the three equations. Empty phases are included.

Choose a missing allocation (x, y) of total $c_0(F) - 1 = k + E - 1$ with $x \le y$; it exists by Lemma 23. Since $k \ge 3E - 1$ and $\lambda \le E - 2$,

$x \le (k + E - 1)/2 \le k - E + 1 \le r$.

Let L be the induced graph on S and the first two phases of an arbitrary process. Its complement contains its actual terminal, of order $r + h \ge r \ge x$. The bound Lemma 7 therefore gives

$\nu(L) \ge |L| + c_0(F) - |F| = m - p$.

Restoration of the complementary vertex set gives

$c_1(L) \ge m - p - \epsilon$, $c_2(L) \ge m - p - \delta$.

All degree-five roots form an independent set: they initially have degree at most five in F-S, so a neighbor among earlier roots would prevent deletion at degree five. Each has exactly one neighbor in S. Two such roots cannot share this neighbor, since replacing that one S vertex by the two nonadjacent roots would give an independent set larger than a. Thus S and these roots induce a matching and isolates. Its first row is zero by Lemma 61. Restoring the remaining vertices in F gives $c_1(F) \le 2k + 1 - a - C$, or

$C \le p + \epsilon$.

Likewise the degree-four roots are independent in the graph left after the degree-five phase, and each has exactly two earlier neighbors in S and the five-roots. Hence $e(L) = C + 2B = 2m - C$. Nonnegative slack and the three preceding lower bounds yield

$2m - C \ge 3m - 3p - \epsilon - \delta$,

so $m + C \le 3p + \epsilon + \delta$. These two bounds hold for every complete process.

Now take any native row A and the component choices in the statement. Directly on each listed path or cycle, S is maximum independent, T_5 is independent with one S-neighbor per vertex, and the complementary T_4 is independent with two neighbors in S union T_5 per vertex. Across components these properties persist. There is one T_5 vertex for each odd cycle and each positive even-order path. Thus $|S| = a$, $|T_5| = p + \epsilon$ and $|T_4| = \beta$. Six-regularity makes every vertex of T_5 degree five after S, and its independence permits every ordering. If another degree-five vertex remained afterwards, completing that process would violate $C \le p + \epsilon$. Thus T_5 exhausts the phase.

After that phase every T_4 vertex has degree four. Independence permits every ordering of T_4. Any completion has $C = p + \epsilon$ and $m \ge 2p + \delta$, while $m + C \le 3p + \epsilon + \delta$ forces $m \le 2p + \delta$. Thus T_4 exhausts its phase as well. The boundary equations now give $c_5(F) = a$, $c_4(F) = a + p + \epsilon$ and $c_3(F) = a + 2p + \delta = r$. Since $r - \beta = a + p + \epsilon$, the target cost difference is obtained with the independently defined r and β.

These same boundary equations force $C = p + \epsilon$ and $m = 2p + \delta$ in every process. Its early pool L then has order r and lower bounds

$\nu(L) \ge p + \delta$, $c_1(L) \ge \beta$, $c_2(L) \ge p$.

Their sum equals $e(L) = 3p + 2\delta - \epsilon$. The slack identity, with all higher row costs nonnegative, forces equality in each of the three bounds, $ps(L) = 0$ and every row of index at least three zero. A zero certificate at capacities (3,0) gives $\Delta(L) \leq 3$, since the degree-zero and matching-budget-zero parts contain no edges. Also $\alpha(L) = a$ because S is contained in L.

Apply Lemma 141 to L, and let Z be its cubic vertex set. Its exact independence formula gives

$a = \alpha(L) = r - (p + \delta) - p + \alpha(L[Z]) = a + \alpha(L[Z])$.

Thus $\alpha(L[Z]) = 0$, which implies Z is empty. Hence $\Delta(L) \leq 2$ and L is native.

For any native A its replay is an actual prefix, so $J = F - V(A)$ is subcubic and the prefix identities give

$\nu(J) + ps(J) = c_0(F) - r = M$,

$c_1(J) = c_1(F) - r = M - \epsilon$, $c_2(J) = c_2(F) - r = M - \delta$.

Its complement A has at least x vertices. Applying Lemma 7 to $V(J)$ gives $\nu(J) \geq |J| + c_0(F) - |F| = M$. Nonnegative slack forces $\nu(J) = M$ and $ps(J) = 0$. The full degree-four phase ended at a covered cubic graph or a graph of smaller degree, so $c_3(J) = 0$; higher rows vanish by monotonicity. The displayed row-zero, row-one and row-two costs of F follow from the definitions and $M = p + \delta + s$.

The strong tail gives $c_i(F) = 0$ for $i \geq 7$. A zero certificate at capacities (5,1) would partition all edges of six-regular F into a graph of maximum degree five and a matching. Every vertex would need an incident edge of this matching, impossible on odd order. Thus $c_6(F) \geq 1$. For any vertex v, a perfect matching of F-v covers every degree-six vertex there; Lemma 61 gives $c_6(F - v) = 0$. Restoration gives $c_6(F) \leq 1$. This proves the full table. At every asserted cost K, the finite threshold definition gives every budget split for every capacity split. The maximum over the finitely many capacity splits is attained; at an attaining split a positive K cannot admit every allocation of K-1. This proves both certificate directions on F.

Finally, for an arbitrary process its remaining root count after the two fixed phases is

$b - (2p + \delta) = s - h$.

This is nonnegative, so $0 \leq h \leq s$. Its early row, remaining roots X, and terminal R are disjoint and partition F. The graph after the early row is subcubic. Any remaining root selected at degree three has lost no neighbor to an earlier root, so X is independent and each root has exactly three neighbors in R. Its other three F-neighbors lie in A by six-regularity. The terminal profiles were derived at the start, as were the exact zero-slack subcubic complement profiles and $\alpha(F - V(R)) = a$. These assertions keep the actual terminal loss h throughout and make no claim that it vanishes. □

Remark. A preserving cubic deletion in the terminal complement first constructs a row. Canonical component choices then attain universal bounds on the first two source phases, determining all source costs without assuming the original terminal preserves independence.

Depends on. Lemma 7, Lemma 23, Lemma 57, Lemma 61, Lemma 72, Lemma 105, Lemma 141.

Lemma 194 (Original-profile restrictions at exact potential). *Let $s, t \geq 1$ be integers and let $n_1 \geq \cdots \geq n_s \geq 1$ and $m_1 \geq \cdots \geq m_t \geq 1$ be integer lists. Let F_1, F_2 be their respective vertex-disjoint unions of stars, prescribed red and blue. Copies are ordinary subgraphs with vertex-disjoint target components. A graph arrows an ordered pair when every red-blue edge coloring contains its entire red or blue target in the prescribed color. Let u, v be the numbers of one-edge components of F_1, F_2, and assume $u \geq v$. Put*

$M = s + t - 1$, $\ell_k = max\{n_i + m_j - 1 : 1 \leq i \leq s, 1 \leq j \leq t, i + j = k\}$

for every integer $2 \leq k \leq M + 1$.

Fix an integer $0 \leq q < M$. Let A_q be the entire family of ordered nonempty suffix pairs obtained by deleting α red and β blue leading components, with integers $0 \leq \alpha < s$, $0 \leq \beta < t$ and $\alpha + \beta = q$. Put $K = M - q$,

$P_q = \sum_{k=q+2}^{M+1} (\ell_k - 1)$, $D = \ell_{q+2} - 1$,

and assume $D \geq 3$. Let R be the length of the initial run of entries equal to $D + 1$ in ℓ_{q+2},..., ℓ_{M+1}. Write $\Phi(H) = e(H) - \nu(H)$ for the edge count minus maximum matching cardinality of a finite simple graph H.

Suppose $\Delta(H) \le D$, H arrows every member of A_q, and $\Phi(H) = P_q$. Define $A = s - u$, $B = t - v$, and $b = \sum_{j=1}^{B}(m_j - 2)$. Then $A, B \ge 1$, $u > v$, and

$q \le \min(s-2, B-2)$,

$B - q \ge R + e(F_1) - s + 1$,

$b + q + R \le s - 1$,

$m_1 \le D$, $n_1 + m_B - 1 \le D$.

In particular $D \ge n_1 + 1$, $q < \min(s,t)$, and H contains each whole original forest F_1, F_2 as an uncolored subgraph. Both suffix allocations (q,0) and (0,q) are legal.

For $u = v$ the stronger statement holds without assuming potential equality: every H satisfying just the displayed arrowing and degree conditions has $\Phi(H) \ge P_q + 1$.

All lengths, indices, degrees, component counts, edge counts, matching sizes and displayed parameters are integers in their indicated nonnegative or positive domains. The conclusions impose no matching-deficit, binary-target, connectedness, or parity hypothesis. Every original unit component and repeated size is retained.

Proof. Choose original indices i, j with $i + j = q + 2$ and $n_i + m_j - 1 = D + 1$. Their leading cuts i-1, j-1 are legal and total q. The all-red coloring of H forces the red suffix to occur in H, and the all-blue coloring forces the blue suffix. Hence $n_i, m_j \le D$. Since $n_i + m_j = D + 2$, both are at least two. Thus A, $B \ge 1$, $i \le A$, $j \le B$, and

$q \le A + B - 2$. (1)

This argument uses only the arrowing and degree assumptions.

We use Lemma 138 in the following precise form. For a legal actual pair, let F be its suffix in a color having the smaller original unit count $U = \min(u, v)$, and suppose F still has a nonunit. Let J be its entire opposite suffix and $c(F)$, $c(J)$ their component counts. Put $N = K - U$. If either $e(J) \ge N - R + 1$ or $e(F) - c(F) \ge N - R + 1$, that theorem gives $\Phi(H) \ge P_q + 1$. Its hypotheses include both original forests having nonunits, which was proved in (1). Moreover such an F retains all U original units.

First suppose $u = v$. If $q \ge A - 1$, choose the red cut A-1 and the blue cut $q - A + 1$. They are nonnegative, and (1) makes the latter at most B-1. Both cuts are legal. The red suffix F has exactly one nonunit and all u units. The opposite suffix J has $K - u = N$ components, so $e(J) \ge N \ge N - R + 1$ because $R \ge 1$. The first test gives $\Phi(H) \ge P_q + 1$. Exchanging the colors proves the same conclusion when $q \ge B - 1$.

In the remaining equal-unit case, $q < A - 1$ and $q < B - 1$. Both endpoint allocations (0,q) and (q,0) are legal, and their shortened-side suffixes still contain nonunits. The originals satisfy

$e(F_1) + e(F_2) \ge 2A + 2B + 2u = 2N + 2q + 2$.

At least one original has at least $N + q + 1$ edges. Use the endpoint whose whole opposite suffix is that original. Its first test again applies, since $N + q + 1 \ge N - R + 1$. This proves the asserted strict conclusion in every equal-unit case. Under the remaining equality hypothesis $\Phi(H) = P_q$, it follows that $u > v$.

Now put $N = K - v = s + B - 1 - q$. Suppose $q \ge B - 1$. Choose the blue cut $\beta = B - 1$ and red cut $\alpha = q - B + 1$. By (1), $\alpha \le A - 1$, so both cuts are legal. The blue suffix F has precisely one nonunit and v units; its opposite red suffix J has exactly N positive-edge components. Thus $e(J) \ge N \ge N - R + 1$, and the first test contradicts equality. Hence $q \le B - 2$.

The actual allocation (0,q) is now legal. Its blue suffix F has B-q nonunits and all v units, and its opposite suffix is the whole F_1. The first test cannot hold, so integrality gives

$e(F_1) \le N - R = s + B - 1 - q - R$.

Rearranging yields

$B - q \ge R + e(F_1) - s + 1$. (2)

Since F_1 has a nonunit, $e(F_1) - s \ge 1$. Thus $p := B - q > R$.

The diagonal sequence ℓ_k is nonincreasing: any candidate on a diagonal after the first has an index greater than one, and decreasing that index gives a candidate of at least the same value on the preceding diagonal.

The candidate $n_1 + m_B - 1$ occurs at diagonal $B + 1$, at position $p = B - q$ of the natural tail. This position is after the initial plateau, by (2). Consequently

$n_1 + m_B - 1 \le \ell_{B+1} \le D$.

Since $m_B \ge 2$, this also proves $D \ge n_1 + 1$.

Suppose $q \ge s - 1$. Choose red cut $\alpha = s - 1$ and blue cut $\beta = q - s + 1$. They are nonnegative and legal: $\alpha < s$ and $\beta \le q \le B - 2 < B \le t$. Since $u > v \ge 0$, the last red component is a unit, so the red suffix consists of one edge. The blue suffix F has

$B - \beta = s + B - 1 - q = N$

nonunits and all v units. Each nonunit contributes at least one to e(F)-c(F), so $e(F) - c(F) \ge N \ge N - R + 1$. The second test contradicts equality. Therefore $q \le s - 2$.

Both endpoint allocations are now legal, since $q \le s - 2$ and $q \le B - 2 \le t - 2$. Use (q,0), whose blue suffix is the whole F_2. The second test must fail, and integrality gives $e(F_2) - t \le N - R$. By the definition of b,

$e(F_2) - t = B + b$.

Substitute $N = s + B - 1 - q$ and cancel B to obtain $b + q + R \le s - 1$.

Finally, apply the all-blue coloring to the (q,0) arrowing assertion and the all-red coloring to the (0,q) assertion. They respectively force whole uncolored copies of F_2 and F_1 in H. In particular $m_1 \le \Delta(H) \le D$. This proves every stated bound and every whole-forest conclusion. □

Remark. A legal cut leaving one nonunit forces a large opposite matching projection. Excluding that possibility makes both whole-original endpoints legal; applying the two projection tests there yields the original component and head bounds.

Depends on. Lemma 138.

Lemma 195 (A smaller vertex pool from a zero-slack independent complement). *Use the finite integer certificate costs $c_i(G)$, matching number $\nu(G)$, independence number $\alpha(G)$, maximum degree $\Delta(G)$, and slack $ps(G) = e(G) - \nu(G) - \sum_{i\ge1} c_i(G)$ of Lemma 23. All graphs are finite and simple, and vertex deletion is induced. Call a deletion process covered if it repeatedly deletes a vertex of current maximum degree at least three, each deletion parent has a matching covering all vertices of that degree, and its terminal graph has maximum degree at most two. The process may have length zero.*

Let $p \ge 1$ and $d \ge 4p + 4$ be integers with d even. Let H be a nonempty d-regular graph, let S be any maximum independent set, and put $a = |S|$ and $G = H - S$. Suppose

$ps(G) = 0$ *and* $|V(G)| - 2\nu(G) = a - 1 - 2p$,

and suppose G admits a covered deletion process.

Then $\nu(G) \le 3p + 1$. Every such process has length $t \le 2p + 1$. For every such process there are a terminal vertex g and a set R of exactly $t + 2$ original neighbors of g in S such that, writing T for its deleted vertex set and

$U = (SminusR)$ *union* T *union* $\{g\}$,

one has $|U| = a - 1$, $\Delta(H - U) \le d - 1$, and a matching of H-U covers every vertex of degree d-1.

Consequently, for every nonnegative integer pair $A + B = d - 1$ and every nonnegative integer pair $x + y = a - 1$, all original H-edges have a certificate whose degree-bounded parts partition $E(H - U)$ and whose matching-bounded parts partition all edges meeting U. The set U and a covering matching may be fixed before all these choices. In particular $c_{d-1}(H) \le a - 1$.

As a consequence, suppose instead that H is connected and factor-critical, meaning that deleting any vertex leaves a perfect matching, and $ps(H) = p$. For every maximum independent set S whose complement G admits a covered deletion process, the integer gain

$\eta = \nu(G) - (\nu(H) + 1 - |S|)$

satisfies $0 \le \eta \le p - 1$. In this consequence zero slack of G and its displayed deficiency are not assumed.

No hereditary-gap or source-minimality hypothesis is imposed. All orders, degrees, indices, capacities, budgets, costs, matching cardinalities, p, a, t, and set sizes are integers in their displayed nonnegative or positive domains.

Proof. Use Lemma 61 for nonnegative slack, the covered-degree zero row, the zero tail above maximum degree, and the covered deletion inequality. Use Lemma 72 for vertex-set restriction and extension.

Fix the graph and set in the first assertion, and fix any of the stipulated covered processes. Write its induced graphs as $G_0 = G$, G_1,..., $G_t = B$, its deleted vertices as v_1,..., v_t, and $T = \{v_1, \dots, v_t\}$. Put $m = \nu(G)$ and $f = |V(G)| - 2m = a - 1 - 2p$. A matching uses two vertices per edge, so $f \geq 0$ and $a \geq 2p + 1$.

If a parent J has zero slack and a covered maximum degree, deleting the chosen vertex v gives

$0 = ps(J) \geq ps(J - v) + 1 - (\nu(J) - \nu(J - v))$.

Nonnegative slack and the matching loss belonging to {0,1} force $ps(J - v) = 0$ and $\nu(J - v) = \nu(J) - 1$. Induction gives zero slack at every stage, $\nu(G_j) = m - j$, and deficiency $|V(G_j)| - 2\nu(G_j) = f + j$. The unmatched vertices of a maximum matching are independent: an edge between two of them would enlarge that matching. Thus $f + j \leq \alpha(G_j) \leq a$, since G_j is induced in H. In particular

$t \leq a - f = 2p + 1$. (1)

Maximality of S gives $\Delta(G) \leq d - 1$, and $\alpha(G) \leq a$. The density bound of Lemma 146 applies to G with $\cap$ $D = d - 1$ and this same process: its zero slack and all covering hypotheses are supplied explicitly. Dropping its nonnegative refinement gives

$e(G) \leq 3m + (d - 4)(\alpha(G) - f) \leq 3m + (d - 4)(2p + 1)$. (2)

Here the first expression is its $B_D(G)$ after substituting $|V(G)| = 2m + f$. Independence of S and regularity of H give

$e(G) = d(|V(G)| - a)/2 = d(m - p - 1/2)$.

Combining with (2) yields

$(d - 3)m \leq (3d - 8)(p + 1/2)$.

Since $d - 3 \geq 4p + 1$ and $p \geq 1$,

$m \leq 3p + 3/2 + (p + 1/2)/(d - 3) < 3p + 2$.

The strict last inequality follows from $2p + 1 < 4p + 1$. Integrality proves

$m \leq 3p + 1$. (3)

The terminal B is nonempty. The set S is nonempty because H is nonempty, and any vertex of S has d distinct neighbors, all in G. Hence $|V(G)| \geq d > 2p + 1 \geq t$.

We next find g in B with at least $t + 2$ original neighbors in S. If $t \leq 2p$, take any g in B. It has at most two neighbors in B and at most t in T, so

$|N_H(g)$ intersect $S| \geq d - 2 - t \geq 4p + 2 - t \geq t + 2$. (4)

If $t = 2p + 1$, then the terminal deficiency is $f + t = a$. The preceding independent-unmatched-set argument and $\alpha(B) \leq a$ imply $\alpha(B) = a$. Since $\Delta(B) \leq 2$, every connected component is a path or a cycle, including singletons as paths. This classification follows by following successive edges from a vertex of degree at most one, or, when every degree is two, around the resulting cycle. On a path of order l the matching and independence numbers are $\lfloor l/2 \rfloor$ and $\lceil l/2 \rceil$; on a cycle of order l both are $\lfloor l/2 \rfloor$, by alternating choices and the bound obtained by pairing consecutive vertices, or sending selected cycle vertices to their distinct unselected successors.

For each such component C, $\alpha(C) - (|V(C)| - 2\nu(C))$ is nonnegative. It is zero only for a singleton or a triangle: for paths of orders 2h and $2h + 1$ its values are h and h, respectively; for cycles of orders 2h and $2h + 1$ they are h and h-1. The component sums equal $\alpha(B) - a = 0$, so B consists of triangles and singletons. Its number of components is a and its number of triangles is

$\nu(B) = m - t \leq p$

by (3). Since $a \geq 2p + 1 > p$, it has a singleton g. All G-neighbors of g lie in T, giving

$|N_H(g)$ intersect $S| \geq d - t \geq 2p + 3 = t + 2$. (5)

Choose any $t + 2$ neighbors R in S supplied by (4) or (5), and define U as in the statement. The three displayed sets defining U are disjoint, so $|U| = a - (t + 2) + t + 1 = a - 1$. The retained graph $J = H - U$ has vertex set R disjoint union (V(B) minus {g}). Every vertex of R loses its original neighbor g, so its J-degree is at most d-1. Every vertex outside R has at most two neighbors in B-g and at most $t + 2$ in R, so its J-degree is at most $t + 4 \leq 2p + 5$.

If $p \geq 2$, then $2p + 5 < 4p + 3 \leq d - 1$. If $p = 1$ and $d \geq 10$, the same strict bound holds because $2p + 5 = 7 < d - 1$. In these cases every vertex of J-degree d-1 lies in R, giving at most $t + 2 \leq 2p + 3$ such vertices. The only remaining parameter pair is $p = 1$, $d = 8$. If $t \leq 2$, vertices outside R have degree at most six, and there are at most $|R| \leq 4$ vertices of degree seven. If $t = 3$, the terminal is triangles and singletons with at most one triangle, as proved above. Only its at most three triangle vertices can attain degree seven outside R. Together with $|R| = 5$ this gives at most eight degree-seven vertices. Thus in every case $\Delta(J) \leq d - 1$ and J has at most $2d + 1$ vertices of degree d-1.

Apply Lemma 86 with the odd integer $r = d - 1 \geq 7$. Its excluded odd-order r-regular component cannot occur at odd r, and its covering threshold is $2(r + 2) - 1 = 2d + 1$. Therefore J has a matching covering every degree-r vertex. The same fact extends this fixed residual and matching over U, supplying every asserted original-edge certificate for every capacity pair and matching-budget pair. It includes edges internal to U and allows empty certificate parts and zero coordinates. This proves the first assertion and $c_{d-1}(H) \leq a - 1$.

For the consequence, fix the indicated maximum independent set S and put $k = \nu(H)$. Factor-criticality gives $|V(H)| = 2k + 1$. Choosing v in S and a perfect matching of H-v, then deleting the other $|S| - 1$ vertices of S, leaves at least $k + 1 - a$ matching edges in G. Hence $\eta \geq 0$. Again $\Delta(G) \leq d - 1$.

We have $c_d(H) = 1$. Indeed H-v has a perfect matching, hence zero row d by Lemma 61; restoring v gives $c_d(H) \leq 1$. If $c_d(H) = 0$, the capacity pair (1,d-1) and zero matching budgets would partition all H-edges into bounded parts. Regularity forces degree exactly one in the first part at every vertex, a perfect matching on the odd-order H, which is impossible. The higher rows of H, and all rows of G at indices at least d, vanish.

For integers $1 \leq i \leq d - 1$ put $r_i = c_i(H) - c_i(G)$. Restriction and extension give $0 \leq r_i \leq a$. Set $\rho = \sum_{i=1}^{d-1}(a - r_i) \geq 0$. Subtracting the slack definitions, using $e(H) - e(G) = da$, $\nu(G) = k + 1 - a + \eta$, and $c_d(H) = 1$, gives

$ps(G) = p - \eta - \rho$. (6)

Nonnegative slack implies $\eta \leq p$. If $\eta = p$, equality (6) forces $ps(G) = 0$ and $\rho = 0$. Thus $r_i = a$ for all those indices, and

$|V(G)| - 2\nu(G) = a - 1 - 2p$.

The first assertion applies with the stipulated process and gives $c_{d-1}(H) \leq a - 1$.

On the other hand $c_{d-1}(G) = 0$. If $\Delta(G) < d - 1$, the empty matching covers its empty degree-(d-1) set. If $\Delta(G) = d - 1$, the first parent of the stipulated process is covered, since $d - 1 \geq 7$. In both cases Lemma 61 gives the zero row. Since $\rho = 0$ also gives $r_{d-1} = a$, we obtain $c_{d-1}(H) = a$, a contradiction. Therefore η cannot equal p, and the integral bound is $0 \leq \eta \leq p - 1$ as claimed. □

Remark. The exact zero-slack density bound gives a smaller residual matching number than the older subcubic estimate. Only the longest deletion process requires a special choice; its terminal has more components than triangles and therefore an isolate with enough original independent-set neighbors. Keeping those neighbors replaces the deleted pool by a smaller one.

Depends on. Lemma 23, Lemma 61, Lemma 72, Lemma 86, Lemma 146.

Lemma 196 (Parity interpolation and integer slopes for color overflow). *Let $D \geq 3$ be an integer and G a finite simple graph with $\Delta(G) \leq D$. Let ρ be the minimum number of ambient degree-D vertices left uncovered by a matching. For positive integers A, B with $A + B = D$, let $\mu = \mu_{A,B}(G)$ be the minimum over red-blue edge colorings of*

$\sum_{v \in V(G)} [\max(d_{red}(v) - A, 0) + \max(d_{blue}(v) - B, 0)]$.

For disjoint $S, T \subseteq V(G)$, let $q_1(S,T)$ count the components C of G-(S union T) consisting wholly of vertices of degree D in G with $|V(C)| + e_G(C,T)$ odd. For real x in [0,D], put

$b_x(S,T) = q_1(S,T) + e_G(S,T) - x|S| - (D-x)|T|,$

$F(x) = \max_{S,T disjoint} b_x(S,T).$

Here $e_G(X,Y)$ counts edges joining disjoint vertex sets X, Y. The function F is convex; $F(a) = \mu_{a,D-a}(G)$ for every odd integer a with $1 \le a \le D-1$; and $F(1) = \rho$. For each fixed S, T, the number $z = |S| - |T|$ is an integer and $b_x(S,T) - b_y(S,T) = (y-x)z$.

If D is odd, let O and E be respectively the odd and even members of {A, B}. Then $F(D) = 0$ and

$\mu \le E\ \rho/(D-1)$, *and*

$\rho \ge \mu + (O-1)\lceil \mu/E \rceil$.

Thus $\mu > 0$ implies $\rho \ge \mu + O - 1$.

If D is even and A, B are odd, let κ be the number of whole connected components of G that have odd order and are D-regular. Put $m = \min(A,B)$ and $w = D/2 - m$. Then $F(D-1) = \rho$, $F(D/2) = \kappa$, and

$\kappa \le \mu \le \rho - [2(m-1)/(D-2)](\rho - \kappa)$.

If $w = 0$, then $\mu = \kappa$. If $w > 0$, then

$\rho \ge \mu + (m-1)ceil((\mu - \kappa)/w)$.

In particular, either $\mu = \kappa$ or $\rho \ge \mu + m - 1$. If A, B ≥ 3 and $\mu = \rho$, then $\rho = \kappa$; every counted odd regular component has a matching leaving exactly one vertex uncovered, and the union of the other components has a matching covering all its ambient degree-D vertices. No factor-criticality is assumed. Empty graphs and isolated vertices are allowed.

Proof. Apply the exact constant-cost overflow theorem Lemma 143. For an odd integer a in the stated range, its parity condition $a|V(C)| + e_G(C,T)$ odd is the one defining q_1. Hence $F(a) = \mu_{a,D-a}(G)$. Its endpoint identity gives $F(1) = \rho$. All extrema over pairs and matchings are attained, since the graph is finite.

For real x, y in [0,D] and $0 \le \theta \le 1$, every fixed pair satisfies

$b_(\theta x + (1-\theta)y)(S,T) = \theta\, b_x(S,T) + (1-\theta) b_y(S,T)$.

Bounding the two terms by their respective maxima and then maximizing the left side proves

$F(\theta x + (1-\theta)y) \le \theta\, F(x) + (1-\theta)F(y)$.

Thus F is convex. Subtracting the definitions for one fixed pair gives

$b_x(S,T) - b_y(S,T) = (y-x)(|S| - |T|)$.

This is an identity for that same pair, whose difference of cardinalities is an integer.

Suppose D is odd. Fix S, T and put $U = V(G) - (SunionT)$. For every component C counted by q_1, the ambient degree sum is

$D|V(C)| = 2e_G(C) + e_G(C,S) + e_G(C,T)$,

where $e_G(C)$ counts edges with both endpoints in C. Taking parity and using odd D and the definition of q_1 shows that $e_G(C,S)$ is odd, hence positive. Different counted components have disjoint joining edge sets into S. Therefore

$q_1(S,T) \le e_G(U,S)$.

Summing degrees in S gives

$E_G(U,S) + e_G(S,T) \le D|S| - 2e_G(S) \le D|S|$.

It follows that $b_D(S,T) \le 0$. At $S = T = empty$, no component is counted: a whole D-regular component has even order when D is odd, by the degree-sum identity. This pair gives zero, so $F(D) = 0$.

Interchanging the names of the colors is a bijection on colorings preserving overflow after interchanging capacities. We may therefore place O in the red position. Since O is odd, $\mu = F(O)$. Convexity between 1 and D gives

$Mu = F(O) \le [(d-o)/(d-1)]F(1) + [(o-1)/(d-1)]F(D)$

$= E\,\rho/(D-1)$.

For the integer improvement, if $\mu = 0$ the asserted inequality is $\rho \ge 0$. Otherwise take any attaining pair S, T with $b_O(S,T) = \mu$, and set $z = |S| - |T|$. From $b_D(S,T) \le 0$ and the affine identity,

$\mu = b_D(S,T) + E\,z \le E\,z$.

Since E is a positive even integer and z is integral, $z \ge \lceil \mu/E \rceil$. Also $b_1(S,T) \le F(1) = \rho$. Using the same pair again gives

$\rho \ge b_1(S,T) = \mu + (O-1)z$

$\ge \mu + (O-1)\lceil \mu/E \rceil$.

The argument includes $O = 1$. For positive μ the ceiling is at least one, proving the stated consequence.

Now suppose D is even. Because D-1 is odd, the constant-cost formula and color symmetry give $F(D-1) = \mu_{D-1,1}(G) = \rho$. We prove the midpoint identity for the same function F even if $D/2$ is not odd. Fix disjoint S, T, put $X = S$ union T and $U = V(G) - X$. A counted component C with a nonempty boundary into X has even boundary size, since

$D|V(C)| = 2e_G(C) + e_G(C,X)$

and D is even. That boundary therefore contains at least two edges. A counted component with empty boundary is a whole connected D-regular component of G; its parity condition then says that its order is odd. At most κ such components occur. Distinct remaining counted components have disjoint boundary edge sets, so

$q_1(S,T) \le \kappa + e_G(X,U)/2$.

On the other hand, summing ambient degrees over X and using $\Delta(G) \le D$ gives

$D|X|/2 - e_G(S,T)$

$\ge e_G(S) + e_G(T) + e_G(X,U)/2$

$\ge e_G(X,U)/2$.

Combining these inequalities yields $b_{D/2}(S,T) \le \kappa$. The empty pair $S = T = empty$ counts exactly the κ whole odd D-regular components and gives equality. Thus $F(D/2) = \kappa$.

For odd capacities A, B we may again interchange colors, so the smaller odd capacity m occupies the red position. Then $\mu = F(m)$, where $1 \le m \le D/2$. Convexity between 1 and $D/2$, whose difference is $(D-2)/2 > 0$, gives

$\mu \le \rho - [2(m-1)/(D-2)](\rho - \kappa)$.

Each whole odd D-regular component contributes at least one to ρ: no matching can cover its odd number of vertices. Thus $\rho \ge \kappa$. Each also contributes at least one to μ for odd A, B. Indeed, zero overflow on that component would force the red degree to be exactly A at every vertex because total degree is $A + B$; the red degree sum would then equal the odd integer $A|V(C)|$, impossible for a graph. The overflow is a nonnegative integer and is additive over whole components, so $\mu \ge \kappa$.

When $w = D/2 - m = 0$, the midpoint identity gives $\mu = \kappa$. Suppose $w > 0$. If $\mu = \kappa$, then $\rho \ge \kappa$ proves the displayed ceiling inequality since its ceiling term is zero. If $\mu > \kappa$, choose any pair attaining $b_m(S,T) = \mu$ and put $z = |S| - |T|$. The midpoint upper bound and the affine identity give

$\mu = b_{D/2}(S,T) + w\,z \le \kappa + w\,z$.

The positive integer w and integral z imply $z \ge ceil((\mu - \kappa)/w)$. At the same pair,

$\rho \ge b_1(S,T) = \mu + (m-1)z$

$\ge \mu + (m-1)ceil((\mu - \kappa)/w)$.

If $\mu > \kappa$ the ceiling is at least one, proving the simpler alternative.

Finally, if A, $B \geq 3$ and $\mu = \rho$, then $m \geq 3$ and the positive coefficient $2(m-1)/(D-2)$ in the convex bound forces $\rho - \kappa = 0$. The minimum exposed degree-D count is additive over connected components, because matchings in separate components can be chosen independently. Every one of the κ odd regular components contributes at least one, and their number equals the total ρ. Hence each has minimum exposed count exactly one, and each remaining component has exposed count zero. In an odd regular component every vertex is degree D, so its attaining matching leaves exactly one vertex uncovered. Uniting attaining matchings of the other components gives the claimed covering matching. This does not assert that a matching exists after deletion of every specified vertex. All arguments allow empty collections of components and zero exposure, which covers empty graphs and isolated vertices. □

Remark. Odd-parity barrier values are affine in capacity with an integer slope. Their matching endpoint and parity midpoint or zero endpoint give both convex interpolation bounds and stronger integral ceiling bounds.

Depends on. Lemma 143.

Lemma 197 (Matching rank in a sextic graph of slack one). *Use the finite integer certificate costs c_i, matching number ν, independence number α, maximum degree Δ, excess $ex = c_0 - \nu$, nonnegative slack ps, and signed $gap = ex - ps$ of Lemma 61. All graphs are finite and simple. A nonempty graph is factor-critical if deleting any vertex leaves a perfect matching. A matching covers a vertex when the vertex is an endpoint of one of its edges.*

Let H be a nonempty connected six-regular factor-critical graph with $ps(H) = 1$ and $gap(H) \leq 1$. Suppose every induced subgraph J of H with $ps(J) = 0$ and $\Delta(J) \geq 3$ has a matching covering all vertices of degree $\Delta(J)$. Then $\nu(H) \leq 2\alpha(H)$.

The covering condition includes H itself if applicable and makes no assertion about positive-slack induced graphs. No gap bound on proper induced subgraphs, minimality assumption, or upper order bound is imposed. All graph orders, degrees, indices, costs, capacities, budgets, matching and independence cardinalities, excesses and slacks are nonnegative integers; gaps may be signed. Every vertex deletion is induced.

Proof. Put $|V(H)| = 2k+1$, $a = \alpha(H)$, and $E = ex(H)$. Factor-criticality gives $\nu(H) = k$, and six-regularity gives $e(H) = 6k+3$ and $k \geq 3$. Since $ps(H) = 1$ and $gap(H) \leq 1$, one has $0 \leq E \leq 2$. Define the integer $\rho = k + 1 - 2a$. The economical cutoff theorem Lemma 123, applied at cutoff $c_0(H)$, gives

$1 = ps(H) \geq 2(|V(H)| - c_0(H)) - 4a = 2(\rho - E)$.

As rho-E is an integer, $\rho \leq E$. If $E \leq 1$, this already gives $k \leq 2a$. We henceforth suppose $E = 2$.

Every zero-slack induced graph J of H admits a complete covered maximum-degree deletion process down to degree at most two. Indeed, whenever its maximum degree is at least three, the stated hypothesis supplies a covering matching. Delete any vertex of that maximum degree. By Lemma 61 the child has slack at most zero and at least zero, and the matching number drops by one. The child is still an induced subgraph of H, so the same argument continues. Finitely many vertex deletions terminate. Thus Lemma 146 applies to every such J at any integer degree $\cap$ at least three and at least $\Delta(J)$.

We first record the exact slack identity for a maximum independent set S of H. Put $G = H - S$ and $\eta = \nu(G) - (k+1-a)$. A perfect matching after deleting one vertex of S loses at most a-1 further edges when the other vertices of S are deleted, so $\eta \geq 0$. Every outside vertex has a neighbor in S by maximality; hence $\Delta(G) \leq 5$.

The sixth row of H is exactly one. For its upper bound delete any vertex and use a perfect matching of the remaining graph. This matching covers all vertices of degree six there, so Lemma 61 gives zero sixth cost on that residual; restoration in Lemma 72 gives $c_6(H) \leq 1$. If $c_6(H) = 0$, a zero-budget certificate at capacities (1,5) partitions H into those two degree bounds. Every vertex has total degree six, so its degree in the first part must be exactly one. That would be a perfect matching on odd order, a contradiction. Thus $c_6(H) = 1$. Rows of H above six and rows of G above five vanish by Lemma 61.

For integers $1 \leq i \leq 5$, restriction and vertex restoration in Lemma 72 give

$0 \leq c_i(H) - c_i(G) \leq a$.

Set $u = \sum_{i=1}^{5}(a - c_i(H) + c_i(G))$, a nonnegative integer. Independence of S means that deleting it removes exactly 6a edges. Subtract the two slack definitions, using $\nu(G) = k + 1 - a + \eta$ and the top rows just determined, to obtain

$ps(G) = 1 - \eta - u$. (1)

Nonnegative slack gives $\eta \le 1$. This identity concerns every maximum independent set and the actual induced complement.

We next exclude $a \le 2$. First, $E = 2$ forces $k \ge 5$. If $k \le 4$, take any nonnegative integer budget split $x + y = k + 1$, relabeled so $x \ge y$. Then $y \le 2$. Choose y vertices, put every incident edge in the part with budget y, and put all remaining edges in the part with budget x. The latter graph has at most

$2k + 1 - y \le 2x + 1$

vertices, since $x = k + 1 - y$ and $y \le 2$. Its matching number is at most x; the former part has matching number at most y because every edge meets the chosen y-set. Both degree-bounded parts are empty. Restoring the original budget labels gives every certificate at total $k + 1$, contradicting $E = 2$. Hence $k \ge 5$.

If $a \le 2$, the previously proved $\rho \le 2$ gives $k \le 2a + 1 \le 5$. Consequently $k = 5$ and $a = 2$. Fix a maximum independent pair S. Its complement G has nine vertices, twenty-one edges, minimum degree at least four, maximum degree at most five, and $\alpha(G) \le 2$. Its matching number is four by the preceding matching lower bound and the order bound, so $\eta = 0$ and (1) gives $ps(G) \le 1$.

The graph G is connected, since each component of minimum degree four needs at least five vertices. It has no cut vertex w. Otherwise the components of G-w each have minimum degree at least three, hence at least four vertices. There are only eight vertices, so there would be exactly two four-vertex components. To have G-degree at least four, each of their vertices would need all three internal neighbors and also w. This gives w degree eight, contradicting $\Delta(G) \le 5$.

Thus G is a connected graph of odd order with $\alpha(G) \le 2$ that remains connected upon deletion of any vertex. By Lemma 26 it is factor-critical. Both degrees four and five occur because its degree sum is forty-two on nine vertices, and all its degrees lie in {4,5}. Choose a degree-four vertex z and a perfect matching of G-z. The same matching in G covers every degree-five vertex. Choose a degree-five vertex w and put $K = G - w$. Factor-criticality gives $\nu(K) = \nu(G) = 4$. The covered deletion inequality of Lemma 61 now gives $ps(K) \le ps(G) - 1 \le 0$; nonnegativity makes $ps(K) = 0$.

This K has eight vertices, sixteen edges, maximum degree at most five and independence number at most two. It is induced in H, so the complete covered process established earlier exists. Apply Lemma 146 with $\cap$ $D = 5$, and put $b = \nu(K) - c_1(K) \ge 0$ as in that theorem. Its refined density bound gives

$16 = e(K) \le -2 \cdot 8 + 7 \cdot 4 + 2\alpha(K) - 4b$

$= 12 + 2\alpha(K) - 4b \le 16.$

All inequalities are equalities. Thus $\alpha(K) = 2$, $b = 0$, and equality holds in the refined density theorem. Its equality conclusion supplies

$2\nu(K) - |V(K)| + \alpha(K) - b = 2$

independent deleted vertices of original K-degree five. Both avoid w in G: a neighbor of w would have G-degree at least six, since its K-degree is five. These two vertices together with w form an independent triple in G, contradicting $\alpha(G) \le 2$. This excludes $a \le 2$, so $a \ge 3$.

Now use Lemma 140 at $d = 6$, $E = 2$ and $a \ge 3$. It gives $\eta \ge 1$ for every maximum independent S. Equation (1) and nonnegative slack force $\eta = 1$, $u = 0$ and $ps(G) = 0$. Write $m = \nu(G) = k + 2 - a$. Then

$|V(G)| - 2m = a - 3$, and $\alpha(G) \le a$.

The inherited covering condition gives a complete covered process on G, whose maximum degree is at most five. Applying Lemma 146 with $D = 5$ and discarding its nonnegative refined correction yields

$e(G) \le -2|V(G)| + 7m + 2\alpha(G)$

$= 3m + 2(\alpha(G) - (|V(G)| - 2m))$

$\le 3m + 6.$

On the other hand independence of S and six-regularity give

$e(G) = 6k + 3 - 6a = 6m - 9.$

Consequently $3m \le 15$ and $m \le 5$. Thus

$k = m + a - 2 \le a + 3 \le 2a,$

where the last inequality uses $a \ge 3$. This proves the required matching-rank bound in the remaining $E = 2$ case and completes the proof. □

Depends on. Lemma 26, Lemma 61, Lemma 72, Lemma 123, Lemma 140, Lemma 146.

(printed in plain notation)

Lemma 198 (Full matching projections at a terminal two-star pair). `Full matching projections at a terminal two-star pair.`

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Let F,J be star forests with complete nonincreasing positive integer edge-size lists
n_1,...,n_s and m_1,...,m_t, where s,t>=1, n_1=D>=3 and m_1<=D. Their components are
vertex-disjoint and copies are ordinary prescribed-color subgraphs. Put M=s+t-1 and
ell_l=max{n_i+m_j-1:1<=i<=s,1<=j<=t,i+j=l} for every integer 2<=l<=s+t. Fix an integer
0<=q<min(s,t), put K=M-q, L_q=sum_(l=q+2)^(s+t)ell_l and P_q=L_q-K, and assume
ell_(q+2)=D+1. Let R be the initial D+1 plateau length of this tail. Suppose valid
terminal indices i_0+j_0=q+R+1 have n_(i_0)=D and m_(j_0)=2. Let U,z,u count the full F
components of size D, of sizes 3 through D-1, and of size one, respectively; let V,v
count the full J nonunits and units.

Let A_q be the family of all ordered pairs of nonempty actual suffixes (F^alpha,J^beta)
for nonnegative integer leading cuts alpha<s, beta<t and alpha+beta=q. For a positive
star forest A, let c(A) be its component count and A^- the forest obtained by shortening
each nonunit once and omitting its units. Write K2 for one edge. Form a tagged family C
from each actual pair (A,B): if A^- is nonempty, include (c(A^-)K2,B) and (c(B)K2,A^-),
tagged F; if B^- is nonempty, include (c(A)K2,B^-) and (c(B^-)K2,A), tagged J. Each
matching is displayed first by exchanging both color names when necessary.

Both tags occur. Every F-tagged member has component budget K-u, and every J-tagged
member has budget K-v, where the budget of (pK2,T) is p+c(T)-1. Thus the maximum budget
is N=K-min(u,v). For each integer 1<=i<=K define lambda_i as the maximum component size
y_j over all members (pK2,T) and all valid integers 1<=a<=p, 1<=j<=c(T), a+j=i+1; an
empty maximum is zero. If R+v>=z, then

lambda_i=ell_(q+i+1)-1 for every 1<=i<=K.

Consequently sum_i lambda_i=P_q, the positive length is N, the head is D and its initial
plateau length is R.

For every finite simple H arrowing all of A_q, and every matching W of H, the graph H-W,
deleting only matching edges, arrows every member of C simultaneously. Now also assume
Delta(H)<=D and delta=K-nu(H)>=1, where nu denotes maximum matching cardinality. Then
R+v>=z holds automatically. Put sigma=e(H)-nu(H)-P_q, let rho be the minimum number of
degree-D vertices uncovered by an H-matching, and let kappa count the whole odd D-regular
factor-critical components of H. Factor-critical means nonempty with a perfect matching
after every one-vertex deletion.

Define alpha*=min(q,U-1), beta*=q-alpha*, h=U-alpha*, and p=V-beta*. Then h,p>=1, p+h-1=R,
and C contains (pK2,F^(alpha*)) of budget K-v. Put g=(v-u)_+. One has sigma>=0, rho>=R and

sigma+g >= rho-R+h ceil(D/2).

For even D this strengthens to

sigma+g >= rho-R+hD/2+min(h,max(R-kappa,0)).

In particular sigma<ceil(D/2) implies v>u. If kappa=0 then sigma+g>=h ceil((D+1)/2).
Equality sigma+g=h ceil(D/2) implies rho=R, and for even D also kappa=R.

All graph orders, degrees, counts, indices, budgets, matching numbers, surpluses and
parameters in these assertions are integers in their displayed domains. The notation
x_+=max(x,0) denotes positive part. Units, repeated sizes, q=0, empty shortened
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projections and isolated host vertices are allowed. No actual suffix is assumed to have a binary side except where that property is proved for the specified suffix.

Proof. For every positive integer a put r_a=|{i:n_i>=a}| and c_a=|{j:m_j>=a}|. For a positive threshold b, the natural q-tail width at b is

T_b=max{(r_x+c_y-q-1)_+:x,y>=1, x+y=b+1, r_x>0,c_y>0}. (1)

Indeed, a size sum is at least b+1 precisely when it dominates positive thresholds x,y with that sum. The corresponding original indices form the rectangle [1,r_x] by [1,c_y], whose index sums fill every integer from 2 through r_x+c_y. Taking the union of these initial intervals proves (1). It also proves that the full diagonal sequence is nonincreasing.

The terminal indices give U+V>=q+R+1. A strict inequality would place a valid size sum at least D+2 beyond the last D+1 entry of the tail, contradicting (1). Thus

U+V=q+R+1. (2)

If c_3>0, the size sum at indices U,c_3 is at least D+3. Criticality and monotonicity therefore give

U+c_3<=q+1. (3)

We verify the projections before computing their envelope. Fix an actual pair (A,B), a matching W of a common arrowing H, and any coloring of Q=H-W. Color all W-edges with the color of A. A forced full B-copy lies in Q. Otherwise a forced full A-copy loses at most one edge at each star center when W is removed. Retain the required one-smaller number of edges at each nonunit center and omit the unit components. This gives the entire A^- in Q, with its components still disjoint. When A^- is nonempty, Q therefore arrows (A^-,B). Selecting one edge from each component of either target gives the two F-tagged matching relaxations, after exchanging colors for the second one. Giving all W-edges the color of B proves the two J-tagged assertions. This argument works for every coloring of the same Q, so all projections hold simultaneously for this fixed W.

If A^- is nonempty, its suffix A still contains a nonunit. The leading cut removed no original F-unit, so c(A^-)=c(A)-u. Both F-tagged budgets are consequently K-u. The J-tagged budgets equal K-v for the same reason. The legal corner with alpha=0 retains the full nonunit F, and the legal corner with beta=0 retains the nonunit J. They ensure that both tags occur and that their budgets are positive. Hence N=K-min(u,v).

A pair (aK2,T) whose full opposite list has d>0 entries at least b has exactly a+d-1 profile entries at least b. If d=0 it has none. For the F-tagged family (c(A^-)K2,B), its width at b, when both relevant suffixes retain these entries, is

r_2+c_b-q-1.

This width occurs precisely when r_2,c_b>0 and r_2+c_b>=q+2. To check the converse, choose alpha=max(0,q-c_b+1), beta=q-alpha. Then alpha<=r_2-1 and beta<=c_b-1, and the two cuts are legal. Thus its envelope width is (r_2+c_b-q-1)_+, omitting the expression when either count is zero. The other three projection types have the same verification with their two threshold counts. Consequently the full envelope width C_b is the maximum of the valid positive parts of

r_2+c_b-q-1, t+r_(b+1)-q-1,

s+c_(b+1)-q-1, r_b+c_2-q-1. (4)

A term is omitted if its displayed threshold count is zero. The two unthresholded counts s,t are positive.

Each term in (4) is a term of T_(b+1) in (1), corresponding to thresholds (2,b),(b+1,1),(1,b+1),(b,2). Thus C_b<=T_(b+1). Assume R+v>=z. For 1<=b<D, every omitted threshold split has both coordinates at least three. Such a split requires c_3>0 and, by (3), has width at most (U+z)+c_3-q-1<=z, also after taking positive part. On the other hand r_(b+1)>=U, so the second term of (4) is present and at least t+U-q-1=R+v>=z. Hence all omitted splits are dominated and C_b=T_(b+1).

For b=D the last term in (4) is U+V-q-1=R. Since T_(D+1)=R, equality holds at this threshold as well. For b>D both widths are zero: the natural head is D+1 and all projected heads are at most D. Each matching profile is nonincreasing, by decreasing a

positive index greater than one in a candidate on a later diagonal. Padding by zeros and taking maxima preserves this property. The nonnegative natural potential list is nonincreasing too. A nonincreasing integer list is determined by its threshold widths, since its i-th entry counts the positive thresholds whose width is at least i. We have therefore proved the stated identity at all K coordinates. Every positive matching profile has its budget as its exact positive length, so the envelope has positive length N. Its sum, head and plateau are as asserted.

Now impose the graph hypotheses. The full-target bound Lemma 89 on H and the same complete A_q yields

t>=q+2delta+2sum_i floor((n_i-1)/2).

Combining this with (2) and t=V+v gives

R+v>=U+2delta+2sum_i floor((n_i-1)/2)-1>=z.

For the last inequality U>=1, delta>=1 and the z lower nonunits each contribute at least two to the doubled sum. Thus the complete envelope identity is available for this H.

We next verify the distinguished projection. The cuts alpha*,beta* are nonnegative and total q. Also alpha*<=U-1<s. If q<U-1, then beta*=0<V, and (3) implies c_3=0. If q>=U-1, then beta*=q-U+1=V-R<V; whenever c_3>0, (3) gives c_3<=beta*. Thus in both cases beta*<V<=t and J^(beta*) consists of p=V-beta*>=1 two-edge stars and all its v units. The suffix F^(alpha*) retains h=U-alpha*>=1 size-D stars. Equation (2) gives p+h-1=R. The J-tagged decrement projection is exactly (pK2,F^(alpha*)); its budget is K-v, with deficit from N equal to g=v-min(u,v)=(v-u)_+.

Apply Lemma 158 to H and its degree-D marked set. It supplies one maximum matching W attaining rho with its simultaneous neighborhood-loss bound. Put I={x:deg_H(x)=D} minus V(W), Q=H-W and T=Q-I. Every vertex of I has Q-degree D, and every other vertex has Q-degree at most D-1: matched vertices lose one incident edge and unmatched lower-degree vertices keep their lower degree. The set I is independent, because an edge between two of its vertices could be added to W. Its size is rho and e(Q)=e(H)-nu(H)=P_q+sigma. This same Q arrows the entire C, as already proved.

In particular it arrows (pK2,F^(alpha*)). If rho<p+h-1, color all Q-edges meeting any min(p-1,rho) vertices of I red and every other edge blue. Red has matching number at most p-1. Every blue size-D star must be centered in the remaining vertices of I, of which there are fewer than h. This avoids the entire opposite forest, a contradiction. Hence rho>=R=p+h-1.

Apply Lemma 78 to the same H,W and finite family C. Its full positive envelope has length N and sum P_q, all its targets are positive, and W minimizes marked exposure. Define

A_D(r)=sum_(i=1)^min(r,N)(D-lambda_i)+(D-1)max(r-N,0).

That theorem gives sigma>=A_D(rho)>=0 and a cover C_Q of Q containing I with

|C_Q|<=N+sigma-A_D(rho). (5)

Because the first R entries are D and every later positive entry is at most D-1, one has A_D(rho)>=rho-R. For rho<=N, count the terms from R+1 through rho. For rho>N, the first N terms give at least N-R and the last term gives at least rho-N.

Write F^(alpha*)=hK_(1,D) disjoint union J_0, where J_0 has t_0>=0 positive components, including all its retained units. Its budget identity says N-g=R+t_0. Put z_0=rho-R>=0. Let J_z be the full suffix of J_0 after omitting its min(z_0,t_0) largest components. It has t_z=max(t_0-z_0,0) components and is empty when t_z=0.

For every U_0 subseteq I of size rho-p+1=h+z_0, there is an h-element subset B_0 of U_0 with pairwise disjoint full Q-neighborhoods such that T-N_Q(B_0) contains J_z. To prove this, color all Q-edges meeting I minus U_0 red. Red is covered by p-1 vertices. A forced full blue F^(alpha*) has its h largest stars centered at vertices B_0 of U_0, since only I has Q-degree D and every vertex of I minus U_0 is blue-isolated. These stars use all neighbors of their centers and are pairwise disjoint. The remaining copied components avoid these stars and I minus U_0. Removing U_0 minus B_0 destroys at most z_0 smaller components. Sorting the surviving sizes shows that they dominate the suffix formed by omitting z_0 largest sizes; trim leaves and discard extras to obtain all of J_z in T-N_Q(B_0). Hence

nu(T-N_Q(B_0))>=t_z. (6)

The argument includes empty J_z.

Exactly kappa vertices of I lie in the whole components counted by kappa. Indeed every maximum matching restricts to a maximum matching in each whole component; a factor-critical odd component has a near-perfect matching and all its vertices are marked, so it contributes one exposed vertex. In particular kappa<=rho.

Choose U_0 with as few of these kappa roots as possible. Its size is rho-p+1, so it contains max(kappa-p+1,0) such roots. Every h-element B_0 obtained above then contains at least

max(h-max(kappa-p+1,0),0)=min(h,max(R-kappa,0))

roots outside them. The equality follows by considering kappa<=p-1, p-1<kappa<R, and kappa>=R. The simultaneous bound of Lemma 158 therefore gives

tau(T)-nu(T-N_Q(B_0))>=h ceil(D/2), (7)

and for even D the right side increases by min(h,max(R-kappa,0)). Here tau is minimum vertex-cover cardinality.

The set C_Q minus I covers T. Combining (5) and (6) bounds the left side of (7) above by

N+sigma-A_D(rho)-rho-t_z

=g+sigma-A_D(rho)-max(z_0-t_0,0)

<=g+sigma-(rho-R).

The equality uses N-g=R+t_0, rho=R+z_0 and t_z=max(t_0-z_0,0). This proves both claimed surplus bounds.

Since h>=1, sigma<ceil(D/2) forces g>0 and hence v>u. If kappa=0, the even-degree extra term equals h; at odd D the two displayed ceilings are equal. This proves the strengthened bound when kappa=0. Finally equality sigma+g=h ceil(D/2) forces rho-R=0. For even D it also forces min(h,max(R-kappa,0))=0, whence kappa>=R because h>=1. Since kappa<=rho=R, one has kappa=R. Every construction retained the same full originals and every selected target's smaller components.

□

Remark.

All four matching-decrement projections recover the natural potential profile once full-target capacity suppresses the remaining nonbinary threshold terms. Their distinguished head-D member forces disjoint exposed neighborhoods, whose matching losses must fit inside the exact cover surplus.

Depends on. Lemma 78, Lemma 89, Lemma 158.

Lemma 199 (Parity and size bounds for tight odd complementary levels). *Use the finite integer certificate costs $c_\theta(H)$ and $c_0(H)$ of Lemma 23. Let $d \geq 4$ be an even integer, and let H be a nonempty connected d-regular finite simple graph such that $H-v$ has a perfect matching for every vertex v. Put $n = |V(H)| = 2k+1$, where $k \geq 1$ is an integer. Let $\theta, \theta' \geq 1$ be odd integers with $\theta + \theta' = d$. If $c_\theta(H) + c_{\theta'}(H) = n+1$, then both costs are odd integers, $\theta, \theta' \geq 3$, $c_{\theta'}(H) \geq 6\theta + 9$, $c_\theta(H) \geq 6\theta' + 9$, and $n \geq 6d + 17$. Consequently $c_\theta(H) + c_{\theta'}(H) \leq n$ holds if $n \leq 6d + 15$ or $c_\theta(H) \leq 6\theta' + 8$ or $c_0(H) \leq 6max(\theta, \theta') + 8$. If $d/2$ and k are odd integers, then $c_{d/2}(H) \leq k$. No assumption on gaps, minimality, or proper induced subgraphs is imposed. All graph orders, degrees, thresholds, costs, matching sizes, and cardinalities are integers in their displayed positive or nonnegative domains.*

Proof. Assume the displayed equality. We use the configurations and weighted potential of Lemma 119 for thresholds θ, theta'. Choose any optimal configuration. Let x be its uncovered vertex, S its part of threshold t, and O the other part of threshold t', so $\{t, t'\} = \{\theta, \theta'\}$. The cited fact gives balance, internal matching edges, odd $|S|$, even $|O|$, and

$c_t(H) = |O| + 1, c_{t'}(H) = |S|$. (1)

In particular both costs in (1) are odd. Write $J = H[S]$. For a vertex u in either part, define $def(u)$ as its part's threshold minus its number of H-neighbors in that part. All deficiencies are nonnegative: in H-M, moving a covered vertex of threshold a to the other part of threshold b would lower the potential if it had at least a internal neighbors, because its total degree is $a + b - 1$. Its internal H-M degree is therefore at most a-1; the internal matching edge raises this by one. The uncovered vertex has internal degree exactly its threshold by balance.

We first establish structural properties that apply to every such optimum. By Lemma 169, decompose J into its exposable vertex set D, its external neighbor set A, and the remaining set C. The graph J has a matching missing exactly x, so J[D] has exactly $|A| + 1$ connected components, each factor-critical, and J[C] has a perfect matching. Every vertex of D has degree t in J by Lemma 119, hence deficiency zero; no J-edge joins D to C.

We will repeatedly exclude a vertex set W of size $|S|$ whose induced graph has maximum degree at most t and has a matching covering every degree-t vertex. Indeed Lemma 35 then supplies every degree splitting of total t on H[W]. Taking the exceptional parts empty gives $c_t(H[W]) = 0$ by Lemma 23. Extension Lemma 72 gives $c_t(H) \le n - |W| = |O|$, contradicting (1). (2)

There is no edge between a deficiency-zero vertex u of S and a deficiency-zero vertex z of O. To prove this, suppose uz is such an edge. If $u = x$, moving x to O changes the potential by zero, since x has t neighbors in S and t' in O. The result is optimal. The vertex z now has t' internal H-M neighbors and t-1 across, so moving z changes the potential by $-1 + (t-1)/t < 0$, a contradiction. If u differs from x, it has t-1 internal H-M neighbors and t' across. Move u to O, changing the potential by $1/t$. Now z has t' internal H-M neighbors and t-1 across, so moving z to S changes it by $-1/t$. The final configuration is optimal. The unchanged matching edge at u now crosses the partition: its partner stayed in S and cannot be z because the original matching edges were internal. This contradicts Lemma 119. (3)

Every deficiency-zero vertex z in O has exactly t neighbors in S, all of positive deficiency and all in C. The number and the positive deficiency follow from regularity and (3), and exclude D. Suppose a neighbor a belongs to A. Choose u_0 in D adjacent to a and a maximum matching N of J missing u_0; such a matching exists by the definition of D. Let b be the N-partner of a. Put $W = (Sminus\{a\})$ union {z} and restrict N by deleting ab. This matching covers all W-vertices except u_0, b, z. All degrees in H[W] are at most t: any vertex gaining a neighbor z had positive deficiency before the change. The degrees of u_0 and z are t-1. The degree of b is at most t-1 as well: it loses its neighbor a; if it gains z, its previous deficiency was at least one, while if it does not gain z, its previous degree was at most t. This contradicts (2), proving the assertion.

Fix any perfect matching N_C of J[C], and write w' for the partner of w in it. For a deficiency-zero z in O and any w in C adjacent to z, we have

$def(w') \le [w' \text{ is adjacent to z}]$, (4)

where the bracket is 1 when its assertion holds and 0 otherwise. By Lemma 169, extend N_C to a maximum matching N of J exposing x, using any incidence choices that expose x; such choices exist because x lies in D. In $W = (Sminus\{x\})$ union {z}, replace ww' by zw in N. This matching leaves only w' uncovered. All degrees are at most t because the old neighbors of z in S have positive deficiency, while z has t neighbors in W. Since no C-vertex is adjacent to x, the degree of w' in W is $t - def(w') + [w' \text{ is adjacent to z}]$. If (4) failed, this degree would be less than t and the matching would cover all degree-t vertices, contradicting (2).

We next count the components of J[D]. For any such component K, all its vertices have J-degree t, so the number b_K of edges from K to A is $b_K = t|V(K)| - 2e(K)$. Both t and $|V(K)|$ are odd, so b_K is an odd nonnegative integer and is therefore at least one. Also $\sum_K b_K \le t|A|$. Let q count components with $b_K \le t-2$ and p the others; since b_K is odd, the others have $b_K \ge t$. The number of components and the edge count give $p + q = |A| + 1$ and $tp + q \le t|A|$. Subtracting gives $t \le (t-1)q$. This excludes $t = 1$; therefore $t \ge 3$ and $q \ge 2$. For any integer r with $1 \le r \le t$, simplicity gives $b_K \ge r(t+1-r) \ge t$ when $|V(K)| = r$, since $r(t+1-r) - t = (r-1)(t-r) \ge 0$. A component of order $t+1$ is impossible because that order is even. Hence each of the q components has at least $t+2$ vertices. Moreover A is nonempty, since otherwise its edge count would be zero despite the positive b_K values. We have proved

$|D| \ge 2t + 4$, $|A| \ge 1$, $|S| \ge 2t + 5 + |C|$. (5)

All these assertions apply to every optimal configuration.

Move x from S to O, which preserves optimality by balance. Its new exposed part is O union {x}, with threshold t'. Let D', A', C' be the corresponding exposable decomposition of $J' = H[O \text{ union } \{x\}]$. The

conclusions already proved apply to this configuration with t and t′ interchanged. In particular, $t' \geq 3$, J′[D′] has $|A'| + 1$ components, at least two of order at least $t' + 2$, and $|A'| \geq 1$. Also x belongs to D′ and has exactly t′ neighbors in J′. Let $N_H[x]$ consist of x and all its H-neighbors.

The set D′ minus $N_H[x]$ contains at least $|A'| + 2$ pairwise nonadjacent vertices. To check this explicitly, let K_x be the D′-component containing x. Every other D′-component avoids $N_H[x]$, since an edge to x would join it to K_x. A D′-component of order at least $t' + 2$ contains two nonadjacent vertices: otherwise its vertices would have J′-degree at least $t' + 1$, although every D′-vertex has degree t′. Choose two large components K_1, K_2. If one of them is K_x, take a nonadjacent pair from the other, one vertex from K_x outside $N_H[x]$, and one vertex from every remaining component. The vertex in K_x exists because that component has at least $t' + 2$ vertices but $N_H[x]$ meets J′ in $t' + 1$ vertices. This yields $2 + 1 + (|A'| - 1) = |A'| + 2$ vertices. If neither large component is K_x, take a nonadjacent pair from each, omit K_x, and take one vertex from every other component. This yields $4 + (|A'| - 2) = |A'| + 2$ vertices. The latter case has at least three components, so $|A'| - 2$ is nonnegative. Vertices from distinct components are nonadjacent. Choose exactly $m = |A'| + 2 \geq 3$ vertices z_1,..., z_m in this manner.

We show that their neighborhoods $W_i = N_H(z_i)$ intersect S are pairwise disjoint subsets of C, each of cardinality t. A vertex z in D′ minus $N_H[x]$ has degree t′ in J′ and is not adjacent to x. Thus it lies in O and has exactly t′ neighbors in O, giving deficiency zero in the original configuration. Its t neighbors in S therefore belong to C by the assertion preceding (4).

There is an optimal configuration exposing this z with threshold-t part $S\cdot = (Sminus\{x\})$ union {z} and other part $O\cdot = (Ominus\{z\})$ union {x}. Indeed J′-z has a perfect matching $N\cdot$, since z is exposable in the odd graph J′ which has a near-perfect matching. Combine $N\cdot$ with the original matching on S minus {x}. The result is a near-perfect matching of H exposing z and internal to $S\cdot$, $O\cdot$. The matching-edge counts in the two parts equal their original values. The induced H-edge counts do also: exchanging x and z loses t and gains t edges on the S side, and loses t′ and gains t′ edges on the O side, because xz is not an edge. Thus the weighted potential is unchanged, and the configuration is optimal.

For any chosen z_i and any vertex w in O with deficiency zero that is adjacent to neither x nor z_i, put $z = z_i$ in this new optimum. The vertex w lies in $O\cdot$ and still has deficiency zero there. Apply the structural conclusion before (4) to this new optimum. Every neighbor of w in $S\cdot$ lies in its remaining set $C\cdot$. But z is exposable in $H[S\cdot]$, so none of its neighbors in $S\cdot$ lies in $C\cdot$, by the definition of that decomposition's external neighbor set. Consequently z and w have no common neighbor in $S\cdot$. Their possible common neighbors in S are the same, since neither is adjacent to x and z is adjacent to neither w nor itself. This proves the disjointness of all W_i.

There is one more deficiency-zero vertex w in O whose neighborhood can be added to this family. Choose y in D outside $N_H[x]$; it exists since $|D| \geq 2t + 4$ while x has exactly t neighbors in S. After moving x to O, the vertex y lies in the other part S minus {x} and still has deficiency zero. Apply the structural conclusion before (4), and then (4), in this moved optimum. Its t′ neighbors all lie in C′ and have positive deficiency in that configuration. Fix a perfect matching of J′[C′]. Since t′ is odd, some one of these neighbors has its matching partner w outside that neighbor set. The moved version of (4) gives deficiency zero at w. Because w belongs to C′, it is distinct from and nonadjacent to x, and it has no neighbor in D′. Thus w is in O with original deficiency zero and is nonadjacent to every z_i. Its t-neighborhood $W\cdot$ in S consists of positive-deficiency vertices of C. The disjointness argument of the previous paragraph applies to z_i and this w, so $W\cdot$ is disjoint from every W_i. We now have $m + 1 = |A'| + 3 \geq 4$ pairwise disjoint t-element neighborhoods in C of deficiency-zero vertices in O.

Use the fixed perfect matching N_C of J[C]. Each of these $m + 1$ neighborhoods has the odd cardinality t, so choose one vertex in each whose matching partner lies outside that neighborhood; otherwise it would be partitioned into matching edges and have even cardinality. Index these neighborhoods by $i = 1$,..., $m + 1$, write w_i for the chosen vertex and w_i' for its partner, and let z_i be the associated deficiency-zero vertex, including the extra vertex w at $i = m + 1$. The partner is not adjacent to its associated z_i by the definition of that neighborhood. Inequality (4) and nonnegative deficiencies imply $def(w_i') = 0$. Every vertex of each of the $m + 1$ neighborhoods has positive deficiency, so w_i' lies outside their entire union. The w_i are distinct by disjointness, and their partners are distinct because N_C is a matching. Therefore $|C| \geq (m + 1)(t + 1) \geq 4(t + 1)$. Combining with (1) and (5) yields $c_{t'}(H) = |S| \geq 2t + 5 + 4(t + 1) = 6t + 9$.

There is an optimum with its exposed vertex in either specified threshold part: moving that vertex across preserves optimality. Apply the last inequality first with $t = \theta$ and then with $t = \theta'$. This proves both claimed cost bounds and θ, $\theta' \geq 3$. Adding the cost bounds to the assumed equality gives $n + 1 \geq 6(\theta + \theta') + 18$, hence $n \geq 6d + 17$.

Finally Lemma 119 always gives a complementary sum at most $n+1$. The order n is odd, so $n \le 6d+15$ excludes the equality case just analyzed. The bound $c_\theta(H) \le 6\theta' + 8$ also contradicts its tightness lower bound. If instead $c_0(H) \le 6max(\theta, \theta') + 8$, choose the orientation whose opposite threshold is the larger and use $c_\theta(H) \le c_0(H)$ from Lemma 23. In each case the integer sum is at most n. Put $n = 2k+1$. If $d/2$ and k are odd, take $\theta = \theta' = d/2$. The complementary bound gives $c_\theta \le k+1$. Equality would make the pair tight and the already proved oddness of c_θ would contradict the even integer $k+1$. Therefore $c_{d/2}(H) \le k$ in this case. □

Depends on. Lemma 23, Lemma 35, Lemma 72, Lemma 119, Lemma 169.

Lemma 200 (Fitting and orientation at a unique plateau). *All graphs are finite and simple. Let F_1, F_2 be nonempty star forests with full nonincreasing positive integer edge-size lists n_1,..., n_s and m_1,..., m_t, where $s, t \ge 1$ are integers. Copies are ordinary subgraphs with vertex-disjoint target components, prescribed red and blue respectively. Put $M = s+t-1$ and $\ell_k = max\{n_i + m_j - 1 : 1 \le i \le s, 1 \le j \le t, i+j = k\}$ for $2 \le k \le s+t$. Fix an integer $0 \le q < M$ and put $K = M - q$, $L_q = \sum_{k=q+2}^{s+t} \ell_k$ and $P_q = L_q - K$. Let A_q consist of every pair (F_1^α, F_2^β) with nonnegative integer cuts $\alpha < s$, $\beta < t$ and $\alpha + \beta = q$, where superscripts delete leading whole components.*

Suppose $D = \ell_{q+2} - 1 \ge 6$ is even and every later natural entry is at most D. Let $z \ge 0$ be an integer with $z + 1 < D/2$. Let H satisfy

$e(H) \le L_q - 1$, $\delta = K - \nu(H) \ge 1$,

$\Phi(H) := e(H) - \nu(H) \le P_q + z$,

where ν is maximum matching cardinality. Suppose C is a whole connected odd D-regular component of H with a matching missing exactly one vertex, and put $\mu = \nu(C)$.

If H arrows every member of A_q, then $q < \min(s,t)$, both full original heads are at most D, and the full original edge totals differ. Orient the full originals and their prescribed colors as F, J with $e(F) < e(J)$, and write S, T for their component counts and B, A for their heads. Exactly F has at most $2\mu+1$ vertices. Put

$U = e(F) - S - (B-1)$, *W=sum_(m in the full list of J)* $\max(m-2, 0)$.

Then

$e(F) + \delta - 1 \le \mu \le e(J) - q - 2$,

$T \ge \mu + \delta + 1$,

$W \le S - 2\delta \le \mu - B - U - 3\delta + 2$,

$B \le D - 1$.

The entire original group of J components of edge size at least $D + 1 - B$ has at most $2\mu+1$ vertices.

If additionally $\delta \ge 2$ and both full original heads are at most D-3, then H has a coloring avoiding both prescribed targets of some member of A_q. The same existence conclusion holds whenever any of the preceding necessary conclusions fails. Every asserted avoiding coloring extends over the original disjoint canonical q-star pool of sizes ℓ_2,..., ℓ_{q+1}, and across every specified actual q-vertex set in any outer graph whose deletion gives H, preserving all H-edge colors and avoiding both full originals.

There is no hypothesis on the full original heads, on degrees outside C, or on any outer matching loss. All sizes, orders, counts, cuts, degrees, matching numbers and allowances are integers in their displayed domains; unit components, repetitions and $q = 0$ are permitted.

Proof. Suppose H arrows every A_q member. The family clause of Lemma 89 applies to these full originals, the same q and positive deficit δ. It gives $q < \min(s,t)$. The selected component has order $2\mu+1$, since its near-perfect matching has maximum possible size. Simplicity and D-regularity imply $D \le 2\mu$, so $\mu \ge D/2$.

Apply Lemma 84 with the full original lists, the same q, D, μ, z and literal component C. Its assumptions require neither a full-head bound nor a positive deficit. Here the natural plateau is unique and $z + 1 < D/2$, as required. Its necessary conclusions under common arrowing show that both full heads are at most D and at least one entire original has order at most $2\mu+1$.

Set $Z = \max(e(F_1), e(F_2))$. The density clause of Lemma 42 now applies: both heads are at most D, $q < \min(s, t)$, the plateau width is one, $e(H) \le L_q - 1$ and $\nu(H) = K - \delta < K$. It gives

$P_q \le (D-1)(Z-q-1)+1.$

Matching number and edge count add over whole components, and a graph has nonnegative potential because every matching is an edge subset. Also $e(C) = Dmu + D/2$. Thus

$(D-1)\mu + D/2 = \Phi(C) \le \Phi(H)$

$\le (D-1)(Z-q-1)+1+z$

$< (D-1)(Z-q-1)+D/2.$

Since μ and Z-q-1 are integers, this implies

$\mu \le Z-q-2.$ (1)

Adjoin to H the original disjoint canonical q-star pool, obtaining G_c. By Lemma 41 (I), G_c arrows both full originals in their prescribed colors. The pool consists of q nonempty stars, each of matching number one, so

$\nu(G_c) = \nu(H) + q = M - \delta.$

C remains a whole component of G_c.

Fix either full original F_i, and let k be its number of positive components. If $\mu \ge k - \delta$, the first, matching-additive localization assertion of Lemma 153 applies with the empty separator and selected component C. The empty separator is matching-additive because maximum matchings unite over whole connected components. After exchanging the complete target and color roles when necessary, it says C arrows $(pK2, F_i)$, where K_2 is one edge and

$p = \mu - k + \delta + 1 \ge 1.$

This uses no half-order premise on any other component. Coloring every edge of C in the matching color forces $p \le \mu$. Choose a vertex set Y in C of exactly p-1 vertices; such a set exists because $p \le \mu$. Color every edge meeting Y in the matching color and all other edges in $F_i's$ color. The former graph has a vertex cover of size p-1 and cannot contain a p-edge matching. Thus C-Y contains the entire F_i, giving

$e(F_i) + k \le 2\mu + 1 - (p-1) = \mu + k + 1 - \delta,$

or

$e(F_i) \le \mu + 1 - \delta.$ (2)

For a maximum-edge original, (1) and (2) would imply

$Z \le \mu + 1 - \delta \le Z - q - 1 - \delta < Z.$

Consequently every maximum-edge original has component count at least $\mu + \delta + 1$. Its order is at least twice that count and exceeds $2\mu + 1$. At least one original does fit this order, by the remnant theorem already applied. Hence the edge totals are distinct, and exactly the smaller-edge original F fits. Write S, T for the component counts and B, A for the heads in this orientation.

Since F has positive components and fits $2\mu + 1$ vertices, $2S \le 2\mu + 1$ and $S \le \mu$. Therefore (2) applies to F and gives $\mu \ge e(F) + \delta - 1$. The maximum-edge original is J, so its preceding count bound is $T \ge \mu + \delta + 1$, and (1) becomes $\mu \le e(J) - q - 2$.

Apply the full-target capacity clause of Lemma 89 to G_c, whose full original deficit is δ. Dropping the nonnegative size of an attaining separator yields

$S \ge 2\delta + 2sum_{m \in J} floor((m-1)/2).$

For each positive integer m one has $\max(m-2, 0) \le 2floor((m-1)/2)$. Therefore $W \le S - 2\delta$. With $U = e(F) - S - (B-1) \ge 0$, the lower bound on μ reads

$\mu \ge S + B + U + \delta - 2.$

Rearranging and combining proves

$W \le S - 2\delta \le \mu - B - U - 3\delta + 2.$

The one-sided fitting consequence of Lemma 84 applies to this same orientation, because $S \le \mu$. It says the entire original J-group of sizes at least $\max(1, D+1-B)$ has at most $2\mu+1$ vertices. We already know $B \le D$. If $B = D$, that group is all of J, contrary to J not fitting. Thus $B \le D-1$ and the threshold is $D+1-B$. This proves every necessary conclusion.

Now impose $\delta \ge 2$ and both original heads at most D-3, and suppose for contradiction that H arrows every A_q member. All the preceding conclusions apply. Put $h = \max(A, B)$, $r = \max(A-B, 0)$, $a = \min(A, B)$, and

$\Gamma = (r+1)U + (r+2)B + r + a + 6.$

The first full diagonal dominates the natural leading diagonal: its value is $A+B-1$, while the latter is $D+1$. Hence $A+B \ge D+2$. Since $h \le D-3$, each head is at least $D+2-h \ge 5$. In particular both are at least four, and $\Gamma > 0$.

The bounds already proved give

$\mu \ge S+B+U,\ T \ge \mu+3,$

$W \le \mu - B - U - 4.$

Thus every numerical hypothesis of Lemma 80 holds for the complete ordered pair (F, J), with their prescribed colors exchanged together if required. Put $b_2 = floor((2\mu+1)/3)$. Since $\mu \ge 1$, $b_2 \le \mu$. Since $D-h-2 \ge 1$ and $D/2 - z - 1 > 0$,

$b_2 \le (D-h-2)\mu + D/2 + \Gamma - z - 1.$

The graph application of that theorem uses the very same C, μ, D, z and original q. It supplies a coloring of H avoiding an original actual A_q member. This contradicts common arrowing and proves the stated head-gap conclusion.

Failure of any proved necessary conclusion also contradicts common arrowing. Negating that finite conjunction means that at least one original actual member admits an avoiding coloring. For every asserted coloring, the original-pool extension and extension across any specified actual outer q-set are Lemma 41 (I), (II). These use its actual legal cuts and retain every H-edge color. No exterior degree or outer matching-loss condition was used. □

Remark. The remnant theorem first forces an original target to fit. Density leaves too little matching capacity for a maximum-edge target to be that target. Localization supplies the exact smaller-target bounds, and a retained high threshold gives the final head restriction.

Depends on. Lemma 41, Lemma 42, Lemma 80, Lemma 84, Lemma 89, Lemma 153.

Lemma 201 (Quartic normalization from proper-subgraph run bounds). *Use the finite integer certificate costs c_i, matching number ν, independence number α, excess $ex = c_0 - \nu$, slack $ps = e - \nu - \sum_{i\ge1} c_i$ and $gap = ex - ps$ of Lemma 113, and the top missing-allocation run tr of Lemma 45. All graphs are finite and simple. Say that a graph satisfies T_3 if positive gap implies gap one and $tr + ex \le 3$. Say that it satisfies T_- if positive gap implies gap one and $tr + ex \le \max(4, \lfloor \Delta/2 \rfloor + 1)$. A proper subgraph may delete vertices, edges, or both, including spanning edge deletions. Vertex deletion is induced.*

Let F be connected and four-regular, and suppose F-v has a perfect matching for every vertex v. Assume $gap(F)$ belongs to {1,2}, F violates T_-, every proper subgraph of F satisfies T_3, and

$c_3(F) \le floor((3\nu(F)+2)/4).$

Then $gap(F) = 1$ and $E = ex(F) \ge 2$. Put $k = \nu(F)$. The graph F is well-covered, meaning that every maximal independent set has maximum cardinality. Put

$a = \alpha(F),\ x = k - a,\ \sigma = 2E - 2 - x,\ g = c_2(F) - a,$

$q = x + 1 - E,\ b = g - 2q,\ m = q + b.$

These integers satisfy $a \ge 5$, $E \le x \le 2E-2$, $q = m - b \ge 1$, $m \ge 1$, $0 \le b < m$, $\sigma \ge b$, and $k \ge 4E - 1$.

For every maximum independent set S and $G = F - S$,

$ps(G) = c_3(G) = 0$,

$c_i(F) = a + c_i(G)$ *for each integer i in {1,2,3}*,

$\nu(G) = x + E$,

$|V(G)| - \nu(G) = a + m - b$,

$c_1(G) = \nu(G) - b$, $c_2(G) = 2m - b$,

$c_0(F) = a + \nu(G)$, *and* $c_2(F) = a + 2m - b$.

There is one pair of nonnegative integers (u, v), independent of S, with $u + v = c_0(F) - 1$ *and* $u \le a - \sigma$ *such that* $E(F)$ *cannot be partitioned into two parts of matching numbers at most u, v. All orders, degrees, costs, indices, matching and independence cardinalities, and displayed parameters are integers in their stated domains. No lower bound on E is imposed as a hypothesis.*

Proof. Write $f = gap(F)$. Factor-criticality gives $|V(F)| = 2k+1$ and $\nu(F-v) = \nu(F) = k$: a perfect matching after deleting a vertex has k edges, and no matching on $2k+1$ vertices can have more. Four-regularity and simplicity give $k \ge 2$. Nonnegative slack from Lemma 61 gives $E = ex(F) \ge f \ge 1$.

First suppose $E = 1$. Then $f = 1$ and $ps(F) = 0$. Since $\Delta(F) = 4$ and F fails T_-, one has $tr(F) + E \ge 5$. Apply Lemma 45 with $D = 4$ and its signed parameter $B = 4$. Its auxiliary integer is zero when $E = 1$, so $k \ge 5$. In particular F is not K_5. The regular-graph part of Lemma 113 gives $c_4(F) = 1$ and all rows above four zero, by choosing any maximal independent set. Lemma 112 gives $c_2(F) \le k$. Also $c_1(F) \le c_0(F) = k+1$, because every zero-capacity certificate remains a certificate at either capacity pair of total one by keeping its degree-bounded parts empty. Therefore

$ps(F) = 4k + 2 - k - c_1(F) - c_2(F) - c_3(F) - 1$

$\ge k - floor((3k+2)/4) = ceil((k-2)/4) \ge 1$,

contradicting $ps(F) = 0$. Thus $E \ge 2$.

Lemma 45 now gives $k \ge 5E - 5 \ge 5$. Again F is not K_5. For any maximal independent set S write $a_S = |S|$ and $G = F - S$. Every vertex outside S has a neighbor in S, so $\Delta(G) \le 3$. The closed neighborhoods of S cover $V(F)$, and each has five vertices; hence $5a_S \ge 2k + 1 \ge 11$ and $a_S \ge 3$.

1. Exact independent deletion and exclusion of gap two.

Put $s = ps(G)$. Restrict a perfect matching of F-z, for any z in S, to G. Since S is independent, deleting S minus {z} removes at most $a_S - 1$ matching edges. Thus $\nu(G) \ge k + 1 - a_S$. Put $\eta = \nu(G) - (k + 1 - a_S) \ge 0$.

Let B be the family of whole components P of G with $floor(|V(P)|/2) \ge a_S + 2$, and set $M = \max(\{0\} union \{ex(P) : P \in B\})$. The arbitrary-deletion assertion of Lemma 113 gives

$c_0(F) \le a_S + \nu(G) + M$,

or $E \le 1 + \eta + M$. Its regular-graph assertions also give $c_4(F) = 1$, all rows above four zero, and $c_i(G) = 0$ for $i \ge 4$.

For $i = 1, 2, 3$ put $r_i = c_i(F) - c_i(G)$. Lemma 72 gives $0 \le r_i \le a_S$. Since exactly $4a_S$ edges are deleted, subtraction of the slack definitions gives

$\epsilon := E - f - s - \eta = 3a_S - (r_1 + r_2 + r_3) \ge 0$.

All these quantities are integers, and $s \ge 0$ by Lemma 61. Consequently

$\epsilon \le M + 1 - f - s$.

If $M = 0$, this forces $f = 1$ and $\epsilon = s = 0$.

Suppose $M \ge 1$. Choose $P.$ in B with $ex(P.) = M$. It is a proper subgraph of F, so its T_3 property implies $gap(P.) \le 1$. Thus $ps(P.) \ge M - 1$. By the component slack inequality of Lemma 72 and nonnegative component slacks, $s \ge M - 1$. It follows that $\epsilon \le 2 - f$.

If $f = 2$ or $\epsilon > 0$, the displayed inequalities force

$s = M - 1$, $E = 1 + \eta + M$.

Indeed, when $f = 2$ they give $\epsilon = 0$ and $s = M - 1$; when $\epsilon > 0$ they give $f = 1$, $\epsilon = 1$ and $s = M - 1$. In either case the definition of ϵ then gives the stated E identity. Equality in the component slack comparison gives

$gap(P.) = 1$, $ps(P.) = M - 1$,

and $ps(Q) = 0$ for every other whole component Q of G.

For each such Q, $ex(Q) = gap(Q)$ because $ps(Q) = 0$. Nonnegative excess and T_3 give $ex(Q)$ in {0,1}. At $ex(Q) = 0$, the zero-capacity allocation set at total $\nu(Q)$ is a full interval. At $ex(Q) = 1$, that total is $c_0(Q) - 1$, its allocation set contains both endpoints, and its missing runs have length at most $tr(Q) \leq 2$. Use the allocation-set composition and interval criterion of Lemma 75, starting with the actual S-pool $[0, a_S]$. Since $a_S \geq 3$, all these other components can be absorbed successively at their matching-number totals, giving the full interval [0,L], where

$L = a_S + \sum_{Q different from P.} \nu(Q)$.

Every step is an actual component partition and a final restoration across S.

Keep $P.$ at total $t = c_0(P.) - 1 = \nu(P.) + M - 1$, whose allocation set Y_t contains 0, t. The same composition theorem gives

$[0, L] + Y_t$ contained in $Y_(c_0(F) - 1)(F)$,

since $L + t = a_S + \nu(G) + M - 1 = k + E - 1 = c_0(F) - 1$.

Here Y denotes the zero-capacity feasible first budgets at the indicated total. Between two consecutive members of Y_t with R missing integers, adding [0,L] leaves $\max(R - L, 0)$ missing integers. There are no endpoint gaps because 0, t belong to Y_t. At least one top allocation of F is missing, by Lemma 45. Hence the inclusion implies

$tr(F) + L \leq tr(P.)$.

But $L \geq a_S \geq 3$ and $tr(F) \geq 1$, whereas $T_3(P.)$ gives $tr(P.) \leq 3 - M \leq 2$. This is a contradiction.

Therefore $f = 1$ and $\epsilon = 0$ in every case. Since S was arbitrary, for every maximal independent S we have

$ps(F) = \eta + s$, $c_0(F) = a_S + \nu(G) + s$,

$c_i(F) = a_S + c_i(G)$ for $i = 1, 2, 3$.

The positive-row identities follow because ϵ is the sum of the three nonnegative shortfalls $a_S - r_i$.

2. Zero complementary slack and well-coveredness.

The connected four-regular graph F differs from K_5, so Lemma 112 gives $c_2(F) \leq k$. Apply Lemma 161 to the subcubic G. It gives $c_1(G) \leq \nu(G)$ and $c_3(G) \leq \lfloor s/2 \rfloor$. The exact identities just proved consequently imply

$c_1(F) \leq k + E - s$, $c_3(F) \leq a_S + \lfloor s/2 \rfloor$.

Expanding $gap(F) = 1$ using the zero tail and $c_4(F) = 1$ gives

$1 = c_0(F) + c_1(F) + c_2(F) + c_3(F) + 1 - (4k + 2)$

$\leq a_S - k + 2E - \lceil s/2 \rceil - 1$.

Thus $a_S \geq k + 2 - 2E + \lceil s/2 \rceil$.

If $s \geq 1$, then $a_S \geq k - 2E + 3$. Lemma 45 applies with $D = 4$, $E \geq 2$ and gives

$a_S + \nu(F - S) \geq k + E$.

The exact zero-row identity instead gives $a_S + \nu(F - S) = k + E - s$, a contradiction. Therefore $s = 0$ for every maximal S. The cubic bound then gives $c_3(G) = 0$. Exact third-row deletion says $a_S = c_3(F)$, independent of S. A maximum independent set exists and is maximal, so every maximal set has size $a = \alpha(F) = c_3(F)$. This proves well-coveredness and the asserted complementary slack and row identities. Also $c_0(F) = a + \nu(G)$ gives $\nu(G) = k + E - a = x + E$.

3. The positive odd-cycle count and normalization.

The lower bound just proved gives $a \geq k+2-2E$, or $x \leq 2E-2$. The complement of a maximum independent set is a vertex cover of size $2k+1-a$. Assigning its vertices to the two budgets partitions every edge by a fixed priority on those vertices and bounds each matching number by its assigned vertex count. Thus $c_0(F) \leq 2k+1-a$, giving $x \geq E-1$.

Now $f=1$ and T_- fails, so $tr(F)+E \geq 5$. Apply the signed-B assertion of Lemma 45 with $B=4$, $D=4$ and auxiliary integer E-2. It gives $k \geq 4E-1$. The assumed weak top-row estimate and $a=c_3(F)$ give $k \leq 4x+2$. Hence $4x+2 \geq 4E-1$, and integrality forces $x \geq E$. Thus $q=x+1-E \geq 1$, $\sigma = 2E-2-x \geq 0$, and

$a = k-x \geq 4E-1-(2E-2) = 2E+1 \geq 5$.

Exact second-row deletion gives $g=c_2(G)$, uniformly in S, and $c_2(F) \leq k$ gives $g \leq x$. Since $e(G)=4x+2$, zero slack and its zero rows at indices at least three give

$c_1(G) = 3x+2-E-g$.

Therefore $b = g-2q = \nu(G)-c_1(G) \geq 0$, again by the subcubic bound. Set $m=q+b$. Then $m \geq 1$, $b<m$ and $\sigma - b = x-g \geq 0$. Direct substitution yields

$c_2(G) = g = 2q+b = 2m-b$,

$|V(G)| - \nu(G) = k+1-E = a+q = a+m-b$,

and $c_2(F) = a+2m-b$.

Finally the missing-allocation interval of Lemma 45 supplies one nonnegative pair (u, v) at total $c_0(F)-1$ with

$u \leq k-E-(E-2) = k-2E+2 = a-\sigma$.

Choose this pair once in F; it does not depend on S. This proves every remaining assertion. □

Depends on. Lemma 45, Lemma 61, Lemma 72, Lemma 75, Lemma 112, Lemma 113, Lemma 161.

Lemma 202 (Slack forced by a nonempty marked barrier). *Use the finite integer certificate costs $c_i(G)$, matching number $\nu(G)$, maximum degree $\Delta(G)$, excess $ex(G) = c_0(G)-\nu(G)$, slack $ps(G) = e(G)-\nu(G)-\sum_{i\geq 1} c_i(G)$, and signed gap $gap(G) = ex(G)-ps(G)$ of Lemma 23. A matching is a set of edges with pairwise disjoint endpoints.*

Let $D \geq 4$ be an integer and let H be a connected finite simple graph with $\Delta(H)=D$. Set $S = \{vinV(H) : deg_H(v) = D\}$, and let $\kappa \geq 1$ be the minimum number of vertices of S left uncovered by a matching of H. For $W \subseteq V(H)$, let H-W denote induced vertex deletion and let $q_S(H-W)$ count its odd-order connected components whose entire vertex sets belong to S.

Suppose X is a nonempty vertex set of maximum cardinality among all sets W satisfying $q_S(H-W)-|W| = \kappa$. Assume $gap(Q) \leq 1$ for every connected component Q of H-X. Put $\epsilon = 1$ when D is odd and $\epsilon = 2$ when D is even. Then

$ps(H) + c_D(H) \geq D(\kappa-1)+\epsilon+1$,

and

$ps(H) \geq (D-1)(\kappa-1)+\epsilon$.

In particular $ps(H)=0$ is impossible under these hypotheses. If $ps(H) \leq 1$, then D is odd, $\kappa=1$, and $ps(H) = c_D(H) = 1$.

No sign assumption on $gap(H)$ is required. All degrees, cardinalities, costs, matching budgets, component counts, and κ are integers in their stated domains. Empty sums are zero.

Proof. Nonnegativity and monotonicity of the integer costs follow from Lemma 23. We use Lemma 61 for $ps(G) \geq 0$ on every finite simple graph and for the zero cost tail above its maximum degree.

Write $b = |X| \geq 1$, let calQ be all components of H-X, let calK be those counted by q_S, and let Y be the union of the vertex sets of calQ minus calK. For K in calK put $d(K) = e_H(V(K), X)$, where $e_H(A,B)$ counts H-edges between disjoint vertex sets A, B. Lemma 104 gives

$|calK| = b+\kappa$, $\nu(H) = b + \sum_{Q \in calQ} \nu(Q)$, and $c_D(H) \leq \kappa$. (1)

It also gives $d(K) > 0$ and $d(K)$ congruent to D modulo two. Every edge leaving K ends in X, so

$d(K) = \sum_(v \in V(K))(D - deg_K(v)) \geq \epsilon$. (2)

We check the maximality needed for factor-criticality. Choose a matching leaving exactly κ vertices of S uncovered; such a matching exists because the matching family is finite and nonempty. For any W, each component counted by $q_S(H - W)$ either contains an uncovered S-vertex or sends a matching edge to W, by its odd order. Distinct components use distinct such vertices or distinct matching edges, and the matching edges have different endpoints in W. Hence $q_S(H - W) \leq \kappa + |W|$. The supplied equality at X therefore attains the maximum of $q_S(H - W) - |W|$. Its maximum-cardinality property implies inclusionwise maximality among maximizers. Lemma 55 makes every K in calK factor-critical. In particular deleting any vertex of K leaves a perfect matching.

Put $T = floor((D - 1)/2)$, so $D = 2T + \epsilon$ and $T \geq 1$. Choose a component Z maximizing $\nu(Q)$ and then maximizing

$w(Q) = |\{i : 1 \leq i \leq Tandc_i(Q) > \nu(Q)\}|$.

The family calQ is finite and nonempty by (1), so this choice exists. Write $k = \nu(Z)$.

For each K in calK with $d(K) < D$, we apply Lemma 135 with degree bound D and actual total defect $d(K)$. Its finite simple factor-critical and defect hypotheses follow from (2) and the preceding paragraph. Its component gap assumption is supplied in the statement. All its additional credit conditions hold. At $D = 4$, fact Lemma 131 gives

$p_{cap}(K) := e(K) - \nu(K) - \sum_{i \geq 1} \min(c_i(K), \nu(K)) \geq 4 - d(K)$.

At $D = 5$ the same required bound is given by Lemma 152. These two applications are legitimate because K is a nonempty connected factor-critical component with positive defect. For odd $D \geq 7$ and $d(K) = 1$, fact Lemma 167 gives

$c_{D-1}(K) \leq \nu(K) - (D - 1)/2$,

which is exactly the required tail bound with $u = (D - d(K))/2$. No additional condition is required by Lemma 135 in the other cases.

It follows that each such K has nonnegative integer credits $a_i(K)$, $1 \leq i < D$, with

$\sum_i a_i(K) = \delta_K := \max(D - d(K) - ps(K), 0)$,

$0 \leq a_i(K) \leq \max(c_i(K) - \nu(K), 0)$,

$a_1(K) \leq (D - d(K))/2 \leq T$,

$a_i(K) \leq 1$ for $i \geq 2$, and $a_i(K) = 0$ for $i > T$. (3)

For K with $d(K) \geq D$, nonnegative slack makes $\delta_K = 0$; give it zero credits. Give zero credits also to every component outside calK, and override the credits of Z to zero. Thus the total-credit condition in (3) is required only for K in calK different from Z, exactly as in Lemma 104.

For every integer $1 \leq i < D$ put $\mu_i(Q) = c_i(Q) - a_i(Q)$ and $M_i = \max_{Q \in calQ} a_i(Q)$. Run the procedures of Lemma 104, choosing arbitrarily among eligible components. At index i the initial value is

$L = b + \sum_(Q : a_i(Q) = 0)c_i(Q)$;

the unprocessed components have positive credit. A component Q is eligible when $\nu(Q) \leq L + 1$; absorbing it increases L by $\mu_i(Q)$. These reduced entries are nonnegative by the credit support. Hence every running value satisfies $L \geq b + c_i(Z)$, and each procedure terminates after at most $|calQ|$ absorptions. Fix any collection of its permitted choices. Let E be the sum of M_i over the indices whose procedures finish with unprocessed components. All indices above T are finished from the start, and (3) gives

$E \leq \sum_{i=1}^{T} M_i \leq 2T - 1$. (4)

Retain the top-row correction

$\zeta = \kappa - c_D(H) + \sum_{Q \in calQ} c_D(Q)$

from Lemma 104. Define

$J = e(H[X]) + e_H(X, Y)$

$+ \sum_{Qnot\in calK, Q\neq Z} ps(Q)$

$+ \sum_{K\in calK, K\neq Z} \max(ps(K) - (D - d(K)), 0).$

Every summand in J is nonnegative. Converting the gap inequality of that fact using $gap = ex - ps$ cancels the excesses and gives

$ps(H) \geq ps(Z) - E + (D-1)\kappa - 1_{Z\in calK}(D - d(Z)) + J + \zeta.$

Here the indicator term is zero if Z is not in calK, so $d(Z)$ is only used when defined. Substitute the displayed definition of ζ and move $c_D(H)$ to the left:

$ps(H) + c_D(H) \geq ps(Z) - E + Dkappa$

$-1_{Z\in calK}(D - d(Z)) + J + \sum_{Q\in calQ} c_D(Q).$ (5)

Suppose for a contradiction that the integer

$\beta = ps(H) + c_D(H) - D(\kappa - 1)$

satisfies $\beta \leq \epsilon$. If Z is not in calK, or if it is in calK with $d(Z) \geq D$, then (5) and nonnegative correction terms imply

$ps(Z) \leq \beta + E - D \leq \epsilon + 2T - 1 - D = -1,$

contradicting nonnegative slack. Thus Z belongs to calK and $\epsilon \leq d := d(Z) < D$. The unconditional profile conclusions of Lemma 135 now give, with $u = (D - d)/2$,

$s := ps(Z) \geq 1, e(Z) = Dk + u, c_i(Z) = 0$ for $i \geq D,$

and $c_i(Z) + c_{D-i}(Z) \leq 2k + 1$ for every $1 \leq i < D.$

Consequently

$s = (D-1)k + u - \sum_{i=1}^{D-1} c_i(Z).$ (6)

Equation (5), still for the same fixed collection of procedures, gives

$s \leq \beta + E - d \leq D - 1 - d.$ (7)

If index one is unfinished, choose a remaining component R and write $r = \nu(R)$. Its failed eligibility test and integer values give

$r \geq L + 2 \geq b + c_1(Z) + 2.$

The choice of Z gives $r \leq k$, so $c_1(Z) \leq k - b - 2$. Monotonicity makes every entry in (6) at most k-b-2, whence

$s \geq u + (D-1)(b+2) > D - 1 - d.$

This contradicts (7), proving that index one is finished.

If an index remains unfinished, let j be the first. Then $2 \leq j \leq T$, so this case cannot occur when $T = 1$. Put $q = T - j \geq 0$. All indices below j are finished and each M_i with $j \leq i \leq T$ is at most one. Thus $E \leq q + 1$ and

$s \leq \beta + q + 1 - d.$ (8)

For any component R remaining at index j, write $r = \nu(R)$. The failed test gives $r \geq b + c_j(Z) + 2$, and its positive credit gives $c_j(R) > r$. We have $r \leq k - 1$. Indeed, if $r = k$ then $c_j(Z) < k$, so monotonicity gives $w(Z) \leq j - 1$, while $c_j(R) > k$ gives $w(R) \geq j$. This contradicts the tie-break among components of rank k. It follows that

$c_j(Z) \leq k - b - 3.$ (9)

Pair the indices i, D-i for $1 \leq i \leq j - 1$. They are disjoint because $j \leq T$. Their paired bounds sum to at most $(j-1)(2k+1)$. The remaining indices j,..., D-j number $D - 2j + 1$ and have entries at most k-b-3 by monotonicity and (9). Inserting their summed upper bound in (6) gives

$s \geq u - j + 1 + (D - 2j + 1)(b + 3).$

The difference between this lower bound and the upper bound in (8) equals

$(d+\epsilon)/2-\beta+(\epsilon+2q+1)(b+3)>0.$

The first term is nonnegative because $d\geq\epsilon$ and $\beta\leq\epsilon$, and the second is positive because $b\geq 1$, $q\geq 0$, and $\epsilon\geq 1$. This contradiction excludes every unfinished index after one.

All procedures in the fixed collection therefore finish, so $E=0$. Inequality (7) before its last relaxation now gives $s\leq\beta-d\leq\epsilon-d\leq 0$, contrary to $s\geq 1$. Thus $\beta\leq\epsilon$ is impossible. Since β is an integer, $\beta\geq\epsilon+1$, proving the first claimed inequality. Using $c_D(H)\leq\kappa$ from (1) in that inequality proves the second.

The second inequality gives $ps(H)\geq\epsilon\geq 1$. If $ps(H)\leq 1$, it forces $\epsilon=1$ and $\kappa=1$, since $D-1\geq 3$ and κ is a positive integer. Hence D is odd and $ps(H)=1$. The first inequality then gives $c_D(H)\geq 1$; (1) bounds it above by one. This proves every stated consequence. □

Remark. The actual top-row cost remains in the boundary accounting. A hypothetical shortfall forces the selected component to have positive slack but very small row costs. The first unfinished absorption index makes the complementary-row sum incompatible with that allowance.

Depends on. Lemma 23, Lemma 55, Lemma 61, Lemma 104, Lemma 131, Lemma 135, Lemma 152, Lemma 167.

Lemma 203 (Strict cubic potential from canonical matching remainders). *All graphs are finite and simple. Put $P_3=K_{1,2}$ and $K_2=K_{1,1}$, where $K_{1,j}$ denotes a star with j edges. Let p, r be positive integers and let h, u, v be nonnegative integers with $1\leq h\leq u$ and $v<u$. Let*

$F=pP_3$ *disjoint union* uK_2,

$J=rK_{1,3}$ *disjoint union* hP_3 *disjoint union* vK_2

be the entire prescribed red and blue target forests. Every union is vertex-disjoint and copies are ordinary subgraphs. Put $s=p+u$, $t=r+h+v$, $M=s+t-1$, and $R=p+r-1$. Let $n_1\geq\cdots\geq n_s$ and $m_1\geq\cdots\geq m_t$ be their positive integer edge-size lists. Define

$\ell_k=max\{n_i+m_j-1:1\leq i\leq s,1\leq j\leq t,i+j=k\}$

for integers $2\leq k\leq s+t$, and put $L=\sum_{k=2}^{s+t}\ell_k$ and $P=L-M$.

Then $P=3R+2u+h$.

Write $e(G)$, $\nu(G)$, $\Delta(G)$ for edge count, maximum matching cardinality, and maximum degree, and put $\Phi(G)=e(G)-\nu(G)$. Every graph G with $\Delta(G)\leq 3$ arrowing (F, J) satisfies

$\Phi(G)\geq P+1=3(p+r-1)+2u+h+1.$

Thus every graph under this degree ∩ with $\Phi(G)\leq P$ has an edge coloring avoiding both entire targets. The result also holds with the prescribed colors interchanged. All counts, indices, degrees and cardinalities are integers in the displayed domains. There is no matching-deficit, connectedness, regularity, or minimum-degree hypothesis.

Proof. We first prove the precise equality property needed later. Let k, b, w be arbitrary positive integers, and suppose T arrows $(kK_2, bP_3 disjoint union wK_2)$.

Then $e(T)\geq 2(k+b-1)+w$.

Indeed, when $k=1$ the all-blue coloring forces the full opposite forest, which has $2b+w$ edges. For $k\geq 2$, repeatedly delete k-1 maximum-degree vertices. At each step Lemma 34 preserves arrowing of a smaller nonempty matching against the same opposite forest. Its all-blue coloring forces a P_3, so the next maximum degree is at least two and the vertex exists. The final graph has at least $2b+w$ edges. The disjoint accounting of edges at their first deleted endpoint gives the asserted lower bound.

If equality holds, we claim that T consists of $k+b-1$ whole P_3 components and w whole K_2 components, together with isolates. Fix b, w and induct on k. For $k=1$ the all-blue opposite copy uses exactly all edges, proving the claim. Let $k\geq 2$. Every vertex deletion leaves an arrowing host for $((k-1)K_2, bP_3 disjoint union wK_2)$. The lower bound just proved shows that deleting any vertex removes at most two edges. The all-blue coloring

forces a P_3, so $\Delta(T) = 2$. Choose a degree-two vertex z. Equality holds for T-z, and induction identifies it as $k + b - 2$ disjoint P_3s and w disjoint edges, apart from isolates.

Let I_0 be the centers of those P_3s and let B be the union of the w single-edge components. No center in I_0 can be adjacent to z, since it already has degree two in T-z. Therefore $I = I_0$ union {z} is independent and each of its vertices has degree two in T.

Choose any k-1 vertices of I_0 and color every edge incident with them red and all remaining edges blue. The red graph has a vertex cover of size k-1, so it has no matching of k edges. The blue edge set is exactly the two edges at z, the b-1 retained old P_3s, and B. These constituent stars have disjoint edge sets, exactly $2b + w$ edges in total, and exactly $b + w$ nonempty constituents. If two constituents meet, their union has fewer than $b + w$ nonisolated connected components. A blue copy of the opposite forest would use all $2b + w$ edges and have exactly $b + w$ components, which is then impossible. Arrowing therefore forces these constituent stars to have disjoint vertex sets for every selection of the red centers.

In particular the star at z is disjoint from B. If $b \geq 2$, any given old P_3 can be retained, because there are $k + b - 3 \geq k - 1$ other old centers from which to choose the red set. Thus the star at z is disjoint from every old component, both its neighbors were isolated in T-z, and the canonical description follows.

Suppose $b = 1$. Now $|I| = k$, every edge outside B is incident with I, and the w edges in B remain whole components. If centers x, y in I share a neighbor a, let c, d be their respective other neighbors. All three neighbor vertices lie outside I. If $c \neq d$, color every edge incident with I minus {x, y} and the two edges xa, ya red; color the rest blue. Red has the cover (I minus {x, y}) union {a} of size k-1. Blue consists of the two disjoint edges xc, yd and the matching B, so it has no P_3. This contradicts arrowing. Hence $c = d$ whenever centers share a neighbor. The two centers and their two common neighbors then form a whole C_4 component, because all four vertices already have degree two. Thus the edges incident with I form disjoint P_3 and C_4 components.

If a C_4 occurs, choose one edge e of B; this exists since $w \geq 1$. Color all edges of that C_4 and the other w-1 edges of B blue, and every remaining edge red. The k-2 centers outside the cycle, together with one endpoint of e, cover all red edges, so red has no k-edge matching. A blue P_3 must use three of the C_4 vertices. The remaining cycle vertex supplies no further disjoint blue edge, leaving only w-1 single-edge components, too few for the entire blue P_3 disjoint union wK_2. This is another avoiding coloring. No C_4 can occur, and the equality claim is proved for $b = 1$ as well.

Return to the stated original pair. Its complete diagonal list is

(4 repeated R times,3 repeated u times,2 repeated h times,1 repeated v times).

Value four is attained exactly for sums two through $p + r$. A value at least three requires a red two with a blue nonunit or a red unit with a blue three; their last possible sums are $p + r + h$ and $p + u + r$, with the latter at least as large. Every sum k from $p + r + 1$ through $p + r + u$ attains three using $j = r$ and $i = k - r$. A value at least two requires either a red two or a blue nonunit; its last possible sum is $\max(p + r + h + v, p + u + r + h) = p + u + r + h$, since $v < u$. For each next sum through $p + u + r + h$ use $i = p + u$ and $j = k - i$; these blue indices run from $r + 1$ through $r + h$. Every later pair consists of units. This verifies the list and $P = 3R + 2u + h$.

The theorem Lemma 56 gives $\Phi(G) \geq P$ for any arrowing G. Suppose $\Delta(G) \leq 3$ and equality holds. Fix any maximum matching W, put $Q = G - W$ by edge deletion, and let I be the degree-three vertices of Q. Covered vertices lose one incident edge and uncovered vertices lose none, so I consists exactly of the degree-three vertices of G missed by W. It is independent, because an edge between uncovered vertices could be added to W.

Apply Lemma 95 to the original single-label pair and this fixed W. The first R maximum-degree deletions in Q have degree three, and the following positive degrees are u twos and h ones. Since I is independent, deleting its members leaves the degrees of the other members unchanged; vertices initially outside I cannot gain degree. Therefore $|I| = R$. This holds for every maximum matching.

Let $H = Q - I$. Its maximum degree is at most two, and $e(H) = \Phi(G) - 3R = 2u + h$. By matching deletion Lemma 34,

$Q\ arrows((p + u)K_2, rP_3 disjoint union h K_2)$.

Set $a = \min(R, p+u-1)$ and $b_0 = R - a$. These are legal cuts totaling $|I| = R$. The retained red matching size is $k = p + u - a \geq 1$, and the retained blue P_3 multiplicity is $r_0 = r - b_0 \geq 1$: if $b_0 > 0$, then $b_0 = r - u \leq r - 1$ because $u \geq 1$. All h blue units remain. The vertex-deletion part of the same fact yields

$H\ arrows(kK_2, r_0P_3 disjointunion hK_2),\ k + r_0 - 1 = u.$

The equality property proved above applies with $b = r_0$ and $w = h \geq 1$, because $e(H) = 2(k + r_0 - 1) + h$. Thus H consists exactly of u whole P_3 components, h whole single edges, and isolates. Its degree-two vertices are independent.

Finally apply Lemma 148 at $D = 3$ to the original G. Let S be its degree-three vertex set and δ the minimum number of S-vertices missed by any matching. That theorem guarantees a nonempty family of maximum matchings attaining δ. Every maximum matching misses exactly R vertices of S by the preceding argument, so $\delta = R$. For every maximum matching, its remainder (G-W)-(S minus V(W)) is exactly the H just proved canonical, and its degree-(D-1) vertices are independent. In particular this holds for every maximum matching attaining δ. The odd-degree conclusion of Lemma 148 then gives $\delta = 0$. This contradicts $R = p + r - 1 \geq 1$.

Equality $\Phi(G) = P$ is impossible, and the integer lower bound therefore strengthens to $\Phi(G) \geq P + 1$. Negating arrowing gives the stated full-target avoiding coloring. Renaming red and blue in the argument proves the prescribed-color reversal. □

Remark. The positive number of lower unit components removes the four-cycle exception in matching equality. The fixed-matching remainder is consequently a union of short stars for every maximum matching. Its independent degree-two vertices trigger the odd-degree coverage theorem, contradicting the exposed cubic roots forced by equality.

Depends on. Lemma 34, Lemma 56, Lemma 95, Lemma 148.

Lemma 204 (An improving certificate under the failed run bound). *Use the finite integer certificate families $C_{A_c,B_c}(G; x, y)$, costs $c_i(G)$, matching number $\nu(G)$, independence number $\alpha(G)$, signed slack $ps(G) = e(G) - \nu(G) - \sum_{i\geq 1} c_i(G)$, and top missing-allocation run $tr(G)$ of Lemma 23. An element of C is an ordered partition of all G-edges into two parts of maximum degrees at most A_c, B_c and two parts of matching numbers at most x, y. All four bounds are nonnegative integers. The quantity $tr(G)$ is the longest consecutive run of missing first matching budgets at total $c_0(G) - 1$, and is zero if $c_0(G) = 0$. All graphs are finite and simple. Write $V(G)$, $E(G)$, $d_G(v)$, and $\Delta(G)$ for the vertex set, edge set, vertex degree, and maximum degree. Vertex deletion takes an induced graph; matching deletion removes only edges.*

For a finite simple G and nonnegative integer q, define $mathfrak C_q(G)$ to be the family of tables

$Q = (T, (Q_{a,x})_{0\leq a\leq q, 0\leq x\leq T}),$

where T is a nonnegative integer and $Q_{a,x}$ belongs to $C_{a,q-a}(G; x, T - x)$ for every indicated integer a, x. Put $cost(Q) = T$. This family is nonempty and its minimum cost is $c_q(G)$.

Let d belong to {8,10,12}. Let F be connected, d-regular, factor-critical and well-covered. Factor-critical means that F-v has a perfect matching for each vertex v; well-covered means that every maximal independent set has maximum cardinality. Write $|V(F)| = 2k + 1$ with integer $k \geq 1$, put $a_F = \alpha(F)$, and assume

$E = c_0(F) - k \geq 1,\ tr(F) + E \geq d/2 + 2.$

Define the integers

$\lambda = 2E - 3 - (k - a_F),\ \delta = c_0(F) - c_2(F),\ \epsilon = c_0(F) - c_1(F),$

$p = E - 2 - \lambda,\ m = p + \delta,\ \beta = m - \epsilon,$

$r = k - \lambda - 1 + \delta,\ s = 2\lambda + 3 - 2\delta,\ u = 2k + 1 - r.$

A complete process chooses a maximum independent set S_0 and then repeatedly deletes any vertex of current maximum degree from $F - S_0$ while that degree is at least three, ending at an induced graph of maximum degree at most two. Suppose:

(a) For every maximum independent S_0 and integer $1 \leq i < d$,

$c_i(F - S_0) = c_i(F) - a_F$ *and* $c_0(F) = a_F + \nu(F - S_0) + ps(F - S_0).$

(b) In every complete process, each deletion parent G of current maximum degree $D \geq 3$ has a matching covering every degree-D vertex; deleting the chosen vertex lowers c_i by exactly one for every integer $1 \leq i < D$. Every terminal has zero slack.

Call an induced subgraph A of F a native row if

$|V(A)| = r$, $\Delta(A) \leq 2$, $ps(A) = 0$,

$\nu(A) = m$, $c_1(A) = \beta$, $c_2(A) = p$.

Native rows exist. Fix any native row A and any complete continuation on F-V(A), repeatedly deleting a current maximum-degree vertex while that degree is at least three. Let R be its terminal, let T_j be its full degree-j phase for each integer $3 \leq j \leq d-3$, and put $X = \bigcup_{j=3}^{d-3} T_j$. Let D2 $= \{zinV(R) : d_R(z) = 2\}$. Choose any maximum independent set T_0 of R[D2] and any shortest whole odd-cycle component C of R, where whole means an entire connected component and cycle order counts vertices. These choices exist. Set

$\ell = (|V(C)| + 1)/2$, $Y = T0$ *union* $V(C)$,

$H = F[V(A)$ *union* X *union* $Y]$, $N = |V(F)$ *minus* $V(H)|$.

Starting with $P_0 = H$, choose any maximum matching M_i of P_i and put $P_{i+1} = P_i - M_i$, retaining all vertices, for every integer $0 \leq i \leq d-3$. Every such successive sequence exists, and its entire final edge set is

$E(P_{d-2}) = E(C)$.

All vertices outside the same original C are isolated in this residual. If $E \leq d/2$, then C is a triangle.

There is an explicit Q in mathfrak$C_{d-2}(F)$ whose degree-bounded parts at every table entry have union precisely $\bigcup_{i=0}^{d-3} M_i$, whose matching-bounded parts contain every original edge with an endpoint outside H, and whose cost is

$cost(Q) = N + \ell - 1 = r - \beta - 1 = c_{d-2}(F) - 1$.

Consequently $c_{d-2}(F) \leq cost(Q) = c_{d-2}(F) - 1$, so the displayed hypotheses cannot hold.

All cardinalities, degrees, phase lengths, matching sizes, capacities, budgets and indices are integers in their stated nonnegative or positive domains; differences are integers. In particular the derived parameters satisfy $p \geq 2$, $0 \leq \epsilon \leq \delta \leq \lambda + 1$, $\beta \geq p$, $r \geq 0$, s positive odd and $2 \leq \ell \leq 4$. Empty degree phases, $\epsilon = 0$ and $\lambda = -1$ are included whenever the hypotheses allow them.

Proof. We use Lemma 23 for finite integer certificates, Lemma 105 for terminal independence, Lemma 173 for exact costs and native rows, Lemma 171 for cycle selection, Lemma 159 for successive matchings, and Lemma 15 for restoration. Their finite integer costs have the same definitions.

1. All terminal rows and all native-row continuations.

Factor-criticality gives $\nu(F) = k$: a perfect matching of F-v has k edges and the order $2k+1$ bounds every matching by k. A maximum independent set exists because F is finite. Each subsequent root deletion reduces the number of vertices, so every legal partial process has a complete continuation.

The final process conclusion of Lemma 105 applies to every complete process here. Indeed $d \geq 8$ is even, F is connected and regular of order $2k+1$ with $\nu(F) = k$ and $E \geq 1$, and F is well-covered. Its two initial positive-row identities and initial row-zero identity are part of (a). Its coverage, lower-row loss and zero terminal slack assumptions are precisely part of (b). Hence every terminal has independence number a_F. In particular at least one such terminal exists.

Now apply Lemma 173. Its hypotheses (a), (b) are exactly those here, and its hypothesis (c) follows from the preceding paragraph. It gives native rows and the identities

$p \geq 0$, $0 \leq \epsilon \leq \delta \leq \lambda + 1$, $\beta \geq p$,

$r = a_F + 2p + \delta$, s positive odd, $u = r + s$, $|V(F)| = 2r + s$,

$c_0(F) = u + m$, $c_1(F) = u + \beta$, $c_2(F) = u + p$,

$c_{d-3}(F) = r$, $c_{d-2}(F) = r - \beta$, $c_{d-1}(F) = a_F$. (1)

Every native row has exactly p odd-cycle components and ϵ positive even-order path components, and its whole components belong to P_1, P_2, P_3, P_4, P_6, C_3, C_4, C_5, C_7.

Let Rcal be the family of all native rows of F. It is finite and nonempty. For any B in Rcal the same row theorem provides a replay of B as an initial maximum independent set followed by the entire degree-(d-1) and degree-(d-2) phases. Its complement has maximum degree at most d-3. Append any complete maximum-degree continuation to this replay. It is a complete process covered by (a), (b), so its terminal has independence a_F by Lemma 105. The terminal formulas of Lemma 173 then give order r, matching number m, first cost β and second cost p. Its slack is zero and its maximum degree is at most two. Thus it belongs to Rcal.

In that arbitrary continuation let n_j be the number of roots deleted at degree j, for $3 \le j \le d-3$. The full process has first two phase counts $p+\epsilon$ and β. They sum to $2p+\delta = r - a_F$. Subtracting these from the full process cost formulas in Lemma 173 gives

$c_i(F) = r + \sum_{j=i+1}^{d-3} n_j$

for every integer $3 \le i \le d-3$. (2)

This is established for every B in Rcal and every complete continuation. Its terminal order r and complement order $u = r+s$ also give exactly s remaining roots. In particular these conclusions hold for the fixed A, R, X in the statement.

2. One selected cycle and one compatible matching sequence.

Let Bcal be the family of all nonnegative integer pairs (b_R, b_B) of total $c_0(F)-1$ that are infeasible as two matching-bounded edge parts. At degree capacities zero both degree-bounded parts are empty, so this is the missing family in Lemma 23 and in Lemma 171.

Apply Lemma 171 to the fixed terminal native row R. The graph F is connected, regular and factor-critical, with $E \ge 1$. The top-cost equality in (1) gives $c_{d-1}(F) \ge a_F$, and the failed-run inequality is an explicit hypothesis here. Its integer parameter definitions agree with ours; the required inequalities follow from (1). The row R has the order, degree, matching number, odd-cycle count and positive even-path count supplied by step 1, and every whole odd cycle has order at most seven. Its total $u+m-1$ equals $c_0(F)-1$ by (1). This checks every hypothesis on the same F and R.

The selector proves $p \ge 2$ and that Bcal is nonempty. It permits every maximum T_0 and every tied shortest C in the statement, with both chosen before any pair in Bcal. Write $b = |Y|$ and $h = |X \text{ union } Y|$. Its conclusions give

$b = \beta + \ell, h = s + \beta + \ell, N = r - \beta - \ell,$

$2 \le \ell \le 4, E(R[Y]) = E(C),$

$N \ge \max(0, \beta + h - 2), N \ge \ell - 2,$

$\min(b_R, b_B) \le N$ for every pair in Bcal. (3)

All vertices of R[Y] outside C are isolated. When $E \le d/2$ it gives $\ell = 2$, so C is a triangle. Its retained graph is exactly $H = F[V(A) \text{ union } X \text{ union } Y]$, because $V(A)$, X, $V(R)$ partition $V(F)$. Thus no graph or cycle has changed.

Choose any one pair (b_R, b_B) in the nonempty Bcal. Apply Lemma 159, part (ii), with $D = d$, the family Rcal, the fixed A and continuation, and $x = b_R$, $y = b_B$. Equation (1) supplies the graph order, low costs and missing total $x + y + 1 = u + m$. Its initial bound $\min(x,y) \le r$ follows from $\min(x,y) \le N$ and $N = r - b \le r$. Step 1 proves, for every row in Rcal and every continuation, both terminal membership in Rcal and the exact costs (2). Thus its universal process hypotheses hold, not merely the chosen-process versions.

The subset Y consists of degree-two vertices of R. The actual h, N agree with that theorem's definitions, and (3) is precisely its omission and smaller-budget bound. All hypotheses of its part (ii) now hold. It supplies the first three maximum matchings and every compatible maximum continuation, for exactly d-2 matchings M_0,..., M_{d-3}. Its entire residual is $E(R[Y]) = E(C)$, with every exterior vertex isolated, and it preserves every original edge of R[Y]. Finiteness guarantees a maximum at every stage, including empty residuals. Hence the stated conclusion holds for every successive choice on the same H.

3. The universal certificate table.

Put $q = d-2$, $w = \ell - 1$, $T = N + w$, and $K = \bigcup_{i=0}^{q-1} M_i$. These q matchings are pairwise edge-disjoint because each was selected in the preceding edge residual. The preceding step says that $E(H)$ minus K is

exactly $E(C)$. Lemma 15 applies with $D = q$, $g = 0$, the same omitted set, the same cycle, and its prescribed matchings equal in order to M_0,..., M_{q-1}. Its zero-surcharge hypothesis is the bound $N \geq \ell - 2$ in (3). Thus its universal total is T.

Here is the construction for every table entry. Order the N omitted vertices as z_1,..., z_N and let U_j contain each original edge whose earliest omitted endpoint is z_j. Each U_j is a possibly empty star. These stars partition every edge outside H, including edges with two omitted endpoints.

For integers $0 \leq a \leq q$ and $0 \leq x \leq T$, put the first a prescribed matchings in Q_R and the remaining q-a in Q_B. If $x \geq w$, put C and the first x-w stars in P_R and the remaining stars in P_B. This is defined because $0 \leq x - w \leq N$. Restriction of any matching to these disjoint edge pieces bounds the two matching numbers by $w + (x - w) = x$ and $N - (x - w) = T - x$.

If $x < w$, put the first x stars in P_R and the cycle and the remaining stars in P_B. The bound $x \leq w - 1 = \ell - 2 \leq N$ makes this defined. Their matching numbers are at most x and $w + N - x = T - x$. The restriction argument permits pieces to share vertices.

The two degree bounds hold because each prescribed matching contributes at most one edge at any vertex. The parts are disjoint and exhaust $E(F)$: the matchings and C partition $E(H)$, and the omitted-vertex stars partition every other edge. Thus their ordered tuple belongs to $C_{a,q-a}(F; x, T - x)$. This proves feasibility for every capacity and budget entry, including the zero endpoints, with degree-bounded union exactly K.

For any finite graph G and nonnegative q, the threshold characterization in Lemma 23 shows that all entries at total T exist exactly when $T \geq c_{a,q-a}(G)$ for every integer a in [0,q], equivalently $T \geq c_q(G)$. A finite set of entries can be selected independently. The finite nonnegative integer $c_q(G)$ is attained, giving the general table interpretation and nonemptiness in the statement.

For the table just constructed on F, equation (3) gives

$cost(Q) = T = N + \ell - 1 = r - \beta - 1.$

Therefore $c_{d-2}(F) \leq r - \beta - 1$, whereas (1) gives $c_{d-2}(F) = r - \beta$ using the same original graph and cost definition. This is the asserted contradiction. □

Remark. The source run bound supplies the exact omission reserve even at small excess. Universal terminal preservation makes every native row compatible with every continuation, so the same cycle survives all successive maximum matchings and its matching rank saves one unit in the universal certificate.

Depends on. Lemma 15, Lemma 23, Lemma 105, Lemma 159, Lemma 171, Lemma 173.

Lemma 205 (Factor-critical graphs with zero slack). *Use the finite integer certificate costs $c_i(Q)$, matching number $\nu(Q)$, and slack $ps(Q) = e(Q) - \nu(Q) - \sum_{i \geq 1} c_i(Q)$ of Lemma 23. For an integer $n \geq 3$, let C_n denote the simple cycle on n vertices. A graph is factor-critical if deleting any one vertex leaves a perfect matching.*

For every finite simple factor-critical graph Q of integer order at least three,

$ps(Q) = 0$ if and only if Q is isomorphic to C_3, C_5, or C_7.

Every zero-slack graph in this domain also satisfies $c_0(Q) \leq \nu(Q) + 1$. Every factor-critical graph in the stated domain that is not one of those three cycles satisfies $ps(Q) \geq 1$. In particular every connected regular factor-critical graph of integer degree at least four has positive slack. No degree bound, order bound, independence-number bound, or hereditary-gap hypothesis is imposed.

Proof. Suppose $ps(Q) = 0$. Apply Lemma 156. Its notation gives a positive even regular degree D, a maximum independent set S of size s, and $G = Q - S$. It supplies

$c_{D-1}(Q) = s + c_{D-1}(G)$

and one of its four explicit structural forms. If $D = 2$, its connectedness makes Q an odd cycle: follow consecutive edges to close a cycle; every cycle vertex has both incident edges there, so connectedness excludes any other vertices, and factor-criticality gives odd order. The degree-two classification Lemma 57 then leaves only C_3, C_5, C_7. It remains to exclude $D \geq 4$.

The small-independence conclusions of Lemma 156 give $s \geq 3$ in this case. For a positive integer r, call an induced residual r-good when its maximum degree is at most r and a matching covers every degree-r vertex, as in Lemma 86. That fact gives $c_r(Q) \leq |U|$ whenever Q-U is r-good: its vertex restoration supplies certificates on every original Q-edge for every capacity and matching-budget allocation at that total. At $r = 3$ it also says that a graph of maximum degree at most three with at most nine vertices of degree three is 3-good. Indeed the odd-degree exclusion is automatic, $b_3 = 5$, and nine equals $2b_3 - 1$.

We will prove $c_{D-1}(G) = 0$ and construct a set U with $|U| = s - 1$ such that Q-U is (D-1)-good. These two statements contradict the displayed exact row identity.

First consider form (a) or (c) of Lemma 156. Here G is respectively a triangle with isolates and $D = 6$, or a star with $D/2$ edges and isolates. Its maximum degree is at most D-2. It is therefore (D-1)-good with the empty matching, and $c_{D-1}(G) = 0$.

Let z be any triangle vertex in the first case, or the star center in the second case. Its neighbors in S number $D - deg_G(z)$, which is four in the triangle case and $D/2 \geq 2$ in the star case. Choose distinct a, b among those neighbors and set

$U = (Sminus\{a, b\})$ union {z}, and $J = Q - U$.

Then $|U| = s - 1$. Each of a, b has exactly D-1 neighbors in J, since its only deleted neighbor outside the independent set S is z. They are nonadjacent. In the star case every other J-vertex has degree at most two, because G-z is edgeless and only a, b are retained from S. In the triangle case every other J-vertex has degree at most three, because G-z is one edge and isolates; here $D - 1 = 5$. Thus J has maximum degree D-1 and its only vertices of that degree are a, b. Each has at least three neighbors, all outside {a, b}. Choose one neighbor of a and a different neighbor of b; the two resulting edges form a matching covering a, b. Hence J is (D-1)-good. This contradicts $c_{D-1}(Q) = s$.

The remaining forms have $D = 4$ and $n = 2s + 3$. First take form (b), so G consists of two triangles T_1, T_2 and isolates. Every triangle vertex has exactly two neighbors in S. Choose v_1 in T_1 and v_2 in T_2 with different S-neighborhoods. Such a choice exists: otherwise all six triangle vertices would have the same two S-neighbors, each of which would have degree at least six in Q, contrary to degree four. Their neighborhood union has at least three elements. Choose a three-element subset R of that union and put

$P = \{v_1, v_2\}$, $U = (SminusR)$ union P, $J = Q - U$.

Then $|U| = s - 1$. Each vertex of R loses a neighbor in P, so has degree at most three in J. A remaining triangle vertex has one neighbor within G-P and at most two neighbors in R. Every other retained vertex outside R has no neighbor within G-P. Thus J is subcubic. Each cubic vertex outside R has at least two neighbors in R. Since the sum of degrees at R is at most nine, at most four cubic vertices lie outside R. There are at most seven cubic vertices in total, so J is 3-good by Lemma 86. The graph G itself has maximum degree two, hence $c_3(G) = 0$. Again $c_3(Q) \leq s - 1$ contradicts $c_3(Q) = s$.

Finally consider form (d). Write $X = \{x\}$, let T be the triangle of G-x, and let Z be its set of s-1 isolated vertices. The only edges of G are those of T and edges incident with x; moreover $deg_G(x) = 3$ and x has a unique S-neighbor a. Put

$r = |N_G(x)$ intersect $V(T)|$ in {0,1,2,3}.

Here $N_G(x)$ is the set of neighbors of x in G. Then x has 3-r neighbors in Z. Write $N_S(w)$ for the neighbors in S of any w outside S. Regularity gives

$|N_S(v)| = 2 - 1_{xvisanedge}$ for v in $V(T)$,

$|N_S(z)| = 4 - 1_{xzisanedge}$ for z in Z.

The only possible cubic vertices of G are x and the at most three triangle vertices adjacent to it. Thus G is 3-good and $c_3(G) = 0$.

We construct a two-element set P outside S and a three-element set R contained in the union $N_S(P)$ of the S-neighborhoods of P. As before set $U = (SminusR)$ union P and $J = Q - U$. Every vertex of R loses at least one neighbor, giving degree at most three in J, and $|U| = s - 1$. Two forms of P, R suffice.

In the first form, $P = \{x, v\}$ for v in T, and R is any three-element subset of $N_S(P)$. Then G-P consists of one edge and isolates. Each remaining triangle vertex has at most two S-neighbors, so its degree in J is at most three. The other vertices outside R have degree at most three as well. Each cubic vertex outside R has at least

two neighbors in R. The degree sum at R is at most nine, so at most seven vertices of J are cubic. Thus J is 3-good.

In the second form, $P = \{v, z\}$ with v in T and z in Z, and x is adjacent to at least one of v, z. Require also that R differs from $N_S(w)$ for every w in Z minus {z} adjacent to x; each such neighborhood has three elements.

For this second form, x retains at most two neighbors in G-P and at most its one S-neighbor, so has degree at most three in J. A retained triangle vertex w has internal degree $1 + 1_{xwisanedge}$ and at most $2 - 1_{xwisanedge}$ neighbors in R, so also has degree at most three. A retained Z-vertex adjacent to x keeps at most two S-neighbors, because R differs from its three-element S-neighborhood; together with x its degree is at most three. A Z-vertex not adjacent to x has no internal neighbor and at most three neighbors in R. Thus J is subcubic.

Only x and the two retained triangle vertices can have internal degree two in G-P. Every cubic vertex outside R has at least two R-neighbors, except possibly these three vertices, each of which still has at least one. If h counts the cubic vertices outside R, their incidences with R number at least 2h-3. This is at most nine, so $h \leq 6$. Adding the three vertices of R leaves at most nine cubic vertices. Hence J is 3-good in the second form too.

It remains to choose one of these forms, respecting all its requirements.

If $r = 0$ and some v in T has a not in $N_S(v)$, take $P = \{x, v\}$ and $R = \{a\}$ union $N_S(v)$. The set $N_S(v)$ has two elements, so this is the first form.

If $r = 0$ and a belongs to $N_S(v)$ for every v in T, then the four neighbors of a are x and the three triangle vertices. Therefore no vertex of Z is adjacent to a. Choose z among the three Z-neighbors of x, choose two distinct elements b, c of its three-element S-neighborhood, and choose any v in T. Put $P = \{v, z\}$ and $R = \{a, b, c\}$. The set R lies in $N_S(P)$, and x is adjacent to z. Every S-neighborhood of a Z-neighbor of x avoids a, so none equals R. This is the second form.

Suppose $r \geq 1$ and some z in Z is not adjacent to x. Choose v in T adjacent to x. The set $N_S(z)$ has four elements, so $N_S(v)$ union $N_S(z)$ has at least four distinct three-element subsets. There are at most $3 - r \leq 2$ forbidden sets $N_S(w)$, from Z-neighbors of x. Choose a three-element subset R of that union unequal to all forbidden sets. With $P = \{v, z\}$, this is the second form.

The only remaining case has $r \geq 1$ and every vertex of Z adjacent to x. Since $|Z| = s - 1 = 3 - r$ and $s \geq 3$, it follows that $s = 3$ and $r = 1$. There are two Z-vertices, and each is adjacent to every vertex of S. There are also two triangle vertices not adjacent to x. They cannot both be adjacent to a, since then a would have the five distinct neighbors x, the two Z-vertices, and these two triangle vertices. Choose one of them, v, not adjacent to a. Its two-element S-neighborhood is S minus {a}. Take $P = \{x, v\}$ and $R = S$. This is the first form.

These cases exhaust all r and all incidences with Z. In every case J is an actual induced 3-good residual after deleting s-1 vertices. Therefore $c_3(Q) \leq s - 1$, contradicting $c_3(Q) = s + c_3(G) = s$. No graph of degree at least four survives the four structural forms, so a zero-slack Q is C_3, C_5, or C_7.

Conversely, deletion of any vertex of each of these three odd cycles leaves an even path. Pairing consecutive path vertices gives a perfect matching, so each cycle is factor-critical. Lemma 57 gives $ps = 0$ for each of them. An odd cycle on $2k + 1$ vertices has matching number k and an independent set of k alternating vertices; their complementary $k + 1$ vertices form a vertex cover. The bound $c_0 \leq \tau$ from Lemma 23, with τ the minimum vertex-cover size, therefore gives $c_0 \leq k + 1 = \nu + 1$ for every cycle in the classification. Finally slack is a nonnegative integer by Lemma 61. Every factor-critical graph in the stated domain that is not one of these cycles therefore has slack at least one, including every regular one of degree at least four. □

Remark. Each zero-slack cut would make its penultimate row cost equal to the independent-set size. The small remainder permits replacing two or three independent vertices while deleting one fewer vertex. The resulting bounded-degree graph has a covering matching, which saves one unit in that same row and rules out every degree above two.

Depends on. Lemma 23, Lemma 57, Lemma 61, Lemma 86, Lemma 156.

Lemma 206 (Tight endpoint barriers and actual separator deletion). *Let $D \geq 2$ be an integer and let G be a finite simple graph with $\Delta(G) \leq D$. Put $A = \{vinV(G) : d_G(v) = D\}$, let η be the minimum number of A-vertices uncovered by a matching, and let $\nu(G)$ be maximum matching cardinality. For $X \subseteq V(G)$, let $q_A(G - X)$ count odd-order connected components of the induced graph G-X wholly contained in A; call X attaining when $q_A(G - X) - |X| = \eta$. For disjoint $S, T \subseteq V(G)$, put $U = V(G)$ minus (S union T), and let q count the components C of $G[U]$ wholly in A with $|V(C)| + e_G(C, T)$ odd. Here $e_G(X, Y)$ counts edges between disjoint sets and $e_G(X)$ counts edges internal to X. Assume*

$q + e_G(S, T) - |S| - (D - 1)|T| = \eta.$

Every maximum matching W exposing exactly η A-vertices covers S, has no edge internal to S, and contains every edge internal to T. Every vertex of T minus A has degree D-1 and is uncovered by W. If v belongs to T minus A, choose any neighbor z of v and let w be the W-mate of z. Then w belongs to A and $W' = (Wminus\{zw\})$ union $\{vz\}$ is a maximum matching exposing exactly $\eta + 1$ A-vertices. Its exposed A-set I is independent and satisfies $\nu(G - I) = \nu(G)$.

If $T \subseteq A$, put $J = G - S$ and $A' = A$ minus S. The minimum A'-exposure in J is $\eta + |S|$, and S is contained in an attaining set X. If additionally $D \geq 3$, G is connected and not D-regular, and S intersects A, there is an independent $I \subseteq A$ of size $\eta + 2$ with $\nu(G - I) \geq \nu(G) - 1$.

Suppose $T \subseteq A$ and every attaining set is disjoint from A. Then S is disjoint from A and is itself attaining. The set T is independent, has no edge to any component counted by q, and every vertex of T has at most one neighbor outside S. In particular $|S| \leq D - 2$ forces T empty. Every odd component of G-S wholly in A is factor-critical, meaning deletion of any vertex leaves a perfect matching, and every other component has an internal matching covering its A-vertices. If moreover $D \geq 3$ and G is connected and not D-regular, G-S has a matching covering all vertices of degree D in G-S. For every pair of nonnegative integers a, b with $a + b = D$ it consequently admits an edge coloring with red degrees at most a and blue degrees at most b.

All counts, degrees, matching cardinalities and capacities are integers in the stated domains. Empty graphs, isolated vertices, $\eta = 0$, and empty S or T are allowed subject to the explicit hypotheses.

Proof. The exact marked deficiency theorem Lemma 46 gives the maximum formula for η, finite attainment, and a maximum matching attaining η. Write b_1 for the assumed endpoint expression.

We first obtain one identity for an arbitrary matching N. For sets among S, T, U, let $m(X, Y)$ and $m(X)$ count its crossing and internal edges. Let u_X be its total uncovered count in X, h_X its uncovered A-count there, and $h = h_S + h_T + h_U$. Put $l_T = u_T - h_T$ and $d_T = \sum_{v \in T}(D - d_G(v))$. For a counted component C let h_C be its uncovered count. Matching parity shows that

$t_C = h_C + m(C, S) + e_G(C, T) - m(C, T)$

is a nonnegative odd integer: its parity is $|V(C)| + e_G(C, T)$, and all its terms are nonnegative. Thus $t_C \geq 1$. Summing and enlarging to all of U gives

$q \leq \sum_{Ccounted} t_C \leq h_U + m(S, U) + e_G(T, U) - m(T, U).$

Define $\epsilon \geq 0$ by subtracting q from the last expression. If any $t_C \geq 3$, then $\epsilon \geq 2$.

Use $e_G(T, U) = D|T| - d_T - 2e_G(T) - e_G(S, T)$ and the matching counts

$|S| = u_S + 2m(S) + m(S, T) + m(S, U),$

$|T| = u_T + 2m(T) + m(S, T) + m(T, U).$

Substitution in b_1 gives the exact identity

$b_1 = h + l_T - d_T - u_S - h_S - 2m(S) - 2(e_G(T) - m(T)) - \epsilon.$ (1)

Every vertex outside A has degree deficit at least one, so $l_T \leq |T$ minus $A| \leq d_T$. Every other subtracted term in (1) is nonnegative.

Apply (1) to any maximum matching W with $h = \eta = b_1$. All these nonnegative losses vanish. In particular $u_S = m(S) = 0$, $e_G(T) = m(T)$, and $l_T = |T$ minus $A| = d_T$. Thus S is covered, there are no W-edges inside S, all T-edges belong to W, and each low T-vertex is uncovered and has degree D-1.

For such a low vertex v, $D - 1 \geq 1$ gives a neighbor z. Since W is maximum and v is uncovered, z must have a W-mate w; otherwise vz could be added. The one-edge exchange $W' = (Wminus\{zw\})$ union {vz} is a matching of the same maximum size. If w were outside A, the A-exposure would remain η, but v would be

covered, contradicting the property just proved for every such matching. Hence w lies in A, so W' has exactly $\eta + 1$ exposed A-vertices. Uncovered vertices of a maximum matching are independent, because an edge between two would augment it. Its exposed A-set I is therefore independent; W' survives deletion of I, so $\nu(G - I) = \nu(G)$.

Now assume $T \subseteq$ A. For any matching N of $J = G - S$, regard N as a matching of G. Then all of S is uncovered, $m(S) = 0$ and $l_T = d_T = 0$. If h' is its A'-exposure in J, (1) gives $b_1 \leq h' - |S|$, so $h' \geq \eta + |S|$. Conversely restrict a maximum eta-exposure matching W to J. It covered S and had no edge internal to S. Deleting S creates at most $|S|$ new exposed A'-vertices and deletes no exposed A-vertex. The restriction thus has A'-exposure at most $\eta + |S|$. This proves the claimed equality. By Lemma 46 choose $Y \subseteq V(J)$ attaining that deficiency. For $X = S$ union Y, the remaining components are the same, and containment in A' is equivalent to containment in A. Therefore

$q_A(G - X) - |X| = (\eta + |S|) - |S| = \eta.$

This X contains S. Under the additional connected, non-D-regular, $D \geq 3$ hypotheses, if S contains a high vertex, apply Lemma 150 to X and that specified vertex. Its independent exposed set gives the asserted size $\eta + 2$ and matching-loss bound.

For the rest assume every attaining set is disjoint from A. The containing set X just found shows S intersect A is empty. For each w in A we first find a matching leaving w uncovered with total A-exposure at most $\eta + 1$. For any $Y \subseteq V(G)$ minus {w}, the set Y union {w} is not attaining. The integer maximum formula in Lemma 46 consequently gives

$q_A(G - (Yunion\{w\})) - |Y \text{ union } \{w\}| \leq \eta - 1.$

Equivalently the marked deficiency expression for G-w, marked by A minus {w}, is at most η for every Y. The same fact supplies a matching of G-w exposing at most η of those marked vertices. In G it leaves w uncovered and has total A-exposure at most $\eta + 1$.

If there were an edge inside T, choose its endpoint w and the last matching. The edge is unused, so $e_G(T) - m(T) \geq 1$. Equation (1), with $l_T = d_T = 0$, would imply $b_1 \leq \eta - 1$, a contradiction. Hence T is independent. If an edge joined T to a counted component C, choose its endpoint w in C and again use a matching exposing w with at most $\eta + 1$ total A-exposures. Then $h_C \geq 1$ and $e_G(C, T) - m(C, T) \geq 1$. The odd number t_C is at least three, giving $\epsilon \geq 2$. Equation (1) gives the same contradiction. Thus no counted component is adjacent to T.

Take a matching of minimum A-exposure η. Equation (1) forces $\epsilon = 0$. Since no counted component has an edge to T,

$q \leq \sum_{Ccounted} t_C = \sum_{Ccounted}(h_C + m(C, S)) \leq h_U + m(S, U).$

But $\epsilon = 0$ means $q = h_U + m(S, U) + e_G(T, U) - m(T, U)$. Since the last difference is nonnegative, it is zero. Every graph edge joining T to U therefore belongs to this matching. Each T-vertex has at most one neighbor in U and none in T. It thus has at most one neighbor outside S. A nonempty T would require at least D-1 distinct neighbors in S for each of its vertices, proving that $|S| \leq D - 2$ forces T empty.

We can now prove attainment by S directly. Each of the q counted components remains an odd all-A component of G-S, because it has no edge to T. Put $t = |T|$. Since T is independent and each T-vertex has zero or one neighbor in U, exactly $t - e_G(T, U)$ of its vertices are isolated in G-S. Each is an additional odd all-A component, distinct from the q counted ones. Summing degrees in T gives $e_G(S, T) = Dt - e_G(T, U)$. Consequently

$q_A(G - S) \geq q + t - e_G(T, U)$

$= q + e_G(S, T) - (D - 1)t = \eta + |S|.$

The marked deficiency maximum bounds the left side minus $|S|$ by η. Equality follows, so S is attaining.

We show that the stated component properties hold for every attaining set Z under the present no-high-attainer hypothesis. Let C be an odd all-A component of G-Z. Inside C, use Lemma 107 with all its vertices marked and an inclusionwise maximal attaining separator W_C. Its deficiency is $\delta_C = |V(C)| - 2\nu(C) \geq 1$ by odd order and Lemma 46. Replacing C by the components of $C - W_C$ gives

$q_A(G - (ZunionW_C)) - |Z \text{ union } W_C| = \eta - 1 + \delta_C \geq \eta.$

The global maximum formula forces equality. Thus Z union W_C is attaining. Since W_C lies in A while every attaining set avoids A, W_C is empty. The factor-critical component conclusion of Lemma 107, applied inside connected C with this empty separator, proves that C is factor-critical.

For the other components, choose any matching attaining η in G. Write $p = q_A(G - Z) = |Z| + \eta$ and let l be the number of those required components receiving a matching edge from Z. There are at most $|Z|$ such components. Each of the p-l others has at least one exposed A-vertex by odd parity. Hence

$\eta \geq p - l \geq p - |Z| = \eta$.

Equality forces all $|Z|$ separator vertices to be matched to distinct required components, one edge each, and no marked vertex outside the unused required components to be exposed. Therefore no matching edge joins a nonrequired component to Z, and its internal restriction covers every A-vertex in it. Apply these properties with $Z = S$.

Finally assume $D \geq 3$, G is connected and not D-regular. If S is nonempty, each component of G-S has a neighbor in S by connectedness. In every required odd component choose as exposed root a vertex having such a neighbor, and take a perfect matching after deleting that root. The root has degree less than D in G-S. In every other component take the internal A-covering matching just proved. Their union covers every vertex of degree D in G-S: such a vertex was in A, and all exposed A-roots lost an edge to S. If S is empty, connectedness and non-D-regularity give $q_A(G) = 0$. Since S attains, $\eta = 0$, and a minimum-exposure matching already covers every degree-D vertex. In both cases apply Lemma 35 to the actual graph G-S and this covering matching. It yields all the stated nonnegative integer degree splits. Every set and matching was chosen from a finite graph; the arguments include zero deficiency and the allowed empty sets. □

Remark. One exact matching-slack identity controls both sides of a tight endpoint barrier. Once T is independent and has at most one neighbor outside S, its isolated vertices account for the entire remaining deficiency, proving that S itself is an actual attaining separator.

Depends on. Lemma 35, Lemma 46, Lemma 107, Lemma 150.

Lemma 207 (Independent-set rigidity of positive-gap quartic graphs). *Use the finite integer certificate costs c_i, matching number ν, independence number α, excess $ex = c_0 - \nu$, slack $ps = e - \nu - \sum_{i\geq 1} c_i$, and $gap = ex - ps$ of Lemma 113. All graphs are finite and simple. Vertex deletion is induced. A nonempty graph is factor-critical if deleting any one vertex leaves a perfect matching. All graph orders, degrees, costs, indices, and cardinalities are nonnegative integers.*

Let F be a nonempty connected four-regular factor-critical graph with $gap(F) \geq 1$ and integer $E = ex(F) \geq 2$. Put $k = \nu(F)$. Then $|V(F)| = 2k + 1$, $k \geq 5E - 5$, and $gap(F) = 1$. The graph F is well-covered: every inclusionwise maximal independent set has cardinality $a = \alpha(F) = c_3(F) \geq 3$.

For every such set S and $G = F - S$, one has

$\Delta(G) \leq 3$, $ps(G) = c_3(G) = 0$,

$c_i(F) = a + c_i(G)$ *for* $i = 1, 2, 3$,

$c_0(F) = a + \nu(G)$, $\nu(G) = k + E - a$,

$c_4(F) = 1$, $c_i(F) = 0$ *for* $i \geq 5$, *and* $c_i(G) = 0$ *for* $i \geq 3$.

Every connected component of G has zero slack.

Put $x = k - a$ and $g = c_2(F) - a$. These integers obey

$E - 1 \leq x \leq 2E - 2$,

$2x + 2 - 2E \leq g \leq x$.

For every maximum independent S the same g equals $c_2(F - S)$, and

$e(F - S) = 4x + 2$, $c_1(F - S) = 3x + 2 - E - g$.

No minimality or hereditary-gap assumption on F, and no upper bound on $c_3(F)$, is imposed.

Proof. Factor-criticality gives $|V(F)| = 2k+1$, and four-regularity gives $e(F) = 4k+2$. Apply Lemma 45 with $H = F$ and $D = 4$. All its hypotheses hold: F is connected, has the stated odd order and matching number, has maximum degree four, and $E \geq 2$. Its auxiliary integer is E-2, so $k \geq 3E - 1 + 2(E-2) = 5E - 5 \geq 5$. In particular F is not K_5.

Fix any inclusionwise maximal independent set S. Write $a_S = |S|$, $G = F - S$, $s = ps(G)$, and $f = gap(F)$. Every vertex outside S has a neighbor in S. Thus $\Delta(G) \leq 3$, and the closed neighborhoods of S cover F. Each such closed neighborhood has five vertices, so $5a_S \geq 2k+1 \geq 11$ and $a_S \geq 3$. By Lemma 147, every subgraph J of G satisfies: positive gap implies gap one and $tr(J) + ex(J) \leq 3$. Indeed every such J has maximum degree at most three. In particular $gap(J) \leq 1$.

Choose v in S and restrict a perfect matching of F-v to G. Deleting the $a_S - 1$ other vertices of S removes at most $a_S - 1$ matching edges, so

$\eta = \nu(G) - (k + 1 - a_S) \geq 0.$

Let B be the family of components P of G satisfying $floor(|V(P)|/2) \geq a_S + 2$, and put $M = \max(\{0\} union \{ex(P) : P \in B\})$. The arbitrary-deletion assertion of Lemma 113 gives

$c_0(F) \leq a_S + \nu(G) + M,$

or $E \leq 1 + \eta + M$.

Its regular-graph assertions apply because F is connected, four-regular, factor-critical, and not K_5. They give $c_4(F) = 1$ and all rows above four zero; the rows of G at and above four are zero.

For $i = 1, 2, 3$ put $r_i = c_i(F) - c_i(G)$. By Lemma 72, $0 \leq r_i \leq a_S$. Exactly $4a_S$ edges are deleted, because S is independent. Subtracting the slack definitions, with $ps(F) = E - f$, gives

$\epsilon := E - f - s - \eta = 3a_S - r_1 - r_2 - r_3 \geq 0.$

Consequently

$\epsilon \leq M + 1 - f - s.$ (1)

Nonnegative slack follows from Lemma 61. If $M = 0$, (1), $f \geq 1$, and $s \geq 0$ force $f = 1$, $\epsilon = 0$, and $s = 0$.

Suppose $M \geq 1$ and choose $P.$ in B with $ex(P.) = M$. Its subcubic gap bound gives $ps(P.) \geq M - 1$. The component slack inequality in Lemma 72 and nonnegativity give $s \geq M - 1$. Equation (1) now implies $\epsilon \leq 2 - f$. Thus f is either one or two.

If $f = 2$ or $\epsilon > 0$, integrality and (1) force $s = M - 1$ and $E = 1 + \eta + M$. More explicitly, $f = 2$ forces $\epsilon = 0$ and $s = M - 1$; $\epsilon > 0$ forces $f = 1$, $\epsilon = 1$, and $s = M - 1$. In either case the definition of ϵ gives $E = 1 + \eta + M$. Equality in the component slack comparison then gives

$gap(P.) = 1$, $ps(P.) = M - 1,$

and $ps(Q) = 0$ for every other component Q of G.

We show that these equalities would give a certificate of F at total $c_0(F) - 1$. Use zero-capacity allocation sets $Y_t(J)$ from Lemma 75. For every other component Q, $ex(Q) = gap(Q)$ since $ps(Q) = 0$. Nonnegative excess follows directly from $c_0(Q) \geq \nu(Q)$: the allocation with one budget zero requires the other budget to cover a maximum matching of Q. Hence its subcubic gap bound gives $ex(Q)$ in {0,1}. If $ex(Q) = 0$, $Y_{\nu(Q)}(Q)$ is a full interval. If $ex(Q) = 1$, the same total equals $c_0(Q) - 1$; its allocation set contains both endpoints and its missing runs have length at most $tr(Q) \leq 2$.

Start with the actual S-pool interval $[0, a_S]$. Its length is at least three, so the interval criterion of Lemma 75 absorbs every component Q different from $P.$ at total $\nu(Q)$. The full interval obtained has length

$L = a_S + \sum_{Q different from P.} \nu(Q) \geq 3.$

On $P.$ use total $t = c_0(P.) - 1 = \nu(P.) + M - 1$. This is at least $\nu(P.)$, so both endpoints occur. Since $gap(P.) = 1$, its subcubic bound gives $tr(P.) \leq 3 - M \leq 2$. The same interval criterion absorbs $P.$ as well, yielding every allocation at

$L + t = a_S + \nu(G) + M - 1 = k + E - 1 = c_0(F) - 1.$

The component unions and the actual vertex-pool extension assign every original edge of F once, including all edges incident with S. The claimed total is nonnegative. Its universal feasibility contradicts the definition of $c_0(F)$. Therefore neither $f = 2$ nor $\epsilon > 0$ is possible.

We have proved for every maximal S that $f = 1$ and $\epsilon = 0$. Since ϵ is a sum of three nonnegative row shortfalls, each $r_i = a_S$. Also $E = 1 + s + \eta$, so

$c_0(F) = a_S + \nu(G) + s.$ (2)

The argument did not fix one maximal set for the final assertion; S was arbitrary.

The middle-row theorem Lemma 112 applies to F and gives $c_2(F) \le k$. By Lemma 161, the subcubic G satisfies $c_1(G) \le \nu(G)$ and $c_3(G) \le \lfloor s/2 \rfloor$. Together with exact row restoration and (2), these yield

$c_1(F) \le k + E - s,\ c_3(F) \le a_S + \lfloor s/2 \rfloor.$

Expand $gap(F) = 1$ using $c_4(F) = 1$ and the zero tail. The three row bounds give

$1 = c_0(F) + c_1(F) + c_2(F) + c_3(F) + 1 - (4k + 2)$

$\le a_S - k + 2E - \lceil s/2 \rceil - 1.$

Thus $a_S \ge k + 2 - 2E + \lceil s/2 \rceil$.

If $s \ge 1$, then $a_S \ge k - 2E + 3$. The independent-set conclusion of Lemma 45, with the same F and $E \ge 2$, gives

$a_S + \nu(F - S) \ge k + E.$

Equation (2) instead gives $a_S + \nu(F - S) = k + E - s$, a contradiction. Therefore $s = 0$. The subcubic third-row bound gives $c_3(G) = 0$, and exact third-row restoration gives $a_S = c_3(F)$. A maximum independent set exists in the finite graph F and is maximal. Hence every maximal set has the same cardinality $a = \alpha(F) = c_3(F)$. All the displayed complement and row identities now follow, and each component of G has zero slack by superadditivity and nonnegativity.

The bound $a \ge k + 2 - 2E$ gives $x = k - a \le 2E - 2$. Conversely, $V(F)$ minus a maximum independent set is a vertex cover of cardinality $2k + 1 - a$. For every split of this cardinality, partition that cover into parts of the two prescribed sizes and give each edge to the first part if it meets that part, and to the second otherwise. The matching numbers are bounded by the two cover-part sizes. Thus $c_0(F) \le 2k + 1 - a$, giving $x \ge E - 1$.

Exact second-row restoration gives $c_2(G) = c_2(F) - a = g$, independent of the chosen S. Since $c_2(F) \le k$, we have $g \le x$. Independent deletion gives $e(G) = 4x + 2$ and (2) gives $\nu(G) = x + E$. Its zero slack and vanishing rows from three onward imply $c_1(G) = 3x + 2 - E - g$. Finally $c_1(G) \le \nu(G) = x + E$ gives $g \ge 2x + 2 - 2E$. These are all the asserted bounds. □

Depends on. Lemma 45, Lemma 61, Lemma 72, Lemma 75, Lemma 112, Lemma 113, Lemma 147, Lemma 161.

Lemma 208 (The order of an odd regular equimatchable graph). *Let $D \ge 6$ be an even integer and G a finite connected simple D-regular graph of odd order. A matching is a set of edges with pairwise disjoint endpoints; it is maximal if no edge can be added and maximum if its cardinality is largest. Suppose G is equimatchable, meaning every maximal matching is maximum. Let $\alpha(G)$ denote the largest cardinality of an independent vertex set, and let $\nu(G)$ denote the largest matching cardinality. Then $|V(G)| \le 2D + 1$ and $\nu(G) \le D$. In particular, for $D = 6$ the order is at most 13 and the matching cardinality at most 6.*

More precisely, define F_D by taking the complete bipartite graph $K_{D,D}$, deleting the edges of a matching of size $D/2$, and adjoining a new vertex adjacent to all D endpoints of the deleted matching. If $\alpha(G) \ge 3$, then G is isomorphic to F_D, and $|V(G)| = 2D + 1$, $\alpha(G) = D$, $\nu(G) = D$. The graph is factor-critical: deleting any vertex leaves a perfect matching. For every maximum independent set I, every v in I, and every perfect matching Z of G-v, let W be the Z-mates of I minus$\{v\}$, and M_2 the edges of Z disjoint from I. Then M_2 consists of exactly one edge uu'. Its endpoints can be named so that u is the unique vertex outside I union W with at least two neighbors in W and u' has no neighbor in W. The two sets I and W union $\{u'\}$ form an equal independent bipartition of G-u.

Conversely F_D is connected, simple, D-regular, factor-critical and equimatchable, with the displayed order, independence number and matching cardinality. All orders, degrees, independence and matching cardinalities and parameters are integers in the specified domains. Vertex deletion is induced.

Proof. Put $\alpha = \alpha(G)$. Suppose first that $\alpha(G) \geq 3$. By Lemma 170 the connected regular equimatchable odd graph G is factor-critical: the other alternatives in that fact have even order. Lemma 67 supplies 2-connectivity. Fix any maximum independent set I, vertex v in I, and perfect matching Z of G-v. Use Lemma 77 for this choice. It gives independence of W with $|W| = \alpha - 1$ and maximum cardinality of M_2 as a matching of G-I. Every vertex outside I union W is an endpoint of M_2. There is exactly one such endpoint u with at least two W-neighbors. There is at most one by Lemma 77. If there were none, Lemma 127 would give a partition I, X, W_0, with $W0 = W$ minus{w0}, satisfying every hypothesis of Lemma 98: $\alpha \geq D$, independent W_0 of size alpha-2 anticomplete to X, connected odd X of independence at most two, exact boundary 2D, and the three specified shapes when X is 2-connected. That fact excludes the partition. Thus the endpoint exists and is unique. Denote its M2-mate by u′. The augmenting-path observation in Lemma 77 gives $N_G(u')$ intersect W empty: if u′ had a W-neighbor, choose a different W-neighbor of u and augment along the resulting three-edge path. Thus {uu′} is precisely the class M_{22} of that decomposition. All other M_2 pairs have maximum endpoint W-degree zero or one.

Retain the definitions from Lemma 77: T″ is the union of endpoint pairs with both endpoints anticomplete to W, and T_w consists of w and all endpoint pairs whose maximum W-degree is one and which are assigned to w. These odd connected blocks are disjoint and anticomplete to W minus{w}. All vertices of M_2 having no W-neighbor form a clique. In particular T″ union {u′} is a clique.

We show for each w in W that T_w is a clique and u′ is anticomplete to T_w. Choose a vertex wbar in $N_G(u)$ $intersect(Wminus\{w\})$, possible because u has at least two W-neighbors. For any pair xy of M_2 assigned to w with y adjacent to w, the vertex x is not adjacent to u′. Otherwise the simple path wbar, u, u′, x, y, w would alternate between nonmatching and matching edges, with matching edges uu′ and xy. Replacing these two by wbar-u, u′-x and y-w increases M_2 by one edge in G-I, a contradiction. If a pair assigned to w had exactly one endpoint y adjacent to w, its other endpoint x would consequently be nonadjacent both to w and to u′. The three vertices w, x, u′ would be independent. All three are anticomplete to W minus{w}: the first because W is independent, the second by assignment of its pair, and the third because u′ has no W-neighbor. Adjoining W minus{w} of size alpha-2 would give an independent set of size $\alpha + 1$ in G, a contradiction. Therefore both endpoints of every pair assigned to w are adjacent to w. Applying the preceding alternating-path exclusion in both orientations shows u′ adjacent to neither endpoint, and u′ also misses w. Hence u′ is anticomplete to T_w. If T_w contained two nonadjacent vertices, they together with u′ and W minus{w} would again make an independent set of size $\alpha + 1$. Thus T_w is a clique, including the singleton case.

Next T″ is anticomplete to every T_w. This holds for the root w by definition. If there were an edge zy with z in T″ and y in a nontrivial T_w, choose wbar as above. The sets $C0 = T''$ union {u′} and T_w are disjoint odd cliques. Their union has a perfect matching containing zy, obtained by perfectly matching each even clique after deleting the endpoint of zy. Adjoin the edge wbar-u. These edges replace exactly the M_2 edges in T″, in T_w minus{w}, and the edge uu′, and cover in addition the two W-vertices w, wbar. All remaining M_2 edges have both endpoints outside these replaced sets and can be retained. The resulting matching of G-I is larger than M_2 by one edge, a contradiction. Thus there is no such edge.

Put $I' = W$ union {u′}. It is independent of size α and disjoint from I, and u lies outside both. The matching M_2 minus{uu′} perfectly matches the graph $R = G - (IunionI'union\{u\})$. We check explicitly the additional neighborhood hypothesis of Lemma 67. Each vertex of R lies either in T″ or in T_w minus{w} for some w. A vertex in T″ has no W-neighbor and has exactly the I′-neighbor u′, since T″ union {u′} is a clique. A vertex in T_w minus{w} has exactly the one W-neighbor w and no edge to u′, as just proved. Consequently every vertex of R has exactly one I′-neighbor, in particular at most one.

If T″ is nonempty, then $C0 = T''$ union {u′} is an odd clique of order at least 3, since T″ is an even union of matching pairs. It meets I′ exactly in u′. It is disjoint from I union {u}. All its external neighbors lie in I union {u}: neither u′ nor T″ has a W-neighbor, and both are anticomplete to all the remaining paired blocks by the preceding arguments. Lemma 67 therefore forbids C_0. Hence T″ is empty.

If some T_w is nontrivial, it is now an odd clique of order at least 3 meeting I′ exactly in w. It is anticomplete to u′, and is anticomplete to every other T'_w because distinct complete blocks are anticomplete by Lemma 77. There are no remaining T″ vertices. Thus its external neighbors all lie in I union {u}. The same Lemma 67 hypotheses hold with the already checked I, I′, u, R, so this odd clique is also forbidden. Hence every T_w is a singleton.

There are consequently no M_2 edges except uu′. The vertex partition is $V(G) = I$ disjoint union W disjoint union {u, u′}, and deleting u leaves the equal independent sets I and $I' = W$ union {u′}. Lemma 8 applies exactly to this connected D-regular equimatchable graph and this equal bipartition. It gives $|I| = |I'| = D$

and G isomorphic to F_D with apex u. It also gives order $2D+1$ and matching number D, proving the necessity and all assertions about the arbitrary chosen decomposition.

Conversely Lemma 8 proves that the stated F_D is finite, simple, connected, D-regular, equimatchable, factor-critical, has $2D+1$ vertices and matching number D. One of its D-element bipartition parts is independent even after the apex is added, so its independence number is at least D. For any independent set A, degree counting gives $D|A| \leq D(2D+1-|A|)$, because all its incident edges go to its complement. Hence $2|A| \leq 2D+1$ and $|A| \leq D$. Thus its independence number is exactly $D \geq 3$, which verifies all hypotheses required for the converse.

It remains to prove the order bound when $\alpha(G) \leq 2$. If G is complete, its common degree gives $|V(G)| = D+1$. Otherwise choose distinct nonadjacent x, y. Every vertex outside{x, y} neighbors x or y, since an omitted vertex would form an independent triple. Neither open neighborhood contains x or y, so $|V(G)| = 2 + |N_G(x)$ union $N_G(y)| \leq 2+2D$. Since the order is odd, $|V(G)| \leq 2D+1$. In all cases every matching has at most $floor(|V(G)|/2) \leq D$ edges. This proves the order and matching bounds without an independence restriction. Substituting $D = 6$ gives order at most 13 and matching cardinality at most 6. □

Remark. A maximum independent set and a perfect matching off one of its vertices leave a unique matching endpoint seeing several partners. Every other pair is forced into an odd clique and then excluded, leaving a balanced bipartite graph and one apex. When independence is at most two, a direct neighborhood count gives the same order ∩.

Depends on. Lemma 8, Lemma 67, Lemma 77, Lemma 98, Lemma 127, Lemma 170.

Lemma 209 (Independent cuts and absorption in a graph of defect one). *Use the finite integer certificate costs $c_i(G)$, matching number $\nu(G)$, independence number $\alpha(G)$, edge count $e(G)$, maximum degree $\Delta(G)$, excess $ex(G) = c_0(G) - \nu(G)$, and positive-level slack $ps(G) = e(G) - \nu(G) - \sum_{i\geq 1} c_i(G)$ of Lemma 23. Thus $c_0(G)$ is the least nonnegative integer total l such that every nonnegative integer pair $x+y=l$ permits a partition of $E(G)$ into parts with matching numbers at most x, y. Put $gap(G) = ex(G) - ps(G)$. All graphs are finite and simple, and vertex deletion is induced.*

Let $D \geq 5$ be an odd integer. Let Z be a nonempty factor-critical graph, meaning that Z-z has a perfect matching for every vertex z. Suppose $\Delta(Z) \leq D$ and

$\sum_{(} z \in V(Z))(D - deg_Z(z)) = 1.$

Put $k = \nu(Z)$, $s = ps(Z)$, and $E = ex(Z)$. There is a unique vertex v of degree D-1, every other vertex has degree D, and

$|V(Z)| = 2k+1 > D$, $k \geq 1$, $s \geq 1$, $E \geq 0$.

For every nonempty maximal independent set S of Z, define

$a = |S|$, $t = 1$ *if v belongs to S and* $t = 0$ *otherwise,* $J = Z - S$,

$\eta = \nu(J) - (k+1-a)$, $q = c_{D-1}(J)$,

$d_i = c_i(Z) - c_i(J)$ *for integers* $1 \leq i < D$,

$\rho = \sum_{i=1}^{D-1}(a - d_i)$.

Here maximal means maximal under inclusion. Then η, q, ρ are nonnegative integers, $0 \leq d_i \leq a$,

$\Delta(J) \leq D-1$, $e(J) = e(Z) - Da + t$,

$ps(J) = s + t - 1 - \eta - \rho$,

and

$k + 1 - s - E \leq c_{D-1}(Z) \leq a + q$.

In particular,

$k - a \leq s + E + q - 1$.

For all remaining conclusions, fix this same S and assume the additional local inequality $q \leq ps(J)$. Then

$k - a \le 2s + E + t - 2 - \eta - \rho$.

Under this same local inequality, let H be any finite simple graph and let I be any vertex subset with $H - I = Z$. Put $w = |I|$. If

$w \ge \max(0, a + E - s + t - \eta - \rho)$,

then every nonnegative integer pair $x + y = w + k + s - 1$ is realized by a partition of all edges of H into parts with matching numbers at most x, y. Hence

$c_0(H) \le w + k + s - 1$.

In particular, under $q \le ps(J)$, the conditions $gap(Z) \le 0$ and $w \ge a + 1$ suffice. One may take S to be a maximum independent set, in which case $a = \alpha(Z)$.

Orders, degrees, cardinalities, matching numbers, costs, indices, x, y, w, k, s, E, a, t, η, q, ρ and d_i are integers in the stated nonnegative or positive domains. The gap is a signed integer.

Proof. Apply Lemma 135 with graph $K = Z$ and total defect $d = 1$. Its unconditional conclusions apply because $D \ge 5$ and $0 < 1 < D$. They give $|V(Z)| = 2k + 1 > D$, $k \ge 1$, $s \ge 1$,

$e(Z) = Dk + (D - 1)/2$,

$c_i(Z) = 0$ for $i \ge D$,

and $c_i(Z) + c_{D-i}(Z) \le 2k + 1$ for every integer $1 \le i < D$. No additional condition from that fact's credit construction is used. The nonnegative integer degree deficits sum to one, so precisely one vertex v has deficit one and all the others have deficit zero. These are the asserted degrees. Nonnegativity of E follows from Lemma 23.

The slack definition and the zero rows give

$\sum_{i=1}^{D-1} c_i(Z) = (D - 1)k + (D - 1)/2 - s$.

Because D is odd, indices 2,..., D-2 partition into $(D - 3)/2$ disjoint pairs {i, D-i}. Their paired bounds imply

$\sum_{i=2}^{D-2} c_i(Z) \le (D - 3)k + (D - 3)/2$.

Subtracting this inequality from the preceding equality yields

$c_1(Z) + c_{D-1}(Z) \ge 2k + 1 - s$.

The row monotonicity in Lemma 23 gives $c_1(Z) \le c_0(Z) = k + E$. Consequently

$c_{D-1}(Z) \ge k + 1 - s - E$.

Now fix any nonempty maximal independent set S with the parameters in the statement. Choose a vertex z in S and a perfect matching of Z-z. It has k edges. Removing the other a-1 vertices of S destroys at most a-1 matching edges. The remaining matching lies in $J = Z - S$ and has at least $k + 1 - a$ edges. Thus $\eta \ge 0$. This argument does not require S to have maximum cardinality.

Every vertex outside S has a neighbor in S, because a vertex without such a neighbor could be added to S. Thus $\Delta(J) \le D - 1$. Since S is independent, summing its vertex degrees counts each edge incident with S exactly once. This count is Da-t, proving $e(J) = e(Z) - Da + t$. The restriction and extension inequalities of Lemma 72 give $0 \le d_i \le a$ at each index. In particular ρ is a nonnegative integer. The nonnegative-slack and zero-tail conclusions of Lemma 61 give $ps(J) \ge 0$ and $c_i(J) = 0$ for $i \ge D$.

Subtracting the slack formulas for Z and J, with

$\nu(J) = k + 1 - a + \eta$

and the edge identity just proved, gives

$ps(J) = s - (D - 1)a + t - 1 - \eta + \sum_{i=1}^{D-1} d_i$

$= s + t - 1 - \eta - \rho$.

At index D-1, the same extension inequality gives

$c_{D-1}(Z) \le a + c_{D-1}(J) = a + q$.

Combining this with the lower bound already proved yields $k - a \le s + E + q - 1$.

Assume next that $q \le ps(J)$. Substituting the exact slack identity into the last inequality gives

$k - a \le s + E + (s + t - 1 - \eta - \rho) - 1$

$= 2s + E + t - 2 - \eta - \rho$.

All quantities here refer to the same fixed S and J.

We verify the original-graph absorption independently of this cost comparison. Set $\delta = s - 1$, a nonnegative integer, and $l = k + \delta = k + s - 1$. Let $Y_l(Z)$ be the integers z in [0,l] for which $E(Z)$ has a partition with matching bounds z, l-z. By the local order construction of Lemma 48, the equality $|V(Z)| = 2k + 1$ and $\nu(Z) = k$ imply that $0, l$ belong to $Y_l(Z)$, and every consecutive run missing from this set has length at most

$\max(0, k - 3\delta - 1) = \max(0, k - 3s + 2)$.

The rank inequality gives

$k - 3s + 2 \le a + E - s + t - \eta - \rho$.

Under the asserted bound on w, every such missing run therefore has length at most w. To see that every integer x in $[0, w + l]$ has the form $h + z$ with $0 \le h \le w$ and z in $Y_l(Z)$, list the elements of $Y_l(Z)$ increasingly. Consecutive elements differ by at most $w + 1$. The integer intervals of length w starting at these elements have no missing integer between consecutive intervals. Their union starts at zero and ends at $w + l$. This proves the required representation. The endpoint-only and singleton lists obey the same argument.

Fix $x + y = w + l$ and such a representation $x = h + z$. Choose any h-element subset I_R of I and put $I_B = I$ minus I_R. Choose a two-part partition of $E(Z)$ realizing z, l-z. Retain these colors on $E(Z)$. Color an edge of H meeting I red if it meets I_R, and blue otherwise. Every blue edge meeting I has an endpoint in I_B. Each edge with both endpoints in I receives exactly one color by this priority rule. As $H - I = Z$ is induced, these assignments account for every H-edge.

A red matching restricts inside Z to at most z edges. Each of its remaining edges meets I_R, and those matching edges use distinct vertices of I_R, so there are at most h of them. Its total cardinality is at most $z + h = x$. A blue matching has at most l-z edges inside Z and at most w-h other edges, so its total cardinality is at most $l - z + w - h = y$. Thus every required allocation is feasible on H, including every edge internal to I or between I and Z. The definition of c_0 now proves $c_0(H) \le w + k + s - 1$.

Finally $gap(Z) \le 0$ means $E \le s$ by Lemma 23. Since $t \le 1$ and $\eta, \rho \ge 0$, it implies

$a + E - s + t - \eta - \rho \le a + 1$.

Hence $w \ge a + 1$ is sufficient for the stated threshold. Every maximum independent set is maximal under inclusion, so the specialization $a = \alpha(Z)$ follows. □

Remark. An independent cut records exactly how much of its permitted row loss is unused. The complementary rows force a lower bound on the last positive row; comparing that bound with the same cut gives a rank bound. That rank bound makes the missing matching allocations short enough for the omitted vertices to fill.

Depends on. Lemma 23, Lemma 48, Lemma 61, Lemma 72, Lemma 135.

Lemma 210 (Cover bounds for exposed vertices at every degree). *All graphs are finite and simple. Write $\nu(Q)$, $\tau(Q)$, $\alpha(Q)$ and $\Delta(Q)$ for the maximum matching cardinality, minimum vertex-cover cardinality, maximum independent-set cardinality and maximum degree of a graph Q. A matching has pairwise disjoint endpoints, and $V(M)$ is its set of covered vertices. A graph is factor-critical if it is nonempty and deleting any vertex leaves a perfect matching; the empty graph has its empty perfect matching.*

Let $D \ge 3$ be an integer. For every Q with $\Delta(Q) \le D$ put $S_D(Q) = \{vinV(Q) : deg_Q(v) = D\}$, and let $\theta_D(Q)$ be the minimum of nu((V(Q), A)) over edge sets $A \subseteq E(Q)$ incident with every vertex of $S_D(Q)$. The empty edge set is allowed when $S_D(Q)$ is empty. This definition uses degrees internal to Q. If Q has odd order $2\mu + 1$, where $\mu \ge 0$ is an integer, then

$\theta_D(Q) \le \max(0, \mu - 1)$.

Consequently every connected factor-critical T of order $2\mu+1$ satisfying $\theta_D(T)=\alpha(T)$ has $\alpha(T)\le\mu-1$.

Let H have $\Delta(H)\le D$, set $S=S_D(H)$ and $\alpha_S=\alpha(H[S])$, and let $\rho=\min_M|S \text{ minus } V(M)|$ over all matchings of H. Let $q_S(H-Y)$ count the odd connected components of H-Y wholly contained in S. Choose any inclusionwise maximal Y among the maximizers of $q_S(H-Y)-|Y|$; such a choice exists. Let calR be the components counted by $q_S(H-Y)$. For T in calR write $|V(T)|=2\mu_T+1$, let b count those T with $\theta_D(T)=\alpha(T)$, and put

$\Lambda=\sum_(\ T\in calR, |V(T)|>1)(\mu_T-\alpha(T))$.

Then

$\Lambda\ge b\ge\max(0,\theta_D(H)+\rho-\alpha_S)$.

Every vertex cover C of H satisfies

$|C|-\nu(H)\ge\rho+\Lambda$

and

$2e(H[C])\ge D(\rho+2b)\ge D(\rho+2max(0,\theta_D(H)+\rho-\alpha_S))$.

In particular

$\tau(H)-\nu(H)\ge\rho+\max(0,\theta_D(H)+\rho-\alpha_S)$,

$\theta_D(H)+2\rho\le\tau(H)-\nu(H)+\alpha_S$.

If $\theta_D(H)=\alpha_S$, then $\tau(H)-\nu(H)\ge2\rho$ and every vertex cover C satisfies $2e(H[C])\ge3D\ \rho$.

Here vertex deletion and bracketed subgraphs are induced, and e counts edges. All orders, degrees, component counts, matching sizes, cover sizes, independence sizes and extrema have integer domains. Empty H, empty S, isolated vertices, $\rho=0$ and empty component families are permitted.

Proof. All defining extrema are attained because their feasible families are finite and nonempty. In particular the entire edge set is an incidence cover. A set Z whose closed neighborhood contains $S_D(Q)$ supplies an incidence cover by taking every edge with an endpoint in Z: a required vertex outside Z has a neighbor in Z, and a required vertex in Z has positive degree D. Its matching number is at most $|Z|$, since Z meets every selected edge.

We first prove the odd-order bound. When $D\ge5$, the first assertion of Lemma 165 gives $\theta_D(Q)\le\max(0,\mu-floor((D-1)/2))$, which implies the required bound.

Suppose $D=3$. The cubic domination theorem Lemma 3 supplies the finite-order hypothesis of Lemma 11 with $N=|V(Q)|$. Apply the latter theorem with required set $S_3(Q)$. Each required vertex has weight three; every optional vertex has weight zero or two. Thus there is a set Z whose closed neighborhood contains $S_3(Q)$ and whose size is at most $3(2\mu+1)/8$. If $\mu\ge2$ this last number is strictly less than μ, so integrality gives $|Z|\le\mu-1$. If $\mu=0$ or 1, simplicity and $|V(Q)|\le3$ imply $S_3(Q)$ is empty, and the empty incidence cover suffices.

Suppose $D=4$, and put $n=2\mu+1$. For $n\le3$ there are no degree-four vertices. For n in {5,7,9,11,13}, choose a set Z of size mu-1 uniformly among all such subsets of $V(Q)$. Every degree-four vertex has a closed neighborhood of size five, so the expected number of degree-four vertices not in the closed neighborhood of Z is at most

$n\ binom(n-5,\mu-1)/binom(n,\mu-1)$.

For $n=5,7,9,11,13$ these upper bounds are respectively $0,1/3,3/7,1/2,56/99$, all less than one; a binomial coefficient with lower index larger than its nonnegative upper index is zero. A nonnegative integer random variable with expectation less than one is zero for some choice. That Z therefore dominates the degree-four vertices.

For odd $n\ge15$ choose each vertex independently with probability $1/3$, forming Z_0. Add to Z_0 every degree-four vertex whose closed neighborhood misses Z_0, obtaining Z. Its closed neighborhood contains all degree-four vertices. A particular such vertex is added with probability $(2/3)^5$. Consequently

$E|Z|\le n/3+n(2/3)^5=113n/243<(n-1)/2=\mu$,

where the strict inequality is equivalent to $17n > 243$ and follows from $n \geq 15$. Some outcome has integer size at most mu-1. In every degree-four case the incidence cover obtained from Z has the desired matching bound.

For completeness $\theta_D(Q) \leq \alpha(Q)$ for every capped Q: choose a maximal independent set I of Q. Every vertex outside I has an I-neighbor, so I has a closed neighborhood containing all vertices and the incidence construction above has matching number at most $|I| \leq \alpha(Q)$. If nonempty odd T satisfies $\theta_D(T) = \alpha(T)$, then $\alpha(T) \geq 1$. Its odd-order bound cannot have right side zero. Thus $\alpha(T) \leq \mu - 1$, proving the stated factor-critical specialization.

Apply Lemma 107 to H with marked set S and the specified Y. Writing $q = |calR|$ and letting R be the induced union of the other components of H-Y, it gives

$q = |Y| + \rho$,

$\nu(H) = |Y| + \sum_{T \in calR} \mu_T + \nu(R)$, (1)

and factor-criticality of every T in calR.

If T in calR is nontrivial, it is connected and has an edge. Choose a maximum independent set I of T and a vertex z outside I. In a perfect matching of T-z, every I-vertex is paired with a different vertex outside I. Hence $2\alpha(T) \leq |V(T)| - 1$ and $\alpha(T) \leq \mu_T$. Thus Λ has nonnegative summands. A singleton T has $\theta_D(T) = 0$ and $\alpha(T) = 1$, so is not counted by b. Each component counted by b satisfies the odd-order specialization and contributes at least one to Λ. This proves $\Lambda \geq b$.

For each component Q of H-Y put $S_Q = S$ intersect $V(Q)$, and put $A_0 = \sum_Q \alpha(H[S_Q])$. Independent sets from the mutually nonadjacent components unite to an independent subset of H[S], so $A_0 \leq \alpha_S$. In a component T in calR choose an incidence cover attaining $\theta_D(T)$. Its matching number is $\alpha(T)$ if counted by b and at most alpha(T)-1 otherwise, by $\theta_D(T) \leq \alpha(T)$ and integrality. For a component Q not in calR choose a maximal independent set I_Q in the graph induced by its internal degree-D vertices, and select every internal edge meeting I_Q. This covers every internal degree-D vertex, and its matching number is at most $|I_Q| \leq \alpha(H[S_Q])$. The latter comparison holds because every internal degree-D vertex also belongs to S. If there are no internal degree-D vertices take an empty selection.

Add every edge of H meeting Y. The union covers S: a marked vertex in Y has positive degree; a marked vertex outside Y with a Y-neighbor is covered by that edge; and every other marked vertex has internal degree D in its component. Any matching of the union uses at most $|Y|$ edges meeting Y and otherwise uses the separate internal selections. As $S_T = V(T)$ for T in calR, this gives

$\theta_D(H) \leq |Y| + A_0 - (q - b) = A_0 - \rho + b \leq \alpha_S - \rho + b$.

Combine this with $b \geq 0$ to obtain $b \geq \max(0, \theta_D(H) + \rho - \alpha_S)$.

Fix a vertex cover C of H and put $C_Y = C$ intersect Y. For each nontrivial T in calR, its vertices outside C are independent, and hence

$|C \text{ intersect } V(T)| \geq 2\mu_T + 1 - \alpha(T) = \mu_T + 1 + (\mu_T - \alpha(T))$.

Let t_0 count singleton members of calR whose vertex is outside C. All other singleton members contribute one vertex. Summing over calR yields

$\sum_T |C \text{ intersect } V(T)| \geq \sum_T \mu_T + q + \Lambda - t_0$. (2)

Every omitted singleton has ambient degree D and all its neighbors in Y. Since C covers every edge, all these neighbors lie in C_Y. Degree counting gives $D\ t_0 \leq D|C_Y|$ and $t_0 \leq |C_Y|$. Also C intersect $V(R)$ is a cover of R and has size at least $\nu(R)$. Add these bounds to (2), and use (1) and $q = |Y| + \rho$, to get $|C| - \nu(H) \geq \rho + \Lambda$. Minimizing over C proves the cover-size inequality, and rearranging after $\Lambda \geq b$ gives $\theta_D(H) + 2\rho \leq \tau(H) - \nu(H) + \alpha_S$.

For the edge inequality write $C_T = C$ intersect $V(T)$, $O_T = V(T)$ minus C, $e_T = e(H[C_T])$, and let β_T count the edges between $V(T)$ and Y. All vertices of T have ambient degree D, there are no edges from T to other components of H-Y, and O_T is independent. Subtracting the degree sums on C_T and O_T gives

$D(|C_T| - |O_T|) \leq 2e_T + \beta_T$. (3)

For every nontrivial T, $|O_T| \le \alpha(T) \le \mu_T$, so $|C_T| - |O_T| \ge 1$. If T is counted by b, $\alpha(T) \le \mu_T - 1$ improves this to at least three. For a singleton, $e_T = 0$ and $\beta_T = D$, so $2e_T + \beta_T \ge D$ whether or not it lies in C. Sum (3) and the singleton bounds to obtain

$D(q + 2b) \le 2sum_T\, e_T + \sum_T \beta_T$.

The first sum is at most $e(H[C])$; the boundary sum is at most the degree sum on Y, hence at most $D|Y|$. Substitute $q = |Y| + \rho$ to conclude $2e(H[C]) \ge D(\rho + 2b)$. The displayed specializations follow by setting $\theta_D(H) = \alpha_S$. All constructions allow empty sets and zero sums, so the boundary cases stated in the theorem are included. □

Remark. An odd graph admits an incidence cover with a spare matching rank. At a marked barrier, each component without that saving must instead pay an independence deficit. Counting a vertex cover across the barrier converts those deficits into cover size and internal edges.

Depends on. Lemma 3, Lemma 11, Lemma 107, Lemma 165.

Lemma 211 (Matching number forced by a dense target forest). *Let s, $t \ge 1$ be integers and let F_1, F_2 be nonempty vertex-disjoint unions of s, t positive-edge stars, respectively. Their component edge sizes are positive integers. Copies are ordinary subgraphs with vertex-disjoint target components, with F_1 prescribed red and F_2 prescribed blue. Write e for edge count and ν for maximum matching cardinality, and put $E = e(F_2)$ and $M = s + t - 1$.*

If $E \ge s + t$ and $2E > 3t - 2$, then every finite simple graph G arrowing (F_1, F_2) satisfies $\nu(G) \ge M$. Equivalently, every finite simple graph with $\nu(G) \le M - 1$ has a red-blue edge coloring containing neither the entire red F_1 nor the entire blue F_2. The assertion remains valid after exchanging both entire targets and their prescribed colors, with the same hypotheses in that orientation.

No order relation between s and t, host edge bound, degree bound, connectivity condition, or host component restriction is imposed. Every unit component is retained. All target component counts, sizes, graph orders, matching numbers and edge counts are nonnegative integers, with the explicitly stated positive domains.

Proof. Suppose G arrows the stated pair and $\nu(G) \le M - 1$. Let $b = M - 1 - \nu(G)$, a nonnegative integer, and adjoin b disjoint single-edge components on new vertices to obtain G'. A matching in a disjoint union restricts to each component, while the union of component maximum matchings attains the sum. Hence $\nu(G') = M - 1 = s + t - 2$. Every coloring of G' restricts to one of G, so G' arrows the same entire pair. Put $N = \nu(G') = s + t - 2$.

Order the positive edge sizes within each target nonincreasingly, which changes neither graph. Apply Lemma 89 at matching deficit one. It supplies a half-order attaining separator S of G', meaning

$N = |S| + \sum_C floor(|V(C)|/2)$,

where C ranges over all connected components of G'-S. Put $h = |S|$. The same fact gives $h \le \min(s, t) - 2$. Thus s, $t \ge 2$. It also gives

$N \ge T(F_2) + floor((s - 1 + h)/2)$,

where $T(F_2)$ is the sum of $\lceil a/2 \rceil$ over the positive edge sizes a of its t components, so $T(F_2) \ge t$.

Lemma 153 makes every complementary component's matching number equal to its half-order floor. Put

$W = \sum_C \nu(C) = N - h$.

The preceding capacity inequality and $h \le s - 2$ imply

$W \ge T(F_2) + floor((s - 1 - h)/2) \ge t$.

In particular there is a nonempty collection of these finitely many components whose matching numbers sum to at least t-1.

Choose such a collection inclusionwise minimal, let H be its union, let l be its number of components, let $\mu = \nu(H)$ be their matching-number sum, and let o be the number of its odd-order components. This choice exists in a finite nonempty family, and $l \ge 1$. At deficit one, Lemma 153 applied to H gives both an entire copy of F_2 in H and

$E \le \mu + o - 1 \le \mu + l - 1$. (1)

Selecting one edge from each of the t positive stars of that copy yields a matching, so $\mu \ge t$. Also $\mu \le W \le N$.

If $l = 1$, (1) gives $E \le \mu \le N$, contradicting $E \ge s + t = N + 2$. Therefore $l \ge 2$. Minimality says that removing any chosen component leaves matching-number sum at most t-2. Add these l inequalities. Each component weight occurs exactly l-1 times, so

$(l-1)\mu \le l(t-2)$. (2)

Since $\mu \ge t$, (2) implies $2l \le t$.

If $l = 2$, (1) gives $E \le \mu + 1 \le N + 1$, again contradicting $E \ge N + 2$. Suppose $l \ge 3$. Then $3 \le l \le t/2$, and consequently

$(3t/2)(l-1) - l(t-2) - l(l-1)$

$= (l-3)(t/2 - l) \ge 0.$

Combining this inequality with (2), and dividing by $l - 1 > 0$, yields $\mu + l \le 3t/2$. Now (1) gives $E \le 3t/2 - 1$, contradicting the strict hypothesis $2E > 3t - 2$. Every possible positive integer l has been excluded.

Thus no such arrowing G exists. Negating the universal-coloring definition of arrowing gives the asserted avoiding coloring whenever $\nu(G) \le M - 1$. Exchanging red with blue and the two whole targets repeats the argument in the other orientation. Padding the host and ordering the target lists never removes a target component or changes a prescribed copy. □

Remark. Localization bounds the edge total of the entire dense target using a smallest component union that reaches its matching threshold. One or two components violate the total host matching budget; three or more violate the strict edge-density threshold.

Depends on. Lemma 89, Lemma 153.

Lemma 212 (Local covers at equality for a matching family). *All graphs are finite and simple. Let calA be a finite nonempty set of labels. For each a in calA let p_a, $t_a \ge 1$ be integers and let J_a be a star forest with the full nonincreasing positive integer edge-size list m^a_1,..., $m^a_{t_a}$. Let K_2 be a single edge. Copies are ordinary subgraphs with vertex-disjoint target components; in each pair (p_aK2, J_a), the matching is prescribed red and the full forest blue. Put*

$N = \max_a(p_a + t_a - 1)$,

$\lambda_i = max\{m^a_j : a \in calA, 1 \le b \le p_a, 1 \le j \le t_a, b + j = i + 1\}$

for every integer $1 \le i \le N$, and $P = \sum_{i=1}^{N} \lambda_i$. Suppose $D = \lambda_1 \ge 4$ is even, and let R be the number of initial entries equal to D.

Let H have maximum degree at most D. For every matching W of H, suppose H-W arrows every pair (p_aK2, J_a); H-W deletes only the matching edges. Write e, ν, τ for edge count, maximum matching cardinality and minimum vertex-cover cardinality. Put $\sigma = e(H) - \nu(H) - P$.

Suppose the family contains (pK2,J), where J has t positive components, exactly $h \ge 1$ of size D, and

$p + h - 1 = R$, $g := N - (p + t - 1) \ge 0$, $\sigma + g = hD/2$.

All the displayed parameters are integers.

The minimum number of ambient degree-D vertices missed by a matching of H is R. There are exactly R whole odd D-regular connected components C_1,..., C_R, and each is factor-critical, meaning every one-vertex deletion has a perfect matching.

For each j write $|V(C_j)| = 2\mu_j + 1$. For every vertex x of C_j and every perfect matching W_j of $C_j - x$, put $U = (C_j - W_j) - x$. Then

$\tau(U) = \mu_j$, $\nu(U - N_{C_j}(x)) = \mu_j - D/2$.

In particular $(C_j - N_{C_j}[x]) - W_j$ has a perfect matching, where $N_{C_j}(x)$ is the open neighborhood and $N_{C_j}[x]$ includes x.

For each j put

`h_j=max{mu_j-|Z|:` *Z is an inclusionwise maximal matching of* `C_j}`.

Then $\sum_{j=1}^{R} h_j \le \sigma$. *If* $D = 4$ *and k*5 *counts all whole* K_5 *components of H, then* $R - k5 \le \sigma$.

Empty complementary graphs, empty lower parts of J, unit components and repeated sizes are permitted. Vertex deletion is induced; a perfect matching of the empty graph is empty. No natural-suffix, positive matching-deficit or edge-budget hypothesis is imposed.

Proof. The complete matching-family envelope has the domains of Lemma 21. That fact gives its positivity and monotonicity. In particular $1 \le R \le N$, $\lambda_i = D$ for $i \le R$, $\lambda_i \le D - 1$ for $i > R$, and

$P \le DN - (N - R)$. (1)

The chosen member satisfies $1 \le h \le R$. Write $t_0 = t - h$. Its budget and plateau equalities give

$N - R = t_0 + g$. (2)

We first establish the asserted component structure. Let S be the vertices of H-degree D and $\rho = \min_M |S \text{ minus } V(M)|$, over all matchings M of H. Apply Lemma 158 to H with this S and D. It supplies one maximum matching W attaining ρ whose neighborhood-loss estimate holds simultaneously for every eligible subset of $I = S$ minus $V(W)$. Put $Q = H - W$ and $T = Q - I$. A maximum matching is maximal, so I is independent: an edge between uncovered vertices could be added. Exactly I has Q-degree D, since every other vertex of ambient degree D loses a matching edge. Hence

$e(Q) = P + \sigma$, $e(T) = P + \sigma - D\,\rho$.

The family arrows Q, so Lemma 21 gives $\sigma \ge 0$.

We have $\rho \ge R$. Otherwise color the edges meeting any $\min(p - 1, \rho)$ vertices of I red and all other edges blue. The red graph has a cover of size at most p-1. The number of remaining possible blue D-star centers is $\max(\rho - p + 1, 0) < h$, so this coloring avoids the selected full pair, a contradiction.

Put $z = \rho - R \ge 0$. Choose any subset A of I of size $h + z = \rho - p + 1$, color edges meeting I minus A red and all other edges blue. A forced blue J-copy has its h D-star centers in a set $B \subseteq$ A. Each such star uses the entire Q-neighborhood of its center, and these neighborhoods are pairwise disjoint. Of its t_0 other components, at most z meet A minus B. The others avoid all I and all neighbors of B. Taking one edge from each of these components gives

$\nu(T - N_Q(B)) \ge \max(t_0 - z, 0)$.

If $\rho \le N$, delete N-rho maximum-degree vertices from T. The actual-set degree bound of Lemma 21 forces at least $\sum_{i=\rho+1}^{N} \lambda_i$ edges to be removed, leaving at most

$\kappa := \sigma - D\,\rho + \sum_{i=1}^{\rho} \lambda_i \le \sigma - z$

edges. Covering those edges and using the deleted vertices gives

$\tau(T) \le N - \rho + \sigma - z$.

Together with (2) this implies

$\tau(T) - \nu(T - N_Q(B)) \le g + \sigma - z - \max(z - t_0, 0) \le g + \sigma - z$.

If $\rho > N$, instead use $\tau(T) \le e(T)$, (1), and nonnegativity of matching size to get

$\tau(T) - \nu(T - N_Q(B)) \le \sigma - z - (D - 1)(\rho - N) \le g + \sigma - z$.

Both bounds include empty deletion sequences.

The simultaneous estimate in Lemma 158 applies to this B and the same W, giving a lower bound $hD/2$. Since $g + \sigma = hD/2$, the preceding upper bound forces $z = 0$, so $\rho = R$. Now A has exactly h vertices and $B = A$. If any root in I were outside all whole odd D-regular factor-critical components, choose A to include it. The even-D improvement in that same fact would give a lower bound $hD/2 + 1$, a contradiction. Thus every root lies in such a component.

A whole odd D-regular factor-critical component of order $2\mu + 1$ has matching number μ. The restriction of a maximum matching of H to each whole component is maximum there: otherwise replacing it by a larger internal matching enlarges the ambient matching. Therefore every such component contributes exactly one root to I. Since all roots lie in these components, there are exactly R of them, denoted C_1,..., C_R. Put $H_0 = H - \bigcup_j V(C_j)$ and let W_0 be the restriction of W to H_0. It is maximum there and covers every ambient degree-D vertex there. In particular no other whole odd D-regular component can exist: W_0 would restrict to a perfect matching on its odd vertex set.

Fix this W_0. In each C_j now choose an arbitrary root x_j and arbitrary perfect matching W_j of $C_j - x_j$. Factor-criticality permits all these choices. Each W_j has μ_j edges and is maximum in its odd component. The union $W = W_0$ union $\bigcup_j W_j$ is again maximum in H by whole-component additivity.

Let $I = \{x_1, \ldots, x_R\}$, $Q = H - W$ and $T = Q - I$. The set I is independent, and its vertices have Q-degree D. These are exactly the Q-vertices of degree D: every other ambient degree-D vertex is covered by W and loses one edge, while all lower ambient degrees remain below D. Thus

$e(Q) = P + \sigma, e(T) = P + \sigma - DR.$ (3)

The graph Q arrows every family member by hypothesis. Every matching leaves at least one degree-D vertex uncovered in each C_j, and this W misses exactly its R roots among the degree-D vertices, consistently with the value of ρ proved above.

We need an upper bound for $\tau(T)$. Apply the degree-forcing assertion of Lemma 21 to Q. After deleting I and any further $b < N - R$ vertices, the next maximum degree is at least $\lambda_{R+b+1} > 0$. Delete N-R maximum-degree vertices from T. Each step exists; their total removed edges are at least $\sum_{i=R+1}^{N} \lambda_i = P - DR$. By (3) the final graph has at most σ edges. The deleted vertices together with one endpoint of each remaining edge cover T. Consequently, using (2) and the assumed equality,

$\tau(T) \le N - R + \sigma = t_0 + g + \sigma = t_0 + hD/2.$ (4)

This construction includes $N = R$, when no additional vertex is deleted.

Take any h-element subset B of I. Color the Q-edges meeting I minus B red and every other Q-edge blue. The red graph has a vertex cover of size $R - h = p - 1$, so contains no matching with p edges. The selected full pair therefore forces a blue J-copy. Only I has Q-degree D. Vertices in I minus B have no incident blue edge, so the h D-star centers of that copy are exactly B. Each of these stars uses its center's entire Q-neighborhood, and these neighborhoods are disjoint because they belong to distinct components of the same copy.

All t_0 remaining positive components of that same blue copy avoid B and its neighbors. They also avoid I minus B, since every vertex in a positive star has an incident edge. They lie entirely in $T - N_Q(B)$, where $N_Q(B)$ is the union of the Q-neighborhoods of B. Selecting one edge of each gives

$\nu(T - N_Q(B)) \ge t_0.$ (5)

For $t_0 = 0$, this is the empty matching. Equations (4)-(5) imply

$\tau(T) - \nu(T - N_Q(B)) \le hD/2.$ (6)

Fix a selected component C_j, with root $x = x_j$, perfect matching W_j and order $2\mu_j + 1$. Write $U_j = (C_j - W_j) - x$. It has $2\mu_j$ vertices. Its D neighbors of x have degree D-2 in U_j, and every other vertex has degree D-1: every retained vertex loses its matching edge and loses the edge to x precisely when it neighbors x.

Let A be any vertex cover of U_j and let O be its complementary independent set. Counting edges from O to A gives

$(D-1)|O| - D \le \sum_{v \in O} deg_{U_j}(v) \le \sum_{v \in A} deg_{U_j}(v) \le (D-1)|A|.$

If $|A| \le \mu_j - 1$, then $|O| - |A| \ge 2$, forcing $2(D-1) \le D$, contrary to $D \ge 4$. Hence $\tau(U_j) \ge \mu_j$. The graph $U_j - N_{C_j}(x)$ has $2\mu_j - D$ vertices, so its matching number is at most $\mu_j - D/2$. Therefore

$\tau(U_j) - \nu(U_j - N_{C_j}(x)) \ge D/2.$ (7)

The graph T is the disjoint union of $H_0 - W_0$ and all the U_j. For B chosen above, $T - N_Q(B)$ replaces exactly the U_j whose roots belong to B by $U_j - N_{C_j}(x_j)$. Matching and cover numbers add over disjoint component

unions: restricting them gives the respective bounds, and uniting optimal choices attains those bounds. Thus the difference in (6) is the sum of the h tested differences in (7), together with $\tau(U_j) - \nu(U_j)$ for each untested component and $\tau(H_0 - W_0) - \nu(H_0 - W_0)$. The latter differences are nonnegative because every vertex cover meets all edges of any matching at distinct vertices.

The h tested differences sum to at least $hD/2$, and (6) bounds the entire sum by that number. Equality holds in every tested (7). Its two nonnegative deviations $\tau(U_j) - \mu_j$ and $\mu_j - D/2 - \nu(U_j - N_{C_j}(x_j))$ therefore both vanish. Each selected component can be included in an h-set B, since $1 \le h \le R$. The roots and perfect matchings were arbitrary. This proves both asserted universal local equalities. The matching in the neighborhood-deleted residual has half its order and is consequently perfect.

For the aggregate bound, in each C_j choose a maximal matching Z_j of minimum cardinality. Such a choice exists by finiteness, and $|Z_j| = \mu_j - h_j$ with $h_j \ge 0$. Put $a_0 = \sum_j h_j$ and $Z = W_0$ union $\bigcup_j Z_j$. This is a matching of H with $|Z| = \nu(H) - a_0$. In each selected component, vertices uncovered by Z_j are independent: an edge between two could be added to Z_j. Their union I′ is independent in H because the components are whole. All its vertices have ambient degree D, and

$k := |I'| = R + 2a_0$.

There are no other uncovered degree-D vertices, because W_0 covers those in H_0.

Put $Q' = H - Z$. The hypothesis for every matching, including this possibly nonmaximum Z, gives that Q′ arrows every member. The independent set I′ retains degree D in Q′, so

$e(Q' - I') = P + \sigma + a_0 - Dk$. (8)

If $k \le N$, Lemma 21 applied to Q′ and the actual k-set I′ gives $e(Q' - I') \ge \sum_{i=k+1}^{N} \lambda_i$, since that fact also gives $\tau(Q') \ge N$. Combining with (8) and $k \ge R$ yields

$\sigma + a_0 \ge Dk - \sum_{i=1}^{k} \lambda_i \ge k - R = 2a_0$.

If $k > N$, nonnegativity of (8) and (1) instead give

$\sigma + a_0 \ge Dk - P \ge (k - R) + (D - 1)(k - N) \ge 2a_0$.

Thus $a_0 \le \sigma$ in both cases.

Finally suppose $D = 4$. Each C_j satisfies the universal root and matching hypotheses of Lemma 66 by the equalities just proved. If C_j is not K_5, that fact supplies a maximal matching of size at most $\mu_j - 1$, so $h_j \ge 1$. Every whole K_5 component of H is among the selected components, since it is odd and four-regular and no other such whole component exists. Exactly R-k5 selected components are not K_5. Consequently $R - k5 \le \sum_j h_j \le \sigma$, completing the proof. □

Remark. At equality, simultaneous neighborhood losses force every exposed top-degree vertex into a whole regular factor-critical component. The same full family then forces equality in every local cover count and charges maximal-matching losses to the global surplus.

Depends on. Lemma 21, Lemma 66, Lemma 158.

Lemma 213 (Quartic independent-cut rank bounds). *Use the finite integer certificate costs and graph quantities of Lemma 23: $c_i(G) = \max_{A+B=i} c_{A,B}(G)$, $ex(G) = c_0(G) - \nu(G)$, $ps(G) = e(G) - \nu(G) - \sum_{i\ge1} c_i(G)$, and $gap(G) = ex(G) - ps(G)$. All graphs are finite and simple, and all capacities, budgets, indices, costs, matching cardinalities, and set cardinalities are nonnegative integers. A graph is factor-critical if deleting any one vertex leaves a perfect matching.*

Let Q be a nonempty connected four-regular factor-critical graph, and put $m = \nu(Q)$, $s = ps(Q)$, and $E = ex(Q)$. Then $|V(Q)| = 2m + 1$, $c_2(Q) \le m$, $c_4(Q) = 1$, and

$c_3(Q) \ge m + 1 - s - E$.

For every maximal independent set I of Q, put $a = |I|$, $J = Q - I$, and

$\eta = \nu(J) - (m + 1 - a)$,

$\rho = \sum_{i=1}^{3} (a - c_i(Q) + c_i(J))$.

Then I is nonempty, η and ρ are nonnegative integers, $\Delta(J) \le 3$, and

$ps(J) = s - \eta - \rho \ge 0$,

$c_3(J) \le floor((s - \eta - \rho)/2)$.

Consequently, on this same cut,

$m - a \le s + E - 1 + floor((s - \eta - \rho)/2)$

$\le s + E - 1 + floor((s - \eta)/2)$.

In particular,

$m - \alpha(Q) \le s + E - 1 + \lfloor s/2 \rfloor$.

If $gap(Q) \le 0$, then

$m - \alpha(Q) \le 2s + \lfloor s/2 \rfloor - 1$.

No bound on the gap of a proper subgraph is assumed.

Proof. Factor-criticality gives a perfect matching after deleting any chosen vertex. Hence Q has odd order $2m + 1$ and maximum matching size m. Four-regularity then gives $e(Q) = 4m + 2$. Nonnegative slack and the vanishing of every row above the maximum degree follow from Lemma 61.

First prove $c_2(Q) \le m$, including the complete graph case. If Q is not the complete graph K_5, the factor-critical specialization of Lemma 112 applies: Q is connected, has degree four at every vertex, and is not K_5. It gives $c_2(Q) \le m$.

For $Q = K_5$, label the vertices 1, 2, 3, 4, 5, so $m = 2$. There are three ordered capacity pairs of total two. At capacities (2,0), retain the triangle on vertices 1, 2, 3 wholly in Q_R and put Q_B empty. This is a zero-budget certificate on that triangle. The other two vertices cover all missing edges; restoring them by the actual vertex-cover construction of Lemma 72 gives every matching-budget allocation of total two. The same construction with the bounded-degree colors exchanged treats capacities (0,2).

At capacities (1,1), the allocations (2,0) and (0,2) are obtained by placing every edge of K_5 in the exceptional part with budget two. For allocation (1,1), take the exceptional red triangle with edges {13,14,34}, the exceptional blue triangle with edges {12,15,25}, the bounded red matching {23,45}, and the bounded blue matching {24,35}. Here ij denotes the edge with endpoints i, j. These four pairwise disjoint edge sets contain all ten edges of K_5. Each exceptional triangle has matching number one, and each bounded part has maximum degree one. This certifies the remaining allocation and completes the proof that $c_2(K_5) \le 2$.

Next, fix a vertex v. A perfect matching in Q-v covers every vertex there, so in particular every vertex of degree four. Its maximum degree is at most four; the covered-degree assertion of Lemma 61 gives $c_4(Q - v) = 0$. Restoring v by Lemma 72 gives $c_4(Q) \le 1$. If $c_4(Q) = 0$, capacities (1,3) would admit an edge partition with zero exceptional budgets. Both exceptional parts would be empty. At each vertex the degree-four condition and the two capacity bounds would force degree exactly one in the first bounded part and exactly three in the second. The first part would thus be a perfect matching of the odd-order graph Q, a contradiction. Therefore $c_4(Q) = 1$.

The positive rows above four vanish. Substituting the edge count and fourth row into the definition of slack gives

$s = 3m + 1 - c_1(Q) - c_2(Q) - c_3(Q)$.

By row monotonicity from Lemma 23, $c_1(Q) \le c_0(Q) = m + E$. Combining this with $c_2(Q) \le m$ yields

$c_3(Q) \ge m + 1 - s - E$.

Fix now any maximal independent set I. Since Q is nonempty, I is nonempty. Every vertex outside I has a neighbor in I, since otherwise it could be added to I. Thus $J = Q - I$ has maximum degree at most three. Choose v in I and a perfect matching of Q-v. Deleting the remaining a-1 vertices of I removes at most a-1 matching edges, so

$\nu(J) \ge m - (a - 1) = m + 1 - a$.

This proves $\eta \geq 0$.

For $i = 1, 2, 3$, put $d_i = c_i(Q) - c_i(J)$. The restriction and vertex-set extension estimates in Lemma 72 give $0 \leq d_i \leq a$. Hence $\rho = 3a - \sum_{i=1}^{3} d_i$ is a nonnegative integer. Independence of I and four-regularity give $e(J) = e(Q) - 4a$. All rows of J above three vanish by Lemma 61. Since $\nu(J) = m + 1 - a + \eta$, subtracting the two slack expressions, using $c_4(Q) = 1$, gives

$ps(J) - s = -4a - (1 - a + \eta) + \sum_{i=1}^{3} d_i + 1$

$= -\eta - \rho.$

Nonnegative slack therefore proves $ps(J) = s - \eta - \rho \geq 0$.

Apply Lemma 161 to J, whose maximum degree is at most three. Its cubic certificate inequality gives

$c_3(J) \leq floor(ps(J)/2) = floor((s - \eta - \rho)/2).$

The same fixed set I restores row three with cost at most a, so

$c_3(Q) \leq a + c_3(J).$

Comparison with the already proved lower bound on $c_3(Q)$ gives the first asserted rank bound. Removing the nonnegative ρ from the numerator only increases its right side, proving the second bound.

A maximum independent set exists because Q is finite, and it is maximal. Apply the same inequality to one of cardinality $\alpha(Q)$, then remove the nonnegative η and ρ to obtain $m - \alpha(Q) \leq s + E - 1 + \lfloor s/2 \rfloor$. Finally, $gap(Q) = E - s$ by Lemma 23. The additional assumption $gap(Q) \leq 0$ therefore gives $E \leq s$, and substitution proves the last inequality. □

Depends on. Lemma 23, Lemma 61, Lemma 72, Lemma 112, Lemma 161.

Lemma 214 (Strict potential from a projection at the next lower degree). *Let $s, t \geq 1$ be integers and let $n_1 \geq \cdots \geq n_s \geq 1$ and $m_1 \geq \cdots \geq m_t \geq 1$ be integer lists. Let F_1, F_2 be their respective vertex-disjoint unions of stars, prescribed red and blue. All graphs are finite and simple. Copies are ordinary subgraphs with vertex-disjoint target components. A graph arrows an ordered pair if every red-blue edge coloring contains its entire first target in red or its entire second target in blue.*

Put $M = s + t - 1$ and $\ell_j = max\{n_i + m_b - 1 : 1 \leq i \leq s, 1 \leq b \leq t, i + b = j\}$ for every integer $2 \leq j \leq M + 1$. Fix an integer $0 \leq q < M$. Let A_q be the family of all ordered pairs of nonempty actual suffixes obtained by deleting α leading components of F_1 and β leading components of F_2, where α, β are nonnegative integers with $\alpha < s$, $\beta < t$, and $\alpha + \beta = q$. Put

$K = M - q$, $\lambda_i = \ell_{q+i+1} - 1$ *for* $1 \leq i \leq K$,

$P_q = \sum_{i=1}^{K} \lambda_i$, $D = \lambda_1$,

and suppose $D \geq 4$. Let N, R, a be the numbers of entries of λ that are positive, equal to D, and equal to D-1, respectively.

For a positive-edge star forest F, let $c(F)$ be its component count, $r(F)$ its number of components of size at least two, and F^- the forest obtained by shortening each such component by one edge and discarding unit components. Write K_2 for one edge. Form a family C_q by taking both orders (F, J_1) of each A_q member: include $(r(F)K2, J_1)$ when $r(F) \geq 1$ and include $(c(F)K2, J_1^-)$ when J_1^- is nonempty. Displayed matching edges have disjoint endpoints.

Suppose $a \geq 1$ and C_q contains a pair (pK2,J), where $p \geq 1$ is an integer, the largest star size of J is D-1 with integer multiplicity $f \geq 1$, and

$p + c(J) - 1 = N$, $p + f - 1 = R + a$.

Then every graph H with $\Delta(H) \leq D$ which arrows every member of A_q satisfies

$\Phi(H) := e(H) - \nu(H) \geq P_q + 1,$

where $e(H)$, $\nu(H)$, $\Delta(H)$ are its edge count, maximum matching cardinality, and maximum degree.

All indices, sizes, degrees, cardinalities, matching sizes and displayed parameters are integers in their positive or nonnegative domains. Unit stars, repeated sizes, isolated host vertices, $q = 0$ and an empty forest of components of

J smaller than D-1 are allowed. No binary-target, matching-deficit, lower-profile, or host-component assumption is imposed.

Proof. The family A_q is finite and nonempty. Indeed, the integer interval

$[\max(0, q-t+1), \min(q, s-1)]$

is nonempty for $q \le s+t-2$; choosing α there and $\beta = q-\alpha$ gives legal cuts. Every member has component budget K. Its full family envelope is ℓ_{q+2},..., ℓ_{M+1}. A local candidate with indices i, b and cuts α, β has original indices $I = \alpha + i$, $B = \beta + b$, whose sum is $q + i + b$. Conversely, given original indices I, B with $I + B = q + k$ and $k \ge 2$, set $\alpha = \min(q, I-1)$, $\beta = q - \alpha$. Then $\alpha \le I-1$ and $\beta \le B-1$, so the cuts are legal and recover the same candidate at local diagonal k. This proves equality of the complete candidate sets.

The full diagonal profile is nonincreasing: decreasing an index greater than one in a candidate from a later diagonal gives a candidate of at least that size on the preceding diagonal. Its tail λ is therefore nonincreasing and nonnegative. Its first R entries are D, its next a entries are D-1, and its first N entries are exactly the positive ones. In particular $R \ge 1$ and $R + a \le N$.

Apply Lemma 56 to the same finite family A_q and its envelope just proved. It gives $\Phi(H) \ge P_q$. Suppose equality. Fix any maximum matching W of H and put $Q = H - W$, deleting only the matching edges and retaining every vertex. By Lemma 95, applied to this family with component budget K, every sequence of maximum-degree vertex deletions in Q has its first K degrees λ_1,..., λ_K. After N steps the graph is edgeless, since the remaining edge count is the sum of the remaining λ entries, which is zero. All first N prescribed degrees are positive.

Let $S = \{v : d_H(v) = D\}$ and $I = S$ minus $V(W)$. The set I is independent: an edge joining two W-uncovered vertices could be added to W. Its vertices are exactly the degree-D vertices of Q, since every matched vertex loses one incident edge and $\Delta(H) \le D$. Deleting vertices of I in any order keeps the degree of each remaining I-vertex equal to D. If $|I| < R$, exhausting I contradicts the next prescribed degree D. If $|I| > R$, deleting R of its vertices leaves a degree-D vertex, contradicting the next prescribed degree D-1, which exists because $a \ge 1$. Thus $|I| = R$. Put $T = Q - I$. Every greedy sequence in T now has N-R positive-degree steps, starting with precisely a degrees D-1 and then degrees at most D-2. Its chosen vertices form a cover of T of size N-R.

We check all matching projections on this same Q. Take either orientation (F, J_1) of an A_q member and any coloring of Q. Color every W-edge red. A blue J_1 in H lies in Q. A red F loses at most one edge at each copied star center when W is removed. Shortening each nonunit once and omitting the unit components leaves a copy of F^- in Q; selecting one edge from each of its components gives $r(F)$ disjoint edges. Thus Q arrows $(r(F)K2, J_1)$ when $r(F) \ge 1$. Coloring W blue instead proves Q arrows $(c(F)K2, J_1^-)$ when J_1^- is nonempty: a red full F in Q contains $c(F)$ disjoint edges, and a blue full J_1 shortens as just described. Exchanging both target roles and colors permits both orientations. In particular Q arrows the selected (pK2,J).

Write $J = fK_{1,D-1}$ disjoint union J_0, where J_0 has $t_0 = c(J) - f \ge 0$ components of sizes less than D-1. The budget equations give $N = R + a + t_0$. Put $b = \min(R, p-1)$, $c = R - b$. Then $0 \le b < p$ and $0 \le c < f$. For the latter, if $R \le p-1$ then $c = 0$; otherwise $c = R - p + 1 = f - a \le f - 1$. Set $p' = p - b$ and $f' = f - c$. These are positive integers with $p' + f' - 1 = a$.

The graph T arrows $(p'K2, f'K_{1,D-1} disjoint union J_0)$. To prove this, suppose a coloring of T avoided that full pair. Partition I into parts of sizes b, c and color all edges incident with the first part red and all edges incident with the second blue. Independence of I makes these prescriptions compatible and colors every Q-edge outside T. A red p-edge matching has at most b edges meeting I, so leaves p′ red matching edges in T. A full blue J has at most c components meeting I, since each uses a different vertex in its blue part. Removing those components leaves a collection dominating the sorted suffix obtained by omitting c largest components of J. Since $c < f$, this suffix is exactly $f'K_{1,D-1}$ disjoint union J_0. Trimming larger remaining stars gives the forbidden blue target in T. This contradiction proves the asserted residual arrowing.

Every independent set of degree-(D-1) vertices of T has size at most a. Otherwise $a+1$ such vertices could be deleted first, each retaining degree D-1 while the others are deleted, contradicting the inherited greedy degree sequence. Fix a cover C of T from one complete greedy sequence. It has size $N - R = a + t_0$. Apply Lemma 36 to T with degree $d = D - 1 \ge 3$, matching size p′, largest-star multiplicity f′, smaller forest J_0 and cover C. It gives an independent set X of a vertices of degree D-1 with pairwise disjoint closed T-neighborhoods. For $B = T - X$ it gives $\Delta(B) \le D - 2$ and

$\nu(B) = \tau(B) = t_0$,

where τ is minimum vertex-cover cardinality.

The vertices of X may be deleted first in T because they are independent and have degree D-1. Every greedy continuation in B therefore has exactly t_0 positive-degree deletions before it becomes edgeless. The final sufficient condition of Lemma 36 applies, including when $t_0 = 0$. It says that X is the entire degree-(D-1) vertex set of T. Thus this whole set is independent and its closed neighborhoods are pairwise disjoint.

This conclusion holds for every initially chosen maximum matching W. Use Lemma 148 with $G = H$ and the same D. Its family Wcal of maximum matchings minimizing uncovered S-vertices is nonempty. Since every maximum matching exposes exactly R vertices of S, its minimum is R. For each member of Wcal the theorem's remainder T_W is our T, and its entire set U_W of degree-(D-1) vertices is our X.

If D is odd, the independence condition in that theorem supplies a maximum matching covering all of S, contradicting $R \geq 1$. Suppose D is even. The same theorem gives exactly R whole K_{D+1} components. Choose one, and fix a member W of Wcal. Its unique uncovered vertex x in this component belongs to I. The theorem further gives a whole component of $B = T - X$ isomorphic to K_D with a perfect matching removed.

First let $D \geq 6$. Put $m = D/2 \geq 3$ and label the removed pairs $\{x_i, y_i\}$, $1 \leq i \leq m$. The edges $x_i\, y_{i+1}$, with indices read cyclically, are present and form a perfect matching, so this component's matching number is m. Its independence number is exactly two: a removed pair is independent, and each vertex has only its paired vertex as a nonneighbor among the other vertices. Complements of covers are independent and complements of independent sets are covers, so its minimum cover size is $D - 2 > D/2$. For any graph, matching number is at most cover size because disjoint matching edges use different vertices of a cover. Both matching number and minimum cover size add over whole components: restriction bounds a global choice componentwise, and the union of componentwise optimal choices attains the sum. Thus one component with strict inequality forces $\nu(B) < \tau(B)$, contrary to the equality already proved.

It remains $D = 4$. There is a whole K_5 component. Replace one such component using Lemma 22 by disjoint stars of sizes five, four and one. That theorem preserves arrowing of every member of the same A_q simultaneously and reduces Φ by one. The resulting H_{star} has $\Phi(H_{star}) = P_q - 1$. The potential bound Lemma 56 applies without a maximum-degree restriction and gives $\Phi(H_{star}) \geq P_q$, a contradiction.

Every possible $D \geq 4$ has excluded equality. Since $\Phi(H)$ and P_q are integers and the non-strict bound holds, $\Phi(H) \geq P_q + 1$. □

Remark. An exact greedy profile and a full matching projection force the whole next-degree layer to be a separated block. Matching exposure then localizes to odd cliques. Their lower remainders violate the forced matching-cover equality, except at degree four, where a simultaneous star replacement lowers the potential.

Depends on. Lemma 22, Lemma 36, Lemma 56, Lemma 95, Lemma 148.

Lemma 215 (Exact matching-envelope gap at a terminal plateau). *All graphs are finite and simple. Let F, J be the vertex-disjoint unions of stars with complete nonincreasing positive integer edge-size lists $n_1,\ldots, n_s$ and $m_1,\ldots, m_t$, where $s, t \geq 1$ are integers. Copies are ordinary prescribed-color subgraphs; the stars within each target have disjoint vertex sets. Put $M = s + t - 1$ and*

$$\ell_l = max\{n_i + m_j - 1 : 1 \leq i \leq s, 1 \leq j \leq t, i + j = l\}$$

for every integer $2 \leq l \leq s + t$. Fix an integer $0 \leq q < M$ and put $K = M - q$, $L_q = \sum_{l=q+2}^{s+t} \ell_l$, and $D = n_1$. Suppose $m_1 \leq D$ and $\ell_{q+2} = D + 1$. Let R be the length of the initial $D + 1$ plateau of this complete tail. Suppose there are valid original indices i_0, j_0 with

$$i_0 + j_0 = q + R + 1,\ a = n_{i_0} \geq 4,\ b = m_{j_0} \geq 3, \text{ and } a + b = D + 2.$$

In particular $D \geq 5$ and $1 \leq R \leq K$.

For each pair of nonnegative integer cuts $\alpha < s$, $\beta < t$ with $\alpha + \beta = q$, let (F^{α}, J^{β}) be the pair obtained by deleting those numbers of leading whole components. Let A_q be the family of all these pairs. Let Γ_q contain both matching relaxations of every member of A_q, each oriented with the matching first:

$$((s - \alpha)K2, J^{\beta}) \text{ and } ((t - \beta)K2, F^{\alpha}),$$

where K_2 is a single edge. For a generic member (pK2,Q), write $d_1 \ge \cdots \ge d_c \ge 1$ for the complete size list of Q. Its length-K profile has l-th entry $max\{d_j : 1 \le h \le p, 1 \le j \le c, h + j = l + 1\}$. Let η_l be the maximum of these entries over Γ_q, for every integer $1 \le l \le K$, and put $E = \sum_{l=1}^{K} \eta_l$.

Suppose H arrows every entire A_q member and

$\delta = K - \nu(H) \ge 1, e(H) \le L_q,$

where ν and e are maximum matching cardinality and edge count. Then $q < \min(s, t)$, the complete envelope η is already the maximum of the profiles of the two members ((t-q)K2,F) and ((s-q)K2,J), and

$L_q - E = R.$

Writing $\tau(H)$ for minimum vertex-cover cardinality and $\gamma = \tau(H) - K$, one has $0 \le \gamma \le R$, and $\gamma \le R - 1$ when $e(H) < L_q$. For every vertex set X of integer size $0 \le z \le K$,

$e(H - X) \ge E - \sum_{l=1}^{z} \eta_l + \gamma,$

where deletion is induced and an empty sum is zero.

If $\gamma = R$, then $e(H) = L_q$. Every sequence of K successive maximum-degree vertex deletions exists, removes exactly η_l edges at step l, and leaves exactly R pairwise vertex-disjoint edges and any number of isolated vertices. All target sizes, counts, indices, widths, deficits, degrees, budgets and cardinalities have the integer domains stated above. No maximum-degree bound on H is assumed.

If in addition the maximum degree of H is at most D, put

$\sigma = e(H) - \nu(H) - (L_q - K)$ *and* $k_D = floor((D - 1)/2).$

Then

$R + (k_D - 1)(\delta + \gamma) \le k_D\, \sigma,$

or equivalently

$\sigma \ge R + (k_D - 1)(L_q - e(H) + \gamma).$

In particular $\sigma \ge R$. The extra maximum-degree hypothesis applies only to these final inequalities.

Proof. Apply Lemma 89 to the entire family. Its positive-deficit conclusion gives $q < \min(s, t)$ and

$t \ge q + 2\delta + 2sum_i\ floor((n_i - 1)/2).$ (1)

Indeed an attaining separator exists and has nonnegative size, which can be dropped from that inequality. Thus both corner members displayed in the statement are legal.

We first record the relevant threshold counts of the complete lists. For positive integer h put $r_h = |\{i : n_i \ge h\}|$ and $c_h = |\{j : m_j \ge h\}|$. Put

$U = r_a, V = c_b, d = r_D, z = r_3 - U, k = r_2 - r_3, j = c_2 - V,$

$f = s - U, g_0 = t - V, u = s - r_2, v = t - c_2.$

All these numbers are nonnegative integers and $d \ge 1$.

For every positive integer h, let N_h be the number of entries in the natural q-tail that are at least h. Then

$N_h = max\{(r_x + c_y - q - 1)_+ : x, y \ge 1, x + y = h + 1, r_x > 0, c_y > 0\},$ (2)

with zero for an empty maximum and $w_+ = \max(w, 0)$. To prove this, a diagonal has an entry at least h precisely when it contains indices from one such rectangle $1 \le i \le r_x, 1 \le j \le c_y$. If a pair of sizes has sum at least $h + 1$, thresholds x, y with $x + y = h + 1$ can be chosen below those sizes. Index sums in each nonempty rectangle fill all integers from 2 to $r_x + c_y$. Hence the original diagonal sequence is nonincreasing, and after removing the first q diagonals its threshold width is exactly (2).

The terminal indices give $U + V \ge q + R + 1$. A strict inequality would give a diagonal of value at least $D + 1$ beyond the end of its initial natural plateau. Therefore

$U + V = q + R + 1.$ (3)

The thresholds (D,2) have positive counts d, c_2 and size sum $D+2$, so $d+c_2 \le U+V$. Hence $0 \le j \le U-d$. The thresholds (D,3) have positive counts, since $c_3 \ge V > 0$, and value $D+2$. Their rectangle must end before the first natural diagonal $q+2$, whose value is only $D+1$. Thus

$d + c_3 \le q + 1 = U + V - R$, and $U \ge R + d$. (4)

Also $K = f + g_0 + R$, $f = z + k + u$, and $v = g_0 - j$.

Every one of the U red sizes at least a contributes at least a-2 to twice the floor sum in (1). Each of the z red sizes from 3 through a-1 contributes at least two. Substitution of $q = U + V - R - 1$ in (1) yields

$g_0 \ge (a-1)U + 2z + 2\delta - R - 1.$

Since $j \le U - d$ and $r_3 = U + z$, this gives

$v - r_3 \ge (a-3)U + z + d + 2\delta - R - 1$

$\ge (a-4)R + (a-2)d + z + 2\delta - 1 > 0.$ (5)

Here the second comparison uses $U \ge R + d$, and the strict positivity uses $a \ge 4$, $d \ge 1$ and $\delta \ge 1$. In particular $v \ge r_3$.

We next compare the entire two families of profiles, rather than a single suffix profile. A matching-first pair with matching demand p and opposite threshold multiplicity $c > 0$ has exactly $p + c - 1$ entries at that threshold; if $c = 0$ it has none. For a legal suffix pair with cuts α, β, the relaxation retaining J^β has threshold width $s + c_h - q - 1$ when $c_h > \beta$, and zero otherwise. The corner $(\alpha, \beta) = (q, 0)$ realizes $s + c_h - q - 1$ whenever $c_h > 0$. The other relaxation has width $t + r_h - q - 1$ when $r_h > \alpha$, and zero otherwise; its full width is attained by $(\alpha, \beta) = (0, q)$. All these profiles are nonincreasing and have length K, so equality of their widths at every positive threshold proves the two-corner envelope assertion.

Let W_h count the entries of η that are at least h. For $1 \le h \le D$, $r_h \ge d > 0$ and

$W_h = \max(g_0 - U + r_h + R, f - V + c_h + R if c_h > 0).$ (6)

The first term equals $t - q + r_h - 1$ and is positive. There are no matching-envelope entries above D. Formula (2), using thresholds (h,1) and (1,h), gives $N_h \ge W_h$. For $h = 1, 2$ these exhaust all threshold pairs, so $N_h = W_h$. There is exactly one nonzero natural width above D: $N_{D+1} = R$.

For $3 \le h \le D$, any pair in (2) with $x, y \ge 3$ has width before taking its positive part at most $z + j + R$. The first term in (6) is at least $g_0 - U + d + R$, which exceeds this bound by $v - r_3 + d \ge d > 0$. For $h \ge 4$ the pair (h-1,2) exceeds the first term of (6) by

$r_{h-1} - r_h - v \le r_3 - v \le 0,$

because the conjugate difference counts red entries of size $h - 1 \ge 3$. Thus the only pair that can add to W_h is (2,h-1), including (2,2) when $h = 3$.

Put $x_0 = k - g_0$ and $B_h = r_h - x_0 - z - U + V$. The untruncated width for (2,h-1) is $k + z + c_{h-1} - V + R$. Subtract $k + z - V + R$ from the two terms in (6). They become B_h and $c_h + u$. Consequently, if $c_h > 0$,

$N_h - W_h = (c_{h-1} - \max(B_h, c_h + u))_+.$ (7)

If $c_h = 0 < c_{h-1}$, the blue term is absent and

$N_h - W_h = (c_{h-1} - B_h)_+.$ (8)

If $c_{h-1} = 0$ the pair is absent and the difference is zero.

Interpret (7) as the length of the real interval $(\max(B_h, c_h + u), c_{h-1}]$ when its endpoints are increasing, and otherwise as zero. Interpret (8) in the same way. These intervals have pairwise disjoint interiors: each interval with $c_h > 0$ has lower endpoint at least c_h, whereas every later interval has upper endpoint at most c_h. There is at most one interval of type (8), and it is last. Each B_h is at least B_D because $r_h \ge r_D = d$. Every upper endpoint is at most c_2. Their total length is therefore at most

$(c_2 - B_D)_+ = (x_0 + z + U + j - d)_+.$

Set $T = z + U + j - d$. Summing all threshold differences gives

$R \le L_q - E \le R + (x_0 + T)_+.$ (9)

The sums of widths equal the sums of their positive integer entries, since an entry of size w is counted once at each threshold 1,..., w.

For a final comparison put $A_0 = e(F) - f - k$. Using $f = z + k + u$ to collect separately the red sizes at least a, those from 3 through a-1, the twos and the units, gives

$A_0 = \sum_{i \le U} n_i + \sum_{U < i \le r_3} (n_i - 1)$

$\ge aU + (D - a)d + 2z.$

Also $T \le z + 2U - 2d$ because $j \le U - d$. Consequently

$A_0 - T \ge (a - 2)U + z + (D - a + 2)d$

$\ge (a - 2)R + Dd + z \ge 2R + Dd + z.$ (10)

The amplified cover bound Lemma 65, on this same H and full originals, gives $\gamma \ge 0$ and

$\gamma \ge e(F) - K + 2\delta - 1 = A_0 + x_0 + 2\delta - R - 1.$ (11)

The graph H arrows every member of Γ_q. Replacing a prescribed forest by one edge from each of its positive, disjoint components can only weaken that target, and exchanging the two color names permits the matching-first orientation. All those members have component budget K. Apply Lemma 21 to obtain

$e(H) \ge E + \gamma.$ (12)

Since $e(H) \le L_q$, formulas (9) and (12) imply $\gamma \le R + (x_0 + T)_+$.

If $x_0 + T > 0$, combine this last bound with (11) and cancel x_0. The result is

$0 \ge A_0 - T + 2\delta - 2R - 1.$

By (10), its right side is at least $Dd + z + 2\delta - 1 > 0$, a contradiction. Therefore $x_0 + T \le 0$. Formula (9) now gives $L_q - E = R$. Formula (12) gives $0 \le \gamma \le R$; when $e(H) < L_q$, integrality strengthens the upper bound to R-1.

The arbitrary vertex-deletion inequality in Lemma 21, with its maximum member budget $N = K$, gives exactly the claimed residual bound for every actual set X, including $z = K$. Finally $\gamma = R$ and (12) force $e(H) = E + R = L_q$ and $\tau(H) = K + R$. The equality assertion of that same cited result, with parameter $r = R$, supplies every maximum-degree sequence and its terminal R-edge matching. It explicitly counts an edge joining two deleted vertices only at its earlier endpoint and terminates after K steps. This proves the endpoint statement without a host-degree assumption.

Now assume the additional $\cap\, \Delta(H) \le D$. Write S for the vertices of H-degree D, ρ for their minimum matching exposure, θ for the minimum matching number of a spanning edge set incident with every S-vertex, and $a_S = \alpha(H[S])$. These are the quantities of Lemma 165. We prove the bounds needed to apply its weighted inequality on this same H.

First $\rho \ge R$. Choose a matching leaving exactly ρ vertices of S uncovered, and choose one incident edge at each of these uncovered vertices. Their union B_0 has at most ρ edges and is disjoint from the matching. In $H - B_0$, every vertex still having degree D is covered by that matching. Thus $c_D(H - B_0) = 0$ by Lemma 61. For each capacity split $A + B = D$, partition the zero-cost residual into its two bounded-degree parts. At any nonnegative integer budgets $x + y = \rho$, assign $\min(x, |B_0|)$ of the removed edges to the red exceptional part and the rest to blue. Their cardinalities, hence matching numbers, are at most x, y. This gives $c_D(H) \le \rho$. On the other hand Lemma 64 gives $c_D(H) \ge R$, since precisely the first R natural-tail entries exceed D. Consequently $\rho \ge R$.

Put $R_D = |\{l : \eta_l = D\}|$ and $h = e(H) - E$. The two-corner identity already proved and $n_1 = D$ show $R_D \ge 1$. Choose a member (pK2,Q) of Γ_q whose complete matching profile attains the last D position. Its opposite head D has some positive multiplicity f and $R_D = p + f - 1$.

We show $\theta \ge p + f - 1$. Choose an incidence edge set of rank θ and then an inclusion-minimal incidence subset B_1 of it. This retains rank θ, because it is still feasible for the defining minimum. Every edge of B_1 has an endpoint in S whose $B_1 - \textit{degree}$ is one; otherwise removing it would preserve incidence with all of S. In particular no edge joins two vertices of $B_1 - \textit{degree}$ at least two. Each nonempty component of B_1 is therefore a star: if it has a vertex with degree at least two all its neighbors have degree one and exhaust the component; otherwise it is one edge. It follows that B_1 has exactly θ nonempty components.

If $\theta < p + f - 1$, color all edges in $\min(\theta, p-1)$ of these components red and every other H-edge blue. The red matching number is at most p-1. Fix one center in each remaining $B_1 - star$, choosing either endpoint when it is one edge. The center of a blue D-edge star has ambient degree D, so it belongs to S and uses all its H-incident edges in that copy. It cannot lie in a red $B_1 - component$. In its remaining $B_1 - component$ it is either the fixed center or a leaf adjacent to that center. Its copied blue star therefore includes that fixed center. Distinct vertex-disjoint blue D-stars require distinct remaining components. There are at most f-1 of them, so an entire blue Q is impossible. This contradicts arrowing of the chosen Γ_q member and proves $\theta \geq R_D$.

Let I be a maximum independent set of H[S], with $|I| = a_S$. Its incident edges cover S, because maximality supplies an I-neighbor to every unchosen S-vertex, and vertices of I have positive degree. Hence $\theta \leq a_S$. We also prove $a_S \leq \nu(H)$. In the bipartite graph of H-edges between I and its complement take a maximum matching. If a vertex of I is unmatched, follow alternating paths from all unmatched I-vertices, directing nonmatching edges toward the complement and matching edges back to I. Let U and W be the reached subsets of I and its complement. No unmatched complement vertex is reached, since reversing a shortest such alternating path would increase the matching. Thus every W-vertex is matched back into U, and the matching pairs W with exactly the matched vertices of U. Since U contains an unmatched vertex, $|U| > |W|$. Every neighbor of U is in W: nonmatching edges extend reachability and matching partners were already reached. But U is independent and each of its vertices has degree D, whereas every W-vertex has degree at most D. Counting crossing edges gives $D|U| \leq D|W|$, a contradiction. A matching saturating I therefore exists, proving $a_S \leq \nu(H) < K$.

Deleting I removes exactly $D\ a_S$ edges because it is independent. The already proved arbitrary-set matching-envelope bound at I gives

$D\ a_S - \sum_{l=1}^{a_S} \eta_l \leq h - \gamma$.

All matching-envelope entries are at most D, precisely the first R_D equal D, and every later one is at most D-1. As $a_S \geq \theta \geq R_D$, the left side is at least $a_S - R_D$. Consequently $a_S - \theta \leq h - \gamma$.

Use Lemma 165 with $k = k_D$ to get

$(k_D + 1)\rho \leq \tau(H) - \nu(H) + k_D(a_S - \theta)$

$\leq \delta + \gamma + k_D(h - \gamma)$.

Substitute $\rho \geq R$, $\tau(H) - \nu(H) = \delta + \gamma$, and $h = R + \sigma - \delta$, the last identity following from $L_q - E = R$. After subtracting $k_D\ R$ the result is

$R + (k_D - 1)(\delta + \gamma) \leq k_D\ \sigma$.

Finally $\delta - \sigma = L_q - e(H)$ gives the equivalent displayed form. The assumptions $e(H) \leq L_q$ and $\gamma \geq 0$ then yield $\sigma \geq R$. Every auxiliary edge set, matching and vertex set was chosen in a finite graph, and all incidence and deletion estimates refer to that same H. □

Remark. Full-target half-order capacity supplies enough opposite units that only a disjoint interval correction can enlarge the natural-to-matching profile gap. Amplified cover capacity rules out that correction, leaving the exact plateau width. Incidence covers then give the weighted surplus estimate on the same capped graph.

Depends on. Lemma 21, Lemma 61, Lemma 64, Lemma 65, Lemma 89, Lemma 165.

Lemma 216 (Degree and independence bounds at gap two). *Use the finite integer certificate costs $c_i(J)$, matching number $\nu(J)$, independence number $\alpha(J)$, maximum degree $\Delta(J)$, excess $ex(J) = c_0(J) - \nu(J)$, slack $ps(J)$, and $gap(J) = ex(J) - ps(J)$ of Lemma 23. For a nonnegative integer t, let $Y_t(J)$ consist of the integers x in [0,t] for which the edges of J can be partitioned into two parts of matching numbers at most x and t-x. Write $run_t(Y_t(J))$ for the longest consecutive run of missing integers in [0,t], with value zero when none are missing. Put $tr(J) = run_($ $c_0(J) - 1)(Y_($ $c_0(J) - 1)(J))$ if $c_0(J) > 0$, and $tr(J) = 0$ otherwise. For nonnegative integers D put $h(D) = \max(4, \lfloor D/2 \rfloor + 1)$. A graph J satisfies T_- when $gap(J) > 0$ implies both $gap(J) = 1$ and $tr(J) + ex(J) \leq h(\Delta(J))$.*

Let H be a connected finite simple d-regular graph on $2k+1$ vertices, where $d \geq 4$ is an even integer and $k \geq 2$ is an integer. Suppose deleting any vertex of H leaves a perfect matching, and $gap(H) = 2$. Assume every proper induced subgraph J of H satisfies T_- and the following implications:

$ps(J) = 0 \Rightarrow run_{(}\nu(J))(Y_{(}\nu(J))(J)) \leq 2$;

$gap(J) > 0$ *and* $ex(J) = 1 \Rightarrow tr(J) \leq 2$;

$gap(J) > 0$ *and* $ex(J) = 2 \Rightarrow tr(J) \leq 4$;

$gap(J) > 0$ *and* $ex(J) = 3 \Rightarrow tr(J) \leq 5$.

Then $\alpha(H) \geq 5$ *and* $d \geq 20$. *Proper induced subgraphs here include the empty graph. All orders, degrees, costs, budgets, matching and independence cardinalities, excesses, run lengths, and indices are integers in their stated nonnegative domains; generic gaps may be signed integers.*

Proof. Put $n = 2k + 1$, $a = \alpha(H)$, $E = ex(H)$, $c = c_0(H)$, and $s = ps(H)$. Factor-criticality and the order bound on matchings give $\nu(H) = k$, so $c = k + E$ and $s = E - 2$. The cutoff theorem Lemma 123, applied at cutoff c in an economical complete maximum-degree deletion sequence, gives

$s \geq \max(0, 2(n - c) - 4a)$.

Consequently $E \geq 2$ and

$2k \leq 3E + 4a - 4$. (1)

Since $c > 0$, the integer threshold definition in Lemma 23 supplies nonnegative integers x, y with $x + y = c - 1$ for which no two-part edge partition has matching bounds x, y. The order construction in Lemma 48 would produce such a partition if $n \leq x + y + \max(x, y) + 1 = c + \max(x, y)$. The missing partition therefore implies

$\max(x, y) \leq n - c - 1 = k - E$.

It follows that $k + E - 1 = x + y \leq 2k - 2E$, and hence

$k \geq 3E - 1$. (2)

Combining (1) and (2) yields

$3E \leq 4a - 2$. (3)

If $a = 1$, this contradicts $E \geq 2$.

Suppose $a \geq 2$ and choose any maximum independent set S. It is inclusionwise maximal. Every component of $G = H - S$ is a proper induced subgraph of H, so it satisfies T_{-}. Its zero-slack matching-total run is at most $2 \leq a$, and its positive-gap runs at excess one, two, and three have the bounds in the statement. Thus all hypotheses of Lemma 160 hold for this same H, S, G. With

$m = \max(\{0\} union \{ex(P) : P is a component of G with floor(|V(P)|/2) \geq a + 2\})$

and $\eta = \nu(G) - (k + 1 - a)$, that fact gives

$E = 1 + m + \eta, \eta \geq 0$. (4)

It also gives the following necessary bounds:

$a = 2 \Rightarrow m \geq 2$ and $d \geq 10$;

$a = 3 \Rightarrow m \geq 2$ and $d \geq 12$;

$a = 4 \Rightarrow m \geq 3$ and $d \geq 16$;

$a \geq 5 \Rightarrow m \geq 4$ and $d \geq 20$.

These conclusions hold separately for every maximum independent set S, with its actual complement and its own m, η.

For $a = 2$, (3) gives $E \leq 2$ whereas (4) gives $E \geq 3$. Thus $a = 2$ is impossible.

For $a = 3$, (3) and (4) force $E = 3$, $m = 2$, $\eta = 0$. Equations (1) and (2) give $8 \leq k \leq 8$, so $n = 17$, $c = 11$, $s = 1$, and $d \geq 12$. The graph H is noncomplete because $a \geq 2$. Apply Lemma 142 with $q = 10$. Its minimum-degree condition holds since $d \geq 12 = 17 - 10/2$. Here $n - q = 7 \leq c$ and

$b = floor(\lceil 7/2 \rceil / 2) = 2$.

Its slack bound gives $s \geq 2 + \max(0, 2(17-11) - 4 \cdot 3) = 2$, contradicting $s = 1$.

For $a = 4$, (3) and (4) force $E = 4$, $m = 3$, $\eta = 0$. Equations (1) and (2) give $11 \leq k \leq 12$, and $d \geq 16$. Since $\eta = 0$ holds for every maximum independent set S, we have

$\nu(H - S) = k - 3$ for every such S. (5)

Also $c = k + 4$ and $s = 2$.

If $k = 12$, then $n = 25$ and $c = 16$. Apply Lemma 142 with $q = 18$: the minimum-degree condition is $d \geq 16 = 25 - 18/2$, the prefix length is $7 \leq c$, and $b = 2$. Its bound gives

$s \geq 2 + \max(0, 2(25-16) - 4 \cdot 4) = 4$,

contrary to $s = 2$. Hence $k = 11$, $n = 23$, and $c = 15$.

If $d \geq 18$, use $q = 10$ in Lemma 142. The minimum-degree condition is $d \geq 18 = 23 - 10/2$, the prefix length is $13 \leq c$, and $b = floor(\lceil 13/2 \rceil / 2) = 3$. The slack is then at least three, again a contradiction. Because d is even and $d \geq 16$, the only remaining case has $d = 16$.

Apply Lemma 142 with $q = 14$ in this remaining case. Its minimum-degree condition is $16 = 23 - 14/2$. Its prefix length is nine and its covered matching-preserving count is at least $floor(\lceil 9/2 \rceil / 2) = 2$. Choose one economical complete maximum-degree deletion sequence, whose existence is supplied by Lemma 123, and let R be its residual at cutoff $c = 15$. The residual has eight vertices. The covered matching-preserving count z_{15} in this same prefix is at least two. The cutoff inequality gives

$2 = s \geq z_{15} + \beta_{15} + \Phi(R) \geq 2 + \Phi(R)$,

where $\beta_{15} \geq 0$ and $\Phi(R) = e(R) - \nu(R) \geq 0$. Thus $\Phi(R) = 0$.

Equality $e(R) = \nu(R)$ says that all edges of R belong to one matching: a maximum matching already contains as many edges as the whole graph. Therefore R consists of disjoint edges and isolated vertices. If $r = e(R)$, then $r \leq 4$ and $\alpha(R) = 8 - r$. Since R is induced in H, $\alpha(R) \leq a = 4$; consequently $r = 4$. Thus R consists of exactly four disjoint edges and has no isolated vertices.

Let $U = V(H)$ minus $V(R)$, so $|U| = 15$. Each vertex of R has one neighbor in R and degree sixteen in H, and hence is adjacent to all fifteen vertices of U. Every vertex of U is therefore adjacent to all eight vertices of R. The induced graph $J = H[U]$ is eight-regular on fifteen vertices and is noncomplete. Apply Lemma 142 to J with $q = 14$: its minimum-degree condition is $8 = 15 - 14/2$. The inheritance conclusion makes J factor-critical. In particular J has a matching M of seven edges.

Choose one endpoint from each of the four R-edges and let S be the set of the four chosen vertices. The set S is independent in H and has size $\alpha(H) = 4$, so it is maximum. The four vertices of R-S are all adjacent to every vertex of J. Delete two edges of M, match their four distinct endpoints bijectively to these four vertices of R-S, and retain the other five M-edges. This is a matching in H-S of size nine. It contradicts (5), which in this case states $\nu(H - S) = k - 3 = 8$.

We have excluded $a = 1, 2, 3, 4$. Thus $a \geq 5$. Choose any maximum independent set once more and apply the last case of Lemma 160, whose local hypotheses were checked above. It gives $d \geq 20$, completing both conclusions. □

Remark. A missing allocation and the residual potential sharply limit the order at small independence. Dense induced perfect matchings then force too much slack, except at one exact residual; regularity there constructs a matching contradicting the component equality.

Depends on. Lemma 23, Lemma 48, Lemma 123, Lemma 142, Lemma 160.

Lemma 217 (Exclusion of the small octic terminal configuration). *Use the finite integer certificate costs c_i, matching number ν, independence number α, maximum degree Δ, edge count e and slack $ps = e - \nu - \sum_{i \geq 1} c_i$ from Lemma 61. All graphs are finite and simple, and vertex deletion is induced. There is no eight-regular graph H on 27 vertices with all the following properties:*

(i) Its independence number and minimum dominating-set cardinality are both seven, and every maximal independent set has seven vertices.

(ii) For every maximum independent set S, its complement $G = H - S$ has twenty vertices, 52 edges, matching number ten, maximum degree at most seven and slack zero. Moreover $c_2(H) = 7 + c_2(G)$.

(iii) Every proper induced subgraph J of H with $\Delta(J) \le 7$, $ps(J) = 0$ and $\Delta(J) \ge 3$ has a matching covering all vertices of degree $\Delta(J)$.

A set is dominating if every vertex outside it has a neighbor in it. The word maximal refers to inclusion, and maximum refers to cardinality. No gap, excess, factor-criticality or minimality assumption is made. All orders, degrees, cardinalities, costs, row indices and matching sizes are nonnegative integers with the displayed values and bounds.

Proof. Suppose H has the stated properties. By Lemma 61, at every covered maximum-degree deletion from a zero-slack graph, the child has zero slack and the matching number decreases by one. Condition (iii) therefore allows arbitrary choices of a maximum-degree vertex at every stage of any proper induced zero-slack graph of maximum degree at most seven. Such choices terminate when the maximum degree is at most two. These are complete covered processes to which Lemma 120 and Lemma 146 apply.

Fix any maximum independent set S and put $G = H - S$. Write $a_G = \alpha(G)$ and $b = \nu(G) - c_1(G)$. The density theorem Lemma 146 gives $b \ge 0$ and

$52 = e(G) \le -4 \cdot 20 + 11 \cdot 10 + 4a_G - 6b = 30 + 4a_G - 6b.$

Since $a_G \le \alpha(H) = 7$, it follows that a_G is six or seven and b is zero or one. If $b = 1$, equality forces $a_G = 7$. The equality conclusion of that theorem gives an induced terminal with three triangles, one edge and three isolates. By the third row of Lemma 30, H has a dominating set of at most six vertices, contrary to (i). Thus $b = 0$ for every maximum independent complement.

Fix any complete covered process on this G. Let t be its number of deletions, d_1,..., d_t their degrees, and R its induced terminal. The row-loss theorem Lemma 120 gives $|V(R)| = 20 - t$, $\nu(R) = 10 - t$, $c_2(G) = t + o_R$, where o_R is the number of odd-cycle components of R. It also gives exactly $b = 0$ positive even-order path components. Put $u = a_G - \alpha(R) \ge 0$. The density identity of Lemma 146 at $\cap$ seven is

$4a_G - 22 = \sum_{j=1}^{t}(7 - d_j) + 4u + \sum_C F_7(C)$, (1)

where C ranges over the whole components of R and $F_7(C) = 7|V(C)| - e(C) - 11(|V(C)| - \nu(C)) + 4\alpha(C)$. All terms on the right are nonnegative. On the allowed terminal components with no positive even-order paths, F_7 is zero for a triangle or isolate, five for a three-vertex path or a five-cycle, and at least ten for every other component. These values follow from the explicit component formulas in Lemma 146.

The left side of (1) is two if $a_G = 6$ and six if $a_G = 7$. Hence $\alpha(R)$ is six or seven. Indeed $u = 0$ when $a_G = 6$ and $u \le 1$ when $a_G = 7$. If $\alpha(R) = 6$, at most two of the right-hand units remain after the independence term, so every terminal component is a triangle or isolate. If their counts are q and z, then $q + z = 6$, $3q + z = 20 - t$, $q = 10 - t$. Solving gives $t = 6$, $q = 4$, $z = 2$. Thus R is four triangles and two isolates, which Lemma 9 forbids in the well-covered H of (i).

We may therefore assume $\alpha(R) = a_G = 7$. The surplus in (1) is at most six. If every component has zero surplus, R consists of triangles and isolates, and the same equations with $q + z = 7$ give $t = 7$, $q = 3$, $z = 4$. Then $o_R = 3$, so $c_2(G) = 7 + 3 = 10$. Property (ii) gives $c_2(H) = 17$. But the four isolates of this induced thirteen-vertex terminal have their entire neighborhood in its fourteen-vertex exterior. The certificate theorem Lemma 47 gives $c_2(H) \le 16$, a contradiction.

If a terminal component has positive surplus, there is exactly one, and it is a three-vertex path or a five-cycle; all other components are triangles or isolates. For a three-vertex path, whose order, matching number and independence are 3, 1, 2, respectively, the equations are $3 + 3q + z = 20 - t$, $1 + q = 10 - t$, $2 + q + z = 7$. They give $t = 6$, $q = 3$, $z = 2$. The resulting induced terminal of three triangles, a three-vertex path and two isolates is forbidden by Lemma 9.

For a five-cycle the corresponding order, matching and independence values are 5, 2, 2. The equations give $t = 6$, $q = 2$, $z = 3$. Consequently the only remaining terminal form is $R = 2C3 + C5 + 3K1$. (2) Here C_3 and C_5 are cycles on three and five vertices, and K_1 is an isolate; sums mean disjoint unions with no edges between summands. Its surplus is five, so (1) gives $\sum_j (7 - d_j) = 1$. There are six deletions, each of degree at most seven and at least three. Since their integer deficits sum to one, five deletion degrees are seven and one is six. The maximum degrees cannot increase during induced deletion, so their order is (7,7,7,7,7,6). (3) The initial S and all process choices were arbitrary. We have proved that every complete process from every maximum independent complement must have both (2) and (3).

Choose one such R and let $W = V(H)$ minus $V(R)$. The terminal has fourteen vertices and eleven edges, so W has thirteen vertices. Eight-regularity and $e(H) = 108$ give $e(H[W]) = 108 - 8 \cdot 14 + 11 = 7$. Every vertex cover of H[W] has size at least $13 - \alpha(H) = 6$. For a vertex of internal W-degree d, that vertex together with one endpoint of every edge not incident with it forms a cover of size at most $1 + 7 - d$. Thus $d \le 2$, proving $\Delta(H[W]) \le 2$.

Choose an independent set S′ inside R by taking its three isolates, one vertex from each of its two triangles, and two nonadjacent vertices of its five-cycle. It has seven vertices, so it is a maximum independent set of H. Put $G' = H - S'$. The graph induced on $U = V(R)$ minus S′ consists of three independent edge components and one isolate. Each of the six edge endpoints has exactly one neighbor in S′, and the isolated vertex has exactly two: on a five-cycle, deleting two nonadjacent vertices leaves an edge and an isolate with these incidences, and each triangle loses its one selected vertex. Since R is induced and S′ is contained in R, these are the entire S′-neighborhoods of those vertices. Their degrees in G′ are respectively seven and six.

Begin a process on G′ by deleting one endpoint from each of the three edges of H[U]. These three vertices are pairwise nonadjacent. Each has degree seven when deleted, so each choice is a legal maximum-degree step. After these three steps the four remaining U-vertices are independent, and each has degree six: the three retained partners have each lost one neighbor, and the isolate lost none. Every vertex in W has at most two neighbors in W and at most four neighbors in the remaining U. It too has degree at most six. Thus this process has made exactly three degree-seven deletions, and no subsequent deletion can have degree seven.

Property (ii) gives zero slack for G′, and (iii) and the initial construction extend this prescribed prefix to a complete covered process. But (3), already proved for every such maximum independent complement and every complete process, requires five degree-seven deletions. This contradiction eliminates (2) and completes the nonexistence proof. □

Remark. The density loss permits four terminal forms. Three fail by short induced-subgraph or certificate bounds. The last produces another maximum independent set whose complement admits only three top-degree deletions, contradicting the five demanded of every such process.

Depends on. Lemma 9, Lemma 30, Lemma 47, Lemma 61, Lemma 120, Lemma 146.

Lemma 218 (The first odd-degree slope forces a single low separator). *Let s, t be positive integers and let F_1, F_2 be the nonempty star forests with positive nonincreasing integer edge-size lists $n_1,\dots, n_s$ and $m_1,\dots, m_t$, prescribed red and blue. Copies are ordinary subgraphs with vertex-disjoint target components. Put $M = s + t - 1$ and*

$$\ell_k = max\{n_i + m_j - 1 : 1 \le i \le s, 1 \le j \le t, i + j = k\}$$

for every integer $2 \le k \le s + t$. For integers $0 \le q \le M$ put $P_q = \sum_{k=q+2}^{s+t}(\ell_k - 1)$. For $0 \le q < M$ let A_q consist of all ordered nonempty actual suffix pairs after cuts α, β with $0 \le \alpha < s$, $0 \le \beta < t$, $\alpha + \beta = q$.

Fix $0 \le q < M$, put $D = \ell_{q+2} - 1$, and let R be the initial plateau length at $D + 1$ in this natural tail. Suppose valid original indices i_0, j_0 satisfy $i_0 + j_0 = q + R + 1$, $n_{i_0} = a \ge 4$ even, $m_{j_0} = b \ge 3$ odd, and $a + b = D + 2$. Put $r = a - 2$. Let H be finite simple, let ν be maximum matching cardinality, put $\Phi(H) = e(H) - \nu(H)$, and assume $\Delta(H) \le D$ and $\Phi(H) \le P_q + r$.

If H arrows every prescribed pair in A_q, then $R = 1$ and $\Phi(H) = P_q + r$. Exactly one whole connected component $C.$ has positive minimum total overflow at capacities a-1, b-1, where overflow is the sum over vertices of $\max(d_{red} - a + 1, 0) + \max(d_{blue} - b + 1, 0)$. This $C.$ is nonregular, its minimum overflow is one, and its minimum ambient-degree-D matching exposure is a-1. Every maximum matching of $C.$ exposes exactly a-1 such vertices, and every maximum matching in the union of the other whole components covers all its ambient-degree-D vertices.

There is a vertex y in $C.$ with $d_H(y) \le D - 1$ such that $C. - y$ has exactly a odd components $C_1,\dots, C_a$ wholly of ambient H-degree D. They are factor-critical and equimatchable, are adjacent to y, and are actual components of H-y. Factor-critical means that deleting any vertex leaves a perfect matching; equimatchable means that every inclusionwise maximal matching has maximum cardinality. Every maximum matching in any other component of H-y covers its ambient-degree-D vertices. The graph H-y has an edge coloring with red degrees at most a-1 and blue degrees at most b-1, and

$$\nu(H) = 1 + \sum_C \nu(C),$$

where C ranges over all actual components of H-y.

Write $|V(C_i)| = 2\mu_i + 1$ and $b_i = e_H(V(C_i), \{y\})$. Then $\nu(C_i) = \mu_i \geq \lceil D/2 \rceil$, each b_i is positive and odd,

$\sum_i b_i \leq d_H(y) \leq D - 1$, *and* $(2\mu_i + 1)(D - 2\mu_i) \leq b_i$.

For distinct i, j, $d_H(y) - (b_i + b_j)/2 \geq a - 1$. For every specified C_i there is a maximum matching of H using no $C_i - -y$ edge, with exactly μ_i internal edges in C_i and exactly a-2 exposed ambient-degree-D vertices outside C_i, none adjacent to C_i.

Failure of any asserted necessary conclusion, including $R \geq 2$ or $\Phi(H) < P_q + r$, gives an avoiding coloring for some entire prescribed pair in A_q. Every such coloring extends over the canonical q-star pool of capacities ℓ_2,..., ℓ_{q+1}, and across any actual outer q-vertex set X when $H = G - X$, to avoidance of the unchanged original targets while preserving every H-edge color. No matching-deficit, original-head bound, strict-cut, smaller-target, or $q < \min(s, t)$ hypothesis is imposed. All sizes, indices, degrees, counts, matching numbers and capacities are integers in the stated domains. All units and repeated sizes are retained, and the assertion permits simultaneous exchange of full target roles and colors.

Proof. Assume H arrows every member of A_q. Put $K = M - q$ and $\sigma = \Phi(H) - P_q$. The maximum component budget of this finite family is K. Its complete envelope at local diagonal h is ℓ_{q+h}: each suffix candidate has that original index sum; conversely, for original $i + j = q + h$ choose $\alpha = \max(0, q - j + 1)$ and $\beta = q - \alpha$. They are nonnegative, sum to q, and satisfy $\alpha \leq i - 1$, $\beta \leq j - 1$, so the components survive and their local indices sum to h. This proves both directions of the envelope identity. The family is nonempty because $0 \leq q \leq s + t - 2$ allows such legal cuts. Its envelope is nonincreasing: lower an index greater than one in a candidate on any later diagonal.

Here $D = a + b - 2$ is odd and at least five. Apply Lemma 99 to this entire family. It gives $\sigma \geq 0$, minimum high exposure $\rho \geq R$, and an upper bound $R + \sigma$ on the high exposure of every maximum matching. For later use it also gives, for every independent set I of $k \geq R$ vertices of H-degree D, with actual loss $j = \nu(H) - \nu(H - I)$,

$\sigma \geq k - R - j$. (1)

The fuller inequality in that fact has an additional nonnegative term, which is dropped here.

Choose the legal cuts $\alpha = \max(0, q - j_0 + 1)$, $\beta = q - \alpha$. The terminal components survive, and their local indices sum to $R + 1$. By Lemma 143 their whole target pair forces total minimum overflow $\omega \geq R$ at capacities a-1, b-1. For each whole component C let ω_C be its minimum overflow and η_C its minimum number of ambient-degree-D vertices exposed by a matching. These quantities add over whole components, since their colorings and matchings are independent choices. Thus $\omega = \sum_C \omega_C$ and $\rho = \sum_C \eta_C$. The odd-capacity integer bound in Lemma 196 gives

$\eta_C \geq \omega_C + r\ ceil(\omega_C/(b - 1))$.

Because $r \geq 2$, at least one ω_C is positive, and $\sigma \leq r$,

$R + r \geq R + \sigma \geq \rho \geq \omega + r|\{C : \omega_C > 0\}| \geq R + r$.

All inequalities are equalities. Consequently $\sigma = r$, there is exactly one positive-overflow component $C_\cdot$, $\omega_{C_\cdot} = \omega = R$, $\eta_{C_\cdot} = \rho = R + r$, all other $\omega_C = \eta_C = 0$, and $1 \leq R \leq b - 1$.

Every exposure-minimizing matching can be augmented to maximum cardinality without uncovering an already covered vertex. Indeed, its symmetric difference with a larger matching has an augmenting alternating path; reversing that path preserves the covered internal vertices and covers its two previously uncovered endpoints. Iteration increases cardinality and ends finitely at maximum. Exposure stays minimal. Combine such a maximum matching of $C_\cdot$ with arbitrary maximum matchings outside. The global upper bound $R + r$ forces every outside choice to expose no high vertex. Conversely, combine any maximum matching of $C_\cdot$ with exposure-minimizing maxima outside. Its exposure is at least $R + r$ and at most the global upper bound. Thus every maximum matching of $C_\cdot$ has exactly $R + r$ high exposures, and every outside maximum has zero.

The component $C_\cdot$ cannot be D-regular. For any maximal matching W there put $j = \nu(C_\cdot) - |W|$. The whole-regular-union inequality in Lemma 99, with complement matched maximally in cardinality, has comparison exposure $k_0 = R + r$ and gives $r \geq k_0 - R + j = r + j$. Thus $j = 0$ for every such W, making $C_\cdot$ equimatchable. Lemma 170 at odd degree D makes it $K_{D,D}$ or K_{D+1}. Both have perfect matchings, by

pairing opposite parts or pairing the even number $D+1$ of vertices, contradicting $\eta_{C.} > 0$. Since this positive exposure also ensures $C.$ contains a degree-D vertex, failure of D-regularity means $C.$ is nonregular.

Put $A = \{v \in V(C.) : d_H(v) = D\}$. No set attaining the marked deficiency $\eta_{C.}$ in $C.$ can meet A. Otherwise Lemma 150 gives an independent I $\subseteq$ A with $|I| = \rho + 2 = R + r + 2$ and $\nu(C. - I) \geq \nu(C.) - 1$. Whole-component additivity gives actual matching loss at most one in H. Applying (1) yields $r \geq |I| - R - 1 = r + 1$, impossible.

By the constant-cost formula in Lemma 143 choose an attaining pair S, T in $C.$ at capacities a-1, b-1. Let $U = V(C.)$ minus (S union T), let Q count components of $C.[U]$ wholly in A with $|V(C)| + e_H(C, T)$ odd, and define

$B_x = Q + e_H(S, T) - x|S| - (D - x)|T|$, $z = |S| - |T|$.

Then $B_{a-1} = R$. The common odd-parity family of Lemma 196 gives $B_D \leq 0$ and $B_1 \leq \eta_{C.} = R + r$. Direct subtraction gives $B_x - B_y = (y - x)z$. Thus

$R = B_D + (b-1)z \leq (b-1)z$, and $B_1 = R + rz \leq R + r$.

Since z is integral and R, r are positive, $z = 1$ and $B_1 = R + r = \eta_{C.}$.

Every component counted by Q sends a positive odd number of edges to S: its ambient degree sum and odd D turn the counted parity into odd $e_H(C, S)$. Consequently $Q \leq e_H(S, U)$. Put $def(S) = \sum_{v \in S}(D - d_H(v))$. The degree sum on S and the equality $B_{a-1} = R$, using $|S| = |T| + 1$, yield

$2e_H(S) + def(S) \leq D - (R + a - 1) = b - 1 - R$. (2)

If T had a vertex outside A, Lemma 206 applied to $B_1 = \eta_{C.}$ would give a maximum matching of $C.$ with one additional high exposure, contrary to the constant exposure just proved. Thus $T \subseteq$ A. That same theorem puts S in an attaining separator, so S is disjoint from A by the preceding paragraph. Every member of S contributes at least one to $def(S)$; therefore

$|S| \leq b - 1 - R \leq b - 2 \leq D - 2$.

The no-high-attainer conclusion of Lemma 206 now forces T empty. Since $z = 1$, $S = \{y\}$ for one vertex of degree at most D-1.

The same endpoint theorem says {y} attains the marked deficiency, every odd all-A component of $C. - y$ is factor-critical, and $C. - y$ has the proper degree split at capacities a-1, b-1. Each component outside $C.$ has zero minimum overflow and therefore has a split at these same capacities. Combining them gives the asserted degree coloring of H-y.

Let $N_a = |\{i : n_i \geq a\}|$ and $T_b = |\{j : m_j \geq b\}|$ for the full original lists. The terminal indices imply $N_a + T_b - 1 \geq q + R$. If the inequality were strict, the valid candidate at original indices N_a, T_b would give an entry at least $D + 1$ strictly after the natural plateau, contradicting the nonincreasing profile and its plateau length. If the plateau reaches the end, such valid indices beyond it cannot exist. Thus

$N_a + T_b - 1 = q + R$. (3)

Suppose $R \geq 2$. Choose $\alpha' = \min(N_a - 1, q + 1)$ and $\beta' = q + 1 - \alpha'$. Equation (3) gives $0 \leq \alpha' < N_a \leq s$, $0 \leq \beta' < T_b \leq t$, and $\alpha' + \beta' = q + 1$. The coloring of H-y avoids the actual suffix pair at these cuts, since those suffixes still contain an a-edge and a b-edge star respectively. At least one cut is positive. Decrease a positive cut by one, obtaining a legal A_q member, and apply the actual single-vertex extension of Lemma 41 to restore y with its corresponding color. This yields an avoiding coloring of an A_q member, a contradiction. Hence $R = 1$ and $\eta_{C.} = a - 1$.

Attainment by {y} now gives exactly a odd all-A components C_1,..., C_a in $C. - y$, each factor-critical. They are actual components of H-y because $C.$ is whole; connectedness of $C.$ makes each adjacent to y. Put $|V(C_i)| = 2\mu_i + 1$. Factor-criticality gives $\nu(C_i) = \mu_i$. Match y to any boundary vertex in one chosen required component, use a perfect matching after deleting that vertex there, and use maximum matchings in all other components of H-y. The result has $1 + \sum_C \nu(C)$ edges. Every matching of H has at most one edge at y and at most $\nu(C)$ other edges internal to each such component, proving the rank identity.

This construction already exposes one high vertex in each of the a-1 required components not reached from y. If any maximum matching in any other component of H-y exposed a high vertex, insert that choice into the construction. The rank identity makes the result maximum in H, but its high exposure would exceed

a-1, a contradiction. Thus all such component maxima cover their high vertices. For any specified C_i choose the edge at y into a different required piece; $a \geq 4$ ensures one exists. This gives a maximum matching with μ_i internal edges in C_i, no edge between C_i and y, and exactly a-2 exposed high vertices outside C_i. These vertices lie in other components of H-y and are nonadjacent to C_i.

If C_i had a maximal matching short of maximum by $j \geq 1$, substitute it into the last construction. The resulting matching has nu(H)-j edges. Its exposed high set I has $k = a - 1 + 2j$ vertices: $1 + 2j$ in C_i and one in each of a-2 other unreached required pieces. They are independent within C_i by maximality, within the other pieces by maximality of their maximum matchings, and between distinct components of H-y by their separation; y is covered. The matching survives H-I, so its actual loss is at most j. Equation (1), with $R = 1$, yields

$r \geq k - 1 - j = a - 2 + j = r + j$,

a contradiction. Therefore every C_i is equimatchable.

Each boundary b_i is positive and their sum is at most $d_H(y) \leq D - 1$. The ambient degree sum is $D(2\mu_i + 1) = 2e(C_i) + b_i$, proving b_i odd and, by simplicity, $(2\mu_i + 1)(D - 2\mu_i) \leq b_i$. If the odd order $n = 2\mu_i + 1$ were at most D, then $n(D - n + 1) \geq D$, because $n(D - n + 1) - D = (n - 1)(D - n) \geq 0$. This would contradict $b_i < D$. Hence $n > D$; both n and D are odd, so $n \geq D + 2$ and $\mu_i \geq \lceil D/2 \rceil$. For distinct i, j put $b' = b_i + b_j$. The other a-2 positive boundaries give $b' \leq d_H(y) - (a - 2)$, and $d_H(y) \geq a$. It follows that $d_H(y) - b'/2 \geq (d_H(y) + a - 2)/2 \geq a - 1$.

We have derived every claimed necessary conclusion from simultaneous arrowing. Its failure means that at least one entire prescribed A_q member has an avoiding coloring. Lemma 41 extends any such coloring across the canonical pool or an arbitrary specified actual outer q-set, preserving all residual colors. Every application used complete actual suffixes, including their units and repeated sizes. Interchanging both full targets and their prescribed colors preserves the profile and every step, proving the symmetric assertion. □

Remark. One integer barrier slope consumes the entire allowed surplus. The endpoint equality then yields an actual low singleton separator and compatible factor-critical pieces, while independent high-exposure tests force each piece to be equimatchable.

Depends on. Lemma 41, Lemma 99, Lemma 143, Lemma 150, Lemma 170, Lemma 196, Lemma 206.

Lemma 219 (Degree twelve at excess at most six). *Use the finite integer certificate costs $c_i(H)$, matching number $\nu(H)$, excess $ex(H) = c_0(H) - \nu(H)$, and signed gap $gap(H) = \sum_{i \geq 0} c_i(H) - e(H)$ of Lemma 23. Let H be a nonempty connected 12-regular finite simple graph such that deleting every vertex leaves a perfect matching. Suppose $E = ex(H) \leq 6$, and write $|V(H)| = 2k + 1$, where $k = \nu(H) \geq 6$. Then either $gap(H) \leq 0$, or all the following hold:*

$k = 14$, $E = 5$, $gap(H) = 1$,

$c_0(H) = c_1(H) = c_2(H) = 19$, $c_{10}(H) = 9$, $c_{11}(H) = 5$,

$c_6(H) = 15$, $c_{12}(H) = 1$,

$c_3(H) + c_9(H) = 29$, $c_4(H) + c_8(H) = 30$, $c_5(H) + c_7(H) = 29$.

In particular, every such graph with $E = 6$ is LP-good, meaning $gap(H) \leq 0$, and the same is true when $E \leq 4$. Every such graph with $E = 5$ and order different from 29 is LP-good. The exceptional profile is a necessary condition for positive gap, not an existence assertion. There is no hereditary or source-minimality hypothesis. All graph orders, degrees, costs, capacities, budgets, matching cardinalities, indices, k, and E are integers in their displayed nonnegative or positive domains; gaps may be signed.

Proof. Write $c_i = c_i(H)$ and $n = 2k + 1$. Factor-criticality and the order bound on matchings give $\nu(H) = k$; simplicity gives $k \geq 6$. Regularity gives $e(H) = 12k + 6$. By Lemma 23, $E \geq 0$ and $c_i \leq k + E$. By Lemma 87, rows above 12 vanish, $c_{12} \leq 1$, $c_6 \leq k + 1$, and each pair {i,12-i} for $i = 3, 4, 5$ has cost sum at most $n + 1$. Thus whenever $c_j \leq k + a_j$ for $j = 1, 2$,

$gap(H) \leq E + a_1 + a_2 - 2k + 2 + c_{11} + c_{10}$. (1)

This enumerates exactly rows 0, 1, 2, 10, 11, 12, 6 and the three pairs.

We establish low-row bounds that retain the integer parity. For integers r in {10,11} and c in {0,1}, put

$h(r,c) = ceil(((r-1)k+2c)/(3r+1))$.

Set $g(r,c) = h(r,c) - 1$ if $h(r,c) \geq 1$, $k + h(r,c)$ is even, and

$(3r+1)(h(r,c)-1) \geq (r-1)(k-1) + 2c$;

otherwise set $g(r,c) = h(r,c)$. For $g = g(r,c)$ and $x_\cdot = floor((k+g)/2)$,

$(r-1)x_\cdot + c \leq 2rg$. (2)

For $g = h$ the left side minus the right side is at most

$((r-1)k + 2c - (3r+1)g)/2 \leq 0$.

For $g = h - 1$ the integer $k + g$ is odd, and that difference equals

$((r-1)(k-1) + 2c - (3r+1)g)/2 \leq 0$.

We claim

$E \leq g(11,1)$, $c_1 \leq k + g(11,0)$, $c_2 \leq k + g(10,0)$. (3)

To prove the first bound, construct an independent set starting with one vertex and adding a vertex at distance two from the set until it dominates H. Connectedness supplies such a vertex whenever the independent set is not dominating. Each addition shares an old neighbor and adds at most eleven new neighbors. A t-vertex prefix therefore has at most $11t + 1$ external neighbors. If its final cardinality is a, domination gives $n \leq 12a + 1$, so all prefix sizes $t \leq \lceil 2k/12 \rceil$ are available.

Put $g = g(11,1)$ and fix a nonnegative integer budget split $x + y = k + g$, arranged with $x \leq y$. If $y \geq k$, all edges may go into the larger exceptional part. Otherwise $x \leq x_\cdot$ and $y < k$. Set $t = x - 2g$. If $t \leq 0$, choose any x-set W and assign edges meeting W to the smaller exceptional part and the remaining edges to the larger part. Their matching numbers are at most x and y, since $n - x \leq 2y + 1$. If $t \geq 1$, (2) gives $11t + 1 \leq x$. Moreover $3x + t = 4x - 2g \leq 2k$, so $34t + 3 \leq 2k$. In particular the independent prefix of size t is available. Enlarge its external neighborhood to an x-set W disjoint from the prefix; there are $n - t = x + 2y + 1$ vertices outside it, so this is possible. All remaining edges are supported outside W and the independent prefix, on $2y + 1$ vertices. This gives the same two matching bounds. The bounded-degree parts are empty. Every split at total $k + g$ is thus certified, proving $c_0 \leq k + g$.

For $j = 1, 2$ put $r = 12 - j$, $g = g(r,0)$, $x = floor((k+g)/2)$, and $t = x - 2g$. If $t \geq 1$, (2) gives $rt \leq x$ and hence $(3r+1)t \leq 3x + t \leq 2k$. Since $r \geq 10$, $t \leq 2k/31 < n/13$. A maximum independent set dominates H, and each closed neighborhood has thirteen vertices, so $\alpha(H) \geq \lceil n/13 \rceil > t$. Also $3j \leq 13$. The final private-neighbor criterion of Lemma 54 therefore applies with trial excess g, proving $c_j \leq k + g$. It applies without an independent-set condition when $t \leq 0$. This proves (3).

We next give a fixed-cardinality sampling bound. For integers j in {1,2} and $0 \leq r \leq n - 1$, put $\theta = 12 - j$ and $m = \theta + 1$ when θ is even, $\theta + 2$ when θ is odd. Use $binom(A,B) = 0$ when the integers satisfy $A \geq 0$ and either $B < 0$ or $B > A$. Define

$Q_j(n,r) = r + (n-r)/binom(n-1,r) \cdot$

$[\sum_{i=0}^{j-1}$ binom(12,i)binom(n-13,r-i)

$+ binom(12,j)binom(n-13,r-j)/m]$. (4)

Then $c_{12-j} \leq floor(Q_j(n,r))$. Indeed choose uniformly an r-set U. Delete also every vertex outside U having fewer than j neighbors in U; call that additional set D, and put $Z = U$ union D. Every vertex outside Z has residual degree at most θ. If it has residual degree θ, it has exactly j neighbors in U. Let T′ be the set of vertices outside U with exactly j neighbors in U. The deletion inequality of Lemma 92 gives

$c_\theta \leq \lfloor r + |D| + |T'|/m \rfloor$

for every outcome. Conditional on v outside U, the probability of exactly i neighbors of v in U is

$binom(12,i)binom(n-13,r-i)/binom(n-1,r)$.

The expectation of $r + |D| + |T'|/m$ is therefore $Q_j(n, r)$. The finite nonempty sample space has an outcome not exceeding its expectation. Its actual deletion set proves (4), for all the required capacity and matching-budget splits.

For all orders we also use the Bernoulli version of Lemma 92. Its choices $j = 1$, $p = 1/5$ and $j = 2$, $p = 3/10$ give respectively

$q_1 = 1/5 + (4/5)^{13} + (12/13)(1/5)(4/5)^{12}$

$= 4247569949/15869140625 < 268/1000,$

$q_2 = 3/10 + (71/20)(7/10)^{11}$

$= 740390198753/2000000000000 < 371/1000,$

and $c_{11} \le \lfloor nq_1 \rfloor$, $c_{10} \le \lfloor nq_2 \rfloor$. The two strict differences are $10719477/31738281250$ and $1609801247/2000000000000$, respectively. Taking $a_1 = a_2 = E \le 6$ in (1) gives

$gap(H) < 20639/1000 - 361k/500.$ (5)

At $k = 28$ the right side is $423/1000 < 1$, and it decreases with k. Integrality proves goodness for every $k \ge 28$.

Now $6 \le k \le 27$. Since $n \le 55 \le 6 \cdot 12 + 15$, the odd-pair bound Lemma 199 saves one unit for each of {3,9} and {5,7}. If k is odd, Lemma 119 saves a further unit at the middle row: $c_6 = k + 1$ would give $c_6 + c_6 = n + 1$, so that fact would make c_6 odd, whereas $k + 1$ is even. Thus $c_6 \le k$ when k is odd. Write $o_k = 1$ for odd k and 0 for even k, and put

$E_0 = \min(6, g(11, 1))$, $a_1 = \min(E_0, g(11, 0))$, $a_2 = \min(E_0, g(10, 0))$.

For displayed choices r_1, r_2 put $b_j = floor(Q_j(n, r_j))$. Equations (1)-(4) now give

$gap(H) \le L_k := E_0 + a_1 + a_2 - 2k + b_1 + b_2 - o_k.$ (6)

Here is the full finite table. Each b_j is the exact floor in (4); all the other entries are exact evaluations of their displayed integer formulas.

k E_0 a_1 a_2 r_1 b_1 r_2 b_2 L_k

6 2 2 2 1 1 2 3 -2

7 2 2 2 2 2 3 3 -4

8 3 3 3 2 3 4 4 0

9 3 3 3 3 3 5 5 -2

10 3 3 3 3 4 5 6 -1

11 4 4 4 4 4 6 7 0

12 4 4 4 4 5 7 7 0

13 4 4 4 5 5 7 8 -2

14 5 5 5 5 6 8 9 2

15 5 5 5 5 7 8 10 1

16 5 5 5 6 7 9 10 0

17 6 5 5 6 8 10 11 0

18 6 5 5 7 8 10 12 0

19 6 6 6 7 9 11 13 1

20 6 6 6 7 9 11 13 0

21 6 6 6 8 10 12 14 -1

22 6 6 6 8 10 13 15 -1

23 6 6 6 9 11 13 16 -2

24 6 6 6 9 11 14 16 -3

25 6 6 6 10 12 14 17 -4

26 6 6 6 10 12 15 18 -4

27 6 6 6 10 13 16 19 -5

It remains to examine $k = 14, 15, 19$. At these ranks $a_1 = a_2 = E_0$. If $E \le E_0 - 1$, monotonicity gives c_1, $c_2 \le k + E$, reducing (6) by at least three. Since $L_k \le 2$, the gap is nonpositive. Hence positive gap forces $E = E_0$.

We improve c_{11} at these three ranks. A greedy dominating-set construction in any 12-regular n-vertex graph has undominated-count bound $u_0 = n$ and

$u_{i+1} = \max(0, u_i - ceil(13u_i/(n-i)))$.

To justify it, every currently undominated vertex has thirteen closed neighbors outside the i already selected vertices. Incidence counting supplies an unselected vertex dominating at least $ceil(13w/(n-i))$ of the w undominated vertices. The function $\max(0, w - ceil(13w/(n-i)))$ is nondecreasing on the nonnegative integers: when $13 \le n - i$ its ceiling increases by at most one per increment of w, and when $13 > n - i$ the function is identically zero. Induction therefore gives the stated upper counts, padding the selected set arbitrarily if domination is achieved earlier. The exact chains are

$n = 29 : 29, 16, 8, 4, 2, 0$;

$n = 31 : 31, 18, 10, 5, 2, 1, 0$;

$n = 39 : 39, 26, 17, 11, 7, 4, 2, 1, 0$.

Thus H has a dominating set of size at most $z_0 = 5, 6, 8$ respectively.

Choose a minimum dominating set Z with actual cardinality $z \le z_0$ and put $e = E_0 - 1$. Apply Lemma 151. Here $d = 12$, and at $k = 14, 15, 19$ respectively one has $e = 4, 4, 5$ and $t_0 = floor((2e-1)/10) = 0$. The first quantity $2floor((k+e)/2) - 4e - 1$ is 1, 1, 3, each at most 13. The second requirement in its condition (i) is $6e + 4 \le n + 12$, namely $28 \le 41$, $28 \le 43$, $34 \le 51$. The remaining requirement in (i) holds because $t_0 = 0$.

For condition (ii), put $a = z + 9 - 2e$. If $a \le 0$, its first alternative $2e + 3 \ge z + 12$ holds. If $a \ge 1$, it is at most 6, 7, 7 in the three rows. Consequently $12ceil(11/a) \ge 24$, whereas $n + z - 4e - 5$ is at most 13, 16, 22. These are the required strict inequalities. For (iii), $6e + 20$ is 44, 44, 50, greater than $n = 29, 31, 39$. For (iv), $8e + 11$ is 43, 43, 51, greater even than n and thus greater than n-z. All conditions hold for the same actual set Z. The cited theorem gives $c_{11} \le z \le z_0$.

Replacing b_1 by z_0 in (6) gives the bounds 1, 0, 0 at $k = 14, 15, 19$, respectively. This proves goodness except possibly $k = 14$, $E = 5$. In that case the gap is at most one. If it is positive, it equals one. The improved sum is

$\sum_{i \ge 0} c_i \le 19 + 19 + 19 + 29 + 30 + 29 + 15 + 9 + 5 + 1 = 175$,

while $e(H) = 174$. Each displayed term is the bound for one of the disjoint rows or pairs enumerated in (1). A gap of one makes the total equal 175, so every individual upper bound in this sum is an equality. These are precisely all the exceptional-profile identities in the statement. The preceding cases and (5) prove every asserted conclusion. □

Remark. Sampling a fixed number of vertices improves the two highest positive rows enough to avoid the former large structural classification. Only the degree-twelve graph on twenty-nine vertices can retain a positive gap at excess at most six.

Depends on. Lemma 23, Lemma 54, Lemma 87, Lemma 92, Lemma 119, Lemma 151, Lemma 199.

Lemma 220 (Exposed regular components shorten the natural plateau). *Let s, $t \geq 1$ be integers and let F_1, F_2 be star forests with nonincreasing positive integer edge-size lists $n_1,\dots, n_s$ and $m_1,\dots, m_t$, prescribed red and blue. Copies are ordinary subgraphs with vertex-disjoint target components. A graph arrows an ordered pair when every red-blue edge coloring contains its entire first forest in red or its entire second forest in blue. Put $M = s + t - 1$ and*

$$\ell_k = max\{n_i + m_j - 1 : 1 \leq i \leq s, 1 \leq j \leq t, i + j = k\}$$

for every integer $2 \leq k \leq s + t$. For each integer $0 \leq z \leq M$ let $P_z = \sum_{k=z+2}^{s+t}(\ell_k - 1)$, with empty sum zero. For $0 \leq z < M$ let A_z be all ordered nonempty actual suffix pairs after nonnegative integer leading cuts $\alpha < s$, $\beta < t$ with $\alpha + \beta = z$.

Fix an integer $0 \leq q < M$. Put $D = \ell_{q+2} - 1$, suppose $D \geq 6$ is even, and let R be the initial plateau length at $D + 1$ in this natural tail. Let H be a finite simple graph. Write e and ν for edge count and maximum matching cardinality, and put $\Phi = e - \nu$. Suppose the integers $\sigma \geq 0$ and $\delta = M - q - \nu(H) \geq 1$ satisfy $\Phi(H) \leq P_q + \sigma$.

Select $\kappa \geq 1$ distinct whole connected components of H, each odd, D-regular, and possessing a near-perfect matching, which leaves one vertex uncovered. Suppose at least one selected component C_0 has $\nu(C_0) \leq D$. Define the nonnegative integer

$$h_\cdot = \min(\kappa - 1, \delta - 1, R - 1).$$

If H arrows every member of A_q, then

$$R + \sigma \geq D/2 + h_\cdot.$$

More precisely, for every integer $0 \leq h \leq h_\cdot$, choose any h selected components other than C_0 and, in each, any vertex exposed by a near-perfect matching. Let I_h be these vertices and put $H_h = H - I_h$. Then $q + h < M$,

$$\nu(H_h) = \nu(H), \Phi(H_h) = \Phi(H) - hD,$$

$$P_{q+h} = P_q - hD, \text{ and } M - q - h - \nu(H_h) = \delta - h \geq 1.$$

The natural tail at $q + h$ still has degree D and has plateau length R-h. The component C_0 remains whole and unchanged. If H arrows every A_q member, H_h arrows every A_{q+h} member. These conclusions hold for every indicated choice; $h = 0$ means I_h is empty.

In particular, suppose also that integers $a \geq 4$ and $w \geq 2$ satisfy $D = 2(a + w - 1)$, $\sigma \leq a - 2$ and $R = \kappa + 1$. Simultaneous arrowing then implies $R \geq \delta + w$. If $\delta \geq a - 1$ as well, it implies $R \geq D/2$ and $\kappa \geq D/2 - 1$.

Failure of the first displayed necessary inequality supplies a coloring of H avoiding both entire prescribed targets of some A_q member. Every such coloring extends over the q canonical stars of sizes $\ell_2,\dots, \ell_{q+1}$, and across every specified actual q-vertex set X in any finite simple G with $G - X = H$, to avoidance of the unchanged full F_1, F_2, preserving all H-edge colors. No degree restriction is imposed on the other components, and only C_0 has a matching-rank upper bound. All sizes, counts, degrees, indices, deficits and allowances are integers in the stated domains. Unit stars, repeated sizes and simultaneous exchange of targets and colors are included.

Proof. We use the small-component theorem Lemma 186 and the full actual-vertex and canonical-pool extensions Lemma 41.

Fix any integer h with $0 \leq h \leq h_\cdot$. There are at least h selected components other than C_0 because $h \leq \kappa - 1$. In each choose any near-perfect matching and its exposed vertex. A near-perfect matching of an odd component is maximum: it has $(|V(C)| - 1)/2$ edges, the largest cardinality allowed by its order. Maximum matching cardinalities add over whole components, because every matching splits over those components and their maximum matchings can be united. The chosen matchings therefore combine with maxima on all other components to a maximum matching of H avoiding I_h. That matching survives deletion of I_h, and every matching in $H - I_h$ is a matching in H. Thus $\nu(H_h) = \nu(H)$.

The selected vertices lie in distinct whole components, so I_h is independent and every one of its vertices has degree D. Exactly hD edges have an endpoint in I_h: no edge is counted twice, since I_h is independent. Hence $\Phi(H_h) = \Phi(H) - hD$. The unchanged component C_0 is still a whole component of H_h and retains its near-perfect matching and bound $\nu(C_0) \leq D$.

The tail has R entries before its first possible change, with $1 \leq R \leq M - q$. Since $h \leq R - 1$, $q + h < M$ and precisely h entries of value D are removed from the potential sum. Thus $P_{q+h} = P_q - hD$ and the remaining

initial plateau has length $R - h \geq 1$ at the same degree D. The matching identity gives the residual deficit $\delta - h \geq 1$ since $h \leq \delta - 1$. In particular

$\Phi(H_h) \leq P_{q+h} + \sigma$. (1)

These arguments include $h = 0$, with no deletion.

Assume now that H arrows every member of A_q. Fix any legal deeper cuts alpha′, beta′ with $\alpha' + \beta' = q + h$. Choose $\alpha = \min(\alpha', q)$ and $\beta = q - \alpha$. Then $0 \leq \alpha \leq \alpha'$, $0 \leq \beta \leq \beta'$, $\alpha < s$ and $\beta < t$. The last comparison $\beta \leq \beta'$ follows from $q \leq \alpha' + \beta'$. The corresponding A_q member is nonempty. Inside that member the differences alpha′-alpha, beta′-beta are legal cuts totaling h, and its component budget is $M - q > h$.

If a coloring of H_h avoided this deeper pair, Lemma 41 applied to that entire A_q member and the actual h-set I_h would extend it to an avoiding coloring of H, a contradiction. Therefore H_h arrows every member of A_{q+h}. This transfer uses the same chosen vertex set and works for every deeper allocation, including zero cuts and $h = 0$.

Take $h = h_.$. The graph H_h and distinguished component C_0 now satisfy all hypotheses of Lemma 186: the full originals at the legal prefix $q + h$ have even degree $D \geq 6$ and plateau R-h; equation (1) gives their potential bound; delta-h is positive; and C_0 is a whole connected odd D-regular component with a near-perfect matching and matching cardinality at most D. That theorem would give avoidance of an A_{q+h} member if $R - h + \sigma < D/2$. Such avoidance contradicts the preceding family arrowing. Consequently $R - h + \sigma \geq D/2$, which is the claimed inequality.

For the specialization $R = \kappa + 1$ and $\kappa \geq 1$ imply $R \geq 2$ and

$h_. = \min(R - 2, \delta - 1)$.

Substitute $D/2 = a + w - 1$ and $\sigma \leq a - 2$ into the proved inequality. It gives

$R - 1 - h_. \geq w$. (2)

If $h_. = R - 2$, the left side is one, contrary to $w \geq 2$. Therefore $h_. = \delta - 1$, and (2) becomes $R - \delta \geq w$. This proves $R \geq \delta + w$. With $\delta \geq a - 1$ it yields $R \geq a + w - 1 = D/2$ and $\kappa = R - 1 \geq D/2 - 1$. No target-size interpretation of the auxiliary integers a, w was required for this arithmetic consequence.

If the necessary inequality fails, H cannot arrow every A_q member. This finite family is nonempty because $q \leq s + t - 2$, so negating its simultaneous arrowing gives a member with an avoiding coloring of H. Both stated full-original color-preserving extensions are Lemma 41. All intermediate transfers and the small-component theorem use entire actual suffixes; units and repeated sizes are retained. Exchanging both prescribed colors and full targets preserves the constructions and conclusions. □

Remark. Deleting exposed vertices from whole regular components preserves matching number and pays the same complete degree in both graph potential and the natural plateau. Keeping one small component and one unit of matching deficit forces a precise lower bound on how much plateau can remain.

Depends on. Lemma 41, Lemma 186.

Lemma 221 (A triangle in the exceptional zero-deficit sextic row). *All graphs are finite and simple. Write $V(G)$, $E(G)$, $\Delta(G)$, $\nu(G)$, and $\alpha(G)$ for vertex set, edge set, maximum degree, matching number, and independence number. A graph is well-covered if every inclusionwise maximal independent set has size $\alpha(G)$. For nonnegative integer capacities a_c, b_c, let $c_{a_c,b_c}(G)$ be the least nonnegative integer T such that every nonnegative integer pair $x + y = T$ admits an ordered partition of $E(G)$ into four parts with maximum degrees at most a_c, b_c and matching numbers at most x, y, respectively. Put $c_i(G) = \max_{a_c+b_c=i} c_{a_c,b_c}(G)$ and $ps(G) = |E(G)| - \nu(G) - \sum_{i\geq 1} c_i(G)$, over nonnegative integer indices and capacities.*

Let F be six-regular and well-covered, with $|V(F)| = 27$ and $\alpha(F) = 9$. Let δ be an integer with $0 \leq \delta \leq 2$. Assume $c_0(F) = 17$, $c_1(F) \geq 17$, $c_2(F) \geq 17 - \delta$, and that $E(F)$ has no partition into two parts each of matching number at most eight.

Let A be an induced subgraph and $J = F - V(A)$ its full induced complement. Suppose

$|V(A)| = 11 + \delta$, $\Delta(A) \leq 2$, $ps(A) = 0$,

$\nu(A) = c_1(A) = 1 + \delta$, $c_2(A) = 1$,

$\Delta(J) \le 3$, $ps(J) = 0$,

$\nu(J) = c_1(J) = 6 - \delta$, $c_2(J) = 6 - 2\delta$.

Then A has a whole triangle component. More precisely, either $\delta = 0$ and A is the disjoint union of a triangle and eight isolated vertices, or $\delta = 2$ and A is the disjoint union of a triangle, a four-cycle, and six isolated vertices. A whole component means an entire connected component; graph sums are vertex-disjoint unions. All orders, degrees, cardinalities, component counts, capacities, budgets, costs, and indices are integers in the stated nonnegative domains. No connectedness, factor-criticality, source-minimality, or deletion-process assumption is imposed.

Proof. Use Lemma 57 for degree-two costs and components, Lemma 88 for the exclusion of three- and six-vertex paths, Lemma 192 for whole-triangle contacts, and Lemma 17 for independence forced by a missing allocation.

The degree bound and zero slack give the component list

P_1, P_2, P_3, P_4, P_6, C_3, C_4, C_5, C_7

for A, where P_t and C_t denote a path and cycle on t vertices. Exactly one component is an odd cycle, since $c_2(A) = 1$. Let ℓ count all nonisolated odd-order components. Each has matching number at least one, and matching numbers add across components, so $\ell \le \nu(A) = 1 + \delta \le 3$.

Put $u = |V(J)| = 16 - \delta$ and $m = \nu(A) = 1 + \delta$. The missing pair (8,8) has total $u + m - 1 = 16$, and

$j = u - 8 = 8 - \delta \ge 6 \ge \ell + 1$.

Also $c_2(F) \ge 17 - \delta = u + 1$. Thus Lemma 88 applies to A and removes P_3 and P_6. The identity $c_1(A) = \nu(A)$, together with Lemma 57, says that A has no positive even-order path component. Consequently its only components are isolates and cycles of orders three, four, five, or seven. The same degree-two identities give

$\alpha(A) = |V(A)| - \nu(A) - c_2(A) = 9 = \alpha(F)$.

The hypotheses of Lemma 192 now hold for A: $c_0(F) = u + m = 17$, $c_1(F) \ge u + m$, $c_2(F) \ge u + 1$, and $j \ge 6 \ge 1 + 4$. It follows that any exterior vertex with exactly three A-neighbors has those neighbors in one whole A-triangle.

Suppose A has no triangle. Its unique odd cycle is then C_5 or C_7. If $\delta = 0$, its matching number one is too small for either. If $\delta = 1$, its matching number two forces exactly C_5 and seven isolates. If $\delta = 2$, its matching number three forces exactly C_7 and six isolates: C_5 would leave matching number one to be supplied by other components, while only isolates and C_4 are available and contribute zero and two, respectively. Thus there are just two cases.

Every J-vertex of J-degree three has exactly three neighbors in A, by six-regularity of F. Purity would give a triangle in A. Hence there is no such vertex. Since $\Delta(J) \le 3$, it follows that $\Delta(J) \le 2$ in both cases.

First let $\delta = 2$. The complement J has order fourteen, matching number four, zero slack, and $c_2(J) = 2$. The degree-two formula gives $\alpha(J) = 14 - 4 - 2 = 8$. Apply Lemma 17 with row J and exterior $V(A)$, of size thirteen. Every component of J is a path or cycle and has matching number half its order rounded down. The missing total satisfies $8 + 8 = 13 + 4 - 1$, and the retention reserve is $13 - 8 = 5$. Protect all odd-order path components by their odd-indexed vertices. The number of unprotected odd components is the number of odd cycles, namely two; thus the required inequality is $5 \ge 2 + 1$. That fact, with well-coveredness of F, gives $\alpha(J) = \alpha(F) = 9$, a contradiction.

It remains to treat $\delta = 1$. Here

$A = C_5$ disjoint union $7P_1$,

$|V(J)| = 15$, $\nu(J) = c_1(J) = 5$, $c_2(J) = 4$, $ps(J) = 0$.

The degree-two formulas imply that J has exactly four odd cycles, no positive even-order paths, and only components P_1, P_3, C_3, C_4, C_5, C_7. Four odd cycles already require matching number at least four. There is exactly one remaining matching unit. It is used either by lengthening one triangle to a pentagon or by one P_3. A four-cycle or a seven-cycle would require at least two further units. The vertex count therefore leaves exactly the following alternatives:

$J = C_5$ disjoint union $3C_3$ disjoint union P_1,

or $J = P_3$ disjoint union $4C_3$.

Both have $\alpha(J) = 15 - 5 - 4 = 6$.

For any maximum independent set I of J, let $B(I)$ be the set of vertices of A with no neighbor in I. A maximal independent extension of I in F exists because F is finite. Its size is nine by well-coveredness. No vertex of J outside I can be added, since I is already maximal in J. The extension therefore adds three vertices from $B(I)$, proving $|B(I)| \geq 3$ for every such I.

We contradict this by explicit finite random choices of I. On a triangle choose one vertex uniformly. If a prescribed set contains t of its vertices, the probability of avoiding that set is $(3 - t)/3$, which is at most $(2/3)^t$ for each integer $0 \leq t \leq 3$. On a pentagon with cyclic labels modulo five, choose uniformly one of the five independent pairs $\{i, i + 2\}$. For a prescribed t-element subset, the number of avoiding pairs is at most 5, 3, 2, 1, 0, 0 for $t = 0, 1, 2, 3, 4, 5$, respectively. For $t = 1$ each vertex belongs to two of the five pairs; for $t = 2$ the three remaining vertices span at least one cycle edge and hence contain at most two independent pairs; for $t = 3$ at most one pair remains; the other entries follow from cardinality. Dividing by five gives an avoidance probability at most $(2/3)^t$ in every case.

In the first alternative for J, make these choices independently on its four cycles and include its isolate. Their union is a maximum independent set I of size six. If a fixed A-vertex is adjacent to the included isolate, it is never in $B(I)$. Otherwise, independence of the choices on different components and the preceding avoidance bounds show that its probability of belonging to $B(I)$ is at most $(2/3)^t$, where t is its total number of J-neighbors. The five cycle vertices of A have four J-neighbors each, and its seven isolates have six each, by six-regularity. Linearity of expectation therefore gives

$E[|B(I)|] \leq 5(2/3)^4 + 7(2/3)^6 = 1168/729 < 3.$

This contradicts $|B(I)| \geq 3$ for every possible I.

In the second alternative for J, take both endpoints of P_3 and independently one uniform vertex from each of its four triangles. This is again a maximum independent set of size six. An A-vertex adjacent to either selected path endpoint has probability zero of being in $B(I)$. For any other A-vertex, at most one J-neighbor can lie in P_3, namely its unselected center. Thus a vertex with t J-neighbors has at least t-1 neighbors on the triangles, and its probability of being undominated is at most $(2/3)^{t-1}$. The same original degrees give

$E[|B(I)|] \leq 5(2/3)^3 + 7(2/3)^5 = 584/243 < 3,$

again a contradiction. These probabilities are over explicitly described finite sets; the argument uses no enumeration of ambient graphs.

Both triangle-free cases are impossible, so A contains a whole triangle. It has exactly one odd cycle, so every other nonisolated component is C_4. If w denotes the number of such four-cycles, matching additivity gives $1 + 2w = 1 + \delta$, or $\delta = 2w$. Since $0 \leq \delta \leq 2$, either $\delta = 0$, $w = 0$ or $\delta = 2$, $w = 1$. The order $11 + \delta$ then fixes the isolate counts to eight and six, respectively. This proves the stated alternatives. □

Remark. Purity makes a triangle-free row complement have degree at most two. The heptagon case violates a forced independence equality; the pentagon case leaves too few vertices undominated on average to extend to the required maximal independent set.

Depends on. Lemma 17, Lemma 57, Lemma 88, Lemma 192.

Lemma 222 (Degree-five coverage with at most one unit of slack). *Use the finite integer certificate costs c_i, matching number ν, maximum degree Δ, slack ps, and signed gap gap of Lemma 61. Let H be any finite simple graph with $\Delta(H) \leq 5$ and $ps(H) \leq 1$. Assume $gap(J) \leq 1$ for every proper induced subgraph J of H. Then a maximum matching of H covers every vertex of degree five, and $c_5(H) = 0$. The empty graph and isolated vertices are permitted. Degrees, costs, matching sizes, and slack bounds are integers in the displayed nonnegative domains. No hypothesis on $gap(H)$ itself, connectedness, regularity, or order is imposed.*

Proof. The row costs are nonincreasing. Indeed, from a capacity pair of total $i+1$ reduce one positive capacity by one. A certificate for the resulting pair of total i is valid for the original pair, so $c_{i+1} \le c_i$ after taking the maximum over pairs.

We first establish the zero-slack assertion needed inside the proof. Let B be a finite simple graph with $\Delta(B) \le 5$, $ps(B) = 0$, and $gap(J) \le 1$ for every proper induced subgraph J of B. When $d = \Delta(B)$ belongs to {4,5}, a matching covers every degree-d vertex. To prove this, suppose a connected component U has positive exposure for its degree-d vertices. Slack superadditivity Lemma 72 and nonnegativity Lemma 61 give $ps(U) = 0$. Choose a maximum-cardinality marked maximizing set X in U. It exists and has the properties in Lemma 107. If X is empty, U is odd-order d-regular and factor-critical, of exposure one. Odd degree $d = 5$ is ruled out by degree-sum parity; $d = 4$ is ruled out by Lemma 205, which gives positive slack. If X is nonempty, all components of U-X are proper induced subgraphs of B, so Lemma 202 gives $ps(U) \ge 1$, again a contradiction. Thus each component containing degree-d vertices has a covering matching, and their union covers them in B.

Repeatedly delete any current maximum-degree vertex while the degree is at least four. Every current graph has zero slack and inherits the proper-induced gap bound. Indeed, at a covered parent of degree d, write h for the matching-number loss and a_i for the loss in cost c_i. Lemma 61 gives h, a_i in {0,1}, $a_d = 0$, and

$0 - ps(B - v) = d - h - \sum_{i=1}^{d} a_i \ge 1 - h \ge 0.$

Nonnegative child slack forces $h = 1$ and child slack zero; then all d-1 losses a_1,..., a_{d-1} equal one. In particular $\nu - c_1$ is unchanged. The finite deletion process ends at maximum degree at most three, where Lemma 161 gives $c_1 \le \nu$. Thus $c_1(B) \le \nu(B)$ for every B in this zero-slack domain. This includes B initially of maximum degree at most three, by the same cited cubic bound.

Now suppose the conclusion for H fails. If every component containing degree-five vertices had a covering matching, the union of those matchings would cover them in H. Thus there is an uncovered connected component U of actual maximum degree five. Nonnegative and superadditive slack give $ps(U) \le ps(H) \le 1$.

Let A be the degree-five vertices of U, let $\kappa \ge 1$ be their minimum matching exposure, and choose X of maximum cardinality among sets maximizing $q_A(U - X) - |X|$. Here q_A counts odd components wholly contained in A. Lemma 107 gives the exact exposure and factor-critical components. The set X is nonempty: otherwise U itself would be odd-order five-regular, contradicting degree-sum parity. Put $b = |X| \ge 1$. Every component of U-X is a proper induced subgraph of H and has gap at most one.

Apply Lemma 202 with $D = 5$. Its two bounds give $ps(U) \ge 4(\kappa - 1) + 1$ and $ps(U) + c_5(U) \ge 5(\kappa - 1) + 2$. The top-row bound $c_5(U) \le \kappa$ from Lemma 104 therefore yields $ps(U) = \kappa = c_5(U) = 1$.

Write calQ for all components of U-X, calK for the odd components wholly contained in A, and Y for the union of the remaining components. There are $b + 1$ components in calK. For K in calK let $d(K)$ be its boundary size to X. Connectedness and degree-sum parity give positive odd $d(K)$. Choose a component Z in calQ with largest matching number m. No tie-breaking rule is required.

For every K in calK other than Z with $d(K) < 5$, its internal total defect at $\cap$ five is exactly $d(K)$, it is factor-critical, and it has gap at most one. Apply Lemma 179 to give supported credits $a_i(K)$, whose sum is max(5-d(K)-ps(K),0), with $a_1(K) \le 2$, $a_2(K) \le 1$, and all other credits zero. Give zero credits to every other component, including Z. For $d(K) \ge 5$ the required credit is zero by nonnegative slack.

Run the procedures of Lemma 104. If E denotes the sum of maximum credits at unfinished indices, then $0 \le E \le 3$. Define

J=e(U[X])+e_U(X, Y)+sum_(Q not in calK, Q!=Z)ps(Q)+sum_(K in calK, K!=Z)max(ps(K)-(5-d(K)),0),

where $e_U(X, Y)$ counts edges between those disjoint sets. Every term is nonnegative. Its exact retained-top inequality, with $ps(U) = c_5(U) = \kappa = 1$, gives

$ps(Z) + J + \sum_{Q \in calQ} c_5(Q) \le 2 + E - 5 + 1_{Z \in calK}(5 - d(Z))$. (1)

This follows directly by substituting $gap = ex - ps$ into the fact's gap inequality; its top correction equals $\sum_Q c_5(Q)$. The indicator is zero when Z is not marked.

If Z is not in calK, it lies in Y and connectedness gives $e_U(X, Y) \ge 1$. The left side of (1) is positive, while the right side is $E - 3 \le 0$, a contradiction. Thus Z belongs to calK. Put $d = d(Z)$; (1) implies $ps(Z) + J \le 2 + E - d$. (2)

If $d \geq 5$, then (2) and $E \leq 3$ force $d = 5$, $ps(Z) = 0$, $J = 0$ and $E = 3$. By Lemma 205, a factor-critical zero-slack Z is a singleton or one of C_3, C_5, C_7. For a cycle its cap-five defect is three times its order, at least nine, inconsistent with $d = 5$. Thus Z is a singleton and $m = 0$. Since Z has largest matching number, every component of U-X is a singleton. Each has $c_i = 0$ and supports no positive credit, contrary to $E = 3$.

Consequently d is one or three. Lemma 179 applied to Z gives $u = (5 - d)/2$, $ps(Z) \geq 1$, $ps(Z) = 4m + u - \sum_{i=1}^{4} c_i(Z)$, and $c_i(Z) + c_{5-i}(Z) \leq 2m + 1$ for $1 \leq i < 5$.

If the procedure at index one fails, an unabsorbed component Q has $\nu(Q) \geq L + 2 \geq b + c_1(Z) + 2$, where L is the current full interval length. Maximality of m gives $c_1(Z) \leq m - b - 2$. The costs of Z are nonincreasing, so $ps(Z) \geq u + 4(b + 2) > 5 - d$. This contradicts (2), whose right side is at most $2 + 3 - d = 5 - d$. Index one therefore finishes.

If index two fails, then $E \leq 1$ and (2) gives $ps(Z) \leq 3 - d$. Again failure and maximality of m give $c_2(Z) \leq m - b - 2$. Pair $c_1(Z)$ with $c_4(Z)$ and use $c_3(Z) \leq c_2(Z)$ to obtain

$ps(Z) \geq 4m + u - [(2m + 1) + 2(m - b - 2)] = u + 2b + 3.$

Since $u \geq 1$ and $b \geq 1$, this is at least six, whereas $3 - d \leq 2$. This is a contradiction. All higher-index credits are zero, so all procedures finish.

Now $E = 0$ in (2). The inequalities $ps(Z) \geq 1$ and $d \geq 1$ force $d = 1$ and $ps(Z) = 1$. Thus Z has exactly one vertex of internal degree four and every other vertex has degree five. It is nonregular and factor-critical. Its order is $2m + 1 > 5$ by Lemma 179. Deleting its degree-four vertex leaves a perfect matching covering all degree-five vertices of Z. Choose any degree-five vertex v. Factor-criticality gives $\nu(Z - v) = m$, and Lemma 61 gives $ps(Z) \geq ps(Z - v) + 1 \geq 1$. Thus $ps(Z - v) = 0$. This graph inherits the proper-induced gap bound from H and has maximum degree at most five. The zero-slack first-row argument above gives $c_1(Z - v) \leq \nu(Z - v) = m$. The single-vertex extension inequality Lemma 72 now gives $c_1(Z) \leq m + 1$.

Apply the supported-credit conclusion of Lemma 179 once more, now to Z itself with $D = 5$, $d = 1$, $ps(Z) = 1$. It requires total credit $5 - 1 - 1 = 3$ on indices one and two. Its first credit is at most $\max(c_1(Z) - m, 0) \leq 1$ and its second credit is at most one. Their sum is at most two, a contradiction.

No uncovered component U exists. Hence a matching of H covers all degree-five vertices. Lemma 107 guarantees a maximum matching attaining the same zero exposure. Finally Lemma 61, with degree $\cap$ five, gives $c_5(H) = 0$. □

Remark. A slack-one uncovered degree-five graph would force a one-boundary factor-critical component. Deleting a maximum-degree vertex leaves zero slack and limits its first-row overflow, while the component's supported credits require more overflow than its two available indices can provide.

Depends on. Lemma 61, Lemma 72, Lemma 104, Lemma 107, Lemma 161, Lemma 179, Lemma 202, Lemma 205.

Lemma 223 (Matching-envelope surcharge at a critical plateau). *All graphs are finite and simple. Let $s, t \geq 1$ be integers, and let F, J be star forests with full nonincreasing positive integer edge-size lists $n_1,\ldots, n_s$ and $m_1,\ldots, m_t$. Copies are ordinary prescribed-color subgraphs, with disjoint target components, F red and J blue. Put $M = s + t - 1$ and*

$\ell_k = max\{n_i + m_j - 1 : 1 \leq i \leq s, 1 \leq j \leq t, i + j = k\}$

for every integer $2 \leq k \leq s + t$. Fix an integer $0 \leq q < \min(s, t)$, put $K = M - q$ and $L_q = \sum_{k=q+2}^{s+t} \ell_k$, and assume

$D = \ell_{q+2} - 1 = \max(n_1, m_1) \geq 3.$

Let R be the length of the initial $D + 1$ plateau in this entire natural tail.

Let A_q contain every pair (F^{α}, J^{β}) with nonnegative integer cuts $\alpha < s$, $\beta < t$, $\alpha + \beta = q$, where superscripts delete leading whole components. For integers $1 \leq j \leq D$ put

$N_j = |\{i : n_i \geq j\}|$, $T_j = |\{i : m_i \geq j\}|$,

$W_j = \max(\{t + N_j - 1 : N_j > 0\} union \{s + T_j - 1 : T_j > 0\}),$

$E_q = \sum_{j=1}^{D} \max(W_j - q, 0)$.

Thus E_q is the complete envelope sum of both matching relaxations of all A_q members, where one target is replaced by one disjoint edge per component and the matching is put first by exchanging colors.

Write e, ν, τ for edge count, maximum matching cardinality and minimum vertex-cover cardinality. Suppose H arrows every A_q member and has maximum degree at most D. Put $S = \{v : deg_H(v) = D\}$, $\rho = \min_W |S \text{ minus } V(W)|$ over all matchings W of H, and $\delta = K - \nu(H)$.

If $\delta \geq 1$, put $\gamma = \tau(H) - K$. Then

$\gamma \geq 0$, $R \leq \rho \leq \tau(H) - \nu(H)$, $e(H) - E_q \geq \max(\gamma, 1, 2\rho - 2)$.

If $e(H) < L_q$, without a separate assumption on δ, then $\delta \geq L_q - e(H) \geq 1$ and

$L_q - E_q \geq \max(2, 2R - 1)$.

Consequently, if the full lists satisfy $L_q - E_q = R$, every finite simple graph H with maximum degree at most D and $e(H) < L_q$ admits one coloring avoiding both entire targets of some member of A_q. Every such coloring extends across any specified actual q-vertex set X in any outer graph G with $G - X = H$ to avoidance of both full originals, preserving every H-edge color. All units and lower components are retained. All counts, indices, degrees, matching sizes and cover sizes are integers in their stated domains; $q = 0$ and isolated vertices are allowed.

Proof. We first identify the complete matching envelope. Let Γ contain both matching relaxations of every actual A_q member, oriented with the matching first. Every member has positive matching demand p, a positive opposite component count c, and $p + c - 1 = K$. A matching-first pair whose opposite forest has $d > 0$ components of size at least j has exactly $p + d - 1$ profile entries at least j; when $d = 0$ its width is zero. This follows directly because its profile repeats its opposite head p times and then lists the other component sizes.

For the relaxation $((t - \beta)K2, F^{\alpha})$, its threshold-j width, if $N_j > \alpha$, is $t - \beta + N_j - \alpha - 1 = t + N_j - q - 1$. Otherwise its width is zero. Because $q < t$, the legal corner $(\alpha, \beta) = (0, q)$ attains this width whenever $N_j > 0$. The other orientation has maximum width $s + T_j - q - 1$, attained at (q,0). All these widths are positive when their threshold count is positive, since $q < \min(s, t)$. Thus the full nonincreasing envelope η_1,..., η_K has width $\max(W_j - q, 0)$ at level j and sum E_q. Its entries are positive, its head is D, and no entry exceeds D. Put $R_D = |\{i : \eta_i = D\}|$, so $1 \leq R_D \leq K$. Selecting one edge from every positive target component proves that H arrows every Γ member; exchanging colors is a bijection on all colorings.

The natural envelope of A_q is precisely ℓ_{q+2},..., ℓ_{s+t}. A candidate on local coordinate k is an original candidate with index sum $q + k + 1$. Conversely, for original indices i, j with $i + j = q + k + 1$, choose $\alpha = \min(q, i - 1)$, $\beta = q - \alpha$. Then $\alpha \leq i - 1$ and $\beta \leq j - 1$, since $i + j - 2 = q + k - 1 \geq q$. These are legal cuts, and the candidate occurs at local coordinate k. This proves both inequalities between the envelopes, including $k = 1$ and $k = K$.

Apply Lemma 99 to this complete natural family, of common budget K and critical $\cap D$. It gives

$\rho \geq R$, $e(H) - \nu(H) \geq L_q - K$. (1)

In particular if $e(H) < L_q$, then $\delta \geq L_q - e(H) \geq 1$. For the remainder assume $\delta \geq 1$ as in the first assertion.

Write $E = E_q$, $f = e(H) - E$, $\gamma = \tau(H) - K$. Lemma 21 applied to Γ gives $\gamma \geq 0$ and $f \geq \gamma$. If $e(H) = E$, fact Lemma 74 applies to this same common-budget family and gives $\nu(H) = K$, contrary to $\delta \geq 1$. Since f is a nonnegative integer, we have

$f \geq 1$. (2)

Define θ as the minimum matching number of a spanning edge subset incident with every vertex of S, and put $a = \alpha(H[S])$, the largest cardinality of an independent subset of S. These are the quantities in Lemma 210. We establish

$R_D \leq \theta \leq a \leq \nu(H) < K$, and $a - \theta \leq f - \gamma$. (3)

For $\theta \geq R_D$, choose an incidence edge subset with minimum matching number θ and make it inclusionwise minimal while retaining incidence with all S. Its rank remains θ by the defining minimum. Every one of its edges has an endpoint in S of degree one within that edge subset; otherwise the edge could be removed. Thus

no edge joins two vertices of degree at least two, and every nonempty component is a star. Their number is θ. Choose a Γ member (pK2,U) attaining the final D entry, and let r be its number of D-star components. Then $p + r - 1 = R_D$.

If $\theta < R_D$, color the edges of $\min(\theta, p-1)$ of those incidence stars red and every other H-edge blue. Red matching number is at most p-1. Fix a center in each remaining incidence star, choosing an endpoint for a single edge. A blue D-star center must be in S and must use its entire ambient neighborhood. It cannot belong to a red incidence star. In its remaining incidence star it is either the fixed center or a leaf adjacent to that center. Its copied star therefore contains that fixed center. Disjoint blue D-stars require different remaining incidence stars, of which there are at most r-1. The full blue U is impossible, contradicting arrowing. This proves $\theta \geq R_D$.

Take a maximum independent set I of H[S], of size a. Its incident edges cover S: maximality supplies an I-neighbor to every other S-vertex. Their matching number is at most a, so $\theta \leq a$. There is a matching saturating I. To prove this, choose a maximum matching in the bipartite graph between I and its complement. If it misses an I-vertex, follow alternating paths from all missed I-vertices. Let U and V be the reached sets in the two parts. No missed opposite vertex is reached, since reversing a shortest such path would augment the matching. Every reached opposite vertex is matched back into U, and every matched U-vertex has its partner in V. Hence $|U| > |V|$. All neighbors of U belong to V. Each U-vertex has degree D and each V-vertex has degree at most D, giving $D|U| \leq D|V|$, a contradiction. Therefore $a \leq \nu(H) < K$.

Deleting I removes D a edges. The actual-set bound in Lemma 21 gives

$D\ a - \sum_{i=1}^{a} \eta_i \leq f - \gamma$.

Since $a \geq \theta \geq R_D$, the left side is at least $a - R_D$: every entry after the initial R_D entries is at most D-1. Thus $a - \theta \leq a - R_D \leq f - \gamma$, proving (3).

We strengthen the marked cover bound using the arrowing hypothesis. Put $z = \tau(H) - \nu(H)$. Choose a maximum-cardinality set Y among the maximizers of $q_S(H - Y) - |Y|$, where q_S counts odd components wholly contained in S. It is inclusionwise maximal among maximizers. Lemma 107 gives a family calR of $q_S(H - Y) = |Y| + \rho$ odd factor-critical components and, with H_0 the union of all other components,

$\nu(H) = |Y| + \sum_{C \in calR} \mu_C + \nu(H_0)$, (4)

where $|V(C)| = 2\mu_C + 1$ and $\nu(C) = \mu_C$.

For each nontrivial C in calR put $\lambda_C = \mu_C - \alpha(C)$. These numbers are nonnegative: a maximum independent set omits some vertex of a connected graph with an edge, and a perfect matching after deleting that vertex pairs every independent-set vertex outside the set. Put $\Lambda = sum\ \lambda_C$, and let b count those C in calR whose internal incidence rank $\theta_D(C)$ equals $\alpha(C)$. By Lemma 210,

$\Lambda \geq b \geq \max(0, \theta + \rho - a), z \geq \rho + \Lambda$. (5)

Every counted component contributes at least one to Λ by that fact's odd-order bound, and no singleton is counted.

Let C_0 be a minimum vertex cover of H, let $C_Y = C_0$ intersect Y, and let t_0 count singleton components in calR whose vertex is not in C_0. In each nonsingleton component its cover part has size at least $\mu_C + 1 + \lambda_C$, because its complement is independent. Every other singleton contributes one. Summing these contributions gives at least

$\sum_{C \in calR} \mu_C + |Y| + \rho + \Lambda - t_0$.

Each omitted singleton has D neighbors, all in C_Y. Counting these edges at C_Y gives $D\ t_0 \leq D|C_Y|$. The part of C_0 in H_0 covers H_0. Adding its size, then subtracting (4), yields

$\tau(H_0) - \nu(H_0) \leq z - \rho - \Lambda$. (6)

The same difference is nonnegative because a cover meets each edge of a matching at a distinct vertex.

Matching and cover numbers add over component unions, by restricting and uniting optimal choices. Thus (6) bounds the cover excess of each nonrequired component by $z - \rho - \Lambda \leq z - \rho - b$. A nonsingleton required C has cover excess $1 + \lambda_C$. If C is counted by b, the other b-1 counted components contribute at least b-1 to Λ. If it is not counted, all b counted components are different from C. Therefore in either case

$\tau(C) - \nu(C) \leq \Lambda - b + 2 \leq z - \rho - b + 2$.

A singleton has both matching and cover numbers zero. Consequently every component of H-Y has cover excess at most

$l = z - \rho - b + 2 \geq 2$. (7)

Take any actual A_q member with red and blue component counts p, t, so $p + t - 1 = K$. We show that $\delta \geq l + 1$ would give an avoiding coloring. Write $y = |Y|$ and $\mu = \nu(H)$. If $\mu < p$, color all edges red. Otherwise $\mu \geq p$. If $p - 1 \leq y$, choose p-1 vertices of Y and color precisely their incident edges red. Red matching number is at most p-1; blue matching number is at most $y - (p-1) + \sum_C \nu(C) = \mu - p + 1 \leq t - 1$, by (4) and $\delta \geq l + 1$.

In the remaining case put $r = p - 1 - y > 0$. Order the positive-matching components of H-Y, and take a longest initial segment whose matching numbers sum to $w \leq r$. Since $\mu \geq p$, the total complementary matching number exceeds r. If $w = r$, color all edges meeting Y and all edges in the prefix red, and all others blue. The matching bounds are p-1 and $\mu - p + 1 \leq t - 1$.

If $w < r$, the next component C has $\mu_C > r - w =: x \geq 1$. Choose x vertices from a minimum cover B of C, possible since $|B| \geq \mu_C > x$. Color edges meeting Y, all edges in the prefix, and all edges inside C meeting those x chosen vertices red. Color the remaining edges blue. Red matching number is at most $y + w + x = p - 1$. The blue part of C is covered by the $|B| - x$ unchosen cover vertices, so blue matching number is at most

$\sum_C \nu(C) - w - \mu_C + \tau(C) - x \leq \mu - p + 1 + l \leq t - 1$.

Every edge is assigned, including all attachments to Y, and zero-matching components have no internal edges. Each entire positive star forest contains a matching of its component count. These colorings therefore contradict arrowing.

It follows that $\delta \leq l$. Equations (5)-(7) imply

$\gamma = z - \delta \geq \rho + b - 2 \geq \theta + 2\rho - a - 2$.

Using (3) gives $\gamma \geq 2\rho - (f - \gamma) - 2$, hence

$f \geq 2\rho - 2$. (8)

Together with (1)-(2) and $z \geq \rho$ from (5), this proves the first assertion.

If $e(H) < L_q$, (1) already supplied $\delta \geq 1$. Integrality and (2), (8) give

$L_q - E_q \geq 1 + e(H) - E_q \geq \max(2, 2\rho - 1) \geq \max(2, 2R - 1)$.

If $L_q - E_q = R$, common arrowing would therefore require both $R \geq 2$ and $R \geq 2R - 1$, which force $R \geq 2$ and $R \leq 1$. This is impossible. Negating common arrowing provides one actual suffix member and a coloring avoiding both its full targets.

For any specified outer q-set X, partition it according to that member's two legal cuts and apply Lemma 41 (II). It preserves all colors on H and colors every attachment and every internal X-edge, excluding both full originals even when a copied component would meet X through a leaf. At $q = 0$ this is the original coloring. This proves the extension and completes the theorem. □

Remark. A full matching family limits how far high-set independence exceeds incidence-cover rank. The marked separator turns each exposed high vertex into cover excess; allocating components between colors converts a positive matching deficit into a two-edge surcharge per exposed vertex.

Depends on. Lemma 21, Lemma 41, Lemma 74, Lemma 99, Lemma 107, Lemma 210.

Lemma 224 (Degree ten at excess at most five). *Use the finite integer certificate costs $c_i(H)$, matching number $\nu(H)$, excess $ex(H) = c_0(H) - \nu(H)$, and signed gap $gap(H) = \sum_{i \geq 0} c_i(H) - e(H)$ of Lemma 23. Let H be a nonempty connected 10-regular finite simple graph such that deleting every vertex leaves a perfect matching. Suppose $E = ex(H) \leq 5$, and write $|V(H)| = 2k + 1$, where $k = \nu(H) \geq 5$. Then either $gap(H) \leq 0$ or E, k, $c_8(H)$, $c_9(H)$ form one of the following six rows:*

$E\ k\ c_8(H)\ c_9(H)$

4 11 8 4

4 12 9 5

4 13 10 6

5 16 12 7

5 17 13 8

5 18 14 9.

In every positive-gap case one also has $gap(H) = 1$,

$c_0(H) = c_1(H) = c_2(H) = k + E$, $c_5(H) = k$, $c_{10}(H) = 1$,

$c_3(H) + c_7(H) = 2k + 1$, $c_4(H) + c_6(H) = 2k + 2$,

and $c_i(H) = 0$ *for every integer* $i \geq 11$.

Consequently every such graph with $E \leq 3$ *is LP-good, meaning* $gap(H) \leq 0$. *With* $E = 4$ *only orders 23, 25, 27 can have positive gap, and with* $E = 5$ *only orders 33, 35, 37 can have positive gap. The six profiles are necessary conditions, not existence assertions. No hereditary or source-minimality hypothesis is imposed. All graph orders, degrees, costs, capacities, budgets, matching cardinalities, indices, k, and E are integers in their displayed nonnegative or positive domains; gaps may be signed.*

Proof. Write $c_i = c_i(H)$ and $n = 2k + 1$. Factor-criticality gives $\nu(H) = k$; simplicity gives $k \geq 5$; regularity gives $e(H) = 10k + 5$. Lemma 23 gives $E \geq 0$ and $c_i \leq k + E$. Lemma 87 gives $c_i = 0$ for $i \geq 11$, $c_{10} \leq 1$, $c_5 \leq k + 1$, and $c_3 + c_7$, $c_4 + c_6 \leq 2k + 2$. Hence bounds $c_1 \leq k + a_1$ and $c_2 \leq k + a_2$ imply

$gap(H) \leq E + a_1 + a_2 - 2k + 1 + c_9 + c_8$. (1)

The sum enumerates exactly rows 0, 1, 2, 8, 9, 10, the middle row 5, and the pairs {3,7}, {4,6}.

We first bound the three low rows, retaining integer parity. For r in {8,9} and c in {0,1} put

$h(r, c) = ceil(((r - 1)k + 2c)/(3r + 1))$.

Set $g(r, c) = h(r, c) - 1$ if $h(r, c) \geq 1$, $k + h(r, c)$ is even, and

$(3r + 1)(h(r, c) - 1) \geq (r - 1)(k - 1) + 2c$;

otherwise set $g(r, c) = h(r, c)$. For $g = g(r, c)$ and $x. = floor((k + g)/2)$ one has

$(r - 1)x. + c \leq 2rg$. (2)

When $g = h$, the left side minus the right side is at most

$((r - 1)k + 2c - (3r + 1)g)/2 \leq 0$.

When $g = h - 1$, $k + g$ is odd, so the same difference equals

$((r - 1)(k - 1) + 2c - (3r + 1)g)/2 \leq 0$.

We claim

$E \leq g(9, 1)$, $c_1 \leq k + g(9, 0)$, $c_2 \leq k + g(8, 0)$. (3)

For the first claim grow an independent set by starting at any vertex and adding a vertex at distance two from the current set until it dominates H. When it does not dominate, connectedness supplies an edge from its closed neighborhood to the outside, so the addition is possible. Each new vertex shares an old neighbor and adds at most nine new neighbors. A prefix of size t therefore has at most $9t + 1$ external neighbors. If the terminal size is a, domination gives $n \leq 10a + 1$. Thus prefixes of every integer size $t \leq \lceil 2k/10 \rceil$ exist.

Put $g = g(9, 1)$. Fix a nonnegative integer split $x + y = k + g$ and exchange the exceptional-part names if necessary so that $x \leq y$. If $y \geq k$, all edges go into the larger exceptional part. Otherwise $x \leq floor((k + g)/2)$. Set $t = x - 2g$. If $t \leq 0$, choose any x-set W. Edges meeting W have matching number at most x, while the other edges are supported on $n - x \leq 2y + 1$ vertices and have matching number at most y. If $t \geq 1$, (2) gives $9t + 1 \leq x$. Also $3x + t = 4x - 2g \leq 2k$, so $28t + 3 \leq 2k$. The independent prefix of size t is therefore available. Enlarge its external neighborhood to an x-set W avoiding the prefix, possible because $n - t = x + 2y + 1 \geq x$. Every edge not meeting W is supported outside both W and the now isolated prefix, on $2y + 1$ vertices. The

same two matching bounds hold. The bounded-degree parts are empty. This covers every split at total $k+g$ and proves the excess bound in (3).

For $j = 1, 2$ put $r = 10 - j$, $g = g(r, 0)$, $x = floor((k+g)/2)$, and $t = x - 2g$. If $t \geq 1$, (2) gives $rt \leq x$ and then

$(3r+1)t \leq 3x + t \leq 2k$.

A maximum independent set dominates H, so $\alpha(H) \geq \lceil n/11 \rceil$. Since $r \geq 8$, one has $t \leq 2k/25 < n/11 \leq \alpha(H)$. Also $3j \leq 11$. The last assertion of Lemma 54, with trial excess g and $\theta = j$, applies because $(10-j)t = rt \leq x$, and proves $c_j \leq k + g$. When $t \leq 0$ that assertion has no independent-set requirement. This establishes (3).

For an all-order bound, use Lemma 92 at $(d, j, p) = (10, 1, 2/9)$ and $(10, 2, 1/3)$. Its two coefficients are

$q_1 = 2/9 + (7/9)^{11} + (20/99)(7/9)^{10}$

$= 104109355975/345191655699 < 304/1000$,

$q_2 = 1/3 + (2/3)^9(29/9)$

$= 73897/177147 < 418/1000$.

The respective positive differences from the displayed upper fractions are

103613419687/43148956962375 and 75223/88573500.

Thus $c_9 \leq \lfloor nq_1 \rfloor$ and $c_8 \leq \lfloor nq_2 \rfloor$. In (1) take $a_1 = a_2 = E \leq 5$ to obtain

$gap(H) < 8361/500 - 139k/250$. (4)

For $k = 29$ the right side is $299/500 < 1$ and it decreases thereafter. Since the gap is an integer, H is LP-good for every $k \geq 29$.

Now $5 \leq k \leq 28$, so $n \leq 57 < 6 \cdot 10 + 17$. Lemma 199 applies both to $(\theta, \theta') = (3, 7)$ and to (5,5). It gives

$c_3 + c_7 \leq 2k + 1$, $2c_5 \leq 2k + 1$,

and hence $c_5 \leq k$ at both parities of k. These save two units in (1).

We give two finite-order top bounds. First fix a vertex v and a perfect matching M of H-v. The graph H-M has degree 10 at v and degree 9 elsewhere. Apply Lemma 37 with $d = 10$. With

$w_0 = 2k - 10$, $w_{i+1} = \max(0, w_i - ceil(10w_i/(2k-i)))$,

let s be the first zero index. The resulting set Z contains v, dominates H-M, and has size at most $b_9 := 1 + s$. Lemma 49 with $D = 10$, $j = 1$ therefore gives $c_9 \leq b_9$ using this same M and Z.

For the second bound, use $binom(A, B) = 0$ for integers $A \geq 0$ when $B < 0$ or $B > A$. For each integer r with $0 \leq r \leq n - 1$ define

$Q(n, r) = r + (n - r)/binom(n-1, r) \cdot$

$[binom(n-11, r) + 10binom(n-11, r-1) + 5binom(n-11, r-2)]$.

Then $c_8 \leq floor(Q(n, r))$. To prove this, choose uniformly an r-set U. Let D be the vertices outside U with at most one neighbor in U and put $Z = U$ union D. Every vertex outside Z has residual degree at most eight. Every vertex of residual degree eight has exactly two neighbors in U; let T′ be the set of all vertices outside U with exactly two such neighbors. The deletion inequality in Lemma 92, with $\theta = 8$ and $m_\theta = 9$, gives

$c_8 \leq \lfloor r + |D| + |T'|/9 \rfloor$

for each outcome. Conditional on a vertex lying outside U, its number of U-neighbors has probabilities $binom(10, i)binom(n-11, r-i)/binom(n-1, r)$. Since $binom(10, 2)/9 = 5$, the expected value inside the floor is $Q(n, r)$. Some outcome of this finite sample space has value at most its expectation. Its actual vertex set proves the bound for the integer certificate cost.

Put $E_0 = \min(5, g(9, 1))$, $a_1 = \min(E_0, g(9, 0))$, $a_2 = \min(E_0, g(8, 0))$. For the sample size r in each row below put $b_8 = floor(Q(n, r))$. The improved form of (1) is

$gap(H) \leq L_k := E_0 + a_1 + a_2 - 2k - 1 + b_9 + b_8$. (5)

Here is the complete finite table. The b_9 entries are $1 + s$ from the stated recurrence, and each b_8 is the exact floor of the stated rational expression.

k E_0 a_1 a_2 r b_8 b_9 L_k

5 2 2 2 3 3 1 -1

6 2 2 2 4 4 2 -1

7 2 2 2 4 4 3 -2

8 3 3 3 5 5 3 0

9 3 3 3 6 6 4 0

10 3 3 3 6 7 4 -1

11 4 4 4 7 8 5 2

12 4 4 4 8 9 6 2

13 4 4 4 8 10 6 1

14 5 4 4 9 10 6 0

15 5 4 4 10 11 7 0

16 5 5 5 10 12 7 1

17 5 5 5 11 13 8 1

18 5 5 5 12 14 9 1

19 5 5 5 12 15 9 0

20 5 5 5 13 15 10 -1

21 5 5 5 14 16 10 -2

22 5 5 5 14 17 11 -2

23 5 5 5 15 18 11 -3

24 5 5 5 16 19 12 -3

25 5 5 5 16 20 12 -4

26 5 5 5 17 20 13 -5

27 5 5 5 18 21 13 -6

28 5 5 5 18 22 14 -6

Only $k = 11, 12, 13, 16, 17, 18$ have positive L_k. At each of these ranks $a_1 = a_2 = E_0$. If $E \le E_0 - 1$, the three rows 0, 1, 2 together decrease by at least three from (5), so the gap is nonpositive. Thus positive gap forces $E = 4$ for $k = 11, 12, 13$ and $E = 5$ for $k = 16, 17, 18$.

We improve c_9 by one at $k = 11, 12$ with $E = 4$. A greedy dominating set in a 10-regular n-vertex graph has undominated-count bounds

$u_0 = n$, $u_{i+1} = \max(0, u_i - ceil(11u_i/(n-i)))$.

Indeed every undominated vertex has its eleven closed neighbors outside the i selected vertices. Incidence counting supplies an unselected vertex covering at least $ceil(11w/(n-i))$ of the w currently undominated vertices. The map $w \to \max(0, w - ceil(11w/(n-i)))$ is nondecreasing on nonnegative integers: if $11 \le n - i$ its ceiling increases by at most one for each increment of w, and if $11 > n - i$ the map is identically zero. Induction therefore gives the printed count bound. The exact two chains are

$n = 23$: 23, 12, 6, 2, 0; $n = 25$: 25, 14, 7, 3, 1, 0.

They give dominating sets of sizes at most $z_0 = 4, 5$, respectively.

Choose a minimum dominating set Z and write $z = |Z| \le z_0$. Check Lemma 151 with $d = 10$, $E = 4$, $e = 3$, $t_0 = \lfloor 5/8 \rfloor = 0$. Its first condition has

$2floor((k+3)/2) - 13 = 1$ for both $k = 11, 12$,

which is at most 11; also $6e + 4 = 22 \le n + 10$ and $t_0 = 0 \le ceil((n-1)/10)$.

For its second condition the denominator $z + d - 3 - 2e = z + 1$ is positive and at most six. Thus

$10ceil(9/(z+1)) \ge 20$,

whereas $n + z - 4e - 5 = n + z - 17$ is at most 10, 13 in the two cases.

The remaining strict conditions are $6e + d + 8 = 36 > n$ and $8e + 11 = 35 > n - z$. All hypotheses hold for the same Z. Its matching and certificate conclusion gives $c_9 \le z \le 4, 5$.

After these two improvements every remaining bound in (5) is one. A positive gap must therefore equal one. In each of its six rows, equality holds in the sum of the individual bounds for rows 0, 1, 2, 8, 9, 10, 5 and the two pairs. Each cost and pair has its own upper bound; if any were strict, the total gap would be at most zero. Thus all are equalities, giving exactly every profile identity in the statement. The finite table and (4) cover all integer $k \ge 5$ and prove the claimed conclusions. □

Remark. Marked greedy domination and fixed-cardinality parity correction leave only six exact small profiles. Oddness of the middle threshold supplies a full unit at both parities of the matching number.

Depends on. Lemma 23, Lemma 37, Lemma 49, Lemma 54, Lemma 87, Lemma 92, Lemma 151, Lemma 199.

Lemma 225 (Regular components at potential surplus at most one). *Let mathcal A be a finite nonempty family of ordered pairs of nonempty star forests. For a member γ, let s_γ, t_γ be its positive integer component counts and let its positive nonincreasing integer component edge-size lists be n_1^γ,..., $n_{s_\gamma}^\gamma$ and m_1^γ,..., $m_{t_\gamma}^\gamma$. Put*

$N = \max_\gamma(s_\gamma + t_\gamma - 1)$,

$\ell_h = max\{n_i^\gamma + m_j^\gamma - 1 : \gamma \in mathcal A, i + j = h, 1 \le i \le s_\gamma, 1 \le j \le t_\gamma\}$

for $2 \le h \le N + 1$. Define $D = \ell_2 - 1 \ge 3$, $P = \sum_{h=2}^{N+1}(\ell_h - 1)$, and let R be the number of initial entries of this envelope equal to $D + 1$.

Let H be finite simple, with $\Delta(H) \le D$, and suppose every red-blue coloring of H has a red copy of the entire first forest or a blue copy of the entire second forest for each prescribed member of mathcal A. Copies are ordinary subgraphs. Write ν for maximum matching cardinality, put $\sigma = e(H) - \nu(H) - P$, and let ρ be the minimum number of degree-D vertices left uncovered by a matching. Then σ is a nonnegative integer,

$R \le \rho \le R + \sigma$,

and every maximum matching of H leaves between R and $R + \sigma$ degree-D vertices uncovered.

A terminal maximizing triple is a choice γ, i, j with $i + j = R + 1$ and $n_i^\gamma + m_j^\gamma = D + 2$; denote its sizes by $a = n_i^\gamma$, $b = m_j^\gamma$. If σ is 0 or 1, every such triple has either both a, b positive even integers, or D is odd and $\{a, b\} = \{2, D\}$.

Suppose additionally that σ is 0 or 1 and at least one terminal maximizing triple has $a, b \ge 3$. Let κ count the whole connected components of H that are D-regular and have odd order, and let Q be the union of all remaining connected components. Then D is even and $D \ge 6$,

$R \le \kappa \le R + \sigma$.

Each counted odd regular component has a near-perfect matching, meaning a matching leaving exactly one vertex uncovered; every whole even D-regular component has a perfect matching. Every maximum matching of Q leaves at most $R + \sigma - \kappa \le \sigma$ vertices of ambient H-degree D uncovered. For each terminal maximizing triple with $a, b \ge 3$, Q admits an edge coloring with red maximum degree at most a-1 and blue maximum degree at most b-1; this coloring may depend on the triple. No edge-count upper bound or matching-deficit hypothesis is imposed.

Proof. The family and its envelope satisfy exactly the hypotheses of fact Lemma 99. In particular the envelope is defined and has $1 \le R \le N$: a member attaining budget N supplies each diagonal, and decreasing an index greater than one from a later diagonal proves that the envelope is nonincreasing. That fact gives nonnegative integral σ, the interval $R \le \rho \le R + \sigma$, and the same interval for the number of degree-D vertices exposed by every maximum matching. These assertions do not require $\sigma \le 1$.

Now assume σ in {0,1}, and fix any terminal maximizing triple γ, i, j. Each of the two whole targets of γ occurs uncolored in H: coloring every edge red forces its first target, while coloring every edge blue forces its second target. Thus a, $b \le \Delta(H) \le D$. Since $a + b = D + 2$, both a, b are at least two. Put $A = a - 1$ and $B = b - 1$, which are positive integers with $A + B = D$, and let μ be the minimum total degree overflow for these capacities. Lemma 143 applied to member γ and its actual indices i, j gives

$\mu \ge i + j - 1 = R$.

If a, b were both odd, the capacities A, B would both be even. Lemma 133 would give an edge coloring of H with maximum red degree at most A and maximum blue degree at most B. It would contain neither a red a-edge star nor a blue b-edge star, and would therefore avoid both entire targets of γ. This contradicts its arrowing assumption. Hence a, b are not both odd.

If a, b have opposite parity, write $e.$ for the even size and $o.$ for the odd size. Then D is odd. The odd capacity is $e. - 1$ and the positive even capacity is $o. - 1$. The odd-degree integer-slope bound in fact Lemma 196 gives

$\rho \ge \mu + (e. - 2)ceil(\mu/(o. - 1))$.

Since $\mu \ge R \ge 1$, the ceiling is at least one. The upper bound $\rho \le R + \sigma$ and $\mu \ge R$ imply $e. - 2 \le \sigma \le 1$. As $e.$ is an even integer at least two, $e. = 2$. The other size is then D, which is odd. This proves the asserted alternatives for every terminal triple.

Assume the joint additional hypothesis that some terminal triple has a, $b \ge 3$. The proved alternatives force both sizes of every such triple to be even and at least four. In particular D is even and at least six. Fix one of these triples; its capacities A, B are odd and at least three. Lemma 196 gives $\mu \ge \kappa$ and the alternative

$\mu = \kappa$ or $\rho \ge \mu + \min(A, B) - 1$.

The second possibility would give $\rho \ge R + 2$, contrary to $\rho \le R + \sigma \le R + 1$. Therefore $\mu = \kappa$. Since $\mu \ge R$ and every matching leaves at least one degree-D vertex uncovered in each counted odd regular component, we obtain

$R \le \kappa \le \rho \le R + \sigma$.

In a whole odd regular component C, a maximum matching leaves $|V(C)| - 2\nu(C)$ vertices uncovered. This number is positive and odd. If it were at least three for one counted component, componentwise maximum matchings, whose union is maximum in H, would leave at least $\kappa + 2 \ge R + 2$ degree-D vertices uncovered. This contradicts the bound $R + \sigma \le R + 1$. Thus each counted component has a near-perfect matching. If a whole even D-regular component had no perfect matching, its maximum matching would leave at least two vertices uncovered, in addition to the at least κ uncovered vertices of the counted odd components. This gives the same contradiction and proves its perfect matching.

Take any maximum matching of Q and combine it with near-perfect matchings of the κ removed components. Matchings in whole connected components are independent choices, so this union is maximum in H. Exactly κ of its uncovered degree-D vertices lie in the removed components. The general upper bound $R + \sigma$ therefore leaves at most $R + \sigma - \kappa$ for Q. Since $\kappa \ge R$, this number is at most σ. The maximum matching of Q was arbitrary.

Finally repeat the argument $\mu = \kappa$ for any terminal triple with a, $b \ge 3$, using its prescribed capacities $A = a - 1$, $B = b - 1$. Minimum overflow is additive over whole components, because both the coloring choices and their objective sums separate. Each counted odd regular component has positive overflow: zero overflow would force red degree A at each vertex, since the two capacities sum to its ambient degree D, but the red degree sum $A|V(C)|$ would then be odd. Overflow is a nonnegative integer, so each such component contributes at least one. These κ contributions exhaust $\mu = \kappa$. Every component of Q therefore has minimum overflow zero. Choosing a minimizing coloring on each component of Q gives the required red and blue degree bounds. This proves the conclusion separately for every eligible triple, without requiring a single coloring to serve all triples. Every step after the general exposure interval used only the explicitly stated joint low-surplus and nonbinary hypotheses; no edge-budget or matching-deficit assumption was introduced. □

Remark. At surplus zero or one the integer barrier slope of a nonregular odd-capacity obstruction is too costly. All overflow is accounted for by whole odd regular components, and arbitrary maximum matchings and complementary colorings follow from component additivity.

Depends on. Lemma 99, Lemma 133, Lemma 143, Lemma 196.

Lemma 226 (Subcubic hosts and suffix tails of height at most four). *Let $s, t \geq 1$ be integers, let $n_1 \geq \cdots \geq n_s \geq 1$ and $m_1 \geq \cdots \geq m_t \geq 1$ be integer lists, and let F_1, F_2 be their vertex-disjoint unions of stars with those edge sizes. Prescribe F_1 red and F_2 blue. Copies are ordinary subgraphs with vertex-disjoint components within each target. A finite simple graph arrows an ordered pair when every red-blue edge coloring contains its entire first target in red or its entire second target in blue. Write $e(H)$ and $\Delta(H)$ for edge count and maximum degree; an edgeless graph has maximum degree zero.*

Put $M = s + t - 1$,

$\ell_k = max\{n_i + m_j - 1 : 1 \leq i \leq s, 1 \leq j \leq t, i + j = k\}$ for $2 \leq k \leq s + t$,

$L = \sum_{k=2}^{s+t} \ell_k$,

$T_q = \sum_{k=2}^{q+1} \ell_k$, and $L_q = L - T_q$ for integers $0 \leq q \leq M$.

For every integer $0 \leq q < M$ put $K = M - q$ and let A_q consist of all ordered nonempty actual suffix pairs $(F_1^{\alpha}, F_2^{\beta})$, with integer cuts $0 \leq \alpha < s$, $0 \leq \beta < t$ and $\alpha + \beta = q$; the superscripts delete exactly the indicated leading whole components. Let S_q be the disjoint canonical prefix stars with sizes ℓ_2,..., ℓ_{q+1}, and let U_q be the disjoint canonical tail stars with sizes ℓ_{q+2},..., ℓ_{s+t}, all on fresh vertices.

(I) Every finite simple H with $\Delta(H) \leq 3$ and $e(H) < L_q$ has a coloring avoiding both entire targets of some member of A_q. Equivalently, every such subcubic H arrowing every member of A_q has at least L_q edges.

(II) If $\ell_{q+2} \leq 4$, the minimum edge count of an unrestricted finite simple graph arrowing every member of A_q equals L_q, attained by U_q. Every H with fewer than L_q edges has a coloring avoiding both entire targets of some member of A_q.

Every avoiding coloring supplied in (I) or (II) extends, preserving all H-edge colors, over the disjoint prefix S_q and across any specified actual q-vertex set X in any finite simple G with $H = G - X$, to avoidance of both unchanged full originals. Vertex deletion is induced.

(III) Suppose G arrows (F_1, F_2) and $e(G) < L$. For every actual vertex set X with $q = |X| < M$, put $H = G - X$. If $e(G) - e(H) \geq T_q$ and $\Delta(H) \leq \ell_{q+2} - 1$, then $e(H) < L_q$ and $\Delta(H) = \ell_{q+2} - 1 \geq 4$. In particular $\ell_{q+2} \geq 5$. This assertion holds for every such X.

All counts, sizes, indices, cuts, degrees and edge budgets are integers in their displayed nonnegative or positive domains. Empty prefixes, isolated host vertices, $q = 0$, $K = 1$, all repeated sizes and every unit component are included. No restriction on either original largest size, component multiplicity, matching deficit or potential is imposed.

Proof. Use the actual-family critical-degree theorem Lemma 177 and the pool and vertex extension theorem Lemma 41.

First the positive diagonal sequence is nonincreasing. Given valid indices i, j with $i + j = k + 1 \geq 3$, at least one index exceeds one. Decreasing that index by one gives a valid pair on diagonal k, and nonincreasing input lists ensure that its value is at least $n_i + m_j - 1$. Taking maxima gives $\ell_k \geq \ell_{k+1}$. Each diagonal in the stated range has a valid pair, and its value is positive.

We prove the lower bound in (II) by induction on the positive integer $K = M - q$, uniformly over all original lists and legal q with $\ell_{q+2} \leq 4$. Suppose H arrows every member of A_q and $e(H) < L_q$. Put $r = \ell_{q+2}$, so $1 \leq r \leq 4$.

If $K = 1$, then $L_q = r$. Thus $\Delta(H) \leq e(H) \leq r - 1$. The first assertion of Lemma 177 applies with $D_q = r - 1 \leq 3$ and rules out this common arrowing. This proves the base case.

Let $K \geq 2$ and assume the lower bound at every smaller remaining component budget. If $\Delta(H) \leq r - 1$, the same assertion again gives a contradiction. Otherwise choose a vertex v with degree at least r. Then

$e(H - v) = e(H) - d_H(v) < L_q - r = L_{q+1}$.

Here $q+1 < M$ and the next tail height is at most $r \le 4$.

We check common arrowing on this same H-v. Fix any legal cuts α, β with $\alpha+\beta = q+1$. If $\alpha \ge 1$, the pair $(F_1^{\alpha-1}, F_2^{\beta})$ belongs to A_q and is arrowed by H. Its red forest has at least two components because $\alpha < s$. Apply part (II) of Lemma 41 to this pair with the actual one-vertex set {v} and local cuts (1,0). These cuts are legal: the local remaining component budget before deletion is $K \ge 2$. If H-v had a coloring avoiding $(F_1^{\alpha}, F_2^{\beta})$, that theorem would extend it to a coloring of H avoiding the preceding pair, a contradiction. Thus H-v arrows the chosen A_{q+1} member. If $\alpha = 0$, then $\beta \ge 1$; apply that same argument to $(F_1, F_2^{\beta-1})$ with local cuts (0,1). Its blue forest has at least two components because $\beta < t$. This proves common A_{q+1} arrowing for every legal pair. The induction hypothesis at budget K-1 contradicts $e(H-v) < L_{q+1}$. The lower bound in (II) follows.

To prove attainment, the family A_{M-1} has the single pair of stars with edge sizes n_s, m_t. The star of size $\ell_{M+1} = n_s + m_t - 1$ arrows that pair: otherwise its red and blue edge counts would be at most $n_s - 1$ and $m_t - 1$, whose sum is too small. The pool equivalence in Lemma 41 at depth M-1 therefore shows that the full canonical star union arrows the originals. This union is U_q disjoint union S_q for every q. The same equivalence at depth q proves that U_q arrows every member of A_q. It has exactly L_q edges. This proves attainment, including $M = 1$, and completes (II).

Now suppose H is subcubic and $e(H) < L_q$. If $r = \ell_{q+2} \le 3$, (II) supplies the required avoiding coloring. If $r \ge 4$ and H arrows every A_q member, then $\Delta(H) \le 3 \le r-1$. The first assertion of Lemma 177 applies and gives $\Delta(H) = r - 1 \ge 4$, contradicting the subcubic bound. Hence common arrowing fails. Its negation is the existence of one member and one coloring avoiding both entire prescribed targets, which proves (I).

For every resulting avoiding coloring, parts (I) and (II) of Lemma 41 give respectively its extension over S_q and across the specified actual X. They preserve every edge color in H, retain all original components, and allow the chosen cuts to be zero. Thus these extensions apply to the same coloring and same actual suffix member supplied above.

Finally impose (III). Part (II) of Lemma 41 implies that $H = G - X$ arrows every A_q member, since G arrows the originals and $q < M$. The assumed physical edge payment gives

$$e(H) \le e(G) - T_q < L - T_q = L_q.$$

Together with $\Delta(H) \le \ell_{q+2} - 1$, these are precisely the hypotheses of the first assertion of Lemma 177. It yields $\Delta(H) = \ell_{q+2} - 1 \ge 4$. The reasoning used an arbitrary specified X, proving the universal quantifier. No vertex set or residual graph was replaced. □

Remark. A high-degree vertex pays the next small tail height; otherwise the exact critical-degree theorem rules out common arrowing. Canonical pool equivalence supplies attainment and the unchanged-color extensions.

Depends on. Lemma 41, Lemma 177.

Lemma 227 (Exclusion of sextic processes with excess at most three). *All graphs are finite and simple, and vertex deletion is induced. Write $V(G)$, $E(G)$, $\Delta(G)$, $\nu(G)$, and $\alpha(G)$ for vertex set, edge set, maximum degree, matching number, and independence number. A graph is well-covered if every inclusionwise maximal independent set has cardinality $\alpha(G)$, and factor-critical if deleting any vertex leaves a perfect matching.*

For nonnegative integer capacities a_c, b_c, define $c_{a_c,b_c}(G)$ as the least nonnegative integer T such that every nonnegative integer split $x + y = T$ admits an ordered partition of $E(G)$ into four parts with maximum degrees at most a_c, b_c and matching numbers at most x, y, respectively. Put $c_i(G) = \max_{a_c+b_c=i} c_{a_c,b_c}(G)$ and $ps(G) = |E(G)| - \nu(G) - \sum_{i\ge1} c_i(G)$, over nonnegative integer capacities and indices.

For $c_0(G) \ge 1$, call an integer x with $0 \le x \le c_0(G) - 1$ missing if $E(G)$ admits no two-part partition with matching numbers at most x and $c_0(G) - 1 - x$. Let $tr(G)$ be the greatest number of consecutive missing integers, with value zero if none are missing.

There is no connected six-regular factor-critical well-covered graph F of order $2k+1$, for an integer $k \ge 1$, satisfying the following conditions. Put $a = \alpha(F)$ and $E = c_0(F) - k$. Require $1 \le E \le 3$ and $tr(F) + E \ge 5$.

(a) For every maximum independent set S of F,

$c_i(F-S) = c_i(F) - a$ *for every integer* $1 \le i \le 5$,

and $c_0(F) = a + \nu(F-S) + ps(F-S)$.

(b) A complete process chooses any such S, then repeatedly deletes a current maximum-degree vertex while that degree is at least three, stopping at a graph of maximum degree at most two. At every deletion parent H in every such process, with $D = \Delta(H)$, there is a matching covering all degree-D vertices. Deleting the chosen vertex lowers c_i by exactly one for every integer $1 \le i < D$. Every terminal of every complete process has zero slack ps.

All orders, degrees, cardinalities, capacities, budgets, costs, row indices, run lengths, excesses and process lengths are integers in the stated nonnegative or positive domains. No independence-preserving terminal or proper-subgraph minimality is assumed.

Proof. Use Lemma 193 for actual sextic rows and costs, Lemma 106 for the top-cost inequality, Lemma 40 for missing allocations and run bounds, Lemma 88 for row paths and independence, Lemma 192 for whole-triangle contacts, Lemma 57 for degree-two costs, Lemma 60 for successive matchings, and Lemma 15 for restoration.

Suppose F exists. Factor-criticality and its odd order give $\nu(F) = k$. Define integers

$g = k - a$, $\lambda = 2E - 3 - g$,

$\delta = c_0(F) - c_2(F)$, $\epsilon = c_0(F) - c_1(F)$,

$p = E - 2 - \lambda$, $m = p + \delta$, $\beta = m - \epsilon$,

$r = k - \lambda - 1 + \delta$, $s = 2\lambda + 3 - 2\delta$, $u = r + s$.

The hypotheses (a), (b) permit Lemma 193. It gives

$p \ge 0$, $0 \le \epsilon \le \delta \le \lambda + 1$,

$-1 \le \lambda \le E - 2$, $s \ge 1$,

$c_1(F) = u + \beta$, $c_2(F) = u + p$, $c_3(F) = r$,

$c_4(F) = r - \beta$, $c_5(F) = a$.

For any fixed complete process there is an actual partition into its early row A, its terminal R, and its remaining independent cubic root set X. Write $\tau = a - \alpha(R)$. Then

$|V(A)| = r$, $|V(R)| = r + \tau$, $|X| = s - \tau$, $0 \le \tau \le s$.

Every X-vertex has exactly three R-neighbors. The native A has components in

P_1, P_2, P_3, P_4, P_6, C_3, C_4, C_5, C_7,

where P_j, C_j denote paths and cycles on j vertices. It has matching number m, odd-cycle count p, positive even-order path count ϵ, $ps(A) = 0$, $\alpha(A) = a$, $c_1(A) = \beta$ and $c_2(A) = p$. The actual terminal R has $\Delta(R) \le 2$ by the stopping rule and $ps(R) = 0$ by (b). Its numerical profiles from Lemma 193 are $\nu(R) = m + \tau$, $c_1(R) = \beta + \tau$ and $c_2(R) = p + \tau$. Apply Lemma 57 directly to R: it gives the same nine-component catalogue, exactly $p + \tau$ odd-cycle components, and $\nu(R) - c_1(R) = m - \beta = \epsilon$ positive even-order path components. No assertion that R is native is needed. The full complement $J = F - V(A)$ is subcubic with zero slack and

$\nu(J) = M = p + \delta + s$, $c_1(J) = M - \epsilon$, $c_2(J) = M - \delta$.

Since $c_5(F) = a$, the top-cost inequality in Lemma 106 gives

$3100k \le 10000g + 2644 = 10000(2E - 3 - \lambda) + 2644$. (1)

At zero capacities the degree-bounded edge parts are empty, so the two-part c_0 in Lemma 40 is the same cost. Apply that fact with $D = 6$. For $1 \le E \le 3$ its parameter $\max(0, floor((2E-3)/4))$ is zero. The assumed $tr(F) + E \ge 5$, with its integer $B = 4$, gives

$k \ge 2E + 3$. (2)

That fact also ensures a missing pair. By symmetry choose one and name its coordinates $x \le y$. Then

$x + y = k + E - 1 = u + m - 1$, $x \le floor((k + E - 1)/2)$. (3)

Fix this pair and a complete process for the rest of the proof.

We first reduce the integer parameters.

For $E = 1$, the range $-1 \le \lambda \le E - 2$ forces $\lambda = -1$ and $g = 0$. Inequality (1) gives $3100k \le 2644$, impossible for $k \ge 1$.

For $E = 2$, one has λ in {-1,0} and $k \ge 7$. If $\lambda = 0$, then $g = 1$ and (1) gives $3100k \le 12644 < 3100 \cdot 5$, contradicting $k \ge 7$. Hence $\lambda = -1$, $g = 2$, and $3100k \le 22644 < 3100 \cdot 8$. Therefore $k = 7$, $p = 1$, and $\delta = \epsilon = 0$.

For $E = 3$, one has λ in {-1,0,1} and $k \ge 9$. If $\lambda = 1$, then $g = 2$ and (1) gives $k \le 7$, impossible. If $\lambda = 0$, then $g = 3$ and $3100k \le 32644 < 3100 \cdot 11$, so k is nine or ten, $p = 1$, and $0 \le \epsilon \le \delta \le 1$. If $\lambda = -1$, then $g = 4$ and $3100k \le 42644 < 3100 \cdot 14$, so $9 \le k \le 13$, $p = 2$, and $\delta = \epsilon = 0$. All divisions here use integer k.

One of these possibilities cannot occur:

$E = 3$, $\lambda = 0$, $\delta = 1$, $\epsilon = 0$, k in {9,10}. (4)

In (4), $r = k$, $u = k + 1$, $m = 2$, $\beta = 2$, $p = 1$, $a = k - 3$. The actual row A has one odd cycle, $\nu(A) = c_1(A) = 2$, and zero slack. The row identities count no positive even-order path components. Its possible whole components therefore belong to P_1, P_3, C_3, C_4, C_5, C_7.

Let ℓ_0 count its nonisolated odd-order components. Each contributes at least one matching edge, so $\ell_0 \le 2$. By (3),

$j = u - x = k + 1 - x \ge 5$,

and $c_2(F) = u + 1$. Thus Lemma 88 applies with the same missing pair and removes every P_3. Matching number two and exactly one odd cycle then force

$A = C_5$ disjoint union $(k - 5)P_1$.

A triangle would leave one matching unit to be supplied, while every other permitted nonisolated component would be a four-cycle supplying two.

Apply Lemma 192 to this row. Its α equality is already given; $c_0(F) = u + 2$, $c_1(F) = u + 2$, $c_2(F) = u + 1$, and the reserve satisfies $j \ge 5 = p + 4$. Consequently every exterior vertex with three A-neighbors would own a whole A-triangle. No triangle exists. Each degree-three vertex of J would have exactly three A-neighbors by six-regularity, so J has none. Since J is subcubic, $\Delta(J) \le 2$.

Now $|V(J)| = k + 1$, $\nu(J) = 3$ and $c_2(J) = 2$. The degree bound and zero slack allow Lemma 57, which gives the nine permitted component types, exactly two odd cycles, and $\alpha(J) = (k + 1) - 3 - 2 = k - 4$.

On the other hand, apply Lemma 88 to row J and its exterior $V(A)$ of size k. Its missing total is $x + y = k + 2 = k + \nu(J) - 1$, and its reserve k-x is at least four for $k = 9$, 10 by (3). The number of nonisolated odd-order J-components is at most $\nu(J) = 3$, since each contributes at least one matching edge. Also $c_2(F) = k + 2$ is the exterior order plus the two odd J-cycles. Thus every hypothesis of that row theorem holds; its independence conclusion gives $\alpha(J) = \alpha(F) = k - 3$. This contradicts $\alpha(J) = k - 4$ and eliminates (4).

In every remaining case A contains a triangle. This follows directly from the row matching and component counts. When $p = m = 1$, the only odd cycle is C_3 and all other components are isolates. When $p = m = 2$, its two odd cycles are both C_3 and all other components are isolates. In the remaining case $p = 1$, $m = 2$, $\epsilon = 1$, the odd cycle consumes at least one matching unit and the even path consumes at least one, so they are C_3 and P_2 and all other components are isolates.

Thus the remaining cases, with b and N defined below, have the following exact values:

$(E, \lambda, \delta, \epsilon; k) = (2, -1, 0, 0; 7)$:

$r = 7$, $s = 1$, $\beta = 1$, $b = 3$, $N = 4$;

$(E, \lambda, \delta, \epsilon) = (3, -1, 0, 0)$, $9 \le k \le 13$:

$r = k$, $s = 1$, $\beta = 2$, $b = 4$, $N = k - 4$;

$(E, \lambda, \delta, \epsilon) = (3, 0, 0, 0)$, k in {9,10}:

$r = k-1, s = 3, \beta = 1, b = 3, N = k-4$;

$(E, \lambda, \delta, \epsilon) = (3, 0, 1, 1)$, k in {9,10}:

$r = k, s = 1, \beta = 1, b = 3, N = k-3$.

Choose one of the actual whole triangles C of A. The degree-two vertices of A are precisely its triangle vertices: all other components in the preceding list are isolates or P_2. Choose one vertex on each triangle, obtaining a maximum independent set D_0 of the eligible graph of size β. Put $Y = D_0$ union $V(C)$. Then every Y-vertex has degree two in A,

$b = |Y| = \beta + 2, N = r - b$,

and $E(A[Y])$ is exactly $E(C)$; selected vertices on other triangles are isolated in A[Y].

We check both matching margins for the single missing pair fixed in (3). In the $E = 2$ case, $x \le floor((7 + 2 - 1)/2) = 4 = N$. For $E = 3$ and $k \ge 9$,

$floor((k+2)/2) \le k-4$,

because $(k+2)/2 < k-3$ when $k > 8$ and the right side is an integer. Thus $x \le N$ in each $E = 3$ case. The quantities $s + \beta + b - 2$ in the four displayed rows are respectively 3, 5, 5, 3. Their respective N values are 4, at least 5, at least 5, and at least 6. Hence

$\min(x, y) \le N, N \ge \max(0, s + \beta + b - 2)$.

Apply Lemma 60 to this actual A, R, X, Y. The row orders, their matching and cycle counts, the common even-path count, and the independent roots with three R-neighbors were established from the fixed complete process. Equation (3) gives its missing-total identity, and the original costs $c_1(F) = r + s + \beta, c_2(F) = r + s + p$, $c_3(F) = r$ give all three lower bounds. The two reserve inequalities have just been checked. It follows that the full induced graph $H = F[V(R)$ union X union Y], which omits exactly N vertices, has four successive maximum matchings whose removal leaves the entire triangle C and isolated vertices. Every matching is chosen in the residual left by its cited results, so the four are compatible and pairwise edge-disjoint.

Apply Lemma 15 with these four matchings, $\ell = 2$, $g = 0$ and the same omitted vertices. Its surcharge is zero because $N \ge \ell - 2 = 0$. It constructs every integer capacity split at total four and every matching-budget split at total $N + 1$ on the entire original graph, restoring all original omitted edges. Therefore

$c_4(F) \le N + 1 = r - \beta - 1$,

contrary to $c_4(F) = r - \beta$. This proves the claimed nonexistence for all three excess values. □

Remark. The original failed-run inequality leaves only a few small integer rows. Their only triangle-free row contradicts independence in its full complement. Every surviving row admits one compatible triangle selection, four successive maximum matchings and a strict original-graph certificate.

Depends on. Lemma 15, Lemma 40, Lemma 57, Lemma 60, Lemma 88, Lemma 106, Lemma 192, Lemma 193.

Lemma 228 (Unit counts and surplus at a terminal two-star pair). *Let F, J be vertex-disjoint unions of stars with full nonincreasing positive integer size lists $n_1,..., n_s$ and $m_1,..., m_t$, where $s, t \ge 1$, $n_1 = D \ge 3$ and $m_1 \le D$. The first target is prescribed red and the second blue; copies are ordinary subgraphs with disjoint target components. Put $M = s + t - 1$ and $\ell_l = max\{n_i + m_j - 1 : 1 \le i \le s, 1 \le j \le t, i + j = l\}$ for each integer $2 \le l \le s + t$. Fix an integer $0 \le q < \min(s, t)$, put $K = M - q$, $L_q = \sum_{l=q+2}^{s+t} \ell_l$ and $P_q = L_q - K$, and assume $\ell_{q+2} = D + 1$. Let R be the initial plateau length at $D + 1$ in this tail. Suppose valid original indices $i_0 + j_0 = q + R + 1$ satisfy $n_{i_0} = D$ and $m_{j_0} = 2$.*

Let U, z, k, u be the numbers of F components of sizes D, from 3 through D-1, two and one, respectively, and let V, v be the numbers of J nonunits and units. Put $B = DU + \sum_{i:3 \le n_i < D}(n_i - 2)$. Let A_q be all nonempty ordered actual suffix pairs (F^α, J^β) with nonnegative integer leading cuts $\alpha < s$, $\beta < t$, $\alpha + \beta = q$. Form Γ from both full component-count matching relaxations of every member: $((s - \alpha)K2, J^\beta)$ and $((t - \beta)K2, F^\alpha)$, exchanging both prescribed colors when necessary to display the matching first. For a member (pK2,T), K_2 denotes one edge and $y_1,..., y_c$ is T's full sorted size list. Define $\eta_i = max\{y_j : (pK2, T) \in \Gamma, 1 \le a \le p, 1 \le j \le c, a + j = i + 1\}$ for $1 \le i \le K$, and $E = \sum_i \eta_i$.

Suppose a finite simple H arrows every A_q member, has maximum degree at most D, and $\delta = K - \nu(H) \geq 1$, where ν is maximum matching cardinality and e is edge count. Put $\sigma = e(H) - \nu(H) - P_q$, $G = L_q - E$, and $\rho = \min_W |\{x : deg_H(x) = D\}$ minus $V(W)|$, the minimum over all matchings. Then

$2\rho \leq G + \sigma$, $\rho \geq R$, *and*

$\sigma \geq \min(R, \lceil D/2 \rceil)$.

Moreover, if $\sigma < \min(R, DU)$, then

$u > v$, $e(F) \geq K + B - \sigma \geq K + 1$, *and* $2e(F) \leq 3s - 2$.

The last inequality is equivalent to

$u \geq (2D - 3)U + \sum_{i:3 \leq n_i < D}(2n_i - 3) + k + 2$.

In particular, for an integer $0 \leq r < \min(R, DU)$, a graph H with the stated degree bound, $\nu(H) < K$ and $e(H) - \nu(H) \leq P_q + r$ has one coloring avoiding both full targets of some A_q member if $u \leq v$ or $2e(F) > 3s - 2$. Every such coloring extends over the disjoint canonical prefix of q stars with edge sizes ℓ_2,..., ℓ_{q+1}, and across any specified actual q-vertex set X of an outer finite simple graph G_0 with $H = G_0 - X$, to avoidance of both unchanged originals, retaining all H-edge colors. All counts, sizes, degrees, indices, extrema, matching numbers and surpluses have integer domains. Units, repeated sizes, $q = 0$ and isolated host vertices are included.

Proof. Write $r_h = |\{i : n_i \geq h\}|$ and $c_h = |\{j : m_j \geq h\}|$ for positive integer thresholds h. The width of the natural tail at h is

$N_h = max\{(r_x + c_y - q - 1)_+ : x, y \geq 1, x + y = h + 1, r_x > 0, c_y > 0\}$,

where an empty maximum is zero and $a_+ = \max(a, 0)$. Indeed each threshold pair defines a rectangle of indices whose sums fill every integer from 2 through $r_x + c_y$. A size sum at least $h + 1$ dominates a threshold pair with exactly that sum. This proves the formula and monotonicity of the natural list.

The terminal pair gives $U + V \geq q + R + 1$. Strict inequality would give a natural entry at least $D + 1$ beyond its initial plateau. Thus

$U + V = q + R + 1$, $K = R + z + k + u + v$. (1)

If $c_3 > 0$, the rectangle at thresholds D, 3 forces $U + c_3 \leq q + 1$, since otherwise the natural head would exceed $D + 1$. By Lemma 198, applied to precisely this complete actual family and positive matching deficit, $R + v \geq z$, $\sigma \geq 0$ and $\rho \geq R$.

We establish two bounds for G using the entire original lists. A matching-first pair with p matching components and $d > 0$ opposite components of size at least h has $p + d - 1$ entries at threshold h; if $d = 0$ it has none. Retaining F^α and relaxing J^β gives width $t - q + r_h - 1$ whenever $r_h > \alpha$, and zero otherwise. The legal corner $\alpha = 0$, $\beta = q$ simultaneously attains every such positive width. The other orientation is attained by $\alpha = q$, $\beta = 0$. Thus Gamma's envelope is already supplied by ((t-q)K2,F) and ((s-q)K2,J), and its threshold widths are

$W_h = \max(\{t - q + r_h - 1 : r_h > 0\} union \{s - q + c_h - 1 : c_h > 0\})$.

Both targets in every pair are nonempty and its component budget is K, so η is positive, nonincreasing and of length K. Its head is D. The thresholds (h,1), (1,h) show $N_h \geq W_h$, with equality at $h = 1, 2$. Also $N_{D+1} = R$, and all widths above $D + 1$ vanish. Hence $G = R + \sum_{h=3}^{D}(N_h - W_h)$.

For $1 \leq h \leq D$, $r_h \geq U$ and (1) rewrites the matching width as

$W_h = R + \max(\{v + r_h - U\} union \{u + z + k + c_h - V : c_h > 0\})$. (2)

Any remaining threshold split with both coordinates at least 3 has width at most z: when it is present, $U + c_3 \leq q + 1$ and $r_x \leq U + z$. It is dominated by $W_h \geq R + v \geq z$. Put

$Z_h = \max(\{r_h - U\} union \{z + k + c_h - V : c_h > 0\})$, $2 \leq h \leq D$.

This is a nonnegative nonincreasing sequence and $Z_2 = z + k$. The only undominated splits at $3 \leq h \leq D$ have one coordinate 2, so

$N_h = \max(W_h, R + Z_{h-1})$.

With $w = \min(u, v)$, (2) gives $W_h \geq R + Z_h + w$. The nonnegative drops $a_h = Z_{h-1} - Z_h$ therefore give

$0 \leq N_h - W_h \leq (a_h - w)_+$.

If some $a_h > w$, summing over those indices subtracts at least w from the total drop; if none does, the sum is zero. Thus

$G \leq R + (z + k - \min(u, v))_+$. (3)

When $v \geq z$ we need the sharper estimate

$G \leq R + (z + k - v)_+$. (4)

For $h \geq 4$, the split (h-1,2) has excess over the first term of (2) equal to $r_{h-1} - r_h - v$, which is nonpositive because the difference counts a group among the z lower nonbinary components. Splits with both coordinates at least 3 are already dominated. Thus only (2,h-1) can contribute, including $h = 3$. Put $x_0 = k - v$ and $B_h = r_h - x_0 - z - U + V$. For $c_h > 0$ its excess is

$N_h - W_h = (c_{h-1} - \max(B_h, c_h + u))_+$.

When $c_h = 0 < c_{h-1}$, it is $(c_{h-1} - B_h)_+$; there is at most one such final threshold. Interpret positive contributions as lengths of the intervals $(\max(B_h, c_h + u), c_{h-1}]$ or the possible final $(B_h, c_{h-1}]$. They are pairwise disjoint: each nonfinal lower endpoint is at least c_h and every later upper endpoint is at most c_h. All upper endpoints are at most V and all lower endpoints are at least $V - x_0 - z$ because $r_h \geq U$. The total length is therefore at most $(x_0 + z)_+$, proving (4). A final lower endpoint may be negative; the same interval bound still applies.

We now show $2\rho \leq G + \sigma$. Write $S = \{x : deg_H(x) = D\}$, $a = \alpha(H[S])$, $\theta = \theta_D(H)$ as defined in Lemma 210, $\gamma = \tau(H) - K$, and $h_0 = e(H) - E$. Here α and τ denote maximum independent-set and minimum vertex-cover cardinalities. Every full actual pair implies both matching relaxations on the same H. Thus Lemma 21 gives $\gamma \geq 0$, $h_0 \geq \gamma$, and its arbitrary-set edge inequality for Γ. Let R_D be the initial D plateau length of η.

Choose a Γ member (pK2,T) whose own profile attains the last D position. If f is its head-D multiplicity then $p + f - 1 = R_D$. We claim $\theta \geq R_D$. Take an incidence edge set of rank θ and then an inclusion-minimal incidence subset B_1. Its rank remains θ by minimality of θ. Each $B_1 - edge$ has an S-endpoint of $B_1 - degree$ one, because otherwise that edge could be removed. Thus no edge joins two vertices of $B_1 - degree$ at least two. Each nonempty component of B_1 is a star, and there are exactly θ such components. If $\theta < p + f - 1$, color all edges of $\min(\theta, p - 1)$ of these components red and all remaining H-edges blue. Red has matching number at most p-1. Fix one center in each remaining $B_1 - star$. Every blue D-star is centered at an S-vertex, uses its full H-neighborhood and cannot meet a red $B_1 - edge$ at its center. In its uncolored $B_1 - component$ it either is the fixed center or includes that fixed center as a leaf. Disjoint blue D-stars therefore require distinct remaining $B_1 - components$, of which there are at most f-1. This avoids the full blue T, a contradiction. Hence $\theta \geq R_D$.

Let I be a maximum independent subset of S, of size a. All its incident edges cover S, since I is maximal in H[S] and every vertex of I has positive degree. Their matching number is at most a, so $\theta \leq a$. We prove $a - \theta \leq h_0 - \gamma$ without any extra restriction on a. If $a \leq K$, independence gives $e(H) - e(H - I) = Da$. Apply Lemma 21 to this actual a-set. It yields $Da - \sum_{i=1}^{a} \eta_i \leq h_0 - \gamma$. Because $a \geq \theta \geq R_D$, the left side is at least $a - R_D$, and hence at least a-theta. If $a > K$, then I together with one endpoint per edge of H-I is a cover. Therefore $e(H) - \tau(H) \geq (D - 1)a$. Since $E \leq DR_D + (D - 1)(K - R_D) = (D - 1)K + R_D$, we get

$h_0 - \gamma = e(H) - \tau(H) - E + K$

$\geq (D - 1)a - (D - 2)K - R_D \geq a - R_D \geq a - \theta$.

This covers every integer a.

Apply the all-degree cover inequality of Lemma 210 to the same H, S, θ, ρ, a. It gives $\theta + 2\rho \leq \tau(H) - \nu(H) + a$. Consequently

$2\rho \leq \delta + \gamma + (a - \theta) \leq \delta + h_0 = G + \sigma$. (5)

No host edge bound has been used.

Suppose $\sigma < \min(R, DU)$. From $\rho \geq R$ and (5),

$G \geq 2R - \sigma > R$. (6)

Direct subtraction of the full F edge sum using (1) gives

$e(F) - K = B + z + k - v - R$. (7)

Assume $u \le v$. If $v \ge z$, (4) and (6) force $z + k - v \ge R - \sigma > 0$. Thus $e(F) - K \ge B - \sigma \ge DU - \sigma > 0$, and $k - u \ge R - \sigma - z$. If $v < z$, (3) and (6) instead force $z + k - u \ge R - \sigma > 0$. Equation (7) gives $e(F) - K \ge B + u - v - \sigma$. Every lower nonbinary component contributes at least one to B-DU, so $B \ge DU + z$ and the integer relation $v < z$ implies $B - v \ge DU + 1$. Again $e(F) - K > 0$, and $k - u \ge R - \sigma - z$.

In both cases integrality gives $e(F) \ge K + 1$. Also

2e(F)-(3s-2)

$= (2D - 3)U + \sum_{i:3\le n_i<D}(2n_i - 3) + k - u + 2$

$\ge (2D - 3)U + \sum_{i:3\le n_i<D}(2n_i - 4) + R - \sigma + 2 > 0.$

The actual corner (F, J^q) is legal and has total component count $K + 1$. Apply Lemma 211 with the entire targets and colors exchanged, so its dense target is F. The hypotheses just proved are $e(F) \ge K + 1$ and $2e(F) > 3s - 2$; its conclusion $\nu(H) \ge K$ contradicts $\delta \ge 1$. Therefore $u > v$.

Now (3) uses $\min(u, v) = v$. Equations (6)-(7) give $e(F) \ge K + B - \sigma \ge K + 1$. If $2e(F) > 3s - 2$, the same application of Lemma 211 again contradicts $\delta \ge 1$. Thus $2e(F) \le 3s - 2$. Expanding $e(F) = DU + \sum_{3\le n_i<D} n_i + 2k + u$ and $s = U + z + k + u$ gives exactly the displayed lower bound on u.

If $\sigma < \min(R, \lceil D/2 \rceil)$, then $\lceil D/2 \rceil \le D \le DU$ allows the conclusion $u > v$ just proved. But Lemma 198 on the same full lists and host gives $\sigma + (v - u)_+ \ge \lceil D/2 \rceil$. Its positive part is zero, a contradiction. Hence $\sigma \ge \min(R, \lceil D/2 \rceil)$.

Finally suppose the stated $\cap$, matching and surplus bounds at r hold and either target-only obstruction holds. Common arrowing would give $\sigma \le r < \min(R, DU)$, contradicting the necessary unit or density inequality. Negating common arrowing supplies one coloring avoiding both full targets of an actual member. The two extension assertions follow from Lemma 41 for that same actual pair and its unchanged coloring. Its canonical-prefix construction and its extension across any partition of the actual q-set retain all H-edges and exclude copies meeting the deleted vertices at leaves as well as at centers. All original units are retained. □

Remark. The full matching envelope charges at least two units of edge mass per exposed high vertex. Threshold intervals force a dense full target unless its unit count exceeds the opposite count. Matching-decrement projections force the reverse order at smaller surplus.

Depends on. Lemma 21, Lemma 41, Lemma 198, Lemma 210, Lemma 211.

Lemma 229 (A sextic certificate from two odd cycles). *Use the finite integer certificate families $C_{A_c,B_c}(G; x, y)$, costs $c_i(G)$, matching number $\nu(G)$, independence number $\alpha(G)$, and signed slack $ps(G) = e(G) - \nu(G) - \sum_{i\ge1} c_i(G)$ of Lemma 23. An element of C is an ordered partition of all G-edges into parts of maximum degrees at most A_c, B_c and matching numbers at most x, y. The capacities and budgets are nonnegative integers. All graphs are finite and simple. Write $V(G)$, $E(G)$, $d_G(v)$, and $\Delta(G)$ for the vertex set, edge set, vertex degree, and maximum degree. Vertex deletion takes an induced graph; matching deletion removes only edges.*

For a finite simple graph G and nonnegative integer q, let $mathfrakC_q(G)$ be the family of tables

$Q = (T, (Q_{a,x})_{0\le a\le q, 0\le x\le T})$,

where T is a nonnegative integer and each $Q_{a,x}$ belongs to $C_{a,q-a}(G; x, T - x)$, with a, x integers. Define $cost(Q) = T$. This family is nonempty and $\min_(Q \in mathfrakC_q(G))cost(Q) = c_q(G)$.

Let F be connected, six-regular and factor-critical, meaning that every F-v has a perfect matching. Write $|V(F)| = 2k + 1$ with integer $k \ge 1$, put $a_F = \alpha(F)$, and assume $E = c_0(F) - k \ge 4$. Define the integers

$\lambda = 2E - 3 - (k - a_F)$, $\delta = c_0(F) - c_2(F)$, $\epsilon = c_0(F) - c_1(F)$,

$p = E - 2 - \lambda$, $m = p + \delta$, $\beta = m - \epsilon$,

$r = k - \lambda - 1 + \delta$, $s = 2\lambda + 3 - 2\delta$, $u = 2k + 1 - r$.

Assume $p \geq 2$.

A complete process chooses a maximum independent set S_0 of F and then repeatedly deletes a vertex of current maximum degree from $F - S_0$ while that degree is at least three, ending at an induced graph R of maximum degree at most two. Suppose:

(a) For every maximum independent S_0 and every integer $1 \leq i \leq 5$,

$c_i(F - S_0) = c_i(F) - a_F$ *and* $c_0(F) = a_F + \nu(F - S_0) + ps(F - S_0)$.

(b) In every complete process, each deletion parent G of current maximum degree $D \geq 3$ has a matching covering every degree-D vertex; deleting the chosen vertex lowers c_i by exactly one for each integer $1 \leq i < D$. Every terminal R has $ps(R) = 0$.

Fix any complete process. Let A be the induced graph on its S_0 together with all roots from its entire degree-five and degree-four phases, let X be its remaining roots, and let R be its actual terminal. Set $\tau = a_F - \alpha(R)$. Then $0 \leq \tau \leq s$. Let $D2 = \{zinV(A) : d_A(z) = 2\}$, choose any maximum independent set T_0 of $A[D2]$, and choose any shortest whole odd-cycle component C of A. Here whole means an entire connected component and cycle order counts vertices. These choices exist. Put

$\ell = (|V(C)| + 1)/2$, $Y = T0$ *union* $V(C)$,

$H = F[V(R)$ *union* X *union* $Y]$, $N = |V(F)$ *minus* $V(H)|$.

Starting with $P_0 = H$, for each integer $i = 0, 1, 2, 3$ choose any maximum matching M_i of P_i and put $P_{i+1} = P_i - M_i$, retaining all vertices. Every such sequence exists and satisfies

$|M_0| = m + s + \beta + \ell$, $|M_1| = s + 2\beta + \ell$,

$|M_2| = p + s + \beta + \ell$, $|M_3| = \beta + \ell$,

$E(P_4) = E(C)$.

All vertices of P_4 outside that same original C are isolated.

There is an explicit Q in mathfrak$C_4(F)$ for which every table entry uses $\bigcup_{i=0}^{3} M_i$ as precisely the union of its degree-bounded parts, every original edge with an endpoint outside H belongs to a matching-bounded part, and

$cost(Q) = N + \ell - 1 = r - \beta - 1 = c_4(F) - 1$.

Consequently $c_4(F) \leq cost(Q) = c_4(F) - 1$, so the displayed hypotheses cannot hold.

All orders, cardinalities, degrees, matching sizes, process lengths, capacities, budgets and table indices are integers in their stated nonnegative or positive domains; differences are integers. In particular the derived parameters satisfy $p \geq 2$, $0 \leq \epsilon \leq \delta \leq \lambda + 1$, $\beta \geq p$, $r \geq 0$, s positive odd, $0 \leq \tau \leq s$ and $2 \leq \ell \leq 4$. The conclusion includes $\epsilon = 0$ and every actual terminal loss in the indicated interval.

Proof. The seven inputs are Lemma 23, Lemma 57, Lemma 40, Lemma 193, Lemma 91, Lemma 60, and Lemma 15. Their costs all use the same nonnegative integer capacities and matching budgets.

1. The actual pair of rows.

Factor-criticality gives $\nu(F) = k$: a perfect matching of F-v has k edges, and a matching on $2k + 1$ vertices has at most k edges. Finite vertex sets admit maximum independent sets, and each subsequent deletion reduces the number of vertices. Thus complete processes exist.

Apply Lemma 193 with its $a = a_F$ and with the two process assumptions exactly (a), (b) here. It gives

$0 \leq \epsilon \leq \delta \leq \lambda + 1$, $p \geq 0$, $\beta \geq p$, $r = a_F + 2p + \delta$,

$u = r + s$, $|V(F)| = 2r + s$,

$c_0(F) = u + m$, $c_1(F) = u + \beta$, $c_2(F) = u + p$,

$c_3(F) = r$, $c_4(F) = r - \beta$, $c_5(F) = a_F$.

The set of the actual early row A, the actual terminal R, and the actual remaining roots X partition $V(F)$, with

$|V(A)| = r$, $|V(R)| = r + \tau$, $|X| = s - \tau$, $0 \leq \tau \leq s$.

The row A has maximum degree at most two, matching number m, exactly ϵ positive even-order path components and p odd-cycle components. Its whole components are among P_1, P_2, P_3, P_4, P_6, C_3, C_4, C_5, C_7. The set X is independent, and every vertex of X has exactly three neighbors in each of A and R.

The same theorem gives on the actual R

$\nu(R) = m + \tau$, $c_1(R) = \beta + \tau$, $c_2(R) = p + \tau$.

The definition of a terminal gives $\Delta(R) \le 2$ and (b) gives $ps(R) = 0$. Apply Lemma 57 to this R. It has the same permitted component list, precisely $c_2(R) = p + \tau$ odd cycles, and precisely

$\nu(R) - c_1(R) = m - \beta = \epsilon$

positive even-order path components. This checks the larger row's component counts on the actual terminal.

The assumed $p \ge 2$ now says that A has at least two whole odd cycles. The remaining parameter domains asserted in the statement follow from the row theorem and these component counts.

2. The selection and all missing allocations.

Let Bcal be the set of all nonnegative integer pairs (b_R, b_B) with $b_R + b_B = c_0(F) - 1 = u + m - 1$ for which F has no partition into two parts of matching numbers at most b_R, b_B. At capacity pair (0,0) the degree-bounded parts are empty, so this is precisely the missing family in Lemma 23. It is nonempty because $c_0(F) = k + E > 0$.

Set $v = floor((2E - 3)/4)$, which is positive since $E \ge 4$. Apply Lemma 40 with degree bound $D = 6$. Its graph is connected, has order $2k + 1$, has matching number k, and has $E \ge 1$; hence every coordinate b of every pair in Bcal satisfies $b \le k - E - v$. The identity

$u - (k - E - v) = s + m + v + 1$

holds because $u = r + s$ and $r - k + E = m + 1$. Thus every such coordinate obeys $u - b \ge s + m + v + 1$.

Apply Lemma 91 at $d = 6$ with its specified induced row R replaced by our smaller row A, and with all numerical parameters unchanged. Its parameter conditions are satisfied: $E \ge 4$ is $E \ge d/2 + 1$, $p \ge 2 = (d - 2)/2$, $r \ge 0$, and the row theorem gives $0 \le \epsilon \le \delta \le \lambda + 1$. The exact row order and matching number, component counts, and odd-cycle orders at most seven were checked in step 1. Its nonempty missing family and both coordinate bounds were just verified.

It allows every choice of T_0 and tied shortest C stated here. For these same choices it gives

$b := |Y| = \beta + \ell$, $2 \le \ell \le 4$, $E(A[Y]) = E(C)$,

$N = r - \beta - \ell$, $h := |V(H)| - r = s + \beta + \ell = s + b$,

$N \ge \max(0, \beta + h - 2) = \max(0, s + \beta + b - 2)$, $N \ge \ell - 2$,

$\min(b_R, b_B) \le N$ for every pair in Bcal. (1)

The retained graph in that theorem is exactly our H, since $V(F)$ minus $V(A) = V(R)$ union X. Its h differs from the actual terminal loss τ; (1) uses only the retained order. The choices of T_0, C, H precede the choice of a missing pair.

3. Four compatible matchings.

Choose one pair (b_R, b_B) in the nonempty Bcal. Apply Lemma 60 with the actual A, R, X, with its r, s, τ unchanged, and with its $m = \nu(A)$, $e_A = \epsilon$, $p = p$, $\beta = m - \epsilon$. The row component domain in step 1 is contained in its domain of isolates, P_3, positive even-order paths and cycles. The larger row has exactly the shifted matching and component counts required there. The independence of X and its three R-neighbors per vertex were also established.

The missing total is $b_R + b_B + 1 = r + s + m$. Its three cost lower bounds are equalities:

$c_1(F) = r + s + \beta$, $c_2(F) = r + s + p$, $c_3(F) = r$.

The set Y consists of degree-two vertices of A. Its omitted count is $N = r - b$, and both omission conditions are exactly (1). Every premise of the matching theorem therefore holds on the same induced H.

That theorem applies to every successive maximum-matching choice. It gives the first three ranks $m + s + b$, $\beta + s + b$, $p + s + b$ and the final rank b, which are the ranks in the statement after substituting $b = \beta + \ell$.

Its full edge conclusion is $E(P_4) = E(A[Y]) = E(C)$, and none of the four matchings removes an A[Y]-edge. Hence the same original cycle survives. Each matching is chosen in the preceding residual, so the four matchings are pairwise edge-disjoint. In particular the final cover rank is

$|M_3| = b = r - N$,

as forced by $c_3(F) = r = N + b$; the actual terminal increment τ has canceled.

4. A table on every original edge.

Put $w = \ell - 1$, $K = \bigcup_{i=0}^{3} M_i$, and $T = N + w$. Apply Lemma 15 with $D = 4$, $g = 0$, its prescribed matchings M_{i+1} equal to our M_i, and the same F, H, C and omitted set. Step 3 supplies its entire odd-cycle remainder; (1) gives its zero-surcharge condition $N \geq \ell - 2$. Thus its universal cost is T. We spell out the resulting table.

Order the omitted vertices as z_1,..., z_N. Let U_j consist of all F-edges whose earliest omitted endpoint in this order is z_j. These possibly empty stars partition $E(F)$ minus $E(H)$, including edges with two omitted endpoints. Each U_j has matching number at most one.

For any integers $0 \leq a \leq 4$ and $0 \leq x \leq T$, put the first a of the four prescribed matchings in Q_R and the remaining 4-a in Q_B. Empty intervals give empty parts. If $x \geq w$, put the whole cycle and the first x-w stars in P_R and all other stars in P_B. Here $0 \leq x - w \leq N$, and their matching numbers are at most $w + (x - w) = x$ and $N - (x - w) = T - x$.

If $x < w$, put the first x stars in P_R, and the whole cycle and all remaining stars in P_B. Since $x \leq w - 1 = \ell - 2 \leq N$, this is defined. The matching numbers are at most x and $w + N - x = T - x$. The matching bounds follow by restricting any matching to the disjoint edge pieces, even if pieces share vertices.

Each degree part is the union of at most its allowed number of matchings and therefore satisfies its degree bound. The four parts are disjoint and cover every F-edge: K and C partition $E(H)$, and the U_j partition all other edges. Their ordered tuple $Q_{a,x}$ is consequently in $C_{a,4-a}(F; x, T - x)$ for every entry, including zero capacities and budgets. All edges with an omitted endpoint are in matching-bounded parts, and the degree-bounded union at every entry is exactly K.

This finite table Q has cost

$T = N + \ell - 1 = (r - \beta - \ell) + \ell - 1 = r - \beta - 1$. (2)

For completeness, on any finite G and nonnegative integer q, the threshold characterization of Lemma 23 says that all budget allocations at total T are feasible for capacities a, q-a exactly when $T \geq c_{a,q-a}(G)$. A table of total T exists exactly when this holds for each of the finitely many integers a, equivalently when $T \geq c_q(G)$. Independent selection of finitely many entries is legitimate, and the finite nonnegative integer cost $c_q(G)$ is attained. This proves the representation of $mathfrakC_q$ in the statement.

Apply that representation to the table constructed on F. Equation (2) gives $c_4(F) \leq r - \beta - 1$, while step 1 gives $c_4(F) = r - \beta$ using the identical cost definition. This proves the asserted one-unit contradiction. Every choice and bound concerns the original graph, and the proof includes every integer allocation and every actual process allowed in (a), (b). □

Remark. The selector already covers excess four once the early row contains at least two odd cycles. Reversing the unequal rows cancels the terminal loss from all matching ranks, and restoring the same omitted vertices supplies every integer allocation at a cost lower by one.

Depends on. Lemma 15, Lemma 23, Lemma 40, Lemma 57, Lemma 60, Lemma 91, Lemma 193.

Lemma 230 (Strict cubic potential for every binary-versus-cubic star forest). *All graphs are finite and simple. Write $P_3 = K_{1,2}$ and $K_2 = K_{1,1}$, where $K_{1,d}$ is a star with d edges. Let p, r be positive integers and let u, h, v be arbitrary nonnegative integers. Form the entire ordered star forests*

$F = pP_3$ disjoint union uK_2,

$J = rK_{1,3}$ disjoint union hP_3 disjoint union vK_2.

All components within a target are vertex-disjoint. Their positive nonincreasing component edge-size lists are n_1,..., n_s and m_1,..., m_t, with $s = p + u$ and $t = r + h + v$. Put $M = s + t - 1$, $R = p + r - 1$, and

$\ell_k = max\{n_i + m_j - 1 : 1 \le i \le s, 1 \le j \le t, i + j = k\}$

for every integer $2 \le k \le s + t$. *Put* $L = \sum_{k=2}^{s+t} \ell_k$ *and* $P = L - M$.

Then

$P = 3R + h + \max(h, u) + \max(u, v)$.

For a graph G *write* $e(G)$, $\nu(G)$, $\Delta(G)$ *for its edge count, maximum matching cardinality, and maximum degree. If* $\Delta(G) \le 3$ *and* G *arrows* (F, J), *meaning that every red-blue edge coloring contains a red ordinary copy of the entire* F *or a blue ordinary copy of the entire* J, *then*

$e(G) - \nu(G) \ge P + 1$.

Equivalently, every graph of maximum degree at most three with $e(G) - \nu(G) \le P$ *has an edge coloring avoiding both entire targets. The result also holds with the two forests and their prescribed colors interchanged. It permits every relative order and every zero value among* u, h, v, *arbitrary repeated component sizes, and disconnected or irregular hosts. No matching-deficit, minimum-degree, source, or normalization assumption is imposed. All indices, counts, degrees, edge numbers, and matching cardinalities are integers in the stated domains.*

Proof. Put $A = \max(h, u)$, $B = h + \max(u, v) - A$, and $C = \min(u, v)$. These are nonnegative integers: $h + \max(u, v)$ is at least both h and u. We first check that the full diagonal list is

(4 repeated R times,3 repeated A times,2 repeated B times,1 repeated C times).

The entries are all between one and four. A value four is possible exactly from a red two-edge component and a blue three-edge component. Their index rectangle gives every sum from two through $p + r$, yielding R entries. A value at least three comes from a red two and a blue nonunit, or a red component of either size and a blue three. The corresponding rectangles give initial index-sum intervals ending at $p + r + h$ and $p + u + r$. Their union ends at $p + r + A$. Likewise a value at least two comes from a red two or a blue nonunit. Its two rectangles end at $p + r + h + v$ and $p + u + r + h$, so their union ends at $p + r + h + \max(u, v)$. Each rectangle gives all index sums from two to its endpoint, so there are no omitted intermediate sums. The remaining entries equal one, and their number is C. Since $R + A + B + C = M$, this accounts for the entire list. Subtracting one from each entry gives

$P = 3R + 2A + B = 3R + h + \max(h, u) + \max(u, v)$.

In particular $\ell_2 - 1 = 3$ and the initial plateau of entries equal to four has length R.

Let G have maximum degree at most three and arrow the full ordered pair. We split the parameter domain into exhaustive cases.

If $v \ge u$, apply Lemma 138 to the original pair with cut $q = 0$. The actual suffix family then has only this pair. Both forests have nonunit components, because $p, r \ge 1$. The minimum original unit count is $U = u$. Choose its red forest F as the theorem's selected side, with p positive nonunit components. In that theorem's notation,

$N = M - U = R + h + v$, $N - R + 1 = h + v + 1$.

The entire opposite forest has $e(J) = 3r + 2h + v \ge h + v + 1$ because $3r + h \ge 1$. Thus the first of its two numerical projection tests holds. Its critical degree is three and its natural potential at $q = 0$ is precisely P, so it gives $e(G) - \nu(G) \ge P + 1$. This includes $u = v$ and $u = v = 0$.

Now suppose $u > v$. If $h = 0$, use the same theorem with $q = 0$, selecting the original blue J as its side with minimum original unit count $U = v$ and its opposite side as the original red F. The selected side has $r \ge 1$ nonunit components. Here

$N = M - v = R + u$, $N - R + 1 = u + 1$,

and the entire opposite forest has $e(F) = 2p + u \ge u + 1$. The first numerical projection test again applies, in the allowed selected color orientation, and proves the same strict bound.

It remains that $u > v$ and $h \ge 1$. If $h \le u$, the hypotheses of Lemma 203 are exactly satisfied: p, r are positive integers, $1 \le h \le u$, and $v < u$. That theorem gives

$e(G) - \nu(G) \ge 3R + 2u + h + 1 = P + 1$,

where the equality uses $\max(h, u) = \max(u, v) = u$.

If instead $h > u$, then $h > u > v$ and Lemma 130 applies, giving

$e(G) - \nu(G) \geq 3R + 2h + u + 1 = P + 1$,

where $\max(h, u) = h$ and $\max(u, v) = u$.

These cases exhaust all nonnegative u, h, v and prove the strict inequality.

A graph with $e(G) - \nu(G) \leq P$ therefore cannot arrow the displayed pair. The negation of that universal coloring statement supplies a single coloring avoiding both entire targets. Interchanging red and blue exchanges the target order without changing any graph quantities or diagonal maxima, proving the final color-reversal assertion. □

Remark. The full projection bound handles the cases with enough opposite-color edges, including every zero lower multiplicity. The remaining two positive-multiplicity ranges have complementary equality obstructions: a full packing in one range and canonical matching remainders in the other.

Depends on. Lemma 130, Lemma 138, Lemma 203.

Lemma 231 (Avoidance at the full-head boundary). *Let s, $t \geq 1$ be integers and let F_1, F_2 be the vertex-disjoint unions of stars with complete nonincreasing positive integer edge-size lists n_1,..., n_s and m_1,..., m_t, prescribed red and blue. Copies are ordinary subgraphs with vertex-disjoint target components. Put $M = s + t - 1$ and $\ell_k = max\{n_i + m_j - 1 : 1 \leq i \leq s, 1 \leq j \leq t, i + j = k\}$ for every integer $2 \leq k \leq s + t$. Fix an integer $0 \leq q < M$, and put $K = M - q$, $L_q = \sum_{k=q+2}^{s+t} \ell_k$ and $P_q = L_q - K$. Let A_q consist of every ordered pair of nonempty actual suffix forests obtained by deleting α and β leading whole components, with integers $0 \leq \alpha < s$, $0 \leq \beta < t$ and $\alpha + \beta = q$.*

Suppose $D = \ell_{q+2} - 1 = \max(n_1, m_1)$ is even and $D \geq 6$. Let R be the initial run length of $D + 1$ in the natural tail ℓ_{q+2},..., ℓ_{s+t}. Suppose valid original indices i_0, j_0 satisfy $i_0 + j_0 = q + R + 1$, $n_{i_0} = a \geq 3$, $m_{j_0} = b \geq 3$ and $a + b = D + 2$. Let r be an integer with $0 \leq r \leq 2$. Every finite simple graph H satisfying $\Delta(H) \leq D$, $\Phi(H) := e(H) - \nu(H) \leq P_q + r$, $\delta := K - \nu(H) \geq r + 1$, where e, ν, Δ denote edge count, maximum matching cardinality and maximum degree, has an edge coloring avoiding both entire prescribed targets of some member of A_q.

Every such avoiding coloring extends over the disjoint canonical q-star pool with sizes ℓ_2,..., ℓ_{q+1}, and across any specified actual q-vertex set X in any finite simple graph G with $H = G - X$, preserving all H-edge colors and avoiding the unchanged full originals. Vertex deletion is induced. All sizes, counts, indices, degrees, deficits and allowances are integers in their displayed domains. All original unit components and repeated sizes are retained. No host connectivity, regularity, component-order or short-prefix hypothesis is imposed.

Proof. The family A_q is finite and nonempty: $\alpha = \min(q, s - 1)$, $\beta = q - \alpha$ are legal cuts because $q \leq s + t - 2$. The original diagonal sequence is nonincreasing. Indeed, from a valid candidate on diagonal $k + 1 \geq 3$ decrease a positive index greater than one; the resulting valid candidate on diagonal k has at least the same value. Taking a maximizing candidate proves $\ell_k \geq \ell_{k+1}$. Thus $1 \leq R \leq K$ and the indicated initial run is defined.

Since $a + b = D + 2$ is even, a, b have the same parity. If both are odd, apply Lemma 133 with the nonnegative even capacities a-1, b-1, whose sum is D. It gives a coloring of H with the respective red and blue degree bounds. Set $\alpha = \min(q, i_0 - 1)$, $\beta = q - \alpha$. The relation $(i_0 - 1) + (j_0 - 1) = q + R - 1 \geq q$ gives $\beta \leq j_0 - 1$, so these are legal cuts retaining the indicated terminal stars. Their red suffix contains an a-edge star and their blue suffix contains a b-edge star. Neither entire suffix can occur in the degree-bounded coloring. This proves avoidance in this case.

It remains that a, b are even, hence both at least four. Suppose for contradiction that H arrows every A_q member. Put $\sigma = \Phi(H) - P_q$, a signed integer initially. The hypotheses give $\sigma \leq r$ and $\delta \geq r + 1$, so $e(H) = L_q + \sigma - \delta \leq L_q - 1$. Exchange both full originals and their prescribed colors, if necessary, so that $n_1 = D$. Both terminal sizes remain at least four and the opposite full head is at most D.

Apply Lemma 215 to these full originals, the same H and the same natural tail. Every hypothesis holds: the positive matching deficit is $\delta \geq 1$, the edge count is below L_q, the terminal size on the head-D side is at least four, the opposite terminal size is at least three, and $\Delta(H) \leq D$. Write E for its complete envelope sum of both matching relaxations of all A_q members, $\gamma = \tau(H) - K$ for its cover surplus, and $k = floor((D - 1)/2)$. Here τ is minimum vertex-cover cardinality. The cited fact gives

$L_q - E = R$, $\gamma \geq 0$, $e(H) \geq E + \gamma$,

$R + (k-1)(\delta + \gamma) \leq k\,\sigma$.

Since D is even and at least six, $k \geq 2$. Using $\delta \geq r+1$ and $\sigma \leq r$ in the last inequality gives

$1 \leq R \leq k\,r - (k-1)(r+1) = r - k + 1$.

Because $r \leq 2$, the only possibility is $k = 2$, $r = 2$ and $R = 1$. Consequently $E = L_q - 1$. The bounds $e(H) \geq E + \gamma \geq E$ and $e(H) \leq L_q - 1$ force $e(H) = E$.

The matching-relaxation family has the common positive component budget K. Explicitly, from cuts α, β its two members, oriented with the matching first, are $((s-\alpha)K_2, F_2^{\beta})$ and $((t-\beta)K_2, F_1^{\alpha})$, where K_2 is one edge. Their matching demand plus opposite component count minus one is $s + t - \alpha - \beta - 1 = K$. Each member is arrowed by H. A copy of either original positive forest gives a matching by choosing one edge from each component, and the second orientation exchanges the two color names. The family is finite and nonempty, and its envelope sum is exactly E by its definition in Lemma 215. Lemma 74 therefore applies at $e(H) = E$ and yields $\nu(H) = K$. This contradicts $\delta = K - \nu(H) \geq 3$.

Thus H cannot arrow every member, so an actual member has an avoiding coloring. The odd-size case already supplied one explicitly. In either case Lemma 41 (I), (II) extends the chosen coloring over the original canonical pool and across every specified actual outer set X. These extensions use its actual legal cuts and preserve every H-edge color, including when a copied component meets X only through a leaf. The argument includes $q = 0$ and all unit and repeated original components. □

Remark. The envelope inequality reduces the full-head boundary to a single equality case. Equality for the same common-budget matching family forces the missing matching capacity, a contradiction.

Depends on. Lemma 41, Lemma 74, Lemma 133, Lemma 215.

Lemma 232 (Collapse of an octic independent-cut window). *Use finite integer certificate costs c_i, matching number ν, independence number α, maximum degree Δ, excess $ex = c_0 - \nu$, slack ps, $gap = ex - ps$, and $\Sigma = \sum_{i \geq 0} c_i$ from Lemma 61. All graphs are finite and simple. Put $f(J) = |V(J)| - 2\nu(J)$, $q(J) = \alpha(J) - f(J)$, and $E_7(J) = e(J) + 4|V(J)| - 11\nu(J) - 4\alpha(J)$. A graph is factor-critical if deleting every vertex leaves a perfect matching.*

Let H be nonempty, connected, eight-regular and factor-critical, with $ex(H) = 4$ and $gap(H) = 1$. Put $k = \nu(H)$, $a = \alpha(H)$, and let o_k be one if k is odd and zero otherwise. Assume

(k, a) belongs to {(14,7),(14,8),(15,8),(16,9),(17,10)}

and $2k - 10 + o_k - a \leq c_6(H) \leq k - 1$.

Assume every proper induced subgraph J of H with $\Delta(J) \leq 7$ satisfies $gap(J) \leq 1$ and the following conditions:

(a) if $ps(J) = 1$ and $q(J) \leq 5$, then $E_7(J) \leq 11$;

(b) if J is connected, $\Delta(J) \geq 3$, $ps(J) \leq 1$, and no matching covers all its maximum-degree vertices, then J is regular of positive even degree, is factor-critical, and has $ps(J) = 1$.

For every maximum independent set S of H, assume that $G = H - S$ and $\eta = \nu(G) - (k + 1 - a)$ satisfy

η in {2,3}, $ps(G) = 3 - \eta$, $\Delta(G) = 7$,

that some matching of G covers all degree-seven vertices, and that

$c_i(H) = a + c_i(G)$ for every integer $1 \leq i \leq 7$.

If $k - a = 7$, assume in addition $\alpha(G) = a$ for every such S.

Then $|V(H)| = 29$, $k = 14$, $a = 8$, $e(H) = 116$, $c_0(H) = 18$, $ps(H) = 3$, $\Sigma(H) = 117$ and $c_7(H) = 8$. For every maximum independent S, its actual complement G has 21 vertices, 52 edges, matching number 10, slack zero, independence number in {7,8}, maximum degree seven and a matching covering that degree. Moreover $c_i(H) = 8 + c_i(G)$ for $1 \leq i \leq 7$ and $2 \leq c_6(G) \leq 5$.

The finite window, induced density bound and independent-cut assumptions are premises. No existence of H, and no deduction of these premises from a hereditary top-run property, is asserted. All orders, degrees, indices, cardinalities,

costs, matching and independence numbers, excesses, slack values, k, a, o_k and η are integers in their displayed nonnegative domains; E_7 and gap may be signed.

Proof. Use Lemma 61 for nonnegative slack, zero tails, covered-degree vanishing and covered deletion, and Lemma 72 for restoration and component slack superadditivity. The zero-slack and four-edge arguments below concern proper induced subgraphs of H of maximum degree at most seven. Induced deletion preserves this domain. A maximum matching leaves an independent set of unmatched vertices, so $q(J) \geq 0$ for every graph J.

Every proper induced zero-slack J of H with $\Delta(J) \leq 7$ is covered at its actual maximum degree when that degree is at least three. Indeed, every component has slack zero, and condition (b) excludes an uncovered component of that degree. Unite the covering matchings of the relevant components. Repeatedly deleting a current maximum-degree vertex of degree at least three preserves zero slack and lowers matching number by one, by the covered deletion inequality. This supplies a complete covered process to degree two. Consequently Lemma 146 applies to each such J with every $\cap\, D \geq \max(3, \Delta(J))$. In particular, for these zero-slack graphs with $\Delta(J) \leq 7$, at $\cap$ seven,

$E_7(J) \leq 0$. (1)

If $\Delta(J) \leq 7$ and $E_7(J)$ is -1 or zero, its small-deficit conclusion applies with deficit $-E_7(J) \leq 1 \leq 7 - 4$: every complete covered process has exactly $q(J)$ deletions, preserves independence number, and terminates in $|V(J)| - \nu(J) - \alpha(J)$ triangles and $2\alpha(J) - |V(J)| + \nu(J)$ isolated vertices.

We need a covering observation for a later low-slack stage. Suppose J is a proper induced subgraph, $\Delta(J) \leq 7$, $ps(J) = 1$, $q(J) \leq 1$, and $E_7(J) \geq 10$. We claim J is covered at its actual maximum degree, which is at least three.

For maximum degree at most two, paths and cycles satisfy $e \leq 3\nu$, and $q \geq 0$ gives $E_7 = e - 3\nu - 4q \leq 0$. Now suppose J is uncovered at degree $d \geq 3$. An uncovered whole component U of degree d has slack one and is regular and factor-critical of positive even degree by (b). Thus d is four or six. The union W of the other components has zero slack, so $E_7(W) \leq 0$ by (1). Additivity and $q(W) \geq 0$ give $q(U) \leq 1$. Put $\alpha(U) = b$ and $\nu(U) = m$; factor-criticality gives $f(U) = 1$, so $b \leq 2$.

At degree four, the cutoff Lemma 123 and the inherited gap bound give

$1 \geq 2(|V(U)| - c_0(U)) - 4b = 2(|V(U)| - m - 2b - ex(U))$.

The integer $|V(U)| - m - 2b$ is therefore at most $ex(U) \leq 2$. Hence

$E_7(U) = m + 6 - 4b = (|V(U)| - m - 2b) + 5 - 2b \leq 7 - 2b \leq 5$.

Then $E_7(J) \leq 5$, a contradiction.

At degree six, condition (a) applies to U and gives $3m + 7 - 4b = E_7(U) \leq 11$. Regularity requires $m \geq 3$. If $b = 1$ this is impossible. If $b = 2$ it gives $m \leq 4$; at $m = 3$ a six-regular graph on seven vertices is complete and has independence one. It remains to exclude a six-regular factor-critical U on nine vertices with slack one.

We prove directly that every such nine-vertex U has slack at least two. Its edge and matching counts are 27 and four. One has $c_0(U) \leq 5$: for any budget split $x + y = 5$ with $x \geq y$, choose y vertices, place all their incident edges in one part, and put the remaining edges in the other. Since $x \geq 3$, the remaining graph has $9 - y = 4 + x \leq 2x + 1$ vertices and matching number at most x. Reverse colors if $x < y$. Thus $c_1, c_2 \leq 5$. By Lemma 190 at thresholds three and three, $2c_3(U) \leq 9$, so $c_3(U) \leq 4$.

Also $c_4(U) \leq 4$. The graph U is noncomplete; retain any five vertices containing a nonedge and call their induced graph R. The degree-four vertices of R are exactly its universal vertices. There are at most three of them, because both ends of the nonedge fail to be universal. Pair universal vertices with each other; if one remains, match it to a vertex outside the universal set. This gives a matching covering every degree-four vertex of R, so $c_4(R) = 0$. Restoring the four omitted vertices gives the claim.

To obtain $c_5(U) \leq 2$, fix a vertex v and a perfect matching M of U-v, and remove the edges of M to obtain L. This L has degree six at v and five elsewhere. Its closed neighborhood of v has seven vertices, leaving two undominated vertices. The closed neighborhood of each of these has six vertices, all among the other eight vertices. The two closed neighborhoods intersect, since $6 + 6 > 8$. Choose w in their intersection. Then $Z = \{v, w\}$ dominates L. Every degree-five vertex of U-Z has an edge of the restricted matching M, since its

degree in L-Z is at most four. Thus $c_5(U - Z) = 0$ and restoration gives $c_5(U) \le 2$. A perfect matching after deleting any vertex also gives $c_6(U) \le 1$, and higher rows vanish. Therefore

$ps(U) \ge 27 - 4 - (5 + 5 + 4 + 4 + 2 + 1) = 2,$

contradicting slack one. This proves the covering observation.

We now exclude four disjoint induced edges among degree-seven vertices of any proper induced J with $\Delta(J) \le 7$ in either of the following states:

$(ps(J), E_7(J), q(J)) = (0, -1, 7)$ or (1,10,5). (2)

Suppose their endpoints are x_i, y_i for $1 \le i \le 4$, with precisely the four edges $x_i\, y_i$ among these eight vertices. The vertices x_i are pairwise nonadjacent. After any of them have been deleted, the remaining ones still have degree seven and are legal maximum-degree choices. Once all four are deleted, every y_i has degree exactly six. A complete process to maximum degree two that begins with these four deletions and has seven deletions in total is impossible: only three additional vertices could be deleted, whereas every surviving y_i would retain degree at least $6 - 3 = 3$. All four y_i would have to be among three deleted vertices.

In the zero-slack state of (2), the small-deficit conclusion preceding the observation supplies exactly $q(J) = 7$ deletions and permits the specified four-vertex prefix. This proves the exclusion in that state.

For the slack-one state, delete the four x_i in order unless slack becomes zero earlier. At any single maximum-degree deletion from K to $K' = K - v$, put $d = deg_K(v)$, $t = \nu(K) - \nu(K')$, and $u = \alpha(K) - \alpha(K')$, both losses in {0,1}. Direct calculation gives

$q(K') = q(K) + 1 - 2t - u,$

$E_7(K') = E_7(K) - d - 4 + 11t + 4u.$ (3)

When covered, the deletion also satisfies $ps(K') \le ps(K) - 1 + t$.

While choosing the x_i at degree seven, a slack-one stage is covered by (b): an uncovered whole component of degree seven could not have the required even regular degree. Suppose the current state has $E_7 = 10$ and $q \le 5$. If $t = 1$, its child's E_7 is $10 + 4u$. The child cannot have zero slack by (1), and its slack is at most one, so it has slack one. Its new q is smaller and still at most five. Condition (a) forces $E_7 \le 11$, hence $u = 0$. Thus this step preserves $E_7 = 10$ and decreases q by one.

If instead $t = 0$ at the j-th deletion, the child has slack zero and $E_7 = -1 + 4u$. Inequality (1) gives $u = 0$, $E_7 = -1$, and $q = 7 - j$. The zero-slack small-deficit conclusion completes the process in exactly 7-j further deletions. Any remaining designated x_i may be chosen first, since they still have degree seven. The resulting seven-deletion process contradicts the four partners as above.

It remains that all four designated deletions had $t = 1$, $u = 0$. The current state is $ps = 1$, $E_7 = 10$, $q = 1$. The covering observation applies. Every subsequent state with $ps = 1$, $E_7 \ge 10$ and $q \le 1$ is again covered and has maximum degree at least three. At such a deletion, if $t = 1$, equation (3) and nonnegative q force $q = 1$, $u = 0$ and $q' = 0$. The child's E_7 is $E_7 + 7 - d \ge 10$, so its slack is one by (1), and (a) bounds E_7 by eleven. A second $t = 1$ step is impossible at $q = 0$. Therefore within at most two such steps a deletion with $t = 0$ occurs.

For this first $t = 0$ step, the parent has $10 \le E_7 \le 11$ and $d \le 7$. The child has slack zero and $E_7' = E_7 - d - 4 + 4u$. By (1), $E_7' \le 0$. These inequalities force $u = 0$ and $-1 \le E_7' \le 0$. If r is the total number of preceding $t = 1$ deletions, then r is four or five and each preserved α. The parent's q is 5-r, and the child's q is 6-r. The zero-slack small-deficit conclusion completes the process in exactly 6-r further deletions. Its total length is $r + 1 + (6 - r) = 7$. This final contradiction proves the exclusion for both states in (2).

Apply this result to the complements in the octic H. Every $G = H - S$ has maximum degree seven by hypothesis and is proper induced because S is nonempty. Thus G and all its induced descendants meet the auxiliary degree $\cap$. The order of H is $2k + 1$ and its edge count $8k + 4$. A maximum independent set S removes exactly 8a edges, so for $G = H - S$

$|V(G)| = 2k + 1 - a, e(G) = 8(k - a) + 4, \nu(G) = k + 1 - a + \eta.$ (4)

Consider one of the four candidate pairs with $k - a = 7$. Then $|V(G)| = a + 15$, $e(G) = 60$ and $\alpha(G) = a$ for every such cut. If any cut has $\eta = 3$, then $\nu(G) = 11$, $ps(G) = 0$, $q(G) = 7$ and $E_7(G) = -1$. The small-deficit conclusion gives a terminal induced graph R consisting of four triangles and a-4 isolated vertices, with

$\alpha(R) = a$. Select one vertex from each triangle and every isolate to form S′. This is an independent set of H of size a, hence maximum.

The eight unchosen triangle vertices induce four disjoint edges in H-S′. Each has exactly one neighbor in S′, since R is induced and has no other edges among its components. Each therefore has degree seven in H-S′. The hypotheses concerning every maximum independent set apply to S′ as well. If its gain is three, its complement has the first state of (2); if its gain is two, it has matching number ten, slack one, $q = 5$ and $E_7 = 10$, the second state. Both were excluded. Thus every maximum independent cut in a $k - a = 7$ candidate has $\eta = 2$.

Fix such a cut and choose a maximum-cardinality ordinary barrier X in G, with $b = |X|$. The ordinary specialization of Lemma 107 makes every component of G-X odd and factor-critical. To see that no even component remains, adjoining one of its vertices to X would create at least one odd component and would not decrease the maximizing objective, contradicting maximal cardinality. The number of components is $|V(G)| - 2\nu(G) + b = a + b - 5$.

The large-component conclusion of Lemma 140 applies to H with $ex(H) = 4$ and $\eta = 2$. Some component Q of G-X has order at least $2a + 2b + 5$. Counting X, this component, and at least one vertex in each remaining component gives

$$a + 15 = |V(G)| \geq b + (2a + 2b + 5) + (a + b - 6) = 3a + 4b - 1.$$

Thus $a + 2b \leq 8$. The four candidate pairs have a in {7,8,9,10}. Values nine and ten are impossible; in the other two cases $b = 0$.

Hence every whole component of G is factor-critical. By hypothesis G has actual degree seven and a matching covering that degree. Delete any degree-seven vertex v. Factor-criticality of its component gives $\nu(G - v) = \nu(G) = 10$. The child J has slack zero by covered deletion. Since $E_7(G) = 10$, equation (3) gives $E_7(J) = -1 + 4u$, and (1) forces $u = 0$. Thus $\alpha(J) = a$ and $E_7(J) = -1$. Its small-deficit terminal again consists of four triangles and a-4 isolates. Selecting one vertex from each triangle and every isolate gives another maximum independent S′ of H. Its complement contains four induced degree-seven edges and has one of the two states in (2), a contradiction. All four $k - a = 7$ pairs have been excluded.

We are left with $k = 14$ and $a = 8$. If some maximum independent cut had $\eta = 2$, its G would have order 21 and matching number nine. At an ordinary barrier X of size b it would have $3 + b$ odd components. The same large-component conclusion gives one of order at least $21 + 2b$. Counting as before yields

$$21 \geq b + (21 + 2b) + (2 + b) = 23 + 4b,$$

which is impossible. Thus every such cut has $\eta = 3$ and $\nu(G) = 10$, $ps(G) = 0$. Formula (4) gives its order 21 and edge count 52. The zero-slack cap-seven density bound gives

$$52 \leq -4 \cdot 21 + 11 \cdot 10 + 4\alpha(G) = 26 + 4\alpha(G).$$

Therefore $\alpha(G) \geq 7$, while $\alpha(G) \leq \alpha(H) = 8$.

The other conclusions now follow on the same graphs. Regularity gives $|V(H)| = 29$ and $e(H) = 116$; excess four gives $c_0(H) = 18$; gap one gives $ps(H) = 3$ and $\Sigma(H) = 117$. The covering matching in G gives $c_7(G) = 0$, so its exact row relation yields $c_7(H) = 8$. The supplied bounds at $k = 14$, $a = 8$ are $10 \leq c_6(H) \leq 13$. Subtracting the exact row loss eight gives $2 \leq c_6(G) \leq 5$. All other row identities and degree-covering assertions were hypotheses for every maximum independent set and remain valid for every such S. This proves the result. □

Remark. A nearly tight degree-seven complement cannot contain four disjoint induced edges at maximum degree: they require more deletions than its density permits. Terminal triangles in the four large-gap parameter pairs create exactly this forbidden configuration after choosing another maximum independent set. An ordinary barrier then rules out the smaller matching gain in the sole remaining pair.

Depends on. Lemma 61, Lemma 72, Lemma 107, Lemma 123, Lemma 140, Lemma 146, Lemma 190.

Lemma 233 (A boundary-one component from slack and a vertex pool). *Use the finite integer certificate costs c_i, matching number ν, excess $ex = c_0 - \nu$, slack ps, $gap = ex - ps$, and top missing-allocation run tr of Lemma 23. Put $h(j) = \max(4, \lfloor j/2 \rfloor + 1)$ for every nonnegative integer j. A graph satisfies T_- when positive gap implies gap one and $tr + ex \le h(\Delta)$.*

Let $D \ge 5$ be an odd integer, let H be a finite simple graph, and let W be a vertex subset with $w = |W| \ge h(D) - 1$. Put $R = H - W$, with induced vertex deletion. Assume $\Delta(R) = D$ and no matching of R covers all its degree-D vertices; every induced subgraph Q of R, including R, satisfies T_-; $ex(R) = ps(R) = s \ge 4$, $c_0(H) = w + \nu(R) + s$, and $|V(R)| < 4w + 6s + 7$. Assume additionally that either $s \ge h(D) - 1$, or every induced subgraph Q of R with $ex(Q) \ge 4$ has $gap(Q) \le 0$.

Then R has a unique connected component C with $ex(C) \ge 4$. Put $P = R - V(C)$. One has $ps(P) = 0$, $ex(C) = ps(C) = s$, $gap(C) = 0$, $\Delta(C) = D$, $|V(C)| = 2\nu(C) + 1$, and $c_D(C) = c_D(R) = 1$. The minimum number of degree-D vertices of C missed by a matching is one.

Let A be the degree-D vertices of C. For $X \subseteq V(C)$, let $q_A(C - X)$ count the odd-order components of C-X whose vertex sets are contained in A. Choose any X of maximum cardinality among the sets maximizing $q_A(C - X) - |X|$, and put $b = |X|$. Then $b \ge 1$, X is independent, and C-X consists of exactly $b + 1$ odd factor-critical components, all with vertex sets contained in A. Factor-critical means that deleting any vertex leaves a perfect matching.

Choose any component Z of C-X having largest matching number m. Its boundary to X has exactly one edge, and $ex(Z) = ps(Z) = s$, $gap(Z) = 0$, $m \ge w + b + \nu(P) + 3s - 1$. Every other component Q of C-X satisfies $\nu(Q) \le w + b + 1$ and $ps(Q) + e_C(V(Q), X) \le D$, where $e_C(U, V)$ counts edges between disjoint vertex sets.

For D in {5,7,9}, the extra alternative imposes no restriction beyond $s \ge 4$: h(D)-1 equals 3, 3, 4, respectively. For $D = 11$ it imposes no extra restriction when $s \ge 5$. All orders, degrees, costs, matching numbers, counts, indices and parameters D, w, s, b, m are integers in the displayed domains.

Proof. We use nonnegative slack from Lemma 61, its zero-tail and covering-matching conclusions, and the component slack and row inequalities of Lemma 72. Feasible row-zero allocations at total l form the set Y_l defined in Lemma 75.

Choose a component C of R with largest matching number. It supplies a full interval at total $c_0(C) \ge \nu(C)$. Every other component supplies endpoints at its matching number and has missing runs at most nu(C)-1, or zero if this expression is negative. Successive component absorption in Lemma 75 gives $c_0(R) \le \nu(R) + ex(C)$. Thus $ex(C) \ge s \ge 4$. We claim $gap(C) \le 0$. This follows from the extra induced-excess condition when that alternative holds. Otherwise $s \ge h(D) - 1$. If $gap(C) > 0$, hereditary T_- gives $ex(C) \le h(\Delta(C)) - 1 \le h(D) - 1 \le s$. Therefore $ex(C) = s$ and $tr(C) \le h(D) - s \le 1$. The actual W-pool fills the missing allocations of C at total $c_0(C) - 1 = \nu(C) + s - 1$. The resulting full interval has length $w + \nu(C) + s - 1 \ge \nu(C)$, so it absorbs each other component at its matching number. This gives all allocations on H at total $w + \nu(R) + s - 1$, contrary to $c_0(H)$. Thus $gap(C) \le 0$ under either alternative, and $ps(C) \ge ex(C) \ge s$. Slack superadditivity and nonnegativity give $ps(C) = ex(C) = s$ and $ps(P) = 0$. Every other component U has $ps(U) = 0$; if $ex(U) > 0$, its T_- implication gives $ex(U) = gap(U) = 1$. Hence no other component has excess at least four, proving uniqueness.

Every degree-D vertex of P can be covered by one matching of P. Otherwise a connected component U of P, of actual maximum degree D, would have positive degree-D matching deficiency. It has $ps(U) = 0$ by superadditivity. Choose a maximum-cardinality marked barrier in U using Lemma 107. If it were empty, connectedness and positive deficiency would make U an odd-order D-regular graph, impossible because D is odd and its degree sum must be even. The barrier is therefore nonempty. Every component after its deletion has gap at most one by hereditary T_-. The bound Lemma 202 gives $ps(U) \ge 1$, a contradiction. Component covering matchings may be united, proving the assertion about P and hence $c_D(P) = 0$. Since R is uncovered, its component C has actual degree D and is uncovered.

Let $\kappa \ge 1$ be C's degree-D deficiency. Fix any X as in the statement. Existence and factor-criticality of the marked odd components follow from Lemma 107; maximum cardinality implies inclusionwise maximality among maximizers. The same odd-degree argument makes X nonempty. Write calQ for all components of C-X, calK for its marked odd components, and Y for the union of the vertex sets of the other components. By Lemma 104, $|calK| = b + \kappa$, $\nu(C) = b + \sum_{Q \in calQ} \nu(Q)$, $c_D(C) \le \kappa$. Every K in calK has a positive odd boundary $d(K) = e_C(V(K), X)$. This is its total internal degree-D defect, since all its vertices have C-degree D.

Choose any Z in calQ of largest matching number m. Its full-cost interval, together with the actual X-pool, absorbs every other component at its matching number. The same allocation theorem gives $c_0(C) \leq \nu(C) + ex(Z)$. Hence $ex(Z) \geq s$. We claim $gap(Z) \leq 0$. Again this follows under the induced-excess alternative. Otherwise $s \geq h(D) - 1$, and positive gap would give $ex(Z) = s$ and $tr(Z) \leq h(D) - s \leq 1$. To rule this out, first absorb P at its matching number using the actual W union X pool. This is possible because $ps(P) = 0$: either $ex(P) = 0$ and that allocation set is full, or T_- gives $ex(P) = gap(P) = 1$ and $tr(P) \leq h(D) - 1 \leq w$. The resulting full interval has length $w + b + \nu(P)$. It fills the missing allocations of Z at total $c_0(Z) - 1 = m + s - 1$. The resulting interval then absorbs every other Q at its matching number, since $\nu(Q) \leq m$. Restoring every edge meeting W union X gives all allocations on H at total $w + \nu(R) + s - 1$, again a contradiction. Thus $gap(Z) \leq 0$ and $ps(Z) \geq ex(Z) \geq s$ under either alternative.

We set up the precise credit inequality before using any completion claim. Put $T = (D-1)/2$. For every marked K different from Z with $d(K) < D$, apply Lemma 179: K is connected and factor-critical, its defect is $d(K)$, and $gap(K) \leq 1$ by heredity. It gives credits $a_i(K)$ with total max(D-d(K)-ps(K),0), supported on $1 \leq i \leq T$, bounded by $\max(c_i(K) - \nu(K), 0)$, with $a_1(K) \leq T$ and $a_i(K) \leq 1$ for $i \geq 2$. Assign zero credits to Z, to non marked components, and to marked components with $d(K) \geq D$; their required total is zero by nonnegative slack.

Run the finite procedures of Lemma 104 with these credits. At index i the initial interval length is $b + \sum_{(} Q : a_i(Q) = 0) c_i(Q)$, hence at least $b + c_i(Z)$. Fix any permitted choices of eligible components. Let E_{rem} be the sum of maximum credits at unfinished indices. Then $E_{rem} \leq 2T - 1 = D - 2$. Define J=e(C[X])+e_C(X, Y)+sum_(Q not in calK, Q!=Z)ps(Q)+sum_(K in calK, K!=Z)max(ps(K)+d(K)-D,0). All its terms are nonnegative. Retaining the top-row correction in that theorem and using $gap(C) = 0$ gives

$ps(Z) + J + \sum_{Q \in calQ} c_D(Q) \leq \beta + E_{rem} - D + 1_{Z \in calK}(D - d(Z))$, (1)

where $\beta = s + c_D(C) - D(\kappa - 1) \leq s + D - (D-1)\kappa \leq s + 1$. The indicator term is zero when Z is not marked. To check this conversion, replace gap(Z)-(ex(Z)-ex(C)) by s-ps(Z) in the theorem's gap bound, then substitute its correction $\kappa - c_D(C) + \sum_Q c_D(Q)$.

If Z is not marked, or is marked with $d(Z) \geq D$, the right side of (1) is at most s-1. This contradicts $ps(Z) \geq s$. Therefore Z is marked with $1 \leq d := d(Z) < D$. Applying Lemma 179 to this same Z gives $|V(Z)| = 2m + 1$, $u = (D - d)/2 > 0$, zero rows at and above D, and

$ps(Z) = (D-1)m + u - \sum_{i=1}^{D-1} c_i(Z)$,

$c_i(Z) + c_{D-i}(Z) \leq 2m + 1$ for $1 \leq i < D$. (2)

Inequality (1) now has right side $\beta + E_{rem} - d$.

We next force a large matching number using actual allocations on H. Since $ps(P) = 0$, either $ex(P) = 0$ and $Y_{(}\nu(P))(P)$ is full, or hereditary T_- gives $ex(P) = gap(P) = 1$ and $tr(P) \leq h(\Delta(P)) - 1 \leq h(D) - 1 \leq w$. Thus the actual deleted set W union X absorbs P at its matching number and supplies a full interval of length $L = w + b + \nu(P)$.

At total $l = m + s - 1$ on Z, every missing allocation has both coordinates at least 2s-1. Indeed a coordinate at least m is feasible by putting all edges in its color. If a coordinate $x < m$ satisfies $x \leq 2s - 2$, choose any x vertices of Z and color all incident edges with that color. Its matching number is at most x. The induced remainder has $2m + 1 - x$ vertices, at most $2(l - x) + 1$, so its matching number is at most l-x. This supplies the allocation. The argument applies to either coordinate. Consequently every missing first budget lies in [2s-1,m-s], and its longest run has length at most $\max(0, m - 3s + 2)$.

If $m \leq L + 3s - 2$, the interval [0,L] would fill this allocation set. Every other Q has $\nu(Q) \leq m$, so the resulting full interval also absorbs all those components at their matching numbers. By Lemma 75 these partitions and the actual W union X extension give every allocation on H at total $w + b + \nu(P) + m + s - 1 + \sum_{Q \neq Z} \nu(Q) = w + \nu(R) + s - 1$. This contradicts the assumed exact $c_0(H)$. Hence

$m \geq w + b + \nu(P) + 3s - 1$. (3)

This reasoning includes P empty and assigns every edge meeting W union X.

If another Q had $\nu(Q) \geq w + b + 2$, then its vertex order is at least twice that number. Equations (2) and (3) would give $|V(R)| \geq |V(Z)| + |V(Q)| \geq 4(w + b) + 2\nu(P) + 6s + 3 \geq 4w + 6s + 7$, contradicting the order hypothesis because $b \geq 1$. Therefore every other Q has $\nu(Q) \leq w + b + 1$.

We can now finish every credit procedure. An unfinished index j has a remaining positive-credit component Q with $\nu(Q) \geq b + c_j(Z) + 2$. The preceding bound forces $c_j(Z) \leq w - 1$. Combining this with (3) gives

$m - c_j(Z) \geq b + \nu(P) + 3s$. (4)

If index one is unfinished, monotonicity bounds all D-1 entries in (2) by $c_1(Z)$, giving $ps(Z) \geq u + 3(D-1)s$. But (1) gives $ps(Z) \leq s + D - 1 - d$. The lower bound exceeds the upper by at least $u + (3D-4)s - D + 1 + d > 0$ for $D \geq 5$, $s \geq 4$, $d \geq 1$. Hence index one is finished.

If any later index is unfinished, let j be the first. Then $2 \leq j \leq T$ and $E_{rem} \leq T - j + 1$. Pair the 2(j-1) indices i, D-i for $1 \leq i < j$ using (2). All remaining $D - 2j + 1$ entries have index at least j and are bounded by $c_j(Z)$. From (2) and (4), $ps(Z) \geq u - j + 1 + (D - 2j + 1)(b + \nu(P) + 3s)$. Inequality (1) gives $ps(Z) \leq s + T - j + 2 - d$. The difference between these bounds is $(d-1)/2 + (D - 2j + 1)(b + \nu(P) + 3s) - s > 0$: d is positive odd, $b \geq 1$, and $j \leq T$ makes $D - 2j + 1 \geq 2$. This is another contradiction. Thus every procedure finishes and $E_{rem} = 0$.

Return to (1). Its left side is at least s and its right side is beta-d. If $\kappa \geq 2$, $\beta \leq s + 2 - D < s$, impossible. Hence $\kappa = 1$. Now $ps(Z) \geq s$, $d \geq 1$ and $c_D(C) \leq 1$ force equality everywhere: $ps(Z) = ex(Z) = s$, $d = 1$, $c_D(C) = 1$, $J = 0$. The equality $ex(Z) = s$ uses $ex(Z) \leq ps(Z)$ already proved. The zero correction makes X independent and gives no edges from X to a non marked component. Connectedness of C excludes any such component. Hence $calQ = calK$ has $b + 1$ factor-critical members, and $J = 0$ gives $ps(Q) + d(Q) \leq D$ for every member other than Z. Counting their odd orders with the matching identity gives $|V(C)| = 2\nu(C) + 1$. This proves all component and barrier conclusions for the arbitrary choices of X and Z.

Finally $c_D(R) \leq c_D(C) + c_D(P) = 1$ by component subadditivity. Cost zero would provide, at capacities (D-1,1) and zero matching budgets, a matching meeting every degree-D vertex of R, contrary to the hypothesis. Thus $c_D(R) = 1$. For $D = 5, 7, 9$, the value h(D)-1 is respectively 3, 3, 4, so $s \geq 4$ always implies the first extra alternative. For $D = 11$ the value is five, so the same holds when $s \geq 5$. This verifies the stated specializations. □

Remark. The actual omitted vertices can fill a short top allocation run. Above the degree-dependent slack threshold this rules out positive gap in both selected large components, leaving the same exact boundary-credit and order argument to force one boundary edge.

Depends on. Lemma 23, Lemma 61, Lemma 72, Lemma 75, Lemma 104, Lemma 107, Lemma 179, Lemma 202.

Lemma 234 (Two units of slack in an octic closed graph). *Use the finite integer certificate costs $c_i(H)$, matching number $\nu(H)$, edge count $e(H)$, excess $ex(H) = c_0(H) - \nu(H)$, slack $ps(H) = e(H) - \nu(H) - \sum_{i \geq 1} c_i(H)$, and signed gap $gap(H) = ex(H) - ps(H)$ of Lemma 23. All graphs are finite and simple. A nonempty graph is factor-critical if deleting any vertex leaves a perfect matching.*

Let H be a nonempty connected eight-regular factor-critical graph. If $ex(H) \leq 2$, then $ps(H) \geq 2$, and therefore $gap(H) \leq 0$. Consequently $gap(H) \leq 1$ implies $ps(H) \geq 2$. In particular the same slack bound holds when every induced subgraph of H, including H itself, has gap at most one.

There is no upper bound on the order of H. No hereditary hypothesis is required for either of the first two implications. Graph orders, degrees, indices, costs, capacities, budgets, matching cardinalities and excesses are nonnegative integers; the slack and gap are initially defined as signed integers.

Proof. Write $n = |V(H)| = 2k + 1$ and $E = ex(H)$. Factor-criticality gives $\nu(H) = k$, while regularity gives $e(H) = 8k + 4$. Simplicity implies $k \geq 4$. The cost definitions and monotonicity in Lemma 23 give $c_i(H) \leq k + E$ for every nonnegative integer i. We first assume $E \leq 2$ and prove $ps(H) \geq 2$.

1. The two top rows and the complete row sum.

The degree-eight specialization of Lemma 53 gives

$c_7(H) \leq floor((800k - 180)/1300)$. (1)

By Lemma 92 with $d = 8$, $j = 2$ and sampling probability $2/5$,

$c_6(H) \leq floor((2k + 1)q)$, where

$q = 2/5 + (3/5)^9 + 8(2/5)(3/5)^8 + 4(2/5)^2(3/5)^7$

$= 940901/1953125 < 241/500$. (2)

The last comparison follows from

$241 \cdot 1953125 - 500 \cdot 940901 = 252625 > 0$.

Fix a vertex v and a perfect matching M of H-v. Applying Lemma 49 with $D = 8$, $j = 0$ and $Z = \{v\}$ gives $c_8(H) \le 1$. For every integer $i \ge 9$ the degree-i set of H is empty and $\Delta(H) \le i$. The first inequality of Lemma 92 therefore gives $c_i(H) = 0$.

Apply Lemma 119 to the complementary thresholds three and five, and then to four and four. It gives $c_3(H) + c_5(H) \le 2k + 2$ and $c_4(H) \le k + 1$. Put $o_k = 1$ for odd k and $o_k = 0$ for even k. When k is odd, equality $c_4(H) = k + 1$ would make the two tight complementary costs even, contrary to the odd-cost conclusion of that theorem. Hence $c_4(H) \le k + 1 - o_k$. Grouping all positive rows as {1,2}, {3,5}, {4}, {6,7}, {8}, and using the zero tail gives

$ps(H) \ge 2k - 2E + o_k - c_6(H) - c_7(H)$. (3)

If $n \le 53$, the order clause of Lemma 190 applies at $d = 8$ to the odd thresholds three and five, improving their sum to $2k + 1$. The same grouping then gives

$ps(H) \ge 2k - 2E + 1 + o_k - c_6(H) - c_7(H)$. (4)

For every $k \ge 13$, equations (1)–(3), $E \le 2$ and $o_k \ge 0$ yield

$ps(H) > 2k - 4 - (482k + 241)/500 - (800k - 180)/1300$

$= (2734k - 28233)/6500$

$\ge 7309/6500 > 1$.

The slack is an integer, so $ps(H) \ge 2$. It remains to consider the nine integers $4 \le k \le 12$; equation (4) applies throughout this range.

2. Three finite sampling improvements for the sixth row.

Here and below $binom(a, b) = 0$ when the integer b is outside [0,a], for a nonnegative integer a. Let U be uniformly distributed among the r-element subsets of $V(H)$, where $0 \le r \le n$. For a vertex outside U, let X be the number of its neighbors in U. Let B consist of vertices outside U with $X \le 1$, and let T consist of vertices outside U with $X = 2$. Put $Z = U$ union B. Each vertex outside Z has at least two neighbors in U, so $\Delta(H - Z) \le 6$. Every vertex of degree six in H-Z belongs to T: it has exactly two neighbors in Z, and hence exactly two in U. The deletion inequality of Lemma 92, with $\theta = 6$ and $m_\theta = 7$, gives for every sample

$c_6(H) \le r + |B| + \lfloor |T|/7 \rfloor$

$= \lfloor r + |B| + |T|/7 \rfloor$.

A specified vertex has eight neighbors and n-9 other vertices distinct from itself. The counts of samples excluding that vertex and meeting its neighborhood in zero, one, or two vertices are respectively

$binom(n - 9, r)$, 8binom(n-9,r-1), 28binom(n-9,r-2).

Linearity of expectation therefore gives

$E(|B| + |T|/7)$

$= n[binom(n - 9, r) + 8binom(n - 9, r - 1) + 4binom(n - 9, r - 2)]/binom(n, r)$. (5)

The sample space is finite and nonempty. Some sample has $|B| + |T|/7$ at most this average. Its actual deletion set Z proves the corresponding integer cost bound after taking the floor.

We use the following three instances; the middle two columns are the exact numerator and denominator of (5):

n | r | numerator | denominator | bound for $c_6(H)$

11 | 3 | 176 | 165 | 4

17 | 6 | 12852 | 12376 | 7

21 | 8 | 221067 | 203490 | 9.

For the first row, $176 < 2 \cdot 165$; for the second, $12852 < 2 \cdot 12376$; for the third, $221067 < 2 \cdot 203490$. Thus each floor is at most $r + 1$, as stated. These bounds include all original edges through the deletion inequality.

3. The thirteen-vertex sixth row.

Suppose $n = 13$. Choose uniformly a four-element subset Z of $V(H)$. A vertex outside Z has fewer than two neighbors in Z only if Z contains zero or one of its eight neighbors. There are four vertices that are neither that vertex nor its neighbors. The number of failing samples for a fixed vertex is consequently

$binom(4,4) + 8binom(4,3) = 33$.

The expected number of failing vertices is $13 \cdot 33/binom(13,4) = 429/715 < 1$. Since that number is an integer, some Z has no failures.

For this fixed four-set let $Q = H - Z$ be induced vertex deletion. Then $|V(Q)| = 9$ and $4 \le d_Q(u) \le 6$ for every vertex u of Q. Each connected component has at least five vertices, so Q is connected. It cannot be six-regular: if it were, precisely eighteen edges would join Q to Z. The degree sum over the four vertices of Z is thirty-two, so H[Z] would have $(32 - 18)/2 = 7$ edges, whereas a simple graph on four vertices has at most six.

Thus Q has no odd-order six-regular connected component. Its degree-six set has size at most nine, which is at most $2(6 + 1) - 1$. The component-excluding assertion of Lemma 86, at $r = 6$, supplies a matching covering every degree-six vertex of Q. Its vertex-correction conclusion applied to the same fixed Z gives $c_6(H) \le 4$. In particular the certificate restores all original edges meeting Z and is not merely a certificate on Q.

4. The remaining small-order values.

For $n = 9$, keep any vertex v and perfect matching M of H-v, and write $J = H - M$, deleting edges only. Simplicity and eight-regularity make H complete on nine vertices. The vertex v has degree eight in J and every other vertex has degree seven. The set {v} dominates J, giving $c_7(H) \le 1$ by Lemma 49 with $j = 1$. Every three-element set Z containing v gives each outside vertex at least two J-neighbors in Z: that vertex has only one nonneighbor in J among the other eight vertices. The same theorem with $j = 2$ gives $c_6(H) \le 3$.

At $n = 9$ one also has $c_0(H) \le 5$, hence $E \le 1$. To see this for every integer budget split of total five, let x be the larger budget and y the smaller. Then $x \ge 3$ and $x + y = 5$. Choose $\min(9, 2x + 1)$ vertices and put their internal edges in the part with budget x; put every remaining edge in the part with budget y. The first part has matching number at most x. Every edge of the other part meets the unchosen set, whose size $\max(0, 8 - 2x)$ is at most $y = 5 - x$ for x in {3,4,5}. Its matching number is therefore at most y. The degree-bounded parts are empty. Relabeling the two parts according to the given budget order proves universal feasibility at total five.

For $k = 7$, use $J = H - M$ on $n = 15$ vertices with its marked vertex v of degree eight and all other vertices of degree seven. The recurrence of Lemma 37, with $d = 8$, starts at six and has successive values

6, 2, 0,

because $\lceil 48/14 \rceil = 4$ and $\lceil 16/13 \rceil = 2$. It supplies a marked dominating set of size at most three. By Lemma 49 with $j = 1$ this gives $c_7(H) \le 3$.

For all other small k use (1) for the seventh row. For the sixth row use (2), except for the five improvements already proved at $n = 9, 11, 13, 17, 21$. The resulting bounds and the lower bounds from (4) are

k | bound for E | bound for $c_7(H)$ | bound for $c_6(H)$ | bound for $ps(H)$

4 | 1 | 1 | 3 | 3

5 | 2 | 2 | 4 | 2

6 | 2 | 3 | 4 | 2

7 | 2 | 3 | 7 | 2

8 | 2 | 4 | 7 | 2

9 | 2 | 5 | 9 | 2

10 | 2 | 6 | 9 | 2

11 | 2 | 6 | 11 | 3

12 | 2 | 7 | 12 | 2.

For clarity, the direct sixth-row uses of (2) occur at $k = 7, 9, 11, 12$. The inequalities

$(2k+1) \cdot 241/500 < k+1$

hold respectively as $3615 < 4000$, $4579 < 5000$, $5543 < 6000$, and $6025 < 6500$; hence these integer costs are at most k. The remaining entries come from the explicit constructions. Substituting each row into (4), with its displayed parity and excess bound, gives exactly the last column. This proves $ps(H) \geq 2$ also for $4 \leq k \leq 12$.

We have proved the first implication for every order. Since $E \leq 2$ then gives $gap(H) = E - ps(H) \leq 0$, its stated goodness conclusion follows. Finally suppose $gap(H) \leq 1$. If $E \leq 2$, the implication already proved gives $ps(H) \geq 2$. Otherwise the nonnegative integer E is at least three, and the identity $ps(H) = E - gap(H)$ gives $ps(H) \geq E - 1 \geq 2$. The hereditary condition in the last assertion includes $gap(H) \leq 1$ for H itself, so it implies the same conclusion. □

Depends on. Lemma 23, Lemma 37, Lemma 49, Lemma 53, Lemma 86, Lemma 92, Lemma 119, Lemma 190.

Lemma 235 (The possible positive-gap row at small degree-fourteen orders). *Use the finite integer certificate costs $c_i(F)$, matching number $\nu(F)$, excess $ex(F) = c_0(F) - \nu(F)$, and signed gap $gap(F) = \sum_{i \geq 0} c_i(F) - e(F)$ of Lemma 23. Let $7 \leq k \leq 76$ be an integer and let F be a nonempty connected 14-regular finite simple graph on $n = 2k+1$ vertices such that deleting every vertex leaves a perfect matching. If $gap(F) > 0$, then $k = 17$, $ex(F) = 6$, and $gap(F) = 1$. No existence of a graph with these exceptional parameters is asserted. There is no hereditary or minimality hypothesis. All orders, degrees, costs, cardinalities, capacities and matching budgets are integers in their stated domains.*

Proof. Put $E = ex(F)$. Factor-criticality gives $\nu(F) = k$, and regularity gives $e(F) = 14k + 7$. The definitions in Lemma 23 give $E \geq 0$ and $c_j(F) \leq k + E$. Lemma 87 gives $c_i = 0$ for $i \geq 15$, $c_{14} \leq 1$, $c_7 \leq k+1$, and $c_i + c_{14-i} \leq 2k+2$ for $i = 3, 4, 5, 6$. Suppose u of the pairs with $i = 3, 5$ have the stronger bound $2k+1$, and $c_7 \leq k + 1 - \iota$ where ι is zero or one. For any bounds $c_1 \leq k + a_1$, $c_2 \leq k + a_2$, counting these four pairs and the remaining rows separately gives

$gap(F) \leq E + a_1 + a_2 - 2k + 3 + c_{13}(F) + c_{12}(F) - u - \iota$. (1)

We first obtain exact trial bounds for the low rows. For r in {9,11,12,13} and c in {0,1}, put

$h(r,c) = ceil(((r-1)k + 2c)/(3r+1))$.

Put $g(r,c) = h(r,c) - 1$ if $h(r,c) \geq 1$, $k + h(r,c)$ is even, and

$(3r+1)(h(r,c) - 1) \geq (r-1)(k-1) + 2c$;

otherwise put $g(r,c) = h(r,c)$. For $g = g(r,c)$ and $x = floor((k+g)/2)$ one has

$(r-1)x + c \leq 2rg$. (2)

When $g = h$ this follows from $((r-1)k + 2c - (3r+1)g)/2 \leq 0$; when $g = h - 1$ the difference of the two sides of (2) equals $((r-1)(k-1) + 2c - (3r+1)g)/2 \leq 0$, using the odd integer $k + g$.

Define $E_0 = g(13,1)$, and $g_j = g(14-j, 0)$ for $j = 1, 2, 3, 5$. Then

$E \leq E_0$, $c_j(F) \leq k + g_j$ for $j = 1, 2, 3, 5$. (3)

To verify (3), start an independent set with one vertex. Until it dominates F, connectedness supplies a vertex at distance two that can be added while sharing a neighbor with the old set. Every addition enlarges its neighborhood by at most 13, so a t-vertex prefix has at most $13t + 1$ neighbors. The final dominating prefix satisfies $n \leq 14t + 1$. Thus every integer $t \leq \lceil 2k/14 \rceil$ is available.

Set $g = E_0$ and fix any nonnegative matching budgets $x + y = k + g$, with $x \leq y$. If $y \geq k$, put all edges into that exceptional part. Otherwise $x \leq floor((k+g)/2)$ and put $t = x - 2g$. If $t \leq 0$, put all edges meeting any x-set into the smaller exceptional part, and the remaining edges into the larger part. Their matching bounds are x and y because $n - x \leq 2y + 1$. If $t \geq 1$, (2) gives $13t + 1 \leq x$. Also $3x + t \leq 2k$, so

$40t + 3 \le 2k$ and $t \le \lceil 2k/14 \rceil$. Choose the independent t-set just constructed and enlarge its neighborhood to an x-set W disjoint from it. This is possible because $n - t = x + 2y + 1$. Edges meeting W form the smaller exceptional part; every remaining edge lies outside both W and the independent set and therefore on at most $n - x - t = 2y + 1$ vertices. Both bounded-degree parts are empty. Swapping exceptional parts handles the reversed budgets, proving $c_0 \le k + E_0$.

For each $j = 1, 2, 3, 5$, put $r = 14 - j$, $g = g_j$, $x = floor((k + g)/2)$, $t = x - 2g$. Inequality (2) gives $rt \le x$. If $t \ge 1$, then $(3r + 1)t \le 3x + t \le 2k$, so $t \le 2k/28 < n/15$. A maximum independent set dominates F and each of its closed neighborhoods has at most 15 vertices; hence $\alpha(F) \ge \lceil n/15 \rceil > t$. Since $3j \le 15$, the final assertion of Lemma 54 now proves $c_j \le k + g_j$. It also covers $t \le 0$, completing (3).

For δ in {13,14}, start $u_0(\delta) = n$ and, while $u_i(\delta) > 0$ and $(\delta + 1)u_i(\delta) > 2(n - i)$, use

$u_{i+1}(\delta) = \max(0, u_i(\delta) - ceil((\delta + 1)u_i(\delta)/(n - i)))$.

Let s_δ be the stopping index. Define

$z = floor((42s_{14} + 14u_{s_{14}}(14) + 2n)/44)$,

$b_1 = 1 + floor((39s_{13} + 13u_{s_{13}}(13) + 2n)/41)$,

$b_2 = \lfloor 33358n/100000 \rfloor$.

These satisfy $\gamma(F) \le z$, $c_{13}(F) \le b_1$, $c_{12}(F) \le b_2$. (4)

Here γ is the minimum cardinality of a dominating set. The first bound follows from Lemma 162 at $\delta = 14$. For the second, fix a vertex v and a perfect matching M of F-v. Apply that fact at $\delta = 13$ to F-M, and add v to the resulting dominating set. It contains the only vertex uncovered by M and dominates that same F-M. Lemma 49 with $D = 14$, $j = 1$ supplies the asserted original-edge certificate. This explains the added unit in b_1. For b_2 use Lemma 92 with $d = 14$, $j = 2$, $p = 27/100$, $m_\theta = 13$. Its expression Q obeys

$13 \cdot 100^{15} \cdot Q$

$= 13 \cdot 27 \cdot 100^{14} + 13 \cdot 73^{15} + 13 \cdot 14 \cdot 27 \cdot 73^{14} + 91 \cdot 27^2 \cdot 73^{13}$

$= 433645874178603081029831871154$

$< 433654000000000000000000000000 = 13 \cdot 33358 \cdot 10^{25}$.

This gives the integer bound in (4).

The stronger complementary conclusions in Lemma 199 yield the following simultaneous numerical savings:

$u = [n \le 99 \text{ or } k + \min(E_0, g_3) \le 74] + [n \le 99 \text{ or } k + \min(E_0, g_5) \le 62]$,

$\iota = [k \text{ is odd or } n \le 99]$. (5)

Brackets are zero-one indicators. The terms in u concern pairs 3/11 and 5/9, with sufficient cost bounds $6 \cdot 11 + 8 = 74$ and $6 \cdot 9 + 8 = 62$ and order condition $n \le 6 \cdot 14 + 15 = 99$. For ι, its 7/7 bound at $n \le 99$ gives $2c_7 \le n$ and $c_7 \le k$; its parity conclusion gives $c_7 \le k$ when k is odd.

Put

$A = \min(E_0, floor((2k - 3 - b_1 - b_2 + u + \iota)/3))$,

$G_A = 3A - 2k + 3 + b_1 + b_2 - u - \iota$,

$G_B = E_0 + \min(E_0, g_1) + \min(E_0, g_2) - 2k + 3 + z + b_2 - u - \iota$.

The following table gives in order

k, E_0, g_1, g_2, g_3, g_5, u, ι, A, s_{14}, $u_{s_{14}}(14)$, z, s_{13}, $u_{s_{13}}(13)$, b_1, b_2, G_A, G_B.

All entries are exact values of the displayed rational floor and ceiling formulas and stopping recurrences, whose denominators are positive at each step by Lemma 162.

7, 2, 2, 2, 2, 2, 2, 1, 2, 1, 0, 1, 1, 1, 3, 5, 0, -2

8, 3, 3, 3, 3, 3, 2, 1, 2, 1, 2, 2, 2, 0, 3, 5, -2, 0

9, 3, 3, 3, 3, 3, 2, 1, 2, 2, 0, 2, 2, 1, 4, 6, -2, -1

10, 3, 3, 3, 3, 3, 2, 1, 3, 2, 1, 3, 2, 2, 4, 7, 0, -1
11, 4, 4, 4, 4, 4, 2, 1, 3, 2, 2, 3, 2, 3, 4, 7, -2, 0
12, 4, 4, 4, 4, 4, 2, 1, 3, 2, 3, 4, 3, 1, 5, 8, -2, 0
13, 4, 4, 4, 4, 4, 2, 1, 4, 3, 2, 4, 3, 2, 5, 9, 0, -1
14, 5, 5, 5, 5, 4, 2, 1, 4, 3, 2, 4, 3, 3, 6, 9, -1, 0
15, 5, 5, 5, 5, 4, 2, 1, 4, 3, 3, 5, 3, 4, 6, 10, -2, 0
16, 5, 5, 5, 5, 5, 2, 1, 4, 3, 4, 5, 4, 2, 7, 11, -2, -1
17, 6, 6, 6, 5, 5, 2, 1, 5, 4, 3, 6, 4, 3, 7, 11, -1, 1
18, 6, 6, 6, 5, 5, 2, 1, 5, 4, 3, 6, 4, 4, 7, 12, -2, 0
19, 6, 6, 6, 6, 6, 2, 1, 5, 4, 4, 6, 4, 5, 8, 13, -2, -1
20, 7, 6, 6, 6, 6, 2, 1, 6, 5, 2, 7, 5, 3, 8, 13, -1, -1
21, 7, 6, 6, 6, 6, 2, 1, 6, 5, 3, 7, 5, 4, 9, 14, -1, -2
22, 7, 7, 7, 7, 7, 2, 1, 6, 5, 4, 8, 5, 5, 9, 15, -2, 0
23, 7, 7, 7, 7, 7, 2, 1, 7, 5, 5, 8, 5, 6, 9, 15, -1, -2
24, 7, 7, 7, 7, 7, 2, 1, 7, 6, 3, 8, 6, 4, 10, 16, -1, -3
25, 8, 8, 8, 8, 8, 2, 1, 7, 6, 4, 9, 6, 5, 10, 17, -2, 0
26, 8, 8, 8, 8, 8, 2, 1, 8, 6, 6, 10, 7, 4, 11, 17, 0, -1
27, 8, 8, 8, 8, 8, 2, 1, 8, 6, 6, 10, 6, 7, 11, 18, -1, -2
28, 9, 9, 9, 9, 8, 2, 1, 8, 7, 4, 10, 7, 5, 12, 19, -1, 0
29, 9, 9, 9, 9, 8, 2, 1, 9, 6, 7, 10, 7, 6, 12, 19, 0, -2
30, 9, 9, 9, 9, 9, 2, 1, 9, 7, 6, 11, 8, 5, 13, 20, 0, -2
31, 10, 10, 10, 10, 9, 2, 1, 9, 7, 7, 11, 8, 6, 13, 21, -1, 0
32, 10, 10, 10, 10, 9, 2, 1, 10, 8, 5, 12, 8, 6, 13, 21, 0, -1
33, 10, 10, 10, 10, 10, 2, 1, 10, 8, 6, 12, 8, 7, 14, 22, 0, -2
34, 11, 11, 11, 10, 10, 2, 1, 10, 8, 6, 12, 8, 8, 14, 23, -1, 0
35, 11, 11, 11, 10, 10, 2, 1, 11, 8, 8, 13, 8, 9, 14, 23, 0, -1
36, 11, 11, 11, 11, 11, 2, 1, 11, 9, 6, 13, 9, 8, 15, 24, 0, -2
37, 12, 12, 11, 11, 11, 2, 1, 11, 9, 7, 14, 9, 8, 15, 25, -1, 0
38, 12, 12, 11, 11, 11, 2, 1, 11, 9, 8, 14, 9, 9, 16, 25, -2, -2
39, 12, 12, 12, 12, 12, 2, 1, 12, 9, 8, 14, 9, 10, 16, 26, 0, -2
40, 13, 12, 12, 12, 12, 2, 1, 12, 9, 9, 15, 10, 8, 17, 27, 0, -1
41, 13, 12, 12, 12, 12, 2, 1, 12, 10, 7, 15, 10, 10, 17, 27, -2, -3
42, 13, 13, 13, 13, 12, 2, 1, 12, 10, 9, 16, 11, 8, 18, 28, -2, -1
43, 13, 13, 13, 13, 12, 2, 1, 13, 10, 9, 16, 10, 11, 18, 29, 0, -2
44, 13, 13, 13, 13, 13, 2, 1, 13, 10, 10, 16, 11, 9, 18, 29, -2, -4
45, 14, 14, 14, 14, 13, 2, 1, 13, 11, 8, 17, 11, 10, 19, 30, -2, -1
46, 14, 14, 14, 14, 13, 2, 1, 14, 11, 9, 17, 11, 11, 19, 31, 0, -2
47, 14, 14, 14, 14, 14, 2, 1, 14, 11, 10, 18, 11, 12, 19, 31, -2, -3

48, 15, 15, 15, 15, 14, 2, 1, 14, 11, 11, 18, 12, 10, 20, 32, -2, -1

49, 15, 15, 15, 15, 14, 2, 1, 14, 12, 10, 19, 12, 12, 21, 33, -2, -1

50, 15, 15, 15, 15, 15, 1, 0, 14, 12, 10, 19, 13, 10, 21, 33, -2, -1

51, 16, 16, 16, 15, 15, 1, 1, 15, 12, 11, 19, 12, 13, 21, 34, -1, 0

52, 16, 16, 16, 15, 15, 1, 0, 15, 12, 12, 20, 13, 11, 21, 35, -1, 1

53, 16, 16, 16, 16, 16, 1, 1, 16, 13, 10, 20, 13, 12, 22, 35, 0, -2

54, 17, 17, 17, 16, 16, 1, 0, 16, 13, 10, 20, 13, 13, 22, 36, 0, 1

55, 17, 17, 17, 16, 16, 1, 1, 16, 13, 11, 20, 13, 14, 23, 37, -1, -1

56, 17, 17, 17, 17, 16, 1, 0, 16, 13, 13, 21, 14, 12, 23, 37, -2, -1

57, 18, 18, 17, 17, 16, 1, 1, 17, 14, 11, 22, 14, 13, 24, 38, 0, 0

58, 18, 18, 17, 17, 17, 0, 0, 16, 14, 12, 22, 14, 14, 24, 39, -2, 1

59, 18, 18, 18, 18, 17, 0, 1, 17, 14, 13, 22, 14, 15, 24, 39, -2, -1

60, 19, 18, 18, 18, 17, 0, 0, 17, 14, 14, 23, 15, 14, 25, 40, -1, 1

61, 19, 18, 18, 18, 18, 0, 1, 18, 15, 12, 23, 15, 14, 25, 41, 0, -1

62, 19, 19, 19, 19, 18, 0, 0, 18, 15, 13, 24, 15, 15, 26, 41, 0, 1

63, 19, 19, 19, 19, 18, 0, 1, 18, 15, 14, 24, 15, 16, 26, 42, -2, -1

64, 19, 19, 19, 19, 19, 0, 0, 18, 15, 14, 24, 16, 14, 26, 43, -2, -1

65, 20, 20, 20, 20, 19, 0, 1, 19, 15, 15, 25, 16, 16, 27, 43, -1, 0

66, 20, 20, 20, 20, 19, 0, 0, 19, 16, 14, 25, 16, 16, 27, 44, -1, 0

67, 20, 20, 20, 20, 20, 0, 1, 19, 16, 15, 26, 16, 17, 28, 45, -2, -1

68, 21, 21, 21, 20, 20, 0, 0, 20, 16, 15, 26, 17, 15, 28, 45, 0, 1

69, 21, 21, 21, 20, 20, 0, 1, 20, 16, 16, 26, 17, 16, 29, 46, -1, -1

70, 21, 21, 21, 21, 20, 0, 0, 20, 17, 14, 27, 18, 15, 29, 47, -1, 0

71, 22, 22, 22, 21, 20, 0, 1, 21, 17, 16, 27, 17, 18, 29, 47, -1, 0

72, 22, 22, 22, 21, 21, 0, 0, 21, 17, 17, 28, 18, 17, 30, 48, 0, 1

73, 22, 22, 22, 22, 21, 0, 1, 21, 17, 17, 28, 18, 17, 30, 49, -2, -1

74, 23, 23, 22, 22, 21, 0, 0, 21, 18, 15, 28, 18, 18, 31, 49, -2, 0

75, 23, 23, 22, 22, 22, 0, 1, 22, 18, 16, 29, 19, 17, 31, 50, -1, -1

76, 23, 23, 23, 23, 22, 0, 0, 22, 18, 16, 29, 19, 18, 32, 51, 0, 0

The table has $A \geq 2$ and $G_A \leq 0$ throughout. If $E \leq A$, (1), (4), and $c_1, c_2 \leq k + E$ give $gap(F) \leq G_A \leq 0$. For every row with $A < E_0$ we will show

$c_{13}(F) \leq \gamma(F) \leq z$ whenever $A + 1 \leq E \leq E_0$. (6)

Choose a minimum dominating set Z, of cardinality $z' \leq z$, and put $e = E - 1$. Thus $A \leq e \leq E_0 - 1$. We apply Lemma 151 with $d = 14$ to this actual Z. To verify all its conditions, put

$t_A = floor((2A - 1)/12), t_B = floor((2E_0 - 3)/12)$.

For the rows with $A < E_0$, direct substitution in the printed table gives the following minimum margins:

$16 - 2floor((k + A)/2) + 4A \geq 6$,

$n + 10 - 6(E_0 - 1) - 2t_B \geq 15$,

$6A + 2t_A + 22 - n \geq 5$,

$8A + 4t_A + 11 - n \geq 8$. (7)

Also, for every such row either $z + 11 - 2A \leq 0$, or, putting $a = z + 11 - 2A \geq 1$,

$14\,ceil(13/a) - (n + z - 4A - 5) \geq 4$. (8)

The first condition in (i) of Lemma 151 is hardest at the smallest $e = A$, and the second is hardest at the largest $e = E_0 - 1$, so the first two margins in (7) prove them. Its remaining t_0 condition follows from $E_0 \leq ceil((12k+2)/40)$, which gives $e \leq (12k+1)/40$, and hence $t_0 = floor((2e-1)/12) \leq e/6 < k/7 \leq \lceil 2k/14 \rceil$.

Condition (ii) holds at $e = A$, $z' = z$ by (8): $a \leq 0$ is exactly its first alternative, and $a > 0$ uses its second alternative. Increasing e or decreasing z′ preserves this disjunction. While the denominator $z' + 11 - 2e$ remains positive, its ceiling quotient does not decrease, while $n + z' - 4e - 5$ does not increase; once the denominator is nonpositive the first alternative holds. The last two margins in (7) prove conditions (iii), (iv) at every $e \geq A$ and every $z' \geq 0$, including the strongest case $z' = 0$ for (iv). Thus every condition of the cited theorem holds, and it proves (6).

For $E \geq A + 1$ insert (3), (5), (6) in (1). The bound

$E + \min(E, g_1) + \min(E, g_2) - 2k + 3 + z + b_2 - u - \iota$

increases by at least one when integer E increases by one. It is at most G_B. The table shows $G_B \leq 0$ except for k in {17,52,54,58,60,62,68,72}, where it is one. Positive gap at any of these orders therefore forces $E = E_0$ and $gap(F) = 1$. It also forces $c_{13}(F) = z$, because decreasing that single term by one in (1) makes the upper bound zero. By (6), $\gamma(F) = z$. Since every maximal independent set dominates F, $\alpha(F) \geq z$, and an independent set S of cardinality $a = z$ exists.

The seven exceptional orders other than 17 are eliminated with Lemma 54. In that fact use $e' = E_0 - 1$, the independent set of size $a = z$, and the value of θ in the following table. Here $b = n - a$, $q = \lfloor 14a/b \rfloor$, $r = 14a - qb$, $x. = floor((k + e')/2)$, $t. = x. - 2e'$. The final two lists give $\lfloor U_t \rfloor$ and $W_t(\theta)$ for every integer $1 \leq t \leq t.$.

k, E_0, a, θ, b, q, r, e′, $x.$, $t.$; $\lfloor U_t \rfloor$; $W_t(\theta)$

52, 16, 20, 2, 85, 3, 25, 15, 33, 3; 14, 26, 36; 12, 22, 30

54, 17, 20, 2, 89, 3, 13, 16, 35, 3; 14, 26, 37; 12, 22, 31

58, 18, 22, 0, 95, 3, 23, 17, 37, 3; 14, 26, 37; 14, 26, 37

60, 19, 23, 0, 98, 3, 28, 18, 39, 3; 14, 26, 37; 14, 26, 37

62, 19, 24, 2, 101, 3, 33, 18, 40, 4; 14, 26, 37, 47; 12, 22, 31, 39

68, 21, 26, 2, 111, 3, 31, 20, 44, 4; 14, 26, 38, 48; 12, 22, 32, 40

72, 22, 28, 2, 117, 3, 41, 21, 46, 4; 14, 26, 38, 48; 12, 22, 32, 40

For direct verification the numerator of U_t is

$(b - r)[binom(a, t) - binom(a - 3, t)] + r[binom(a, t) - binom(a - 4, t)]$

and its denominator is $binom(a, t)$. At $\theta = 0$, $W_t = \lfloor U_t \rfloor$; at $\theta = 2$ its maximum uses $m_0 = 0$, $m_l = \max(13, \lceil 13l/2 \rceil)$. Every displayed W_t is at most $t + 2e'$, so the cited fact applies. At $k = 58$, 60 it gives $c_0 \leq k + E_0 - 1$, contradicting the actual excess $E = E_0$. In the other five rows it gives $c_2 \leq k + E_0 - 1$, saving one against the previously used value $k + \min(E_0, g_2) = k + E_0$. This contradicts positive gap by (1).

Only $k = 17$ remains. Its row gives $E_0 = 6$, and the preceding monotonicity argument gives $E = 6$ and $gap(F) = 1$ whenever the gap is positive. This proves precisely the stated implication, without asserting that such a graph exists. □

Remark. Odd-pair savings and domination leave eight unit-gap possibilities. Averaged independent neighborhoods save one unit in seven of them, leaving only the stated necessary row.

Depends on. Lemma 23, Lemma 49, Lemma 54, Lemma 87, Lemma 92, Lemma 151, Lemma 162, Lemma 199.

Lemma 236 (Two units of slack in a ten-regular factor-critical graph). *Use the finite integer certificate costs $c_i(H)$, matching number $\nu(H)$, edge count $e(H)$, excess $ex(H) = c_0(H) - \nu(H)$, slack $ps(H) = e(H) - \nu(H) - \sum_{i\geq 1} c_i(H)$, and signed gap $gap(H) = ex(H) - ps(H)$ of Lemma 23. All graphs are finite and simple. A nonempty graph is factor-critical if deleting any vertex leaves a perfect matching.*

Let H be a nonempty connected ten-regular factor-critical graph. If $ex(H) \leq 2$, then $ps(H) \geq 2$ and $gap(H) \leq 0$. Consequently $gap(H) \leq 1$ implies $ps(H) \geq 2$. In particular, if every induced subgraph of H, including H itself, has gap at most one, then $ps(H) \geq 2$.

No order bound, independence bound, minimality, or hereditary hypothesis is required for the first two implications. Graph orders, degrees, costs, row indices, matching cardinalities and excesses are nonnegative integers; slack and gap are initially defined as signed integers.

Proof. Put $n = |V(H)| = 2k+1$ and $E = ex(H)$. Factor-criticality gives $\nu(H) = k$, regularity gives $e(H) = 10k+5$, and simplicity gives $k \geq 5$. First assume $E \leq 2$. By Lemma 23, each $c_i(H) \leq c_0(H) = k+E$.

Lemma 61 gives nonnegative integer slack and $c_i(H) = 0$ for $i \geq 11$. Fix any vertex v. A perfect matching of H-v covers every vertex of its degree-ten set, so the covered-degree assertion of that fact gives $c_{10}(H-v) = 0$. Its one-vertex row bound then gives $c_{10}(H) \leq 1$.

The complementary inequalities of Lemma 119, at the pairs (3,7), (4,6), and (5,5), give

$c_3(H) + c_7(H) \leq 2k+2$,

$c_4(H) + c_6(H) \leq 2k+2$,

$c_5(H) \leq k+1$.

Together with the bounds for rows one, two, and ten, they yield

$ps(H) \geq 2k - 2E - 1 - c_8(H) - c_9(H)$. (1)

If $n \leq 63$, apply Lemma 190 to the odd pairs (3,7) and (5,5). Their sums are at most $n = 2k+1$; integrality in the second pair gives $c_5(H) \leq k$. These two savings improve (1) to

$ps(H) \geq 2k - 2E + 1 - c_8(H) - c_9(H)$. (2)

The degree-ten instance of Lemma 53 gives

$c_9(H) \leq floor((560k - 430)/1000)$. (3)

Apply Lemma 92 with $d = 10$, $j = 2$, $\theta = 8$, $m_\theta = 9$, and sampling probability $p = 1/3$. Expanding the binomial probabilities in its statement gives

$c_8(H) \leq floor((2k+1)q)$,

$q = 1/3 + (2/3)^{11} + (10/3)(2/3)^{10} + (5/9)(2/3)^9$

$= 73897/177147 < 21/50$. (4)

Indeed $21 \cdot 177147 - 50 \cdot 73897 = 25237 > 0$. The coefficient $5/9$ in (4) is $binom(10,2)p^2/m_\theta = 45/(9 \cdot 9)$. Thus the application uses the correct even-threshold denominator.

For $k \geq 10$, use $E \leq 2$ in (1), then (3) and the strict upper bound in (4):

$ps(H) > 2k - 5 - 21(2k+1)/50 - (560k - 430)/1000$

$= (60k - 499)/100$

$\geq 101/100 > 1$.

Since $ps(H)$ is an integer, $ps(H) \geq 2$ throughout this range.

For the four integers $6 \leq k \leq 9$, one has $n \leq 19 \leq 63$, so (2) applies. Equations (3)–(4), using the rational upper bound $21/50$, give the following complete table:

k | upper bound for $c_9(H)$ | upper bound for $c_8(H)$ | lower bound for $ps(H)$

6 | 2 | 5 | 2

7 | 3 | 6 | 2

8 | 4 | 7 | 2

9 | 4 | 7 | 4.

To check each entry explicitly, the numerators 560k-430 in (3) are 2930, 3490, 4050, 4610, whose floors after division by 1000 are 2, 3, 4, 4. The numerators $21(2k+1)$ are 273, 315, 357, 399, whose floors after division by 50 are 5, 6, 7, 7. Substituting these integer upper bounds and $E \le 2$ into (2) gives the last column.

The remaining integer is $k = 5$. Here $n = 11$, and ten-regularity and simplicity make H the complete graph K_{11}. The graph H-v is K_{10}; it has a perfect matching, so Lemma 61 gives $c_9(H - v) = 0$ and its one-vertex row bound gives $c_9(H) \le 1$. Delete any three vertices, obtaining K_8 of degree seven. The deletion inequality of Lemma 92 at $\theta = 8$ has an empty degree-eight set and gives $c_8(H) \le 3$. Equation (2) therefore gives $ps(H) \ge 10 - 4 + 1 - 3 - 1 = 3$. This proves $ps(H) \ge 2$ for every possible k when $E \le 2$, and $gap(H) = E - ps(H) \le 0$ follows.

Finally suppose $gap(H) \le 1$. If $E \le 2$, the implication just proved applies. Otherwise the nonnegative integer E satisfies $E \ge 3$, and the identity $ps(H) = E - gap(H)$ gives $ps(H) \ge E - 1 \ge 2$. A hereditary gap bound including H itself implies this last hypothesis. This proves every assertion. □

Remark. Two top-row estimates dominate the whole slack sum at large order. Tight odd complementary rows supply the two units needed at the few small orders.

Depends on. Lemma 23, Lemma 53, Lemma 61, Lemma 92, Lemma 119, Lemma 190.

Lemma 237 (Closed lumps of degree sixteen are LP-good). *Use the finite integer certificate costs $c_i(H)$, matching number $\nu(H)$, excess $ex(H) = c_0(H) - \nu(H)$, and signed gap $gap(H) = \sum_{i\ge 0} c_i(H) - e(H)$ of Lemma 23. Let H be any nonempty connected 16-regular finite simple graph such that deleting every vertex leaves a perfect matching. Then $gap(H) \le 0$. More precisely, writing $|V(H)| = 2k+1$, the integer $k = \nu(H)$ satisfies $k \ge 8$. For $k \ge 180$, $gap(H) < 17606557/2300000 - 43443k/1150000 < 1$; the proof gives nonpositive bounds for every integer $8 \le k \le 179$. There is no hereditary, minimality, excess, top-run, or upper-order hypothesis. All orders, degrees, costs, capacities, budgets, matching sizes and indices are integers in their stated nonnegative or positive domains; gaps may be signed.*

Proof. Put $n = 2k+1$ and $E = ex(H)$. A perfect matching of each vertex deletion and the order bound on matchings give $\nu(H) = k$; simplicity gives $k \ge 8$, and regularity gives $e(H) = 16k + 8$. By Lemma 23, $E \ge 0$ and $c_j(H) \le k + E$ for every j. By Lemma 87, the rows above 16 vanish, $c_{16} \le 1$, $c_8 \le k+1$, and each pair {i,16-i}, $3 \le i \le 7$, has cost sum at most $2k+2$. If u of its three odd-index pairs have sum at most $2k+1$, and $c_j \le k + a_j$ for $j = 1, 2$, then enumerating rows 0, 1, 2, 14, 15, 16, 8 and those five pairs gives

$gap(H) \le E + a_1 + a_2 - 2k + 4 + c_{15}(H) + c_{14}(H) - u$. (1)

All these indices are distinct and together exhaust 0 through 16.

For integers r in {14,15} and c in {0,1}, define

$h(r,c) = ceil(((r-1)k + 2c)/(3r+1))$.

Define $g(r,c) = h(r,c) - 1$ when $h(r,c) \ge 1$, $k + h(r,c)$ is even, and

$(3r+1)(h(r,c) - 1) \ge (r-1)(k-1) + 2c$;

otherwise define $g(r,c) = h(r,c)$. For $g = g(r,c)$ and $x. = floor((k+g)/2)$,

$(r-1)x. + c \le 2rg$. (2)

If $g = h$, the difference of these sides is at most

$((r-1)k + 2c - (3r+1)g)/2 \le 0$.

If $g = h - 1$, then $k + g$ is odd and the difference equals

$((r-1)(k-1) + 2c - (3r+1)g)/2 \le 0$.

Put $E_0 = g(15,1)$, $g_1 = g(15,0)$, and $g_2 = g(14,0)$. We next verify

$E \le E_0$, $c_1(H) \le k + g_1$, $c_2(H) \le k + g_2$. (3)

Start an independent set with one vertex. Until it dominates H, connectedness supplies a new vertex at distance two from the old set. It shares a neighbor with the old set and increases the neighborhood by at most fifteen. Thus a t-vertex prefix has at most $15t+1$ neighbors. At the final dominating prefix, $n \le 16t+1$, so every prefix cardinality $t \le \lceil 2k/16 \rceil$ is available.

Fix nonnegative integer budgets $x+y = k+E_0$, arranged so $x \le y$. If $y \ge k$, put all edges into its exceptional part. Otherwise $x \le floor((k+E_0)/2)$ and $t = x - 2E_0$. If $t \le 0$, choose any x-set W, put all edges meeting it into the smaller exceptional part, and put the remaining edges into the larger part. The two matching numbers are at most x and y because $n - x \le 2y+1$. If $t \ge 1$, (2) and $x \le x.$ give $15t+1 \le x$. Also $3x+t \le 2k$, hence $46t+3 \le 2k$ and $t \le \lceil 2k/16 \rceil$. Choose the independent t-set just constructed and enlarge its neighborhood to an x-set W disjoint from it; this is possible because $n - t = x + 2y + 1$. The edges meeting W have matching number at most x, and every remaining edge lies outside W and the independent set, on at most $n - x - t = 2y + 1$ vertices. The bounded-degree parts are empty. This proves feasibility of every budget split at total $k + E_0$ and gives the first bound in (3).

For $j = 1, 2$, put $r = 16 - j$, $g = g_j$, $x = floor((k+g)/2)$, and $t = x - 2g$. Inequality (2) with $c = 0$ gives $rt \le x$. If $t \ge 1$, then $(3r+1)t \le 3x + t \le 2k$ and $r \ge 14$, so $t \le 2k/43 < n/17$. A maximum independent set dominates H and each closed neighborhood has at most 17 vertices, giving $\alpha(H) \ge \lceil n/17 \rceil > t$. Since $3j \le 17$, the crude private-neighbor corollary in Lemma 54 applies with degree 16 and trial excess g. It also applies directly when $t \le 0$. It gives both positive-row bounds in (3). In particular $E_0 \le ceil((14k+2)/46)$.

The parity-domination theorem Lemma 92 gives

$c_{15}(H) \le \lfloor 2203n/10000 \rfloor$, $c_{14}(H) \le \lfloor 30429n/100000 \rfloor$. (4)

The second is its stated degree-sixteen specialization. For the first, use $d = 16$, $j = 1$, $\theta = 15$, $m_\theta = 17$, $p = 1/6$. Its expression is

$q = 1/6 + (5/6)^{17} + (16/17)(1/6)(5/6)^{16}$

$= 63370245379877/287753210560512 < 2203/10000$,

which proves the first integer bound. Taking $a_1 = a_2 = E$ and $u = 0$ in (1), and using (3)-(4), yields

$gap(H) < 3((14k+2)/46 + 1) - 2k + 4 + (2203/10000 + 30429/100000)(2k+1)$

$= 17606557/2300000 - 43443k/1150000$. (5)

At $k = 180$ this equals $1967077/2300000 < 1$, and it decreases with k. Integrality proves the assertion for all $k \ge 180$.

It remains to consider $8 \le k \le 179$. For δ in {15,16}, start $u_0(\delta) = n$ and, while $u_i(\delta) > 0$ and $(\delta+1)u_i(\delta) > 2(n-i)$, put

$u_{i+1}(\delta) = \max(0, u_i(\delta) - ceil((\delta+1)u_i(\delta)/(n-i)))$.

Let s_δ be the first stopping index and put

$z = floor((48s_{16} + 16u_{s_{16}}(16) + 2n)/50)$,

$b_1 = \min(1 + floor((45s_{15} + 15u_{s_{15}}(15) + 2n)/47), \lfloor 2203n/10000 \rfloor)$,

$b_2 = \lfloor 30429n/100000 \rfloor$.

The greedy domination theorem Lemma 162 guarantees termination with positive denominators and gives a dominating set in H of size at most z. Fix a vertex v and a perfect matching M of H-v. The graph H-M has minimum degree at least 15. Apply the same theorem with $\delta = 15$, and add v to its dominating set. This set contains the sole vertex missed by M and dominates H-M. Thus Lemma 49 with $D = 16$, $j = 1$ gives the first alternative in b_1 as an original-edge cost bound. Together with (4), we have

$c_{15}(H) \le b_1$, $c_{14}(H) \le b_2$, $\gamma(H) \le z$, (6)

where $\gamma(H)$ is the smallest dominating cardinality.

Define

$u = |\{i \in \{3,5,7\} : n \le 93 or k + E_0 \le 5(16-i) + 7\}|$.

Since $c_0(H) = k + E \le k + E_0$, the tight odd-pair bound Lemma 190 applies to exactly these pairs and saves u units in (1). This uses its order cutoff $5 \cdot 16 + 13 = 93$ and its opposite-threshold cost bound, with the larger threshold 16-i. Set

$L_k = E_0 + \min(E_0, g_1) + \min(E_0, g_2) - 2k + 4 + b_1 + b_2 - u.$

By (1), (3), and (6), $gap(H) \le L_k$. The next table lists the exact values of L_k for every integer $8 \le k \le 179$. The values are given in increasing k within each displayed interval, so each list has exactly one value per integer in that interval. Every entry is an equality obtained from the displayed rational floors, ceilings, parity rule and stopping recurrences.

8..19: 2, 0, 0, 1, 0, 0, 1, 1, 0, 1, 1, -1

20..39: 1, 1, -1, 0, -2, -1, -1, -3, 0, -2, -3, 0, -2, -2, -1, -2, -2, -1, -1, -2

40..59: -1, -2, -3, -2, -2, -3, -3, -3, -1, -2, -3, -1, -2, -3, -1, -2, -2, 0, -1, -2

60..79: 0, -1, -2, 0, -1, -2, -1, -2, -2, -2, -3, -2, -3, -4, -2, -4, -4, -2, -3, -4

80..99: -3, -3, -4, -2, -3, -4, -4, -4, -5, -3, -4, -5, -4, -6, -4, -5, -7, -4, -6, -6

100..119: -4, -6, -6, -5, -6, -6, -5, -5, -6, -6, -6, -7, -6, -6, -7, -6, -8, -7, -7, -9

120..139: -6, -8, -9, -7, -8, -9, -7, -8, -9, -8, -8, -8, -8, -9, -9, -8, -8, -9, -9, -10

140..159: -9, -10, -11, -9, -11, -11, -9, -10, -11, -9, -10, -11, -10, -11, -11, -10, -11, -11, -10, -11

160..179: -11, -11, -13, -11, -12, -13, -11, -13, -13, -11, -13, -13, -13, -14, -13, -13, -13, -13, -12, -13

The only positive entries are k in {8,11,14,15,17,18,20,21}. Their remaining exact data are as follows; $g_1 = g_2 = E_0$ and $u = 3$ in each row.

k, E_0, z, b_1, b_2, L_k, $E_0 + g_1 + g_2 - 2k + 4 + z + b_2 - u$

8, 3, 1, 3, 5, 2, 0

11, 4, 3, 4, 6, 1, 0

14, 5, 4, 5, 8, 1, 0

15, 5, 4, 6, 9, 1, -1

17, 6, 5, 6, 10, 1, 0

18, 6, 5, 7, 11, 1, -1

20, 7, 6, 7, 12, 1, 0

21, 7, 6, 8, 13, 1, -1

If $E \le E_0 - 1$ in one of these rows, then $c_1, c_2 \le k + E$ give, by (1) and (6),

$gap(H) \le 3(E_0 - 1) - 2k + 4 + b_1 + b_2 - u = L_k - 3 \le -1.$

Thus any positive-gap graph in these eight rows must have $E = E_0$.

Choose a minimum dominating set Z of H, of actual cardinality $z' \le z$, and put $e = E - 1 = E_0 - 1$. The table gives $2 \le e \le 6$, $z' \le z \le e$, and k in $\{3e + 2, 3e + 3\}$, so n in $\{6e + 5, 6e + 7\}$. We apply Lemma 151 with $d = 16$ to this actual set. Its parameter $t_0 = floor((2e - 1)/14)$ equals zero. The first condition in (i) holds because

$2floor((k + e)/2) - 4e - 1 \le k - 3e - 1 \le 2 < 17 = d + 1.$

Its second condition holds because $6e + 2t_0 + 4 = 6e + 4 \le n + d$, and its third holds because $t_0 = 0 \le ceil((n - 1)/16)$.

For (ii), if $a = z' + 13 - 2e \le 0$, then $2e + 3 \ge z' + 16$, which is its first alternative. Otherwise $1 \le a \le 13 - e \le 11$, so

$16ceil(15/a) \ge 32 > 3e + 2 \ge n + z' - 4e - 5;$

this is its second alternative. For (iii), $6e + 2t_0 + d + 8 = 6e + 24 > n$. For (iv), $8e + 4t_0 + 11 = 8e + 11 > 6e + 7 \ge n - z'$. Every hypothesis is therefore satisfied on the same H and Z.

The cited theorem gives $c_{15}(H) \le |Z| = z' \le z$. Substitute this improved bound in (1) with $E = E_0$, $g_1 = g_2 = E_0$ and $u = 3$. The final column of the eight-row table is nonpositive, contradicting positive gap in every remaining case. Combined with (5) and the complete finite table, this proves $gap(H) \le 0$ at every admissible order. □

Remark. Exact low-row parity and domination settle all but eight small orders. Their shared rank relation makes the matching-barrier criterion uniform and removes the last unit from the top row.

Depends on. Lemma 23, Lemma 49, Lemma 54, Lemma 87, Lemma 92, Lemma 151, Lemma 162, Lemma 190.

Lemma 238 (The exceptional allocation run of a quartic component). *Use the finite integer certificate costs, matching number ν, independence number α, excess $ex = c_0 - \nu$, slack ps, and $gap = ex - ps$ of Lemma 213. Use the zero-capacity allocation set $Y_l(U) = Y_l^{0,0}(U)$ and its missing-run length run_l from Lemma 75. Define $tr(U) = run_{c_0(U)-1}(Y_{c_0(U)-1}(U))$ when $c_0(U) > 0$, and zero otherwise. The condition $T_-(U)$ means that positive gap implies gap one and $tr(U) + ex(U) \le \max(4, floor(\Delta(U)/2) + 1)$.*

Let U be a nonempty finite simple connected four-regular factor-critical graph, meaning that deleting any vertex leaves a perfect matching. Suppose $T_-(U)$ and $p = ps(U) \ge 1$. Put $m = \nu(U)$, $a = \alpha(U)$, $E = ex(U)$, and $L = m + p - 1$. Then $m \ge 2$, $|V(U)| = 2m + 1$, and 0, L belong to $Y_L(U)$. Its longest missing run $R = run_L(Y_L(U))$ satisfies

$R \le \max(0, m - 3p + 2) \le a + 1$.

If $R > a$, then necessarily

$p = 1$, $E = 2$, $m = a + 2$, $R = a + 1 = m - 1$, *and* $Y_m(U) = \{0, m\}$.

Consequently, for any nonnegative integer $w \ge a$, if $R > w$ then $w = a$ and all these exceptional equalities hold. No hereditary condition on proper subgraphs is imposed. All orders, degrees, costs, matching numbers, independence numbers, totals, budgets, and run lengths are integers in the displayed nonnegative or positive domains.

Proof. Factor-criticality gives $|V(U)| = 2m + 1$ and $\nu(U) = m$, and simplicity with degree four gives $m \ge 2$. Since $L = m + p - 1 \ge m$, assigning all edges to either exceptional part proves 0, L in $Y_L(U)$, as also stated in Lemma 75.

We first construct every allocation with smaller coordinate at most 2p-2. Write z for that smaller coordinate. If $z \ge m$, put every edge into the part with budget z. Otherwise $z < m < 2m + 1$, so a z-element vertex set Z exists. Put every edge meeting Z into the exceptional part with budget z and all other edges into the other part. A matching in the first part uses distinct vertices of Z, so its size is at most z. The second part is the graph induced on the $2m + 1 - z$ remaining vertices. The inequality $z \le 2p - 2$ gives

$2m + 1 - z \le 2(L - z) + 1$,

so every matching in the second part has at most L-z edges. This is an actual partition of all edges, including those internal to Z. Exchanging the two exceptional parts treats either choice of smaller coordinate.

Every missing first budget at total L must therefore lie in the integer interval

$[2p - 1, L - (2p - 1)] = [2p - 1, m - p]$.

Its number of integers, if positive, is $m - 3p + 2$. This proves $R \le \max(0, m - 3p + 2)$.

We now compare this number with a. Excess is a nonnegative integer. Since $T_-(U)$ implies $gap(U) \le 1$, $E \le p + 1$. If $E > p$, then $gap(U) = E - p$ is positive, so $T_-(U)$ gives $gap(U) = 1$ and $tr(U) + E \le 4$. The top missing set is nonempty at positive $c_0(U)$, hence $tr(U) \ge 1$ and $E \le 3$. Consequently $E \le p$ whenever $p \ge 3$.

Lemma 213 gives

$m - a \le p + E - 1 + \lfloor p/2 \rfloor$.

When $p = 1$ and $E \le 1$, this is $m - a \le 1$; hence $m - 3p + 2 = m - 1 \le a$.

When $p = 2$ and $E \le 2$, it gives $m - a \le 4$; hence $m - 3p + 2 = m - 4 \le a$.

When $p \geq 3$, the bound $E \leq p$ gives $m - a \leq 2p - 1 + \lfloor p/2 \rfloor$. Hence $m - 3p + 2 \leq a + 1 - \lceil p/2 \rceil \leq a - 1$.

The number a is at least one because U is nonempty, so taking a maximum with zero preserves the corresponding upper bounds by a.

The remaining parameter choices are $(p, E) = (1, 2)$ and $(p, E) = (2, 3)$. In both cases $gap(U) = 1$ and $E \geq 2$, so Lemma 207 applies to this same connected four-regular factor-critical graph. It gives $m - a \leq 2E - 2$. At $(p, E) = (2, 3)$, this yields $m - 4 \leq a$, so $R \leq a$. At $(p, E) = (1, 2)$, it yields $m - 1 \leq a + 1$, so $R \leq a + 1$.

Suppose $R > a$. All cases except $(p, E) = (1, 2)$ have been excluded. Since the quantities are integers and

$a < R \leq m - 1 \leq a + 1$,

one has $R = m - 1 = a + 1$ and $m = a + 2$. Here $L = m$, both endpoints 0, m are feasible, and the entire interior interval [1,m-1] has exactly m-1 integers. A missing run of this length must be that entire interval, giving $Y_m(U) = \{0, m\}$.

Finally let the integer $w \geq a$ and suppose $R > w$. The preceding conclusions give $R = a + 1$; thus $a \leq w < a + 1$ and integrality forces $w = a$. This proves the final assertion and every exceptional equality. □

Depends on. Lemma 75, Lemma 207, Lemma 213.

Lemma 239 (Transfer under the component gap bound). *Use the finite integer certificate costs $c_i(G)$, matching number $\nu(G)$, excess $ex(G)$, slack $ps(G)$, and $gap(G) = ex(G) - ps(G)$ from Lemma 61, and the top missing-allocation run $tr(G)$ from Lemma 81. Let $D \geq 4$ be an integer and H a connected finite simple graph with maximum degree D. Put $S = \{vinV(H) : deg_H(v) = D\}$, and let $\kappa \geq 1$ be the minimum number of S-vertices left uncovered by a matching of H. For any vertex set W, let $q_S(H - W)$ count the odd-order connected components of H-W whose entire vertex sets belong to S. Supply a nonempty vertex set X of maximum cardinality among all sets satisfying $q_S(H - X) - |X| = \kappa$.*

Let calQ be all connected components of H-X, let calK be those counted by q_S, and put $b = |X|$ and $T = floor((D - 1)/2)$. For K in calK define $d(K) = e_H(V(K), X)$, the number of H-edges from K to X. Suppose every Q in calQ satisfies $gap(Q) \leq 1$. Choose any component Z maximizing $\nu(Z)$, and among those choices maximizing $w_T(Z) = |\{i : 1 \leq i \leq T, c_i(Z) > \nu(Z)\}|$.

If D is even, then $gap(H) \leq 0$. If D is odd and $gap(H) \geq 1$, then $\kappa = 1$, Z belongs to calK and is factor-critical, $d(Z) = 1$,

$gap(H) = gap(Z) = 1$, $ex(H) = ex(Z)$, *and* $tr(H) + b \leq tr(Z)$.

Here factor-critical means that deleting any vertex leaves a perfect matching. In particular $gap(H) \leq 1$ for every D. The assertions hold for every indicated extremal choice of Z. There is no additional assumption on component excess or order.

More precisely, define the following credits and finite procedures. Set $a_i(Q) = 0$ for Z and for all Q outside calK. For each K in calK different from Z, put $\delta_K = \max(D - d(K) - ps(K), 0)$ and $h_i(K) = \max(c_i(K) - \nu(K), 0)$. If $d(K) \geq D$, set all its credits to zero. Otherwise put $a_1(K) = \min(\delta_K, h_1(K))$, put $a_i(K) = 1$ when $2 \leq i \leq T$ and $i \leq 1 + \delta_K - a_1(K)$, and put $a_i(K) = 0$ at every other integer index $2 \leq i < D$. All credits are nonnegative integers, satisfy $a_i(Q) \leq \max(c_i(Q) - \nu(Q), 0)$, and $\sum_i a_i(K) = \delta_K$.

For each integer $1 \leq i < D$ put $\mu_i(Q) = c_i(Q) - a_i(Q)$. Start with $L = b + \sum_(Q : a_i(Q) = 0) c_i(Q)$, leaving the components with $a_i(Q) > 0$ unprocessed. While an unprocessed Q satisfies $\nu(Q) \leq L + 1$, choose any such Q, remove it from the unprocessed family, and increase L by $\mu_i(Q)$. Stop when none is eligible. Each procedure terminates after at most $|calQ|$ absorptions. If $gap(H) \geq 1$, every procedure finishes with no unprocessed component, regardless of its allowed choices, and

$c_i(H) \leq b + \sum_{Q \in calQ} \mu_i(Q)$

for every integer $1 \leq i < D$.

All degrees, orders, ranks, costs, matching budgets, defects, indices and iteration counts are integers in their specified positive or nonnegative domains; gaps may be signed. Empty sums and zero credits are allowed.

Proof. Fix a matching leaving exactly κ marked vertices uncovered. For any vertex set W, each marked odd component of H-W either contains an uncovered vertex or has a matching edge to W: otherwise all its

vertices would be paired internally, contrary to odd order. At most κ components use uncovered marked vertices and at most $|W|$ use distinct matching edges into W. Hence $q_S(H-W)-|W|\le\kappa$. The supplied X attains this upper bound and is inclusionwise maximal among maximizers. Lemma 55 makes every member of calK factor-critical.

By Lemma 104, $|calK| = b+\kappa$, every $d(K)$ is positive and has the parity of D, and

$\nu(H) = b + \sum_{Q\in calQ}\nu(Q)$.

Thus calQ is nonempty. Put $\epsilon = D-2T$, equal to one for odd D and two for even D, so $d(K)\ge\epsilon$. Lemma 61 gives $ps(Q)\ge 0$ for every component.

We first justify the credits. If $d(K)\ge D$ then $\delta_K = 0$. Otherwise its total internal defect equals $d(K)$, since each K-vertex has H-degree D and every edge leaving K ends in X. Apply Lemma 179 to this connected factor-critical K with $0<d(K)<D$ and $gap(K)\le 1$. Its explicit formula is exactly the one in the statement. It gives support and the exact total, together with

$a_1(K)\le (D-d(K))/2\le T$, $a_i(K)\le 1$ for $i\ge 2$, $a_i(K)=0$ for $i>T$,

$a_i(K)>0$ implies $ex(K)\le D-d(K)-i+1$. (1)

For every $j\ge 2$ with $a_j(K)>0$ it also gives

$0\le \nu(K)-c_{D-1}(K)\le (D-d(K))/2+T-j$. (2)

The zero credits on Z agree with the convention in Lemma 104. All that theorem's credit hypotheses are now verified on the actual components.

Fix an arbitrary allowed choice of each procedure. Put $M_i = \max_Q a_i(Q)$, let I be the indices whose procedures finish, and put $E_{rem} = \sum_{inot\in I} M_i$. The boundary theorem Lemma 104 supplies their termination and row bounds. It also supplies the following complete accounting. Let Y be the union of vertices of the components outside calK, put $m = ex(H)$, and define

$\zeta = \kappa - c_D(H) + \sum_Q c_D(Q)$,

J_c or $r = e(H[X]) + e_H(X,Y) + \zeta$

$+\sum_{Qnot\in calK, Q\ne Z} ps(Q)$

$+\sum_{K\in calK, K\ne Z}\max(ps(K)-(D-d(K)),0)$.

All terms in J_c or r are nonnegative, and

$gap(H)\le gap(Z)-(ex(Z)-m)+E_{rem}-(D-1)\kappa$

$+1_{Zincal K}(D-d(Z))-J_c$ or r. (3)

The indicator term is zero when Z is not in calK. The row bounds in that theorem are actual integer certificates on all edges of H, including edges inside X.

It remains to analyze $gap(H)\ge 1$, which we assume until the conclusion. Nonnegative slack gives $m\ge 1$. Put $k=\nu(Z)$. The matching identity and Lemma 81 imply $ex(Z)\ge m$. From (1),

$E_{rem}\le T+(T-1) = 2T-1 = D-1-\epsilon$. (4)

If $\kappa\ge 2$, the right side of (3) is at most

$1+(D-1-\epsilon)-2(D-1)+(D-\epsilon) = 2-2\epsilon\le 0$.

This upper bound also holds when Z is not in calK, because its indicator term is then zero. If $\kappa=1$ and Z is not in calK, or is in calK with $d(Z)\ge D$, the right side is at most $1-\epsilon\le 0$. Each alternative contradicts positive gap. Therefore

$\kappa=1$, Z in calK, $\epsilon\le d := d(Z)<D$. (5)

Put $u=(D-d)/2$ and $\psi = ps(Z)$. The local profile Lemma 179 gives $e(Z) = Dk+u$, zero rows at $i\ge D$, and

$\psi = (D-1)k+u-\sum_{i=1}^{D-1} c_i(Z)$. (6)

Equation (3) now becomes $gap(H) \le m - \psi + E_{rem} + 1 - d - J_c$ or r, so

$\psi \le m + E_{rem} - d \le m + 2T - 1 - d$. (7)

The procedure at index one always finishes. Suppose otherwise and choose a remaining component R with rank $r = \nu(R)$. Its failed eligibility and the fact that L never decreases imply $r \ge b + c_1(Z) + 2$, while maximal rank gives $r \le k$. Hence $c_1(Z) \le k - b - 2$. Its positive credit implies $ex(R) \le D - d(R) \le 2T$ by (1). If $m \le 2T$, monotonicity of the rows in (6) gives

$\psi \ge u + (D-1)(b+2) \ge u + 6T > 4T - 1 - d$,

contradicting (7). If $m \ge 2T + 1$, then $ex(R) < m$. Rank separation in Lemma 81 gives $k \ge b + r + m + 1$, so $c_1(Z) \le k - 2b - m - 3$. This time (6) gives

$\psi \ge u + (D-1)(2b + m + 3) \ge u + 2T(m+5)$.

Subtracting the upper bound in (7) leaves at least

$(2T-1)m + 8T + u + 1 + d > 0$,

since $T \ge 1$. This is the other contradiction. The two integer cases prove that index one finishes, including $T = 1$.

If any procedure fails, let j be its smallest index. The zero credits above T and the preceding result give $2 \le j \le T$. Put $q = T - j \ge 0$. Every failed index is at least j and has maximum credit at most one; hence $E_{rem} \le q + 1$ and

$\psi \le m + q + 1 - d$. (8)

For any remaining R at index j, put $r = \nu(R)$. Failed eligibility and positive credit give $r \ge b + c_j(Z) + 2$ and $c_j(R) > r$. The extremal-component implication in Lemma 81 therefore yields

$r \le k - 1$, $c_j(Z) \le k - b - 3$. (9)

For any positive integer C with $c_j(Z) \le k - C$, pair indices i, D-i for $1 \le i < j$. The paired estimate in Lemma 179 bounds each sum by $2k + 1$. The remaining $D - 2j + 1$ indices j,..., D-j have rows at most k-C by monotonicity. Substitution in (6) gives

$\psi \ge u - j + 1 + (D - 2j + 1)C$. (10)

If $ex(R) < m$, rank separation strengthens (9) to $c_j(Z) \le k - 2b - m - 3$. For $C = 2b + m + 3$, the lower bound (10) minus the upper bound (8) equals

$(d + \epsilon)/2 + (\epsilon + 1 + 2q)(2b + m + 3) - m > 0$.

Indeed $b, m \ge 1$ and $\epsilon + 1 + 2q \ge 2$. This contradiction proves $ex(R) \ge m$.

Use $C = b + 3$ in (10) and compare with (8). The result is

$m \ge (d + \epsilon)/2 + (\epsilon + 1 + 2q)(b + 3)$. (11)

For odd D the right side is at least $9 + 8q$; for even D it is at least $14 + 8q$. In particular $m \ge 9 + 8q$. For this same active R, (1) gives

$m \le ex(R) \le D - d(R) - j + 1 \le T + q + 1$. (12)

Lemma 102 applied to the graph R with the positive integer m gives $r \ge 3m - 1$. Equation (9) then gives $k \ge 3m$.

We now obtain upper bounds on two distinct tails. Define $s = k - c_{D-1}(Z) \ge 0$, $h_i = \max(c_i(Z) - k, 0)$, and $t_i = \max(k - c_i(Z), 0)$. Equation (9) gives $h_i = 0$ for $i \ge j$. For every $2 \le i < j$, the paired inequality implies $t_{D-i} \ge \max(h_i - 1, 0)$. These reflected indices are distinct and exclude D-1. Expanding (6) therefore gives

$\psi = u + \sum_i t_i - \sum_i h_i \ge u + s - h_1 - (j - 2)$.

Monotonicity gives $h_1 \le ex(Z)$, and the sole component assumption $gap(Z) \le 1$ gives $ex(Z) \le \psi + 1$. Consequently

$s \le 2\psi - u + j - 1$

$\le 2m + q + 1 - (\epsilon + 3d)/2$

$\le 2m + q - 1$, (13)

using (8), $j = T - q$, $u = T + (\epsilon - d)/2$, and $d \ge \epsilon \ge 1$.

For the active component R put $t = r - c_{D-1}(R)$. Its separate local bound (2) gives

$0 \le t \le (D - d(R))/2 + q \le T + q$. (14)

For either factor-critical graph Z or R of matching number a, its order is $2a + 1$ and its maximum degree is at most D. Apply Lemma 86 with degree $\cap D$. Its weighted count $N_D + N_{D-1}/D$ is at most $2a + 1$. Its induced-deletion construction and certificate consequence give, for every real p in [0,1],

$c_{D-1} \le (2a + 1)[p + (1 - p)^D]$.

Thus with $f_D(p) = p + (1 - p)^D$ and $B_D(a, p) = a - (2a + 1)f_D(p)$, we have

$s \ge B_D(k, 1/10)$, $t \ge B_D(r, 1/10)$, $t \ge B_D(r, 1/6)$. (15)

The integers D, m, k, r, q satisfy every hypothesis of Lemma 13 by (11), (12) and the rank bounds. That theorem makes (13)–(15) impossible. No procedure can fail.

We have proved completion for arbitrary allowed choices, so $E_{rem} = 0$ and every stronger row bound in the statement follows from Lemma 104. Return to (3) and (5):

$gap(H) \le gap(Z) - (ex(Z) - m) + 1 - d - J_c$ or $r \le 2 - d$.

If D is even, then $d \ge 2$, contradicting our positive-gap assumption. Hence even D always gives $gap(H) \le 0$. If D is odd and $gap(H) \ge 1$, positivity and integrality force $d = 1$ and $gap(H) = 1$. Equality in the displayed estimate forces $gap(Z) = 1$, $ex(Z) = m$, and J_c or $r = 0$. The required factor-criticality of Z was established before its selection. Its maximum rank and equal positive excess permit the run contraction in Lemma 81, giving $tr(H) + b \le tr(Z)$. Integer gaps below one are at most zero; together with these positive-gap cases this proves $gap(H) \le 1$ for every D. The initial extremal choice of Z and every procedural choice were arbitrary, proving all asserted quantifiers. □

Depends on. Lemma 13, Lemma 55, Lemma 61, Lemma 81, Lemma 86, Lemma 102, Lemma 104, Lemma 179.

Lemma 240 (Covering odd-degree vertices at slack at most one). *Use the finite integer certificate costs $c_i(G)$, matching number $\nu(G)$, maximum degree $\Delta(G)$, excess $ex(G) = c_0(G) - \nu(G)$, slack $ps(G) = e(G) - \nu(G) - \sum_{i \ge 1} c_i(G)$, and signed gap $gap(G) = ex(G) - ps(G)$ of Lemma 23.*

Let $D \ge 7$ be an odd integer and let H be a finite simple graph with $\Delta(H) \le D$ and $ps(H) \le 1$. Suppose every proper induced subgraph J of H satisfies $gap(J) \le 1$. Then H has a maximum matching covering every vertex of degree D, and $c_D(H) = 0$. No sign or upper bound on $gap(H)$ is assumed.

All degrees, costs, matching budgets, graph orders, and cardinalities are integers. Empty graphs, isolated vertices, and an empty degree-D set are permitted.

Proof. Use Lemma 61 for nonnegative slack and the zero cost row when a matching covers a degree threshold. Use Lemma 23 for finite nonnegative costs, row monotonicity, and $gap = ex - ps$.

Suppose no matching of H covers all its degree-D vertices. There is then a connected component U with actual maximum degree D in which no matching covers all degree-D vertices. Otherwise, the union of a covering matching in each component having degree-D vertices would cover them all in H. By component slack superadditivity in Lemma 72 and nonnegative slack,

$0 \le ps(U) \le ps(H) \le 1$. (1)

Every proper induced subgraph of U is a proper induced subgraph of H and has gap at most one.

Put $S = \{vinV(U) : deg_U(v) = D\}$, and let $\kappa \ge 1$ be the minimum number of vertices of S missed by a matching of U. For $W \subseteq V(U)$, let $q_S(U - W)$ count the odd components of U-W contained in S. By Lemma 107, the maximum of $q_S(U - W) - |W|$ equals κ and is attained. Choose a maximum-cardinality maximizing set X. If X were empty, connectedness and $q_S(U) = \kappa > 0$ would force U itself to be odd and

$S = V(U)$. It would be an odd-order D-regular graph with odd D, contradicting the even degree sum 2e(U). Thus $b := |X| \geq 1$.

The component assumptions of Lemma 202 hold, because every component of U-X is proper induced in H. Apply that theorem to U, X and use (1). Its slack-at-most-one consequence gives

$ps(U) = 1, \kappa = 1$, and $c_D(U) = 1$. (2)

Let calQ be all components of U-X, let calK be the odd components contained in S, and let Y be the union of the vertices of the other components. By Lemma 107, each K in calK is factor-critical: deleting any vertex leaves a perfect matching. Also $|calK| = b + 1$. All edges leaving K end in X. Put $d(K) = e_U(V(K), X)$, where $e_U(A, B)$ counts edges between disjoint vertex sets. Connectedness makes $d(K) > 0$. Since every vertex of K has U-degree D and K has odd order,

$d(K) = D|V(K)| - 2e(K)$

is odd and is its total internal degree defect from D.

Set $T = (D-1)/2 \geq 3$. Choose Z in calQ maximizing $\nu(Q)$ and then maximizing

$w(Q) = |\{i : 1 \leq i \leq Tandc_i(Q) > \nu(Q)\}|$.

This finite nonempty choice exists. Write $k = \nu(Z)$.

For each K in calK with $d(K) < D$, apply Lemma 179. The graph K is connected, factor-critical, has maximum degree at most D, and its actual defect lies strictly between zero and D. The additional $gap(K) \leq 1$ holds because it is proper induced in H. That theorem supplies nonnegative integer credits $a_i(K)$, $1 \leq i < D$, of total

$\delta_K = \max(D - d(K) - ps(K), 0)$,

supported by $a_i(K) \leq \max(c_i(K) - \nu(K), 0)$, with $a_1(K) \leq T$, $a_i(K) \leq 1$ for $i \geq 2$, and $a_i(K) = 0$ for $i > T$. If $d(K) \geq D$, nonnegative slack gives $\delta_K = 0$ and we use zero credits. Give zero credits to every unmarked component and override all credits of Z to zero.

Apply Lemma 104 to these actual credits. At each index i, start with

$L = b + \sum_(Q : a_i(Q) = 0)c_i(Q)$

and absorb a remaining positive-credit component Q whenever $\nu(Q) \leq L + 1$, adding $\mu_i(Q) = c_i(Q) - a_i(Q)$. Its support makes $\mu_i(Q) \geq 0$. Fix any collection of permitted choices. Every procedure terminates, and every running value satisfies $L \geq b + c_i(Z)$. Put $M_i = \max_Q a_i(Q)$, and let E be the sum of M_i over unfinished indices. The indices above T are finished initially, so

$0 \leq E \leq T + (T-1) = D - 2$. (3)

In the exact boundary formula define the nonnegative correction

$J = e(U[X]) + e_U(X, Y)$

$+ \sum_{Qnot\in calK, Q\neq Z} ps(Q)$

$+ \sum_{K\in calK, K\neq Z} \max(ps(K) - (D - d(K)), 0)$

$+ \sum_{Q\in calQ} c_D(Q)$.

Indeed every term is nonnegative by the cost definitions and nonnegative slack. By (2), the top correction ζ of Lemma 104 is precisely $\sum_Q c_D(Q)$. Converting that fact's gap inequality to slack therefore gives

$ps(Z) + J \leq 2 + E - D + 1_{Z\in calK}(D - d(Z))$. (4)

The indicator term is zero when Z is not in calK; its boundary is only used when defined.

If Z is unmarked, it is a component contained in Y. It sends at least one edge to X because U is connected and X is nonempty. Hence $J \geq e_U(X, Y) \geq 1$, but the right side of (4) is at most $2 + (D-2) - D = 0$. This is impossible. Thus Z belongs to calK.

Suppose $d(Z) \geq D$. By (3) and (4),

$0 \leq ps(Z) + J \leq D - d(Z) \leq 0$.

Consequently $d(Z) = D$, $ps(Z) = 0$, $J = 0$, and $E = D - 2$. We show that this forces Z to be a singleton.

First, Z must be regular at its actual maximum degree. If it were not regular, choose a vertex z of degree below its maximum ρ and a vertex v of degree ρ. A perfect matching of Z-z covers every degree-rho vertex. Factor-criticality also gives $\nu(Z - v) = \nu(Z)$. The covered maximum-degree deletion inequality of Lemma 61 would then give $ps(Z) \geq ps(Z - v) + 1 \geq 1$, a contradiction. Thus Z is rho-regular for a nonnegative integer ρ.

Put $n = |V(Z)|$. Its total defect equation is

$(D - \rho)n = D$.

If $n > 1$, positivity and integrality of D-rho give $n \leq D$. Simplicity gives $\rho \leq n - 1$; substituting $\rho = D - D/n$ yields $D \leq n$. Hence $n = D$ and $\rho = D - 1$, so Z is the complete graph K_D. The odd-clique bound Lemma 110 applies because D is odd and at least seven, and gives $ps(Z) \geq 1$, another contradiction. Therefore $n = 1$ and $k = \nu(Z) = 0$.

By the maximum-rank choice of Z, all components of U-X now have matching number zero. They have no edges, since one edge would itself be a matching. Their certificate costs are all zero by the empty edge partition, so the supported credits are all zero. This gives $E = 0$, contradicting $E = D - 2 > 0$. We have proved

$1 \leq d := d(Z) < D$. (5)

Apply the unconditional part of Lemma 179 to Z. With $u = (D - d)/2$ and $s = ps(Z)$, it gives

$s \geq 1$, $e(Z) = Dk + u$, $c_i(Z) = 0$ for $i \geq D$,

and $c_i(Z) + c_{D-i}(Z) \leq 2k + 1$ for every $1 \leq i < D$.

Thus

$s = (D - 1)k + u - \sum_{i=1}^{D-1} c_i(Z)$. (6)

Equation (4), for the same fixed procedures, yields

$s \leq 2 + E - d \leq D - d$. (7)

If index one is unfinished, a remaining component R of rank $r = \nu(R)$ satisfies

$r \geq L + 2 \geq b + c_1(Z) + 2$.

The maximum-rank choice gives $r \leq k$, so monotonicity bounds every row of Z by k-b-2. Equation (6) then gives $s \geq u + (D - 1)(b + 2) > D - d$, contradicting (7). Thus index one is finished.

If a later index is unfinished, let j be the first and put $q = T - j \geq 0$. We have $2 \leq j \leq T$. The earlier indices are finished, and the remaining M_i are at most one, so $E \leq q + 1$ and

$s \leq q + 3 - d$. (8)

A component R remaining at j has $r = \nu(R) \geq b + c_j(Z) + 2$ and $c_j(R) > r$. If $r = k$, then $c_j(Z) < k$ gives $w(Z) \leq j - 1$, while $c_j(R) > k$ gives $w(R) \geq j$. This contradicts the fixed tie-break. Hence $r \leq k - 1$ and $c_j(Z) \leq k - b - 3$.

The complementary pairs i, D-i for $1 \leq i \leq j - 1$ are disjoint. Bound their total by $(j - 1)(2k + 1)$ and the $D - 2j + 1$ remaining entries by k-b-3. Substitution in (6) gives

$s \geq u - j + 1 + (D - 2j + 1)(b + 3)$.

Subtracting the upper bound in (8), and using $D = 2T + 1$ and $j = T - q$, gives

$(d + 1)/2 - 2 + (2 + 2q)(b + 3) \geq -1 + 8 = 7 > 0$.

This contradiction excludes every unfinished index. Thus $E = 0$.

Now (4) gives $s + d + J \leq 2$. The positive integer s and the positive odd integer d must both equal one. In particular $ps(Z) = 1$ and $d(Z) = 1$. The additional $gap(Z) \leq 1$ is available, so the lower slack bound in Lemma 179 gives

$ps(Z) \geq ceil((D - d - T)/2) = \lceil T/2 \rceil \geq 2$,

because $T \geq 3$. This final contradiction excludes U.

Therefore H has a matching covering every degree-D vertex. Lemma 107, applied with those vertices as its prescribed set, supplies a maximum matching attaining the same zero exposure. Lemma 61 gives $c_D(H) = 0$. If there are no degree-D vertices, the empty matching already covers the prescribed set and the same conclusions follow. Nonnegative integer slack makes the contrapositive precisely $ps(H) \geq 2$, completing the proof. □

Remark. Slack one forces equality in the nonempty-barrier accounting. The only apparent high-defect escape is a regular component that must be a singleton or an odd clique; both are excluded directly. The remaining component has defect one, whose local credit bound requires at least two slack units once the odd degree reaches seven.

Depends on. Lemma 23, Lemma 61, Lemma 72, Lemma 104, Lemma 107, Lemma 110, Lemma 179, Lemma 202.

Lemma 241 (A finite numerical window at octic excess four). *Use the finite integer certificate costs $c_i(H)$, matching number $\nu(H)$, independence number $\alpha(H)$, excess $ex(H) = c_0(H) - \nu(H)$, and signed gap $gap(H) = \sum_{i\geq 0} c_i(H) - e(H)$ of Lemma 23. All graphs are finite and simple. A graph is factor-critical if deleting any vertex leaves a perfect matching.*

Let H be a nonempty connected eight-regular factor-critical graph with $ex(H) = 4$ and $gap(H) \geq 1$. Put $k = \nu(H)$ and $a = \alpha(H)$, and assume the integer lower bound $k \geq 14$. Then $|V(H)| = 2k+1$, $c_6(H) \leq k-1$, and k belongs to

{14,15,16,17,18,19,20,22}.

In this order the corresponding upper bounds for $c_7(H)$ are

8, 8, 9, 10, 10, 11, 12, 13.

Writing $o_k = 1$ when k is odd and $o_k = 0$ when k is even, one also has

$c_6(H) \geq 2k - 10 + o_k - c_7(H)$.

If additionally $c_7(H) \geq a$ and $k - a \leq 7$, then (k, a) belongs to

{(14,7),(14,8),(15,8),(16,9),(17,10)}.

If moreover $c_7(H) = a$, then, in this displayed order,

$c_6(H) - a \geq 4, 2, 5, 4, 5$, *respectively,*

and $c_6(H) - a \leq k - 1 - a$ in every case.

No hereditary gap condition, source minimality, independent-deletion identity, matching-gain bound, or deletion-process hypothesis is imposed. The lower bound $k \geq 14$ and the two additional hypotheses in the final clauses are explicit assumptions. All orders, degrees, indices, costs, matching and independence cardinalities, k, a and o_k are integers in their stated nonnegative or positive domains; gaps may be signed on other graphs.

Proof. Factor-criticality gives $|V(H)| = 2k+1$: deleting any vertex leaves a perfect matching with k edges, and the odd order forbids a larger matching. Regularity gives $e(H) = 8k+4$, and $ex(H) = 4$ gives $c_0(H) = k+4$.

First apply Lemma 92 with $d = 8$, $j = 2$, $\theta = 6$, $m_\theta = 7$, and sampling probability $2/5$. Its explicit binomial expression is

$Q_{8,2}(2/5)$

$= 2/5 + (3/5)^9 + 8(2/5)(3/5)^8 + 4(2/5)^2(3/5)^7$

$= 2/5 + (73/25)(3/5)^7$

$= 940901/1953125$

$< 241/500$.

For the strict comparison, $241 \cdot 1953125 - 500 \cdot 940901 = 252625 > 0$. The cited result therefore gives

$c_6(H) \le floor((2k+1)\cdot 940901/1953125)$.

For every integer $k \ge 14$,

$500k - 241(2k+1) = 18k - 241 \ge 11 > 0$.

Thus the quantity inside this floor is strictly less than k, and

$c_6(H) \le k-1$. (1)

The eight-regular case of Lemma 53 gives

$c_7(H) \le floor((800k-180)/1300)$. (2)

These are integer certificate bounds on this same graph H.

The row sum has no terms beyond index eight. In fact, for every integer $i \ge 9$, H has maximum degree at most i and no vertex of degree i. The first inequality of Lemma 92 with $\theta = i$ gives $c_i(H) = 0$. To bound the eighth row, fix a vertex v and a perfect matching M of H-v. Apply Lemma 49 with $D = 8$, $j = 0$, and $Z = \{v\}$. Its neighbor condition is empty, and Z contains the only uncovered vertex. Consequently $c_8(H) \le 1$.

By monotonicity in Lemma 23, each of $c_0(H)$, $c_1(H)$, $c_2(H)$ is at most $k+4$. Applying Lemma 119 with complementary thresholds three and five gives $c_3(H) + c_5(H) \le 2k+2$. Applying it with thresholds four and four gives $c_4(H) \le k+1$. When k is odd, equality in the latter inequality would make both complementary costs equal the even integer $k+1$, whereas the equality clause of that cited result makes both costs odd. Therefore

$c_4(H) \le k+1-o_k$. (3)

Here o_k is the parity indicator in the statement.

Sum the bounds for rows 0, 1, 2; the pair 3, 5; the midpoint 4; the separate rows 6, 7; and row 8. These disjoint index groups exhaust the nonzero row range. Subtracting $e(H) = 8k+4$ gives

$gap(H) \le 12 - o_k - 2k + c_6(H) + c_7(H)$.

Since $gap(H) \ge 1$, equation (1) implies

$k \le 10 - o_k + c_7(H) \le 10 + (800k-180)/1300$.

Multiplying the last inequality by 1300 and rearranging gives

$500k \le 12820$.

Thus $k \le 25$.

It follows that $|V(H)| = 2k+1 \le 51 \le 5\cdot 8 + 13$. The order clause of Lemma 190 now applies to the odd complementary thresholds three and five and gives the stronger bound $c_3(H) + c_5(H) \le 2k+1$. Repeating the same complete row sum with this improvement gives

$gap(H) \le 11 - o_k - 2k + c_6(H) + c_7(H)$. (4)

The lower bound $gap(H) \ge 1$ proves the displayed lower bound on $c_6(H)$. Using (1) and then (2) in (4) also gives

$k \le 9 - o_k + c_7(H) \le 9 + (800k-180)/1300$.

Thus $500k \le 11520$ and $k \le 23$.

If $k = 23$, then $o_k = 1$ and (2) gives $c_7(H) \le 14$, contradicting $k \le 9 - o_k + c_7(H) = 8 + c_7(H) \le 22$. If $k = 21$, then (2) gives $c_7(H) \le 12$, and the same inequality would give $21 \le 20$. Together with the explicit lower bound $k \ge 14$, these exclusions leave precisely

k in {14,15,16,17,18,19,20,22}.

Substituting these integers in (2) gives the claimed top-row bounds except that it gives nine rather than eight at $k = 15$.

We prove the sharper $k = 15$ bound using one marked domination construction. Choose any vertex v and a perfect matching M of H-v, and let J be the spanning graph formed by deleting only the matching edges.

It has $n = 31$ vertices, degree eight at v, and degree seven everywhere else. Apply Lemma 37 with its parameters $n = 31$ and $d = 8$. Its recurrence starts at $r_0 = 22$ and is

22, 16, 11, 7, 4, 2, 1, 0.

The successive subtracted ceilings are 6, 5, 4, 3, 2, 1, 1, with positive denominators 30, 29, 28, 27, 26, 25, 24. More explicitly, these are the ceilings of

176/30, 128/29, 88/28, 56/27, 32/26, 16/25, 8/24.

The recurrence therefore produces a set Z containing v, of size at most eight, that dominates J. For every vertex outside Z there is at least one neighbor in Z in $J = H - M$. Applying Lemma 49 to the same H, M, Z with $D = 8$ and $j = 1$ gives $c_7(H) \le |Z| \le 8$. This completes the top-row table. Both matching deletion and the domination construction have kept their stipulated graph; no numerical graph search is used.

Finally suppose $c_7(H) \ge a$ and $k - a \le 7$. Then $k - 7 \le a \le c_7(H)$. At $k = 18, 19, 20, 22$ the table bounds $c_7(H)$ by 10, 11, 12, 13 respectively, all strictly below k-7. These four ranks are impossible. At the remaining ranks 14, 15, 16, 17, the same interval for the integer a gives respectively {7,8}, {8}, {9}, {10}. These are exactly the five pairs in the statement.

If $c_7(H) = a$, the lower bound already obtained from (4) gives

$c_6(H) - a \ge 2k - 10 + o_k - 2a$.

Substitution of those five pairs yields respectively 4, 2, 5, 4, 5. Equation (1) gives the upper bound $c_6(H) - a \le k - 1 - a$. Every clause has therefore been proved with the stated additional hypotheses. □

Depends on. Lemma 23, Lemma 37, Lemma 49, Lemma 53, Lemma 92, Lemma 119, Lemma 190.

Lemma 242 (Degree-seven density at slack one). *Use the finite integer certificate costs c_i, matching number ν, independence number α, maximum degree Δ, excess $ex = c_0 - \nu$, slack ps, and $gap = ex - ps$ of Lemma 61. All graphs are finite and simple. For a graph J define*

$f(J) = |V(J)| - 2\nu(J)$, *and*

$E_7(J) = e(J) + 4|V(J)| - 11\nu(J) - 4\alpha(J)$.

A graph is factor-critical when deleting every vertex leaves a perfect matching.

Let H have maximum degree at most seven. Suppose every induced subgraph J of H, including H itself, has $gap(J) \le 1$. Suppose also that every connected induced subgraph J of H with $\Delta(J) \ge 3$ and $ps(J) \le 1$ for which no matching covers all maximum-degree vertices is regular of positive even degree, factor-critical, and has $ps(J) = 1$.

If $ps(H) = 1$ and $\alpha(H) - f(H) \le 5$, then $E_7(H) \le 11$.

In particular, this conclusion holds whenever H is connected, six-regular, factor-critical, $ps(H) = 1$, $\alpha(H) \le 6$, and satisfies the two stated hereditary conditions.

The condition on uncovered connected induced subgraphs is an explicit premise. No property of positive-slack subgraphs other than that condition is assumed. Empty induced subgraphs and isolated vertices are permitted. All orders, degrees, indices, cardinalities, costs, matching and independence numbers, excesses and slack values are integers in their displayed nonnegative domains; E_7 and gap may be signed integers.

Proof. Use Lemma 61 for nonnegative slack, zero tails, covered-degree vanishing and covered vertex deletion, and Lemma 72 for restriction, restoration and component slack superadditivity. Put

$\rho(J) = |V(J)| - \nu(J) - 2\alpha(J)$.

All the hypotheses concerning induced subgraphs pass to every induced subgraph.

First, every zero-slack induced J of maximum degree at least three has a matching covering its maximum-degree vertices. Each whole component has slack zero. Any component of that actual maximum degree must be covered, since the stated uncovered-component condition would otherwise give it slack one. Unite the covering matchings of these components. Repeated deletion of any current maximum-degree vertex while its degree is at least three now produces a complete covered process. The deletion inequality shows that every stage has slack zero and that each deletion lowers the matching number by one. Thus Lemma 146 applies

to any zero-slack induced subgraph, with any integer $\cap$ at least its maximum degree and at least three. In particular,

$ps(J) = 0$ implies $E_7(J) \le 0$. (1)

We will use an ordinary-barrier consequence of Lemma 107. Choose a maximum-cardinality maximizing set X with every vertex marked. Every component of J-X is odd and factor-critical, and their number is

$q = |X| + |V(J)| - 2\nu(J)$.

Indeed the cited theorem gives the formula and factor-criticality for the odd components. An even component cannot remain: adjoining any one of its vertices to X creates at least one new odd component and costs one separator vertex, so the objective cannot decrease. It cannot increase by maximality of the objective, and equality contradicts maximal cardinality of X. Choosing one vertex from each component also gives $q \le \alpha(J)$.

The economical cutoff Lemma 123 yields, for every induced J of slack one and excess E,

$1 \ge 2(|V(J)| - c_0(J)) - 4\alpha(J) = 2(\rho(J) - E)$.

The difference rho(J)-E is integral. Our gap hypothesis therefore gives

$\rho(J) \le E \le 2$. (2)

We next bound a connected six-regular factor-critical induced graph Q with $ps(Q) = 1$ and $a = \alpha(Q) \le 6$. Write $|V(Q)| = 2k + 1$, so $\nu(Q) = k$ and $e(Q) = 6k + 3$. Its density error is

$E_7(Q) = 3k + 7 - 4a$. (3)

Its sixth cost is one: after deleting a vertex a perfect matching covers all remaining degree-six vertices, so covered-degree vanishing and restoration give $c_6(Q) \le 1$. A zero-budget partition at capacities (1,5) in a six-regular graph would force the first part to be a perfect matching, impossible on odd order. Hence $c_6(Q) = 1$.

For any maximum independent set S, put $G = Q - S$ and $\eta = \nu(G) - (k + 1 - a)$. Factor-criticality gives $\eta \ge 0$: delete one vertex of S, use a perfect matching, and remove the matching edges meeting the other a-1 vertices of S. Maximality of S gives $\Delta(G) \le 5$. At each row $1 \le i \le 5$ restoration gives $0 \le c_i(Q) - c_i(G) \le a$. Subtract the slack definitions, using that S removes exactly 6a edges and that all higher rows except $c_6(Q)$ vanish. With

$u = \sum_{i=1}^{5} (a - c_i(Q) + c_i(G)) \ge 0$,

the exact identity is

$ps(G) = 1 - \eta - u$. (4)

Consequently every such gain belongs to {0,1}.

We first exclude $k = 5$, $a = 2$. For an independent pair S, its complement G has nine vertices, twenty-one edges, minimum degree at least four, maximum degree at most five, and independence at most two. Its matching number is four, by the lower bound above and its order. Thus $\eta = 0$ and $ps(G) \le 1$.

The graph G is connected, because two components of minimum degree four would require at least ten vertices. It has no cut vertex w: components after deleting w have minimum degree at least three and hence at least four vertices each. A disconnection on eight vertices would therefore consist of two sets of four. Every vertex in those sets needs all three internal neighbors and the neighbor w, forcing degree eight at w, contrary to the maximum degree five. For any vertex w, if G-w had no perfect matching, its ordinary barrier would have $q \ge |X| + 2$ odd components. Since $\alpha(G) \le 2$, the preceding barrier consequence would force X empty and $q = 2$, making w a cut vertex. Thus G is factor-critical.

The degrees of G are four or five, and its order and edge count exclude regularity. Choose a vertex of degree four and use a perfect matching after deleting it; this covers all degree-five vertices of G. Choose now a degree-five vertex w and put $K = G - w$. Factor-criticality gives $\nu(K) = \nu(G) = 4$. The covered deletion inequality and $ps(G) \le 1$ give $ps(K) = 0$. Moreover K has eight vertices, sixteen edges and independence at most two. Apply Lemma 146 with $\cap$ five and a complete covered process on K. Its refined density bound gives

$16 = e(K) \le 12 + 2\alpha(K) - 4(4 - c_1(K)) \le 16$.

Thus $\alpha(K) = 2$ and $c_1(K) = 4$, and equality holds. The equality conclusion gives two independent deleted vertices of original K-degree five. Both avoid w in G, whose maximum degree is five. Together with w they form an independent triple in G, a contradiction. This excludes $k = 5$, $a = 2$.

Suppose $E = ex(Q) = 2$. Then $k \geq 5$. For completeness, at $k \leq 4$ every nonnegative budget split $x + y = k + 1$ is feasible with zero capacities. If $x \geq y$, choose y vertices and put all edges meeting them in the part with budget y; the remaining graph has at most $2x + 1$ vertices since

$2k + 1 \leq k + 2 + ceil((k + 1)/2) \leq x + y + x + 1.$

It has matching number at most x. Exchange the two colors when $x < y$. This would give $c_0(Q) \leq k + 1$, contrary to $E = 2$.

By (2), $k + 1 - 2a \leq 2$. If $a \leq 2$ this forces $k = 5$, $a = 2$, which was just excluded. Hence $a \geq 3$. The degree-six consequence of Lemma 140 gives $\eta \geq 1$ for every maximum independent S because $E = 2$. Identity (4) forces $\eta = 1$ for every such S. The zero-slack coverage established above permits Lemma 178 with $p = 1$, $d = 6$. It gives $\nu(Q - S) \leq 4$. Thus $k \leq a + 2$, and (3) gives

$E_7(Q) \leq 13 - a \leq 10.$ (5)

It remains to treat $E \leq 1$. Inequality (2), before its final bound by two, gives $\rho(Q) \leq E \leq 1$, hence $k \leq 2a \leq 12$. The tight complementary-level bound Lemma 190 at thresholds three and three applies, since $|V(Q)| \leq 25 \leq 43$. It gives $2c_3(Q) \leq 2k + 1$, and therefore $c_3(Q) \leq k$. Each of c_1, c_2, c_4 is at most $c_0(Q) \leq k + 1$, while $c_6(Q) = 1$ and higher rows vanish. The slack identity yields

$1 = ps(Q) \geq k - 1 - c_5(Q).$ (6)

We give actual row-five certificates ruling out $8 \leq k \leq 12$. Fix v in Q and a perfect matching M of Q-v. The graph L obtained by deleting the edges of M has degree six at v and five elsewhere. Start $Z = \{v\}$; exactly 2k-6 vertices are undominated in L. If i additional vertices have been chosen and r vertices remain undominated, each such vertex has six vertices in its closed L-neighborhood, all among the 2k-i unselected vertices. Some new vertex therefore dominates at least $ceil(6r/(2k - i))$ of them. Greedily selecting such a vertex bounds the undominated counts by

$r_0 = 2k - 6, r_{i+1} = \max(0, r_i - ceil(6r_i/(2k - i))).$

The update is nondecreasing as a function of the nonnegative integer input: when the denominator is at least six the ceiling rises by at most one per unit increase, and when it is smaller the truncated update is zero. Thus this recurrence bounds the actual process. Its complete chains are

$k = 8$: 10, 6, 3, 1, 0;

$k = 9$: 12, 8, 5, 3, 1, 0;

$k = 10$: 14, 9, 6, 4, 2, 1, 0;

$k = 11$: 16, 11, 7, 4, 2, 1, 0;

$k = 12$: 18, 13, 9, 6, 4, 2, 1, 0.

These give dominating sets Z in the same L, containing v, of size at most 5, 6, 7, 7, 8 respectively. The induced graph Q-Z has degree at most five. Every vertex of degree five there is covered by the restriction of M, because its degree in L-Z is at most four. Thus its fifth row vanishes, and restoration on the actual original edges gives $c_5(Q) \leq |Z|$. Substitution in (6) gives $ps(Q) \geq 2, 2, 2, 3, 3$, respectively, a contradiction. Consequently $k \leq 7$.

Two remaining pairs require attention: $(k, a) = (6, 3)$ and (7,4). In either pair, if all maximum independent cuts had gain one, Lemma 178 would give $\nu(Q - S) \leq 4$, whereas the gain formula gives five. By (4) some maximum independent S therefore has gain zero. Choose its ordinary barrier X, and put $b = |X|$. The complement G has deficiency a-1, so G-X has $q = a + b - 1$ odd components. Since $q \leq \alpha(G) \leq a$, $b \leq 1$. Every component has minimum internal degree at least 6-a-b.

For $(k, a) = (6, 3)$, G has ten vertices and twenty-one edges. If $b = 0$, its two odd components each have at least five vertices, hence both have order five and together at most twenty edges. If $b = 1$, its three odd components have at least three vertices each, totaling nine, so they are three triangles; the one separator vertex contributes at most five edges, giving at most fourteen edges. Both possibilities contradict $e(G) = 21$.

For $(k,a) = (7,4)$, G has eleven vertices. If $b = 1$, its four odd components have internal minimum degree at least one, so their total order is at least twelve, exceeding $|V(G - X)| = 10$. If $b = 0$, its three odd components have minimum degree at least two and total order eleven, hence their orders are 3, 3, 5. The first two are triangles. Each of their six vertices has two neighbors in G and therefore all four vertices of S as its other Q-neighbors. Each vertex of S then has its full degree six used on the triangles. The order-five component has no edge to S or to either triangle, contradicting connectedness of Q. Both pairs are excluded.

Finally, $k \geq 3$ by six-regularity. At $k = 3$ the graph is K_7, whose independence number one contradicts $k \leq 2a$. At $k = 4$ or 5 that same inequality gives $a \geq 2$ or $a \geq 3$, respectively, and (3) gives $E_7(Q) \leq 11$ or ≤ 10. At $k = 6$ or 7, the excluded pairs force $a \geq 4$ or $a \geq 5$, respectively, and (3) gives $E_7(Q) \leq 9$ or ≤ 8. Together with (5), this proves the asserted sextic bound.

We finish the general theorem by induction on $|V(H)|$. If the maximum degree is at most two, each component is a path or cycle. The quantity $-E_7$ equals $5r + 1$ on a path of order 2r, 5r on a path of order $2r + 1$ or a cycle of order 2r, and 5(r-1) on a cycle of order $2r + 1$. These values are nonnegative in their respective domains and include isolated vertices. Additivity gives $E_7(H) \leq 0$.

Suppose H is uncovered of actual degree $d \geq 3$. An uncovered whole component U of degree d has slack one by the premise. Superadditivity and nonnegativity make the union W of the other components have slack zero, so $E_7(W) \leq 0$ by (1). Maximum-matching unmatched vertices are independent in every graph, giving $f(W) \leq \alpha(W)$. Additivity yields $\alpha(U) - f(U) \leq \alpha(H) - f(H) \leq 5$. The premise makes U regular and factor-critical of even degree d, so d is four or six. At degree six, $f(U) = 1$ and $\alpha(U) \leq 6$, and the proved sextic bound applies. At degree four, writing $a = \alpha(U) \geq 1$ and $\nu(U) = k$ gives

$$E_7(U) = k + 6 - 4a = \rho(U) + 5 - 2a \leq 7 - 2a \leq 5$$

by (2). In either case $E_7(H) = E_7(U) + E_7(W) \leq 11$.

Otherwise a matching covers the actual maximum degree $d \geq 3$ of H. Choose a degree-d vertex v and put $K = H - v$, $h = \nu(H) - \nu(K)$ and $u = \alpha(H) - \alpha(K)$, both in {0,1}. Covered deletion gives $ps(K) \leq h$. Direct subtraction gives

$$E_7(H) - E_7(K) = d + 4 - 11h - 4u.$$

If $h = 0$, (1) applies to K, so $E_7(H) \leq d + 4 \leq 11$. If $h = 1$, then $f(K) = f(H) + 1$ and $\alpha(K) - f(K) = \alpha(H) - f(H) - 1 - u \leq 5$. Also $E_7(H) \leq E_7(K)$ because $d \leq 7$. If $ps(K) = 0$ use (1); if $ps(K) = 1$ apply induction to the smaller graph K, which inherits both hereditary conditions. In all cases $E_7(H) \leq 11$. This completes the induction and the proof. □

Remark. The cutoff already controls sextic order at excess at most one. At excess two, uniform independent-cut gain gives a stronger order bound. A small ordinary-barrier analysis removes the two remaining exceptional pairs, and covered deletion propagates the component density bound to every graph in the stated class.

Depends on. Lemma 61, Lemma 72, Lemma 107, Lemma 123, Lemma 140, Lemma 146, Lemma 178, Lemma 190.

Lemma 243 (Slack pays for uncovered maximum-degree vertices). *Use the finite integer certificate costs $c_i(H)$, matching number $\nu(H)$, maximum degree $\Delta(H)$, slack $ps(H)$, and signed gap $gap(H)$ of Lemma 23. Let H be a finite simple graph of actual maximum degree $D \geq 3$. Let κ be the minimum number of its degree-D vertices left uncovered by a matching. If $D \geq 4$, assume $gap(J) \leq 1$ for every proper induced subgraph J of H; no hereditary hypothesis is required for $D = 3$. Then $c_D(H) \leq \kappa \leq ps(H)$. When $D = 3$, the stronger inequality $2\kappa \leq ps(H)$ holds.*

For $D \geq 4$, let t be the number of connected components U with positive degree-D exposure κ_U. In each such component let S_U be its set of degree-D vertices, and choose any set X_U of maximum cardinality among the maximizers of $q_{S_U}(U - X_U) - |X_U|$, where q_{S_U} counts odd-order components wholly contained in S_U. Let s count those U with X_U nonempty, and let ϵ be one for odd D and two for even D. Then $ps(H) \geq \kappa + (D-2)(\kappa - t) + (\epsilon - 1)s$. The empty-barrier uncovered components are whole D-regular factor-critical components with $\kappa_U = 1$. Factor-critical means that deleting any one vertex leaves a perfect matching.

Consequently, if $ps(H) = 0$, some maximum matching covers every degree-D vertex and $c_D(H) = 0$. If $D \geq 4$ is even, $ps(H) \leq 1$ and $\kappa > 0$, then $\kappa = 1$ and H has exactly one uncovered connected component. It is a whole D-regular factor-critical component with slack one; every other connected component has slack zero.

All orders, degrees, costs, cardinalities, matching numbers, counts D, κ, κ_U, t, s, and ϵ are integers in the displayed positive or nonnegative domains.

Proof. The finite matching family has a minimum exposure. Lemma 86 gives $c_D(H) \le \kappa$, and Lemma 107 ensures that this exposure is attained by a maximum matching. For $D = 3$, all assertions concerning that degree follow from Lemma 161.

Assume $D \ge 4$. Matchings restrict independently to whole connected components, and unions of component matchings are matchings of H. It follows that κ is the sum of the component exposures. Only components containing degree-D vertices can contribute. Write U for the t components with positive exposure κ_U; then each has actual maximum degree D and $\kappa = \sum_U \kappa_U \ge t$.

For each such U, the required maximum-cardinality maximizing set X_U exists because U has finitely many vertex subsets. Lemma 107, with marked set S_U, gives $q_{S_U}(U - X_U) - |X_U| = \kappa_U$ and factor-criticality of every counted odd component.

If X_U is empty, connectedness and positive exposure imply that U itself is the only counted component. Thus U has odd order, every vertex has degree D, $\kappa_U = 1$, and U is factor-critical. Odd order and the equality $D|V(U)| = 2e(U)$ force D even. Simplicity and $D \ge 4$ give $|V(U)| \ge 5$. By Lemma 205 this regular factor-critical graph is not a zero-slack cycle, so $ps(U) \ge 1$.

If X_U is nonempty, every component Q of $U - X_U$ is a proper induced subgraph of H, including when $U = H$. Thus $gap(Q) \le 1$ by hypothesis. All hypotheses of Lemma 202 apply to U, its degree-D marked set, and this X_U. It gives $ps(U) \ge (D-1)(\kappa_U - 1) + \epsilon$.

Slack is nonnegative by Lemma 61 and superadditive over whole components by Lemma 72. Put K equal to the sum of κ_U over the s components with nonempty X_U. The t-s empty-barrier components each have exposure one, so $\kappa = (t - s) + K$. Summing the preceding lower bounds gives

$ps(H) \ge (t - s) + (D - 1)(K - s) + \epsilon\, s$

$= \kappa + (D - 2)(\kappa - t) + (\epsilon - 1)s.$

Every term beyond κ is nonnegative. This proves the stated bound, including the case $t = s = \kappa = 0$, where it is exactly nonnegativity.

If $ps(H) = 0$, the bound forces $\kappa = 0$. The maximum matching attaining the exposure therefore covers every degree-D vertex, and $c_D(H) \le \kappa$ makes that cost zero. The same conclusion for $D = 3$ follows from its stronger bound.

Suppose $D \ge 4$ is even, $ps(H) \le 1$ and $\kappa > 0$. Here $\epsilon = 2$, so $1 \ge ps(H) \ge \kappa + (D - 2)(\kappa - t) + s$. The nonnegative integer terms force $\kappa = 1$, $t = 1$, and $s = 0$. The unique uncovered component therefore has empty X_U and is regular and factor-critical as already proved. Its slack is at least one. Superadditivity and nonnegative component slacks force this component's slack to be one and every other component's slack to be zero. □

Remark. Each uncovered whole component pays for its exposure. A nonempty marked barrier pays the quantitative rate; an empty barrier is regular and factor-critical and contributes a positive slack unit.

Depends on. Lemma 23, Lemma 61, Lemma 72, Lemma 86, Lemma 107, Lemma 161, Lemma 202, Lemma 205.

Lemma 244 (A vertex pool bounds the odd-degree exposure at large slack). *Use the finite integer costs c_i, matching number ν, maximum degree Δ, excess ex, slack ps, $gap = ex - ps$, and top missing-allocation run tr of Lemma 23. For every nonnegative integer j put $h(j) = \max(4, \lfloor j/2 \rfloor + 1)$. A graph satisfies T_- if positive gap implies gap one and $tr + ex \le h(\Delta)$.*

Let $D \ge 5$ be an odd integer, let H be a finite simple graph, and let W be a vertex subset of size $w \ge h(D) - 2$. Put $R = H - W$, with induced deletion. Suppose $\Delta(R) \le D$, every induced subgraph of R including R satisfies T_-, $ex(R) = ps(R) = s \ge h(D) - 1$, and $c_0(H) = w + \nu(R) + s$. Then the minimum number of degree-D vertices of R missed by a matching is at most one, and $c_D(R) \le 1$. There is no order bound or independence hypothesis on H, W, or R. All degrees, orders, cardinalities, costs, matching numbers, and parameters are integers.

Proof. Nonnegative slack, zero cost above actual degree, and zero top cost for a covered degree threshold follow from Lemma 61. If a matching of R covers its degree-D vertices, both conclusions follow. This includes $\Delta(R) < D$. Suppose instead that $\Delta(R) = D$ and no such matching exists. We use component slack superadditivity and row subadditivity from Lemma 72, and the actual vertex-pool allocation construction Lemma 75.

Choose a whole component C of R of largest matching number, and put $P = R - V(C)$. Its full allocation interval at $c_0(C) \geq \nu(C)$ absorbs all other components at their matching numbers, so $c_0(R) \leq \nu(R) + ex(C)$, and $ex(C) \geq s$. Suppose $gap(C) > 0$. Hereditary T_- and the positive top run give

$s \leq ex(C) \leq h(\Delta(C)) - 1 \leq h(D) - 1 \leq s.$

Thus $ex(C) = s$ and $tr(C) \leq h(D) - s \leq 1$. Since $s \geq 3$, the total $c_0(C) - 1 = \nu(C) + s - 1$ is at least $\nu(C)$; both endpoints are feasible. The actual W-pool fills its missing runs. The resulting full interval has length $w + \nu(C) + s - 1 \geq \nu(C)$, so it absorbs every other component at its matching number. The vertex-pool construction gives every allocation on H at total $w + \nu(R) + s - 1$, contradicting its stated cost. Consequently $gap(C) \leq 0$. Nonnegative slack and superadditivity now force

$ps(C) = ex(C) = s, gap(C) = 0, ps(P) = 0.$ (1)

The degree-D vertices of P have a covering matching. To prove this, otherwise take an uncovered connected component U of P of actual degree D. It has $ps(U) = 0$. Choose, using Lemma 107, a maximum-cardinality set X_U maximizing the number of odd components of $U - X_U$ wholly contained in its degree-D set, minus $|X_U|$. This maximum is the positive minimum degree-D exposure. If X_U were empty, connectedness would make U itself odd-order and D-regular, impossible because odd D would give an odd degree sum. Thus X_U is nonempty. Its component gaps are at most one by heredity, so Lemma 202 gives $ps(U) \geq 1$, a contradiction. Component covering matchings unite on P, and Lemma 61 gives $c_D(P) = 0$. Hence C has actual degree D and is uncovered.

Let $\kappa_C \geq 1$ be its minimum degree-D exposure and A its degree-D set. Choose X of maximum cardinality among sets maximizing $q_A(C - X) - |X|$, where q_A counts odd components wholly contained in A, and put $b = |X|$. Again odd-degree parity gives $b \geq 1$. Let calQ be all components of C-X, calK the odd components contained in A, and Y the union of all other component vertex sets. By Lemma 107 and Lemma 104, each K in calK is factor-critical, $|calK| = b + \kappa_C$,

$\nu(C) = b + \sum_{Q \in calQ} \nu(Q)$, and $c_D(C) \leq \kappa_C$.

Each such K has positive odd boundary $d(K) = e_C(V(K), X)$, equal to its total internal degree-D defect. Here $e_C(U, V)$ counts edges between disjoint vertex sets.

Choose Z in calQ of largest matching number m. Its full-cost interval, together with the actual X-pool, absorbs all other Q at their matching numbers. The same composition theorem gives $c_0(C) \leq \nu(C) + ex(Z)$, so $ex(Z) \geq s$. If $gap(Z) > 0$, the preceding chain of inequalities using T_- gives $ex(Z) = s$ and $tr(Z) \leq 1$. The actual W union X pool first absorbs P at $\nu(P)$: by (1), either $ex(P) = 0$ and that allocation interval is full, or T_- gives $ex(P) = gap(P) = 1$ with $tr(P) \leq h(D) - 1 \leq w + b$, since $b \geq 1$. The resulting full interval of length $w + b + \nu(P)$ fills Z's allocations at total $c_0(Z) - 1 = m + s - 1$, whose endpoints are feasible. Afterwards its length is at least m, and it absorbs every other Q at $\nu(Q)$. Matching additivity yields total

$w + b + \nu(P) + m + s - 1 + \sum_{Q \neq Z} \nu(Q) = w + \nu(R) + s - 1$

on the full graph H, again a contradiction. We conclude

$gap(Z) \leq 0$ and $ps(Z) \geq ex(Z) \geq s.$ (2)

It remains to bound the exposure from the retained boundary inequality; no completion assertion is needed. Put $T = (D - 1)/2$. For every K in calK different from Z with $d(K) < D$, the factor-criticality, positive defect, and hereditary gap bound verify all hypotheses of Lemma 179. Choose its supported credits $a_i(K)$ of total max(D-d(K)-ps(K),0). They vanish outside $1 \leq i \leq T$, satisfy $a_1(K) \leq T$ and $a_i(K) \leq 1$ for $i \geq 2$, and are bounded by $\max(c_i(K) - \nu(K), 0)$. Give zero credits to Z, to non marked components, and to marked components with $d(K) \geq D$. In the last case the required total is zero by nonnegative slack.

Apply the finite procedures of Lemma 104 to these credits, with initial total $b + \sum_{(} Q : a_i(Q) = 0) c_i(Q)$ at each index. Let E_{rem} be its sum of maximum credits over unfinished indices. It satisfies

$0 \leq E_{rem} \leq T + (T - 1) = D - 2.$ (3)

Define the nonnegative correction

$J = e(C[X]) + e_C(X, Y)$

$+ \sum_{Qnot\in calK, Q\neq Z} ps(Q)$

$+ \sum_{K\in calK, K\neq Z} \max(ps(K) + d(K) - D, 0).$

The retained top-row formula of that fact, using $gap(C) = 0$ and (1), gives

$ps(Z) + J + \sum_{Q\in calQ} c_D(Q)$

$\leq \beta + E_{rem} - D + 1_{Z\in calK}(D - d(Z))$, (4)

where

$\beta = s + c_D(C) - D(\kappa_C - 1) \leq s + D - (D-1)\kappa_C.$

The indicator term is zero when Z is not marked. Explicitly, the conversion substitutes

$gap(Z) - (ex(Z) - ex(C)) = s - ps(Z)$

in the gap formula and retains its correction

$\kappa_C - c_D(C) + \sum_{Q\in calQ} c_D(Q).$

Since $\kappa_C \geq 1$, $\beta \leq s + 1$. If Z is not marked, or is marked with $d(Z) \geq D$, (3) bounds the right side of (4) by s-1, contradicting (2). Hence Z is marked with $1 \leq d(Z) < D$. In this case (4), after dropping its nonnegative correction terms and using (3), gives

$s \leq ps(Z) \leq s + D - (D-1)\kappa_C + D - 2 - d(Z).$

If $\kappa_C \geq 2$, the right side is at most $s - d(Z) < s$, a contradiction. Therefore $\kappa_C = 1$.

A matching attaining this exposure on C united with a covering matching of P shows that R has minimum degree-D exposure one. Finally component subadditivity gives $c_D(R) \leq c_D(C) + c_D(P) \leq 1$. Every use of a deleted-vertex pool above is the construction of Lemma 75, so every edge meeting W union X, including internal edges, is assigned in its certificate. All choices and procedures are finite. □

Remark. The actual deleted-vertex pool excludes a positive-gap component carrying all the excess. The largest barrier component then carries at least all the slack. Two exposed vertices would leave too little room for that component in the retained boundary inequality.

Depends on. Lemma 23, Lemma 61, Lemma 72, Lemma 75, Lemma 104, Lemma 107, Lemma 179, Lemma 202.

Lemma 245 (Low-excess goodness from zero-slack coverage). *Use the finite integer certificate costs $c_i(G)$, matching number $\nu(G)$, independence number $\alpha(G)$, maximum degree $\Delta(G)$, excess $ex(G) = c_0(G) - \nu(G)$, slack $ps(G)$, and $gap(G) = ex(G) - ps(G)$ of Lemma 23. All graphs are finite and simple. A graph is factor-critical if deleting any vertex leaves a perfect matching.*

Say that a graph H has zero-slack coverage when every induced subgraph J of H, including H itself, with $ps(J) = 0$ and $\Delta(J) \geq 3$ has a matching covering every vertex of degree $\Delta(J)$. This condition makes no assertion about induced subgraphs of positive slack.

Every factor-critical graph Q with zero-slack coverage, $\alpha(Q) \leq 2$, and $|V(Q)| \geq 11$ satisfies $ps(Q) \geq 2$.

Let d belong to {8,10}, and let H be a nonempty connected d-regular factor-critical graph with zero-slack coverage. If $ex(H) \leq 2$, then $gap(H) \leq 0$.

No source-minimality, top-run, or hereditary-gap assumption is imposed. All graph orders, degrees, cardinalities, indices, capacities, budgets, matching and independence numbers, excesses and slack values are integers in their displayed nonnegative or positive domains; gaps may be signed integers.

Proof. Use Lemma 61 for nonnegative slack, covered-degree zero rows, zero tails, and maximum-degree deletion accounting; use Lemma 72 for restriction, vertex restoration and component slack superadditivity. Zero-slack coverage is inherited by every induced subgraph.

We first give the zero-slack consequence used below. Suppose J has zero-slack coverage and $ps(J) = 0$. Starting at J, delete any current maximum-degree vertex while the maximum degree is at least three. The coverage hypothesis and the deletion inequality give

$0 \geq ps(J - v) + 1 - (\nu(J) - \nu(J - v)) \geq 0.$

Thus each step leaves zero slack and lowers the matching number by exactly one. The process terminates by strict decrease of the vertex count, and every parent is covered. Put $n = |V(J)|$, $m = \nu(J)$, $a = \alpha(J)$, and $f = n - 2m$. After t steps its terminal B has order n-t, matching number m-t, maximum degree at most two, and deficiency $f + t$. Unmatched vertices in a maximum matching are independent, since an edge between two would enlarge the matching; hence $f + t \leq \alpha(B) \leq a$. Every graph of maximum degree at most two is a union of paths and cycles. Alternating vertices give an independent set of at least one third of its order, including the triangle and singleton endpoints. Thus $n - t \leq 3\alpha(B) \leq 3a$. Combining these inequalities yields

$n - m \leq 2a$. (1)

The same process supplies every covering hypothesis of Lemma 146. Consequently, for every integer $\cap$ $D \geq \max(3, \Delta(J))$,

$e(J) \leq 3m + (D - 3)(a - f)$. (2)

Its equality description is available on this very process, not on a separately chosen residual.

We record an elementary allocation bound. On a graph of order n, nonnegative integer budgets $x \geq y$ with $n \leq 2x + y + 1$ can be realized by assigning all edges meeting any y-element vertex set to the part with budget y and all remaining edges to the part with budget x. A matching in the first part uses distinct chosen vertices; the remaining graph has at most $2x + 1$ vertices. If y exceeds n, every edge can instead be placed in that first part. Exchanging the colors handles $x < y$. In particular $c_0(K_7) \leq 4$.

The complete graph K_7 has $c_1, c_2 \leq 4$ by row monotonicity. Deleting three vertices leaves K_4, whose perfect matching covers its degree-three vertices; hence $c_3(K_7) \leq 3$, and $c_4(K_7) \leq 3$ by monotonicity. Deleting one vertex leaves K_6 with a perfect matching, giving $c_5(K_7) \leq 1$ and $c_6(K_7) \leq 1$. Higher rows vanish. Its edge count is 21 and matching number three, so $ps(K_7) \geq 21 - 3 - 16 = 2$. For every integer $l \geq 8$, deleting a vertex of K_l cannot increase slack: when l is odd the deletion preserves matching number, and when l is even its perfect matching covers every maximum-degree vertex. Induction gives

$ps(K_l) \geq 2$ for every integer $l \geq 7$. (3)

We now prove the asserted independence-two bound. Suppose Q is factor-critical, has zero-slack coverage, has independence at most two and odd order $n = 2k + 1 \geq 11$, and satisfies $ps(Q) \leq 1$. Factor-criticality gives $\nu(Q) = k$.

If Q is nonregular, expose a vertex below maximum degree using a perfect matching after deleting it. This covers every maximum-degree vertex. Deleting a vertex w of maximum degree preserves the matching number by factor-criticality, so

$ps(Q) \geq 1 + ps(Q - w)$.

Nonnegative slack forces $ps(Q - w) = 0$. Apply (1) to Q-w: its order is 2k, its matching number is k, and its independence number is at most two. This gives $k \leq 4$, contrary to $k \geq 5$. We have therefore excluded all nonregular Q in this domain.

Suppose Q is d-regular. Odd order forces d even. A complete Q is excluded by (3). If $d \leq 4$, repeatedly selecting a vertex and removing its closed neighborhood constructs an independent set of cardinality at least $ceil(n/(d + 1)) \geq 3$, since each selection removes at most $d + 1$ vertices. This contradicts $\alpha(Q) \leq 2$. Hence $d \geq 6$ and Q is noncomplete.

Its complement L is a positive r-regular triangle-free graph, where $r = n - 1 - d$. Triangle-freeness follows from $\alpha(Q) \leq 2$. If L were bipartite, in each component regularity and the count of edges between the two parts would make those parts have equal cardinalities. As $r > 0$ there are no isolated vertices, so n would be even. Thus L has a shortest odd cycle, of some odd length $l \geq 5$.

Every vertex outside this cycle has at most two neighbors on it. Otherwise the cyclic gaps between its consecutive neighbors are all at least two by triangle-freeness. There are at least three gaps and their sum l is odd, so an odd gap has length at most l-4. The path along that gap together with the two edges to the

outside vertex would be a shorter odd cycle, a contradiction. The chosen cycle has no chord, since a chord divides an odd cycle into two cycles, one odd and shorter. Summing degrees at its vertices therefore gives $lr \le 2l + 2(n-l) = 2n$. It follows that $r \le 2n/5$, or

$3n \le 5d + 5$.

For $d = 6$, the odd integer n is at most $11 = 2d - 1$. For $d \ge 8$, the inequality $5d + 5 \le 6d - 3$ gives $n \le 2d - 1$ as well.

Choose an independent pair {u, v} in Q and put $F = Q - \{u, v\}$. It has odd order $n - 2 \le 2d - 3$. Every vertex loses at most two neighbors and at least one, the latter because otherwise it would form an independent triple with u, v. Thus every F-degree lies between d-2 and d-1.

The graph F is connected: each of two components would have at least d-1 vertices, exceeding its order bound. It has no cut vertex z either. Every component of F-z would have at least d-2 vertices. Since $|V(F - z)| \le 2d - 4$, a cut vertex would leave exactly two components, each of order d-2. Every vertex of either component must then be adjacent to z and to all other vertices of its component to attain degree at least d-2. This forces $deg_F(z) \ge 2d - 4 > d - 1$, a contradiction.

The ordinary matching barrier in Lemma 107 proves that F is factor-critical. Indeed, if F-w failed to have a perfect matching, its even order would give an attaining set X with $q - |X| \ge 2$, where q counts the odd components of F-w-X. Choosing one vertex from each such component gives $q \le \alpha(F - w) \le 2$. Thus X is empty and $q = 2$, so F-w is disconnected, contrary to the absence of a cut vertex.

The graph F is nonregular. If it were regular of degree d', both d and d' would be even by odd order. Every retained vertex loses exactly d-d' neighbors among u, v, so that number is zero or two. Zero would make u have no neighbors, since uv is absent. Two would make u adjacent to all n-2 remaining vertices, giving the odd degree n-2. Both contradict positive even degree d.

We check that $ps(F) \le ps(Q) \le 1$. First $c_d(Q) = 1$: exposing a vertex gives $c_d(Q) \le 1$ by covered-degree vanishing and restoration, while a zero-budget certificate at capacities (1,d-1) would give a perfect matching of odd-order Q and is impossible. Rows of Q above d and of F at and above d vanish. Put $a = 2$ and $r_i = c_i(Q) - c_i(F)$, with $0 \le r_i \le a$ by restriction and restoration. Factor-criticality of F gives $\nu(F) = k - 1 = k + 1 - a$. Since deleting this independent pair removes da edges, subtraction of slack definitions gives

$ps(F) = ps(Q) - \sum_{i=1}^{d-1}(a - r_i) \le ps(Q)$. (4)

If $n \ge 13$, F has order at least eleven, is nonregular and factor-critical, has independence at most two, and inherits zero-slack coverage. The nonregular case already proved contradicts (4). Therefore $n = 11$. The even degree $d \ge 6$ is at most eight because Q is noncomplete.

If $d = 8$, then F has nine vertices and 28 edges. It is nonregular with degrees six or seven, so its maximum degree is seven. Delete a degree-seven vertex w. Nonregular factor-criticality makes that parent covered and preserves its matching number, so $K = F - w$ has zero slack, eight vertices, 21 edges and matching number four. It inherits zero-slack coverage, and $\alpha(K) \le 2$. Formula (2) at $\cap$ seven gives $e(K) \le 12 + 4\alpha(K) \le 20$, a contradiction.

If $d = 6$, F has nine vertices and 21 edges, and its nonregular degrees are four or five. Deleting a degree-five vertex w again gives a zero-slack graph K with eight vertices, matching number four and 16 edges. Applying the refined density bound of Lemma 146 at $\cap$ five to a covered process on K gives

$16 \le 12 + 2\alpha(K) - 4(4 - c_1(K)) \le 16$.

Here $4 - c_1(K) \ge 0$ is supplied by that fact. Equality forces $\alpha(K) = 2$ and $c_1(K) = 4$. Its equality description has precisely $2 \cdot 4 - 8 + 2 = 2$ deleted vertices, independent in K and both of original K-degree five. Neither is adjacent to w in F, because $\Delta(F) \le 5$. Together with w they form an independent triple in F, contradicting $\alpha(F) \le 2$. All cases contradict $ps(Q) \le 1$, proving $ps(Q) \ge 2$.

Finally let H have degree d in {8,10} as stated, write $|V(H)| = 2k + 1$, and suppose $gap(H) > 0$ and $ex(H) \le 2$. If $ps(H) = 0$, zero-slack coverage would give a matching covering every degree-d vertex, namely a perfect matching of the odd-order graph H. Thus $ps(H) \ge 1$. Nonnegative integral excess and slack, together with $gap = ex - ps > 0$, force

$ex(H) = 2$ and $ps(H) = gap(H) = 1$.

If $k \le 4$, the elementary allocation construction above gives $c_0(H) \le k+1$: for every split of total $k+1$, its larger coordinate is at least $ceil((k+1)/2)$, and $2k+1 \le k+2+ceil((k+1)/2)$ for these integers. This would give $ex(H) \le 1$. Hence $k \ge 5$ and $|V(H)| \ge 11$. The independence-two bound just proved forces $a = \alpha(H) \ge 3$.

Fix a maximum independent set S of size a and put $G = H - S$. Factor-criticality, by restricting a perfect matching after deleting a vertex of S, gives $\eta = \nu(G) - (k+1-a) \ge 0$. Also $\Delta(G) \le d-1$. The same calculation as (4), now with $c_d(H) = 1$ and this arbitrary a, yields

$ps(G) = 1 - \eta - \rho$, where $\rho = \sum_{i=1}^{d-1}(a - c_i(H) + c_i(G)) \ge 0$.

Consequently η is zero or one. If $\eta = 1$, then $ps(G) = 0$, so its zero-slack coverage supplies a covered process. The consequence in Lemma 195 applies with $p = 1$ and $d \ge 8$ and forces $\eta \le 0$, a contradiction.

Assume $\eta = 0$. Choose X of maximum cardinality among the ordinary attaining sets in Lemma 107, and write $b = |X|$. Every component of G-X is odd and factor-critical. To check the absence of an even component, adjoining any vertex from one to X creates at least one odd component and increases $|X|$ by one; it would still attain the maximum and have larger cardinality. The matching identity therefore gives exactly $a+b-1$ components. Their independent sets can be united, so $a+b-1 \le \alpha(G-X) \le a$ and $b \le 1$.

If $b = 1$, the a components each have independence one, hence are odd cliques. Their maximum degrees are at most $d-1 \le 9$, so their odd orders are at most nine and their matching numbers at most four. If $b = 0$, there are a-1 whole components of G and each has independence at most two. Component slack superadditivity and $ps(G) \le 1$ imply that each has slack at most one. Each inherits zero-slack coverage, so the independence-two bound makes its order at most nine, again giving matching number at most four.

In either case every component C of H-(S union X) has $\nu(C) \le 4 \le a+b+1$. Apply the whole-block absorption in Lemma 51 at row zero, with deleted set S union X, edgeless retained graph, and these components as blocks. The initial full interval has length $a+b$; every block rank is at most that length plus one, and each absorption only increases the length. Thus it supplies every matching allocation at total

$a + b + \sum_C \nu(C) = a + \nu(G) = k+1$.

All original edges meeting S union X are included by that theorem. This contradicts $c_0(H) = k+2$. The contradiction proves $gap(H) \le 0$ for both displayed degrees. □

Remark. Zero slack with covering deletions limits order relative to matching and independence. Independence two then reduces to one nine-vertex residual, whose density equality forces an independent triple. The independent-cut barrier has only small pieces unless it spends its one slack unit in matching gain; absorption excludes the first option and the smaller-pool exchange excludes the second.

Depends on. Lemma 23, Lemma 51, Lemma 61, Lemma 72, Lemma 107, Lemma 146, Lemma 195.

Lemma 246 (Exceptional even-degree overflow has a single low separator). *Let $s, t \ge 1$ be integers and let F_1, F_2 be the star forests with nonincreasing positive integer edge-size lists $n_1,\dots, n_s$ and $m_1,\dots, m_t$, prescribed red and blue. Copies are ordinary subgraphs with vertex-disjoint target components. Arrowing an ordered pair means that every red-blue edge coloring contains its entire first target in red or its entire second target in blue. Put $M = s+t-1$ and*

$\ell_k = max\{n_i + m_j - 1 : 1 \le i \le s, 1 \le j \le t, i+j = k\}$

for every integer $2 \le k \le s+t$. For each integer $0 \le z \le M$ put $P_z = \sum_{k=z+2}^{s+t}(\ell_k - 1)$, with empty sum zero. For $0 \le z < M$ let A_z be the family of all ordered nonempty actual suffix pairs after deleting α and β leading components, where $0 \le \alpha < s$, $0 \le \beta < t$ and $\alpha + \beta = z$ are integers.

Fix an integer $0 \le q < M$, put $D = \ell_{q+2} - 1$, and let R be the initial plateau length at $D+1$ in the natural q-tail. Suppose valid original indices i_0, j_0 satisfy $i_0 + j_0 = q + R + 1$, $n_{i_0} = a$, $m_{j_0} = b$ and $a + b = D + 2$, where $4 \le a \le b$ are even integers. Put $r = a - 2$. Let H be finite and simple with $\Delta(H) \le D$ and $\Phi(H) := e(H) - \nu(H) \le P_q + r$, where e is edge count and ν is maximum matching cardinality.

For each whole connected component C of H let ω_C be its minimum, over all red-blue edge colorings, of the sum of $\max(d_{red}(v) - (a-1), 0) + \max(d_{blue}(v) - (b-1), 0)$ over its vertices. Let η_C be the minimum number of vertices of ambient H-degree D left uncovered by a matching of C, and let κ_C be one if C is odd and D-regular and zero otherwise. Put $\omega = \sum_C \omega_C$, $\eta = \sum_C \eta_C$ and $\kappa = \sum_C \kappa_C$.

If H arrows every member of A_q, exactly one of the following alternatives holds:

(I) $\omega = \kappa \geq R$.

(II) $\omega > \kappa$. In this case $b \geq a + 4$, $D \geq 10$, $\Phi(H) = P_q + r$, $\omega = R$, $\eta = R + r$ and $\kappa = R - 1$. Every whole D-regular component is equimatchable; every odd one is factor-critical and has matching cardinality at least $D/2$, and every even one is $K_{D,D}$. Equimatchable means that every inclusionwise maximal matching is maximum. Factor-critical means that deletion of any vertex leaves a perfect matching.

There is exactly one component $C_\cdot$ with $\omega_C > \kappa_C$. It is not D-regular, $\omega_{C_\cdot} = 1$ and $\eta_{C_\cdot} = a - 1$. Every maximum matching in every whole component C leaves exactly η_C ambient-degree-D vertices uncovered. For C other than $C_\cdot$ one has $\eta_C = \omega_C = \kappa_C$.

There is y in $V(C_\cdot)$ with $d_H(y) \leq D - 1$ such that $C_\cdot - y$ has exactly a odd components C_1,..., C_a wholly of ambient H-degree D. They are adjacent to y, factor-critical, equimatchable, and actual components of H-y. Every maximum matching in any other component of $C_\cdot - y$ covers its ambient-degree-D vertices. Writing $|V(C_i)| = 2\mu_i + 1$ and $b_i = e_H(V(C_i), \{y\})$, one has

$\nu(C_i) = \mu_i \geq D/2$, $b_i \geq 2$ *even*, $\sum_i b_i \leq d_H(y) \leq D - 1$,

and $\nu(H) = 1 + \sum_J \nu(J)$, with the sum over all actual components J of H-y.

Choose in each of the κ whole odd D-regular components any near-perfect matching and its exposed vertex, and let I be the set of these vertices. A near-perfect matching leaves one vertex uncovered; when $\kappa = 0$ take I empty. Then I is independent, $q + \kappa < M$, and $H' = H - I$ satisfies

$\nu(H') = \nu(H)$, $\Phi(H') = \Phi(H) - \kappa\, D = P_{q+\kappa} + r$,

and $P_{q+\kappa} = P_q - \kappa\, D$.

The graph H′ arrows every member of $A_{q+\kappa}$, has no whole odd D-regular component, and H′-y has a coloring of red maximum degree at most a-1 and blue maximum degree at most b-1. The natural tail at $q + \kappa$ has plateau length one and the same terminal sizes a, b.

Failure of both alternatives gives an avoiding coloring for some entire prescribed pair in A_q. Every such coloring extends over the q canonical stars of sizes ℓ_2,..., ℓ_{q+1}, and across every specified actual q-vertex set X when $H = G - X$, to avoidance of the unchanged F_1, F_2 while preserving all H-edge colors. No matching-deficit, original-head bound or further restriction on q is assumed. All orders, cardinalities, indices and degrees are integers in the stated domains. Unit stars, repeated sizes, and simultaneous exchange of full targets and colors are included.

Proof. Assume H arrows every member of A_q, and put $\sigma = \Phi(H) - P_q$. The complete envelope of this family is the natural q-tail. Indeed each retained local candidate on diagonal h comes from original indices summing to $q + h$. Conversely, for such original indices i, j choose $\alpha = \max(0, q - j + 1)$ and $\beta = q - \alpha$. These cuts are nonnegative, total q and satisfy $\alpha \leq i - 1$ and $\beta \leq j - 1$; they retain both components on local diagonal h. The family is nonempty because $q \leq s + t - 2$. Its component budget is M-q. The full profile is nonincreasing, since decreasing an index greater than one in any candidate gives an at-least-as-large candidate on the preceding diagonal.

Apply Lemma 99 to this family. It gives $\sigma \geq 0$ and bounds the high exposure of every maximum matching between R and $R + \sigma$. It also gives, for every independent set L of $k \geq R$ ambient-degree-D vertices, with $j = \nu(H) - \nu(H - L)$,

$\sigma \geq k - R - j$. (1)

The additional term in its full inequality is nonnegative. Choose the legal cuts $\alpha = \max(0, q - j_0 + 1)$, $\beta = q - \alpha$. They retain the terminal pair on local diagonal $R + 1$. The whole-pair overflow bound Lemma 143 therefore gives $\omega \geq R$.

Exposure and overflow minima add over whole components, because all their matchings and colorings can be chosen independently. An exposure-minimizing matching can be augmented to maximum size without uncovering previously covered vertices: its symmetric difference with any larger matching has an augmenting alternating path, and reversal of that path preserves coverage of all internal vertices and covers its two previously uncovered endpoints. Iteration terminates as cardinality strictly increases. Thus exposure-minimizing maximum matchings exist, and

$R \leq \omega \leq \eta \leq R + \sigma \leq R + r$. (2)

Here $\omega_C \le \eta_C$ and $\omega_C \ge \kappa_C$ follow componentwise from Lemma 196 at the odd capacities a-1, b-1.

If $\omega = \kappa$, (2) gives (I). Suppose $\omega > \kappa$ and put $w = (b-a)/2$. By Lemma 196 the case $w = 0$ has $\omega = \kappa$, so $w > 0$. Every component with $\omega_C > \kappa_C$ satisfies $\eta_C \ge \omega_C + r$. If z_0 counts these components, then

$R + r \ge \eta \ge \omega + r\, z_0 \ge R + r$.

All inequalities are equalities. Therefore $\sigma = r$, $\omega = R$, $\eta = R + r$, there is exactly one exceptional component $C_\cdot$, and $\eta_{C_\cdot} = \omega_{C_\cdot} + r$. On every other component $\eta_C = \omega_C = \kappa_C$. Combining an arbitrary component maximum matching with exposure-minimizing maxima elsewhere and using the global upper bound $R + r$ shows that every component maximum matching has exactly its minimum high exposure.

Every whole D-regular component is equimatchable. For such a component C and any maximal matching W of it put $j = \nu(C) - |W|$. The regular-union inequality of Lemma 99, using maximum matchings on the complementary components, has comparison exposure $k_0 = \eta = R + r$. It gives $r \ge k_0 - R + j = r + j$, so $j = 0$. Since D is even, Lemma 170 now makes each odd regular component factor-critical and each even regular component $K_{D,D}$. Their high exposures are respectively one and zero. Simplicity and odd factor-critical order give matching cardinality at least $D/2$ for the former. Neither regular possibility can satisfy $\eta_{C_\cdot} = \omega_{C_\cdot} + r$ with $\omega_{C_\cdot} > \kappa_{C_\cdot}$, so $C_\cdot$ is not D-regular. Write $g = \omega_{C_\cdot} = R - \kappa \ge 1$; then $\eta_{C_\cdot} = g + r$.

Put $A = \{v \in V(C_\cdot) : d_H(v) = D\}$. For $Z \subseteq V(C_\cdot)$, call Z attaining if the number of odd components of $C_\cdot - Z$ wholly in A, minus $|Z|$, equals $\eta_{C_\cdot}$. No attaining set meets A. Otherwise Lemma 150 supplies an independent $L_\cdot \subseteq$ A of size $\eta_{C_\cdot} + 2$ and a matching surviving its deletion of size $\nu(C_\cdot) - 1$. Add maximum matchings on all other whole components, and include their exposed high vertices in L. Those outside vertices number κ, since their component exposures equal κ_C. They are independent and have no edges to $C_\cdot$. Thus L is independent, has $k = \eta + 2 = R + r + 2$ vertices, and $\nu(H - L) \ge \nu(H) - 1$. Equation (1) gives $r \ge k - R - 1 = r + 1$, a contradiction.

By Lemma 143 choose a constant-cost pair S, T attaining overflow g on $C_\cdot$. Set $U = V(C_\cdot)$ minus (S union T), let Q count components of $C_\cdot[U]$ wholly in A with $|V(C)| + e_H(C, T)$ odd, and define

$B_x = Q + e_H(S, T) - x|S| - (D - x)|T|$, $z = |S| - |T|$.

Then $B_{a-1} = g$. Every counted component has a nonempty boundary into S union T, because $C_\cdot$ is connected and not D-regular. Its boundary has positive even size, by the even degree sum on its all-high vertices. Hence $Q \le e_H(SunionT, U)/2$. Put $\epsilon = \sum_{v \in SunionT}(D - d_H(v))$. Degree counting gives

$B_{D/2} \le -e_H(S) - e_H(T) - \epsilon/2$. (3)

The odd-parity family and endpoint identity in Lemma 196 give $B_1 \le \eta_{C_\cdot}$. The same pair satisfies $B_x - B_y = (y - x)z$. Therefore

$0 < g \le wz$, and $B_1 = g + rz \le g + r$.

Integrality and positivity force $z = 1$ and $B_1 = \eta_{C_\cdot}$. Equation (3) consequently yields

$2e_H(S) + 2e_H(T) + \epsilon \le b - a - 2g$. (4)

If T contained a vertex outside A, the tight-endpoint theorem Lemma 206 would produce a maximum matching of $C_\cdot$ with $\eta_{C_\cdot} + 1$ high exposures, contradicting the already proved constant exposure. Thus $T \subseteq$ A. That theorem places S in an attaining set, so S is disjoint from A. Each vertex of S contributes at least one to ϵ, and (4) gives $|S| \le b - a - 2g \le D - 2$. The no-high-attainer clause of Lemma 206 now forces T empty. As $z = 1$, $S = \{y\}$ for a vertex y with $d_H(y) \le D - 1$. The same theorem proves that {y} attains, every required odd all-A component of $C_\cdot - y$ is factor-critical, and $C_\cdot - y$ admits a proper degree split at capacities a-1, b-1.

In each whole odd D-regular component choose any near-perfect matching and its exposed vertex, and let I be the resulting kappa-element set. These components are whole, different from $C_\cdot$, and pairwise disjoint, so I is independent. Its deletion leaves a perfect matching in each selected component. By Lemma 196, matching exposure zero at degree bound D implies zero overflow there at the same capacities. Each other whole component outside $C_\cdot$ also has zero overflow, since its κ_C is zero. Combine their colorings with the split on $C_\cdot - y$. This gives a proper split of H-(I union {y}).

Let N_a and T_b count all original red sizes at least a and blue sizes at least b. The terminal indices give $N_a + T_b - 1 \ge q + R$. Strict inequality would place the valid candidate at indices N_a, T_b after the initial natural plateau while its value is at least $D + 1$, contradicting the nonincreasing profile. If the plateau reaches the last position, those later valid indices cannot exist. Thus

$N_a + T_b - 1 = q + R$. (5)

Suppose $R \geq \kappa + 2$. Put $l = \kappa + 1$, so $q + l \leq q + R - 1 < M$. Choose $\alpha' = \min(N_a - 1, q + l)$ and $\beta' = q + l - \alpha'$. By (5), both cuts are legal and smaller than N_a, T_b. The proper split just constructed avoids this deeper actual pair, because its suffixes still contain an a-edge red star and a b-edge blue star. Choose nonnegative $\alpha \leq \alpha'$, $\beta \leq \beta'$ with $\alpha + \beta = q$; such a choice exists since the upper bounds total $q + l$. The differences are cuts totaling l inside this nonempty A_q member. Apply Lemma 41 across the actual l-set I union {y}. It extends the same coloring to avoidance of that A_q member, a contradiction. Since $g = R - \kappa \geq 1$, it follows that $R = \kappa + 1$ and $g = 1$. Thus $\eta_{C.} = a - 1$.

Attainment by {y} now gives exactly a required odd all-A components C_1,..., C_a of $C. - y$. They are actual components of H-y and each is adjacent to y, by connectedness of $C.$. They are factor-critical. Put $|V(C_i)| = 2\mu_i + 1$, so $\nu(C_i) = \mu_i$. Match y to any neighbor in one such component, perfectly match that component after deleting the neighbor, and take maximum matchings in all other components of H-y. This constructs $1 + \sum_J \nu(J)$ edges. Every matching has at most one edge at y and otherwise at most $\nu(J)$ internal edges in each J, proving the matching identity.

This construction exposes one high vertex in each required piece not reached from y, as well as one in each of the κ whole odd regular components. Their total is $a - 1 + \kappa = R + r$. The global maximum-exposure bound implies that every maximum matching of every nonrequired component of $C. - y$ covers its high vertices: substitute any such matching in the construction, and an extra exposure would violate the bound.

Suppose a required C_i had a maximal matching short of maximum by $j \geq 1$. Match y into a different required piece, possible since $a \geq 4$, and substitute the deficient matching in C_i in the preceding construction. The resulting matching has nu(H)-j edges. Its entire exposed high set L has $k = R + r + 2j$ vertices, is independent within C_i by maximality, and is independent elsewhere and between the components of H-y because y is covered. It survives deletion of L, so the actual matching loss is at most j. Equation (1) gives $r \geq k - R - j = r + j$, a contradiction. Thus every C_i is equimatchable.

Each boundary b_i is positive, and $D(2\mu_i + 1) = 2e(C_i) + b_i$ makes it even. Therefore $b_i \geq 2$ and $\sum_i b_i \leq d_H(y) \leq D - 1$. Simplicity gives $n(D - n + 1) \leq b_i$ for $n = 2\mu_i + 1$. If $n \leq D$, then $n(D - n + 1) - D = (n - 1)(D - n) \geq 0$, contradicting $b_i < D$. Thus $n > D$ and $\mu_i \geq D/2$. Also $2a \leq D - 1 = a + b - 3$; since a, b are even, this implies $b \geq a + 4$ and $D \geq 10$.

The arbitrary selected near-perfect matchings and maximum matchings elsewhere combine to a maximum matching of H avoiding I. Thus $\nu(H - I) = \nu(H)$. Every I-vertex has degree D, and independence makes the edge loss exactly κ D. Since $\kappa = R - 1 < M - q$, $q + \kappa < M$ and the first κ potential entries are D. This proves every claimed potential identity and leaves plateau length one at the same terminal original indices.

No component newly created by deleting the selected vertex from a connected odd regular component is D-regular: every such new component has an edge to that vertex in the original graph, by connectedness, and hence has a vertex losing degree. Unchanged components were nonregular or even regular. Thus $H' = H - I$ has no odd D-regular component; its asserted split after deleting y was constructed above. For any legal deeper cuts alpha', beta' totaling $q + \kappa$, choose subcuts α, β totaling q. If H' avoided that deeper pair, Lemma 41 applied to the corresponding A_q member across exactly I would give avoidance on H. This proves arrowing of every $A_{q+\kappa}$ member, including $\kappa = 0$ with the empty extension.

We have proved (II) under $\omega > \kappa$ and (I) otherwise, so they are exclusive and exhaustive under arrowing. Failure of both negates the finite conjunction of arrowing every A_q member and supplies avoidance for an entire actual pair. Its two color-preserving extensions are Lemma 41. Every transfer retains the full sorted suffixes, including units and repetitions. Simultaneously interchanging the full targets and their colors preserves all conclusions. □

Remark. At the first even-degree slope, one exceptional component uses all surplus. Its tight endpoint barrier is an actual low singleton, and the compatible exposed vertices of the regular components shorten the whole suffix profile to one plateau position.

Depends on. Lemma 41, Lemma 99, Lemma 143, Lemma 150, Lemma 170, Lemma 196, Lemma 206.

Lemma 247 (At most one uncovered odd-degree vertex at small slack). *Use the finite integer costs $c_i(H)$, matching number $\nu(H)$, maximum degree $\Delta(H)$, excess $ex(H)$, slack $ps(H)$, and $gap(H) = ex(H) - ps(H)$ from Lemma 61. Let $D \geq 5$ be an odd integer and let H be a finite simple graph with $\Delta(H) \leq D$. Suppose $0 \leq ps(H) \leq D-1$, $gap(H) \geq 0$, and $gap(J) \leq 1$ for every proper induced subgraph J of H. Let κ be the minimum number of degree-D vertices left uncovered by a matching of H. Then $\kappa \leq 1$ and $c_D(H) \leq 1$. The empty graph and empty degree-D set are allowed. All orders, degrees, costs, matching cardinalities, and parameters are integers.*

Proof. If H has no degree-D vertex, its minimum exposure is zero and $c_D(H) = 0$ by Lemma 61. Assume henceforth that $\Delta(H) = D$, and put $s = ps(H)$. The same fact gives nonnegative slack on every graph. Choose a connected component C of H with largest matching number, and put $P = H - V(C)$. This choice exists because H is nonempty.

The component-absorption theorem Lemma 75 at capacities zero gives

$c_0(H) \leq c_0(C) + \nu(P)$.

Indeed each other component Q has $\nu(Q) \leq \nu(C) \leq c_0(C)$, so every required rank is at most the initial full interval plus one. Matching numbers add over components, and therefore

$ex(C) \geq ex(H) \geq s$.

If P is nonempty, C is a proper induced subgraph, so $gap(C) \leq 1$ gives $ps(C) \geq ex(C) - 1 \geq s - 1$. Component slack superadditivity Lemma 72 then gives

$0 \leq ps(P) \leq s - ps(C) \leq 1$.

If P is empty, $ps(P) = 0$ and the same upper bound holds.

Every proper induced subgraph of P is a proper induced subgraph of H. For $D = 5$ apply Lemma 222 to P, and for odd $D \geq 7$ apply Lemma 240 to P. The hypotheses include $\Delta(P) \leq D$ and $ps(P) \leq 1$, with no sign assumption on its own gap. Thus P has a matching covering all its degree-D vertices.

If $\Delta(C) < D$, or if C has a matching covering all its degree-D vertices, the union with this matching of P covers the degree-D set of H, giving $\kappa = 0$. Otherwise let $\kappa_C \geq 1$ be the minimum degree-D exposure in C. Every proper induced subgraph of C has gap at most one. Apply Lemma 243 to C. In its component notation $t = 1$, since C is connected and uncovered. As D is odd, its ϵ equals one, regardless of the chosen marked barrier. The stated stronger bound is therefore

$ps(C) \geq \kappa_C + (D-2)(\kappa_C - 1)$.

On the other hand, nonnegative slack and Lemma 72 give $ps(C) \leq ps(H) \leq D - 1$. If $\kappa_C \geq 2$, the preceding lower bound is at least D, a contradiction. Hence $\kappa_C = 1$. Matchings attaining the finite minima on C and P can be united, so $\kappa \leq 1$.

Finally apply Lemma 243 directly to H, which has actual maximum degree D and the stated proper-induced gap hypothesis. It gives $c_D(H) \leq \kappa \leq 1$. All component choices and exposure minima were over finite nonempty sets; the cases with no degree-D vertices were handled at the outset. □

Remark. A maximum-rank component holds all but at most one unit of slack. The exterior is covered, while two exposed vertices in that one component would require at least D slack units.

Depends on. Lemma 61, Lemma 72, Lemma 75, Lemma 222, Lemma 240, Lemma 243.

Lemma 248 (Avoiding colorings for a binary star forest at nonpositive potential surplus). *All graphs are finite and simple. For a positive integer j, let $K_{1,j}$ be the star with j edges, and put $P_3 = K_{1,2}$ and $K_2 = K_{1,1}$. Let p, t be positive integers and u a nonnegative integer. Let $F = pP_3$ disjoint union uK_2, and let J be the vertex-disjoint union of stars with positive nonincreasing integer edge sizes*

$m_1 \geq \cdots \geq m_t \geq 1$.

Put $s = p + u$ and let n_1,..., n_s be p twos followed by u ones. Define

$M = s + t - 1$,

$\ell_k = max\{n_i + m_j - 1 : 1 \leq i \leq s, 1 \leq j \leq t, i + j = k\}$

for every integer $2 \le k \le M+1$, and $L = \sum_{k=2}^{M+1} \ell_k$. Write $e(G)$, $\nu(G)$, $\Delta(G)$ for edge count, maximum matching cardinality, and maximum degree, and put $\Phi(G) = e(G) - \nu(G)$. An avoiding coloring of G is a red-blue edge coloring containing neither an ordinary red copy of the entire F nor an ordinary blue copy of the entire J. Components within a copied target have disjoint vertex sets.

Every G satisfying

$\Phi(G) \le L - M$ *and* $\nu(G) < M$

has an avoiding coloring, with no bound on its maximum degree or restrictions on its connected components. In particular this holds when $e(G) = L - 1$ and $\nu(G) = M - 1$.

More precisely, let $d = M - \nu(G) > 0$ and $\rho = \Phi(G) - (L - M) \le 0$. The graph H formed by adjoining d-1 disjoint copies of K_2 to G satisfies

$\nu(H) = M - 1$ *and* $e(H) = L + \rho - 1$.

It too has an avoiding coloring, whose restriction to G avoids the same full targets.

For any such G and any edge f in $E(G)$, let Ω_f be the set of avoiding colorings of G-f, where only f is deleted. Define completion counts $C_R(f,\chi)$, $C_B(f,\chi)$ using either of the following two conventions, fixed throughout: count all injective embeddings of the corresponding entirely labelled target into G, or count all actual target subgraphs of G. In either convention the counted image must contain f and every other image edge must have the prescribed color under χ. In the labelled convention every target vertex has a distinct fixed label, including vertices in repeated and unit components. Let $\Psi(f,\chi) = C_R(f,\chi)C_B(f,\chi)$.

The set Ω_f is finite and nonempty, and $\min_{\chi\in\Omega_f} \Psi(f,\chi) = 0$. Fix any order on $E(G)$, put red before blue, and let τ be the first avoiding color word of G in the resulting lexicographic order. For every χ in Ω_f with $\Psi(f,\chi) > 0$, simultaneously overwrite exactly those edges of G-f on which χ differs from τ. The resulting coloring $\chi' = \tau$ restricted to G-f belongs to Ω_f and has $\Psi(f,\chi') = 0$, so this global recoloring strictly decreases the product and requires one step. For any χ in Ω_f with a zero completion count in a prescribed color, assigning that color to f extends χ to an avoiding coloring of G. These conclusions hold in particular when $e(G) = L - 1$, $\nu(G) = M - 1$ and $\nu(G - f) = \nu(G)$. All counts, sizes, indices, deficits, and graph quantities are integers in the stated domains, and ρ is an integer that may be negative. The targets and their prescribed colors may be interchanged together.

Proof. Write $P = L - M$. Suppose first that $\Phi(G) < P$. Apply the single-pair case of Lemma 56 to the entire original lists. Every arrowing host has $\Phi \ge P$, so this G cannot arrow the pair and has an avoiding coloring.

It remains to consider $\Phi(G) = P$ and $\nu(G) < M$. Suppose, to obtain a contradiction, that G arrows (F, J). Apply Lemma 185 to this same G and these entire targets. Its hypotheses hold with positive deficit M-nu(G). It supplies an actual induced residual Q, a positive integer p′, and an actual nonempty suffix J′ of J such that Q arrows

$(p'P_3 disjoint union u K_2, J')$.

If L′, M′ are this retained pair's complete diagonal sum and component budget, the same residual satisfies

$\Phi(Q) = L' - M'$, $\Delta(Q) = D$,

where D is the largest component edge size in J′ and $D \ge 3$. No original component is shortened in forming this actual suffix.

If $D = 3$, write $J' = r'K_{1,3}$ disjoint union $h'P_3$ disjoint union $v'K_2$, where $r' \ge 1$ and h′, $v' \ge 0$ are integers counting the retained component sizes. The integers p′, r′ are positive and u, h′, v′ are nonnegative. The full cubic theorem Lemma 230 applies to Q and this actual pair with $\Delta(Q) = 3$. It gives $\Phi(Q) \ge L' - M' + 1$, contradicting the displayed equality.

Now let $D \ge 4$. Write

$J' = rK_{1,D}$ disjoint union disjoint $\bigcup_{i=1}^{b} K_{1,z_i}$ disjoint union vK_2,

where $r \ge 1$, b, $v \ge 0$ and $2 \le z_i \le D - 1$ are integers. The sum over i is empty when $b = 0$. Let a be the number of these z_i equal to D-1 and put $R = p' + r - 1$. In the full potential list of this retained pair, the largest entry is D and it occurs exactly R times. Indeed, such a value requires a red two and a blue D, whose index rectangle gives every index sum from two through $p' + r$.

We will also need two complete profile counts. A potential value at least D-1 requires either a red two with a blue size at least D-1, or any red component with a blue D. Their index rectangles end at coordinates $R + a$ and $R + u$, respectively, where coordinate k-1 corresponds to diagonal k. Each rectangle gives every coordinate from one through its endpoint. Hence there are exactly $\max(a, u)$ entries equal to D-1. A positive potential value requires a nonunit on at least one side. The red-nonunit rectangle ends at coordinate M′-u, and the blue-nonunit rectangle ends at coordinate M′-v. Both are nonempty. Thus the number of positive entries is $N = M' - \min(u, v)$.

If $u \le v$, apply Lemma 138 at cut zero to this actual pair, selecting its binary red side as the side with minimum original unit count. Both sides have nonunits. Here $N = M' - u = R + b + v$ and

$e(J') = Dr + \sum_{i=1}^{b} z_i + v \ge b + v + 1 = N - R + 1.$

The theorem's full-opposite edge test holds, its critical degree is D, and its potential is exactly L′-M′. It contradicts $\Phi(Q) = L' - M'$ by giving $\Phi(Q) \ge L' - M' + 1$.

Suppose $u > v$, so $N = M' - v = R + b + u$. Define the integer

$E_0 = (D - 1)r + \sum_{i=1}^{b}(z_i - 2).$

The complete blue forest has $c(J') = r + b + v$ components and satisfies $e(J') - c(J') = E_0 + b$. If $u \le E_0 - 1$, apply Lemma 138 at cut zero selecting J′ as the minimum-unit side, in its permitted color orientation. Its shortening test holds because

$e(J') - c(J') = E_0 + b \ge u + b + 1 = N - R + 1.$

It gives the same strict contradiction on Q.

In the remaining integer case $u \ge E_0$, one has

$E_0 \ge (D - 1)r + (D - 3)a > a,$

since $r \ge 1$ and $D \ge 4$. Thus $u > a$ and $u \ge 1$. The profile counts proved above now give exactly R entries D, exactly u entries D-1, and N positive entries. Form $(J')^-$ by shortening each blue nonunit once and discarding the blue units. It is nonempty, has $r + b$ components, and its largest size D-1 has multiplicity exactly r, because every $z_i \le D - 1$.

In the projection family of Lemma 214 for this actual pair and cut zero, choose

$((p' + u)K_2, (J')^-).$

This member is supplied by the component-count projection of the binary red forest; the matching size $p' + u$ is positive. Its component budget and largest-star multiplicity satisfy

$(p' + u) + (r + b) - 1 = N,$

$(p' + u) + r - 1 = R + u.$

Together with $u \ge 1$, these are precisely the two required equalities for that theorem, whose next-degree count is the present u. The cut-zero actual family has only our entire retained pair and is arrowed by Q; the maximum-degree bound is $\Delta(Q) = D \ge 4$. Thus Lemma 214 gives $\Phi(Q) \ge L' - M' + 1$, again contradicting equality. These cases cover every $D \ge 3$, and prove the avoidance assertion for the full range $\Phi(G) \le P$.

The boundary conditions give $\Phi(G) = (L - 1) - (M - 1) = P$ and positive deficit one, so they are included.

For the padding assertion, d-1 is a nonnegative integer. A matching in a disjoint union restricts to a matching in each component, while the union of componentwise maximum matchings is a matching. Consequently

$\nu(H) = \nu(G) + d - 1 = M - 1.$

Edge counts add, and the definition of ρ gives $e(G) = L + \rho - d$. Hence $e(H) = L + \rho - 1$ and $\Phi(H) = P + \rho \le P$. The first assertion applies to H with deficit one. Any full monochromatic target in the restriction of its avoiding coloring to G would be that same target in H, so the restriction avoids both targets.

Finally fix any G in the first assertion and an edge f. There are finitely many edge colorings of G, and the first assertion makes the set of avoiding ones nonempty. The indicated order therefore selects a well-defined τ. Its restriction chi′ to G-f avoids both targets, since deleting an edge creates no monochromatic subgraph

copy. Thus Ω_f is nonempty; it is finite because G-f has finitely many edges. Both completion counts are finite nonnegative integers in either convention, since the targets and host have finite vertex sets.

Let c be tau's color on f. If a completion in color c were counted at (f, chi'), assigning f its color c would make all image edges of that entire target monochromatic in τ. This contradicts avoidance. The argument applies to every injective labelled embedding and every actual full target subgraph, without selecting a component, center, unit-edge orientation, or placement of repeated components. Therefore the completion count in color c is zero and $\Psi(f, \chi') = 0$.

Every Ψ is nonnegative and this value is attained in Ω_f, proving that its minimum is zero. Over writing exactly the edges where χ differs from τ produces this same chi', leaves the distinguished edge f fixed, and strictly decreases Ψ whenever the original product was positive. This is one simultaneous recoloring of the whole edge set off f; no condition on intermediate partial over write s is asserted or needed.

For the extension assertion, let χ in Ω_f have zero completion count in color c and give f that color. A forbidden target in the other color cannot use f, so would already be in χ. A forbidden target in color c either avoids f, again contradicting chi's avoidance, or uses f and supplies a counted completion, contradicting its zero count. In the actual-subgraph convention this is itself a counted image; in the labelled convention any isomorphism from the fixed labelled target to that image supplies a counted injective embedding. Thus extension avoids both targets. The argument applies to every edge, and therefore also to a matching-preserving edge. Renaming red and blue throughout proves simultaneous color reversal. □

Remark. The first deletion preserving matching number retains an actual binary pair. Cubic equality is excluded by the full cubic theorem. At every larger degree, the unit counts either satisfy a direct edge-count test or force an exact next-degree projection. One full avoiding coloring then eliminates every completion in the color assigned to the missing edge.

Depends on. Lemma 56, Lemma 138, Lemma 185, Lemma 214, Lemma 230.

Lemma 249 (Nonbinary terminal exclusion at surplus one). *All graphs are finite and simple. Let $s, t \geq 1$ be integers and let F_1, F_2 be nonempty star forests with complete nonincreasing positive integer edge-size lists n_1,..., n_s and m_1,..., m_t. Copies are ordinary subgraphs with vertex-disjoint target components; the targets are prescribed red and blue. Put $M = s + t - 1$ and $\ell_k = max\{n_i + m_j - 1 : 1 \leq i \leq s, 1 \leq j \leq t, i + j = k\}$ for every integer $2 \leq k \leq s + t$.*

Fix an integer $0 \leq q < M$, put $K = M - q$, $L_q = \sum_{k=q+2}^{s+t} \ell_k$, $P_q = L_q - K$, and $D = \ell_{q+2} - 1 \geq 3$. Let R be the number of initial entries equal to $D + 1$ in this natural tail. Let A_q contain every pair $(F_1^{\alpha}, F_2^{\beta})$ for nonnegative integer cuts $\alpha < s$, $\beta < t$ and $\alpha + \beta = q$, where superscripts remove that many leading whole components.

Suppose valid original indices i_0, j_0 satisfy $i_0 + j_0 = q + R + 1$, $n_{i_0} + m_{j_0} = D + 2$, and both indicated sizes are at least three. Every graph H with

$$\Delta(H) \leq D,\ e(H) - \nu(H) \leq P_q + 1,\ K - \nu(H) \geq 2$$

has a coloring avoiding both entire targets of some member of A_q. Here e, ν, Δ denote edge count, maximum matching cardinality and maximum degree.

Every such avoiding coloring extends over the disjoint original canonical q-star pool of edge sizes ℓ_2,..., ℓ_{q+1}, and across every specified actual q-vertex set X in any graph G with $H = G - X$, to a coloring avoiding the full originals and retaining all H-edge colors.

Consequently, if a graph H satisfying the same three inequalities arrows every member of A_q, then every terminal maximizing original pair has unordered sizes {2,D}, and $\max(n_1, m_1) = D$. No original-head bound, strict-cut, unit-count, repetition, connectivity, component structure, or smaller-target induction hypothesis is imposed.

Proof. Assume that H arrows every member of A_q and has a stipulated nonbinary terminal pair. Write $\sigma = e(H) - \nu(H) - P_q$ and $\delta = K - \nu(H) \geq 2$. The family A_q is finite and nonempty. Every member has component budget K. Its envelope at local diagonal h is ℓ_{q+h}: a retained candidate has that full index sum; conversely any full i, j with $i + j = q + h$, $h \geq 2$, admit $\alpha = \min(q, i - 1)$, $\beta = q - \alpha \leq j - 1$, giving legal cuts retaining that candidate. In particular its critical degree, natural potential and plateau width are D, P_q, R.

The same allocation applied to i_0, j_0 represents the stipulated terminal pair in this family, at local index sum $R+1$.

Apply Lemma 225 to this family and H. It gives $\sigma \geq 0$, so σ is zero or one, and gives even $D \geq 6$. The stipulated terminal sizes are both even and at least four. There are $\kappa \geq R$ whole connected odd D-regular components, each with a near-perfect matching. The graph's edge bound is

$e(H) = L_q + \sigma - \delta \leq L_q - 1$. (1)

The actual-family clause of Lemma 89 gives $q < \min(s,t)$. Thus (F_1, F_2^q) and (F_1^q, F_2) are actual family members. Coloring all H-edges red in the first pair forces an entire F_1 copy, and coloring all edges blue in the second forces an entire F_2 copy. Hence both full original heads are at most D.

Suppose one full head equals D and orient that original as F and the opposite one as J, exchanging their prescribed colors together. The stipulated terminal pair has sizes $a \geq 4$ on F and $b \geq 4$ on J. All hypotheses of Lemma 215 hold: its maximum head is D, its critical natural entry and terminal indices agree, the deficit is positive, (1) gives $e(H) \leq L_q$, and the $\cap$ is D. With $\gamma = \tau(H) - K \geq 0$, where τ denotes minimum vertex-cover size, and $k_D = floor((D-1)/2) \geq 2$, that theorem gives

$R + (k_D - 1)(\delta + \gamma) \leq k_D\, \sigma \leq k_D$.

The left side is at least $1 + 2(k_D - 1) = 2k_D - 1 > k_D$. This contradiction proves both full heads are at most D-1. Each is at least four because it dominates its component in the stipulated pair.

Apply Lemma 176 with allowance $z = 1$. The inequalities $D \geq 6$ and $z \leq D/2 - 1$ hold, and the $\kappa \geq R$ components supplied above are exactly whole odd regular near-perfect components of the required degree. For every such component C, writing $\mu = \nu(C)$, it gives

$\mu \geq n_1 + m_1 - 4 + 3\delta \geq D - 2 + 3\delta \geq D + 4$. (2)

The middle step uses $n_1 + m_1 \geq D + 2$, since the first full diagonal dominates the first natural-tail diagonal.

Every one of these components is non-equimatchable. Indeed, if all its inclusionwise maximal matchings were maximum, Lemma 208 would apply to the connected odd D-regular graph C and give $\nu(C) \leq D$, contradicting (2). For a whole D-regular component define $g(C)$ as its maximum loss $\nu(C) - |W|$ over inclusionwise maximal matchings W. Its finite domain is nonempty, and $g(C) \geq 1$ for each of the κ odd components. The complete-family theorem Lemma 99, applied to the same A_q and H, bounds the sum of $g(C)$ over all whole D-regular components by σ. Therefore

$1 \leq R \leq \kappa \leq \sum_C g(C) \leq \sigma \leq 1$.

All are equal. In particular $R = 1$ and there is a unique whole odd D-regular component C, with its near-perfect matching. Fix this actual C and put $\mu = \nu(C)$.

The $R = 1$ clause of Lemma 176 applies to the same objects. The full original edge totals differ. Orient the full originals as F, J with $e(F) < e(J)$, and let S, T be their component counts and B, A their heads. It gives

$\mu \geq e(F) + \delta - 1 \geq e(F) + 1$, $T \geq \mu + \delta + 1 \geq \mu + 3$. (3)

Both heads remain in [4,D-1]. Since a matching in C is a matching in H,

$q + \mu \leq q + \nu(H) = M - \delta < M$. (4)

Apply the actual-family clause of Lemma 89 again in this orientation. An attaining separator exists and has nonnegative cardinality. Dropping that cardinality yields

$Wplus := 2sum_{minJ} floor((m-1)/2)$

$\leq S - q - 2\delta \leq S - 4$. (5)

If $B = D - 1$, the last necessary conclusion of Lemma 84 gives the additional fitting premise needed below. To check that application, the plateau is unique, $D \geq 6$ is even, $\mu \geq D/2$ follows from (2), the same C is whole, and its allowance $z = 1$ satisfies $z + 1 = 2 < D/2$. Moreover $S \leq \mu$ follows from (3) and positivity of the original components. Thus its one-sided consequence with this full orientation says that the entire original J-group of sizes at least $\max(1, D + 1 - B) = 2$ has at most $2\mu + 1$ vertices. If $B \leq D - 2$, no such extra fitting condition is required.

All hypotheses of Lemma 90 now hold on the same full pair F, J, q, D, μ, H and C. Equations (3)-(5) give its full component and half-order bounds and its unexhausted depth; $R = 1$ gives its unique plateau. Equation (1) is its edge bound, and $\Phi(H) \le P_q + 1$ implies its allowance-two potential bound. In the only case requiring a nonunit fitting condition it was just supplied on the same component capacity. That theorem constructs a coloring avoiding an entire actual member, contradicting the assumed common arrowing. Therefore a stipulated nonbinary terminal pair implies the asserted avoiding coloring.

We verify both extensions for any actual member avoided by a coloring of H, with legal cuts α, β. Color α of the disjoint pool stars red and the other β blue. A positive connected copied component in a pool star uses that star's center, so at most one vertex-disjoint target component can occupy each pool star. Thus at most α components of a red full target or β components of a blue full target could use the pool. The remaining components would contain the corresponding suffix in H, a contradiction.

For the actual-set extension, partition $X = X_R$ disjoint union X_B with sizes α, β. Color every edge meeting X_R red, and color all other edges meeting X_B blue. Every positive red connected subgraph meeting X contains a vertex of X_R; every such blue subgraph contains a vertex of X_B. Distinct copied target components consume distinct assigned-color vertices. Consequently at most α red or β blue components of a full copied target meet X. If s-alpha red components remain, order their original indices $i_1 < \cdots < i_{s-\alpha}$; then $i_r \le \alpha + r$, because otherwise too few later indices would remain. Their sizes therefore dominate $n_{\alpha+r}$. Trimming them supplies F_1^{α} inside H. The corresponding t-beta blue indices give F_2^{β}. Both contradict the original avoiding coloring of H. This includes $q = 0$, units, and components meeting X through leaves.

Finally suppose H satisfies the three displayed inequalities and arrows every actual member, without assuming a nonbinary terminal. The result already proved excludes every terminal pair with both sizes at least three. The legal-corner and monochromatic arguments using Lemma 89 still give both full heads at most D. A terminal pair consists of positive sizes summing to $D + 2$. A size one would force a size $D + 1$, exceeding that head $\cap$. Thus one size is two and the other D. At least one terminal maximizing pair exists because the last plateau entry is an attained finite maximum. Hence a full original contains a size-D star, giving $\max(n_1, m_1) = D$. □

Remark. The head-D envelope inequality first removes the boundary head. Every surviving regular component is too large to be equimatchable, so the single available unit of surplus leaves exactly one component and one plateau entry. Full-target fitting then supplies the common exact remnant that pays for removing this component.

Depends on. Lemma 84, Lemma 89, Lemma 90, Lemma 99, Lemma 176, Lemma 208, Lemma 215, Lemma 225.

Lemma 250 (Independent cuts in even regular factor-critical graphs). *Use the finite integer certificate costs $c_i(G)$, matching number $\nu(G)$, maximum degree $\Delta(G)$, independence number $\alpha(G)$, slack $ps(G) = e(G) - \nu(G) - \sum_{i\ge1} c_i(G)$, excess $ex(G) = c_0(G) - \nu(G)$, and signed gap $gap(G) = ex(G) - ps(G)$ from Lemma 61. All graphs are finite and simple, and vertex deletion is induced.*

Let $D \ge 4$ be an even integer, and let C be a connected D-regular factor-critical graph, meaning that C-v has a perfect matching for every vertex v. Suppose $gap(Q) \le 1$ for every proper induced subgraph Q of C, including the empty graph. Put $m = \nu(C)$, $s = ps(C)$, and $g = gap(C)$.

For every inclusionwise maximal independent set I of C put $b = |I|$, $J = C - I$, and

$\eta = \nu(J) - (m + 1 - b)$,

$\rho = \sum_{i=1}^{D-1}(b - c_i(C) + c_i(J))$,

$q = c_{D-1}(J)$.

Then $m \ge D/2$, $b \ge 1$, $|V(C)| = 2m + 1$, $\Delta(J) \le D - 1$, $c_D(C) = 1$, $\eta \ge 0$, $\rho \ge 0$, and

$ps(J) = s - \eta - \rho \ge 0$,

$0 \le q \le s - \eta - \rho$,

$m - b \le 2s + g + D/2 - 2 + q \le 3s + g + D/2 - 2 - \eta - \rho$.

When $D = 4$ one has the stronger bounds

$2q \le s - \eta - \rho$,

$m - b \le 2s + g + floor((s - \eta - \rho)/2)$.

In particular, choosing I maximum gives

$\nu(C) - \alpha(C) \le 3ps(C) + gap(C) + D/2 - 2$,

and at $D = 4$ the right side can be replaced by $2ps(C) + gap(C) + floor(ps(C)/2)$.

All orders, degrees, indices, costs, matching and independent-set cardinalities, slack values, η, ρ and q are integers in the displayed nonnegative or positive domains; g is a signed integer. The inequalities apply to every maximal independent set, with no well-coveredness, source-minimality, or sign assumption on g.

Proof. Since C has positive degree, it is nonempty. Deleting any vertex leaves a perfect matching, so $|V(C)|$ is odd and, with $m = \nu(C)$, its order is $2m + 1$. Simplicity and D-regularity give $2m + 1 \ge D + 1$ and $m \ge D/2$. Fix any inclusionwise maximal independent set I. It is nonempty; put $b = |I|$ and $J = C - I$. Every vertex outside I has a neighbor in I, since otherwise it could be added to I. Thus $\Delta(J) \le D - 1$. Independence and regularity give

$e(J) = e(C) - Db$. (1)

For any v in I, choose a perfect matching of C-v. Deleting the other b-1 vertices of I removes at most b-1 matching edges, leaving a matching of J with at least $m + 1 - b$ edges. Therefore $\eta = \nu(J) - (m + 1 - b) \ge 0$.

We first determine the top row of C. A perfect matching of C-v covers every vertex of C-v, so Lemma 61 gives $c_D(C - v) = 0$. Vertex extension from Lemma 72 gives $c_D(C) \le 1$. If $c_D(C) = 0$, take degree capacities (1,D-1) and exceptional matching budgets (0,0). The exceptional parts have no edges. At every vertex the first bounded part has degree at most one and the second at most D-1. Their degrees sum to the original degree D, so the first part has degree exactly one at every vertex. It would be a perfect matching of C, impossible on its odd number of vertices. Hence

$c_D(C) = 1$. (2)

For integers $1 \le i < D$ put $d_i = c_i(C) - c_i(J)$. By restriction and vertex extension in Lemma 72, one has $0 \le d_i \le b$. Consequently $\rho = \sum_{i=1}^{D-1}(b - d_i) \ge 0$. The zero-tail conclusion of Lemma 61 gives $c_i(C) = 0$ for $i \ge D + 1$ and $c_i(J) = 0$ for $i \ge D$. Using (1)-(2), the definitions of slack and η, and $\sum_{i=1}^{D-1} d_i = (D - 1)b - \rho$, one obtains

$ps(J)$

$= e(C) - Db - [m + 1 - b + \eta] - \sum_{i=1}^{D-1}[c_i(C) - d_i]$

$= e(C) - m - 1 - \sum_{i=1}^{D-1} c_i(C) - \eta - \rho$

$= s - \eta - \rho$. (3)

Nonnegative slack in Lemma 61 proves that this value is nonnegative.

Put $q = c_{D-1}(J)$. If $\Delta(J) < D - 1$, its zero tail gives $q = 0$. Otherwise $\Delta(J) = D - 1 \ge 3$. When $D = 4$, the degree-three conclusion of Lemma 243 gives $q \le \kappa$ and $2\kappa \le ps(J)$, where κ is the minimum number of degree-three vertices of J missed by a matching. Thus $2q \le ps(J)$. When $D \ge 6$, every proper induced subgraph of J is also proper induced in C and has gap at most one. The same cited result at actual degree D-1 gives $q \le ps(J)$. These statements and (3) establish every claimed bound on q. Vertex extension also gives

$c_{D-1}(C) \le b + q$. (4)

The zero-capacity cost bounds the row of total degree capacity one:

$c_1(C) \le c_0(C) = m + s + g$. (5)

Indeed, for each nonnegative budget split of total $c_0(C)$, its zero-capacity certificate has empty degree-bounded parts and remains a valid certificate at either capacity split of total one.

Apply Lemma 119 to this same connected even-regular factor-critical graph C. It gives

$c_i(C) + c_{D-i}(C) \le 2m + 2$ for $1 \le i < D$.

Pair the indices 2,..., D-2 about $D/2$; the midpoint obeys $c_{D/2}(C) \le m+1$. Thus

$\sum_{i=2}^{D-2} c_i(C) \le (D-3)(m+1)$. (6)

For $D = 4$ the sum consists just of that midpoint, so the same formula holds. Regularity gives $e(C) = D(2m+1)/2$. With (2) and the zero tail, the slack identity therefore reads

$\sum_{i=1}^{D-1} c_i(C) = (D-1)m + D/2 - 1 - s$.

Subtract (6) and then use (4)-(5). The result is

$2m - D/2 + 2 - s$

$\le c_1(C) + c_{D-1}(C)$

$\le m + s + g + b + q$.

Rearranging proves

$m - b \le 2s + g + D/2 - 2 + q$.

The general estimate $q \le s - \eta - \rho$ from (3) proves the second rank inequality. If $D = 4$, use instead $q \le floor((s - \eta - \rho)/2)$. Finally a maximum independent set is inclusionwise maximal, so setting $b = \alpha(C)$ and using $\eta, \rho \ge 0$ gives the two stated bounds involving $\alpha(C)$. Every matching and certificate used above is on the specified induced graph; no simultaneous choice for different independent sets is required. □

Depends on. Lemma 61, Lemma 72, Lemma 119, Lemma 243.

Lemma 251 (Degree-ten goodness at excess at most four). *Use the finite integer certificate costs $c_i(H)$, matching number $\nu(H)$, independence number $\alpha(H)$, excess $ex(H) = c_0(H) - \nu(H)$, nonnegative slack $ps(H) = e(H) - \nu(H) - \sum_{i\ge1} c_i(H)$, and signed gap $gap(H) = ex(H) - ps(H)$ of Lemma 61. All graphs are finite and simple, and vertex deletion is induced.*

Let H be a nonempty connected ten-regular graph such that deleting any vertex leaves a perfect matching. Suppose $ex(H) \le 4$ and every proper induced subgraph J of H, including the empty graph, satisfies $gap(J) \le 1$. Then $gap(H) \le 0$.

In particular the conclusion holds when every proper induced subgraph satisfies the implication

$gap(J) > 0 \Rightarrow [gap(J) = 1$ *and* $tr(J) + ex(J) \le \max(4, floor(\Delta(J)/2) + 1)]$,

Here $tr(J) = 0$ if $c_0(J) = 0$; otherwise $tr(J)$ is the largest number of consecutive integers x in $[0, c_0(J) - 1]$ for which the edges of J cannot be partitioned into two parts with matching numbers at most x and $c_0(J) - 1 - x$.

No minimality with respect to edge deletion is assumed. All graph orders, degrees, costs, matching and independence cardinalities, and run lengths are integers in their stated nonnegative domains; gaps may be signed integers.

Proof. Suppose $gap(H) > 0$. Factor-criticality gives $|V(H)| = 2k+1$ and $\nu(H) = k$ for an integer k. Apply Lemma 224. Since $ex(H) \le 4$, it gives

$ex(H) = 4$, $gap(H) = 1$, $ps(H) = 3$,

k in {11,12,13}, $c_9(H) = k - 7$, $c_{10}(H) = 1$,

and $c_i(H) = 0$ for all $i \ge 11$. (1)

Every hypothesis of Lemma 149 now holds: H has minimum degree ten, hence at least eight, and has the required order, matching number and excess. In particular every induced graph of order at least 2k-6 has its floor-half matching number, and every odd-order induced graph of order at least 2k-5 is factor-critical.

We first record the exact density input available on the proper induced graphs used below. For a graph J of maximum degree at most nine define

$B_9(J) = -6|V(J)| + 15\nu(J) + 6\alpha(J)$.

If J is proper induced in H and $ps(J) = 0$, there is a complete covered maximum-degree deletion sequence on J down to degree at most two. Indeed, at a stage of actual degree $D \ge 3$, Lemma 243 supplies coverage

of all degree-D vertices. Its proper-induced gap premise for $D \geq 4$ follows because every such descendant is proper induced in H; its cubic case needs no gap premise. If v is a current maximum-degree vertex and $h = \nu(J) - \nu(J - v)$, then h is zero or one. By Lemma 61,

$0 = ps(J) \geq ps(J - v) + 1 - h \geq 0.$

Thus $ps(J - v) = 0$ and $h = 1$. Induction provides the entire covered sequence, which terminates because each step removes a vertex.

Apply Lemma 146 to this sequence with degree ∩ nine. Its nonnegative refined deficit gives

$ps(J) = 0$ implies $e(J) \leq B_9(J)$. (2)

We will also use its equality and prefix conclusion: if J has a perfect matching and $e(J) = B_9(J)$, and J is obtained from a graph G of maximum degree at most nine by q vertex deletions each made at degree nine, then

$q + \alpha(J) \leq \alpha(G)$. (3)

To check (3), its unrefined deficit L is zero, so its number of subsequent deletions is $t = 2\nu(J) - |V(J)| + \alpha(J) = \alpha(J)$. Its prefix clause gives $q + t \leq \alpha(G)$. The covered sequence needed by that clause was just supplied.

Choose any maximum independent set S of H. Put $a = |S|$, $G = H - S$, and $\eta = \nu(G) - (k + 1 - a)$. The set S is nonempty and dominates H, since a vertex outside its closed neighborhood could be added to it. Consequently $\Delta(G) \leq 9$. Since S is independent and H is ten-regular,

$|V(G)| = 2k + 1 - a$, $e(G) = 10(k - a) + 5$, $\alpha(G) \leq a$. (4)

For $1 \leq i \leq 9$ put $r_i = c_i(H) - c_i(G)$. Induced restriction and vertex restoration in Lemma 72 give $0 \leq r_i \leq a$. The vanishing rows in Lemma 61 give $c_i(G) = 0$ for $i \geq 10$. Subtracting the slack definitions, using (1) and (4), gives the exact identity

$ps(G) = 3 - \eta - \epsilon$, where $\epsilon = \sum_{i=1}^{9}(a - r_i) \geq 0$. (5)

All terms are integers.

If $a \geq 7$, Lemma 149 gives $\eta \geq 3$. Nonnegative slack in (5) forces $\eta = 3$ and $\epsilon = 0$. Then $r_9 = a$, whence $c_9(H) \geq a \geq 7$, contradicting (1), where $c_9(H) \leq 6$. Therefore $a \leq 6$. The exact rank conclusion of Lemma 149 gives

$\nu(G) = k - \lfloor a/2 \rfloor$, $\eta = floor((a - 1)/2)$. (6)

If G has even order it has a perfect matching. If G has odd order it is factor-critical. In either case $c_9(G) = 0$. For even order, a perfect matching covers every degree-nine vertex. For odd order with $\Delta(G) = 9$, the degree sum on its odd vertex set prevents every degree being nine; choose a vertex of smaller degree and a perfect matching after its deletion, which covers every degree-nine vertex. These coverings imply $c_9(G) = 0$ by Lemma 61. If $\Delta(G) \leq 8$, the same conclusion is its vanishing-row property.

The restoration bound now gives $c_9(H) \leq a$, and $r_9 = c_9(H)$. Thus $\epsilon \geq a - c_9(H)$. Combining this with (1), (5), and (6) leaves the following exhaustive table:

k a $|V(G)|$ $\nu(G)$ $e(G)$ upper bound for $ps(G)$

11 4 19 9 75 2

11 5 18 9 65 0

11 6 17 8 55 -1

12 5 20 10 75 1

12 6 19 9 65 0

13 6 21 10 75 1. (7)

The negative entry is impossible. In the two zero entries, nonnegative slack makes $ps(G) = 0$. At $(k, a) = (11, 5)$, (2) and (4) give $e(G) \leq -6 \cdot 18 + 15 \cdot 9 + 6 \cdot 5 = 57$, contradicting $e(G) = 65$. At $(k, a) = (12, 6)$, they give $e(G) \leq -6 \cdot 19 + 15 \cdot 9 + 6 \cdot 6 = 57$, again contradicting 65.

It remains to exclude the other three rows. They have

$a = k - 7$, $e(G) = 75$, $q := 7 - a = 14 - k$ in {3,2,1}. (8)

We construct q successive maximum-degree vertex deletions in G, all covered, and denote the last graph by K. Every prefix has resulted from deleting at most $a + q = 7$ vertices of H. By Lemma 149 each even-order prefix has a perfect matching and each odd-order prefix is factor-critical. We check coverage at each odd parent explicitly; at an even parent its perfect matching supplies coverage.

An odd-order regular graph has even degree, since its degree sum is even, and therefore its number of edges is divisible by its order. Thus any odd factor-critical graph whose edge count is not divisible by its order is nonregular. It has a vertex below its maximum degree, and a perfect matching after deleting that vertex covers all maximum-degree vertices.

For $k = 11$, G has order nineteen and seventy-five edges. Its maximum degree is eight or nine: it is at most nine, and twice its edge count exceeds seven times its order. It is nonregular because nineteen does not divide seventy-five. Delete any maximum-degree vertex. The eighteen-vertex child has sixty-six or sixty-seven edges, is perfectly matchable, and again has maximum degree eight or nine. Delete any maximum-degree vertex from this child. The next graph has seventeen vertices and between fifty-seven and fifty-nine edges. It is factor-critical and nonregular, since none of those three edge counts is divisible by seventeen. Its maximum degree is at least seven, so a third maximum-degree deletion is permitted and covered.

For $k = 12$, G has twenty vertices and seventy-five edges, is perfectly matchable, and has maximum degree eight or nine. After any maximum-degree deletion its nineteen-vertex child has sixty-six or sixty-seven edges. It is factor-critical and nonregular because neither edge count is divisible by nineteen. Its maximum degree is at least seven. A second maximum-degree deletion is therefore permitted and covered.

For $k = 13$, G has twenty-one vertices and seventy-five edges. It is factor-critical and nonregular because twenty-one does not divide seventy-five. Its maximum degree is at least eight. One maximum-degree deletion is permitted and covered.

All deletion parents just described have degree at least three. At an odd parent of order $2r + 1$ its matching number and its even child's matching number both equal r, so this deletion preserves matching number. At an even parent of order 2r its matching number is r and its odd child's is r-1, so this deletion lowers it by one. For $k = 11$, 12, 13 the respective counts of matching-preserving steps are two, one, one. They are exactly the three slack upper bounds in the remaining rows of (7). Apply the covered deletion inequality of Lemma 61 at each step and sum it. Nonnegative final slack then forces $ps(K) = 0$.

The final graph K has

$|V(K)| = 2k - 6$, $\nu(K) = k - 3$, $\alpha(K) \le \alpha(G) \le a$.

Each removed vertex had degree at most nine in its current graph, so

$e(K) \ge 75 - 9q$.

On the other hand, (2) and the perfect matching of K give

$e(K) \le B_9(K) = 3(k - 3) + 6\alpha(K) \le 3(k - 3) + 6a$.

By (8), both extreme expressions equal 9k-51. Equality must hold throughout. In particular $\alpha(K) = a$, $e(K) = B_9(K)$, and the q deletion degrees sum to 9q. Since each is at most nine, every one equals nine. The prefix conclusion (3), applied with this same G and K, yields

$7 = q + a = q + \alpha(K) \le \alpha(G) \le a \le 6$,

a contradiction.

Every row of (7) has now been excluded. This contradicts $gap(H) > 0$ and proves $gap(H) \le 0$. Finally, the stated positive-gap implication on a proper induced graph forces its gap to be at most one, whether its gap is positive or nonpositive. It therefore implies precisely the hereditary gap hypothesis used above, proving the last assertion. □

Remark. The exact six-profile reduction and robust induced matchings leave three graphs with seventy-five edges. At most three covered deletions exhaust their slack. Equality in the zero-slack density bound would then extend an independent set beyond the ambient maximum.

Depends on. Lemma 61, Lemma 72, Lemma 146, Lemma 149, Lemma 224, Lemma 243.

Lemma 252 (Absorption of an uncovered odd-degree remainder with small slack). *Use the finite integer certificate costs c_i, matching number ν, independence number α, maximum degree Δ, excess $ex = c_0 - \nu$, slack ps, $gap = ex - ps$, and top missing-allocation run tr of Lemma 23. Put $h(j) = \max(4, \lfloor j/2 \rfloor + 1)$ for nonnegative integers j. A graph satisfies T_- when positive gap implies gap one and $tr + ex \le h(\Delta)$.*

Let H be any finite simple graph, W any vertex subset, $w = |W|$, and $R = H - W$ its induced remainder. Suppose $D \ge 5$ is an odd integer, $\Delta(R) = D$, $1 \le s = ps(R) \le 4$, $w \ge h(D) - 1$, and $\alpha(R) \le w + 1$. Suppose every induced subgraph of R, including R, satisfies T_-. If no matching of R covers all its degree-D vertices, then $c_0(H) \le w + \nu(R) + s - 1$. More explicitly, for every pair of nonnegative integers x, y with $x + y = w + \nu(R) + s - 1$, all edges of H admit a partition into two parts whose matching numbers are at most x, y.

Consequently, under the same hypotheses except noncoverage, $c_0(H) \ge w + \nu(R) + s$ implies that a matching of R covers every degree-D vertex and $c_D(R) = 0$. No independence of W, order bound, regularity, minimality, or sign condition on $gap(R)$ is assumed. All degrees, orders, cardinalities, matching numbers, costs and parameters D, w, s are integers in the displayed domains.

Proof. Use nonnegative slack and the covering-matching tail theorem of Lemma 61, row restriction and component slack superadditivity from Lemma 72, and integer allocation composition from Lemma 75.

Choose an uncovered connected component U of R of actual degree D. Such a component exists, because component covering matchings could otherwise be united. Put $P = R - V(U)$ and $p = ps(U)$. Then $0 \le p \le s \le 4$ and $ps(P) \le s - p$.

Let A be the degree-D vertices of U and let $\kappa \ge 1$ be their minimum matching exposure. Choose X of maximum cardinality among the maximizers of $q_A(U - X) - |X|$, where q_A counts odd components wholly contained in A. Lemma 107 gives the exact exposure, factor-critical required components, and a maximum matching attaining it. The set X is nonempty: an empty one would make the connected U an odd-order D-regular graph, contradicting degree-sum parity. Put $b = |X| \ge 1$.

Write calQ for all components of U-X, calK for the required odd components wholly in A, and Y for the union of the other components' vertices. There are $b + \kappa$ required components. For K in calK let $d(K)$ be the number of edges to X. These are positive odd integers; they are also the total internal degree-D defects. Their sum is at most Db by degree summation at X, so some $d(K) < D$. Choose Z with largest matching number m among these small-boundary components, and put $d = d(Z)$, $u = (D - d)/2$, and $\psi = ps(Z)$. No rank tie-break is imposed. All positive-credit components used below will have matching number at most m.

For each small-boundary K different from Z, apply Lemma 179 with $\cap D$ and defect $d(K)$. Factor-criticality comes from the marked barrier, and $gap(K) \le 1$ follows from hereditary T_-. It supplies credits $a_i(K)$ of total max(D-d(K)-ps(K),0), supported on $1 \le i \le T = (D-1)/2$, with $a_1(K) \le T$, $a_i(K) \le 1$ for $i \ge 2$, and $a_i(K) \le \max(c_i(K) - \nu(K), 0)$. Give zero credits to Z, to nonrequired components, and to required components of boundary at least D. In the last case the required total is zero.

Run the procedures in Lemma 104. Their initial full interval at index i has length at least $b + c_i(Z)$. Let E_{rem} be the sum of maximum credits at unfinished indices. Initially $E_{rem} \le 2T - 1 = D - 2$. Define

J=e(U[X])+e_U(X, Y)+sum_(Q not in calK)ps(Q)+sum_(K in calK, K!=Z)max(ps(K)+d(K)-D,0).

Every term is nonnegative. Here $e_U(X, Y)$ counts edges between the two disjoint sets. Substituting $gap = ex - ps$ in that fact's retained-top inequality gives

$\psi + d + J + \sum_{Q \in calQ} c_D(Q) \le \beta + E_{rem}$, (1)

where $\beta = p + c_D(U) - D(\kappa - 1) \le p + D - (D-1)\kappa \le p + 1$. This computation uses $c_D(U) \le \kappa$ from the same fact.

Applied to Z, Lemma 179 gives $\psi \ge 1$, $|V(Z)| = 2m + 1$, $e(Z) = Dm + u$, zero rows at and above D, and

$\psi = (D-1)m + u - \sum_{i=1}^{D-1} c_i(Z)$,

$c_i(Z) + c_{D-i}(Z) \le 2m + 1$ for $1 \le i < D$. (2)

If the first-index procedure fails, a remaining positive-credit component Q has $\nu(Q) \ge b + c_1(Z) + 2$. Its rank is at most m, so $c_1(Z) \le m - b - 2$. Monotonicity of the costs then gives $\psi \ge u + (D-1)(b+2)$, whereas (1) gives $\psi \le p + D - 1 - d$. The lower bound exceeds the upper by $u + d + (D-1)(b+1) - p > 0$, since $D \ge 5$, $b \ge 1$, $p \le 4$. Thus index one finishes.

If a later index first fails at j, then $2 \le j \le T$ and $E_{rem} \le T - j + 1$. Failure again gives $c_j(Z) \le m - b - 2$. Pair indices i, D-i for $1 \le i < j$ in (2) and bound all remaining $D - 2j + 1$ entries by $c_j(Z)$. This gives $\psi \ge u - j + 1 + (D - 2j + 1)(b + 2)$. The upper bound from (1) is $\beta + T - j + 1 - d$. The difference is

$(d + 1)/2 - \beta + (D - 2j + 1)(b + 2) \ge 1 - (p + 1) + 6 = 6 - p > 0$.

Here $D - 2j + 1 \ge 2$. Hence every procedure finishes.

With $E_{rem} = 0$, (1) implies $\psi + d + J \le p + D - (D - 1)\kappa$. If $\kappa \ge 2$, the right side is at most $p - D + 2 \le 1$, whereas $\psi + d \ge 2$. Thus $\kappa = 1$, and

$\psi + d + J \le p + 1$. (3)

In particular $p \ge 1$, and calK has exactly $b + 1$ members.

There are at least two noncomplete members of calK. Indeed a complete member of odd order l has $l \le D$, and boundary $l(D + 1 - l) \ge D$, since $l(D + 1 - l) - D = (l - 1)(D - l) \ge 0$. If b of the $b + 1$ members were complete, their boundaries would total at least Db and the remaining positive boundary would exceed the capacity Db. Also Z is noncomplete because its boundary is below D. Choose a maximum independent set S of Z of size a. From one other noncomplete required component choose two nonadjacent vertices, and from each of the other b-1 required components choose one vertex. There are no edges between distinct components of U-X. Their union with S is therefore independent in R, giving

$a + b + 1 \le \alpha(R) \le w + 1$, or $a \le w - b$. (4)

We derive a rank bound for the same component Z, allowing its odd defect d. Put $F = Z - S$, $E = ex(Z)$, and $t = \sum_{v \in S}(D - deg_Z(v))$, so $0 \le t \le d$. Maximality of S gives $\Delta(F) \le D - 1$, and independence gives $e(F) = e(Z) - Da + t$. Choose one vertex of S and a perfect matching after deleting it from Z. Removing the other a-1 vertices destroys at most a-1 matching edges, so $\eta = \nu(F) - (m + 1 - a) \ge 0$. Set $d_i = c_i(Z) - c_i(F)$ for $1 \le i < D$ and $\rho = \sum_{i=1}^{D-1}(a - d_i) \ge 0$; these inequalities follow from Lemma 72. The zero tails and slack definitions give exactly

$ps(F) = \psi + t - 1 - \eta - \rho$. (5)

Let $q = c_{D-1}(F)$. If $\Delta(F) < D - 1$, the empty matching covers its empty degree-(D-1) set, and Lemma 61 gives $q = 0 \le ps(F)$. Otherwise every proper induced subgraph of F inherits $gap \le 1$ from R, and Lemma 243 applied at actual degree $D - 1 \ge 4$ gives $q \le ps(F)$.

The middle indices 2,..., D-2 form $(D - 3)/2$ disjoint pairs in (2). Subtracting their upper bounds from the full positive-row sum gives $c_1(Z) + c_{D-1}(Z) \ge 2m + (3 - d)/2 - \psi$. Since $c_1(Z) \le c_0(Z) = m + E$ and $c_{D-1}(Z) \le a + q$, it follows that $m - a \le \psi + E + q + (d - 3)/2$. Using $q \le ps(F)$ and (5) yields

$m - a \le 2\psi + E + t + (d - 5)/2 - \eta - \rho \le 3\psi + (3d - 3)/2$. (6)

For the last inequality, $t \le d$ and $E \le \psi + 1$ hold, the latter because $gap(Z) \le 1$.

Put $\delta = \psi + d - 2$, a nonnegative integer. At total $l = m + \delta$ on Z, every missing allocation lies in $[2\delta + 1, m - \delta - 1]$. To prove this, a coordinate at least m is feasible by putting every edge in that color. If a coordinate $x < m$ satisfies $x \le 2\delta$, choose any x vertices, color all edges incident with them in that color, and give the remaining edges the other color. The first matching number is at most x. The remaining induced graph has $2m + 1 - x$ vertices, at most $2(l - x) + 1$, so its matching number is at most l-x. Applying this to either coordinate proves the interval claim. Both endpoint allocations are feasible.

Consequently the maximum missing run at this total is at most $\max(0, m - 3\delta - 1)$. By (6) and (4),

$m - 3\delta - 1 \le a + (7 - 3d)/2 \le a + 2 \le w + b$. (7)

The last inequality uses $b \ge 1$.

We now construct the allocation on all of H. For any induced piece B of R, the actual interval of length at least w absorbs B at total $\nu(B) + ps(B)$. If $gap(B) \le 0$, this total is at least $c_0(B)$, so its allocation set is full. Otherwise T_- gives $gap(B) = 1$, $ex(B) = ps(B) + 1$ and $tr(B) \le h(\Delta(B)) - ex(B) \le h(D) - 1 \le w$. Then $\nu(B) + ps(B) = c_0(B) - 1$, and its missing run is at most w. Both endpoints occur because the total is at least $\nu(B)$. In either case Lemma 75 gives the claimed interval absorption.

Start with the actual vertex pool W union X, of size $w + b$. Absorb P at total $\nu(P) + ps(P)$. Then absorb each nonrequired component and each required component K with $d(K) \ge D$ at total $\nu(K) + ps(K)$. Let Bcal

denote that component family. Its total slack is at most J: nonrequired slacks occur directly in J, and for a required large-boundary component the corresponding term $ps(K) + d(K) - D$ is at least $ps(K)$. The current interval remains full and has length at least $w + b$.

By (7), absorb Z at total $m + \delta$. The interval now has length at least m. Every remaining component is required, has boundary below D, and has matching number at most m by the choice of Z. Its matching-total allocation set has both endpoints and missing run at most max(0,nu(K)-1), by Lemma 75. Absorb each remaining component at that matching total.

Matching additivity at the marked barrier, from Lemma 104, gives $\nu(U) = b + \sum_{Q \in calQ} \nu(Q)$, and $\nu(R) = \nu(U) + \nu(P)$. The final full interval on H therefore has total

$w + \nu(R) + ps(P) + \sum_{B \in Bcal} ps(B) + \delta$.

Its excess over $w + \nu(R)$ is at most $(s - p) + J + \psi + d - 2 \le s - 1$ by (3). The vertex-pool construction in Lemma 75 assigns every edge meeting W union X, including edges internal to that pool, and the remaining pieces are exactly the components just absorbed. Thus no original edge is omitted. The threshold and padding property in Lemma 23 supplies every allocation at the possibly larger total $w + \nu(R) + s - 1$.

This proves the asserted $c_0(H)$ bound. Its stated covering consequence follows by contradiction, and Lemma 61 gives $c_D(R) = 0$ once a covering matching exists. □

Remark. A small-slack marked barrier leaves a small-boundary component and a limited budget for all other pieces. Its degree defect controls a cut by an independent set, making its missing allocation interval short enough for the actual vertex pool. Spending the retained correction on the other pieces saves one unit on the original graph.

Depends on. Lemma 23, Lemma 61, Lemma 72, Lemma 75, Lemma 104, Lemma 107, Lemma 179, Lemma 243.

Lemma 253 (Density of even regular components at slack one). *Use the integer certificate costs c_i, matching number ν, independence number α, excess $ex = c_0 - \nu$, slack $ps = e - \nu - \sum_{i \ge 1} c_i$, and signed $gap = ex - ps$ of Lemma 123. All graphs are finite and simple. A nonempty graph is factor-critical if deletion of every vertex leaves a perfect matching. Define*

$E_{11}(J) = e(J) + 8|V(J)| - 19\nu(J) - 8\alpha(J)$.

Let d belong to {4,6,8,10}, and let H be a nonempty connected d-regular factor-critical graph satisfying $ps(H) = 1$ and $gap(H) \le 1$. Then d belongs to {4,6}, and $E_{11}(H) \le 10$. More precisely, if $d = 4$ then $E_{11}(H) \le 5$; if $d = 6$ then $\alpha(H) \ge 2$ and $E_{11}(H) \le 10$.

No hereditary condition, independence upper bound, or order bound is imposed. Graph orders, degrees, costs, matching and independence cardinalities, and d are integers in the stated domains; E_{11} and gap may be signed integers.

Proof. Put $n = |V(H)| = 2k + 1$, $a = \alpha(H)$, and $E = ex(H)$. Factor-criticality gives $\nu(H) = k$, and regularity gives $e(H) = dk + d/2$. The gap identity and the assumptions imply $E = 1 + gap(H) \le 2$.

The degree-eight possibility contradicts Lemma 234, which gives $ps(H) \ge 2$ under $gap(H) \le 1$. The degree-ten possibility contradicts Lemma 236 with the same hypothesis. Therefore $d = 4$ or $d = 6$.

Choose an economical sequence from Lemma 123 and apply its inequality at the integer cutoff $c_0(H)$. This cutoff is permitted because that fact gives a sequence of length at least $c_0(H)$. The nonnegative counting terms and its residual-potential bound yield

$1 = ps(H) \ge 2(n - c_0(H)) - 4a$

$= 2(k + 1 - E - 2a)$.

The quantity $k + 1 - E - 2a$ is an integer, so it is at most zero. Consequently

$k \le 2a + E - 1 \le 2a + 1$. (1)

Substitution in the definition of E_{11} gives

$E_{11}(H) = (d - 3)k - 8a + d/2 + 8$.

For $d = 4$, (1) implies $E_{11}(H) \le 11 - 6a \le 5$, since $a \ge 1$.

Suppose $d = 6$. We first exclude $a = 1$. Independence number one means that every pair of distinct vertices is adjacent; six-regularity then makes $H = K_7$ and $k = 3$. Every nonnegative integer matching-budget pair of total four is feasible for the cost $c_0(K_7)$. To verify this, call the larger budget x and the smaller y, so $x \ge 2$ and $x + y = 4$. Choose $\min(7, 2x + 1)$ vertices. Put all edges internal to this chosen set in the part with budget x and all remaining edges in the part with budget y; the two degree-bounded parts are empty. The first part has matching number at most x by its vertex count. Every edge in the second part meets the unchosen set, of size $\max(0, 6 - 2x)$. This number is at most $y = 4 - x$ for each possible $x = 2, 3, 4$. Thus its matching number is at most y. Relabel the parts according to the original budget order. This proves $c_0(K_7) \le 4$ and $E \le 1$.

But the earlier integer inequality $k + 1 - E - 2a \le 0$, with $k = 3$ and $a = 1$, would give $E \ge 2$. This contradiction proves $a \ge 2$. Finally, for $d = 6$, (1) gives

$E_{11}(H) \le 3(2a + 1) - 8a + 11 = 14 - 2a \le 10$.

All alternatives have been covered. □

Remark. The two largest possible even degrees require at least two slack units. At degrees four and six, the economical cutoff controls matching rank closely enough to bound density.

Depends on. Lemma 123, Lemma 234, Lemma 236.

Lemma 254 (Degree-nine density and exact rows at slack one). *Use the finite integer certificate costs c_i, matching number ν, independence number α, edge count e, maximum degree Δ, excess $ex = c_0 - \nu$, nonnegative slack $ps = e - \nu - \sum_{i \ge 1} c_i$, and signed $gap = ex - ps$ of Lemma 61. All graphs are finite and simple, and every vertex deletion is induced. For a graph J put*

$B_9(J) = -6|V(J)| + 15\nu(J) + 6\alpha(J)$,

$E_9(J) = e(J) - B_9(J)$, *and* $f(J) = |V(J)| - 2\nu(J)$.

A graph of positive maximum degree is covered if a matching covers every vertex of that degree.

Let H satisfy $\Delta(H) \le 9$, $ps(H) = 1$, and $gap(J) \le 1$ for every induced subgraph J of H, including H itself and the empty graph. Then $E_9(H) \le 15$.

Suppose $E = E_9(H) \ge 10$. Put $a = \alpha(H)$, $k = \nu(H)$, and $f = f(H)$. Start at $H_0 = H$, repeatedly delete an arbitrary vertex v_i of current maximum degree $d_i = \Delta(H_{i-1})$ whenever that degree is at least three, and stop at $H_T = R$ with $\Delta(R) \le 2$. For every such process, every deletion parent is covered. Exactly one deletion preserves matching number; every other deletion lowers it by one. Independence number stays equal to a at every step, and R is a disjoint union of triangles and isolated vertices. Moreover

$T = a + 2 - f$, $\sum_{i=1}^{T} (9 - d_i) = 15 - E$,

$c_1(H) = k + 1$,

$c_8(H) = |\{i : 1 \le i \le T, d_i = 9\}| \ge a - f + E - 13$,

$E \le f + 13$.

The vertices deleted at degree nine have original H-degree nine and are pairwise nonadjacent. At each step every row with integer index $1 \le j < d_i$ drops by exactly one, and every row with $j \ge d_i$ vanishes on both parent and child.

No bound on order, independence number or matching deficiency is imposed. Every graph order, degree, cost, matching or independence cardinality, index, count and the parameters T, a, k, f are nonnegative integers in the displayed domains. The error E_9 and gap may be signed integers. Empty induced subgraphs, isolated vertices and empty sums are permitted.

Proof. Every induced subgraph of an induced subgraph of H is again induced in H. Thus the entire hereditary gap hypothesis passes to each graph considered below. Use Lemma 61 for nonnegative slack and its covered maximum-degree deletion inequality.

First let J be any induced subgraph of H with $ps(J) = 0$. If its actual maximum degree is at least three, Lemma 243 gives a matching covering every vertex of that degree. For degrees at least four, its required

proper-induced gap hypothesis follows from heredity; the cubic clause requires no gap hypothesis. Choose any current maximum-degree vertex v and put $h = \nu(J) - \nu(J - v)$, which is zero or one. The covered deletion inequality gives

$0 = ps(J) \geq ps(J - v) + 1 - h \geq 0.$

Consequently $ps(J - v) = 0$ and $h = 1$. Repeating this argument supplies a complete covered process down to maximum degree at most two. It terminates because each step decreases the number of vertices. Apply Lemma 82 with degree $\cap D = 9$ and initial slack zero to this process. Its density bound yields

$ps(J) = 0$ implies $E_9(J) \leq 0$. (1)

The same argument proves that every zero-slack induced subgraph of any component considered below is covered at its maximum degree whenever that degree is at least three.

Next let J be any induced subgraph of H with $ps(J) = 1$, actual maximum degree $d \geq 3$, and no covering matching. If $d = 3$, Lemma 243 gives $2\kappa \leq ps(J) = 1$, where κ is its positive integer minimum exposure, a contradiction. If $d = 5$, Lemma 222 gives coverage. If d is seven or nine, Lemma 240 gives coverage. In each of the latter applications every proper induced subgraph of J has gap at most one. These exclude all odd possibilities, so d belongs to {4,6,8}.

The even-degree consequence of Lemma 243 now gives exactly one uncovered whole connected component U of J. It is d-regular, factor-critical, and has $ps(U) = 1$. Factor-critical means that deleting any vertex leaves a perfect matching. Let W be the induced union of the remaining whole components, allowing W to be empty. Component slack superadditivity from Lemma 72, applied to the two pieces U and W, and nonnegative slack give

$1 = ps(J) \geq ps(U) + ps(W) = 1 + ps(W),$

so $ps(W) = 0$. Hence (1) gives $E_9(W) \leq 0$. The induced component U also satisfies $gap(U) \leq 1$.

Degree $d = 8$ is impossible by Lemma 234: its own-gap implication gives $ps(U) \geq 2$ for this nonempty connected eight-regular factor-critical U.

Suppose $d = 6$. The zero-slack coverage already established for all induced subgraphs of H supplies the entire covering premise of Lemma 197 on U. Its other hypotheses are $ps(U) = 1$, $gap(U) \leq 1$, connectedness, six-regularity and factor-criticality, all just proved. Put $m = \nu(U)$ and $b = \alpha(U)$. The matching-rank conclusion gives $m \leq 2b$. Factor-criticality and regularity give $|V(U)| = 2m + 1$ and $e(U) = 6m + 3$. Therefore

$E_9(U) = 6m + 3 + 6(2m + 1) - 15m - 6b$

$= 3m + 9 - 6b \leq 9.$ (2)

Suppose $d = 4$, and again put $m = \nu(U)$ and $b = \alpha(U)$. A maximum independent set is maximal, so every vertex outside it has a neighbor in it. The closed neighborhoods of its b vertices cover U, and each has five vertices by four-regularity. Thus $2m + 1 = |V(U)| \leq 5b$. Since U is nonempty, $b \geq 1$. Using $e(U) = 4m + 2$ gives

$E_9(U) = 4m + 2 + 6(2m + 1) - 15m - 6b$

$= m + 8 - 6b \leq (15 - 7b)/2 \leq 4.$ (3)

Orders and edge counts add over whole components. Matching and independence numbers also add: restriction gives one inequality and the union of optimal component objects gives the reverse. Thus $E_9(J) = E_9(U) + E_9(W)$. Equations (1)–(3) prove, for every induced J in the specified slack-one uncovered domain,

$ps(J) = 1, \Delta(J) \geq 3, J$ uncovered implies $E_9(J) \leq 9.$ (4)

We have now verified both local conditions in the last assertion of Lemma 82 with $D = 9$: every zero-slack induced graph of maximum degree at least three is covered, and every uncovered slack-one induced graph in that degree range has density error at most nine. The condition includes H itself, since our hereditary hypothesis did. Its criterion gives $E_9(H) \leq 2 \cdot 9 - 3 = 15$.

If $E_9(H) \geq 10$, the same criterion shows that every arbitrary maximum-degree deletion process to degree at most two is complete and covered. Apply all the slack-one conclusions of that fact to each such process, with $D = 9$, $a = \alpha(H)$, $k = \nu(H)$ and $f = |V(H)| - 2\nu(H)$. They give exactly one matching-preserving deletion, preservation of independence at every step, a triangle-and-isolate terminal, and

$T = a + 2 - f$, $\sum_i (9 - d_i) = 15 - E_9(H)$.

They also give $c_1(H) = k + 1$ and

$c_8(H) = |\{i : d_i = 9\}| \geq a - f + E_9(H) - 2 \cdot 9 + 5 = a - f + E_9(H) - 13$,

together with $E_9(H) \leq f + 2 \cdot 9 - 5 = f + 13$. The same fact supplies original degree nine and pairwise nonadjacency of those deleted vertices, every stated row drop, and the zero rows on each parent and child. Its matching count says all other deletions lower matching number by one. These conclusions hold for every process separately; no compatible matching choices across distinct processes are assumed. All hypotheses used to obtain these processes have been discharged above. □

Remark. At slack one, an uncovered graph can have only one responsible even-regular component. The octic component is impossible, the sextic rank bound controls its density, and a dominating independent set controls the quartic component. The general deletion identity then forces every high-error process to have the same exact losses.

Depends on. Lemma 61, Lemma 72, Lemma 82, Lemma 197, Lemma 222, Lemma 234, Lemma 240, Lemma 243.

Lemma 255 (Covered-prefix accounting and the first uncovered degree). *Use the finite integer certificate costs $c_i(G)$, matching number $\nu(G)$, edge count $e(G)$, maximum degree $\Delta(G)$, slack $ps(G)$, and $gap(G) = c_0(G) - \nu(G) - ps(G)$ of Lemma 23. All graphs are finite and simple. All vertex deletions and restrictions are induced. Call a graph of positive maximum degree covered if one matching covers every vertex of that maximum degree.*

Let $d \geq 4$ be an integer, let F be d-regular of order n, and let S be a nonempty inclusionwise maximal independent set of cardinality a. Put $H_0 = F - S$, $p_0 = ps(H_0)$, and $g_0 = gap(H_0)$. Suppose

$c_0(F) = a + \nu(H_0) + p_0$, $g_0 \geq 0$,

and $gap(J) \leq 1$ for every induced subgraph J of H_0, including H_0 and the empty graph. Neither maximum cardinality of S nor source minimality is assumed.

Fix any integer $\ell \geq 0$ and a sequence $H_j = H_{j-1} - v_j$ for $1 \leq j \leq \ell$ such that each parent H_{j-1} is covered and v_j has its current maximum degree $D_j \geq 3$. For $0 \leq i \leq \ell$ put $W_i = S$ union $\{v_1, \ldots, v_i\}$, $w_i = a + i$, and $p_i = ps(H_i)$. For $1 \leq j \leq \ell$ and $0 \leq \theta < D_j$ put

$A_j(\theta) = c_\theta(H_{j-1}) - c_\theta(H_j)$,

$\epsilon_j = 1 - A_j(0)$, $z_j = \sum_{\theta=1}^{D_j - 1} (1 - A_j(\theta))$.

Put $Z_i = \sum_{j=1}^{i} z_j$, and let M_i be the number of j in {1,...,i} for which $\nu(H_{j-1}) = \nu(H_j)$. Empty sums and M_0 are zero. Then every $A_j(\theta)$ and ϵ_j belongs to {0,1}, and

$gap(H_i) = g_0 + \sum_{j=1}^{i} \epsilon_j + Z_i$ *in {0,1}*,

$p_i = p_0 - M_i - Z_i$,

$c_0(F) = w_i + \nu(H_i) + p_i + Z_i$.

Thus at most $1 - g_0$ pairs (j, theta), with $1 \leq j \leq \ell$ and $0 \leq \theta < D_j$, have $A_j(\theta) = 0$. Every row of index at least D_j is zero on both graphs at step j. In particular, if $g_0 = 1$, every displayed lower-row loss equals one.

The actual deleted vertices satisfy

$|N_F(v_j)$ *intersect* $W_{j-1}| = d - D_j$,

$e(F[W_i]) = \sum_{j=1}^{i} (d - D_j) \geq i$,

$e(F) - e(H_i) = da + \sum_{j=1}^{i} D_j$.

For disjoint vertex sets X, Y let $e_F(X, Y)$ count their joining edges, and for Q contained in $V(J)$ let $boundary_J(Q)$ count edges of J with exactly one endpoint in Q. If $Q \subseteq V(H_i)$ and every vertex of Q has H_i − degree r, then, for every $0 \leq j \leq i$,

$boundary_F(Q) = boundary_{H_i}(Q) + (d-r)|Q|$,

$e_F(S,Q) + \sum_{h=1}^{i} e_F(\{v_h\},Q) = (d-r)|Q|$,

$boundary_{H_j}(Q) = boundary_{H_i}(Q) + \sum_{h=j+1}^{i} e_F(\{v_h\},Q)$.

If the endpoint $R = H_\ell$ is uncovered and has maximum degree $r \geq 3$, put $w = w_\ell$ and $b = e(F[W_\ell])$. Then $ps(R) \geq 1$, and R has a whole connected component C with $\nu(C) \geq w+2$. Consequently

$3w \leq n-4$,

$(2r-d)w \geq 4(d-r) + 2b > 0$,

$d/2 < r \leq d-1$.

If $\ell \geq 1$, then $r < D_\ell$. Thus a first uncovered degree can only be 3 when $d = 4$, 4 or 5 when $d = 6$, 5, 6, 7 when $d = 8$, 6, 7, 8, 9 when $d = 10$, and 7, 8, 9, 10, 11 when $d = 12$.

Every greedy process which repeatedly deletes an arbitrary current maximum-degree vertex while the maximum degree is at least three terminates at degree at most two. If a covered prefix as above has reached $w_\ell > floor((n-4)/3)$, or $\Delta(H_\ell) \leq d/2$, or $p_\ell = 0$, then every such continuation from H_ℓ is covered at every deletion parent. Empty prefixes and empty terminal graphs are allowed. All orders, degrees, costs, row indices, matching cardinalities, process lengths and displayed counts are integers in their stated nonnegative or positive domains; g_0 belongs to {0,1}.

Proof. The certificate quantities and finite thresholds are those of Lemma 23. Use Lemma 72 for restriction and the one-vertex extension bound, Lemma 61 for nonnegative slack, the strong cost tail and the local deletion identity, and Lemma 243 for the hereditary maximum-degree exposure bound.

Maximality of S means that every vertex outside S has an F-neighbor in S. Hence $\Delta(H_0) \leq d-1$. Degrees cannot increase under induced deletion, so every D_j and every later maximum degree are at most d-1. The hereditary hypothesis applies to H_i and every one of its induced subgraphs, because each is induced in H_0. It also gives $g_0 \leq 1$; with the nonnegative integer hypothesis, g_0 is zero or one.

Fix a step j. The one-vertex bounds give $A_j(\theta)$ in {0,1}. Coverage and Lemma 61 give $c_{D_j}(H_{j-1}) = 0$; restriction gives the same row zero on H_j. Every higher row is zero by the strong tail. Exactly D_j edges are deleted, so subtracting the finite sums defining the two gaps gives

$gap(H_j) - gap(H_{j-1})$

$= D_j - \sum_{\theta=0}^{D_j-1} A_j(\theta)$

$= \epsilon_j + z_j$.

All summands in the last expression are nonnegative integers. Summing over a prefix and using $gap(H_i) \leq 1$ proves the asserted gap formula and the bound of $1 - g_0$ on the number of failed unit losses. This also proves the $g_0 = 1$ consequence.

Put $\mu_j = \nu(H_{j-1}) - \nu(H_j)$. It is zero or one, since restricting a maximum matching loses at most the edge at v_j, and every matching of H_j is a matching of its parent. The local slack identity in Lemma 61 becomes

$p_{j-1} - p_j$

$= D_j - \mu_j - \sum_{\theta=1}^{D_j-1} A_j(\theta)$

$= 1 - \mu_j + z_j$.

Summing gives $p_i = p_0 - M_i - Z_i$. In particular the slack cannot increase during a covered deletion. Adding the matching-number loss to the slack loss shows that $\nu(H_{j-1}) + p_{j-1}$ exceeds $\nu(H_j) + p_j$ by exactly $1 + z_j$. Since $w_j - w_{j-1} = 1$ and $Z_j - Z_{j-1} = z_j$, the quantity

$w_i + \nu(H_i) + p_i + Z_i$

is independent of i. At $i = 0$ it is $c_0(F)$ by hypothesis. This proves the complete cost identity without assuming that any complete process is covered or that the initial gap is zero.

At deletion time, precisely D_j neighbors of v_j remain; its other $d - D_j$ neighbors lie in W_{j-1}. This proves the first neighbor identity. Initially S has no internal edges. When a root is adjoined to the deleted set, exactly its edges to the earlier deleted vertices become new internal edges. Each internal edge is counted at its later inserted endpoint, giving $e(F[W_i]) = \sum_{j \le i}(d - D_j)$. Since $D_j \le d - 1$, every summand is at least one. Deleting S removes da edges because F is d-regular and S is independent. Each subsequent root deletion removes its D_j current incident edges; an edge between two roots is removed exactly at its earlier deleted endpoint. This proves the asserted edge-loss formula.

For the boundary assertions, inducedness gives $F[Q] = H_i[Q]$. Degree summation over Q therefore yields

$boundary_F(Q) = d|Q| - 2e(F[Q])$,

$boundary_{H_i}(Q) = r|Q| - 2e(F[Q])$.

Their difference is $(d - r)|Q|$. All edges in that difference join Q to W_i, whose disjoint parts are S and the individual deleted roots. This proves the second boundary identity as well. Restoring the roots v_{j+1},..., v_i restores to the boundary of Q exactly the edges from these vertices to Q. Edges between restored vertices do not meet Q. This proves the third identity, including empty Q and empty root ranges.

Now suppose $R = H_\ell$ is uncovered and $r = \Delta(R) \ge 3$. Let κ be the minimum number of its degree-r vertices missed by a matching. The finite matching family attains this minimum, and uncoveredness means $\kappa \ge 1$. If $r \ge 4$, every proper induced subgraph of R has gap at most one by the hereditary hypothesis; if $r = 3$ no such hypothesis is needed. In either case Lemma 243 gives $ps(R) \ge \kappa \ge 1$. The established host identity and $Z_\ell \ge 0$ thus imply

$c_0(F) > w + \nu(R)$. (1)

We prove that (1) forces the claimed large component by an explicit certificate on F. Suppose every whole component C of R has $\nu(C) \le w + 1$. List their matching numbers as m_1,..., m_t; they are nonnegative integers and their sum is $\nu(R)$, since maximum matchings on distinct whole components can be united.

Every integer x in $[0, w + \sum_h m_h]$ can be written as

$x = z + \sum_{h \in B} m_h$

with $0 \le z \le w$ and B a subset of {1,...,t}. To prove this, start with the interval [0,w]. If the first q components produce every integer in [0,L], then allowing the next weight produces [0,L] union $[m_{q+1}, L + m_{q+1}]$. Since $m_{q+1} \le w + 1 \le L + 1$, these two integer intervals have no missing integer between them and cover $[0, L + m_{q+1}]$. Induction proves the assertion, including the endpoint $m_{q+1} = w + 1$, zero weights and no components.

Given any nonnegative integers $x + y = w + \nu(R)$, choose this representation and partition the actual deleted set W_ℓ into W_R, W_B of sizes z, w-z. Assign all edges of components indexed by B to a red part and all other edges internal to R to a blue part. Every other F-edge has an endpoint in W_ℓ. Assign it red if it meets W_R, and blue otherwise. Each added blue edge meets W_B. Thus each edge, including an edge with two endpoints in W_ℓ, is assigned exactly once. A red matching has at most $\sum_{h \in B} m_h$ edges internal to R and at most z other edges, since those other matching edges meet distinct W_R vertices. Its size is at most x. The blue matching bound is at most $\sum_{hnot \in B} m_h + w - z = y$. Taking the bounded-degree parts empty gives the required row-zero certificate for every allocation at this total. The threshold definition in Lemma 23 implies $c_0(F) \le w + \nu(R)$, contrary to (1). Therefore some whole component C has $\nu(C) \ge w + 2$.

Such a component has at least $2w + 4$ vertices, so $n \ge w + |V(C)| \ge 3w + 4$. Each vertex in C has internal R-degree at most r and ambient F-degree d. Because C is a whole component of the induced remainder, all its exterior F-edges end in W_ℓ. Hence its boundary has at least $(d - r)|V(C)|$ edges. The entire cut between W_ℓ and R has exactly dw-2b edges by regularity. Therefore

$(d - r)(2w + 4) \le dw - 2b$,

which is equivalent to $(2r - d)w \ge 4(d - r) + 2b$. Here $r \le d - 1$, $b \ge 0$ and $w \ge a \ge 1$, so the right side is positive and $r > d/2$. Combining these strict and weak bounds gives exactly the stated finite degree lists.

For completeness, if K is covered with maximum degree D and v has degree D, restrict one covering matching to K-v. Any vertex still of degree D in K-v was not adjacent to v; its covering edge was therefore not deleted. Thus K-v remains covered whenever its maximum degree is still D. Applied to the last parent, this proves $r < D_\ell$ whenever $\ell \ge 1$ and R is uncovered.

Finally every specified greedy continuation exists: a nonterminal finite graph has a maximum-degree vertex, and every deletion decreases its order. It therefore ends at degree at most two after finitely many steps. If a continuation from a covered prefix first becomes uncovered, all roots before that endpoint form a covered prefix from H_0, so the inequalities just proved apply there. Its deleted-pool size is at least w_ℓ. Consequently $w_\ell > floor((n-4)/3)$ is inconsistent with $3w \le n-4$ at that endpoint. Its maximum degree is at most $\Delta(H_\ell)$, so $\Delta(H_\ell) \le d/2$ is inconsistent with the strict bound $r > d/2$. If $p_\ell = 0$, the local slack inequality and nonnegative child slack show by induction that every preceding covered parent and child still have slack zero; the first uncovered endpoint would then contradict $ps(R) \ge 1$. Each of the three conditions therefore rules out every such first failure, proving the universal continuation statement without selecting a special process. □

Remark. Covered deletions consume at most the hereditary one-unit gap allowance. The exact host cost forces a large surviving component at any failure; fitting that component's actual boundary into the deleted vertices' cut forces a degree greater than half the original one.

Depends on. Lemma 23, Lemma 61, Lemma 72, Lemma 243.

Lemma 256 (Matching gain in an octic graph of excess four). *Use the finite integer certificate costs c_i, matching number ν, independence number α, maximum degree Δ, excess $ex = c_0 - \nu$, slack ps, signed $gap = ex - ps$, and longest missing-allocation run tr from Lemma 61 and Lemma 172. Put $f(J) = |V(J)| - 2\nu(J)$, $q(J) = \alpha(J) - f(J)$, and $E_7(J) = e(J) + 4|V(J)| - 11\nu(J) - 4\alpha(J)$. Write $T_-(J)$ for the implication that $gap(J) > 0$ entails $gap(J) = 1$ and $tr(J) + ex(J) \le \max(4, floor(\Delta(J)/2) + 1)$.*

Let H be a nonempty connected eight-regular finite simple factor-critical graph, where factor-critical means that deletion of any vertex leaves a perfect matching. Suppose $ex(H) = 4$, $gap(H) = 1$, and every proper induced subgraph satisfies T_-. Put $k = \nu(H)$, $a = \alpha(H)$, and assume $k \ge 14$.

For every proper induced subgraph J of H with $\Delta(J) \le 7$, assume:

(a) if $ps(J) = 1$ and $q(J) \le 5$, then $E_7(J) \le 11$;

(b) if J is connected, $\Delta(J) \ge 3$, $ps(J) \le 1$, and no matching covers all maximum-degree vertices, then J is regular of positive even degree, factor-critical, and has slack one.

Then (k, a) belongs to {(14,7),(14,8),(15,8),(16,9),(17,10)}. For every maximum independent set S of H, let $G = H - S$ and $\eta = \nu(G) - (k + 1 - a)$. One has η in {2,3}, $ps(G) = 3 - \eta$, $\Delta(G) = 7$, and a matching of G covers all degree-seven vertices. Moreover

$|V(G)| = 2k + 1 - a$, $e(G) = 8(k - a) + 4$, $\nu(G) = k + 1 - a + \eta$,

$c_7(H) = a$, and $c_i(H) = a + c_i(G)$ for every integer $1 \le i \le 7$.

If $k - a = 7$, then $\alpha(G) = a$ for every such S. Writing $o_k = 1$ for odd k and zero for even k, one has

$2k - 10 + o_k - a \le c_6(H) \le k - 1$.

For the five displayed pairs in their displayed order, $c_6(G)$ is at least 4, 2, 5, 4, 5 respectively, and is always at most k-1-a.

The order lower bound $k \ge 14$ and the two induced-subgraph interfaces are explicit premises. All graph orders, degrees, indices, costs, cardinalities, matching and independence numbers, excesses, slacks, k, a, o_k and η are integers in their displayed nonnegative domains; gaps and E_7 may be signed.

Proof. By Lemma 172, H has order $2k + 1$, edge count $8k + 4$, cost $c_0(H) = k + 4$ and slack three. For every maximum independent S with $|S| = a$, its complement G has maximum degree at most seven, gain $0 \le \eta \le 3$, slack 3-eta, and exact positive-row losses a at indices one through seven. Independence removes 8a edges. Consequently, writing $x = k - a$,

$|V(G)| = 2k + 1 - a$, $e(G) = 8x + 4$, $\nu(G) = x + 1 + \eta$,

$q(G) = \alpha(G) - a + 1 + 2\eta \le 1 + 2\eta$,

$E_7(G) = 5x - 3 - 11\eta + 4(a - \alpha(G))$. (1)

Apply the cutoff Lemma 123 at $c_0(H)$. It gives

$3 \geq 2(|V(H)| - c_0(H)) - 4a = 2k - 6 - 4a$.

Integrality implies $k \leq 2a + 4$. Since $k \geq 14$, $a \geq 5$.

The numerical theorem Lemma 241 gives

k in {14,15,16,17,18,19,20,22},

with respective bounds $c_7(H) \leq 8, 8, 9, 10, 10, 11, 12, 13$, and

$c_6(H) \geq 2k - 10 + o_k - c_7(H)$, $c_6(H) \leq k - 1$. (2)

Here o_k is one when k is odd and zero otherwise. Combining (2) with the exact row loss gives

$c_6(G) \geq 2k - 10 + o_k - c_7(H) - a$. (3)

We record the density tools on the actual induced graphs of maximum degree at most seven. The quantities q and E_7 add over whole connected components, and $q \geq 0$ because the unmatched vertices of a maximum matching are independent. By Lemma 61 and Lemma 72, slack is nonnegative and superadditive over whole components.

A zero-slack proper induced graph J is covered at every actual maximum degree at least three. Its whole components have slack zero, and (b) excludes an uncovered component at that degree. Unite their covering matchings. A maximum-degree deletion then preserves zero slack and loses one matching edge, by Lemma 61. Repeating to degree two supplies the complete covered process required by Lemma 146. At $\cap$ seven that fact gives $E_7(J) \leq 0$. Together with premise (a), this proves

$ps(J) \leq 1$ and $q(J) \leq 5$ imply $E_7(J) \leq 11$ (4)

for every such J; at zero slack the q restriction is unnecessary. Every induced deletion below remains proper in H and has maximum degree at most seven.

Choose a maximum-cardinality ordinary matching barrier X in G and put $b = |X|$. The all-marked specialization of Lemma 107 makes its odd components factor-critical and gives their number as $b + |V(G)| - 2\nu(G)$. There are no even components: adjoining a vertex of an even component to X would produce at least one odd component and would not decrease the maximizing objective, contradicting maximal cardinality. The matching formula is

$\nu(G) = b + \sum_Q (|V(Q)| - 1)/2$. (5)

If $\eta = 0$, the independent-cut theorem Lemma 140, applied with degree eight and excess four, gives $2a + 9b \leq 9$. This contradicts $a \geq 5$. Thus $\eta \geq 1$.

Suppose $\eta = 1$. Then $ps(G) = 2$ and $q(G) \leq 3$. The same theorem gives $2a + 9b \leq 23$, so b is zero or one. It also gives a component Q of G-X with order at least $2a + 2b + 5$. The total number of components is $a + b - 3$.

First take $b = 0$. Counting Q and at least one vertex in each other component gives

$2k + 1 - a \geq 2a + 5 + (a - 4) = 3a + 1$,

so $k \geq 2a$ and $a \leq \lfloor k/2 \rfloor$. Inserting this bound and the table (2) into (3), and dropping the nonnegative o_k, gives respectively

$c_6(G) \geq 3, 5, 5, 6, 7, 8, 8, 10$

at the eight allowed ranks. In particular $c_6(G) \geq 3$.

If $\Delta(G) \leq 6$, each whole component C is odd and factor-critical by (5), and $c_6(C) \leq 1$: delete a vertex, use a perfect matching to cover all remaining degree-six vertices, apply covered-degree vanishing, and restore the vertex. If $ps(C) = 0$, then $c_6(C) = 0$; this follows from zero-slack coverage when its degree is six and from zero tails when its degree is smaller. Thus $c_6(C) \leq ps(C)$ in every case, since positive slack is an integer. Component composition gives

$c_6(G) \leq \sum_C c_6(C) \leq \sum_C ps(C) \leq ps(G) = 2$,

a contradiction. Therefore $\Delta(G) = 7$.

Every component having degree-seven vertices is an odd factor-critical graph of maximum degree seven. It is nonregular, since a seven-regular graph cannot have odd order. A perfect matching after deleting a vertex of degree at most six therefore covers all its degree-seven vertices. Uniting these matchings shows that G is

covered at degree seven. Choose any degree-seven vertex v. Factor-criticality of its whole component gives $\nu(G-v)=\nu(G)$. Put $J=G-v$ and $u=\alpha(G)-\alpha(J)$ in {0,1}. Covered deletion yields $ps(J)\le 1$, while

$q(J)=q(G)+1-u\le 4$,

$E_7(J)=E_7(G)-11+4u$.

Using (4) gives $E_7(G)\le 22$. By (1), $5(k-a)-14\le 22$, so $k-a\le 7$.

Together $k\ge 2a$ and $k-a\le 7$ imply $a\le 7$. If a is five or six, $k\ge 14$ gives $k-a\ge 8$, a contradiction. If $a=7$, the inequalities force $k=14$. In that case G has four odd components, one of order at least nineteen. Each component has minimum degree at least $8-a=1$, hence positive odd order at least three. Thus G has at least $19+3+3+3=28$ vertices, whereas its actual order is $2k+1-a=22$. This excludes $b=0$.

Now take $b=1$ and write $X=\{z\}$. The inequality $2a+9\le 23$ leaves a in {5,6,7}. Every component of G-z has minimum internal degree at least 7-a and is odd.

If $a=5$, the cutoff $k\le 2a+4$ and $k\ge 14$ force $k=14$. There are three components. The large one has order at least seventeen, and the others have order at least three. Their total order is $|V(G-z)|=23$, so their orders are exactly 17, 3, 3. They have at most $\lfloor 7\cdot 17/2\rfloor+3+3=65$ edges. The vertex z has at most seven incident edges, giving $e(G)\le 72$. But (1) gives $e(G)=8(14-5)+4=76$, a contradiction.

If $a=6$, there are four components, each of odd order at least three, with one of order at least nineteen. Their total order is at least $19+3+3+3=28$, whereas $k\le 16$ gives $|V(G-z)|=2k-6\le 26$. This is impossible.

It remains that $a=7$. There are five components, one Q of order at least twenty-one. Since $k\le 18$ and their total order is 2k-7, one must have k in {16,17,18}. The four other components have total order at most eight. Each therefore has order at most five, and at least two are singletons: starting with four odd positive orders, three nonsingletons would already require total order at least ten. Each singleton is adjacent to z in G, because it has degree eight in H and has no possible neighbors outside S union {z}, a set of eight vertices.

For $k=16$, 17, 18, inequality (3) and the table give $c_6(G)\ge 6$, 8, 9 respectively. On the other hand, restoring z and composing components yields $c_6(G)\le 1+c_6(Q)$, since all the other components have order at most five and zero sixth row. If $\Delta(Q)\le 6$, factor-criticality and deletion of one vertex give $c_6(Q)\le 1$ as before. This contradicts the displayed lower bounds. Thus Q has a vertex v of internal degree seven. Its G-degree is also seven, and it is not adjacent to z because $\Delta(G)\le 7$.

There is a matching of G-v with cardinality $\nu(G)$. Match z to one singleton component, take a perfect matching of Q-v, and take a near-perfect matching in every remaining component. The factor-critical components and formula (5) verify its size. Hence $\nu(G-v)=\nu(G)$.

There is also a matching of G covering every degree-seven vertex. If z has a neighbor w in Q, use zw and a perfect matching of Q-w. This covers all of Q union {z}; every outside vertex has degree at most five in G. If z has no neighbor in Q, choose a vertex w in Q of internal degree at most six, which exists because its order is odd. Take a perfect matching of Q-w and match z to a singleton. The only uncovered vertex in Q has G-degree at most six, and every other uncovered vertex has degree at most five. This proves covering in both cases.

Delete the chosen internal degree-seven v and put $J=G-v$, $u=\alpha(G)-\alpha(J)$. Covered deletion gives $ps(J)\le 1$, and $q(J)=q(G)+1-u\le 4$. But (1) gives $E_7(G)\ge 5(k-7)-14\ge 31$, so

$E_7(J)=E_7(G)-11+4u\ge 20$,

contrary to (4). This excludes $b=1$ and therefore $\eta=1$.

Every maximum independent cut consequently has η in {2,3}. At $\eta=2$, $ps(G)=1$ and $q(G)\le 5$, so (a) and (1) give

$5(k-a)-25+4(a-\alpha(G))\le 11$.

At $\eta=3$, $ps(G)=0$, and the zero-slack density bound gives

$5(k-a)-36+4(a-\alpha(G))\le 0$.

In both cases $k-a\le 7$. If $k-a=7$, the respective inequalities become $10+4(a-\alpha(G))\le 11$ and $-1+4(a-\alpha(G))\le 0$. The nonnegative integer a-alpha(G) must be zero.

At $\eta = 3$, the zero-slack covering argument covers degree seven whenever it occurs. At $\eta = 2$, an uncovered connected component of degree seven would violate (b), since that component has slack at most one. Thus degree-seven vertices are covered in that case as well. When $\Delta(G) \le 6$ there are no degree-seven vertices and the seventh row vanishes by zero tails. Therefore in every case $c_7(G) = 0$. The exact row loss gives $c_7(H) = a$.

Now apply the last clauses of Lemma 241. Its hypotheses $c_7(H) \ge a$ and $k - a \le 7$ have been proved. They give exactly the five displayed pairs. The equality $c_7(H) = a$ gives the claimed lower and upper bounds on $c_6(H)$, and the exact row loss gives the five lower bounds and common upper bound on $c_6(G)$.

It remains to verify that the degree of G is actually seven. The just-proved lower bound is at least two in every row. If $\Delta(G) \le 5$, its sixth row is zero. If $\Delta(G) = 6$ and it is covered, the sixth row again vanishes. If it is uncovered, some whole component U is uncovered at degree six. Since $ps(G) \le 1$, premise (b) makes U six-regular, factor-critical and of slack one, and all other components have slack zero. The perfect-after-deletion matching gives $c_6(U) \le 1$; zero-slack coverage or zero tails give sixth cost zero on every other component. Composition yields $c_6(G) \le 1$. Each alternative contradicts $c_6(G) \ge 2$. Hence $\Delta(G) = 7$, and the already proved coverage is at its actual maximum degree.

All arguments began with an arbitrary maximum independent set S, and every application retained that same actual complement. Formula (1), the row identities from Lemma 172, and the conclusions above establish every asserted property for every such S. □

Remark. At matching gain one, an ordinary barrier has at most one vertex. The numerical sixth-row lower bound forces an internal degree-seven vertex whose deletion preserves matching number. Its child has at most one slack unit and too large a density error. This excludes gain one without a separate slack-two rank classification.

Depends on. Lemma 61, Lemma 72, Lemma 107, Lemma 123, Lemma 140, Lemma 146, Lemma 172, Lemma 241.

Lemma 257 (Original head bounds at exact potential). *Let $s, t \ge 1$ be integers and let $n_1 \ge \cdots \ge n_s \ge 1$ and $m_1 \ge \cdots \ge m_t \ge 1$ be integer lists. Let F_1, F_2 be their respective vertex-disjoint unions of stars, prescribed red and blue. Copies are ordinary subgraphs with vertex-disjoint target components. A finite simple graph arrows an ordered pair if every red-blue coloring of its edges contains the entire first target in red or the entire second target in blue. Let u, v be the respective numbers of one-edge components, and assume $u \ge v$. Put*

$$M = s + t - 1,$$

$$\ell_k = max\{n_i + m_j - 1 : 1 \le i \le s, 1 \le j \le t, i + j = k\}$$

for every integer $2 \le k \le M + 1$.

Fix an integer $0 \le q < M$. Let A_q consist of every ordered pair of nonempty actual suffixes obtained by deleting α red and β blue leading components, for integers $0 \le \alpha < s$, $0 \le \beta < t$ and $\alpha + \beta = q$. Define

$$K = M - q,\ P_q = \sum_{k=q+2}^{M+1} (\ell_k - 1),\ D = \ell_{q+2} - 1,$$

and suppose $D \ge 5$. Let R be the length of the initial run of entries equal to $D + 1$ in ℓ_{q+2},..., ℓ_{M+1}. For a graph H write $e(H)$, $\nu(H)$, $\Delta(H)$ for its edge count, maximum matching cardinality and maximum degree, and put $\Phi(H) = e(H) - \nu(H)$.

Suppose $\Delta(H) \le D$, H arrows every member of A_q, and $\Phi(H) = P_q$. Then

$$u > v,\ n_1 \le D - 2,\ m_1 \le D - 1.$$

Moreover D is even and at least six. Every terminal maximizing pair has both sizes even and at least four: this means any legal cuts α, β and valid local indices i, j with $i + j = R + 1$ and $n_{\alpha+i} + m_{\beta+j} = D + 2$.

The graph H has exactly R whole connected components that have odd order and are D-regular. Each admits a near-perfect matching, meaning a matching leaving exactly one vertex uncovered. Every maximum matching of the union of the other whole components covers every vertex of ambient H-degree D.

Consequently, for fixed original lists and q with the displayed domains and $u \ge v$, if $n_1 \ge D - 1$ or $m_1 \ge D$, every finite simple H with $\Delta(H) \le D$ which arrows every member of A_q satisfies $\Phi(H) \ge P_q + 1$. All indices, sizes,

multiplicities, degrees and matching cardinalities are integers. The statements allow $q = 0$, repeated component sizes, original unit components and isolated host vertices, and impose no matching-deficit or binary-target hypothesis.

Proof. The family A_q is finite and nonempty: the interval

$[\max(0, q - t + 1), \min(q, s - 1)]$

contains an integer α, and $\beta = q - \alpha$ completes legal cuts. Every member has component budget K. Its full family envelope is ℓ_{q+2},..., ℓ_{M+1}. Indeed a local candidate at indices i, j in the pair with cuts α, β comes from original indices $I = \alpha + i$, $J = \beta + j$ with $I + J = q + i + j$. Conversely, for original indices I, J with $I + J = q + h$ and $h \geq 2$, choose $\alpha = \min(q, I - 1)$, $\beta = q - \alpha$. Then $\alpha \leq I - 1$ and $\beta \leq J - 1$, so the cuts are legal and the same candidate occurs at local diagonal h. Thus both candidate sets agree on every diagonal. Applying Lemma 225 to this family, with its maximum component budget equal to K, gives $\Phi(H) \geq P_q$ without assuming equality.

Assume $\Phi(H) = P_q$. Apply Lemma 194 to the full original lists and the same q, H. Define $A_2 = s - u$, $B_2 = t - v$, $\epsilon = e(F_1) - s$, and $b = \sum_{j=1}^{B_2}(m_j - 2)$. Its conclusions give

$A_2, B_2 \geq 1$, $u > v$, $q \leq \min(s - 2, B_2 - 2)$,

$n_1 \leq D - 1$, $m_1 \leq D$,

$B_2 - q \geq R + \epsilon + 1$, $b + q + R \leq s - 1$.

Every red nonunit contributes at least one to ϵ, so $\epsilon \geq A_2$. Put $E = q + R$. Then

$B_2 \geq E + \epsilon + 1$, $E + b \leq s - 1$. (1)

The full potential entries $\ell_{k+1} - 1$, $1 \leq k \leq M$, are nonnegative and nonincreasing. To see monotonicity, a candidate on any diagonal after the first has an index greater than one; decreasing such an index yields a candidate of at least the same value on the preceding diagonal. Since the entry at $q + 1$ is D and its initial tail plateau has length R, E is the last original coordinate whose potential entry is at least D. Consequently every candidate $n_i + m_j - 2 \geq D$ satisfies $i + j - 1 \leq E$.

We identify the positive tail length and two actual endpoint projections. A potential candidate is positive precisely when at least one size is at least two. The last coordinates in these two index rectangles are $A_2 + t - 1$ and $s + B_2 - 1$. The latter exceeds the former by $u - v > 0$. Thus exactly

$N = s + B_2 - 1 - q$

entries of the potential tail are positive. Both endpoint cuts (0,q) and (q,0) are legal by the displayed bounds on q. In the projection family C_q defined in Lemma 214, the first endpoint, oriented with its blue suffix first, supplies

$((B_2 - q)K2, F_1)$.

Here K_2 is one edge and the blue suffix has $B_2 - q \geq 2$ nonunits. The second endpoint supplies

$((s - q)K2, F_2^-)$,

where F_2^- shortens each blue nonunit by one edge and omits blue units. It is nonempty because $B_2 \geq 1$. Each matching size is positive, and both projected pairs have component budget N, since F_1 has s components and F_2^- has B_2 components. Each projection retains the entire displayed opposite forest.

First suppose $m_1 = D$. Let $f \geq 1$ be the multiplicity of that blue head, and let $x \geq 0$ count the original red entries equal to D-1. Define $w_B = s + f - 1$ and, when $x > 0$, $w_R = x + B_2 - 1$. We claim that the last original potential coordinate at least D-1 is

$W = \max(w_B, w_R)$ if $x > 0$, and $W = w_B$ if $x = 0$. (2)

Take a candidate $n_i + m_j - 2 \geq D - 1$. A blue unit would require $n_i \geq D$ and is impossible. If $m_j = 2$, then $n_i = D - 1$, so $x > 0$ and $i + j - 1 \leq w_R$. If $m_j \geq 3$ and $n_i = 1$, the candidate condition and $m_1 = D$ force $j \leq f$, giving $i + j - 1 \leq w_B$. In the remaining case $m_j \geq 3$ and $n_i \geq 2$, the pair (i, f) has potential $n_i + D - 2 \geq D$ and hence $i + f - 1 \leq E$. Also $j - f \leq b$: if $j \leq f$ this follows from $b \geq 0$; if $j > f$, each entry at indices $f + 1$,..., j has size at least three and contributes at least one to b. Therefore

$i + j - 1 = (i + f - 1) + (j - f) \le E + b \le s - 1 < w_B$.

These cases give the upper bound in (2). For the reverse bound, $u > v \ge 0$ implies $n_s = 1$, and (s, f) realizes potential D-1 at w_B. When $x > 0$, (x, B_2) has sizes D-1 and at least two, so realizes a potential at least D-1 at w_R. This proves (2).

By (1), $w_B = s + f - 1 \ge E + b + f > E$. Hence $a := W - E \ge 1$. Monotonicity and integrality show that a is exactly the number of tail entries equal to D-1, and $R + a = W - q$. The blue-shortened endpoint projection has largest size D-1 with multiplicity f and matching-plus-head width

$(s - q) + f - 1 = w_B - q$.

When $x > 0$, the full-red endpoint projection has largest size D-1 with multiplicity x and width

$(B_2 - q) + x - 1 = w_R - q$.

By (2), at least one of these budget-N projections has width exactly $R + a$. All hypotheses of Lemma 214 are satisfied by the same original profiles, complete actual family, q, D, N, R, a and H. That fact gives $\Phi(H) \ge P_q + 1$, contradicting equality. Thus $m_1 \le D - 1$.

Now suppose $n_1 = D - 1$ and let $x \ge 1$ be its multiplicity. Let B_3 be the number of original blue entries of size at least three. It is positive: otherwise every original potential candidate would be at most $(D - 1) + 2 - 2 = D - 1$, contradicting the tail leading value D. The candidate at (x, B_3) has potential at least D, so $E \ge x + B_3 - 1$. Together with (1) this gives $B_2 \ge x + B_3 + \epsilon$. Set $w = x + B_2 - 1$. Then

$w - (A_2 + B_3 - 1) \ge 2x + \epsilon - A_2 \ge 2x > 0$. (3)

A candidate of potential at least D-1 cannot use a blue unit. If it uses a blue size two, it must use red size D-1, giving coordinate at most w. If it uses blue size at least three, its red size cannot be one because $m_1 \le D - 1$; thus its coordinate is at most $A_2 + B_3 - 1 < w$ by (3). Conversely (x, B_2) has potential at least D-1 and coordinate w. Therefore the last original coordinate at least D-1 is $W = w$. By (1),

$w = x + B_2 - 1 \ge x + E + \epsilon > E$.

Thus $a = W - E \ge 1$ again counts exactly the tail entries equal to D-1, and $R + a = W - q$. The actual full-red endpoint projection $((B_2 - q)K2, F_1)$ has component budget N, largest size D-1 with multiplicity x, and width $(B_2 - q) + x - 1 = W - q = R + a$. Applying Lemma 214 contradicts equality once more. Hence $n_1 \le D - 2$.

Apply Lemma 225 at surplus zero to the same full family and the envelope established in the first paragraph. Each terminal maximizing pair has either both sizes positive even integers, or D is odd and its unordered sizes are {2,D}. Neither original list now contains an entry of size D. The second alternative is therefore impossible; in the first alternative neither size can be two, since their sum is $D + 2$. Both sizes are even and at least four. Such a terminal pair exists because the finite envelope is attained at its final leading coordinate. Its sum $D + 2$ implies that D is even and at least six.

The additional hypothesis in the structural part of Lemma 225 is now satisfied. At surplus zero its count κ of whole odd-order D-regular components obeys $R \le \kappa \le R$, and each has a near-perfect matching. Every maximum matching of the union of the remaining components exposes at most $R - \kappa = 0$ vertices of ambient H-degree D. These are exactly the claimed component and matching conclusions.

Finally fix profiles satisfying $n_1 \ge D - 1$ or $m_1 \ge D$. The first paragraph gives $\Phi(H) \ge P_q$ under just the stated arrowing and degree hypotheses. Equality would contradict the two head bounds just proved. Integrality yields $\Phi(H) \ge P_q + 1$. □

Remark. At either boundary head size, every interior candidate for the next potential level lies before an endpoint matching projection. That full projection excludes equality. The remaining heads remove the binary terminal exception, leaving only even terminal sizes and whole odd regular components.

Depends on. Lemma 194, Lemma 214, Lemma 225.

Lemma 258 (The first even-degree slope has only regular overflow). *Let $s, t \geq 1$ be integers and let F_1, F_2 be the nonempty star forests with full positive nonincreasing integer edge-size lists $n_1,\ldots, n_s$ and $m_1,\ldots, m_t$, prescribed red and blue. Copies are ordinary subgraphs with vertex-disjoint target components. Put $M = s + t - 1$ and $\ell_k = max\{n_i + m_j - 1 : 1 \leq i \leq s, 1 \leq j \leq t, i + j = k\}$ for every integer $2 \leq k \leq s + t$. For every integer $0 \leq q < M$ let A_q consist of all ordered nonempty actual suffix pairs after legal leading cuts α, β with $0 \leq \alpha < s$, $0 \leq \beta < t$ and $\alpha + \beta = q$. Put $K = M - q$, $P_q = \sum_{k=q+2}^{s+t}(\ell_k - 1)$, $D = \ell_{q+2} - 1$, and let R be the initial plateau length at $D + 1$ in this natural tail.*

Suppose valid original indices i_0, j_0 satisfy $i_0 + j_0 = q + R + 1$, $n_{i_0} = a$, $m_{j_0} = b$ and $a + b = D + 2$, where $4 \leq a \leq b$ are even integers. Let $0 \leq r \leq a - 2$ be an integer. Let H be finite simple, write ν for maximum matching cardinality, put $\Phi(H) = e(H) - \nu(H)$, and assume $\Delta(H) \leq D$, $\Phi(H) \leq P_q + r$, and $\delta = K - \nu(H) \geq r + 1$. Let κ count the whole connected odd D-regular components of H. Define ω as the minimum, over all red-blue edge colorings of H, of

$$\sum_{v \in V(H)} [\max(d_{red}(v) - (a-1), 0) + \max(d_{blue}(v) - (b-1), 0)],$$

where $d_{red}(v)$, $d_{blue}(v)$ are the respective colored degrees.

If H arrows every member of A_q, then $\omega = \kappa \geq R$. Consequently $\omega > \kappa$ or $\kappa < R$ supplies a coloring avoiding both entire prescribed targets of some member of A_q. Every such coloring extends over the original canonical q-star pool with capacities $\ell_2,\ldots, \ell_{q+1}$, and across any specified actual outer q-set X when $H = G - X$, to avoidance of the unchanged full originals while preserving all H-edge colors. No original-head $\cap$, short-prefix restriction, strict-cut condition, smaller-target lower bound or upper bound on D, R, s, t is assumed. Every unit, repeated component and full actual suffix is retained; both full target roles and prescribed colors may be exchanged. All sizes, indices, cuts, degrees, matching cardinalities, counts, deficits and allowances are integers in the stated domains.

Proof. Suppose H arrows every member of A_q. The minimum defining ω exists because the finite edge set has finitely many colorings. Colorings in different whole components can be chosen independently, so ω equals the sum of their separate minimum overflows. Hence Lemma 246 applies at allowance a-2. If its first alternative holds, the assertion follows.

Suppose its exceptional alternative holds. The exact potential equality and $r \leq a - 2$ force $r = a - 2$ and $\Phi(H) = P_q + r$. Put $w = (b - a)/2$. That alternative gives $w \geq 2$, $D \geq 10$, $\omega = R$, $\kappa = R - 1$, and an actual low vertex y with exactly a adjacent odd factor-critical components of H-y wholly of ambient H-degree D. Every whole odd D-regular component of H is equimatchable and factor-critical. By Lemma 208, each has matching number at most D. Each also has a near-perfect matching, by its factor-criticality.

Since $\delta \geq r + 1 \geq 1$, Lemma 33 gives $q < \min(s, t)$, uncolored full-original copies in H, and both original heads at most D. The two heads are at least a and b, respectively, so both are at least four. If $\kappa = 0$, then $R = 1$. Apply Lemma 181 to the same H, y and the a specified pieces, with $p = a$. Its hypotheses are $D \geq 7$, full heads in [4,D], natural plateau one, $r = a - 2 = p - 2$ and $\delta \geq a - 1 = p - 1$. It gives an avoiding A_q coloring, a contradiction. Thus $\kappa \geq 1$.

Apply Lemma 220 to all κ selected whole odd regular components, distinguishing any one of them as C_0. Every selected component has a near-perfect matching and $\nu(C_0) \leq D$; $\kappa \geq 1$, $\delta \geq a - 1 \geq 1$, $D = 2(a + w - 1)$, $r = a - 2$ and $R = \kappa + 1$ have all been established. Its particular consequence gives $R \geq \delta + w$, $R \geq D/2$ and $\kappa \geq D/2 - 1$. The same theorem supplies the actual exposed-vertex deletions and natural-tail identities supporting this inequality; its component remains whole throughout.

Let N_a, T_b count the full original components of sizes at least a, b. They are positive and satisfy

$$N_a + T_b - 1 = q + R.$$

For the lower bound, the stipulated terminal indices satisfy $i_0 \leq N_a$ and $j_0 \leq T_b$. If the count sum were larger, the valid original candidate (N_a, T_b) would give a diagonal value at least $a + b - 1 = D + 1$ strictly after the initial natural plateau. This is impossible. To justify this last statement at every endpoint, the full diagonal profile is nonincreasing: from any candidate on a noninitial diagonal, lower an index greater than one to obtain a candidate on the preceding diagonal of at least the same size. If the plateau ends at the last diagonal, no later valid index sum exists. This proves the identity in either case.

No whole component of H is K_{D+1}. Otherwise Lemma 137 applies with this same terminal rectangle and allowance $r = a - 2$: $D \geq 10$ is even, $4 \leq a < b$ are even, $\delta \geq 1$ and $\Phi(H) \leq P_q + r$. It gives an avoiding A_q coloring, a contradiction.

Put $d = D/2$ and $k_0 = \max(0, ceil((D-q-1)/3))$. Since $d \geq 5$ and $q \geq 0$,

$k_0 \leq ceil((2d-1)/3) \leq d-2$.

The second inequality follows from $2d-1 \leq 3d-6$ and integrality of d-2. Thus $k_0 < \kappa$. Select all κ whole odd D-regular components. They are noncliques, possess near-perfect matchings and all have matching number at most D. In Lemma 132 their number of large components is therefore zero. Its remaining hypotheses hold: $k_0 \geq 0$, $q + 3k_0 + 1 \geq D$ by the definition of k_0, $\delta \geq 1$, and $R = \kappa + 1 \leq 3\kappa$ because $\kappa \geq 1$. That theorem gives

$r = \Phi(H) - P_q \geq (d-1)(\kappa - k_0) + \kappa - R$

$= (d-1)(\kappa - k_0) - 1 \geq d-2$.

But $d = a + w - 1$ and $w \geq 2$, so $d - 2 = a + w - 3 > a - 2 = r$. This contradiction excludes the exceptional alternative.

Therefore $\omega = \kappa \geq R$ whenever H arrows the entire family. If $\omega > \kappa$ or $\kappa < R$, negating that finite conjunction yields an avoiding coloring for some entire actual A_q pair. Lemma 41 gives both the original canonical-pool extension and the extension across every specified actual q-set X, preserving all H-colors and handling target components meeting X only at leaves. All steps used the full originals or legal complete suffixes, and all arguments remain valid under simultaneous exchange of their roles and colors. □

Remark. Exceptional overflow forces many small equimatchable whole regular components. Exact exposed-root deletion makes the plateau long, while the terminal clique exclusion and sparse-triple transfer force more potential than the allowed first slope.

Depends on. Lemma 33, Lemma 41, Lemma 132, Lemma 137, Lemma 181, Lemma 208, Lemma 220, Lemma 246.

Lemma 259 (Closed regular witnesses for excess-dependent run bounds). *Use the finite integer certificate costs $c_i(G)$, matching number $\nu(G)$, nonnegative slack $ps(G)$, excess $ex(G) = c_0(G) - \nu(G)$, and signed gap $gap(G) = ex(G) - ps(G)$ of Lemma 61. Use the top missing-allocation run $tr(G)$ from Lemma 68. Thus, for every nonnegative integer t, $Y_t(G)$ consists of all integers x in [0,t] for which all edges of G partition into two parts of matching numbers at most x and t-x. If $c_0(G) > 0$, $tr(G)$ is the longest consecutive integer run missing from $Y_(c_0(G)-1)(G)$, and otherwise $tr(G) = 0$. All graphs are finite and simple, and vertex deletion is induced. The empty graph has maximum degree zero.*

Let $b \geq 1$ be an integer, and prescribe nonnegative integers $L(1)$,..., $L(b)$ such that

$L(e) \geq \max(0, 3-e)$ *for every integer* $1 \leq e \leq b$,

$L(e-1) \leq L(e) + 1$ *for every integer* $2 \leq e \leq b$.

Let H be a finite simple graph such that every induced subgraph J of H, including H itself and the empty graph, satisfies $gap(J) \leq 1$. Suppose

$gap(H) > 0$, $1 \leq ex(H) \leq b$, *and* $tr(H) > L(ex(H))$.

Then H contains an induced subgraph F satisfying all of the following:

F is connected and D-regular for an even integer $D \geq 4$;

F-v has a perfect matching for every vertex v of F;

$gap(F) = 1$, $1 \leq ex(F) \leq ex(H)$, *and* $tr(F) > L(ex(F))$;

$tr(F) \geq tr(H)$ *and* $tr(F) + ex(F) \geq tr(H) + ex(H)$.

The graph F may equal H. In particular $|V(F)| = 2\nu(F) + 1$.

Consequently, if every connected even-regular factor-critical induced subgraph F of H of degree at least four, with positive gap and excess at most b, satisfies $tr(F) \leq L(ex(F))$, then every induced subgraph J of H with positive gap and $1 \leq ex(J) \leq b$ satisfies the same inequality. Here factor-critical means that deleting any vertex leaves a perfect matching. The converse is immediate by restricting the family of tested induced subgraphs.

For any integer $B \geq 3$, there is also the following specialization without a prescribed upper bound on excess: if H has the displayed hereditary gap bound, positive gap, and $tr(H) + ex(H) > B$, it contains an F of the same connected regular factor-critical type with

$1 \leq ex(F) \leq ex(H)$ *and* $tr(F) + ex(F) \geq tr(H) + ex(H) > B$.

All orders, degrees, costs, matching sizes, excesses, run lengths, cutoffs, indices and interval endpoints are integers in their stated nonnegative or positive domains. Gaps may be signed integers.

Proof. Write $E_H = ex(H)$ and $R_H = tr(H)$. Every graph of positive gap under the hereditary assumption has gap one. Lemma 61 gives nonnegative slack, so its excess is positive.

Consider the family of induced subgraphs F of H satisfying

$gap(F) > 0, 1 \leq ex(F) \leq E_H, tr(F) > L(ex(F))$,

$tr(F) \geq R_H$, and $tr(F) + ex(F) \geq R_H + E_H$. (1)

This is a nonempty finite family: H belongs to it, and a finite vertex set has only finitely many subsets. Choose F with the fewest vertices. This proves existence of the extremal choice before using it. Put $E = ex(F)$ and $R = tr(F)$. The graph F has gap one and $ps(F) = E - 1$. Every induced subgraph of F is induced in H, so has gap at most one.

Suppose F is disconnected. Choose a component Z of largest excess m, and among those components choose one of largest top run. By Lemma 68 with empty deleted set, $m \geq E$. By the slack superadditivity in Lemma 72 and nonnegative slack in Lemma 61, $ps(Z) \leq ps(F) = E - 1$. Hence

$gap(Z) = m - ps(Z) \geq m - E + 1 \geq 1$.

The inherited gap bound forces equality throughout. Therefore $gap(Z) = 1$, $m = E$, and $ps(Z) = E - 1$. The equal-excess conclusion of Lemma 68, with the stated extremal choice of Z, gives $tr(Z) \geq R$. Thus Z satisfies every condition in (1) and has fewer vertices than F, a contradiction. This proves connectedness.

Put $D = \Delta(F)$. An edgeless graph has all costs zero, because its empty partition satisfies every capacity and matching-budget bound at total zero. It therefore has gap zero. Thus F has an edge. If $D \leq 3$, then Lemma 147 gives $R + E \leq 3$. But $L(E) \geq 3 - E$, so $R \leq L(E)$, contrary to (1). Consequently $D \geq 4$.

Suppose a matching covers every vertex of F-degree D. Choose a vertex v of degree D and put $J = F - v$. For each integer $0 \leq i \leq D$ define $a_i = c_i(F) - c_i(J)$. The restriction and extension bounds in Lemma 72 give a_i in {0,1}. Set $\epsilon = \nu(F) - \nu(J)$; restricting a maximum matching loses at most one edge, so ϵ is zero or one. By Lemma 61 the two graphs have zero rows above D. Its covering conclusion gives $c_D(F) = 0$, and restriction gives $c_D(J) = 0$. Let g be the number of indices i in {1,...,D} with $a_i = 0$; then $g \geq 1$. Deleting v removes exactly D edges, so subtracting the finite cost sums gives

$gap(F) - gap(J) = a_0 - g$.

Since $gap(F) = 1$ and $a_0 \leq 1$, $gap(J) \geq 1$. The hereditary bound makes $gap(J) = 1$ and $a_0 = g = 1$. In particular

$c_0(J) = c_0(F) - 1, ex(J) = E - 1 + \epsilon$. (2)

We verify the run increase on this same graph J. Put $c = c_0(F)$ and $k = \nu(F)$, so $c = k + E$. The graph F has an edge, hence $k \geq 1$ and $c \geq 2$. At total c-1, every first budget $x \leq E - 1$ is feasible by putting all edges in the second part, and every $x \geq k$ is feasible by putting all edges in the first part. A longest missing run, say $[u, u + R - 1]$, therefore lies in [E, k-1], which is contained in [1,c-2].

A feasible first budget y for J at total c-2 gives feasible first budgets y and $y + 1$ for F at total c-1. Indeed, keep its two edge parts and put every edge incident with v in the second or first part, respectively. A matching contains at most one such added edge. Thus for each integer x in the missing F-run, both x-1 and x are missing for J at total c-2. The whole interval $[u - 1, u + R - 1]$ is missing, lies in [0,c-2], and has length $R + 1$. By (2), this is J's top level, so

$tr(J) \geq R + 1$. (3)

Nonnegative slack and $gap(J) = 1$ exclude $ex(J) = 0$. Equations (2)-(3) give $1 \leq ex(J) \leq E \leq E_H$, $tr(J) \geq R_H$, and

$tr(J) + ex(J) \geq R + E \geq R_H + E_H$.

If $ex(J) = E$, then $tr(J) > L(E)$. If $ex(J) = E - 1$, then $E \geq 2$ and the integer inequalities $R > L(E)$ and (3) imply

$tr(J) \geq L(E) + 2 > L(E - 1)$,

where the last inequality uses $L(E - 1) \leq L(E) + 1$. Thus J satisfies every condition in (1) and has one fewer vertex, contradicting extremality. No matching covers all degree-D vertices.

Let S be the set of degree-D vertices and let $\kappa \geq 1$ be the least number of vertices of S left uncovered by a matching. For a vertex subset X, let $q_S(F - X)$ count the odd-order components of F-X whose entire vertex sets belong to S. Choose X of maximum cardinality among the maximizers of $q_S(F - X) - |X|$. The set of all vertex subsets is finite, so this choice exists. Lemma 107, with marked set S, gives

$q_S(F - X) - |X| = \kappa$

and makes every counted component factor-critical. The selected X is also inclusionwise maximal among the maximizers, as required there.

If X is nonempty, every component of F-X is a proper induced subgraph with gap at most one. The same D, F, S, κ, X therefore satisfy every hypothesis of Lemma 239. Its finite component family is nonempty, since $q_S(F - X) = |X| + \kappa > 0$. Choose a component Z by that theorem's prescribed order: largest matching number, then largest number of rows above that matching number among indices 1 through $floor((D - 1)/2)$. Both maximizations exist over a nonempty finite set. If D is even the theorem gives $gap(F) \leq 0$, a contradiction. If D is odd, it gives

$gap(Z) = 1, ex(Z) = E, tr(Z) \geq R + |X|$.

This Z satisfies every condition in (1) and has fewer vertices because X is nonempty, again a contradiction. Hence X is empty.

The connected graph F is its sole component. The equality $q_S(F) = \kappa \geq 1$ forces that component to have odd order and all its vertices to lie in S. Thus F is D-regular and is one of the components made factor-critical by Lemma 107. Counting the two endpoints of every edge shows $D|V(F)| = 2e(F)$; since $|V(F)|$ is odd, D is even. We already proved $D \geq 4$. A perfect matching of F-v has $(|V(F)| - 1)/2$ edges, and no matching of the odd-order graph F has more. Therefore $|V(F)| = 2\nu(F) + 1$. All remaining asserted inequalities are conditions (1), so the extraction is proved.

For the consequence, suppose an induced J of H of positive gap and excess between one and b violated its cutoff. Every induced subgraph of J inherits the gap bound. Applying the extraction with J in place of H gives an induced connected regular factor-critical F of H, with positive gap and excess between one and b, violating its cutoff. This contradicts the proposed bound on those F. Conversely a bound on all tested induced J includes every F in the smaller specified family.

Finally let $B \geq 3$ and suppose $R_H + E_H > B$. Positive gap and nonnegative slack give $E_H \geq 1$. Set $b = E_H$ and $L(e) = \max(0, B - e)$ for $1 \leq e \leq b$. These are nonnegative integers, $L(e) \geq \max(0, 3 - e)$, and $L(e - 1) \leq L(e) + 1$. If $E_H \leq B$, the assumed sum inequality gives $R_H > L(E_H)$. If $E_H > B$, then $L(E_H) = 0$ and the defining minimum positive cost has a nonempty top missing set, so $R_H \geq 1 > L(E_H)$. The extraction therefore applies and supplies exactly the stated specialized conclusion. □

Remark. Component extraction and covered deletion preserve an excessive run. A nonempty marked barrier would give a smaller such graph, so an extremal induced witness is regular and factor-critical. The cutoff condition accounts for the possible unit decrease in excess.

Depends on. Lemma 61, Lemma 68, Lemma 72, Lemma 107, Lemma 147, Lemma 239.

Lemma 260 (Dense remainders of the degree-twelve exceptional profile). *Use the finite integer certificate row costs c_i, matching number ν, independence number α, maximum degree Δ, edge count e, excess $ex = c_0 - \nu$, nonnegative slack $ps = e - \nu - \sum_{i \geq 1} c_i$, and signed $gap = ex - ps$ of Lemma 61. Use the top missing-allocation run tr of Lemma 125. For every nonnegative integer D put $h(D) = \max(4, \lfloor D/2 \rfloor + 1)$. A graph satisfies T_- when positive gap implies gap one and $tr + ex \leq h(\Delta)$. All graphs are finite and simple. Vertex deletion is induced. A nonempty graph is factor-critical if deleting any vertex leaves a perfect matching.*

Let H be a nonempty connected twelve-regular factor-critical graph with $ex(H) \le 6$ and $gap(H) > 0$. Suppose every proper induced subgraph of H satisfies T_-, including the empty graph. Then H has 29 vertices,

$\nu(H) = 14$, $ex(H) = 5$, $gap(H) = 1$, $ps(H) = 4$, $\alpha(H) = 5$,

$c_0(H) = c_1(H) = c_2(H) = 19$, $c_{10}(H) = 9$, $c_{11}(H) = 5$, $c_{12}(H) = 1$,

and its only missing matching-budget allocation at total 18 is (9,9).

For every maximum independent set S of H, put $G = H - S$. Then

$|V(G)| = 24$, $e(G) = 114$, $\nu(G) = 12$, $\Delta(G) = 11$, $ps(G) = 2$,

$c_i(G) = c_i(H) - 5$ *for every integer* $1 \le i \le 11$,

$c_i(G) = 0$ *for every integer* $i \ge 12$,

$gap(G)$ *belongs to {0,1}, and* $c_0(G) = 14 + gap(G)$.

Every vertex of G has degree at least seven, and G has a perfect matching.

For every vertex v of degree eleven in this same G, put $J = G - v$. Then J is nonregular and factor-critical, and

$|V(J)| = 23$, $e(J) = 103$, $\nu(J) = 11$, $\alpha(J) \le 5$,

$\Delta(J) \le 11$, $ps(J) \le 2$.

Every induced subgraph of J, including J itself, satisfies T_-. All graph orders, degrees, row indices, costs, capacities, budgets, matching and independence cardinalities, excesses and slack values are nonnegative integers in their displayed domains; gaps may be signed.

Proof. By Lemma 219, positive gap forces order 29, matching number 14, excess 5, gap 1 and all the displayed costs on H. Thus $ps(H) = 4$ and $e(H) = 174$. Apply Lemma 125 to the same connected H, with its maximum degree 12 and matching number 14. It gives the unique missing total 18 allocation (9,9), $tr(H) = 1$, perfect matchings on every even induced order at least 20, and factor-criticality on every odd induced order at least 21.

We first prove $\alpha(H) \ge 5$ by retaining a covered deletion in the cutoff inequality of Lemma 123. Choose an economical complete maximum-degree deletion sequence supplied by that fact. Its length is at least $c_0(H) = 19$, so its first three stages and cutoff 19 exist. The first stage deletes a vertex of degree 12. In the remaining 28-vertex graph, the sixteen vertices originally nonadjacent to that vertex still have degree 12. Thus the second stage also deletes a degree 12 vertex. The graph after the first two deletions has 27 vertices and $174 - 24 = 150$ edges. It is factor-critical by the preceding robust matching conclusion. It is not regular: a regular graph on 27 vertices has even degree by the degree sum identity, and hence has edge count divisible by 27, whereas 150 is not.

For any nonregular factor-critical graph, choose a vertex of degree below its maximum and a perfect matching after deleting that vertex. The matching covers every maximum-degree vertex in the original graph. Deleting any maximum-degree vertex also preserves matching number, since the graph is factor-critical. Therefore the third stage of our sequence is covered and preserves matching number. In the notation of Lemma 123, $z_{19} \ge 1$ and $\beta_{19} \ge 0$. Its cutoff inequality gives

$$4 = ps(H) \ge 1 + 2(29 - 19) - 4\alpha(H) = 21 - 4\alpha(H).$$

Integrality implies $\alpha(H) \ge 5$.

Fix a maximum independent set S, write $a = |S|$ and $G = H - S$. Such a set is nonempty and is maximal under inclusion. Every vertex outside S therefore has an S-neighbor, so $\Delta(G) \le 11$. For integers $0 \le i \le 11$ define

$$b_i = a - c_i(H) + c_i(G).$$

Certificate restriction and the one-vertex extension proved in Lemma 61, iterated over S, give $0 \le b_i \le a$. The row 12 of G and every higher row vanish by $\Delta(G) \le 11$ and that fact. The numerical profile gives $c_{12}(H) = 1$; all higher H-rows vanish. Independence and regularity give $e(H) - e(G) = 12a$. Subtracting the finite cost sums in the gap definition therefore yields

$$gap(G) = 1 - [12a - \textstyle\sum_{i=0}^{11} b_i + 1] + 12a$$

$= \sum_{i=0}^{11} b_i$.

Since G is proper induced in H and satisfies T_-, this nonnegative integer sum is at most one. In particular $b_{11} = a - 5 + c_{11}(G) \geq a - 5$, so $a \leq 6$. We have proved a in {5,6}. The robust matching formula now gives $\nu(G) = floor((29 - a)/2)$, equal to 12 for $a5$ and 11 for $a6$.

Suppose some b_i is positive at an index $1 \leq i \leq 11$. Then sum $b_i = 1$, $b_0 = 0$ and $gap(G) = 1$. Hence $c_0(G) = 19 - a$ and $ex(G) = 2$ in either of the two possible values of a. Put $t = c_0(G) - 1 = 18 - a$, which is at least $\nu(G)$. For a graph Q and a nonnegative integer l, let $Y_l(Q)$ be the feasible first matching budgets at total l. Both 0 and t belong to $Y_t(G)$, by assigning every G-edge to a single part.

We prove $Y_t(G) + [0, a]$ is contained in $Y_{18}(H)$, with all intervals consisting of integers. For any feasible budget x in $Y_t(G)$ and any integer $0 \leq r \leq a$, keep a witnessing edge partition of G. Choose an r-element subset S_R of S, assign every restored edge meeting S_R to the first part, and assign all other restored edges to the second part. The first added graph has vertex cover S_R and matching number at most r; the second has vertex cover S minus S_R and matching number at most a-r. A matching in either union splits into matchings on its old and added edges, so the resulting two bounds are $x + r$ and $t - x + a - r$. Every original edge is assigned exactly once. This proves the containment.

List $Y_t(G)$ increasingly, including its endpoints 0, t. Adding [0,a] replaces every run of r missing integers between consecutive list members by a run of length $\max(0, r - a)$; there are no endpoint gaps. Since the integer 9 is missing from $Y_{18}(H)$, it is also missing from that sumset. Thus some original missing run has length at least $a + 1$, giving $tr(G) \geq a + 1$. Consequently

$tr(G) + ex(G) \geq a + 3 \geq 8$.

But G satisfies T_-, $gap(G) = 1$ and $\Delta(G) \leq 11$, so $tr(G) + ex(G) \leq h(\Delta(G)) \leq 6$. This contradiction proves $b_i = 0$ at every positive index $1 \leq i \leq 11$.

It follows that $c_{11}(G) = 5 - a \geq 0$, so $a \leq 5$. Combined with the lower bound, $a = \alpha(H) = 5$. We have every positive-row identity asserted in the statement. Also $gap(G) = b_0$ belongs to {0,1} and $c_0(G) = 14 + b_0$. The order is 24, the robust matching number is 12, and $ex(G) = 2 + b_0$; hence $ps(G) = 2$. Deleting the independent five-set removes 60 edges, giving $e(G) = 114$. Each remaining vertex loses at most five of its twelve neighbors, so its G-degree is at least seven.

In fact $\Delta(G) = 11$. We already know the upper bound 11. If $\Delta(G) \leq 10$, the perfect matching of G covers every vertex of degree 10, and Lemma 61 gives $c_{10}(G) = 0$. This contradicts $c_{10}(G) = c_{10}(H) - 5 = 4$. Therefore degree 11 vertices exist.

Choose any such vertex v and put $J = G - v$. It has 23 vertices and $114 - 11 = 103$ edges. The robust matching conclusion makes it factor-critical, with $\nu(J) = 11$. The perfect matching of G covers all its degree 11 vertices, and $\nu(G) - \nu(J) = 1$. The covered-deletion inequality in Lemma 61 therefore gives $ps(G) - ps(J) \geq 1 - 1 = 0$, so $ps(J) \leq 2$. Induced deletion gives $\alpha(J) \leq \alpha(H) = 5$ and $\Delta(J) \leq 11$. A regular graph on 23 vertices would have even degree and edge count divisible by 23. Since 103 is not divisible by 23, J is nonregular. Finally every induced subgraph of J is a proper induced subgraph of H, so it satisfies the assumed T_-. The argument used arbitrary S and arbitrary degree 11 v, proving all stated quantifiers. □

Remark. The robust perfect matchings force an early covered matching-preserving deletion, improving the independence cutoff by one unit. A missing-allocation interval then forces exact independent-cut rows, leaving a fixed dense nonregular factor-critical remainder.

Depends on. Lemma 61, Lemma 123, Lemma 125, Lemma 219.

Lemma 261 (Well-coveredness of the octic 29-vertex profile). *Use the finite integer certificate costs c_i, matching number ν, independence number α, maximum degree Δ, excess $ex = c_0 - \nu$, slack ps, signed $gap = ex - ps$, and top missing-allocation run tr of Lemma 172. A graph satisfies T_- when positive gap implies gap one and $tr + ex \leq \max(4, \lfloor \Delta/2 \rfloor + 1)$.*

Let H be a connected eight-regular finite simple graph on 29 vertices such that deleting any vertex leaves a perfect matching. Suppose $ex(H) = 4$, $gap(H) = 1$, $c_7(H) = 8$, and every proper induced subgraph of H satisfies T_-. Then every inclusionwise maximal independent set of H has eight vertices. In particular $\alpha(H) = 8$ and H is well-covered, meaning that every maximal independent set is maximum. All graph orders, degrees, costs, matching and independence cardinalities, and row indices are integers in their stated nonnegative or positive domains.

Proof. Fix an arbitrary maximal independent set I, put $a = |I|$, and let $J = H - I$. Eight-regularity gives $e(H) = 116$. Lemma 172 applies with exactly the stated connectedness, regularity, factor-criticality, excess, gap and proper-induced T_- hypotheses. It gives $\Delta(J) \le 7$, $ps(J) \le 3$ and $c_7(J) = c_7(H) - a = 8 - a$. Put $q = 8 - a$. Costs are nonnegative, so $q \ge 0$. Deleting the independent set I removes exactly 8a edges and therefore

$e(J) = 116 - 8a = 52 + 8q.$

Suppose $q \ge 1$. The zero-tail conclusion of Lemma 61 shows that $\Delta(J) = 7$, since otherwise $c_7(J) = 0$. Every proper induced subgraph of J is a proper induced subgraph of H, so T_- gives its gap at most one. Let κ be the minimum number of degree-seven vertices of J left uncovered by a matching, and let h count the whole connected components of J with positive degree-seven exposure. Apply Lemma 243 at actual degree seven. It gives

$1 \le q \le \kappa \le ps(J) \le 3,$

$ps(J) \ge \kappa + 5(\kappa - h).$

Exposure is additive over whole connected components: a matching is the disjoint union of its component restrictions, and minimum component exposures can be attained independently and united. Thus κ is the sum of the h positive integer component exposures and $\kappa \ge h$. If $\kappa - h \ge 1$, the displayed lower bound is at least six, contradicting $ps(J) \le 3$. Consequently $\kappa = h$ and every one of those h components has exposure exactly one.

We show that each such component U has at least 63 edges. Its actual maximum degree is seven. Let M be its set of degree-seven vertices. Choose an inclusionwise maximal set X maximizing the number of odd components of U-X wholly contained in M, minus $|X|$. This maximum exists because U has finitely many vertex subsets. By Lemma 107 applied to U and its marked set M, if $b = |X|$ then there are exactly $b + 1$ such odd components. The set X is nonempty: otherwise positive exposure and connectedness would say that U itself has odd order and every vertex of U has degree seven. This contradicts $7|V(U)| = 2e(U)$.

Every counted component C has odd order n_C, and all its vertices have degree seven in U. All edges leaving C have their other endpoint in X, since C is a component of U-X. Write d_C for their number. Connectedness of U and the nonempty disjoint set X give $d_C \ge 1$. The degree sum gives

$d_C = 7n_C - 2e(C),$

so d_C is odd. If $n_C \le 7$, simplicity gives

$d_C \ge 7n_C - n_C(n_C - 1) = n_C(8 - n_C) \ge 7,$

as n_C belongs to {1,3,5,7}. If $n_C \ge 9$, the positive lower bound $d_C \ge 1$ still holds.

Assume at most one of the $b + 1$ counted components has order at least nine. If none does, their total boundary is at least $7(b + 1) > 7b$. If exactly one does, the other b components contribute at least 7b edges and the remaining component contributes at least one, again a total greater than 7b. But every one of these edges meets X, and the sum of the degrees of its b vertices is at most 7b. Both alternatives are impossible. Hence at least two counted components have order at least nine. They contain at least eighteen distinct vertices of degree seven in U. Summing degrees in U now gives $2e(U) \ge 18 \cdot 7 = 126$, proving $e(U) \ge 63$.

The h uncovered whole components of J are vertex-disjoint and have no edges between them. Thus

$52 + 8q = e(J) \ge 63h = 63\kappa \ge 63q.$

This implies $55q \le 52$, impossible for the positive integer q. It follows that $q = 0$ and $a = 8$. The maximal independent set I was arbitrary. Every maximum independent set is maximal, so its size is also eight. This proves both conclusions. □

Remark. A maximal independent set smaller than eight forces positive degree-seven exposure. Low slack restricts each uncovered component to exposure one, while the marked barrier forces at least eighteen degree-seven vertices per such component, exceeding the available edge count.

Depends on. Lemma 61, Lemma 107, Lemma 172, Lemma 243.

Lemma 262 (Terminal mass forces a surviving vertex cover). *Let $s, t \geq 1$ be integers and let F, J be star forests with full nonincreasing positive integer edge-size lists $n_1,\ldots, n_s$ and $m_1,\ldots, m_t$. All graphs are finite and simple, and copies are ordinary prescribed-color subgraphs with vertex-disjoint target components, F red and J blue. Put $M = s + t - 1$ and*

$\ell_l = max\{n_i + m_j - 1 : 1 \leq i \leq s, 1 \leq j \leq t, i + j = l\}$

for every integer $2 \leq l \leq s + t$. Fix an integer $0 \leq q < M$, put $K = M - q$, $L_q = \sum_{l=q+2}^{s+t} \ell_l$, and assume

$D = \ell_{q+2} - 1 = n_1 \geq 3, m_1 \leq D.$

Let R be the number of initial entries equal to $D + 1$ in the full natural q-tail. Suppose valid original indices i_0, j_0 satisfy

$i_0 + j_0 = q + R + 1, n_{i_0} = D, m_{j_0} = 2.$

Let U count the full F components of size D, and put

$B = D\,U + \sum_{i:3\leq n_i<D}(n_i - 2).$

Let A_q contain every pair (F^α, J^β) with nonnegative integer cuts $\alpha < s$, $\beta < t$, $\alpha + \beta = q$, where superscripts delete leading whole components. Suppose a finite simple graph H arrows every entire A_q member, has maximum degree at most D, and satisfies $e(H) < L_q$. Write ν, τ, e for maximum matching cardinality, minimum vertex-cover cardinality and edge count. Put

$\delta = K - \nu(H), \epsilon = L_q - e(H).$

Then $D \geq 4$, $q < \min(s, t)$, $\delta \geq \epsilon \geq 1$, and

$R \geq \delta + ceil((B + \epsilon - 1)/2),$

$\tau(H) \geq K + R - \delta.$

For every actual vertex set Y with integer cardinality $0 \leq a \leq K$,

$\tau(H - Y) \geq K - a + R - \delta.$

In particular every actual K-vertex deletion leaves vertex-cover and edge numbers at least

$R - \delta \geq ceil((B + \epsilon - 1)/2) \geq \lceil D/2 \rceil.$

When $e(H) = L_q - 1$, these conclusions become $R \geq \delta + \lceil B/2 \rceil$ and the corresponding bound $\tau(H - Y) \geq K - a + R - \delta$, without requiring $\delta \geq 2$. If H occurs after an outer sequence of vertex deletions, its surviving edges after the additional deletion of Y are disjoint from all edges removed at every preceding step. Each removed edge is counted exactly at its earliest deleted endpoint. The full originals may have arbitrary units, repeated sizes and lower nonbinary components. No potential-surplus, unit-count, component-structure or prescribed-matching hypothesis is imposed.

Proof. The first assertion of Lemma 177 applies directly to the stipulated common host, actual family, strict edge budget and critical degree $\cap$. It gives $D \geq 4$.

We first obtain positive matching deficit without assuming a small cut. The complete natural q-tail is the envelope of A_q: for a candidate with original indices i, j and local coordinate k satisfying $i + j = q + k + 1$, the cuts $\alpha = \min(q, i - 1)$, $\beta = q - \alpha$ are legal and retain it, because $\beta \leq j - 1$. Every local candidate has the corresponding original index sum, giving the reverse inclusion.

Apply Lemma 99 to this full family, with budget K and critical $\cap$ D. It gives $e(H) - \nu(H) \geq L_q - K$, so $\delta \geq \epsilon \geq 1$. The actual-family assertion in Lemma 89 gives $q < \min(s, t)$ and

$t \geq q + 2\delta + T, T := 2sum_{i=1}^{s}\, floor((n_i - 1)/2).$ (1)

The nonnegative attaining-separator size was dropped from its bound.

Apply Lemma 223 to these same originals and H. Let E be the complete both-orientation matching-envelope sum, and put $G = L_q - E$, $\gamma = \tau(H) - K$. It gives

$G \geq \max(2, 2R - 1), \gamma \geq 0, \gamma \leq G - \epsilon,$

and $\gamma \geq R - \delta$. (2)

In particular G cannot equal R. The last inequality follows there from $R \leq \rho \leq \tau(H) - \nu(H) = \gamma + \delta$.

Write V for the number of full J nonunits, $g = t - V$ for its unit count, z for the number of F components of sizes 3 through D-1, k for its size-two count, and u for its unit count. Put $f = s - U = z + k + u$ and $X = k - g + z$.

The terminal pair implies $U + V \geq q + R + 1$. If strict, the original indices U, V would supply an entry at least $D + 1$ beyond the final position of the initial plateau, a contradiction. Therefore

$U + V = q + R + 1$, $K = f + g + R$, $e(F) - K = B + X - R$. (3)

The full-original amplified cover bound Lemma 65 gives

$\gamma \geq e(F) - K + 2\delta - 1 = B + X - R + 2\delta - 1$. (4)

Suppose first $g < z$. Using $t = V + g$ and $q = U + V - R - 1$ in (1), followed by $g \leq z - 1$, gives

$R \geq U + 2\delta + T - z$.

For each integer $d \geq 3$, $4floor((d-1)/2) \geq d$: if $d = 2a \geq 4$ this is $4a - 4 \geq 2a$, and if $d = 2a + 1 \geq 3$ it is $4a \geq 2a + 1$. Thus $2T \geq B + 2z$, since unit and two-edge stars contribute zero to T. Consequently

$2R \geq B + 2U + 4\delta \geq B + 2\delta + \epsilon - 1$,

using $U \geq 1$ and $\delta \geq \epsilon$. This proves the desired plateau bound in this case.

It remains that $g \geq z$. We derive directly the interval estimate

$R \leq G \leq R + \max(X, 0)$. (5)

For each positive integer h put $r_h = |\{i : n_i \geq h\}|$ and $c_h = |\{j : m_j \geq h\}|$. The number A_h of natural-tail entries at least h is

$A_h = max\{(r_x + c_y - q - 1)_+ : x, y \geq 1, x + y = h + 1, r_x > 0, c_y > 0\}$. (6)

Indeed each threshold split gives a rectangle of original indices whose sums fill the interval from 2 through $r_x + c_y$; a size sum at least $h + 1$ admits such a split below its two sizes. Here $w_+ = \max(w, 0)$, and an empty maximum is zero.

For $1 \leq h \leq D$, the complete matching envelope has threshold width

$C_h = \max(g - U + r_h + R, f - V + c_h + Rifc_h > 0)$, (7)

because its two full-corner profiles are ((t-q)K2,F) and ((s-q)K2,J). The first term is positive, as $r_h \geq U \geq 1$. Formula (7) also follows by subtracting each legal cut from its corresponding threshold count, as in Lemma 223. The splits (h,1), (1,h) in (6) give these terms. Hence $A_h \geq C_h$, with equality at $h = 1, 2$. At $h = D + 1$ the natural width is R and the matching width is zero; all larger widths vanish.

For $3 \leq h \leq D$, a split with $x, y \geq 3$ has untruncated width at most $z + R$, because $r_x \leq U + z$ and $c_y \leq V$. This is at most $g + R$ and hence at most the first term of (7). For $h \geq 4$ the split (h-1,2) exceeds that same first term by $r_{h-1} - r_h - g \leq z - g \leq 0$. Thus the only possible additional term is (2,h-1), including (2,2) when $h = 3$, and it is absent when $c_{h-1} = 0$.

Put $x_0 = k - g$ and $B_h = r_h - x_0 - z - U + V$. The untruncated width of this remaining split is $z + k + c_{h-1} - V + R$. Subtracting $z + k - V + R$ from the two matching terms (7) gives B_h and $c_h + u$. Therefore, when $c_h > 0$,

$A_h - C_h = (c_{h-1} - \max(B_h, c_h + u))_+$.

When $c_h = 0 < c_{h-1}$, the difference is $(c_{h-1} - B_h)_+$; this can occur only at the last nonzero blue threshold.

Each positive difference is the length of the interval whose upper endpoint is c_{h-1} and whose lower endpoint is $\max(B_h, c_h + u)$, or B_h in the possible final case. These intervals have disjoint interiors: a nonfinal lower endpoint is at least c_h, while every later upper endpoint is at most c_h. All upper endpoints are at most $V = c_2$, and all lower endpoints are at least $B_D = V - x_0 - z$ because $r_h \geq U$. Their summed length is at most $(x_0 + z)_+ = \max(X, 0)$. Sums of threshold widths equal sums of the profile entries, since an entry of size d contributes once at each integer level 1,..., d. Including the R contribution at $D + 1$ proves (5).

If $X \le 0$, (5) gives $G = R$, contrary to (2). Hence $X > 0$. Combining (2), (4) and (5) yields

$B + X - R + 2\delta - 1 \le \gamma \le R + X - \epsilon$.

Cancel X and rearrange to obtain $2R \ge B + 2\delta + \epsilon - 1$. Together with the preceding $g < z$ case and integer rounding, this proves

$R \ge \delta + ceil((B + \epsilon - 1)/2)$.

The host cover bound is the last assertion in (2).

For any actual a-set Y, the set Y together with a minimum cover of H-Y covers H, so $\tau(H - Y) \ge \tau(H) - a \ge K - a + R - \delta$. At $a = K$ this gives the stated surviving cover. A graph has a vertex cover obtained by selecting one endpoint of each edge, so its edge count is at least its cover number. Since $U \ge 1$ and $B \ge DU \ge D$ and $\epsilon \ge 1$, the lower bound is at least $\lceil D/2 \rceil$.

An edge remaining after all the stated vertex deletions has no deleted endpoint. Every removed edge has a unique earliest-deleted endpoint and is charged at that step, including an edge between two deleted vertices. Thus all surviving-edge and earlier-deletion supports are disjoint. Setting $\epsilon = 1$ gives the exact-edge specialization, completing the proof. □

Remark. The terminal plateau fixes the full *high/nonunit* rectangle. Comparing its conjugate threshold widths to the matching envelope bounds their gap by one disjoint family of intervals. The amplified cover inequality cancels the interval length, forcing enough plateau mass to survive every vertex deletion.

Depends on. Lemma 65, Lemma 89, Lemma 99, Lemma 177, Lemma 223.

Lemma 263 (Regularity and factor-criticality of minimal failures). *Use the finite integer certificate costs $c_i(G)$, matching number $\nu(G)$, excess $ex(G)$, positive-level slack $ps(G)$, signed gap $gap(G)$, and top missing-allocation run $tr(G)$ from Lemma 23. All graphs are finite and simple, including the empty graph, whose maximum degree is zero. Let h be a nondecreasing integer-valued function on the nonnegative integers with $h(D) \ge 3$ for every nonnegative integer D. Say that G satisfies T_h if*

$gap(G) > 0$ *implies* $[gap(G) = 1$ *and* $tr(G) + ex(G) \le h(\Delta(G))]$.

Suppose F violates T_h and every proper induced subgraph of F satisfies T_h, where these are all $F[U]$ with U a proper subset of $V(F)$, including U empty. Then F is connected and d-regular for an even integer $d \ge 4$, has odd order $2k + 1$ with $k = \nu(F)$, and is factor-critical: deleting any vertex leaves a perfect matching. It is totally critical in the following exact form: for every vertex v and every integer i with $1 \le i \le d$,

$c_i(F - v) = c_i(F) - 1$.

Its gap belongs to {1,2}. If $gap(F) = 2$, then every vertex v satisfies

$c_0(F - v) = c_0(F) - 1$, $gap(F - v) = 1$, *and* $ex(F - v) = ex(F) - 1$.

For each integer d in {4,6,8,10,12}, define M_d to be the entire class of finite simple graphs F with $\Delta(F) = d$ that violate T_- while every proper subgraph satisfies T_-, where $T_- = T_h$ for $h(D) = \max(4, \lfloor D/2 \rfloor + 1)$. A proper subgraph may delete vertices, edges, or both, including spanning edge deletions. Every member of every M_d has all the preceding structural and row-drop properties. These class definitions impose no additional condition on order, rank, excess, regularity, connectedness, or host origin. All degrees, orders, ranks, costs, indices and finite cardinalities are integers in the displayed nonnegative or positive domains; gaps may be signed.

Proof. Apply Lemma 121 to F and the stated function h. It gives connectedness, an edge, $gap(F) \ge 1$, and the absence of a matching covering all maximum-degree vertices. Put $d = \Delta(F)$. If $d \le 3$, then Lemma 147 gives $gap(F) = 1$ and $tr(F) + ex(F) \le 3 \le h(d)$, contrary to failure of T_h. Therefore $d \ge 4$.

Let S be the degree-d vertices, and let $\kappa \ge 1$ be the minimum number of S-vertices left uncovered by a matching of F. For $X \subseteq V(F)$, let $q_S(F - X)$ count odd components of F-X whose vertex sets are wholly contained in S. Choose a set X of maximum cardinality among maximizers of $q_S(F - X) - |X|$. This is a successive maximization over a finite nonempty collection, and therefore exists. It is inclusionwise maximal among maximizers. Lemma 107, with marked set S, gives $q_S(F - X) = |X| + \kappa$ and makes every counted component factor-critical.

Suppose X is nonempty. Every component Q of F-X is a proper induced subgraph of F, so satisfies T_h and hence has $gap(Q) \le 1$. Apply Lemma 239 to this connected F, its actual degree $d \ge 4$, the marked set S, and this maximum-cardinality attaining X. These are exactly the hypotheses of that theorem. At even d it gives $gap(F) \le 0$, a contradiction. At odd d positive gap gives a component Z of F-X with

$gap(F) = gap(Z) = 1$, $ex(F) = ex(Z)$, and $tr(F) + |X| \le tr(Z)$.

This Z is a proper induced subgraph, so $T_h(Z)$ and monotonicity of h give

$tr(F) + ex(F) \le tr(Z) + ex(Z) - |X|$

$\le h(\Delta(Z)) - |X| \le h(d) - 1$.

Together with $gap(F) = 1$, this asserts $T_h(F)$, another contradiction. Hence X is empty.

Connectedness and $q_S(F) = \kappa \ge 1$ force the only component F to have odd order and all vertices in S. Thus F is d-regular. Its degree sum $d|V(F)| = 2e(F)$, with odd $|V(F)|$, forces d even. Since X empty is a maximal maximizer, its counted component F is factor-critical by Lemma 107. Write $|V(F)| = 2k + 1$. A perfect matching of F-v has k edges, while no matching on $2k + 1$ vertices can have more than k edges; therefore $\nu(F - v) = \nu(F) = k$ for every v.

Every vertex now has maximum degree d, so the exact criticality conclusion of Lemma 121 applies to every v and every $1 \le i \le d$. Fix any v and put $a_0 = c_0(F) - c_0(F - v)$. The single-vertex restriction and extension in Lemma 72 give a_0 in {0,1}. Lemma 61 makes both graphs' rows above d zero. Deleting v removes exactly d edges and exactly d units of positive-row cost, by the established row drops. Hence the finite cost-sum identity gives

$gap(F) - gap(F - v) = a_0$.

The graph F-v satisfies T_h, so $gap(F - v) \le 1$. Because $gap(F) \ge 1$ and $a_0 \le 1$, this proves $1 \le gap(F) \le 2$. If $gap(F) = 2$, the same identity forces $a_0 = 1$ and $gap(F - v) = 1$. The equality $\nu(F - v) = \nu(F)$ then gives $ex(F - v) = ex(F) - 1$. The vertex v was arbitrary.

Finally every proper induced subgraph is a proper subgraph under the displayed broader definition. Every F in M_d therefore has the induced-minimality hypothesis used above. Its specified function $h(D) = \max(4, \lfloor D/2 \rfloor + 1)$ is nondecreasing, integer-valued, and at least three at every nonnegative integer D. Substitution proves every stated property for the entire class M_d, retaining arbitrary edge deletions in its defining minimality. □

Depends on. Lemma 23, Lemma 61, Lemma 72, Lemma 107, Lemma 121, Lemma 147, Lemma 239.

Lemma 264 (Top runs at excess one, two, and three). *Use the finite integer certificate costs $c_i(G)$, matching number $\nu(G)$, excess $ex(G) = c_0(G) - \nu(G)$, nonnegative slack $ps(G)$, and signed gap $gap(G) = ex(G) - ps(G)$ from Lemma 61. Use the top missing-allocation run $tr(G)$ and the sets $Y_t(G)$ from Lemma 68: $Y_t(G)$ consists of the integers x in [0,t] for which $E(G)$ partitions into two parts with matching numbers at most x and t-x. If $c_0(G) > 0$, $tr(G)$ is the longest consecutive integer run missing from $Y_{(}c_0(G) - 1)(G)$; otherwise $tr(G) = 0$.*

For every nonnegative integer D put $h(D) = \max(4, \lfloor D/2 \rfloor + 1)$. Say that G satisfies T_- if positive gap implies both $gap(G) = 1$ and $tr(G) + ex(G) \le h(\Delta(G))$. All graphs are finite and simple; the empty graph has maximum degree zero.

Let H be any such graph for which every induced subgraph, including H itself, satisfies T_-. If $gap(H) > 0$, then

$ex(H) = 1$ *implies* $tr(H) \le 2$,

$ex(H) = 2$ *implies* $tr(H) \le 4$,

$ex(H) = 3$ *implies* $tr(H) \le 5$.

Consequently, if the same hereditary hypothesis holds and $ps(H) = 0$, then $ex(H)$ belongs to {0,1}, and the longest consecutive integer run missing from $Y_{(}\nu(H))(H)$ has length at most two. At $ex(H) = 0$ this allocation set is the entire integer interval [0,nu(H)]. No degree, order, connectedness, regularity, or matching-deficiency restriction is imposed. All capacities, budgets, orders, degrees, matching numbers, costs, excesses, and run lengths are nonnegative integers; gaps may be signed.

Proof. Write $L_1 = 2$, $L_2 = 4$, and $L_3 = 5$. We prove the assertions for excess $E = 1, 2, 3$ in that order. At a given value E, the assertions at smaller positive excess are induction hypotheses. Suppose the assertion at E fails. Choose a graph H of minimum vertex cardinality among those satisfying the hereditary hypothesis, $gap(H) > 0$, $ex(H) = E$, and $tr(H) > L_E$. This minimum exists because the nonempty set of counterexample orders is a set of nonnegative integers. Every induced subgraph of H inherits the hereditary hypothesis. Since H itself satisfies T_-, $gap(H) = 1$, and hence $ps(H) = E - 1$.

The graph H has an edge. An edgeless graph has an empty-edge certificate at total zero and has gap zero. Suppose H is disconnected. Choose a connected component Z of largest excess m, and among such components choose one of largest top run. These choices exist in the finite component family. Lemma 68 with empty deleted set gives $m \geq E$. Lemma 72 and the nonnegative slack in Lemma 61 give $ps(Z) \leq ps(H) = E - 1$. Thus

$gap(Z) = m - ps(Z) \geq m - E + 1 \geq 1$.

The hereditary hypothesis gives $gap(Z) \leq 1$. Equality therefore holds throughout, and $m = E$, $ps(Z) = E - 1$. The equal-excess run contraction in Lemma 68 gives $tr(Z) \geq tr(H)$. This proper induced component is a smaller counterexample at the same excess, which contradicts the choice of H. Therefore H is connected.

Put $D = \Delta(H)$. If $D \leq 3$, fact Lemma 147 gives $tr(H) + E \leq 3$. For $E = 1, 2, 3$ this implies $tr(H) \leq 2, 1, 0$ respectively, each contradicting $tr(H) > L_E$. Hence $D \geq 4$.

Suppose a matching covers every degree-D vertex of H. Choose any vertex v of degree D, and put $J = H - v$. For each integer i with $0 \leq i \leq D$ let $a_i = c_i(H) - c_i(J)$. By Lemma 72, a_i is zero or one. Let $\epsilon = \nu(H) - \nu(J)$, also zero or one because deletion of a vertex destroys at most one edge of a maximum matching. By Lemma 61, both cost sequences vanish above D, and the covering matching gives $c_D(H) = 0$. Restriction makes $c_D(J) = 0$, so $a_D = 0$.

Let g be the number of indices i in {1,...,D} with $a_i = 0$; then $g \geq 1$. Deleting v removes D edges, so subtracting the finite cost sums gives

$gap(H) - gap(J) = a_0 - g$.

Since $gap(H) = 1$ and $a_0 \leq 1$, this implies $gap(J) \geq 1$. The inherited T_- property gives $gap(J) = 1$. Thus $a_0 = g = 1$. Consequently

$c_0(J) = c_0(H) - 1$, and $ex(J) = E - 1 + \epsilon$ in {E-1,E}. (1)

We verify the run increase in this same deletion. Put $c = c_0(H)$, $r = \nu(H)$, and $R = tr(H)$. Here $r \geq 1$, $c = r + E \geq 2$, and $R > 0$. At total c-1, every first budget $x \leq E - 1$ is feasible by putting all H-edges in the second part. Every $x \geq r$ is feasible by putting all edges in the first part. Therefore a longest missing run, written $[u, u + R - 1]$, lies in [E, r-1] and hence in [1,c-2].

Any feasible first budget y of J at total c-2 yields feasible first budgets y and $y + 1$ of H at total c-1: put all edges incident with v into the second or first part, respectively. A matching uses at most one of these added edges. Therefore both x-1 and x are missing in J whenever x belongs to the chosen missing H-run. The integer interval $[u - 1, u + R - 1]$ is missing in J, lies in [0,c-2], and has length $R + 1$. By $c_0(J) = c - 1$, this gives

$tr(J) \geq tr(H) + 1$. (2)

If $ex(J) = E$, (1)-(2) make J a smaller counterexample at the same excess. If $ex(J) = E - 1 \geq 1$, they contradict the already proved bound at that smaller excess, since $L_{E-1} < L_E$. If $ex(J) = 0$, $gap(J) = ex(J) - ps(J) \leq 0$ by nonnegative slack, contrary to $gap(J) = 1$. These alternatives exhaust (1), so no matching covers all degree-D vertices of H.

Let S be the degree-D vertex set, and let $\kappa \geq 1$ be the least number of S-vertices left uncovered by a matching. For $X \subseteq V(H)$, let $q_S(H - X)$ count the odd-order connected components whose vertices all lie in S. Choose X of maximum cardinality among the maximizers of $q_S(H - X) - |X|$. A choice exists because there are finitely many vertex subsets. It is inclusionwise maximal among the maximizers. Lemma 107 gives

$q_S(H - X) - |X| = \kappa$

and makes every counted component factor-critical.

If X is nonempty, every component of H-X is a proper induced subgraph and satisfies T_-, so has gap at most one. All hypotheses of Lemma 239 hold for this same H, S, X and D. If D is even, it gives $gap(H) \leq 0$, a contradiction. If D is odd, it gives a component Z of H-X satisfying

$gap(Z) = 1$, $ex(Z) = E$, and $tr(Z) \geq tr(H) + |X|$.

This proper induced component is a smaller counterexample. Hence X is empty.

Connectedness and $q_S(H) = \kappa \geq 1$ now force H itself to have odd order and every vertex to belong to S. Thus H is D-regular. The degree sum $D|V(H)| = 2e(H)$ forces D even. The factor-critical conclusion of Lemma 107 applies to its counted component H, so deleting every vertex leaves a perfect matching.

If $E = 1$, $ps(H) = 0$. Lemma 205 says that every regular factor-critical graph of degree at least four has positive slack. This contradicts $D \geq 4$.

If $E = 2$, the integer inequality $tr(H) > 4$ and the own T_- property give $h(D) \geq tr(H) + 2 \geq 7$. Since D is even and $h(D) = \max(4, D/2 + 1)$, one has $D \geq 12$. Lemma 175 (i), at excess two, gives $gap(H) \leq 0$, a contradiction.

If $E = 3$, $tr(H) > 5$ gives $h(D) \geq tr(H) + 3 \geq 9$, and therefore $D \geq 16$. Lemma 175 (ii), at excess three, gives the same contradiction. This completes the induction and proves all three run bounds.

Finally suppose $ps(H) = 0$ under the hereditary hypothesis. Then $gap(H) = ex(H) \geq 0$. If it is positive, T_- gives $gap(H) = 1$ and hence $ex(H) = 1$; the bound just proved gives $tr(H) \leq 2$. Since $c_0(H) - 1 = \nu(H)$ in this case, this is exactly the claimed run bound at total $\nu(H)$. If $ex(H) = 0$, $c_0(H) = \nu(H)$, and the defining universal threshold supplies every budget in [0,nu(H)]. These cases also include the empty graph and establish the last paragraph. □

Remark. A smallest excessive top run survives neither component extraction nor covered vertex deletion. The remaining marked-barrier obstruction is regular and factor-critical; its own degree threshold puts low excess in a range where goodness is already known.

Depends on. Lemma 61, Lemma 68, Lemma 72, Lemma 107, Lemma 147, Lemma 175, Lemma 205, Lemma 239.

Lemma 265 (The run bound in proper subgraphs of a four-regular graph). *Use the finite integer certificate costs $c_i(G)$, matching number $\nu(G)$, excess $ex(G) = c_0(G) - \nu(G)$, $gap(G) = \sum_{i \geq 0} c_i(G) - e(G)$, and top missing-allocation run $tr(G)$ from Lemma 23. A graph is LP-good if $gap(G) \leq 0$. Write T_3 for the implication $gap(G) > 0$ implies $[gap(G) = 1$ and $tr(G) + ex(G) \leq 3]$, and T_- for the implication $gap(G) > 0$ implies $[gap(G) = 1$ and $tr(G) + ex(G) \leq \max(4, floor(\Delta(G)/2) + 1)]$. All graphs are finite and simple; the empty graph has maximum degree zero. All degrees, orders, cardinalities, matching numbers, costs, capacities and row indices are nonnegative integers, and gaps may be signed.*

Let Q be any nonempty connected four-regular graph. Every proper subgraph J of Q satisfies T_3. Proper subgraphs permit arbitrary vertex deletions, edge deletions, or both, including spanning edge deletions and the empty subgraph. Consequently every proper subgraph J with $ex(J) \geq 3$ is LP-good, and every proper subgraph of Q satisfies T_-.

In particular, if Q additionally has odd order and every vertex deletion leaves a perfect matching, $gap(Q) = 1$, and $tr(Q) + ex(Q) \geq 5$, then Q violates T_- and every proper subgraph satisfies T_-. Thus it belongs to the entire class M_4 of finite simple graphs of maximum degree exactly four violating T_- with every proper subgraph satisfying T_-. No minimality, factor-criticality, excess or gap assumption on Q is required for the proper-subgraph conclusions of the preceding paragraph.

Proof. Fix a proper subgraph J of Q. Suppose J fails T_3. Among all induced subgraphs of J failing T_3 choose F of minimum vertex cardinality. This is possible because J itself belongs to that finite nonempty family. Every proper induced subgraph of F has fewer vertices and therefore satisfies T_3.

Apply Lemma 121 with the constant function $h(D) = 3$. It gives that F is connected, $gap(F) \geq 1$, and no matching covers all vertices of maximum degree in F. Since F is a subgraph of Q, its maximum degree is at most four. If it were at most three, Lemma 147 would give $T_3(F)$, a contradiction. Hence $\Delta(F) = 4$.

Let S be the degree-four vertex set of F and let $\kappa \geq 1$ be the minimum number of S-vertices uncovered by a matching. For a vertex set X let $q_S(F - X)$ count odd components of F-X whose vertex sets are contained in

S. Choose X of maximum cardinality among maximizers of $q_S(F - X) - |X|$. Both successive maxima exist over finite nonempty sets, and this X is inclusionwise maximal among maximizers. Lemma 107 applied with marked set S gives $q_S(F - X) = |X| + \kappa$.

If X were nonempty, every component R of F-X would be a proper induced subgraph of F. Thus $T_3(R)$ holds and $gap(R) \leq 1$. Lemma 239 applies to the connected graph F, actual even degree $D = 4$, prescribed set S, and this nonempty maximum-cardinality attaining X. Its even-degree conclusion gives $gap(F) \leq 0$, contradicting $gap(F) \geq 1$. Therefore X is empty. Connectedness and $q_S(F) = \kappa \geq 1$ force the one component F to have odd order with every vertex in S. In particular F is nonempty and four-regular.

The graph F is a subgraph of Q, even though J need not be induced in Q. At every vertex u of F, its four incident F-edges are Q-edges and exhaust all four Q-edges incident with u. No Q-edge leaves $V(F)$. Since Q is connected and $V(F)$ is nonempty, $V(F) = V(Q)$. The same incident-edge equality then gives $E(F) = E(Q)$. But F is a subgraph of J and J is a subgraph of Q, so these two equalities force $J = Q$. This contradicts the properness of J. Thus every proper J satisfies T_3.

If such a J has $ex(J) \geq 3$ and is not LP-good, its integer gap is positive. The result just proved gives $gap(J) = 1$ and $tr(J) + ex(J) \leq 3$. Since $c_0(J) = \nu(J) + ex(J) \geq 3$, the positive-cost definition of tr in Lemma 23 gives $tr(J) \geq 1$. Then $tr(J) + ex(J) \geq 4$, a contradiction. Hence every proper J with $ex(J) \geq 3$ is LP-good. A proper J with nonpositive gap satisfies T_- vacuously; one with positive gap satisfies it because T_3 gives the same gap-one conclusion and $3 \leq \max(4, floor(\Delta(J)/2) + 1)$.

Finally $\Delta(Q) = 4$ makes the T_- threshold equal to four. The additional assumptions $gap(Q) = 1$ and $tr(Q) + ex(Q) \geq 5$ therefore make Q violate T_-. The proper-subgraph assertion already proved shows that every proper subgraph satisfies T_-, establishing exactly membership in the displayed entire class M_4. The extra odd-order and perfect-matching assumptions are permitted here but were not needed for any proper-subgraph step. □

Depends on. Lemma 23, Lemma 107, Lemma 121, Lemma 147, Lemma 239.

Lemma 266 (Exclusion of the exceptional sextic deletion process). *All graphs are finite and simple, and vertex deletion is induced. Write $V(G)$, $E(G)$, $\Delta(G)$, $\nu(G)$, and $\alpha(G)$ for vertex set, edge set, maximum degree, matching number, and independence number. A graph is well-covered if every inclusionwise maximal independent set has size $\alpha(G)$, and factor-critical if deleting any one vertex leaves a perfect matching. For nonnegative integer capacities a_c, b_c, let $c_{a_c,b_c}(G)$ be the least nonnegative integer T such that every nonnegative integer split $x + y = T$ admits a partition of $E(G)$ into four parts with maximum degrees at most a_c, b_c and matching numbers at most x, y, respectively. Define $c_i(G) = \max_{a_c+b_c=i} c_{a_c,b_c}(G)$ and $ps(G) = |E(G)| - \nu(G) - \sum_{i\geq 1} c_i(G)$, over nonnegative integer indices and capacities.*

There is no connected, six-regular, factor-critical, well-covered graph F satisfying $|V(F)| = 27$, $\alpha(F) = 9$, $c_0(F) = 17$ and the following two conditions.

(a) For every maximum independent set S of F,

$c_i(F - S) = c_i(F) - 9$ for every integer $1 \leq i \leq 5$,

and $c_0(F) = 9 + \nu(F - S) + ps(F - S)$.

(b) A complete process chooses any such S, then repeatedly deletes a vertex of current maximum degree whenever that degree is at least three, stopping at a graph of maximum degree at most two. In every complete process, every deletion parent H of current maximum degree D has a matching covering all its degree-D vertices, and deleting the chosen vertex lowers c_i by exactly one for every integer $1 \leq i < D$. Every terminal of every such complete process has zero slack ps.

All orders, degrees, cardinalities, capacities, budgets, costs, row indices, and process lengths are nonnegative integers. The conditions concern the actual original graph and every indicated complete process; no independence-preserving terminal or proper-subgraph minimality is assumed.

Proof. Use Lemma 193 for the actual sextic rows and their costs, Lemma 57 for the classification of the actual terminal, Lemma 145 for the complete missing-budget set at the exceptional parameters, Lemma 221 for a triangle in a zero-deficit row, Lemma 60 for successive matchings on the unequal rows, and Lemma 15 for full restoration of their odd-cycle remainder.

Suppose F exists. Factor-criticality gives $\nu(F) = 13$, since deleting any vertex leaves a matching of that size and twenty-seven vertices allow no larger one. Put $k = 13$, $a = 9$ and $E = c_0(F) - k = 4$. The hypotheses (a), (b) are exactly those of Lemma 193. Its parameters specialize to

$\lambda = 2E - 3 - (k - a) = 1$, $p = E - 2 - \lambda = 1$,

$\delta = c_0(F) - c_2(F)$, $\epsilon = c_0(F) - c_1(F)$,

$0 \le \epsilon \le \delta \le 2$,

$r = 11 + \delta$, $s = 5 - 2\delta$, $\beta = 1 + \delta - \epsilon$, $u = 16 - \delta$.

It gives, on the original graph,

$c_1(F) = 17 - \epsilon$, $c_2(F) = 17 - \delta$,

$c_3(F) = r$, $c_4(F) = 10 + \epsilon$, $c_5(F) = 9$.

In particular $c_5(F) \ge \alpha(F)$. All hypotheses of Lemma 145 hold. Its integer $p = k - E + 1 - a$ is one, so its exceptional alternative applies and the entire missing-pair set at total sixteen is exactly {(8,8)}.

Choose a complete process, which exists by finite deletion, and keep all its choices fixed. Let A be its early row formed from the initial independent set and the complete degree-five and degree-four phases. Let R be its terminal and X its remaining cubic root set. Write $\tau = 9 - \alpha(R)$. The row theorem gives one actual partition

$V(F) = V(A)$ disjoint union $V(R)$ disjoint union X,

$|V(A)| = r$, $|V(R)| = r + \tau$, $|X| = s - \tau$, $0 \le \tau \le s$.

The set X is independent and each of its vertices has exactly three R-neighbors. The native row A has components among

P_1, P_2, P_3, P_4, P_6, C_3, C_4, C_5, C_7,

where subscripts give vertex counts. It has matching number $m = 1 + \delta$, one odd-cycle component, exactly ϵ positive even-order path components, zero slack, $c_1(A) = \beta$ and $c_2(A) = 1$.

The actual terminal R has $\Delta(R) \le 2$ by the stopping rule and $ps(R) = 0$ by (b). Its numerical profiles from Lemma 193 are $\nu(R) = m + \tau$, $c_1(R) = \beta + \tau$ and $c_2(R) = 1 + \tau$. Apply Lemma 57 to this same R. It gives precisely the displayed nine-component catalogue for R, exactly $c_2(R) = 1 + \tau$ odd-cycle components, and $\nu(R) - c_1(R) = m - \beta = \epsilon$ positive even-order path components. This verifies the terminal's component conditions without asserting that R is native.

The full complement $J = F - V(A)$ satisfies

$\Delta(J) \le 3$, $ps(J) = 0$,

$\nu(J) = c_1(J) + \epsilon = 6 - \delta$, $c_2(J) = 6 - 2\delta$.

There is exactly one whole odd cycle C in A. Write its order as 2ell-1. The component catalogue gives an integer $2 \le \ell \le 4$.

If $\epsilon = 0$, the original costs become $c_0(F) = c_1(F) = 17$ and $c_2(F) = 17 - \delta$. The preceding A and J identities give every row and complement hypothesis of Lemma 221, with the same missing pair (8,8), order twenty-seven, and independence number nine. That theorem makes C a triangle, so $\ell = 2$.

If $\epsilon \ge 1$, the cycle contributes ell-1 to $\nu(A)$, and each of the ϵ positive even-order paths contributes at least one. Matching numbers add over the row components, so

$\ell - 1 + \epsilon \le \nu(A) = 1 + \delta$.

Thus $\ell \le 2 + \delta - \epsilon \le 4 - \epsilon$. In particular $\epsilon = 1$ implies $\ell \le 3$, and $\epsilon = 2$ implies $\ell \le 2$.

Let Z be the set of vertices having degree two within A. Choose a maximum independent set D_0 in the induced graph A[Z]. Such a choice exists because the graph is finite. Its size is β. Indeed the contributions to this independence number from

P_1, P_2, P_3, P_4, P_6, C_3, C_4, C_5, C_7

are respectively

0, 0, 1, 1, 2, 1, 2, 2, 3.

For a path these are the independence numbers of its induced internal path: the nonempty cases are a single vertex, an edge, and a four-vertex path. For a cycle all its vertices are eligible and alternating vertices attain half its order rounded down, which is also an upper bound by the successor injection. Each displayed contribution is that component's matching number minus one for a positive even-order path and minus zero otherwise. Summing gives $|D_0| = \nu(A) - \epsilon = \beta$. Since different row components have no edges between them, maximal cardinality also forces D_0 to be maximum on C, with $|D_0 \text{ intersect } V(C)| = \ell - 1$.

Put $Y = D_0$ union $V(C)$, $b = |Y|$, $N = r - b$. Every Y-vertex has A-degree two. The cycle is a whole component and D_0 is independent off it, so the entire induced edge set $E(A[Y])$ is precisely $E(C)$. Moreover

$b = \beta + \ell$, $N = 10 + \epsilon - \ell$.

If $\epsilon = 0$, then $\ell = 2$ and $N = 8$. If $\epsilon = 1$, then $\ell \leq 3$ and $N \geq 8$. If $\epsilon = 2$, then $\ell \leq 2$ and $N \geq 10$. These exhaust the allowed ϵ values. Thus $N \geq 8$ and $N \geq 0$. The other matching margin is

$N - (s + \beta + b - 2) = 5 + 3\epsilon - 2\ell$.

This is one when $\epsilon = 0$, at least two when $\epsilon = 1$, and at least seven when $\epsilon = 2$. Therefore

$\min(8, 8) \leq N$, $N \geq \max(0, s + \beta + b - 2)$.

Apply Lemma 60 to this actual partition, the selected Y, and the missing pair (8,8). Its row-order and component hypotheses, common even-path count, increased matching and odd-cycle counts, and independent X with three R-neighbors were all supplied above by the single fixed complete process. Its missing-total identity is

$8 + 8 + 1 = r + s + m = 17$.

Its original-cost lower bounds hold with equality:

$c_1(F) = r + s + \beta$, $c_2(F) = r + s + 1$, $c_3(F) = r$.

The two required N inequalities have just been verified. Consequently on the full induced graph

$H = F[V(R) \text{ union } X \text{ union } Y] = F - (V(A) minus Y)$

there exist four successive maximum matchings M_0, M_1, M_2, M_3 in their actual successive spanning edge residuals. They are pairwise edge-disjoint, and after their deletion the entire residual is the same cycle C together with isolated vertices. This conclusion permits $\tau = s$ and an empty X; it does not require $\tau = 0$.

The omitted set has exactly N vertices. Use Lemma 15 with $D = 4$ prescribed matchings, $g = 0$ single-edge components and this cycle of order 2ell-1. Since $N \geq 8$ and $\ell \leq 4$, its surcharge is zero: $N + g \geq \ell - 2$. It supplies, for every nonnegative integer capacity split $a_c + b_c = 4$ and every nonnegative integer budget split of total $N + \ell - 1$, a certificate of the original F. Every original edge incident with an omitted vertex is restored, including edges with both endpoints omitted. Hence the same original cost satisfies

$c_4(F) \leq N + \ell - 1 = 9 + \epsilon$.

This contradicts $c_4(F) = 10 + \epsilon$. Therefore no F satisfying the printed hypotheses exists. □

Remark. The actual terminal's zero-slack classification supplies its component conditions explicitly. At the exceptional numerical point every allowed early row admits a short-cycle selection with both required reserves, and four compatible matchings leave a cycle that saves one certificate unit.

Depends on. Lemma 15, Lemma 57, Lemma 60, Lemma 145, Lemma 193, Lemma 221.

Lemma 267 (Matching allocations in a zero-slack exterior). *Use the finite integer certificate costs $c_i(H)$, matching number $\nu(H)$, excess $ex(H) = c_0(H) - \nu(H)$, slack $ps(H)$, signed $gap(H) = ex(H) - ps(H)$, and top missing-allocation run $tr(H)$ of Lemma 23. All graphs are finite and simple. Vertex deletion is induced.*

For any nonnegative integer l, let $Y_l(H)$ be the set of integers x in [0,l] such that all edges of H can be partitioned into two parts whose matching numbers are at most x and l-x. Let $run_l(Y)$ be the largest cardinality of a consecutive integer interval contained in [0,l] and disjoint from Y, or zero when no such interval exists.

Suppose every induced subgraph J of H, including H itself, satisfies $gap(J) \leq 1$, and $ps(H) = 0$. Then

$0 \le ex(H) \le 1$, *and* $run_{(}\nu(H))(Y_{(}\nu(H))(H)) \le 2$.

If $ex(H) = 0$, *then* $Y_{(}\nu(H))(H) = [0, \nu(H)]$. *If* $ex(H) = 1$, *then this missing-run length equals* $tr(H)$. *If also* $\Delta(H) \ge 3$, *then* $ex(H) = 1$ *implies* $tr(H) = 1$.

If $\Delta(H) \ge 3$, *a matching covers every vertex of degree* $\Delta(H)$, *and* $c_{(}\Delta(H))(H) = 0$. *The empty graph, isolated vertices, zero matching number, and zero excess are permitted. All orders, degrees, costs, matching cardinalities, totals, budgets, interval endpoints, and run lengths are integers in the stated nonnegative domains.*

Proof. The definitions and nonnegative excess from Lemma 23 give

$0 \le ex(H) = gap(H) \le 1$,

since $ps(H) = 0$ and the hereditary gap hypothesis includes H. If $ex(H) = 0$, then $c_0(H) = \nu(H)$; the defining universal threshold gives every allocation at total $\nu(H)$, proving all allocation claims in that case. Empty and edgeless graphs belong to this case.

The exposure bound Lemma 243 applies when $D = \Delta(H) \ge 3$: every proper induced subgraph of H has gap at most one by the stated hereditary assumption. Its minimum number of degree-D vertices missed by a matching is at most $ps(H) = 0$. The finite matching family has an attaining member, so such a matching covers every degree-D vertex. The same fact gives $c_D(H) = 0$. This proves the covering conclusion independently of the allocation induction.

We prove by strong induction on $n = |V(H)|$ that, under the stated hypotheses and $ex(H) = 1$, $tr(H) \le 2$, and that $tr(H) \le 1$ when $\Delta(H) \ge 3$. The assertion is empty at $n = 0$, since the empty graph has zero excess. For any graph with $ex(H) = 1$ one has $c_0(H) - 1 = \nu(H)$, so its top run is exactly $run_{(}\nu(H))(Y_{(}\nu(H))(H))$.

If $\Delta(H) \le 2$, the exact degree-two allocation theorem Lemma 103 applies. Its positive-gap conclusion gives $tr(H) + ex(H) \le 3$, hence $tr(H) \le 2$. This low-degree application does not need connectedness.

It remains to consider $D = \Delta(H) \ge 3$. Choose any vertex v of degree D and put $K = H - v$. The preceding covering conclusion and the exact deletion inequality in Lemma 61 give, with $h = \nu(H) - \nu(K)$ in {0,1},

$0 = ps(H) \ge ps(K) + 1 - h \ge 1 - h \ge 0$.

Here $ps(K) \ge 0$ is also supplied by Lemma 61. Therefore $h = 1$ and $ps(K) = 0$. Every induced subgraph of K is an induced subgraph of H, so K inherits the entire hereditary gap hypothesis.

Put $m = \nu(K)$, so $\nu(H) = m + 1$. Since $ex(H) = 1$,

$c_0(H) = m + 2$.

Vertex restoration in Lemma 72 gives $c_0(H) \le c_0(K) + 1$, hence $c_0(K) \ge m + 1$. On the other hand,

$ex(K) = gap(K) \le 1$

because $ps(K) = 0$ and K satisfies the hereditary hypothesis. Thus $c_0(K) = m + 1$ and $ex(K) = 1$. The induction hypothesis applies to K, which has n-1 vertices, and gives

$run_m(Y_m(K)) = tr(K) \le 2$.

Every x in $Y_m(K)$ yields both x and $x + 1$ in $Y_{m+1}(H)$. To obtain the first, retain its edge partition on K and assign every edge incident with v to the second part. These added edges form a star. An arbitrary matching in their union with the old second part contains at most one added edge, so the new bounds are x and $m - x + 1$. For the second allocation put the entire star in the first part, giving bounds $x + 1$ and m-x. Every original edge is assigned exactly once. Therefore

$Y_{m+1}(H)$ contains $Y_m(K)$ union $\{x + 1 : xinY_m(K)\}$.

Both endpoints 0, $m + 1$ belong to $Y_{m+1}(H)$, by putting every edge in one part, since the other budget is $\nu(H) = m + 1$. Let [a, b] be any nonempty consecutive missing interval at this total. Then $1 \le a \le b \le m$. The displayed inclusion shows that every integer in [a-1,b] is missing from $Y_m(K)$: the interval [a, b] is excluded by the unshifted set, and [a-1,b-1] by the shifted set. The interval [a-1,b] lies within [0,m] and has cardinality $b - a + 2$. Consequently

$b - a + 2 \le run_m(Y_m(K)) \le 2$,

so $b - a + 1 \le 1$.

This proves $tr(H) \leq 1$ in the induction step and hence completes the induction at all orders.

Finally, when $ex(H) = 1$, the total $c_0(H) - 1$ is not universally feasible by the definition of c_0. Thus $tr(H) \geq 1$, and the strengthened bound at $\Delta(H) \geq 3$ is exactly $tr(H) = 1$. Combining this with the $ex(H) = 0$ case proves all assertions. □

Remark. A covered maximum-degree deletion preserves zero slack and lowers the matching number by one. Restoring its star fills one position in every missing allocation interval. Induction therefore reduces the largest possible gap to the explicitly known low-degree cases.

Depends on. Lemma 23, Lemma 61, Lemma 72, Lemma 103, Lemma 243.

Lemma 268 (Density at small slack below degree eight). *Use the finite integer certificate costs c_i, matching number ν, independence number α, maximum degree Δ, slack ps and signed gap of Lemma 242. All graphs are finite and simple. For a graph J put*

$f(J) = |V(J)| - 2\nu(J)$,

$B_7(J) = -4|V(J)| + 11\nu(J) + 4\alpha(J)$,

$E_7(J) = e(J) - B_7(J)$.

A graph is factor-critical when deleting any vertex leaves a perfect matching.

Let H have maximum degree at most seven, and suppose every induced subgraph of H, including H itself and the empty graph, has gap at most one. Every induced subgraph J of H has the following properties.

(i) If $ps(J) = 0$, then $e(J) \leq B_7(J)$, $c_7(J) = 0$, and at every parent of every complete maximum-degree vertex-deletion process stopped at maximum degree at most two, a maximum matching covers all vertices of the current maximum degree and the current slack remains zero.

(ii) If $ps(J) = 1$ and $\alpha(J) - f(J) \leq 5$, then $E_7(J) \leq 11$. If $ps(J) \leq 1$, then $c_7(J) = 0$.

(iii) If J is connected, nonregular, factor-critical and $ps(J) = 1$, then

$e(J) \leq 3\nu(J) + 4\alpha(J) + 6$.

(iv) Let t be the number of whole connected components Q of J which are factor-critical and satisfy $\Delta(Q) \geq 3$. Then $t \leq ps(J)$.

Here nonregular means that not all vertex degrees are equal. A complete process deletes any vertex of current maximum degree while that degree is at least three, and stops when that degree is at most two. All orders, degrees, indices, cardinalities, costs, matching and independence numbers and t are nonnegative integers; E_7 and gap may be signed integers.

Proof. Every induced subgraph of an induced subgraph of H remains induced in H. Thus all hereditary gap bounds below include each graph being considered and every one of its induced descendants.

We first establish the needed local coverage facts. If J has $ps(J) = 0$ and actual maximum degree $D \geq 3$, Lemma 243 gives a maximum matching covering all degree-D vertices. At $D = 3$ it requires no hereditary hypothesis; at $D \geq 4$ the required proper-induced gap bound follows from the preceding paragraph. If instead J is connected, $ps(J) \leq 1$, $D \geq 3$ and no matching covers all degree-D vertices, its actual degree is four or six and J is regular, factor-critical and has slack one. Indeed, $D = 3$ is impossible because $2\kappa \leq ps(J) \leq 1$ in Lemma 243 contradicts its positive integer exposure κ. At $D = 5$, Lemma 222 supplies coverage. At $D = 7$, Lemma 240 supplies coverage. These cases exhaust the odd degrees from three to seven. At $D = 4$ or six, the even-degree conclusion of Lemma 243 identifies a unique uncovered whole regular factor-critical component of slack one, which is all of J because J is connected. Also Lemma 240 with degree bound seven gives $c_7(J) = 0$ for every J with $ps(J) \leq 1$, even if its actual maximum degree is smaller.

Now let $ps(J) = 0$ and fix any complete maximum-degree deletion process. At a parent K of slack zero and maximum degree $D \geq 3$, the preceding coverage assertion applies. If the process chooses v, put $h = \nu(K) - \nu(K - v)$, which belongs to {0,1}. The covered-degree inequality in Lemma 61 gives

$ps(K) - ps(K - v) \geq 1 - h \geq 0$.

Nonnegative slack from that same fact forces $ps(K - v) = 0$. Induction proves coverage and zero slack at every parent of every allowed process. Such a process exists and terminates because each deletion removes one vertex. Apply Lemma 146 at degree bound seven to this process. Its nonnegative density identity gives $e(J) \le B_7(J) - 6b \le B_7(J)$, where $b = \nu(J) - c_1(J) \ge 0$. Together with the already proved $c_7(J) = 0$, this proves (i).

For any fixed induced J, each connected induced descendant with maximum degree at least three and slack at most one satisfies the uncovered-component implication established above. Its descendants also have gap at most one. These are precisely the two hereditary premises of Lemma 242. Applying that fact to J proves $E_7(J) \le 11$ whenever $ps(J) = 1$ and $\alpha(J) - f(J) \le 5$. The previously established seventh-row assertion completes (ii).

Suppose J has the hypotheses of (iii), and put $r = \nu(J)$, $a = \alpha(J)$. Factor-criticality gives $|V(J)| = 2r + 1$ and $\nu(J - v) = r$ for every vertex v. Its actual maximum degree D is at least three. To see this, a connected graph of maximum degree at most two is a path or a cycle, or a singleton. A factor-critical cycle has odd order and is regular; a singleton is regular. A path of order at least two is not factor-critical: its order would have to be odd, and deleting its second vertex leaves its first vertex isolated and at least one other vertex. Thus a connected nonregular factor-critical graph cannot have maximum degree at most two.

If J had no matching covering its maximum-degree vertices, the local implication proved above would make J regular, contrary to its hypothesis. It therefore has such a matching. Choose any degree-D vertex v and put $K = J - v$. Here the matching-number loss is zero. The covered-degree inequality in Lemma 61 gives $ps(J) - ps(K) \ge 1$, so $ps(K) = 0$. Part (i) gives

$e(K) \le -4(2r) + 11r + 4\alpha(K) = 3r + 4\alpha(K) \le 3r + 4a.$

Since $D \le 7$, it follows that $e(J) \le 3r + 4a + 7$.

If equality held, we would have $D = 7$, $\alpha(K) = a$ and $e(K) = B_7(K)$. In the refined density inequality of Lemma 146, this forces $b = 0$ and equality. Its equality statement, applied to any complete covered process supplied by (i), gives an independent set T of exactly $2\nu(K) - |V(K)| + \alpha(K) = a$ deleted vertices, each of original K-degree seven. Their degrees in J are at most seven. Hence no vertex of T is adjacent to v. The set T union {v} is then independent in J and has $a + 1$ vertices, contradicting $a = \alpha(J)$. Equality is impossible, and integrality gives $e(J) \le 3r + 4a + 6$. This proves (iii).

Finally, consider a component Q counted in (iv). If $ps(Q) = 0$, the local coverage assertion applies at its actual maximum degree, which is at least three. Deleting a maximum-degree vertex has matching-number loss zero by factor-criticality. The covered-degree inequality would therefore give $ps(Q) - ps(Q - v) \ge 1$, contradicting nonnegative slack. Thus every counted component has $ps(Q) \ge 1$. Lemma 72 gives $ps(J) \ge \sum_Q ps(Q)$, summing over all whole components, and Lemma 61 makes all these summands nonnegative. The t counted components contribute at least t, giving (iv). □

Remark. Low slack supplies the matching coverage needed to apply the exact density identity. Equality after a factor-critical vertex deletion would create an independent set that can be enlarged by the deleted vertex.

Depends on. Lemma 61, Lemma 72, Lemma 146, Lemma 222, Lemma 240, Lemma 242, Lemma 243.

Lemma 269 (Closed lumps of degree fourteen are LP-good). *Use the finite integer certificate row costs $c_i(G)$ of Lemma 87, and let $e(G)$ and $\nu(G)$ be the edge count and matching number of G. Put $ex(G) = c_0(G) - \nu(G)$ and $gap(G) = \sum_{i \ge 0} c_i(G) - e(G)$. A graph is LP-good when its gap is nonpositive. Let H be any nonempty connected 14-regular finite simple graph such that deleting every vertex leaves a perfect matching. Then $gap(H) \le 0$. No hereditary, minimality, excess, top-run, or upper-order hypothesis is imposed. All orders, degrees, costs, capacities, budgets, matching sizes and indices are integers in their stated nonnegative or positive domains; gaps may be signed.*

Proof. A perfect matching after a vertex deletion shows that $|V(H)| = 2k + 1$ and $\nu(H) = k$ for an integer $k \ge 0$. Simplicity and 14-regularity imply $2k \ge 14$, so $k \ge 7$. Suppose for a contradiction that $gap(H) > 0$. Lemma 184 excludes every integer $77 \le k \le 147$, and Lemma 182 excludes every $k \ge 148$. Hence $7 \le k \le 76$. Lemma 235 then gives $k = 17$, $ex(H) = 6$ and $gap(H) = 1$. In particular $|V(H)| = 35$ and $c_0(H) = 23$. We will construct better upper rows at this single remaining order.

First H has a dominating set of at most six vertices. Recall that a set dominates if every vertex outside it has a neighbor in it. Suppose i vertices have already been chosen and u vertices remain undominated. Each undominated vertex has a closed neighborhood of size fifteen, disjoint from all the i chosen vertices. Count incidences between these u closed neighborhoods and the 35-i available vertices. Some available vertex lies in at least $ceil(15u/(35-i))$ of the neighborhoods. Choosing it leaves at most

$u - ceil(15u/(35-i)) = floor((20-i)u/(35-i))$

undominated vertices. Starting from $u = 35$, after four choices the successive upper bounds are 20, 11, 6, 3. If the construction ends sooner, its chosen set already dominates.

If at most two vertices remain undominated after the fourth choice, choosing those vertices finishes with at most six choices. Otherwise exactly three remain. Their three closed neighborhoods, each of size fifteen, are subsets of the remaining 31 vertices. Since $45 > 31$, two of these neighborhoods intersect. Choose a vertex in their intersection; it dominates at least two of the three, and a sixth choice can dominate the remaining vertex if needed. Thus $\gamma(H) \le 6$, where γ denotes minimum dominating cardinality.

Choose a minimum dominating set Z, of cardinality $z \le 6$. We apply Lemma 151 with $d = 14$, $k = 17$, $E = ex(H) = 6$ and $e = E - 1 = 5$. Its parameter $t_0 = floor((2e-1)/(d-2))$ equals 0. The three requirements in condition (i) hold:

$15 \ge 2floor((17+5)/2) - 4 \cdot 5 - 1 = 1,$

$6 \cdot 5 + 2 \cdot 0 + 4 = 34 \le 35 + 14 = 49,$

$0 \le \lceil 34/14 \rceil.$

For condition (ii), its denominator is $a = z + d - 3 - 2e = z + 1$, so $1 \le a \le 7$. Consequently

$14ceil(13/a) \ge 28 > 10 + z = 35 + z - 4 \cdot 5 - 5,$

which gives its second alternative. Condition (iii) is $6 \cdot 5 + 2 \cdot 0 + 14 + 8 = 52 > 35$. Condition (iv) is $8 \cdot 5 + 4 \cdot 0 + 11 = 51 > 35 - z$. Every condition holds for this actual H and Z, so the cited theorem gives

$c_{13}(H) \le z \le 6.$ (1)

We now obtain $c_{12}(H) \le 10$ by a fixed-cardinality construction. Choose an eight-element subset U of $V(H)$ uniformly from all such subsets. For each vertex v outside U put $X_v = |N_H(v)$ intersect $U|$. Let D be the set of vertices outside U with $X_v \le 1$, let T′ be those with $X_v = 2$, and put $W = U$ union D. Every vertex outside W has at least two neighbors in U, so $\Delta(H - W) \le 12$. A vertex of degree exactly 12 in H-W has exactly two H-neighbors in W. It is outside D and hence has at least two neighbors in U; since U is contained in W, it therefore belongs to T′. The deletion inequality in Lemma 92, with $\theta = 12$ and $m_\theta = 13$, gives for every outcome

$c_{12}(H) \le 8 + |D| + \lfloor |T'|/13 \rfloor$

$= \lfloor 8 + |D| + |T'|/13 \rfloor.$ (2)

Conditional on v not belonging to U, the set U is a uniform eight-element subset of the other 34 vertices. Among those vertices, v has 14 neighbors and 20 nonneighbors. Hence, for $i = 0, 1, 2,$

$Pr(X_v = i | vnotinU) = binom(14,i)binom(20,8-i)/binom(34,8).$

Also $Pr(vnotinU) = 27/35$. Linearity of expectation, with no independence between different vertices required, gives

$E[8 + |D| + |T'|/13]$

$= 8 + 27[binom(20,8) + 14binom(20,7) + (91/13)binom(20,6)]/binom(34,8)$

$= 2408627286/236030652 < 11,$

where $binom(34,8) = 18156204$ and

$11 \cdot 236030652 - 2408627286 = 187709886 > 0.$

The sample space is finite and nonempty, so some actual eight-set U makes the quantity inside the expectation smaller than eleven. For its associated W, (2) yields the integer bound

$c_{12}(H) \le 10.$ (3)

The parity deletion inequality supplies actual certificates, so (2)-(3) refer to the original integer row costs.

Every zero-capacity certificate remains a certificate when the degree capacities are increased, since its two degree-bounded edge parts are empty. Thus the definition of the row costs gives $c_1(H), c_2(H) \le c_0(H) = 23$. Lemma 87 gives $c_i(H) = 0$ for $i \ge 15$, $c_{14}(H) \le 1$, and the bounds

$c_4(H) + c_{10}(H) \le 36$, $c_6(H) + c_8(H) \le 36$.

By Lemma 199, the two odd complementary pairs $3/11$ and $5/9$ have sums at most 35: the order 35 is at most $6 \cdot 14 + 15$. The same fact gives $c_7(H) \le 17$ because $14/2 = 7$ and $k = 17$ are both odd. Combining these disjoint rows with (1)-(3) gives

$\sum_{i \ge 0} c_i(H)$

$\le 3 \cdot 23 + 6 + 10 + 2 \cdot 35 + 2 \cdot 36 + 17 + 1$

$= 245$.

Regularity gives $e(H) = 14 \cdot 35/2 = 245$. Hence $gap(H) \le 0$, contradicting the assumed positive gap. The three order ranges exhaust every admissible integer k, proving the theorem. □

Remark. All earlier order bounds leave one possible positive-gap row. A fixed eight-vertex sample saves a unit at level twelve, while a six-vertex dominating set controls level thirteen and closes that row.

Depends on. Lemma 87, Lemma 92, Lemma 151, Lemma 182, Lemma 184, Lemma 199, Lemma 235.

Lemma 270 (Matching coverage from a large slack and a small independent set). *Use finite integer certificate costs c_i, matching number ν, independence number α, maximum degree Δ, excess ex, slack ps, $gap = ex - ps$, and top missing-allocation run tr from Lemma 23. Put $h(j) = \max(4, \lfloor j/2 \rfloor + 1)$ for nonnegative integers j. Property T_- means that positive gap implies gap one and $tr + ex \le h(\Delta)$.*

Let H be a finite simple graph, W a vertex subset, $w = |W|$, and $R = H - W$ an induced subgraph. Suppose $D \ge 5$ is odd, $\Delta(R) = D$, $w \ge h(D) - 1$, $ex(R) = ps(R) = s \ge 4$, $c_0(H) = w + \nu(R) + s$, $|V(R)| < 4w + 6s + 7$, and $\alpha(R) \le w + 1$. Suppose every induced subgraph of R, including R, satisfies T_-. Assume in addition that either $s \ge h(D) - 1$, or every induced subgraph Q of R with $ex(Q) \ge 4$ has $gap(Q) \le 0$.

Then a matching of R covers every vertex of degree D, and $c_D(R) = 0$. In particular the extra alternative is automatic for D in {5,7,9}, or for $D = 11$ and $s \ge 5$. All graph orders, degrees, cardinalities, costs, matching numbers, and parameters D, w, s are integers in the displayed nonnegative or positive domains.

Proof. Suppose R has no matching covering its degree-D vertices. Apply Lemma 233 with the displayed H, W, R, D, w, s. It supplies a whole component C, the graph $P = R - V(C)$, and a maximum-cardinality marked barrier X of C, with $b = |X| \ge 1$. The graph C-X consists of exactly $b + 1$ odd factor-critical components. Choose one, Z, of largest matching number m. Every vertex of Z has degree D in C, its only external edges go to X, and exactly one such edge exists. Thus $\sum_(v \in V(Z))(D - deg_Z(v)) = 1$. Also $ex(Z) = ps(Z) = s$ and

$m \ge w + b + \nu(P) + 3s - 1$. (1)

Choose a maximum independent set S of Z, set $a = |S|$, and put $J = Z - S$ and $q = c_{D-1}(J)$. Such S exists and is nonempty since Z is a nonempty finite graph. The unconditional part of Lemma 209 gives $\Delta(J) \le D - 1$. We check its additional local condition $q \le ps(J)$. If $\Delta(J) < D - 1$, the empty matching covers the empty set of degree-(D-1) vertices. By Lemma 61, $q = 0 \le ps(J)$. If $\Delta(J) = D - 1 \ge 4$, every proper induced subgraph of J is an induced subgraph of R and hence has gap at most one by T_-. Apply Lemma 243 to J at its actual degree D-1. It gives $q \le ps(J)$.

We may therefore use the conditional rank bound of Lemma 209 with this same S and J. In its notation $E = ex(Z) = s$, the indicator t is at most one, and η, ρ are nonnegative. Hence

$m - a \le 2s + E + t - 2 - \eta - \rho \le 3s - 1$. (2)

For each of the other b components of C-X choose one vertex. Their union with S is independent in R, since distinct components of C-X have no edges between them and R adds no edges between its whole components. Consequently $a + b \le \alpha(R) \le w + 1$. Combining this with (2) gives $m \le w + 3s - b$. Together with (1), this yields $2b + \nu(P) \le 1$, impossible because $b \ge 1$ and $\nu(P) \ge 0$.

Thus a covering matching exists in R. The zero-top-row conclusion of Lemma 61 gives $c_D(R) = 0$. The final degree specializations follow from $h(5) - 1 = h(7) - 1 = 3$, $h(9) - 1 = 4$ and $h(11) - 1 = 5$. □

Remark. The carrier forces a component whose matching number is too large for the available independent set. The local top-row bound applies to the same independent cut and makes the two rank inequalities incompatible.

Depends on. Lemma 23, Lemma 61, Lemma 209, Lemma 233, Lemma 243.

Lemma 271 (Avoidance at the first odd-degree slope). *Let $s, t \geq 1$ be integers and let F_1, F_2 be the star forests with full positive nonincreasing integer edge-size lists $n_1,\ldots, n_s$ and $m_1,\ldots, m_t$. Prescribe F_1 red and F_2 blue; copies are ordinary subgraphs with vertex-disjoint target components. Put $M = s + t - 1$ and $\ell_k = max\{n_i + m_j - 1 : 1 \leq i \leq s, 1 \leq j \leq t, i + j = k\}$ for every integer $2 \leq k \leq s + t$. Fix an integer $0 \leq q < M$. Let A_q contain every ordered nonempty actual suffix pair after leading cuts α, β with $0 \leq \alpha < s$, $0 \leq \beta < t$ and $\alpha + \beta = q$. Put $K = M - q$, $P_q = \sum_{k=q+2}^{s+t}(\ell_k - 1)$, $D = \ell_{q+2} - 1$, and let R be the initial plateau length at $D + 1$ in this natural tail. Terminal maximizing indices are valid original indices i_0, j_0 with $i_0 + j_0 = q + R + 1$ and $n_{i_0} + m_{j_0} = D + 2$.*

(I) Suppose a terminal maximizing pair has unordered sizes $\{a, b\}$, with $a \geq 4$ even and $b \geq 3$ odd. Let $0 \leq r \leq a - 2$ be an integer. Every finite simple graph H satisfying $\Delta(H) \leq D$, $\Phi(H) := e(H) - \nu(H) \leq P_q + r$, $\delta := K - \nu(H) \geq r + 1$, where ν denotes maximum matching cardinality, has a coloring avoiding both entire prescribed targets of some member of A_q.

(II) In particular, if D is odd and a terminal maximizing pair has both sizes at least three, the same conclusion holds for every integer $0 \leq r \leq 2$ under the displayed degree, potential and deficit hypotheses.

Every asserted coloring extends over the original canonical q-star pool with capacities $\ell_2,\ldots, \ell_{q+1}$, and across any specified actual outer q-set X with $H = G - X$, to avoidance of the unchanged original targets while preserving every H-edge color. No short-prefix condition $q < \min(s, t)$, original-head bound, strict-cut condition, or smaller-target lower bound is assumed. All original units, repeated sizes and complete actual suffixes are retained, and the assertions are invariant under exchanging full target roles and colors. All list lengths, sizes, indices, cuts, degrees, matching numbers, deficits and allowances are integers in the stated domains.

Proof. For (I), exchange the full target roles and their prescribed colors if necessary so that $n_{i_0} = a$ and $m_{j_0} = b$. This preserves the full diagonal profile and every graph hypothesis. Suppose H arrows every member of A_q. Since $r \leq a - 2$, Lemma 218 applies with threshold allowance a-2. It gives $R = 1$ and $\Phi(H) = P_q + a - 2$, so the assumed potential bound forces $r = a - 2$. It also supplies one actual vertex y of degree at most D-1 and exactly a odd factor-critical components of H-y, each adjacent to y and wholly of ambient H-degree D.

The assumed deficit now satisfies $\delta \geq a - 1 \geq 3$. Apply Lemma 33 to this same H and entire original q-family. It gives $q < \min(s, t)$ and uncolored copies of both full original targets in H. Therefore their full heads are at most $\Delta(H) \leq D$. Their heads are also at least a and b respectively, because those terminal components occur in the full lists. This retains the original full forests without any change to q or the natural potential.

Here $D = a + b - 2$ is odd and at least five. If $D = 5$, the only permitted pair is $a = 4$, $b = 3$. The full original heads consequently belong to [3,5]. We have $r = 2$, $\delta \geq 3$, natural plateau one, and four required odd factor-critical ambient-degree-five components behind a vertex of degree at most four. Lemma 180 applies to the same H and original lists and supplies an avoiding A_q coloring, a contradiction.

If $D \geq 7$, put $p = a$. Then $r = p - 2$ and $\delta \geq p - 1$. When $b \geq 5$, both full original heads are at least four, so the first head regime of Lemma 181 applies. When $b = 3$, the inequality $D \geq 7$ forces $a \geq 6$, and both full heads are at least three, so its second regime applies with $p \geq 6$. In either case the same a actual components, same y, same original targets and same q satisfy every condition of that theorem. It yields an avoiding A_q coloring, again a contradiction. These cases exhaust the odd integer $D \geq 5$ and prove (I).

For (II), the terminal sum $D + 2$ is odd, so exactly one size is even and the other odd. Since both are at least three, the even size a is at least four and the odd size b at least three. Every integer $0 \leq r \leq 2$ satisfies $r \leq a - 2$. Thus (I) applies directly.

Each obtained coloring avoids an entire member with original legal cuts α, β summing to q. Lemma 41 extends it over the original canonical pool and across any specified actual q-set, preserving every H-edge

color. Its extension includes copies meeting deleted vertices only at leaves and permits all unit and repeated components. Undoing the simultaneous target-color exchange proves the assertions in the original orientation. □

Depends on. Lemma 33, Lemma 41, Lemma 180, Lemma 181, Lemma 218.

Lemma 272 (Whole odd cliques at an even natural degree). *Let $s, t \geq 1$ be integers and let $n_1 \geq \cdots \geq n_s \geq 1$ and $m_1 \geq \cdots \geq m_t \geq 1$ be integer lists with $n_1, m_1 \geq 4$. Let F_1, F_2 be their vertex-disjoint unions of stars, prescribed red and blue. Copies are ordinary subgraphs with vertex-disjoint target components. A finite simple graph arrows an ordered pair if every red-blue coloring of its edges contains the entire first target in red or the entire second target in blue.*

Put $M = s + t - 1$ and $\ell_k = max\{n_i + m_j - 1 : 1 \leq i \leq s, 1 \leq j \leq t, i + j = k\}$ for every integer $2 \leq k \leq M + 1$. For every integer $0 \leq z \leq M$ put

$P_z = \sum_{k=z+2}^{M+1} (\ell_k - 1)$,

where an empty sum is zero. For $0 \leq z < M$ let A_z contain all ordered pairs of nonempty suffix forests after deleting α red and β blue leading components, where α, β are nonnegative integers with $\alpha < s$, $\beta < t$ and $\alpha + \beta = z$.

Fix an integer $0 \leq q < M$ and suppose $D = \ell_{q+2} - 1$ is even and $D \geq 4$. Write $e(H)$, $\nu(H)$, $\Delta(H)$ for the edge count, maximum matching cardinality and maximum degree of H, and put $\Phi(H) = e(H) - \nu(H)$. There is no finite simple graph H which simultaneously arrows every member of A_q, satisfies $\Delta(H) \leq D$ and $\Phi(H) = P_q$, and has a whole connected component isomorphic to K_{D+1}, the complete graph on $D + 1$ vertices.

All indices, list sizes, component counts, degrees, edge counts and matching cardinalities are integers in the displayed domains. Original unit components, repeated sizes, $q = 0$ and isolated host vertices are allowed. No matching-deficit, lower-multiplicity or individual-envelope-attainment condition is imposed.

Proof. Orient the two full targets and their prescribed colors together so that the original red unit count u is at least the original blue unit count v. The hypotheses are symmetric under this exchange, and both original heads remain at least four. At any stage with the full natural A_z family, $\ell_{z+2} - 1 = 4$, maximum degree at most four and $\Phi = P_z$, fact Lemma 194 would give $4 \geq n_1 + 1 \geq 5$. Thus such a stage is impossible. This already proves the assertion when $D = 4$. Assume $D \geq 6$ and suppose the stated H exists.

Put $L = \sum_{k=2}^{M+1} \ell_k$. Take, disjoint from H, the canonical pool S_q of stars with sizes ℓ_2,..., ℓ_{q+1}. By Lemma 41 (I), $G = H$ disjoint union S_q arrows the full original pair. Edge count and matching cardinality are additive across disjoint components: every matching restricts to each component, and componentwise maximum matchings unite. Each pool star has matching cardinality one. Therefore

$\Phi(G) = P_q + \sum_{k=2}^{q+1} (\ell_k - 1) = L - M.$

Choose any maximum matching W of H and one edge in each pool star. Their union $W^{\cdot}$ is a maximum matching of G.

Apply Lemma 95 with the single original ordered pair and this prescribed $W^{\cdot}$. A legal step deletes a vertex of maximum degree in the complement of the surviving $W^{\cdot}$-edges. For every legal prefix of z deletions, let H_z be the remaining induced graph, let W_z be the $W^{\cdot}$-edges with both endpoints remaining, and let $J_z = H_z - W_z$, where only the matching edges are removed. That fact gives

W_z maximum in H_z, $\Phi(H_z) = e(J_z) = P_z$

for $0 \leq z \leq M$, and $\Delta(J_z) = \ell_{z+2} - 1$ for $z < M$, while $e(J_M) = 0$. Moreover H_z arrows every member of the full original A_z family when $z < M$, by Lemma 41 (II) applied to the actual z deleted vertices of G. All original lists therefore remain fixed.

The sequence ℓ_k is nonincreasing. A candidate on a diagonal after the first has an index greater than one; reducing that index gives a candidate on the preceding diagonal of at least the same size. Thus each of the first q pool centers, taken in order, has complement degree $\ell_k - 1$ at least D. Remaining pool leaves have degree at most one, and the matching complement in H has degree at most D. The q centers may consequently be selected first as legal greedy choices. Afterward H_q is H together with isolated pool leaves, and its prescribed matching is W. In particular $\Delta(H_q) \leq D$ and its specified whole clique persists.

We show how to decrease a positive even degree by two while preserving a whole odd clique. Suppose a legal stage $z < M$ has

$\Delta(H_z) \le d, \Delta(J_z) = \ell_{z+2} - 1 = d,$

where $d \ge 6$ is even, and has a whole component $C = K_{d+1}$. The restriction of W_z to a whole component must be maximum there; otherwise a larger component matching, united with its unchanged outside edges, enlarges W_z. A complete graph of odd order $d+1$ has a matching of size $d/2$ and no larger one, so precisely one vertex x of C is uncovered.

First delete every vertex of J_z of degree d. Such vertices are exactly the $W_z - uncovered$ vertices of $H_z - degree\ d$, since a covered vertex loses one edge from ambient degree at most d. Uncovered vertices are independent in H_z: an edge joining two of them could be added to W_z. They therefore keep complement degree d until they are selected. Deletions cannot create new vertices of that degree. All these selections are legal; only x is selected in C. The remaining whole clique is K_d, with a perfect restriction of $W^{\cdot}$. Its matching complement is regular of degree d-2. The global complement now has degree at most d-1 and at least d-2. None of these stages can reach M while this positive-degree clique complement persists, because $e(J_M) = 0$.

While the complement maximum is d-1, choose a covered vertex attaining it whenever one exists. Every such vertex lies outside K_d, so the clique and its matching persist. If the complement maximum were d-1 with no covered vertex attaining it, the current actual graph would have maximum degree at most d-1: covered vertices have complement degree at most d-2 and gain one matching edge; uncovered vertices have the same degree in both graphs. This stage still has exact potential P_w and arrows every A_w member. Since $\ell_{w+2} - 1 = d - 1 \ge 5$, fact Lemma 257 would force d-1 to be even, a contradiction.

Thus the covered choices can continue until the complement maximum is d-2. Finiteness guarantees termination: each choice removes a vertex, the clique is intact, and reaching M is impossible while its complement has an edge. At the resulting stage $w < M$, select any vertex y of the even clique K_d. It is covered, has complement degree d-2, and is a legal maximum-degree choice. Let y′ be its $W \cdot -partner$. The new whole clique $C' = K_{d-1}$ retains exactly that matching with y′s edge removed. Its sole uncovered vertex y′ has complement degree d-2, and its other vertices have complement degree d-3. The global complement maximum is still d-2.

Continue deleting covered vertices attaining complement degree d-2 as long as one exists. Each lies outside C′, since its only vertex of that complement degree is the uncovered y′. Thus C′ and y′ persist, and the complement maximum remains d-2. There are finitely many available vertices, and the process cannot reach M with y′ still of positive degree. It ends at a legal stage $z' < M$ with no covered vertex of complement degree d-2. Covered vertices then have actual degree at most d-2, and uncovered vertices also have actual degree at most d-2. Consequently

$\Delta(H_{z'}) \le d - 2, \Delta(J_{z'}) = \ell_{z'+2} - 1 = d - 2,$

$\Phi(H_{z'}) = P_{z'},$

and $H_{z'}$ arrows the complete $A_{z'}$ family and has the whole clique $C' = K_(d-2)+1)$. These are the same inductive conditions with d replaced by d-2. At every step the matching is the restriction of the one initially chosen $W^{\cdot}$, not a replacement matching.

The initial conditions hold at $z = q$ with $d = D$. Repeatedly apply the step for $d \ge 6$. Every application either gives the stated contradiction at an odd intermediate degree or decreases d by two while keeping all conditions. A finite number of such decreases reaches $d = 4$. The original heads are unchanged and at least four, so the degree-four contradiction in the first paragraph applies. This proves the theorem. □

Remark. Deleting the uncovered clique vertex leaves an even clique with its perfect matching. One later covered deletion restores an odd clique two orders smaller. Covered deletions outside it cannot destroy this witness, and the exact profile excludes odd intervening degrees.

Depends on. Lemma 41, Lemma 95, Lemma 194, Lemma 257.

Lemma 273 (Even-degree barrier absorption with small vertex pools). *Use the finite integer certificate costs c_i, matching number ν, independence number α, maximum degree Δ, excess $ex = c_0 - \nu$, slack ps, $gap = ex - ps$, and top missing-allocation run tr of Lemma 264. Put $h(j) = \max(4, \lfloor j/2 \rfloor + 1)$ for every nonnegative integer j. A graph satisfies T_- if positive gap implies gap one and $tr + ex \le h(\Delta)$. A graph is factor-critical if deleting any one vertex leaves a perfect matching.*

Let H be a finite simple graph, W any vertex subset, $w = |W|$, and $R = H - W$ the induced remainder. Suppose $D \ge 4$ is even, $\Delta(R) = D$, $1 \le s = ps(R) \le 9$, $w \ge 2$, $\alpha(R) \le w + 1$, and every induced subgraph of R, including R, satisfies T_-. Assume also that either $D \le 12$ or $w \ge h(D) - 1$.

Suppose U is a connected component of R having degree-D vertices which no matching of U covers simultaneously. Let A be those vertices, and let $q_A(U - X)$ count the odd-order connected components of U-X whose entire vertex sets belong to A. Suppose there exists a nonempty X of maximum cardinality among the sets maximizing $q_A(U - X) - |X|$. Then for every pair of nonnegative integers x, y with $x + y = w + \nu(R) + s - 1$, the entire edge set of H has a partition into two parts with matching numbers at most x, y. In particular $c_0(H) \le w + \nu(R) + s - 1$.

Consequently, under the hypotheses on H, W, R, D, s and the opposite bound $c_0(H) \ge w + \nu(R) + s$, every connected component of R whose degree-D vertices have no covering matching is a whole D-regular factor-critical graph. No independence of W, order bound, minimality, or sign assumption on $gap(R)$ is imposed. All degrees, orders, costs, matching numbers, component counts, indices and parameters are integers in the stated domains.

Proof. Use the integer costs and gap identities defined in Lemma 264 and Lemma 61. Restriction, vertex-set extension and component slack superadditivity are supplied by Lemma 72. Allocation sets, their missing-run criterion and actual-pool restoration are those of Lemma 75. Rows are nonincreasing: for any capacities $A + B = i \ge j$ choose $A' \le A$ and $B' \le B$ with $A' + B' = j$; every certificate for the smaller capacities is valid for A, B, so $c_i \le c_j$. In particular $c_i \le c_0$.

Fix the supplied U and X. Put $P = R - V(U)$, $p = ps(U)$, $b = |X|$, and let $\kappa \ge 1$ be the minimum number of A-vertices missed by a matching. Then $0 \le p \le s \le 9$ and $ps(P) \le s - p$. The maximum-cardinality maximizing set X is inclusionwise maximal, so Lemma 107 gives $q_A(U - X) = b + \kappa$, factor-criticality of all counted components, and the matching identity. Write calQ for all components of U-X, calK for the counted components, and Y for the vertex union of the other components. For K in calK put $d(K) = e_U(V(K), X)$, the number of edges to X. All its vertices have U-degree D, so $d(K)$ is its total internal degree-D defect. By Lemma 104 these boundaries are positive and even, their sum is at most Db, and

$\nu(U) = b + \sum_{Q \in calQ} \nu(Q)$, $c_D(U) \le \kappa$. (1)

There is a member of calK with $d(K) < D$, since otherwise their $b + \kappa$ boundaries total more than Db. Choose Z of largest matching number m among those members. Put $d = d(Z)$, $u = (D - d)/2$ and $\psi = ps(Z)$. Thus $2 \le d \le D - 2$.

For each required component K different from Z with $d(K) < D$, use Lemma 179 to assign its credits $a_i(K)$. Its factor-criticality was just established, and its gap is at most one by hereditary T_-. The total credit is max(D-d(K)-ps(K),0). It is supported on indices 1 through $T = (D - 2)/2$, with $a_1(K) \le T$ and $a_i(K) \le 1$ for $i \ge 2$. Each credit is at most $\max(c_i(K) - \nu(K), 0)$. Give zero credits to Z, to nonrequired components, and to required components with boundary at least D. Such a required component has required credit zero, since ps is nonnegative.

Perform the finite credit procedures of Lemma 104 at all indices $1 \le i < D$. Their initial interval has length at least $b + c_i(Z)$, and every positive-credit component has rank at most m. Let E_{rem} be the sum of maximum credits at unfinished indices. Initially $E_{rem} \le T + (T - 1) = D - 3$; this includes $D = 4$, $T = 1$. Set

$C = e(U[X]) + e_U(X, Y) + \sum_{Q not \in calK} ps(Q)$

$+ \sum_{K \in calK, K \ne Z} \max(ps(K) + d(K) - D, 0)$.

All terms are nonnegative. Substituting $gap = ex - ps$ into the exact retained-top inequality of that fact gives

$\psi + d + C + \sum_{Q \in calQ} c_D(Q) \le \beta + E_{rem}$, (2)

where

$\beta = p + c_D(U) - D(\kappa - 1) \le p + D - (D - 1)\kappa \le p + 1$.

Indeed its top correction is $\kappa - c_D(U) + \sum_Q c_D(Q)$; retain this entire correction when rearranging.

For the same Z, Lemma 179 gives $\psi \geq 1$, $|V(Z)| = 2m+1$, $e(Z) = Dm+u$, zero rows at and above D, and

$\psi = (D-1)m + u - \sum_{i=1}^{D-1} c_i(Z)$,

$c_i(Z) + c_{D-i}(Z) \leq 2m+1$ for $1 \leq i < D$,

$c_{D/2}(Z) \leq m$. (3)

Suppose the first-index procedure stops with a component Q remaining. Then $\nu(Q) \geq b + c_1(Z) + 2$, whereas $\nu(Q) \leq m$. Hence $c_1(Z) \leq m - b - 2$. Row monotonicity in (3) implies $\psi \geq u + (D-1)(b+2)$. But (2) implies $\psi \leq p + D - 2 - d$. The difference between these lower and upper bounds is at least

$u + (D-1)b + D + d - p \geq 2D + 2 - p \geq 10 - p > 0$,

using $u \geq 1$, $b \geq 1$, $d \geq 2$, $D \geq 4$ and $p \leq 9$. Thus index one finishes.

If any later index fails, let j be the first such index. Only $2 \leq j \leq T$ can fail, and the unfinished error is at most $T - j + 1$, since the earlier indices have finished and each later maximum credit is at most one. The failure again gives $c_j(Z) \leq m - b - 2$. In (3), pair indices i, D-i for $1 \leq i < j$. Each remaining one of the $D - 2j + 1$ indices is at least j and has cost at most $c_j(Z)$. Consequently

$\psi \geq u - j + 1 + (D - 2j + 1)(b+2)$.

Its upper bound from (2) is $\beta + T - j + 1 - d$. The difference equals

$1 + d/2 - \beta + (D - 2j + 1)(b+2) \geq 1 + 1 - (p+1) + 9 = 10 - p > 0$,

because $D - 2j + 1 \geq 3$. This is a contradiction. Every procedure therefore finishes, including all zero-credit indices, and (2) yields

$\psi + d + C \leq p + D - (D-1)\kappa \leq p + 1$. (4)

Put $\delta = \psi + d - 2$. It is an integer at least one, and (4) gives $\delta + C \leq p - 1$.

We next bound the rank of Z relative to an independent set, using the same Z throughout. At least two members of calK are noncomplete. To see this, a complete member of order l has positive boundary, hence $l \leq D$, and its boundary is $l(D+1-l) \geq D$, since the difference from D is $(l-1)(D-l) \geq 0$. If all but one of the $b + \kappa$ members were complete, their boundaries would total at least Db and the remaining positive boundary would exceed the total available Db. Also Z is noncomplete because $d < D$. Choose a maximum independent set S of Z, of size a. Choose two nonadjacent vertices from another noncomplete required component and one vertex from each remaining required component. There are no edges between these distinct components of U-X, so

$a + b + \kappa \leq \alpha(R) \leq w + 1$, and thus $a \leq w - b$. (5)

Since Z is noncomplete, $a \geq 2$. As b, $\kappa \geq 1$, (5) excludes $w = 2$. If $w = 3$, it forces $a = 2$ and $b = \kappa = 1$. There are exactly two required components, each containing two independent vertices. Any vertex of P or of a nonrequired component is nonadjacent to these four vertices, giving an independent set of size five in R and contradicting $\alpha(R) \leq 4$. Thus P and Y are empty. Each of the two required boundaries is positive and their sum is at most D; both are therefore below D. There are consequently no nonrequired or large-boundary components when $w = 3$.

Let $F = Z - S$, $E = ex(Z)$, and $t = \sum_{v \in S}(D - deg_Z(v))$, so $0 \leq t \leq d$. Maximality of S ensures $\Delta(F) \leq D - 1$. Independence gives $e(F) = e(Z) - Da + t$. The set S is nonempty, and a perfect matching of Z after deleting one chosen vertex of S loses at most a-1 edges when the rest of S is removed. Thus $\eta = \nu(F) - (m + 1 - a) \geq 0$. Define $d_i = c_i(Z) - c_i(F)$ for $1 \leq i < D$ and $\rho = \sum_{i=1}^{D-1}(a - d_i) \geq 0$. The inequalities follow from restriction and extension by the actual set S in Lemma 72. Both graphs have zero rows at and above D, so their slack definitions give the exact identity

$ps(F) = \psi + t - 1 - \eta - \rho$. (6)

Put $q = c_{D-1}(F)$. If $\Delta(F) < D - 1$, the empty matching covers its empty degree-(D-1) set and Lemma 61 gives $q = 0 \leq ps(F)$. Otherwise Lemma 243 at actual degree $D - 1 \geq 3$ gives $q \leq ps(F)$. At $D - 1 = 3$ it requires no hereditary hypothesis; at larger degrees all proper induced subgraphs of F inherit gap at most one from R.

The middle rows with indices 2,..., D-2 consist of $(D-4)/2$ disjoint pairs in (3) and the midpoint row. Their sum is at most $(D-3)m+(D-4)/2$. Subtracting from the full positive-row sum in (3) gives

$c_1(Z)+c_{D-1}(Z)\geq 2m+2-d/2-\psi$.

Since $c_1(Z)\leq c_0(Z)=m+E$ and $c_{D-1}(Z)\leq a+q$, it follows that

$m-a\leq \psi+E+q+d/2-2$

$\leq 2\psi+E+t+d/2-3-\eta-\rho$

$\leq 3\psi+3d/2-2$. (7)

The second inequality uses (6) and $q\leq ps(F)$; the third uses $t\leq d$, $\eta,\rho\geq 0$, and $E\leq \psi+1$, the last inequality following from $gap(Z)\leq 1$. The midpoint argument also covers $D=4$, where there are zero paired middle rows.

At total $l=m+\delta$, every missing red allocation on Z lies in $[2\delta+1, m-\delta-1]$. Here and below all intervals contain integers. For a direct construction, any coordinate at least m is feasible by putting every edge in that color. If a coordinate $x<m$ satisfies $x\leq 2\delta$, choose any x vertices and put all edges incident with them into that color. Its matching number is at most x. The remaining graph has $2m+1-x\leq 2(l-x)+1$ vertices, so its matching number is at most l-x. This realizes the split. Applying the argument to either coordinate proves the interval claim; both endpoint allocations are feasible as $l\geq m$. Therefore its longest missing run is at most $\max(0, m-3\delta-1)$. By (7), $d\geq 2$, and (5),

$m-3\delta-1\leq a+3-3d/2\leq a\leq w-b\leq w+b$. (8)

For any induced piece B of R with $gap(B)\leq 0$, the total $\nu(B)+ps(B)$ is at least $c_0(B)$, so its allocation set is full. If $gap(B)>0$, T_{-} gives $gap(B)=1$ and $ex(B)=ps(B)+1$. The same total is $c_0(B)-1$, whose missing run is $tr(B)$; both endpoints occur since the total is at least $\nu(B)$. When $D>12$, our hypothesis gives $tr(B)\leq h(\Delta(B))-ex(B)\leq h(D)-1\leq w$. When $D\leq 12$, hereditary T_{-} holds on every induced subgraph of B. Lemma 264 bounds $tr(B)$ by 2, 4, 5 at excess 1, 2, 3 respectively. At excess at least four, T_{-} gives $tr(B)\leq h(12)-4=3$. Thus $tr(B)\leq 5$ in every case with $D\leq 12$. For $w\geq 4$ the actual pool has $w+b\geq 5$; when $w=3$ the previous paragraph showed that P and all companions to be charged slack are absent. The interval criterion of Lemma 75 therefore allows every needed slack-charged piece to be absorbed by the actual pool W union X. Empty pieces require no budget.

Start with the interval $[0, w+b]$ for the actual vertex pool W union X. Absorb P at total $\nu(P)+ps(P)$. Next absorb every nonrequired component of U-X and every required component with boundary at least D at its matching number plus slack. Let calB be this family. Its total slack is at most C: the nonrequired slacks occur directly there, and a required large-boundary component contributes $\max(ps(K)+d(K)-D, 0)\geq ps(K)$. The interval remains full and has length at least $w+b$.

Use (8) to absorb Z at total $m+\delta$. The full interval now has length at least m. Every remaining component is required, has boundary below D, and has matching number at most m. At its matching total its allocation set contains both endpoints and has longest missing run at most max(0,nu(K)-1), by Lemma 75. Absorb these finitely many components, one at a time, at their matching totals.

By (1) and matching additivity across U and P, the final full interval has total

$w+\nu(R)+ps(P)+\sum_{B\in calB} ps(B)+\delta$

$\leq w+\nu(R)+(s-p)+C+\delta\leq w+\nu(R)+s-1$. (9)

Every point of this interval is a sum of feasible component allocations and a contribution from $[0, w+b]$. Choose the corresponding edge partitions on the components of H-(W union X), and restore the pool by Lemma 75: for its assigned red budget choose that many vertices of W union X, put all edges meeting these vertices in the red exceptional part, and put every remaining edge meeting the pool in the blue exceptional part. The new matching bounds are paid by distinct pool vertices. This assigns every original edge once, including edges inside the pool. The two degree-bounded parts at capacities zero are empty, so it is the asserted two-part matching-bounded partition of H. For a larger target split $x+y$, choose integers $x'\leq x$ and $y'\leq y$ summing to the constructed total; for example set $x'=\min(x, total)$ and $y'=total-x'$. Its certificate also obeys the larger bounds. Thus (9) gives every split at the specified total.

Finally assume the opposite $c_0(H)$ bound. In every uncovered connected component U, choose a maximum-cardinality marked maximizer. If it were nonempty the construction would contradict that bound. It is

therefore empty. By Lemma 107 the marked odd-component count at this empty set is the positive exposure. Connectedness forces U itself to be the single counted component. Thus every vertex of U has degree D, U has odd order, and it is factor-critical, as claimed. □

Remark. The even boundary leaves enough credit to pay the rank deficit of one factor-critical piece. Completing all credit procedures funds its extra allocation budget and the other slack charges. The actual pool and the hereditary low-excess run bounds handle small omitted sets, saving one unit on the original graph.

Depends on. Lemma 61, Lemma 72, Lemma 75, Lemma 104, Lemma 107, Lemma 179, Lemma 243, Lemma 264.

Lemma 274 (Large-slack coverage with a two-vertex pool). *Use the finite integer certificate costs c_i, matching number ν, maximum degree Δ, excess $ex = c_0 - \nu$, nonnegative slack ps, signed gap $gap = ex - ps$, and top missing-allocation run tr from Lemma 264. Write α for maximum independent-set cardinality. For every nonnegative integer j put $h(j) = \max(4, \lfloor j/2 \rfloor + 1)$. A graph satisfies T_- when positive gap implies gap one and $tr + ex \le h(\Delta)$.*

Let $D \ge 5$ be an odd integer, let H be a finite simple graph, let W be a vertex subset with $w = |W| \ge 2$, and put $R = H - W$, with induced vertex deletion. Suppose $\Delta(R) = D$, every induced subgraph of R including R itself satisfies T_-,

$ex(R) = ps(R) = s \ge \max(4, h(D) - 1)$,

$c_0(H) = w + \nu(R) + s$,

$|V(R)| < 4w + 6s + 7$,

and $\alpha(R) \le w + 1$.

Then a matching of R covers every vertex of degree D, and $c_D(R) = 0$. No independence of W, regularity, source minimality, or additional excess bound is assumed. All orders, degrees, costs, matching numbers, cardinalities, budgets and parameters D, w, s are integers in their displayed domains.

Proof. Suppose for a contradiction that no matching of R covers all its degree-D vertices. We use nonnegative slack from Lemma 61, its zero-tail and covering-matching conclusions, and the component slack and row inequalities of Lemma 72. The costs are nonincreasing in the row index: reduce one positive degree capacity by one and retain its certificate. Also $c_0(Q) \ge \nu(Q)$, since allocation $(0, c_0(Q))$ places every edge in the second part. Feasible row-zero allocations at total l form the set Y_l defined in Lemma 75.

Choose a component C of R with largest matching number and put $P = R - V(C)$. The choice exists because $\Delta(R) = D \ge 5$ makes R nonempty. It supplies a full interval at total $c_0(C) \ge \nu(C)$. Every other component supplies endpoints at its matching number and has missing runs at most nu(C)-1, or zero if this expression is negative. Successive component absorption in Lemma 75 gives $c_0(R) \le \nu(R) + ex(C)$. Thus $ex(C) \ge s \ge 4$. We claim $gap(C) \le 0$. By hypothesis $s \ge h(D) - 1$. If $gap(C) > 0$, hereditary T_- gives $ex(C) \le h(\Delta(C)) - 1 \le h(D) - 1 \le s$. Therefore $ex(C) = s$ and $tr(C) \le h(D) - s \le 1$. The actual W-pool fills the missing allocations of C at total $c_0(C) - 1 = \nu(C) + s - 1$. The resulting full interval has length $w + \nu(C) + s - 1 \ge \nu(C)$, so it absorbs each other component at its matching number. This gives all allocations on H at total $w + \nu(R) + s - 1$, contrary to $c_0(H)$. Thus $gap(C) \le 0$, and $ps(C) \ge ex(C) \ge s$. Slack superadditivity and nonnegativity give $ps(C) = ex(C) = s$ and $ps(P) = 0$. Every other component U has $ps(U) = 0$; if $ex(U) > 0$, its T_- implication gives $ex(U) = gap(U) = 1$. Hence no other component has excess at least four, proving uniqueness.

Every degree-D vertex of P can be covered by one matching. If $\Delta(P) < D$, this is the empty-set condition. If $\Delta(P) = D$, apply Lemma 243: every proper induced subgraph of P is induced in R and has gap at most one. Since $ps(P) = 0$, that fact gives coverage. Hence $c_D(P) = 0$ by Lemma 61. Since R is uncovered, C has actual degree D and is uncovered.

Let A be the set of degree-D vertices of C, and let $\kappa \ge 1$ be their minimum matching exposure. For $X \subseteq V(C)$, let $q_A(C - X)$ count the odd-order components of C-X whose entire vertex sets are contained in A. Choose X of maximum cardinality among the maximizers of $q_A(C - X) - |X|$, and put $b = |X|$. This choice exists because C is finite. Existence and factor-criticality of the marked odd components follow from Lemma 107; maximum cardinality implies inclusionwise maximality among maximizers. If X were empty, connectedness

and positive exposure would make C odd-order and D-regular, which is impossible by degree-sum parity. Thus X is nonempty. Write calQ for all components of C-X, calK for its marked odd components, and Y for the union of the vertex sets of the other components. By Lemma 104, $|calK| = b + \kappa$, $\nu(C) = b + \sum_{Q \in calQ} \nu(Q)$, $c_D(C) \le \kappa$. Every K in calK has a positive odd boundary $d(K) = e_C(V(K), X)$. This is its total internal degree-D defect, since all its vertices have C-degree D.

Choose any Z in calQ of largest matching number m. Its full-cost interval, together with the actual X-pool, absorbs every other component at its matching number. The same allocation theorem gives $c_0(C) \le \nu(C) + ex(Z)$. Hence $ex(Z) \ge s$. We claim $gap(Z) \le 0$. Positive gap, hereditary T_- and $s \ge h(D) - 1$ would give $ex(Z) = s$ and $tr(Z) \le h(D) - s \le 1$. To rule this out, first absorb P at its matching number using the actual W union X pool. This is possible because $ps(P) = 0$ and P inherits hereditary T_-. Lemma 264 bounds the missing run at total $\nu(P)$ by two, which is at most $w + b$. The resulting full interval has length $w + b + \nu(P)$. It fills the missing allocations of Z at total $c_0(Z) - 1 = m + s - 1$. The resulting interval then absorbs every other Q at its matching number, since $\nu(Q) \le m$. Restoring every edge meeting W union X gives all allocations on H at total $w + \nu(R) + s - 1$, again a contradiction. Thus $gap(Z) \le 0$ and $ps(Z) \ge ex(Z) \ge s$.

We set up the precise credit inequality before using any completion claim. Put $T = (D-1)/2$. For every marked K different from Z with $d(K) < D$, apply Lemma 179: K is connected and factor-critical, its defect is $d(K)$, and $gap(K) \le 1$ by heredity. It gives credits $a_i(K)$ with total max(D-d(K)-ps(K),0), supported on $1 \le i \le T$, bounded by $\max(c_i(K) - \nu(K), 0)$, with $a_1(K) \le T$ and $a_i(K) \le 1$ for $i \ge 2$. Assign zero credits to Z, to non marked components, and to marked components with $d(K) \ge D$; their required total is zero by nonnegative slack.

Run the finite procedures of Lemma 104 with these credits. At index i the initial interval length is $b + \sum_{(} Q : a_i(Q) = 0) c_i(Q)$, hence at least $b + c_i(Z)$. Fix any permitted choices of eligible components. Let E_{rem} be the sum of maximum credits at unfinished indices. Then $E_{rem} \le 2T - 1 = D - 2$. Define J=e(C[X])+e_C(X, Y)+sum_(Q not in calK, Q!=Z)ps(Q)+sum_(K in calK, K!=Z)max(ps(K)+d(K)-D,0). All its terms are nonnegative. Retaining the top-row correction in that theorem and using $gap(C) = 0$ gives

$ps(Z) + J + \sum_{Q \in calQ} c_D(Q) \le \beta + E_{rem} - D + 1_{Z \in calK}(D - d(Z))$, (1)

where $\beta = s + c_D(C) - D(\kappa - 1) \le s + D - (D-1)\kappa \le s + 1$. The indicator term is zero when Z is not marked. To check this conversion, replace gap(Z)-(ex(Z)-ex(C)) by s-ps(Z) in the theorem's gap bound, then substitute its correction $\kappa - c_D(C) + \sum_Q c_D(Q)$.

If Z is not marked, or is marked with $d(Z) \ge D$, the right side of (1) is at most s-1. This contradicts $ps(Z) \ge s$. Therefore Z is marked with $1 \le d := d(Z) < D$. Applying Lemma 179 to this same Z gives $|V(Z)| = 2m + 1$, $u = (D - d)/2 > 0$, zero rows at and above D, and

$ps(Z) = (D-1)m + u - \sum_{i=1}^{D-1} c_i(Z)$,

$c_i(Z) + c_{D-i}(Z) \le 2m + 1$ for $1 \le i < D$. (2)

Inequality (1) now has right side $\beta + E_{rem} - d$.

We next force a large matching number using actual allocations on H. Since $ps(P) = 0$ and P is hereditarily T_-, Lemma 264 again bounds the missing run in $Y_{(}\nu(P))(P)$ by two, at most $w + b$. Thus the actual deleted set W union X absorbs P at its matching number and supplies a full interval of length $L = w + b + \nu(P)$.

At total $l = m + s - 1$ on Z, every missing allocation has both coordinates at least 2s-1. Indeed a coordinate at least m is feasible by putting all edges in its color. If a coordinate $x < m$ satisfies $x \le 2s - 2$, choose any x vertices of Z and color all incident edges with that color. Its matching number is at most x. The induced remainder has $2m + 1 - x$ vertices, at most $2(l - x) + 1$, so its matching number is at most l-x. This supplies the allocation. The argument applies to either coordinate. Consequently every missing first budget lies in [2s-1,m-s], and its longest run has length at most $\max(0, m - 3s + 2)$.

If $m \le L + 3s - 2$, the interval [0,L] would fill this allocation set. Every other Q has $\nu(Q) \le m$, so the resulting full interval also absorbs all those components at their matching numbers. By Lemma 75 these partitions and the actual W union X extension give every allocation on H at total $w + b + \nu(P) + m + s - 1 + \sum_{Q \ne Z} \nu(Q) = w + \nu(R) + s - 1$. This contradicts the assumed exact $c_0(H)$. Hence

$m \ge w + b + \nu(P) + 3s - 1$. (3)

This reasoning includes P empty and assigns every edge meeting W union X.

If another Q had $\nu(Q) \geq w + b + 2$, then its vertex order is at least twice that number. Equations (2) and (3) would give $|V(R)| \geq |V(Z)| + |V(Q)| \geq 4(w + b) + 2\nu(P) + 6s + 3 \geq 4w + 6s + 7$, contradicting the order hypothesis because $b \geq 1$. Therefore every other Q has $\nu(Q) \leq w + b + 1$.

We can now finish every credit procedure. An unfinished index j has a remaining positive-credit component Q with $\nu(Q) \geq b + c_j(Z) + 2$. The preceding bound forces $c_j(Z) \leq w - 1$. Combining this with (3) gives

$m - c_j(Z) \geq b + \nu(P) + 3s$. (4)

If index one is unfinished, monotonicity bounds all D-1 entries in (2) by $c_1(Z)$, giving $ps(Z) \geq u + 3(D-1)s$. But (1) gives $ps(Z) \leq s + D - 1 - d$. The lower bound exceeds the upper by at least $u + (3D-4)s - D + 1 + d > 0$ for $D \geq 5$, $s \geq 4$, $d \geq 1$. Hence index one is finished.

If any later index is unfinished, let j be the first. Then $2 \leq j \leq T$ and $E_{rem} \leq T - j + 1$. Pair the 2(j-1) indices i, D-i for $1 \leq i < j$ using (2). All remaining $D - 2j + 1$ entries have index at least j and are bounded by $c_j(Z)$. From (2) and (4), $ps(Z) \geq u - j + 1 + (D - 2j + 1)(b + \nu(P) + 3s)$. Inequality (1) gives $ps(Z) \leq s + T - j + 2 - d$. The difference between these bounds is $(d-1)/2 + (D - 2j + 1)(b + \nu(P) + 3s) - s > 0$: d is positive odd, $b \geq 1$, and $j \leq T$ makes $D - 2j + 1 \geq 2$. This is another contradiction. Thus every procedure finishes and $E_{rem} = 0$.

Return to (1). Its left side is at least s and its right side is beta-d. If $\kappa \geq 2$, $\beta \leq s + 2 - D < s$, impossible. Hence $\kappa = 1$. Now $ps(Z) \geq s$, $d \geq 1$ and $c_D(C) \leq 1$ force equality everywhere: $ps(Z) = ex(Z) = s$, $d = 1$, $c_D(C) = 1$, $J = 0$. The equality $ex(Z) = s$ uses $ex(Z) \leq ps(Z)$ already proved. The zero correction makes X independent and gives no edges from X to a non marked component. Connectedness of C excludes any such component. Hence $calQ = calK$ has $b + 1$ factor-critical members, and $J = 0$ gives $ps(Q) + d(Q) \leq D$ for every member other than Z. Counting their odd orders with the matching identity gives $|V(C)| = 2\nu(C) + 1$. All these conclusions concern the same chosen X and Z.

We now use the independent-set hypothesis to contradict (3). Choose a maximum independent set S of Z, write $a = |S|$, and put $J_0 = Z - S$. The defect-one conclusion just established means exactly one vertex of Z has internal degree D-1, and all others have internal degree D. Let t be one or zero according as that deficient vertex belongs to S or not. Maximality of S gives $\Delta(J_0) \leq D - 1$. Restrict a perfect matching of Z after deleting one vertex of S, then delete the other a-1 vertices of S. At most a-1 matching edges disappear, so $\eta = \nu(J_0) - (m + 1 - a) \geq 0$.

For $1 \leq i < D$ put $d_i = c_i(Z) - c_i(J_0)$ and $\rho = \sum_{i=1}^{D-1}(a - d_i)$. Vertex restriction and extension give $0 \leq d_i \leq a$ and $\rho \geq 0$. Independence of S gives $e(J_0) = e(Z) - Da + t$. Using the zero tails at index D and above, subtraction of the two slack definitions yields

$ps(J_0) = s + t - 1 - \eta - \rho \leq s$. (5)

Let $q = c_{D-1}(J_0)$. If $\Delta(J_0) < D - 1$, then $q = 0 \leq ps(J_0)$. Otherwise Lemma 243 applies at actual degree $D - 1 \geq 4$, since every proper induced subgraph of J_0 is induced in R and has gap at most one. It gives $q \leq ps(J_0)$. Thus $q \leq s$ in either case.

For Z with defect one, (2) gives $\sum_{i=1}^{D-1} c_i(Z) = (D-1)m + (D-1)/2 - s$. The indices 2 through D-2 are $(D-3)/2$ complementary pairs, whose sum is at most $(D-3)m + (D-3)/2$. Hence $c_1(Z) + c_{D-1}(Z) \geq 2m + 1 - s$. Since $c_1(Z) \leq c_0(Z) = m + s$, we get $c_{D-1}(Z) \geq m + 1 - 2s$. Vertex extension from J_0 gives $c_{D-1}(Z) \leq a + q \leq a + s$. Therefore

$m - a \leq 3s - 1$. (6)

The graph C-X has b other nonempty components besides Z, with no edges between distinct components. Add one vertex from each of them to S. This independent set lies in R, so $a + b \leq \alpha(R) \leq w + 1$. Equation (6) gives $m \leq w + 3s - b$. Combining with (3) forces $2b + \nu(P) \leq 1$, impossible since $b \geq 1$.

We conclude that R has a matching covering every degree-D vertex. Lemma 61 gives $c_D(R) = 0$. All uses of vertex pools above are the actual construction in Lemma 75 and restore every edge of the original H, including edges internal to the pool. Every set choice and credit procedure is finite. □

Remark. The zero-slack exterior has allocation runs of length at most two. This allows a two-vertex pool to force a large defect-one component, whose matching rank then contradicts the available independent-set size.

Depends on. Lemma 61, Lemma 72, Lemma 75, Lemma 104, Lemma 107, Lemma 179, Lemma 243, Lemma 264.

Lemma 275 (Small-slack absorption with a two-vertex pool). *Use the finite integer certificate costs c_i, matching number ν, maximum degree Δ, excess $ex = c_0 - \nu$, nonnegative slack ps, signed gap $gap = ex - ps$, and top missing-allocation run tr of Lemma 264. Write α for maximum independent-set cardinality. Put $h(j) = \max(4, \lfloor j/2 \rfloor + 1)$ for nonnegative integers j, and say that a graph satisfies T_- when positive gap implies gap one and $tr + ex \le h(\Delta)$.*

Let H be a finite simple graph, W a vertex subset with $w = |W| \ge 2$, and $R = H - W$ the induced remainder. Let D belong to {5,7,9,11}. Suppose $\Delta(R) = D$, $1 \le s = ps(R) \le 4$, $\alpha(R) \le w + 1$, and every induced subgraph of R, including R itself, satisfies T_-. If no matching of R covers every vertex of degree D, then $c_0(H) \le w + \nu(R) + s - 1$. More explicitly, every pair of nonnegative integers x, y with $x + y = w + \nu(R) + s - 1$ admits a partition of all H-edges into parts of matching numbers at most x, y.

Consequently, under these hypotheses except noncoverage, $c_0(H) \ge w + \nu(R) + s$ implies that R has a matching covering every degree-D vertex and $c_D(R) = 0$. No independence of W, order bound, sign condition on $gap(R)$, regularity, or minimality hypothesis is imposed. All orders, degrees, matching numbers, costs, cardinalities, budgets and parameters are integers in their stated domains.

Proof. Use Lemma 61 for nonnegative slack and the covering-matching zero tail, Lemma 72 for restriction, vertex extension and component slack superadditivity, and Lemma 75 for actual allocation-set composition. Costs are nonincreasing in their index: reduce one positive capacity of a pair of total $i+1$, and every certificate for the reduced pair is valid for the original pair.

Choose an uncovered connected component U of R of actual degree D; otherwise unions of covering component matchings would cover R. Put $P = R - V(U)$, $p = ps(U)$. Then $0 \le p \le s \le 4$ and $ps(P) \le s - p$. Let A be U's degree-D vertex set and $\kappa \ge 1$ its minimum matching exposure. Choose X of maximum cardinality among the maximizers of $q_A(U - X) - |X|$, where q_A counts odd components wholly contained in A. Lemma 107 gives the exposure equality and factor-criticality of the counted components. The set X is nonempty: otherwise connectedness makes U odd-order and D-regular, contrary to degree-sum parity. Put $b = |X| \ge 1$.

Let calQ be all components of U-X, calK the counted odd components, and Y the union of all non counted components. For K in calK put $d(K) = e_U(V(K), X)$, where e_U counts edges between disjoint sets. These positive odd numbers equal the internal degree-D defects. Their sum is at most Db and $|calK| = b + \kappa$, so at least one is below D. Choose Z with largest matching number m among the components of calK with $d(K) < D$. Put $d = d(Z)$, $u = (D - d)/2$ and $\psi = ps(Z)$.

For every small-boundary K different from Z apply Lemma 179. Its factor-critical, actual-defect and $gap \le 1$ hypotheses hold by the marked barrier and hereditary T_-. It gives supported credits $a_i(K)$ of total max(D-d(K)-ps(K),0), supported on $1 \le i \le T = (D-1)/2$, with $a_1(K) \le T$ and $a_i(K) \le 1$ for $i \ge 2$. They are at most $\max(c_i(K) - \nu(K), 0)$. Use zero credits on Z and on all other components; a large-boundary component's required total is zero.

Run the finite procedures of Lemma 104, and let E_{rem} be their unfinished maximum-credit sum. Then $E_{rem} \le D - 2$. Define the nonnegative correction

J=e(U[X])+e_U(X, Y)+sum_(Q not in calK)ps(Q)+sum_(K in calK, K!=Z)max(ps(K)+d(K)-D,0).

That fact's retained top-row inequality, after substituting $gap = ex - ps$, is

$\psi + d + J + \sum_{Q \in calQ} c_D(Q) \le \beta + E_{rem}$, (1)

where $\beta = p + c_D(U) - D(\kappa - 1) \le p + D - (D-1)\kappa \le p + 1$.

Here $c_D(U) \le \kappa$ is supplied by the same fact. The top correction $\kappa - c_D(U) + \sum_Q c_D(Q)$ has been retained.

Applied to Z, Lemma 179 gives $\psi \ge 1$, $|V(Z)| = 2m + 1$, zero rows at and above D, and

$\psi = (D-1)m + u - \sum_{i=1}^{D-1} c_i(Z)$,

$c_i(Z) + c_{D-i}(Z) \le 2m + 1$ for $1 \le i < D$. (2)

If index one is unfinished, a remaining positive-credit Q satisfies $\nu(Q) \ge b + c_1(Z) + 2$. Its rank is at most m by choice of Z. Monotonicity gives $\psi \ge u + (D-1)(b+2)$, while (1) gives $\psi \le p + D - 1 - d$. Their difference is $u + d + (D-1)(b+1) - p > 0$ since $D \ge 5$, $b \ge 1$, $p \le 4$. This contradiction makes index one finished.

If a later index first fails at j, then $2 \le j \le T$, $E_{rem} \le T - j + 1$ and $c_j(Z) \le m - b - 2$. Pair the indices i, D-i for $1 \le i < j$ in (2), and bound the remaining $D - 2j + 1$ entries by $c_j(Z)$. Thus $\psi \ge u - j + 1 + (D - 2j + 1)(b+2)$. The upper bound in (1) is $\beta + T - j + 1 - d$. The difference is

$(d+1)/2 - \beta + (D - 2j + 1)(b+2) \ge 6 - p > 0$.

Therefore all procedures finish. With $E_{rem} = 0$, (1) gives $\psi + d + J \le p + D - (D-1)\kappa$. If $\kappa \ge 2$ the right side is at most $p - D + 2 \le 1$, while $\psi + d \ge 2$. Consequently

$\kappa = 1$, $p \ge 1$, $|calK| = b + 1$, $\psi + d + J \le p + 1$. (3)

At least two components of calK are noncomplete. A complete odd component of order l has $l \le D$ and boundary $l(D + 1 - l) \ge D$, since the difference from D is $(l-1)(D-l) \ge 0$. If b of the $b+1$ components were complete, their boundaries and the remaining positive boundary would exceed Db. The small-boundary Z is noncomplete. Choose a maximum independent set S in Z with $a = |S|$, two nonadjacent vertices in one other noncomplete component, and one vertex in each of the remaining b-1 marked components. Distinct components of U-X have no edges between them, so

$a + b + 1 \le \alpha(R) \le w + 1$, hence $a \le w - b$. (4)

In particular $a \ge 2$ implies $\alpha(R) \ge b + 3 \ge 4$. Thus $w = 2$ is impossible.

For a rank estimate on this same Z and S, put $F = Z - S$, $E = ex(Z)$, $t = \sum_{v \in S}(D - deg_Z(v))$ and $\eta = \nu(F) - (m + 1 - a)$. Then $0 \le t \le d$ and $\eta \ge 0$: restrict a perfect matching of Z after deleting one vertex of S. Maximality gives $\Delta(F) \le D - 1$, and independence gives $e(F) = e(Z) - Da + t$. Put $d_i = c_i(Z) - c_i(F)$, $\rho = \sum_{i=1}^{D-1}(a - d_i)$. Vertex extension gives $0 \le d_i \le a$ and $\rho \ge 0$. Subtracting the slack definitions gives

$ps(F) = \psi + t - 1 - \eta - \rho$. (5)

Let $q = c_{D-1}(F)$. If $\Delta(F) < D - 1$, then $q = 0$ by Lemma 61. Otherwise Lemma 243 applies at degree $D - 1 \ge 4$, since every proper induced subgraph of F inherits $gap \le 1$. It gives $q \le ps(F)$. Thus this inequality holds in both cases.

Pair the middle indices 2,..., D-2 in (2). Subtracting their sum gives $c_1(Z) + c_{D-1}(Z) \ge 2m + (3-d)/2 - \psi$. Using $c_1(Z) \le c_0(Z) = m + E$ and $c_{D-1}(Z) \le a + q$, then (5), $E \le \psi + 1$ from $gap(Z) \le 1$ and $t \le d$, yields

$m - a \le \psi + E + q + (d-3)/2 \le 3\psi + (3d-3)/2$. (6)

Put $\delta = \psi + d - 2 \ge 0$ and $l = m + \delta$. Both endpoint allocations at total l are feasible. In fact every missing first budget lies in $[2\delta + 1, m - \delta - 1]$. A coordinate at least m is feasible by putting every edge in that color. If a coordinate $x < m$ satisfies $x \le 2\delta$, choose x vertices of Z and give all incident edges that color. Its matching number is at most x. The remaining $2m + 1 - x$ vertices number at most $2(l - x) + 1$, so their matching number is at most l-x. This proves the interval assertion for either coordinate. The maximum missing run at total l is at most $\max(0, m - 3\delta - 1)$. By (6) and (4),

$m - 3\delta - 1 \le a + (7 - 3d)/2 \le a + 2 \le w + b$. (7)

We now absorb the pieces with the actual pool W union X of size $w + b$. If $w = 3$, the bound $\alpha(R) \le 4$ and (4) force $b = 1$, $a = 2$, exactly two noncomplete marked components, and no vertex in P or Y: such a vertex could be added to two independent pairs from the two marked components. Their two positive boundaries sum to at most D, so both are below D. Thus in this case P and the family of non marked or large-boundary components are empty.

For $w \ge 4$, the starting pool has size at least five. The exterior P has $ps(P) \le s - p \le 3$ by (3). Let Bcal consist of non marked components and marked components with boundary at least D. Their total slack is at most J, because each such summand is included or bounded by its term in J. By (3), $J \le p - 1 \le 3$. Hence every piece B among P and Bcal has slack at most three. These are induced subgraphs of R and inherit hereditary T_-. At total $\nu(B) + ps(B)$, if $gap(B) \le 0$ its allocation set is full. If $gap(B) > 0$, then $gap(B) = 1$ and $ex(B) = ps(B) + 1$ is one of 1, 2, 3, 4. For excess one, two, three, Lemma 264 bounds its top missing

run by two, four, five respectively. At excess four, T_- bounds it by $h(\Delta(B)) - 4 \le h(11) - 4 = 2$. Its total is $c_0(B) - 1$ and is at least $\nu(B)$, so both endpoints are feasible. Thus every such piece can be absorbed into the current full interval by Lemma 75, whose length remains at least five.

After these absorptions, or immediately in the $w = 3$ case, the full interval has length at least $w + b$. By (7) it absorbs Z at total $m + \delta$. Its length then is at least m. Every remaining component is marked with boundary below D and rank at most m. Its allocation set at its matching number has endpoints and missing run at most max(0,nu(K)-1), so each is absorbed at that total.

The matching identity of Lemma 104 gives $\nu(U) = b + \sum_{Q \in calQ} \nu(Q)$, and $\nu(R) = \nu(U) + \nu(P)$. The resulting full interval on H has total

$w + \nu(R) + ps(P) + \sum_{B \in Bcal} ps(B) + \delta$.

Its excess above $w + \nu(R)$ is at most $(s - p) + J + \psi + d - 2 \le s - 1$ by (3). Every edge meeting W union X, including internal edges, is restored by the actual vertex-pool construction Lemma 75. All remaining edges lie in exactly one absorbed piece. Padding budgets supplies every allocation at total $w + \nu(R) + s - 1$. This proves the asserted cost bound. Its covering consequence follows by contradiction, and Lemma 61 gives $c_D(R) = 0$ when coverage holds. □

Remark. The actual inner pool needs to fill only runs of length at most five. Very small pools force the barrier to have exactly two noncomplete components and no companions, eliminating the only place where the larger pool was used.

Depends on. Lemma 61, Lemma 72, Lemma 75, Lemma 104, Lemma 107, Lemma 179, Lemma 243, Lemma 264.

Lemma 276 (Large-degree goodness and the minimal-obstruction criterion). *Use the finite integer certificate costs $c_i(G)$, matching number $\nu(G)$, excess $ex(G)$, slack $ps(G)$, signed gap $gap(G)$, and top missing-allocation run $tr(G)$ of Lemma 263. All graphs are finite and simple, including the empty graph with maximum degree zero. For every nonnegative integer D put $h(D) = \max(4, \lfloor D/2 \rfloor + 1)$. Property T_- is the implication*

$gap(G) > 0 \Rightarrow [gap(G) = 1$ *and* $tr(G) + ex(G) \le h(\Delta(G))]$.

For a nonnegative integer d, a closed d-lump is a nonempty connected d-regular graph such that deleting every vertex leaves a perfect matching. Every closed d-lump of even degree $d \ge 14$ is LP-good, meaning its gap is nonpositive.

Let F violate T_- while every proper induced subgraph of F satisfies T_-, including the empty induced subgraph. Then F is connected and d-regular for an even integer d in {4,6,8,10,12}, has odd order $2k + 1$ with $k = \nu(F) \ge 2$, and deleting every vertex leaves a perfect matching. Moreover

$gap(F) = 1$, $tr(F) + ex(F) \ge h(d) + 1$,

$c_i(F - v) = c_i(F) - 1$ *for every v in $V(F)$ and every integer $1 \le i \le d$.*

Every finite simple graph G violating T_- contains such a graph F as an induced subgraph; equality $F = G$ is allowed.

For each d in {4,6,8,10,12}, let M_d be the entire class of finite simple graphs F of maximum degree exactly d which violate T_- while every proper subgraph satisfies T_-. A proper subgraph may delete vertices, edges, or both, including a spanning edge deletion. These are exactly the classes of Lemma 263, with no further rank, excess, order, process, or host-origin condition. Every finite simple graph violating T_- contains a subgraph in one of these five classes; this last contained subgraph is not required to be induced. Every member of every class has all the preceding structural and exact cost conclusions. In particular every finite simple graph satisfies T_- if and only if all five classes M_d are empty. All degrees, orders, costs, indices, matching numbers, excesses, and run lengths are integers in their displayed nonnegative or positive domains; gaps may be signed.

Proof. First let H be a closed d-lump of even degree $d \ge 14$. If $d = 14$, Lemma 269 gives $gap(H) \le 0$. If $d = 16$, Lemma 237 gives the same conclusion. If $d \ge 18$, Lemma 183 gives it. These cases exhaust the even integer domain. Every cited theorem applies to exactly the stated connected, regular, factor-critical graph, with no order, excess or hereditary hypothesis. This proves the large-degree assertion.

The function h is nondecreasing and takes integer values at least four. Applying Lemma 263 with this function gives connectedness, even regular degree $d \ge 4$, odd order $2k + 1$, $k = \nu(F)$, perfect matchings after

every vertex deletion, and the displayed positive-row losses. It also gives $gap(F)$ in {1,2}. Simplicity gives $2k \geq d \geq 4$ and $k \geq 2$.

Fix any proper induced subgraph J of F. Every induced subgraph of J, including J itself, is a proper induced subgraph of F and hence satisfies T_-. Lemma 264 therefore applies to J. It bounds $tr(J)$ by two, four, and five when J has positive gap and respective excess one, two, and three. At $ps(J) = 0$ it also bounds by two the longest missing-allocation run at total $\nu(J)$. Thus all local hypotheses in Lemma 216 hold for every proper induced J.

If $gap(F) = 2$, fact Lemma 216 applies to the same connected regular factor-critical F and gives $d \geq 20$. The large-degree assertion just proved gives $gap(F) \leq 0$, a contradiction. Therefore $gap(F) = 1$. A failure of T_- with this gap must satisfy the integer inequality $tr(F) + ex(F) \geq h(d) + 1$.

If $d \geq 14$, the large-degree assertion again contradicts the positive gap. Since d is an even integer at least four, the remaining possibilities are exactly {4,6,8,10,12}. This proves every asserted conclusion for a minimal induced failure.

Now let G be any finite simple graph violating T_-. Among its induced subgraphs which violate T_-, choose one F of minimum vertex cardinality. This is a nonempty finite family because G itself belongs. Every proper induced subgraph of F has fewer vertices and is induced in G, so satisfies T_-. The result just proved applies.

For the proper-subgraph assertion, choose instead a T_- violating subgraph F of G minimizing the nonnegative integer $|V(F)| + e(F)$. This choice exists because the finite collection contains G. Every proper subgraph of F has strictly smaller value of that sum and remains a subgraph of G, so satisfies T_-. In particular every proper induced subgraph satisfies T_-. Applying the established classification shows that $\Delta(F)$ is one of the five displayed degrees. The complete proper-subgraph hypothesis puts this same F in its corresponding class M_d. Conversely every member of any M_d satisfies the stronger proper-subgraph minimality, hence all the conclusions proved for induced minimal failures.

If every finite simple graph satisfies T_-, no graph can belong to a class M_d by its defining failure clause. If all five classes are empty and some finite simple graph violates T_-, the preceding minimum-subgraph choice produces a member of one of them, a contradiction. This establishes the final equivalence. □

Remark. Large-degree goodness removes every even degree at least fourteen. Hereditary low-excess run bounds exclude gap two, and finite minimal choices yield the two precise containment forms.

Depends on. Lemma 183, Lemma 216, Lemma 237, Lemma 263, Lemma 264, Lemma 269.

Lemma 277 (Low critical degree at nonpositive potential surplus). *Let $s, t \geq 1$ be integers and let $n_1 \geq \cdots \geq n_s \geq 1$ and $m_1 \geq \cdots \geq m_t \geq 1$ be integer lists. Let F_1, F_2 be the respective vertex-disjoint unions of stars with these edge sizes, prescribed red and blue. A star of size j has j edges. Copies are ordinary subgraphs, with vertex-disjoint components within each target. A graph arrows an ordered pair if every red-blue edge coloring contains its entire first target in red or its entire second target in blue.*

Put $M = s + t - 1$ and

$$\ell_k = max\{n_i + m_j - 1 : 1 \leq i \leq s, 1 \leq j \leq t, i + j = k\}$$

for every integer $2 \leq k \leq M + 1$. Put $L = \sum_{k=2}^{M+1} \ell_k$. Fix an integer $0 \leq q < M$. Let A_q be the entire family of ordered nonempty actual suffix pairs obtained by deleting α leading components of F_1 and β leading components of F_2, with nonnegative integers $\alpha < s$, $\beta < t$ and $\alpha + \beta = q$. Put

$$K = M - q,\ P_q = \sum_{k=q+2}^{M+1} (\ell_k - 1),\ D = \ell_{q+2} - 1,$$

and suppose D is either three or four.

For a finite simple graph H, let $e(H)$, $\nu(H)$, $\Delta(H)$ be its edge count, maximum matching cardinality, and maximum degree, with value zero for the empty graph, and put $\Phi(H) = e(H) - \nu(H)$. If

$$\Delta(H) \leq D,\ \nu(H) < K,$$

and H arrows every member of A_q, then $\Phi(H) \geq P_q + 1$.

Equivalently, every H with $\Delta(H) \le D$, $\nu(H) < K$ and $\Phi(H) \le P_q$ has a coloring avoiding both entire prescribed targets of some member of A_q. Every such coloring extends, preserving all H-edge colors, to avoidance of both full originals on the disjoint union of H and the q stars of sizes ℓ_2,..., ℓ_{q+1}. It also extends across any specified q-vertex set X in a finite simple graph G with $G - X = H$. All list lengths, indices, sizes, degrees, matching cardinalities and displayed parameters are integers in the stated domains. Original unit components, repeated sizes, $q = 0$, isolated host vertices and arbitrary positive matching deficits are included.

Proof. Suppose H satisfies the arrowing, degree and matching hypotheses but $\Phi(H) \le P_q$. Form a disjoint pool S_q of stars with sizes ℓ_2,..., ℓ_{q+1}, with S_0 empty, and put $G_{star} = H$ disjoint union S_q. Each size is positive. By Lemma 41, G_{star} arrows the entire original pair. Additivity of matching number and edge count over these disjoint components gives

$\nu(G_{star}) = \nu(H) + q < M$,

$\Phi(G_{star}) = \Phi(H) + \sum_{k=2}^{q+1}(\ell_k - 1) \le L - M$.

The single-pair bound in Lemma 56 gives the reverse potential inequality. Hence $\Phi(G_{star}) = L - M$ and $\Phi(H) = P_q$.

Interchange the entire targets and their color names, if needed, so that their original unit counts u, v satisfy $u \ge v$. This leaves all diagonal values, q, and the common-arrowing assertion unchanged. Apply Lemma 194 to the same full original lists, q and H. Both originals have nonunit components, $u > v$, and

$2 \le n_1 \le D - 1, 2 \le m_1 \le D$.

Let B_2 be the original blue nonunit count, let

$\xi = \sum_{j=1}^{B_2}(m_j - 2)$,

and let R be the length of the initial run of $D + 1$ in the natural diagonal tail. The same fact gives

$B_2 - q \ge R + e(F_1) - s + 1, \xi + q + R \le s - 1$. (1)

If $n_1 = 2$, the entire red original is p copies of the two-edge star together with u single edges, for an integer $p \ge 1$. The graph G_{star} has potential L-M and matching number less than M, so Lemma 248 supplies a coloring avoiding both entire originals. This contradicts its arrowing. Thus $n_1 \ge 3$. The bound $n_1 \le D - 1$ and D in {3,4} now force $n_1 = 3$ and $D = 4$.

By Lemma 133 with the even capacities 2, 2, H has a coloring in which both color degrees are at most two. Consequently no actual suffix pair can have both heads at least three: that coloring would avoid both leading stars and hence both entire suffix targets of that pair, contradicting common arrowing.

Choose valid original indices i, j attaining $\ell_{q+2} = 5$, so $i + j = q + 2$ and $n_i + m_j = 6$. Their cuts i-1, j-1 form a legal actual pair. Since $n_i \le 3$ and $m_j \le 4$, its head sizes are either (3,3) or (2,4). The former is excluded by the preceding paragraph. The latter therefore occurs. In particular the full original lists have the form

$(3^a, 2^p, 1^u), (4^r, 3^b, 2^h, 1^v)$,

where exponents denote repetition, a, p, $r \ge 1$ and b, $h \ge 0$ are integers. The previously chosen u, v still satisfy $u > v$.

We show the exact cut interval. If $q \le a + r + b - 2$, set

$\alpha = \max(0, q - (r + b - 1)), \beta = q - \alpha$.

Then $0 \le \alpha \le a - 1$ and $0 \le \beta \le r + b - 1$. Both cuts are legal, because $\alpha < a \le s$ and $\beta < r + b \le t$. Their suffix heads are three and at least three, which was excluded. Thus $q \ge a + r + b - 1$. The attaining (2,4) candidate above has $i \le a + p$ and $j \le r$, so $q \le a + p + r - 2$. Comparing the two bounds gives $p > b$.

Put $E = a + p + r - 1$ and index the full potential profile by $k = i + j - 1$, so its value is $\ell_{k+1} - 1$. It is nonincreasing: from a candidate on any diagonal after the first, decrease an index greater than one; the corresponding size cannot decrease. A value at least four requires either a red size at least three and a blue size at least three, or a red size at least two and a blue size at least four. These two nonempty index rectangles have last coordinates $a + r + b - 1$ and E. Since $p > b$, E is their largest coordinate and is attained. The retained profile starts at value four. Its initial run of fours consequently ends at coordinate E, and

$R = E - q$. (2)

For the displayed originals, $B_2 = r + b + h$, $\xi = 2r + b$, $s = a + p + u$ and $e(F_1) - s = 2a + p$. Substituting these identities and (2) into (1) gives

$2r + b + E \le a + p + u - 1$,

$r + b + h \ge E + 2a + p + 1$.

Using $E = a + p + r - 1$, these inequalities become

$u \ge 3r + b$, $h \ge 3a + 2p - b$.

Together with the cut interval, a, $r \ge 1$, $p > b$ and $u > v$, these are all the hypotheses of Lemma 43.

Apply that fact to the unchanged full originals and their same cut q. It gives exactly R entries equal to four in the natural potential tail, exactly

$A_3 = \max(u, b + h - p) \ge 1$

entries equal to three, and exactly $N = R + u + b + h$ positive entries. It also supplies the following actual projection. For a positive-edge forest T, let $c(T)$ be its component count, $r(T)$ its nonunit count, and T^- its forest obtained by shortening each nonunit component by one and discarding its unit components. From both orders (T, J_1) of every actual A_q member, form the family C_q containing $(r(T)K_2, J_1)$ when $r(T) \ge 1$ and $(c(T)K_2, J_1^-)$ when J_1^- is nonempty; K_2 is a single edge. The supplied member (mK_2, J) has $m \ge 1$, largest star size three with multiplicity $f \ge 1$, and

$m + c(J) - 1 = N$, $m + f - 1 = R + A_3$.

These are the full projection family, positive next-degree count and two exact cardinality identities required by Lemma 189. Apply it with the matching-size parameter m and next-degree count A_3. Its host is this same H, of maximum degree at most four and arrowing every unchanged A_q member. It yields $\Phi(H) \ge P_q + 1$, contradicting $\Phi(H) = P_q$.

Thus no common-arrowing H in the stated degree and matching domain has $\Phi(H) \le P_q$. All potentials are integers, which proves the asserted strict bound. Negating common arrowing gives an avoiding coloring for some actual pair when $\Phi(H) \le P_q$. The canonical-pool and actual-vertex extensions, preserving every H-edge color, are precisely the two extension constructions in Lemma 41. They apply to every such coloring and include $q = 0$. □

Remark. Original-profile restrictions force a binary red side at critical degree three. At degree four, the even two-two split leaves exactly the three-versus-four profiles whose next-degree matching projection forces strict potential. Canonical completion applies the binary theorem while keeping every original component and matching deficit.

Depends on. Lemma 41, Lemma 43, Lemma 56, Lemma 133, Lemma 189, Lemma 194, Lemma 248.

Lemma 278 (The octic excess-three run bound). *Use the finite integer certificate costs c_i, matching number ν, independence number α, maximum degree Δ, excess $ex = c_0 - \nu$, slack ps, signed $gap = ex - ps$, and top missing-allocation run tr from Lemma 45 and Lemma 250. A finite simple graph J satisfies T_- when positive gap implies both gap one and*

$tr(J) + ex(J) \le \max(4, floor(\Delta(J)/2) + 1)$.

Let H be a nonempty connected eight-regular finite simple graph such that deleting any vertex leaves a perfect matching. Suppose $gap(H) = 1$, $ex(H) = 3$, and every proper induced subgraph of H satisfies T_-. Then

$|V(H)| = 17$, *$\alpha(H)$ belongs to {3,4}, and $tr(H) = 1$.*

All graph orders, degrees, costs, indices, matching and independence cardinalities, excesses and run lengths are integers in their stated nonnegative or positive domains.

Proof. Put $k = \nu(H)$ and $a = \alpha(H)$. Factor-criticality gives $|V(H)| = 2k + 1$; the excess and gap give $c_0(H) = k + 3$ and $ps(H) = 2$. Apply Lemma 45 at $D = 8$ and $E = 3$. Its integer $v = \max(0, floor((2E - 3)/(D - 2)))$ is zero, so

$k \geq 8$ and $1 \leq tr(H) \leq k-7$. (1)

The cutoff $c_0(H)$ in Lemma 123 is available in an economical sequence, and gives

$2 = ps(H) \geq 2(|V(H)| - c_0(H)) - 4a = 2k - 4 - 4a$.

Consequently $k \leq 2a+3$ and $a \geq 3$.

For every proper induced subgraph of H, all its induced descendants are proper induced in H and have gap at most one. In particular Lemma 268 applies to every such graph whose maximum degree is at most seven. We will use its density bounds and factor-critical component count on the actual complements under consideration.

Suppose $k \geq 9$. Fix any maximum independent set S and put $G = H - S$, $g = k - a$ and $\eta = \nu(G) - (k+1-a)$. Lemma 250 gives a nonnegative integer row shortfall ρ with

$\Delta(G) \leq 7$, $ps(G) = 2 - \eta - \rho \geq 0$.

Also $|V(G)| = 2k+1-a$, $e(G) = 8g+4$, and $\alpha(G) \leq a$. Thus η is zero, one or two.

Choose a maximum-cardinality maximizing ordinary matching barrier X in G as in Lemma 140, and put $b = |X|$. That fact gives $p = a + b - 1 - 2\eta$ odd factor-critical components of G-X and the matching identity

$\nu(G) = b + \sum_Q \nu(Q)$.

When $\eta \leq 1$, the inequality $ex(H) = 3 \geq \eta + 2$ ensures a component Q with $\nu(Q) \geq a + b + 2$.

First suppose $\eta = 0$. Lemma 140 then gives $b = 0$ and $a \leq 4$, so a is three or four. These are whole components of G, each of minimum degree at least 8-a: a vertex lost at most its a possible neighbors in S. If $a = 3$, there are two odd components, one on at least $2a+5 = 11$ vertices and the other on at least seven vertices, since its minimum degree is at least five and its order is odd. They require at least eighteen vertices. But $k \leq 2a+3 = 9$ gives $|V(G)| = 2k-2 \leq 16$. If $a = 4$, there are three components, one on at least thirteen vertices and the other two each on at least five vertices. They require at least 23 vertices, while $|V(G)| = 2k-3 \leq 19$ by $k \leq 11$. Both cases are impossible.

Now suppose $\eta = 1$. Then $ps(G)$ is zero or one. The large component and the matching identity imply

$k + 2 - a = \nu(G) \geq b + (a + b + 2)$, hence $k \geq 2a + 2b$. (2)

If $ps(G) = 0$, Lemma 268 (i) gives

$8g + 4 = e(G) \leq -4(2k+1-a) + 11(k+2-a) + 4a = 3g + 18$.

Thus $g \leq 2$, contrary to (2), which gives $g \geq a + 2b \geq 3$.

If $ps(G) = 1$, its matching deficiency is

$f(G) = |V(G)| - 2\nu(G) = a - 3$.

Therefore $\alpha(G) - f(G) \leq 3 \leq 5$, and Lemma 268 (ii) applies. Since $\alpha(G) \leq a$,

$11 \geq E_7(G) = e(G) + 4|V(G)| - 11\nu(G) - 4\alpha(G) \geq 5g - 14$.

Hence $g \leq 5$. Combining this with (2) gives $a + 2b \leq 5$. For $\eta = 1$ and $a \geq 3$ the barrier fact also gives $b \leq 1$. If $b = 1$, then $a = 3$ and $k \leq a + 5 = 8$, contrary to $k \geq 9$. Thus $b = 0$. We have $3 \leq a \leq 5$, $2a \leq k \leq a + 5$ and $k \geq 9$, leaving only $(a, k) = (4, 9)$ or (5,10).

At $(a, k) = (4, 9)$, the component count $p = a - 3$ is one, so G is connected and factor-critical. It has fifteen vertices, 44 edges, matching number seven, and $\alpha(G) \leq 4$. It is nonregular because a common integer degree would have to equal $88/15$. The strict inequality Lemma 268 (iii) gives

$e(G) \leq 3 \cdot 7 + 4 \cdot 4 + 6 = 43$,

a contradiction. At $(a, k) = (5, 10)$, there are two whole factor-critical components of G, and both have minimum degree at least $8 - a = 3$. Part (iv) of the same fact gives $2 \leq ps(G) = 1$, a contradiction. This excludes $\eta = 1$.

The independent set S was arbitrary, so $\eta = 2$ for every maximum independent set when $k \geq 9$. We check the zero-slack coverage hypothesis needed by Lemma 178. Let J be any zero-slack induced subgraph of H with maximum degree at least three. It is proper because $ps(H) = 2$. If $\Delta(J) \leq 7$, Lemma 268 (i) supplies

coverage. If $\Delta(J) = 8$, write $J = H - U$ with U nonempty, choose v in U, and restrict a perfect matching of H-v to J. Any surviving unmatched vertex had its matching partner in U minus {v}, so has an H-neighbor in U and degree at most seven in J. The restricted matching therefore covers every degree-eight vertex of J. This proves coverage for every required J.

Apply Lemma 178 with $d = 8$ and $p = 2$. All its hypotheses have now been supplied, including gain two for every maximum independent set. It gives $ps(G) = 0$, $\nu(G) \le 7$ and exact positive-row losses. In particular

$k + 3 - a \le 7$, so $a \ge k - 4$,

and $c_7(H) = a + c_7(G) = a$,

where $c_7(G) = 0$ follows from Lemma 268 (i). The weighted certificate bound of Lemma 53 gives

$a = c_7(H) \le floor((800k - 180)/1300)$. (3)

For every $k \ge 11$, one has $800k - 180 < 1300(k - 4)$, since $500k > 5020$. This contradicts $a \ge k - 4$. It remains to exclude $k = 9$ and ten.

We construct a dominating set of size at most four when $|V(H)| = 19$ and at most five when $|V(H)| = 21$. Suppose i vertices have already been chosen and r vertices remain outside their closed neighborhoods. Each undominated vertex has its entire closed neighborhood of nine vertices among the n-i unchosen vertices. Counting incidences therefore gives an unchosen vertex whose closed neighborhood meets at least $ceil(9r/(n - i))$ undominated vertices. Choose one such vertex. Starting at $r = n$, the following lists are consequently upper bounds for successive undominated counts:

$n = 19$: 19, 10, 5, 2, 0;

$n = 21$: 21, 12, 6, 3, 1, 0.

The recurrence $r \to \max(0, r - ceil(9r/(n - i)))$ is nondecreasing in integer r in these steps, since $n - i \ge 9$, so the listed upper bounds remain valid if an earlier step covers more vertices. Pad the resulting dominating sets, if necessary, to sets Z of exactly four or five vertices. Their induced complements have orders fifteen and sixteen, respectively, and maximum degree at most seven. Lemma 73 at odd degree bound seven gives $c_7(H) \le 4$ or ≤ 5. These contradict $c_7(H) = a \ge k - 4$, which is at least five or six, respectively. Hence $k \ge 9$ is impossible.

Therefore $k = 8$ and $|V(H)| = 17$. We already have $a \ge 3$. Suppose $a \ge 5$. For every maximum independent S the same independent-cut ledger gives $\eta \le 2$. The barrier fact rules out $\eta = 0$ because it would force $a \le 4$. If $\eta = 1$, its large component and matching identity again give $k \ge 2a + 2b \ge 10$, impossible at $k = 8$. Thus $\eta = 2$ for every maximum independent set. The zero-slack coverage verification above did not depend on $k \ge 9$ and applies unchanged. Lemma 178 and the cap-seven zero row therefore give $c_7(H) = a$. But (3) at $k = 8$ gives $c_7(H) \le \lfloor 6220/1300 \rfloor = 4$, contradicting $a \ge 5$. Consequently a belongs to {3,4}. Finally (1) at $k = 8$ gives $1 \le tr(H) \le 1$. This proves every asserted conclusion. □

Remark. The slack-one density inequality and the large component forced by a matching barrier leave only two complements; strictness or component slack excludes both. Uniform gain then forces too large an independent set for domination certificates above order seventeen.

Depends on. Lemma 45, Lemma 53, Lemma 73, Lemma 123, Lemma 140, Lemma 178, Lemma 250, Lemma 268.

Lemma 279 (Gap one and bounded degree for minimal failures). *Use the finite integer certificate costs $c_i(G)$, matching number $\nu(G)$, excess $ex(G)$, slack $ps(G)$, signed gap $gap(G)$, and top missing-allocation run $tr(G)$ of Lemma 263. For every nonnegative integer D put $h(D) = \max(4, \lfloor D/2 \rfloor + 1)$. A graph G satisfies T_- if*

$gap(G) > 0$ implies $[gap(G) = 1$ and $tr(G) + ex(G) \le h(\Delta(G))]$.

All graphs are finite and simple.

Let F violate T_- while every proper induced subgraph of F satisfies T_-. Then F is connected, has odd order $2k + 1$ with $k = \nu(F) \ge 2$, and is d-regular for an integer d in {4,6,8,10,12,14,16}. Deleting any vertex of F leaves a perfect matching. Moreover $gap(F) = 1$,

$tr(F) + ex(F) \ge h(d) + 1$,

and for every vertex v and every integer $1 \le i \le d$,

$c_i(F - v) = c_i(F) - 1$.

Every finite simple graph violating T_- contains such an F as an induced subgraph.

The same conclusions hold under the stronger hypothesis that every proper subgraph of F satisfies T_-, where a proper subgraph may delete vertices, edges, or both, including a spanning edge deletion. In particular they hold for every member of each entire class M_d, d in {4,6,8,10,12}, defined in Lemma 263. These classes retain exactly their maximum-degree and proper-subgraph minimality definition, with no added excess, rank, order, process, or host-origin condition.

All degrees, orders, indices, costs, matching numbers, excesses and run lengths are integers in their displayed positive or nonnegative domains; gaps may be signed integers.

Proof. The integer function $h(D) = \max(4, \lfloor D/2 \rfloor + 1)$ is nondecreasing and at least three. Apply Lemma 263 to F with this function. It gives connectedness, even regular degree $d \ge 4$, odd order $2k+1$, $k = \nu(F)$, factor-criticality, and the exact displayed positive-row losses at every vertex. It also gives $gap(F)$ in {1,2}. Since F is simple and d-regular, $2k \ge d \ge 4$, so $k \ge 2$.

Fix any proper induced subgraph J of F. Every induced subgraph of J is also a proper induced subgraph of F, including J itself and the empty graph. All of them satisfy T_-. Therefore Lemma 264 applies to J. It gives $tr(J) \le 2, 4, 5$ when $gap(J) > 0$ and $ex(J) = 1, 2, 3$, respectively. Its zero-slack consequence also gives the matching-total missing-run bound at most two when $ps(J) = 0$. Thus every proper induced J supplies every local implication required in Lemma 216.

For every connected even-regular factor-critical graph of degree at least eighteen, the earlier goodness theorems give nonpositive gap: Lemma 164 covers degree eighteen, Lemma 174 degree twenty, Lemma 134 degree twenty-two, Lemma 157 degree twenty-four, and Lemma 129 every even degree at least twenty-six. These cases exhaust all even integers at least eighteen. Each theorem requires exactly the finite simple, connected, regular, factor-critical hypotheses already established for F.

If $gap(F) = 2$, apply Lemma 216 to this same F, using its already established connectedness, even regularity, odd order and perfect matchings after all vertex deletions. It gives $d \ge 20$. The applicable goodness theorem from the preceding exhaustive list then gives $gap(F) \le 0$, a contradiction. Consequently $gap(F) = 1$.

If $d \ge 18$, the same goodness cases again give $gap(F) \le 0$, contradicting $gap(F) = 1$. Hence d is one of 4, 6, 8, 10, 12, 14, 16. Since F fails T_- at gap one, its integer run and excess obey $tr(F) + ex(F) \ge h(d) + 1$.

If a finite simple graph G violates T_-, choose an induced subgraph F of G with the fewest vertices among those violating T_-. This finite collection is nonempty because it includes G. Every proper induced subgraph of F is induced in G and has fewer vertices, so satisfies T_-. The conclusions just proved apply to this F.

Finally every proper induced subgraph is a proper subgraph in the broader stated sense. Proper-subgraph minimality therefore implies all hypotheses just used. The class definitions of M_d in Lemma 263 impose exactly this stronger minimality and the specified maximum degree; no further assumption has been introduced. □

Remark. Hereditary low-excess run bounds force any gap-two source into degree at least twenty. Unconditional goodness for every even degree at least eighteen excludes that alternative and bounds the remaining gap-one sources.

Depends on. Lemma 129, Lemma 134, Lemma 157, Lemma 164, Lemma 174, Lemma 216, Lemma 263, Lemma 264.

Lemma 280 (The octic excess-four window under induced heredity). *Use the finite integer certificate costs c_i, matching number ν, independence number α, maximum degree Δ, excess $ex = c_0 - \nu$, slack $ps = e - \nu - \sum_{i\ge1} c_i$, signed $gap = ex - ps$, and top missing-allocation run tr from Lemma 256. Write $\Sigma = \sum_{i\ge0} c_i$. A graph satisfies T_- when positive gap implies gap one and $tr + ex \le \max(4, \lfloor \Delta/2 \rfloor + 1)$.*

Let H be a nonempty connected eight-regular finite simple graph such that deleting any vertex leaves a perfect matching. Suppose $ex(H) = 4$, $gap(H) = 1$, and every proper induced subgraph of H satisfies T_-. Then

$|V(H)| = 29$, $\nu(H) = 14$, $\alpha(H) = 8$, $e(H) = 116$,

$c_0(H) = 18$, $ps(H) = 3$, $\Sigma(H) = 117$, $c_7(H) = 8$.

For every maximum independent set S of H, the actual induced complement $G = H - S$ has 21 vertices, 52 edges, matching number ten, slack zero, independence number in {7,8}, maximum degree seven, and a matching covering every degree-seven vertex. Moreover

$c_i(H) = 8 + c_i(G)$ *for every integer* $1 \le i \le 7$,

and $2 \le c_6(G) \le 5$.

All orders, degrees, costs, row indices, matching and independence cardinalities and displayed numerical parameters are integers in their displayed nonnegative or positive domains.

Proof. Put $k = \nu(H)$. Factor-criticality gives $|V(H)| = 2k + 1$, and excess four gives $c_0(H) = k + 4$. First $k \ge 11$. By the integer certificate definition, some nonnegative integer allocation $x + y = k + 3$ fails at zero degree capacities. Exchange its colors so that $x \le y$. If $x \le 6$, choose any x vertices W, put all edges meeting W in the exceptional part of budget x, and put all remaining edges in the part of budget y. The former has matching number at most x. The latter has at most

$floor((2k + 1 - x)/2) = k - \lfloor x/2 \rfloor \le k + 3 - x = y$

matching edges, since $\lceil x/2 \rceil \le 3$. These vertices can be chosen: eight-regularity gives $k \ge 4$, while $x \le floor((k + 3)/2) < 2k + 1$. This would realize the allegedly failing allocation. Therefore $x \ge 7$ and $y \ge 7$, so $k + 3 \ge 14$ and $k \ge 11$.

We establish the local hypotheses needed by the window theorems. Every proper induced graph J in H has $gap(J) \le 1$ by T_-. Every induced subgraph of such J, including J itself, remains proper induced in H.

If J has maximum degree at least three, maximum degree at most seven, and slack zero, apply Lemma 243 at its actual maximum degree. The required proper-induced gap bounds have just been checked, and its zero-slack consequence gives a matching covering all vertices of that degree.

Now let J be connected, proper induced in H, of degree d between three and seven and slack at most one, and suppose it has no matching covering all its maximum-degree vertices. At $d = 3$, the inequality $2\kappa \le ps(J) \le 1$ in Lemma 243 is incompatible with its positive integer exposure κ. At $d = 5$, Lemma 222 supplies the covering maximum matching, again a contradiction. At $d = 7$, Lemma 240 with $D = 7$ does the same. At $d = 4$ or 6, the even-degree consequence of Lemma 243 gives a unique uncovered whole regular factor-critical component of slack one. Since J is connected and uncovered, that component is J itself. Thus every such uncovered J is regular of positive even degree, factor-critical and has slack one.

Fix any proper induced J of H with maximum degree at most seven. Each induced subgraph K of J has $gap(K) \le 1$, and each connected one satisfies the uncovered-component implication in the preceding paragraph. These are exactly the two hereditary hypotheses of Lemma 242 applied to J. Consequently, putting

$f(J) = |V(J)| - 2\nu(J)$, $q(J) = \alpha(J) - f(J)$,

$E_7(J) = e(J) + 4|V(J)| - 11\nu(J) - 4\alpha(J)$,

one has $E_7(J) \le 11$ whenever $ps(J) = 1$ and $q(J) \le 5$. This proves both the required cap-seven density condition and the local uncovered-component implication without additional assumptions.

Suppose k is eleven, twelve or thirteen. Apply Lemma 188. Its original-graph hypotheses are precisely those stated here, and its two local assumptions are the zero-slack coverage and slack-one density just established. It follows that $k = 13$, $\alpha(H) = 7$, H is well-covered with minimum dominating-set size seven, and every maximum independent complement has twenty vertices, 52 edges, matching number ten, maximum degree at most seven and slack zero. Its exact row identities include $c_2(H) = 7 + c_2(H - S)$. The zero-slack coverage applies to all its proper induced descendants. These are all three hypotheses of Lemma 217, which excludes such a graph. Hence $k \ge 14$.

Apply Lemma 256 to this same H. Its order assumption now holds. The cap-seven density and uncovered-component assumptions were established above on every proper induced subgraph with the required degree $\cap$. The theorem gives, with $a = \alpha(H)$,

(k, a) in {(14,7),(14,8),(15,8),(16,9),(17,10)}.

For every maximum independent S and $G = H - S$, it gives gain $\eta = \nu(G) - (k + 1 - a)$ in {2,3}, slack $ps(G) = 3 - \eta$, actual degree seven and a matching covering that degree. It gives $c_i(H) = a + c_i(G)$ for $1 \le i \le 7$, and $\alpha(G) = a$ when $k - a = 7$. Finally, with o_k the indicator that k is odd, it gives

$2k - 10 + o_k - a \le c_6(H) \le k - 1.$

All hypotheses of Lemma 232 now hold simultaneously. The finite numerical window and sixth-row interval are the ones just obtained. Its proper-induced gap bound comes from T_-, and its density and local uncovered-component assumptions were proved above. Its hypotheses for every maximum independent complement are supplied by Lemma 256, including the conditional independence equality. Therefore Lemma 232 gives exactly the order, matching number, independence number, edge count, costs, slack and certificate sum stated for H, together with every displayed property of every actual complement H-S. In each application S is arbitrary; no separately chosen complement or matching is substituted into another cut's identity. This proves every conclusion. □

Remark. Hereditary coverage supplies the local density interface. The small-order reduction and terminal exclusion remove orders 23 through 27, after which the large numerical window collapses to order 29.

Depends on. Lemma 188, Lemma 217, Lemma 222, Lemma 232, Lemma 240, Lemma 242, Lemma 243, Lemma 256.

Lemma 281 (Low-slack density at maximum degree eleven). *Use the finite integer certificate costs c_i, edge count e, matching number ν, independence number α, maximum degree Δ, excess $ex = c_0 - \nu$, nonnegative slack $ps = e - \nu - \sum_{i \ge 1} c_i$, and signed $gap = ex - ps$ of Lemma 61. All graphs are finite and simple, and all vertex deletions are induced. A graph of positive maximum degree is covered if a matching covers every vertex of that degree. Put*

$E_{11}(J) = e(J) + 8|V(J)| - 19\nu(J) - 8\alpha(J),$

$f(J) = |V(J)| - 2\nu(J).$

Let H have $\Delta(H) \le 11$, and suppose every induced subgraph J of H, including H itself and the empty graph, satisfies $gap(J) \le 1$. Then

$ps(H) = 0$ *implies* $E_{11}(H) \le 0,$

$ps(H) = 1$ *implies* $E_{11}(H) \le \min(19, f(H) + 17).$

Moreover, if $ps(H) = 1$ and $E_{11}(H) \ge 12$, every sequence deleting an arbitrary current maximum-degree vertex while that degree is at least three is a complete covered process, terminating at maximum degree at most two.

Consequently, let Q be any nonempty nonregular factor-critical graph with $\Delta(Q) \le 11$, $ps(Q) \le 2$, and $gap(J) \le 1$ for every induced subgraph J of Q, including Q. Here factor-critical means that deleting any vertex leaves a perfect matching, and nonregular means that not all vertex degrees are equal. Then

$e(Q) \le 3\nu(Q) + 8\alpha(Q) + 28.$

No independence bound, order bound, source-minimality condition, prescribed deletion process, or top-run hypothesis is imposed. Empty graphs and isolated vertices are allowed in the assertion for H. All graph orders, degrees, costs, cardinalities, matching and independence numbers, and deficiencies f are nonnegative integers; gap and E_{11} may be signed integers.

Proof. We first verify both local hypotheses of Lemma 82 with the fixed degree $\cap\, D = 11$. The induced heredity of the gap bound will be retained throughout. Use Lemma 61 for nonnegative integer slack, covered deletion, and one-vertex row bounds; use Lemma 72 for superadditivity of slack over whole components.

Every induced graph J of H with $ps(J) = 0$ and actual maximum degree $d \ge 3$ is covered. Indeed, Lemma 243 bounds the minimum number of uncovered degree-d vertices by $ps(J) = 0$. For $d \ge 4$ its proper-induced gap hypothesis follows from the assumption on H; for $d = 3$ no such hypothesis is needed.

Such a zero-slack J has a complete covered process: delete any current maximum-degree vertex while its degree is at least three. At each step coverage and Lemma 61 imply

$ps(parent) - ps(child) \ge 1 - h,$

where h is the matching-number loss and belongs to {0,1}. Since the parent has zero slack and the child has nonnegative slack, the child also has zero slack and $h = 1$. Thus coverage continues. Vertex count decreases at every step, giving termination at maximum degree at most two. Apply the complete-process density inequality of Lemma 82 with $D = 11$ and initial slack zero. It gives $E_{11}(J) \le 0$. This argument includes an initially degree-at-most-two graph by the empty process. It proves the first assertion and the local zero-slack coverage condition.

Now let J be an induced graph of H with $ps(J) = 1$ and actual maximum degree $d \ge 3$, and suppose J is not covered. A matching covering every degree-d vertex exists exactly when such matchings exist independently in every whole component containing degree-d vertices: their unions are matchings, and any matching restricts to each component. Therefore choose a whole component U of J with $\Delta(U) = d$ which is not covered.

Component slack is nonnegative and its sum is at most one. If $ps(U) = 0$, the zero-slack coverage argument would cover U, a contradiction. Thus $ps(U) = 1$. Let $W = J - V(U)$, the induced union of all the other components. Applying superadditivity to U and W gives $ps(W) = 0$. Hence $E_{11}(W) \le 0$ by the result just proved.

Degree $d = 3$ is impossible: Lemma 243 gives twice the positive integer degree-three exposure at most $ps(U) = 1$. Degree $d = 5$ is impossible by Lemma 222, and every odd $d \ge 7$ is impossible by Lemma 240. The proper-induced gap conditions in both facts hold for U, including when $U = J$ or $U = H$. Thus d is even and belongs to {4,6,8,10}.

Apply the even-degree slack-one conclusion of Lemma 243 to U. Since U is connected and uncovered, it is itself d-regular and factor-critical. It satisfies $gap(U) \le 1$ because it is an induced subgraph of H. Lemma 253 now gives $E_{11}(U) \le 10$. The quantities e, vertex count, ν, and α are additive on whole components: optimal matchings and independent sets restrict to components, and unions of component optima attain the sums. Consequently

$E_{11}(J) = E_{11}(U) + E_{11}(W) \le 10 \le 11.$

This proves the second local hypothesis of Lemma 82 for every induced J, with no connectedness assumption on J.

If $ps(H) = 1$, the criterion in that fact therefore gives $E_{11}(H) \le 19$. When $E_{11}(H) \ge 12$ it supplies every claimed complete covered process. Its refined inequality gives

$E_{11}(H) \le f(H) + 2 \cdot 11 - 5 = f(H) + 17.$

When $E_{11}(H) \le 11$ the same inequality holds because $f(H) \ge 0$: a matching uses two distinct vertices per edge, so $2\nu(H) \le |V(H)|$. Thus $E_{11}(H) \le \min(19, f(H) + 17)$ in every slack-one case. This proves all assertions for H.

For the consequence, put $k = \nu(Q)$, choose any maximum-degree vertex v of Q, and set $J = Q - v$. Nonregularity supplies a vertex u of degree strictly below $\Delta(Q)$. A perfect matching of Q-u covers all vertices of maximum degree in Q, so Q is covered. Factor-criticality gives $|V(Q)| = 2k + 1$ and gives J a perfect matching of size k. In particular deleting v has matching loss zero. The covered deletion inequality in Lemma 61 now gives

$0 \le ps(J) \le ps(Q) - 1 \le 1.$

The graph J inherits the maximum-degree $\cap$ and the full induced gap bound, so the already proved assertion applies. Its order is 2k and $f(J) = 0$. If $ps(J) = 0$ it gives

$e(J) \le 3k + 8\alpha(J);$

if $ps(J) = 1$ it gives

$e(J) \le 3k + 8\alpha(J) + 17.$

Since $\alpha(J) \le \alpha(Q)$ and $deg_Q(v) \le 11$, both cases imply

$e(Q) = e(J) + deg_Q(v) \le 3k + 8\alpha(Q) + 28.$

Every step concerns the same induced child J and the same original edge count of Q, as required. □

Remark. At slack one, an uncovered component must be even-regular and its density error is already small. All larger density errors therefore force covered deletion, and one such deletion turns a nonregular factor-critical graph with two slack units into a perfect-matching graph with at most one.

Depends on. Lemma 61, Lemma 72, Lemma 82, Lemma 222, Lemma 240, Lemma 243, Lemma 253.

Lemma 282 (Concentration of an even-degree matching obstruction). *Use the finite integer certificate costs c_i, matching number ν, maximum degree Δ, excess $ex = c_0 - \nu$, nonnegative slack ps, signed gap $gap = ex - ps$, and top missing-allocation run tr of Lemma 264. For every nonnegative integer j put $h(j) = \max(4, \lfloor j/2 \rfloor + 1)$. A graph satisfies T_- if positive gap implies gap one and $tr + ex \le h(\Delta)$. All graphs are finite and simple, and vertex deletion is induced.*

Let $D \ge 4$ be an even integer, let H be a graph, and let W be a vertex subset of cardinality $w \ge 2$. Put $R = H - W$. Suppose $\Delta(R) = D$, every induced subgraph of R including R itself satisfies T_-, $ex(R) = ps(R) = s \ge \max(4, h(D) - 1)$, $c_0(H) = w + \nu(R) + s$, and $|V(R)| < 4w + 6s + 7$. If no matching of R covers every degree-D vertex, there is a unique whole connected component C of R having excess at least four. It is D-regular and factor-critical, meaning that deleting any one vertex leaves a perfect matching. With $P = R - V(C)$, one has $ex(C) = ps(C) = s$, $gap(C) = 0$, $ps(P) = 0$, $c_D(C) = 1$, $c_D(P) = 0$, $|V(C)| = 2\nu(C) + 1$, and $\nu(C) \ge w + \nu(P) + 3s - 1$. There are no edges between C and P, and every degree-D vertex in P can be covered by a single matching.

No independence, regularity, source minimality, or deletion-process hypothesis is imposed on H or W. The graph P may be empty or have isolated vertices. All orders, degrees, costs, cardinalities, matching numbers, capacities, budgets, and parameters are integers in their displayed nonnegative or positive domains; gaps may be signed integers.

Proof. Assume R is uncovered as in the statement. Use nonnegative slack and zero tails from Lemma 61, and component slack superadditivity and vertex extension from Lemma 72. Rows are nonincreasing: a certificate with smaller degree capacities is valid with larger ones, and every smaller total capacity can be obtained by decreasing the two larger capacities. Also $c_0(Q) \ge \nu(Q)$, since allocation $(0, c_0(Q))$ puts every edge in the second part. Write $Y_l(Q)$ for the row-zero feasible first budgets at total l. Lemma 75 supplies these sets and the following rule: a full interval [0,L] absorbs $Y_l(Q)$ if its endpoints occur and its longest missing integer run has length at most L.

Choose C among the whole components of R with largest matching number, and put $P = R - V(C)$. The choice exists since $\Delta(R) > 0$. The full interval at total $c_0(C) \ge \nu(C)$ absorbs all other components at their matching numbers: their endpoint allocations have missing runs at most $\max(0, \nu(Q) - 1) \le \nu(C)$. Therefore $c_0(R) \le \nu(R) + ex(C)$, whence $ex(C) \ge s$. If $gap(C) > 0$, hereditary T_- and $tr(C) \ge 1$ give $ex(C) \le h(\Delta(C)) - 1 \le h(D) - 1 \le s$. Thus $ex(C) = s$ and $tr(C) \le 1$. The actual W-pool fills the allocation set of C at total $c_0(C) - 1 = \nu(C) + s - 1$. The resulting full interval is long enough to absorb all other components at their matching numbers. Restoring W would give all allocations on H at total $w + \nu(R) + s - 1$, a contradiction. It follows that $gap(C) \le 0$ and $ps(C) \ge ex(C) \ge s$. Slack superadditivity gives $ps(C) = ex(C) = s$ and $ps(P) = 0$. Every other whole component Q has $ps(Q) = 0$. If its excess is positive, its T_- implication gives $ex(Q) = gap(Q) = 1$. This proves uniqueness of the component with excess at least four.

If $\Delta(P) = D$, Lemma 243 applies because every proper induced subgraph of P inherits gap at most one. Its zero-slack conclusion gives a matching covering all degree-D vertices of P. If $\Delta(P) < D$, this is an empty-set condition. In both cases Lemma 61 gives $c_D(P) = 0$. Since R is uncovered, C has actual degree D and is uncovered.

Let A be the set of degree-D vertices of C, and let $\kappa \ge 1$ be the minimum number missed by a matching. For $X \subseteq V(C)$, let $q_A(C - X)$ count odd components whose vertices all belong to A. Choose X of maximum cardinality among maximizers of $q_A(C - X) - |X|$. All choices are finite. Lemma 107 gives $q_A(C - X) = |X| + \kappa$ and factor-criticality of every counted component. We prove that X is empty.

Suppose X is nonempty, and put $b = |X|$. Let calQ be all components of C-X, calK the counted ones, and Y the union of the other component vertex sets. Lemma 104 gives $|calK| = b + \kappa$, $\nu(C) = b + \sum_{Q \in calQ} \nu(Q)$, and $c_D(C) \le \kappa$. For K in calK put $d(K) = e_C(V(K), X)$, where $e_C(U, V)$ counts edges between disjoint sets. Every $d(K)$ is positive and even. It is exactly the internal degree-D defect of K.

Choose Z in calQ with largest matching number m. The actual X-pool and the full interval at total $c_0(Z)$ absorb all other Q at their matching numbers, so $c_0(C) \le \nu(C) + ex(Z)$ and $ex(Z) \ge s$. If $gap(Z) > 0$,

hereditary T_{-} gives $ex(Z) = s$ and $tr(Z) \leq 1$. Since $ps(P) = 0$, Lemma 264 bounds the missing run of $Y_{(}\nu(P))(P)$ by two. The W union X pool has size $w + b \geq 3$, and therefore absorbs P at this matching total, then Z at total $m + s - 1$, then all other Q at their matching numbers. Their ranks are at most m. By the matching identity this constructs all allocations on H at total $w + \nu(R) + s - 1$, a contradiction. Consequently $gap(Z) \leq 0$ and $ps(Z) \geq ex(Z) \geq s$.

Put $T = (D-2)/2$. For K in calK different from Z with $d(K) < D$, apply Lemma 179 to its positive defect and its factor-criticality. Heredity gives $gap(K) \leq 1$. Its credits $a_i(K)$ sum to max(D-d(K)-ps(K),0), are supported on $1 \leq i \leq T$, obey $a_1(K) \leq T$ and $a_i(K) \leq 1$ for $i \geq 2$, and are bounded by $\max(c_i(K) - \nu(K), 0)$. Give zero credits to Z and all other components. A required component with $d(K) \geq D$ has zero required credit by nonnegative slack.

Perform the finite credit procedures of Lemma 104. Their initial interval at index i has length at least $b + c_i(Z)$. Let E_{rem} be the sum of maximum credits at unfinished indices. Then $E_{rem} \leq 2T - 1 = D - 3$. Define the nonnegative integer

J=e(C[X])+e_C(X, Y)+sum_(Q not in calK, Q!=Z)ps(Q)+sum_(K in calK, K!=Z)max(ps(K)+d(K)-D,0).

Retaining the top-row correction in that fact gives

$ps(Z) + J + \sum_{Q \in calQ} c_D(Q) \leq \beta + E_{rem} - D + 1_{Z \in calK}(D - d(Z))$, (1)

where $\beta = s + c_D(C) - D(\kappa - 1) \leq s + 1$. Indeed $gap(C) = 0$, $ex(C) = s$, and the difference gap(Z)-(ex(Z)-ex(C)) is s-ps(Z); substituting $\kappa - c_D(C) + \sum_Q c_D(Q)$ gives (1). The indicator term is zero for an uncounted Z.

If Z is uncounted, or counted with $d(Z) \geq D$, the right side of (1) is at most s-2, contradicting $ps(Z) \geq s$. Thus Z is counted and its even defect $d = d(Z)$ satisfies $2 \leq d \leq D - 2$. Lemma 179 now gives, with $u = (D-d)/2 \geq 1$,

$|V(Z)| = 2m + 1$, $ps(Z) = (D-1)m + u - \sum_{i=1}^{D-1} c_i(Z)$,

$c_i(Z) + c_{D-i}(Z) \leq 2m + 1$ for $1 \leq i < D$. (2)

The right side of (1) becomes $\beta + E_{rem} - d$.

We need an explicit allocation bound on the same Z. The W union X pool absorbs P at total $\nu(P)$, as already proved, giving a full interval of length $L = w + b + \nu(P)$. Put $l = m + s - 1$. Any allocation on Z with one coordinate at least m is feasible by putting all edges in that part. If a coordinate $x < m$ obeys $x \leq 2s - 2$, choose x vertices and put all their incident edges in that part. Its matching number is at most x. The other part has $2m + 1 - x$ vertices, which is at most $2(l - x) + 1$, so its matching number is at most l-x. This treats either coordinate. Thus every missing first budget lies in [2s-1,m-s], and the longest missing run is at most $\max(0, m - 3s + 2)$. If $m \leq L + 3s - 2$, the full interval of length L absorbs Z at total l and then every other Q at its matching number. The matching identity and actual-pool extension again give all allocations on H at total $w + \nu(R) + s - 1$. Therefore

$m \geq w + b + \nu(P) + 3s - 1$. (3)

If another Q in calQ has $\nu(Q) \geq w + b + 2$, then its order is at least $2(w + b + 2)$. Together with (2)-(3) this gives

$|V(R)| \geq |V(Z)| + |V(Q)| \geq 4(w + b) + 2\nu(P) + 6s + 3 \geq 4w + 6s + 7$,

contrary to the strict order hypothesis. Hence every other Q has $\nu(Q) \leq w + b + 1$.

If index j is unfinished, an unabsorbed positive-credit component has rank at least $b + c_j(Z) + 2$, by the stopping condition in Lemma 104 and integrality. The preceding upper bound implies $c_j(Z) \leq w - 1$; (3) yields

$m - c_j(Z) \geq b + \nu(P) + 3s$. (4)

If index one is unfinished, monotonicity and (2) imply $ps(Z) \geq u + 3(D-1)s$. On the other hand (1) gives $ps(Z) \leq s + D - 2 - d$. The lower minus the upper is at least $u + (3D - 4)s - D + 2 + d > 0$ for $D \geq 4$, $s \geq 4$, $d \geq 2$. Thus index one finishes.

If a later index fails, let j be the first such index. Then $2 \le j \le T$ and $E_{rem} \le T - j + 1$. Pair the indices i, D-i for $1 \le i < j$ using (2). Bound each of the remaining $D - 2j + 1$ rows by $c_j(Z)$. Equations (2)-(4) give

$ps(Z) \ge u - j + 1 + (D - 2j + 1)(b + \nu(P) + 3s)$.

Equation (1) instead gives $ps(Z) \le s + T - j + 2 - d$. The difference of these bounds is

$d/2 + (D - 2j + 1)(b + \nu(P) + 3s) - s > 0$,

since $d \ge 2$, $b \ge 1$, and $D - 2j + 1 \ge 3$. This is impossible. When $D = 4$, $T = 1$ and there is no later index to check. All procedures therefore finish. Equation (1) now gives $ps(Z) \le \beta - d \le s - 1$, contrary to $ps(Z) \ge s$. This proves that X is empty.

Connectedness and $q_A(C) = \kappa \ge 1$ make C its single counted component. Thus C is D-regular, has odd order, and is factor-critical by Lemma 107. Put $m = \nu(C)$. Then $|V(C)| = 2m + 1$. Deleting any vertex v leaves a perfect matching, so Lemma 61 gives $c_D(C - v) = 0$. Vertex extension gives $c_D(C) \le 1$. If $c_D(C) = 0$, use capacities (1,D-1) and exceptional matching budgets (0,0). The exceptional edge sets are empty. Regularity and the total capacity D force every vertex to have degree one in the first bounded part, giving a perfect matching on an odd-order graph. This contradiction proves $c_D(C) = 1$.

Finally the W-pool absorbs P at total $\nu(P)$, since its missing run is at most two, giving $L = w + \nu(P)$. The same explicit allocation construction on the odd graph C at total $m + s - 1$ bounds its missing run by $\max(0, m - 3s + 2)$. If $m \le L + 3s - 2$, restoring W would give all allocations on H at total $w + \nu(R) + s - 1$, a contradiction. Thus $m \ge w + \nu(P) + 3s - 1$, as claimed. Every use of pool restoration is the operation in Lemma 75: select its assigned number of red vertices, put all edges meeting them in the red exceptional part, and put every remaining edge meeting the pool in the blue part. Distinct matching edges use distinct assigned pool vertices. This restores every original H-edge exactly once, including edges internal to the pool. Empty exteriors are included. □

Remark. A large-slack matching obstruction concentrates in one whole component. The order bound completes every boundary-credit procedure; the even boundary then forbids a nonempty barrier, leaving a regular factor-critical component.

Depends on. Lemma 61, Lemma 72, Lemma 75, Lemma 104, Lemma 107, Lemma 179, Lemma 243, Lemma 264.

Lemma 283 (Degree-ten goodness at excess five). *Use the finite integer certificate costs c_i, matching number ν, independence number α, excess $ex = c_0 - \nu$, nonnegative slack $ps = e - \nu - \sum_{i \ge 1} c_i$, signed $gap = \sum_{i \ge 0} c_i - e = ex - ps$, and top missing-allocation run tr of Lemma 154. A finite simple graph J satisfies T_- if positive $gap(J)$ implies $gap(J) = 1$ and $tr(J) + ex(J) \le \max(4, floor(\Delta(J)/2) + 1)$, where Δ is maximum degree and the empty graph has maximum degree zero. A graph is LP-good if its gap is nonpositive.*

Let H be a nonempty connected ten-regular finite simple graph such that deleting any vertex leaves a perfect matching. If $ex(H) = 5$ and every proper induced subgraph of H satisfies T_-, then H is LP-good.

All graph orders, degrees, indices, certificate capacities, matching budgets, costs, matching and independence cardinalities, excesses, slacks and run lengths are integers in their displayed nonnegative or positive domains. Gaps may be signed integers. The graph H itself need not satisfy T_-, and no minimality under edge deletion is assumed.

Proof. Suppose for contradiction that $gap(H) > 0$. By Lemma 224, writing $|V(H)| = 2k + 1$ with $k = \nu(H)$, one has

k in {16,17,18}, $gap(H) = 1$, $c_0(H) = k + 5$,

$c_8(H) = k - 4$, $c_9(H) = k - 9$. (1)

We first prove that every induced subgraph of H on 2k-6 vertices has a perfect matching. Choose any seven-element vertex set W, and put $K = H - W$. Suppose K has no perfect matching, and write $r = \nu(K) \le k - 4$. Choose a maximum-cardinality set B among the maximizers of $o(K - B) - |B|$, where o counts odd-order connected components. This choice exists because K is finite; put $b_0 = |B|$. Apply Lemma 107 to K with every vertex marked. The minimum number of uncovered vertices is $|V(K)| - 2\nu(K) = 2k - 6 - 2r$. Hence K-B has

$p = b_0 + 2k - 6 - 2r \ge b_0 + 2$

counted odd factor-critical components.

There are no even components of K-B. Otherwise adjoin any vertex of an even component to B. The remaining vertices from that component have odd total cardinality and therefore form at least one odd component; all previously counted odd components stay unchanged. The odd-component count increases by at least one and the separator size by exactly one. The maximizing objective cannot decrease and cannot exceed its maximum. Equality would contradict the maximum-cardinality choice of B. Thus all components are the counted odd ones. By the matching formula in Lemma 107,

$r = b_0 + \sum_C \nu(C)$,

and every component C has order $2\nu(C) + 1$.

Choose a component Q with largest matching number $m = \nu(Q)$. There is another component because $p \geq 2$. For each companion C different from Q, its rank $r_C = \nu(C)$ is at most m, and $m + r_C \leq r - b_0$. Consequently

$r_C \leq floor((r - b_0)/2) \leq floor((k - 4)/2) \leq 7$. (2)

Use Lemma 63 on the original H with omitted set $X = W$ union B of cardinality $7 + b_0$. Take Q as the distinguished odd graph, all its companion components as the ordered paid edge blocks, and both degree capacities and the retained bounded-degree edge set to be zero. These blocks, together with Q, partition all edges of H-X. Every ordered absorption inequality holds by (2), since the initial pool already has size at least seven. Put

$\delta = k - 3 - r \geq 1$,

$L = |X| + \sum_{C different from Q} \nu(C) = 7 + r - m$.

The proposed total is $L + m + \delta = k + 4 < c_0(H)$. The failed-absorption conclusion of that fact therefore gives

$L \leq m - 3\delta - 2$,

which simplifies to $2(m + r) \geq 3k$. (3)

We contradict (3) by counting ranks outside Q. Every K-vertex has degree at least three, since H is ten-regular and W has seven vertices. Each vertex of a component of K-B has internal degree at least $3 - b_0$. If $b_0 = 0$, every odd component has order at least five and rank at least two. There is a companion, so $m \leq r - 2$. If $b_0 = 1$, there are at least three odd components, each of order at least three and rank at least one. The two companions give $m \leq r - 3$. If $b_0 = 2$, there are at least four odd components, again of order at least three; the three companions give $m \leq r - 5$. If $b_0 \geq 3$, nonnegative companion ranks give $m \leq r - b_0 \leq r - 3$. In every case

$m + r \leq 2r - 2 \leq 2k - 10 < 3k/2$,

where the last strict inequality uses $k \leq 18 < 20$. This contradicts (3). Thus every induced graph of order 2k-6 has a perfect matching.

The same conclusion holds at every larger even order. Induct on that even order. A larger induced graph contains an induced (2k-6)-vertex graph and therefore an edge uv, because $2k - 6 \geq 26$. Delete u, v, use the induction hypothesis on the remaining induced graph, and add uv to its perfect matching. Every odd-order induced graph of order at least 2k-5 is factor-critical, since each vertex deletion has an even order in the proved range.

Choose a maximum independent set S, and put $a = |S|$ and $G = H - S$. The set S is nonempty and is inclusionwise maximal. Every vertex outside S has a neighbor in S, because otherwise it could be added to S. Hence $\Delta(G) \leq 9$. We claim $a \geq 7$. If $a \leq 6$, the induced matching conclusion just proved makes G perfectly matchable at even order and factor-critical at odd order. At even order its perfect matching covers all degree-nine vertices. At odd order, not all vertices can have degree nine, since their degree sum would be odd. Choose a vertex of degree at most eight and a perfect matching after its deletion; this again covers every degree-nine vertex. Thus Lemma 61 with degree bound nine gives $c_9(G) = 0$ in both cases, also permitting an empty degree-nine set.

Delete the vertices of S one at a time. Since S is independent, every remaining S-vertex retains degree ten at each stage and is a current maximum-degree vertex. The one-vertex row bound in Lemma 61 therefore

bounds each ninth-row decrease by one. Summing gives $c_9(H) - c_9(G) \le a$. This contradicts (1), since $c_9(H) = k - 9 \ge 7 > a$. Therefore $a \ge 7$.

Put $b = \alpha(G) \le \alpha(H) = a$. Apply Lemma 154 to this same maximal independent set S, and define $\eta = \nu(G) - (k + 1 - a)$. Its hypotheses hold by (1), the assumed proper-induced T_- property and $a \ge 7$. It gives

$0 \le \eta \le \min(4, floor((a-1)/2))$, $ps(G) = 4 - \eta$,

$c_8(G) = k - 4 - a$, $c_9(G) = k - 9 - a$.

Thus $a \le k - 9 \le 9$ and $k - a \ge 9$. Choose a seven-element subset S' of S. The graph H-S' has a perfect matching of k-3 edges. Deleting the other a-7 vertices of S removes at most a-7 edges from this matching, so $\nu(G) \ge k + 4 - a$ and $\eta \ge 3$. Consequently η is three or four.

Independence and ten-regularity give $|V(G)| = 2k + 1 - a$ and $e(G) = 10k + 5 - 10a$. Every induced subgraph J of G, including G itself and the empty graph, is a proper induced subgraph of H and satisfies $gap(J) \le 1$. We now exclude the two possible gains.

We first exclude $\eta = 3$. In this case $ps(G) = 1$, $\nu(G) = k + 4 - a$, and the quantities of Lemma 254 are

$f(G) = a - 7$,

$E_9(G) = e(G) + 6|V(G)| - 15\nu(G) - 6b$

$= 7k - 49 - a - 6b$

$\ge 7(k - a) - 49 \ge 14$.

All the hereditary gap and maximum-degree hypotheses of that fact have been checked on G. Since $E_9(G) \ge 10$, its eighth-row bound gives

$c_8(G) \ge b - f(G) + E_9(G) - 13$

$= 7k - 55 - 2a - 5b$

$\ge 7k - 55 - 7a$.

Substitute $c_8(G) = k - 4 - a$. It follows that $6(k - a) \le 51$, whereas $k - a \ge 9$ gives $6(k - a) \ge 54$. This is impossible.

Next exclude $\eta = 4$. The bound $\eta \le floor((a-1)/2)$ gives $a \ge 9$. Since $a \le k - 9 \le 9$, necessarily $k = 18$ and $a = 9$. The same formulas give

$|V(G)| = 28$, $\nu(G) = 14$, $e(G) = 95$, $ps(G) = 0$,

$b \le 9$, $c_8(G) = 5$, $E_9(G) = 53 - 6b$.

We construct a complete covered deletion process in G. At any current induced graph J with $ps(J) = 0$ and actual maximum degree $d \ge 3$, fact Lemma 243 supplies a matching covering every degree-d vertex. For $d \ge 4$ its proper-induced gap assumption follows from the hereditary hypothesis already checked; for $d = 3$ that assumption is unnecessary. Choose any vertex v of degree d and write $h = \nu(J) - \nu(J - v)$. Lemma 61 gives h in {0,1} and

$0 = ps(J) \ge ps(J - v) + 1 - h$.

Nonnegative slack implies $h = 1$ and $ps(J - v) = 0$. Repeat until the maximum degree is at most two. Vertex count decreases at every step, so the process terminates; write its terminal graph as R, its length as t, and its successive deletion degrees as d_1,..., d_t. It is a complete covered process with every deletion lowering matching number by one.

Apply the first density identity of Lemma 82 with degree ∩ nine and initial slack zero. Its nonnegative terms give $E_9(G) \le 0$. Hence $53 - 6b \le 0$ and $b \le 9$ force $b = 9$ and $E_9(G) = -1$. There are no matching-preserving deletions, so its exact identity reads

$1 = \sum_{i=1}^{t}(9 - d_i) + 6(9 - \alpha(R)) + B_9(R) - e(R)$,

where $B_9(R) = -6|V(R)| + 15\nu(R) + 6\alpha(R)$.

Each term is nonnegative. Every terminal component other than a triangle or isolated vertex contributes at least seven by the component formulas in that fact. Thus R consists of triangles and isolated vertices, $\alpha(R) = 9$, and $\sum_i (9 - d_i) = 1$.

Such a terminal satisfies $|V(R)| = 2\nu(R) + \alpha(R)$: a triangle contributes (3,1,1) to order, matching number and independence number, and an isolated vertex contributes (1,0,1); these quantities add over components. Since $|V(R)| = 28 - t$ and $\nu(R) = 14 - t$, we obtain $t = 9$. Each $d_i \le 9$, and the nine nonnegative integers $9 - d_i$ have sum one. Exactly eight deletions therefore have degree nine and one has degree eight.

The row losses at these zero-slack steps can be read directly from Lemma 61. Put $a_j = c_j(J) - c_j(J - v)$. Each a_j is zero or one. Coverage makes $c_d(J) = c_d(J - v) = 0$, and both graphs have zero rows above d. The exact slack identity and $h = 1$ give

$0 = d - 1 - \sum_{j=1}^{d-1} a_j$.

All d-1 terms are at most one, so every a_j with $1 \le j < d$ equals one. Consequently row eight drops by one exactly at a degree-nine deletion and vanishes on both parent and child at a degree-eight deletion. Its terminal value is zero because $\Delta(R) \le 2$. Telescoping proves $c_8(G) = 8$, contradicting $c_8(G) = 5$. This excludes $\eta = 4$.

Both possible values of η have been excluded. This contradicts $gap(H) > 0$ and proves that H is LP-good. □

Remark. Seven-vertex deletion robustness makes every maximum independent cut have matching gain three or four. The exact eighth row contradicts the density identities at both gains.

Depends on. Lemma 61, Lemma 63, Lemma 82, Lemma 107, Lemma 154, Lemma 224, Lemma 243, Lemma 254.

Lemma 284 (Saturated frames for every quartic minimal obstruction). *Use the finite integer certificate costs c_i, matching number ν, independence number α, excess $ex = c_0 - \nu$, slack $ps = e - \nu - \sum_{i \ge 1} c_i$, $gap = ex - ps$, and top missing-allocation run tr of Lemma 263. All graphs are finite and simple. A graph satisfies T_- if positive gap implies gap one and $tr + ex \le \max(4, \lfloor \Delta/2 \rfloor + 1)$. Let M_4 consist of all graphs of maximum degree exactly four violating T_- while every proper subgraph satisfies T_-. Proper subgraphs allow arbitrary vertex deletions, edge deletions, or both, including spanning edge deletions. Vertex deletion is induced.*

Let F be any member of M_4.

Put $k = \nu(F)$, $E = ex(F)$, $a = \alpha(F)$, $x = k - a$, $\sigma = 2E - 2 - x$, $g = c_2(F) - a$, $q = x + 1 - E$, $b = g - 2q$, and $m = q + b$.

Then F is connected and four-regular. It is factor-critical, meaning that every vertex deletion leaves a perfect matching, and well-covered, meaning that every maximal independent set has maximum cardinality. Also $c_3(F) \le floor((3k + 2)/4)$, and

$gap(F) = 1$, $E \ge 2$, $a \ge 5$, $E \le x \le 2E - 2$, $k \ge 4E - 1$,

$q = m - b \ge 1$, $m \ge 1$, $0 \le b < m$, *and* $\sigma \ge b$.

For every maximum independent set S, let $C(S)$ be the graph induced by the degree-three vertices of F-S. Then

$\max_S \alpha(C(S)) = m$.

A frame is a pair (S, U) with S maximum independent and U an independent m-element subset of V(C(S)). At least one frame exists. For every frame put $W = S$ union U, $R = F - W$, and $r = \nu(R)$. Then:

(i) F[W] consists of m disjoint edges and a-m isolates. One has $\Delta(R) \le 2$, $ps(R) = 0$, $\alpha(R) = \alpha(F - S) = a$, $\nu(R) = \nu(F - S) - m$, $c_1(R) = r - b$, and $c_2(R) = q$.

(ii) The whole components of R are exactly q odd cycles from C_3, C_5, C_7, exactly b even-order paths from P_2, P_4, P_6, and otherwise P_3, C_4, or isolates. Here P_j and C_j have j vertices.

(iii) Every orientation of the m edges of F[W], taking one endpoint of each edge and all frame isolates as the independent side, gives another frame with the same W, R. For every frame-edge endpoint p, the induced graph $K_p = F[V(R)$ union $\{p\}]$ satisfies $\nu(K_p) = r + 1$ and $c_1(K_p) = r + 1 - b$.

(iv) The original graph satisfies $c_2(F) = a + 2m - b$. There is one fixed pair of nonnegative integers (u, v), independent of the frame, such that $E(F)$ cannot be partitioned into parts of matching numbers at most u, v, and every frame satisfies

$u + v = c_0(F) - 1 = |W| + r - 1$,

$|W| - u \geq q + b + \sigma$.

In particular $|W| - u \geq q + 2$ when $b \geq 1$. At $b = 0$ no stronger shortage bound is asserted.

All orders, degrees, costs, indices, matching and independence cardinalities, component counts, and displayed parameters are integers in the stated domains. No weak cost bound, gap, excess, order, rank, saturation, or frame hypothesis is imposed in addition to membership in M_4.

Proof. Proper-subgraph minimality implies proper-induced-subgraph minimality: each graph induced by a proper vertex subset is one of the permitted proper subgraphs. Apply Lemma 263 with $h(D) = \max(4, \lfloor D/2 \rfloor + 1)$ for nonnegative integers D. This function is nondecreasing and has values at least four, so it meets that theorem's requirements. The original definition of M_4 is exactly its degree-four class. It gives connectedness, four-regularity, factor-criticality, and $gap(F)$ in {1,2}.

We derive the needed third-row bound on this same F. Write $|V(F)| = 2k + 1$, with integer $k = \nu(F) \geq 2$; odd order and this matching identity follow from Lemma 263, and $k \geq 2$ follows from simplicity and degree four. Choose a vertex z and a perfect matching M of F-z. Define $J = (F - z) - E(M)$, deleting the edges of M while retaining the vertex set $V(F)$ minus {z}. Its four vertices adjacent to z in F have degree two, and every other vertex has degree three. Thus $|V(J)| = 2k$, $e(J) = 3k - 2$, and J has no degree-one vertex.

Lemma 3 gives $8\gamma(C) \leq 3|V(C)|$ for every finite simple cubic graph C, where γ is the minimum size of a set whose closed neighborhoods cover every vertex. In particular this holds for every such C of order at most 2k. Apply Lemma 11 with its order bound $N = 2k$ and with exactly the cubic vertices of J required. It yields a vertex set D whose closed neighborhoods cover all those vertices and whose cardinality satisfies $4|D| \leq e(J)$. This minimum is attained because the available vertex sets form a finite nonempty family.

Put $O = D$ union {z}, $t = |O| = 1 + |D|$, and $H = J - D$. Any surviving vertex originally cubic in J has a neighbor in D, so $\Delta(H) \leq 2$. Let M_0 consist of the edges of M with both endpoints outside O. Then M_0 is a matching, and $E(F - O)$ is the disjoint union of M_0 and $E(H)$. For capacities $(A_c, B_c) = (1, 2)$ or (2,1), put M_0 in the capacity-one part and H in the capacity-two part. For capacities (0,3) or (3,0), put their entire union in the capacity-three part; its maximum degree is at most three. These are all nonnegative integer capacity splits of total three.

Order O and assign each original edge having an endpoint in O to its earliest endpoint in that order. This partitions the edges outside F-O into t labelled stars, allowing empty stars and counting an edge with both endpoints in O exactly once. For any nonnegative integer matching budgets $x_c + y_c = t$, give x_c of those stars to the first matching-bounded part and all remaining stars to the second. A matching uses at most one edge from each star, so the required bounds hold. Together with the two degree-bounded parts this is a certificate on all of F. It works at every capacity and budget split just specified. Therefore

$c_3(F) \leq t \leq 1 + floor((3k - 2)/4) = floor((3k + 2)/4)$.

The structure used in this construction was obtained directly from membership in M_4.

The graph F is a nonempty connected four-regular graph. Thus Lemma 265 applies with $Q = F$ and gives T_3 for every proper subgraph of F, where T_3 means that positive gap implies gap one and $tr + ex \leq 3$. This includes arbitrary edge deletions and does not change the source F or its minimality convention.

Apply Lemma 201 to this F. The structure and gap range have just been established, F fails T_- by membership in M_4, the proper-subgraph T_3 condition was just supplied, and the weak third-row inequality has just been proved on the same F. It gives well-coveredness, gap one, $E \geq 2$, and all the stated parameter bounds. In particular $a \geq 5$, $m \geq 1$, $0 \leq b < m$, and $\sigma \geq b$. For every maximum independent set S, writing $G = F - S$, it also gives

$ps(G) = c_3(G) = 0$,

$|V(G)| - \nu(G) = a + m - b$,

$c_1(G) = \nu(G) - b$, $c_2(G) = 2m - b$,

$c_0(F) = a + \nu(G)$, $c_2(F) = a + c_2(G)$.

It supplies one missing pair (u, v) at total $c_0(F) - 1$ with $u \le a - \sigma$. Fix that pair once.

These are every hypothesis of Lemma 191, with its matching-budget names (x, y) replaced by (u, v) and its graph and normalized parameters unchanged. Apply that theorem. The finite maximum $h = \max_S$ alpha(C(S)) is attained and satisfies $h \le m$. Put $d = m - h$. For every attaining pair (S, U), with U an independent h-set in $C(S)$, its terminal R satisfies

$\alpha(R) = \alpha(F - S) = a - d$.

The terminal has exactly $h + 2d - b$ whole odd cycles from C_3, C_5, C_7, exactly b whole even paths from P_2, P_4, P_6, and otherwise P_3, C_4, or isolates. The same theorem proves that $d > 0$ implies $d \ge b$.

Apply Lemma 31 to precisely this F, a, h, m, b. The graph is four-regular and well-covered, h is the same maximum over all maximum independent complements, and $h \le m$. The positive-deficit implication and both uniform terminal-independence equalities have just been established. Every required terminal component count also holds for every attaining pair. Therefore $h = m$.

An attaining pair exists by the finite maximization already proved. Since $h = m$, it is a frame as defined in the statement. Conversely, every frame in that definition is attaining. The $h = m$ conclusions of Lemma 191 consequently apply to all frames and give (i), (ii), (iii), and $c_2(F) = a + 2m - b$. They apply to arbitrary orientations on the same W and to the actual induced graphs K_p, including $m = 1$.

That application retains the already fixed pair (u, v), so it gives $u + v = |W| + r - 1$ and $|W| - u \ge q + b + \sigma$ in every frame. Here $q = m - b$ by the definitions. If $b \ge 1$, $\sigma \ge b$ yields $b + \sigma \ge 2$, proving the stronger $q + 2$ bound in that case. No stronger bound at $b = 0$ is used. This establishes all conclusions for the original F. □

Remark. Original minimality supplies factor-criticality; a domination-based certificate proves the weak cost bound. Exact source normalization and the saturation inequality then construct the required uniform frame family.

Depends on. Lemma 3, Lemma 11, Lemma 31, Lemma 191, Lemma 201, Lemma 263, Lemma 265.

Lemma 285 (The two-run bound at excess one and two). *Use the finite integer certificate costs $c_i(G)$, matching number $\nu(G)$, excess $ex(G) = c_0(G) - \nu(G)$, nonnegative slack $ps(G)$, signed gap $gap(G) = ex(G) - ps(G)$, and top missing-allocation run $tr(G)$ of Lemma 259. Thus, when $c_0(G) > 0$, $tr(G)$ is the greatest length of a consecutive integer run of first budgets x in $[0, c_0(G) - 1]$ for which all edges of G cannot be partitioned into two parts of matching numbers at most x and $c_0(G) - 1 - x$; when $c_0(G) = 0$ it is zero.*

All graphs are finite and simple. For each nonnegative integer D put $h(D) = \max(4, \lfloor D/2 \rfloor + 1)$, and define T_- by the implication

$gap(G) > 0 \Rightarrow [gap(G) = 1$ *and* $tr(G) + ex(G) \le h(\Delta(G))]$.

The empty graph has maximum degree zero. Let H be any graph such that every induced subgraph J of H, including H itself and the empty graph, satisfies T_-. If $gap(H) > 0$ and $ex(H)$ belongs to {1,2}, then

$1 \le tr(H) \le 2$.

There is no connectedness, degree, order, matching-deficiency, or source-minimality restriction. All degrees, graph orders, costs, capacities, budgets, matching numbers, excesses and run lengths are integers in their nonnegative domains; gaps may be signed integers.

Proof. Every induced subgraph J of H has $gap(J) \le 1$: this is automatic for nonpositive gap and follows from T_- for positive gap. Every induced subgraph of any such J is again induced in H.

We first check the zero-slack covering premise needed later. Let J be any induced subgraph of H with $ps(J) = 0$ and actual maximum degree $D \ge 3$. If $D \ge 4$, every proper induced subgraph of J has gap at most one by the preceding hereditary observation. For $D = 3$ no such extra hypothesis is needed. In both cases Lemma 243 applies and supplies a maximum matching covering every degree-D vertex. Thus H, and each of its induced subgraphs, has zero-slack coverage in the precise sense of Lemma 245.

Suppose $gap(H) > 0$, $ex(H)$ belongs to {1,2}, and $tr(H) \ge 3$. Apply Lemma 259 with $b = 2$ and $L(1) = L(2) = 2$. The two cutoffs satisfy $L(e) \ge \max(0, 3 - e)$, and $L(1) \le L(2) + 1$. Its hereditary gap hypothesis

was checked. It therefore gives an actual induced subgraph F of H that is connected and D-regular for an even integer $D \geq 4$, is factor-critical, has odd order, and satisfies

$gap(F) = 1, 1 \leq ex(F) \leq ex(H) \leq 2, tr(F) \geq 3.$

Here factor-critical means that every vertex deletion leaves a perfect matching. The graph F itself satisfies T_-, because it is induced in H.

If $ex(F) = 1$, the identity $gap(F) = ex(F) - ps(F)$ gives $ps(F) = 0$. The already proved zero-slack coverage gives a matching covering every degree-D vertex of F. Since F is D-regular, this matching is perfect. An odd-order graph cannot have a perfect matching, because each matching edge covers exactly two vertices. Thus $ex(F)$ is not one.

It remains that $ex(F) = 2$. If D is four or six, $h(D) = 4$, whereas $tr(F) + ex(F) \geq 5$; this contradicts T_- on F. If D is eight or ten, apply Lemma 245 to the same F. Its connectedness, regularity, factor-criticality, excess at most two, and zero-slack coverage have all been checked. It gives $gap(F) \leq 0$, contradicting $gap(F) = 1$. If $D \geq 12$, apply Lemma 175 (i) to F; every even degree in this range and every excess at most two are covered without a hereditary assumption. It also gives $gap(F) \leq 0$. These cases exhaust all even integers $D \geq 4$.

There is consequently no graph meeting the hypotheses with $tr(H) \geq 3$. Since $tr(H)$ is a nonnegative integer, $tr(H) \leq 2$. Finally $c_0(H) = \nu(H) + ex(H) \geq 1$. By the definition of $c_0(H)$ as the least universal total, the smaller total $c_0(H) - 1$ has a missing first budget. Hence its longest missing run has length at least one. This proves both inequalities. □

Remark. A longer run descends to an odd regular graph with no larger excess. Zero-slack coverage excludes excess one; the own run threshold and the established small-excess goodness bounds exclude every degree at excess two.

Depends on. Lemma 175, Lemma 243, Lemma 245, Lemma 259.

Lemma 286 (Regular components and matching gaps at surplus at most two). *Let $s, t \geq 1$ be integers and let F_1, F_2 be the nonempty star forests with full positive nonincreasing integer edge-size lists $n_1,\ldots, n_s$ and $m_1,\ldots, m_t$, prescribed red and blue. Copies are ordinary subgraphs with vertex-disjoint target components. Put $M = s + t - 1$ and $\ell_k = max\{n_i + m_j - 1 : 1 \leq i \leq s, 1 \leq j \leq t, i + j = k\}$ for every integer $2 \leq k \leq s + t$. Fix an integer $0 \leq q < M$. Let A_q consist of all ordered nonempty actual suffix pairs after legal leading cuts α, β with $0 \leq \alpha < s$, $0 \leq \beta < t$ and $\alpha + \beta = q$. Put $K = M - q$, $P_q = \sum_{k=q+2}^{s+t}(\ell_k - 1)$, $D = \ell_{q+2} - 1$, and let R be the initial plateau length at $D + 1$ in this natural tail.*

Suppose valid original terminal indices i_0, j_0 satisfy $i_0 + j_0 = q + R + 1$, $n_{i_0} = a$, $m_{j_0} = b$ and $a + b = D + 2$, with integer $a, b \geq 3$. Let $0 \leq r \leq 2$ be an integer. Let H be finite simple, write ν for maximum matching cardinality, put $\Phi(H) = e(H) - \nu(H)$, and suppose $\Delta(H) \leq D$, $\Phi(H) \leq P_q + r$ and $\delta = K - \nu(H) \geq r + 1$.

If H arrows every member of A_q, then a, b are even and at least four, D is even and at least six, and $\sigma = \Phi(H) - P_q$ is an integer in [0,r]. The minimum total degree overflow at capacities a-1, b-1 in the prescribed colors,

$\omega = \min_{coloring} \sum_{v \in V(H)} [\max(d_{red}(v) - (a-1), 0) + \max(d_{blue}(v) - (b-1), 0)],$

equals the number κ of whole connected odd D-regular components, with $R \leq \kappa \leq R + \sigma$. Every whole odd D-regular component has a near-perfect matching, and every whole even D-regular component has a perfect matching. A near-perfect matching leaves exactly one vertex uncovered. After all whole odd D-regular components are removed, the remaining graph has an edge coloring with red maximum degree at most a-1 and blue maximum degree at most b-1.

For each whole connected D-regular component C define

`g(C)=max{nu(C)-|W|:` *W is an inclusionwise maximal matching of C}*,

and let $\zeta = \sum_C g(C)$, summing over all whole D-regular components. Then $\zeta \leq R + \sigma - \kappa \leq 2$. At most $R + \sigma - \kappa$ of these components fail to be equimatchable, meaning that every inclusionwise maximal matching is maximum. Every maximum matching in the graph remaining after all whole D-regular components are deleted exposes at most $R + \sigma - \kappa - \zeta$ vertices of ambient H-degree D.

Failure of any asserted necessary condition supplies a coloring avoiding both entire prescribed targets of some A_q member. This coloring extends over the canonical q-star pool of capacities $\ell_2,\ldots, \ell_{q+1}$, and across any specified actual

outer q-set X when $H = G - X$, to avoidance of the unchanged full originals while preserving every H-edge color. No original-head $\cap$, short-prefix, strict-cut, smaller-target or target-count hypothesis is imposed. Every unit, repeated component and complete actual suffix is retained. Full target roles and their prescribed colors may be exchanged. All sizes, indices, cuts, degrees, matching numbers, gaps, exposures and allowances are integers in their stated domains.

Proof. Assume H arrows every A_q member. This finite family is nonempty: since $q \le s + t - 2$, one may take $\alpha = \min(q, s-1)$, $\beta = q - \alpha < t$. Each member has component budget K. Its complete diagonal envelope is the original natural tail. In fact a candidate at local index sum k comes from original index sum $q + k$. Conversely, for valid original i, j with $i + j = q + k$ and $k \ge 2$, set $\alpha = \min(q, i-1)$, $\beta = q - \alpha$. Then $\beta \le j - 1$, so both cuts are legal and leave those components at local index sum k. This proves the envelope identity for every candidate, including all units and repeated sizes.

Here $D = a + b - 2 \ge 4$. Apply Lemma 99 to this entire family. It gives $\sigma \ge 0$ and ensures that every maximum matching exposes between R and $R + \sigma$ vertices of ambient H-degree D. The assumed potential bound gives $\sigma \le r \le 2$.

If D is odd, Lemma 271 applies to the same terminal pair and to the displayed degree, allowance and deficit hypotheses. It gives an avoiding A_q coloring, a contradiction. Thus D is even and a, b have the same parity.

If a, b were both odd, their capacities a-1, b-1 would be nonnegative even integers summing to D. Lemma 133 gives a coloring of H with these two maximum-degree bounds. There are legal total-q cuts retaining the terminal stars: take $\alpha = \min(q, i_0 - 1)$, $\beta = q - \alpha$; then $\beta \le j_0 - 1$ because $(i_0 - 1) + (j_0 - 1) = q + R - 1 \ge q$. A copy of the red suffix would contain the retained a-edge star, and a blue suffix would contain the retained b-edge star. The degree bounds prohibit both. This again contradicts common arrowing. Therefore a, b are even and at least four, and $D \ge 6$.

If necessary exchange the full target roles and colors to make $a \le b$. Since $r \le 2 \le a - 2$, Lemma 258 gives $\omega = \kappa \ge R$. Undoing the exchange returns the capacities to their original prescribed colors. Each odd regular component exposes at least one ambient degree-D vertex in every matching, so the preceding maximum-matching upper bound gives $\kappa \le R + \sigma$.

We show that every whole D-regular component has a matching of its half-order. Suppose a component C fails this assertion. Its maximum matching leaves at least two additional vertices beyond the parity minimum: at least three for odd order, or at least two for even order. Consequently a maximum matching of H exposes at least $\kappa + 2$ degree-D vertices.

The connected regular classification Lemma 170 implies that an equimatchable connected D-regular graph is odd and factor-critical or is $K_{D,D}$; in both cases a half-order matching exists. Thus this particular C is not equimatchable. Choose an inclusionwise maximal matching W of C with $g = \nu(C) - |W| \ge 1$, and any maximum matching Z of H-V(C). In the regular-union inequality of Lemma 99, take $Q = C$. Its comparison exposure k_0 is the exposure of a maximum C-matching together with Z, and hence $k_0 \ge \kappa + 2$. Discarding its nonnegative last term gives

$$\sigma \ge k_0 - R + g \ge \kappa + 3 - R \ge 3,$$

contrary to $\sigma \le 2$. This proves all near-perfect and perfect matching claims.

The minimum overflow is additive over whole components: each finite component has a minimizing coloring, and those colorings may be combined independently. Every odd D-regular component contributes at least one. Otherwise its zero-overflow coloring would have red degree at most a-1 and blue degree at most b-1 everywhere. Their sum equals $D = a + b - 2$, so every red degree would equal the odd integer a-1. The sum of red degrees over an odd number of vertices would then be odd, contrary to its being twice the red edge count. These κ contributions exhaust the total minimum $\omega = \kappa$. Every other whole component therefore has minimum overflow zero. Combining its minimizing coloring with those of the other remaining components gives the required bounded-degree coloring outside all odd regular components.

Each matching family is finite and contains an inclusionwise maximal matching, so every $g(C)$ is an attained nonnegative integer. The established half-order matchings imply

$$\sum_C \left(|V(C)| - 2\nu(C)\right) = \kappa$$

over all whole D-regular components. Apply the final quantitative clause of Lemma 99 to their union. For every maximum matching of its complement with ambient-degree-D exposure u, that clause gives

$$u \le R + \sigma - \kappa - \zeta.$$

As $u \geq 0$, $\zeta \leq R + \sigma - \kappa \leq \sigma \leq 2$. A component fails equimatchability exactly when its nonnegative integer gap is positive, so their number is at most ζ and hence at most $R + \sigma - \kappa$. The complement may be empty, in which case its unique matching has $u = 0$.

Every condition has been derived from simultaneous arrowing of the finite nonempty family A_q. Failure of any therefore yields an actual member with an avoiding coloring. Lemma 41 supplies its original canonical-pool extension and its extension across any specified actual q-set, retaining all core colors and handling target components meeting deleted vertices only through leaves. No target list has been replaced by a threshold truncation. □

Remark. Once the first-slope theorem makes total overflow equal the odd regular count, a regular component without a half-order matching would consume three surplus units. The same independent-exposure accounting bounds every maximal-matching gap and the high exposure outside the regular components.

Depends on. Lemma 41, Lemma 99, Lemma 133, Lemma 170, Lemma 258, Lemma 271.

Lemma 287 (Positive slack in nonregular graphs of maximum degree four). *Use the finite integer certificate costs $c_i(G)$, matching number $\nu(G)$, vertex-cover number $\tau(G)$, excess $ex(G) = c_0(G) - \nu(G)$, slack $ps(G) = e(G) - \nu(G) - \sum_{i\geq 1} c_i(G)$, $gap(G) = ex(G) - ps(G)$, and top missing-allocation run $tr(G)$ of Lemma 23. Put $T_3(G)$ for the implication $gap(G) > 0$ implies $[gap(G) = 1$ and $tr(G) + ex(G) \leq 3]$. All graphs are finite and simple. Vertex deletion is induced, and a proper subgraph may delete vertices, edges, or both. All orders, degrees, costs, indices, matching numbers, defects, and finite cardinalities are nonnegative integers; gaps may be signed.*

Let J be a nonempty connected graph with $\Delta(J) \leq 4$ which is not four-regular. Every subgraph of J, including J itself and the empty graph, satisfies T_3. If $ps(J) \geq 1$, then $gap(J) \leq 0$. Equivalently, if $gap(J) > 0$, then $gap(J) = ex(J) = 1$ and $ps(J) = 0$. No hereditary-gap, minimality, or matching hypothesis is imposed on J.

In particular, let K be a nonempty graph such that deleting any vertex leaves a perfect matching, $\Delta(K) \leq 4$, and $\sum_{v\in V(K)}(4 - deg_K(v)) = 2$. Then K is connected and nonregular, $\Delta(K) = 4$, and for the integer $m = \nu(K) \geq 2$,

$|V(K)| = 2m + 1$, $e(K) = 4m + 1$, $c_i(K) = 0$ *for every integer* $i \geq 4$,

$ps(K) = 3m + 1 - c_1(K) - c_2(K) - c_3(K) \geq 1$, *and* $gap(K) \leq 0$.

Proof. First, J contains no nonempty four-regular subgraph B. Every B-vertex already has four incident B-edges, and its degree in J is at most four. Thus every J-edge incident with a B-vertex is already in B. No J-edge joins $V(B)$ to its complement. Connectedness forces $V(B) = V(J)$, and the same degree saturation forces $E(B) = E(J)$. This would make J four-regular, contrary to the hypothesis.

Suppose a subgraph L of J violates T_3. Among the finitely many induced subgraphs of L which violate T_3 choose F with the fewest vertices. Every proper induced subgraph of F satisfies T_3. Apply Lemma 263 with the constant integer function $h(D) = 3$ for all nonnegative integers D. It gives a connected d-regular F for an even integer $d \geq 4$. Since F is a subgraph of J, $d \leq 4$, hence $d = 4$. This contradicts the preceding saturation argument. Consequently every subgraph of J satisfies T_3.

We record the precise consequence of a covered deletion within J. Let H be any induced subgraph of J with $gap(H) > 0$, and suppose H has an edge and a matching covering every vertex of its actual maximum degree D. Choose any vertex v of degree D and put $H' = H - v$. Since H and H′ satisfy T_3, $gap(H) = 1$ and $gap(H') \leq 1$. By Lemma 61, $c_D(H) = 0$ and all rows above D vanish in both graphs. Restriction makes $c_D(H') = 0$ as well. For every integer i with $0 \leq i < D$ put $a_i = c_i(H) - c_i(H')$. Lemma 72 gives a_i in {0,1}. Deleting v removes D edges, so

$gap(H') - gap(H) = D - \sum_{i=0}^{D-1} a_i \geq 0$.

Therefore $gap(H') = 1$, every $a_i = 1$, and $c_0(H') = c_0(H) - 1$.

Also $tr(H') \geq tr(H) + 1$. To prove this from actual partitions, put $c = c_0(H)$, $n = \nu(H)$, and $R = tr(H)$. Positive gap and nonnegative slack from Lemma 61 give $ex(H) \geq 1$; H has an edge, so $n \geq 1$ and $c \geq 2$. By Lemma 23 there is a longest nonempty missing run x_0,..., $x_0 + R - 1$ at total c-1. Budgets $x \leq ex(H) - 1$ are feasible by placing all edges in the second part, and budgets $x \geq n$ are feasible by placing them all in the first. Thus

$1 \le x_0 \le x_0 + R - 1 \le n - 1 \le c - 2$.

Any feasible first budget u of H′ at total c-2 becomes feasible on H at both first budgets u and $u+1$: put all edges incident with v into the second or first exceptional part respectively. Each matching uses at most one added edge. It follows that the entire interval $[x_0 - 1, x_0 + R - 1]$ is missing on H′ at total c-2. Its $R+1$ integers lie in [0,c-2]. Since $c_0(H') = c - 1$, this is its top missing level and proves the assertion. This argument establishes the deletion implication for every such induced H, including disconnected ones.

Now suppose $ps(J) \ge 1$ and $gap(J) > 0$. Write $E = ex(J)$ and $p = ps(J)$. Nonnegative integer slack and T_3 give $gap(J) = 1$, $E = p + 1 \ge 2$, $tr(J) \ge 1$, and $tr(J) + E \le 3$. Hence $E = 2$, $p = 1$, and $tr(J) = 1$.

If $\Delta(J) \le 2$, the degree-two assertion of Lemma 147 would give $ps(J) = 0$, a contradiction. Thus $\Delta(J)$ is three or four. When it is three, Lemma 243 gives twice the minimum number of uncovered degree-three vertices at most $p = 1$, so a matching covers all of them. When it is four, every proper induced subgraph has gap at most one by the T_3 conclusion already proved. If no matching covered all degree-four vertices, the even-degree slack-one assertion of Lemma 243 would give a whole connected four-regular factor-critical component of J. Since J is connected, it would be four-regular, a contradiction. Thus J has a covering matching in this case too.

Choose a maximum-degree vertex v of J and put $G = J - v$. The covered-deletion implication gives $gap(G) = 1$, $c_0(G) = c_0(J) - 1$, and $tr(G) \ge 2$. Set $h = \nu(J) - \nu(G)$. Restricting any maximum matching removes at most its edge incident with v, so h is either zero or one. The excess identity gives $ex(G) = E - 1 + h = 1 + h$. Since G satisfies T_3, $tr(G) \ge 2$ forces $ex(G) \le 1$. Therefore $h = 0$, $ex(G) = 1$, $ps(G) = 0$, and $tr(G) = 2$.

If $\Delta(G) \ge 3$, it is at most four and every proper induced subgraph of G has gap at most one. The zero-slack conclusion of Lemma 243 gives a matching covering all its maximum-degree vertices. Apply the covered-deletion implication again to any maximum-degree vertex w of G. It yields $gap(G - w) = 1$ and $tr(G - w) \ge 3$. Nonnegative slack implies $ex(G - w) \ge 1$, contradicting $T_3(G - w)$. Hence $\Delta(G) \le 2$.

Every component of a finite simple graph of maximum degree at most two is a path, a cycle, or an isolated vertex. This follows by following edges from an endpoint when one exists, or around a cycle when all component degrees are two; the degree bound permits no branching and connectedness leaves no extra vertices.

There is no whole odd-cycle component O of G. If one existed, connectedness of J would give a neighbor z of v on O: all edges from O to outside its component in J must use the only deleted vertex v. The even path O-z has a perfect matching of cardinality $\nu(O)$. Together with vz and maximum matchings of the other components of G, it gives a matching in J of size $\nu(G) + 1$. Here matching numbers add over whole components, by restriction and unions of maximum matchings. This contradicts $h = 0$.

Thus every component of G is a path, an even cycle, or an isolated vertex. A path on q vertices has a matching and a vertex cover both of size $\lfloor q/2 \rfloor$, using disjoint consecutive pairs and alternate vertices. An even cycle has an alternating matching and alternating vertex cover, both of half its order. An isolated vertex contributes zero. Their unions give $\tau(G) = \nu(G)$. Lemma 23 then gives $c_0(G) \le \tau(G) = \nu(G)$, contradicting $ex(G) = 1$. This proves $gap(J) \le 0$ when $ps(J) \ge 1$. If instead $gap(J) > 0$, the proved conclusion and nonnegative integer slack force $ps(J) = 0$, and T_3 gives $gap(J) = ex(J) = 1$.

Finally apply Lemma 109 with degree $\cap D = 4$ and total defect $d = 2$ to K. It proves connectedness, nonregularity, $\Delta(K) = 4$, $|V(K)| = 2m + 1 > 4$, $e(K) = 4m + 1$, the zero rows at indices at least four, and the stated positive slack identity. In particular $m \ge 2$. The conclusion just proved applies to this K and yields $gap(K) \le 0$. No hereditary hypothesis is needed. □

Depends on. Lemma 23, Lemma 61, Lemma 72, Lemma 109, Lemma 147, Lemma 243, Lemma 263.

Theorem 288 (Exclusion of degree-four minimal obstructions). *For a finite simple graph G, write $V(G)$, $E(G)$, $e(G)$, $\nu(G)$, $\alpha(G)$, $\Delta(G)$ for its vertex set, edge set, edge count, matching number, independence number, and maximum degree. For nonnegative integer degree capacities A_c, B_c, define $c_{A_c,B_c}(G)$ to be the least nonnegative integer total t such that every nonnegative integer pair x_c, y_c with $x_c + y_c = t$ admits an edge partition into four parts having maximum degrees at most A_c, B_c and matching numbers at most x_c, y_c, respectively. For every nonnegative integer i set $c_i(G) = \max_{A_c+B_c=i} c_{A_c,B_c}(G)$, with the maximum over nonnegative integer capacities. Put $c_0 = c_0$, $ex = c_0 - \nu$, and signed integer $gap(G) = \sum_{i \ge 0} c_i(G) - e(G)$; the sum is finite.*

Define $tr(G) = 0$ if $c_0(G) = 0$. Otherwise a first matching budget x in the integer interval [0,c0(G)-1] is missing if no partition of $E(G)$ into two parts has matching numbers at most x and c0(G)-1-x, respectively. Let $tr(G)$ be the greatest nonnegative integer length of a consecutive run of missing first matching budgets, with value 0 when no budget is missing. Property T_- is the implication

$gap(G) > 0 \Rightarrow [gap(G) = 1$ *and* $tr(G) + ex(G) \leq \max(4, floor(\Delta(G)/2) + 1)]$.

Let M_4 be the entire class of finite simple graphs F of maximum degree exactly four which violate T_- while every proper subgraph satisfies T_-. Proper subgraphs allow arbitrary vertex deletions, edge deletions, or both. Let N_4 denote the assertion that M_4 is empty.

Then N_4 holds: M_4 is empty. Thus there is no finite simple graph of maximum degree exactly four violating T_- with every proper subgraph satisfying T_-. The conclusion imposes no source-order, rank, excess, deficit, sector, saturation, extremal-frame, terminal-component, or process assumption beyond the displayed definition of M_4.

All degree capacities, matching budgets, least universal totals, cost indices, degrees, cardinalities, matching sizes, and finite choice domains are nonnegative integers. Generic gaps and parameter differences may be signed integers.

Proof. Suppose F belongs to M_4. The finite integer certificate costs, gap, top missing-allocation run and T_- in the statement agree with those in Lemma 284. Both definitions of M_4 allow every proper subgraph, including spanning edge deletions. Apply that fact to F. It gives connectedness, four-regularity, well-coveredness, and $a = \alpha(F) \geq 5$. Its integer parameters m, b, q, σ satisfy

$m \geq 1, 0 \leq b < m, q = m - b \geq 1, \sigma \geq b,$

and its exact second cost is $c_2(F) = a + 2m - b$.

For a maximum independent set S, let $C(S)$ be the graph induced by the degree-three vertices of F-S. The same fact gives $\max_S \alpha(C(S)) = m$. Define a frame to be such an S together with an independent m-element set U in $C(S)$. At least one frame exists, and every frame satisfies the following conclusions. Put $W = S$ union U, $R = F - W$ and $r = \nu(R)$. The graph F[W] is m disjoint edges and a-m isolates, and $\alpha(R) = a$. The whole components of R are exactly q odd cycles from C_3, C_5, C_7, exactly b even paths from P_2, P_4, P_6, and otherwise P_3, C_4 or isolates, with subscripts denoting vertex counts. For every frame-edge endpoint p, the actual induced graph $K_p = F[V(R)$ union {p}] satisfies

$\nu(K_p) = r + 1, c_1(K_p) = r + 1 - b.$

There is one fixed pair of nonnegative integer budgets (u, v), independent of the frame, missing as a two-part matching allocation on F, with

$u + v = |W| + r - 1, |W| - u \geq q + b + \sigma.$

If $b = 0$, then $q = m$. The terminal has exactly m of the listed odd cycles and otherwise P_3, C_4 or isolates, and each K_p has $\nu(K_p) = c_1(K_p) = r + 1$. The graph F is connected, four-regular and well-covered with $a \geq 4$; every $C(S)$ has independence number at most m. Thus all uniform frame hypotheses of Lemma 111 hold. Its conclusion that no frame exists contradicts the existence just obtained.

Hence $b \geq 1$. Since $\sigma \geq b$, the same missing pair satisfies $|W| - u \geq q + 2$ in every frame. Apply Lemma 58 with these F, a, m, b and with its budgets $(x, y) = (u, v)$. Its connectedness, regularity, well-coveredness and integer hypotheses hold, and all its terminal, endpoint-cost and missing-allocation hypotheses were established for every frame. Therefore no terminal contains P_6. Writing P for the frame-edge endpoints and Z for the frame isolates, every p in P has at most one degree-two neighbor in R and cannot meet both the center and a leaf of one whole P_3. Each whole odd cycle has at most one edge to P.

That fact also supplies, for every $T \subseteq W$ with $|T| \leq q + 2$, edges tv_t into pairwise distinct whole odd-order components C_t of R, with $C_t - v_t$ having a perfect matching. Such a matching has $(|V(C_t)| - 1)/2$ edges. Since no matching of the odd-order C_t can be larger, $\nu(C_t - v_t) = \nu(C_t)$.

The terminal catalogue with P_6 removed, the two endpoint-neighbor restrictions, and these simultaneous assignments are precisely conditions (i), (ii) and (iii) of Lemma 69. They hold for every frame of the same well-covered four-regular F. Its remaining cost hypothesis is the already established identity $c_2(F) = a + 2m - b$. That theorem excludes C_3 from every frame terminal.

The same catalogue, both endpoint-neighbor restrictions, the at-most-one edge from each odd cycle to P, and that same second-cost identity supply every hypothesis of Lemma 71, with $a \geq 4$, $m \geq 1$ and $0 \leq b < m$. It excludes C_5 and C_7 from every frame terminal.

Finally choose an existing frame. Its terminal has exactly $q \geq 1$ whole odd cycles, each C_3, C_5 or C_7, while all three possibilities have been excluded. This contradiction proves M_4 is empty. □

Remark. The original source-to-frame theorem supplies a uniform saturated frame family. The zero-deficit and exhaustive cycle exclusions contradict its existence.

Depends on. Lemma 58, Lemma 69, Lemma 71, Lemma 111, Lemma 284.

Lemma 289 (Positive slack and excess at least two in minimal failures). *Use the finite integer certificate costs $c_i(G)$, matching number $\nu(G)$, excess $ex(G) = c_0(G) - \nu(G)$, slack $ps(G)$, signed gap $gap(G) = ex(G) - ps(G)$, maximum degree $\Delta(G)$, and top missing-allocation run $tr(G)$ of Lemma 279. For each nonnegative integer D put $h(D) = \max(4, \lfloor D/2 \rfloor + 1)$. A graph G satisfies T_- if $gap(G) > 0$ implies both $gap(G) = 1$ and $tr(G) + ex(G) \leq h(\Delta(G))$. All graphs are finite and simple.*

If F violates T_- and every proper induced subgraph of F satisfies T_-, including the empty induced subgraph, then

$ps(F) \geq 1$ *and* $ex(F) \geq 2$.

The same conclusions hold if every proper subgraph of F satisfies T_-, where proper subgraphs may delete vertices, edges, or both, including spanning edge deletions. In particular no additional order, degree, well-coveredness, deletion-process or excess hypothesis is imposed. Costs, degrees, matching numbers, excesses and run lengths are nonnegative integers; gaps may be signed integers.

Proof. Use Lemma 279 for the structure and exact positive-row losses of an induced-minimal failure, Lemma 61 for nonnegative slack, and Lemma 267 for maximum-degree coverage at zero slack.

Let F satisfy the first stated hypothesis and put $D = \Delta(F)$. By Lemma 279, F is nonempty and D-regular, D is an even integer at least four, $gap(F) = 1$, and for every vertex v of F,

$c_D(F - v) = c_D(F) - 1$. (1)

For every proper induced subgraph J of F, the assumed property T_- implies $gap(J) \leq 1$: positive gap must equal one, and nonpositive gap already satisfies that inequality. Together with $gap(F) = 1$ this proves $gap(J) \leq 1$ for every induced subgraph J of F, including F itself and the empty graph.

By Lemma 61, $ps(F)$ is a nonnegative integer. Suppose $ps(F) = 0$. The preceding hereditary gap bound and $D \geq 4$ meet all hypotheses of Lemma 267, applied to $H = F$. Its maximum-degree conclusion gives $c_D(F) = 0$. Choose any vertex v of the nonempty graph F. Equation (1) would then give $c_D(F - v) = -1$, contradicting the nonnegativity of the finite integer certificate cost. Therefore $ps(F)$ is not zero, and its integrality yields $ps(F) \geq 1$. Since $gap(F) = 1$ and $gap = ex - ps$, one has $ex(F) = ps(F) + 1 \geq 2$.

Finally, every proper induced subgraph is a proper subgraph in the broader sense stated. The stronger proper-subgraph hypothesis therefore implies the first hypothesis, and the same proof gives both conclusions without adding any other assumption. □

Remark. A zero-slack graph with hereditary gap at most one has a vanishing top certificate row. Minimal failure makes that same row drop by one under vertex deletion, which cannot happen from zero.

Depends on. Lemma 61, Lemma 267, Lemma 279.

Lemma 290 (Nonbinary terminal avoidance outside degree six). *Let $s, t \geq 1$ be integers and let F_1, F_2 be the vertex-disjoint unions of stars with complete nonincreasing positive integer edge-size lists $n_1,\ldots, n_s$ and $m_1,\ldots, m_t$, prescribed red and blue. Copies are ordinary subgraphs with vertex-disjoint target components. Put $M = s + t - 1$ and $\ell_k = max\{n_i + m_j - 1 : 1 \leq i \leq s, 1 \leq j \leq t, i + j = k\}$ for every integer $2 \leq k \leq s + t$. Fix an integer $0 \leq q < M$, and put $K = M - q$, $L_q = \sum_{k=q+2}^{s+t} \ell_k$ and $P_q = L_q - K$. Let A_q consist of all nonempty ordered actual suffix pairs after leading whole-component cuts α, β with integers $0 \leq \alpha < s$, $0 \leq \beta < t$ and $\alpha + \beta = q$.*

Put $D = \ell_{q+2} - 1$, and assume $D \geq 3$ and $D \neq 6$. Let R be the initial run length of $D+1$ in the complete natural tail ℓ_{q+2},..., ℓ_{s+t}. Suppose valid original indices i_0, j_0 satisfy $i_0 + j_0 = q + R + 1$, $n_{i_0} = a \geq 3$, $m_{j_0} = b \geq 3$ and $a + b = D + 2$. Let r be an integer with $0 \leq r \leq 2$. Then every finite simple graph H with $\Delta(H) \leq D$, $\Phi(H) := e(H) - \nu(H) \leq P_q + r$, $\delta := K - \nu(H) \geq r + 1$, where e, ν, Δ denote edge count, maximum matching cardinality and maximum degree, has an edge coloring avoiding both entire prescribed targets of some member of A_q.

Every such avoiding coloring extends over the original disjoint canonical q-star pool of sizes ℓ_2,..., ℓ_{q+1}, and across any specified actual q-vertex set X in any finite simple graph G with $H = G - X$, preserving every H-edge color and avoiding both unchanged full originals. Vertex deletion is induced. No full-head bound, short-prefix, strict-cut, component-structure, unit-count, repetition or smaller-target hypothesis is imposed. All sizes, counts, indices, degrees, matching cardinalities, deficits and allowances are integers in their stated domains.

Proof. Suppose for contradiction that H arrows every member of A_q. Apply Lemma 286 to these same full original lists, terminal indices, allowance r and matching deficit δ. Its hypotheses hold exactly as stated here. Put $\sigma = \Phi(H) - P_q$. The result gives $0 \leq \sigma \leq r$, even terminal sizes a, $b \geq 4$, and even $D \geq 6$. Since $D \neq 6$, we have $D \geq 8$. Let κ be the number of whole connected odd D-regular components of H. Each has a near-perfect matching, $\kappa \geq R \geq 1$, and

$\zeta \leq R + \sigma - \kappa$, (1)

where ζ is the sum over all whole connected D-regular components C of the attained nonnegative integers

`g(C)=max{nu(C)-|W|:` W is an inclusionwise maximal matching of C}.

The positive-deficit family clause of Lemma 89 gives $q < \min(s, t)$. Therefore the corner pairs (F_1, F_2^q) and (F_1^q, F_2) are legal and arrowed by H. An all-red coloring of the first contains the entire F_1, because its positive blue target is absent; an all-blue coloring of the second contains the entire F_2. Consequently both original heads are at most $\Delta(H) \leq D$. They are at least the respective terminal sizes a, b, and thus at least four. Also

$e(H) = L_q + \sigma - \delta \leq L_q - 1$. (2)

If either full head equals D, exchange both full original targets and prescribed colors so that this is the first head. Its terminal size is still at least four and the opposite terminal size at least four. Lemma 215 applies to this full family: the positive deficit and edge bound follow above, both heads are at most D, the leading natural value is $D + 1$, and the same H has maximum degree at most D. With $k = floor((D-1)/2) \geq 3$ and its nonnegative cover surplus γ it gives

$R + (k-1)(\delta + \gamma) \leq k\,\sigma$.

Using $\delta \geq r + 1$ and $\sigma \leq r$ yields

$1 \leq R \leq k\,r - (k-1)(r+1) = r - k + 1 \leq 0$,

a contradiction. Hence both original heads lie in [4,D-1].

Apply Lemma 176 with allowance $z = r$. Its requirements hold: D is even and at least eight, both full heads are below D, $0 \leq r \leq 2 \leq D/2 - 1$, the deficit is positive, and at least R whole odd D-regular near-perfect components exist. Every such component C therefore satisfies

$\nu(C) \geq D - 2 + 3\delta \geq D + 1$. (3)

If C were equimatchable, Lemma 208 would give $\nu(C) \leq D$. Thus all κ of these components fail equimatchability and contribute at least one to ζ. From (1) we obtain

$2\kappa - R \leq \sigma \leq r \leq 2, 1 \leq R \leq \kappa$.

These integer inequalities force either $R = \kappa = 1$ or $R = \kappa = 2$. Indeed $\kappa \leq 2$, and $\kappa = 2$, $R = 1$ violates the first inequality.

If $R = \kappa = 2$, the same inequalities force $\sigma = r = 2$, so $\delta \geq 3$. The two-component conclusion of Lemma 176 now applies: both heads are in [4,D-1], the plateau length is two, the potential is at most $P_q + 2$, and two whole odd regular near-perfect components exist. It supplies an avoiding A_q member, contradicting common arrowing. Therefore $R = \kappa = 1$. The inequalities also give $r \geq 1$, and hence $\delta \geq 2$.

Let C be this unique whole odd D-regular component and put $\mu = \nu(C)$. Apply Lemma 200 to the same H, C, full lists and q, with allowance $z = r$. Its strict numerical requirement holds because $r + 1 \leq 3 < D/2$. The

plateau is unique, (2) gives $e(H) \le L_q - 1$, δ is positive, and C has a near-perfect matching. Orient the full originals and colors as F, J with $e(F) < e(J)$, and write S, T for their component counts and B, A for their heads. The theorem gives

$\mu \ge e(F) + \delta - 1 \ge e(F) + 1$, $T \ge \mu + \delta + 1 \ge \mu + 3$. (4)

If $B = D - 1$, it also says that the entire original group of J-components of size at least $D + 1 - B = 2$ has at most $2\mu + 1$ vertices.

Define Wplus=2sum_(m in the full J-list)floor((m-1)/2). Choose an attaining separator of H; one exists by Lemma 89, and let $h \ge 0$ be its size. The family bound of that fact, in the present full orientation, gives

$q + 2\delta + h \le S - Wplus$.

Thus $Wplus \le S - q - 2\delta - h \le S - 4$. This uses the full opposite list, without discarding any units or intermediate sizes. Since C is a subgraph, $\mu \le \nu(H)$, so

$q + \mu \le q + \nu(H) = M - \delta < M$. (5)

All hypotheses of Lemma 90 are now verified for these same full originals and the actual C: even $D \ge 8$; both heads in [4,D-1]; the unique initial $D + 1$ entry; integer capacity μ with (4), (5), and $Wplus \le S - 4$; and, when $B = D - 1$, the full nonunit-group fitting condition just obtained. Equation (2) and the assumed potential bound give the graph premises $e(H) \le L_q - 1$ and $\Phi(H) \le P_q + 2$. Its graph conclusion supplies a coloring avoiding both entire targets of an actual total-q pair, with C wholly in J's prescribed color. Undo any full color-role exchange to obtain an avoiding member of the original A_q. This is the final contradiction.

Hence H does not arrow every A_q member. The family is finite and nonempty: $\alpha = \min(q, s - 1)$, $\beta = q - \alpha$ are legal because $q \le s + t - 2$. Negating common arrowing therefore gives a coloring avoiding both entire targets of some member. Its original canonical-pool extension and extension across every specified actual X are Lemma 41 (I), (II). Those extensions preserve all H-edge colors and retain every original component, including copies meeting X through leaves. Every step used the same full original lists or an explicit simultaneous exchange of their roles and colors. □

Remark. The envelope removes a full critical head. Fitting makes every odd regular component too large to be equimatchable, and the total matching-gap budget leaves only one after the two-component exit. Full-target half-order capacity then licenses one exact remnant on that same component.

Depends on. Lemma 41, Lemma 89, Lemma 90, Lemma 176, Lemma 200, Lemma 208, Lemma 215, Lemma 286.

Lemma 291 (Closed octic graphs of excess four are LP-good under induced heredity). *Use the finite integer certificate costs $c_i(H)$, matching number $\nu(H)$, excess $ex(H) = c_0(H) - \nu(H)$, slack $ps(H)$, signed $gap(H) = \sum_{i \ge 0} c_i(H) - e(H) = ex(H) - ps(H)$, and top missing-allocation run $tr(H)$ from Lemma 276. A finite simple graph J satisfies T_- when*

$gap(J) > 0$ *implies* $[gap(J) = 1$ *and* $tr(J) + ex(J) \le \max(4, floor(\Delta(J)/2) + 1)]$.

Here Δ is maximum degree, with Δ of the empty graph zero.

Let H be a nonempty connected eight-regular finite simple graph such that deleting any vertex leaves a perfect matching. Suppose $ex(H) = 4$ and every proper induced subgraph of H satisfies T_-. Then H is LP-good:

$\sum_{i \ge 0} c_i(H) \le e(H)$.

All orders, degrees, indices, costs, matching cardinalities, excesses and run lengths are integers in their displayed nonnegative or positive domains; gaps may be signed integers.

Proof. Suppose $gap(H) > 0$. We first have $gap(H) = 1$. Otherwise H fails T_- because its positive gap is not one, while by hypothesis every proper induced subgraph satisfies T_-. The minimal-induced-failure assertion of Lemma 276 would then give $gap(H) = 1$, a contradiction. Thus the positive integer gap is one.

Apply Lemma 280 to this same H. Its hypotheses now hold exactly. It gives $|V(H)| = 29$, $\alpha(H) = 8$ and $c_7(H) = 8$. For every maximum independent set S, the actual complement $G = H - S$ has $\nu(G) = 10$, $ps(G) = 0$, and

$c_i(H) = 8 + c_i(G)$ for every integer $1 \le i \le 7$.

The graph H also satisfies all hypotheses of Lemma 261: its order and seventh row were just established and its connectedness, regularity, factor-criticality, excess, gap and proper-induced property are unchanged. Therefore every maximal independent set of H has eight vertices.

We verify the universal matching-coverage hypothesis of Lemma 187. Fix any maximum independent set S and any process from $G = H - S$ that repeatedly deletes a vertex of current maximum degree while that degree is at least three. All its parent graphs are proper induced subgraphs of H. Every induced subgraph of such a parent, including the parent itself, remains proper induced in H and therefore has gap at most one by T_-.

Initially $ps(G) = 0$. Suppose a current parent J has $ps(J) = 0$ and actual maximum degree $D \ge 3$. Apply Lemma 243 at this D, using the proper-induced gap bound just checked if $D \ge 4$. Its zero-slack consequence gives a maximum matching covering all degree-D vertices. Let v be whichever maximum-degree vertex this process chooses. Put $h = \nu(J) - \nu(J - v)$, which is zero or one: restricting a maximum matching removes at most one edge, and every matching of J-v remains a matching of J. The covered-degree deletion inequality of Lemma 61 gives

$ps(J) - ps(J - v) \ge 1 - h \ge 0$.

That fact also gives $ps(J - v) \ge 0$. Since $ps(J) = 0$, these inequalities force $ps(J - v) = 0$. This proves by induction that every deletion parent of the arbitrary process has zero slack and the required covering matching. The process terminates, because each step removes one vertex, and stops at maximum degree at most two. No common choice of covering matching across parents was required.

All hypotheses of Lemma 187 are now supplied for every maximum independent set and every complete process. Consequently $c_2(H) = 18$, every such process has seven deletions with degrees

(7,7,7,6,6,6,4),

and every terminal is the disjoint union of three triangles and five isolated vertices. Choose one maximum independent set and one complete process, which exist by finiteness, and let R be its actual terminal vertex set. The same fact gives a vertex x and a set A such that

$V(H) = R$ disjoint union {x} disjoint union A,

both induced sides H[R], H[A] consist of three triangles and five isolates, and x has four neighbors in each side.

We can therefore apply Lemma 79 to these same H, R, x, A. Its first hypothesis holds because $c_2(H) = 18$ and every maximal independent set has size eight. Its actual partition and all side and center conditions were supplied by the preceding paragraph. Every independent eight-set is maximum because $\alpha(H) = 8$, so its universal complete-process premise follows from the universal seven-deletion word just proved, whose first six entries are 7, 7, 7, 6, 6, 6. The exclusion gives a contradiction.

Thus $gap(H) > 0$ is impossible. The identity $gap(H) = \sum_{i \ge 0} c_i(H) - e(H)$ yields the asserted LP-goodness. □

Remark. The order window and the independent-set argument supply an actual well-covered 29-vertex graph. Zero slack guarantees coverage along every deletion history, forcing the frame that the finite octic exclusion rules out.

Depends on. Lemma 61, Lemma 79, Lemma 187, Lemma 243, Lemma 261, Lemma 276, Lemma 280.

Lemma 292 (Low-excess goodness for closed lumps of degree at least twelve). *Use the finite integer certificate costs $c_i(G)$, edge count $e(G)$, matching number $\nu(G)$, independence number $\alpha(G)$, maximum degree $\Delta(G)$, excess $ex(G) = c_0(G) - \nu(G)$, nonnegative slack $ps(G) = e(G) - \nu(G) - \sum_{i \ge 1} c_i(G)$, signed $gap(G) = \sum_{i \ge 0} c_i(G) - e(G)$, and top missing-allocation run $tr(G)$ of Lemma 260. For every nonnegative integer D put $h(D) = \max(4, \lfloor D/2 \rfloor + 1)$. A graph satisfies T_- if positive gap implies gap one and $tr + ex \le h(\Delta)$. All graphs are finite and simple. A closed d-lump is a nonempty connected d-regular graph of odd order for which deleting any vertex leaves a perfect matching. A graph is LP-good if its gap is nonpositive.*

For every even integer $d \geq 12$ and every closed d-lump H, if $ex(H) \leq 6$ and every proper induced subgraph of H satisfies T_-, including the empty graph, then H is LP-good.

Consequently, let F be a finite simple graph with $\Delta(F) \geq 12$ which violates T_- while every proper induced subgraph satisfies T_-. Then $\Delta(F) = 12$, F is a closed twelve-lump, and $ex(F) \geq 7$. The same conclusion holds when every proper subgraph satisfies T_-, where proper subgraphs may delete arbitrary vertices, edges, or both, including spanning edge deletions. All graph orders, degrees, capacities, matching budgets, costs, indices, matching cardinalities, excesses and run lengths are nonnegative integers in their stated domains; gaps may be signed.

Proof. Let d and H satisfy the first assertion. If $d \geq 14$, the unconditional large-degree theorem in Lemma 276 gives $gap(H) \leq 0$. Its definition of a closed lump agrees with ours: nonempty connected regularity and a perfect matching after every vertex deletion. This case does not require the excess bound or the hereditary assumption.

It remains to treat $d = 12$. Suppose $gap(H) > 0$. Lemma 260 applies to this same connected twelve-regular factor-critical H with excess at most six and T_- on every proper induced subgraph. It gives $\alpha(H) = 5$ and, for every maximum independent set S, a graph $G = H - S$ with maximum degree exactly eleven. Choose a maximum independent set S; it exists because the finite graph has finitely many vertex subsets. Choose a vertex v of degree eleven in G; such a vertex exists by that exact maximum degree.

For the actual induced graph $J = G - v$, the same fact gives nonregular factor-criticality and

$|V(J)| = 23$, $e(J) = 103$, $\nu(J) = 11$, $\alpha(J) \leq 5$,

$\Delta(J) \leq 11$, $ps(J) \leq 2$.

Every induced subgraph of J, including J itself and the empty graph, satisfies T_-. Therefore each such subgraph has gap at most one: a nonpositive gap already has that bound, and a positive gap equals one by T_-.

Apply the nonregular factor-critical consequence of Lemma 281 to precisely this J. It is nonempty, has maximum degree at most eleven and slack at most two, and has the full hereditary gap bound just checked. There is no additional order, rank, top-run or local-coverage premise in that consequence. It yields

$103 = e(J) \leq 3\nu(J) + 8\alpha(J) + 28 \leq 3 \cdot 11 + 8 \cdot 5 + 28 = 101$,

a contradiction. This proves the first assertion at $d = 12$ and hence at every indicated even degree.

For the consequence, apply Lemma 276 to the stated induced-minimal failure F. It gives connectedness, factor-criticality, positive gap one, and regular degree in {4,6,8,10,12}. The hypothesis $\Delta(F) \geq 12$ therefore makes its degree exactly twelve. Thus F is a closed twelve-lump with T_- on every proper induced subgraph. If $ex(F) \leq 6$, the first assertion gives $gap(F) \leq 0$, contradicting $gap(F) = 1$. The excess is an integer, so $ex(F) \geq 7$. Finally every proper induced subgraph is a proper subgraph, so the stronger proper-subgraph hypothesis implies the same induced-minimal hypotheses. This also includes the specified spanning edge deletions without changing the graph or its degree. □

Remark. The unique remaining degree-twelve profile has a dense factor-critical induced remainder. The low-slack density estimate on that same remainder is smaller than its exact edge count.

Depends on. Lemma 260, Lemma 276, Lemma 281.

Lemma 293 (Independent-set sizes and rank bounds in minimal failures). *Use the finite integer certificate costs c_i, matching number ν, independence number α, maximum degree Δ, excess $ex = c_0 - \nu$, slack ps, $gap = ex - ps$, and top missing-allocation run tr of Lemma 23. For every nonnegative integer j put $h(j) = \max(4, \lfloor j/2 \rfloor + 1)$. A graph satisfies T_- when positive gap implies gap one and $tr + ex \leq h(\Delta)$.*

Let d belong to {6,8,10,12}, and let F be a finite simple graph of maximum degree d which violates T_- while every proper induced subgraph satisfies T_-. Put $k = \nu(F)$, $E = ex(F)$, $a = \alpha(F)$, $z = c_{d-1}(F) - a$, $u = \max(0, floor((2E - 3)/(d - 2)))$, and $\lambda = 2E - 3 - (k - a)$.

Then F is connected, d-regular and factor-critical, meaning that deleting any vertex leaves a perfect matching. It satisfies $|V(F)| = 2k + 1$, $gap(F) = 1$, $E \geq 1$, $k \geq 3E - 1 + 2u$, z in {0,1}, and

$k - a \leq 2E + d/2 - 3 + z$, equivalently $\lambda \geq -d/2 - z$.

Every inclusionwise maximal independent set I has cardinality w satisfying $(d+1)w \geq 2k+1$ and $w \geq a+z-1$. Thus every such cardinality is a or a-1, and $z = 1$ implies that F is well-covered, meaning all maximal independent sets have cardinality a.

For every such I, put $R = F - I$ and $s = ps(R)$. The minimum number of degree-(d-1) vertices of R missed by a matching is at most one,

$c_{d-1}(R) = a + z - w$ *in {0,1},*

$gap(R)$ *in {0,1},* $c_0(F) = w + \nu(R) + s$, *and*

$1 \leq E - s = w + \nu(R) - k \leq \lceil w/2 \rceil$.

If $c_{d-1}(R) = 1$, then $\Delta(R) = d-1$ and that minimum exposure is exactly one. Every maximum independent set S satisfies $c_{d-1}(F - S) = z$.

The same conclusions hold if every proper subgraph satisfies T_-, where proper subgraphs allow vertex deletions, edge deletions, or both, including spanning edge deletions. No excess floor, rank restriction or deletion process is assumed. All orders, degrees, costs, indices, cardinalities, matching numbers, and parameters are integers in the displayed domains; λ may be negative.

Proof. Lemma 279 applies to the stated minimal induced failure. It gives connectedness, d-regularity, factor-criticality, odd order $2k+1$ and $gap(F) = 1$. The function h is nondecreasing on the nonnegative integers, so Lemma 115 applies with $T_h = T_-$. It gives $E \geq 1$, $c_d(F) = 1$, all higher costs zero, and all the stated independent-cut identities except the new bounds on the top cost and exposure. Simplicity and d-regularity imply $k \geq d/2 \geq 3$. The hypotheses of Lemma 45 apply to F with degree bound d. With the stated nonnegative integer u, that fact gives

$k \geq 3E - 1 + 2u$. (1)

Fix an arbitrary maximal independent set I. It is nonempty. Every vertex outside I has a neighbor in I, because otherwise that vertex could be added to I. Put $w = |I|$ and $R = F - I$. Then $\Delta(R) \leq d-1$, and the union of the closed neighborhoods of I covers F. Each of these neighborhoods has $d+1$ vertices, so

$(d+1)w \geq 2k+1 \geq 6E - 1$.

Write $D = d-1$ and $s = ps(R)$. Lemma 115 gives $s \geq 0$, $gap(R)$ in {0,1}, the exact identity $c_0(F) = w + \nu(R) + s$, and the stated bounds on E-s. Every induced subgraph of R is proper induced in F and therefore satisfies T_-.

If $s \leq D - 1$, apply Lemma 247 to R with this odd $D \geq 5$. Its maximum degree is at most D, its gap is nonnegative, and each proper induced subgraph has gap at most one. It gives exposure at most one and $c_D(R) \leq 1$.

Suppose instead $s \geq D$. The independent-cut inequality $E - s \geq 1$ gives $E \geq D + 1 = d$. Thus $(d+1)w \geq 6d - 1$, which implies $w \geq 5$ throughout d in {6,8,10,12}. Since $h(D) - 2 \leq 4$ in this range, $w \geq h(D) - 2$. If $gap(R) = 1$, then $ex(R) = s + 1$ and the positive top run satisfies $tr(R) \geq 1$ by Lemma 23. Property T_- would give $s + 2 \leq h(\Delta(R)) \leq h(D) \leq D$, a contradiction. Thus $gap(R) = 0$ and $ex(R) = ps(R) = s$. We may apply Lemma 244 with $H = F$ and $W = I$: the actual pool size satisfies its lower bound, its exact cost is the one just established, $s \geq D \geq h(D) - 1$, and R is hereditarily T_-. That fact again gives exposure at most one and $c_D(R) \leq 1$.

The independent-cut identity is $c_D(R) = c_D(F) - w$. Taking I to be a maximum independent set proves $0 \leq z \leq 1$; z is an integer. For the original arbitrary maximal I, this same identity and $c_D(R) \leq 1$ give $w \geq a + z - 1$. Since $w \leq a$ by the definition of $\alpha(F)$, its only possible values are a-1 and a. If $z = 1$ this forces $w = a$. The exact expression $c_D(R) = a + z - w$ follows. If that cost is one, $\Delta(R) < D$ would contradict the zero tail in Lemma 61. A matching covering all its degree-D vertices would also force that cost zero by the same fact. Its degree is therefore D and its exposure is positive, hence exactly one. In particular every maximum independent complement has cost z.

It remains to prove the rank inequality. Lemma 119 applies to F and any pair of positive integer thresholds summing to d. It gives

$c_i(F) + c_{d-i}(F) \leq 2k + 2$ for $1 \leq i < d$.

For the middle index it gives $c_{d/2}(F) \le k+1$. The integer indices 2 through d-2 consist of the pairs i, d-i for $2 \le i \le d/2-1$ and the singleton $d/2$; there are $d/2-2$ pairs. Thus their summed costs are at most $(d-3)(k+1)$, including the endpoint $d=6$. The remaining costs are $c_0(F) = k+E$, $c_1(F) \le k+E$ by row monotonicity in Lemma 23, $c_{d-1}(F) = a+z$, $c_d(F) = 1$, and the zero higher tail. Consequently

$\sum_{i\ge 0} c_i(F) \le 2(k+E) + (d-3)(k+1) + a + z + 1.$

Regularity gives $e(F) = d(2k+1)/2 = dk + d/2$. Subtracting this quantity and using $gap(F) = 1$ yields

$1 \le 2E + a + z - k + d/2 - 2,$

which rearranges to $k - a \le 2E + d/2 - 3 + z$. Substitution in the definition of λ gives its equivalent lower bound. Every independent set used in a cut above is the same set that supplies its actual pool and exact cost identity. Finally, the stronger proper-subgraph hypothesis implies proper-induced minimality, so all conclusions apply on that domain as well. □

Remark. A maximal independent complement has at most one uncovered top-degree vertex at either small or large slack. Exact independent deletion converts that exposure bound into near-equality of all maximal independent-set sizes; complementary cost bounds then control the source rank.

Depends on. Lemma 23, Lemma 45, Lemma 61, Lemma 115, Lemma 119, Lemma 244, Lemma 247, Lemma 279.

Lemma 294 (Independent cuts and absorption at quartic defect two). *Use the finite integer certificate costs $c_i(G)$, matching number $\nu(G)$, independence number $\alpha(G)$, maximum degree $\Delta(G)$, excess $ex(G) = c_0(G) - \nu(G)$, slack $ps(G) = e(G) - \nu(G) - \sum_{i\ge 1} c_i(G)$, and $gap(G) = ex(G) - ps(G)$ of Lemma 287. For a nonnegative integer L, let $Y_L(G)$ consist of the integers x in [0,L] for which every edge of G can be assigned to one of two parts of matching numbers at most x, L-x. Let $run_L(Y_L(G))$ be the largest cardinality of a consecutive integer interval missing from $Y_L(G)$, or zero if none is missing.*

Let Z be a nonempty finite simple graph such that Z-v has a perfect matching for every vertex v, $\Delta(Z) \le 4$, and $\sum_(v \in V(Z))(4 - deg_Z(v)) = 2$. Put $m = \nu(Z)$, $s = ps(Z)$, and $E = ex(Z)$. Then Z is connected, $|V(Z)| = 2m+1$, $m \ge 2$, $s \ge 1$, and $0 \le E \le s$.

For every maximal independent set S, where maximal means maximal under inclusion, put $a = |S|$, $J = Z - S$,

$t = \sum_{v\in S}(4 - deg_Z(v))$,

$\eta = \nu(J) - (m+1-a)$, *and*

$\rho = \sum_{i=1}^{3}(a - c_i(Z) + c_i(J))$.

Then $0 \le t \le 2$, $\eta, \rho \ge 0$, $\Delta(J) \le 3$, and

$ps(J) = s + t - 1 - \eta - \rho \ge 0,$

$m - a \le s + E - 1 + floor((s + t - 1 - \eta - \rho)/2).$

In particular,

$m - \alpha(Z) \le 2s - 1 + floor((s+1)/2).$

Put $L = m + s - 1$ and $B = \max(0, \alpha(Z) + 1 - \lfloor s/2 \rfloor)$. Then 0, L belong to $Y_L(Z)$, and

$run_L(Y_L(Z)) \le \max(0, m - 3s + 2) \le B.$

For every finite simple graph H and every vertex subset W with $H - W = Z$, where deletion is induced, if $w = |W| \ge B$, then every pair of nonnegative integers x, y with $x + y = w + m + s - 1$ admits a partition of all H-edges into parts of matching numbers at most x, y. Consequently $c_0(H) \le w + m + s - 1$. All edges meeting W, including edges inside W, occur in that partition.

Thus $w \ge \alpha(Z) + 1$ always suffices; $w \ge \alpha(Z)$ suffices if $s \ge 2$; and $w \ge \max(0, \alpha(Z) - 1)$ suffices if $s \ge 4$. No hereditary-gap, source-minimality, or independence hypothesis on W is imposed. Every order, degree, cost, index, cardinality, defect, matching number, budget, total and run length is an integer in the stated nonnegative or positive domain.

Proof. The defect-two conclusion of Lemma 287 applies to Z without a connectedness assumption. It gives connectedness, $|V(Z)| = 2m+1$ with $m \geq 2$, $e(Z) = 4m+1$, zero rows at every index at least four, $s \geq 1$, and $gap(Z) \leq 0$. The latter means $E \leq s$. Nonnegative excess is part of the imported certificate definitions.

Apply Lemma 179 to this same graph with degree $\cap$ four and total defect two. Its unconditional upper-half bound gives $c_2(Z) \leq m$. Row monotonicity gives $c_1(Z) \leq c_0(Z) = m+E$. The slack identity is

$s = 3m+1-c_1(Z)-c_2(Z)-c_3(Z)$.

Therefore

$c_3(Z) \geq m+1-s-E$. (1)

Fix any maximal independent set S. The nonempty finite graph Z has no empty maximal independent set, so $a \geq 1$. Every vertex outside S has a neighbor in S, since otherwise S could be enlarged. Thus $\Delta(J) \leq 3$. All vertex deficits $4 - deg_Z(v)$ are nonnegative and sum to two; their sum t over S therefore lies in {0,1,2}.

Choose a vertex v of S and a perfect matching in Z-v. It has m edges. Removing the other a-1 vertices of S destroys at most a-1 edges of that matching, leaving a matching in J of size at least $m+1-a$. Hence $\eta \geq 0$. For $i = 1, 2, 3$ let $d_i = c_i(Z) - c_i(J)$. The restriction and vertex-set extension estimates of Lemma 72 give $0 \leq d_i \leq a$, so $\rho = 3a - \sum_{i=1}^{3} d_i \geq 0$.

Independence of S implies that its incident edges are counted exactly once by summing its vertex degrees. Thus $e(J) = 4m+1-4a+t$. By Lemma 161 the subcubic J has no positive rows beyond three. Substituting $\nu(J) = m+1-a+\eta$ into its slack formula gives

$ps(J) = 3m-3a+t-\eta-\sum_{i=1}^{3} c_i(J)$

$= s+t-1-\eta-\rho$.

The same cubic theorem gives $ps(J) \geq 0$ and $c_3(J) \leq floor(ps(J)/2)$. Restoring S with Lemma 72 gives $c_3(Z) \leq a + c_3(J)$. Comparing with (1) proves the asserted bound on m-a for this same S, J, t, η, ρ.

Choose S maximum, which is possible by finiteness. Since $E \leq s$, $t \leq 2$, and $\eta, \rho \geq 0$, the bound just proved gives

$m - \alpha(Z) \leq 2s-1+floor((s+1)/2)$. (2)

We next construct the allocation interval at total $L = m+s-1$. This is at least m, so assigning every edge to either part gives both endpoints. Consider any nonnegative pair with this total and denote its smaller coordinate by z. If $z \geq m$, every edge can be assigned to that part. If $z < m$ and $z \leq 2s-2$, choose a z-element vertex set A of Z, put every edge meeting A in the part of budget z, and put the other edges in the remaining part. The first part has matching number at most z because every matching edge there uses a distinct A-vertex. The second part is induced on $2m+1-z$ vertices. The inequality $z \leq 2s-2$ implies

$2m+1-z \leq 2(L-z)+1$,

so every matching in that induced graph has at most L-z edges. This partitions all edges, including edges inside A, and exchanging the two parts handles either location of the smaller coordinate.

If a first budget is missing, its smaller coordinate must consequently be at least 2s-1. Both coordinates are then at least 2s-1, so every missing first budget lies in

$[2s-1, L-(2s-1)] = [2s-1, m-s]$.

This interval has $\max(0, m-3s+2)$ integers. Thus it bounds the length of every missing run. By (2),

$m-3s+2 \leq \alpha(Z)+1-s+floor((s+1)/2)$

$= \alpha(Z)+1-\lfloor s/2 \rfloor$.

Taking maxima with zero proves the displayed bound by B. This reasoning also covers an empty missing interval and arbitrarily large s; no bound on s relative to m was used.

Finally fix H, W as in the statement. The allocation-set composition theorem Lemma 75 applies to the actual induced remainder $Z = H - W$ at capacities zero. Since $Y_L(Z)$ contains its endpoints and its missing runs have length at most $B \leq w$, that theorem gives

$\{0, \ldots, w\} + Y_L(Z) = \{0, \ldots, w+L\}$.

For any first budget x in this last interval, choose $x = z_W + x_Z$ with $0 \le z_W \le w$ and x_Z in $Y_L(Z)$. Choose a certificate on Z for x_Z, $L - x_Z$ and a $z_W - element$ subset W_R of W. Retain the two parts on Z. Assign each added edge meeting W_R to the first part and every remaining added edge to the second part. Each latter edge meets W minus W_R. A matching contains at most z_W new first-part edges and at most $w - z_W$ new second-part edges, so the total matching bounds are x and $w + L - x$. Every edge of H is assigned exactly once, including an edge internal to W. This proves the full allocation assertion and its cost consequence by the definition of c_0.

For $s \ge 1$ one has $B \le \alpha(Z) + 1$. For $s \ge 2$, $\lfloor s/2 \rfloor \ge 1$ gives $B \le \alpha(Z)$. For $s \ge 4$, $\lfloor s/2 \rfloor \ge 2$ gives $B \le \max(0, \alpha(Z) - 1)$. These inequalities prove all three final sufficient conditions, including zero-size pools when permitted. □

Depends on. Lemma 72, Lemma 75, Lemma 161, Lemma 179, Lemma 287.

(printed in plain notation)

Lemma 295 (Strict potential for the complete natural suffix family). Strict potential for the complete natural suffix family.

Let s,t>=1 be integers and let n_1>=...>=n_s>=1 and m_1>=...>=m_t>=1 be integer lists. Let F_1,F_2 be their respective vertex-disjoint unions of stars, prescribed red and blue; a star of size j has j edges. Copies are ordinary subgraphs with vertex-disjoint target components. A finite simple graph arrows an ordered pair if every red-blue edge coloring contains the entire first target in red or the entire second target in blue.

Put M=s+t-1 and

ell_k=max{n_i+m_j-1:1<=i<=s,1<=j<=t,i+j=k}

for every integer 2<=k<=M+1. Fix an integer 0<=q<M. Let A_q be the complete family of ordered nonempty suffix pairs obtained by deleting alpha red and beta blue leading components, for integers 0<=alpha<s, 0<=beta<t and alpha+beta=q. Put

P_q=sum_(k=q+2)^(M+1)(ell_k-1), D=ell_(q+2)-1.

Assume either D>=5, or D>=3 and both original heads satisfy n_1,m_1>=4. Then every finite simple graph H with maximum degree Delta(H)<=D which arrows every member of A_q satisfies

e(H)-nu(H)>=P_q+1,

where e(H) is its edge count and nu(H) is its maximum matching cardinality.

All sizes, indices, cuts, degrees and graph cardinalities are integers in the displayed domains. The assertion imposes no matching-deficit, connectedness or parity hypothesis and allows q=0, original unit components, repeated component sizes and isolated host vertices. Every target in A_q retains its entire actual suffix.

Proof. Put N=M-q>=1. The family A_q is finite and nonempty: alpha=min(q,s-1), beta=q-alpha gives legal cuts because q<=s+t-2 implies beta<=t-1. Every member has component budget N.

For every integer 2<=h<=N+1, the full family envelope at local diagonal h equals ell_(q+h). Indeed, a suffix candidate with local indices i,j and cuts alpha,beta comes from original indices I=alpha+i,J=beta+j with I+J=q+h. Conversely, for any original candidate with I+J=q+h, choose alpha=min(q,I-1), beta=q-alpha. If alpha=q then beta=0; otherwise alpha=I-1 and beta=q-I+1=J-h+1<=J-1. Thus both cuts are nonnegative integers with alpha<=I-1<s, beta<=J-1<t, and the positive local indices I-alpha,J-beta total h. This proves equality of candidate sets. Write bar_ell_h=ell_(q+h) for this family envelope. Its potential baseline is P_q and its degree parameter is D.

Apply Lemma 99 to A_q and H. It gives Phi(H):=e(H)-nu(H)>=P_q. If equality holds, every whole D-regular component is equimatchable: each inclusionwise maximal matching in that component has maximum cardinality. Suppose henceforth that Phi(H)=P_q.

Let u,v be the numbers of unit components in the two original forests. Exchange both full targets and their prescribed colors if necessary so that u>=v. This preserves every diagonal, the full family arrowing assertion and both alternative head hypotheses of the statement. If D is three or four, the original heads are both at least four. Fact Lemma 194 , on these same originals, q and H, gives D>=n_1+1>=5, a contradiction.

It remains to exclude equality when D>=5. Let R be the positive length of the leading D+1 plateau of the natural tail. Fact Lemma 257 gives even D>=6 and exactly R whole connected components of H that are odd and D-regular, each with a near-perfect matching. It also says that every terminal maximizing pair has both sizes even and at least four. Such a pair exists because the finite family envelope at local diagonal R+1 attains D+1. Both sizes occur in the original lists, so n_1,m_1>=4.

Choose one of these whole components C. It is finite, connected, simple, odd, D-regular, admits a near-perfect matching and, by the earlier application of Lemma 99 , is equimatchable. Write alpha(C) for its maximum independent-set cardinality.

We first obtain the matching-complement restriction needed in both cases. Fix a maximum matching W_out of H-V(C). For any vertex x for which C-x has a perfect matching Z, the union W=W_out union Z is maximum in H: Z attains the odd component's upper bound (|V(C)|-1)/2 and matching numbers add across whole components. Apply Lemma 95 to A_q, H and this fixed W. Every sequence of N maximum-degree vertex deletions in the edge-deleted graph H-W has degree sequence bar_ell_2-1,...,bar_ell_(N+1)-1, and the terminal matching complement is edgeless. Let a_D be the number of entries equal to D-1 in this prescribed sequence.

In H-W the degree-D vertices are exactly the W-uncovered vertices of ambient degree D. They are independent, because an edge between two uncovered vertices could be added to W. Delete all of them first. They retain degree D until selected, so all these choices are legal. Precisely x is deleted in C, and the deleted vertices outside C depend only on W_out. In the remaining C-part, let A_x=V(C) minus({x} union N_C(x)). Its vertices have complement degree D-1, while the neighbors of x have complement degree D-2: every surviving vertex lost its Z-edge, and only the neighbors also lost an edge to x.

Next delete degree-(D-1) vertices outside C until none remain, using a fixed rule on the exterior graph. These are legal maximum-degree choices, and their number b_D is independent of x,Z. For any inclusionwise maximal independent set I of (C-Z)[A_x], delete the vertices of I in any order. Here C-Z removes matching edges only and the brackets mean induced subgraph. Independence means each retains degree D-1 until selected. Maximality means every vertex of A_x minus I loses a neighbor, so after these deletions all remaining degrees are at most D-2. No described positive-degree phase can pass the N-th step: its first N choices would be allowed by Lemma 95 , whose N-th complement is edgeless. Thus the described prefix extends to N steps and accounts for all degree-(D-1) choices. It follows that b_D+|I|=a_D. An eligible x exists by the near-perfect matching of C, and maximal independent sets exist by finiteness, so kappa_C=a_D-b_D is a nonnegative integer. It is the common cardinality of every such I for every eligible x,Z, including an empty A_x.

First suppose alpha(C)<=2. We prove that C must be complete. If it is already complete, its degree makes it K_(D+1). Otherwise put n_C=|V(C)| and fix a vertex x. Write B=N_C(x) for its open neighborhood and A=V(C) minus(B union {x}) for its nonneighbors, excluding x. The set A is a clique, since a nonadjacent pair in A together with x would be an independent triple. Thus |A|<=D+1. Its cardinality n_C-D-1 is even, so |A|<=D and n_C<=2D+1.

Equality n_C=2D+1 is impossible. At equality |A|=|B|=D. Each a in A has D-1 neighbors in A, none equal to x, and exactly one neighbor b(a) in B. For b in B let k_b be its number of A-neighbors. Its degree within B is D-1-k_b. Consequently the graph L on B whose edges are exactly the missing pairs of C[B] has degree k_b at b, and e(L)=sum_b k_b/2=D/2>=3. For every a in A the set B minus {b(a)} is a clique: a missing pair there would form an independent triple with a. Every L-edge must therefore meet b(a). At least two different values b(a) occur, since a single vertex adjacent to all D vertices of A and to x would have degree at least D+1. If b,c are two such values, every L-edge must have both b and c as endpoints. There would be at most one L-edge, contradicting e(L)>=3. Since n_C is odd, we have proved n_C<=2D-1.

For every three-element set X in V(C), the induced graph J=C-X has a perfect matching. To prove this, suppose otherwise and apply Lemma 10 with graph J and prescribed set U=V(J). It yields Y subseteq V(J) with o(J-Y)>|Y|, where o counts odd-order components. Since |V(J)|=n_C-3 is even, o(J-Y)-|Y| is positive and even, hence at least two. An induced subgraph of C has at most two nonempty components, because vertices from three components would be independent. Thus o(J-Y)<=2, forcing Y empty and exactly two odd components of J. Every vertex of J has degree at least D-3. Each of its two odd components therefore has at least D-2 vertices and, since D-2 is even, at least D-1 vertices. This implies n_C-3>=2D-2, contrary to n_C<=2D-1. The contradiction proves the perfect matching

assertion. In particular, for every edge ab of C-x, take a perfect matching of C-{x,a,b} and adjoin ab. This gives an actual perfect matching of C-x containing that prescribed edge.

Return to the fixed vertex x. Because C is not complete and is D-regular, n_C>D+1. Hence a=|A|=n_C-D-1 is even with 2<=a<=D-2. Every A-vertex has exactly D-a+1>=3 neighbors in B, so an A-to-B edge exists.

The common cardinality kappa_C established above now applies to every perfect matching of C-x and every maximal independent set in its induced matching complement on A.

If a=2, write A={y,z}. The edge yz extends to a perfect matching Z_1 of C-x by the three-deletion argument. Then (C-Z_1)[A] is edgeless and its unique maximal independent set has size two. An A-to-B edge extends to a perfect matching Z_2 of the same C-x. Its incident A-vertex cannot also be matched by yz, so (C-Z_2)[A] is a complete graph on two vertices, with maximal independent sets of size one. This contradicts the common kappa_C.

Suppose a>=4. Choose a perfect matching Z of C-x containing an A-to-B edge. Its internal A-edges form a matching which does not cover A. If it is nonempty, the clique with those matching edges removed has a universal vertex, namely any vertex not covered by an internal edge. That singleton is maximal independent. The endpoints of any removed internal edge form a maximal independent pair: every other A-vertex is adjacent to both endpoints, since only matching edges were removed. The sizes one and two again contradict kappa_C.

If Z has no internal A-edge, it matches the a vertices of A to distinct partners in B. Among these partners some b_1,b_2 are adjacent, since otherwise any three partners would be an independent triple. Let a_1,a_2 be their distinct partners in A. Replace a_1b_1,a_2b_2 by a_1a_2,b_1b_2. The new edges exist, the four endpoints are distinct, and retaining every other edge of Z yields a perfect matching of C-x. It has exactly one internal A-edge, which does not cover A because a>=4. The preceding singleton and pair contradiction applies to this actual matching. This exhausts all possibilities and proves C=K_(D+1) when alpha(C)<=2.

Now Lemma 272 excludes this whole K_(D+1) component in H. Its hypotheses concern the same original lists and q, both original heads are at least four as proved above, D is the same even degree, and H arrows the same entire A_q with Phi(H)=P_q. Thus the case alpha(C)<=2 is impossible.

If alpha(C)>=3, apply Lemma 208 to the connected odd D-regular equimatchable graph C, with even D>=6. It makes C isomorphic to F_D: start with K_(D,D), delete a matching of size k=D/2, and adjoin an apex x adjacent to its D endpoints. In the two old shores let A_0,B_0 be the k-element endpoint sets and A_1,B_1 their k-element complements. Label A_0={a_1,...,a_k}, B_0={b_1,...,b_k} so that the removed edges are a_i b_i. The edges a_i b_(i+1), with indices read modulo k, are present and perfectly match A_0 to B_0, since k>=3. Add any perfect matching of the complete bipartite graph on A_1,B_1. Their union Z is a perfect matching of C-x.

For this root, A_x=A_1 union B_1 and (C-Z)[A_x] is K_(k,k) minus a perfect matching. The whole side A_1 is maximal independent of size k, since every B_1-vertex still has k-1 neighbors in A_1. The two endpoints of a removed A_1-to-B_1 matching edge form a maximal independent pair: every other vertex of either shore neighbors its opposite endpoint. These sizes k>=3 and two contradict kappa_C. Thus alpha(C)>=3 is also impossible.

The two independence-number cases exhaust C, so equality Phi(H)=P_q is impossible. Since both quantities are integers and the non-strict bound was proved first, Phi(H)>=P_q+1 follows in every stated case.

□

Remark.

Exact potential fixes every greedy matching-complement phase. That rigidity excludes the low-independence regular component by explicit matching switches and the high-independence component by its bipartite-apex structure.

Depends on. Lemma 10, Lemma 95, Lemma 99, Lemma 194, Lemma 208, Lemma 257, Lemma 272.

Lemma 296 (The hereditary run-plus-excess bound). *Use the finite integer certificate costs $c_i(G)$, matching number $\nu(G)$, maximum degree $\Delta(G)$, nonnegative slack $ps(G)$, excess $ex(G) = c_0(G) - \nu(G)$, and signed gap $gap(G) = ex(G) - ps(G)$ of Lemma 259. The top missing-allocation run $tr(G)$ is zero when $c_0(G) = 0$; otherwise it is the greatest length of a consecutive run of integers x in $[0, c_0(G) - 1]$ for which the edges of G cannot be partitioned into two parts with matching numbers at most x and $c_0(G) - 1 - x$.*

All graphs are finite and simple; the empty graph has maximum degree zero. For each nonnegative integer D put $h(D) = \max(4, \lfloor D/2 \rfloor + 1)$. A graph satisfies T_- if

$gap(G) > 0$ implies $[gap(G) = 1$ and $tr(G) + ex(G) \le h(\Delta(G))]$.

A graph is LP-good if $gap(G) \le 0$.

Let H be any finite simple graph such that every induced subgraph of H, including H itself and the empty graph, satisfies T_-. If $gap(H) > 0$, then

$gap(H) = 1$, $1 \le ex(H) \le 3$, and $tr(H) + ex(H) \le 4$.

Consequently every such H with $ex(H) \ge 4$ is LP-good.

In particular, let F be any finite simple graph such that every proper subgraph satisfies T_-. Then every proper subgraph J of F with $ex(J) \ge 4$ is LP-good. A proper subgraph may delete arbitrary vertices, edges, or both, including spanning edge deletions. This last assertion imposes no degree, regularity, connectedness or failure condition on F.

All orders, degrees, costs, matching cardinalities, excesses, run lengths and budget indices are integers in their stated nonnegative domains; gaps may be signed integers.

Proof. Every induced subgraph J of H has $gap(J) \le 1$: a nonpositive gap already satisfies this inequality, and a positive gap equals one by T_-. Positive gap and nonnegative slack also imply positive excess.

Suppose $gap(H) > 0$ and $tr(H) + ex(H) > 4$. Apply the final specialization of Lemma 259 with the integer $B = 4$. Its hereditary gap hypothesis was just checked. It gives an actual induced subgraph F of H which is nonempty, connected, and D-regular for an even integer $D \ge 4$, such that deleting every vertex of F leaves a perfect matching, and

$gap(F) = 1, ex(F) \ge 1, tr(F) + ex(F) \ge tr(H) + ex(H) > 4$. (1)

Every induced subgraph of F is induced in H, so F and all its induced subgraphs satisfy T_-.

Write $E = ex(F)$ and $R = tr(F)$. Since $c_0(F) = \nu(F) + E \ge 1$, minimality of the universal integer certificate total $c_0(F)$ implies that at total $c_0(F) - 1$ at least one first matching budget is missing. Thus $R \ge 1$. The T_- property on F itself gives

$R + E \le \max(4, D/2 + 1)$. (2)

We exclude all possible even values of D.

If D is four or six, (2) gives $R + E \le 4$, contrary to (1).

If $D = 8$, (2) and $R \ge 1$ give $1 \le E \le 4$. If E is one or two, Lemma 285 applies to F because every induced subgraph of F, including F, satisfies T_-. It gives $R \le 2$ and hence $R + E \le 4$. If $E = 3$, the graph F meets every hypothesis of Lemma 278: it is nonempty, connected, eight-regular, has a perfect matching after every vertex deletion, has gap one and excess three, and all its proper induced subgraphs satisfy T_-. That theorem gives $R = 1$, again making $R + E = 4$. If $E = 4$, Lemma 291 gives $gap(F) \le 0$ on this same graph with the same proper-induced hypothesis. Each case contradicts (1).

If $D = 10$, (2) and $R \ge 1$ give $1 \le E \le 5$. If $E \le 4$, apply Lemma 251. Its proper-induced gap bound follows from the T_- property already established on every induced subgraph of F. Its other hypotheses are the connected regular factor-critical properties in (1). It gives $gap(F) \le 0$. If $E = 5$, apply Lemma 283 to this same nonempty connected ten-regular factor-critical F with its proper-induced T_- hypothesis. It also gives $gap(F) \le 0$. Both conclusions contradict (1).

If $D = 12$, (2) gives $E \le 6$. Lemma 292 applies to this connected twelve-regular factor-critical graph with excess at most six and T_- on every proper induced subgraph. It gives $gap(F) \le 0$, a contradiction.

Finally, if $D \geq 14$, the unconditional large-degree assertion of Lemma 276 applies to this same nonempty connected even-regular factor-critical graph and gives $gap(F) \leq 0$. This contradicts (1) without any bound on E.

The cases exhaust all even integers $D \geq 4$. Therefore every positive-gap H in the stated hereditary family satisfies $tr(H) + ex(H) \leq 4$. Its own T_- property gives $gap(H) = 1$, and the identity $gap = ex - ps$ with nonnegative slack gives $ex(H) \geq 1$. Consequently $c_0(H) \geq 1$, so the same minimal-total argument used above gives $tr(H) \geq 1$. It follows that $ex(H) \leq 3$. If such an H has $ex(H) \geq 4$, positive gap is therefore impossible, proving LP-goodness.

For the final assertion, fix a proper subgraph J of the specified F. Every induced subgraph Q of J is a subgraph of F and is proper: equality $Q = F$ would force F to be a subgraph of J and thus $J = F$, contrary to the choice of J. This argument also covers J that retains every F-vertex but deletes an edge. Hence every induced subgraph of J, including J itself and the empty graph, satisfies T_-. The preceding result with J in place of H proves that $ex(J) \geq 4$ implies $gap(J) \leq 0$, as required. □

Remark. Any hereditary violation of the bound descends to a closed even-regular witness. Its own degree threshold and the complete small-excess goodness theorems exclude every degree.

Depends on. Lemma 251, Lemma 259, Lemma 276, Lemma 278, Lemma 283, Lemma 285, Lemma 291, Lemma 292.

Lemma 297 (Avoiding colorings at nonpositive potential surplus). *Let $s, t \geq 1$ be integers and let $n_1 \geq \cdots \geq n_s \geq 1$ and $m_1 \geq \cdots \geq m_t \geq 1$ be arbitrary integer lists. Let F_1, F_2 be the respective vertex-disjoint unions of stars with these edge sizes, prescribed red and blue. A star of size j has j edges. Copies are ordinary subgraphs with vertex-disjoint components within each target. A finite simple graph arrows (F_1, F_2) if every red-blue edge coloring contains the entire F_1 in red or the entire F_2 in blue.*

Put $M = s + t - 1$,

$\ell_k = max\{n_i + m_j - 1 : 1 \leq i \leq s, 1 \leq j \leq t, i + j = k\}$

for every integer $2 \leq k \leq M + 1$, and put

$L = \sum_{k=2}^{M+1} \ell_k$, $P = L - M$.

For a finite simple graph G, let $e(G)$ be its edge count, $\nu(G)$ its maximum matching cardinality, and $\Phi(G) = e(G) - \nu(G)$. Define the signed integers $\sigma(G) = \Phi(G) - P$ and $\delta(G) = M - \nu(G)$.

Every G arrowing (F_1, F_2) with $e(G) < L$ satisfies

$1 \leq \sigma(G) \leq \delta(G) - 1$ *and* $\delta(G) \geq 2$.

Every finite simple G satisfying $\Phi(G) \leq L - M$ and $\nu(G) < M$ has a red-blue edge coloring avoiding both entire prescribed targets. In particular, $e(G) = L - 1$ and $\nu(G) = M - 1$ imply that G does not arrow (F_1, F_2). These conclusions allow every positive matching deficit, all original unit components, repeated sizes and parities, and arbitrary host degrees and components, including isolated vertices and empty hosts. All list lengths, indices, edge sizes, graph orders, edge counts and matching cardinalities are integers in the stated nonnegative or positive domains.

Proof. We first prove the assertion about an arrowing graph G with $e(G) < L$. Suppose, to obtain a contradiction, that $\sigma(G) \leq 0$. Apply Lemma 177 to this same G and the unchanged full original lists. It gives an integer $0 \leq q < M$ and an actual induced residual H which arrows every ordered pair of nonempty actual suffix forests obtained by deleting α red and β blue leading components with $\alpha + \beta = q$. Write A_q for this complete family, and put

$K = M - q$, $P_q = \sum_{k=q+2}^{M+1} (\ell_k - 1)$, $D = \ell_{q+2} - 1$.

The same output satisfies

$\Delta(H) = D \geq 4$, $K - \nu(H) \geq 1$,

$\Phi(H) - P_q = \sigma(G) - \epsilon - j \leq 0$,

where ϵ, j are nonnegative integers supplied by that normalization. Here Δ is maximum degree, and the displayed Φ uses the same edge count and matching number as in the statement.

If $D = 4$, apply Lemma 277 to the unchanged full originals, this q, and this same H. Its natural-degree, degree-cap, common-family arrowing and positive matching-deficit hypotheses all hold. It gives $\Phi(H) \geq P_q + 1$, contradicting $\Phi(H) \leq P_q$.

If $D \geq 5$, apply Lemma 295 to the same full originals, q and H. Its first degree alternative holds, and it requires no original-head or matching-deficit condition. The degree $\cap$ and common arrowing again give $\Phi(H) \geq P_q + 1$, the same contradiction. These cases cover every integer $D \geq 4$.

Therefore $\sigma(G) > 0$. Since it is an integer, $\sigma(G) \geq 1$. Directly from the definitions,

$e(G) - L = \sigma(G) - \delta(G)$.

The left side is an integer at most -1. Hence $\delta(G) \geq \sigma(G) + 1 \geq 2$, proving the entire first assertion.

Now let G satisfy $\Phi(G) \leq L - M$ and $\nu(G) < M$. Integrality gives

$e(G) = \Phi(G) + \nu(G) \leq L - M + (M - 1) = L - 1$.

If G arrowed the entire original pair, the assertion just proved would force $\sigma(G) \geq 1$, contrary to $\Phi(G) \leq L - M$. Thus G does not arrow the pair, which by the definition of arrowing means that an avoiding red-blue coloring exists.

At the stated exact boundary, $\Phi(G) = (L - 1) - (M - 1) = L - M$ and $\nu(G) = M - 1 < M$. It is therefore included in this avoidance assertion. Every sum used above is the complete original sum through $M + 1 = s + t$; no unit entry or repeated component has been removed. □

Remark. Actual-suffix normalization preserves nonpositive surplus and positive matching deficit. The two strictness inputs exclude its possible critical degrees, forcing positive surplus in every below-budget arrower.

Depends on. Lemma 177, Lemma 277, Lemma 295.

Lemma 298 (Quartic remainder absorption at every slack). *Use the finite integer certificate costs c_i, matching number ν, independence number α, maximum degree Δ, excess $ex = c_0 - \nu$, slack $ps = e - \nu - \sum_{i \geq 1} c_i$, and $gap = ex - ps$ of Lemma 243. For nonnegative integer L, write $Y_L(G)$ for the set of first matching budgets x in [0,L] admitting a partition of all G-edges into parts of matching numbers at most x, L-x. Write $run_L(Y_L(G))$ for the longest consecutive missing integer run, with value zero if none is missing. Define $tr(G) = run_(c_0(G) - 1)(Y_(c_0(G) - 1)(G))$ when $c_0(G) > 0$ and zero otherwise. Say that G satisfies T_- if positive gap implies gap one and $tr(G) + ex(G) \leq \max(4, floor(\Delta(G)/2) + 1)$.*

Let H be a finite simple graph, W a vertex subset, $w = |W|$, and $R = H - W$ the induced remainder. Suppose $\Delta(R) = 4$, every induced subgraph of R including R satisfies T_-, and $w \geq \max(3, \alpha(R))$. If no matching of R covers every degree-four vertex, then $s = ps(R) \geq 1$ and at least one of the following holds:

(i) Every nonnegative integer pair x, y with $x + y = w + \nu(R) + s - 1$ admits a partition of all H-edges into parts of matching numbers at most x, y. In particular $c_0(H) \leq w + \nu(R) + s - 1$.

(ii) The graph R is connected, four-regular and factor-critical, meaning that deleting any vertex leaves a perfect matching; $s = 1$, $ex(R) = 2$, $w = \alpha(R)$, $\nu(R) = w + 2$, and $Y_(\nu(R))(R) = \{0, \nu(R)\}$.

Consequently, if $s \geq 2$ or $w > \alpha(R)$, conclusion (i) holds whenever R is uncovered. If instead $c_0(H) \geq w + \nu(R) + s$, uncoveredness forces precisely the necessary conditions in (ii). The conclusion does not assert existence of a graph satisfying (ii). No upper slack bound, order bound, regularity or minimality of H, or independence of W is assumed. All graph orders, degrees, costs, indices, cardinalities, matching numbers, capacities, budgets, totals and run lengths are integers in their stated nonnegative or positive domains.

Proof. We use Lemma 75 for allocation intervals, component unions and restoration of an actual vertex set. Matching numbers add over whole components, by restriction of a matching and union of maximum component matchings. Component cost subadditivity follows from the same allocation theorem for each capacity pair and hence for each row. Subtracting these inequalities from the additive edge and matching counts gives slack superadditivity over whole components.

All induced graphs and components used below have nonnegative slack. Here is a justification from the declared inputs. A connected graph of maximum degree at most four that is not four-regular satisfies the conclusions of Lemma 287. If its gap is nonpositive, its slack is at least its nonnegative excess. If its gap is positive, that theorem gives slack zero. For a four-regular connected graph induced in R, the hereditary gap-at-most-one hypothesis holds, so Lemma 243 gives nonnegative slack. Component superadditivity then treats disconnected graphs and the empty graph.

Choose an uncovered whole component U of R. Such a component has actual maximum degree four, since covering matchings on all degree-four components could otherwise be united. Put $p = ps(U)$, $m = \nu(U)$, and $P = R - V(U)$. The exposure bound Lemma 243 applies to U with its inherited proper-induced gap bound and gives $p \geq 1$. Hence $s = ps(R) \geq p + ps(P) \geq 1$.

Put $L_P = \nu(P) + ps(P)$. The set $Y_{L_P}(P)$ contains both endpoints. If $gap(P) \leq 0$ it is a full interval. Otherwise the inherited T_- property gives $gap(P) = 1$, $ex(P) = ps(P) + 1 \geq 1$ and $tr(P) \leq 4 - ex(P) \leq 3$; now $L_P = c_0(P) - 1$, so its missing runs have length at most three. Since $w \geq 3$, the interval criterion gives

$[0, w] + Y_{L_P}(P) = [0, w + L_P]$. (1)

The empty graph P is included. If $c_0(U) \leq m + p - 1$, this interval absorbs the full interval on U at total $m + p - 1$. The resulting total is

$w + L_P + m + p - 1 \leq w + \nu(R) + s - 1$.

Upward feasibility of the integer cost threshold gives (i). We may therefore suppose $ex(U) \geq p$.

Choose a set X of maximum cardinality among maximizers of the degree-four marked odd-component expression in U, and let $\kappa \geq 1$ be its matching exposure. Its existence and the factor-criticality of its required odd components follow from Lemma 107, using the degree-four vertices as the marked set.

If X is empty, connectedness and positive exposure force U itself to have odd order with every vertex of degree four, and the same marked-barrier theorem makes it factor-critical. Apply Lemma 238 to U. The allocation set at total $m + p - 1$ has endpoints and missing runs of length at most $\alpha(U) + 1$. If its run is at most $w + L_P$, (1) absorbs it and proves (i).

Otherwise that run exceeds $w + L_P$. Since

$w + L_P \geq w \geq \alpha(R) = \alpha(U) + \alpha(P) \geq \alpha(U)$,

Lemma 238 forces $p = 1$, $ex(U) = 2$, $m = \alpha(U) + 2$ and a missing run of exactly $\alpha(U) + 1$. Integrality now forces $w + L_P = w = \alpha(U)$. Thus $\alpha(P) = 0$ and P is empty. Hence $R = U$, $s = 1$, $w = \alpha(R)$, and $Y_m(R) = \{0, m\}$, which is precisely (ii). This settles the empty-X case.

Suppose $b = |X| \geq 1$. Let calQ be all whole components of U-X, calK the required odd components lying wholly in the degree-four marked set, and Y the union of the other components. For K in calK let $d(K)$ be its number of U-edges to X. Choose Z in calQ with largest matching number $q = \nu(Z)$. The marked barrier and Lemma 104 give

$|calK| = b + \kappa$, $\nu(U) = b + \sum_{Q \in calQ} \nu(Q)$,

and every $d(K)$ is a positive even integer. Starting with X and the full-cost allocation interval of Z gives length $b + c_0(Z) \geq q$. Every other Q has rank at most q and its matching-total allocation set has missing run at most max(0,nu(Q)-1). Thus Lemma 75 absorbs all of them and gives

$c_0(U) \leq b + c_0(Z) + \sum_{Q different from Z} \nu(Q)$.

Consequently $ex(Z) \geq ex(U) \geq p$. (2)

We apply the boundary-credit inequality Lemma 104. Every required K of boundary two is connected and factor-critical with total degree-four defect two. Lemma 179 gives $ps(K) \geq 1$ and the required credit max(2-ps(K),0), which is at most one and is supported only at index one. Assign this credit to each such K other than Z. Give zero credits to Z, the nonrequired components, and required K with $d(K) \geq 4$; the required credit for this last group is zero. These credits meet every support and exact-total condition of the boundary theorem. All procedures at indices two and three are finished; write E_{rem} for the unfinished index-one maximum credit, so $0 \leq E_{rem} \leq 1$.

Define

$C = e(U[X]) + e_U(X, Y)$

$+ \sum_{Qnot\in calK, Q\neq Z} ps(Q)$

$+ \sum_{K\in calK, K\neq Z} \max(ps(K) - (4 - d(K)), 0).$

Here $e_U(X, Y)$ counts edges between the displayed disjoint vertex sets. Every term is nonnegative. The boundary theorem also gives the nonnegative top correction $\zeta = \kappa - c_4(U) + \sum_Q c_4(Q)$. Substituting $gap = ex - ps$ into its gap bound cancels $ex(U)$ and gives

$ps(Z) + C + \zeta \leq p + E_{rem} - 3\kappa$

$+1_{ZincalK}(4 - d(Z))$. (3)

The indicator term is zero when Z is not required.

If Z is not required, or $d(Z) \geq 4$, (3) gives $ps(Z) \leq p - 2$. The inherited $gap(Z) \leq 1$ would then give $ex(Z) \leq p - 1$, contradicting (2). Thus Z is required with $d(Z) = 2$. Lemma 287 applies to this same factor-critical defect-two graph and gives $ex(Z) \leq ps(Z)$. Equations (2) and (3) now give

$p \leq ex(Z) \leq ps(Z) \leq p + E_{rem} - 3\kappa + 2 \leq p.$

Equality throughout forces

$ex(U) = ex(Z) = ps(Z) = p, \kappa = 1, E_{rem} = 1, C = \zeta = 0.$ (4)

In particular X is independent and no edge joins X to Y. Since U is connected and X is nonempty, Y must be empty: any component in Y would have no edge to X or to a different component of U-X. Thus $calQ = calK$ consists of exactly $b + 1$ odd factor-critical components.

For each Q different from Z, the zero correction gives $ps(Q) \leq 4 - d(Q)$. Nonnegative slack and positive even boundary imply either $d(Q) = 2$ and $1 \leq ps(Q) \leq 2$, or $d(Q) = 4$ and $ps(Q) = 0$.

In the boundary-two case, Lemma 294 gives

$\nu(Q) - \alpha(Q) \leq 2ps(Q) - 1 + floor((ps(Q) + 1)/2) \leq 4.$

At total $\nu(Q)$, the elementary allocation bound of Lemma 75 therefore gives missing run at most $\alpha(Q) + 3$. In the boundary-four case Q is connected and nonregular of maximum degree at most four. By Lemma 287 it satisfies T_3: positive gap implies gap one and $tr + ex \leq 3$. Since $ps(Q) = 0$ and excess is nonnegative, either $ex(Q) = 0$ and its matching-total interval is full, or $ex(Q) = 1$ and that total has missing run at most two.

The selected Z is not complete. It has odd order at least five and maximum degree four; completeness would force order five and total defect zero, contrary to defect two. Hence $\alpha(Z) \geq 2$. Independence numbers add on U-X, and an independent set there is independent in U, so

$w \geq \alpha(R) \geq \alpha(U) \geq \sum_{Q\in calQ} \alpha(Q).$

For any Q different from Z the other b components include Z and b-1 nonempty components. Thus

$w + b \geq \alpha(Q) + (b + 1) + b \geq \alpha(Q) + 3.$

Also $w + b \geq 4$. It follows that the actual pool W union X, together with P at total L_P, absorbs every Q different from Z at its matching number. Indeed (1) first absorbs P; adding [0,b] for X preserves a full interval, whose length is at least $w + b$, and each succeeding absorption only increases that length.

For Z itself, (4) and Lemma 294 give endpoints and a missing-run bound at total $q + p - 1$ of at most

$\max(0, \alpha(Z) + 1 - \lfloor p/2 \rfloor) \leq \alpha(Z) + 1.$

The other b nonempty components imply $w + b \geq \alpha(Z) + 2b \geq \alpha(Z) + 2$, so the same full interval absorbs Z at this total.

The graph H-(W union X) has exactly the whole pieces P and calQ. The total just constructed is

$w + b + L_P + \sum_{Qdifferentfrom Z} \nu(Q) + (q + p - 1)$

$= w + \nu(R) + ps(P) + p - 1$

$\leq w + \nu(R) + s - 1.$

For every allocation at this total choose the corresponding piece allocations, unite their parts, and restore each edge meeting the actual set W union X by assigning it according to a partition of that pool. This is the vertex-set construction of Lemma 75; each new edge is assigned once and the extra matching sizes are bounded by the two pool cardinalities. It includes edges inside the pool. Upward feasibility gives every allocation at $w + \nu(R) + s - 1$, proving (i).

We have proved the alternatives for every uncovered R. Alternative (ii) requires $s = 1$ and $w = \alpha(R)$, so either $s \geq 2$ or $w > \alpha(R)$ excludes it. Finally $c_0(H) \geq w + \nu(R) + s$ is incompatible with (i), forcing (ii) under uncoveredness. These are exactly the remaining assertions. □

Depends on. Lemma 75, Lemma 104, Lemma 107, Lemma 179, Lemma 238, Lemma 243, Lemma 287, Lemma 294.

Lemma 299 (Odd-degree coverage along source deletion processes). *Use the finite integer certificate costs $c_i(G)$, matching number $\nu(G)$, independence number $\alpha(G)$, maximum degree $\Delta(G)$, excess $ex(G) = c_0(G) - \nu(G)$, slack $ps(G)$, signed gap $gap(G) = ex(G) - ps(G)$, and top missing-allocation run $tr(G)$ of Lemma 293. For every nonnegative integer D put $h(D) = \max(4, \lfloor D/2 \rfloor + 1)$. A finite simple graph G satisfies T_- if $gap(G) > 0$ implies $gap(G) = 1$ and $tr(G) + ex(G) \leq h(\Delta(G))$. A graph of positive maximum degree is covered when some matching covers every vertex of that maximum degree.*

Let d belong to {6,8,10,12}, and let F be a finite simple graph of maximum degree d which violates T_- while every proper induced subgraph satisfies T_-, including the empty graph. Let S be any inclusionwise maximal independent set of F and put $H_0 = F - S$. Let $\ell \geq 0$ be an integer and let

$H_i = H_{i-1} - v_i$ *for* $1 \leq i \leq \ell$

be induced vertex deletions such that every parent H_{i-1} is covered and v_i has its current maximum degree $D_i = \Delta(H_{i-1}) \geq 3$.

If the endpoint H_ℓ has odd maximum degree at least three, then it is covered. Empty prefixes are included. Consequently, in any complete process from F-S which repeatedly deletes an arbitrary current maximum-degree vertex while that degree is at least three, the first uncovered deletion parent, if one exists, has degree in the following set:

$d = 6$*: {4};*

$d = 8$*: {6};*

$d = 10$*: {6,8};*

$d = 12$*: {8,10}.*

No coverage assertion is made for these remaining even degrees.

The same conclusions hold under the stronger hypothesis that every proper subgraph of F satisfies T_-, where proper subgraphs may delete arbitrary vertices, edges, or both, including spanning edge deletions. There is no assumed excess floor, source-order bound, well-coveredness, or complete covered process. All graph orders, degrees, costs, matching and independence numbers, cardinalities, indices and process lengths are integers in their stated nonnegative domains; gaps may be signed.

Proof. We use Lemma 293, Lemma 255, Lemma 275, and Lemma 274.

Put $j = |S|$, $a = \alpha(F)$, $k = \nu(F)$, $E = ex(F)$, and $n = |V(F)|$. Lemma 293 gives that F is connected and d-regular, $n = 2k + 1$, and, with $z = c_{d-1}(F) - a$ in {0,1},

$k - a \leq 2E + d/2 - 3 + z$, $j \geq a + z - 1$.

Adding the first inequality to $a - j \leq 1 - z$ gives

$k - j \leq 2E + d/2 - 2$.

Since $z \geq 0$, the second inequality also gives $a \leq j + 1$. The same fact supplies

$gap(H_0)$ in {0,1},

$c_0(F) = j + \nu(H_0) + ps(H_0)$.

The graph F is nonempty because its maximum degree is $d \geq 6$. Maximality therefore implies that S is nonempty. Every induced subgraph of H_0 is proper induced in F and satisfies T_-. Its gap is at most one, whether that gap is positive or nonpositive. These statements verify all initial hypotheses of Lemma 255.

Fix the prescribed prefix, put $R = H_\ell$ and $W = S$ union $\{v_1, \ldots, v_\ell\}$, and write $w = |W| = j + \ell$ and $s = ps(R)$. Suppose that R is uncovered and has odd maximum degree $D \geq 3$. The prefix theorem applies to this exact prefix, including $\ell = 0$. It gives

$d/2 < D \leq d - 1, s \geq 1,$

$gap(R)$ in {0,1},

$c_0(F) = w + \nu(R) + s + Z,$

where Z is the nonnegative integer sum of the deficits

$1 - c_\theta(H_{i-1}) + c_\theta(H_i)$

over $1 \leq i \leq \ell$ and $1 \leq \theta < D_i$. Its gap formula also gives $gap(R) \geq Z$, because the initial gap and all remaining deficits are nonnegative. As d belongs to {6,8,10,12}, the odd integer D belongs to {5,7,9,11}.

We check the actual pool size before invoking either absorption theorem. If $w = 1$, then $j = 1$ and $\ell = 0$. Maximality makes the sole vertex of S adjacent to every other vertex of F. Its degree d therefore gives $n = d + 1$. Every vertex of this simple graph has degree $d = n - 1$ by regularity, so F is complete. Thus $R = F - S$ is complete on the even number d of vertices. Partitioning those vertices into pairs gives a perfect matching, contradicting uncoveredness. Therefore $w \geq 2$. Every independent set in the induced graph R is independent in F, so

$\alpha(R) \leq a \leq j + 1 \leq w + 1.$

Every induced subgraph of R, including R itself, is proper induced in F and satisfies T_-.

If $1 \leq s \leq 4$, apply Lemma 275 with $H = F$, the same actual deleted set W, remainder R and degree D. Its requirements $w \geq 2$, D in {5,7,9,11}, $\alpha(R) \leq w + 1$, and hereditary T_- have all been checked. Its noncoverage conclusion gives

$c_0(F) \leq w + \nu(R) + s - 1.$

This contradicts $c_0(F) = w + \nu(R) + s + Z$ with $Z \geq 0$.

It remains to treat the integer range $s \geq 5$. If $gap(R) = 1$, then $ex(R) = s + 1 \geq 6$. Consequently $c_0(R) = \nu(R) + ex(R) > 0$. By the definition of this least certificate total, at least one allocation at total $c_0(R) - 1$ is missing; if all were feasible, that smaller total would contradict minimality of $c_0(R)$. Thus $tr(R) \geq 1$. But T_- on R would require

$7 \leq tr(R) + ex(R) \leq h(D) \leq h(11) = 6,$

a contradiction. Therefore $gap(R) = 0$. The inequality $gap(R) \geq Z$ and nonnegativity of Z give $Z = 0$. We have proved on the same remainder that

$ex(R) = ps(R) = s,$

$c_0(F) = w + \nu(R) + s.$

We now verify the order hypothesis needed at large slack. A matching of R uses two vertices per edge, so

$\nu(R) \leq floor((2k + 1 - w)/2).$

Together with $k + E = c_0(F) = w + \nu(R) + s$, this gives

$E \leq s + \lceil w/2 \rceil.$

Indeed $w + floor((2k + 1 - w)/2) - k$ equals $\lceil w/2 \rceil$, for both even and odd w. The source rank bound and $j \leq w$ yield

$|V(R)| = 2k + 1 - w$

$\leq 2j + 4E + d - 3 - w$

$\leq w + 4E + d - 3$

$\leq 3w + 4s + d - 1$.

In the last line we used $4ceil(w/2) \leq 2w + 2$. Finally

$(4w + 6s + 7) - (3w + 4s + d - 1) = w + 2s + 8 - d \geq 8 > 0$,

because $w \geq 2$, $s \geq 5$ and $d \leq 12$. Thus $|V(R)| < 4w + 6s + 7$.

Apply Lemma 274 to $H = F$, the same W and R, and this odd actual degree D. The graph R is hereditarily T_-, its excess and slack are equal to s, and its exact full-host cost, order bound and independence bound have just been verified. Since $D \leq 11$, $h(D) - 1 \leq 5$, so $s \geq 5$ implies $s \geq \max(4, h(D) - 1)$. The pool size is at least two. Every hypothesis of that theorem therefore holds, and it gives a matching covering all degree-D vertices of R. This contradicts the assumed uncoveredness. Both slack ranges have been excluded, proving the endpoint assertion.

Any complete greedy process is finite because each deletion reduces the vertex count. If it has a first uncovered deletion parent, all its preceding parents are covered, so the result just proved excludes every odd degree there. Applying Lemma 255 to that same preceding prefix also gives $d/2 < D \leq d - 1$. The even integers in this interval are exactly {4}, {6}, {6,8}, and {8,10} for $d = 6$, 8, 10, 12 respectively. This proves the listed possibilities without choosing a special process. Proper-subgraph minimality implies the proper-induced hypothesis used throughout and therefore gives the same conclusion. □

Remark. Small slack is paid by the actual deleted pool. At large slack the hereditary source inequality forces zero remainder gap, making the full-host cost exact; the source rank bound then supplies the order bound required by large-slack absorption.

Depends on. Lemma 255, Lemma 274, Lemma 275, Lemma 293.

Lemma 300 (A whole regular component at a small-slack first failure). *Use the finite integer certificate costs c_i, matching number ν, independence number α, maximum degree Δ, excess $ex = c_0 - \nu$, nonnegative slack ps, signed gap $gap = ex - ps$, and top missing-allocation run tr of Lemma 293. For every nonnegative integer j put $h(j) = \max(4, \lfloor j/2 \rfloor + 1)$. Property T_- means that positive gap implies gap one and $tr + ex \leq h(\Delta)$.*

Let d belong to {6,8,10,12}. Let F be a finite simple graph of maximum degree exactly d which violates T_-, while every proper induced subgraph, including the empty graph, satisfies T_-. Fix any maximum independent set S of F. A greedy deletion process from F-S repeatedly deletes any current maximum-degree vertex while that degree is at least three, and stops at degree at most two. A deletion parent is covered if one matching simultaneously covers all vertices of its actual maximum degree.

Suppose R is the first uncovered parent of any such process, after ℓ preceding deletions. Let W be S together with those ℓ deleted vertices, $w = |W|$, $D = \Delta(R)$, and $s = ps(R)$. If D is even and $1 \leq s \leq 9$, then $d/2 < D \leq d - 2$, $w \geq 2$, and R has exactly one whole connected component C with positive slack. With $P = R - V(C)$, one has $ps(C) = s$, $ps(P) = 0$, $ex(C)$ in $\{s, s+1\}$, $c_0(F) = w + \nu(R) + s$, $c_D(C) = 1$, and $c_D(P) = 0$. The graph C is D-regular and factor-critical, meaning that deleting any one vertex leaves a perfect matching. Every degree-D vertex of P can be covered by one matching, and $|V(C)| = 2\nu(C) + 1$, $\nu(C) \geq w + \nu(P) + 3s - 1$. There are no edges between C and P. If $ex(C) = s + 1$, then $gap(C) = 1$, $s \leq h(D) - 2$, and $tr(C) \leq h(D) - s - 1$. In particular $ex(C) = s$ whenever $s \geq h(D) - 1$.

These conclusions also hold when every proper subgraph of F satisfies T_-, where proper subgraphs may delete arbitrary vertices, edges, or both, including spanning edge deletions. There is no imposed source excess, rank, order, independent-set size, or complete-covered-process hypothesis. Empty prefixes and empty exteriors P are allowed. All orders, degrees, costs, cardinalities, matching numbers, budgets, process indices, and displayed parameters are integers in their stated nonnegative or positive domains; gaps may be signed integers.

Proof. Put $k = \nu(F)$, $E = ex(F)$, $a = |S|$, and $n = |V(F)|$. Lemma 293 gives connectedness, d-regularity, factor-criticality, $n = 2k + 1$, $gap(F) = 1$, and $E \geq 1$. Since F violates T_- and has gap one, integrality gives $tr(F) + E \geq h(d) + 1$. The same fact gives, for $H_0 = F - S$, $gap(H_0)$ in {0,1} and $c_0(F) = a + \nu(H_0) + ps(H_0)$. Every induced subgraph of H_0 is proper induced in F. Thus Lemma 255 applies to the actual covered prefix. It gives $d/2 < D \leq d - 1$ and a nonnegative integer Z such that $gap(R)$ in {0,1}, $Z \leq 1$, and

$c_0(F) = k + E = w + \nu(R) + s + Z$. (1)

Here Z is its cumulative number of failed unit positive-row losses. Evenness of D and d gives $D \le d-2$ and $D \ge 4$.

The set S is nonempty because F is nonempty. If $w = 1$, then $a = 1$ and $\ell = 0$. Maximum independence one makes F complete; its regular degree makes it K_{d+1}. Then $R = F - S$ is K_d on an even number of vertices. Pairing those vertices gives a perfect matching and covers every maximum-degree vertex, a contradiction. Hence $w \ge 2$. Every independent set in R is independent in F, so $\alpha(R) \le a \le w$. Every induced subgraph of R satisfies T_-. Since a matching of R uses at most n-w vertices, $\nu(R) \le floor((2k+1-w)/2)$. With $Q = s + Z$, (1) therefore gives

$E - Q = w + \nu(R) - k \le \lceil w/2 \rceil$. (2)

Use Lemma 61 for nonnegative slack and zero tails, Lemma 72 for restriction, vertex extension and component slack superadditivity, and Lemma 75 for finite allocation sets and their unions. Choose a whole component C of R having largest matching number m, and put $P = R - V(C)$. Such a component exists since $D \ge 4$. The full interval at total $c_0(C) \ge m$ absorbs every other whole component at its matching number: their ranks are at most m and their endpoint allocations have missing runs of length at most $\max(0, m-1)$. Hence $c_0(R) \le \nu(R) + ex(C)$. Vertex extension over W and (1) give

$ex(C) \ge Q = s + Z$. (3)

We rule out the case $ex(C) = Q$ and $gap(C) = 1$ by tracking actual top allocations. Assume this case and put $l = m + Q - 1 = c_0(C) - 1$. As $Q \ge s \ge 1$, $l \ge m$ and the set $A = Y_l(C)$ contains $0, l$. Here $Y_l(J)$ consists of the first matching budgets of partitions of all J-edges into two parts at total l. Its longest missing run is $tr(C)$.

For each other whole component of R, of matching number r, add its feasible endpoint set {0,r} to A, successively. This operation does not increase the longest missing run. Indeed, if the current set B lies in [0,L] and contains both endpoints, then $B + \{0, r\} = B$ union $(B + r)$. Here $r \le m \le l \le L$. Its intervals [0,L] and $[r, L + r]$ overlap, and r and L belong to the union. A consecutive missing run cannot cross either of these attained points, so it is contained in one of the two intervals and bounded by the corresponding translated missing-run bound. The case $r = 0$ changes nothing. The resulting set B has endpoints $0, N$, where $N = \nu(R) + Q - 1$, and every missing run has length at most $tr(C)$. Every member of B is an actual feasible allocation on R, by component union.

Restoring W gives $B + [0, w] \subseteq Y_{N+w}(F)$. Equation (1) makes $N + w = c_0(F) - 1$. Let $[u, u + t - 1]$ be any nonempty consecutive run missing from this last allocation set. Since B contains 0, N, the sumset contains [0,w] and $[N, N + w]$, so $u > w$ and $u + t - 1 < N$. Every integer in $[u - w, u + t - 1]$ is missing from B: if some b in this interval belonged to B, the intervals $[b, b + w]$ and $[u, u + t - 1]$ would intersect, contradicting the missing run and $B + [0, w] \subseteq Y_{N+w}(F)$. Thus $t + w \le tr(C)$. The top allocation set of F has a nonempty missing run by definition of $c_0(F)$, so

$tr(F) + w \le tr(C)$. (4)

The component C satisfies T_- and has positive gap, so $tr(C) + ex(C) \le h(\Delta(C)) \le h(D)$. Equations (2)-(4) and $ex(C) = Q$ imply

$tr(F) + E \le h(D) - Q - w + E \le h(D) + \lceil w/2 \rceil - w \le h(D) \le h(d)$,

contradicting $tr(F) + E \ge h(d) + 1$. Therefore the case $ex(C) = Q$, $gap(C) = 1$ is impossible.

Heredity always gives $gap(C) \le 1$. If $ex(C) = Q$, the excluded case and integrality give $gap(C) \le 0$, so $ps(C) \ge Q = s + Z$. Nonnegative component slack and superadditivity bound $ps(C) \le ps(R) = s$. Hence $Z = 0$, $ps(C) = ex(C) = s$, $gap(C) = 0$, and $ps(P) = 0$. If $ex(C) \ge Q + 1$, then $ps(C) \ge ex(C) - 1 \ge Q = s + Z$. The same upper bound forces $Z = 0$, $ps(C) = s$, $ps(P) = 0$, and $ex(C) = s + 1$. In both cases (1) becomes the stated exact host cost, and C is the only component of positive slack. Every whole component of P has slack zero.

If $\Delta(P) = D$, apply Lemma 243 to P: its proper induced subgraphs have gap at most one, and $ps(P) = 0$. It supplies a matching covering all degree-D vertices and gives $c_D(P) = 0$. If $\Delta(P) < D$, the degree-D set is empty and Lemma 61 gives the same zero row. The uncovered R therefore forces C to have actual degree D and to be uncovered.

Apply the consequence of Lemma 273 to this same F, W, R. Its even-degree, hereditary T_-, slack range 1..9, $w \ge 2$, and $\alpha(R) \le w + 1$ hypotheses were all established. Its additional degree alternative holds because $D \le d - 2 \le 10 \le 12$. Its opposite cost bound is the proved equality $c_0(F) = w + \nu(R) + s$. It follows that C is

a whole D-regular factor-critical component. In particular $|V(C)| = 2m+1$. A perfect matching of C-v covers all its vertices, so Lemma 61 and vertex extension give $c_D(C) \le 1$. If this row were zero, capacities (1,D-1) and exceptional matching budgets (0,0) would partition all C-edges into bounded-degree parts. Regularity forces the first part to have degree exactly one at every vertex, which is impossible on $2m+1$ vertices. Thus $c_D(C) = 1$.

By Lemma 264, the zero-slack hereditary graph P has longest missing run at most two at total $\nu(P)$. The actual pool W, with $w \ge 2$, therefore absorbs P, giving a full interval of length $L = w + \nu(P)$. At total $l = m + s - 1$ on C, each coordinate at least m is feasible by putting all edges in that part. If a coordinate $x < m$ satisfies $x \le 2s-2$, choose x vertices and put all edges meeting them in that part. Its matching number is at most x. The other part is induced on $2m+1-x \le 2(l-x)+1$ vertices and has matching number at most l-x. This applies to either coordinate, so the longest missing run at total l is at most $\max(0, m-3s+2)$. If $m \le L + 3s - 2$, the full interval [0,L] absorbs these allocations too, and actual-pool restoration from Lemma 75 gives every allocation on F at total $w + \nu(R) + s - 1$. This contradicts the exact $c_0(F)$. Thus $m \ge w + \nu(P) + 3s - 1$. The restoration assigns every edge meeting W once, including edges with both endpoints in W, by choosing the required number of red pool vertices and using blue for every remaining edge meeting the pool.

Finally, if $ex(C) = s+1$, then $gap(C) = 1$ and $T_-(C)$ gives $tr(C) + s + 1 \le h(D)$. Its positive excess implies $c_0(C) > 0$ and $tr(C) \ge 1$. Hence $s \le h(D) - 2$ and $tr(C) \le h(D) - s - 1$. This proves every assertion. The stronger proper-subgraph hypothesis implies the proper-induced hypothesis because each proper induced subgraph is among the allowed proper subgraphs. □

Remark. The source's own excessive top run excludes a positive-gap component at exactly the source's residual allowance. Slack superadditivity then concentrates every slack unit, and the small-slack even barrier theorem forces a whole regular factor-critical component.

Depends on. Lemma 61, Lemma 72, Lemma 75, Lemma 243, Lemma 255, Lemma 264, Lemma 273, Lemma 293.

Lemma 301 (Well-coveredness and initial matching coverage of minimal failures). *Use the finite integer certificate costs $c_i(G)$, matching number $\nu(G)$, independence number $\alpha(G)$, maximum degree $\Delta(G)$, excess $ex(G) = c_0(G) - \nu(G)$, nonnegative slack $ps(G)$, signed gap $gap(G) = ex(G) - ps(G)$, and top missing-allocation run $tr(G)$ of Lemma 23. For each nonnegative integer j put $h(j) = \max(4, \lfloor j/2 \rfloor + 1)$. Say that G satisfies T_- if positive gap implies gap one and $tr(G) + ex(G) \le h(\Delta(G))$.*

Let d belong to {4,6,8,10,12}, and let F be a finite simple graph of maximum degree exactly d which violates T_- while every proper induced subgraph satisfies T_-, including the empty graph. Then F is well-covered: every inclusionwise maximal independent set has cardinality $a = \alpha(F)$. Moreover $c_{d-1}(F) = a$. For every maximal independent set I, put $R = F - I$. Then $\Delta(R) \le d-1$, a matching of R covers every vertex of R-degree d-1, $c_{d-1}(R) = 0$, and $c_0(F) = a + \nu(R) + ps(R)$. When $d = 4$, one also has $ps(R) = 0$.

In particular these conclusions hold for every member of each entire class M_d, defined to consist of all finite simple graphs of maximum degree exactly d violating T_- while every proper subgraph satisfies T_-. Here proper subgraphs may delete arbitrary vertices, edges, or both, including spanning edge deletions. No excess floor, rank restriction, order bound, or deletion-process hypothesis is imposed. All orders, degrees, costs, matching and independence cardinalities, row indices, and run lengths are integers in their stated nonnegative or positive domains; gaps may be signed integers.

Proof. Put $k = \nu(F)$, $E = ex(F)$, $n = |V(F)|$, and $a = \alpha(F)$. Lemma 279 gives connectedness, d-regularity, factor-criticality, $n = 2k+1$, and $gap(F) = 1$. Lemma 289 gives $E \ge 2$. In particular F is nonempty.

If $d = 4$, all hypotheses of Lemma 207 hold on this same F. That fact gives well-coveredness, $c_3(F) = a$, and for every maximal independent I and $R = F - I$, the conclusions $\Delta(R) \le 3$, $ps(R) = 0$, $c_3(R) = 0$, and $c_0(F) = a + \nu(R)$. If $\Delta(R) = 3$, the degree-three case of Lemma 243 gives a matching covering all degree-three vertices because $ps(R) = 0$. If $\Delta(R) < 3$, the empty matching covers the empty degree-three set. This proves the entire assertion at $d = 4$.

Assume d belongs to {6,8,10,12}. Set $z = c_{d-1}(F) - a$ and $u = \max(0, floor((2E-3)/(d-2)))$. Lemma 293 gives

z in {0,1}, $k \ge 3E - 1 + 2u$, and $k - a \le 2E + d/2 - 3 + z$. (1)

Fix an arbitrary maximal independent set I, and put $w = |I|$, $R = F - I$, $D = d - 1$, and $s = ps(R)$. The same fact gives

$w \geq a + z - 1$, $c_D(R) = a + z - w$ in {0,1},

$gap(R)$ in {0,1}, $c_0(F) = w + \nu(R) + s$, and $E - s \geq 1$. (2)

Maximality makes every vertex outside I adjacent to I, so $\Delta(R) \leq D$. Also $\alpha(R) \leq a \leq w + 1$. Every induced subgraph of R is proper induced in F and hence satisfies T_-.

We prove that a matching of R covers every degree-D vertex. Suppose otherwise. The degree-D set is then nonempty, so $\Delta(R) = D$. Since F is nonempty, maximality implies $w \geq 1$. If $w = 1$, its sole vertex is adjacent to every other vertex. Its degree d makes $n = d + 1$, and d-regularity makes F the complete graph on $d + 1$ vertices. Thus R is complete on the even number d of vertices. Pairing its vertices gives a perfect matching, a contradiction. Therefore $w \geq 2$.

If $s = 0$, apply Lemma 243 to R at its actual degree $D \geq 5$. Every proper induced subgraph has gap at most one by T_-, so its zero-slack conclusion gives coverage, a contradiction. If $1 \leq s \leq 4$, all hypotheses of Lemma 275 hold with $H = F$ and $W = I$: the same actual pool has $w \geq 2$, D is in {5,7,9,11}, $\alpha(R) \leq w + 1$, and the same R is hereditarily T_-. Its covering consequence and the exact cost identity in (2) again contradict noncoverage.

It remains that $s \geq 5$. If $gap(R) = 1$, then $ex(R) = s + 1$ and $c_0(R) > 0$. The definition in Lemma 23 gives $tr(R) \geq 1$. Property T_- would imply $s + 2 \leq tr(R) + ex(R) \leq h(D) \leq 6$, which is impossible. Thus $gap(R) = 0$ and $ex(R) = ps(R) = s$. Moreover $s \geq 5 \geq h(D) - 1$ and $E \geq s + 1 \geq 6$.

We check the order bound required by Lemma 274. Combining (1) with $w \geq a + z - 1$ gives $k - w \leq 2E + d/2 - 2$. Since $n = 2k + 1$, it follows that

$n - 5w = -3k + 5(k - w) + 1 \leq E + 5d/2 - 6 - 6u$.

Here $E \geq 6$ makes $u = floor((2E - 3)/(d - 2))$. The remainder when the integer 2E-3 is divided by d-2 is at most d-3, so $u \geq (2E - d)/(d - 2)$. Substitution bounds n-5w by 18-2E when $d = 6$, 22-E when $d = 8$, $(53 - E)/2$ when $d = 10$, and $(156 - E)/5$ when $d = 12$. For $E \geq 6$ all four quantities are at most 30. Consequently

$|V(R)| = n - w \leq 4w + 30 < 4w + 6s + 7$.

We may now apply Lemma 274 to the same $H = F$, $W = I$, R, D, w and s. Its actual degree, pool size, common excess and slack, hereditary T_-, exact cost, independence, and strict order hypotheses have all been proved. It gives a matching covering every degree-D vertex, the final contradiction. Coverage therefore holds for this arbitrary maximal independent set.

If $\Delta(R) = D$, Lemma 243 bounds $c_D(R)$ by the minimum degree-D matching exposure, which is now zero. If $\Delta(R) < D$, the implication in Lemma 293 that $c_D(R) = 1$ forces actual degree D, together with (2), also gives $c_D(R) = 0$. Equation (2) now yields $w = a + z = c_{d-1}(F)$. Taking I to be a maximum independent set, which exists because F is finite, gives $z = 0$. Hence every maximal independent set has size a, and $c_{d-1}(F) = a$. Substitution in the exact cost identity of (2) proves the remaining assertion for every such I.

Finally, every proper induced subgraph is a proper subgraph in the broader sense in the definition of M_d. Its stronger minimality hypothesis therefore implies the hypothesis already proved, with exactly the same F and degree d. □

Remark. Every maximal independent complement has exact cost and row losses. Its one-unit width bound makes the independent set large enough relative to the remainder for both small-slack and large-slack absorption. Matching coverage then forces every maximal set to have the same cardinality.

Depends on. Lemma 23, Lemma 207, Lemma 243, Lemma 274, Lemma 275, Lemma 279, Lemma 289, Lemma 293.

Lemma 302 (Maximal independent sets in sextic minimal failures). *Use the finite integer certificate costs $c_i(G)$, matching number $\nu(G)$, independence number $\alpha(G)$, maximum degree $\Delta(G)$, excess $ex(G) = c_0(G) - \nu(G)$, slack $ps(G)$, signed gap $gap(G) = ex(G) - ps(G)$, and top missing-allocation run $tr(G)$ of Lemma 293. For every nonnegative integer D put $h(D) = \max(4, \lfloor D/2 \rfloor + 1)$. A graph G satisfies T_- if $gap(G) > 0$ implies $gap(G) = 1$ and $tr(G) + ex(G) \le h(\Delta(G))$.*

Let F be a finite simple graph of maximum degree exactly six which violates T_- while every proper induced subgraph satisfies T_-, including the empty graph. Then F is connected, six-regular and factor-critical, meaning that deleting any vertex leaves a perfect matching. It has $gap(F) = 1$ and $ex(F) \ge 1$. It is well-covered: every inclusionwise maximal independent set has cardinality $\alpha(F)$. Moreover $c_5(F) = \alpha(F)$.

For every maximal independent set I, the induced complement $R = F - I$ satisfies $c_5(R) = 0$ and has a matching covering all its vertices of degree five. This is coverage of the degree-five set; if that set is empty the requirement is vacuous.

The same conclusions hold under the stronger condition that every proper subgraph satisfies T_-, where proper subgraphs may delete arbitrary vertices, edges, or both, including spanning edge deletions. No excess restriction, order bound, deletion-process hypothesis, or terminal-independence assumption is imposed. All orders, degrees, costs, matching and independence cardinalities, excesses and run lengths are nonnegative integers; gaps may be signed integers.

Proof. Apply Lemma 293 with $d = 6$. Put $k = \nu(F)$, $a = \alpha(F)$, $E = ex(F)$, and $z = c_5(F) - a$. It gives connectedness, six-regularity, factor-criticality, $|V(F)| = 2k + 1$, $gap(F) = 1$, $E \ge 1$, and z in {0,1}. Its rank bound specializes to $k - a \le 2E + z$. For every maximal independent set I, write $w = |I|$, $R = F - I$ and $s = ps(R)$. The same fact supplies

$w \ge a + z - 1 \ge a - 1$,

$c_5(R) = a + z - w$ in {0,1},

$gap(R)$ in {0,1},

$c_0(F) = w + \nu(R) + s$,

$1 \le E - s \le \lceil w/2 \rceil$.

In particular $s \ge 0$ and $s \le E - 1$. If $c_5(R) = 1$, it also gives $\Delta(R) = 5$.

Fix any such I. It is nonempty because F is nonempty. Maximality means that every vertex outside I has a neighbor in I, hence $\Delta(R) \le 5$. Every induced subgraph J of R is a proper induced subgraph of F, since it omits I. It therefore satisfies T_-, including $J = R$ and the empty graph, and consequently has $gap(J) \le 1$. Every independent set of R is independent in F, so

$\alpha(R) \le a \le w + 1$. (1)

If $\Delta(R) < 5$, the implication $c_5(R) = 1 \Rightarrow \Delta(R) = 5$ and the dichotomy $c_5(R)$ in {0,1} give $c_5(R) = 0$. The empty matching covers its empty degree-five set. Suppose henceforth $\Delta(R) = 5$.

When $s = 0$, apply Lemma 243 to R at its actual degree five. Every proper induced subgraph has gap at most one. Its bound $c_5(R) \le \kappa \le ps(R)$, where κ denotes the minimum exposure of the degree-five set, yields $\kappa = 0$ and $c_5(R) = 0$, and supplies a maximum matching covering that set.

Suppose $s \ge 1$. The inequality $E - s \ge 1$ gives $E \ge 2$. Source failure together with $gap(F) = 1$ implies $tr(F) + E \ge 5$. Apply Lemma 40 with $H = F$, $D = 6$ and its integer $B = 4$. Its connectedness, order and matching hypotheses hold, and its two-part zero-row cost is the present $c_0(F)$, since both degree-zero edge parts are empty. It gives

$k \ge 2E + 3 + 2max(0, floor((2E - 3)/4)) \ge 2E + 3 \ge 7$.

The closed neighborhoods of I cover F, and each has size seven, so $7w \ge 2k + 1 \ge 15$. Integrality gives $w \ge 3 = h(5) - 1$.

If $1 \le s \le 4$, apply Lemma 252 to $H = F$, $W = I$, this same R and $D = 5$. We have checked its actual degree, slack range, pool size, independence bound (1), and hereditary T_- condition. Its original-host inequality is the exact identity $c_0(F) = w + \nu(R) + s$. Its covering consequence gives the required matching and $c_5(R) = 0$.

It remains to treat $s \ge 5$. First $gap(R) = 0$. Otherwise the dichotomy above gives $gap(R) = 1$, and $ex(R) = s + 1 \ge 6$. Then $c_0(R) = \nu(R) + ex(R) > 0$. At total $c_0(R) - 1$ at least one allocation must be

missing; if all were feasible, it would contradict the least universal total defining $c_0(R)$. Thus $tr(R) \geq 1$. Since R satisfies T_- and $\Delta(R) = 5$, this would give $tr(R) + ex(R) \leq h(5) = 4$, contrary to $ex(R) \geq 6$. Hence $ex(R) = ps(R) = s$.

We check the strict order bound on this very same remainder. The source rank inequality and $w \geq a + z - 1$ imply

$k \leq a + 2E + z \leq w + 2E + 1$.

Put $b = E - s$, an integer with $1 \leq b \leq \lceil w/2 \rceil$. It follows that

$|V(R)| = 2k + 1 - w \leq w + 4E + 3 = w + 4s + 4b + 3$

$\leq 3w + 4s + 5 < 4w + 6s + 7$.

Here $4ceil(w/2) \leq 2w + 2$ for every integer w, and the final difference is $w + 2s + 2 > 0$.

Apply Lemma 270 with $H = F$, $W = I$ and this same R, $D = 5$ and slack s. The actual degree is five; $w \geq 3 = h(5) - 1$; $ex(R) = ps(R) = s \geq 5$; $c_0(F) = w + \nu(R) + s$; the strict order bound was just proved; and (1) gives $\alpha(R) \leq w + 1$. Every induced subgraph of R satisfies T_-. Its last alternative holds directly because $s \geq 5 \geq h(5) - 1$. All hypotheses are therefore met on the original graph and actual removed set. The theorem gives a matching covering every degree-five vertex and $c_5(R) = 0$.

The degree and slack cases exhaust every maximal independent set I. Thus $c_5(F - I) = 0$ for all of them. The exact row identity now gives $|I| = c_5(F)$. Choosing a maximum independent set, which exists in the finite graph and is maximal, identifies this common value with $a = \alpha(F)$. This proves well-coveredness and the stated cost identity, together with the degree-five coverage already proved. Finally the stronger proper-subgraph hypothesis implies the proper-induced hypothesis, so all conclusions hold on that domain too. □

Remark. The same maximal independent cut controls both the source rank and its matching surplus. These imply the strict remainder-order bound needed for large-slack absorption; small-slack absorption handles the other positive values. Every complement then has zero fifth cost, forcing all maximal independent sets to have one size.

Depends on. Lemma 40, Lemma 243, Lemma 252, Lemma 270, Lemma 293.

Lemma 303 (Whole regular components at first failures). *Use the finite integer certificate costs $c_i(G)$, matching number $\nu(G)$, independence number $\alpha(G)$, maximum degree $\Delta(G)$, excess $ex(G) = c_0(G) - \nu(G)$, slack $ps(G)$, signed gap $gap(G) = ex(G) - ps(G)$, and top missing-allocation run $tr(G)$ of Lemma 23. For each nonnegative integer j put $h(j) = \max(4, \lfloor j/2 \rfloor + 1)$. A graph satisfies T_- if positive gap implies gap one and $tr + ex \leq h(\Delta)$. For each d in {8,10,12}, let M_d consist of all finite simple graphs of maximum degree exactly d which violate T_- while every proper subgraph satisfies T_-. Proper subgraphs may delete arbitrary vertices, edges, or both, including spanning edge deletions.*

Fix d in {8,10,12}, F in M_d, and any maximum independent set S. Put $n = |V(F)|$, $k = \nu(F)$, $a = |S| = \alpha(F)$, $E = ex(F)$, and $H_0 = F - S$. A graph of positive maximum degree is covered when one matching covers every vertex of that maximum degree. Fix any integer $\ell \geq 0$ and induced deletions $H_j = H_{j-1} - v_j$, for $1 \leq j \leq \ell$, such that each parent is covered and v_j has its current maximum degree $D_j \geq 3$. Suppose $R = H_\ell$ is uncovered and $D = \Delta(R) \geq 3$. Put $W_j = S$ union $\{v_1, \ldots, v_j\}$, $W = W_\ell$, $w = |W| = a + \ell$, and $s = ps(R)$. Define

$Z_j = \sum_{q=1}^{j} \sum_{i=1}^{D_q - 1} (1 - c_i(H_{q-1}) + c_i(H_q))$

for $0 \leq j \leq \ell$, with empty sums equal to zero.

Then F is connected, d-regular and factor-critical, $n = 2k + 1$, and

(d, D) belongs to {(8,6),(10,6),(10,8),(12,8),(12,10)}.

Here factor-critical means that deleting any vertex leaves a perfect matching. One has $w \geq 2$, $1 \leq s \leq E - 1$, and

$Z_j = 0$, $c_0(F) = a + j + \nu(H_j) + ps(H_j)$ for $0 \leq j \leq \ell$.

Every positive lower row has unit loss:

$c_i(H_{j-1}) - c_i(H_j) = 1$ *for* $1 \le j \le \ell$ *and* $1 \le i < D_j$.

In particular

$c_0(F) = w + \nu(R) + s$, $E \le s + \lceil w/2 \rceil$,

$|V(R)| \le 3w + 4s + d - 1$.

There is a whole connected component C of R with positive slack. It is D-regular and factor-critical. Put $P = R - V(C)$, $m = \nu(C)$, $t = \nu(P)$, *and* $g = gap(C)$. *Then*

$ps(C) = s$, $ps(P) = 0$, g *in {0,1}*, $ex(C) = s + g$,

$c_D(C) = 1$, $c_D(P) = 0$, $|V(C)| = 2m + 1$,

$m \ge w + t + 3s - 1$,

$m - \alpha(C) \le 3s + g + D/2 - 2$,

$\ell + t + \alpha(P) \le g + D/2 - 1$.

There are no edges between C and P, and one matching of P covers every degree-D vertex. Every vertex of C has exactly d-D neighbors in W; consequently the number of F-edges joining C to W is $(d - D)(2m + 1)$. *The graph P may be empty or contain isolated vertices.*

For every $0 \le j \le \ell$, *one has* $0 \le gap(H_j) \le g$. *If* $g = 0$, *then all these gaps vanish and the row-zero losses also equal one:*

$c_0(H_{j-1}) - c_0(H_j) = 1$ *for* $1 \le j \le \ell$.

If $g = 1$, *then* $s \le h(D) - 2$ *and* $tr(C) \le h(D) - s - 1$. *Thus the possible positive-gap slack ranges are s in {1,2} at* $D = 6$, *s in {1,2,3} at* $D = 8$, *and s in {1,2,3,4} at* $D = 10$. *In all cases* $\ell \le D/2$; *if* $g = 0$, $\ell \le D/2 - 1$. *These are necessary conclusions, not existence assertions. Empty prefixes* $\ell = 0$ *remain included. All orders, degrees, costs, indices, cardinalities, matching numbers, slack values, and process lengths are integers in the displayed nonnegative or positive domains; generic gaps may be signed integers.*

Proof. Proper-subgraph minimality implies proper-induced-subgraph minimality, including the empty graph. Thus Lemma 293 applies to F. It gives connectedness, d-regularity, factor-criticality, $n = 2k + 1$, $E \ge 1$ and, with $z = c_{d-1}(F) - a$ in {0,1},

$k - a \le 2E + d/2 - 3 + z \le 2E + d/2 - 2$. (1)

For the fixed maximum independent set S it also gives

$gap(H_0)$ in {0,1}, $c_0(F) = a + \nu(H_0) + ps(H_0)$,

$ps(H_0) \le E - 1$. (2)

The set S is nonempty because F has positive degree. It is inclusionwise maximal, and every induced subgraph of H_0 is proper induced in F. Such a subgraph satisfies T_- and hence has gap at most one. These observations verify every initial hypothesis of Lemma 255.

Apply that theorem to the actual prefix. It gives $s \ge 1$, nonincreasing slack, and

$gap(H_j) = gap(H_0) + \sum_{q=1}^{j} \epsilon_q + Z_j$ in {0,1},

$c_0(F) = a + j + \nu(H_j) + ps(H_j) + Z_j$, (3)

where $\epsilon_q = 1 - c_0(H_{q-1}) + c_0(H_q)$. Each ϵ_q and each summand in Z_j is a nonnegative integer. In particular $s \le E - 1$ and $0 \le Z_\ell \le gap(R) \le 1$.

The endpoint conclusion of Lemma 299 applies to the same source, maximal initial set and covered prefix. It excludes every odd endpoint degree and supplies exactly the five displayed pairs (d, D). To regard this prefix as part of a complete greedy process, continue choosing a maximum-degree vertex whenever that degree is at least three. This is possible in each nonterminal finite graph and reduces the vertex count at every step, so the continuation terminates. Its first uncovered parent is the prescribed R. No smaller-degree initial coverage is inferred from coverage of the degree-(d-1) set.

If $w = 1$, then $a = 1$ and $\ell = 0$. Independence number one means F is complete: any two nonadjacent vertices would be an independent pair. Since F is d-regular, it would be the complete graph on $d + 1$ vertices. Then $R = F - S$ is complete on the even number d of vertices. Partitioning them into pairs gives a perfect matching, contrary to uncoveredness. Thus $w \geq 2$. Also $\alpha(R) \leq a$, because R is induced in F.

First suppose $s \geq \max(4, h(D) - 1)$. If $gap(R) = 1$, then $ex(R) = s + 1$ and $c_0(R) > 0$. The finite threshold definition in Lemma 23 gives $tr(R) \geq 1$. Property T_- on R would yield

$s + 2 \leq tr(R) + ex(R) \leq h(D)$,

contrary to $s \geq h(D) - 1$. Hence $gap(R) = 0$. The first identity in (3) at $j = \ell$ is a sum of nonnegative integers, so all its summands vanish. In particular $Z_\ell = 0$ and

$k + E = c_0(F) = w + \nu(R) + s$. (4)

The matching number is bounded by half the order:

$\nu(R) \leq floor((2k + 1 - w)/2)$.

Combining this with (4) and evaluating even and odd integer w gives

$E \leq s + w + floor((2k + 1 - w)/2) - k = s + \lceil w/2 \rceil$. (5)

From (1), $a \leq w$, and $4ceil(w/2) \leq 2w + 2$, one obtains

$|V(R)| = 2k + 1 - w$

$\leq 2a + 4E + d - 3 - w$

$\leq w + 4E + d - 3$

$\leq 3w + 4s + d - 1$. (6)

Moreover

$(4w + 6s + 7) - (3w + 4s + d - 1) = w + 2s + 8 - d \geq 6$,

because $w \geq 2$, $s \geq 4$ and $d \leq 12$. The induced graph R and all its induced subgraphs satisfy T_-. Therefore Lemma 282 applies with $H = F$ and this same W, R, D, s: its even degree, pool bound, equal excess and slack, exact host cost, and strict order bound have all been proved. It gives the required D-regular factor-critical whole component C, $ps(C) = ex(C) = s > 0$, zero-slack exterior P, the two top costs, covering matching in P, and $m \geq w + \nu(P) + 3s - 1$.

In the remaining case $1 \leq s < \max(4, h(D) - 1)$, the possible degrees $D = 6, 8, 10$ make $s \leq 4$. The completed greedy process constructed above, its first uncovered parent R, and the original proper-subgraph hypothesis therefore meet every condition of Lemma 300. That theorem supplies the same conclusions about a unique positive-slack whole component C, its regularity and factor-criticality, P, the top costs, and the lower bound on m. It also gives $ex(C)$ in $\{s, s + 1\}$ and the exact equality (4). Thus, in both cases, $g = gap(C)$ belongs to {0,1}, $ex(C) = s + g$, and all claimed component conclusions except the upper rank bound have been established.

Equation (4) compared with (3) gives $Z_\ell = 0$. Since Z_ℓ is a sum of nonnegative integer deficits, every deficit vanishes. Hence $Z_j = 0$ for all j, every positive lower-row loss is one, and (3) gives the exact cost at every prefix. Derivations (5)-(6) depended only on (1), (4), and matching order, so they now hold in both slack ranges.

We next show that $gap(R) \leq g$ by an explicit row-zero certificate. Put $t = \nu(P)$ and $L = c_0(C) = m + s + g$. For each integer x in [0,L], the threshold definition in Lemma 23 supplies an edge partition of C with matching bounds x, L-x. Putting all P-edges in one exceptional part gives feasible first budgets 0 and t at total t. Since C and P are disjoint whole components, unions of these partitions have matching numbers equal to the sums of their component matching numbers. Thus every integer in [0,L] union $[t, L + t]$ is feasible on R at total $L + t$. The lower bound $m \geq w + t + 3s - 1$ implies $t \leq L + 1$, so these two integer intervals cover $[0, L + t]$, including $t = 0$. Therefore

$c_0(R) \leq L + t = \nu(R) + s + g$.

As $ps(R) = s$, this proves $gap(R) \le g$. Equation (3) says that gaps along the prefix are nonnegative and nondecreasing, so $0 \le gap(H_j) \le g$ for every j. If $g = 0$, all these gaps are zero. Equation (3) then forces every $\epsilon_q = 0$, which is precisely the asserted row-zero unit loss.

Every proper induced subgraph of C is proper induced in F and therefore has gap at most one. The degree D is even and at least six, and C is connected, D-regular and factor-critical. These are all the hypotheses of Lemma 250. Choosing a maximum independent set in that theorem gives

$$m - \alpha(C) \le 3s + g + D/2 - 2. \quad (7)$$

Maximum independent sets of C and P unite to an independent set in F because there are no edges between the two whole components. Consequently $\alpha(C) + \alpha(P) \le a$. Combining (7) with $m \ge w + t + 3s - 1$ yields

$$w + t - \alpha(C) \le g + D/2 - 1.$$

Since $w = a + \ell$, its left side is at least $\ell + t + \alpha(P)$, proving the desired bound. All three summands are nonnegative integers, so $\ell \le D/2$, and $g = 0$ gives $\ell \le D/2 - 1$.

Each vertex of C has F-degree d and C-degree D. There are no R-edges from C to P, and $R = F - W$ is induced. Thus its remaining d-D neighbors all lie in W. Summing over the $2m + 1$ vertices of C counts each C-to-W edge once and gives $(d - D)(2m + 1)$.

Finally C is a proper induced subgraph of F, since W is nonempty. If $g = 1$, property T_- gives $tr(C) + s + 1 \le h(D)$. Its positive excess implies $c_0(C) > 0$, so Lemma 23 gives $tr(C) \ge 1$. Therefore $s \le h(D) - 2$ and $tr(C) \le h(D) - s - 1$. Substituting $h(6) = 4$, $h(8) = 5$, and $h(10) = 6$ gives exactly the listed positive-gap ranges. Every argument used the specified S, prefix, W, R, C and full exterior P. Empty prefixes and empty exteriors satisfy all the same formulas. □

Remark. The small- and large-slack concentration theorems apply to the same first uncovered remainder. Their exact cost identity eliminates every positive-row deficit. Comparing the regular component's lower matching bound with its independent-cut upper bound then restricts both the earlier prefix and the full exterior.

Depends on. Lemma 23, Lemma 250, Lemma 255, Lemma 282, Lemma 293, Lemma 299, Lemma 300.

Lemma 304 (Surplus-two exclusion at a terminal two-star pair). *Let $s, t \ge 1$ be integers and let $n_1 \ge \cdots \ge n_s \ge 1$ and $m_1 \ge \cdots \ge m_t \ge 1$ be integer lists. Let F_1, F_2 be their vertex-disjoint star unions, prescribed red and blue. Copies are ordinary subgraphs with vertex-disjoint target components. All host graphs are finite and simple. A host arrows an ordered pair if every red-blue edge coloring contains the entire first target in red or the entire second target in blue. A common host for a family arrows every member. Put $M = s + t - 1$,*

$$\ell_l = max\{n_i + m_j - 1 : 1 \le i \le s, 1 \le j \le t, i + j = l\} \text{ for } 2 \le l \le s + t,$$

and $L = \sum_{l=2}^{s+t} \ell_l$.

For each integer $0 \le q < M$ let A_q contain every pair (F_1^α, F_2^β) with integer cuts $0 \le \alpha < s$, $0 \le \beta < t$, $\alpha + \beta = q$, where superscripts delete leading whole components. Put $K = M - q$, $L_q = \sum_{l=q+2}^{s+t} \ell_l$, $P_q = L_q - K$. Write e, ν, Δ for edge count, maximum matching cardinality and maximum degree, and $\Phi = e - \nu$.

(I) Suppose $D = \ell_{q+2} - 1 \ge 3$, let R be the number of initial $D + 1$ entries of the full natural q-tail, and suppose a valid terminal maximizing pair has indices $i_0 + j_0 = q + R + 1$ and unordered sizes $\{n_{i_0}, m_{j_0}\} = \{D, 2\}$. Every H with

$$\Delta(H) \le D, e(H) < L_q, \Phi(H) \le P_q + 2$$

has one coloring avoiding both entire targets of some member of A_q.

(II) For arbitrary original lists and legal q, every H with

$$\Delta(H) \le 4, e(H) < L_q, \Phi(H) \le P_q + 2$$

has one such actual-member avoiding coloring, without a terminal-pair or natural-head assumption.

(III) In either the degree and profile class of (I) or that of (II), every common-arrowing H with $\nu(H) \geq K-3$ satisfies $e(H) \geq L_q$. Every maximum matching Q then satisfies

$e(H-Q) \geq P_q + K - \nu(H)$,

where H-Q deletes only the matching edges and retains every vertex.

Every avoiding coloring in (I) or (II) extends over the disjoint canonical q-star prefix of sizes ℓ_2,..., ℓ_{q+1}, and across any specified actual q-vertex set X in any outer graph G with $G - X = H$, to avoidance of both full originals, preserving every H-edge color.

In particular let G arrow the full originals, and let $G_0 = G$, $G_i = G_{i-1} - v_i$ for $1 \leq i \leq q < M$ be any actual sequence of maximum-degree vertex deletions satisfying $d_i = deg_{G_{i-1}}(v_i) \geq \ell_{i+1}$. Put $H = G_q$,

$Pprefix_q = \sum_{i=1}^{q}(d_i - \ell_{i+1})$,

$\delta = K - \nu(H)$, $\sigma = \Phi(H) - P_q$.

If $\delta \leq 3$ and H belongs to either degree and profile class in (I), (II), then

$\sigma \geq \delta$, $Pprefix_q + \sigma \geq \delta$, $e(G) \geq L$.

Thus a history with $Pprefix_q = 0$, $\delta = 3$, $\sigma = 2$ is impossible in either class. At $q = M$ the same inequalities and edge lower bound hold unconditionally, with $K = L_M = P_M = 0$, $\delta = -\nu(H)$, $\sigma = \Phi(H)$.

Each removed edge is charged at its earliest deleted endpoint, and surviving edges are disjoint from all prefix charges. All counts, sizes, indices, degrees and cardinalities are integers in their displayed nonnegative or positive domains; δ and σ may be signed where stated. Units, repeated sizes, arbitrary host components and $q = 0$ are included. No conclusion is asserted for higher-degree residuals having only nonbinary terminal maximizing pairs.

Proof. We begin with a consequence that applies to any common A_q host H with $e(H) < L_q$ and $\Phi(H) \leq P_q + 2$. Put $\sigma = \Phi(H) - P_q$ and $\delta = K - \nu(H)$. Let S_q be the disjoint canonical prefix of q stars of sizes ℓ_2,..., ℓ_{q+1}, on fresh vertices, and put $Z = H$ disjoint union S_q. The pool equivalence in Lemma 41 makes Z an arrower of the unchanged full originals. Each positive star has matching number one, and matching and edge numbers add over disjoint components. Therefore

$e(Z) = e(H) + L - L_q < L$,

$\Phi(Z) - (L - M) = \sigma$, $M - \nu(Z) = \delta$.

The first assertion of Lemma 297 on this same Z gives

$1 \leq \sigma \leq 2$, $\delta \geq \sigma + 1 \geq 2$. (1)

We derive the full original-head bounds on the actual H. The cuts $\alpha = \max(0, q - t + 1)$, $\beta = q - \alpha$ are legal because $q \leq s + t - 2$ and retain exactly $\min(s, K)$ red components. In the all-red coloring their full red target supplies a matching with that many edges. Thus $\nu(H) \geq \min(s, K)$. The reversed cuts $\beta = \max(0, q - s + 1)$, $\alpha = q - \beta$ and the all-blue coloring give $\nu(H) \geq \min(t, K)$. By (1), $\nu(H) < K$, so s, $t < K$ and $\nu(H) \geq s$, t. Consequently $q < \min(s, t)$. The legal corners (q,0) and (0,q), in all-blue and all-red respectively, give uncolored copies of both full originals in H. Both original heads are therefore at most $\Delta(H)$.

Now suppose H belongs to class (I) and arrows every A_q member. Interchange both full targets and their colors, if necessary, so the terminal D-star occurs in F and the size-two star in J. The head of F is exactly D and the head of J is at most D. Let U be the full D-star count of F, put $B = D\,U + \sum_{F components with 3 \leq d < D}(d-2)$, and let u, v be the full unit counts of F, J. Put $\epsilon = L_q - e(H) \geq 1$. The complete terminal-mass theorem Lemma 262 applies to this same H and gives

$D \geq 4$, $R \geq \delta + ceil((B + \epsilon - 1)/2) \geq \delta + \lceil D/2 \rceil$. (2)

In particular $R \geq 4$, since $\delta \geq 2$ and $B \geq D$.

The quantitative terminal bound Lemma 228 gives

$\sigma \geq \min(R, \lceil D/2 \rceil)$.

If $D \geq 5$, both terms in this minimum are at least three, contrary to (1). Thus $D = 4$, and the same inequality gives $\sigma \geq 2$. Hence $\sigma = 2$ and $\delta \geq 3$. Equation (2) now gives $R \geq 5$. Since $U \geq 1$, we have $\sigma = 2 < \min(R, 4U)$, so the unit conclusion of that same fact gives $u > v$.

Use Lemma 198 on these unchanged full originals and H. Its complete tagged matching-first family C has maximum component budget $N = K - \min(u, v) = K - v$ and full positive envelope of sum P_q, head $D = 4$ and leading multiplicity R. For every matching W of H, the graph H-W obtained by deleting only its matching edges arrows every entire C member. The theorem also supplies a full member $(pK2, F^{\alpha\cdot})$ of budget $K - v = N$ with $h \geq 1$ head-D components and $p + h - 1 = R$. Its budget shortfall is $g = (v - u)_+ = 0$. The projection inequality is

$2 = \sigma \geq h \lceil D/2 \rceil = 2h.$

It forces $h = 1$ and $\sigma + g = hD/2$.

Thus Lemma 212 applies to this same H, the entire C and its selected full member. Its every-matching hypothesis has been proved, with no restriction to maximum matchings. Its quartic conclusion is

$R - k5 \leq \sigma = 2,$ (3)

where $k5$ counts the whole connected K_5 components of H. Since $R \geq 5$, there are at least three such components.

Choose exactly two distinct whole K_5 components. Apply Lemma 22 with $k = 2$, replacing each by separate disjoint stars of sizes 5, 4, 1, and call the resulting graph Hstar. It arrows the same entire A_q family, and

$e(Hstar) = e(H), \nu(Hstar) = \nu(H) + 2,$

$\Phi(Hstar) = \Phi(H) - 2 = P_q.$

Its remaining deficit $\delta\cdot = K - \nu(Hstar) = \delta - 2$ is at least one. Adjoin the same original canonical prefix S_q on fresh vertices, giving Zstar. By Lemma 41 it arrows both unchanged originals, while

$\Phi(Zstar) = P_q + L - L_q - q = L - M,$

$\nu(Zstar) = \nu(Hstar) + q = M - \delta\cdot < M.$

The unrestricted nonpositive-potential assertion of Lemma 297 gives a coloring avoiding both originals, a contradiction. This proves (I). The two replacements are made simultaneously in whole components; no residual degree $\cap$ is required after replacement.

We record an elementary edge splitting for use in (II). Every finite simple graph of maximum degree at most four has an edge partition into two graphs of maximum degree at most two. Take two disjoint copies, and for each corresponding vertex pair add 4-d parallel edges between them, where d is their original common degree. The resulting loopless multigraph is four-regular. Every component has an Euler closed trail: follow unused edges until returning to the starting vertex, which parity forces to be the only possible stopping vertex; whenever unused edges remain in that component, start another closed trail at a visited vertex and splice it in. Finiteness terminates this construction. A component with n vertices has 2n edges, so its Euler trail has even length. Color its edges alternately red and blue around the closed trail. Each entrance-exit pair has one edge of each color, giving degree two in each color at every vertex. Restriction to the first original copy proves the asserted splitting.

Now suppose a host H in (II) arrows every A_q member. The opening canonical-completion argument gives (1), and the following monochromatic argument gives $q < \min(s, t)$ and both full original heads at most four. These arguments concern this same H and do not require a critical natural degree.

If $\ell_{q+2} \leq 4$, fact Lemma 226 (II) contradicts $e(H) < L_q$. If $\ell_{q+2} \geq 6$, a pair of original indices attaining this first remaining diagonal has size sum at least seven. As each size is at most four, both are at least three. The cuts i-1, j-1 are a legal A_q pair with both heads at least three. The just-proved two-two split avoids both its entire targets, contradicting common arrowing. It follows that $\ell_{q+2} = 5$.

Let i, j attain the final entry of its leading five plateau of width R. Then $i + j = q + R + 1$ and $n_i + m_j = 6$. Neither size exceeds four. The possibility (3,3) is excluded as follows. Choose $\alpha = \min(q, i - 1)$, $\beta = q - \alpha$. Their sum is q, $\alpha \leq i - 1$ and $\beta \leq j - 1$ because $i + j - 2 = q + R - 1 \geq q$. These legal cuts retain heads at least three on both sides, and the two-two split again avoids that entire actual pair. Thus the terminal sizes are {4,2}. Part (I), with $D = 4$, applies to H and gives a contradiction. This proves (II).

For (III), if a common host in either degree and profile class had $\nu(H) \geq K-3$ and $e(H) < L_q$, then

$\Phi(H) \leq L_q - 1 - (K-3) = P_q + 2$.

The applicable part (I) or (II) would contradict common arrowing. Hence $e(H) \geq L_q$. Every maximum matching Q has $\nu(H)$ edges, so

$e(H-Q) = e(H) - \nu(H) \geq L_q - \nu(H) = P_q + K - \nu(H)$.

Every avoiding coloring from (I) or (II) specifies legal cuts α, β. The two constructions in Lemma 41 extend that same coloring over the canonical prefix and across any specified actual q-set, preserving every residual color and handling components meeting deleted vertices through leaves.

Finally consider the actual history in the statement. The contrapositive of the actual-vertex extension in Lemma 41 makes H a common A_q host. The bound $\delta \leq 3$ is exactly $\nu(H) \geq K-3$, so (III) gives $e(H) \geq L_q$. Consequently

$\sigma - \delta = e(H) - L_q \geq 0$.

Each successful deleted degree gives a nonnegative summand in $Pprefix_q$. Every original edge with a deleted endpoint is counted at its first deleted endpoint, while each remaining edge is in H. Thus

$e(G) = \sum_{i=1}^{q} d_i + e(H)$,

$e(G) - L = Pprefix_q + \sigma - \delta \geq 0$.

This proves all the payment claims. At $q = M$, the full successful prefix has natural sum L, and $\sigma - \delta = e(H) \geq 0$ directly because $K = L_M = P_M = 0$. The same disjoint edge partition gives all the assertions without a residual degree or profile condition. Empty prefixes give zero sums and the same identities. □

Remark. Canonical completion makes the residual surplus positive. Terminal mass and projections force surplus two and at least three whole complete five-vertex components. Replacing exactly two lowers the potential to its baseline while keeping positive matching deficit, contradicting nonpositive-potential avoidance.

Depends on. Lemma 22, Lemma 41, Lemma 198, Lemma 212, Lemma 226, Lemma 228, Lemma 262, Lemma 297.

Lemma 305 (Nonbinary terminal avoidance at degree six). *Let $s, t \geq 1$ be integers and let F_1, F_2 be star forests with complete nonincreasing positive integer edge-size lists $n_1,\ldots, n_s$ and $m_1,\ldots, m_t$. Each forest is the vertex-disjoint union of its stars, prescribed red and blue respectively. Copies are ordinary subgraphs with vertex-disjoint target components. Put $M = s + t - 1$ and $\ell_k = max\{n_i + m_j - 1 : 1 \leq i \leq s, 1 \leq j \leq t, i + j = k\}$ for every integer $2 \leq k \leq s + t$.*

Fix an integer $0 \leq q < M$ with $\ell_{q+2} = 7$. Put $K = M - q$, $L_q = \sum_{k=q+2}^{s+t} \ell_k$ and $P_q = L_q - K$. Let R be the number of initial entries equal to seven in this natural tail. Let A_q consist of every pair $(F_1^{\alpha}, F_2^{\beta})$ for nonnegative integer cuts $\alpha < s$, $\beta < t$ and $\alpha + \beta = q$, where superscripts delete that many leading whole components. Suppose valid original indices i_0, j_0 satisfy $i_0 + j_0 = q + R + 1$, $n_{i_0} + m_{j_0} = 8$, and both indicated sizes are at least three.

Every finite simple graph H with

$\Delta(H) \leq 6$, $\Phi(H) := e(H) - \nu(H) \leq P_q + 2$, $\delta := K - \nu(H) \geq 3$

has a red-blue coloring avoiding both entire prescribed targets of some member of A_q. Here e, ν, Δ are edge count, maximum matching cardinality and maximum degree, with value zero on an empty graph.

Every such avoiding coloring extends over the disjoint original canonical q-star pool of sizes $\ell_2,\ldots, \ell_{q+1}$, and across any specified actual q-vertex set X in any finite simple graph G with $H = G - X$, preserving all H-edge colors and avoiding both full originals. Vertex deletion is induced. No original-head bound, plateau-length restriction, strict-cut, unit-count, host-component or smaller-target hypothesis is imposed. All sizes, indices, cuts, orders, degrees and matching cardinalities are integers in the displayed domains; every unit component and repeated size is retained, and $q = 0$ is included.

Proof. Suppose H arrows every member of A_q. This family is finite and nonempty: $\alpha = \min(q, s-1)$, $\beta = q - \alpha$ are legal because $q \le s + t - 2$. Put $\sigma = \Phi(H) - P_q$. Apply Lemma 286 with allowance $r = 2$ and critical degree $D = 6$ to the unchanged full originals, q, H and the indicated terminal pair. It gives σ in {0,1,2}, both terminal sizes equal to four, and $\kappa \ge R$ whole connected odd six-regular components, all with near-perfect matchings. For each whole six-regular component C let $g(C)$ be the maximum of $\nu(C) - |W|$ over inclusionwise maximal matchings W of C, and put $\zeta = \sum_C g(C)$. The same theorem gives

$\zeta \le R + \sigma - \kappa$. (1)

The definitions also give

$e(H) = L_q + \sigma - \delta \le L_q - 1$. (2)

The actual-family clause of Lemma 89 gives $q < \min(s, t)$. The two corner pairs (F_1, F_2^q) and (F_1^q, F_2) are therefore legal. Their all-red and all-blue colorings force copies of the entire respective originals in H. Thus both full original heads are at most six, and each is at least four because of the stipulated terminal sizes. If one head were six, Lemma 231 with $D = 6$, $r = 2$ would contradict common arrowing. Both heads consequently lie in {4,5}.

Apply Lemma 176 with $D = 6$, $z = 2$. Its allowance condition $z \le D/2 - 1$ holds at equality. The required R whole odd regular near-perfect components are those already obtained. Every such component C satisfies

$\nu(C) \ge n_1 + m_1 - 4 + 3\delta \ge 4 + 3\delta \ge 13$.

It cannot be equimatchable: otherwise Lemma 208, applied to the same connected odd six-regular graph, would give $\nu(C) \le 6$. Hence $g(C) \ge 1$ for each of the κ counted components. Equation (1) implies

$\kappa \le \zeta \le R + 2 - \kappa$.

As $\kappa \ge R \ge 1$, this gives $R \le 2$. The $R = 2$ clause of Lemma 176 now excludes $R = 2$: both heads belong to [4,5], and its potential and deficit hypotheses are precisely those here. Thus $R = 1$. The displayed inequality then gives $\kappa = 1$. Fix that actual whole odd six-regular component C and put $\mu = \nu(C)$. Its near-perfect matching gives $|V(C)| = 2\mu + 1$.

The $R = 1$ conclusion of Lemma 176 says the full original edge totals differ. Orient the full originals and their prescribed colors as F, J with $e(F) < e(J)$, and write S, T for their counts and B, A for their heads. It supplies

$\mu \ge e(F) + \delta - 1 \ge e(F) + 2$, $T \ge \mu + \delta + 1 \ge \mu + 4$. (3)

Here A, B belong to {4,5}. In particular $S \le \mu$, and $\mu \le \nu(H)$ gives $q + \mu \le M - \delta < M$. The actual-family capacity inequality in Lemma 89, dropping the nonnegative separator cardinality, gives

Wplus:=2sum_(m in the full list of J)floor((m-1)/2)

$\le S - q - 2\delta \le S - 6$. (4)

We show that when $B = 5$ the entire group of nonunit components of J has at most $2\mu + 1$ vertices. Suppose it does not. For every integer $j \ge 1$ let N_j count the original J components of size at least j. Let a_j be the largest cardinality of a subcollection with sizes in [j,6] using at most $2\mu + 1$ vertices, and set

$c_j = \max_{z \ge j} \max(N_z - a_z, 0)$.

These finite maxima include the empty subcollection; thresholds above the full head contribute zero. Let U have $c_j - c_{j+1}$ components of size j. By the construction in Lemma 84, U is a componentwise subforest of J. Every beta-suffix copy of J in C disjoint union Q leaves a copy of U^β in Q whenever $0 \le \beta < c_1$ and Q is disjoint from C.

All original J sizes are at most five. The failure of the nonunit group to fit means $a_2 < N_2$, so $c_2 > 0$. Also $a_1 \le \mu$ since each positive component uses at least two vertices, and $c_1 \ge T - a_1$. Put $b = T - c_1$. Componentwise containment gives $b \ge 0$, while the last inequality gives $b \le a_1 \le \mu$. In particular U is nonempty. Its full pair with F has component budget

$M' = S + c_1 - 1 = M - b$,

so $M' - q = K - b \ge K - \mu \ge \delta \ge 3$. Its natural q-tail is therefore unexhausted.

Apply only the one-sided numerical estimate of Lemma 84 to the full pair (F, J), the same q, $D = 6$, μ and U. Its hypotheses are $R = 1$, $\mu \geq 3$, $S \leq \mu$ and $c_h > 0$ for $h = \max(1, 7 - B) = 2$, all checked above. Denote by P'_q the own natural q-tail potential of the new full pair (F, U). It gives

$0 \leq P_q - P'_q \leq 5\mu + 1$. (5)

No graph-allowance conclusion of that theorem is used.

Put $Q = H - V(C)$. The component is whole, so edge counts and maximum matching cardinalities add. Summing its degrees gives $e(C) = 6\mu + 3$ and $\Phi(C) = 5\mu + 3$. Equations (5) and the original allowance imply

$\Phi(Q) \leq P_q + 2 - (5\mu + 3) \leq P'_q$,

$(M' - q) - \nu(Q) = \delta + \mu - b \geq \delta > 0$. (6)

We explain explicitly the avoiding-coloring consequence of (6). Write ℓ'_k for the complete diagonal profile of (F, U), through $k = M' + 1$, and set $L' = \sum_{k=2}^{M'+1} \ell'_k$ and $T'_q = \sum_{k=2}^{q+1} \ell'_k$. Adjoin to Q a disjoint union of q stars of sizes ℓ'_2,..., ℓ'_{q+1}, obtaining $Q^{\cdot}$. The empty union is used at $q = 0$. Its matching number is q and its edge count is T'_q, so

$\Phi(Q^{\cdot}) \leq P'_q + T'_q - q = L' - M'$, $\nu(Q^{\cdot}) < M'$.

Both targets F, U are nonempty positive star forests with their complete lists. Thus Lemma 297 supplies a coloring of $Q^{\cdot}$ avoiding F in red and U in blue.

Scan its q adjoined stars in order. Before the r-th star, let α red heads and β blue heads of (F, U) have been selected in earlier stars, so $\alpha + \beta = r - 1$. Neither list can have been exhausted, since the selected stars would already form a forbidden full target. The current pool star has at least the sum of the two current head sizes minus one edges: their indices sum to $r + 1$, and its size is ℓ'_{r+1}. It therefore contains a copy of one current head in its prescribed color. Select that head. Selections in different pool components are vertex-disjoint. Avoidance prevents exhaustion at every step. At the end $\alpha + \beta = q$, $\alpha < S$ and $\beta < c_1$. A red F^{α} in Q or blue U^{β} in Q would join the selected heads to form a forbidden full target in $Q^{\cdot}$. Its restricted coloring consequently avoids this entire suffix pair.

Keep those Q-edge colors and color C wholly blue. The red F^{α} remains absent because all red edges lie in Q. A blue J^{β} would leave a forbidden U^{β} in Q by the every-prefix property already stated for this same C and U. Since $\beta < c_1 \leq T$, these are legal actual cuts of the original oriented pair. This contradicts common A_q arrowing. Therefore the whole nonunit group of J fits $2\mu + 1$ vertices whenever $B = 5$.

We can now apply Lemma 90 to the same F, J, q, H, C and μ at $D = 6$. Its full heads are in [4,5]. Put $U_0 = e(F) - S - (B - 1) \geq 0$. Equation (3) gives $\mu \geq e(F) + 1 = S + B + U_0$ and $T \geq \mu + 3$; equation (4) gives $Wplus \leq S - 4$. We checked $q + \mu < M$, and $R = 1$ means every natural entry after the first seven is at most six. If $B = D - 1 = 5$, the required fit of the entire opposite nonunit group was just proved. If $B = 4$ that extra hypothesis is absent. Equation (2) is its strict edge bound, while the initial potential bound is exactly its allowed $P_q + 2$. All premises refer to the same whole regular near-perfect component C. That theorem constructs an avoiding coloring of an entire original actual member, again contradicting common arrowing. This proves the asserted existence.

For completeness, both advertised extensions work for any avoided actual member, with its original cuts α, β. Color α whole stars of the original disjoint q-pool red and the other β blue. A vertex-disjoint collection of positive copied components inside one host star has at most one member because every edge uses its center. Thus a red full original uses at most α pool components, and a blue one at most β.

For an actual outer set X, choose X_R, X_B partitioning X with sizes α, β. Color every edge meeting X_R red and all remaining edges meeting X_B blue. All other edges already lie in H. Every positive connected red subgraph meeting X contains a vertex of X_R: a red edge incident to X_B has its other endpoint in X_R. Every positive connected blue subgraph meeting X contains X_B. Hence at most α red or β blue vertex-disjoint target components can meet X, including through leaves.

In either extension the surviving red components in H contain F_1^{α}. Indeed, among any s-alpha surviving original components, their sorted original indices $i_1 < \cdots < i_{s-\alpha}$ satisfy $i_j \leq \alpha + j$, so their sizes dominate

$n_{\alpha+j}$; trim surplus spokes. The blue components give F_2^{β} by the same index argument. Either full target would therefore contradict the avoided suffix on H. This proves both extensions, preserving all H-edge colors, including the empty-extension case $q = 0$. □

Remark. Large matching rank limits the number of regular components to one or two, and the two-component remnant already has a strict saving. For the remaining component, the degree-six fitting failure gives equality in the remnant potential. Its matching deficit remains positive, so the nonpositive-potential theorem supplies the missing avoiding coloring.

Depends on. Lemma 84, Lemma 89, Lemma 90, Lemma 176, Lemma 208, Lemma 231, Lemma 286, Lemma 297.

Lemma 306 (Deletion processes in sextic minimal failures). *Use the finite integer certificate costs $c_i(G)$, matching number $\nu(G)$, independence number $\alpha(G)$, maximum degree $\Delta(G)$, excess $ex(G) = c_0(G) - \nu(G)$, slack $ps(G)$, signed gap $gap(G) = ex(G) - ps(G)$, and top missing-allocation run $tr(G)$ of Lemma 302. For each nonnegative integer D put $h(D) = \max(4, \lfloor D/2 \rfloor + 1)$. Property T_- means that $gap(G) > 0$ implies $gap(G) = 1$ and $tr(G) + ex(G) \le h(\Delta(G))$. All graphs are finite and simple, and all vertex deletions below are induced.*

Let F have maximum degree exactly six, violate T_-, and have every proper induced subgraph satisfy T_-, including the empty graph. Then F is connected, six-regular, factor-critical and well-covered. Here factor-critical means that deleting any vertex leaves a perfect matching; well-covered means that every inclusionwise maximal independent set has size $\alpha(F)$. Put $k = \nu(F)$, $a = \alpha(F)$ and $E = ex(F)$. Then $|V(F)| = 2k + 1$, $k \ge 1$, $E \ge 1$, $gap(F) = 1$, $tr(F) + E \ge 5$ and $c_5(F) = a$.

For every maximum independent set S, put $H_0 = F - S$. One has

$c_i(H_0) = c_i(F) - a$ *for every integer* $1 \le i \le 5$,

$c_0(F) = a + \nu(H_0) + ps(H_0)$.

A complete process chooses any such S and repeatedly deletes an arbitrary vertex of current maximum degree while that degree is at least three. Every such process terminates. Write it as $H_j = H_{j-1} - v_j$ for $1 \le j \le t$, where $t \ge 0$ is its integer length, $D_j = \Delta(H_{j-1}) \ge 3$, and $\Delta(H_t) \le 2$. At every deletion parent a matching covers every degree $- D_j$ vertex. For every integer $1 \le j \le t$,

$c_i(H_j) = c_i(H_{j-1}) - 1$ *for every integer* $1 \le i < D_j$,

and every row of integer index $i \ge D_j$ is zero on both graphs. Moreover $ps(H_t) = 0$. These assertions hold for every choice of S and every sequence of permitted roots; the covering matching may depend on the parent.

All these conclusions hold, in particular, for the entire class M_6 of graphs of maximum degree exactly six violating T_- while every proper subgraph satisfies T_-. Proper subgraphs may delete arbitrary vertices, edges, or both, including spanning edge deletions. No excess ceiling, order bound, row hypothesis, or pre-existing covered process is assumed. All orders, degrees, cardinalities, costs, matching numbers, indices, run lengths and process lengths are integers in the displayed nonnegative or positive domains; generic gaps may be signed.

Proof. By Lemma 302, F is connected, six-regular, factor-critical and well-covered, $gap(F) = 1$, $E \ge 1$, and $c_5(F) = a$. Factor-criticality implies that $|V(F)|$ is odd and that a perfect matching in F-v has $(|V(F)| - 1)/2$ edges. No matching in an odd-order graph is larger. Thus $|V(F)| = 2k + 1$. Since a simple six-regular graph has at least seven vertices, $k \ge 3$, which implies the asserted $k \ge 1$. Failure of T_-, with $gap(F) = 1$ and $h(6) = 4$, gives the integer inequality $tr(F) + E \ge 5$.

Fix an arbitrary maximum independent set S. It is nonempty because F is nonempty, and is also maximal under inclusion. The function h is nondecreasing on nonnegative integers, so Lemma 115 applies to this same F with $d = 6$ and $I = S$. It supplies every displayed initial positive-row identity and

$c_0(F) = a + \nu(H_0) + ps(H_0)$, $gap(H_0)$ in {0,1}. (1)

Every induced subgraph J of H_0 omits S and is therefore a proper induced subgraph of F. It satisfies T_- and consequently $gap(J) \le 1$, including H_0 and the empty graph.

Every permitted process exists and terminates: whenever its finite current graph has maximum degree at least three, there is a vertex attaining that degree, and deleting it decreases the order by one. The stopping

graph has maximum degree at most two. Fix any one such process. Maximality of S gives every vertex of H_0 a neighbor in S, so $\Delta(H_0) \le 5$. Induced deletion cannot increase maximum degree.

Suppose that this process has an uncovered deletion parent, where uncovered means that no matching covers all its vertices of current maximum degree. Choose the first such parent and denote it by $R = H_\ell$, with integer $0 \le \ell < t$ and $r = \Delta(R) \ge 3$. Every parent before R is covered. Apply Lemma 255 to F, S, and this covered prefix. Its regularity, nonempty maximal initial set, exact initial cost, nonnegative initial gap, hereditary gap bound, root-degree requirements and parent coverage have all been checked.

Put $W = S$ union $\{v_1, \dots, v_\ell\}$, $w = |W| = a + \ell$, $b = e(F[W])$, and $p = ps(R)$. The prefix theorem gives a nonnegative integer Z such that

$c_0(F) = w + \nu(R) + p + Z$, (2)

as well as

$3 < r \le 5$, and $(2r - 6)w \ge 4(6 - r) + 2b$. (3)

If $\ell \ge 1$, it further gives $r < D_\ell \le 5$. Thus an uncovered degree-five parent could occur only when $\ell = 0$. That case is impossible: Lemma 302 supplies a matching in F-S covering its entire degree-five set. Hence $r = 4$.

Substituting $r = 4$ into (3) gives $2w \ge 8 + 2b$, and therefore $w \ge 4$ since $b \ge 0$. Every independent set in R is independent in F, so $\alpha(R) \le a \le w$. Each induced subgraph of R satisfies T_- by the hereditary observation following (1). The all-slack quartic absorption theorem Lemma 298 consequently applies to the original host F, actual omitted set W, and this same induced R: its maximum degree is four, its hereditary condition holds, and $w \ge \max(3, \alpha(R))$.

The first absorption alternative would give

$c_0(F) \le w + \nu(R) + p - 1$,

contradicting (2), because $Z \ge 0$. Its remaining alternative therefore says that R is connected, four-regular and factor-critical, and

$p = 1$, $w = \alpha(R)$, $m = \nu(R) = w + 2$. (4)

Since $\alpha(R) \le a \le w$, these equalities force $a = w$. The finite containment $S \subseteq W$ then gives $W = S$ and $\ell = 0$, so $R = H_0$. Equation (1) now becomes $c_0(F) = a + m + 1$. Factor-criticality of this very R gives $|V(R)| = 2m + 1$.

Apply Lemma 94 to F with its actual maximum degree $D = 6$ and to the subset S just fixed. Its connectedness, order $2k + 1$, matching number k, positive excess E and strict inequality $tr(F) + E > h(6)$ were established above. Its additional same-complement assumptions are exactly $|V(F - S)| = 2m + 1$, $\nu(F - S) = m$ and $c_0(F) = a + m + 1$, all now proved. It follows that $m \ge a + 3$, contradicting $m = w + 2 = a + 2$ from (4). This rules out every possible first uncovered parent.

The fixed process is therefore covered at each deletion parent. Apply Lemma 118 to F, the same maximum independent set S and the same complete process. Its initial identity is (1); completeness and universal parent coverage have just been established. Its covered-process conclusion gives $ps(H_t) = 0$, every lower positive-row unit loss, and the zero top and higher rows exactly as stated. It includes the empty process $t = 0$. Since S and the complete process were arbitrary, all conclusions have the asserted universal quantifiers.

Finally, proper-subgraph minimality implies the proper-induced-subgraph hypothesis, including the empty graph. The identical proof therefore applies to every member of the stated entire M_6 class without any additional assumption. □

Remark. The first loss of matching coverage must lower the degree. Degree five is initially covered, degree three is forbidden by the boundary inequality, and the only possible quartic absorption exception contradicts the same original source's rank bound.

Depends on. Lemma 94, Lemma 115, Lemma 118, Lemma 255, Lemma 298, Lemma 302.

Lemma 307 (Surplus-two avoidance and matching-deficit payment). *Let $s, t \geq 1$ be integers and let $n_1 \geq \cdots \geq n_s \geq 1$ and $m_1 \geq \cdots \geq m_t \geq 1$ be integer lists. Let F_1, F_2 be the respective vertex-disjoint unions of stars with these edge sizes, prescribed red and blue. Copies are ordinary subgraphs with vertex-disjoint components within each target. A graph arrows a prescribed pair when every red-blue edge coloring contains at least one entire target in its prescribed color. Put $M = s + t - 1$, $\ell_k = max\{n_i + m_j - 1 : 1 \leq i \leq s, 1 \leq j \leq t, i + j = k\}$ for every integer $2 \leq k \leq s + t$, $L = \sum_{k=2}^{s+t} \ell_k$ and $P = L - M$. For a finite simple graph G, write $e(G)$, $\nu(G)$ and $\Delta(G)$ for its edge count, maximum matching cardinality and maximum degree, and put $\Phi(G) = e(G) - \nu(G)$. These graph quantities are zero on the empty graph.*

(I) Every finite simple graph G with $e(G) < L$ and $\Phi(G) \leq P + 2$ has a red-blue edge coloring avoiding both entire prescribed targets.

More generally, fix any integer $0 \leq q < M$. Let A_q consist of every pair $(F_1^{\alpha}, F_2^{\beta})$ with nonnegative integer cuts $\alpha < s$, $\beta < t$ and $\alpha + \beta = q$, where a superscript deletes that many leading whole components from the full sorted list. Put $K_q = M - q$, $L_q = \sum_{k=q+2}^{s+t} \ell_k$ and $P_q = L_q - K_q$. Every finite simple graph H with $e(H) < L_q$ and $\Phi(H) \leq P_q + 2$ has a coloring avoiding both entire targets of some member of A_q.

Every such avoiding coloring of H extends over the disjoint canonical q-star pool of sizes ℓ_2,..., ℓ_{q+1}, and across any specified actual q-vertex set X in any finite simple outer graph G_0 with $H = G_0 - X$, preserving every H-edge color and avoiding both unchanged full originals. Vertex deletion is induced, and the pool is empty when $q = 0$.

(II) For every integer $0 \leq q < M$ and every finite simple H arrowing every member of A_q, define the signed integers $\delta = K_q - \nu(H)$ and $\sigma = \Phi(H) - P_q$. Then

$\sigma \geq \min(3, \delta)$.

For every maximum matching Q of H, deleting its edges while retaining every vertex gives

$e(H - E(Q)) \geq P_q + \min(3, \delta)$.

In particular, $\nu(H) \geq K_q - 3$ implies $e(H) \geq L_q$ and $e(H - E(Q)) \geq P_q + \delta$ for every such Q. At $q = 0$ these assertions apply to every original-arrowing G, with $K_0 = M$, $L_0 = L$ and $P_0 = P$.

(III) Let $G_0 = G$ arrow the full originals, and take any ordered actual vertex-deletion history $G_i = G_{i-1} - v_i$ for $1 \leq i \leq q$, where $0 \leq q \leq M$ is an integer, v_i belongs to $V(G_{i-1})$, and each current degree $d_i = d_{G_{i-1}}(v_i)$ satisfies $d_i \geq \ell_{i+1}$. Put $H = G_q$,

$\epsilon = \sum_{i=1}^{q} (d_i - \ell_{i+1})$, $K_q = M - q$,

$\delta = K_q - \nu(H)$, $\sigma = \Phi(H) - P_q$,

where $L_M = P_M = 0$ and empty sums are zero. When $q < M$ one has $\sigma \geq \min(3, \delta)$. Consequently, if $\delta \leq \epsilon + 3$, then $\epsilon + \sigma \geq \delta$ and $e(G) \geq L$; if $\delta \leq 3$, one also has $\sigma \geq \delta$. Thus no original-arrowing history has $\epsilon = 0$, $\delta = 3$, $\sigma = 2$. If $e(G) < L$ and $q < M$, necessarily

$\delta \geq \epsilon + 4$, $3 \leq \sigma \leq \delta - \epsilon - 1$.

At $q = M$ one has $\sigma \geq \delta$, $\epsilon + \sigma \geq \delta$ and $e(G) \geq L$ unconditionally.

The edge sets removed at successive deletions are pairwise disjoint: an edge joining two deleted vertices is charged exactly once, at its earlier deleted endpoint. They are disjoint from $E(H)$. For every maximum matching Q in H, its edges and the edges of H-E(Q) partition $E(H)$, so both are also disjoint from every prefix charge. The history may use any eligible vertex at each step; maximum-degree choices are permitted but not required.

There is no restriction on host degree, original heads, unit counts, repeated sizes, host components or connectivity, and no smaller-target hypothesis. All list lengths, sizes, cuts, indices, graph orders, edge counts, matching cardinalities and prefix over payments are integers in the stated nonnegative or positive domains; δ and σ may initially be signed integers. No unrestricted payment when $\delta > \epsilon + 3$ is asserted.

Proof. We first exclude a capped common-family host. Fix the full original lists and any integer $0 \leq q < M$, and suppose that H arrows every member of A_q, that $e(H) < L_q$ and $\Phi(H) \leq P_q + 2$, and that $\Delta(H) \leq D := \ell_{q+2} - 1$. The finite family A_q is nonempty: $\alpha = \min(q, s - 1)$, $\beta = q - \alpha$ are legal because $q \leq s + t - 2$.

Put $\sigma = \Phi(H) - P_q$ and $\delta = K_q - \nu(H)$. Adjoin the original canonical q-star pool S_q disjoint from H, and call the completed graph C_q. Its stars have positive sizes and matching number one. The exact canonical correspondence of Lemma 41 (I) gives arrowing of the full originals on C_q, while edge and matching additivity give

$e(C_q) - L = e(H) - L_q < 0,$

$\Phi(C_q) - P = \sigma,\ M - \nu(C_q) = \delta.$

Apply Lemma 297 to this same C_q and the unchanged full pair. Its original-arrowing assertion yields $1 \le \sigma \le \delta - 1$. Since $\sigma \le 2$, one has σ in {1,2} and $\delta \ge \sigma + 1$. If $D \le 4$, then $\Delta(H) \le 4$ and the cap-four clause of Lemma 304 already contradicts common arrowing. We may assume $D \ge 5$.

We check the full-head bounds without changing the original lists. If $q \ge s - 1$, choose $\alpha = s - 1$ and $\beta = q - s + 1$. These cuts are legal: $\beta \ge 0$ and $\beta \le t - 1$ because $q \le s + t - 2$. The blue target F_2^β has $t - \beta = M - q = K_q$ positive components. Color all H-edges blue. Common arrowing for this member forces a blue F_2^β, which supplies a matching of cardinality K_q by taking one edge from each disjoint component. This contradicts $\nu(H) = K_q - \delta < K_q$. Hence $q < s - 1$. Exchanging the two complete targets and colors proves $q < t - 1$. In particular the corner pairs (F_1, F_2^q) and (F_1^q, F_2) are legal. The all-red coloring of the first forces the entire F_1 in H, and the all-blue coloring of the second forces the entire F_2. Therefore $n_1, m_1 \le \Delta(H) \le D$.

The full diagonal sequence is nonincreasing. For a candidate on diagonal $k + 1$, where $k \ge 2$, at least one of its positive indices exceeds one. Decreasing that index gives a valid candidate on diagonal k whose star-size sum is no smaller, because the corresponding original list is nonincreasing. Taking a maximizing candidate on diagonal $k + 1$ proves $\ell_k \ge \ell_{k+1}$. Let R be the positive length of the initial $D + 1$ plateau in the complete natural q-tail. It satisfies $1 \le R \le K_q$. At the last diagonal of that plateau choose valid original indices i_0, j_0 with $i_0 + j_0 = q + R + 1$ attaining $\ell_{q+R+1} = D + 1$, and put $a = n_{i_0}$, $b = m_{j_0}$. Then $a + b = D + 2$. Each is at most its full original head, and hence at most D, so a, $b \ge 2$.

If $a = 2$ or $b = 2$, the unordered terminal pair is {D,2}. The terminal clause of Lemma 304 applies: $D \ge 5 \ge 3$, the same terminal indices lie at the end of the original leading plateau, and the same H satisfies its degree, edge and potential bounds. It supplies an avoided entire A_q member, a contradiction.

Otherwise a, $b \ge 3$. If $\sigma = 1$, then $\delta \ge 2$, so Lemma 249 applies to this same nonbinary terminal pair, the unchanged full originals and H. Its degree, potential and deficit conditions are $D \ge 3$, $\Delta(H) \le D$, $\Phi(H) \le P_q + 1$ and $K_q - \nu(H) \ge 2$, all established above. It supplies an avoided entire A_q member.

The remaining possibility is $\sigma = 2$ and $\delta \ge 3$. If $D \ne 6$, apply Lemma 290 with allowance $r = 2$ to the same originals, q, H, R, i_0, j_0. Its conditions $D \ge 3$, $D \ne 6$, $a + b = D + 2$, $\Delta(H) \le D$, $\Phi(H) \le P_q + 2$ and $\delta \ge 3$ have all been established. If $D = 6$, apply Lemma 305 instead: $\ell_{q+2} = 7$, the terminal sum is eight and both terminal entries are at least three. It has the same graph allowance and matching-deficit hypotheses. Each case gives an avoided A_q member. This exhausts all terminal pairs and both possible surpluses, proving the capped impossibility.

Now suppose a finite simple G satisfies $e(G) < L$ and $\Phi(G) \le P + 2$ but arrows the entire original pair. Apply Lemma 41 (III) to this very G and the unchanged full original lists. It produces an actual induced residual H at a depth $q < M$ which arrows every A_q member and satisfies

$\Delta(H) \le \ell_{q+2} - 1,\ e(H) < L_q,$

$\Phi(H) - P_q = \Phi(G) - P - \epsilon - j \le 2,$

where ϵ and j are nonnegative integers. This is precisely the capped common-family host excluded above. Thus G has the asserted avoiding coloring.

For the arbitrary-q assertion, take H with $e(H) < L_q$ and $\Phi(H) \le P_q + 2$. Adjoin its original canonical q-pool S_q, disjoint from H, and write $B = H$ disjoint union S_q. Every pool star has positive size and matching number one, so

$e(S_q) = L - L_q,\ \nu(S_q) = q.$

Edge counts and maximum matching cardinalities add on disjoint unions. Consequently

$e(B) = e(H) + L - L_q < L,$

$\Phi(B) = \Phi(H) + L - L_q - q \le P + 2$.

The original-host assertion just proved gives a coloring of B avoiding both full originals. By Lemma 41 (I), its unchanged restriction to H avoids both targets of some member of the entire original A_q family. This proves the family assertion without a host-degree restriction.

Finally Lemma 41 (I), (II), applied to any such avoiding H-coloring and its legal cuts, give respectively the original canonical-pool extension and the extension across any specified actual outer set X. Both preserve every H-edge color, including target copies meeting X through leaves. The $q = 0$ arguments use an empty pool and no deleted vertices; the $M = 1$ case is included in the finite normalization theorem and the nonempty-family convention. Every proof step retained the complete positive original lists.

For (II), fix a common A_q host H and put $\delta = K_q - \nu(H)$, $\sigma = \Phi(H) - P_q$, without imposing a sign on either. If $\sigma < \min(3, \delta)$, integrality gives $\sigma \le 2$ and

$e(H) - L_q = \sigma - \delta < 0$.

Part (I), already proved for these exact original lists and this actual family, then gives an avoiding member, a contradiction. Hence $\sigma \ge \min(3, \delta)$. Every maximum matching Q has exactly $\nu(H)$ edges. Deleting precisely those edges leaves

$e(H - E(Q)) = e(H) - \nu(H) = \Phi(H) \ge P_q + \min(3, \delta)$.

If $\nu(H) \ge K_q - 3$, then $\delta \le 3$ and the minimum equals δ. The identity $e(H) - L_q = \sigma - \delta$ consequently gives $e(H) \ge L_q$ and the stated stronger matching-residual bound. At $q = 0$ the actual family consists of the original pair, proving the original-host specialization.

Consider a history in (III). Its prefix overpayment ϵ is nonnegative since every $d_i \ge \ell_{i+1}$. Let $E_i = E(G_{i-1})$ minus $E(G_i)$. Actual induced deletion makes these sets pairwise disjoint with $|E_i| = d_i$. Each original edge meeting the deleted set belongs precisely to E_i for its earliest deleted endpoint; all other edges lie in $E(H)$. Thus these sets partition $E(G)$, including when $q = 0$. Since the full diagonal sum ends at $M + 1$, they give the exact identity

$e(G) - L = \epsilon + e(H) - L_q = \epsilon + \sigma - \delta$. (1)

For $q < M$, the actual-set extension of Lemma 41 (II) implies that H arrows every A_q member: an avoided member would extend across these very deleted vertices to an avoided original coloring of G. Applying (II) proves $\sigma \ge \min(3, \delta)$ on this same H. If $\delta \le 3$, this is $\sigma \ge \delta$, and $\epsilon \ge 0$ gives $\epsilon + \sigma \ge \delta$. If instead $3 < \delta \le \epsilon + 3$, then $\sigma \ge 3$ and $\epsilon + \sigma \ge \epsilon + 3 \ge \delta$. Equation (1) yields $e(G) \ge L$ in both cases, proving the full payment range and excluding $\epsilon = 0$, $\delta = 3$, $\sigma = 2$.

Suppose $e(G) < L$ and $q < M$. The proved payment excludes $\delta \le \epsilon + 3$, so integrality gives $\delta \ge \epsilon + 4 \ge 4$. The surplus bound therefore gives $\sigma \ge 3$. Equation (1) and the integer strict edge bound give $\sigma \le \delta - \epsilon - 1$, as asserted. This derives both inequalities without any degree $\cap$ or preferred deletion choice.

At $q = M$ one has $K_M = L_M = P_M = 0$ and therefore $\delta = -\nu(H)$, $\sigma = \Phi(H)$. Directly,

$\sigma - \delta = e(H) \ge 0$.

Together with $\epsilon \ge 0$ and (1) this yields all three endpoint conclusions. No family at $q = M$ and no empty target have been used. Finally $E(Q)$ and $E(H)$ minus $E(Q)$ partition the surviving edge set for every maximum matching Q in H. They are disjoint from every E_i, establishing the stated residual and prefix edge accounting. □

Remark. Finite degree reduction preserves surplus at most two. Binary and nonbinary terminal exclusions rule out the reduced host; canonical completion restores every actual family. The exact edge partition then turns avoidance into signed-deficit payment on the unchanged residual graph.

Depends on. Lemma 41, Lemma 249, Lemma 290, Lemma 297, Lemma 304, Lemma 305.

Theorem 308 (Exclusion of degree-six minimal obstructions). *For a finite simple graph G write $\nu(G)$, $\alpha(G)$, $e(G)$, $\Delta(G)$ for its matching number, independence number, edge count, and maximum degree. For nonnegative integer degree capacities A_c, B_c, define $c_{A_c,B_c}(G)$ as the least nonnegative integer total T_c such that every pair of nonnegative integer matching budgets x_c, y_c with $x_c + y_c = T_c$ admits an edge partition into four parts of maximum degrees at most A_c, B_c and matching numbers at most x_c, y_c, respectively. For every nonnegative integer index i put $c_i(G) = \max_{A_c+B_c=i} c_{A_c,B_c}(G)$, over nonnegative integer capacities, and put $c_0 = c_0$, $ex = c_0 - \nu$, $ps = e - \nu - \sum_{i\geq 1} c_i$, and signed integer $gap = \sum_{i\geq 0} c_i - e$. Put $tr(G) = 0$ if $c_0(G) = 0$; otherwise $tr(G)$ is the greatest nonnegative integer length of a consecutive run of x in {0,...,c0(G)-1} for which $E(G)$ has no partition into two parts of matching numbers at most x and c0(G)-1-x. Property T_- means $gap(G) > 0$ implies $[gap(G) = 1$ and $tr(G) + ex(G) \leq \max(4, floor(\Delta(G)/2) + 1)]$. Let M_6 be the entire class of finite simple graphs F with maximum degree exactly six that violate T_- while every proper subgraph satisfies T_-, including arbitrary vertex deletions, edge deletions, or both. Then M_6 is empty. This conclusion imposes no deficit, excess, rank, order, row, component, owner, pool, process, normalization, or reachability hypothesis.*

Proof. Suppose that F belongs to M_6. Every proper induced subgraph of F is a proper subgraph, so Lemma 306 applies. We first identify its quantities with the ones in the statement. The certificate definitions use the identical nonnegative integer capacities and budgets, the same four edge parts, and every edge of the original graph. The finite row sum is well defined: whenever $i \geq 2\Delta(G)$, at least one capacity in every split of i is at least $\Delta(G)$, so placing every edge in that degree-bounded part gives $c_i(G) = 0$. The identity

$ex(G) - ps(G) = c_0(G) - \nu(G) - [e(G) - \nu(G) - \sum_{i\geq 1} c_i(G)]$

$= \sum_{i\geq 0} c_i(G) - e(G)$

shows that its signed gap is exactly the one here. At capacities zero both degree-bounded parts are empty, so the two-part missing-allocation definition of tr also agrees, including its zero-cost convention. Thus the original minimality and failure conditions are precisely the hypotheses of that source theorem.

Put $k = \nu(F)$, $a = \alpha(F)$, and $E = c_0(F) - k$. By Lemma 306, F is connected, six-regular, factor-critical and well-covered, with $|V(F)| = 2k + 1$, integer $k \geq 1$, integer $E \geq 1$, $tr(F) + E \geq 5$, and $c_5(F) = a$. On this same F, for every maximum independent S,

$c_i(F - S) = c_i(F) - a$ for every integer $1 \leq i \leq 5$,

$c_0(F) = a + \nu(F - S) + ps(F - S)$. (1)

It also proves the following universal process property: after any such S is deleted, every complete process deleting arbitrary current maximum-degree vertices while their degree is at least three has a covering matching at each deletion parent; each positive row below that parent's degree drops by exactly one, and every terminal has zero slack. (2)

Both assertions concern every permitted choice on the original F.

If $1 \leq E \leq 3$, the structural conclusions, strict run bound, (1) and (2) are all the hypotheses of Lemma 227, with its parameters k, a, E unchanged. That theorem rules out F.

It remains to consider $E \geq 4$. Define integers

$g = k - a$, $p = g - E + 1$, $\lambda = 2E - 3 - g$.

Apply Lemma 145. Its connectedness, six-regularity, factor-criticality, odd order and positive integer k have been established, and its top-cost assumption holds because $c_5(F) = a = \alpha(F)$. Therefore either $p \geq 2$ or

$|V(F)| = 27$, $\alpha(F) = 9$, $c_0(F) = 17$. (3)

In the first alternative, $p = E - 2 - \lambda$ by its definition. Set

$\delta = c_0(F) - c_2(F)$, $\epsilon = c_0(F) - c_1(F)$,

$\beta = p + \delta - \epsilon$, $r = k - \lambda - 1 + \delta$.

These are exactly the parameters of Lemma 229, with its $a_F = a$. Its assumptions $E \geq 4$ and $p \geq 2$ hold in this alternative; its structural assumptions hold by the source theorem; its complete-process definition and universal hypotheses (a), (b) are precisely (1), (2). Hence it supplies a nonnegative integer

$T = r - \beta - 1$

and a table Q whose entry at every nonnegative integer capacity split $A_c + B_c = 4$ and every nonnegative integer budget split $x_c + y_c = T$ is a feasible four-part partition of all F-edges with those bounds. It also gives the exact identity $c_4(F) = r - \beta$. The table is on the original F, including every edge incident with its omitted vertices. By the least-universal-total definition, $c_{A_c,B_c}(F) \le T$ for each capacity split. Taking the maximum over those splits gives

$c_4(F) \le cost(Q) = T = r - \beta - 1 = c_4(F) - 1,$

where $cost(Q)$ denotes the table total T. This is impossible for the finite integer $c_4(F)$.

In alternative (3), the graph is still connected, six-regular, factor-critical and well-covered. Specializing $a = 9$ in (1) gives exactly condition (a) of Lemma 266, and (2) is its universal condition (b) without any change of process or terminal. Its three numerical hypotheses are (3). That theorem therefore rules out F in this alternative too.

The integer cases $1 \le E \le 3$ and $E \ge 4$ are exhaustive, and the latter's two alternatives are exhaustive. Every assumed member of the entire M_6 class has been contradicted. Thus M_6 is empty, with no additional source hypothesis. □

Remark. Universal source processes make the three existing certificate branches exhaustive. Every branch contradicts the finite integer certificate cost on the original graph, so no additional parameter or process restriction remains.

Depends on. Lemma 145, Lemma 227, Lemma 229, Lemma 266, Lemma 306.

Lemma 309 (Deletion processes in higher-degree minimal failures). *Use the finite integer certificate costs $c_i(G)$, matching number $\nu(G)$, independence number $\alpha(G)$, maximum degree $\Delta(G)$, excess $ex(G) = c_0(G) - \nu(G)$, slack $ps(G)$, signed gap $gap(G) = ex(G) - ps(G)$, and top missing-allocation run $tr(G)$ of Lemma 115. For every nonnegative integer D put $h(D) = \max(4, \lfloor D/2 \rfloor + 1)$. Property T_- means that $gap(G) > 0$ implies $gap(G) = 1$ and $tr(G) + ex(G) \le h(\Delta(G))$. All graphs are finite and simple, and every vertex deletion below is induced.*

Let d belong to {8,10,12}, and let F be a finite simple graph of maximum degree exactly d which violates T_- while every proper subgraph satisfies T_-. A proper subgraph may delete arbitrary vertices, edges, or both, including spanning edge deletions and the empty graph. Then F is connected, d-regular, factor-critical and well-covered. Here factor-critical means that deleting any vertex leaves a perfect matching, and well-covered means that every inclusionwise maximal independent set has cardinality $\alpha(F)$.

Put $k = \nu(F)$, $a = \alpha(F)$, and $E = ex(F)$. Then $|V(F)| = 2k + 1$, $k \ge 1$, $E \ge 1$, $gap(F) = 1$, $tr(F) + E \ge d/2 + 2$, and $c_{d-1}(F) = a$. For every maximum independent set S, put $H_0 = F - S$. One has

$c_i(H_0) = c_i(F) - a$ *for every integer* $1 \le i < d$,

$c_0(F) = a + \nu(H_0) + ps(H_0)$.

A complete process chooses any such S and repeatedly deletes an arbitrary vertex of current maximum degree while that degree is at least three. Every such process terminates. Write it as $H_j = H_{j-1} - v_j$ for integers $1 \le j \le t$, where $t \ge 0$ is its length, $D_j = \Delta(H_{j-1}) \ge 3$, and $\Delta(H_t) \le 2$. At every deletion parent a matching covers all vertices of degree D_j. For every $1 \le j \le t$ and every integer $1 \le i < D_j$,

$c_i(H_j) = c_i(H_{j-1}) - 1$.

Every row of integer index $i \ge D_j$ is zero on both H_{j-1} and H_j. Moreover $ps(H_t) = 0$.

For $0 \le j \le t$ put $W_j = S$ union $\{v_1, \dots, v_j\}$, with $W_0 = S$, and put $p = ps(H_0)$. Then

$c_0(F) = |W_j| + \nu(H_j) + ps(H_j)$.

Exactly p of the t deletions preserve matching number, and $\nu(H_t) = c_0(F) - a - t$. All conclusions hold for every choice of S and every sequence of permitted roots. The covering matching may depend on the parent.

Thus these conclusions hold for every member of each entire class M_d, d in {8,10,12}, defined by exactly the maximum-degree, failure, and proper-subgraph conditions just stated. No excess ceiling, order bound, row hypothesis, selected process, or pre-existing covered process is assumed. All graph orders, degrees, costs, matching and independence

cardinalities, indices, run lengths and process lengths are integers in their displayed nonnegative or positive domains; generic gaps may be signed.

Proof. We use Lemma 279 for the structure of the same minimal failure, Lemma 115 for independent deletion, Lemma 301 for well-coveredness and the top cost, Lemma 303 for the first uncovered remainder, Lemma 144 and Lemma 139 for its two possible degree separations, and Lemma 118 for complete-process accounting.

Every proper induced subgraph of F is among its proper subgraphs, so the hypotheses imply proper-induced minimality as well. Lemma 279 therefore applies to F and gives connectedness, d-regularity, factor-criticality, $|V(F)| = 2k+1$, $k \geq 2$, $gap(F) = 1$, and $tr(F) + E \geq h(d) + 1$. For the three allowed degrees $h(d) = d/2 + 1$, so the last inequality is $tr(F) + E \geq d/2 + 2$. Lemma 301 gives well-coveredness and $c_{d-1}(F) = a$.

The integer function h is nondecreasing. Thus Lemma 115 applies to this same connected, even-regular, factor-critical F with gap one and the original minimal failure of $T_h = T_-$. It gives $E \geq 1$. The graph F is nonempty because its maximum degree is positive, so every maximum independent set S is nonempty. Applying the independent-deletion theorem to this S gives exactly the initial positive-row identities and $c_0(F) = a + \nu(H_0) + ps(H_0)$ in the statement.

Fix any maximum independent S and any complete process. Both choices exist: a maximum independent set exists in a finite graph, and each nonterminal finite remainder has a maximum-degree vertex. Every root deletion reduces the vertex count, so every permitted sequence terminates after finitely many deletions at degree at most two. It remains to prove coverage for this arbitrary process.

Suppose that some deletion parent is uncovered, and take the first such parent R. Let $\ell \geq 0$ be the number of preceding roots, W their union with S, $w = |W| = a + \ell$, and $D = \Delta(R)$. The covered prefix may be empty. Lemma 303 applies to the same source F, initial set S, process and parent R. It gives

(d, D) in {(8,6),(10,6),(10,8),(12,8),(12,10)}

and a whole D-regular factor-critical component C of R. Put $P = R - V(C)$, $m = \nu(C)$, $t_P = \nu(P)$, and $s = ps(C)$. Every matching of R restricts to matchings of C and P, and maximum matchings in those two whole parts can be united. Hence $\nu(R) = m + t_P$. The positive degree of C gives $m \geq 1$. The first-failure conclusions therefore give

$s \geq 1$, $gap(C)$ in {0,1}, $c_D(P) = 0$,

$c_0(F) = w + m + t_P + s$,

$m \geq w + t_P + 3s - 1$.

The sets C and P are whole complementary parts of the same induced remainder, with no edges between them.

If $D = d - 4$, the displayed list forces $(d, D) = (10, 6)$ or (12,8). Apply Lemma 144 with exactly these F, W, C, P and their matching numbers. The structure of F, $gap(F) = 1$, regularity and factor-criticality of C, positive component slack, $gap(C) \leq 1$, exterior top cost zero, exact source cost, and rank bound have all been established. The conclusion excludes this tuple.

It remains that $D = d - 2$. We check the internal-edge premise of Lemma 139 on the actual set W. Maximality of S implies $\Delta(F - S) \leq d - 1$: every vertex outside S has a neighbor in S. Maximum degrees cannot increase under induced deletion.

We claim that every preceding root has deletion degree d-1. If the prefix is empty, there is nothing to prove. Otherwise its last parent is covered. For any covered graph K of maximum degree q and any vertex v of degree q, fix a matching covering all degree-q vertices. If $\Delta(K - v) = q$, every vertex still of degree q was not adjacent to v. Its covering edge survives restriction of the matching to K-v. Thus K-v is covered whenever its maximum degree remains q. Since the last root's child R is uncovered, its parent degree is strictly greater than D. It is at most d-1, so it equals d-1. All earlier deletion degrees are at least this last degree and at most d-1, proving the claim.

Every preceding root therefore has exactly one neighbor among the vertices deleted before it: its degree in F is d and its degree at deletion time is d-1. Starting from the independent set S, insert the preceding roots in their deletion order to form W. Each induced edge inside W is counted at its later inserted root endpoint. Each root adds exactly one edge, and S starts with none. Hence $e(F[W]) = \ell$. This identity also holds for $\ell = 0$, when $W = S$. The top cost established earlier now gives

$c_{d-1}(F) + e(F[W]) = a + \ell = w$.

All hypotheses of Lemma 139 therefore hold on this same F, W, C, P. In particular its internal-edge count is exactly the one just computed, and its two allowed component gaps are the ones supplied by Lemma 303. That theorem excludes this tuple as well.

Both possible degree separations have led to contradictions. The arbitrary complete process consequently has a covering matching at every deletion parent.

Apply Lemma 118 to this same process, starting from the same maximum independent set S. Its initial cost assumption is the identity from Lemma 115 proved earlier. The process is complete by its choice, and all its parents are now proved covered. The accounting theorem gives unit loss in every positive row below the current degree, zero current top and higher rows on each parent and child, and zero terminal slack. It also gives the exact displayed cost at every prefix, exactly $ps(H_0)$ matching-preserving deletions, and $\nu(H_t) = c_0(F) - a - t$. Its conclusions include $t = 0$. Because both S and the permitted process were arbitrary, all universal quantifiers in the statement follow.

The class M_d imposes exactly the maximum-degree, failure, and proper-subgraph hypotheses with which the proof began. Hence the conclusions apply to the entire class, without any further source restriction. □

Remark. Every first uncovered parent has a whole regular component. The two possible degree separations are excluded on the same actual deleted set; universal coverage then forces exact process costs.

Depends on. Lemma 115, Lemma 118, Lemma 139, Lemma 144, Lemma 279, Lemma 301, Lemma 303.

Theorem 310 (Exclusion of degree-twelve minimal obstructions). *For a finite simple graph G, write $V(G)$, $E(G)$, $e(G)$, $\nu(G)$, $\alpha(G)$, $\Delta(G)$ for its vertex set, edge set, edge count, maximum matching cardinality, maximum independent-set cardinality, and maximum degree. For nonnegative integer degree capacities A_c, B_c, define $c_{A_c,B_c}(G)$ as the least nonnegative integer total t such that every nonnegative integer pair x, y with $x + y = t$ admits a partition of $E(G)$ into four edge sets Q_R, Q_B, P_R, P_B satisfying $\Delta(Q_R) \le A_c$, $\Delta(Q_B) \le B_c$, $\nu(P_R) \le x$, $\nu(P_B) \le y$. Quantities of an edge set mean those of its spanning graph. For every nonnegative integer i put $c_i(G) = \max_{A_c+B_c=i} c_{A_c,B_c}(G)$, over nonnegative integer capacities, and put $c_0 = c_0$. Define $ex(G) = c_0(G) - \nu(G)$, $ps(G) = e(G) - \nu(G) - \sum_{i\ge1} c_i(G)$, and signed integer $gap(G) = \sum_{i\ge0} c_i(G) - e(G)$; the sums are finite. If $c_0(G) = 0$ put $tr(G) = 0$; otherwise let $tr(G)$ be the greatest length of a consecutive integer run of x in {0,...,c0(G)-1} for which $E(G)$ has no two-part partition with matching numbers at most x and c0(G)-1-x. Property T_- means that $gap(G) > 0$ implies both $gap(G) = 1$ and $tr(G) + ex(G) \le \max(4, floor(\Delta(G)/2) + 1)$.*

Let M_{12} be the entire class of finite simple graphs F with maximum degree exactly twelve that violate T_- while every proper subgraph satisfies T_-, where proper subgraphs allow arbitrary vertex deletions, edge deletions, or both. Let N_{12} denote the proposition M_{12} is empty. There is no additional restriction on source order, excess, rank, cost sector, native rows, selectors, matching prefixes, omission margins, or phase lengths.

Then M_{12} is empty: N_{12} is a theorem. Equivalently, every finite simple graph F of maximum degree exactly twelve whose every proper subgraph satisfies T_- also satisfies T_-.

Proof. The certificate definitions in Lemma 309 and Lemma 204 agree with those here: they use partitions of all edges, nonnegative integer degree capacities and matching budgets, universal feasibility at each total, and then the maximum over capacity splits. At capacities zero both degree-bounded parts are empty, so their missing-allocation run is exactly $tr(G)$ as defined here, including the zero-cost convention. Write $ps(G) = e(G) - \nu(G) - \sum_{i\ge1} c_i(G)$ for their finite slack. Then $ex(G) - ps(G) = \sum_{i\ge0} c_i(G) - e(G)$, so their gap and property T_- also agree with those here.

Suppose F belongs to M_{12} and put $d = 12$. It has maximum degree exactly d, violates T_-, and every proper subgraph satisfies T_-, with arbitrary vertex and edge deletions allowed. These are precisely the original-source hypotheses of Lemma 309, so that theorem applies to this same F.

Lemma 309 gives connectedness, d-regularity, factor-criticality and well-coveredness. Put $k = \nu(F)$, $a = \alpha(F)$, and $E = ex(F)$. It gives $|V(F)| = 2k + 1$ with $k \ge 1$, $E \ge 1$, and $gap(F) = 1$. Since F violates T_- and has gap one, its run bound fails. For this degree $\max(4, \lfloor d/2 \rfloor + 1) = d/2 + 1$, and integrality gives $tr(F) + E \ge d/2 + 2$.

For every maximum independent set S, the same source-process theorem gives $c_i(F-S) = c_i(F) - a$ for each integer $1 \le i < d$ and $c_0(F) = a + \nu(F-S) + ps(F-S)$. In every complete process starting at F-S, every deletion parent has a matching covering all vertices of its actual maximum degree $D \ge 3$; deletion lowers every row of index $1 \le i < D$ by one; and every terminal has zero slack. Thus both universal conditions (a) and (b) of Lemma 204 hold, with no choice of a special independent set or process.

Apply Lemma 204 at this value of d. It constructs a table Q at a nonnegative integer total $T = c_{d-2}(F) - 1$. For every nonnegative integer capacity split $A + B = d - 2$ and every nonnegative integer budget split $x + y = T$, the corresponding table entry is an actual certificate of the entire same F. The definition of $c_{d-2}(F)$ therefore gives $c_{d-2}(F) \le T = c_{d-2}(F) - 1$, a contradiction. Hence M_{12} is empty.

If a degree-twelve graph whose every proper subgraph satisfies T_- failed T_-, it would belong to the empty class M_{12}. Conversely, the displayed universal implication rules out every member of M_{12} by its definition. This proves the final equivalence. □

Remark. The universal source theorem supplies the complete construction hypotheses. Its degree-ten capacity certificate table improves the original graph's minimum cost by one, excluding every original degree-twelve source.

Depends on. Lemma 204, Lemma 309.

Theorem 311 (Exclusion of degree-eight minimal obstructions). *For a finite simple graph G, write $E(G)$, $e(G)$, $\nu(G)$, $\alpha(G)$, and $\Delta(G)$ for its edge set, edge count, matching number, independence number, and maximum degree. For nonnegative integer degree capacities A_c, B_c, define $c_{A_c,B_c}(G)$ to be the least nonnegative integer total T such that every pair of nonnegative integer matching budgets x, y with $x + y = T$ permits a partition of $E(G)$ into four parts: two with maximum degrees at most A_c, B_c and two with matching numbers at most x, y, respectively. For each nonnegative integer cost index i, define $c_i(G) = \max_{A_c+B_c=i} c_{A_c,B_c}(G)$, with the maximum over nonnegative integer capacities, and write $c_0 = c_0$. Define $ex(G) = c_0(G) - \nu(G)$ and $gap(G) = \sum_{i\ge 0} c_i(G) - e(G)$, where gap is a signed integer. The cost sum is finite. Put $tr(G) = 0$ if $c_0(G) = 0$. Otherwise let $tr(G)$ be the greatest nonnegative integer length of a consecutive run of values x in {0,...,c0(G)-1} for which $E(G)$ cannot be partitioned into two edge parts of matching numbers at most x and c0(G)-1-x. Property T_- is the implication*

$gap(G) > 0 \Rightarrow [gap(G) = 1$ *and* $tr(G) + ex(G) \le \max(4, floor(\Delta(G)/2) + 1)]$.

Let M_8 be the entire class of finite simple graphs F with maximum degree exactly eight such that F violates T_- and every proper subgraph of F satisfies T_-. Here a proper subgraph allows arbitrary vertex deletions, edge deletions, or both. Every degree capacity, matching budget, least universal total, cost index, cardinality, matching size, component count, phase count, and finite choice domain is over the nonnegative integers unless a different integer range is explicitly stated; generic gaps and parameter differences may be signed integers.

Then M_8 is empty. Thus N_8, the assertion of emptiness of this entire class, holds with no additional excess, deficit, rank, order, native-row, replay, selector, port-assignment, matching-prefix, or omission-margin hypothesis.

Proof. The certificate definitions in Lemma 309 and Lemma 204 agree with those here: they use partitions of all edges, nonnegative integer degree capacities and matching budgets, universal feasibility at each total, and then the maximum over capacity splits. At capacities zero both degree-bounded parts are empty, so their missing-allocation run is exactly $tr(G)$ as defined here, including the zero-cost convention. Write $ps(G) = e(G) - \nu(G) - \sum_{i\ge 1} c_i(G)$ for their finite slack. Then $ex(G) - ps(G) = \sum_{i\ge 0} c_i(G) - e(G)$, so their gap and property T_- also agree with those here.

Suppose F belongs to M_8 and put $d = 8$. It has maximum degree exactly d, violates T_-, and every proper subgraph satisfies T_-, with arbitrary vertex and edge deletions allowed. These are precisely the original-source hypotheses of Lemma 309, so that theorem applies to this same F.

Lemma 309 gives connectedness, d-regularity, factor-criticality and well-coveredness. Put $k = \nu(F)$, $a = \alpha(F)$, and $E = ex(F)$. It gives $|V(F)| = 2k + 1$ with $k \ge 1$, $E \ge 1$, and $gap(F) = 1$. Since F violates T_- and has gap one, its run bound fails. For this degree $\max(4, \lfloor d/2 \rfloor + 1) = d/2 + 1$, and integrality gives $tr(F) + E \ge d/2 + 2$.

For every maximum independent set S, the same source-process theorem gives $c_i(F-S) = c_i(F) - a$ for each integer $1 \le i < d$ and $c_0(F) = a + \nu(F-S) + ps(F-S)$. In every complete process starting at F-S, every

deletion parent has a matching covering all vertices of its actual maximum degree $D \geq 3$; deletion lowers every row of index $1 \leq i < D$ by one; and every terminal has zero slack. Thus both universal conditions (a) and (b) of Lemma 204 hold, with no choice of a special independent set or process.

Apply Lemma 204 at this value of d. It constructs a table Q at a nonnegative integer total $T = c_{d-2}(F) - 1$. For every nonnegative integer capacity split $A + B = d - 2$ and every nonnegative integer budget split $x + y = T$, the corresponding table entry is an actual certificate of the entire same F. The definition of $c_{d-2}(F)$ therefore gives $c_{d-2}(F) \leq T = c_{d-2}(F) - 1$, a contradiction. Hence M_8 is empty. □

Remark. The universal deletion-process theorem supplies every hypothesis of the improving certificate construction on the original graph. The resulting table has cost one below the minimum defining the same certificate quantity.

Depends on. Lemma 204, Lemma 309.

Theorem 312 (Contained minimal obstructions and conditional avoidance). *Let s, t be positive integers, and let F_1, F_2 be arbitrary nonempty star forests with full nonincreasing positive integer component edge-size lists $n_1,\ldots, n_s$ and $m_1,\ldots, m_t$. Components within each target are vertex-disjoint, copies are ordinary subgraphs, and F_1 is prescribed red and F_2 blue. Put $M = s + t - 1$ and*

$\ell_k = max\{n_i + m_j - 1 : 1 \leq i \leq s, 1 \leq j \leq t, i + j = k\}$

for every integer $2 \leq k \leq s + t$. For every integer $0 \leq q < M$ put $K = M - q$ and $L_q = \sum_{k=q+2}^{s+t} \ell_k$. Let A_q contain every ordered nonempty actual suffix pair $(F_1^{\alpha}, F_2^{\beta})$, where the nonnegative integer cuts satisfy $\alpha < s$, $\beta < t$ and $\alpha + \beta = q$, and each superscript deletes exactly that many leading whole components.

Use the finite integer universal-allocation certificate costs $c_{A,B}$, their row maxima $c_i = c_i^{\cdot}$, the finite sum $\Sigma = \sum_{i \geq 0} c_i$, matching number ν, excess $ex = c_0 - \nu$ with $c_0 = c_{0,0}$, signed $gap = \Sigma - e$, and top missing-allocation run tr from the definition fact Lemma 23. All degree capacities, matching budgets, row indices and certificate totals are nonnegative integers. For every nonnegative integer D put $h_-(D) = \max(4, \lfloor D/2 \rfloor + 1)$; a graph satisfies T_- if positive gap implies both gap one and $tr + ex \leq h_-(\Delta)$, where Δ denotes maximum degree and the empty graph has maximum degree zero. A closed d-lump is a nonempty connected d-regular odd-order finite simple graph for which deleting any vertex leaves a perfect matching. It is totally critical if every vertex deletion strictly lowers every positive-threshold row through d. A proper subgraph may delete vertices, edges, or both and need not be induced. For each integer d in {4,6,8,10,12}, let M_d be the entire class of finite simple graphs of maximum degree exactly d which violate T_- while every proper subgraph satisfies T_-; no additional condition is part of this class definition.

(I) If a finite simple graph H arrows every entire prescribed-color member of A_q and $e(H) < L_q$, then there are an actual connected component C of H and a subgraph F of C such that

$ex(C) \geq K - \nu(H) \geq 4$ *and* $gap(C) \geq 1$,

while F is a totally critical closed d-lump with d in {4,6,8,10,12}, $d \leq \Delta(H)$, $gap(F) = 1$, and

$tr(F) + ex(F) \geq h_-(d) + 1$.

Every proper subgraph of F satisfies T_-, so F belongs to M_d, and every proper subgraph J with $ex(J) \geq 4$ is LP-good, meaning $\Sigma(J) \leq e(J)$. The graph F need not be induced in H, and its excess is not asserted to be at least K-nu(H).

Write $|V(F)| = 2k + 1$, with integer $k = \nu(F) \geq 2$. If $ex(F) \geq 4$, then for every edge e of F and every integer $\theta \geq 0$ the row drop

$r_\theta(e) = c_\theta(F) - c_\theta(F - e)$

lies in {0,1}. It is zero for $\theta \geq d + 1$ and equals one for $\theta = d$. Moreover

$\nu(F - e) = k$, $ex(F - e) = ex(F) - r_0(e)$,

$R(e) := \sum_{\theta=0}^{d} r_\theta(e) \geq 2$,

and $gap(F - e) = 2 - R(e) \leq 0$.

In particular every edge strictly lowers some row below d, though that row may depend on the edge. If $d = 4$, every proper subgraph J of F satisfies positive gap implies gap one and $tr(J) + ex(J) \leq 3$, and every proper J with $ex(J) \geq 3$ is LP-good.

(II) Let H be any finite simple graph. Suppose every totally critical closed d-lump subgraph F of H with d in {4,6,8,10,12}, $gap(F) = 1$ and every proper subgraph satisfying T_- obeys $tr(F) + ex(F) \leq h_-(d)$. Then $e(H) < L_q$ implies that H has a coloring avoiding both entire prescribed targets of some actual member of A_q. Every such avoiding coloring extends over the unchanged canonical q-star pool, the disjoint stars of edge sizes ℓ_2,..., ℓ_{q+1}, and across any specified actual outer q-vertex set X in any finite simple graph G with $H = G - X$, to avoidance of both unchanged full originals, preserving every residual H-edge color. The pool is empty when $q = 0$.

No natural-degree, head, potential, exact-boundary or equality-profile restriction on H is imposed. All graph orders, edge counts, degrees, matching cardinalities, list entries, cuts and row indices are integers in their displayed nonnegative or positive domains; gaps may be signed integers. This theorem asserts contained obstructions and conditional avoidance; it does not assert universal emptiness of the five classes or degree preservation in the containment.

Proof. Fix the original lists, q and H in (I). Lemma 307 (II), with $K_q = K$, states that $\nu(H) \geq K - 3$ would imply $e(H) \geq L_q$ for this same common A_q host. Since $e(H) < L_q$, integer contraposition gives $K - \nu(H) \geq 4$ without changing H or q. Lemma 166 applies to the same lists and same common arrowing host. Choose any connected component C of H of largest matching number; the host has an edge by that fact, so its finite component family is nonempty. Its simultaneous component conclusions give

$ex(C) \geq ex(H) \geq K - \nu(H) \geq 4$ and $gap(C) \geq 1$. (1)

The graph C has an induced subgraph violating T_-. Otherwise every induced subgraph of C, including C itself, would satisfy T_-, and Lemma 296 would make C LP-good because $ex(C) \geq 4$. This would contradict $gap(C) \geq 1$ in (1).

Consider all subgraphs of this actual C which violate T_-. This finite family is nonempty by the preceding paragraph. Choose F first with the fewest edges and then, among those, with the fewest vertices. Every proper subgraph J of F satisfies T_-: if an edge is deleted, $e(J) < e(F)$; if no edge is deleted, properness deletes at least one vertex and gives $e(J) = e(F)$, $|V(J)| < |V(F)|$. In either case a violating J would contradict the specified finite minimum. This includes spanning edge deletions and deletion of isolated vertices. The candidates were all subgraphs of C, without a fixed degree, regularity or inducedness condition.

Lemma 276 now applies to this same F. Full proper-subgraph minimality implies its proper-induced hypothesis. It gives connectedness, even regular degree d in {4,6,8,10,12}, odd order $2k + 1$ with $k = \nu(F) \geq 2$, a perfect matching after every vertex deletion, $gap(F) = 1$ and

$tr(F) + ex(F) \geq h_-(d) + 1$.

For every vertex v of F and every integer $1 \leq i \leq d$ it also gives $c_i(F - v) = c_i(F) - 1$. Thus F is totally critical in the stated sense and is a closed d-lump. Its maximum degree is $d \leq \Delta(H)$, since F is a subgraph of the component C of H. Its failure and full proper-subgraph minimality put it in precisely M_d.

The proper-subgraph consequence of Lemma 296 applies to F and gives LP-goodness of every proper J with $ex(J) \geq 4$. This claim concerns the actual proper subgraphs of F, including graphs produced by deleting only edges. It does not transfer the lower bound (1) to $ex(F)$.

Assume now $ex(F) \geq 4$. All hypotheses of Lemma 97 hold on F with $b0 = 4$: it is regular and factor-critical, has gap one, and every proper subgraph of excess at least four is LP-good. That theorem gives $c_d(F) = 1$ and, for every edge e and every integer $\theta \geq 0$, exactly the asserted row-drop bounds, matching preservation, excess identity, total-drop bound and gap identity. It also gives the strictly lowered row below d for every edge. All costs and graph arguments are the same as in the statement.

If $d = 4$, apply Lemma 265 to the connected four-regular F. It gives the claimed bound $tr(J) + ex(J) \leq 3$ at positive gap on every proper subgraph J, including every spanning edge deletion, and LP-goodness of each such J with excess at least three. This proves all of (I).

For (II), fix H with the stated universal condition on its totally critical closed subgraphs and suppose $e(H) < L_q$. If H arrowed every member of A_q, part (I) would produce F contained in H with all the hypotheses of that universal condition but with $tr(F) + ex(F) \geq h_-(d) + 1$. This contradicts its asserted bound by $h_-(d)$. Therefore H does not arrow every member of A_q. By the definition of arrowing, there is an actual member of A_q and a red-blue coloring of H avoiding both of its entire prescribed targets.

Apply Lemma 41 (I) to any such coloring. Its lists, suffix family and canonical q-star pool coincide with those in this statement. It extends the coloring to the union with that pool, avoiding both unchanged full originals and preserving all H-edge colors. For any specified actual q-vertex set X with $H = G - X$, part (II) of that same fact extends the given H-coloring over precisely X to G with the same two original targets and the same retained H-edge colors. These extension assertions apply to every avoiding suffix coloring, not just a separately chosen one. No degree or potential hypothesis is required by either extension. The $q = 0$ case uses an empty pool and no added vertices. This proves (II). □

Remark. A low-deficit payment theorem first forces matching deficit at least four on the original host. Component localization and the hereditary run-plus-excess bound then supply a failing component. A finite lexicographic choice over all its subgraphs gives full proper-subgraph minimality, after which the source criterion and edge-deletion refinements apply on that same witness.

Depends on. Lemma 23, Lemma 41, Lemma 97, Lemma 166, Lemma 265, Lemma 276, Lemma 296, Lemma 307.

Theorem 313 (Exclusion of degree-ten minimal obstructions). *For a finite simple graph G write $V(G)$, $E(G)$, $e(G)$, $\nu(G)$, $\alpha(G)$, $\Delta(G)$ for its vertex set, edge set, edge count, matching number, independence number, and maximum degree. For nonnegative integer degree capacities A_c, B_c, define $c_{A_c,B_c}(G)$ as the least nonnegative integer total T such that every nonnegative integer pair x_c, y_c with $x_c + y_c = T$ admits a partition of $E(G)$ into four parts with maximum degrees at most A_c, B_c and matching numbers at most x_c, y_c, respectively. For every nonnegative integer i let $c_i(G) = \max_{A_c+B_c=i} c_{A_c,B_c}(G)$, where capacities are nonnegative integers. Put $c_0 = c_0$, $ex = c_0 - \nu$, $ps = e - \nu - \sum_{i\geq 1} c_i$, and signed integer $gap = \sum_{i\geq 0} c_i - e$. These sums are finite. Set $tr(G) = 0$ if $c_0(G) = 0$; otherwise $tr(G)$ is the greatest nonnegative integer length of a consecutive run of first budgets x in {0,...,c0(G)-1} for which no partition of $E(G)$ into two parts has matching numbers at most x, c0(G)-1-x. Property T_- is the implication $gap(G) > 0 \Rightarrow [gap(G) = 1$ and $tr(G) + ex(G) \leq \max(4, floor(\Delta(G)/2) + 1)]$.*

For each d in {4,6,8,10,12}, let M_d be the entire class of finite simple graphs F with maximum degree exactly d which violate T_- while every proper subgraph satisfies T_-. Proper subgraphs allow arbitrary vertex deletions, edge deletions, or both. Let N_d mean M_d is empty. Every capacity, budget, least universal total, cost index, degree, cardinality, matching size, and finite choice domain is a nonnegative integer. Generic gaps and displayed parameter differences may be signed integers.

Then N_{10} holds: the entire class M_{10} is empty. In particular degree ten is excluded in both equality and strict cost sectors, at every order, excess, rank, deficit, native row, process, selector, phase profile, matching-prefix, and omission count. No additional source hypothesis is imposed.

Proof. The certificate definitions in Lemma 309 and Lemma 204 agree with those here: they use partitions of all edges, nonnegative integer degree capacities and matching budgets, universal feasibility at each total, and then the maximum over capacity splits. At capacities zero both degree-bounded parts are empty, so their missing-allocation run is exactly $tr(G)$ as defined here, including the zero-cost convention. Write $ps(G) = e(G) - \nu(G) - \sum_{i\geq 1} c_i(G)$ for their finite slack. Then $ex(G) - ps(G) = \sum_{i\geq 0} c_i(G) - e(G)$, so their gap and property T_- also agree with those here.

Suppose F belongs to M_{10} and put $d = 10$. It has maximum degree exactly d, violates T_-, and every proper subgraph satisfies T_-, with arbitrary vertex and edge deletions allowed. These are precisely the original-source hypotheses of Lemma 309, so that theorem applies to this same F.

Lemma 309 gives connectedness, d-regularity, factor-criticality and well-coveredness. Put $k = \nu(F)$, $a = \alpha(F)$, and $E = ex(F)$. It gives $|V(F)| = 2k + 1$ with $k \geq 1$, $E \geq 1$, and $gap(F) = 1$. Since F violates T_- and has gap one, its run bound fails. For this degree $\max(4, \lfloor d/2 \rfloor + 1) = d/2 + 1$, and integrality gives $tr(F) + E \geq d/2 + 2$.

For every maximum independent set S, the same source-process theorem gives $c_i(F - S) = c_i(F) - a$ for each integer $1 \leq i < d$ and $c_0(F) = a + \nu(F - S) + ps(F - S)$. In every complete process starting at F-S, every deletion parent has a matching covering all vertices of its actual maximum degree $D \geq 3$; deletion lowers every row of index $1 \leq i < D$ by one; and every terminal has zero slack. Thus both universal conditions (a) and (b) of Lemma 204 hold, with no choice of a special independent set or process.

Apply Lemma 204 at this value of d. It constructs a table Q at a nonnegative integer total $T = c_{d-2}(F) - 1$. For every nonnegative integer capacity split $A + B = d - 2$ and every nonnegative integer budget split

$x + y = T$, the corresponding table entry is an actual certificate of the entire same F. The definition of $c_{d-2}(F)$ therefore gives $c_{d-2}(F) \le T = c_{d-2}(F) - 1$, a contradiction. Hence M_{10} is empty. □

Remark. Universal source-process conclusions permit the same integral improving-certificate construction at degree ten. The resulting universal total is one less than the cost it defines on the original graph.

Depends on. Lemma 204, Lemma 309.

Theorem 314 (Lower bound for two star forests). *Let s, t be positive integers, and let $n_1 \ge \cdots \ge n_s \ge 1$ and $m_1 \ge \cdots \ge m_t \ge 1$ be nonincreasing lists of positive integers. For a positive integer h, let $K_{1,h}$ be the finite simple graph with one center, h leaves, and exactly the h edges joining the center to its leaves. Let F_1 and F_2 be the vertex-disjoint unions of K_{1,n_i}, $1 \le i \le s$, and K_{1,m_j}, $1 \le j \le t$, respectively. For each integer $2 \le k \le s + t$ put $\ell_k = max\{n_i + m_j - 1 : 1 \le i \le s, 1 \le j \le t, i + j = k\}$, and put $B = \sum_{k=2}^{s+t} \ell_k$.*
Every finite simple graph G for which every red/blue coloring of $E(G)$ contains a red ordinary subgraph isomorphic to F_1 or a blue ordinary subgraph isomorphic to F_2 has $e(G) \ge B$, where $e(G) = |E(G)|$.
The copies need not be induced.

Proof. Fix the two lists and such a graph G. Suppose $e(G) < B$. Set $M = s + t - 1$ and $q = 0$. The positivity of s and t gives $M \ge 1$, hence $0 \le q < M$.
With the notation of the contained-obstruction theorem Theorem 312, $K = M - q = M$ and $L_0 = \sum_{k=2}^{s+t} \ell_k = B$. The only pair of nonnegative integers α, β with $\alpha + \beta = 0$ is $\alpha = \beta = 0$. Both are legal leading deletions because s, $t \ge 1$. Thus A_0 consists exactly of (F_1, F_2), and G arrows every member of A_0.
Apply the full contained-obstruction theorem Theorem 312 to these same profiles, $q = 0$, and the host G. It gives an ordinary subgraph F of an actual component of G and an integer d in {4,6,8,10,12}. The graph F is d-regular, has $gap(F) = 1$, and satisfies $tr(F) + ex(F) \ge h_-(d) + 1$, where $h_-(d) = \max(4, \lfloor d/2 \rfloor + 1)$.
Every proper subgraph of F satisfies T_-.
Use the certificate quantities and the property T_- with exactly the definitions in the contained-obstruction and whole-degree exclusion theorems. Since F is d-regular and nonempty, $\Delta(F) = d$. Since $gap(F) = 1 > 0$ and $tr(F) + ex(F) > h_-(d)$, F violates T_-. Its proper-subgraph property includes arbitrary vertex deletions, edge deletions, and both; therefore F belongs to the entire class M_d in the corresponding whole-degree exclusion theorem.

For $d = 4$, theorem Theorem 288 asserts M_4 is empty; for $d = 6$, theorem Theorem 308 asserts M_6 is empty; for $d = 8$, theorem Theorem 311 asserts M_8 is empty; for $d = 10$, theorem Theorem 313 asserts M_{10} is empty; and for $d = 12$, theorem Theorem 310 asserts M_{12} is empty. These five cases exhaust the integer d supplied by containment. Each contradicts membership of the same F in M_d. Hence $e(G) \geq B$. □

Depends on. Theorem 288, Theorem 308, Theorem 310, Theorem 311, Theorem 312, Theorem 313.

Theorem 315 (Size Ramsey number of unions of stars). *Let s, t be positive integers, and let $n_1 \geq \cdots \geq n_s \geq 1$ and $m_1 \geq \cdots \geq m_t \geq 1$ be nonincreasing lists of positive integers. For a positive integer h, let $K_{1,h}$ be the finite simple star with one center, h leaves, and exactly the h edges joining its center to its leaves. Let F_1 be the vertex-disjoint union of K_{1,n_i} over $1 \leq i \leq s$, and let F_2 be the vertex-disjoint union of K_{1,m_j} over $1 \leq j \leq t$.*

For finite simple target graphs G_1, G_2, define $\hat{R}(G_1, G_2)$ to be the least nonnegative integer number of edges in a finite simple graph H such that every red/blue coloring of $E(H)$ contains either a red subgraph isomorphic to G_1 or a blue subgraph isomorphic to G_2. Copies are ordinary subgraphs, not required to be induced. Write $\hat{R}(G) = \hat{R}(G, G)$.

For every integer k with $2 \leq k \leq s + t$, set $\ell_k = max\{n_i + m_j - 1 : 1 \leq i \leq s, 1 \leq j \leq t, i + j = k\}$.

Then, for all these lists, $\hat{R}(F_1, F_2) = \sum_{k=2}^{s+t} \ell_k$.

Proof. Fix arbitrary lists in the statement and put $B = \sum_{k=2}^{s+t} \ell_k$. The star-profile theorem Lemma 29 uses these same full positive lists and ordinary prescribed-color copies. Its budget L equals B, since both are the sum of these same diagonal maxima. By the upper-bound theorem Theorem 59, the vertex-disjoint union of K_{1,ℓ_k}, over every integer $2 \leq k \leq s + t$, arrows (F_1, F_2). It has exactly B edges. Thus the set of edge counts of finite simple arrowing hosts is a nonempty set of nonnegative integers and has a least element, with $\hat{R}(F_1, F_2) \leq B$. The lower-bound theorem Theorem 314 applies to every finite simple arrowing host for these same full targets and gives at least B edges. Consequently $\hat{R}(F_1, F_2) \geq B$. Combining the inequalities proves the

displayed equality for the arbitrary lists, including every endpoint in the statement. □

Depends on. Lemma 29, Theorem 59, Theorem 314.